PML-AI
Body of Knowledge

The reference for the PCI Project Management Leader – AI
Leadership · delivery systems · governance · the governed use of AI

FIRST EDITION — DRAFT FOR EDITORIAL AND TECHNICAL REVIEW

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PROJECT CONTROLS INSTITUTE GLOBAL

Finance intelligently. Control predictively. Deliver successfully.

AI proposes; the professional verifies, decides and remains accountable.

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How to use this book

This is a reference, not a course. It is written to be read in order once and consulted out of order thereafter, and the apparatus below exists to make the second use as easy as the first.

How anything is addressed. Content is numbered Domain.KnowledgeArea.Topic — so 7.3.2 is the second topic of the third Knowledge Area of Domain 7, and a cross-reference of the form KA 10.2 is precise. Page numbers are never used for cross-references, because this volume is regenerated from source and its pagination is not stable across editions. Every reference in the text is therefore still valid in the next one.

The worked examples are the spine of the book. Each follows the same five steps — Setup, Formula, Substitution, Result, Interpretation — and the Interpretation is deliberately the longest and the most valuable. It is where the number is turned into a decision: what breaks the result, what the breakeven is, which assumption the conclusion is most sensitive to, and what a reviewer should check. If you read only one part of a worked example, read that one. The four steps before it exist so that you can reproduce the number and disagree with it.

Every number can be checked, and should be. Every figure printed as a result anywhere in this volume — in worked examples, in the text, in multiple-choice options, in exercise solutions, in case studies — is recomputed independently with decimal arithmetic by a verification suite that must pass in full before any chapter is accepted. Appendix C records how many checks stand behind each domain. If you find a number that does not reproduce from its stated method and inputs, that is a defect in this book and worth reporting.

The AI in this KA sections are not an appendix to the subject. Each states three things: where AI genuinely earns its place in that Knowledge Area, where it must not go, and how to verify its output concretely. They exist because the alternative — a single chapter about AI at the back — teaches that the question is separable from the work, and it is not. One principle governs all of them: AI proposes; the professional verifies, decides and remains accountable.

What the sections at the end of each domain are for. Advanced topics extend the domain past the level a candidate is assessed on, and each closes with a list of invariants a reviewer can test. Industry variations say what changes by sector, and are written so that a reader in one sector can see what a reader in another is dealing with. Case studies carry real arithmetic and are the place where several Knowledge Areas have to work together. Executive perspective is what a director cannot delegate. Calculation exercises each carry a full solution and a Common error note, and the common error is frequently the more useful half. The Practitioner's toolkit items are meant to be adopted and then left stable; adapt the headings to your organisation and stop changing them. Exam preparation lists what is assessed, the calculations to be able to do under time pressure, and the traps — each cross-referenced to where it is taught.

Two conventions worth knowing before you meet them. Where a result is established in one domain, the others cite it rather than re-derive it; if two domains appear to derive the same thing independently, that is a defect. And where one word carries two genuinely different concepts, the later use states the collision explicitly rather than leaving you to infer it. Appendix B lists both.

If you are studying for the examination, work the calculation exercises before reading their solutions, and treat the Common error notes as the syllabus they effectively are. If you are using this at work, start from the Practitioner's toolkit of the domain you need and follow its cross-references back into the treatment.

Contents

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PART ONE

Leading projects

Domains 1–4 — the profession, strategy and selection, governance and decision rights, and delivery architecture: what a project leader is for and answerable for.

01

DOMAIN 1

The Project Leadership Profession

Group: Leading projects (Domain 1 of 4 in Part One). **Target:** ~68 pages. **Binds to:** the PCI Book Pattern Specification and the shared registries (`docs/books/registries/`). This domain fixes the book's core vocabulary — accountability, outputs/outcomes/benefits, systems thinking, the responsible-AI principle — that every later domain assumes. British English; USD (+SAR where useful, indicative `USD 1 ≈ SAR 3.75`).

Why this domain exists

Every technique in this book — the networks of Domain 6, the earned value of Domain 7, the risk quantification of Domain 8 — is instrumentation. This domain is about the person holding the instruments: what a project leader is for, what they are answerable for, and how they think. It establishes the delivery landscape and how leading temporary work differs from running a steady-state function (KA 1.1); defines accountability precisely, because the word is used loosely and the distinction decides who answers when things go wrong (KA 1.2); builds the two habits of mind that separate a leader from a coordinator — systems thinking, and the discipline of outputs versus outcomes versus benefits (KA 1.3); and grounds professional ethics and the governed use of AI (KA 1.4). A reader who finishes only this domain should already reason like the profession: know what success actually means, know who owns it, and stay answerable for the judgment even when a machine produced the analysis.

Learning objectives. After this domain a candidate can: distinguish projects, programmes and portfolios and explain what each optimises; **price the delivery verdict and the programme verdict on the same programme and state why the two are not directly commensurable**; contrast project leadership with operational leadership; explain why a temporary organisation makes trust and clarity urgent; distinguish accountability from responsibility and apply the distinction to a delegation; state the leader's obligations to sponsor, team, users and the public, and where they conflict; **compute the value of escalating a problem early and the number of weeks of earliness that funds the whole remedy**; explain the professional standard of care and show what exercising it looked like on a specific priced decision; analyse a project as a system with feedback and delay; **compute the transient trough that follows reinforcing a late team, derive its crossover horizon and state the condition under which no trough exists**; distinguish outputs, outcomes and benefits and demonstrate why output-based claims overstate value; **convert a benefit stream into a simple payback, show that payback scales as the reciprocal of adoption, and invert a payback rule into the adoption commitment it silently makes**; compute the value of time in benefit terms and the adoption level below which paying for speed destroys value; describe the ethical obligations of the role; apply the PCI responsible-AI principle to a realistic delivery decision; and **derive the breakeven error probability that makes a verification proportionate, and name the class of artefact for which that arithmetic must not be the deciding test.**

The master case. One programme runs through this domain and returns in Domains 2, 5, 15 and 16: **Meridian Care Records**, a fictional public-health programme rolling a shared clinical-records system out to **40 clinics** at an approved cost of **USD 2,400,000**. Its full-potential benefit is **USD 979,200** a year, its realistic benefit **USD 685,440** a year at 70 % adoption, and its cost of delay **USD 14,280 a week —** three figures derived in KA 1.3 and used, unchanged, by a dozen later domains. This domain's job is to make them arithmetic rather than rhetorical, and to be explicit about what each rests on.

KNOWLEDGE AREA 1.1

Projects, programmes and portfolios

Topics: 1.1.1 the delivery landscape · 1.1.2 project versus operational leadership · 1.1.3 the temporary organisation.

1.1.1 The delivery landscape

Definitions. A **project** is a temporary endeavour undertaken to produce a defined result. A **programme** is a group of related projects and change activities managed together to achieve **outcomes** no single project could deliver alone. A **portfolio** is the set of projects and programmes an organisation chooses to fund, selected and balanced against strategy and capacity. The three are not sizes; they are different objects of management:

Level	Optimises	Primary success test	This book
Project	Delivery of a defined output to time, cost, quality	Was the thing delivered, fit for purpose?	Domains 4–13
Programme	Coherent outcomes and benefits across components	Did the change actually land?	Domain 15
Portfolio	Value and balance of the whole investment set	Are we doing the right things at all?	Domains 2, 15

The practical consequence is that **the same event can be a project success and a programme failure**: Meridian delivering all 40 clinic installations on time is a project triumph and worth nothing if clinicians do not use the system. The example below prices both verdicts, because a leader who can only assert the distinction will lose the argument to whoever has a number.

WORKED EXAMPLE

Worked example 1.1.1 — the two verdicts on one programme, priced.

1. **Setup.** Meridian's completion report states the delivery verdict: **40 of 40** clinics installed, on time, **3 % under** the approved cost of **USD 2,400,000**. The programme verdict is taken separately, against benefit: the honest case assumed **70 %** adoption and **USD 685,440** a year (derived in 1.3.2); measured adoption is **40 %**, worth **USD 391,680** a year. Both statements are true and they are reported to the same board in the same month.
2. **Formula.** Delivery verdict = approved cost × underspend rate (a one-off capital variance). Programme verdict = potential benefit × (planned adoption − actual adoption), an **annual flow**. The comparison ratio = annual benefit shortfall ÷ capital underspend.
3. **Substitution.** Underspend `2,400,000 × 0.03`; shortfall `685,440 − 391,680`, equivalently `979,200 × (0.70 − 0.40)`; ratio `293,760 / 72,000`.
4. **Result.** Delivery verdict **+USD 72,000**, once. Programme verdict **(USD 293,760)** a year. The shortfall is **4.08 times** the underspend **in the first year alone**. Undiscounted over Domain 2's eight-year appraisal life the shortfall totals **USD 2,350,080**, or **97.92 %** of what the whole programme cost to build; discounted at Domain 2's 7 % (annuity factor **5.971299**) it is **USD 1,754,128.79**, or **73.09 %** of the approved cost.
5. **Interpretation.** Four readings, and the fourth is the one that keeps a leader honest.

The verdicts are not commensurable, and saying so is part of reporting them. One is a stock (a capital variance, banked once); the other is a flow (annual benefit, recurring for as long as the service runs). Dividing one by the other is legitimate only as a scale comparison, which is what the 4.08 is for; converting the flow into a single figure requires a horizon and a discount rate, and those belong to the business case (Domain 2, KA 2.2.2), not to the delivery report. A leader who quotes the undiscounted 2,350,080 without saying it is undiscounted has overstated by **USD 595,951.21** against the discounted figure.

The delivery lever is small because it is bounded and the benefit lever is not. To offset the first year of shortfall through cost alone, the programme would have had to underspend by **12.24 %** — four times what it achieved, on a rollout whose costs are mostly contracted. The asymmetry is structural, not particular to Meridian: on benefit-generating work the accessible range of cost variance is a few per cent of a one-off, while the accessible range of adoption is tens of per cent of a perpetuity. That is the quantitative reason KA 1.3 spends its attention on the outcome link.

The weekly form is the one that changes behaviour. The 293,760 a year is **USD 6,120 a week** — which is exactly **12 clinics × 6 hours × USD 85**, the cost of delay of 1.3.3 applied to the twelve clinics that are not using the system. A board that hears "we are 293,760 light on the annual case" adjourns; a board that hears "the gap is costing 6,120 a week and has been for nine months" acts. Same number, different unit, different decision.

The delivery verdict was provisional and was later reversed. Domain 16's closing account records a final capital outturn of **USD 2,514,000**, against the **USD 2,328,000** implied by the 3 % underspend at installation — a swing of **USD 186,000**, most of it the recovery this domain's case study describes. The professional caution: an underspend reported before the outcome is measured is an interim figure, and a leader who banks it rhetorically ("we came in under") has made a claim the programme may still take back. A reviewer should ask, of any celebrated underspend, what remains to be spent to make the output work.

How the discipline arrived at that distinction. The vocabulary above is recent, and knowing why it exists prevents a candidate from treating it as bureaucratic layering. Four shifts, each of which added an obligation that had not previously been anyone's:

Shift	What was codified	What it added to the leader's duty
Technique	Network scheduling, cost control and configuration control, developed on large capital, defence and aerospace programmes and generalised from there	Produce a plan that can be checked by someone else
Process and profession	Bodies of knowledge, competence frameworks and certification; a common lifecycle vocabulary	Practise to a stated standard, not to personal habit
Governance and outcomes	Sponsorship, decision rights, gates, benefits management; internationally, the ISO 21500 family — context and concepts (ISO 21500), project management guidance (ISO 21502), programme (ISO 21503), portfolio (ISO 21504) and governance (ISO 21505), alongside ISO 31000 for risk and ISO 10006 for quality in projects	Answer for whether the change landed, not only for whether the thing was built
Adaptive and digital delivery	Iterative and flow-based methods generalised out of software; data-centred delivery environments; and management-system standards for artificial intelligence itself, including ISO/IEC 42001 for AI management systems and ISO/IEC 23894 for AI risk	Choose a delivery approach and defend the choice; govern the tools that now draft the work

Those standards are named here, not reproduced: each is described in this book's own words, and a reader who needs the requirements themselves must obtain the standard. The relevant point for Domain 1 is directional. **The discipline has moved its test of success outward** — from "was the plan followed?" to "was the thing delivered?" to "did the change land, and can you show how you knew?" Every one of those moves increased what the leader must be able to evidence, and none of them removed an earlier obligation. That is why a modern project leader is answerable for a benefits chain (1.3.2) that a scheduler of the technique era was never asked about, and why the fourth shift puts a governed AI tool inside the same accountability frame rather than outside it (KA 1.4).

1.1.2 Project versus operational leadership

Operations optimise a repeating process; projects deliver a novel result once. The differences that change a leader's behaviour:

	Operational leadership	Project leadership
Work	Repeating, refined over time	Novel, done once
Team	Stable, known, yours	Assembled, borrowed, temporary
Authority	Usually line authority	Often influence without authority
Time	Continuous	Bounded, with an end that must be planned for
Learning curve	Amortised over many cycles	Paid once, mid-delivery
Failure mode	Drift, inefficiency	Late discovery, then irreversibility

The last row is the one that shapes everything downstream. Operational problems are usually recoverable at similar cost whenever they are found; project problems get monotonically more expensive as commitments harden. That is why this book spends so much on early definition (Domain

5), honest schedules (Domain 6) and early warning (Domains 7–8) — **in temporary work, lead time is the cheapest resource a leader has, and it is spent whether used or not.**

That claim is quantified elsewhere in this volume rather than asserted twice, and a candidate should know where: Domain 5 (KA 5.2.1) prices the rising cost of correcting a requirement defect by the stage at which it is found; Domain 9 (KA 9.2.2) computes the containment economics that make a later layer of checking an order of magnitude dearer than an earlier one; and Domain 3 (KA 3.3.1) values a gate by netting the cost of the delay it imposes against the failure cost it prevents. Domain 1's contribution is the leadership consequence, which is behavioural: because correction cost rises with commitment, **the profession's most valuable act is moving information earlier**, and KA 1.2 prices one instance of exactly that.

1.1.3 The temporary organisation

A project team is a temporary organisation: people arrive with divided loyalties, no shared history, and no accumulated trust to draw on. Three consequences a leader must design for. **Trust has no time to accrue naturally**, so it must be built deliberately and early — the team charter (Domain 12) exists because the informal norms that operations grow over years must here be stated in week one. **Clarity substitutes for familiarity**: roles, decision rights and done-definitions have to be explicit precisely because nobody can infer them (Domains 3–5). **Endings must be engineered**: a temporary organisation dissolves, and the knowledge, the relationships and the operational readiness either transfer deliberately or evaporate (Domain 16).

AI in this domain — first statement of the principle

AI now sits inside the delivery environment itself: drafting plans, summarising status, generating risk lists, answering questions the team used to bring to a senior colleague. The suite's governing principle, stated once here and applied in every later domain:

> **AI proposes; the professional verifies, decides and remains accountable.**

Three things it means concretely for a leader, before any technique in this book is applied. **Verification is a duty, not a preference** — machine output entering a plan, report or decision is checked by a named human against evidence. **Accountability is not transferable** — "the model said so" is never a defence, because accountability cannot be delegated to software (KA 1.2 makes the reason precise). **Disclosure is part of honesty** — material AI use in a deliverable is stated, so colleagues can weigh it. Domain 14 builds the full governance architecture; this domain establishes that the obligation attaches to the person, not the tool.

KEY TERMS — KA 1.1

Term	Meaning
Project	Temporary endeavour producing a defined result.
Programme	Related projects managed together for outcomes no one project delivers.
Portfolio	The funded, balanced set of projects and programmes.
Temporary organisation	A team with no accumulated trust or shared norms; both must be designed.
Irreversibility	The rising cost of correction as commitments harden — projects' defining failure mode.
Delivery verdict	The project-level judgment: was the output delivered to time, cost and quality? A one-off variance.
Programme verdict	The programme-level judgment: did the outcome and its benefit materialise? An annual flow.
Commensurability	Whether two figures may be compared directly; a capital variance and an annual benefit may not, without a horizon and a rate.
Responsible-AI principle	AI proposes; the professional verifies, decides and remains accountable.

SAMPLE MCQS — KA 1.1

MCQ 1.1-A `[1.1.1 · Analysis]` A programme's four projects each deliver on time and to budget, but the intended service change does not materialise. The most accurate statement is: - A. the programme succeeded, since all its projects succeeded - B. the projects succeeded on their own test while the programme failed on its own test — outcomes are not the sum of outputs - C. the projects must therefore have failed - D. programme success cannot be assessed until every project closes

Rationale: Projects are judged on delivered outputs, programmes on outcomes; the two tests are different, which is exactly why the distinction exists. C rewrites history to preserve a single verdict, and D denies the measurement programmes exist to make.

MCQ 1.1-B `[1.1.2 · Analysis]` Why does project leadership place more weight on early definition than operational leadership does? - A. project teams are less capable - B. project problems grow monotonically more expensive as commitments harden, so lead time is the cheapest resource available - C. operations have no need for planning - D. projects have larger budgets

Rationale: Irreversibility is the structural difference (1.1.2). A and D are unfounded; C misstates operational practice, which plans continuously — just with the ability to correct cheaply next cycle.

MCQ 1.1-C `[1.1.3 · Recall]` A team charter is written in week one primarily because: - A. governance requires the document - B. a temporary organisation has no accumulated norms, so what operations grow informally must here be made explicit - C. it replaces the need for a project plan - D. it fixes the team's membership for the duration

Rationale: Explicit clarity substitutes for familiarity the team has had no time to build. A is a by-product, C confuses artefacts, D is false — membership usually changes.

MCQ 1.1-D `[1.1.1 · Evaluation]` A programme delivers 3 % under an approved cost of 2,400,000 while adoption reaches 40 % against a planned 70 % on a full-potential benefit of 979,200 a year. The soundest single sentence for the board is: - A. the programme is 72,000 ahead, so the net position is positive - B. the delivery result is +72,000 once, the benefit result is (293,760) a year — 4.08 times the underspend in year one alone — and the two are a stock and a flow, so no net figure should be quoted without a horizon and a rate - C. the programme has lost 221,760 net in its first year - D. no comparison is possible, since the two figures measure different things

Rationale: Both verdicts must be reported, scaled against each other, and flagged as non-commensurable (1.1.1). A nets a one-off against a perpetuity and keeps only the flattering half; C performs the same illegitimate subtraction (`293,760 − 72,000`) and presents the remainder as a result; D uses non-commensurability as an excuse not to compare at all, when the ratio is precisely what makes the scale visible.

MCQ 1.1-E `[1.1.1 · Comprehension]` Which best describes what the governance-and-outcomes shift added to the project leader's duty? - A. a requirement to follow a certified process - B. an obligation to answer for whether the intended change landed, in addition to whether the output was built - C. the replacement of scheduling technique by benefits management - D. a transfer of delivery accountability to the sponsor

Rationale: Each shift added an obligation without removing an earlier one (1.1.1). A describes the preceding process-and-profession shift; C is the common misreading — technique was retained, not displaced; D inverts KA 1.2, where accountability for delivery stays with the leader.

SELF-CHECK — KA 1.1

1. Can a project succeed while its programme fails? — Yes: outputs delivered, outcomes not realised. The tests differ (1.1.1).
2. Name the defining failure mode of temporary work. — Late discovery followed by irreversibility; correction cost rises as commitments harden.
3. State the responsible-AI principle. — AI proposes; the professional verifies, decides and remains accountable.
4. Why may a 72,000 underspend not simply be netted against a 293,760 annual benefit shortfall? — One is a stock and the other a flow; netting them needs a horizon and a discount rate, which belong to the business case (1.1.1, Domain 2 KA 2.2.2).
5. Name the four shifts in the discipline and the obligation each added. — Technique (a checkable plan), process and profession (practice to a stated standard), governance and outcomes (answer for the change landing), adaptive and digital delivery (defend the approach; govern the tools).

KNOWLEDGE AREA 1.2

The project leader's accountability

Topics: 1.2.1 accountability and responsibility · 1.2.2 the obligation set · 1.2.3 the professional standard of care.

1.2.1 Accountability and responsibility

Definitions. Responsibility is the obligation to do the work — it can be shared and delegated. **Accountability** is the obligation to answer for the outcome — it sits with exactly one named person and **cannot be delegated**, only reassigned by whoever holds the authority to do so. A leader may hand a task to a specialist, an outsourcer or an AI tool; the answer for the result stays where it was.

Two working rules follow.

Delegation transfers responsibility, never accountability. The leader who delegates keeps the duty to specify, to verify at a proportionate depth, and to notice when the work is failing. "I gave it to the vendor" is a description of responsibility, not a defence of accountability.

One accountable person per outcome. Where two people are accountable for the same thing, nobody is: each defers, and the gap is discovered late. This is why Domain 3 builds explicit decision rights and why RACI-style matrices insist on a single "A" per row.

The mechanism deserves stating precisely, because it is not merely untidy — it has a cost and a signature. Two named holders of one outcome each face the same private calculation: acting costs effort and political capital now, while the consequence of not acting is shared, deferred and attributable to the other. Each therefore waits for evidence that the other will not act, and the evidence takes the form of the outcome deteriorating. **Dual accountability does not slow the response by a little; it converts the trigger from a measure into an incident.** Its signature in a report pack is an outcome measure that is reported by both parties, owned by neither, and never accompanied by a proposed action. On Meridian, the two holders were the delivery leader (installations) and the clinical directorate (practice change), the un-owned outcome was adoption, and the incident that finally triggered action was an external audit — twenty-six weeks after the measure first showed it. Worked example 1.2.2 prices those twenty-six weeks.

Accountability is not the same as legal liability. This domain uses accountability in the professional sense defined above. Legal liability — who must compensate whom, on what basis, and subject to what limitation or exclusion — is set by the contract and by the law of the jurisdiction, and it can sit with a party who is not professionally accountable, or be capped for a party who is. The two are routinely conflated, in both directions: a contractual cap is offered as a reason not to answer, or a professional duty is treated as if it created a payment obligation. Keep them separate in your own reasoning, and where a decision turns on which is engaged — an indemnity, a professional-indemnity notification, a duty to warn a third party — take the position from counsel for the applicable jurisdiction rather than from this book (Domain 10, KA 10.4 for the contractual machinery).

The AI corollary is now immediate rather than novel. A tool has no standing to be accountable — it cannot be asked to answer, cannot be sanctioned, cannot hold a duty. So every use of AI in delivery leaves accountability exactly where it was: with the professional who used the output. That is not a policy choice PCI made; it follows from what accountability is.

1.2.2 The obligation set

A project leader owes duties in four directions, and the skill is holding them together honestly rather than pretending they never conflict:

To	Owes	Typical conflict
Sponsor / organisation	Honest forecasts, stewardship of funds, no surprises	Pressure to report reassurance rather than status
Team	Clarity, safety, achievable asks, fair credit	Absorbing schedule pressure by quietly overloading people
Users / customers	A result that is genuinely fit for purpose	Delivering on time by descoping what users needed most
Public / third parties	Safety, legality, honest claims, environmental care	Cost pressure meeting a safety or compliance margin

The escalation duty. When these conflict beyond the leader's authority to resolve, the professional act is to escalate with options — not to absorb the conflict silently and hope. An unescalated conflict becomes a late surprise, and late surprises are the failure mode of §1.1.2.

The honesty asymmetry. Bad news travels badly: it costs the messenger and helps the project. A leader's most consequential cultural act is making early bad news safe — because a schedule or cost problem reported five weeks early is usually recoverable (Domain 6's case study turns on exactly this) and the same problem reported at the deadline is not. The asymmetry is worth making arithmetic, because "escalate early" is advice everyone agrees with and almost nobody prices.

WORKED EXAMPLE

Worked example 1.2.2 — what the timing of one escalation was worth.

1. **Setup.** Meridian measures adoption monthly from go-live. At **week 13** the clinical lead's data shows **16** of 40 clinics in daily use — 40 % against the planned **70 %**, a shortfall of **12 clinics**, each worth `6 × 85 =` **USD 510** a week (1.3.2). Nobody escalates: the measure is reported by two parties and owned by neither (1.2.1). The shortfall surfaces at **week 39**, in an external audit. The remedy, once authorised, is clinical champions in the **11** lowest-adoption clinics at **USD 12,000** each; it lifts adoption to **68 % — 27.2** clinics — closing **11.2** of the 12-clinic gap, and it costs the same whenever it is authorised. (Domain 16's registered clinic-count measure reads **67.50 %** and reconciles the two definitions at KA 16.4.1; the 11.2 clinics used here is therefore the more generous of the two readings.) What did the 26-week silence cost?
2. **Formula.** Shortfall run rate = gap clinics × weekly value per clinic. Recovered run rate = closed clinics × weekly value. Timing value of the escalation = weeks of earliness × recovered run rate. Weeks of earliness that fund the whole remedy = remedy cost ÷ recovered run rate.
3. **Substitution.** Run rate `12 × 510`; recovered `11.2 × 510`; timing value `26 × 5,712`; remedy `11 × 12,000`; funding earliness `132,000 / 5,712`.
4. **Result.** The shortfall ran at **USD 6,120 a week**; the remedy recovers **USD 5,712 a week**. Escalating at week 13 instead of week 39 was worth **USD 148,512** — against a remedy costing **USD 132,000**, so the timing of the escalation was worth **1.13 times** the entire remedy it would have paid for. **23.11 weeks** of earliness funds the remedy outright, so the 26 weeks available carried **2.89 weeks** of margin.
5. **Interpretation.** Five readings, and the last is the one a leader must not skip.

The most valuable act in the sequence cost nothing. The remedy required a business case, funding and eleven appointments; the escalation required one person to send one message. On these figures the message was worth more than the programme it triggered. That is the honesty asymmetry in numbers, and it generalises: wherever a problem has a run rate, the value of early notice rises linearly with earliness while its cost stays fixed and small, so **the benefit-to-cost ratio of escalation is bounded only by how early the measure could have shown** — which is why the behaviour is worth engineering rather than exhorting.

The recovered run rate, not the shortfall run rate, is the honest multiplier. The gap was 12 clinics but the remedy closes 11.2, so valuing the earliness at 6,120 rather than 5,712 would overstate it by **USD 10,608** across the 26 weeks. The general error is pricing early warning at the size of the problem rather than at the size of the fix that early warning brings forward. Note the identity that falls out: 5,712 is exactly **40.00 %** of Meridian's USD 14,280 cost of delay, because 11.2

of 28 adopting clinics is 40 %. Early warning about part of a programme is worth the corresponding fraction of that programme's cost of delay, and no more.

Everything rests on the remedy working. Had clinical champions failed to move adoption, the 148,512 would be zero: earliness is worth exactly the value of the action it enables. The same dependency governs 1.3.3, and a leader who cannot name the action an escalation is meant to trigger does not yet have an escalation — only a complaint.

The sensitivity runs the reassuring way. The calculation assumes a constant 12-clinic gap. In practice an unaddressed adoption gap tends to widen as early enthusiasm decays and paper workarounds harden, so a constant-gap model **understates** the timing value and 148,512 should be read as a floor. If instead the gap had been closing unaided, the timing value would fall — and the test of which world you are in is whether the measure has a trend, which is the reason a single monthly number without a trend line is not yet a measure.

The professional caution: this arithmetic must not be turned on the analyst. The temptation on seeing 148,512 is to ask who failed to escalate. But the analyst reported the data on time; the failures were that no name owned the measure (1.2.1) and no route made raising it safe (1.2.2). The 148,512 is therefore the leader's number. Used to discipline a junior it destroys the next escalation and buys a second incident; used to fund an owned measure and a standing route to the board, it is the strongest business case available for the cultural work this KA describes.

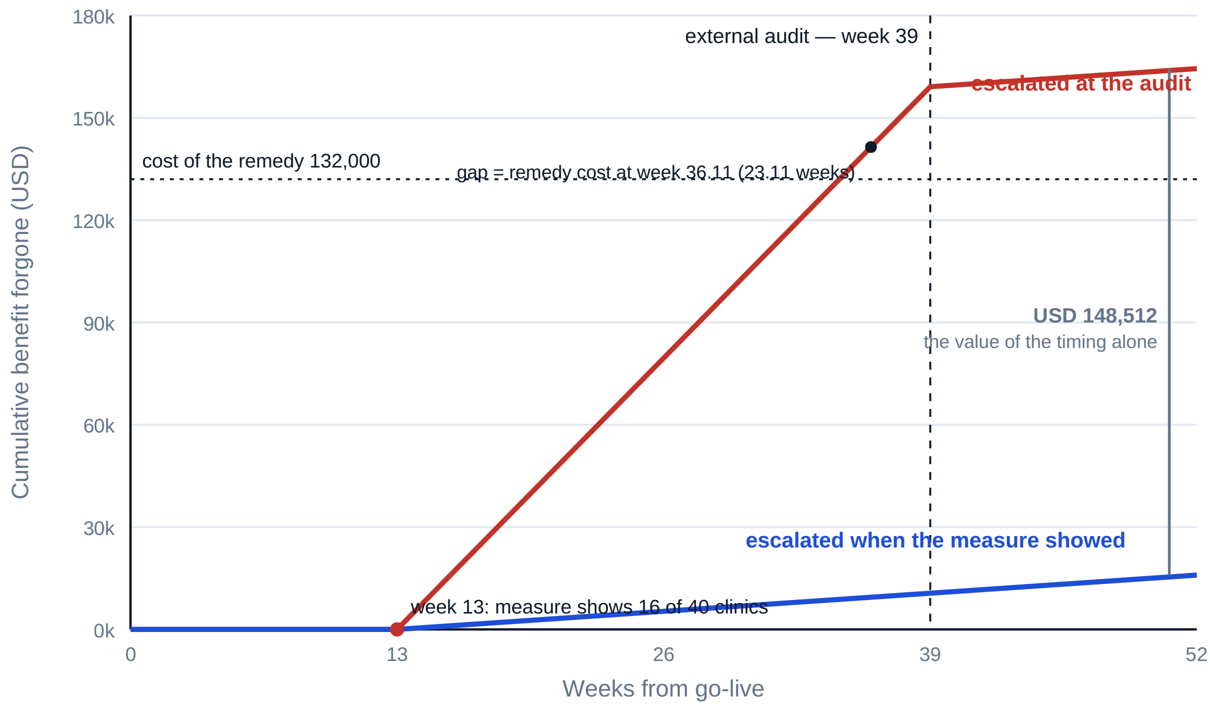


Fig 1.2.1 — The cost of a silent measure

1.2.3 The professional standard of care

Professionals are judged not on outcomes alone but on whether they exercised the **care a competent practitioner would** in the circumstances. Practically, four things constitute care in this discipline: **method appropriate to the stakes** (a two-week internal task and a safety- critical programme do not warrant the same rigour); **evidence proportionate to the claim** (a forecast presented to a board carries its basis — Domain 7's estimate basis sheet); **records that let others**

check the work (decision logs, assumption registers, audit trails); and **candour about uncertainty** (ranges, not false single numbers). A leader who did all four and still missed can defend the work. One who hit the date by luck and cannot show any of them has no defence available when the next project does not get lucky.

What care looks like on one real decision. Abstract lists of this kind are easy to agree with and hard to apply, so take the acceleration decision worked in 1.3.3 — spend USD 60,000 to arrive eight weeks sooner — and ask what each constituent required of the leader who took it.

Constituent	What it required here	What its absence would have looked like
Method proportionate to the stakes	A 60,000 decision does not warrant a simulation; it warrants a run rate, a breakeven and one sensitivity — the three lines of 1.3.3	"The team thinks we can pull it in and it seems worth it"
Evidence proportionate to the claim	The USD 14,280 a week traced to 28 adopting clinics, 6 hours, USD 85 — every factor with a source, and adoption identified as the one factor not yet observed	A weekly benefit figure quoted with no derivation, which cannot be challenged and therefore cannot be trusted
Records others can check	The decision recorded with its inputs, its breakeven of 4.20 weeks, and the adoption assumption named as the thing that would invalidate it	A minute reading "acceleration approved", which tells a later reviewer nothing about what was known
Candour about uncertainty	The 50/70/90 % adoption range carried into the paper, and the statement that below **36.76 %** adoption the spend destroys value	A single-point benefit and a confident recommendation

The leader who did those four things and then watched adoption stall at 40 % has a defensible file: the decision was right on the information available, the invalidating condition was named in advance, and the record shows it. That is the whole practical content of the standard of care — **it is not a promise about outcomes, it is a discipline about evidence, and it is built before the outcome is known, because it cannot be built afterwards.**

Two cautions. First, care is judged **in the circumstances**, which cuts both ways: the same one-page analysis that constitutes care on a 60,000 decision would be negligent on a safety-related one, and a leader who applies a uniform standard is wrong in one direction or the other (the same proportionality argument as KA 1.4.3). Second, where care is **codified** — regulated construction, pharmaceuticals, aviation, nuclear, financial services — the required records, competencies and sign-offs are set by that regime and by the relevant professional-body rules, and what counts as adequate there is a question for the regime and, where liability is engaged, for counsel in the applicable jurisdiction. This book states the professional discipline; it does not state anyone's legal duty.

KEY TERMS — KA 1.2

Term	Meaning
Responsibility	The obligation to do; shareable and delegable.
Accountability	The obligation to answer; single-holder, non-delegable.
Escalation duty	The obligation to raise, with options, a conflict beyond one's authority.
Honesty asymmetry	Bad news costs the messenger and helps the project; leaders must make it safe.
Timing value of an escalation	Weeks of earliness × the run rate the remedy recovers — what the timing alone is worth, separate from the remedy.
Silent measure	An outcome measure reported by two parties, owned by neither, and never accompanied by a proposed action.
Legal liability	Who must compensate whom, set by contract and jurisdiction; distinct from professional accountability and capable of sitting elsewhere.
Standard of care	The care a competent practitioner would exercise in the circumstances.

SAMPLE MCQS — KA 1.2

MCQ 1.2-A `[1.2.1 · Application]` A leader outsources the migration design to a specialist vendor under a fixed-price contract. Accountability for the migration's success now rests with: - A. the vendor, under the contract - B. the leader, who retains the duty to specify, verify proportionately and act on failure - C. jointly and equally with the vendor - D. the sponsor, who approved the contract

Rationale: The contract transfers responsibility (and commercial risk); accountability is non-delegable. C describes the arrangement that guarantees nobody answers (1.2.1); D confuses approving a route with owning the delivery.

MCQ 1.2-B `[1.2.1 · Analysis]` An AI planning tool generates a schedule that the leader submits unchanged; it contains an infeasible dependency that causes a two-month slip. The accountability position is: - A. shared with the tool's vendor, who supplied a defective product - B. the leader's: a tool cannot answer for an outcome, and submitting unverified output is a failure of the verification duty - C. nobody's — the failure was technological - D. the team's, for not catching it

Rationale: Accountability requires a party who can be asked to answer, so it cannot attach to software (1.2.1); the specific lapse is verification (1.1's principle). Vendor recourse (A) is a commercial question and does not relocate the professional answer; D inverts delegation.

MCQ 1.2-C `[1.2.3 · Analysis]` A leader's project overran despite a documented method, ranged forecasts, a decision log and escalated risks. Under the standard of care this is: - A. professional negligence, because the project overran - B. defensible practice — care is judged on the exercise of competent method, not on outcome alone - C. irrelevant, since standards of care apply only to regulated professions - D. defensible only if the overrun was under 10 %

Rationale: The standard is conduct-based (1.2.3). A collapses care into outcome; C is false in substance — the expectation attaches to professional practice generally; D invents a threshold.

MCQ 1.2-D `[1.2.2 · Application]` An adoption shortfall of 12 clinics, each worth USD 510 a week, is visible at week 13 but escalated at week 39. The remedy costs USD 132,000 and recovers 11.2 of the 12 clinics. The value of the 26 weeks of earliness that were not taken is: - A. USD 159,120 - B. USD 148,512 - C. USD 16,512 - D. USD 132,000

Rationale: Earliness is worth the run rate the remedy recovers: `26 × 11.2 × 510 = 148,512` (1.2.2). A values it at the full 12-clinic gap (`26 × 6,120`), crediting the remedy with a recovery it does not achieve; C deducts the remedy cost, which is incurred either way and so does not belong in a timing comparison; D is the remedy cost itself.

MCQ 1.2-E `[1.2.1 · Analysis]` A programme's adoption measure appears in both the delivery report and the clinical directorate's report, with no proposed action in either. The most accurate diagnosis is: - A. duplicated reporting, which should be rationalised for efficiency - B. two accountable names and therefore none — the measure has no owner, and its trigger has become an incident rather than a threshold - C. adequate governance, since two parties are monitoring the measure - D. a data-quality problem in the reporting pipeline

Rationale: Dual accountability converts the trigger from a measure into an incident, and the signature is exactly this: reported twice, owned once by nobody, never with an action (1.2.1). A treats the symptom as an administrative annoyance; C mistakes visibility for ownership; D relocates a governance defect to the tooling.

SELF-CHECK — KA 1.2

1. What can be delegated and what cannot? — Responsibility can; accountability cannot.
2. Why can accountability never attach to an AI tool? — Accountability is the obligation to answer; a tool cannot be asked to answer, sanctioned, or hold a duty.
3. Name the four constituents of care in this discipline. — Proportionate method, evidence matching the claim, checkable records, candour about uncertainty.
4. What is the signature of dual accountability in a report pack? — An outcome measure reported by both parties, owned by neither, never accompanied by a proposed action (1.2.1).
5. At what run rate should the value of early warning be priced? — The run rate the remedy **recovers**, not the size of the problem: 5,712 a week, not 6,120 (1.2.2).
6. Is professional accountability the same as legal liability? — No: liability is set by contract and jurisdiction and may sit elsewhere or be capped; take that position from counsel (1.2.1).

KNOWLEDGE AREA 1.3

Systems thinking and value

Topics: 1.3.1 projects as systems · 1.3.2 outputs, outcomes, benefits, value · 1.3.3 leading under uncertainty.

1.3.1 Projects as systems

A project is a system: parts that interact, with **feedback** and **delay** between cause and visible effect. Three systems behaviours a leader must recognise, because each defeats common-sense management.

Delay hides consequence. Adding people to a late project consumes the productivity of those already on it (onboarding, communication paths) before adding any, so the intervention looks harmful before it looks helpful — and leaders often reverse it in the interval.

Local optimisation degrades the whole. A discipline lead maximising their own efficiency levels their team's workload smoothly and delivers the interface documents late, stalling three other teams.

Every function can be efficient while the project is slow — the constraint logic of Domain 6 (6.A.1) is the schedule expression of this.

Pressure moves, it does not vanish. Squeeze the schedule and the pressure emerges as quality defects (Domain 9), scope disputes (Domain 5), team attrition (Domain 12) or supplier claims (Domain 10). A leader's question on receiving any "we absorbed it" report is where did it go?

The first of those three is the one that most often causes a leader to undo their own correct decision, and it is priced in KA 1.3.3 (Worked example 1.3.3b), where the cost of delay it needs has been derived: adding people to late work produces a measurable **transient** in which progress is worse than it would have been, and the arithmetic gives the week at which the intervention overtakes doing nothing. Domain 12 (KA 12.2.2) treats the steady-state counterpart — the coordination cost of team size, which has a hard capacity ceiling — and the two are different mechanisms with different remedies, so neither substitutes for the other: shortening a ramp does nothing about a mesh, and structuring a team does nothing about a ramp.

1.3.2 Outputs, outcomes, benefits and value

Definitions. An **output** is what the project delivers (a system installed, a road built). An **outcome** is the change in how things work as a result (clinicians actually using the records; journeys actually faster). A **benefit** is the measurable improvement that the outcome produces (hours released, cost avoided, harm reduced). **Value** is benefit weighed against cost and risk, for a stated stakeholder. The chain has a hard property: **each link can fail independently**, so an output claimed as a benefit is an overstatement by construction.

WORKED EXAMPLE

Worked example 1.3.2 — what Meridian actually delivers.

1. **Setup.** Meridian Care Records installs the shared records system in **40 clinics** (the output). Realistic adoption at 12 months is **70 %** of clinics using it in daily practice. In an adopting clinic the system releases **6 clinician-hours per week**, valued at **USD 85 per hour**, over a **48-week** operating year.
2. **Formula.** Annual benefit = clinics × adoption × hours/week × rate × weeks. Contrast with the output-based claim, which counts all 40 clinics.
3. **Substitution.** Adopting clinics `40 × 0.70 = 28`; benefit `28 × 6 × 85 × 48`. Output-based claim `40 × 6 × 85 × 48`.
4. **Result.** Benefit **USD 685,440 per year** (≈ SAR 2.57 million indicatively). The output-based claim gives **USD 979,200** — an overstatement of **USD 293,760**, exactly **30.0 %**, which is the non-adoption rate reappearing as fictitious value.
5. **Interpretation.** The 30 % gap is not an estimating error; it is the outcome link being skipped. This is why a benefits register (Domain 16) tracks adoption as a measure in its own right, and why "40 clinics live" is a milestone, not a benefit. Five things follow.

The overstatement is exactly the non-adoption rate, whatever the other numbers are. Because both figures share the hours, the rate and the weeks, the ratio of the honest benefit to the output-based claim is the adoption term alone: `685,440 / 979,200 = 0.70`. So an output-based claim overstates by `1 − a` — here 30.0 % — **irrespective of how many hours are released, what they are worth, or how long the operating year is.** That is a useful thing to know in a meeting: you do not need the model to say how wrong a claim is, only the adoption assumption it omitted. The corollary is a reviewer's shortcut — if a benefits case contains no adoption term, its overstatement is already known to be the non-adoption rate, and the discussion can move straight to what that rate is.

Sensitivity is the leader's real lever. At 50 % adoption the benefit is USD 489,600; at 90 %, USD 881,280. **A programme of this shape creates more value by moving adoption than by moving its installation date** — which should change where the leader spends personal attention (Domain 11's engagement work, not Domain 6's schedule compression). Per clinic the same arithmetic gives `979,200 / 40 =` **USD 24,480** of full potential and **USD 17,136** realistic (the per-clinic figures Domain 5 uses to price scope decisions at KA 5.1.3), so a single clinic won or lost is worth more than most of the change requests a rollout of this kind argues about.

The definition of adoption is where the number is actually decided. "Using it in daily practice" has to be operationalised — a proportion of encounters recorded in the system, a threshold of active clinicians, a fortnight without reversion to paper — and reasonable definitions differ. On these figures a **5-percentage-point** difference in where the threshold is drawn is worth `0.05 × 979,200 =` **USD 48,960** a year. So the definition belongs in the benefits register with the measure, agreed before the first report, or the programme will spend its second year arguing about the denominator (Domain 16, KA 16.4.1).

Released hours are capacity, not cash — and the distinction is a professional obligation, not a quibble. Valuing 6 clinician-hours at USD 85 states what those hours are worth if they are used for something. They become a cash saving only if headcount or agency spend falls, and a service improvement only if the hours go to patients; if neither happens they are absorbed and the benefit is real to nobody. A benefits case must therefore say which of the two it is claiming and who is accountable for making it happen — the same distinction Domain 2 draws between cost removed and cost moved (KA 2.1.2). A leader who lets a released-hours figure be read as cash has authorised a claim the finance function will later, correctly, refuse to recognise.

What would falsify the whole chain. Three things, in order of likelihood: adoption measured on a definition nobody agreed (above); hours released estimated rather than sampled — Domain 16's measurement found 5.4 hours, not 6.0, which alone removes **USD 66,096** a year at that domain's measured steady adoption of **67.50 %** (`163,200 × 0.675 × 0.6`); and an attribution problem, where hours would have been released anyway by an unrelated change, which is why a comparison cohort matters (Domain 16, KA 16.4.3). None of these is a reason to omit the arithmetic. They are the reasons to publish it with its assumptions attached, so that the argument is about the assumptions rather than about the conclusion.

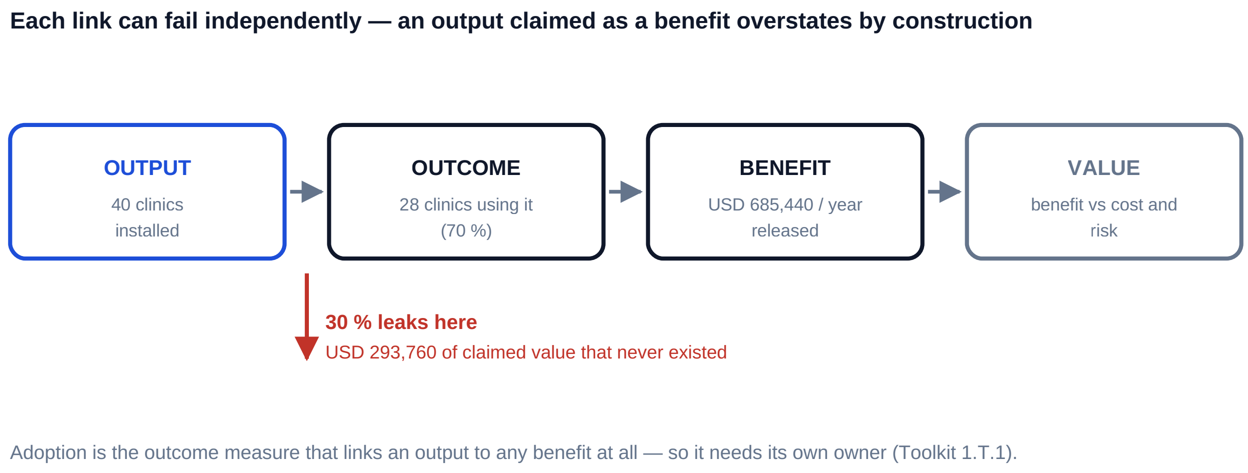


Fig 1.3.1 — The value chain, and where it leaks

The fourth link is the one leaders skip. Benefit is not value: value is benefit weighed against cost and risk, for a stated stakeholder. A programme with a genuine benefit and a bad ratio of benefit to cost is a bad investment, and the leader who reports only the benefit has not yet said anything about value. The simplest instrument that closes the link is **simple payback** — the number of periods of benefit needed to repay the cost — and it repays study here because it is the instrument boards actually use, it makes the adoption dependency arithmetic, and its limitations are instructive.

WORKED EXAMPLE

Worked example 1.3.2b — payback, and the adoption commitment a payback rule silently makes.

1. **Setup.** Meridian's approved cost is **USD 2,400,000**. Full potential benefit is **USD 979,200** a year; the honest benefit at 70 % adoption is **USD 685,440**. The sponsoring organisation applies a **three-year simple payback** rule to discretionary investment. Compute the payback on both benefit figures, and find the adoption the rule actually requires.
2. **Formula.** Simple payback = cost ÷ annual benefit, where annual benefit = potential × adoption. Required adoption for a payback target `T` = cost ÷ (`T` × potential).
3. **Substitution.** Output-based `2,400,000 / 979,200`; honest `2,400,000 / 685,440`; required annual benefit `2,400,000 / 3 = 800,000`; required adoption `800,000 / 979,200`.
4. **Result.** Payback on the output-based claim **2.4510 years**; on the honest benefit **3.5014 years**. The ratio of the two is **1.428571**, which is exactly `1 / 0.70`. A three-year payback rule requires **USD 800,000** a year, and therefore **81.70 %** adoption — **11.70 percentage points**, or **4.68 clinics**, beyond what the case assumed.
5. **Interpretation.** Five readings, and the fourth is the one that changes how a leader reads a board's own rules.

Payback scales as the reciprocal of adoption, exactly. `payback = cost / (potential × a)`, so halving adoption doubles payback and the 1.428571 ratio above is `1/0.70` with no residue. This is worth holding as an identity because it converts a governance question into an arithmetic one: any output-based payback is understated by the factor `1/a`, and a leader who knows the adoption assumption can correct a payback in their head.

A payback rule is an adoption commitment in disguise. Nobody in the approval meeting said "we undertake to reach 81.70 % adoption"; the three-year rule said it for them, and nobody noticed because the rule was expressed in years and the commitment is in percentage points. **Inverting a hurdle into the operational condition it implies is one of the highest-value habits in this book** — it is the same move Domain 2 makes when it reports breakeven adoption instead of NPV (KA 2.2.2), and Domain 3 makes when it converts a delegation threshold into an escalation volume (KA 3.2.3). The professional act on seeing a 3.5014-year payback against a three-year rule is not to argue about the rule but to say what the rule demands and ask who owns delivering it.

The marginal value of adoption is largest where adoption is worst, by a factor of 3.60. Because payback is hyperbolic in adoption, moving from 40 % to 50 % cuts payback by **1.2255 years** while moving from 80 % to 90 % cuts it by **0.3404 years** — the first ten points are worth **3.60 times** the last ten. This contradicts the instinct to chase the final laggards for completeness, and it is the arithmetic behind the recovery in this domain's case study, which funded champions in the **eleven lowest-adoption** clinics rather than pushing the leaders higher. The same reasoning appears in Domain 11's engagement prioritisation: effort goes where the curve is steep.

Simple payback is the wrong instrument for the decision and the right one for the conversation. It ignores the time profile of benefits, the cost of capital, everything after the payback point, and — on Meridian — the operating cost the case omitted. Including Domain 16's measured run cost of **USD 108,000** a year, net benefit falls to **USD 577,440** and payback lengthens to **4.1563 years**, an addition of **0.6549 years** from a line that was simply absent. Domain 2 does the decision properly, discounted over eight years at 7 %, and reports a breakeven sustained adoption of **41.05 %**. Note what the two breakevens are for: **81.70 % is what a three-year payback rule demands; 41.05 % is what creating value demands.** The **40.65-point** gap between them is the price of using a payback rule as a proxy for value — and a programme killed at 70 % adoption for failing a payback rule would have been killed while creating value, which is a governance failure and not an arithmetic one.

The caution. Every figure here is undiscounted, in nominal money, and takes the approved cost as complete. Do not present a simple payback to a board without saying which of those three simplifications is doing the most work; on Meridian it was the third. Where the investment is material, the discounted appraisal is not an optional refinement: Domain 2 is where it belongs.

Fig 1.3.2 — Payback is hyperbolic in adoption, so the first points are worth the most

1.3.3 Leading under uncertainty

Uncertainty is the condition, not an exception to be eliminated before acting. Three disciplines constitute leading well inside it.

Decide at the right moment. Deciding early buys momentum and forecloses options; deciding late preserves options and costs time. The professional question is not "do we have enough information?" but what is the last responsible moment, and what does waiting cost?

Price the time. In benefit-generating work, delay has an arithmetic cost.

WORKED EXAMPLE

Worked example 1.3.3 — what a week of Meridian delay costs.

1. **Setup.** Meridian's benefit runs at the KA 1.3.2 rate once adopted. The team can compress the rollout by **8 weeks** at an incremental cost of **USD 60,000**. Should the leader buy it?
2. **Formula.** Benefit per week = adopting clinics × hours × rate. Net = (weeks × benefit per week) − cost. Breakeven weeks = cost ÷ benefit per week.
3. **Substitution.** Per week `28 × 6 × 85 = 14,280`; eight weeks `114,240`; net `114,240 − 60,000`.
4. **Result.** Benefit per week **USD 14,280**; eight weeks **USD 114,240**; **net +USD 54,240**. Breakeven at **4.20 weeks** — beyond which acceleration pays.
5. **Interpretation.** The decision is not "is 60,000 a lot?" but "does it buy more than it costs?" — and here it clearly does, provided adoption materialises. Five further readings, and the second is the one that separates a priced decision from a lucky one.

The breakeven is the sentence to take to the sponsor, not the net. A net of +54,240 invites an argument about the inputs; a breakeven of 4.20 weeks invites a question that can be answered — can we really pull in more than 4.20 weeks? The identity is `breakeven weeks = cost ÷ cost of delay`, which is scale-free: it does not care about the size of the programme, only about the price of a week. The same identity governs Domain 3's gate durations and Domain 15's escalation latency, which is why the cost of delay is registered once and reused rather than re-derived.

Every figure rests on the 70 % adoption of 1.3.2, and the sensitivity is severe. Spending 60,000 to arrive sooner at an unadopted system buys nothing. Precisely: the acceleration pays for itself over eight weeks at any adoption above **36.76 %** — because 60,000 over 8 weeks needs **USD 7,500** a week, which is **14.7059** clinics at USD 510 each. That is comfortably below the planned 70 %, so the decision is robust in direction. But it is fragile in magnitude: at the adoption Meridian actually achieved, 40 %, the cost of delay is `16 × 510 =` **USD 8,160** a week, the breakeven stretches to **7.3529 weeks** of the 8 available, and the net collapses from **USD 54,240** to **USD 5,280** — a **90.3 %** reduction, from a variable nobody in the meeting was watching. **A decision can be correct and still have almost no margin, and the leader is the person who has to know which.**

The maximum justified spend follows immediately and is worth stating in the paper. Eight weeks at 14,280 is **USD 114,240**, so any compression package priced above that destroys value however attractive it looks; at the measured 40 % adoption the ceiling falls to `8 × 8,160 =` **USD 65,280**. Quoting the ceiling rather than only the recommendation is what stops a 60,000 proposal becoming a 120,000 proposal during procurement without anyone re-testing it.

Sequencing engagement before acceleration is not a preference; it is what the numbers say. Adoption work raises both the benefit and the value of every subsequent timing decision, because the cost of delay is proportional to adoption. Acceleration bought before adoption is secured is leveraged on an unverified assumption; adoption secured first makes the acceleration worth more.

What breaks it. The eight weeks must be genuinely available on the critical path — compression applied to non-critical work buys nothing at all, which Domain 6 (KA 6.4) computes properly, and a compression estimate produced by the same team that owns the date should be treated as an advocacy figure until tested. The 60,000 must be the whole incremental cost, including any overtime premium, additional supervision and the rework that rushed installation tends to generate (Domain 9, KA 9.2.3 prices the rework-capacity effect). And the benefit must actually begin when the output arrives: if go-live is gated by a training window or a licence date, arriving early buys nothing but idle readiness. A reviewer's single best question on any acceleration case is what starts earning on the earlier date, and who has confirmed it can?

The other way to buy time is people, and it does not price the same way. Acceleration bought with money answered cleanly in the example above. The same objective pursued by adding staff behaves differently, because capacity added mid-delivery arrives with a transient attached — which is 1.3.1's "delay hides consequence" property in its most expensive everyday form. The two decisions look alike to a sponsor and reach opposite verdicts, which is the reason to work both.

WORKED EXAMPLE

Worked example 1.3.3b — the reinforcement trough, and when buying time with people is a mistake.

1. **Setup.** Eight clinics remain in Meridian's rollout. The crew of **6** deployment specialists completes **0.1** clinics each per week, so **0.6** a week in total. The programme is late, and the leader can add **3** more specialists at **USD 4,200** per specialist-week. Newcomers take a **4-week** ramp, during which each works at **25 %** of full productivity and absorbs **50 %** of an existing specialist's time in supervision. The cost of delay is the **USD 14,280** a week derived above. Both the ramp length and the supervision load are **locally calibrated planning figures, not constants** — the same discipline Domain 12 applies to its coordination parameter.
2. **Formula.** Ramp-period rate = base rate − (newcomers × supervision × per-head rate) + (newcomers × ramp productivity × per-head rate). Post-ramp rate = base rate + newcomers × per-head rate. Duration = ramp weeks + (remaining work − ramp-period output) ÷ post-ramp rate. Crossover horizon = ramp weeks × (1 + deficit rate ÷ gain rate). Net = weeks saved × cost of delay − incremental specialist-weeks × rate.
3. **Substitution.** Ramp rate `0.6 − (3 × 0.5 × 0.1) + (3 × 0.25 × 0.1) = 0.6 − 0.15 + 0.075`; post-ramp `0.6 + 0.3`; baseline duration `8 / 0.6`; new duration `4 + (8 − 4 × 0.525) / 0.9`; specialist-weeks `9 × 10.5556` against `6 × 13.3333`.
4. **Result.**

	Crew of 6	Crew of 9
Delivery rate, weeks 1–4	0.6 clinics/week	**0.525** clinics/week
Delivery rate thereafter	0.6	**0.9**
Cumulative at week 4	2.4 clinics	**2.1** clinics
Duration for the last 8 clinics	**13.3333** weeks	**10.5556** weeks
Specialist-weeks consumed	80.0	**95.0**

Weeks saved **2.7778**, worth **USD 39,666.67** at the cost of delay. Incremental effort **15.0** specialist-weeks, costing **USD 63,000**. **Net (USD 23,333.33)** — the reinforcement loses money. The crossover horizon is `4 × (1 + 0.075/0.3) =` **5.00 weeks**, and at week 4 the reinforced crew is **0.3 clinics — 12.5 % — behind** where the original crew would have been. 5. **Interpretation.** Four results, of which the third is the one that decides the case and the fourth is the one that keeps the model honest.

The trough is real, measurable and misread. For four weeks every progress metric says the decision was wrong, by 12.5 % at its worst. A leader who reverses in the trough pays the whole supervision cost, receives none of the added capacity, and ends later than either pure option — the single most expensive available move. The counter-measure is procedural, not stoical: **state the**

crossover week when the decision is taken, and put it in the record, so the trough is a forecast being met rather than a surprise being explained.

The crossover horizon is computable in advance, and it is the decision rule. `R × (1 + d/g)` needs only the ramp length, the net deficit rate during the ramp and the gain rate after it — three figures any team can state. Here it is 5.00 weeks: with more than five weeks of work left, reinforcement finishes earlier; with less, it finishes later, and no amount of commitment changes that. The identity also tells you which lever matters. Halving the ramp to two weeks moves the crossover to **2.50** weeks — a gain of 2.50; halving the supervision load instead removes the deficit altogether (`d = 0`) and moves the crossover to **4.00** weeks, the ramp length itself — a gain of 1.00. **Shortening the ramp is worth two and a half times as much as easing supervision**, which is an argument for prepared onboarding material rather than for asking mentors to give newcomers less of their time.

Being earlier is not the same as being better off. The 2.7778 weeks saved are worth 39,666.67; the 15 extra specialist-weeks cost 63,000; the reinforcement therefore destroys **USD 23,333.33** of value while genuinely delivering sooner. It becomes worthwhile only above a cost of delay of **USD 22,680 a week** — and Meridian's maximum conceivable cost of delay, at 100 % adoption, is `40 × 6 × 85 =` **USD 20,400**, which is **11.18 % short** of the breakeven. **No adoption level makes this reinforcement pay on delay grounds.** So if the crew is added anyway, it must be for a reason that is stated and different — a statutory date, a contractual liquidated sum, the crew being needed for the next tranche regardless — and that reason belongs in the decision record (1.2.3). The alternative lever is the one that is cheap: the same 2.7778 weeks bought by shortening the ramp costs nothing like 63,000. The breakeven specialist-week rate is **USD 2,644.44**; at Meridian's 4,200 the arithmetic simply does not clear.

What breaks the model. It assumes the remaining work is divisible and rate-limited by specialist capacity — not by a fixed clinic-by-clinic sequence, a licence window or a single scarce approver, any of which makes added capacity worth nothing at all (Domain 6, KA 6.A.1 on the constraint). It assumes the newcomers are competent in the work and merely unfamiliar with this programme; recruiting into an unfamiliar discipline has a longer ramp and a larger supervision load, both of which push the crossover out. And it prices only duration and effort: quality consequences of a rushed ramp sit in Domain 9's containment economics, and morale consequences in Domain 12. The invariant worth remembering, because it is parameter-free: **a trough exists precisely when the supervision load a newcomer imposes exceeds the newcomer's own ramp productivity.** At 50 % supervision against 25 % productivity there is a trough; at 25 % supervision against 50 % productivity the reinforced rate is 0.675 from day one and there is none. A leader who knows only that inequality already knows whether to expect a dip.

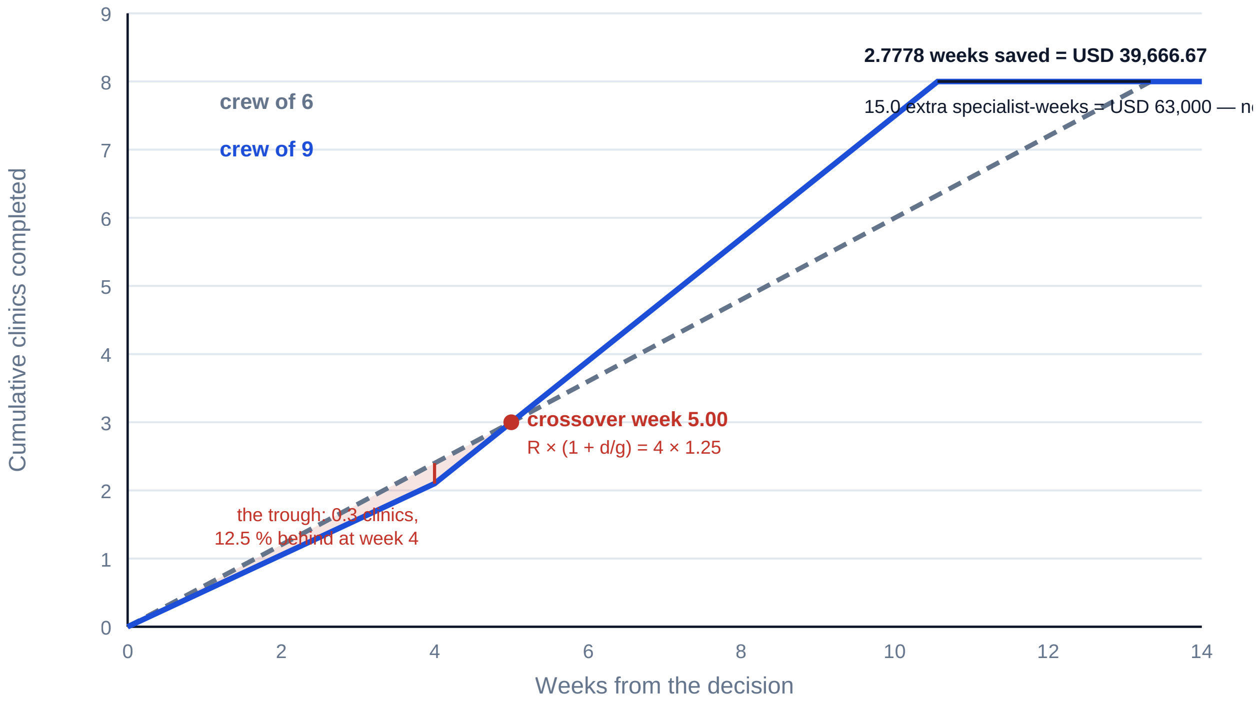


Fig 1.3.3 — The reinforcement trough and its crossover

Act reversibly where you can. Prefer decisions that can be unwound, pilots over commitments, and options kept open at low cost — then commit hard once uncertainty has genuinely reduced. This is the leadership behaviour behind rolling-wave planning (Domain 6, KA 6.3.3) and staged investment (Domain 2).

AI in this KA

Systems and value reasoning is where AI assistance is weakest and most confidently wrong: a model will produce a fluent benefits case that silently treats outputs as benefits, because that is what most documents in its training data do. The governed use is narrow and real — stress-test a benefits chain by asking what would have to be true, generate adoption scenarios, surface second-order effects a team has not considered — followed by human judgment on which link is actually fragile. The check that catches the common failure is one question: for each claimed benefit, which measured outcome produces it, and who owns that measure? (Domain 16 turns that into the benefits register.)

KEY TERMS — KA 1.3

Term	Meaning
Output / outcome / benefit / value	Delivered thing · change in how things work · measurable improvement · benefit against cost and risk.
Adoption	The outcome measure linking an output to any benefit at all.
Feedback and delay	Systems properties that make interventions look wrong before they look right.
Local optimisation	Function-level efficiency that degrades whole-system performance.
Last responsible moment	The latest point a decision can be taken without foreclosing a needed option.
Cost of delay	The benefit forgone per period of lateness.
Simple payback	Cost ÷ annual benefit, in periods; scales as the reciprocal of adoption.
Reinforcement trough	The transient period after adding people in which progress is worse than it would have been; exists when supervision load per newcomer exceeds their ramp productivity.
Crossover horizon	`R × (1 + d/g)` — the remaining work below which reinforcement finishes later, not earlier.
Released hours	Capacity freed, not cash; a benefit only once redeployed to a stated use.
Decision breakeven adoption	The adoption at which one priced decision stops paying — 36.76 % for Meridian's acceleration. Distinct from Domain 2's **breakeven adoption**, which is the level at which the whole investment's NPV is zero (41.05 %).

SAMPLE MCQS — KA 1.3

MCQ 1.3-A `[1.3.2 · Application]` 40 clinics installed; 70 % adoption; 6 hours/week released per adopting clinic at USD 85/hour over 48 weeks. The defensible annual benefit is: - A. USD 979,200 - B. USD 685,440 - C. USD 293,760 - D. USD 14,280

Rationale: `40 × 0.70 × 6 × 85 × 48 = 685,440`. A is the output-based claim that skips adoption; C is the overstatement itself; D is the weekly benefit.

MCQ 1.3-B `[1.3.2 · Analysis]` A programme board is told "40 clinics live — benefits delivered". The soundest challenge is: - A. ask for the installation evidence - B. ask which measured outcome produces the benefit and who owns that measure, since installation is an output and adoption is unverified - C. accept it, since the output target was met - D. ask for the cost variance instead

Rationale: The claim skips the outcome link, which is precisely where 30 % of the value leaked in 1.3.2. A verifies the wrong thing, C repeats the error, D changes the subject.

MCQ 1.3-C `[1.3.3 · Application]` Benefit accrues at USD 14,280 per week once adopted. Compressing the rollout by 8 weeks costs USD 60,000. The decision and its breakeven are: - A. reject — 60,000 exceeds the weekly benefit - B. accept — net +USD 54,240, breakeven at 4.20 weeks - C. accept — net +USD 114,240 - D. indifferent — the two are equal

Rationale: `8 × 14,280 = 114,240`, less the 60,000 cost, nets **+54,240**; breakeven `60,000/14,280 = 4.20` weeks. A compares a total against a weekly rate; C forgets to deduct the cost; D asserts an equality the arithmetic denies.

MCQ 1.3-D `[1.3.1 · Analysis]` A discipline lead reports having "absorbed" a two-week schedule squeeze with no impact. The systems-literate response is: - A. record the recovery and move on - B. ask where the pressure went — into quality, scope, people or suppliers — because pressure moves rather than vanishing - C. squeeze the remaining teams equally - D. treat it as evidence the original estimate was padded

Rationale: Pressure relocates (1.3.1); the leader's job is to find where. A accepts an unverified claim, C compounds it, D leaps to a conclusion the report does not support.

MCQ 1.3-E `[1.3.3 · Evaluation]` Reinforcing a crew saves 2.7778 weeks at a cost of delay of USD 14,280 a week and consumes 15.0 extra specialist-weeks at USD 4,200 each. The programme's cost of delay could not exceed USD 20,400 even at 100 % adoption. The strongest professional conclusion is: - A. approve — the work finishes 2.7778 weeks earlier, which is the objective - B. reject on delay grounds: the breakeven cost of delay is USD 22,680 a week, above the programme's maximum conceivable 20,400, so no adoption level makes it pay and any approval needs a different stated reason - C. approve — 39,666.67 of delay value exceeds the 23,333.33 net loss - D. reject — adding people always makes a late project later

Rationale: `63,000 / 2.7778 = 22,680`, which exceeds the 20,400 ceiling (1.3.3b). A optimises earliness rather than value; C compares the gross benefit against the net result, double-counting the benefit; D over-applies the coordination result of Domain 12 as a law — here reinforcement genuinely does finish earlier, it simply costs more than it saves.

MCQ 1.3-F `[1.3.2 · Application]` A cost of USD 2,400,000 against a full-potential benefit of USD 979,200 a year is assessed under a three-year simple payback rule. The adoption the rule requires is: - A. 70.00 % - B. 81.70 % - C. 245.10 % - D. 41.05 %

Rationale: The rule needs `2,400,000/3 = 800,000` a year, and `800,000/979,200 = 81.70 %` (1.3.2b). A is the case's assumption, which is the thing being tested; C divides the cost by the potential and reads the payback in years as a percentage; D is Domain 2's breakeven adoption for value creation — a different question, and 40.65 points lower.

MCQ 1.3-G `[1.3.3 · Evaluation]` Compressing the rollout by 8 weeks costs USD 60,000. Which statement best assesses the decision's robustness? - A. it is robust: the net is +54,240, comfortably positive - B. it pays at any adoption above 36.76 %, so the direction is robust, but at the 40 % adoption actually achieved the net falls to 5,280 — robust in direction, fragile in magnitude - C. it is not robust, because it depends on an adoption assumption - D. robustness cannot be assessed without a discounted appraisal

Rationale: `60,000/8 = 7,500` a week is 14.7059 clinics, or 36.76 % adoption; at 40 % the weekly benefit is 8,160 and the net is `8 × 8,160 − 60,000 = 5,280` (1.3.3). A quotes the point estimate as though it were the range; C treats any dependency as fatal and so never approves anything; D imports Domain 2's instrument into a decision whose horizon is eight weeks.

SELF-CHECK — KA 1.3

1. Why is an output claimed as a benefit an overstatement by construction? — Because the outcome link can fail independently; unadopted outputs produce no benefit (Meridian's 30 %).
2. What is the leader's question on hearing "we absorbed it"? — Where did the pressure go?
3. What must be true before paying to accelerate Meridian? — That adoption materialises; arriving sooner at an unadopted system buys nothing.
4. By how much does an output-based benefit claim overstate, and what does the answer depend on? — By exactly the non-adoption rate, `1 − a`; it depends on nothing else — not the hours, the rate or the length of the operating year (1.3.2).
5. When does adding people to late work finish later rather than earlier? — Below the crossover horizon `R × (1 + d/g)`; and a trough exists at all only when supervision load per newcomer exceeds their ramp productivity (1.3.3b).

6. What does a three-year payback rule commit an organisation to, on Meridian's figures? — 81.70 % adoption — a commitment expressed in years and never discussed in percentage points (1.3.2b).
7. Are released clinician-hours a benefit? — Not yet: they are capacity, and become a cash saving or a service improvement only when redeployed to a stated use with an owner (1.3.2).

KNOWLEDGE AREA 1.4

Ethics and the responsible use of AI

Topics: 1.4.1 professional ethics in delivery · 1.4.2 the responsible-AI principle applied · 1.4.3 the leader's AI accountability.

1.4.1 Professional ethics in delivery

Ethics in this discipline is rarely dramatic; it is a series of small reporting and allocation decisions taken under pressure. Four recurring tests:

- **Report what is, not what is wanted.** Presenting an optimistic case as the base case is a misrepresentation whatever the spreadsheet says (the same rule PFL-AI applies to forecasts).
- **Do not spend other people's margins silently.** Absorbing pressure into team overtime, a supplier's contingency or a safety margin is a decision someone is entitled to be told about.
- **Claim only what is true.** Milestone and benefit claims are statements of fact to people relying on them (1.3.2).
- **Compete and procure honestly.** Conflicts declared, evaluations run as published, suppliers not used as free consultancies (Domain 10's ethical sourcing).

Conflicts of interest are handled by disclosure and separation: declared before engagement, managed or declined — and tested by daylight, would every party, seeing the full picture, still regard this as impartial? Where the answer wavers, impartiality has already failed.

1.4.2 The responsible-AI principle applied

The principle from KA 1.1 becomes operational through four obligations that recur in every later domain:

Obligation	What it requires of a leader
Appropriate use	Match the tool to the task; prohibit it where the stakes or data forbid (Domain 14)
Verification	Machine output checked against evidence by a named human before reliance
Transparency	Material AI use disclosed in deliverables and decisions
Human decision	The judgment, and its record, belongs to a person

The failure modes to expect, stated honestly: **hallucination** (fluent, false specifics — the invented precedent, the plausible citation); **silent staleness** (confident answers from out-of-date data); **bias** inherited from training data, especially in anything touching people; **confidentiality leakage** (project data entering a tool is a disclosure); and **over-trust through fluency** — the deepest one, because polished output suppresses the scrutiny that rough output invites. None of these is a reason to refuse the technology; each is a reason the verification duty is not optional.

1.4.3 The leader's AI accountability

Three concrete leadership acts, which Domain 14 then systematises:

Name the owner. Every AI-assisted artefact that informs a decision has a human owner who verified it. Anonymous machine output has no place in a governed decision record (Domain 3, KA 3.3.4).

Make verification proportionate and real. A summary for internal reading needs a glance; a forecast in a board paper needs recomputation and a source trace; a safety-relevant analysis needs independent review. Uniform verification is either wasteful or negligent — usually both, in different places. "Proportionate" is only a defensible word if it can be derived, so the example below derives it.

WORKED EXAMPLE

Worked example 1.4.3 — how deep is proportionate, and what actually decides it.

1. **Setup.** Three AI-assisted artefacts on Meridian. A weekly internal **status summary**: a glance costs 0.5 hours at USD 85 (**USD 42.50**); a downstream control — the delivery meeting, where four people who know the work read it aloud — catches a material error with probability **0.90**; an uncaught error costs about **USD 5,000** in a week of misdirected attention. A **board benefits figure**: recomputation plus a source trace costs 6 hours at USD 85 (**USD 510**); once a paper is tabled the chance of anyone catching the error is **0.25**; an uncaught error carries the output-based overstatement of 1.3.2, **USD 293,760** a year. A **clinical-safety impact assessment** for the prescribing module: independent clinical review costs **USD 12,000**; the chance of another control catching the error is **0.05**; the modelled consequence is **USD 4,000,000**.
2. **Formula.** Verification is worth performing when its cost is below the expected loss it removes: `C_v ≤ P(error) × (1 − q) × L`, where `q` is the probability a downstream control catches the error anyway and `L` the loss if it flows through. Rearranged, the **breakeven error probability** is `P* = C_v ÷ [(1 − q) × L]` — the error rate above which the check pays.
3. **Substitution.** Summary `42.50 / (0.10 × 5,000)`; board figure `510 / (0.75 × 293,760)`; safety assessment `12,000 / (0.95 × 4,000,000)`.
4. **Result.**

Artefact	Verification C_v	q	Loss L	Exposed loss (1−q)L	Breakeven P*
Weekly status summary	42.50	0.90	5,000	500.00	**8.5000 %**
Board benefits figure	510	0.25	293,760	220,320	**0.2315 %**
Clinical-safety assessment	12,000	0.05	4,000,000	3,800,000	**0.3158 %**

1. **Interpretation.** Five readings. The third is the counter-intuitive one and the fifth is the limit of the method.

The breakevens differ by a factor of 36.72, and that is what "proportionate" means. A glance at the summary is justified only if such summaries are wrong more than **8.5 %** of the time — which, for AI-drafted narrative over live data, they plainly are, so a glance is warranted and a glance is all that is warranted. The board figure must be verified if the error rate exceeds **0.2315 %**, which is to say: always. Proportionality is not a sliding scale of diligence applied by feel; it is the observation that the same duty produces very different depths once the exposure is written down.

Uniform depth is arithmetically absurd, not merely wasteful. Apply the board paper's six-hour recompute to the weekly summary and the breakeven becomes `510 / 500 =` **1.02** — a required error probability **above 1**, meaning **no error rate whatever could justify it**. That is a sharper statement than "wasteful": it is a check that cannot be justified on any assumption. Meanwhile the same six hours applied to the safety assessment would not merely be too shallow, it would be the wrong kind of check — recomputation cannot find a clinical misjudgement. Depth and kind are separate choices, and conflating them is how organisations produce thick verification files that miss the thing that matters.

The cost of checking barely matters; the exposed loss decides everything. The safety review costs **23.53 times** the board recompute, yet its breakeven is only **1.36 times** higher — the two are effectively in the same band, while the trivial artefact sits two orders of magnitude away. The discriminating variable is `(1 − q) × L`, not `C_v`. The practical consequence is that the familiar objection — "we cannot afford to check everything properly" — is answering the wrong question: on anything with material downstream reliance, the check is cheap relative to the exposure, and the real constraint is attention, not money.

`q` **is where this arithmetic is most often abused.** A high `q` makes any check unnecessary, so a `q` is asserted rather than evidenced far more often than an `L` is. A reviewer's question is therefore not "what is your verification policy?" but which named control would have caught this, and when has it caught anything? On the summary, `q = 0.90` is credible only because the delivery meeting genuinely reads it; remove that meeting and the breakeven falls to `42.50/5,000 =` **0.8500 %**, a tenfold increase in the case for checking, with no change to the artefact at all.

The limit, stated plainly: row three should not be decided this way. Where a statutory duty, an operating licence or patient safety is engaged, the obligation to review is not contingent on an expected-value test, and computing `P*` there has exactly one legitimate use — to show that the review is also cheap, which forecloses the cost argument. It must never be used to conclude that a safety review is not worth performing. What constitutes an adequate assessment, who is competent to sign it, and what must be retained are set by the applicable regulatory regime and the relevant professional body in the jurisdiction; take that from the regulator and from counsel, not from this table (Domain 14, KA 14.4 for the governance architecture, and Domain 9 for the assurance regime). Two further limits: expected-value reasoning is the wrong frame for a single catastrophic tail even where safety is not engaged, and this model prices one artefact at a time, whereas AI-assisted work arrives in volume — sizing a sample across many artefacts is Domain 14's problem (KA 14.3), and it starts where this example stops.

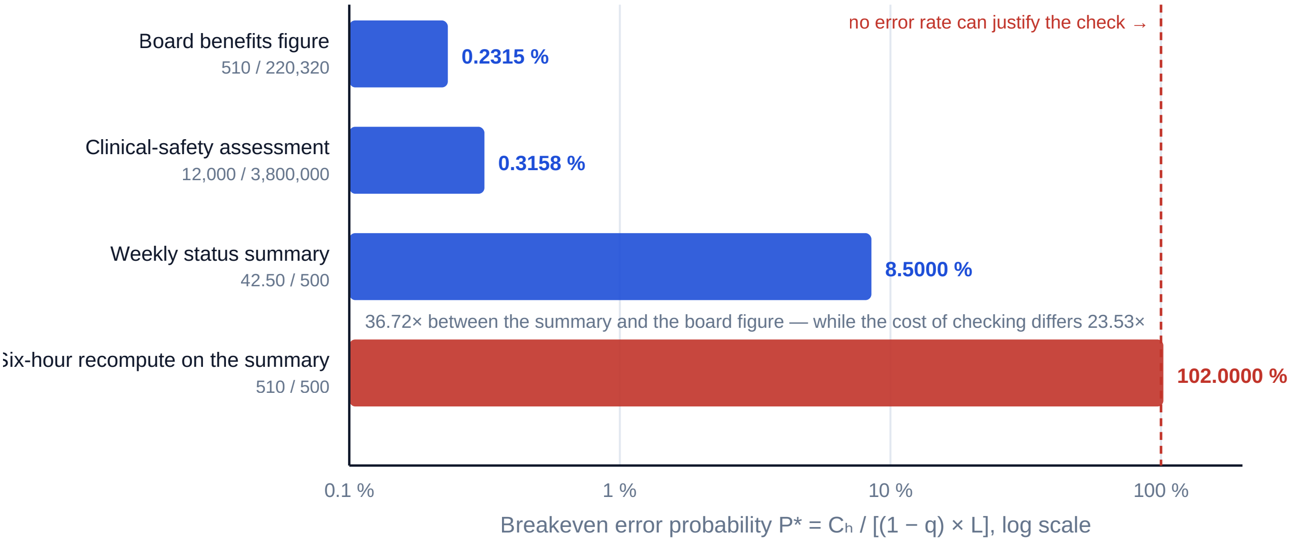


Fig 1.4.1 — Proportionate verification: the breakeven error probability

Protect the team's judgment. The subtler risk is not a wrong output but the atrophy of the skills needed to notice one. A leader who accepts AI-drafted plans no one on the team can critique has traded capability for speed, and will discover the price during the first recovery (Domain 6, KA 6.4).

KEY TERMS — KA 1.4

Term	Meaning
Daylight test	Would every party, seeing the full picture, still regard this as impartial?
Hallucination	Fluent, confidently-stated content that is false.
Over-trust through fluency	Polished output suppressing the scrutiny rough output would attract.
Verification proportionality	Depth of checking matched to the stakes of reliance.
Breakeven error probability (`P*`)	`C_v ÷ [(1 − q) × L]` — the error rate above which a verification pays for itself.
Exposed loss	`(1 − q) × L` — the loss a verification is actually protecting against, after any downstream control.
Depth and kind	Two separate verification choices: how hard to look, and what sort of looking finds this class of error.
Named owner	The human who verified an AI-assisted artefact and answers for it.

SAMPLE MCQS — KA 1.4

MCQ 1.4-A `[1.4.1 · Analysis]` A sponsor asks that the optimistic scenario be presented as the base case "to keep the board confident". The professional response is: - A. comply — scenario labels are presentational - B. decline, and offer the honest base case with sensitivities and the evidence that supports confidence - C. comply while noting the true base case privately - D. present both without indicating which is the base case

Rationale: Candour about status is a duty, not a style (1.4.1). C documents the misrepresentation without preventing it; D abdicates the professional judgment the board is relying on.

MCQ 1.4-B `[1.4.2 · Application]` Which AI failure mode most directly explains an authoritative-sounding schedule that encodes assumptions nobody made? - A. confidentiality leakage - B. over-trust through fluency, compounded by absent verification - C. model bias - D. hardware limitation

Rationale: Polished output invites less scrutiny (1.4.2), and the missing control is verification. A concerns data egress, C concerns systematic skew (real, but not this symptom), D is irrelevant.

MCQ 1.4-C `[1.4.3 · Analysis]` A leader requires the same verification depth for every AI-assisted artefact, from internal summaries to safety analyses. This is: - A. best practice — consistency is the point - B. simultaneously wasteful and negligent: proportionality means depth matched to the stakes of reliance - C. acceptable only if the team is small - D. unnecessary, since approved tools need no verification

Rationale: Uniform depth over-checks the trivial and under-checks the critical (1.4.3). A mistakes uniformity for rigour; D abandons the duty entirely.

MCQ 1.4-D `[1.4.3 · Application]` Recomputing an AI-drafted board benefits figure costs USD 510. An error surviving into the paper would carry an overstatement of USD 293,760, and the chance of anyone catching it after tabling is 0.25. The breakeven error probability above which the recompute pays is: - A. 0.1736 % - B. 0.2315 % - C. 0.6944 % - D. 0.1302 %

Rationale: `510 / (0.75 × 293,760) = 0.2315 %` (1.4.3). A divides by the full loss and forgets that only 75 % of it is exposed; C multiplies by 0.25 instead of (1 − 0.25), using the catch probability as if it were the escape probability; D applies the escape fraction to the verification cost (`510 × 0.75`) rather than to the loss, which is the same slip in the opposite direction.

MCQ 1.4-E `[1.4.3 · Evaluation]` A safety review costing USD 12,000 has a breakeven error probability of 0.3158 %, while a USD 510 recompute has one of 0.2315 %. The most useful professional inference is: - A. the cheaper check is better value and should be preferred - B. the cost of checking is nearly irrelevant to whether to check — a 23.53-fold cost difference produces only a 1.36-fold difference in breakeven, because the exposed loss dominates - C. both checks are unnecessary, since the breakevens are below 1 % - D. the safety review should be dropped, as its expected value is the weaker of the two

Rationale: `P* = C_v/[(1 − q)L]`, and the exposed losses differ by more than the costs do (1.4.3). A compares the wrong quantity — the checks protect different exposures; C inverts the meaning of a breakeven, which is a threshold to exceed, not a hurdle to fail; D applies an expected-value test to a safety decision, which the example expressly rules out.

SELF-CHECK — KA 1.4

1. State the daylight test. — Would every party, seeing the full picture, still regard this as impartial?
2. Which AI failure mode is hardest to counter, and why? — Over-trust through fluency: polish suppresses the scrutiny that rough work attracts.
3. What does "protect the team's judgment" mean in practice? — Do not accept AI output the team cannot critique; capability traded for speed is repaid during recovery.
4. State the breakeven error probability and say which of its inputs decides the answer. — `P* = C_v / [(1 − q) × L]`; the exposed loss `(1 − q)L` dominates, and the cost of checking barely moves it (1.4.3).
5. What does a breakeven above 100 % tell you? — That no error rate could justify that check on that artefact: it is unjustifiable rather than merely uneconomic (1.4.3).

6. Where must this arithmetic not be the deciding test? — Where a statutory duty, licence or safety is engaged; there its only legitimate use is to show the review is also cheap (1.4.3).

Advanced topics — Domain 1

1.A.1 Authority, influence and the borrowed team

Most project leaders hold less authority than accountability — the defining asymmetry of the role. The sources of influence that actually work in temporary organisations: **competence** (visible command of the work), **reciprocity** (having helped others deliver), **information** (being the person with a true picture), **coalition** (sponsor and peer support secured before it is needed), and **fairness** (a record of allocating credit and blame accurately). Positional authority is the weakest of the six in borrowed teams, because the borrowed people's careers are decided elsewhere. The practical implication: influence is built in advance, and a leader who begins building it during a crisis is too late.

1.A.2 Success criteria and the multiple-verdict problem

Projects are judged simultaneously against delivery (time, cost, quality), outcome and benefit realisation, and stakeholder satisfaction — which is why the same project is often described as a success and a failure by different parties, honestly. The professional response is not to pick a flattering verdict but to **agree the criteria in advance, with owners and measures** (Domain 2's benefits mapping, Domain 5's acceptance criteria), and to report against all of them. Unagreed success criteria are settled retrospectively by whoever is most disappointed.

1.A.3 Dissent, compliance and the record that protects everyone

The obligation set of 1.2.2 has a case it does not cover: the leader escalates with options, the accountable authority chooses the option the leader advised against, and the leader must now deliver it. This is common, it is not misconduct by anyone, and handling it badly damages both the project and the leader.

Three positions are available and only one of them is professional. **Silent compliance** protects the relationship and destroys the record: when the risk materialises there is no evidence the question was asked, and the leader who warned privately is indistinguishable from the leader who did not notice. **Obstruction** — delivering the decision unenthusiastically, slowly, or with visible reservation to the team — is a breach of the duty to the sponsor and, because a team reads its leader accurately, converts a difficult decision into a failing one. **Dissent and comply** is the professional position: the disagreement is stated once, in writing, to the person who holds the authority; the reasoning and the condition that would change the answer are recorded in the decision log (Domain 3, KA 3.3.4); the leader then implements the decision as if it were their own, including to the team; and the recorded condition is monitored, so that if it occurs the matter returns automatically rather than as an accusation.

What makes this work is that the record is **neutral in tone and specific in content**: the option chosen, the option advised, the reason, and the observable that would reopen it — for example, on Meridian, "proceed to full rollout without funded adoption support; the writer's advice was to fund champions first; reopen if adoption at week 13 is below 55 %." That sentence would have converted the 26-week silence of 1.2.2 into an automatic trigger, and it costs one line. A record written to apportion blame in advance ("for the avoidance of doubt, I do not accept responsibility for...")

achieves the opposite: it invites the authority to stop asking for advice, and it is usually worth nothing anyway, because accountability is not reallocated by assertion (1.2.1).

Two boundaries. First, dissent and comply applies to decisions within the range of legitimate judgment. It does **not** apply where the instruction is unlawful, where safety or a regulatory obligation would be breached, or where the leader is being asked to make a statement they know to be untrue (1.4.1) — those are refusal-and-escalate situations, and they are the reason a profession has ethics rather than only a service ethic. Second, the protections available to someone who raises such a matter, the routes that count as a protected disclosure, and any obligation to report externally are entirely jurisdiction- and sector-specific, and can turn on details of how and to whom the disclosure was made. Before acting on that class of concern, take advice — from counsel, from the relevant professional body, or from the organisation's designated route — rather than from any general account, including this one.

1.A.4 The reviewer's leadership eye

Invariants an experienced reviewer tests in the first hour on any project: exactly one accountable name per outcome; a benefits chain where each claimed benefit traces to a measured outcome with an owner; decision records that show what was known when; escalations that arrived early enough to matter; forecasts with ranges rather than single numbers; and AI-assisted artefacts carrying a named verifier. Every one of these is cheap to check and expensive to have missing — the same posture Domains 6 and 7 apply to schedules and cost.

Four questions convert that list into arithmetic in the same hour, and each has a one-line answer or a finding. Does the business case contain an adoption term? — if not, the overstatement is already known to be the non-adoption rate (1.3.2). What operational condition does the approval hurdle imply? — a payback rule restated as an adoption percentage often turns out to be a commitment nobody made (1.3.2b). What is the cost of delay per week, and who computed it? — a project that cannot state it cannot price a single timing decision, including its own escalations (1.3.3). For the last AI-assisted artefact that mattered, what was the exposed loss and which named control was relied on? — the answer distinguishes a verification regime from a verification policy (1.4.3). A reviewer who asks those four learns more in ten minutes than a document review yields in a day, because each of them is answerable only by someone who has actually done the arithmetic.

Industry variations — Domain 1

- **Public programmes.** Accountability is doubled — professional and political — and the public obligation of 1.2.2 is enforceable through scrutiny bodies; benefits (not outputs) are the currency of accountability, and announced dates often precede plans (Domain 6's Case B).
- **Regulated industries (pharma, aviation, nuclear).** The standard of care is codified and auditable; the "care exercised" defence of 1.2.3 is a documented reality rather than an argument, and AI use faces explicit validation expectations.
- **Construction and engineering.** Authority is contractual as much as organisational; influence runs through commercial relationships, and the leader's obligation set includes site safety as a first-order duty.
- **Technology and product.** Outcome ownership frequently sits with a product role rather than the delivery leader, making the 1.3.2 chain a shared accountability that must be explicitly split — the commonest source of "who owns adoption?" disputes.

- **Professional services and consulting.** Duties to the client organisation and to the end-users can diverge, and independence (1.4.1) becomes a standing rather than occasional test.

Case study — Domain 1: Meridian under scrutiny (public health)

Situation. Eighteen months in, Meridian Care Records reports 40 of 40 clinics installed, on time, 3 % under budget. The programme is publicly praised. Nine months later a health-service audit finds clinician hours released are running at roughly a third of the business case, and the programme is publicly described as a failure. Both accounts cite true facts.

Analysis. The delivery verdict was sound: the outputs landed. The benefits case had been written against **outputs** — all 40 clinics, no adoption term — so it claimed USD 979,200 per year where the honest figure at 70 % adoption was USD 685,440, and actual adoption reached only about 40 %, giving roughly USD 391,680. Two failures compounded: an arithmetic one (skipping the outcome link, 1.3.2) and an accountability one (**nobody owned adoption** — the delivery leader owned installations, the clinical directorate owned practice change, and no single name owned the measure that connected them, 1.2.1).

The account in numbers. Both public verdicts were arithmetically defensible, which is the whole difficulty.

Line	Figure	Where it comes from
Approved cost	USD 2,400,000	The business case
Delivery verdict — 3 % underspend at installation	**+USD 72,000**, once	WE 1.1.1
Benefit claimed in the case (output basis, no adoption term)	USD 979,200 a year	1.3.2
Honest benefit at the case's own 70 % adoption	USD 685,440 a year	1.3.2
Benefit at the 40 % adoption actually measured	USD 391,680 a year	WE 1.1.1
Programme verdict — annual shortfall against the honest case	**(USD 293,760) a year**	WE 1.1.1
Scale of the two verdicts	shortfall = **4.08 ×** the underspend, in year one alone	WE 1.1.1
Weekly form of the shortfall	**USD 6,120 a week** = 12 clinics × 6 h × USD 85	WE 1.2.2
Simple payback, honest benefit against approved cost	**3.5014 years**	WE 1.3.2b
Simple payback at the measured 40 % adoption	**6.1275 years**	WE 1.3.2b

Two figures in that table explain the collapse of the public narrative. The programme was praised on a line worth 72,000 once and condemned on a line worth 293,760 a year, and **nobody had put the two in the same table** — which is a reporting failure, not a delivery one. And the payback the board believed it had approved, 2.4510 years on the output-based claim, became 6.1275 years on measured adoption: the same programme, the same installations, an investment case two and a half times longer.

What was done. The recovery was not technical. Adoption was made a named accountability with a monthly measure; the benefits case was restated with an adoption term and a sensitivity range (50/70/90 % → 489,600 / 685,440 / 881,280); clinical champions were funded in the eleven lowest-adoption clinics at USD 12,000 each, **USD 132,000** in total; and the programme board's report changed from "clinics live" to "clinics using, and hours released". Adoption reached 68 % within a year, recovering **11.2** of the 12 missing clinics — **USD 5,712 a week**, or **USD 274,176** a year.

What the timing cost. The monthly measure had shown 16 of 40 clinics in daily use at **week 13**; the matter reached the board at **week 39**, through the audit. On WE 1.2.2's arithmetic those 26 weeks were worth **USD 148,512 — 1.13 times** the entire remedy they would have funded, and more than twice the 72,000 underspend the programme had been congratulated for. The remedy needed only **23.11 weeks** of earliness to pay for itself outright.

And the underspend did not survive. Domain 16's closing account records a final capital outturn of **USD 2,514,000** against the **USD 2,328,000** implied by the 3 % underspend — a **USD 186,000** swing, most of it the recovery described above. The celebrated cost result was an interim figure that the programme later took back, which is the general lesson: **an underspend booked before the outcome is measured is a loan against the recovery, not a saving.**

What the domain teaches here. Every element of this domain appears: outputs are not benefits (1.3.2); one accountable name per outcome (1.2.1); success criteria agreed in advance or settled by the disappointed (1.A.2); and the leader's attention belongs where the value actually leaks — which the arithmetic, not instinct, identified.

Case study B — Domain 1: the plan nobody could critique (financial services)

Situation. A platform-migration leader under time pressure used an AI planning assistant to generate the full delivery schedule and dependency network, reviewed it briefly, and issued it. It was detailed, internally consistent and professionally formatted. It also assumed the legacy data extract could run concurrently with the schema freeze — an assumption no one on the team had made, and which the two engineers who would have spotted it never saw.

What happened. The conflict surfaced eleven weeks later during rehearsal, costing a seven-week slip and a weekend of reputational damage with the regulator's supervisory team. The post-incident review found no fault in the tool: the schedule was a reasonable inference from an under-specified prompt. The failures were the leader's — no verification against the discipline leads (the 1.1 principle and 1.4.2 obligation), and no named verifier on the artefact (1.4.3). The compounding finding was cultural: the polish of the output had actively discouraged challenge, and two team members later said they had assumed the network "must have been checked".

What it cost, and what the check would have cost. This programme's cost of delay was **USD 38,000 a week** — dual-running licence and support on the legacy platform at **USD 12,000**, plus **USD 26,000** of deferred decommissioning saving — so the seven-week slip cost **USD 266,000** before any regulatory consequence. The verification that would have found the conflict was two discipline leads reading the dependency network for four hours each: `2 × 4 × 145 =` **USD 1,160**. The loss was therefore **229.31 times** the check. Put through KA 1.4.3's test, the breakeven error probability was `1,160 / 266,000 =` **0.4361 %**: the review paid for itself if an AI-generated dependency network carried one material error more than about four times in a thousand. It is not necessary to know the true rate to make that decision — only to observe that no honest estimate of it is that low.

Two details make this the general case rather than an unlucky one. First, the breakeven sits in the same band as the board-paper and safety-review rows of 1.4.3 (**0.2315 %** and **0.3158 %**), which is what one should expect: wherever an artefact carries material downstream reliance, the check is trivially cheap against the exposure, and the argument that verification is unaffordable does not survive being written down. Second, the eleven weeks between issue and discovery are 1.3.1's delay property doing its work — the seven-week loss was locked in from the day the plan was issued, and the rehearsal that finally revealed it was the only control in the chain, with the schedule freeze already committed by the time it fired.

What the domain teaches here. Fluency is not evidence. Accountability did not move to the vendor or the model — it stayed with the person who issued the plan (MCQ 1.2-B is this case in miniature). And the durable fix is structural, not exhortative: a named verifier per artefact, discipline leads walking their own links, a team kept capable of critique — and a rule that no artefact whose exposed loss exceeds a stated figure is issued without a recorded check, because at 229 to 1 the decision is not close enough to require judgment.

Executive perspective — Domain 1

What a project leader cannot delegate in this domain:

- **The single accountable name.** For every outcome, including the awkward ones like adoption. Where two names appear, the leader's job is to reduce it to one.
- **The benefits chain's honesty.** Each claimed benefit traced to a measured outcome with an owner. A leader who lets outputs be reported as benefits has authorised the failure of Meridian's first eighteen months.
- **The obligation set, held openly.** Sponsor, team, users, public — with conflicts escalated with options rather than absorbed in silence.
- **Bad news made safe.** The single highest-leverage cultural act available, because it converts irreversible surprises into recoverable problems — and on Meridian it was worth **USD 148,512**, 1.13 times the remedy it would have funded, for the price of one message (1.2.2).
- **The inversion of every hurdle the organisation imposes.** A three-year payback rule is an **81.70 %** adoption commitment; a delegation threshold is an escalation volume; a gate is a priced delay. The director's contribution is insisting that hurdles be restated as the operational conditions they imply, because that is the form in which someone can own them (1.3.2b).
- **The AI accountability line.** Named verifiers, proportionate depth, and a team that remains able to critique what the tools produce. Proportionate is a derivable word: the exposed loss `(1 − q) × L` decides the depth, and a check whose breakeven error probability exceeds 100 % is one the organisation should stop performing (1.4.3).

Calculation exercises — Domain 1

> **Exercise 1.1** A programme installs a system in 60 sites; adoption is 65 %; each adopting site releases 5 hours/week at USD 90/hour over 46 weeks. Compute the defensible annual benefit and the overstatement in an output-based claim. Solution. Adopting sites `60 × 0.65 = 39`; benefit `39 × 5 × 90 × 46 =` **USD 807,300**. Output-based `60 × 5 × 90 × 46 =` **USD 1,242,000**; overstatement **USD 434,700** = **35.0 %**, exactly the non-adoption rate. Common error: applying adoption to the rate

rather than the site count — the same product here, but it conceals which link failed and misdirects the fix.

Exercise 1.2 Using Meridian's figures (28 adopting clinics, 6 h/week, USD 85/h), compute the cost of a 5-week delay and the maximum justified acceleration spend for 5 weeks. Solution. Per week $28 \times 6 \times 85 =$ **USD 14,280**; five weeks **USD 71,400** — which is also the maximum justified spend, since above it acceleration destroys value. Common error: comparing the spend against annual benefit rather than the benefit for the weeks actually saved.

Exercise 1.3 Meridian's benefit at 50 %, 70 % and 90 % adoption (40 clinics, 6 h/week, USD 85/h, 48 weeks). What does the spread imply for where the leader spends attention? Solution. **USD 489,600 · 685,440 · 881,280**. The 40-point adoption swing moves annual benefit by **USD 391,680** — far more than plausible schedule compression could add (Exercise 1.2 values 5 weeks at 71,400). Attention belongs on adoption (Domain 11), not acceleration.

Exercise 1.4 A programme is approved at USD 1,800,000 and delivers 2 % under budget. Its full-potential benefit is USD 540,000 a year; the case assumed 75 % adoption and measurement finds 45 %. Report both verdicts and their scale, and state what may not be done with them. Solution. Delivery verdict $1{,}800{,}000 \times 0.02 =$ **+USD 36,000**, once. Benefit at plan $540{,}000 \times 0.75 =$ **USD 405,000**; at outturn $540{,}000 \times 0.45 =$ **USD 243,000**; programme verdict **(USD 162,000) a year**, equivalently $540{,}000 \times (0.75 - 0.45)$. The shortfall is **4.50 times** the underspend in the first year alone. What may not be done: netting them into a single figure — one is a stock, the other a flow, and a net requires a horizon and a discount rate from the business case. Common error: reporting "the programme is 126,000 down" ($162{,}000 - 36{,}000$), which subtracts an annual flow from a one-off variance and produces a number that means nothing.

Exercise 1.5 An investment of USD 3,150,000 has a full-potential benefit of USD 1,050,000 a year and assumes 60 % adoption. The sponsor applies a four-year simple payback rule. Compute the payback as assumed, and the adoption the rule requires. Solution. Benefit $1{,}050{,}000 \times 0.60 =$ **USD 630,000**; payback $3{,}150{,}000 / 630{,}000 =$ **5.00 years**, which fails the rule. The rule needs $3{,}150{,}000 / 4 =$ **USD 787,500** a year, so adoption of $787{,}500 / 1{,}050{,}000 =$ **75.00 %** — fifteen percentage points above the assumption, and the sentence the approval meeting should have heard. Common error: dividing cost by potential benefit ($3{,}150{,}000/1{,}050{,}000 =$ 3.00 years) and reporting a pass; that is the output-based payback, understated by the factor $1/0.60 = 1.6667$.

Exercise 1.6 A crew of 5 completes 0.08 units a week each. Two people are added; they take a 3-week ramp at 20 % productivity and each absorbs 60 % of an existing member's time. Compute the crossover horizon and say whether a trough occurs. Solution. Supervision loss $2 \times 0.6 \times 0.08 = 0.096$; newcomer output $2 \times 0.2 \times 0.08 = 0.032$; deficit $d = 0.064$ a week; post-ramp gain $g = 2 \times 0.08 = 0.16$ a week. Crossover $R \times (1 + d/g) = 3 \times (1 + 0.4) =$ **4.20 weeks**. A trough does occur, because supervision load per newcomer (60 %) exceeds ramp productivity (20 %); with fewer than 4.20 weeks of work remaining the reinforcement finishes later. Common error: comparing capacity before and after (0.40 against 0.56 a week) and concluding the addition helps whenever work remains — which never computes the crossover and so cannot distinguish the two cases.

Exercise 1.7 Verifying an AI-drafted forecast costs USD 720. An error surviving into the decision would cost USD 250,000, and an existing control catches such errors with probability 0.40. Compute the breakeven error probability, and state what would change it most. Solution. Exposed

> loss `(1 − 0.40) × 250,000 =` **USD 150,000**; breakeven `720 / 150,000 =` **0.48 %**. What changes it most is the control: removing it raises the exposed loss to 250,000 and lowers the breakeven to **0.288 %**, while halving the verification cost only moves it to 0.24 % — so evidence for `q` matters about as much as the price of checking. Common error: dividing by the full loss and reporting 0.288 %, which is arithmetically the same as assuming no control exists; the error understates the breakeven and so overstates the case for checking, and it is the mirror of the commoner abuse in the other direction — asserting a high `q` for a control nobody has named.

Practitioner's toolkit — Domain 1

Adoption-ready artefacts; adapt headings to your organisation, then keep them stable.

Toolkit 1.T.1 — Leader's accountability map (one page)

Per outcome: the outcome stated in outcome language (not output language) · **the single accountable name** · responsible parties · the measure that evidences it and its owner · the decision authority and threshold (Domain 3) · review cadence. Rule: if any row has two accountable names or an output in the outcome column, it is not finished.

Toolkit 1.T.2 — Benefits-chain test (run on any benefits claim)

- [ ] Each claimed benefit names the measured **outcome** that produces it.
- [ ] Each outcome measure has an owner and a reporting cadence.
- [ ] Adoption (or the equivalent behavioural change) appears explicitly, with a value.
- [ ] The claim carries a sensitivity range, not a single number.
- [ ] Output milestones are labelled as milestones, never as benefits.
- [ ] Cost of delay per period is stated, so timing decisions can be priced (1.3.3).

Toolkit 1.T.3 — AI-use and verification record

Per AI-assisted artefact: tool and approved environment · data classification cleared · prompt/intent summary · **verification performed and its depth** · **named verifier** · disclosure status in the deliverable · date. Standing agenda item, not an archive: the register's purpose is that the next artefact is verified, not that the last one was.

Toolkit 1.T.4 — Verification-depth card (one per artefact class, not per artefact)

Six fields, filled once for a class of artefact and reused: the class · what relies on it · the **loss** `L` if a material error flows through · the **named downstream control** and its assumed catch rate `q`, with the evidence for that rate · the verification's **cost** `C_v` and **kind** (glance · recompute and source trace · independent review of a different discipline) · the resulting **breakeven** `P* = C_v/[(1−q)L]` · the decision, and who verifies.

Three rules that make it usable rather than decorative. **A `q` with no named control is recorded as zero** — this is the single field that determines whether the card is honest. **A class whose `P*` exceeds 100 % has its check removed, not reduced**, and the removal is recorded, because a check no error rate can justify is consuming attention that a real exposure needs. And **any class where a statutory duty, licence or safety is engaged is marked** `mandatory — not an expected-value decision`, with the `P*` retained only as evidence that the review is cheap (1.4.3). Review the card when the tooling changes, not when the artefact changes.

Exam preparation — Domain 1

The traps. Treating outputs as benefits (1.3.2 — the domain's central trap, and Meridian's whole case study) · saying accountability is "shared" (1.2.1) · assigning accountability to a vendor or a tool (MCQ 1.2-A, 1.2-B) · comparing an acceleration cost against annual rather than saved-period benefit (Exercise 1.2) · reading a delivery success as a programme success (1.1.1) · accepting "we absorbed it" (1.3.1) · uniform AI verification depth (1.4.3) · judging the standard of care by outcome alone (1.2.3) · **netting a one-off capital variance against an annual benefit flow** (WE 1.1.1, Exercise 1.4) · **valuing early warning at the size of the problem rather than at the run rate the remedy recovers** (WE 1.2.2 — 6,120 instead of 5,712) · **dividing cost by potential rather than realistic benefit when computing payback**, which understates it by exactly `1/a` (WE 1.3.2b, Exercise 1.5) · **approving a reinforcement because it finishes earlier, without testing it against the cost of delay** (WE 1.3.3b — earlier and worse off are compatible) · **omitting the downstream control from a breakeven error probability**, or asserting one that has no name (WE 1.4.3, Exercise 1.7).

The calculations to be able to do under time pressure. Annual benefit `= sites × adoption × hours × rate × weeks`, and the overstatement of an output-based claim as `1 − a`. Cost of delay `= adopting sites × hours × rate`, and breakeven weeks `= cost ÷ cost of delay`. Simple payback `= cost ÷ (potential × adoption)`, and the adoption a payback target `T` requires `= cost ÷ (T × potential)`. Crossover horizon `= R × (1 + d/g)`. Breakeven error probability `= C_v ÷ [(1 − q) × L]`. Five formulae, all of them one line, and between them they price every decision this domain contains.

Reflection questions. 1. Take your current project: write the single accountable name for its principal outcome. If you hesitated, or wrote two, what does that predict? 2. For each benefit in your business case, which measured outcome produces it and who owns that measure? What percentage of your claimed value survives the question? 3. Which AI-assisted artefact in your last month of work had no named verifier — and what would have caught it if it had been wrong?

Domain 1 summary

Project leadership is a distinct profession because temporary work fails distinctively: problems discovered late become irreversible, teams have no accumulated trust to draw on, and lead time is the cheapest resource available and is spent whether used or not. The discipline rests on a precise view of accountability — responsibility is delegable, the obligation to answer is not, one name per outcome — held across four directions at once (sponsor, team, users, public) with conflicts escalated rather than absorbed, and judged by the care a competent practitioner would exercise rather than by outcome alone. Two habits of mind make a leader rather than a coordinator: systems thinking, which expects feedback, delay, local optimisation and pressure that relocates instead of vanishing; and the outputs–outcomes–benefits–value chain, which Meridian turns into arithmetic — 40 clinics installed produce USD 685,440 of annual benefit at 70 % adoption, not the USD 979,200 an output-based claim asserts, and the 30 % gap is the outcome link being skipped, a proportion that is exactly the non-adoption rate whatever the hours or the rate. Against the approved cost of USD 2,400,000 that benefit repays in 3.5014 years, and a sponsor's three-year payback rule is therefore an undisclosed commitment to 81.70 % adoption — the habit of inverting a hurdle into the condition it implies being one of the most useful in the book. Timing decisions are priceable (USD 14,280 per week; acceleration breaking even at 4.20 weeks and paying at any adoption above 36.76 %), delay itself is priceable (a measure left silent for 26 weeks cost USD 148,512, more than the remedy it would have funded), and reinforcement is priceable in both directions — 2.7778 weeks earlier and USD 23,333.33 worse off, with a crossover horizon of 5.00 weeks computable before the decision.

Attention belongs where value actually leaks. Around all of it sits professional ethics and the suite's governing principle — **AI proposes; the professional verifies, decides and remains accountable** — with named verifiers, a team kept capable of critique, and a depth of checking that is derived rather than felt: the breakeven error probability `C_v/[(1−q)L]` puts a weekly summary at 8.50 % and a board benefits figure at 0.2315 %, a difference of 36.72 times that is the whole content of the word "proportionate". Domain 2 turns strategy into selected work; Domain 3 gives the accountability of this domain its governance machinery.

02

DOMAIN 2

Strategy, Selection and Business Alignment

Group: Leading projects (Domain 2 of 4 in Part One). **Target:** ~78 pages. **Binds to:** the PCI Book Pattern Specification and the shared registries (`docs/books/registries/`). This domain continues the **Meridian Care Records** programme from Domain 1 into its business case and benefits map, and bridges to the appraisal machinery the finance book builds in detail (PFL-AI, Domain 4). It uses Domain 3's governance-latency formula and Domain 8's expected-value machinery rather than restating them, and supplies the breakeven and assumption-exposure arithmetic that Domain 15 applies at portfolio scale and Domain 16 tests at realisation. British English; USD (+SAR where useful, indicative `USD 1 ≈ SAR 3.75`).

Why this domain exists

Domain 1 established that a project is judged on outcomes and benefits, not delivered outputs, and proved it arithmetically. That leaves the prior question unanswered: **how does work get chosen at all, and how does anyone know it was worth choosing?** This domain answers it. It sets the strategic context and treats alignment as a test that must be re-run rather than a box ticked at approval (KA 2.1); builds the business case as a decision instrument, with the options analysis and appraisal that make it one, and the selection models that rank competing candidates (KA 2.2); works benefits mapping, measurement baselines and the sustainability dimension, plus the assumption and dependency management on which every forecast rests (KA 2.3); and closes with the hardest leadership act in the discipline — **stopping** (KA 2.4). The through-line is that a business case is a promise about the future made to obtain money, and the professional question is always whether anyone will be held to it.

Learning objectives. After this domain a candidate can: explain how strategy becomes a portfolio and why alignment decays; **compute a strategy-portfolio alignment index and the reallocation distance it implies, on both a mapped and a total-spend denominator, and say which denominator the finding belongs to**; **convert an annual supersession hazard into a survival curve, an alignment half-life and an expected misaligned-spend figure, and price the re-test that removes it**; describe the components of a decision-grade business case; construct and appraise a genuine options set including the do-nothing baseline; **quantify a counterfactual do-nothing cost and show how omitting it understates a case, in the opposite direction to the flat-profile error**; build a benefits map from outputs through outcomes to benefits with owners and measures; **build a benefits profile that ramps rather than assuming steady state from day one, and quantify the overstatement a flat assumption produces**; **distinguish flat-equivalent from ramped-basis breakeven adoption and convert between them**; define measurement baselines, **separate an attributable improvement from a raw one with a comparison cohort, and state the invariant that fixes the over-claim share**; avoid double-counting; rank candidates with weighted scoring and under a binding constraint, and state each method's limits; **compute how**

far a criterion weight must move to reverse a ranking, and which criteria can never decide anything; **derive the marginal value of a unit of the binding constraint by enumeration and show why it is lumpy and non-monotone**; manage assumptions and dependencies as live risks, **price an assumption register against the NPV it supports and order the tests by exposure per unit of test cost**; apply sunk-cost-free reasoning to a continuation decision; **distinguish the forward breakeven at a gate from the whole-investment breakeven and calibrate a kill criterion against the cost of its two errors**; **decompose a staged commitment into the price of staging and the value of the abandonment option, and state the probability of bad news at which staging begins to pay**; and set kill criteria that make honest gates possible.

The master programme. Meridian Care Records continues from Domain 1 — the shared clinical-records rollout to **40 clinics**, whose verified benefit figures (full potential **USD 979,200** per year; **USD 685,440** at the realistic 70 % adoption) are now used to build the business case that should have been written for it.

KNOWLEDGE AREA 2.1

Strategic context and alignment

Topics: 2.1.1 from strategy to portfolio · 2.1.2 drivers, constraints and environment · 2.1.3 alignment as a repeated test.

2.1.1 From strategy to portfolio

Strategy states intent; a portfolio is what an organisation is actually doing about it. The gap between the two is where most strategic failure lives, and it is visible in three symptoms a leader should be able to name: **a portfolio that does not reflect the stated priorities** (the strategy says "digital first", the funded work is 80 % maintenance); **more work than capacity**, so everything is slow and nothing is finished (Domain 15's capacity management); and **no route for stopping**, so the portfolio only ever accretes.

The mechanism that closes the gap is a **portfolio process** with real teeth: candidate work described comparably, ranked against explicit criteria, funded to capacity rather than to appetite, and reviewed at intervals that allow re-decision. Where a project leader sits inside this, the obligation is specific — **describe your work honestly enough to be compared**, including its weaknesses, because a portfolio process fed by advocacy rather than evidence selects the best storyteller.

The first symptom is the only one that can be measured directly from the ledger, and measuring it is what converts a complaint into a governance paper.

WORKED EXAMPLE

Worked example 2.1.1 — the alignment index, and the money it says must move.

1. **Setup.** Meridian's parent health authority publishes four strategic objectives with declared weights — **access and capacity 40 %**, **safety and quality 25 %**, **digital foundation 20 %**, **cost efficiency 15 %** — and a discretionary programme budget of **USD 30,000,000** a year. The funded portfolio, read off the finance system and mapped objective by objective, is: access **4,800,000**; safety **6,600,000**; digital **1,800,000**; cost efficiency **4,800,000**; and **12,000,000** of mandatory sustainment (estate compliance, licence renewals, end-of-life replacement) that maps to no strategic objective at all. Compute how aligned this portfolio is, and what the answer implies in money.
2. **Formula.** For each objective, `funded share = spend ÷ denominator`. The **alignment index** is the overlap between the declared and funded distributions, `index = Σ min(declared weight_i, funded share_i)`. The **reallocation distance** is the money that must move to close the gap, `Σ over under-funded objectives of (declared − funded) × denominator`, which is identically `(1 − index) × denominator`.
3. **Substitution.** On the strategically mapped **18,000,000**: shares `4,800,000/18,000,000 = 26.6667 %`, `6,600,000/18,000,000 = 36.6667 %`, `1,800,000/18,000,000 = 10.0000 %`, `4,800,000/18,000,000 = 26.6667 %`. Overlap `min(40, 26.6667) + min(25, 36.6667) + min(20, 10) + min(15, 26.6667)`. On the full **30,000,000**, shares are 16 %, 22 %, 6 %, 16 % and 40 % unmapped, and the declared weight on sustainment is zero.
4. **Result.**

Objective	Declared weight	Funded share of mapped spend	Gap	Money
Access and capacity	40.0000 %	26.6667 %	**−13.3333 pts**	**(2,400,000)**
Safety and quality	25.0000 %	36.6667 %	+11.6667 pts	+2,100,000
Digital foundation	20.0000 %	10.0000 %	**−10.0000 pts**	**(1,800,000)**
Cost efficiency	15.0000 %	26.6667 %	+11.6667 pts	+2,100,000

Alignment index on mapped spend 76.6667 %; **on total discretionary spend 59.0000 %**. Reallocation distance **USD 4,200,000** — and the deficits (2,400,000 + 1,800,000) equal the surpluses (2,100,000 + 2,100,000) exactly, as they must. 5. **Interpretation.** Four readings, and the second is the one that decides whether this paper survives its first meeting.

The number is not the finding; the denominator is. The two indices differ by **17.6667 percentage points**, and the entire difference is the **40.0000 %** of discretionary money that serves no stated objective. Neither figure is wrong and neither is sufficient. Reported on mapped spend alone, the portfolio looks broadly aligned and the sustainment question never gets asked. Reported on total spend alone, the authority appears to be 41 % misaligned, which is unfair — sustainment is not misalignment, it is unstated strategy. The professional report gives both, and its recommendation is about the third figure: **either fund sustainment as a declared objective with a weight, or stop calling the other 60 % "the portfolio".** An organisation that will not put a weight on keeping the lights on has no portfolio process; it has a discretionary fund and a large unexamined subsidy.

The reallocation distance is the sentence that makes this actionable, and it has a scale a board already understands. USD 4,200,000 is **1.75 times** Meridian's entire approved cost of 2,400,000 — and the access deficit alone, 2,400,000, is exactly one Meridian. That is the comparison to put in the paper, because "the portfolio is 76.6667 % aligned" invites a debate about the measure while "the access objective is short by one programme the size of Meridian" invites a decision. The digital objective is funded at **50.00 %** of its declared weight, which is the arithmetic behind the classic symptom named above: the strategy says digital first and the money says otherwise, by a factor of two.

The identity is worth holding. Because the declared weights and the funded shares both sum to one, the positive gaps and the negative gaps are equal in magnitude, so `reallocation distance = (1 − index) × denominator` with no residue — here `0.233333 × 18,000,000 = 4,200,000`. One consequence: an index cannot be improved by adding money to a favoured objective without taking it from another, which is exactly the argument a portfolio board should be having and exactly the argument that "we will fund digital next year as well" avoids.

What the index cannot see, and where it is dangerous. It is a measure of money against declared intent, not of value, deliverability or benefit. A portfolio can score 100 % and consist entirely of failing projects; it can score 60 % and be the best available set because the under-weighted objective has no fundable candidates — which is a **strategic coverage** finding (2.A.2), not an allocation one, and the two are routinely confused. The declared weights are themselves a governance artefact, often written for a strategy document rather than as spending targets, so the first question on seeing a low index is whether the weights were ever intended to bind. And the mapping of programmes to objectives is a judgement made by someone: a programme claiming three objectives can be counted once, split, or double-counted, and the index moves materially with that choice. State the mapping rule with the index, or the number is unauditable. Finally, the index is a **snapshot of commitments already made**; it changes slowly, because most of the portfolio is mid-flight. The lever that actually moves it is intake and stopping (KA 2.4), not exhortation.

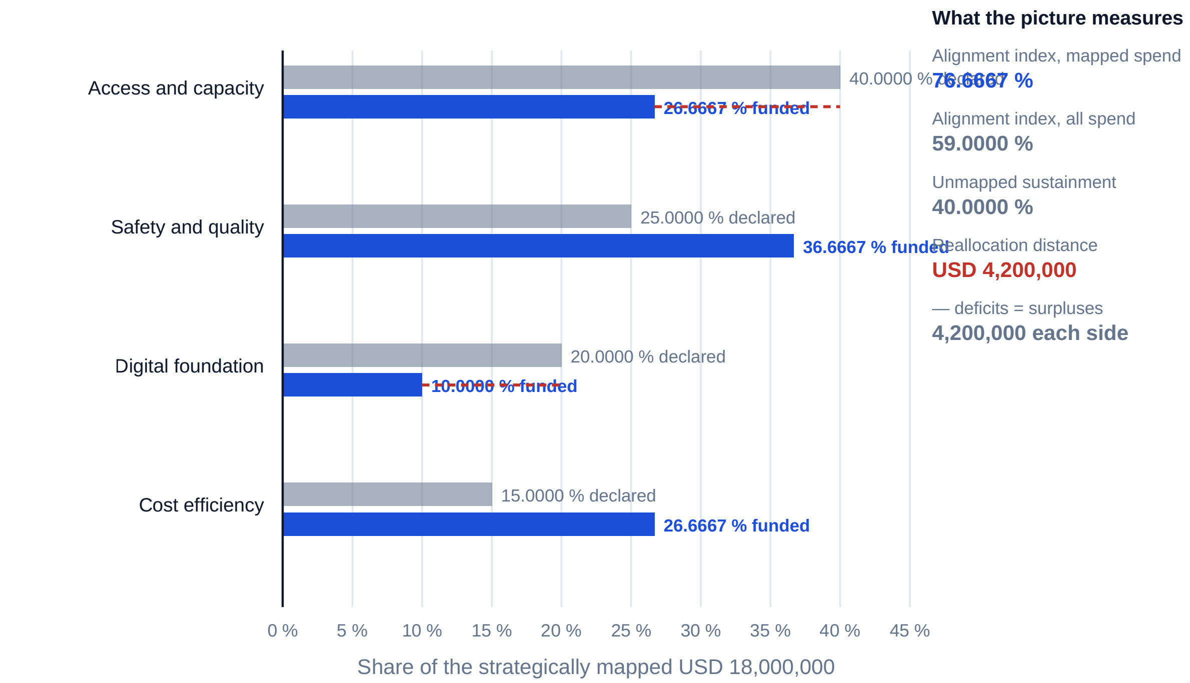


Fig 2.1.1 — The strategy-portfolio alignment index

Intake cadence is the other half of the process, and it has a price. A portfolio board that meets quarterly with a three-week paper deadline makes a candidate wait, on Domain 3's latency formula `E[wait] = M/2 + L` (KA 3.2.3), an expected `13/2 + 3 =` **9.5 weeks** before it can be decided at all; a monthly board with the same deadline makes it wait **5.0 weeks**. For a candidate of Meridian's shape, whose cost of delay Domain 1 fixes at **USD 14,280 a week**, that is a difference of `4.5 × 14,280 =`

USD 64,260 per candidate — paid before any work starts, and invisible in every business case because no case has a line for the time it spent in a queue. The counter-argument is real and must be stated: a board meeting more often decides on thinner evidence, and a bad selection costs far more than 64,260. The resolution is not a cadence but a **threshold** — small candidates decided continuously against pre-agreed criteria, large ones held for the full board — which is Domain 3's delegation design applied to intake rather than to escalation.

2.1.2 Drivers, constraints and environment

A business case that does not name its **driver** cannot be tested. The families, and what each implies:

Driver	Success looks like	Failure mode if misidentified
Compliance / regulatory	The obligation is met by the date	Gold-plating a mandatory minimum
Cost reduction	Measured, sustained cost decrease	Cost moved rather than removed
Revenue / growth	New or protected income	Cannibalised existing revenue counted as new
Capability / enabling	A later benefit becomes possible	Benefits claimed twice — here and downstream
Risk reduction	Exposure measurably lower	Unquantified "insurance" spending
Service / outcome improvement	Users measurably better served	Output delivered, outcome unmeasured (Domain 1)

Constraints — funding envelope, regulatory deadline, capacity, technical dependency — are not the same as objectives, and conflating them corrupts option generation: a "constraint" that is actually a preference silently eliminates the option that would have won. The professional habit is to write constraints down and mark each **hard** (physics, law, contract) or **soft** (preference, convention), because only soft ones can be traded.

2.1.3 Alignment as a repeated test

Alignment is granted at approval and **decays** thereafter: strategy moves, markets move, and the project's own understanding of what it can deliver moves. A project can be perfectly executed and strategically irrelevant by the time it lands — the most expensive form of success available.

Three practices keep alignment live. **Re-test at gates** — each stage gate (Domain 3, KA 3.3.1) asks not only "is delivery on track?" but "is this still the right thing?". **Track the assumption set** rather than the conclusion: business cases fail through their assumptions (KA 2.3.4), and an assumption that has been falsified invalidates the case whether or not delivery is green. **Name a benefits owner outside the project** — usually the operational leader who will run the changed service — because a project that owns its own benefits case will never be the one to report that it has stopped making sense (Domain 1's accountability rule; Domain 16's realisation).

Decay is usually discussed as a mood. It is a hazard, and a hazard has a half-life.

WORKED EXAMPLE

Worked example 2.1.3 — the alignment half-life, and what a re-test is worth.

1. **Setup.** The health authority's planning team, reviewing eight years of its own board minutes, assesses that a stated strategic priority is superseded — replaced, merged or de-prioritised — at a rate of about **15 % a year**. This is a **locally calibrated planning figure, not a constant**; in a regulated utility on a five-year price control it would be far lower, and in a consumer-facing technology business far higher. Meridian is a **three-year** delivery spending **USD 800,000** a year. Compute how long alignment survives, what spend is incurred after the driver has gone, and what an annual alignment re-test at each gate is worth if it detects a superseded driver **80 %** of the time.
2. **Formula.** Survival of alignment `S(t) = (1 − h)ᵗ`. **Alignment half-life** `t½ = ln(0.5) ÷ ln(1 − h)`. Probability the driver is superseded during year `k` is `(1 − h)^(k−1) × h`. Expected misaligned spend is the probability-weighted spend incurred after supersession and before detection; with detection probability `d` at each subsequent gate, a supersession missed at `j` consecutive gates costs the spend of those `j` years.
3. **Substitution.** `S(1) = 0.85`, `S(2) = 0.85² = 0.7225`, `S(3) = 0.614125`. `t½ = ln(0.5)/ln(0.85)`. Without a re-test: `0.15 × 1,600,000 + 0.1275 × 800,000`. With a re-test: year-1 supersession costs `0.8 × 0 + 0.2 × 0.8 × 800,000 + 0.2² × 1,600,000 = 192,000` in expectation; year-2 supersession costs `0.2 × 800,000 = 160,000`.
4. **Result.**

	Year 1	Year 2	Year 3	Year 4	Year 5
Probability the driver is still current	85.00 %	72.25 %	**61.41 %**	52.20 %	44.37 %

Alignment half-life 4.2650 years. Expected misaligned spend with **no** re-test `240,000 + 102,000 =` **USD 342,000 — 14.25 %** of the programme's cost. With an annual alignment re-test at 80 % detection, `0.15 × 192,000 + 0.1275 × 160,000 =` **USD 49,200**. The re-test is therefore worth **USD 292,800**, and its **breakeven cost is USD 97,600 per gate** across the three gates. 5. **Interpretation.** Four readings, and the first is the one to remember because it is parameter-free once the hazard is stated.

The half-life is the sentence that changes how long programmes are allowed to be. At a 15 % annual hazard, alignment is more likely than not to have survived a **four-year** delivery (52.20 %) and more likely than not to have failed a **five-year** one (44.37 %), with the crossing at **4.2650 years**. That is a design constraint on programme architecture, not a commentary on it: a delivery longer than its own alignment half-life is being asked to outlive the reason it exists, and the professional response is to tranche it so that something of value lands and is judged inside the half-life (Domain 15, KA 15.1.2 builds the tranche architecture; Domain 13's incremental delivery is the same instinct at iteration scale). Note what the half-life is not: a forecast that the strategy will change at year 4.27. It is the point at which the odds turn, which is the only thing a design decision needs.

The value of the re-test lives almost entirely in the early years. A driver superseded during year 1 exposes two further years of spend; one superseded during year 3 exposes none. So of the 342,000 of expected misaligned spend, **240,000 — 70.18 % — comes from the first year alone**, and a re-test that only starts at the second gate captures a fraction of the value. This inverts the usual gate emphasis, which tightens as the money grows: on alignment, the early gate is worth the most, because it is the only one with anything left to protect.

Detection quality is worth less than it looks, and this is the honest part. Raising detection from 80 % to a perfect 100 % moves the expected misaligned spend from 49,200 to zero — a further **USD 49,200** — while introducing the re-test at all was worth 292,800, **5.95 times** as much. The large gain is in asking the question; the small gain is in asking it well. A leader who cannot get a rigorous strategic re-test into the gate pack should still get a crude one in: at `d = 0.5` the re-test is still worth **USD 201,000**, which is **58.77 %** of the 342,000 a perfect re-test would save, from a process amounting to one honest agenda item. The corresponding caution: the 80 % is a judgement about whether a committee reading a gate paper would notice that its own strategy had moved, and every organisation should suspect its own number is lower than it thinks.

The machinery is Domain 3's, the loss is not, and the difference matters. Domain 3, KA 3.3.1 prices a gate by its ability to detect a **delivery defect** and computes the remediation saved; this example prices the same gate by its ability to detect a **strategic** change and computes the irrelevant spend avoided. The two are additive — they detect different things and are not substitutes — which is why a gate paper that reports only delivery status is leaving the second value stream entirely uncollected. What the model omits is also worth naming: it assumes supersession is observable at a gate if looked for, that stopping on discovery is possible (Domain 3's decision rights, and the reason 2.4.1's "no route for stopping" symptom converts this value to zero), and that misaligned spend is a total loss, which overstates the case wherever the output has residual use. Where the residual value is material, replace the spend figure with the spend-less-salvage and the arithmetic runs unchanged.

AI in this KA

Portfolio-level analysis is a legitimate AI application: normalising heterogeneous candidate descriptions into comparable form, surfacing dependency clashes across dozens of cases, flagging where a stated driver and the proposed benefits do not match. The governed limits are two. **Strategic weighting is a governance judgment**, not an optimisation output — a model asked to "optimise the portfolio" will faithfully optimise whatever proxy it was given, and the proxy is the decision. And **advocacy is invisible to a model**: an AI reading business cases cannot tell an evidenced benefit from a confident one, so it amplifies whatever the strongest writer submitted. AI proposes; the professional verifies, decides and remains accountable.

Verification, concretely. The alignment index is where a model earns most of its keep and where its output must be checked hardest, because the arithmetic is trivial and the **mapping** is not: a model assigning programmes to objectives from their titles and summaries will produce a defensible-looking index built on classifications nobody agreed. So publish the mapping rule with the index, sample-check a stated proportion of the assignments by hand, and re-run the index under the alternative treatment of any multi-objective programme — if the index moves materially, the finding is about the mapping and not about the portfolio. The index itself is one addition of minima and should be reproduced in a spreadsheet whose formulae are visible. For the decay arithmetic, the supersession hazard must come from an organisation's own decision history, counted, and never from a model's sense of how fast strategies usually change: a plausible-sounding hazard drives a half-life that then drives programme architecture, which is a long way for an invented number to travel.

KEY TERMS — KA 2.1

Term	Meaning
Portfolio process	Comparable description, explicit criteria, funding to capacity, periodic re-decision.
Driver	The reason the work exists; determines what success means and how it is measured.
Hard / soft constraint	Physics, law, contract vs preference and convention; only soft ones are tradeable.
Alignment decay	The erosion of strategic fit after approval, as strategy and understanding move.
Alignment index	`Σ min(declared weight, funded share)` — the overlap between stated priorities and funded spend.
Reallocation distance	The money that must move to close the gaps: `(1 − index) × denominator`; deficits equal surpluses.
Supersession hazard	The annual probability that a stated strategic priority is replaced, merged or de-prioritised.
Alignment half-life	`ln(0.5) ÷ ln(1 − h)` — the delivery duration beyond which alignment is more likely lost than kept.
Benefits owner	The accountable person outside the project who will realise the benefit.

SAMPLE MCQS — KA 2.1

MCQ 2.1-A `[2.1.3 · Analysis]` A project is delivering to plan, but the strategic priority it served has been superseded. The governance-sound response is: - A. continue — the project is on track and the case was approved - B. re-test the case at the next gate and decide on current strategy and remaining cost and benefit - C. continue but reduce scope proportionately - D. transfer the project to another portfolio

Rationale: Alignment decays and gates exist to re-decide (2.1.3), on remaining cost and benefit (KA 2.4.2). A treats approval as permanent; C is an arbitrary compromise that decides nothing; D relocates the question without answering it.

MCQ 2.1-B `[2.1.2 · Application]` A case states its constraint as "must use the existing platform", which reflects an architectural preference rather than a contractual or legal requirement. The correct treatment is: - A. accept it as a constraint, since the architects have stated it - B. record it as a **soft** constraint, so it can be traded in option generation and its cost made visible - C. remove it from the case entirely - D. re-classify it as an objective

Rationale: Only hard constraints (physics, law, contract) are untradeable; a soft one recorded as hard silently eliminates options (2.1.2). C loses real information; D confuses a limit with a goal.

MCQ 2.1-C `[2.1.1 · Application]` Declared weights are 40 / 25 / 20 / 15 %; the strategically mapped spend of USD 18,000,000 is funded 26.6667 / 36.6667 / 10.0000 / 26.6667 %. The reallocation distance is: - A. USD 8,400,000 - B. USD 4,200,000 - C. USD 2,400,000 - D. USD 12,000,000

Rationale: The positive deficits are `0.133333 × 18,000,000 = 2,400,000` and `0.10 × 18,000,000 = 1,800,000`, totalling **4,200,000**, which equals `(1 − 0.766667) × 18,000,000` (2.1.1). A adds the deficits to the surpluses, counting the same movement twice; C is the access deficit alone; D is the unmapped sustainment line, which is a separate finding about the denominator.

MCQ 2.1-D `[2.1.3 · Application]` A strategic priority is superseded at an annual rate of 15 %. The alignment half-life is: - A. 6.6667 years - B. 4.2650 years - C. 3.3333 years - D. 4.6210 years

Rationale: `ln(0.5)/ln(0.85) = 4.2650` (2.1.3). A is `1/h`, the mean waiting time under a different model, not the median; C halves linearly (`0.5/0.15`), ignoring compounding; D uses `ln(0.5)/h`, treating the annual hazard as a continuous rate — a near miss that is systematically too long.

MCQ 2.1-E `[2.1.1 · Evaluation]` An alignment index of 76.6667 % is reported on mapped spend while the index on all discretionary spend is 59.0000 %, the difference being 40 % of spend that maps to no objective. The soundest treatment is: - A. report the higher figure, since sustainment is not discretionary in practice - B. report the lower figure, since 40 % of the money is misaligned - C. report both, and make the recommendation about the unmapped 40 % — either give sustainment a declared weight or stop describing the other 60 % as the portfolio - D. exclude sustainment and re-baseline the declared weights to the mapped spend

Rationale: The denominator is the finding (2.1.1). A hides the largest single item; B misnames unstated strategy as misalignment; D quietly ratifies the omission by rebasing the weights to the spend they were meant to test.

MCQ 2.1-F `[2.1.1 · Comprehension]` A portfolio scores 100 % on the alignment index. It is therefore: - A. well balanced, since every objective is funded to its declared weight - B. correctly allocated against declared intent, which says nothing about risk profile, time to benefit, capability concentration or deliverability - C. optimal, since no money needs to move - D. mis-measured, because a perfect score is not attainable in practice

Rationale: The index measures money against stated intent and nothing else (2.1.1, 2.A.2). A and C read an allocation measure as a balance or optimality measure — a portfolio funded exactly to weight can still be entirely high-risk, entirely long-dated and entirely dependent on one scarce team. D is false: a perfect score simply means the funded shares match the declared weights.

SELF-CHECK — KA 2.1

1. Name the three symptoms of a strategy-portfolio gap. — Funded work not reflecting priorities; more work than capacity; no route for stopping.
2. Why must a benefits owner sit outside the project? — A project owning its own benefits case will not be the one to report that the case has stopped making sense.
3. What distinguishes a hard from a soft constraint? — Physics, law or contract versus preference; only soft constraints can be traded.
4. State the alignment-index identity and why it constrains the argument. — Reallocation distance `= (1 − index) × denominator`, and deficits equal surpluses, so an objective can only be better funded by taking money from another.
5. At a 15 % annual supersession hazard, what is the longest delivery whose alignment is more likely kept than lost? — 4.2650 years; a five-year delivery has a 44.37 % chance of landing under the strategy that authorised it.
6. Where does most of the value of an alignment re-test sit, and why? — In the earliest gate: 70.18 % of Meridian's expected misaligned spend arises from a year-1 supersession, because that is the only case with two further years of spend still to protect.

KNOWLEDGE AREA 2.2

The business case and selection

Topics: 2.2.1 the business case as a decision instrument · 2.2.2 options and appraisal · 2.2.3 selection and prioritisation models.

2.2.1 The business case as a decision instrument

A decision-grade business case answers six questions and nothing else: **why** (the driver and the problem, with evidence); **what options** exist, including doing nothing; **what it costs**, with a range and an accuracy class (Domain 7, KA 7.1.1); **what it delivers**, as outcomes and benefits with owners and measures (KA 2.3); **what could go wrong**, with the exposure quantified (Domain 8); and **how it will be judged**, with success criteria agreed in advance (Domain 1, KA 1.A.2).

Its two failure modes are worth naming precisely. A **advocacy document** answers only "why" and "what it delivers", omitting ranges, options and risk — designed to obtain approval rather than support a decision. A **compliance artefact** answers all six in a template nobody reads, produced after the decision has been taken informally. Both are common; the test that separates them from the real thing is whether the case **could have concluded "no"**.

The living-case principle. The case is not filed at approval; it is the instrument the gates re-test (2.1.3) and the source of the benefits register Domain 16 tracks. A business case whose figures no longer match the project is not a historical document — it is an unmanaged forecast.

2.2.2 Options and appraisal

A genuine options set includes the **do-nothing / do-minimum baseline** (against which all value is measured — without it, benefits are unmeasurable), at least one option that is materially cheaper than the preferred one, and at least one that is materially more ambitious. An options set constructed to make the preferred option look inevitable — the "straw options" pattern — is the advocacy failure mode in structural form, and reviewers spot it by checking whether any option could plausibly have won.

Appraisal applies the discounted machinery PFL-AI Domain 4 builds in full: NPV as the primary value measure, with the range and sensitivity that make it honest. This book uses the results; the finance book derives them. What belongs to a delivery leader is the **benefits profile** — because that is where most business cases are wrong, and the error is systematic.

WORKED EXAMPLE

Worked example 2.2.2 — the Meridian business case, twice.

1. **Setup.** Meridian costs **USD 2,400,000** to deliver. Domain 1 established the arithmetic: full potential at 100 % adoption is **USD 979,200** per year; realistic steady-state adoption is **70 %**, worth **USD 685,440** per year. The benefits are appraised over **8 years** at a **7 %** discount rate. Compare the case as commonly written — full potential from year one — with an honest **ramped** profile: 40 % adoption in year 1, 60 % in year 2, 70 % thereafter.
2. **Formula.** Flat case: `PV = annual benefit × AF(r, n)`. Ramped case: $PV = \Sigma\ (\text{potential} \times \text{adoption}_t) / (1 + r)^t$. NPV = PV − cost.
3. **Substitution.** `AF(0.07, 8) = 5.971299`. Flat: `979,200 × 5.971299`. Ramped: $391{,}680/1.07 + 587{,}520/1.07^2 + 685{,}440 \times$ (years 3–8 discounted).
4. **Result.**

	Year 1	Year 2	Years 3-8	PV of benefits	NPV
Flat, full potential	979,200	979,200	979,200	5,847,096	**+3,447,096**
Ramped, 70 % steady state	391,680	587,520	685,440	3,732,898	**+1,332,898**

The flat case overstates NPV by **USD 2,114,198 — 158.6 %** of the honest figure.

1. **Interpretation.** Both cases approve the programme, which is precisely why the error survives: the decision is unchanged, so nobody checks, and the promise is what gets remembered. Meridian was later judged a failure against benefits it never could have produced (Domain 1's case study), and this table is where that judgement was actually created — at approval, two years before anyone noticed. Five further readings follow, and the third is the one most business cases get subtly wrong even after they have fixed the profile.

The ramp is not pessimism. It is the adoption curve of the same 70 % figure, merely arriving when it actually arrives. Nothing in the honest column is a more cautious assumption than the flat column; the two differ only in when the identical steady state is reached. That is why the flat profile is not a judgement call a reasonable person might defend — it is an arithmetic claim that adoption is instantaneous, which nobody would sign if it were written in words.

The ramp costs a fixed fraction of value, and the fraction is worth knowing. The ramped PV of 3,732,898 is **91.2027 %** of the flat PV at the same 70 % steady state (`0.70 × 5,847,096 = 4,092,967`), so the two ramp years give up **8.7973 %** of the discounted value of the whole benefit stream. Because the profile 40 / 60 / 70 is the steady state scaled by `4/7, 6/7, 1`, that fraction is **independent of the steady-state level**: halve the adoption and both figures halve. A leader can therefore correct any flat case for a two-year ramp of this shape by multiplying by 0.912027, in their head, in the meeting.

Breakeven adoption comes in two flavours, and the difference is 3.9592 points. The **flat-equivalent breakeven is 41.05 %** — `2,400,000 / 5,847,096` — the level a benefit stream flat from year one would have to reach. The **ramped-basis breakeven is 45.01 %** — `2,400,000 / 5,332,711`, where 5,332,711 is the PV of the ramped profile per unit of sustained adoption — the steady state a proportionally ramped profile must reach. Both are correct answers to different questions, and reporting the first while meaning the second understates the requirement by nearly four points. This book uses the flat-equivalent figure as its headline, because it is comparable across profiles, and states the basis every time. **A breakeven without its basis is not a number.**

The breakeven, not the NPV, is the board sentence. "This programme creates value at any sustained flat-equivalent adoption above 41.05 %" states the condition value depends on, in a unit the board can monitor monthly, and it survives every argument about the discount rate that an NPV invites. It also gives the sponsor a falsifiable commitment, which an NPV never does: an NPV can only be wrong in retrospect, while a breakeven can be tested every month from go-live.

What breaks the whole example. Three things. The 8-year horizon is a choice, and a benefit stream that in fact decays — as clinical workflows change and the released hours are re-absorbed — is not captured by a flat tail; where decay is plausible, profile it and let the horizon fall out. The 7 % rate is an input the leader usually does not own, and the industry variations below compute what a point of it is worth. And the cost side is treated as a point estimate, which Domain 7, KA 7.1.1 forbids: the same table drawn with the cost at the top of its accuracy range moves every breakeven, and a case that ranges its benefits but not its costs has ranged the wrong side.

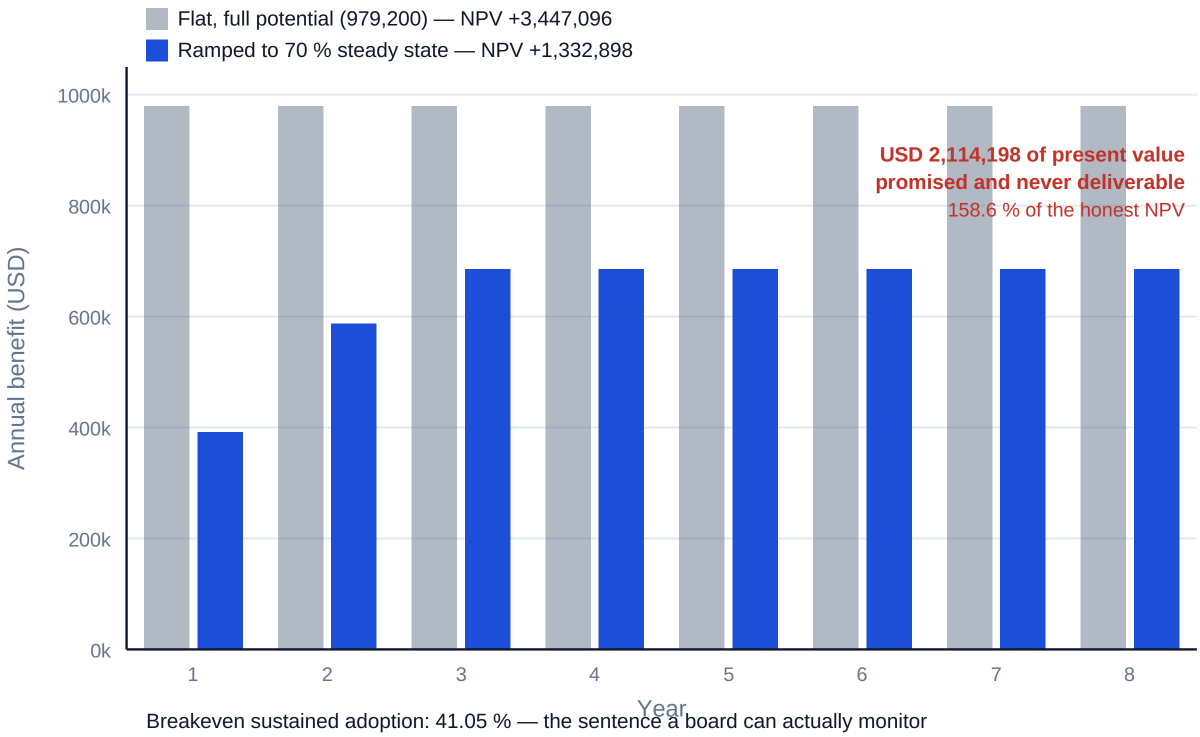


Fig 2.2.1 — Two Meridian business cases from identical facts

The other half of the appraisal is the thing the project is being compared with, and it is almost never zero. "Do nothing" is a course of action with its own cash flows: contracts that expire, equipment that fails, workarounds whose cost grows, obligations that fall due anyway. Where those costs are omitted, the case understates its own value — which makes this the mirror image of the flat-profile error, and the reason a reviewer must look for errors in both directions rather than assuming every case is inflated.

WORKED EXAMPLE

Worked example 2.2.2b — the do-nothing baseline that was not zero.

1. **Setup.** Meridian's approved case compared the programme against a do-nothing option valued at **zero**. Two facts were available at the time and omitted. First, the legacy records system's vendor support ends at the end of **year 3**; continuing without Meridian requires a mandatory re-platforming of the legacy estate costing **USD 600,000**, falling in **year 4**. Second, the manual reconciliation the legacy system requires between clinics grows as clinic numbers rise, adding **USD 40,000 a year in years 5 to 8**. Both are avoided if Meridian proceeds. Same horizon, same 7 % rate. Restate the case.
2. **Formula.** `Incremental NPV = PV(project benefits) − project cost + PV(costs avoided in the do-nothing case)`. Equivalently, the do-nothing option is appraised in its own right and the project's value is the difference: `NPV(project) − NPV(do nothing)`. The flat-equivalent breakeven adoption becomes `(cost − PV of avoided cost) ÷ PV of full-potential benefits`.
3. **Substitution.** $600{,}000 / 1.07^4 = 600{,}000 \times 0.762895$. $40{,}000 \times (AF(0.07,8) - AF(0.07,4)) = 40{,}000 \times (5.971299 - 3.387211) = 40{,}000 \times 2.584087$. Breakeven $(2{,}400{,}000 - 561{,}101) / 5{,}847{,}096$.
4. **Result.**

Component	Undiscounted	Present value
Legacy re-platforming avoided (year 4)	600,000	**457,737**
Manual reconciliation avoided (years 5–8)	160,000	**103,363**
Do-nothing cost, and therefore value created by avoiding it	**760,000**	**561,101**

Incremental NPV rises from **+1,332,898** to **+1,893,998** — an uplift of **USD 561,101**, or **42.10 %** of the stated NPV. Flat-equivalent breakeven adoption falls from **41.0460 %** to **31.4498 %**, an improvement of **9.5962 percentage points**. 5. **Interpretation.** Five readings, and the third is the most consequential sentence in this domain.

A zero do-nothing baseline is a claim, and usually a false one. It asserts that the current arrangement can continue indefinitely at its current cost. That is true of very little: support contracts expire, regulation tightens, volumes grow, and the workaround that costs nothing today costs a post holder tomorrow. The professional discipline is to appraise the do-nothing option with the same rigour as the preferred one — its own cost profile, its own risks, its own owner — and to record that appraisal, because that is what makes the comparison a comparison rather than an assertion.

The error runs in the opposite direction to the flat profile, which is why both must be looked for. The flat profile flattered Meridian by 2,114,198 of present value; the missing counterfactual penalised it by 561,101. A reviewer who only knows that cases are optimistic will find the first and not the second, and will then be wrong in a way that is harder to correct — because arguing that a case is understated costs credibility that arguing the reverse does not. Net across the two corrections, Meridian's honest NPV is `3,732,898 − 2,400,000 + 561,101 =` **+1,893,998**, which is **54.94 %** of the 3,447,096 the approved case advertised. The case was wrong twice and still positive.

Meridian's actual outcome sits between the two breakevens, and that is the whole lesson. At its achieved flat-equivalent adoption of **40 %**, the programme's NPV on the case as written is `0.40 × 5,847,096 − 2,400,000 =` **(USD 61,162)** — a small loss, and the arithmetic that made a public failure verdict defensible. On the honest baseline it is `−61,162 + 561,101 =` **+USD 499,939** — a clear success. **The same programme, the same 40 % adoption, the same cost: value-destroying or value-creating according to whether anyone appraised the do-nothing option.** The judgement passed on Meridian was not a measurement of the programme. It was a measurement of its business case.

Do-minimum is usually the honest baseline, not do-nothing, and the difference is not presentational. Pure inaction is often not permissible — the support contract cannot simply be allowed to lapse on a clinical system — so the correct comparator is the **cheapest compliant alternative**, here the 600,000 re-platforming on its own. Appraised that way, the 600,000 moves from "avoided cost" to "the do-minimum option's cost", and the incremental NPV is identical; what changes is that the options set now contains a genuine third option that could have won on a low adoption forecast. **Framing the counterfactual as an option rather than as an adjustment is what makes the options set genuine** (2.2.2's straw-options test), and it is the better habit for exactly that reason.

What breaks it, and where to be careful. Avoided costs must be genuinely avoided and not merely deferred: if the legacy re-platforming would still be needed for another system on the same estate, nothing has been avoided and the 457,737 is fictitious — the same single-claimant discipline KA 2.3.2 applies to benefits, applied to costs. The avoided cost must not also be counted as a benefit elsewhere in the case, which is the commonest form of the double count here. The timing carries real weight: the 600,000 is worth 457,737 at year 4 and would be worth **560,748** at year 1, so a vaguely dated obligation is a materially different number, and the year is a fact to be evidenced from the contract rather than assumed. And an avoided cost is not cash released unless the budget holding it is actually removed — 2.3.2's cash-releasing test applies with full force, because a finance function that never held a provision for the 600,000 will not recognise its avoidance. **This is an area where the accounting and budgetary treatment differs materially between organisations and jurisdictions; where the case turns on it, agree the treatment with the finance function in writing before the paper is written.**

2.2.3 Selection and prioritisation models

Weighted scoring ranks candidates against criteria carrying explicit weights. Its value is that it forces the criteria and weights into the open, where they can be argued; its limits are that scores are **ordinal judgments** (the arithmetic caution of Domain 8, KA 8.2.1 applies) and that weights are chosen by whoever runs the model — so the model can be steered by anyone who understands it.

WORKED EXAMPLE

Worked example 2.2.3 — four candidates, two rankings.

1. **Setup.** Four candidate programmes scored 1–5 against weighted criteria: strategic fit (0.35), benefit value (0.30), deliverability (0.20), risk — inverse, so higher is safer (0.15).
2. **Formula.** Weighted score = Σ(weight × score). Then, under a binding constraint, rank by **NPV per unit of the scarce resource** instead.
3. **Substitution and result.**

Candidate	Fit	Benefit	Deliver.	Risk	Weighted score
Meridian	4	5	3	3	**3.95**
Beta	5	3	4	4	**4.05**
Gamma	3	4	5	4	**3.85**
Delta	2	2	5	5	**3.05**

Now suppose the binding constraint is **integration-team capacity**, of which only 3 units exist: Meridian needs 3 units for an NPV of 1,693,072; Beta 2 units for 1,200,000; Gamma 1 unit for 900,000.

Candidate	NPV	Capacity	NPV per unit
Meridian	1,693,072	3	**564,357**
Beta	1,200,000	2	**600,000**
Gamma	900,000	1	**900,000**

1. **Interpretation.** Two defensible methods give two different answers, and a leader who can say why commands the selection meeting. Four readings.

The constraint changes the answer, not merely the ordering. The scoring model ranks **Beta first** (4.05 against Meridian's 3.95) on strategic fit and deliverability. Under the capacity constraint, **Beta + Gamma** together consume the 3 units for a combined NPV of **2,100,000**, beating Meridian's 1,693,072 alone — so the constrained answer is to run the two smaller programmes, which neither the raw NPV ranking nor the scoring model would have selected. This is the delivery-side twin of PFL-AI's capital rationing (its Domain 4, KA 4.3.2). The leadership content is that **the binding**

constraint decides the method, and naming that constraint honestly — usually a scarce team, not money — is the whole game.

Enumeration is cheap and greedy ranking is not safe. The feasible sets under 3 units are exactly `{}`, `{Gamma}` = 900,000, `{Beta}` = 1,200,000, `{Meridian}` = 1,693,072 and `{Beta, Gamma}` = **2,100,000**; `{Meridian, Beta}` needs 5 units and `{Meridian, Gamma}` needs 4, so both are infeasible. Here greedy ranking by NPV per unit — Gamma at 900,000, Beta at 600,000, Meridian at 564,357 — happens to reach the optimum, and that is luck rather than method. Had a fifth candidate **Epsilon** needed all 3 units for an NPV of **2,200,000** (733,333 per unit), greedy would still have taken Gamma first and finished with Beta + Gamma at 2,100,000, missing the best set by **USD 100,000**. With n candidates there are 2^n subsets, which for the ten or twenty candidates a real portfolio board considers is a spreadsheet exercise, not a research problem. **Enumerate; rank only to explain the answer afterwards.**

Divisibility is the assumption that decides whether any of this is legitimate. Ranking by value per unit is the correct optimum when candidates are divisible — when half of Beta can be bought for half the money and half the value. Projects are almost never divisible in that sense, which is what makes them lumpy and what makes greedy unsafe. Where a candidate genuinely can be **staged**, it becomes partly divisible and the arithmetic improves; that is one of the underrated benefits of the tranching argued for in 2.1.3 and priced in 2.A.1, and it belongs in the selection paper rather than being discovered later.

What no model does. Scoring and ratios cannot see option value, sequencing dependencies, or the strategic cost of not doing something. They order candidates; they do not decide. A portfolio board that treats a scoring output as the decision has automated its own accountability (Domain 1, KA 1.2.1). Two further blindnesses are specific to the constrained form: it takes the constraint as given, when the most valuable move is often to relax it — priced immediately below — and it assumes the candidates are independent, when Gamma may be the enabler on which Meridian's benefits depend, in which case the sets are not free to be chosen (Domain 15, KA 15.1.3 handles dependent candidates).

WORKED EXAMPLE

Worked example 2.2.3b — what one unit of the binding constraint is worth.

1. **Setup.** The three candidates above compete for one scarce integration team measured in **units**: Meridian needs 3 units for an NPV of 1,693,072; Beta needs 2 for 1,200,000; Gamma needs 1 for 900,000. The portfolio director can grow or shrink the team, and a unit of durable integration capacity — a qualified engineer with the platform knowledge, recruited and retained — costs **USD 400,000** a year all-in. How much capacity should the organisation hold?
2. **Formula.** For each capacity level c, solve $V(c) = \max \sum NPV_i \text{ subject to } \sum units_i \le c$ by enumeration. The **marginal value of the nth unit** is $V(n) - V(n-1)$. Buy k further units when $V(c + k) - V(c) > k \times$ unit cost.
3. **Substitution.** At $c = 3$, the feasible maximum is `{Beta, Gamma}` = 2,100,000. At $c = 4$, the admissible sets include `{Meridian, Gamma}` = $1{,}693{,}072 + 900{,}000$. At $c = 5$, `{Meridian, Beta}` = $1{,}693{,}072 + 1{,}200{,}000$. At $c = 6$, all three fit.
4. **Result.**

Capacity	Optimal set	Portfolio NPV	Marginal value of that unit
1	Gamma	900,000	**900,000**
2	Beta	1,200,000	**300,000**
3	Beta + Gamma	2,100,000	**900,000**
4	Meridian + Gamma	2,593,072	**493,072**
5	Meridian + Beta	2,893,072	**300,000**
6	Meridian + Beta + Gamma	3,793,072	**900,000**

From a base of 3 units, at 400,000 a unit: **+1 unit** gains 493,072 for 400,000 — net **+93,072**; **+2 units** gains 793,072 for 800,000 — net **(6,928)**; **+3 units** gains 1,693,072 for 1,200,000 — net **+493,072**. 5. **Interpretation.** Four readings, and the second contradicts the way capacity decisions are almost always taken.

There is no such thing as "the value of a unit of integration capacity". The marginal value runs 900,000 · 300,000 · 900,000 · 493,072 · 300,000 · 900,000 — it is **lumpy and non-monotone**, and it is a property of the candidate set, not of the team. Anyone who quotes a single figure for what a scarce team is worth has averaged away the only information in the answer. The same applies to the shadow prices a solver reports: they are valid for the set that was solved and change when a candidate is added, withdrawn or re-estimated.

The right answer here is to add one unit or three, and never two — which no incremental process would find. Adding units one at a time, each judged on its own, accepts the first (it gains 493,072 for 400,000, a net **+93,072**) and then rejects the second, which gains only 300,000 for 400,000 — a net **(100,000)** on its own, and (6,928) taken as a block of two. So the process stops there and never reaches the block of three, whose net is **+493,072 — 5.30 times** the gain the incremental process settles for. Capacity decisions are non-convex because the candidates are lumpy, so they must be taken as a **block decision against enumerated plans**, which is precisely what an annual headcount round asking each manager to justify the next post cannot do. This is the arithmetic behind Domain 15's treatment of enterprise capacity as a portfolio-level decision (KA 15.3) rather than a departmental one.

The comparison the portfolio board should actually be shown. Not "should we hire?" but a table of feasible plans with their NPVs and their capacity costs — because on these figures the question "what is the best use of 1,200,000?" has the answer "three integration engineers", which no capital proposal would ever have contained. Note also the direction that is usually forgotten: at `c = 2` the marginal unit is worth only 300,000, **below its 400,000 cost**, so an organisation sitting at 2 units with these candidates should shrink rather than hold — a conclusion the same table produces and no advocacy paper ever reaches.

What breaks it. The unit cost must be the **durable** cost of holding capacity, not a contractor day rate for a peak, and if the capacity can be rented for a season the whole question changes shape (Domain 7, KA 7.4 prices resource acquisition modes; Domain 10 prices the make-or-buy). Recruitment lag matters and is absent here: three engineers who arrive in nine months do not enable a decision taken today, and Domain 1's reinforcement-trough arithmetic prices the transient they bring with them. The NPVs are point estimates on candidates whose own ranges overlap, so a marginal value of 300,000 is not reliably different from one of 493,072. And capacity has value beyond this candidate set — optionality against next year's candidates, and resilience — which this model prices at zero and Domain 15, KA 15.3 prices properly as protective capacity.

WORKED EXAMPLE

Worked example 2.2.3c — how far must a weight move to reverse the ranking?

1. **Setup.** The same four candidates and the same weighted-scoring model: strategic fit 0.35, benefit value 0.30, deliverability 0.20, risk-inverse 0.15, scores on a 1–5 scale. Beta leads Meridian by 4.05 to 3.95. Two people who both accept the scores disagree about the weights: one believes strategic fit should carry less and benefit value more. **How much less?**
2. **Formula.** Shift δ of weight from criterion a to criterion b, holding the rest fixed. A candidate's total changes by $\delta \times (\text{score\_b} - \text{score\_a})$. The ranking flips where $\delta \times [(\text{score}_{\beta}\text{\_b} - \text{score}_{\beta}\text{\_a}) - (\text{score\_M\_b} - \text{score\_M\_a})] =$ the current margin. Separately, the most any criterion can move a total — its **criterion influence** — is $(\text{score range}) \times \text{weight}$.
3. **Substitution.** Meridian gains $\delta(5 - 4) = +\delta$; Beta gains $\delta(3 - 5) = -2\delta$. The margin is 0.10, so $\delta + 2\delta = 0.10$.
4. **Result.** $\delta =$ **0.033333**, i.e. **3.3333 percentage points** of weight. At fit 0.316667 and benefit 0.333333 both candidates score exactly **3.983333**. Criterion influence, on a 1–5 scale (range 4): fit **1.40**, benefit **1.20**, deliverability **0.80**, risk **0.60** — summing to the full 4.00 range of possible totals. The four candidates' actual totals span only **1.00**, which is **25.00 %** of that maximum swing.
5. **Interpretation.** Five readings, and the last is the governance conclusion the arithmetic forces.

A 3.33-point weight shift reverses the portfolio's top candidate, and 3.33 points is inside the noise of any weighting workshop. Nobody can defend 0.35 against 0.3167 for strategic fit on any evidence; the two numbers are indistinguishable as statements of priority. So the model's answer here is not robust, and reporting "Beta ranks first" without reporting that it takes 3.33 points to change that is a material omission. The reviewer's question is exact: **what is the smallest defensible change of weight that reverses this ranking?**

The score side is worse. Meridian needs only **+0.50** on deliverability — 3 to 3.5 — to draw level, because deliverability carries a weight of 0.20 and $0.10/0.20 = 0.50$. On a 1–5 ordinal scale nobody can distinguish 3 from 3.5, which is Domain 8, KA 8.2.1's arithmetic caution about ordinal scales in its sharpest form: **the model is being asked to resolve a difference finer than its own inputs can express.** On the risk criterion, weighted 0.15, the required move is **+0.6667**, and the general rule falls out: the score movement needed to close a gap is $\text{margin} \div \text{weight}$, so the lightly weighted criteria are the ones where scoring error does least damage.

Some weight shifts can never flip a ranking, and knowing which is free information. Moving weight between strategic fit and deliverability changes Meridian's total by $\delta(3 - 4) = -\delta$ and Beta's by $\delta(4 - 5) = -\delta$ — identically — so the 0.10 margin is invariant to that trade no matter how large it is. A weighting argument between those two criteria is therefore not an argument about this decision at all, and a chair who knows it can stop the discussion. The pairs that matter are the ones where the two candidates' score differences differ.

Criterion influence tells you what the model cannot decide. With a 1–5 scale, the risk criterion at 0.15 can move a total by at most **0.60** across its whole range. If the real disagreement in the room is about risk, and the candidates' risk scores differ by 1 point (worth 0.15), the model literally cannot express it: 0.15 against a spread of totals of 1.00. The remedy is not a bigger weight but a different instrument — quantified exposure per Domain 8, KA 8.2.2, reported beside the score rather than compressed into it.

Scoring models compress, and compression is what makes them steerable. The four candidates occupy 25.00 % of the available range, because a good candidate scores well on some criteria and badly on others and weighting averages the differences away. Compression is not a defect to be engineered out — it is the honest consequence of averaging — but it has a governance implication: **when the outputs are close, the weights decide, and the weights are chosen by whoever runs the model.** The countermeasures are procedural and cheap: agree and minute the weights before the candidates are scored, publish the flip point with the ranking, and where two candidates lie within the flip point of each other, report them as **not separated by this method** and decide on something else — constraint economics (2.2.3b), option value (2.A.1) or portfolio balance (2.A.2). A model that cannot separate two candidates has still done useful work by saying so.

AI in this KA

Business-case drafting is now routinely AI-assisted, and this KA identifies the specific hazard: a model will produce a **fluent, complete-looking case with a flat benefits profile**, because that is what most cases in its training data look like. It will also generate plausible options sets that are structurally straw. The verification duty is therefore concrete — check the benefits profile ramps, check at least one option could have won, check every benefit traces to a measured outcome with an owner (Domain 1's benefits-chain test), and check the range and class on the cost. An AI-drafted case that passes those four checks is genuinely useful work; one that has not been checked against them is the advocacy document of 2.2.1 with better formatting.

Four further checks belong on that list once this KA's arithmetic is available, and each catches a defect the four above do not. **A do-nothing option valued at zero** is as characteristic of generated cases as a flat profile, and for the same reason — most cases in the corpus do it — so ask the model explicitly what doing nothing costs in each year of the horizon, and then verify each item against a contract, licence or asset register rather than accepting the list. And **a breakeven quoted without its basis** is a near-certainty in generated text, because the distinction between flat-equivalent and ramped-basis is finer than the language usually carries; recompute it and label it. Where a model has been asked to run a scoring or constrained-selection model, the specific verification is the **flip point** and the **enumeration**: a model will produce a ranking with total confidence and will not volunteer that a 3.33-point weight shift reverses it, and a model asked to "optimise" a constrained selection will frequently return the greedy per-unit answer, which is not the optimum. Both are checkable in a spreadsheet in minutes, and both change recommendations.

KEY TERMS — KA 2.2

Term	Meaning
Do-nothing baseline	The comparison against which all benefits are measured; without it value is unmeasurable.
Do-minimum option	The cheapest compliant alternative where inaction is not permissible; the honest comparator, and an option in its own right.
Counterfactual cost	Cost incurred only in the do-nothing case, and therefore value created by avoiding it.
Straw options	An options set built so the preferred option cannot lose.
Benefits profile	The time-phased benefit stream; ramps in reality, flat in most business cases.
Breakeven adoption	The adoption level at which NPV is zero — a more useful board sentence than NPV.
Flat-equivalent basis	Breakeven stated as the level a profile flat from year one would need (Meridian 41.05 %).
Ramped basis	Breakeven stated as the steady state a proportionally ramped profile must reach (Meridian 45.01 %).
Weighted scoring	Ranking by Σ(weight × ordinal score); forces criteria into the open, steerable by whoever sets weights.
Value per unit of constraint	Ranking basis when a scarce resource binds; a heuristic, not an optimum.
Marginal value of the constraint	`V(n) − V(n−1)` from enumeration; lumpy, non-monotone, and a property of the candidate set.
Flip point	The smallest weight shift that reverses a ranking; report it with the ranking or the ranking is unqualified.
Criterion influence	`(score range) × weight` — the most a criterion can move a total, and therefore what the model cannot decide.

SAMPLE MCQS — KA 2.2

MCQ 2.2-A `[2.2.2 · Application]` Full benefit potential is 979,200 per year; steady-state adoption is 70 %; the profile ramps 40 %/60 %/70 % and is appraised over 8 years at 7 % (`AF` = 5.971299). Against a flat full-potential case, the ramped NPV is lower by: - A. USD 979,200 - B. USD 2,114,198 - C. USD 1,332,898 - D. USD 293,760

Rationale: Flat PV 5,847,096 less ramped PV 3,732,898 = **2,114,198** of overstated present value. C is the honest NPV itself; A is one year's potential; D is Domain 1's single-year output-based overstatement, a different figure.

MCQ 2.2-B `[2.2.2 · Analysis]` Both the flat and ramped Meridian cases produce a positive NPV and the same approval decision. Why does the flat case still matter? - A. it does not — the decision is unchanged, so the error is harmless - B. because the case becomes the promise the programme is later judged against, and benefits that were never deliverable are recorded as failure - C. because the discount rate must be recalculated - D. because the flat case understates cost

Rationale: The approval is identical; the commitment is not (2.2.2, and Domain 1's case study where Meridian was publicly called a failure). A is the reasoning that let the error survive.

MCQ 2.2-C `[2.2.3 · Analysis]` Integration capacity of 3 units is the binding constraint. Meridian needs 3 units for NPV 1,693,072; Beta 2 for 1,200,000; Gamma 1 for 900,000. The value-maximising selection is: - A. Meridian alone — the highest single NPV - B. Beta + Gamma — combined NPV 2,100,000 within the 3-unit constraint - C. Meridian + Gamma — the two highest NPVs - D. all three, phased across two years

Rationale: Beta and Gamma together fit the constraint and beat Meridian's 1,693,072. C needs 4 units; D changes the premise rather than answering under it; A ignores the constraint's implications.

MCQ 2.2-D `[2.2.1 · Recall]` The clearest test that a business case is a decision instrument rather than an advocacy document is: - A. whether it follows the corporate template - B. whether it could have concluded "no" - C. whether it was approved - D. whether its NPV is positive

Rationale: An options set and evidence that permit a negative conclusion are what make it a decision (2.2.1). A is the compliance failure mode; C and D are outcomes, not tests.

MCQ 2.2-E `[2.2.2 · Application]` A case shows NPV +1,332,898 against a do-nothing option valued at zero. In fact doing nothing forces a mandatory 600,000 spend in year 4 and 40,000 a year in years 5–8; the rate is 7 % ($1/1.07^4 = 0.762895$; $AF(0.07,8) - AF(0.07,4) = 2.584087$). The incremental NPV is: - A. USD 1,332,898 - B. USD 1,893,998 - C. USD 2,092,898 - D. USD 771,797

Rationale: Avoided cost PV is $457{,}737 + 103{,}363 = 561{,}101$, added to the project's NPV (2.2.2b). A omits the counterfactual entirely; C adds the **undiscounted** 760,000; D subtracts the avoided cost instead of adding it, treating a cost the project removes as a cost it incurs.

MCQ 2.2-F `[2.2.3 · Analysis]` Beta leads Meridian 4.05 to 3.95. Scores on strategic fit are Beta 5, Meridian 4; on benefit value Beta 3, Meridian 5. Shifting weight from strategic fit to benefit value flips the ranking at a shift of: - A. 10.00 percentage points - B. 3.33 percentage points - C. 2.50 percentage points - D. no shift between these two criteria can flip it

Rationale: Meridian gains δ and Beta loses 2δ, so $3\delta = 0.10$ and $\delta = 0.033333$ (2.2.3c). A removes weight from fit without adding it to benefit, so the weights no longer sum to one; C divides the 0.10 margin by the 4-point score range; D is true of a fit-to-deliverability shift, where both candidates move identically, but not of this pair.

MCQ 2.2-G `[2.2.3 · Evaluation]` With 3 units of integration capacity the optimal set is Beta + Gamma at 2,100,000; with 4 units it is Meridian + Gamma at 2,593,072. The marginal value of the fourth unit is: - A. USD 900,000 - B. USD 1,693,072 - C. USD 493,072 - D. USD 564,357

Rationale: $2{,}593{,}072 - 2{,}100{,}000 = 493{,}072$ (2.2.3b). A applies Gamma's best-in-set per-unit ratio to the marginal unit; B counts the whole NPV of the candidate newly admitted while ignoring that Beta is displaced; D is Meridian's own NPV per unit, which is an average and not a margin.

SELF-CHECK — KA 2.2

1. What must an options set contain to be genuine? — A do-nothing baseline, a materially cheaper option, a materially more ambitious one — and at least one that could plausibly have won.
2. Why is breakeven adoption a better board sentence than NPV? — It states the condition the value depends on, which is what the board can actually influence and monitor.
3. When does ranking by value per unit of constraint fail? — With lumpy candidates: the feasible sets must be enumerated, as Beta + Gamma shows.
4. State Meridian's breakeven adoption on both bases and the conversion between them. — 41.0460 % flat-equivalent, 45.0053 % ramped-basis; the ramped PV is 91.2027 % of the flat PV at the same steady state, and $41.0460 = 0.912027 \times 45.0053$.
5. Which two errors in a business case run in opposite directions? — A flat benefits profile flatters (Meridian by 2,114,198 of PV); a zero do-nothing baseline penalises (by 561,101).

6. Why should a capacity decision be taken as a block? — The marginal value of the constraint is non-monotone, so judging one unit at a time stops at +93,072 and never reaches the +493,072 available from three.
7. What should a scoring model report alongside its ranking? — The flip point, and where two candidates lie within it, the statement that this method does not separate them.

KNOWLEDGE AREA 2.3

Benefits, value and sustainability

Topics: 2.3.1 benefits mapping · 2.3.2 measures and baselines · 2.3.3 sustainability and ESG value · 2.3.4 assumptions and dependencies.

2.3.1 Benefits mapping

A **benefits map** is a chain, drawn explicitly, from what the project delivers to why anyone wanted it: **output → enabling change → outcome → benefit → strategic objective**. Domain 1 established the links; this KA draws them and assigns them.

The **enabling change** is the step most maps omit and most programmes fail on. Meridian's outputs (installed clinics) produce no benefit without training, workflow redesign, clinical-champion support and data migration — none of which are software, and several of which sit outside the project's authority. A benefits map that leaps from output to benefit has hidden exactly the work that determines whether anything is realised, which is why Meridian's adoption stalled at 40 % while installation ran to plan.

Every element on the map carries an **owner** and, for outcomes and benefits, a **measure**. The rule from Domain 1 (KA 1.2.1) applies unchanged: one accountable name per outcome, and the benefits owner sits outside the project (2.1.3).

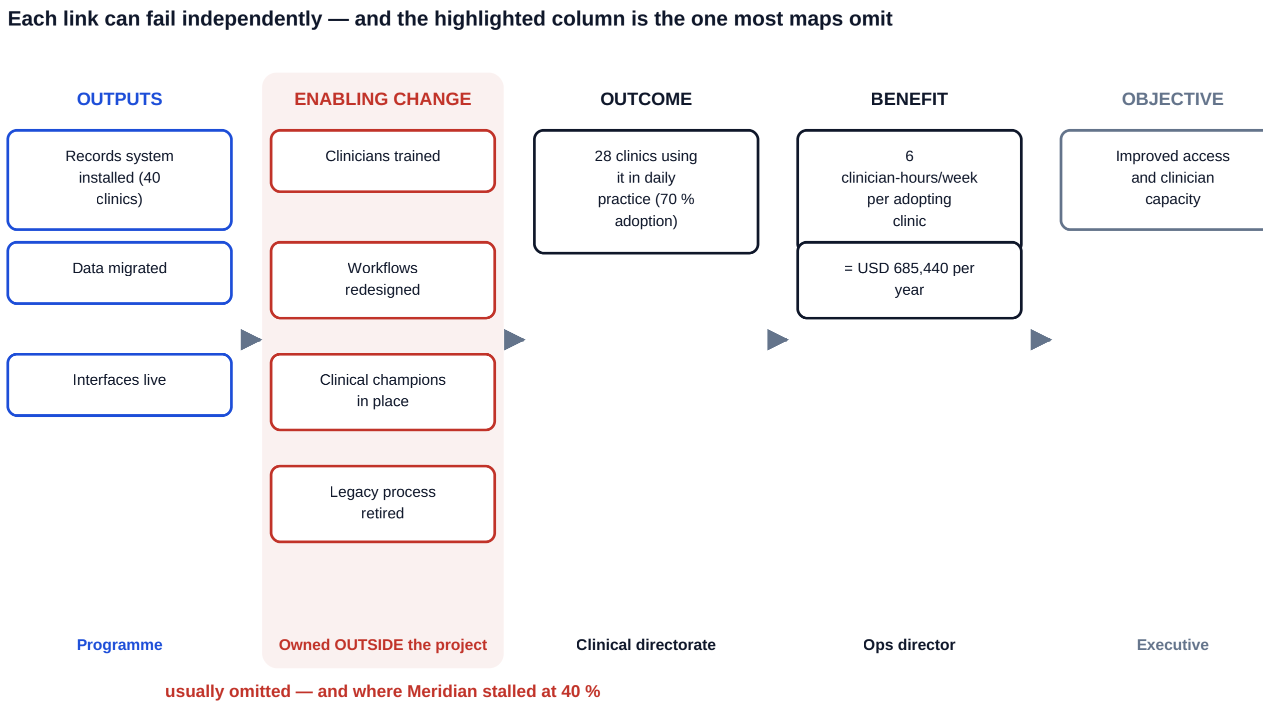


Fig 2.3.1 — Meridian's benefits map, with the enabling change restored

2.3.2 Measures and baselines

A benefit is only claimable against a **baseline** — the measured position before the change. Three disciplines make measurement honest:

- **Baseline before, not after.** Measure the current state before the change lands; a baseline reconstructed afterwards is an estimate shaped by the result it is used to justify.
- **Attribution.** Other things change too. Where a benefit could have other causes, say so and quantify what portion is plausibly attributable — over-claiming is the fastest way to lose the credibility that future cases depend on.
- **No double-counting.** The single commonest audit finding in benefits work: an enabling programme claims the benefit its dependent projects also claim, and the portfolio's total exceeds what the organisation could possibly realise. The portfolio benefits register (Domain 15) exists to catch this, and it only works if each benefit has exactly one claimant.

Cash-releasing versus non-cash-releasing benefits must be distinguished plainly. Meridian releases clinician hours; that becomes cash only if the organisation reduces cost or converts the hours into additional activity that has value. Reporting released hours as savings, when the headcount is unchanged and the hours are absorbed, is the error that makes finance directors distrust benefits cases — and it is avoided by stating the conversion explicitly, or by claiming the benefit in its true unit (capacity) rather than in money it never became.

Attribution is the discipline of the three that is most often waved at and least often done, because doing it requires a **comparison cohort** — a group that did not receive the change — and choosing to keep one feels like withholding a benefit. It is worth the discomfort, and it is worth arithmetic.

WORKED EXAMPLE

Worked example 2.3.2 — the measured benefit and the attributable benefit.

1. **Setup.** Twelve months after go-live, Meridian's benefits owner measures records-administration time per clinic per week. The pre-change baseline, measured before deployment, was **22.0 hours**. In the **28** adopting clinics it is now **15.6 hours**. In the **12** clinics that have not adopted — the comparison cohort — the same measure has fallen from 22.0 to **20.9 hours**, because an unrelated triage redesign was rolled out across the whole authority in the same period. The valuation basis is Domain 1's: **USD 85** an hour over a **48-week** operating year. The programme's business case assumed **6.0 hours** released per clinic per week. What may be claimed?
2. **Formula.** `Raw improvement = baseline − post-change in the adopting group`. `Counterfactual improvement = baseline − post-change in the comparison group`. `Attributable improvement = raw − counterfactual` (the difference of the two differences). `Annual benefit = adopting clinics × attributable hours × rate × weeks`.
3. **Substitution.** Raw `22.0 − 15.6 = 6.4`; counterfactual `22.0 − 20.9 = 1.1`; attributable `6.4 − 1.1 = 5.3`. Claimable `28 × 5.3 × 85 × 48`; the raw claim would be `28 × 6.4 × 85 × 48`.
4. **Result.**

Basis	Hours per clinic per week	Annual benefit, 28 clinics
Raw measured improvement	6.4	**731,136**
Business-case assumption	6.0	685,440
Attributable improvement	**5.3**	**605,472**

The raw claim overstates the attributable benefit by **USD 125,664**, which is **17.1875 %** of the raw claim. 5. **Interpretation.** Five readings, and the first is the one that catches an unwary benefits owner in a good year.

A measurement that beats the case can still fail to support it. The raw figure, 6.4 hours, exceeds the case's 6.0, so the natural report is "benefits exceeding forecast". The attributable figure, 5.3 hours, is **11.67 % below** the case, and the case's 6.0 is **13.21 % higher** than what can honestly be claimed. Raw and attributable are different quantities, and it is the raw one that programmes report because it is the one the programme's own data produces. **The comparison cohort is what converts a measurement into evidence**, and a programme without one cannot distinguish its own effect from everything else that happened that year — in either direction.

The over-claim share is an invariant, and it is worth knowing in a meeting. Because the raw and attributable figures share the clinic count, the rate and the weeks, the over-claim as a fraction of the raw claim is `counterfactual ÷ raw = 1.1/6.4 =` **17.1875 %** — irrespective of the valuation rate, the operating year or how many clinics adopted. It is the exact structural twin of Domain 1's adoption identity, where an output-based claim overstates by `1 − a` whatever the other numbers are (KA 1.3.2). Both identities exist for the same reason and are used the same way: **you do not need the model to say how wrong a claim is, only the term it omitted.**

The correction moves the whole case, not just this year's report. At 5.3 attributable hours, full potential falls from 979,200 to `40 × 5.3 × 85 × 48 =` **USD 864,960**, which is **88.3333 %** of the original — exactly `5.3/6.0`, since nothing else changed. The flat-equivalent breakeven adoption therefore rises from 41.0460 % to `2,400,000 / (864,960 × 5.971299) =` **46.4672 %**, a deterioration of **5.4212 percentage points**. Taken together with the do-nothing correction of 2.2.2b, which improved the breakeven by 9.5962 points, the fully corrected flat-equivalent breakeven is `(2,400,000 − 561,101) / (864,960 × 5.971299) =` **35.6035 %** — and Meridian's achieved 40 % adoption clears it with **4.3965 points** to spare, for an NPV of **+USD 227,074**. **Two errors of opposite sign, and you must fix both: fixing only the flattering one produces a verdict as wrong as fixing neither.**

Choosing a comparison cohort is a real decision with real costs, and it should be made deliberately. The 12 non-adopting clinics were not held back for measurement — they simply had not adopted — which makes them a **convenience cohort**, and convenience cohorts differ systematically from the adopters: the clinics that adopt first are usually the better-led, better-staffed ones, so part of the 5.3 may still be selection rather than software. A deliberately staged rollout gives a defensible cohort as a free by-product of the sequencing decision, which is another benefit of the tranching argued in 2.1.3, and it is one of the strongest arguments a benefits owner has against a simultaneous switch-on. Where withholding a change is not acceptable — and in clinical or safety settings it frequently is not — the honest alternatives are a pre-change trend line extended forwards, or a stated range with the attribution assumption on the face of the report. **What is not acceptable is claiming the raw figure and calling the question unanswerable.**

What breaks it. The two groups must be measured the **same way at the same time**; a post-change measurement method that is more thorough in adopting clinics manufactures an effect. A counterfactual improvement can also be negative — the comparison group getting worse — in which case the attributable figure exceeds the raw one and the honest report is a larger claim, which is the case a leader should be most careful to actually make. Attribution and adoption interact and must not be double-corrected: the 5.3 hours is a per-adopting-clinic figure, and multiplying by 28 already applies the adoption term, so applying 70 % again is a common and severe error. And none of this converts the benefit to cash: 605,472 of attributable released time is **capacity**, and the cash-releasing test above still has to be passed separately.

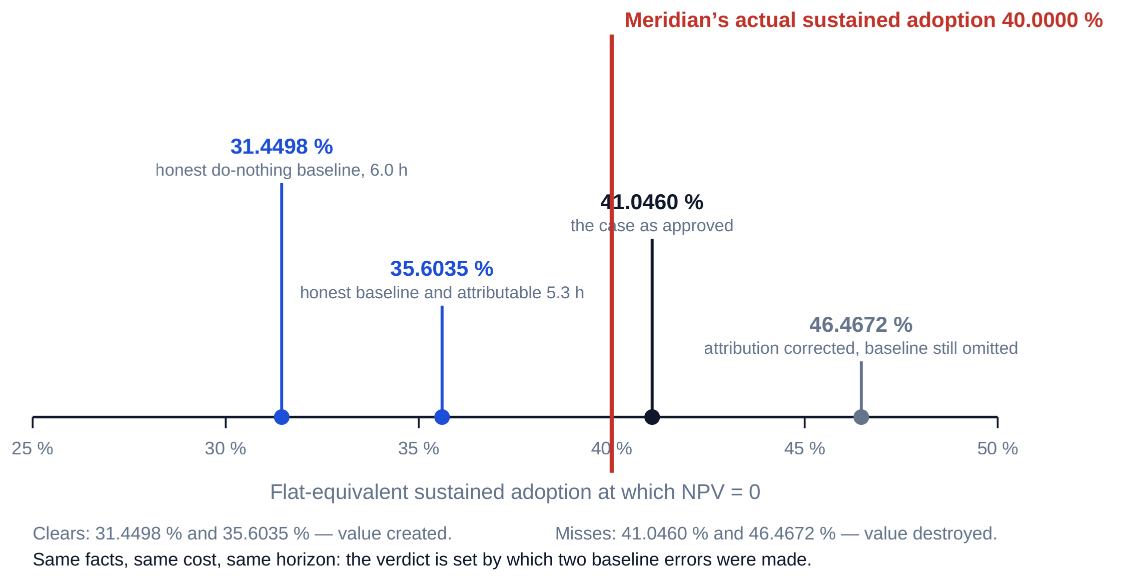


Fig 2.3.2 — Four breakevens from one programme

2.3.3 Sustainability and ESG value

Environmental, social and governance considerations enter this domain in two distinct ways, and conflating them causes most of the confusion.

As constraints and obligations: emissions limits, environmental permits, accessibility law, labour standards in the supply chain (Domain 10's ethical sourcing). These are hard constraints (2.1.2) and belong in the case as requirements, not benefits.

As value: whole-life carbon reduction, energy cost avoided, social value created, resilience to climate risk. These belong in the benefits map with measures and owners like any other benefit — and with the same honesty about attribution and monetisation. Where a carbon reduction has a price (a compliance cost avoided, a levy, an internal carbon price), it can be valued directly; where it does not, it should be reported in its physical unit alongside the financial case rather than converted with an invented price. **Whole-life thinking** is the practical contribution: a cheaper capital solution with higher operating energy and shorter life frequently loses on a whole-life basis, and the appraisal horizon of KA 2.2.2 is what makes that visible (PFL-AI Domain 4's equivalent annual value handles unequal lives).

2.3.4 Assumptions and dependencies

Business cases fail through their assumptions, and the assumptions are usually stated once and never revisited. Meridian's case rested on an adoption assumption that was never tracked as a measure — the direct cause of everything Domain 1's case study describes.

An **assumption register** carries, per assumption: the statement, why it is believed, the impact if false, the owner, the date it will be tested, and the **trigger** that would falsify it. Its relationship to the risk register is exact: **every assumption is a risk in disguise** (Domain 8, KA 8.1.3), so material assumptions get register entries with `EMV` where quantifiable.

Quantifying them is the step that turns the register from a list into an instrument, because it permits two questions no list can answer: is the case's promised value large relative to the exposure carried by its own assumptions? and which assumption should be tested first?

WORKED EXAMPLE

Worked example 2.3.4 — pricing Meridian's assumption register.

1. **Setup.** Meridian's honest case (2.2.2) gives a ramped PV of benefits of **3,732,898** against a cost of 2,400,000 — an NPV of **+1,332,898** — and, per unit of sustained adoption, a PV of **5,332,711**. Five material assumptions underpin it. Each has an assessed impact if false, in present-value terms, and a probability of being false, both **assessed by the programme team and recorded as judgements, not measurements**. The cost of testing each is also known.
2. **Formula.** `EMV = probability × impact`. `Assumption exposure ratio = Σ EMV ÷ NPV`. Test priority ranks by `EMV ÷ cost to test` (infinite where the test is free, so those go first).
3. **Substitution.** A1: adoption falling from 70 % to 45 % costs `0.25 × 5,332,711 = 1,333,178`. A2: attributable hours 6.0 → 5.3 costs `3,732,898 × (1 − 5.3/6.0) = 435,505`. A3: the legacy remediation is not in fact avoided, $600{,}000/1.07^4 = 457{,}737$. A4: the enabling change must be programme-funded, `320,000`. A5: finance values clinician time at 70 rather than 85, `3,732,898 × (1 − 70/85) = 658,747`.
4. **Result.**

	Assumption	Impact if false (PV)	P(false)	EMV	Cost to test	EMV per unit of test cost
A1	Sustained adoption reaches 70 %	1,333,178	0.35	**466,612**	40,000	11.67
A2	6.0 attributable hours per clinic-week	435,505	0.50	**217,752**	15,000	14.52
A3	Legacy remediation genuinely avoided	457,737	0.25	**114,434**	0	— (free)
A4	Enabling change funded outside the programme	320,000	0.45	**144,000**	0	— (free)
A5	Finance accepts USD 85 an hour	658,747	0.20	**131,749**	0	— (free)
	Total	3,205,166		**1,074,548**	55,000	

Total EMV is **USD 1,074,548**, which is **80.62 %** of the NPV the case promises. A1 alone is **35.01 %** of it. The three free tests together carry **USD 390,184 — 36.31 %** of the register's exposure. 5. **Interpretation.** Five readings, and the first is the single most useful sanity test on any business case.

The exposure ratio tells a board how thin its own case is. A positive NPV of 1,332,898 sitting above an assumption exposure of 1,074,548 is a ratio of **0.8062** — the case is right-side-up, but only just, and a board told "NPV +1.33m" hears something quite different from a board told "NPV +1.33m against 1.07m of expected exposure in the assumptions it rests on". The ratio also gives a usable rule of thumb for challenge: **where Σ EMV approaches or exceeds the NPV, the case is not a value proposition but a bet**, and it should be structured as one — staged, with kill criteria (2.4.3) and a real abandonment option (2.A.1) — rather than approved as a commitment.

The test-order rule is where the leader gets something for nothing. Three of the five assumptions cost nothing to test: A3 is a letter to the vendor asking whether support will be extended; A4 is a written funding confirmation from the clinical directorate; A5 is a conversation with the finance business partner about the valuation rate. Together they carry **390,184** of exposure and can be resolved in a week, before approval. Ranking by `EMV ÷ cost to test` rather than by EMV alone is what surfaces that, and it inverts the usual instinct, which is to spend the analysis budget on the biggest number — A1 — and to leave the cheap ones as "assumptions". **The professional habit: sort the register by exposure per unit of test cost, and clear the free rows before the paper is written.**

The one that must be structural, not analytical, is adoption. A1 cannot be resolved by analysis, only by evidence from operation, which is why its test costs 40,000 (a measured pilot cohort) and why it remains the largest residual after the free rows are cleared — 466,612 of a remaining 684,365. An assumption that can only be tested by doing part of the work is the definition of a case for staging, and 2.A.1 prices exactly that trade for Meridian. Note the symmetry with the alignment re-test of 2.1.3: both convert an unmanaged belief into a scheduled observation, and both are worth far more than the sophistication of the analysis around them.

EMVs are not additive as a portfolio loss, and saying they are is a real error. A1 (adoption) and A2 (attributable hours) are both statements about the same clinical behaviour and will fail together more often than independently; A5 (the valuation rate) is largely independent of both. So the 1,074,548 is the correct expected total — expectations add regardless of correlation — but it is **not** the right basis for a downside statement, because the correlated pair makes the tail worse than an independence assumption implies. Domain 8, KA 8.A.1 handles the correlation properly and KA

8.3 sizes contingency from it; the register's job here is to state the expected exposure and flag which entries move together.

What breaks it, and the honest limits. Every probability in the table is a judgement, and a register whose probabilities were set by the case's author will be systematically low — the same advocacy problem 2.2.1 identifies, appearing here as calibration rather than as omission. Impacts assessed as single points hide their own ranges. EMV also treats all failures as equivalent losses, when some are recoverable and some are not: A4 failing costs 320,000 and is survivable, while A1 failing changes what the programme is, and an expected-value table cannot express that difference — Domain 8's irreversibility treatment must be read alongside it. And a register is only alive if the test dates are in someone's calendar with a named owner; an assumption register reviewed at the same cadence as the risk register, by the same forum, with the same escalation, is the version that works, and a separate document nobody opens is the version that exists in most organisations.

Dependencies — on other projects, on suppliers, on decisions not yet taken — are managed by naming them, naming the owner on the other side, and stating the date by which the dependency must be satisfied. Two failure modes recur: **assumed dependencies** (nobody on the other side knows they owe you anything) and **unmanaged reciprocals** (they are depending on you too, on a different date). Domain 15's programme dependency management handles the multi-project case; at project level the discipline is simply that a dependency without a named counterpart is a hope.

AI in this KA

Benefits mapping is judgment work about causation, which is where AI is weakest — a model will happily assert that an output produces a benefit because most documents assert exactly that (Domain 1's KA 1.3 warning). Two legitimate uses. **Challenge**: ask a model to argue that the benefit will not materialise, and the enabling changes it names are often the ones the map omitted. **Cross-checking at portfolio scale**: detecting the same benefit claimed by two cases is pattern-matching over documents, which machines do well and humans do badly across fifty cases. The conclusions stay human, with named benefit owners.

Verification, concretely. Two specific hazards attach to the arithmetic of this KA. A model asked to compute an attributable benefit will happily do the subtraction and will **not** ask whether the comparison group is a fair counterfactual — so the professional check is on the cohort, not on the subtraction: were both groups measured the same way at the same time, and is the non-adopting group systematically different from the adopting one? That is a judgement about selection, and no amount of arithmetic substitutes for it. And a model asked to populate an assumption register will supply **probabilities**, fluently and without provenance, which then drive an exposure ratio that a board reads as analysis. Where the organisation has no history to calibrate from, the correct output is that it has none, plus the range of probability over which the recommendation is unchanged. What is safe to delegate is the reconciliation work: checking that every benefit in the register has exactly one claimant across the portfolio, that no avoided cost appears also as a benefit, and that the adoption term has been applied once rather than twice — all three are mechanical, all three are commonly wrong, and all three are reported with a location so a human can confirm them.

KEY TERMS — KA 2.3

Term	Meaning
Benefits map	Output → enabling change → outcome → benefit → objective, with owners and measures.
Enabling change	The non-project work (training, workflow, adoption support) without which no benefit occurs.
Baseline	The measured pre-change position; must be measured before, not reconstructed after.
Attribution	The portion of a measured improvement plausibly caused by this project.
Comparison cohort	A group that did not receive the change, used to measure the counterfactual improvement.
Attributable improvement	Raw improvement less the counterfactual improvement; the difference of the two differences.
Over-claim share	`counterfactual ÷ raw` — invariant to the valuation rate, the operating year and the number of adopters.
Double-counting	The same benefit claimed by more than one case; one claimant per benefit.
Cash-releasing / non-cash-releasing	Benefits that reduce spend versus those that release capacity.
Assumption register	Statement, basis, impact-if-false, owner, test date, falsifying trigger.
Assumption exposure ratio	`Σ EMV ÷ NPV`; as it approaches one, the case is a bet rather than a value proposition.
Test-order rule	Rank assumptions by `EMV ÷ cost to test`, so the free tests are cleared before the paper is written.

SAMPLE MCQS — KA 2.3

MCQ 2.3-A `[2.3.1 · Analysis]` A benefits map runs directly from "system installed" to "USD 685,440 released per year". Its principal defect is: - A. the benefit figure is too precise - B. the **enabling change** is missing — training, workflow redesign and adoption support, largely owned outside the project, are what convert the output into the outcome - C. it should be expressed in hours rather than money - D. the map should start with the strategic objective

Rationale: Omitting enabling change hides the work that determines realisation (2.3.1) — exactly how Meridian stalled at 40 % adoption with installation on plan. C is a separate (also real) question of unit; D is presentational.

MCQ 2.3-B `[2.3.2 · Application]` A programme releases 6 clinician-hours per week per clinic; headcount is unchanged and the hours are absorbed by existing demand. Reporting this as a cash saving is: - A. correct — released time has a value - B. incorrect: it is a non-cash-releasing (capacity) benefit unless cost is reduced or the capacity is converted to valued activity - C. correct if the hourly rate is documented - D. acceptable provided it is discounted

Rationale: Value released is not cash released (2.3.2); claiming otherwise is the error that discredits benefits cases. A and C mistake a valuation rate for a cash effect; D discounts a figure that was never cash.

MCQ 2.3-C `[2.3.4 · Analysis]` An enabling platform programme and three dependent projects each claim the same USD 4m of downstream benefit. The portfolio total is: - A. USD 16m, since each case is individually valid - B. overstated by double-counting — one benefit, one claimant, with the enabler credited through the dependents or vice versa but not both - C. USD 4m, and the dependent projects have no benefits - D. indeterminate until delivery completes

Rationale: Double-counting is the standing audit finding (2.3.2); the fix is a single claimant per benefit in the portfolio register. C overcorrects by denying the dependents any case; D defers a question answerable now.

MCQ 2.3-D `[2.3.3 · Application]` A carbon reduction has no priced compliance consequence for the organisation. The soundest treatment in the case is: - A. omit it, since it cannot be valued - B. report it in its physical unit alongside the financial case, rather than monetising it with an invented price - C. monetise it using a rate found in a published study - D. record it as a constraint

Rationale: Unpriced benefits are reported honestly in their own unit (2.3.3). A discards real value; C imports a price the organisation does not face; D confuses value with obligation.

MCQ 2.3-E `[2.3.2 · Application]` A pre-change baseline of 22.0 hours a week falls to 15.6 in the 28 adopting clinics and to 20.9 in the non-adopting comparison clinics. At USD 85 an hour over 48 weeks, the attributable annual benefit across the adopters is: - A. USD 731,136 - B. USD 605,472 - C. USD 685,440 - D. USD 125,664

Rationale: Attributable release is `6.4 − 1.1 = 5.3` hours, so `28 × 5.3 × 85 × 48 = 605,472` (2.3.2). A is the raw claim with no counterfactual deducted; C is the business case's 6.0-hour assumption, which is a forecast and not a measurement; D is the over-claim itself, not the benefit.

MCQ 2.3-F `[2.3.2 · Analysis]` A programme measures a raw improvement above its business-case assumption and reports benefits exceeding forecast. The comparison cohort improved by 1.1 of the 6.4 hours. The strongest professional statement is that: - A. benefits exceed forecast, since the measured improvement is larger than assumed - B. the attributable improvement is 5.3 hours — below the assumed 6.0 — so the case is not supported, and the over-claim is 17.1875 % of the raw figure whatever the valuation rate - C. the result is unusable because no randomised comparison exists - D. the counterfactual should be added to the claim, since both improvements are real

Rationale: Raw and attributable are different quantities, and the over-claim share is `counterfactual ÷ raw`, invariant to the rate, weeks and clinic count (2.3.2). A is the error the raw figure invites in a good year; C over-reaches — a convenience cohort is weak evidence, not no evidence; D claims an improvement the programme did not cause.

MCQ 2.3-G `[2.3.4 · Evaluation]` An assumption register carries EMVs of 466,612, 217,752, 114,434, 144,000 and 131,749 against a case NPV of 1,332,898; the last three cost nothing to test. The soundest first action is: - A. commission the 40,000 adoption study, since 466,612 is the largest exposure - B. clear the three free tests, resolving USD 390,184 — 36.31 % of the register's exposure — before the paper is written - C. reduce the case's NPV by the total EMV of 1,074,548 and re-submit - D. no action: an exposure ratio of 0.8062 is below 1, so the case stands

Rationale: Test priority ranks by EMV per unit of test cost, and a free test has no competitor (2.3.4). A is right on magnitude and wrong on order; C double-counts, since the EMVs are exposures around the NPV, not deductions from it; D treats a sanity ratio as an approval threshold.

SELF-CHECK — KA 2.3

1. Which element do most benefits maps omit, and why does it matter most? — The enabling change; it is the work that converts outputs into outcomes and it usually sits outside the project.
2. What makes a baseline trustworthy? — Measurement before the change, not reconstruction after.

3. State the relationship between assumptions and risks. — Every assumption is a risk in disguise; material ones get register entries with quantified exposure.
4. How is an attributable improvement computed, and what is the invariant? — Raw improvement less the counterfactual improvement; the over-claim share is `counterfactual ÷ raw` — 17.1875 % on Meridian — whatever the rate, weeks or number of adopters.
5. Why must both of Meridian's baseline errors be corrected, not one? — The flat profile flatters and the missing counterfactual penalises; correcting only one gives a breakeven of 46.4672 % or 31.4498 %, where the honest figure is **35.6035 %**.
6. What does an assumption exposure ratio approaching one tell a board? — That the case is a bet rather than a value proposition, and should be staged with kill criteria rather than committed.

KNOWLEDGE AREA 2.4

Strategic termination

Topics: 2.4.1 stopping as a strategic decision · 2.4.2 sunk cost and escalation of commitment · 2.4.3 kill criteria and honest gates.

2.4.1 Stopping as a strategic decision

Organisations that cannot stop projects cannot really select them either: a portfolio that only accretes has no capacity for anything new, and its selection process becomes a queue. Stopping is therefore a **portfolio capability**, not an admission of failure — and the leaders who can do it create the capacity that funds everything else.

The barriers are cultural and structural, and both are addressable. **Culturally**, cancellation is read as failure and punished, so nobody proposes it; the counter is to celebrate early stopping explicitly and to distinguish "we learned this will not work" from "we managed this badly". **Structurally**, no gate has authority to stop, or the decision is scheduled after the money is committed; the counter is Domain 3's decision rights with a real stop option and gates placed before irreversible commitments (Domain 1's irreversibility point).

2.4.2 Sunk cost and escalation of commitment

The principle. A continuation decision depends only on **remaining cost and remaining benefit**. Money already spent is irrelevant to it — it cannot be recovered by spending more, and it carries no information about the future beyond what is already in the forecast.

WORKED EXAMPLE

Worked example 2.4.2 — Meridian at the reset point.

1. **Setup.** A hypothetical reset: USD 1,800,000 of the 2,400,000 has been spent. Completion requires a further **USD 900,000** (the original 600,000 plus 300,000 of newly discovered integration work). The benefits case has been re-based on measured adoption, giving a remaining benefit present value of **USD 780,000**. Continue or stop?
2. **Formula.** Forward-looking NPV = remaining benefit PV − remaining cost. Sunk cost excluded.
3. **Substitution.** `780,000 − 900,000`.
4. **Result. Forward NPV = (USD 120,000).** The correct decision is to **stop** (or to descope to a variant whose remaining benefit exceeds its remaining cost).
5. **Interpretation.** Five readings, and the fourth is the one that keeps a leader from over-applying the rule.

The sunk-cost frame does not merely mislead; it multiplies the loss by 7.5. The pull to continue is powerful and entirely arithmetical in appearance: "we have spent 1,800,000 — stopping wastes it." That reasoning adds a sunk 1,800,000 back into a decision it cannot affect, and it converts a **120,000** loss into a **900,000** one. The 1,800,000 is already gone under either choice; the only question is whether the next 900,000 buys 780,000 of value. It does not.

State the decision as a threshold, because a threshold can be checked. The continuation is worth taking at any remaining cost below **780,000**, so the 900,000 estimate exceeds the justified ceiling by **USD 120,000**, or **15.38 %** of it. Equivalently, continuation needs the remaining benefit to be at least 900,000, which is **15.38 %** more than the re-based 780,000. That is a far better gate sentence than a negative NPV, because it tells the meeting exactly what would have to be true — and it is testable: a descope that removes 120,000 of remaining cost without removing benefit turns the decision round, which is why "stop" and "descope" belong in the same paper (Case study B does exactly this at scale).

Stopping releases the money into the constrained selection problem, which is where its real value appears. The 900,000 is redeployable to candidates the portfolio could not previously fund (KA 2.2.3's constraint arithmetic) — and if the released capacity also includes the scarce team, 2.2.3b prices what that is worth, which may exceed the money. **This is why stopping is a strategic act and not merely a loss**, and it is also why the paper must say what the released resources will do next: a board asked to accept a write-off with no redeployment proposal is being asked to take the whole loss and none of the gain.

The rule is about unrecoverable spend, and three things are commonly mis-sorted into it. Sunk cost is spend that cannot be recovered and carries no information about the future beyond what the forecast already holds. Recoverable amounts are not sunk and belong in the forward arithmetic — resale or redeployment value of equipment and licences, recoverable prepayments, and reusable assets such as cleaned data or a qualified team. Contractual exit costs are not sunk either: they are a **forward** cost of stopping, and if terminating the supplier costs 150,000 the comparison is `780,000 − 900,000 = (120,000)` for continuing against `(150,000)` for stopping, and the correct decision reverses. And a cost already incurred does carry information where it revises the forecast — an overspend that reveals the work is harder than believed should change the estimate of the remaining 900,000, which is a legitimate use of history and not the sunk-cost fallacy. **The discipline is to exclude sunk cost from the comparison while using the evidence it generated.**

What breaks it. The remaining-benefit figure must be re-based on measured evidence, not inherited from the original case — using the approval-time benefit here would give `3,732,898 − 900,000` and a spurious continuation. Both sides must be on the same present-value basis and the same horizon. And the decision must be available: where an obligation is contractual or statutory, there is no stop option to exercise and the arithmetic is describing a choice nobody holds, in which case the honest output is the size of the loss and a claim or renegotiation strategy (Domain 10), not a recommendation to stop. **What a termination actually costs, and whether the exit right exists at all, are questions of the particular contract and the governing law: take legal advice on the exit position before the 150,000 goes into a gate paper as a number.**

Escalation of commitment is the behavioural pattern that makes this hard: personal identification with the project, reputational exposure for the approver, and the asymmetry that continuing defers the reckoning while stopping realises it now. Its countermeasures are structural, not exhortative — pre-set kill criteria (2.4.3), decision-makers who did not author the case, and forward-only framing in every gate paper (Domain 8's bias table).

2.4.3 Kill criteria and honest gates

Kill criteria are conditions, agreed **in advance**, under which the project will stop — "if adoption in the pilot cohort is below 50 % at month six", "if the regulatory approval is not granted by Q3", "if remaining cost exceeds remaining benefit at any gate". Their power is entirely in the advance agreement: the same evidence assessed against a pre-agreed threshold is a decision, while assessed against no threshold it is a negotiation, and negotiations are won by whoever has most invested.

That establishes why a criterion works. It says nothing about **where the number goes**, and a criterion set at the wrong level is worse than none, because it stops the wrong projects with full procedural authority. Setting it is an arithmetic problem with two error costs.

WORKED EXAMPLE

Worked example 2.4.3 — calibrating the kill criterion, and the two errors it trades.

1. **Setup.** Meridian's successor programme carries a kill criterion at the **month-6** gate, by which point **USD 900,000** of the 2,400,000 has been committed and **USD 1,500,000** remains. The month-6 signal is early adoption in the first cohort, which classifies a programme as **Weak** or **Strong**. From the authority's own history of comparable rollouts, eventual sustained adoption falls into three states — **Low 25 %** (probability 0.25), **Mid 45 %** (0.45), **High 70 %** (0.30) — and the signal is imperfect: it reads Weak for **80 %** of Low programmes, **35 %** of Mid programmes and **10 %** of High programmes. All figures are **locally calibrated judgements from a small sample and must be labelled as such.** Should the criterion stop programmes flagged Weak?
2. **Formula.** Forward NPV at the gate `= PV of remaining benefit − remaining cost`, where the PV of the remaining benefit at sustained adoption `s` is `5,332,711 × s` (the ramped PV per unit of sustained adoption, 2.2.2). Forward breakeven adoption `= remaining cost ÷ 5,332,711`. Then compare two policies: **A** continue everything, `E[NPV] = Σ P(s) × NPV(s)`; **B** stop on Weak, `E[NPV] = Σ P(s) × P(Strong|s) × NPV(s)`, taking a stopped programme's forward value as zero.
3. **Substitution.** Forward breakeven `1,500,000 / 5,332,711`. `NPV(0.25) = 1,333,178 − 1,500,000`; `NPV(0.45) = 2,399,720 − 1,500,000`; `NPV(0.70) = 3,732,898 − 1,500,000`. Policy B: `0.25 × 0.20 × (−166,822) + 0.45 × 0.65 × 899,720 + 0.30 × 0.90 × 2,232,898`.
4. **Result. Forward breakeven adoption at this gate is 28.1283 %**, against a whole-investment breakeven on the same ramped basis of **45.0053 %**.

Eventual state	Prior	Forward NPV	P(Weak)	Effect of stopping on Weak
Low 25 %	0.25	**(166,822)**	0.80	loss avoided **+33,364**
Mid 45 %	0.45	899,720	0.35	value destroyed **(141,706)**
High 70 %	0.30	2,232,898	0.10	value destroyed **(66,987)**

Policy A (continue everything) **E[NPV] = USD 1,033,038**. Policy B (stop on Weak) **E[NPV] = USD 857,709**. The criterion **destroys USD 175,328** of expected value: it saves 33,364 and costs 208,693. 5. **Interpretation.** Six readings. The first two are the calibration errors that produce most bad criteria; the last reconciles this result with everything 2.4.3 says in favour of kill criteria.

A criterion set at the investment breakeven kills programmes that are still worth completing. The intuitive threshold — "stop unless it will clear the adoption the business case needed" — is **45.0053 %** on this basis. The threshold that maximises value at this gate is **28.1283 %**, because 900,000 is already sunk and the decision is about the next 1,500,000 only. A criterion pitched at the investment breakeven condemns every programme whose eventual adoption lands between 28.1283 % and 45.0053 % — a **16.8770-point** band containing the entire Mid state — even though completing them returns 899,720 each. **The forward breakeven is the only defensible level for a criterion applied at a gate**, and the fact that it moves later in delivery is a feature: a criterion should get harder to trigger as the remaining cost falls, which is the exact opposite of how most escalating governance behaves.

The signal, not the threshold, is what makes this criterion fail. Even at the right threshold, the criterion is applied to a proxy. Of the programmes it flags Weak, most are not Low: the flagged population is `0.25 × 0.80 = 0.20` Low, `0.45 × 0.35 = 0.1575` Mid and `0.30 × 0.10 = 0.03` High, so **48.39 %** of everything it stops would have returned positive value. The arithmetic requirement is severe and worth stating exactly: even a criterion that detected **every** Low programme and never flagged a High one could tolerate a false-flag rate on Mid programmes of no more than `41,706 / 404,874 =` **10.3009 %**. A proxy signal that good does not exist at month six on a two-year rollout, which is the honest reason early adoption thresholds so often do damage.

The asymmetry is structural, not a quirk of these numbers. The loss avoided by stopping a bad programme is bounded by how negative its forward NPV is — here 166,822 — while the value destroyed by stopping a good one is its whole remaining upside, 899,720 or 2,232,898. When the upside is an order of magnitude larger than the downside, a criterion has to be far more accurate than intuition suggests. The corollary, and it is important, is that **the asymmetry reverses where the downside is large** — a programme whose failure carries a regulatory penalty, a safety consequence or a contractual liability has a much more negative forward NPV, and the same signal quality then makes the criterion clearly worth having. **Kill criteria pay where failure is expensive, not where success is uncertain.**

The fix is not to abandon the criterion but to change what it does. Three changes each make it value-positive. Set the threshold on the **decision-relevant quantity** — forward NPV computed from the re-based benefit case — rather than on a proxy, which is what the third of 2.4.3's example criteria ("if remaining cost exceeds remaining benefit at any gate") already does and is why it is the strongest of the three. Where a proxy is unavoidable, make the criterion trigger a **mandatory re-appraisal with a real stop option** rather than an automatic stop: the re-appraisal costs a few tens of thousands and removes the false-stop loss almost entirely, which on these figures is a trade of 208,693 against a fraction of it. And **improve the signal by buying information before the money is committed** — which is precisely the staged commitment 2.A.1 prices, and the reason real options and kill criteria are two halves of one instrument rather than two topics.

What this does not license. Nothing here argues against pre-agreed criteria, and the reasoning must not be borrowed to defend a portfolio that never stops anything. The result is narrower and sharper: **a criterion is an instrument with a computable expected value, and it must be calibrated rather than asserted.** The gate that has never stopped a project (MCQ 2.4-B) is failing a different test — it has no stop option at all — and this arithmetic assumes throughout that the option exists and can be exercised. A leader who cannot compute the two error costs should still insist on criteria; a leader who can should insist on the right ones.

The limits of the model. It assumes a stopped programme has zero forward value, which understates Policy B wherever partial delivery is usable or the released capacity has an alternative use worth more than nothing — reinstating either shrinks the 175,328 and could reverse it, and the honest version of this table has a redeployment value in it. The three states and the signal rates come from a small internal sample, so the conclusion should be tested across a range rather than taken at a point; the direction of the result is robust to plausible variation, its magnitude is not. And it prices only expected value, so it says nothing about a board's willingness to carry the variance — which is a legitimate risk-appetite judgement belonging to the sponsor (Domain 8, KA 8.4).

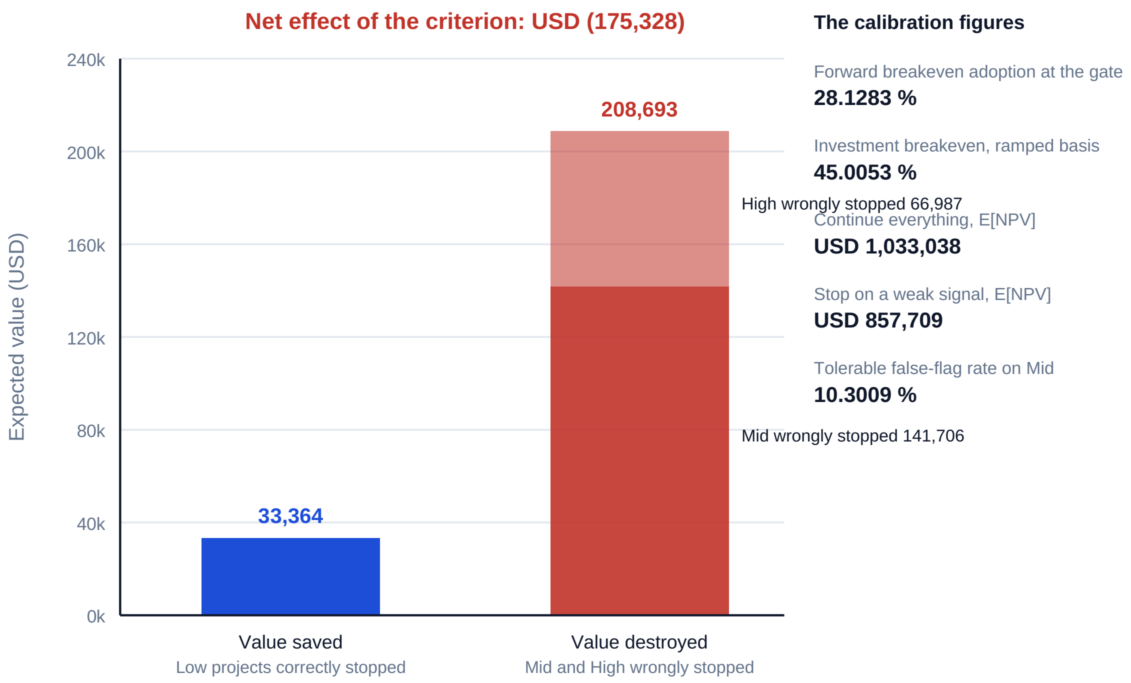


Fig 2.4.1 — Why a kill criterion on a weak signal destroys value

An **honest gate** has four properties: a **real stop option** with someone authorised to exercise it; **evidence prepared to a standard set beforehand**, not assembled to support a conclusion; **decision-makers who did not write the case**; and **a recorded decision with reasons** (Domain 3's auditability). A gate that has never stopped anything is not a control; it is a milestone with a meeting attached.

Stopping well is its own competence: capture what was learned and make it findable (Domain 9's lessons); harvest anything reusable (code, designs, cleaned data, supplier relationships, qualified people); close contracts deliberately rather than by abandonment (Domain 10); redeploy the team with their reputations intact — because how a cancellation is handled determines whether anyone proposes the next one; and tell stakeholders honestly, early (Domain 11).

AI in this KA

The relevant AI use here is monitoring: tracking assumption triggers and kill-criteria thresholds across a portfolio, and flagging when a project's own reported data has crossed one. That is genuinely valuable because the crossing is usually visible in data nobody is reading. What must remain human is the decision and the framing: a model asked whether to continue will optimise whatever objective it was handed, and the escalation-of-commitment bias lives in the humans around it, not in the arithmetic. The leader's use of AI here is to make the trigger impossible to miss — not to outsource the courage.

Verification, concretely. Monitoring is only as good as the threshold it monitors, so the first check is on the criterion itself: is it set at the **forward** breakeven, recomputed at this gate, or at the investment breakeven inherited from approval? A monitoring system faithfully alerting against a mis-calibrated threshold industrialises the error of 2.4.3. Where a model is used to score the two error costs of a candidate criterion, its signal-quality inputs — the false-flag rates — must come from counted outcomes on comparable past projects, and where those do not exist the honest output is the **breakeven signal quality** (10.3009 % on Meridian's successor) rather than a recommendation: it tells a sponsor how good the signal would have to be, which is a decidable question. Reproduce the two policy expectations by hand; each is three multiplications and a sum. And no model output should ever be the trigger of an automatic stop — a stop is an attributable decision by a named authority (Domain 3, KA 3.A.2), and the value of automation here is entirely in making the crossing visible on the day it happens rather than in the quarter it is noticed.

KEY TERMS — KA 2.4

Term	Meaning
Forward-looking NPV	Remaining benefit PV less remaining cost; the only basis for continuation.
Sunk cost	Unrecoverable spend carrying no information beyond the forecast; irrelevant to a continuation decision.
Forward exit cost	The cost of stopping — termination, demobilisation, disposal; a forward cost, never sunk.
Forward breakeven	The adoption or benefit level at which the remaining cost is just repaid; the only defensible level for a gate criterion.
Escalation of commitment	Continuing because of what is invested; countered structurally.
Kill criteria	Pre-agreed stop conditions; powerful only because agreed in advance, and only useful if calibrated.
False stop / false continue	Stopping a project that would have created value; continuing one that will not. The two errors a criterion trades.
Honest gate	Real stop option, pre-set evidence standard, independent deciders, recorded reasons.

SAMPLE MCQS — KA 2.4

MCQ 2.4-A `[2.4.2 · Application]` 1,800,000 is spent; completion needs a further 900,000; remaining benefit PV is 780,000. The correct decision and its basis are: - A. continue — stopping wastes the 1,800,000 already spent - B. stop — forward NPV is (120,000); the sunk 1,800,000 is irrelevant to the remaining choice - C. continue — total spend 2,700,000 against total benefits exceeds the original case - D. continue at reduced pace to spread the cost

Rationale: Only remaining cost and benefit bear on the decision (2.4.2). A is the sunk-cost fallacy stated plainly; C reintroduces sunk cost as "total"; D changes the schedule without changing the negative economics.

MCQ 2.4-B `[2.4.3 · Analysis]` A portfolio's gates have never stopped a project in four years. The soundest inference is: - A. selection is excellent, so no project has needed stopping - B. the gates are not functioning as controls — no real stop option, evidence assembled to support continuation, or deciders too close to the cases - C. the gate criteria are too lenient and should be tightened numerically - D. stopping is unnecessary if delivery is well managed

Rationale: A control that never fires is not demonstrably a control (2.4.3). A is implausible across a whole portfolio; C addresses thresholds when the defect is process and authority; D mistakes delivery quality for strategic validity.

MCQ 2.4-C `[2.4.3 · Recall]` Kill criteria derive their power from: - A. the severity of the thresholds - B. being agreed in advance, so the same evidence produces a decision rather than a negotiation - C. being set by the project manager - D. being confidential until invoked

Rationale: Advance agreement converts assessment into decision (2.4.3); without it, whoever has most invested wins the argument. C removes independence; D prevents the behavioural effect entirely.

MCQ 2.4-D `[2.4.3 · Application]` At a gate, USD 900,000 is spent and USD 1,500,000 remains; the PV of remaining benefit is `5,332,711 × s` for sustained adoption `s`. The adoption at which the remaining spend is just repaid is: - A. 45.0053 % - B. 28.1283 % - C. 41.0460 % - D. 62.5000 %

Rationale: `1,500,000 / 5,332,711 = 28.1283 %` (2.4.3). A is the whole-investment breakeven on the same ramped basis, which reintroduces the sunk 900,000 into a forward decision; C is the flat-equivalent whole-investment breakeven, wrong on both counts; D divides remaining cost by total cost, which is a completion percentage and not a breakeven at all.

MCQ 2.4-E `[2.4.3 · Evaluation]` A criterion flags 80 % of Low-adoption programmes, 35 % of Mid and 10 % of High. Priors are 0.25 / 0.45 / 0.30 and forward NPVs (166,822) / 899,720 / 2,232,898. Stopping on a flag: - A. saves USD 33,364 of expected loss, so it is worth applying - B. destroys USD 175,328 of expected value, because the value of the good programmes it stops exceeds the loss of the bad ones it prevents - C. is value-neutral, since the criterion is applied consistently to every programme - D. cannot be assessed without knowing each programme's actual outcome

Rationale: `33,364 − (141,706 + 66,987) = (175,328)` (2.4.3). A counts only the benefit side of the trade — the standard omission; C confuses procedural consistency with expected value; D is the argument that prevents any criterion from ever being calibrated, since actual outcomes are by definition unavailable at the gate.

MCQ 2.4-F `[2.4.2 · Analysis]` Remaining cost is 900,000 and remaining benefit PV is 780,000, so forward NPV is (120,000). Terminating the principal supplier would cost 150,000. The correct decision is: - A. stop, since forward NPV is negative - B. continue, because the 150,000 exit cost is a forward cost of stopping and exceeds the 120,000 cost of completing - C. stop, and treat the 150,000 as a sunk cost of the original decision - D. continue, because the 1,800,000 already spent would otherwise be wasted

Rationale: Exit costs are forward costs, so the comparison is (120,000) against (150,000) (2.4.2). A applies the rule without completing the comparison; C misclassifies a future payment as sunk; D is the sunk-cost fallacy, which reaches the same answer here for entirely the wrong reason — and would reach the wrong answer if the exit cost were 50,000.

SELF-CHECK — KA 2.4

1. What is the only correct basis for a continuation decision? — Remaining benefit against remaining cost; sunk cost excluded, forward exit costs included.
2. Why is stopping a strategic act rather than a loss? — It releases capacity and funding for candidates the portfolio could not otherwise support.
3. Name two properties of an honest gate. — A real stop option with authority; deciders who did not write the case (also: pre-set evidence standards; recorded reasons).
4. Which three things are commonly misclassified as sunk cost? — Recoverable value (resale, redeployment, reusable assets); forward exit costs; and the information an overspend generated, which should revise the remaining estimate.
5. At what level should a gate criterion be set, and why does it move? — At the forward breakeven — 28.1283 % on Meridian's successor at month six against a 45.0053 % investment breakeven — and it rises as remaining cost falls, so a criterion should get harder to trigger over time.
6. When do kill criteria clearly pay? — Where failure is expensive rather than where success is uncertain: the false-stop loss scales with the upside forgone, so the trade improves as the downside grows.

Advanced topics — Domain 2

2.A.1 Real options thinking in selection

Static NPV values a committed plan; much project value lies in **flexibility** — the option to scale if a pilot succeeds, to defer while uncertainty resolves, to abandon cheaply, to switch technology. Treating these as options changes selection behaviour in a specific way: a small, staged first phase with a negative standalone NPV can be the highest-value choice if it buys the right to a large second phase at a decision point where far more is known. The practical rule for a delivery leader is **structure the work so that decision points exist** — phases sized to reach a genuine learning milestone, contracts that permit exit (Domain 10), architectures that permit switching. Formal option valuation is specialist; the thinking is not, and its absence is why organisations routinely commit fully to things they could have tested for a fraction.

The thinking is also arithmetic, and doing the arithmetic once cures the two opposite errors it attracts — treating a pilot as automatically prudent, and treating it as automatically a delay.

WORKED EXAMPLE

Worked example 2.A.1 — staging Meridian, and what the option to abandon is actually worth.

1. **Setup.** Two ways to acquire Meridian's 40 clinics. Sustained adoption will turn out to be **High, 70 %** (probability 0.60) or **Low, 35 %** (0.40) — an honest statement of the uncertainty the case carried. The PV of the ramped benefit stream per unit of sustained adoption is **5,332,711** (2.2.2), so High is worth 3,732,898 and Low 1,866,449 of benefit PV. **Option F — full commitment:** 2,400,000 now for all 40 clinics, at 60,000 a clinic. **Option S — staged:** a first phase of **12 clinics** now for **840,000** (70,000 a clinic — a 16.6667 % premium, because a small deployment loses the scale of a rollout), which reveals the adoption state at the end of year 1; then either commit the remaining 28 clinics for **1,680,000** at the original unit rate, or abandon. Staging delays the whole estate's benefit stream by **one year**; if abandoned, the 12 installed clinics stay in service and earn their 30 % share at the Low adoption. Rate 7 % throughout.
2. **Formula.** Value each option as a probability-weighted NPV at `t = 0`, discounting the year-1 commitment and the shifted benefit stream by `1/1.07`. Then decompose: **price of staging** `= E[NPV of F] − E[NPV of staging with no abandonment option]`, and **value of the option to abandon** `= E[NPV of staging] − E[NPV of staging committed to complete]`. The verdict is the option's value less its price, which is identically `E[NPV of S] − E[NPV of F]`.
3. **Substitution.** F: `0.60 × (3,732,898 − 2,400,000) + 0.40 × (1,866,449 − 2,400,000)`. S, High branch: `−840,000 − 1,680,000/1.07 + 3,732,898/1.07`. S, Low branch (abandon): `−840,000 + 0.30 × 1,866,449/1.07`. S, Low branch if committed anyway: `−840,000 − 1,680,000/1.07 + 1,866,449/1.07`.
4. **Result.**

	High (0.60)	Low (0.40)	Expected NPV
F — full commitment	+1,332,898	(533,551)	**+586,318**
S — staged, abandon on Low	+1,078,596	(316,697)	**+520,479**
S — staged, committed to complete	+1,078,596	(665,749)	+380,858

Price of staging USD 205,460 (delay plus the 120,000 scale premium). **Value of the option to abandon USD 139,621.** Net **(USD 65,839)** — staging loses money on these figures. Staging becomes the better choice once the probability of the Low state exceeds **53.9740 %**. Priced instead at the programme's own unit rate (720,000 for 12 clinics, no premium), staging is worth **+640,479** and beats full commitment by **54,161** — so the **maximum justifiable first-phase cost is USD 774,161**, a premium ceiling of **7.5223 %** against the 16.6667 % assumed. 5. **Interpretation.** Six readings. The decomposition in the second is the transferable technique; the fourth is the actionable finding.

The option to abandon is real, computable and worth USD 139,621 — and that is not the answer. Everything the received wisdom says about piloting is true: the option has positive value, it is worth more the more uncertain the adoption, and abandoning after 840,000 is enormously better than abandoning after 2,400,000 (the Low branch improves from (665,749) to (316,697), a gain of **349,052**). None of that settles the decision, because the option has a **price**, and here the price is larger.

Always separate the price of staging from the value of the option, because they have different owners and different fixes. The price, 205,460, is delay plus scale premium — both consequences of how the staging is designed, and both reducible by design. The value, 139,621, comes from the uncertainty and the abandonment right — properties of the situation, largely not in the leader's gift. Reporting one net figure hides the only lever that exists. **The professional move is not "should we pilot?" but "what would the pilot have to cost and how long could it take?"**

The whole result turns on 120,000 of scale premium, which is the actionable finding. Remove the premium and staging wins by 54,161; keep it and staging loses by 65,839. The ceiling is **7.5223 %**, so any pilot priced above about a 7.5 % premium over the programme's unit economics destroys value on these probabilities. That converts an abstract exhortation — "structure the work so decision points exist" — into a procurement instruction: **the pilot must be bought at rollout unit rates, or it is not a pilot but a small expensive project.** In practice that means negotiating the whole-estate rate up front with a phased commitment, which is Domain 10's framework-with-call-off pattern (KA 10.3), and it is the single highest-value thing a leader can do to make staging viable.

Delay is the other half of the price, and it is priced by the cost of delay, not by sentiment. Shifting the whole benefit stream by a year costs `3,732,898 × (1 − 1/1.07) =` **244,208** in the High state. That is why staging fails most often on benefit-generating work with a high cost of delay — Meridian's USD 14,280 a week — and succeeds most often where benefits are far off or the downside is catastrophic. **A pilot that does not shorten its own decision point is paying full price for the option and taking delivery late.**

The breakeven probability is the sentence for the board. "Staging pays if we think there is more than a 53.9740 % chance this does not achieve the adoption the case needs" is answerable in a room; "the real option is worth 139,621" is not. And it exposes something uncomfortable and useful: an organisation that wants to pilot everything is asserting that it usually expects failure — which may be true, and if it is, the correct response is to fix the selection process rather than to pay a premium on every programme to hedge it.

What this is not, and the honest limits. This is an **expected-value decision tree** (Domain 8, KA 8.2.2), not an option valuation: it discounts both branches at the same 7 % and takes the probabilities as given, where formal option pricing would treat the risk differently — PFL-AI Domain 4 sets out the appraisal machinery and its assumptions, and a genuinely large staged commitment warrants specialist valuation. It also assumes the pilot **resolves** the uncertainty, which is generous: a 12-clinic pilot with early adopters may say very little about clinic 37, and a pilot that only partly resolves the question is worth proportionately less — the same signal-quality problem 2.4.3 computes for kill criteria, and the reason the two topics belong together. It ignores the learning that makes phase 2 cheaper or better, which biases against staging, and it ignores the organisational cost of restarting a stalled programme, which biases for it. Finally, the two states are a simplification of a continuum; the sensitivity that matters — the premium ceiling — should be recomputed across the plausible probability range before it is quoted.

2.A.2 Portfolio balance, not just portfolio ranking

Ranking selects the best individual candidates; **balance** asks whether the resulting set is survivable and coherent. Four balance questions: **risk profile** (is everything high-risk, or everything safe and incremental?), **time-to-benefit** (does anything land this year, or is all value in years 3–5?), **capability spread** (does the whole set depend on one scarce team — the correlation problem of Domain 8, KA 8.A.1, at portfolio level?), and **strategic coverage** (are some stated objectives served by nothing at all?). A portfolio of individually optimal projects can be collectively unbalanced, and no scoring model detects it — which is why Domain 15 treats portfolio balancing as its own discipline.

The alignment index of 2.1.1 sits on the allocation side of this line and answers none of the four questions: a portfolio funded exactly to weight can still be entirely high-risk, entirely long-dated, and entirely queued behind the one integration team whose marginal value 2.2.3b computes. **Read the index for allocation, then ask the four balance questions separately**, because each has a different remedy and the index cannot see any of them.

2.A.3 The reviewer's business-case eye

Invariants testable in an hour: a do-nothing baseline exists and is quantified; at least one option could plausibly have won; costs carry a range and an accuracy class; **the benefits profile ramps**; every benefit traces to a measured outcome with an owner outside the project; enabling changes are named with their owners; no benefit is claimed by another case in the portfolio; assumptions have test dates and falsifying triggers; risk exposure is quantified and reconciles to the register; kill criteria exist and are numeric; and success criteria were agreed in advance. The single highest-yield check is the benefits profile — it is wrong in most cases, and it is wrong in the direction that flatters.

Six further checks, each of which the domain's arithmetic makes into a one-line test, and each of which a case can fail while passing all of the above.

- **The breakeven carries its basis.** A breakeven adoption stated without "flat-equivalent" or "ramped-basis" is ambiguous by 3.9592 points on Meridian's figures (2.2.2). Ask which.
- **The counterfactual is priced, not asserted as zero.** Ask what doing nothing costs in years 3 to 8, and whether the avoided costs are genuinely avoided rather than deferred or claimed elsewhere (2.2.2b). Meridian's omission was worth 561,101, in the case's favour.
- **The benefit measurement has a comparison cohort, or says it has not.** Raw and attributable are different quantities, and the over-claim share is `counterfactual ÷ raw` (2.3.2).
- **Σ EMV of the assumption register against the NPV.** Meridian's ratio is 0.8062; anything near or above 1 means the case should be staged rather than committed (2.3.4).
- **The ranking's flip point is reported.** If a 3.33-point weight shift reverses the recommendation, the recommendation is a preference (2.2.3c).
- **Gate criteria are set at the forward breakeven, not the investment breakeven.** A criterion pitched at the latter condemns everything in a 16.8770-point band that is still worth completing (2.4.3).

Industry variations — Domain 2

- **Public sector.** Appraisal is often mandated (published guidance, prescribed discount rates, optimism-bias uplifts), social value and distributional effects sit beside financial NPV, and political commitment can precede the case — making 2.4's stopping problem acutely hard and 2.1.3's re-testing acutely necessary. The specific arithmetic consequence is that a mandated uplift is a **breakeven** change, not a presentational one: a 20 % optimism-bias uplift on Meridian's cost takes it to 2,880,000 and the flat-equivalent breakeven adoption from 41.0460 % to **49.2552 %**, a rise of **8.2092 points** that puts the achieved 40 % decisively out of reach. A leader working under a mandated uplift should therefore compute the breakeven after it and manage adoption to that figure from day one, rather than discovering at evaluation that the standard was moved before approval.
- **Regulated utilities.** Investment cases are made to a regulator on a periodic cycle; benefits are defined by the regulatory framework, and the "customer" for the case is external — so the discount rate is set by the regulatory determination rather than chosen. That makes rate sensitivity a live governance figure: on Meridian's shape, one percentage point of discount rate (7 % to 8 %, `AF` 5.971299 to 5.746639) moves the flat-equivalent breakeven from 41.0460 % to

42.6507 %, a rise of **1.6047 points**. It also means the supersession hazard of 2.1.3 is low within a price control and spikes at its boundary, which is where the alignment half-life should be recomputed rather than assumed.

- **Technology and product.** Real options thinking (2.A.1) is the native mode — staged funding, pilots, kill criteria on usage metrics — and benefits are frequently non-cash-releasing capacity or optionality, so 2.3.2's honesty about units matters most here. The premium ceiling is the figure to carry across: staging pays only while the first phase can be bought near rollout unit economics (7.5223 % on Meridian's numbers), which in a product context usually holds, because a smaller first release is genuinely cheaper rather than merely smaller — and that structural difference, not cultural preference, is why staging is the default here and not in construction.
- **Energy and infrastructure.** Whole-life and carbon considerations are first-order value (2.3.3), horizons are decades, and PFL-AI's bankability tests (its Domain 5) run alongside the internal case because external capital must also be convinced. Two of this domain's mechanisms invert here. The alignment half-life is short relative to everything else in the appraisal — a 30-year asset outlives every strategy that will ever be written about it — so alignment is managed by designing for optionality rather than by re-testing fit. And the scale premium on staging is usually prohibitive, because civil works do not divide, which is why the flexibility is bought in the design (spare capacity, provision for a second circuit) rather than in the commitment.
- **Health and social programmes.** Benefits are largely non-financial and attribution is genuinely hard; the professional response is to measure outcomes in their own units with pre-change baselines rather than to force monetisation. Meridian's own attribution correction is the worked illustration — a comparison cohort removed **17.1875 %** of a claim that had looked like an over-performance — and it is representative rather than exceptional: where a whole system is changing at once, the counterfactual is frequently a large fraction of the measured effect. Where withholding the change is not acceptable, a staged rollout gives the cohort as a by-product of the sequencing, which is usually the only defensible route available.

Case study — Domain 2: the Meridian case that should have been written (public health)

Situation. Meridian's approved business case claimed **USD 979,200** of annual benefit from year one — the full-potential figure — supporting an NPV of **+3,447,096** over eight years at 7 % against a delivery cost of 2,400,000. Every number in it was arithmetically correct. No adoption term appeared anywhere, and no benefit had an owner outside the programme.

What the honest case looks like. Same cost, same horizon, same rate, same 70 % steady-state adoption Domain 1 establishes — but profiled as it would actually arrive (40 % / 60 % / 70 %): PV of benefits **3,732,898**, NPV **+1,332,898**. The approved case overstated present value by **USD 2,114,198**, or **158.6 %** of the honest NPV. It also omitted the sentence that would have mattered most: **breakeven sustained adoption is 41.05 %**, so the programme creates value at any adoption above roughly 41 % — a threshold a board can monitor, unlike an NPV.

What followed, and why it was predictable. Both cases approve the programme, so the flat profile changed no decision — and was therefore never challenged. Two years later, delivery complete and adoption at 40 %, the programme was measured against a promise of 979,200 and publicly judged a failure (Domain 1's case study). Actual annual benefit at 40 % adoption was **391,680**, and against the case as approved the programme's NPV at that adoption is `0.40 × 5,847,096 − 2,400,000 =` **(USD 61,162)** — a small negative, and the arithmetic that made the failure verdict defensible.

The verdict does not survive an honest baseline. Two corrections were available at approval and neither was made, and they run in opposite directions.

Treatment of the same facts	Flat-equivalent breakeven adoption	NPV at the achieved 40 %
The case as approved (zero counterfactual, 6.0 h assumed)	41.0460 %	**(61,162)**
Attribution corrected only (5.3 attributable hours)	46.4672 %	(334,026)
Counterfactual priced only (do-nothing costs 561,101 of PV)	31.4498 %	+499,939
Both corrections — the honest case	**35.6035 %**	**+227,074**

The programme was **1.05 percentage points short** of the breakeven it was judged against and **4.3965 points clear** of the breakeven that was true. Meridian at 40 % adoption created **USD 227,074** of value and was publicly recorded as a failure. **The verdict measured the business case, not the programme** — and the two errors that produced it are of opposite sign, which is why a reviewer who only knows that cases are optimistic would have found one of them and made the finding worse.

A third figure was available and would have changed the monthly conversation. At the assumed 70 % adoption the honest case's assumption register carried **USD 1,074,548** of expected exposure against an NPV of 1,332,898 — a ratio of **0.8062** — of which **USD 390,184** could have been resolved for nothing before approval by a letter to the legacy vendor, a written funding confirmation for the enabling change and one conversation with the finance business partner about the valuation rate (KA 2.3.4). None of the three was done.

What was done differently on the successor programme. The case carried a ramped profile with adoption as a tracked measure; a named benefits owner in the clinical directorate; the enabling changes (training, workflow redesign, champions) itemised with their own owners and costs; the breakeven adoption stated on the front page; and a kill criterion at 50 % adoption by month six. The board's monthly question changed from "are clinics live?" to "what is adoption, against 41 %?".

What the domain teaches here. A business case is a promise, and the profile is the promise's shape. An overstatement that changes no decision still changes the standard the programme will be held to — which makes the flat-profile error uniquely dangerous, because nothing at approval time creates any pressure to catch it.

Case study B — Domain 2: the platform that could not be stopped (financial services)

Situation. A core-platform replacement, approved at USD 40m over three years, reached year four having spent USD 62m with roughly 55 % of scope delivered. A re-based assessment put remaining cost at USD 21m and remaining benefit present value at USD 14m — a forward NPV of **(USD 7m)**.

What happened at the gate. The paper presented total investment (62m spent, 21m to go) against total lifetime benefits, showing a positive "programme return", and recommended continuation. Three structural features made that recommendation almost inevitable: the gate had **no stop option** defined; the paper was written by the programme's own director; and there were **no kill criteria** from the original approval to test against. The sponsor who had championed the original case chaired the gate.

How it was resolved. An independent review, commissioned only after a regulator asked about delivery timescales, reframed the decision as forward-only: 21m to buy 14m. The programme was descoped to a variant whose remaining benefit (9m) exceeded its remaining cost (6m), the balance of capacity was redeployed, and the write-off was taken and disclosed. The descoped variant delivered in eleven months.

The arithmetic of the reframing, because it is the transferable part. Continuing as planned had a forward NPV of `14 − 21 =` **(7m)**. The descoped variant has `9 − 6 =` **+3m**. The swing between the two decisions available at that gate is therefore **10m**, which is **25 %** of the programme's original 40m approval and **16.13 %** of the 62m spent — a decision larger than most of the change requests the programme had spent four years processing. Two secondary figures made the descope defensible rather than merely attractive. The **discarded 5m of remaining benefit** costs 15m of remaining spend, a ratio of **0.3333** — value per unit of spend far below the 1.5 the retained scope achieves, which is the same value-per-unit-of-constraint reasoning as KA 2.2.3 applied inside a single programme rather than across candidates. And the **released capacity had a named use**: the 15m and the integration team went to two funded candidates, so the board was not asked to accept a write-off with nothing on the other side of it.

The gate criterion that would have caught it years earlier. A criterion set at the forward breakeven — remaining benefit must exceed remaining cost at every gate — is the one criterion in 2.4.3's list that needs no proxy signal and no calibration, because it is stated in the decision's own units. On this programme it would have been breached well before year four, and it is available to any gate willing to compute two numbers.

What the domain teaches here. Escalation of commitment is structural before it is personal: absent kill criteria, an independent decider and a real stop option, a rational actor presenting a "total investment" frame will continue. Note that the resolution was not "stop everything" — descoping to the part where remaining benefit exceeds remaining cost is the same arithmetic applied with more imagination.

Executive perspective — Domain 2

What a project leader cannot delegate in this domain:

- **The benefits profile.** Not the NPV — the shape. The leader who lets a flat full-potential profile through has set the standard their programme will be judged against, and Meridian is what that costs.
- **The breakeven sentence, with its basis.** Stating the condition on which value depends (41.0460 % flat-equivalent, 45.0053 % ramped-basis) in a form the board can monitor monthly, and never quoting one while meaning the other.
- **The counterfactual.** Insisting that doing nothing is appraised rather than assumed to cost zero. It is the one error that runs in the leader's favour, which is exactly why nobody else will find it — and on Meridian it was the difference between a recorded failure and a 227,074 success.
- **The enabling change and who owns it.** Naming the non-project work that converts outputs into outcomes, and securing owners for it outside the project, before approval rather than after.
- **The honest options set.** Ensuring at least one option could have won — including doing nothing.
- **The priced assumption register.** Two acts nobody else will perform: clearing the zero-cost rows before the paper is written — 390,184 of Meridian's 1,074,548, because after approval they stop being questions and become risks — and then saying the ratio out loud, since "NPV +1.33m

against 1.07m of expected exposure in the assumptions it rests on" is a different sentence from "NPV +1.33m".

- **Forward-only framing.** Refusing every "total investment" frame at every gate, and personally insisting on remaining cost against remaining benefit — including the forward exit cost, which is not sunk.
- **Kill criteria at approval, calibrated rather than asserted.** The leader is the only person who can get them agreed while goodwill is high; after the first slip, nobody will. And the level matters as much as the existence: set at the forward breakeven, not at the investment breakeven, or the criterion will stop programmes that are still worth completing.
- **The price of staging, separated from the value of the option.** Piloting is not automatically prudent. The leader owns the two numbers that decide it — what the first phase costs above rollout unit rates, and how long it delays everything.

Calculation exercises — Domain 2

Exercise 2.1 A programme's full benefit potential is USD 600,000 per year, steady-state adoption 75 %, profiled 30 % / 60 % / 75 % thereafter, appraised over 6 years at 8 %. Delivery costs USD 1,300,000. Compute the ramped NPV and the flat full-potential NPV, and the overstatement. Solution. `AF(0.08, 6) = 4.622880`. Flat: `600,000 × 4.622880 = 2,773,728`; NPV **+1,473,728**. Ramped: `180,000/1.08 + 360,000/1.08² + 450,000 ×` (years 3–6 factors 0.793832 + 0.735030 + 0.680583 + 0.630170 = 2.839615) `= 166,667 + 308,642 + 1,277,827 = 1,753,136`; NPV **+453,136**. Overstatement **USD 1,020,592**. Common error: applying the steady-state percentage to all years including the ramp period.

Exercise 2.2 Using Exercise 2.1's flat form, find the sustained adoption at which NPV is zero. Solution. Required annual benefit `= 1,300,000 / 4.622880 = 281,210`; as a share of the 600,000 potential, **46.87 %**. Common error: dividing cost by total undiscounted benefits (1,300,000 / 3,600,000 = 36.1 %), which ignores discounting and understates the threshold.

Exercise 2.3 Candidates scored 1–5 on strategic fit (0.40), benefit (0.30), deliverability (0.20), risk-inverse (0.10): P = 4/5/2/2; Q = 5/3/4/3. Rank them. Solution. P: `1.60 + 1.50 + 0.40 + 0.20 =` **3.70**. Q: `2.00 + 0.90 + 0.80 + 0.30 =` **4.00**. **Q ranks first.** Common error: unweighted summation (P = 13, Q = 15 — same order here, but it is coincidence, and the method loses the weights that were argued over).

Exercise 2.4 Spent USD 4,200,000; remaining cost USD 1,600,000; remaining benefit PV USD 1,950,000. Decide, and state what the sunk-cost frame would wrongly add. Solution. Forward NPV `1,950,000 − 1,600,000 =` **+USD 350,000 → continue.** The sunk-cost frame would add the irrelevant 4,200,000 to the comparison (as "total spend of 5,800,000 against benefits"), which here happens to reach the same decision — the point being that it reaches it for the wrong reason and would reach the wrong one whenever forward NPV is negative.

Exercise 2.5 A directorate declares strategic weights of 45 % / 30 % / 25 % across three objectives. Its strategically mapped spend is USD 12,000,000, funded 3,600,000 / 5,400,000 / 3,000,000. Compute the alignment index and the reallocation distance. Solution. Funded shares **30.0000 % / 45.0000 % / 25.0000 %**. Index `min(45,30) + min(30,45) + min(25,25) = 30 + 30 + 25 =` **85.0000 %**. Reallocation distance `(1 − 0.85) × 12,000,000 =` **USD 1,800,000**, which is the single deficit on objective one (`0.15 × 12,000,000`) and equals the single surplus on objective two. Common error: adding the deficit and the surplus to report 3,600,000, which counts one movement of money twice.

Exercise 2.6 A programme costs USD 1,500,000 and has a full benefit potential of USD 540,000 a year, appraised over 6 %, 8 years. Doing nothing forces a mandatory USD 450,000 replacement in year 3 and USD 25,000 a year of workaround cost in years 4 to 8. Compute the flat-equivalent breakeven adoption with and without the counterfactual. Solution. `AF(0.06, 8) = 6.209794`, so full-potential PV is `540,000 × 6.209794 = 3,353,289`. Without the counterfactual, breakeven `1,500,000 / 3,353,289 =` **44.7322 %**. Avoided cost PV `450,000/1.06³ = 377,828.68` plus `25,000 × 3.536782 = 88,419.55` (the years 4–8 discount factors), total **466,248**. With it, breakeven `(1,500,000 − 466,248) / 3,353,289 =` **30.8280 %** — an improvement of **13.9042 percentage points**. Common error: treating the avoided costs as additional benefits and adding them to the numerator of the adoption term, which mixes an adoption-dependent stream with one that is not adoption-dependent and understates the breakeven further.

Exercise 2.7 Candidates P (fit 4, benefit 5, deliverability 2, risk-inverse 2) and Q (5, 3, 4, 3) are scored under weights 0.40 / 0.30 / 0.20 / 0.10. Q leads. Find the shift of weight from strategic fit to benefit value that flips the ranking, and state which criterion can never decide the outcome. Solution. Totals P **3.70**, Q **4.00** — a margin of **0.30**. Shifting `δ` from fit to benefit, P gains `δ(5 − 4) = +δ` and Q gains `δ(3 − 5) = −2δ`, so `3δ = 0.30` and `δ =` **0.100000 — 10.0000 percentage points** (weights become fit 0.30, benefit 0.40; both score **3.80**). Criterion influence on a 1–5 scale is `4 × weight`: **1.60 / 1.20 / 0.80 / 0.40**, so the risk criterion can move a total by at most **0.40** across its whole range, and closing the 0.30 margin through it would need a **3.00-point** swing in the risk scores (`0.30 / 0.10`) — three-quarters of the entire scale, which no assessor can defend. Common error: reporting only the ranking. A 10-point flip point is defensible where a 3-point one is not, and the reviewer cannot tell which case they are looking at unless it is stated.

Exercise 2.8 A case with an NPV of USD 1,100,000 rests on four assumptions whose impacts if false are 900,000 (P 0.30), 400,000 (0.45), 250,000 (0.20) and 150,000 (0.60). Compute the assumption exposure ratio and state what it implies. Solution. EMVs **270,000 / 180,000 / 50,000 / 90,000**, total **USD 590,000**. Exposure ratio `590,000 / 1,100,000 =` **53.64 %**. The case is comfortably right-side-up — about two-thirds of Meridian's 0.8062 — so it can be committed rather than staged, though the 270,000 entry still warrants a test date and a falsifying trigger. Common error: deducting the 590,000 from the NPV to report a "risk-adjusted NPV" of 510,000. The EMVs describe exposure around a forecast that already embodies its own expectations; deducting them double-counts, and it also hides the ratio, which is the useful output.

Exercise 2.9 Four candidates compete for a scarce integration team: R needs 3 units for an NPV of 2,100,000; S needs 2 for 1,500,000; T needs 2 for 1,350,000; U needs 1 for 800,000. Capacity is 4 units. Find the optimal set, the shortfall a greedy per-unit ranking incurs, and the marginal value of a fifth unit. Solution. Per-unit values: U **800,000**, S **750,000**, R **700,000**, T **675,000**. Greedy takes U (1 unit), then S (2 units), then cannot fit R or T — total **2,300,000** with one unit idle. Enumerating the feasible sets at 4 units gives `{R,U}` = **2,900,000**, `{S,T}` = 2,850,000, `{S,U}` = 2,300,000, `{T,U}` = 2,150,000, `{R}` = 2,100,000. The optimum is **{R, U} = 2,900,000**, so greedy gives up **USD 600,000 — 20.69 %** of the available value. At 5 units the best set is `{S,T,U}` = **3,650,000**, so the marginal value of the fifth unit is **USD 750,000**. Common error: stopping at the

> per-unit ranking because it is the method the constrained example in KA 2.2.3 introduces. Ranking explains an answer; enumeration finds it.

Practitioner's toolkit — Domain 2

Adoption-ready artefacts; adapt headings to your organisation, then keep them stable.

Toolkit 2.T.1 — Business-case skeleton (the six questions)

Why (driver named from 2.1.2, evidence, do-nothing consequence) · **Options** (including do-nothing and one materially cheaper; why each was rejected) · **Cost** (range, accuracy class, whole-life not just capital) · **Benefits** (map per 2.T.2; **profiled, not flat**; owners outside the project; measures and baselines; cash-releasing or not, stated) · **Risk** (quantified exposure reconciling to the register) · **Judgment** (success criteria agreed now, kill criteria numeric, gate schedule). Front page carries the **breakeven sentence**, not only the NPV.

Toolkit 2.T.2 — Benefits map and register row

Per benefit: description · **the outcome measure that produces it** and its owner (outside the project) · **enabling changes required**, each with an owner and whether funded · baseline value, measurement method and date measured · profile by year (**ramp explicit**) · cash-releasing or capacity · attribution note · single-claimant confirmation against the portfolio register · review cadence.

Toolkit 2.T.3 — Gate decision pack rules

One page of forward-only economics (**remaining cost, remaining benefit, forward NPV**, and the **forward exit cost** of stopping — total spend may be reported for information but never as the decision basis) · the **forward breakeven** at this gate, recomputed, with the current measure against it · kill criteria with current values against thresholds · assumption register extract showing which assumptions have been tested and which falsified · alignment re-test against current strategy · options at this gate, including stop and descope, each with its own forward NPV · decider names (none of whom authored the case) · recorded decision and reasons.

Toolkit 2.T.4 — Assumption register row, and the test-order rule

Per assumption: the statement, in one falsifiable sentence · why it is believed, with the evidence named · **impact if false, in present-value terms** · probability, with who assessed it (never the case's author alone) · **EMV** · **cost to test**, including "nil" where the test is a letter or a conversation · **EMV per unit of test cost**, which sets the order · owner · test date, in a named person's calendar · the **falsifying trigger** — the observation that would settle it · whether the entry is correlated with another (mark the pairs; expectations add, tails do not) · and whether failure is recoverable or changes what the project is.

Two rules make the register an instrument rather than a list. **Sort by EMV per unit of test cost and clear the free rows before the paper is written** — on Meridian that was 390,184 of exposure resolvable in a week. And **report `Σ EMV ÷ NPV` on the case's front page**: as it approaches one, the recommendation should change from "approve" to "stage" (2.3.4, 2.A.1).

Exam preparation — Domain 2

The traps. A flat full-potential benefits profile (2.2.2 — the domain's central error) · applying steady-state adoption to the ramp years (Exercise 2.1) · computing breakeven on undiscounted benefits (Exercise 2.2) · **quoting a flat-equivalent breakeven while meaning a ramped-basis one, or the reverse** (2.2.2 — worth 3.9592 points on Meridian) · **valuing the do-nothing option at zero** (2.2.2b) · **using undiscounted avoided costs** (MCQ 2.2-E) · **subtracting an avoided cost instead of adding it** (MCQ 2.2-E) · reading sunk cost into a continuation decision (2.4.2, Exercise 2.4) · **misclassifying a forward exit cost as sunk** (MCQ 2.4-F) · reporting non-cash-releasing benefits as savings (2.3.2) · **claiming a raw improvement where a comparison cohort shows part of it was happening anyway** (2.3.2) · **applying the adoption term twice, once through the clinic count and once as a percentage** (2.3.2) · double-counting a benefit across an enabler and its dependents (MCQ 2.3-C) · **deducting assumption EMVs from an NPV to produce a "risk-adjusted" figure** (Exercise 2.8) · treating a soft constraint as hard (2.1.2) · ranking by raw NPV when a resource constraint binds (MCQ 2.2-C) · **taking a greedy per-unit ranking for an optimum** (2.2.3, Exercise 2.9) · **judging capacity one unit at a time when the marginal value is non-monotone** (2.2.3b) · **adding deficits to surpluses when reporting a reallocation distance** (Exercise 2.5) · **treating an annual hazard as a continuous rate when computing a half-life** (MCQ 2.1-D) · **setting a gate criterion at the investment breakeven rather than the forward breakeven** (2.4.3) · mistaking a gate that has never stopped anything for a functioning control (2.4.3).

Reflection questions. 1. Take your current business case: is the benefits profile flat? What is the breakeven condition, and could you state it in one sentence to a board — with its basis? 2. What does doing nothing actually cost your organisation over the appraisal horizon? If the answer is "nothing", name the contract, licence or asset that will not need attention in the next five years. 3. Which enabling changes does your case depend on, and does each have a funded owner outside your project? What happens to the benefits if they do not? 4. List your case's five material assumptions with an impact and a probability. What is `Σ EMV ÷ NPV`, and which of the five could you resolve this week for nothing? 5. What are your project's kill criteria — and if there are none, who would have to agree them, and would they agree today? If there are, are they set at the forward breakeven or at the investment breakeven, and what would each of the two errors cost? 6. If your programme were staged instead of committed, what premium would the first phase carry over your rollout unit rates, and how long would it delay the rest? Which of those two numbers could you negotiate away?

Domain 2 summary

Work gets chosen through a portfolio process, and the quality of that choice depends on candidates being described honestly enough to be compared — which makes a business case a decision instrument rather than an advocacy document only when it could have concluded "no". How far the funded work matches the stated strategy is measurable: an **alignment index** of `Σ min(declared, funded)` gave 76.6667 % on the mapped spend of Meridian's parent authority, 59.0000 % on all of it, and a **reallocation distance** of **USD 4,200,000** — 1.75 Meridians — with the whole difference between the two indices being the 40 % of money that serves no declared objective. Alignment granted at approval then **decays**: at a 15 % annual supersession hazard the **alignment half-life is 4.2650 years**, so gates re-test the case and not merely delivery, and an annual re-test was worth **USD 292,800** against 342,000 of expected misaligned spend. The domain's central arithmetic is the benefits profile: Meridian's approved case claimed full potential from year one for an NPV of +3,447,096, where the same facts profiled as adoption actually arrives (40 % / 60 % / 70 % of a 979,200 potential) give **+1,332,898** — an overstatement of **USD 2,114,198**, 158.6 % of the honest figure, which changed no approval decision and set the standard the programme was later judged to have failed. The more useful board sentence is the breakeven, stated with its basis: **41.0460 % flat-equivalent**, or 45.0053 % on the ramped basis. **The opposite error matters equally**: valuing the do-nothing option at zero cost Meridian **USD 561,101** of present value and 9.5962 points of breakeven headroom, while an honest attribution correction — a comparison cohort showing 1.1 of a 6.4-hour improvement was happening anyway, an over-claim of **17.1875 % —** moved it 5.4212 points the other way. Corrected both ways the breakeven is **35.6035 %**, and Meridian at its achieved 40 % adoption created **USD 227,074** of value while being publicly recorded as a failure. Selection methods each have limits — weighted scoring exposes criteria and can be steered, and a **3.3333-point** weight shift reverses this domain's ranking; ranking by value per unit of a binding constraint beats raw NPV (Beta + Gamma's 2,100,000 over Meridian's 1,693,072) but remains a heuristic for lumpy candidates, and the constraint's own marginal value is lumpy too — 900,000 · 300,000 · 900,000 · 493,072 · 300,000 · 900,000 — so capacity is a block decision, add one unit or three and never two. Benefits require maps that include the **enabling change** most omit, baselines measured before the change, comparison cohorts, single claimants, honesty about cash-releasing versus capacity, and sustainability treated as constraint or as value but never confused between them; and an assumption register is priced, not listed — Meridian's carried **USD 1,074,548** of expected exposure against a 1,332,898 NPV, a ratio of **0.8062**, of which 390,184 was resolvable for nothing. And the discipline closes with stopping: continuation depends only on remaining cost against remaining benefit, plus the forward exit cost — 780,000 of value for 900,000 of spend is a stop, whatever the 1,800,000 already gone suggests — made possible by kill criteria agreed in advance, set at the **forward** breakeven (28.1283 % at Meridian's successor's month-6 gate, not the investment breakeven of 45.0053 %) and calibrated against their two errors, since a criterion on a weak proxy signal destroyed **USD 175,328** of expected value by stopping good programmes to prevent bad ones. Where the uncertainty is genuine, stage instead: Meridian's abandonment option was worth **USD 139,621** and its staging cost **USD 205,460**, so staging paid only above a 53.9740 % chance of failure — or at a first-phase premium below **7.5223 %**. Domain 3 supplies the governance and decision rights this domain assumes; Domain 15 scales the selection arithmetic to a portfolio; Domain 16 measures what it promised.

03

DOMAIN 3

Governance, Organization and Decision Rights

Group: Leading projects (Domain 3 of 4 in Part One). **Target:** ~84 pages. **Binds to:** the PCI Book Pattern Specification and the shared registries (`docs/books/registries/`). This domain continues the **Meridian Care Records** programme from Domains 1 and 2 into the structures that decide things, takes its first reading of **Project Auriga** at the point where governance must act on a forecast, and supplies the governance latency formula — with the multi-party, assurance-exposure and action-window extensions of it — used later by Domain 6 (schedule), Domain 7 (change control) and Domain 8 (risk escalation). British English; USD (+SAR where useful, indicative `USD 1 ≈ SAR 3.75`).

Why this domain exists

Domain 1 established what a project leader is answerable for; Domain 2 established how work gets chosen and whether the promise was honest. Both assumed something this domain has to supply: **that someone, somewhere, can actually decide.**

Governance is where most delivery failures are finally located, and almost never in the terms practitioners use for it. Post-project reviews record "poor governance" as though it were a culture problem. It is usually a design problem, and a measurable one: authority set at the wrong level, committees meeting on a cadence that cannot keep up with the work, escalation paths whose total latency nobody has ever added up, and decisions that are made but not recorded, so they are made again. Each of those has an arithmetic consequence, and this domain computes it. That is the domain's central claim: **governance is a delivery variable with a price, not a compliance overhead** — and a leader who cannot price it will be asked to accept a structure that guarantees delay.

The domain proceeds from purpose to mechanism. KA 3.1 asks what governance is for, and separates it from management, assurance and administration — the three things it is most often confused with — then works structures across organisational forms and the specific problem of governing iterative delivery. KA 3.2 covers the two roles that do the deciding: the sponsor, whose obligations are almost always understated, and the steering body, whose failure modes are predictable; it closes with decision authorities and thresholds, and prices the cost of setting them too low. KA 3.3 builds the assurance and gate machinery, designs escalation as a timed pathway rather than an organisation chart, and insists on the decision record — because an unrecorded decision is not a decision, it is a memory.

Learning objectives. After this domain a candidate can: state what governance is for and distinguish it from management and assurance; describe governance structures across functional, matrix, projectised and multi-party organisational forms, and identify the accountability weakness of each; **price governance by unanimity against a defined majority and show why its cost grows**

exponentially in the number of parties; govern iterative and hybrid delivery without either abandoning control or destroying cadence, and **compute what share of an iterative stream's decisions a periodic committee can serve at all**; specify the sponsor role as a set of testable obligations and **size it in diary hours and decision turnaround**; diagnose the standard steering-committee failure modes and compute a committee's average, peak and off-peak utilisation; **compute the expected latency of a committee decision from its meeting interval and paper lead time, and use it to price a governance design**; set delegation thresholds on value, reversibility and externality, and demonstrate arithmetically when a threshold is too low; design stage gates, compute whether a gate is worth its elapsed time, state its breakeven detection rate, and **show how an untracked conditional pass converts a gate's detection rate into an effective one and can invert its value**; distinguish the three lines of assurance, **compute residual exposure across an assurance map and reallocate effort at constant cost**, and stop the lines duplicating; design an escalation path with a stated total latency, an out-of-cycle mechanism and **a decision action window tested against the remaining duration**; set objective escalation triggers on named forecast methods; maintain an auditable decision record, **price its re-decision tax**, and **design a cumulative test that catches clusters without re-centralising the delegated band**; and govern AI-assisted governance analysis without delegating any decision to it.

The master programme, and the master project. Meridian Care Records continues from Domains 1 and 2 — the clinical-records rollout to **40 clinics**, approved cost **USD 2,400,000**, benefits **USD 685,440** per year at the realistic 70 % adoption, and the **cost of delay of USD 14,280 per week** derived in Domain 1. That last figure is what makes this domain quantitative: every week a decision waits has a price, and Meridian's governance design spends it without ever appearing in a budget line. Where governance has to act on a forecast rather than on a proposal, the domain uses the volume's single-project thread, **Project Auriga** — 25 weeks, BAC **USD 4,000,000**, and at week 13 the position PV 2,080,000 / EV 1,920,000 / AC 2,120,000 that Domains 6 and 7 develop — because an escalation trigger has to be written against a forecast method, and 3.3.3 shows that the choice of method decides whether the trigger fires.

KNOWLEDGE AREA 3.1

Governance models

Topics: 3.1.1 what governance is for · 3.1.2 structures across organisational forms · 3.1.3 governance in agile and hybrid environments.

3.1.1 What governance is for

Definition. Project governance is the set of **decision rights, accountabilities and information flows** through which an organisation directs and controls a project: who may decide what, on whose authority, on what information, by when, and how the decision is recorded.

Every word of that carries weight, and the definition earns its keep mainly by exclusion. Governance is **not management** — management runs the work inside the authority granted; governance grants, bounds and withdraws that authority. It is **not assurance** — assurance forms an independent opinion on whether the work is likely to succeed; governance decides what to do about the opinion. And it is **not administration** — a portal, a report template and a monthly slide pack are governance artefacts, and an organisation can have all of them and no governance at all, because no one in the chain can actually decide anything. The confusion is not academic: the commonest remedy applied to a struggling project is more reporting, which adds administration, consumes the scarce attention of the people who could decide, and leaves the decision rights exactly where they were.

What governance exists to produce. Four things, and a governance design should be testable against each:

1. **Decidability.** Every decision the project will face has exactly one accountable decision-maker with sufficient authority. A decision with two owners is undecidable; one with none is unowned. This is a countable property, and KA 3.2 counts it.
2. **Timeliness.** The decision arrives while it can still change the outcome. This is the property most governance designs never examine, and KA 3.2 makes it computable.
3. **Legitimacy.** The decision is made by someone whose authority the organisation recognises, on information it can rely on, through a process it accepts — so the decision sticks rather than being relitigated by whoever dislikes it.
4. **Traceability.** The decision, its basis and its date can be reconstructed afterwards. This is the property that converts a decision from a memory into an institutional fact (3.3.4).

The two failure directions. Governance fails by being too heavy or too light, and both are common in the same organisation at once — heavy on the small decisions that are easy to control and light on the large ones that are uncomfortable. Heavy governance shows as latency: escalation for decisions that could have been delegated, agendas too full to reach the difficult item, and a reporting burden that consumes the delivery capacity it was meant to protect. Light governance shows as drift: decisions made by whoever is present, scope and budget changing without a decision-maker, and a project that no one can stop because no one is entitled to.

The professional discipline is therefore **proportionality, designed deliberately**. A governance structure is a design artefact with inputs (the project's value, novelty, risk, reversibility and external exposure) and outputs (tiers, thresholds, cadences, gates). Copying the structure used for the last programme is not proportionality; it is inheritance.

Governance and the leader's accountability. Domain 1's distinction returns here with force. Governance can allocate responsibility — the doing — freely, and should. It cannot dissolve the project leader's **obligation to answer**. A leader who says "the steering committee decided" has described a fact, not a defence: the obligations to have framed the decision honestly, to have surfaced the material information, to have stated the recommendation and the risk of the alternative, and to have escalated in time all remain the leader's, and every one of them is assessable afterwards from the decision record.

3.1.2 Structures across organisational forms

The same governance intent produces different structures depending on where authority sits in the host organisation, and each form has a characteristic weakness a leader should assume is present.

Functional. Resources and authority stay with functional line managers; the project leader coordinates. Characteristic weakness: the leader has accountability without authority over the people doing the work, so every resource conflict is an escalation. The governance countermeasure is explicit, written resource commitments with named individuals and dated availability, agreed by the functional manager and treated as breached when unilaterally changed — not a request, a commitment.

Matrix (weak, balanced, strong). Authority is shared on a spectrum. Characteristic weakness: **dual reporting without a stated precedence rule**, which does not split authority evenly but transfers it to whichever manager escalates harder. The countermeasure is a precedence rule written in advance for the specific conflicts that will arise — who wins on priority of work, on technical approach, on performance assessment — because a matrix without one is a functional organisation that holds project meetings.

Projectised. The project owns its resources and the leader holds full authority. Characteristic weakness: isolation from the enterprise — locally optimal decisions, divergent standards, and a benefits chain that ends where the project ends, precisely the failure Domain 2's benefits map addresses. The countermeasure is enterprise representation in the governance body with a real veto on the standards that matter, and a receiving-organisation owner for each benefit.

Multi-party (joint ventures, consortia, public-private). Two or more organisations with distinct interests, and often distinct legal duties. Characteristic weakness: **governance by unanimity**, which is indistinguishable from an inability to decide, particularly under stress when interests diverge. The countermeasures are structural and belong in the agreement, not in a terms of reference: a defined majority for defined classes of decision, a reserved-matters list that genuinely requires unanimity and is kept short, a deadlock-breaking mechanism with a deadline, and a single integrating authority for day-to-day direction. PFL-AI Domain 11 treats the risk-allocation side of the same problem; the governance side is that a structure where any party can stop everything will eventually be used that way.

Unanimity has a price, and it is not linear. Suppose a multi-party board meets on interval M with paper lead time L, and that at any given meeting each of n parties independently arrives able to approve with probability p — able, that is, in the practical sense of having its own internal clearance, its own legal sign-off and no unresolved question. Under unanimity the decision passes in a cycle with probability p^n; a failed cycle costs a whole further meeting interval. So

```
E[cycles] = 1 / pⁿ        E[wait] = M/2 + L + (E[cycles] − 1) × M
```

The first expression is the whole argument. Approval probability enters as a **power of the party count**, so the expected wait rises geometrically as parties are added, and it rises much faster still as p falls — which is exactly what happens under stress, when interests diverge and internal clearances become slower and more conditional. A defined majority replaces p^n with the probability that enough parties are ready, which is far less sensitive to both.

WORKED EXAMPLE

Worked example 3.1.2 — pricing Meridian's consortium board: unanimity against a defined majority.

1. **Setup.** Meridian's rollout is governed by a three-party board: the health provider, the clinical systems supplier and the regional commissioner. It meets every **4 weeks** with a **2-week** paper lead time, so a single-authority decision would wait $4/2 + 2 =$ **4.0 weeks** (3.2.3). Reviewed history of the parties' behaviour on comparable items puts each party's per-cycle readiness at $p =$ **0.90** in normal conditions and **0.70** once an item is contested. Cost of delay **USD 14,280** per week. Compare unanimity with a **two-of-three** defined majority.
2. **Formula.** Unanimity pass per cycle $= p^n$. Two-of-three pass per cycle $= 3p^2(1 - p) + p^3$. Then $E[\text{wait}] = M/2 + L + (1/\text{pass} - 1) \times M$, costed at the cost of delay.
3. **Substitution.** Unanimity at $p = 0.90$: $0.90^3 = 0.729$, $1/0.729 = 1.3717$ cycles, $4.0 + 0.3717 \times 4$. Majority at 0.90: $3 \times 0.81 \times 0.10 + 0.729 = 0.243 + 0.729 = 0.972$, $1/0.972 = 1.0288$ cycles, $4.0 + 0.0288 \times 4$. Contested, $p = 0.70$: unanimity 0.343, 2.9155 cycles; majority $3 \times 0.49 \times 0.30 + 0.343 = 0.784$, 1.2755 cycles.
4. **Result.**

Rule	Pass per cycle	Expected cycles	E[wait]	Cost
Single integrating authority	1.000	1.0000	**4.0000 w**	**USD 57,120.00**
Two-of-three majority, p = 0.90	0.9720	1.0288	**4.1152 w**	**USD 58,765.43**
Unanimity, p = 0.90	0.7290	1.3717	**5.4870 w**	**USD 78,353.91**
Two-of-three majority, p = 0.70	0.7840	1.2755	**5.1020 w**	**USD 72,857.14**
Unanimity, p = 0.70	0.3430	2.9155	**11.6618 w**	**USD 166,530.61**

Unanimity costs **USD 19,588.48** more than a two-of-three majority per decision in normal conditions, and **USD 93,673.47** more once the item is contested. 5. **Interpretation.** Read the two rows that matter together. In normal conditions unanimity looks affordable — a **1.37-week** penalty, which is the sort of number a governance review waves through as the price of consensus. Under contest it becomes an **11.66-week** wait, a **2.13-fold** increase from a change in the parties' readiness of only 0.20, while the majority rule moves from 4.12 to 5.10 weeks, a rise of under a week. **The two rules are close in good conditions and divergent in bad ones, and governance is only load-bearing in bad ones.** That asymmetry is the reason the countermeasure is structural rather than behavioural: nothing about a consortium's good intentions in month one predicts its p in month fourteen.

The identity to carry forward is that unanimity's cost is driven by p^n, so **each additional party multiplies the failure probability rather than adding to it.** At p = 0.90 the expected wait runs 4.94 weeks at two parties, 5.49 at three, 6.77 at five and **8.36 at seven** — and at p = 0.70 the same series runs 8.16, 11.66, 23.80 and **48.57 weeks**, at which point the structure has not slowed the decision, it has removed it. This is why the professional prescription is not "avoid unanimity" but **keep the reserved-matters list short**: unanimity is the correct rule for a handful of decisions that genuinely should not be taken over a party's objection — a change of purpose, a change of contribution, admission of a new member — and applying it to the ordinary traffic of a programme imposes p^n on decisions that never needed it.

Three cautions, each of which a reviewer should raise. **Independence is the model's weak assumption.** Parties' positions are correlated: two commissioners aligned against a supplier fail together, so the realised pass probability under contest is worse than p^n in the direction of a bloc, and better than p^n where a single party's readiness genuinely drives the others'. State the assumption and treat the figures as a lower bound on the penalty, not a point forecast. p **is a measured quantity or it is nothing** — it is countable directly from a board's own minutes as the share of items approved at first presentation, and a consortium that has never counted it is guessing at the cost of its own constitution. And **the deadlock-breaking mechanism is what caps the tail**: with an expert-determination or escalation-to-principals route carrying a stated deadline, the series above is truncated rather than geometric, which is worth more than any improvement in p. Deadlock provisions and reserved-matters lists are enforceability-sensitive and sit in the shareholders' or consortium agreement rather than in a terms of reference; take local counsel on both, since what a deadlock clause can compel varies by jurisdiction.

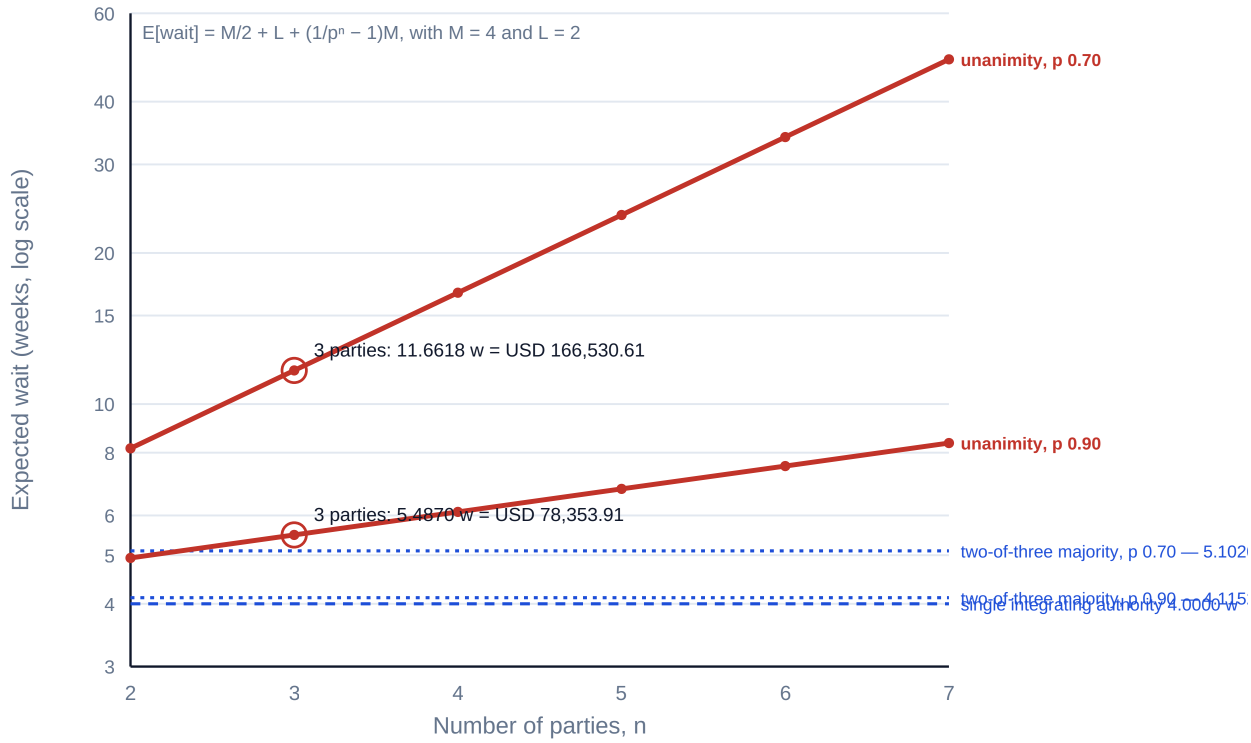


Fig 3.1.1 — Why unanimity is a power law

Programme and portfolio layers. Where a project sits inside a programme inside a portfolio, each layer adds a tier of potential escalation — and, unmanaged, adds its latency to every decision that travels up. KA 3.3 computes what that costs and why the design rule is **the fewest tiers a decision must legitimately pass through**, with the others informed rather than consulted.

A note on outsourced delivery. Where a supplier delivers, governance has to span the contract boundary, and the boundary is where decision rights are most often left undefined. The specific questions to settle before mobilisation: who may approve a change (and at what value on each side — the two thresholds are rarely equal and the mismatch is where change control fails); whose governance body is authoritative when they disagree; what the supplier is contractually obliged to escalate and within what period; and what information the client is entitled to, in what form, on what cadence. Domain 7 handles the commercial mechanics; the governance point is that a contract that does not allocate decision rights has not allocated the work.

3.1.3 Governance in agile and hybrid environments

Iterative delivery does not remove the need for governance; it changes what governance is asked to decide, and a structure designed for sequential delivery applied unchanged to iterative delivery fails in a specific, predictable way.

What changes. In sequential delivery, governance approves a plan and then controls variance against it. In iterative delivery the increment is the unit of decision, the backlog is the plan, and the governance question is not "is the project on plan?" but **"is the value being produced worth the next increment's cost, and is the direction still right?"** That converts governance from variance control to **continuation decisions at a cadence** — which is a more honest posture, and also a more demanding one, because it requires a decision-maker who can say "stop" repeatedly rather than once at a gate.

The characteristic failure. A monthly steering committee governing two-week sprints is always behind, and worse, its latency (computed in KA 3.2) exceeds the cycle it is governing — so the team either waits, destroying flow, or proceeds and seeks retrospective approval, destroying the governance. Both outcomes are attributed to the team. Neither is the team's doing.

The design response, in four moves:

- **Delegate inside a bounded envelope.** The team decides freely within stated bounds — scope within the agreed increment, technical approach within the architecture, spend within a period budget — and the governance body sets and reviews the envelope rather than the decisions inside it. This is the single highest-value governance decision in iterative delivery, and the worked example below shows that it is not an efficiency measure but a **feasibility requirement**.
- **Match cadence to the work.** Governance interaction at the increment's rhythm — increment reviews as the decision point, with the committee attending rather than being reported to.
- **Govern outcomes, not activity.** Working software or a released service demonstrated, with the benefit measure attached (Domain 2's map), rather than percentage-complete against a plan that the method deliberately does not have.
- **Keep the gates that carry real optionality.** Funding tranches, go-live authorisation and regulatory approvals remain genuine gates because they are genuinely irreversible. What should be removed are the gates that merely re-approve a decision already taken.

The test that settles the argument. Envelope width is usually debated as a matter of trust, and it can be settled as a matter of arithmetic. Each decision an iterative team raises has a **need-by time** — the point beyond which the answer no longer changes what the team does, because the work has either been done a different way or abandoned. A periodic body can serve a decision only if its expected latency is shorter than that decision's need-by time. So the governance question is not "how much should we delegate?" but **"what share of the stream's decisions can this body serve at all?"**, and the answer is countable from a single sprint's decision log.

WORKED EXAMPLE

Worked example 3.1.3 — how much of Meridian's iterative stream can a monthly committee serve?

1. **Setup.** Meridian's records application is built in **2-week** increments over a **26-week** build, so **13** increments. The team's decision log records **5** decisions per increment that exceed its current authority — **65** in total. Each was tagged at the time with its need-by time: **41** needed within the current increment (≤ 2 weeks), **9** at 2–3 weeks, **7** at 3–4 weeks and **8** beyond 4 weeks. The steering committee's expected latency is **4.0 weeks** (`M` = 4, `L` = 2). Cost of delay **USD 14,280** per week; **25 %** of escalated decisions sit on the rollout critical path.
2. **Formula.** Decisions servable in time = count whose need-by time exceeds `E[wait]`. Servable share = that count ÷ total. Minimum envelope coverage = the share whose need-by time is inside the increment. Upper-bound wait cost = unservable count × critical-path share × `E[wait]` × cost of delay.
3. **Substitution.** At `E[wait]` = 4.0 only the `> 4` bucket is servable: `8/65`. At 3.0, the `3–4` bucket joins: `15/65`. At 2.0 (a fortnightly committee with a one-week deadline), the `2–3` bucket joins: `24/65`. Minimum envelope `41/65`. Wait cost `57 × 0.25 × 4 × 14,280`.
4. **Result.**

Governance design	E[wait]	Decisions servable in time	Share
Monthly, 2-week papers (as designed)	4.0 w	8 of 65	**12.31 %**
Monthly, 1-week papers	3.0 w	15 of 65	**23.08 %**
Fortnightly, 1-week papers	2.0 w	24 of 65	**36.92 %**

87.69 % of the stream's decisions cannot be served in time by the committee as designed. If the team waited for all 57 of them, the upper bound on delay is `57 × 0.25 = 14.25` decisions delaying 4.0 weeks each — **57.00 delay-weeks**, or **USD 813,960**. The envelope must therefore cover at least the **63.08 %** of decisions needed inside the current increment before the team can keep cadence at all.

5. **Interpretation.** The decisive number is the third column of the table, and specifically what happens to it as the cadence is tuned. **Doubling the committee's meeting frequency and halving its paper deadline together take servable coverage from 12.31 % to 36.92 % — still under two decisions in five.** Cadence tuning cannot solve this problem, because the binding constraint is not the committee's speed but the ratio of its latency to the increment: at `E[wait]` = 4.0 weeks against a 2-week increment the ratio is **2.0**, meaning the answer arrives two increments after the question, by which time the increment that raised it has shipped. Only delegation changes the ratio, which is why the bounded envelope is a feasibility requirement rather than a courtesy.

The **USD 813,960** is an upper bound and should be presented as one, because no team actually pays it. What a team does instead is the second failure of 3.1.3: it proceeds and seeks retrospective approval. So the honest reading is that the figure is the size of the pressure the design creates — and the governance consequence of that pressure is not delay but **the quiet transfer of 57 decisions to people who were never given authority over them**, unrecorded, and therefore invisible until an audit reconstructs them (Case study B is exactly this mechanism at a different scale). A design that cannot serve 87.69 % of its stream's decisions has not delegated them; it has lost track of them.

Three cautions. The upper bound **adds delays that in reality overlap** — two decisions waiting in the same four weeks do not cost eight weeks of programme — so the figure bounds the exposure and does not forecast it; where a defensible point estimate is needed, model the critical-path decisions serially and the rest as concurrent (Domain 6 supplies the machinery). The need-by tags are **the team's own judgement recorded at the time**, which is the right source and a biased one: a team under pressure to keep flow will tag optimistically, so the tag should be captured before the wait, never reconstructed after it. And the residual **8 decisions that the committee can serve are the ones it should keep** — they are, by construction, the decisions with time in them, and they will tend to be the irreversible and externally visible ones. The envelope's boundary is therefore not a value line but a **time-and-reversibility line**, which is the same conclusion 3.2.3 reaches from the other direction.

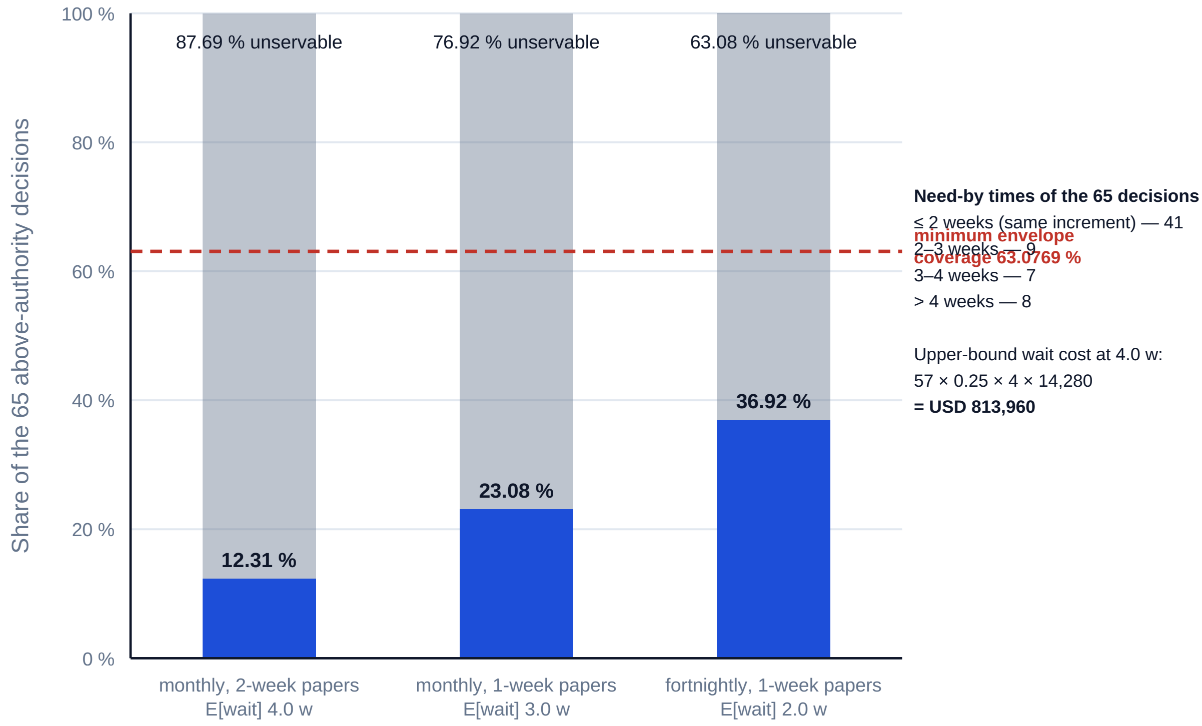


Fig 3.1.2 — What share of an iterative stream a periodic committee can serve

Hybrid honestly. Most real programmes are hybrid — an iterative build inside a sequential infrastructure or regulatory frame — and the governance failure is to apply one model to both parts. Meridian is exactly this shape: an iteratively developed records application, a sequential clinic rollout with immovable estate and training dependencies, and a regulatory approval that is a genuine gate. The workable design governs each part on its own terms and holds a **single integrated view of the whole** for the decisions that span them — which is the topic of Domain 4.

AI in this KA

AI is useful in governance analysis and dangerous in governance decisions, and the line is bright.

Where it earns its place. Reading a set of terms of reference, delegation schedules and contract schedules against each other and listing the decisions with no named owner, two owners, or conflicting thresholds — a document-comparison task at which it is fast and thorough, and one humans do badly because it is tedious. Extracting the decision log from meeting minutes into a structured register, flagging items recorded without an owner or a date. Summarising a decision's history for a board paper. Modelling the latency of a proposed governance design (KA 3.2's arithmetic) across alternative cadences, decision rules and party counts, which is deterministic and verifiable. Tagging a sprint's decision log with need-by times drawn from the decision text and computing the servable share of 3.1.3, which is a counting task nobody performs because it is dull rather than difficult.

Where it must not go. No decision, and no recommendation presented as a decision. Governance authority is conferred on accountable people; it cannot be exercised by a system that cannot be answerable, and an organisation that lets a model's output stand unchallenged as a governance conclusion has created accountability without a holder — the exact defect Domain 1 identified.

The governing principle, applied. AI proposes; the professional verifies, decides and remains accountable. Concretely: every AI-produced governance analysis is reviewed by the named accountable person before it reaches a decision body, its inputs are stated, its conclusions are independently checked on the material items, and the decision record shows a human decision-maker — never a tool — as the author of the decision.

KEY TERMS — KA 3.1

Term	Meaning
Governance	The decision rights, accountabilities and information flows through which an organisation directs and controls a project.
Decision right	The authority to make a specified class of decision, bounded by value, scope and reversibility.
Decidability	The property that every decision has exactly one accountable decision-maker with sufficient authority.
Reserved matter	A decision class that requires the highest (often unanimous) authority; kept deliberately short.
Bounded envelope	Stated limits within which a delivery team decides freely, reviewed rather than approved decision by decision.
Precedence rule	The pre-agreed rule for which authority prevails in a matrix conflict.
Governance artefact	A report, portal or template — evidence of governance, never a substitute for decision rights.
Party readiness (`p`)	The probability that a party arrives at a cycle able to approve; countable as the share of items approved at first presentation.
Need-by time	The point beyond which a decision no longer changes what the team does; the test of whether a body can serve it.
Envelope coverage	The share of a stream's above-authority decisions the delegated envelope contains.
Latency-to-cycle ratio	Governance latency ÷ the delivery cadence it governs; above 1.0 the answer arrives after the increment that asked.

SAMPLE MCQS — KA 3.1

MCQ 3.1-A `[3.1.1 · Comprehension]` Which statement best distinguishes governance from management? - A. governance is performed by senior people and management by junior people - B. governance grants, bounds and withdraws the authority within which management runs the work - C. governance is concerned with reporting and management with delivery - D. governance applies to programmes and management to projects

Rationale: The distinction is about authority, not seniority, reporting or scale (3.1.1). C describes governance artefacts, which an organisation can have in full while having no governance.

MCQ 3.1-B `[3.1.2 · Analysis]` A balanced matrix has no written precedence rule for conflicts between project and functional priorities. The most likely consequence is that authority: - A. is shared evenly, as intended - B. transfers to whichever manager escalates harder - C. defaults to the project leader - D. defaults to the steering committee automatically

Rationale: Undefined precedence does not split authority; it awards it to escalation behaviour (3.1.2). The countermeasure is a precedence rule written before the conflict arises.

MCQ 3.1-C `[3.1.3 · Application]` A monthly steering committee governs two-week sprints. The predictable failure is that: - A. the team will produce lower-quality increments - B. governance latency exceeds the cycle being governed, so the team either waits or proceeds and seeks retrospective approval - C. the committee will meet too often - D. the sprints must be lengthened to a month

Rationale: The mismatch is one of latency against cycle time (3.1.3); the design response is a bounded envelope and cadence matching, not slower delivery.

MCQ 3.1-D `[3.1.2 · Analysis]` In a three-party consortium, the governance provision most likely to produce paralysis under stress is: - A. a defined majority for defined decision classes - B. a short reserved-matters list - C. unanimity for all substantive decisions - D. a deadlock-breaking mechanism with a deadline

Rationale: Unanimity is indistinguishable from an inability to decide once interests diverge (3.1.2); the other three are the countermeasures.

MCQ 3.1-E `[3.1.2 · Application]` A three-party board meets every 4 weeks with a 2-week paper lead time. Each party is independently ready to approve in a given cycle with probability 0.70. Under unanimity, the expected wait for one decision is closest to: - A. 4.0 weeks - B. 5.7 weeks - C. 11.7 weeks - D. 12.0 weeks

Rationale: $p^3 = 0.343$, so `E[cycles] = 1/0.343 = 2.9155` and `E[wait] = 4/2 + 2 + 1.9155 × 4 = 11.6618` weeks (3.1.2). A is the single-authority wait, ignoring the failure probability altogether. B applies the readiness probability once rather than to the power of three (`1/0.70 = 1.4286` cycles → 5.71 weeks) — the commonest error, because it treats the board as one actor. D rounds the 2.9155 expected cycles up to 3 whole cycles and so adds two whole meeting intervals (`4 + 2 × 4 = 12.0`) instead of the expected 1.9155 of them.

MCQ 3.1-F `[3.1.3 · Evaluation]` A monthly committee with a 2-week paper deadline governs 2-week increments. Of 65 above-authority decisions, 41 are needed within the current increment, 9 at 2–3 weeks, 7 at 3–4 weeks and 8 beyond 4 weeks. Management proposes meeting fortnightly with a one-week deadline. The strongest evaluation of that proposal is that it: - A. solves the problem, because the committee will then be faster than the increment - B. raises servable coverage from 12.31 % to 36.92 % and therefore still leaves the majority of decisions unservable — only delegation changes the outcome - C. fails, because committee latency is irrelevant to iterative delivery - D. raises servable coverage to 63.08 %, which is the minimum required

Rationale: At `E[wait]` = 2.0 weeks the servable buckets are the 8, 7 and 9, i.e. 24 of 65 = 36.92 %, against 8 of 65 = 12.31 % as designed (3.1.3). A mistakes latency equal to the increment for latency shorter than it: at `E[wait]` = 2.0 against a 2-week increment the latency-to-cycle ratio is 1.0, and the 41 same-increment decisions need their answer inside the increment, not at the end of it. C overcorrects: latency matters, it is simply not sufficient. D misreads 63.08 % — that is the minimum envelope coverage the same-increment decisions require, not a coverage the committee achieves.

SELF-CHECK — KA 3.1

1. Name the four things a governance design must produce. — Decidability, timeliness, legitimacy, traceability.
2. Why is "more reporting" the wrong remedy for a struggling project's governance? — It adds administration and consumes decision-makers' attention while leaving decision rights unchanged.
3. What does a project leader still owe after a steering committee decides badly? — The obligations to have framed the decision honestly, surfaced the material information, stated the recommendation, and escalated in time — all assessable from the record.

4. Why does unanimity fail suddenly rather than gradually? — Because pass probability is p^n: each party multiplies the failure probability, so a modest fall in readiness produces a large rise in expected wait (3.1.2 — 5.49 weeks at p = 0.90, 11.66 at p = 0.70, three parties).
5. What makes a bounded envelope a feasibility requirement rather than an efficiency measure? — Governance latency of 4.0 weeks against a 2-week increment gives a latency-to-cycle ratio of 2.0, so 87.69 % of Meridian's stream decisions cannot be served in time at any realistic cadence (3.1.3).

KNOWLEDGE AREA 3.2

Sponsorship and steering

Topics: 3.2.1 the sponsor role · 3.2.2 steering committees that work · 3.2.3 decision authorities and thresholds.

3.2.1 The sponsor role

Definition. The sponsor is the individual accountable for the project's **business outcome** — for its being worth doing, remaining worth doing, and being adopted by the organisation that must use it. The project leader is accountable for delivery within the mandate; the sponsor is accountable for the mandate.

That is a demanding role and it is routinely treated as an honorific. The remedy is to state it as **testable obligations**, each of which can be evidenced or found absent:

Obligation	Evidence it is being met
Own the business case	The sponsor can state the benefit, its measure and its owner without notes (Domain 2).
Secure and defend funding	Funding decisions carry the sponsor's name; funding challenges are answered by the sponsor, not the project leader.
Decide within authority, promptly	Decisions escalated to the sponsor have recorded dates in and out; the interval is monitored.
Own the receiving organisation's readiness	The enabling change is resourced and progressing (Domain 2's omitted column).
Resolve conflicts between functions	Escalations that are genuinely inter-functional stop at the sponsor rather than travelling upward.
Hold the project leader to account, and support them	Recorded, dated performance conversations; and visible backing when the project is under pressure.
Be willing to stop it	Kill criteria agreed in advance and revisited at gates (Domain 2, KA 2.4).

The sponsor failure modes, in descending order of frequency and each with its detection test. The absent sponsor — a name on a chart, no diary time; test: the count of sponsor decisions in the last quarter, and their turnaround. The delegating sponsor — a deputy attends everything, so authority is in the room but accountability is not; test: whether the last three material decisions were made by the sponsor or reported to them afterwards. The advocate sponsor — committed to the project rather than the outcome, and therefore unable to stop it; test: whether they can state a condition under which they would recommend stopping (Domain 2's escalation of commitment, at the governance level). The operational sponsor — drawn into managing delivery, which vacates the role that only they can hold; test: whether the sponsor's contributions in the last three meetings were about outcome and mandate, or about task sequence.

The absent sponsor is a capacity problem before it is a commitment problem. "No time" is the universal explanation for the first failure mode and it is almost never tested, which is unfortunate, because the role's time demand is a small sum that anyone can do — and doing it changes the conversation from an accusation about priorities into a question about portfolio load.

WORKED EXAMPLE

Worked example 3.2.1a — what Meridian's sponsor role actually costs in diary time.

1. **Setup.** The seven obligations above imply a specific set of commitments for Meridian: **13** steering meetings a year at **2 hours** (the 4-weekly cadence of 3.2.2); **22** sponsor-level decisions a year at **1.5 hours** each of reading, consultation and decision; **4** gate reviews at **4 hours**; **12** monthly one-to-one sessions with the project leader at **1 hour**; and enabling-change oversight with the receiving organisation at **2 hours** a month. Assume **46** working weeks and a nominal **40-hour** week.
2. **Formula.** Annual hours `= Σ (frequency × duration)`. Weekly load = annual hours ÷ working weeks. Share of a week = weekly load ÷ 40.
3. **Substitution.** `13 × 2 + 22 × 1.5 + 4 × 4 + 12 × 1 + 12 × 2 = 26 + 33 + 16 + 12 + 24`. Then `111/46`, then `2.4130/40`.
4. **Result. 111.0 hours** a year — **2.4130 hours** a week, or **6.03 %** of a nominal working week. Held across **five** such sponsorships the same executive carries **555.0 hours**, **12.0652 hours** a week, **30.16 %** of the week.
5. **Interpretation.** Six per cent is the number that ends the argument. A role costing under two and a half hours a week cannot honestly be declined for want of time, and a sponsor who is absent at that price is making a **priority** choice, which is a legitimate thing to surface to whoever allocates sponsors. But the second half of the result is the more useful half, and it points away from the individual: at 2.4130 hours a week, an executive prepared to commit **20 %** of their week to sponsorship can carry `8/2.4130 =` **3.32**, that is **three** programmes of Meridian's size — and organisations routinely name the same four or five names across a dozen. **Sponsor capacity is a portfolio constraint, not a personal virtue**, and the portfolio body that allocates sponsors without a load model is manufacturing absent sponsors and then complaining about them (Domain 15 treats the allocation; the arithmetic here is what a project leader brings to it).

Two cautions on using this. The estimate is **load, not latency**: 111 hours spread evenly is a different object from 111 hours that arrive in three clusters around the gates, and it is the clustering that produces the missed decision — which is why 3.2.1b measures turnaround separately rather than inferring it from availability. And the model **excludes the unschedulable half of the role** — defending the funding when it is challenged, standing behind the leader when the programme is under pressure, having the conversation about stopping. Those cost little time and most of the role's difficulty, so a sponsor who meets the 111 hours and none of them has met the diary and not the obligation.

Turnaround is the sponsor's own latency, and it has a tail. A sponsor is not a committee, so `M/2 + L` does not apply; what applies is an empirical turnaround distribution, and the obligation "decide within authority, promptly" is only testable once that distribution is counted. It is worth counting because it behaves in a way the mean conceals.

WORKED EXAMPLE

Worked example 3.2.1b — Meridian's sponsor decision turnaround, and where the mean comes from.

1. **Setup.** The decision log records **22** decisions escalated to Meridian's sponsor over twelve months, with dates in and out: **9** returned in **1.0 week**, **8** in **2.5 weeks**, and **5** in **9.0 weeks**. Cost of delay **USD 14,280** per week; **25 %** of escalated decisions sit on the critical path. The steering committee's expected latency, for comparison, is **4.0 weeks** (3.2.3).
2. **Formula.** Mean turnaround `= Σ (count × weeks) ÷ count`. Expected delay-weeks `= critical-path share × Σ (count × weeks)`. Cost = delay-weeks × cost of delay.
3. **Substitution.** `9 × 1.0 + 8 × 2.5 + 5 × 9.0 = 9 + 20 + 45 = 74` decision-weeks; mean `74/22`. Delay-weeks `0.25 × 74`. Committee comparison `0.25 × 22 × 4.0`.
4. **Result.** Mean turnaround **3.3636 weeks**, median **2.5 weeks**. Expected delay **18.500 weeks**, costing **USD 264,180.00**. The same 22 decisions taken by the committee at 4.0 weeks would cost `0.25 × 88 × 14,280 =` **USD 314,160.00**, so the sponsor route is **USD 49,980.00** cheaper. The **5** slow decisions — **22.73 %** of the count — carry **60.81 %** of the total wait. Bringing those five to the middle group's 2.5 weeks takes the total to **41.5** decision-weeks, the mean to **1.8864 weeks** and the cost to **USD 148,155.00**: a saving of **USD 116,025.00**, a **43.92 %** reduction, from changing the handling of five decisions.
5. **Interpretation.** Three results, in ascending order of usefulness. First, the sponsor route is faster than the committee — 3.36 weeks against 4.0 — which is the arithmetic case for routing single-owner decisions to a single owner rather than to a body, and it is worth stating because organisations under stress do the reverse and add the decision to a committee agenda "for visibility". Second, **the mean is a tail phenomenon**: the median sponsor decision returns in 2.5 weeks and the mean is 3.36, and the whole of that gap is five items. Reporting a mean turnaround therefore describes a distribution the sponsor does not experience and the project does not suffer from — the right metric is the **count and cause of decisions beyond a stated service level**, not the average.

Third, and this is the finding that changes behaviour: **the fix is worth 43.92 % of the cost and touches 22.73 % of the decisions**, so a project leader arguing for a change to sponsor handling should argue about five items, not about diligence in general. And the five will have a nameable common cause — in practice one of three: the decision was framed without a recommendation, so the sponsor had to do the analysis; it required a peer's agreement the sponsor had to go and get, which is a decision-rights defect rather than a sponsor defect (3.1.2's precedence rule); or it was one the sponsor did not want to make, which is 3.2.1's advocate failure mode arriving as slowness rather than as refusal. Each has a different remedy, and none of them is "remind the sponsor".

Two cautions. The delay figure **prices the wait as if each decision's weeks were additive**, which bounds rather than forecasts the programme effect, exactly as in 3.1.3 — the defensible presentation is delay-weeks and the assumption alongside the currency figure. And a fast turnaround is not by itself evidence of a good sponsor: a sponsor who returns everything in a day may be approving without reading, which the record will show as decisions carrying no options considered and no versioned information reference (3.3.4). **Turnaround and record quality must be read together**, or the metric selects for the wrong behaviour.

What a project leader does with a weak sponsor. This is a real and common situation, and the professional response is neither to complain nor to absorb the gap silently — absorbing it is attractive, because it feels like competence, and it is how a leader ends up accountable for a mandate they were never given. The workable response is procedural: write the decisions needed and their dates; send them with a recommendation and a stated consequence of non-decision; record the non-decision when it occurs; escalate the pattern rather than the individual instance once it recurs; and keep the record. That is not politics. It is the mechanism by which an organisation's governance failure becomes visible to the organisation while there is still time to fix it — and, incidentally, the only defensible position for the leader afterwards.

3.2.2 Steering committees that work

A steering committee exists to make the decisions that exceed the project leader's authority, in time for them to matter. Almost every observed dysfunction traces to one of five design faults.

Fault 1 — membership without authority. Attendees who must consult before agreeing. The committee's effective authority is the minimum of its members' authority on the decision in hand, not the maximum, so one under-empowered member on a decision class blocks it. Test: for each decision class in the delegation schedule, name the member who can commit.

Fault 2 — membership too large. Beyond roughly six to eight decision-makers the body becomes a briefing audience: contributions become positional, dissent becomes private, and the difficult item gets deferred. Attendance is not membership — the separation of members (who decide) from attendees (who inform) is the cheapest available fix.

Fault 3 — the agenda consumed by reporting. Status occupies the meeting and the decisions arrive in the last ten minutes, which is why the reliable diagnostic of committee health is not attendance but the **share of agenda time spent on decisions**. Status is read in advance; the meeting decides.

Fault 4 — cadence mismatched to the work. Treated quantitatively in 3.2.3 — and it is the fault with the largest measurable price.

Fault 5 — no capacity model. A committee has finite throughput, and a design that routes more decisions to it than it can hear does not slow decisions down evenly: the items that get deferred are the contentious ones, which are the ones that needed the decision. The capacity arithmetic is elementary and almost never done.

WORKED EXAMPLE

Worked example 3.2.2 — does Meridian's steering committee have the capacity to govern?

1. **Setup.** Meridian's steering committee meets every **4 weeks** (**13** meetings a year) and can handle **8** substantive agenda items per meeting. Annual demand: **36** escalated change requests (the count above the current delegation threshold, from 3.2.3), **26** standing reports (two per meeting), and **15** gate and assurance decisions.
2. **Formula.** Capacity = meetings × items per meeting. Utilisation = demand ÷ capacity.
3. **Substitution.** Capacity `13 × 8 = 104`. Demand `36 + 26 + 15 = 77`. `77/104`.
4. **Result.** Capacity **104** item-slots; demand **77**; utilisation **74.0 %**.
5. **Interpretation.** Seventy-four per cent looks comfortable and is not, and the reasons are computable rather than impressionistic.

First, demand is not uniform, and the peak is what a committee experiences. The **51** decision items — 36 changes plus 15 gate and assurance decisions — do not arrive evenly: they cluster at phase boundaries, and Meridian's history puts **40 %** of them in the **3** meetings that straddle the design-completion, first-clinic and full-rollout boundaries. That is `51 × 0.40 = 20.40` items into 3 meetings, or **6.80** decision items per peak meeting, plus the 2 standing reports — **8.80 items against a capacity of 8**, a peak utilisation of **110.00 %**. The remaining **30.60** items spread over 10 meetings give `3.06 + 2 = 5.06`, an off-peak utilisation of **63.25 %**. So the true profile is not 74 % but **110 % at the boundaries and 63 % between them**, and a committee at 110 % does not defer items at random: it defers the ones that will take longest to discuss, which are the contentious ones, which are the ones that needed the meeting. **An average utilisation below capacity is not evidence of sufficient capacity; it is evidence that nobody has looked at the profile.**

Second, a quarter of the scarcest resource in the programme produces no decisions. The 26 standing reports consume `26/104 =` **25.0 %** of total capacity (Fault 3). Moving them to pre-reading releases 26 slots, drops average utilisation to **49.04 %** and — the number that matters — takes **peak** utilisation from 110.00 % to **85.00 %**, which is the first design in this sequence that can actually hear a contentious item at a phase boundary. Raising the delegation threshold as 3.2.3 recommends removes a further 12 items, taking demand to **65** and utilisation to **62.50 %** on the original agenda design, or to **37.50 %** if both changes are made.

Third, the demand figure above is incomplete, and knowing why is the point of 3.3.4. It counts the decisions the committee is asked to make and omits the ones it makes twice; Worked example 3.3.4b counts those at **25** further slots a year and takes real utilisation to **98.08 %**, which is the number that matches what the members report. A capacity model built only from the forward demand estimate will always understate, because re-decisions are invisible to everyone except the decision log.

Two cautions on the model. Capacity in **item-slots assumes items are comparable**, and they are not: a contested threshold change and a routine assurance acceptance are not one slot each, so a committee near its limit should weight items by expected discussion time and re-run the sum. And **raising capacity is the weakest of the three available levers** — adding meetings or lengthening them increases the demand on exactly the people whose attention is scarcest, whereas pre-reading and delegation reduce demand at no cost in scrutiny. The professional point is that committee capacity is a designed quantity with an average, a peak and a hidden component, and a governance structure proposed without this arithmetic has not been designed.

3.2.3 Decision authorities and thresholds

The structure. A delegation schedule states, for each decision class, the level of authority required — typically as a monetary threshold, but the better schedules use three dimensions: **value** (how much), **reversibility** (how hard to undo), and **externality** (who outside the project is affected). A cheap, irreversible, externally visible decision may deserve more authority than an expensive, reversible, internal one; a schedule that reads only on value cannot express that.

Making reversibility a number, so the schedule can read on it. Reversibility is usually left as an adjective, which is why it drops out of practice; it becomes operable as a ratio. Define the **reversal-cost ratio** `ρ = (cost to undo the decision) ÷ (value of the decision)`, and read the required authority on `max(value, cost to undo)` rather than on value. The rule is one line and it catches the decisions a value-only schedule is structurally blind to. On Meridian, a **USD 15,000** change to the clinical template set that has already been deployed to the first cohort of clinics carries a reversal cost of **USD 180,000** — re-configuration, re-validation and re-training across those clinics — so `ρ =` **12**, and the authority appropriate to the decision is that of the **100,001–500,000** band, not the delegated band its price tag puts it in. Externality takes the same treatment as a three-point rating (internal / cross-functional / external or public) with a stated escalation of one authority level per step, because it is the dimension least amenable to a ratio: what an external decision costs is the cost of being wrong in public, which is not estimable in advance and is precisely why it is handled by rule rather than by calculation. Both dimensions, with the cumulative test of Worked example 3.3.4c, are the columns of Toolkit 3.T.2.

Governance latency: the formula this domain contributes. Committees meet periodically and require papers in advance, and those two facts alone determine how long a decision waits. Let

```
M = the meeting interval (e.g. 4 weeks for a monthly committee)
L = the paper lead time — how far before the meeting submissions close
```

For a decision that arises at a uniformly random point in the cycle:

```
E[wait] = M/2 + L
```

The derivation is elementary. A decision arising more than `L` before the next meeting makes that meeting, and waits on average `M/2` of the remaining interval; one arising inside the closed window misses it and waits for the following meeting. Averaging over arrival times gives half the meeting interval plus the whole paper lead time.

Why the formula matters more than it looks. It settles a question organisations habitually get wrong. The two available levers do **not** have equal effect: cutting the paper lead time by one week saves **one full week** of expected wait, while shortening the meeting interval by one week saves only **half a week**. Administrative deadlines, which feel unchangeable and cost nothing to change, are twice as powerful as meeting frequency, which is expensive and resisted. And a governance design's latency can be computed at design time, from two numbers, before anyone has waited.

Meridian's steering committee: `M = 4`, `L = 2`, so `E[wait] = 4/2 + 2 =` **4 weeks**. Note what that says — a monthly committee with a two-week paper deadline imposes an expected wait of a full month, not the fortnight most people assume. At Domain 1's cost of delay of 14,280 per week, one escalated decision on the critical path costs **USD 57,120** in delay alone, before anyone in the room disagrees about anything.

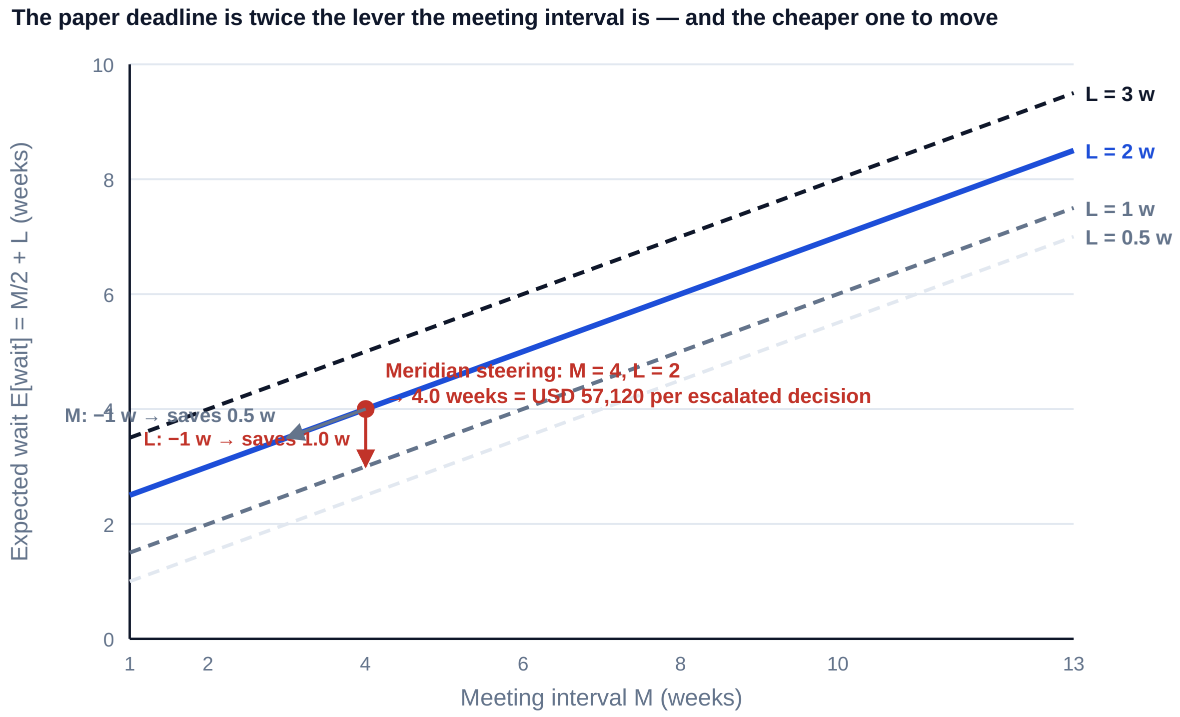


Fig 3.2.1 — Governance latency and its two levers

Setting the threshold. The threshold question is an economic one and can be answered as one. Setting a threshold too high risks decisions taken with insufficient authority or scrutiny; setting it too low guarantees latency on decisions that did not need it. The second cost is certain and computable; the first is probabilistic. Comparing them is the whole of the analysis.

WORKED EXAMPLE

Worked example 3.2.3 — Meridian's delegation threshold, priced.

1. **Setup.** Meridian's project leader may approve changes up to **USD 10,000**; everything above goes to the steering committee (`E[wait]` = **4 weeks**, cost of delay **14,280** per week). Year-one change requests, by value: **≤ 10,000 → 24**; **10,001–25,000 → 12** (average value **17,000**); **25,001–100,000 → 15**; **100,001–500,000 → 7**; **> 500,000 → 2**. Total **60**. Reviewed history shows **25 %** of escalated changes sit on the critical path and therefore convert their wait into programme delay; the rest do not. Should the threshold move to 25,000?
2. **Formula.** Escalated count × critical-path share × `E[wait]` × cost of delay, evaluated at each threshold. Then the worst-case cost of delegating the affected band.
3. **Substitution.** At 10,000: escalated `60 − 24 = 36`, delaying `36 × 0.25 = 9`, each costing `4 × 14,280 = 57,120`. At 25,000: escalated `36 − 12 = 24`, delaying `24 × 0.25 = 6`.
4. **Result.** Annual governance delay cost **USD 514,080** at a 10,000 threshold and **USD 342,720** at 25,000 — a saving of **USD 171,360** a year. The band being delegated is 12 changes worth **204,000** in total. Even in the **worst imaginable case** — the delegate decides every one of the twelve wrongly and destroys **40 %** of the value each time — the loss is `12 × 17,000 × 0.40 =` **USD 81,600**, leaving the delegation ahead by **USD 89,760**. The threshold at which the two are equal is a value destruction of `171,360/204,000 =` **84 %** of every delegated decision.
5. **Interpretation.** The escalation cannot be justified at any plausible error rate — the delegate would have to destroy 84 % of the value of every delegated decision for the committee's involvement to break even, and a delegate who did that would be removed for reasons unrelated to thresholds. That is the general shape of the result, and it is why over-centralised delegation schedules are so common and so expensive: **the cost of escalation is certain, recurring and invisible, while the cost of a delegated error is uncertain, occasional and highly visible.** Organisations optimise against the visible cost. Three professional cautions, however. The calculation prices delay, not scrutiny — where a decision class carries irreversibility or external exposure the value test is the wrong test, which is why the schedule reads on three dimensions and not one. The 25 % critical-path share is an assumption from history and must be stated as one, since the conclusion is proportional to it — though note that it would have to fall below `81,600/(12 × 57,120) =` **11.9 %** before the worst case even competed. And a raised threshold requires the delegate to have the information to decide, which is a real prerequisite: delegation without information is abdication.

AI in this KA

Where it earns its place. Modelling latency across candidate governance designs — the arithmetic above, over dozens of cadence and threshold combinations, is deterministic and a natural fit. Classifying a change-request history into value bands and testing whether a proposed threshold would have altered any actual outcome — a genuinely useful retrospective test, and one nobody has time to do by hand. Drafting a delegation schedule from an existing one plus a set of stated changes, for human review. Detecting the decidability defects of 3.1.1 across a document set — decisions with no owner, two owners, or thresholds that conflict between the contract and the terms of reference.

Where it must not go. Setting a threshold, which is a risk-appetite decision belonging to the sponsor and the accountable body. Estimating the critical-path share or the error rate from plausibility rather than data — a model asked for these will produce confident numbers with no provenance, and they will then drive a real decision. Nor should its latency model be trusted unverified: the formula is two operations, and a leader who cannot reproduce `M/2 + L` on paper should not be quoting it.

Verification, concretely. Reproduce every number by hand, state the assumptions with their sources, and put the sensitivity in the paper — the breakeven error rate and the breakeven critical-path share, not merely the point estimate, because those are what tell a board whether the recommendation is robust or fragile.

KEY TERMS — KA 3.2

Term	Meaning
Sponsor	The individual accountable for the project's business outcome and mandate.
Steering committee	The body that makes decisions exceeding the project leader's authority, in time for them to matter.
Delegation schedule	The statement of authority by decision class, on value, reversibility and externality.
Governance latency	The expected wait for a committee decision: `E[wait] = M/2 + L`.
Meeting interval (`M`)	The period between meetings of a decision body.
Paper lead time (`L`)	How far before a meeting submissions close.
Committee capacity	Meetings per year × substantive items per meeting; compared against decision demand — average, peak and off-peak.
Member vs attendee	Members decide; attendees inform. Conflating them enlarges the body and reduces its authority.
Sponsor turnaround	The measured interval from a decision reaching the sponsor to its return; read as a distribution, not a mean.
Reversal-cost ratio (`ρ`)	Cost to undo a decision ÷ its value; authority reads on `max(value, cost to undo)`.
Sponsor load	The annual diary hours the sponsor obligations imply; a portfolio allocation constraint, not a personal one.

SAMPLE MCQS — KA 3.2

MCQ 3.2-A `[3.2.3 · Application]` A committee meets every 6 weeks and closes papers 2 weeks before each meeting. The expected wait for a decision arising at a random point in the cycle is: - A. 3 weeks - B. 5 weeks - C. 6 weeks - D. 8 weeks

Rationale: `E[wait] = M/2 + L = 6/2 + 2 = 5` weeks (3.2.3). A counts only half the interval and omits the paper lead time; C is the interval itself; D adds the whole interval to the lead time.

MCQ 3.2-B `[3.2.3 · Analysis]` A committee meets every 4 weeks with a 2-week paper lead time. Two options are on the table: cut the paper lead time by one week, or cut the meeting interval by one week. Which reduces expected latency more, and by how much? - A. cutting the meeting interval, by 1.0 week - B. cutting the paper lead time, by 1.0 week — twice the 0.5-week saving from the interval - C. both, equally, by 1.0 week - D. both, equally, by 0.5 weeks

Rationale: `E[wait] = M/2 + L` is 4.0 weeks initially; cutting `L` to 1 gives 3.0 (saves 1.0), cutting `M` to 3 gives 3.5 (saves 0.5) — a one-week cut in the paper lead time always saves a full week, a one-week cut in the meeting interval only half of one (3.2.3). The administrative deadline is also the cheaper lever to move.

MCQ 3.2-C `[3.2.3 · Evaluation]` Raising Meridian's threshold from 10,000 to 25,000 saves 171,360 a year in delay. The band delegated comprises 12 changes worth 204,000. The strongest argument for the change is that: - A. the project leader is experienced - B. the delegate would have to destroy 84 % of the value of every delegated decision for the escalation to break even - C. the committee is too busy - D. 12 changes is a small number

Rationale: The decisive argument is the breakeven value-destruction rate, which makes the comparison explicit and quantitative (3.2.3). A and D are assertions; C is a capacity argument, which is real but secondary.

MCQ 3.2-D `[3.2.2 · Analysis]` A steering committee's effective authority on a given decision class is best described as: - A. the authority of its chair - B. the highest authority among its members - C. the lowest authority among the members whose agreement is required - D. the authority delegated to the project leader

Rationale: One member who must consult before agreeing blocks the class, so the binding constraint is the minimum, not the maximum (3.2.2, Fault 1).

MCQ 3.2-E `[3.2.1 · Analysis]` The most reliable test of whether a sponsor is an advocate sponsor rather than an effective one is whether they: - A. attend every steering committee - B. can state a condition under which they would recommend stopping the project - C. defend the project's funding - D. know the project's schedule in detail

Rationale: An advocate sponsor is committed to the project rather than the outcome and therefore cannot stop it (3.2.1). C is an obligation of the role; D is closer to the operational failure mode.

MCQ 3.2-F `[3.2.1 · Analysis]` A sponsor returned 22 decisions in a year: 9 in 1.0 week, 8 in 2.5 weeks and 5 in 9.0 weeks. Which statement best supports a proposal to change how sponsor decisions are handled? - A. the mean turnaround is 3.3636 weeks, which is too slow - B. the median is 2.5 weeks, so performance is acceptable - C. 22.73 % of the decisions carry 60.81 % of the total wait, so handling five items differently removes 43.92 % of the cost - D. the sponsor is slower than a 4-week committee and the decisions should go to the committee

Rationale: The mean (A) describes a distribution nobody experiences, and the median (B) hides the tail; the decisive presentation is the tail's share of the wait and the saving available from it (3.2.1). D is arithmetically wrong: 3.3636 weeks is faster than the committee's 4.0 weeks. Common error: arguing from the average, which invites a debate about diligence instead of about five items.

MCQ 3.2-G `[3.2.3 · Application]` A delegated change is worth USD 15,000 but would cost USD 180,000 to undo once deployed. Under a schedule that reads on value, reversibility and externality, the authority required is that appropriate to: - A. USD 15,000, since that is the change's value - B. USD 165,000, the net exposure - C. USD 180,000, because authority reads on `max(value, cost to undo)` - D. USD 195,000, the sum of value and reversal cost

Rationale: The reversal-cost ratio is `180,000/15,000 = 12`, and the rule reads authority on the larger of the two figures (3.2.3). A is the value-only failure of Case study B. B nets the change's value off the reversal cost as though the value earned offset the cost of undoing it; there is no such offset — the reversal cost is what it costs to return to the prior state. D adds two quantities that are not additive — the reversal cost already includes undoing the 15,000 of work.

SELF-CHECK — KA 3.2

1. State the governance latency formula and both of its levers. — `E[wait] = M/2 + L`; shorten the paper lead time (a full week per week cut) or the meeting interval (half a week per week cut).
2. Why do organisations systematically set delegation thresholds too low? — The cost of escalation is certain, recurring and invisible; the cost of a delegated error is uncertain, occasional and highly visible, so optimisation runs against the visible cost.
3. What must accompany a raised threshold for it to be delegation rather than abdication? — The information, criteria and support the delegate needs to decide, plus a record of what they decided.
4. Why is average committee utilisation a misleading measure? — Demand clusters at phase boundaries; Meridian's 74.0 % average conceals **110.00 %** at the boundaries and **63.25 %** between them, and it is at the peak that contentious items are deferred (3.2.2).
5. How does a delegation schedule express reversibility? — Through the reversal-cost ratio `ρ = cost to undo ÷ value`, with authority read on `max(value, cost to undo)` — which put a USD 15,000 Meridian template change into the 100,001–500,000 band at ρ = 12 (3.2.3).
6. Roughly how much diary time does a Meridian-sized sponsorship require, and why does the answer matter? — **111.0 hours a year**, about **6.03 %** of a working week; small enough that absence is a priority choice, and large enough that no executive carries more than about three (3.2.1).

KNOWLEDGE AREA 3.3

Assurance, gates and escalation

Topics: 3.3.1 stage gates · 3.3.2 assurance lines · 3.3.3 escalation design · 3.3.4 auditability and the decision record.

3.3.1 Stage gates

Definition. A stage gate is a point at which continuation requires an explicit decision by a named authority against stated criteria, with authority to stop, hold or redirect as well as proceed. A checkpoint without the authority to stop is a milestone with a meeting attached.

What a gate is for. Gates exist to convert **irreversibility into optionality**: before committing the next tranche of cost, an organisation buys the chance to stop or change direction while stopping is still cheap. It follows that gates belong where irreversibility genuinely steps up — before major commitment, before build, before go-live, before regulatory submission — and not at regular calendar intervals, which is how they proliferate into re-approval ceremonies.

Gate criteria. Criteria must be set **in advance**, be **objectively assessable**, and cover both the work and the decision: is the deliverable adequate; is the business case still valid (Domain 2's benefits and assumptions, re-tested rather than restated); are the risks acceptable and owned; is the receiving organisation ready; and is the plan for the next stage credible? Criteria written after the evidence is available are not criteria.

The honest gate. A gate that has never held or stopped anything is not evidence of good delivery, it is evidence of a gate that does not function — and its cost is being paid for nothing. The pattern to watch for is the **conditional pass**: proceed subject to conditions, which are then neither tracked nor enforced. A conditional pass is a genuine and useful instrument, and it is only that if the conditions have owners, dates and a stated consequence of non-completion, verified at the next gate.

Gates cost time, and time has a price. This is the part of gate design that is almost never computed, and it decides whether the gate is worth having.

WORKED EXAMPLE

Worked example 3.3.1 — is Meridian's design gate worth its elapsed time?

1. **Setup.** Meridian's design-completion gate requires **USD 45,000** of review effort and **6 weeks** of elapsed time (cost of delay **14,280** per week). Historical evidence on comparable work: a material design defect is present with probability **0.30**; the gate detects it with probability **0.80**; a defect corrected at design costs **120,000**; the same defect found in build costs **900,000**.
2. **Formula.** Expected total cost with the gate = review + delay + `P(defect) ×` [`P(detect) ×` design-fix `+ P(miss) ×` build-fix]. Without the gate = `P(defect) ×` build-fix.
3. **Substitution.** Without: `0.30 × 900,000`. With: `45,000 + 6 × 14,280 + 0.30 × (0.80 × 120,000
4. 0.20 × 900,000)`.
5. **Result.** Without the gate **USD 270,000**. With the gate `45,000 + 85,680 + 82,800 =` **USD 213,480**. The gate is worth **USD 56,520**.
6. **Interpretation.** The gate pays — and the useful output is not the 56,520 but the two **breakeven points** it implies, because those are what a leader negotiates with. Holding everything else, the gate stops paying once its elapsed time reaches `(270,000 − 45,000 − 82,800)/14,280 =` **9.96 weeks**: a gate that takes ten weeks destroys the value it exists to protect, which is the arithmetic behind the familiar and usually unquantified complaint that assurance has become an obstacle. And at the actual 6 weeks, the gate needs a detection probability above **55.85 %** to be worth holding — so a review staffed by people who cannot competently detect the defect is worse than no review, because it costs the money and the delay and returns nothing. Both results generalise: **gate value is destroyed by elapsed time and by weak detection**, and a leader whose gate is under challenge should compute which of the two is the actual problem before defending or abandoning it.

The conditional pass, priced — and the identity that makes it dangerous. The instrument was described above as real only if its conditions have owners, dates and consequences. That is not a counsel of tidiness; it follows from the gate arithmetic. A gate detects a defect and then directs a remedy, and a directed remedy that is not completed leaves the defect in the work. So if a gate's detection rate is `q` and the share of conditions actually closed is `r`, the gate's **effective detection rate** is

```
q_eff = q × r
```

— an identity worth stating because it converts an administrative failure into the same variable the gate's value already depends on. A gate that detects perfectly and closes half its conditions is a gate that detects half.

WORKED EXAMPLE

Worked example 3.3.1b — Meridian's conditional passes, and the gate that inverted.

1. **Setup.** Meridian's four gates in the last twelve months issued **23** conditions between them. At the following gate: **9** were verified closed, **8** were open past their stated date, and **6** had no owner or date recorded and were never tracked at all. Everything else is Worked example 3.3.1's design gate: review effort **USD 45,000**, elapsed **6 weeks** at **14,280** per week, `P(defect)` **0.30**, nominal detection `q` **0.80**, design fix **120,000**, build fix **900,000**; the gate's nominal value was **USD 56,520**.
2. **Formula.** Closure rate `r = closed ÷ issued`. `q_eff = q × r`. Then re-run 3.3.1's comparison with `q_eff` in place of `q`, and solve `q × r ≥ d*` for the closure rate the gate needs, where `d*` is 3.3.1's breakeven detection probability.
3. **Substitution.** `r = 9/23`. `q_eff = 0.80 × 0.3913`. Expected remediation `0.30 × (0.3130 × 120,000 + 0.6870 × 900,000)`. Total with gate `45,000 + 6 × 14,280 + that`. Breakeven `d* = 130,680/234,000`; `r* = d*/0.80`.
4. **Result.** Closure rate **39.13 %**; effective detection **31.30 %**, against a nominal 80 %. Expected remediation rises from 82,800 to **USD 196,747.83**, so the total with the gate is `45,000 + 85,680 + 196,747.83 =` **USD 327,427.83** against **USD 270,000** without it. **The gate now destroys USD 57,427.83** — a swing of **USD 113,947.83** from its nominal value, caused by no change whatever to the gate itself. The closure rate the gate needs to break even is `d*/q = 0.5585/0.80 =` **69.81 %**, i.e. **17** of the 23 conditions rather than 9; at 16 of 23 (69.57 %) it is still **USD 453.91** underwater, and at 17 of 23 (73.91 %) it returns **USD 7,685.22**.
5. **Interpretation.** The headline is the swing. **A gate worth +56,520 became a gate worth −57,428 without anyone changing its criteria, its staffing or its duration** — the whole of the movement is in a tracking failure that no governance report would have shown, because governance reports count gates held and conditions issued, not conditions closed. That is the practical case for making condition closure a counted field rather than an exhortation, and it is why Toolkit 3.T.3 makes "conditions past their date" one of five monthly integers.

The identity `q_eff = q × r` also relocates the remedy. When a gate is challenged as an obstacle, the instinct is to argue about its criteria or its duration; here neither is the problem. At the observed closure rate the gate's **breakeven elapsed time collapses from 9.96 weeks to 1.9784 weeks**, so a leader defending a six-week gate on this evidence is defending the indefensible — and a leader attacking it would be attacking the wrong object, since the gate at full closure is worth 56,520. **The question to compute first is which of `q`, `r` and elapsed time is the binding constraint**, and only one of the three is cheap to fix.

Note also the composition of the 14 unclosed conditions, because the two halves are different defects. The **8 open past their date** are a follow-through failure: the owner and date existed, so the register can chase them, and a monthly count will close most of them. The **6 with no owner or date** were never conditions at all — they were reservations minuted as conditions, and they are the more serious finding, because a gate that issues them has recorded a decision it did not make. Closing the 8 alone takes `r` to `17/23 =` **73.91 %** and the gate back into value, which is the cheapest available governance intervention in this domain.

Three cautions. The identity assumes a condition's remedy, if completed, restores the detection the gate credited itself with — reasonable for a defined corrective action, wrong for a condition that merely defers the question ("proceed subject to confirming the interface design"), which detects nothing and should never have been a conditional pass. **A conditional pass whose condition is "find out" is a hold recorded as a pass.** The model also treats closure as binary, and partial closure is common; where it matters, weight `r` by the share of each condition's remedy completed rather than by count. And `r` should be computed **per gate and per condition type**, not as one programme number, because an aggregate of 39.13 % can conceal one gate closing everything and another closing nothing — and it is the second that will produce the escape.

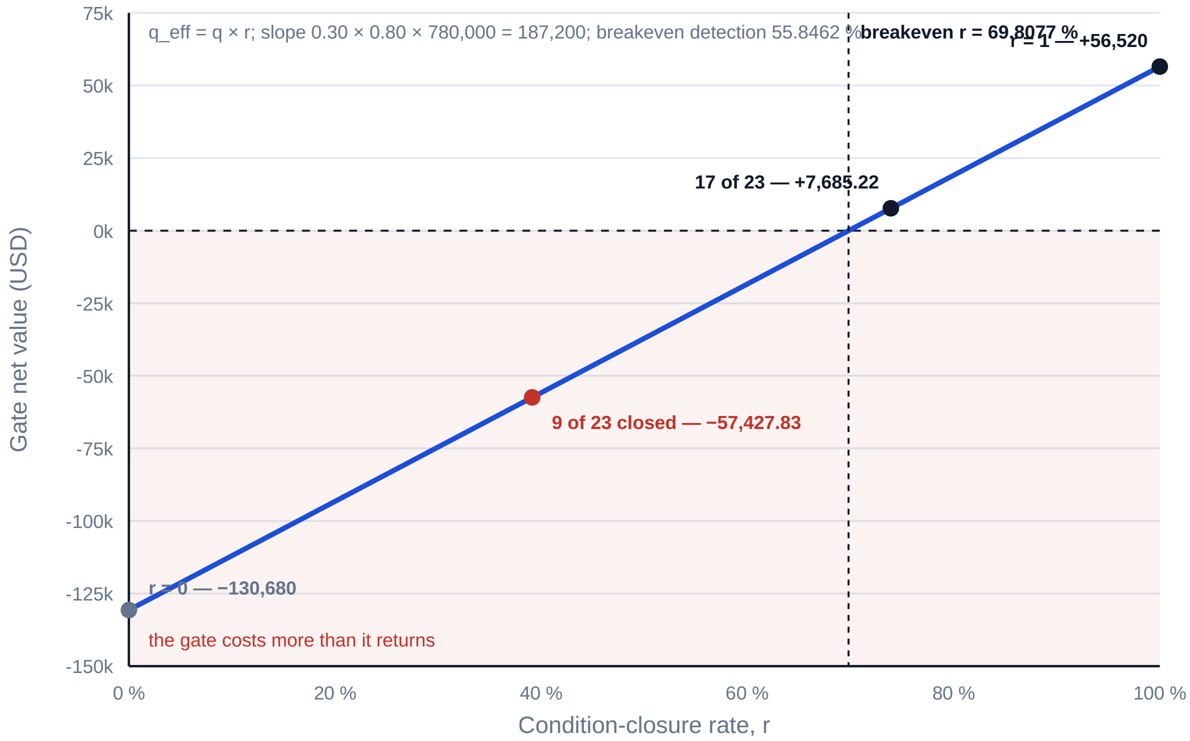


Fig 3.3.2 — The conditional pass: how closure rate decides a gate's value

3.3.2 Assurance lines

The three lines, and what each is for.

- **First line — management.** The project's own controls: reviews, testing, quality checks, reporting. Owned by the project leader, and the only line that can prevent a defect rather than detect it.
- **Second line — oversight function.** A PMO, risk or quality function providing independent challenge while remaining inside management's chain. Its value is comparability across the portfolio and pattern detection no single project can see.
- **Third line — internal audit (and external assurance).** Independent of management, reporting to the audit committee or equivalent, forming an opinion on whether the whole control system works.

The failure modes. Duplication — three lines asking the same questions, which multiplies cost and destroys the project team's willingness to engage. Gap — everyone assumes another line covers a risk. Capture — the second line drafts the plan it later assures, and so cannot challenge it; this is the most damaging and the least visible, because the assurance product still looks independent. Assurance as accountability transfer — a leader treats a favourable assurance opinion as a discharge of their own obligation; Domain 1's principle applies unchanged, an opinion is information and not a transfer of accountability.

The countermeasure is an **assurance map**: risks and controls down the side, lines across the top, and each cell marked as covered, not covered or duplicated — reviewed by the accountable authority. It is a half-day artefact that surfaces both gaps and duplication, and its absence is the usual reason a project is simultaneously over-assured and unassured.

The map becomes a decision tool when the cells carry numbers. A map marked "covered / not covered" identifies gaps and cannot rank them, which is why maps are drawn, admired and not acted on. Give each material risk i a probability p_i that the control failure it describes is present, a consequence u_i if it escapes to where it does damage, and each line j covering it a detection rate q_{ij}, and the map computes the only quantity that ranks its rows:

$$\text{residual exposure} = \Sigma_i \; p_i \times \prod_j (1 - q_{ij}) \times u_i$$

Two properties follow immediately and both are counter-intuitive. Because the miss probabilities **multiply**, the marginal contribution of an additional line to a risk already covered twice is small — the third line on a risk with two 0.50-plus detectors is buying the last slice of a small number. And because uncovered risks contribute $p_i u_i$ in full, **a single gap almost always outranks every duplication in the map**. The reallocation that follows is therefore available at constant cost, which makes it the rare governance improvement that needs no budget decision.

WORKED EXAMPLE

Worked example 3.3.2 — Meridian's assurance map, and the reallocation that costs nothing.

1. **Setup.** Meridian's five material control risks are assured as follows, at a blended assurance day rate of **USD 950**. First line **90 days** (USD 85,500), second line **40 days** (USD 38,000), third line **30 days** (USD 28,500) — **160 days**, **USD 152,000** in total.

Risk	p	u (USD)	1st line q	2nd line q	3rd line q
R1 clinical data migration integrity	0.30	900,000	0.60	0.50	0.40
R2 access control and confidentiality	0.20	1,200,000	0.50	—	—
R3 benefits ownership in the receiving organisation	0.35	685,440	—	—	—
R4 clinic readiness and training	0.40	85,680	0.70	0.40	—
R5 third-party interface conformance	0.25	214,200	0.55	0.45	—

u for R3 is one year of the programme's realistic benefit (USD 685,440); for R4 and R5, six and fifteen weeks of delay at USD 14,280 per week. 2. **Formula.** Residual exposure $= \Sigma\ p_i\ \prod(1 - q_{ij})\ u_i$. Marginal value of a line on a risk $= p_i \times \Delta(\text{miss}) \times u_i$. Breakeven consequence for a line on a risk $= \text{line cost} \div (p_i \times \Delta q)$. 3. **Substitution.** R1 $0.30 \times (0.40 \times 0.50 \times 0.60) \times 900{,}000$; R2 $0.20 \times 0.50 \times 1{,}200{,}000$; R3 $0.35 \times 1.00 \times 685{,}440$; R4 $0.40 \times (0.30 \times 0.60) \times 85{,}680$; R5 $0.25 \times (0.45 \times 0.55) \times 214{,}200$. 4. **Result — as assured.**

Risk	Combined miss	Residual exposure (USD)	Share of residual
R1 (all three lines)	0.1200	32,400.00	**7.87 %**
R2 (first line only)	0.5000	120,000.00	**29.15 %**
R3 (no line)	1.0000	**239,904.00**	**58.27 %**
R4	0.1800	6,168.96	1.50 %
R5	0.2475	13,253.63	3.22 %
Total		**411,726.59**	100 %

Now move the third line's **30 days** off R1 — where all three lines sit — onto R3, at the same detection rate of 0.40 and the same cost. R1's residual rises to **USD 54,000.00**; R3's falls to **USD 143,942.40**. Total residual **USD 337,364.99** — a reduction of **USD 74,361.60** (**18.06 %** of residual, **13.19 %** of residual plus assurance cost) for **no additional spend**. 5. **Interpretation.** The distribution in the fourth column is the finding, and it is the finding in almost every assurance map drawn for the first time. **All three lines sit on the risk carrying 7.87 % of residual exposure, while 87.41 % of it sits in the two risks with the least coverage** — and the largest single row, at **58.27 %**, has no assurance at all. This is not incompetence; it is the predictable result of assurance following auditability. Migration integrity is testable, sampleable and comfortable to review, so three functions review it. Benefits ownership in the receiving organisation is diffuse, politically awkward and owned outside the project, so nobody does — which is precisely Domain 2's omitted-column failure reappearing as an assurance gap. **Assurance concentrates where it is easy, not where residual exposure is, and only the arithmetic makes that visible.**

The reallocation arithmetic explains itself once the marginal values are written down. The third line on R1 was buying `0.30 × (0.20 − 0.12) × 900,000 =` **USD 21,600** of exposure reduction; on R3 the same 30 days buys `0.35 × 0.40 × 685,440 =` **USD 95,961.60 — 4.44 times** as much. Against its **USD 28,500** cost, the third line on R1 was running at a net **loss of USD 6,900**, and on R3 it returns a net **USD 67,461.60**. The breakevens make the point transferable: for the third line to pay on R1, either the consequence would have to exceed `28,500/(0.30 × 0.08) =` **USD 1,187,500** (against R1's 900,000) or the failure probability would have to exceed `28,500/(0.08 × 900,000) =` **39.58 %** (against R1's 0.30). **Neither is close**, and both are the sort of test a reviewer can run on a single row in a minute.

The professional caution that must accompany this, because the arithmetic over-applies. Nothing above justifies deleting the third line. Third-line assurance exists to form an **independent opinion on whether the control system works** — a portfolio-level and constitutional product, reporting outside management, whose value is not the marginal detection it contributes to any one project's risk register. Charging its cost to a single project and testing it on detection economics misprices it by construction, and the honest conclusion from the numbers above is narrower and more useful: **the third line's effort was pointed at the wrong risk on this project.** Where an organisation genuinely wants the detection contribution of a third pass on a single risk, the marginal arithmetic says it will rarely pay, which is an argument about where independent assurance looks, never about whether it should exist.

Three further cautions. The multiplication of miss probabilities assumes the lines are **independent detectors**, and Domain 9, KA 9.2.2 shows why they usually are not: three functions reading the same design document with the same blind spot miss the same defect, so realised coverage on R1 is worse than 88.00 % and closer to the 80.00 % that the first two lines achieve alone. Make consecutive lines methodologically different or do not count them separately. The detection rates are **estimates**, and where no history exists the defensible presentation is a range with the reallocation decision computed across it — note that the R1-to-R3 conclusion survives any q for the third line above about 0.09 on R3, so it is robust in a way the point estimate does not show. And **capture invalidates a cell rather than reducing it**: a line assuring work it helped produce should be entered as $q = 0$, not as a lower q, because its opinion is not weak evidence, it is not evidence.

Proportionality, and how to size it. Assurance effort should scale with novelty, value, irreversibility and external exposure — not uniformly, and not with organisational anxiety. A high-value, low-novelty, reversible project may need less assurance than a small, novel, irreversible, publicly visible one, and a regime that cannot express that will over-assure the first and under-assure the second. The residual-exposure column above is what makes proportionality operable within a project: allocate the next assurance day to the row with the largest $p_i \prod(1 - q_{ij})\ u_i$, and stop when the marginal reduction falls below the day rate. Across projects, the same logic is a portfolio judgement and belongs to the second line, whose comparability across the portfolio is the whole of its value.

3.3.3 Escalation design

The principle. Escalation is a **designed pathway with stated latency**, not an instruction to tell someone senior. A usable design states, for each escalation class: the trigger (objective, not "if concerned"), the destination, the decision required, the **time within which the decision will be made**, and the out-of-cycle mechanism if the ordinary cadence is too slow.

That fourth element is the one usually missing, and its absence has a computable cost.

WORKED EXAMPLE

Worked example 3.3.3 — the total latency of Meridian's escalation path.

1. **Setup.** A decision requiring executive authority passes three tiers: the **project board** (meets every 2 weeks, papers close 1 week ahead), the **programme board** (every 4 weeks, papers 2 weeks ahead), and the **executive committee** (quarterly — every 13 weeks — papers 3 weeks ahead). Cost of delay **14,280** per week.
2. **Formula.** Total expected latency = Σ over tiers of $M/2 + L$. Cost = latency × cost of delay.
3. **Substitution.** Project board $2/2 + 1 = 2.0$; programme board $4/2 + 2 = 4.0$; executive committee $13/2 + 3 = 9.5$. Total $2.0 + 4.0 + 9.5$.
4. **Result. 15.5 weeks** of expected latency for a single decision, costing **USD 221,340** in delay — before anyone in any of the three rooms disagrees with the recommendation.
5. **Interpretation.** Two hundred and twenty-one thousand dollars is the price of an organisation chart, and it is invisible: it appears in no budget, is attributed to no decision, and is generally described afterwards as the project having been slow. Notice the distribution — the quarterly committee alone accounts for **9.5 of the 15.5 weeks** (**61 %**) and **USD 135,660**, which is what makes "add the executive committee to the approval path" such an expensive sentence when spoken casually in a governance review. Two redesigns follow directly. **Reduce the tiers a decision must legitimately pass:** if the programme board can decide with the executive committee informed, latency falls to **4.0 weeks** and **USD 57,120** — a saving of **USD 164,220**, or **74.2 %**, from a change that removes no scrutiny that anyone can name. **Add an out-of-cycle mechanism:** a written-resolution procedure with a five-working-day turnaround makes the single-tier path **1.0 week** and **USD 14,280**, a **93.5 %** reduction against the original. The general rule this supports: **count the tiers, add the latency, price it, and then justify each tier against its cost** — which is a conversation governance reviews almost never have, because until the latency is added up there is nothing to weigh the tier against.

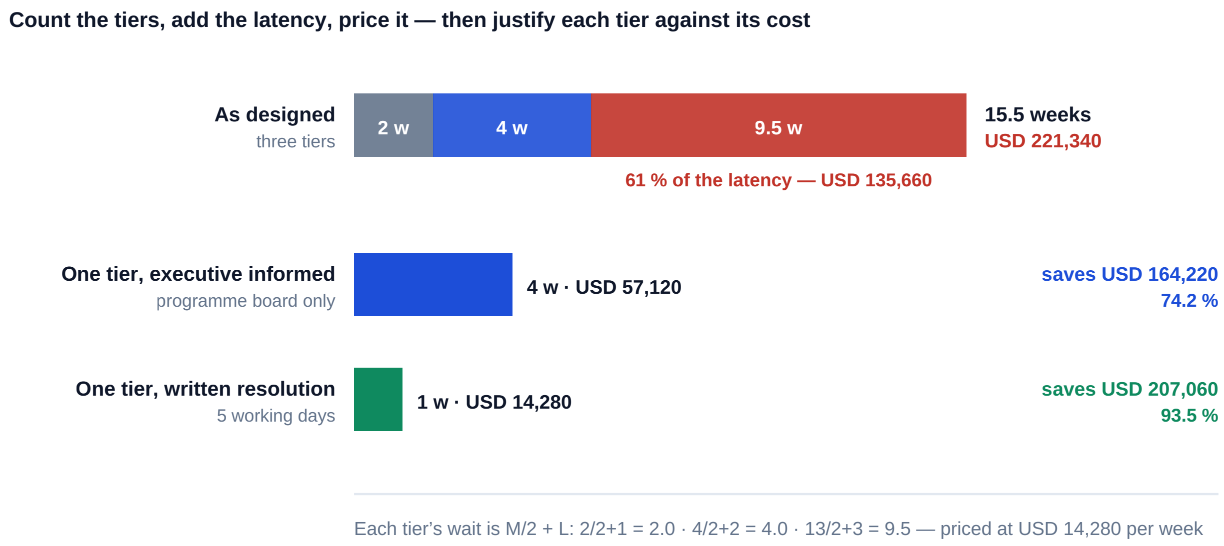


Fig 3.3.1 — The price of an escalation path

Designing the trigger. Objective triggers only: a forecast variance beyond a stated tolerance, a risk exposure above a stated threshold, a decision required above the delegated authority, a dependency breached. "Escalate if concerned" produces two failure modes at once — late escalation by those who fear it and constant escalation by those who fear the alternative — and both are then treated as individual judgement problems rather than as the design defect they are.

But an objective trigger is not yet a well-specified one. "Escalate if the forecast overrun exceeds 10 % of budget" reads as objective and is not, because forecast is not a single number: the earned value family produces several legitimate estimates at completion from the same data (Domain 7, KA 7.3), and they can sit on opposite sides of a tolerance. A trigger must therefore name three things the usual wording omits — **the forecast method**, **the measurement point** and **the confirmation rule** — and the cost of omitting them is not ambiguity in the abstract but a real escalation that either fires on noise or fails to fire at all. The volume's single-project thread makes this concrete.

WORKED EXAMPLE

Worked example 3.3.3b — writing Auriga's escalation trigger, and testing whether it can act.

1. **Setup.** Project Auriga is a **25-week** project with **BAC USD 4,000,000**. At the **week 13** data date: **PV 2,080,000**, **EV 1,920,000**, **AC 2,120,000**, giving **CPI 0.91** and **SPI 0.92** to two places. Domain 7's three estimates at completion are **4,200,000** (remaining work at plan), **4,416,667** (`BAC/CPI`) and **4,608,056** (`AC + (BAC − EV)/(CPI × SPI)`). The governance framework says: escalate to the programme board where the forecast overrun exceeds 10 % of BAC. The escalation path available is 3.3.3's three-tier route at **15.5 weeks**, with a single-tier alternative at **4.0 weeks** and a written-resolution route at **1.0 week**. Planned average burn is `4,000,000/25 =` **USD 160,000** a week.
2. **Formula.** Variance at completion `VAC = EAC − BAC`, expressed as a share of BAC and tested against the tolerance. Then the **decision action window** `= (remaining duration − escalation latency) ÷ remaining duration`, and the last week at which a trigger can still produce an actionable decision `= planned finish − escalation latency`.
3. **Substitution.** `4,200,000 − 4,000,000`; `4,416,666.67 − 4,000,000`; `4,608,055.56 − 4,000,000`; each over 4,000,000. Tolerance `0.10 × 4,000,000 = 400,000`. Remaining duration `25 − 13 = 12`. Windows `(12 − 15.5)/12`, `(12 − 4.0)/12`, `(12 − 1.0)/12`.
4. **Result.**

Forecast method	EAC (USD)	VAC (USD)	VAC as % of BAC	Trigger at 10 %?
Remaining work at plan	4,200,000.00	200,000.00	**5.00 %**	**No**
`BAC/CPI`	4,416,666.67	416,666.67	**10.42 %**	**Yes**
`AC + (BAC − EV)/(CPI × SPI)`	4,608,055.56	608,055.56	**15.20 %**	**Yes**

The three methods span **USD 408,055.56** on identical data, and the tolerance line at 400,000 falls inside that span. And the action windows:

Escalation route	Latency	Weeks left after the decision	Action window
Three tiers as designed	15.5 w	**−3.5**	**−29.17 %**
Single tier, board decides	4.0 w	8.0	**66.67 %**
Written resolution	1.0 w	11.0	**91.67 %**

The as-designed path returns its decision **3.5 weeks after Auriga's planned completion**. Working backwards, the latest week at which that path can still deliver an actionable answer is week `25 − 15.5 =` **9.5** — the first **38.00 %** of the project. 5. **Interpretation.** Two results, and the second is the more important.

The trigger's answer depends entirely on which forecast method it names, and nobody has named one. `BAC/CPI` clears the tolerance by USD 16,666.67 and the plan-based estimate misses it by USD 200,000; the composite clears it by USD 208,055.56. A framework that says "the forecast" has therefore delegated the escalation decision to whoever prepares the report, which is the opposite of what an objective trigger is for — and it does so invisibly, because each of the three numbers is correctly calculated. Worse, the `BAC/CPI` trigger is **marginal to the point of being noise**. The cost performance index at which `BAC/CPI` exactly equals 1.10 × BAC is `1/1.10 =` **0.9091**, and Auriga's CPI is **0.9057** — below it by **0.0034**. In earned-value terms, `EV` would need to be **USD 1,927,272.73**

rather than 1,920,000 for the trigger not to fire: **USD 7,272.73**, or **0.18 %** of BAC, one accrual timing difference. **A trigger stated on a single ratio at a single data date fires on measurement noise**, which is how organisations acquire a reputation for escalating trivia and then stop escalating at all. The corrections are cheap and specific: name the method (`BAC/CPI` is the defensible default because it is the method whose assumption — that present cost performance persists — is the one a governance body should be asked to accept or reject); state the measurement point (a defined data date, not "when noticed"); and add a **confirmation rule** — two consecutive data dates beyond tolerance, or one date beyond a wider secondary tolerance — so that a single period's noise cannot fire it and a sustained trend cannot fail to.

The second result is that on this project the trigger cannot help, whatever it says. An escalation latency of 15.5 weeks against 12 weeks of remaining duration gives a **negative action window**: the decision is delivered after the work it concerns has finished, so the escalation is not slow, it is **ceremonial**. This is the test that belongs beside every escalation path and is almost never applied — is the latency shorter than the remaining duration of the thing being decided? — and its diagnostic form is sharper still: the as-designed path is actionable only during the first **38.00 %** of a 25-week project, so from week 9.5 onward Auriga has an escalation route that can produce decisions and cannot produce useful ones. Note what this does to the two redesigns of Worked example 3.3.3. There they saved **USD 164,220** and **USD 207,060**; here they do something that cannot be expressed in money at all — they convert a decision from impossible to possible, moving the action window from −29.17 % to **66.67 %** and **91.67 %**. **Latency priced in currency understates the case for a shorter path; latency compared with remaining duration is what shows when the path is not a cost but a nullity.**

Three cautions. The action window uses **planned** remaining duration, and at SPI 0.92 Auriga's remaining duration is itself likely longer than 12 weeks, which flatters the as-designed path — the defensible practice is to compute the window on the current schedule forecast (Domain 6) and to state which duration was used. The window is a **necessary and not a sufficient** test: a decision returned inside the window still has to arrive early enough for the response to be implemented, so a fuller form subtracts the implementation lead time as well, and for an irreversible response that lead time can exceed the decision latency. And a negative window is **not an argument for skipping the escalation** — the decision may be needed for reasons that outlive the project, and the honest response is to escalate and record that the ordinary path cannot answer in time, which is the evidence that gets the out-of-cycle mechanism built before the next project needs it.

The escalation culture problem. Escalation is only a functioning mechanism if using it is safe. Where escalation is read as failure, it happens late, which is precisely when the decision can no longer help. The countermeasures are governance ones and belong to the sponsor and the committee: respond to early escalation visibly and well; separate the escalation of an issue from any judgement about the person raising it; and track the **lead time** of escalations — how far before the impact they arrive — because a shortening lead time is the earliest available signal that the mechanism is decaying, and it appears in no standard report.

3.3.4 Auditability and the decision record

The principle. A decision that is not recorded has not been made — it has been remembered, and memories diverge exactly when the stakes rise. The decision record is the mechanism that converts a meeting into an institutional fact.

What a decision record must contain, per decision, and it is short enough that its absence is never a resourcing problem: a unique reference; the date; the decision-maker by name and role (never a committee alone — a body records the decision, a person is accountable for it); the decision, in words that permit only one reading; the options considered and why the chosen one was chosen;

the information relied on, referenced to its version; the conditions attached, with owners and dates; and the review date if the decision is provisional.

Why the "information relied on, referenced to its version" line matters more than it looks. When a decision is examined afterwards — in a lessons review, a dispute, an audit or an inquiry — the question is almost never "was the decision right?" but **"was it reasonable on what was known at the time?"** Only a versioned reference can answer that, and its absence converts a defensible decision into an indefensible one at exactly the moment the defence is needed.

Two governance defects the record exposes, and this is the practical case for keeping it well. The **re-decided decision**: the same question arriving at the same body a third time is a symptom of either an unrecorded decision or an illegitimate one, and the log makes the recurrence visible where minutes do not. And the **decision nobody made**: a change that took effect without an entry, which is how scope and budget move without anyone having decided — the most common finding of a governance audit and the hardest to see from inside.

RACI and its integrity check. Responsibility assignment matrices — Responsible, Accountable, Consulted, Informed — are the standard artefact for decision rights, and their standard failure is being drafted, circulated and never checked. The checks are countable, and a matrix that fails any of them is not a matrix but a document: **exactly one A** per decision (two make it undecidable, none make it unowned); the A holds sufficient authority under the delegation schedule; C and R are distinguished (consultation is not agreement); and the count of C's per decision is bounded, since consultation is where latency accumulates invisibly.

WORKED EXAMPLE

Worked example 3.3.4 — auditing Meridian's decision-rights matrix.

1. **Setup.** Meridian's matrix covers **12** decision classes across 7 roles. On review: **9** classes carry exactly one Accountable, **2** carry two, and **1** carries none.
2. **Formula.** Defect rate = classes failing the single-A test ÷ total classes.
3. **Substitution.** `(2 + 1)/12`.
4. **Result.** **3** of 12 classes are defective — a **25.0 %** defect rate.
5. **Interpretation.** A quarter of the programme's decision classes cannot be decided as documented, and the two failure types behave differently under stress, which is why they are worth distinguishing rather than totalling — and why the 25.0 % headline is the least useful number in the example.

Price them separately. The **two-A** classes will be decided, badly: by whichever holder acts first, or after a delay while the two reconcile, and reconciliation in practice means the item returns to the next meeting. Those two classes generated **11** decisions over the year; each incurring one extra committee cycle at 3.2.3's rates, with the standing **25 %** critical-path share, costs `11 × 0.25 × 4 × 14,280 =` **USD 157,080 — USD 78,540** per defective class. The **zero-A** class behaves differently and worse: nothing forces it to a decision at all, so it drifts until the consequence forces an escalation, and an escalation that arrives without an owner arrives at the top. Its **4** decisions, drifting to the full three-tier path of Worked example 3.3.3 at **15.5 weeks**, cost `4 × 0.25 × 15.5 × 14,280 =` **USD 221,340** — coincidentally exactly one expected three-tier escalation, and **2.82 times** the cost of a two-A class. Total documented cost of a 25.0 % defect rate: **USD 378,420**.

That ordering is the professional content of the example. **A decision class with two accountable roles is expensive; a class with none is nearly three times as expensive**, because the two-A defect delays a decision and the zero-A defect relocates it — upward, late, and to people with no preparation. Yet a reviewer scanning a matrix finds the two-A cases first, since they are visible as duplicated letters, while the zero-A case is visible only as a blank in a row nobody thought to write. The practical instruction is to audit the matrix **by decision class, not by role**: list every decision the project will face, then look for its A — which finds the missing rows that a column-by-column read never will.

Two cautions. The costing assumes each defective class's decisions are **independent occasions of delay**, which bounds rather than forecasts as elsewhere in this domain; present the counts and the delay-weeks alongside the currency. And a single A is **necessary and not sufficient**: the A must also hold authority sufficient for the class under the delegation schedule, which is a second and separate check — a matrix can pass the single-A test in full while every A sits below the threshold its class requires, at which point the matrix is not wrong, it is decorative. The test costs an hour and it is the highest-yield governance check available before mobilisation.

The re-decided decision has a price, and it is the largest hidden number in this domain. 3.3.4 named the defect; counting it is what makes it actionable, and the count comes from the decision log because it comes from nowhere else — minutes record what a meeting decided, never that the meeting had decided it before.

WORKED EXAMPLE

Worked example 3.3.4b — Meridian's re-decision tax.

1. **Setup.** Meridian's decision log holds **148** decisions over twelve months. Clustering the decision text shows **19** questions that arrived at the same body more than once, of which **6** arrived three or more times. Of the 19, **13** carried no versioned reference to the information relied on and **6** carried no named decision-maker. The committee's capacity is **104** item-slots a year against the **77** items of forward demand computed in 3.2.2; `E[wait]` is **4.0 weeks**, cost of delay **14,280** per week, critical-path share **25 %**.
2. **Formula.** Repeat slots consumed = second appearances + third appearances. Corrected demand = forward demand + repeat slots. Delay cost = repeat slots × critical-path share × `E[wait]` × cost of delay.
3. **Substitution.** Repeat slots `19 + 6 = 25`. Corrected utilisation `(77 + 25)/104`. Delay cost `25 × 0.25 × 4 × 14,280`.
4. **Result.** Re-decision rate **12.84 %** of all logged decisions. Repeat slots **25**, which is **24.04 %** of the committee's annual capacity. Corrected demand **102** against capacity 104 — utilisation **98.08 %**, not the 74.0 % of 3.2.2. Delay cost **USD 357,000.00**, which is **2.08 times** the entire annual saving from raising the delegation threshold (USD 171,360).
5. **Interpretation.** Start with the reconciliation, because it resolves a contradiction the domain has been carrying. Worked example 3.2.2 computed 74.0 % utilisation and the members reported a committee that was full; **98.08 %** is why. The forward demand model counted the decisions the committee was asked to make and could not count the ones it made twice, and a capacity model that omits re-decisions will always read low by roughly the re-decision rate. **The single most useful correction to a committee capacity estimate is to add last year's repeat count**, and it takes an afternoon with the log.

Then the size of it. **USD 357,000 a year is spent deciding things that had already been decided** — more than twice the celebrated saving from the threshold change, and unlike that saving it requires no delegation, no risk appetite conversation and no approval from anybody. It is also, uniquely in this domain, a defect whose remedy costs nothing: the causal split says **68.42 %** of the re-decisions lacked a versioned information reference and **31.58 %** lacked a named decision-maker, and both are **fields on a template**. A decision re-arrives because its basis can be disputed ("that was decided on the old numbers") or its authority can be disputed ("the committee noted it, nobody decided it"), and the two fields close both routes. This is the concrete answer to the question 3.3.4 raises and most organisations treat as rhetorical — what is the decision record actually worth? Here, **USD 357,000 a year and 25 committee slots**, from two fields.

Three cautions. Not every repeat is a defect: a **provisional** decision with a stated review date is supposed to return, and a decision returning because the world changed is governance working, not failing — so the count must exclude scheduled reviews and material-change re-openings, and the defensible metric is unplanned re-decisions. The clustering that produces the count is a judgement about whether two differently worded questions are the same question, which is exactly the task AI is good at and exactly the task where a false positive is embarrassing, so every flagged pair is confirmed by a human before it is reported (see the AI treatment below). And the delay cost, as always here, **adds waits that may overlap**; the slot count — 25 of 104 — is the harder number and the one to lead with, because nobody can argue with it.

The cumulative test, designed rather than asserted. Toolkit 3.T.2 and Case study B both call for a rule that aggregates related decisions, and a rule of that shape has two parameters — the aggregate threshold `X` and the **relatedness class** over which decisions are summed within a period `P`. Neither can be set without the other, and setting them badly does more damage than having no rule, because a cumulative test that trips constantly re-centralises the whole delegated band while appearing to be a control improvement.

WORKED EXAMPLE

Worked example 3.3.4c — sizing Meridian's cumulative test without re-centralising the delegation.

1. **Setup.** Meridian has raised its delegation threshold to **USD 25,000** (3.2.3), delegating **36** changes a year: the 24 at or below 10,000 averaging **5,000**, and the 12 in the 10,001–25,000 band averaging **17,000 — USD 324,000** of delegated value a year. Separately, one genuine cluster occurred: **7** related changes to the clinical template set, in sequence **8,400 · 6,200 · 11,500 · 9,800 · 4,600 · 13,200 · 7,300**, none individually above the threshold. The programme has **3** workstreams and **14** deliverable-level change classes. Period `P` = one quarter. Escalating one change costs **4.0 weeks** at 25 % critical-path share, i.e. an expected `0.25 × 57,120 =` **USD 14,280**.
2. **Formula.** Cluster aggregate = Σ of the related changes; the rule trips at the first change whose running total exceeds `X`. Base-rate aggregate per relatedness class per period = delegated value ÷ (classes × periods). Annual cost of the rule = expected trips × expected escalation cost.
3. **Substitution.** Running totals `8,400 → 14,600 → 26,100`. Broad relatedness (workstream): `324,000/(3 × 4)`. Narrow relatedness (deliverable class): `324,000/(14 × 4)`.
4. **Result.** The cluster totals **USD 61,000** and trips at the **third** change (running total **26,100** against `X` = 25,000), leaving **USD 34,900** of the cluster to be authorised at the aggregate's level.

Relatedness class	Classes × periods	Base-rate aggregate per class-period	`X` = 25,000 trips?	Trips a year	Annual cost
Workstream (broad)	3 × 4 = 12	**USD 27,000**	Yes, on the base rate alone	**12**	**USD 171,360**
Deliverable class (narrow)	14 × 4 = 56	**USD 5,785.71**	No	**1**	**USD 14,280**

The broad rule costs **USD 171,360** a year — **exactly** the saving the threshold increase produced — while the narrow rule costs **USD 14,280**, a factor of **12** less, and catches the same cluster. 5. **Interpretation.** The equality in the first row is not a coincidence and is worth naming, because it generalises. Raising the threshold delegated 12 changes and saved `12 × 0.25 × 57,120`; a cumulative rule that trips 12 times a year escalates 12 changes and costs `12 × 0.25 × 57,120`. **A cumulative test cancels a delegation exactly when its annual trip count equals the number of decisions the delegation released.** So the trip count, not the threshold, is the quantity to design against — and the trip count is driven overwhelmingly by the relatedness class, because that is what determines how much unrelated traffic is being summed.

The design rule that falls out is a **window**, and it is the transferable result: `X` must sit comfortably **above** the base-rate aggregate of its relatedness class per period and comfortably **below** the aggregates of the clusters it must catch. At deliverable-class relatedness, `X` = 25,000 sits **4.32 times** above the base rate of 5,785.71 and **2.44 times** below the 61,000 cluster — a working window. Broadening relatedness to the workstream multiplies the base rate by **4.67**, to 27,000, which closes the window from below: every workstream now trips every quarter on ordinary traffic, and the rule has become a threshold reduction wearing a control's clothing. **Widening the relatedness class is the error, and it is the natural error**, because a broad class feels safer and costs nothing to write.

Two further observations belong in the record. The rule catches the cluster **after three of seven changes** — the first 26,100 was already decided under delegated authority, and the rule cannot undo it. So the cumulative test is a **containment** control, not a prevention control, and it must be paired with the provision Case study B actually needed: every change generates a **register entry regardless of value**, which costs no latency at all and is what makes the aggregate visible before it trips. A cumulative test without universal registration is a rule that cannot see its own inputs. And relatedness should be defined by **what the changes touch** — the same deliverable, the same assured control, the same interface — not by who requested them or which budget they hit, because the exposure the rule exists to catch is a coherent change to one thing, arriving in instalments.

AI in this KA

Where it earns its place. Extracting a structured decision register from meeting minutes and flagging entries missing an owner, a date or a versioned information reference — a genuinely tedious task with a clear right answer. Running the RACI integrity checks above across a large matrix. Detecting re-decided decisions by clustering decision text across a long log, which is exactly the pattern humans miss because the recurrences are months apart. Testing gate criteria for assessability and flagging the unmeasurable ones. Modelling gate and escalation economics across alternative designs, as computed above.

Where it must not go. No gate decision, no assurance opinion, and no authorship of a decision record entry — the record must show the accountable human, and a record generated wholesale by a tool cannot evidence that a person applied judgement. Nor should a model's summary of a decision's history be relied on for a dispute or an audit without verification against the source documents: the summary is a convenience, the versioned source is the evidence.

Verification, concretely. Any AI-produced register or audit is checked against the source minutes on a sampled basis with the sample size stated; every flagged defect is confirmed by a human before it is reported as a finding; and the gate and latency arithmetic is reproduced by hand, because all of it is a handful of operations and none of it should ever be taken on trust.

KEY TERMS — KA 3.3

Term	Meaning
Stage gate	A continuation decision by a named authority against pre-set criteria, with power to stop, hold or redirect.
Conditional pass	Proceeding subject to conditions — a real instrument only if the conditions have owners, dates and consequences.
Three lines of assurance	Management controls; independent oversight inside management; independent audit outside it.
Assurance map	Risks and controls against assurance lines, marking coverage, gaps and duplication.
Assurance capture	An assurance function assuring work it helped produce, and therefore unable to challenge it.
Escalation class	A defined trigger, destination, decision, latency and out-of-cycle route.
Out-of-cycle mechanism	A written-resolution or delegated-authority route used when the ordinary cadence is too slow.
Escalation lead time	How far before impact an escalation arrives; a shortening trend signals a decaying mechanism.
Decision record	The versioned, attributable log that converts a decision from a memory into an institutional fact.
Single-A test	The check that each decision class has exactly one Accountable role.
Effective detection (`q_eff`)	A gate's detection rate multiplied by its condition-closure rate: `q_eff = q × r`.
Condition-closure rate (`r`)	The share of a gate's conditions verified closed by the following gate.
Residual exposure	$\Sigma\ p_i\ \prod(1 - q_{ij})\ u_i$ across an assurance map; the only quantity that ranks its rows.
Decision action window	`(remaining duration − escalation latency) ÷ remaining duration`; negative means the escalation is ceremonial.
Confirmation rule	The provision that stops a tolerance trigger firing on one period's measurement noise.
Re-decision tax	The committee slots and delay cost consumed by questions that had already been decided.
Cumulative test	The delegation-schedule provision that related decisions aggregating above `X` within period `P` require the authority appropriate to the aggregate. Setting `X` and `P` is arithmetic, not judgement, and is derived in Domain 4, KA 4.3.3 — a round number chosen without reference to the observed decision rate provides the appearance of a control and none of the function.
Relatedness class	The set over which a cumulative test sums; widening it multiplies false trips and re-centralises the delegation.

SAMPLE MCQS — KA 3.3

MCQ 3.3-A `[3.3.1 · Application]` A gate costs 45,000 in review effort and 6 weeks of elapsed time at a delay cost of 14,280 per week. Expected remediation cost with the gate is 82,800; without the gate it is 270,000. The gate's net value is: - A. USD 141,480 - B. USD 56,520 - C. USD 85,680 - D. USD 187,200

Rationale: `270,000 − (45,000 + 85,680 + 82,800) = 56,520` (3.3.1). A omits the delay cost; C is the delay cost alone; D omits the remediation cost.

MCQ 3.3-B `[3.3.1 · Evaluation]` For the gate above, the elapsed time at which it stops adding value is closest to: - A. 6 weeks - B. 8 weeks - C. 10 weeks - D. 19 weeks

Rationale: `(270,000 − 45,000 − 82,800)/14,280 = 9.96` weeks (3.3.1) — the arithmetic behind the complaint that assurance has become an obstacle.

MCQ 3.3-C `[3.3.3 · Application]` Three tiers must approve a decision: 2-weekly with 1-week papers, 4-weekly with 2-week papers, and 13-weekly with 3-week papers. Total expected latency is: - A. 9.5 weeks - B. 15.5 weeks - C. 19.0 weeks - D. 6.0 weeks

Rationale: `(2/2+1) + (4/2+2) + (13/2+3) = 2.0 + 4.0 + 9.5 = 15.5` (3.3.3). A is the top tier alone; D counts only half the intervals and omits the paper lead times.

MCQ 3.3-D `[3.3.2 · Analysis]` A PMO drafts the delivery plan and later provides second-line assurance on it. The defect is: - A. duplication of first-line controls - B. assurance capture — the function cannot challenge what it produced - C. a proportionality failure - D. an assurance gap

Rationale: Capture is the most damaging and least visible line failure, because the product still looks independent (3.3.2).

MCQ 3.3-E `[3.3.4 · Comprehension]` The decision-record field most often missing and most consequential when a decision is later examined is: - A. the date - B. the versioned reference to the information relied on - C. the decision-maker's role - D. the decision reference number

Rationale: The retrospective question is whether the decision was reasonable on what was known at the time, which only a versioned reference can answer (3.3.4).

MCQ 3.3-F `[3.3.4 · Analysis]` In a 12-class decision-rights matrix, 2 classes carry two Accountable roles and 1 carries none. Which statement is most accurate? - A. the defect rate is 8.3 % and only the zero-A class matters - B. the defect rate is 25.0 %; the two-A classes will be decided late or twice, and the zero-A class will drift until it becomes an escalation - C. the matrix is acceptable because 9 of 12 classes are correct - D. the defect rate is 25.0 % and all three classes fail identically

Rationale: `3/12 = 25.0 %`, and the two failure types behave differently under stress, which is why they are distinguished rather than totalled (3.3.4).

MCQ 3.3-G `[3.3.1 · Evaluation]` A gate with a nominal detection rate of 0.80 has a net value of USD 56,520 and a breakeven detection probability of 55.85 %. Of the 23 conditions it issued, 9 were verified closed. The best evaluation of the gate is that it: - A. remains worthwhile, since its detection rate is well above the breakeven - B. has an effective detection rate of 31.30 %, is therefore below breakeven, and destroys USD 57,427.83 - C. is worthwhile but should be shortened from 6 weeks to 4 - D. has an effective detection rate of 39.13 % and is therefore below breakeven

Rationale: `q_eff = q × r = 0.80 × 9/23 = 0.3130`, which is below the 0.5585 breakeven, inverting the gate's value (3.3.1b). A ignores closure altogether — the commonest error, because governance reports count conditions issued, not closed. C treats elapsed time as the binding constraint when the binding constraint is `r`. D mistakes the closure rate itself for the effective detection rate, omitting the multiplication by `q` — it reaches the right conclusion from the wrong quantity, and will not survive a gate where `q × r` and `r` fall on opposite sides of the breakeven.

MCQ 3.3-H `[3.3.2 · Evaluation]` An assurance map shows all three lines on a risk carrying 7.87 % of residual exposure and no line on the risk carrying 58.27 % of it. The soundest conclusion is that: - A. the third line should be discontinued, since its marginal detection value is negative - B. the third line's effort is pointed at the wrong risk; reallocating it reduces residual exposure at constant cost, but third-line existence is not justified on project-level detection economics - C. the coverage is acceptable because the highest-consequence risk is the one with three lines - D. the uncovered risk should be accepted, since no line has the competence to assure it

Rationale: Reallocating the third line's 30 days from R1 to R3 cut residual exposure by USD 74,361.60 for no extra spend, but third-line assurance exists to opine on the control system across a portfolio and is mispriced by a single project's detection arithmetic (3.3.2). A over-applies the arithmetic — the named error. C confuses consequence with residual exposure, which is consequence after detection. D treats a coverage gap as a capability fact.

MCQ 3.3-I `[3.3.3 · Application]` A 25-week project reaches week 13 and a trigger fires. The available escalation path has a total expected latency of 15.5 weeks. The decision action window is: - A. 66.67 % - B. 0 % - C. −29.17 %, so the decision arrives 3.5 weeks after planned completion - D. 129.17 %

Rationale: $(12 - 15.5)/12 = -0.2917$ (3.3.3b). A is the window for the 4.0-week single-tier path. B assumes the window floors at zero, which hides the finding. D computes latency as a share of remaining duration (15.5/12) rather than the window itself — a real ratio, but not this one.

MCQ 3.3-J `[3.3.4 · Analysis]` A programme delegates 12 changes a year by raising its threshold, saving USD 171,360 in delay. It then adds a cumulative test at the same threshold, summed over workstreams, which trips 12 times a year. The consequence is that: - A. the control improves at negligible cost - B. the trip count equals the number of decisions the delegation released, so the rule cancels the saving exactly - C. the saving falls by about a quarter - D. the rule is sound but the threshold should be lowered again

Rationale: $12 \times 0.25 \times 57{,}120 = 171{,}360$, identical to the saving (3.3.4c). The trip count, driven by the width of the relatedness class, is the quantity to design against. A is the assumption a governance review makes when it does not count trips; C invents a partial figure; D compounds the error.

SELF-CHECK — KA 3.3

1. What does a gate buy, and what does that imply about where gates belong? — Optionality against irreversibility; so gates belong where irreversibility steps up, not at calendar intervals.
2. Name the assurance failure mode that is hardest to see, and why. — Capture: the assurance product still looks independent while the function cannot challenge what it helped produce.
3. Which element of an escalation class is most often missing? — The stated time within which the decision will be made, and with it the out-of-cycle mechanism.
4. State the identity that connects conditional passes to gate value. — $q_{eff} = q \times r$: an untracked condition leaves the defect in the work, so closure rate multiplies detection rate (3.3.1b).
5. Which row of an assurance map should receive the next assurance day? — The one with the largest $p \prod(1 - q)\ u$, which is almost always an uncovered risk rather than a duplicated one (3.3.2).
6. Why must an escalation trigger name a forecast method? — Because the earned-value family produces several legitimate estimates from the same data, and they can straddle the tolerance — Auriga's three sit at 5.00 %, 10.42 % and 15.20 % of BAC against a 10 % trigger (3.3.3b).
7. What are the two parameters of a cumulative test, and which one usually goes wrong? — The aggregate threshold X and the relatedness class; widening the class is the natural error and it re-centralises the delegated band (3.3.4c).

Advanced topics — Domain 3

3.A.1 Governance under stress, and the recovery structure

Governance designed for steady state is tested in crisis, and the failures are consistent: the cadence is too slow for the decision rate, authority fragments as senior figures intervene individually, information becomes contested at the moment it must be relied on, and the decision record — never robust — collapses first, so that a fortnight later nobody can reconstruct why a choice was made.

A **recovery governance** design is a legitimate and different structure, and it should be prepared in advance rather than improvised: a single accountable decision-maker with materially raised authority; a short daily or twice-weekly decision cadence with a standing agenda of decisions rather than reports; a narrowed membership, since crisis governance fails from too many participants before it fails from too few; a single authoritative information source, with disputes about the data resolved before decisions rather than during them; and a decision log kept in the room, in real time, because reconstruction afterwards is precisely what will not be possible.

Two invariants are worth stating because they are what distinguishes recovery governance from an abandonment of governance. Raised authority must remain **bounded and time-limited**, with a stated review date, or it does not revert. And the escalation of bad news must be made safer, not harder, because the mechanism's failure mode under stress is silence — and silence is the one condition under which no governance design of any shape can work.

3.A.2 Governance of AI-assisted delivery

Where a project uses AI in its own delivery — estimating, scheduling, code generation, risk analysis, document review — governance acquires obligations that most existing structures do not express, and the honest position is that they must be added deliberately.

Four are minimal. **A register of AI uses**, stating for each what the tool does, whose decision it informs, and who is accountable for the output — the same decidability test as 3.1.1, applied to tools. **A verification standard proportional to consequence**: an AI-drafted internal summary and an AI-produced estimate that will set a baseline are not the same object and must not carry the same review. **A prohibition on unattributable authorship** in anything that will be relied on — a decision record, an assurance opinion, a board paper conclusion — because Domain 1's accountability principle requires a person to answer for it, and a tool cannot be that person. And **a data and confidentiality boundary** stated before use, not after an incident.

The governance body's own duty here is the one most often skipped: to ask, of any AI-informed recommendation it receives, what was the model's input, who verified its output, and what would change the conclusion? A body that cannot get those three answers is not governing the recommendation; it is ratifying it.

3.A.3 The reviewer's governance eye

Invariants to test on any governance design, each cheap and each diagnostic:

Every decision class has **exactly one** accountable role, and that role's authority is sufficient under the delegation schedule. The delegation schedule reads on **value, reversibility and externality**, not value alone. The **latency of every escalation path is stated**, computed as `Σ (M/2 + L)`, and each tier can be justified against its cost. Every decision body has a **capacity number** and a demand estimate. Committee agendas show the **share of time on decisions** rather than reports. Every gate has criteria set in advance, the authority to stop, and a **computed value** against its elapsed time and detection rate. Conditional passes have owners, dates and consequences, verified at the following gate. The assurance map shows **no gaps and no duplication**. Every escalation class has a stated decision deadline and an out-of-cycle route. The decision log's entries carry a named decision-maker and a **versioned information reference**. And — the one test that subsumes many of the others — the **re-decision count** is tracked, because a question arriving at the same body a third time is proof that something upstream of it is broken.

Six further invariants, each answerable from documents that already exist:

Every decision body's utilisation is stated at **average, peak and off-peak**, and the peak is computed from the actual clustering of change and gate decisions rather than assumed uniform (3.2.2). Every escalation path carries a **decision action window** as well as a latency, tested against the remaining duration of the thing being decided; a negative window is a finding, not a nuance (3.3.3b). Every tolerance trigger names its **forecast method, measurement point and confirmation rule**, and a trigger stated on a single ratio at a single data date is treated as unspecified (3.3.3b). Every gate's **condition-closure rate** is counted and multiplied through its detection rate, because a gate reported as held is not a gate that worked (3.3.1b). The assurance map carries **residual exposure per row**, so that reallocation is arguable at constant cost rather than being a matter of taste (3.3.2). And any **cumulative test** is presented with its expected annual trip count and the base-rate aggregate of its relatedness class, since a rule that trips on ordinary traffic has reversed the delegation it was added to protect (3.3.4c).

Where the structure is multi-party, two more: the board's **first-presentation approval rate** is counted from its own minutes, because that is the p on which unanimity's cost depends; and the **reserved-matters list** is read against it, since every item on that list is priced at p^n (3.1.2).

Industry variations — Domain 3

- **Public sector and government.** Statutory decision rights, published assurance regimes and mandatory gates that are genuinely non-negotiable; latency is high by design and the leader's lever is almost entirely the paper lead time and the out-of-cycle route, not the cadence. Where a statutory path's latency exceeds the remaining duration of what it decides, the **action window is negative** and the honest report says so — because the alternative narrative, that the project was slow, is the one that will otherwise be recorded (3.3.3b).
- **Regulated industries (pharmaceutical, nuclear, aviation, financial services).** Some gates are external and unmovable, and the governance design must place the internal decision before the external submission with enough margin that a failed internal test does not force a submission. That margin is computable rather than a matter of comfort: it is the internal escalation path's action window against the time remaining to the submission date, and a negative window means the internal test cannot change the submission it was built to protect.
- **Construction and infrastructure.** Multi-party governance is the norm; decision rights follow the contract structure, so a governance design inconsistent with the contract loses to the contract every time (PFL-AI Domain 11). The **decision rule in the joint-venture agreement is usually the largest single latency term** in the whole structure: at four parties each 85 % ready, unanimity costs **10.4941 weeks** per decision against **5.7379** under a three-of-four majority (Exercise 3.5), and no schedule shows the difference.
- **Technology and product organisations.** Bounded envelopes and continuous funding decisions dominate; the characteristic risk is light governance on genuinely irreversible choices — architecture, data model, third-party dependency — because they do not present as gates. The

reversal-cost ratio is the instrument that catches them, since it reads authority on `max(value, cost to undo)` and those decisions are cheap to take and expensive to unmake (3.2.3).

- **Healthcare.** Clinical governance runs in parallel with project governance and clinical authority is not delegable to a project body; Meridian's design must place clinical sign-off with clinical authority and integrate rather than absorb it. On the assurance map, clinical sign-off is a **line in its own right with its own detection rate**, not a component of the project's first line — and where the project helped produce what clinicians are asked to sign, the cell is entered as `q = 0` (3.3.2).
- **Energy and resources.** Stage-gated capital processes with substantial assurance at each gate, and the standard pathology is gate proliferation — re-approval ceremonies whose elapsed time exceeds 3.3.1's breakeven. The second pathology, less discussed and equally expensive, is the **unclosed condition**: it reduces a gate's effective detection to `q × r` while reducing its elapsed time by nothing at all, so a stage-gated estate that does not count condition closure is paying full price for partial gates (3.3.1b).

Case study — Domain 3: the four-week month (health, Meridian)

Situation. Twelve weeks into the clinic rollout, Meridian's steering committee was widely described as "the bottleneck". The committee met monthly, was well attended, and had never refused a request. The programme was six weeks behind, and the delay was attributed in the quarterly report to "slower than expected clinic readiness".

What the arithmetic showed. The project leader computed three numbers before the next governance review. Governance latency: `M/2 + L = 4/2 + 2 =` **4 weeks** — the committee that everyone called monthly imposed a full month's expected wait, and no one in the organisation had believed it was more than a fortnight. Escalation volume: with a **10,000** delegation threshold, **36** of the year's **60** change requests required the committee, of which about a quarter sat on the critical path, costing `9 × 4 × 14,280 =` **USD 514,080** a year in delay. Committee capacity: **104** item-slots against **77** items of demand, of which **26** were standing reports producing no decision — so a quarter of the scarcest resource in the programme was being spent on status.

What changed. Three changes, none of which removed any scrutiny anyone could name. The paper lead time went from 2 weeks to 1, taking expected latency to **3.0 weeks** — a full week saved from a change to an administrative deadline. The delegation threshold went from 10,000 to **25,000**, worth **USD 171,360** a year against a worst case of 81,600 even if every delegated decision were decided wrongly and destroyed 40 % of its value. And the standing reports moved to pre-reading, releasing 26 slots and taking utilisation from 74.0 % to **49.0 %**, which is what created room for the contentious item to be reached.

The two findings that were not in the original three. The committee's own members had said the body was full, and the 74.0 % utilisation figure said it was not. Two further counts reconciled them. First, demand was not uniform: **40 %** of the 51 decision items fell in the **3** meetings straddling phase boundaries, giving `51 × 0.40/3 + 2 =` **8.80** items against a capacity of 8 — a peak utilisation of **110.00 %** against **63.25 %** between the boundaries. Second, a clustering of the decision log's **148** entries found **19** questions that had arrived at the same body more than once, **6** of them three or more times: **25** repeat slots, **24.04 %** of the committee's annual capacity, which took real utilisation to **98.08 %** and cost **USD 357,000** a year in delay on the same 25 % critical-path assumption. **68.42 %** of those repeats carried no versioned reference to the information relied on. A fourth change followed, and it was the cheapest of the four: two mandatory fields on the decision-record template.

The outcome, and the part that mattered. Escalated decisions began arriving in about three weeks rather than four, and the peak-month deferrals stopped. But the durable change was to the conversation: "the committee is a bottleneck" is an accusation, while "our governance design costs 514,080 a year in delay, plus 357,000 a year deciding things twice, and here are four changes worth most of it" is a proposal. The second was approved in one meeting. The first had been raised, in various forms, for two quarters. Note the ranking the arithmetic produced, which was not the ranking anyone expected: the largest single item was not the threshold, the cadence or the agenda — it was the **re-decision tax**, at **2.08 times** the entire saving from the threshold change, and it was the only one of the four that required no risk-appetite decision from anybody.

What the domain teaches here. Governance is a delivery variable with a computable price. Until the price is computed, governance complaints are cultural and go nowhere; once computed, they are engineering, and engineering gets approved. And note which lever was largest per unit of organisational pain: the paper deadline — free, unglamorous, and twice as effective per week as meeting more often.

Case study B — Domain 3: the decision nobody made (financial services)

Situation. A payments-platform programme completed build 5 % over budget with no approved change above 50,000 — a clean change-control record. A routine internal audit then found that the delivered scope differed materially from the approved baseline in four respects, none of which appeared in any decision record.

What had happened. Each of the four differences had been assembled from a handful of individual changes — **20** in all, five per respect, averaging **35,000** and so **175,000** per respect and **700,000** in total. Each had been discussed in a working group, agreed in substance, and implemented. None had been raised as a change, because each had been individually below the project leader's **50,000** authority and had therefore, in the team's understanding, required no decision at all. One of the four altered a control the second line had assured on the basis of the original design. The programme's governance was not weak in the ordinary sense: the committee functioned, the gates were held, the delegation schedule existed. The defect was narrower and more instructive — **the schedule read on individual value only**, and the decision log recorded decisions taken by bodies rather than decisions taken.

How it resolved. Three corrections, each mapping to a specific defect. The delegation schedule acquired **reversibility and externality** dimensions, so a change touching an assured control required second-line agreement regardless of value. It acquired a **cumulative test**: related changes aggregating above a threshold within a period required the authority appropriate to the aggregate, which is the provision that would have caught all four. And the decision log was made the **register of record for every change to the baseline**, whoever decided it and at whatever value, so that a decision below a threshold still generated an entry. The audit finding was closed; the platform went live four weeks late.

The parameter the first draft of the fix got wrong. The cumulative rule was initially written at `X` = **50,000**, the same value as the individual threshold, and summed **over each of the programme's four workstreams** each quarter — which sounded conservative and was the opposite. The programme logged **84** sub-threshold changes a year averaging **18,000**, so **USD 1,512,000** of delegated value, and a workstream-quarter therefore carried a base-rate aggregate of `1,512,000/16 =` **USD 94,500** — nearly twice the trip threshold on ordinary unrelated traffic alone. The rule would have tripped in **all 16** workstream-quarters, re-escalating routine change and leaving the delegation nominal. Redrawing the relatedness class at **deliverable level** — **22** classes rather than 4 — dropped the base-rate aggregate to **USD 17,181.82** per class-quarter, put `X` **2.91 times** above it and **3.5 times** below each respect's **175,000** aggregate, and cut the expected trip count from 16 a year to about **4** — one per respect, a **four-fold** reduction that caught every cluster the rule existed to catch. On that sizing each respect trips at its **second** change (running total 70,000 against 50,000), leaving **105,000** of it to be authorised at the aggregate's level. The lesson the audit report recorded was that **the relatedness class, not the threshold, is the parameter that decides whether a cumulative test is a control or a re-centralisation.**

What the domain teaches here. A decision below a threshold is still a decision and still needs a record. A delegation schedule that reads only on value cannot see irreversibility, externality or accumulation — the three ways a small decision becomes a large one — and a decision log that records only what committees decided will systematically miss the changes that matter most, because those are exactly the ones that never reached a committee. And a control added to catch accumulation must be sized against the base rate of the class it sums over, or it will trip on the ordinary traffic it was never aimed at.

Executive perspective — Domain 3

What a programme director cannot delegate in this domain:

- **The latency of your own governance.** Compute `Σ (M/2 + L)` for every escalation path you own, price it at your cost of delay, and be able to state it. An unquantified governance design is one you are not managing (3.2.3, 3.3.3).
- **The delegation schedule, on three dimensions.** Value, reversibility, externality — and a cumulative test. Read on value alone, it will miss exactly the decisions that matter (Case study B).
- **The sponsor's obligations, in writing.** Not the title. The seven testable obligations, agreed and evidenced, with turnaround on sponsor decisions monitored like any other lead time (3.2.1).
- **Whether your gates function.** A gate that has never held anything is not working, and its elapsed time may already exceed the point at which it destroys value (3.3.1).
- **The decision record's integrity.** Named decision-maker, versioned information reference, conditions with owners. This is what makes a defensible decision defensible a year later (3.3.4).
- **Escalation lead time as a leading indicator.** Track how far before impact escalations arrive. A shortening trend is the earliest signal that people no longer believe escalation is safe — and it appears in no standard report (3.3.3).
- **The action window, not just the latency.** For every escalation path you own, compare its latency with the remaining duration of what it decides. Auriga's 15.5-week path on a 25-week project is actionable for the first **38.00 %** of it and ceremonial thereafter, and no currency figure shows that (3.3.3b).
- **Sponsor allocation as a portfolio decision.** A Meridian-sized sponsorship costs **111.0 hours** a year, about **6.03 %** of a working week, so no executive carries more than about **three**. Naming the same four people across a dozen programmes manufactures the absent sponsor you will later diagnose as a behaviour (3.2.1).
- **What your gates' conditions did next.** Count condition closure, not conditions issued. `q_eff = q × r` turned a gate worth **+56,520** into one destroying **57,428** on nothing but a tracking failure, and the swing appears in no governance report (3.3.1b).

- **Where your assurance is pointed.** Require residual exposure per row on the assurance map. All three of Meridian's lines sat on **7.87 %** of it while **58.27 %** sat uncovered; the correction cost nothing (3.3.2).
- **The cost of deciding things twice.** Track unplanned re-decisions. Meridian's ran at **12.84 %** of logged decisions, **24.04 %** of committee capacity and **USD 357,000** a year — more than twice the saving from its threshold change, and remediable with two fields on a template (3.3.4b).

Calculation exercises — Domain 3

Exercise 3.1 A committee meets every 8 weeks and closes papers 2 weeks before each meeting; cost of delay is 14,280 per week. Compute the expected latency and its cost, then compute the saving from (a) halving the meeting interval to 4 weeks and (b) halving the paper lead time to 1 week. Solution. `E[wait] = 8/2 + 2 =` **6.0 weeks**, costing `6 × 14,280 =` **USD 85,680**. (a) `4/2 + 2 =` **4.0 weeks** — saves 2.0 weeks, **USD 28,560**. (b) `8/2 + 1 =` **5.0 weeks** — saves 1.0 week, **USD 14,280**. Note the general rule the two results illustrate: a cut of `x` in `L` always saves `x`; a cut of `x` in `M` saves `x/2`. Halving looks like the bigger intervention here only because `M` is four times `L`. Common error: assuming the expected wait is half the meeting interval, which omits the paper lead time entirely.

Exercise 3.2 An organisation logs 80 change requests a year: **≤ 5,000 → 30**; **5,001–20,000 → 20** (average value 11,000); **20,001–75,000 → 22**; **> 75,000 → 8**. The committee meets every 4 weeks with a 1-week paper lead time; cost of delay is 9,500 per week; 30 % of escalated changes sit on the critical path. Compute the annual governance delay cost at a 5,000 threshold and at a 20,000 threshold, and the breakeven value-destruction rate on the delegated band. Solution. `E[wait] = 4/2 + 1 =` **3.0 weeks**, so each delaying change costs `3 × 9,500 =` **28,500**. At 5,000: escalated `80 − 30 = 50`, delaying `50 × 0.30 = 15`, cost `15 × 28,500 =` **USD 427,500**. At 20,000: escalated `50 − 20 = 30`, delaying **9**, cost **USD 256,500**. Saving **USD 171,000**. The delegated band is worth `20 × 11,000 =` **220,000**, so the escalation breaks even only if the delegate destroys `171,000/220,000 =` **77.7 %** of the value of every delegated decision. Common error: comparing the saving with the total value of the delegated band rather than with a plausible loss on it, which understates the case for delegating by an order of magnitude.

Exercise 3.3 A gate costs 30,000 in review effort and 4 weeks of elapsed time at a delay cost of 9,500 per week. A material defect is present with probability 0.25; the gate detects it with probability 0.75; correction at design costs 80,000 and in build 600,000. Compute the gate's net value, its breakeven elapsed time and its breakeven detection probability. Solution. Without the gate: `0.25 × 600,000 =` **150,000**. With: review 30,000 + delay `4 × 9,500 =` 38,000 + expected remediation `0.25 × (0.75 × 80,000 + 0.25 × 600,000) = 0.25 × 210,000 =` 52,500, total **120,500**. Net value **USD 29,500**. Breakeven elapsed time `(150,000 − 30,000 − 52,500)/9,500 =` **7.105 weeks**. Breakeven detection probability: solve `30,000 + 38,000 + 0.25 × (600,000 − 520,000d) = 150,000` → `d =` **52.31 %**. Common error: omitting the elapsed-time cost, which makes every gate look worthwhile and is the reason gates proliferate.

Exercise 3.4 A decision must pass a delivery board (every 2 weeks, 1-week papers), a portfolio board (every 6 weeks, 2-week papers) and an investment committee (every 12 weeks, 4-week papers). Cost of delay is 9,500 per week. Compute the total expected latency and cost, and the saving from a single-tier path at the portfolio board with a written-resolution procedure of one week. Solution. `2/2 + 1 =` **2.0**; `6/2 + 2 =` **5.0**; `12/2 + 4 =` **10.0**. Total **17.0 weeks**, costing **USD 161,500** — of which the investment committee alone is 10.0 weeks and **95,000**. A one-week written-resolution path costs **USD 9,500**, a saving of **USD 152,000** (**94.1 %**). For comparison, retaining the portfolio board on its ordinary cadence as the sole tier gives 5.0 weeks and **47,500**, saving 114,000. Common error: adding only the meeting intervals and ignoring the paper lead times, which understates latency here by 7 of the 17 weeks.

Exercise 3.5 A four-party consortium board meets every 6 weeks with a 2-week paper lead time. Each party independently arrives able to approve in a given cycle with probability 0.85. Cost of delay is 9,500 per week. Compute the expected wait and cost for one decision under (a) unanimity and (b) a three-of-four majority, and compare both with a single integrating authority. Solution. Base wait `M/2 + L = 6/2 + 2 =` **5.0 weeks**, the single-authority case, costing **USD 47,500**. (a) Unanimity passes with 0.85^4 `=` **0.5220**, so `E[cycles] = 1/0.5220 =` **1.9157** and `E[wait] = 5.0 + 0.9157 × 6 =` **10.4941 weeks**, costing **USD 99,694.09**. (b) Three-of-four passes with $4 \times 0.85^3 \times 0.15 + 0.85^4 = 0.3685 + 0.5220 =$ **0.8905**, so `E[cycles] =` **1.1230** and `E[wait] = 5.0 + 0.1230 × 6 =` **5.7379 weeks**, costing **USD 54,510.33**. Unanimity costs **4.7562 weeks** and **USD 45,183.76** more per decision than the majority rule. Common error: applying the readiness probability once instead of raising it to the power of the party count — that gives `1/0.85 = 1.1765` cycles and 6.06 weeks, understating the unanimity penalty by more than four weeks, and it is the error that makes multi-party boards look survivable on paper.

Exercise 3.6 The gate of Exercise 3.3 (review 30,000; 4 weeks elapsed at 9,500 per week; `P(defect)` 0.25; detection 0.75; design fix 80,000; build fix 600,000; net value 29,500; breakeven detection 52.31 %) issued 18 conditions on a conditional pass, of which 7 were verified closed by the following gate. Compute the effective detection rate, the gate's net value at that closure rate, and the closure rate the gate needs to break even. Solution. `r = 7/18 =` **38.89 %**, so `q_eff = 0.75 × 0.3889 =` **29.17 %**. Expected remediation `0.25 × (0.2917 × 80,000 + 0.7083 × 600,000) =` **112,083.33**, so the total with the gate is `30,000 + 38,000 + 112,083.33 =` **180,083.33** against 150,000 without it: the gate now **destroys USD 30,083.33**, a swing of **USD 59,583.33** from its nominal +29,500. Breakeven closure rate `r* = d*/q = 0.5231/0.75 =` **69.74 %**, i.e. **13** of the 18 conditions — at 12 of 18 the gate is still 3,000 underwater, at 13 it returns **USD 2,416.67**. Common error: treating the closure rate as the effective detection rate and reporting 38.89 %, which omits the multiplication by `q` and makes the gate look closer to breakeven than it is. Note that `r*` here (69.74 %) is close to Meridian's 69.81 % by coincidence, not by rule: `r* = d*/q` and both terms differ between the two gates.

Exercise 3.7 Four material risks are assured by two lines. R1: `p` 0.25, `u` 600,000, first line 0.60, second line 0.50. R2: `p` 0.30, `u` 450,000, first line 0.50 only. R3: `p` 0.40, `u` 250,000, no line. R4: `p` 0.20, `u` 900,000, first line 0.40 only. The second line's contribution is 25 days at USD 800. Compute total residual exposure, then the reduction available from moving the second line's 25 days off R1 to (a) R4 and (b) R3, at the same detection rate of 0.50 and the same cost. Solution. Residual `= Σ p ∏(1 − q) u`: R1 `0.25 × 0.20 × 600,000 =` **30,000**; R2 `0.30 × 0.50 × 450,000 =` **67,500**; R3 `0.40 × 1.00 × 250,000 =` **100,000**; R4 `0.20 × 0.60 × 900,000 =` **108,000**; total **USD 305,500**. (a) To R4: R1 rises to 60,000, R4 falls to `0.20 × 0.30 × 900,000 =` **54,000**; total **USD 281,500**, a reduction of **USD 24,000**. (b) To R3: R3 falls to **50,000**, R4 stays at 108,000; total **USD 285,500**, a reduction of **USD 20,000**. Both reallocations cost nothing extra, and **(a) is better than (b) by USD 4,000 — 16.67 % of the available gain**. Common error: reallocating to the uncovered risk by reflex. The rule is the largest **residual exposure**, not the largest gap: R4 at 108,000 outranks the wholly

> uncovered R3 at 100,000, because a high consequence with one weak detector can leave more exposure than a low consequence with none. (Marginal values: 30,000 on R1, 54,000 on R4, 50,000 on R3, against the line's 20,000 cost.)

> **Exercise 3.8** A 40-week project has BAC 6,000,000. At week 22: PV 3,300,000, EV 3,000,000, AC 3,300,000. The governance framework escalates where "the forecast overrun exceeds 10 % of BAC". The escalation path runs through three tiers: 3-weekly with 1-week papers, 5-weekly with 2-week papers, and 13-weekly with 4-week papers. Compute the three standard estimates at completion and test each against the tolerance; then compute the escalation path's total latency, the decision action window and the last week at which the path can still deliver an actionable decision. Solution. `CPI = 3,000,000/3,300,000 =` **0.9091**; `SPI = 3,000,000/3,300,000 =` **0.9091**. `EAC = AC + (BAC − EV) =` **6,300,000**, VAC 300,000 = **5.00 %** → no trigger. `EAC = BAC/CPI =` **6,600,000**, VAC 600,000 = **10.00 %** → exactly on the tolerance of 600,000. `EAC = AC + (BAC − EV)/(CPI × SPI) =` **6,930,000**, VAC 930,000 = **15.50 %** → trigger. Latency `(3/2 + 1) + (5/2 + 2) + (13/2 + 4) = 2.5 + 4.5 + 10.5 =` **17.5 weeks** against a remaining duration of `40 − 22 =` **18 weeks**, so the action window is `(18 − 17.5)/18 =` **2.78 %** — half a week — and the last week at which the path is actionable at all is `40 − 17.5 =` **week 22.5**, or **56.25 %** of the duration. Common errors: two of them. First, reporting "the forecast" without naming a method, when the three methods here straddle the tolerance and one sits exactly on it — a framework with no named method and no tie-break rule has not specified its trigger. Second, stopping at the latency and calling 17.5 weeks "slow"; it is not slow, it is **0.5 weeks from being impossible**, and only the action window shows that.

Practitioner's toolkit — Domain 3

Adoption-ready artefacts; adapt headings to your organisation, then keep them stable.

Toolkit 3.T.1 — Governance design sheet

One page, completed before mobilisation and reviewed at every gate. Rows: each decision body, with its purpose, members (distinguished from attendees), decision classes it owns, meeting interval `M`, paper lead time `L`, computed `E[wait]`, capacity (meetings × items) and estimated demand. Below it, each escalation path with its tiers, the summed latency `Σ (M/2 + L)`, that latency priced at the project's cost of delay, the **decision action window** against the remaining duration of what the path decides, and the out-of-cycle mechanism. Two columns that are usually omitted and are worth more than most of the others: **peak** utilisation as well as average, computed from the actual clustering of change and gate decisions; and, for a multi-party body, the **first-presentation approval rate** and the decision rule (unanimity / defined majority / single integrating authority), since p^n is what prices the constitution. The sheet's purpose is to make latency and capacity **visible at design time**; a governance design that cannot fill it in has not been designed.

Toolkit 3.T.2 — Delegation schedule with a cumulative test

Columns: decision class · value threshold · **estimated cost to undo** and the reversal-cost ratio `ρ`, with authority read on `max(value, cost to undo)` · externality rating (internal / cross-functional / external or public), escalating one authority level per step · authority required · **cumulative rule** (related decisions aggregating above `X` within period `P` require the authority appropriate to the aggregate) · **relatedness class** for that rule, with its base-rate aggregate per period and the expected annual trip count · information the delegate must hold to decide. The three dimensions prevent the value-only failure of Case study B; the cumulative rule prevents its specific mechanism; the base-rate and trip-count columns are what stop the rule re-centralising the band it was added to protect (Worked example 3.3.4c); and the final column is what distinguishes delegation from abdication. Sizing test to apply before the rule goes live: `X` should sit comfortably above the relatedness class's base-rate aggregate per period and comfortably below the aggregates of the clusters it must catch — if it cannot, narrow the class rather than raising `X`.

Toolkit 3.T.3 — Decision record entry and its integrity checks

Per-decision fields: reference · date · decision-maker (**named person and role**) · decision (one reading only) · options considered and why this one · information relied on **with versions** · conditions with owners and dates · review date if provisional. Monthly integrity checks, each a count: entries missing a named decision-maker; entries missing a versioned reference; conditions past their date; decision classes with zero or multiple Accountable roles in the RACI (the single-A test); and **re-decisions** — the same question arriving at the same body more than twice, counted excluding scheduled reviews of provisional decisions and re-openings forced by a material change, so that the number measures the defect and not governance working properly. Every one of these is a number, and a governance function that reports them monthly will find its defects before an auditor does. Two derived figures belong on the same monthly page because they convert the counts into an argument: the **re-decision tax** (repeat slots × critical-path share × `E[wait]` × cost of delay) and the repeat slots as a **share of committee capacity**, which is what corrects a capacity model that would otherwise read low by the re-decision rate.

Toolkit 3.T.4 — Gate and conditional-pass checklist

Two halves, and the second is the one that is usually missing. **Before the gate:** criteria written and issued in advance, each objectively assessable, covering the deliverable, the business case re-tested (not restated), the risks with named owners, the receiving organisation's readiness and the next stage's plan · the named authority, with the power to stop, hold or redirect stated in writing · the gate's elapsed time and review cost estimated · and the gate's **computed value** against its breakeven elapsed time and breakeven detection probability, so that the decision to hold the gate is itself an evidenced decision.

After the gate, per condition on any conditional pass: the condition in words that permit only one reading (a condition that says "confirm" or "find out" is a hold, and should be recorded as one) · the named owner · the date · the consequence of non-completion · the verification method · and, at the following gate, the **closure state**. The three monthly integers this yields are conditions issued, conditions verified closed and conditions past their date, from which the **closure rate** `r` and the gate's **effective detection rate** `q_eff = q × r` follow — computed per gate and per condition type, never as one programme average, because an aggregate closure rate hides the one gate that closed nothing.

Exam preparation — Domain 3

What is assessed. Governance purpose and its separation from management and assurance; structural weaknesses by organisational form and **the price of a decision rule** (unanimity, defined majority, single integrating authority); governing iterative and hybrid delivery, and **what share of an iterative stream a periodic body can serve**; the sponsor's obligations, failure modes, **diary load**

and turnaround distribution; steering-committee design faults and average, peak and hidden utilisation; **the governance latency formula and its application**; delegation thresholds and their economics on **three dimensions**; gate purpose, criteria, value and **the conditional pass through** `q_eff = q × r`; the three assurance lines, their failure modes and **residual exposure across an assurance map**; escalation design with stated latency, **objective triggers that name a forecast method**, and **the decision action window**; and the decision record with its integrity checks, its **re-decision tax** and its **cumulative test**.

The calculations to be able to do under time pressure. `E[wait] = M/2 + L` for a single body and `Σ (M/2 + L)` for a multi-tier path, priced at a cost of delay. Multi-party latency $M/2 + L + (1/p^n - 1)M$ under unanimity, and the majority-rule comparison. Servable share of a decision stream against need-by times. Sponsor load in annual hours and share of a week; turnaround mean, median and tail share. Escalation volume and delay cost at alternative thresholds, and the breakeven value-destruction rate on a delegated band. Reversal-cost ratio and authority on `max(value, cost to undo)`. Gate net value, breakeven elapsed time, breakeven detection probability, and effective detection `q × r` with the breakeven closure rate `d*/q`. Residual exposure `Σ p ∏(1 − q) u`, the marginal value of a line on a risk, and the reallocation gain at constant cost. VAC as a share of BAC across the EAC family and the tolerance test. Decision action window `(remaining duration − latency)/remaining duration`. Committee capacity, utilisation at average and peak, and repeat slots. RACI single-A defect rate and the cost of each defect type. Cumulative-test base-rate aggregate and expected trip count.

The traps. Taking expected wait as half the meeting interval and forgetting the paper lead time (Exercise 3.1) · adding meeting intervals without paper lead times in a multi-tier path (Exercise 3.4) · applying a multi-party readiness probability once instead of as p^n, which makes unanimity look survivable (Exercise 3.5) · treating governance artefacts as governance (3.1.1) · omitting elapsed-time cost from gate value, which makes every gate look worthwhile (Exercise 3.3) · mistaking a condition-closure rate for an effective detection rate, omitting the multiplication by `q` (Exercise 3.6) · comparing a delegation saving with the total value of the delegated band rather than a plausible loss on it (Exercise 3.2) · reading average committee utilisation as evidence of capacity when demand clusters (3.2.2) · reallocating assurance to the uncovered risk rather than the largest **residual** one (Exercise 3.7) · quoting "the forecast" against a tolerance without naming an EAC method, when the family can straddle the line (Exercise 3.8) · stopping at an escalation path's latency without testing it against the remaining duration (3.3.3b) · assuming a committee's authority is its highest member's rather than its binding minimum (3.2.2) · treating a favourable assurance opinion as a transfer of accountability (3.3.2) · recording only committee decisions and missing the ones taken below a threshold (Case study B) · sizing a cumulative test on its threshold and ignoring the trip count its relatedness class generates (3.3.4c) · reading a delegation schedule on value alone.

How the domain connects. Domain 1 supplies the accountability principle governance must respect and the cost of delay every calculation here is priced at. Domain 2 supplies the business case and kill criteria that gates re-test, and its omitted-column failure is what reappears here as the largest uncovered row on Meridian's assurance map. Domain 4 integrates the governance of parts into a whole. Domain 6 consumes governance latency as schedule input — a decision path of 15.5 weeks is a 15.5-week predecessor, whatever the plan says — and supplies the schedule forecast on which the decision action window should properly be computed. Domain 7's change control is the delegation schedule in operation, and its earned-value forecasting supplies the EAC family that an escalation trigger must name. Domain 8's risk escalation uses the escalation classes designed here. Domain 9's independence rule governs whether two assurance lines may be counted as two detectors. Domain 14, KA 14.3.3 derives the verification standard that 3.A.2 requires here. Domain 15 owns the portfolio allocation of the scarce sponsor capacity that 3.2.1 sizes. And PFL-AI Domain 11 handles the risk-allocation face of the multi-party governance problem whose decision-rule cost 3.1.2 prices.

Domain 3 summary

Governance is the decision rights, accountabilities and information flows through which an organisation directs a project — and it is not management, not assurance, and emphatically not the reporting apparatus that is usually offered in its place. A governance design must produce decidability, timeliness, legitimacy and traceability, and each of those is testable.

The domain's contribution is to make governance **computable**. The expected wait for a committee decision is `M/2 + L`: half the meeting interval plus the whole paper lead time. That single formula prices a governance design at design time, shows that the administrative deadline is twice the lever meeting frequency is, and adds across tiers to reveal what an escalation path actually costs — 15.5 weeks and **USD 221,340** for Meridian's three-tier path, of which the quarterly committee alone is 61 %. Extended by the failure probability of a decision rule, it prices a constitution: $M/2 + L + (1/p^n - 1)M$ puts Meridian's three-party board at **5.4870 weeks** under unanimity against **4.1152** under a two-of-three majority when the parties are 90 % ready, and at **11.6618** against **5.1020** when they are 70 % ready — the two rules are close in good conditions and divergent in bad ones, which is when governance is load-bearing. The same arithmetic prices a delegation threshold: Meridian's 10,000 threshold cost **USD 514,080** a year, and raising it to 25,000 saves **USD 171,360** against a worst case of 81,600 even if every delegated decision were decided wrongly — a breakeven value destruction of **84 %** per decision, which no plausible delegate approaches. And it prices a gate: Meridian's design gate is worth **USD 56,520**, stops paying beyond **9.96 weeks** of elapsed time, and requires a detection probability above **55.85 %** to be worth holding at all — but only if its conditions are closed, because $q_{eff} = q \times r$ turned that same gate into one destroying **USD 57,427.83** at an observed closure rate of **39.13 %**, a swing of **USD 113,947.83** with nothing changed but the tracking.

Three further quantities decide whether a governance design can function at all. **Servable coverage:** at 4.0 weeks of latency against 2-week increments, Meridian's committee can serve **12.31 %** of its iterative stream's decisions in time, and doubling its frequency while halving its paper deadline reaches only **36.92 %** — which is why a bounded envelope is a feasibility requirement and not a courtesy. **Residual exposure:** $\Sigma\ p\ \prod(1 - q)\ u$ showed all three of Meridian's assurance lines sitting on the risk carrying **7.87 %** of residual exposure while **58.27 %** sat wholly uncovered, and reallocating 30 days cut exposure by **USD 74,361.60** at no additional cost. And the **decision action window:** Project Auriga's 15.5-week escalation path against 12 weeks of remaining duration is **−29.17 %** — the decision arrives 3.5 weeks after planned completion, so the path is not slow but ceremonial, and it was actionable only during the first **38.00 %** of the project.

The sponsor's role is a set of testable obligations, not a title — **111.0 hours** a year, about **6.03 %** of a working week, which is small enough that absence is a priority choice and large enough that no executive carries more than about three; and the sponsor's own latency is a distribution with a tail, where **22.73 %** of Meridian's decisions carried **60.81 %** of the wait and handling five of them differently removed **43.92 %** of the cost. Steering committees fail in five predictable ways, four of them designed in, and their true load is not the average: Meridian's 74.0 % concealed **110.00 %** at phase boundaries and, once re-decisions were counted, **98.08 %** overall. Gates buy optionality against irreversibility and belong where irreversibility steps up. Assurance has three lines whose worst failure is capture, and whose effort concentrates where review is comfortable rather than where exposure is. Escalation is a timed pathway with an out-of-cycle route and an objective trigger that **names its forecast method** — Auriga's three estimates at completion sit at **5.00 %**, **10.42 %** and **15.20 %** of BAC against a 10 % tolerance, so an unnamed method delegates the escalation to whoever prepares the report. And a decision that is not recorded, with a named decision-maker and a versioned information reference, has not been made — it has been remembered, which cost Meridian **USD 357,000** a year and **24.04 %** of its committee's capacity in questions decided twice, remediable with two fields on a template. Case study B's four differences were assembled from **20** individually authorised changes totalling **700,000**, which is the whole argument for a delegation schedule that reads on reversibility and externality and aggregates — and its first cumulative rule, summed over four workstreams, would have tripped in all **16** workstream-quarters on ordinary traffic, which is why such a rule must be sized against the base rate of the class it sums over.

The through-line: **governance has a price, the price is computable, and until it is computed governance complaints are cultural and go nowhere.** Compute it, and they become engineering.

04

DOMAIN 4

Integration and Delivery Architecture

Group: Leading projects (Domain 4 of 4 in Part One — the part's closing domain). **Target:** ~70 pages. **Binds to:** the PCI Book Pattern Specification and the shared registries (`docs/books/registries/`). This domain completes Part One by assembling Domains 1–3 into a single delivery architecture, and hands over to Part Two, where **Project Auriga** works the same disciplines at single-project scale. British English; USD (+SAR where useful, indicative `USD 1 ≈ SAR 3.75`).

Why this domain exists

Domain 1 established what a project leader is answerable for. Domain 2 established how work is chosen and whether the promise was honest. Domain 3 established who may decide, how quickly, and at what price. Each produced something real, and none of them produced a **project**.

That is what this domain does, and the word that names the gap is **integration**. A project is not a collection of well-run parts; it is a set of parts that must fit, arrive in an order, share interfaces, and change together. Every one of those relationships is a place where a project fails without any single part having failed — which is why integration failure is the hardest kind to attribute and the most expensive to repair. The specific and recurring form it takes: **the parts are managed and the joins are not.** Scope is controlled, the schedule is controlled, cost is controlled, and nobody owns the fact that a change to one moves all three.

The domain's second claim is that integration is **architectural** — a design choice with a price, made once and paid for continuously. Interfaces do not multiply linearly with components; they multiply combinatorially, and KA 4.2 computes the difference and what buying an integration layer saves. Baselines are not three documents but one three-dimensional statement, and KA 4.3 shows what the arithmetic of the hundred-per-cent rule catches that reading the documents does not. Changes do not cost what a change form says they cost, and KA 4.4 shows that the quoted figure is typically the smallest component of the true one — which is why a delegation threshold applied to a quoted cost is not a control at all.

The through-line: **integration is where the parts become a project, and the joins are where the money is.**

Learning objectives. After this domain a candidate can: state what a charter must contain to confer authority, and distinguish a charter from a plan; assemble a coherent plan of plans, **audit a plan set against itself and price the exposure the audit finds**; derive a rolling-wave horizon from the longest commitment lead time in the plan rather than asserting it; tailor defensibly; build a WBS that satisfies the hundred-per-cent rule and detect an unallocated or omitted element arithmetically; distinguish work from product breakdown, **cross-match the two and separate a genuine orphan from a matrix artefact**; **compute the interface count of a delivery architecture and price what an integration layer buys, including its breakeven cost and its marginal advantage as**

the programme grows; integrate scope, schedule and cost into one baseline and **compute what a stale time-phased baseline does to each earned-value index and which indices it leaves exactly correct**; apply configuration management to a baseline, audit its integrity, and **derive the item-level audit tolerance from an interface-level target through the pairwise exponent**; maintain a baseline through change without losing traceability to the original, and **perform the original-to-current reconciliation and decompose its residual gross rather than net**; design an integrated change-control flow and **price a screening misclassification in both of its directions**; **assess a change's true cost including schedule, rework, interface re-verification and regression, and explain why a threshold set on quoted direct cost fails**; detect and quantify baseline drift by accumulation and set a cumulative test that would actually have caught it; **price an untraced rejection**; and govern AI-assisted impact assessment without letting it decide.

The master programme, and the handover. Meridian Care Records continues from Domains 1–3: the clinical-records rollout to **40 clinics**, approved cost **USD 2,400,000** (Domain 2's business case), benefits **USD 685,440** a year at 70 % adoption, cost of delay **USD 14,280 per week** (Domain 1), and the governance design priced in Domain 3. This domain architects it. Part Two then picks up **Project Auriga** — the 25-week control-systems upgrade — to work planning, cost and risk at the scale of a single project, because the two scales need different illustrations and a book that uses only one has taught only half the discipline.

KNOWLEDGE AREA 4.1

Charter and management plans

Topics: 4.1.1 the charter · 4.1.2 the plan of plans · 4.1.3 tailoring.

4.1.1 The charter

Definition. The charter is the document by which the organisation **authorises the project and confers authority on its leader**. It is not a plan, not a business case, and not a summary of either: the business case argues that the work is worth doing (Domain 2), the plan states how it will be done, and the charter states that it is to be done, by whom, within what bounds.

What a charter must contain to do its job. Each item earns its place by being something a leader cannot proceed without, and the test of a charter is whether its absence would be noticed:

Element	Why it must be there
Purpose and objectives	The measurable statement of what success is (Domain 2's benefits, not activity).
Named sponsor	The accountable owner of the outcome (Domain 3, KA 3.2.1) — a role, and a person.
Named project leader and their authority	The **bounds** within which they decide without asking: spend, scope, resource, commitment (Domain 3's delegation schedule).
Scope boundary, including exclusions	Exclusions are the half most often omitted, and the half that prevents the argument.
Key deliverables and success criteria	What will be handed over, and how acceptance is judged.
Milestones and constraints	The dates that are genuinely fixed, distinguished from those that are planned.
Budget authority	The approved sum and the authority to commit against it.
High-level risks and assumptions	The material ones, carried forward into the risk register (Domain 8).
Governance and decision rights	Bodies, thresholds and escalation, by reference to the design of Domain 3.
Approval	Signature of the authority entitled to confer it.

The two charter failures. The charter that authorises nothing: a document of aspiration with no stated authority, so every decision is an ask, and the leader discovers their authority incident by incident — which is the functional-organisation weakness of Domain 3, KA 3.1.2, arriving through the front door. The charter that is a plan: dozens of pages of approach and schedule, signed once, immediately stale, and thereafter neither read nor updated; its length conceals the absence of the one paragraph that mattered.

The test. A usable charter answers, on one or two pages: what is this for, who owns the outcome, who leads it, what may they decide alone, what is in and out, and what is fixed? A leader who cannot answer the fourth of those from the charter does not have a charter; they have an announcement.

4.1.2 The plan of plans

The concept. The project management plan is not a single narrative but an **integrated set** of subsidiary plans — scope, schedule, cost, quality, resource, communication, risk, procurement, stakeholder, change and configuration — plus the baselines they produce. Its integration is the whole point: the subsidiary plans are individually easy and collectively contradictory unless someone owns the consistency.

The consistency checks that make it a plan rather than a folder, and each is a specific, answerable question:

- Does the **schedule** contain the activities the **scope** requires — every deliverable traceable to work, and every work package to a deliverable?
- Does the **cost** baseline sum to the resourced schedule, at the resource rates the **resource** plan assumes (Domain 7's rate arithmetic)?
- Do the **quality** plan's verification activities appear in the schedule with duration and resource — or are they assumed to happen in gaps?
- Do the **risk** responses have owners, dates, and budget in the cost baseline (Domain 8's contingency)?

- Do the **procurement** lead times appear as schedule activities rather than as assumptions?
- Does the **change** plan's authority match the governance design's delegation schedule, on the same thresholds (Domain 3) — and does it read on assessed impact rather than quoted cost (KA 4.4)?
- Does the **communication** plan's reporting cadence fit the governance bodies' paper lead times (Domain 3's L), so that reports exist when papers close?

That last check is the kind this domain exists to surface: it is invisible in any single plan and obvious the moment the plans are read against each other. A programme whose monthly report is produced three days after the steering committee's papers close will report last month's position every month, for its entire life, and will describe the problem as a reporting delay.

Those seven questions are usually treated as a review formality. They are an arithmetic exercise with a price attached.

WORKED EXAMPLE

Worked example 4.1.2 — auditing Meridian's plan set against itself, and pricing what it finds.

1. **Setup.** Before baseline approval, Meridian's **eleven** subsidiary plans are read against one another using the checks above — two reviewers over three days, **6 person-days** at a blended **USD 880** per person-day (the USD 110 an hour of Domain 14, KA 14.2.1, over an eight-hour day). The plan set covers **172** work packages, **24** quality-verification activities, **19** risk responses with named owners and **7** procured items. Cost of delay **USD 14,280 per week** (Domain 1).
2. **Formula.** Run each check; convert each finding into money — unfunded work at the person-day rate or at its own stated cost, unscheduled elapsed time at the cost of delay, counting only what lands on the critical path. Total, then compare with the cost of the audit.
3. **Substitution.** Check by check. Scope against schedule: **4** deliverables with no scheduled activity — flagged, deliberately **not** priced. Cost against the resourced schedule: the cost baseline sums to the resourced schedule at the resource plan's rates — **this check passes**. Quality: of 24 verification activities, **9** appear nowhere in the schedule, carrying **46 person-days** of resource → `46 × 880`; **4** of those nine sit on the critical path with **12** working days between them → `(12/5) × 14,280`. Risk: of 19 responses, **6** have no line in the cost baseline, together planned at **118,000**. Procurement: **3** items have lead times stated only in the procurement plan; the longest is **14 weeks** against **6** allowed in the schedule → `(14 − 6) × 14,280`. Change: the change plan's threshold reads on quoted direct cost — the defect KA 4.4.2 prices. Communication: the report is produced after papers close; the staleness arithmetic belongs to Domain 14, KA 14.2.1 and is not repeated here.
4. **Result.** Unfunded verification labour **USD 40,480**; unfunded risk responses **USD 118,000**; procurement lead-time exposure `8 weeks =` **USD 114,240**; critical-path verification exposure `2.4 weeks =` **USD 34,272**. Quantified total **USD 306,992 — 12.79 %** of the approved 2,400,000 — found by an audit costing **USD 5,280**, a ratio of **58.14** to one. Three classes of finding are recorded unpriced — the four unscheduled deliverables, the threshold basis and the reporting cadence — and one check passes clean.
5. **Interpretation.** The first thing to be honest about is what the 306,992 is not. It is **not a saving**. Most of it is money that will be spent whatever anyone decides — the verification activities are needed, the risk responses were chosen deliberately, the lead times are what the market quotes. What the audit changes is whether that money is spent inside an authorised baseline or outside it as an unexplained overrun, and an assurance function that reports the figure as a saving will be correctly disbelieved and will lose the argument it was right about. The defensible sentence is: 306,992 of committed but unauthorised cost and schedule moved from invisible to visible, for 5,280.

Set that against WE 4.3.3. Meridian's baseline drifts **291,176** over a year, through 34 individually authorised approvals, and everybody involved works hard for it. The plan-set audit finds **more** — 306,992, a ratio of **1.0543** — in three days, before any of the work has started, without a single change request being raised. **The cheapest change control is the one applied to the plan set before it is baselined**, because at that point a correction is a re-estimate rather than a change: no board, no assessment cost, no traceability obligation, no supplier conversation.

Now notice which check passes. Cost against the resourced schedule is the one organisations habitually test, and it is the one where both sides are owned by the same function and read from the same tool. Every failing check spans two owners: quality and planning, risk and cost, procurement and planning, change and governance. That is the general rule this KA exists to state — **plan-set**

defects live on ownership boundaries, and a check that either party could have run alone has almost certainly already been run. It follows that the audit should be sequenced by boundary rather than by plan, and that the reviewer must have standing with both owners, which is why this work fails when it is delegated to whoever has capacity.

The sensitivity is concentrated in one component. The 114,240 of procurement exposure assumes the eight-week lead-time gap lands on work that gates the rollout. If the affected item has eight weeks of float the component is **zero** and the total falls to **USD 192,752** — still **8.03 %** of the baseline, so the conclusion survives, but the headline drops by **37.21 %**. Every schedule-priced finding must therefore carry its float position (KA 4.4.2's third assessment rule), and an audit that prices delay without stating float has quoted an upper bound as a point estimate.

Finally, a caution about the four unpriced deliverables, because the instinct is to put a placeholder against them so that the total looks complete. Resist it. Those four are the findings a reviewer should care about most: an unpriced finding is one whose magnitude is unknown, and unknown magnitudes are where residual risk actually sits. A total that silently includes invented numbers for them is worse than a total that stops short and says why.

Progressive elaboration and rolling wave. Detail is added as it becomes knowable rather than invented early: the near horizon is planned to work-package level, the far horizon to a coarser level, and the boundary moves forward at a stated cadence. Two disciplines make this honest rather than an excuse. The **horizon must be stated** — "detailed to work-package level for 12 weeks, planning-package level beyond" — so that nobody mistakes a coarse plan for an absent one. And the **far-horizon estimate must carry its uncertainty explicitly**, because a rolling-wave plan whose outer years are stated as point numbers has hidden its uncertainty in exactly the place it is largest (Domain 8's ranges).

Setting the horizon rather than asserting it. The horizon itself is usually chosen for administrative tidiness — a quarter, a phase, twelve weeks — and it has a derivable lower bound: **the horizon must be at least as long as the longest commitment lead time falling inside it**, because a plan that does not yet exist at work-package level cannot support a commitment that must be placed before it does. Meridian illustrates the failure exactly. Its stated horizon is **12 weeks**; the longest procurement lead time WE 4.1.2 found is **14 weeks**. The horizon is **two weeks shorter than its own supply chain**, and no amount of diligence inside the wave can close that gap, because the order must be placed before the work package that specifies it has been written. Two responses are legitimate: lengthen the horizon to at least 14 weeks on the affected path, or plan that one procurement to work-package level ahead of the wave and label it a **partial horizon**. What is not legitimate is to leave a 12-week horizon and a 14-week lead time in the same plan set and describe the consequence as a supplier problem. The test generalises to every commitment class in the plan — procurement, recruitment, regulatory submission, facility access, decommissioning permission — compare each lead time with the horizon, and set the horizon requirement from the longest.

4.1.3 Tailoring

The principle. Every method requires tailoring, and tailoring is a **decision with a rationale and an owner**, not the quiet omission of inconvenient parts. The distinction is procedural: a tailoring decision is recorded, states what was removed or added and why, names who approved it, and is reviewable. An omission is discovered later by someone else.

What drives it. Project size and value; novelty and technical uncertainty; regulatory and contractual obligation; delivery approach (sequential, iterative, hybrid); organisational maturity; and the receiving organisation's capacity to absorb process. A small, novel, publicly visible project may need more assurance and less documentation than a large, familiar, internal one, and a tailoring regime that cannot express that combination is not tailoring but scaling.

The limits. Three things are not tailorable and should be stated as such, because they are what tailoring is most often used to remove: the **decision record** (Domain 3, KA 3.3.4) — nothing about project size makes an unrecorded decision acceptable; **the traceability of a baseline change to an authority** (KA 4.3.3); and any **legal, regulatory or contractual** obligation, which does not scale with convenience. Everything else is a judgement, and every judgement is recorded.

AI in this KA

Where it earns its place. Reading a plan set against itself and listing the inconsistencies above — deliverables with no work packages, quality activities absent from the schedule, risk responses without budget, reporting cadences that miss governance paper deadlines. This is precisely the work integration requires, humans do it badly because it spans documents, and the answers are checkable. Drafting a charter or a subsidiary plan from an approved template plus stated inputs, for human completion. Proposing a tailoring set from project characteristics, as a starting position to be argued with.

Where it must not go. Approving a tailoring decision, which requires an accountable authority. Producing a charter's authority statement, which is a conferral of power and must be authored by the conferring body. And no AI-produced plan should enter a baseline unreviewed: a plan is a commitment, and Domain 1's principle requires a person to stand behind it.

Verification, concretely. Every flagged inconsistency is confirmed against the source documents before it is reported; every AI-drafted plan section is reviewed by the accountable owner of that plan; and the tailoring record states that the proposal was AI-assisted, which is an honesty obligation rather than a disclaimer.

KEY TERMS — KA 4.1

Term	Meaning
Charter	The document authorising the project and conferring bounded authority on its leader.
Authority bounds	The spend, scope, resource and commitment limits within which the leader decides alone.
Plan of plans	The integrated set of subsidiary plans and the baselines they produce.
Consistency check	A specific question testing whether two subsidiary plans agree.
Plan-set consistency audit	Reading the subsidiary plans against one another and pricing the findings, before baseline approval (WE 4.1.2).
Progressive elaboration	Adding detail as it becomes knowable, at a stated horizon and cadence.
Planning package	A far-horizon element planned above work-package level, with its uncertainty stated.
Commitment lead-time rule	The rolling-wave horizon must be at least as long as the longest commitment lead time inside it.
Partial horizon	One element planned to work-package level ahead of the wave because its lead time demands it, labelled as such.
Tailoring	A recorded, owned decision to adapt method — never a silent omission.

SAMPLE MCQS — KA 4.1

MCQ 4.1-A `[4.1.1 · Comprehension]` The element of a charter whose absence most directly disables the project leader is: - A. the list of high-level risks - B. the statement of the leader's authority bounds - C. the milestone schedule - D. the communication approach

Rationale: Without stated bounds every decision becomes an ask and authority is discovered incident by incident (4.1.1). The others are important and none of them confer power.

MCQ 4.1-B `[4.1.2 · Analysis]` A programme's monthly report is produced three days after its steering committee's papers close. The consequence is that: - A. the report is slightly late - B. the committee will systematically consider last month's position, every month - C. the reporting cadence must be made weekly - D. the paper lead time must be abolished

Rationale: The defect is a plan-consistency failure between the communication plan and the governance design's paper lead time (4.1.2), and it recurs for the project's whole life until the cadences are aligned.

MCQ 4.1-C `[4.1.3 · Evaluation]` Which is not legitimately tailorable? - A. the number of subsidiary plans maintained - B. the depth of the schedule beyond the planning horizon - C. the traceability of a baseline change to an approving authority - D. the frequency of progress reporting

Rationale: Traceability of baseline change to authority, the decision record, and legal or contractual obligations are outside tailoring (4.1.3).

MCQ 4.1-D `[4.1.2 · Application]` In WE 4.1.2 the plan-set audit finds unfunded verification labour of 40,480, unfunded risk responses of 118,000, eight weeks of procurement lead-time exposure and 2.4 weeks of critical-path verification exposure, at a cost of delay of 14,280 per week. The quantified total is: - A. USD 158,480 - B. USD 272,720 - C. USD 306,992 - D. USD 392,672

Rationale: `40,480 + 118,000 + (8 × 14,280) + (2.4 × 14,280) = 306,992` (4.1.2). A totals only the two money classes; B omits the critical-path verification exposure; D prices the whole 14-week procurement lead time rather than the 8-week gap against the schedule.

MCQ 4.1-E `[4.1.2 · Evaluation]` A rolling-wave plan states a 12-week detailed horizon; the longest procurement lead time in the plan set is 14 weeks. The correct conclusion is that: - A. the procurement should be expedited to fit the horizon - B. the horizon cannot support its own commitments and must be lengthened on that path, or the item planned ahead of the wave as a stated partial horizon - C. the horizon is a planning convention and the lead time is a supplier matter - D. the plan should carry a risk for late procurement

Rationale: The order has to be placed before the work package specifying it exists, so no diligence inside the wave closes the gap (4.1.2). D converts a structural defect into a risk, which is the commonest way this finding is disposed of without being fixed.

SELF-CHECK — KA 4.1

1. What distinguishes a charter from a plan? — The charter authorises and bounds; the plan states how. The charter's irreplaceable content is the authority statement.
2. Name two plan-consistency checks that only appear when plans are read against each other. — Quality verification activities missing from the schedule; reporting cadence missing the governance paper deadline.
3. Why do plan-set defects cluster on ownership boundaries? — Because a check both parties could run alone has already been run; the failing checks are the ones spanning two owners (WE 4.1.2).
4. What sets the lower bound on a rolling-wave horizon? — The longest commitment lead time inside it, since a commitment cannot be placed from a plan that does not yet exist at work-package level.

5. What makes a tailoring decision legitimate? — It is recorded, states what changed and why, names its approver, and is reviewable.

KNOWLEDGE AREA 4.2

Breakdown structures

Topics: 4.2.1 WBS · 4.2.2 product and deliverable breakdown · 4.2.3 interfaces.

4.2.1 The work breakdown structure

Definition. A WBS is a **deliverable-oriented hierarchical decomposition** of the total scope of work. Two words carry the weight. Deliverable-oriented: it decomposes what will be produced, not who will produce it or when — an organisational chart and a schedule are different artefacts, and a WBS built by team or by phase loses the property that makes it useful. Total: the decomposition covers the whole scope and nothing outside it.

The hundred-per-cent rule. The children of any element sum to exactly that element — no more, no less. More means duplication or scope not authorised; less means work that will be done and is not budgeted. The rule is not a style guideline; it is an arithmetic invariant, and because it is arithmetic it can be **checked**, which is the point of the next worked example.

Work packages. The lowest level, and the unit that is estimated, scheduled, assigned and measured. A work package should have a single accountable owner, a deliverable or verifiable outcome, an estimate the owner endorses, and a duration that permits progress to be assessed meaningfully — sized so that the answer to "is it done?" is not routinely "about half". Above the work packages, **control accounts** are the level at which performance is measured and reported, which is where Domain 7's earned value attaches.

WORKED EXAMPLE

Worked example 4.2.1 — auditing Meridian's WBS against the hundred-per-cent rule.

1. **Setup.** Meridian's approved cost baseline is **USD 2,400,000** (Domain 2). Its five level-2 WBS elements are estimated at: records application build **780,000**; integration layer and interfaces **536,000** (the figure KA 4.2.3 derives); data migration **310,000**; clinic rollout across 40 clinics **520,000**; programme management and assurance **186,000**.
2. **Formula.** Hundred-per-cent test: Σ children − parent. Then re-test with any omitted element restored.
3. **Substitution.** `780,000 + 536,000 + 310,000 + 520,000 + 186,000 = 2,332,000`; against `2,400,000`.
4. **Result.** The children sum to **2,332,000**, leaving **68,000** (**2.83 %**) of the parent unallocated. The review then identifies the omitted element: **clinician training and enabling change**, estimated at **214,000** — the same column Domain 2's benefits map omitted. Restoring it gives an honest baseline of **2,546,000**, which is **146,000 — 6.1 %** — above the approved figure.
5. **Interpretation.** Both halves of that result matter and they are usually confused. The 68,000 of apparently spare budget is not spare; it is the residue of an incomplete decomposition, and in practice it will be consumed early by whatever arrives first, after which the omission becomes visible with no budget left to absorb it. And the omission is not random: it is the **enabling change** — the work that converts an output into an outcome — which Domain 2 identified as the column most benefits maps leave out, and which reappears here as the element most WBSs leave out, for the same reason. It belongs to somebody else, so nobody decomposes it. Note finally what the arithmetic does not say: at 2,546,000 the programme's NPV falls from Domain 2's **+1,332,898** to **+1,186,898**, a reduction of **11.0 %** — still comfortably positive. The omission bought nothing. It was not a decision to descope training in order to make the case work; it was a failure to look, and the honest baseline would have been approved.

4.2.2 Product and deliverable breakdown

The distinction. A **product breakdown structure** decomposes the thing being delivered into its constituent products and components; a WBS decomposes the work. They answer different questions and both are useful — but only one of them is a good starting point.

Why to start with the product. Building the product breakdown first and deriving the work from it prevents the commonest scope defect: **work that produces nothing**, and its mirror, **a product with no work**. If every product traces to work and every work package to a product, the two structures verify each other, and the verification is a matter of matching lists rather than judgement.

Where each belongs. The product breakdown drives configuration management (KA 4.3.2) — because configuration items are products, not activities — and acceptance, since acceptance is of products. The WBS drives estimating, scheduling and performance measurement. Confusing them produces two recognisable pathologies: configuration control applied to activities, which cannot work because activities do not have versions; and estimating against products, which omits the work that has no product (integration, testing, migration, training) and is therefore how the elements of the previous worked example go missing.

WORKED EXAMPLE

Worked example 4.2.2 — cross-matching Meridian's product breakdown against its WBS.

1. **Setup.** Meridian's product breakdown identifies **148** products; its WBS contains **172** work packages. A traceability matrix relates the two — $148 \times 172 =$ **25,456** cells. The test is that every product traces to at least one work package and every work package to at least one product.
2. **Formula.** Product coverage = products with at least one work package ÷ products. Work-package coverage = work packages with at least one product ÷ work packages. Then **classify every exception**, because the raw counts are not the finding.
3. **Substitution.** Raw exceptions: **12** products with no work package and **9** work packages with no product — **21** in all. Classification, item by item, by the owner of each item. Of the 12: **5** are the clinician-training and enabling-change product set — WE 4.2.1's omitted **214,000** — and **7** are produced inside work packages named for their parent product, so they are matrix granularity rather than omission. Of the 9: **7** are legitimately product-free (integration testing, data migration, training delivery, commissioning, interface verification) and **2** are genuine orphans, budgeted at **26,000**, producing nothing anyone has asked for.
4. **Result.** Product coverage $136/148 =$ **91.89 %**; work-package coverage $163/172 =$ **94.77 %**. Of 21 raw exceptions, **14** are legitimate and **7** are real findings — a false-positive rate of **66.67 %**. The real findings are worth **214,000** of omitted work and **26,000** of unnecessary work: **USD 240,000**, exactly **10.00 %** of the approved baseline.
5. **Interpretation.** The classification, not the matrix, is the professional work. Two thirds of the raw exceptions are artefacts, so a leader who takes raw counts to a steering committee will be corrected in the room on the seven granularity artefacts and will lose the five that mattered along with them. State the rule that way round: **a cross-match produces a candidate list, and the finding is the residue after each exception has been ruled on by the owner of the item.** Twenty-one rulings is an afternoon; 25,456 cells is not human work at all, which is exactly the division of labour the AI note below describes.

The two real classes are not symmetric, and only one of them is visible later. **214,000 of products with no work is an omission**: it arrives as an overrun, at the end, in the enabling-change column that Domain 2's benefits map dropped and that WE 4.2.1's WBS dropped again — the third time the same column has gone missing in this programme, which is what a systematic defect looks like as opposed to an accident. **26,000 of work with no product is waste**: it will be executed diligently, reported as complete, and never questioned, because a work package that produces nothing still reports 100 %. Earned value will confirm it. That is the uncomfortable corollary of Domain 7's machinery — performance measurement measures conformance to the plan, so work that should not be in the plan is measured as success.

Note what removing the orphans does to WE 4.2.1's arithmetic, because the direction surprises people. Allocated cost falls from 2,332,000 to **2,306,000**, so the unallocated residue **rises** from 68,000 to **94,000 — 3.92 %** of the parent. Cleaning a WBS usually widens the hundred-per-cent gap rather than closing it, because part of the gap was being filled by work that should not have been there. A leader who expects the 214,000 omission and the 26,000 of waste to offset each other has misread both: one is outside the baseline and one is inside it, and they land on different sides of the sum.

A caution on the legitimate no-product class, which is where this check is most often misunderstood. It is the same class the paragraph above identified as the one a product-derived WBS omits by construction — so it has to be added rather than found, and the honest sequence is therefore: product breakdown first, for completeness of what is delivered; then the deliberate addition of the product-free work, as a decision; then the cross-match. On that sequence the seven legitimate exceptions are the **register of that addition**, not a defect list — and if that class comes back empty, the WBS is a product breakdown with different headings and the integration work has not been planned at all. An empty legitimate class is a worse finding than a large one.

What a reviewer should test is not the coverage percentages but whether each of the 21 exceptions carries a recorded ruling with a named ruler. An unruled exception list is the commonest outcome of this check and is indistinguishable, six months later, from never having run it.

4.2.3 Interfaces — and why they are the expensive part

The definition and the problem. An interface is a defined relationship across a boundary — between components, teams, contracts, organisations or phases — at which something must be agreed and verified. Interfaces are where integration effort actually lives, and they are systematically under-planned for one arithmetic reason: **components grow linearly and interfaces grow combinatorially.** A programme that has doubled its component count has quadrupled its interface count, and the plan that was written for the first count is still in use.

For `n` components each of which may connect to any other, the number of possible pairwise interfaces is:

```
Mesh interfaces      = n(n − 1)/2         — every component to every other
Layered interfaces   = n                   — every component to one integration layer
```

WORKED EXAMPLE

Worked example 4.2.3 — what an integration layer buys Meridian.

1. **Setup.** Meridian must integrate **12** components: records core, patient master index, appointments, prescribing, laboratory results, imaging, billing, the national reporting gateway, identity and access management, the clinical document store, analytics, and the legacy records system during migration. Specifying, building, testing and documenting one interface costs **USD 18,000**. An integration layer would cost **USD 320,000** to build.
2. **Formula.** Mesh count `n(n−1)/2`; layered count `n`. Cost = count × unit cost (+ layer build).
3. **Substitution.** Mesh `12 × 11/2 = 66`; layered `12`. Mesh cost `66 × 18,000`; layered `12 × 18,000 + 320,000`.
4. **Result. 66** point-to-point interfaces costing **USD 1,188,000**, against **12** interfaces plus the layer costing **USD 536,000** — the layer saves **USD 652,000**, or **54.9 %**. It remains worth building while its cost is below `1,188,000 − 216,000 =` **USD 972,000**.
5. **Interpretation.** The saving is large and it is not the most important part of the result. The decisive number is **marginal**: adding a thirteenth component costs `12 × 18,000 =` **USD 216,000** on a mesh architecture and **USD 18,000** on a layered one — a factor of **12**, which grows with every component added. Since programmes acquire components (a new department, an acquired system, a regulatory feed), the architecture is a bet on the future component count, and a mesh is a bet that the count will not grow. Three professional cautions, however, because this arithmetic is easy to over-apply. Real architectures are **partial meshes** — not every pair genuinely needs to connect, and the honest count is of required interfaces, which is why the interface register (Toolkit 4.T.2) is built from need rather than from the formula. The layer introduces a **single point of failure and a throughput constraint** that the point-to-point design does not have, and those are real costs not captured here. And the 18,000 unit cost is an average over interfaces of very different difficulty: an internal API and a national reporting gateway are not the same object, and a leader quoting one figure for both should say so.

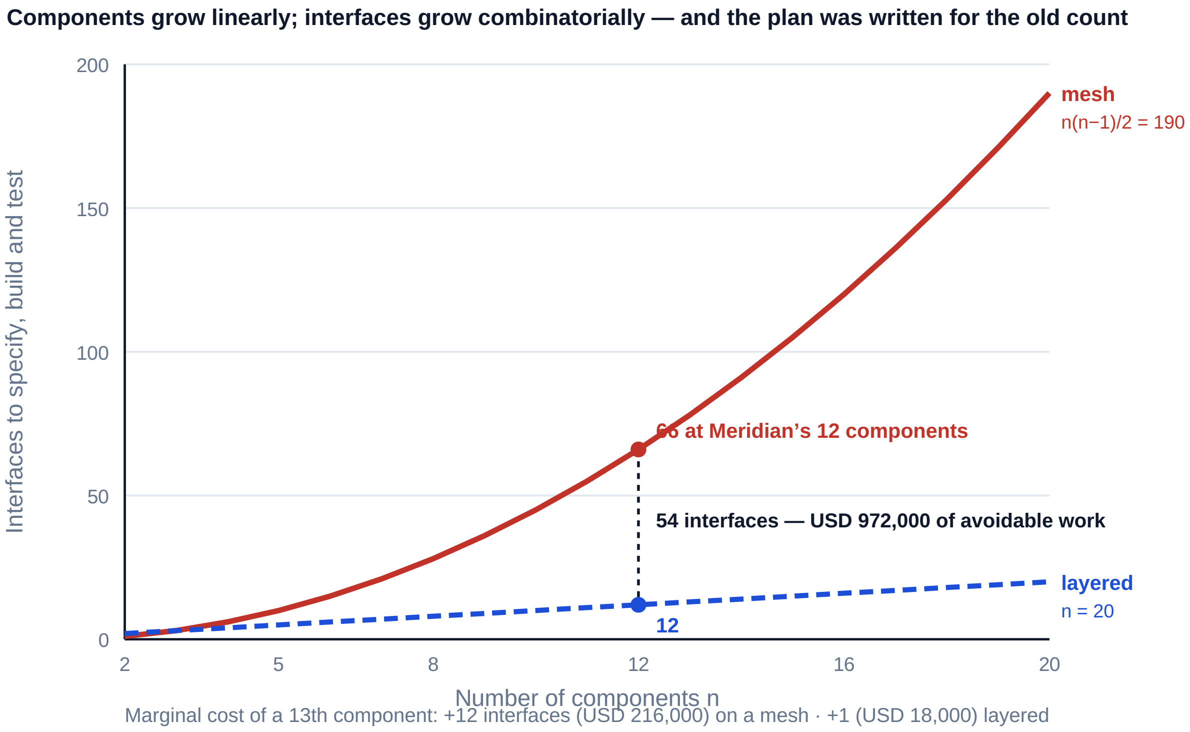


Fig 4.2.1 — Interfaces grow combinatorially; components do not

Managing interfaces once counted. The count is the beginning. Each required interface needs an **interface agreement**: the two parties, the thing exchanged, its format and content, the direction, the timing, the error and exception behaviour, who verifies it and when, and the version it is agreed at. Two disciplines matter more than the document. An interface has **exactly one owner on each side**, named — Domain 3's decidability test applied to a boundary, and the reason unowned interfaces are resolved late and by whoever notices. And interface verification is a **scheduled activity with duration and resource** (KA 4.1.2's consistency check), because the alternative is that verification happens at integration testing, which is where interface defects are most expensive to find and where they consume the schedule float that Domain 6 shows was never there.

AI in this KA

Where it earns its place. Checking the hundred-per-cent rule across a large WBS and reporting every element whose children do not sum — mechanical, exhaustive, and exactly the check humans skip. Cross-matching a product breakdown against a WBS and listing products with no work and work with no product — 25,456 matrix cells in Meridian's case, which is not human work at any level of diligence. Extracting an interface register from architecture and design documents, which is tedious and error-prone by hand. Flagging interfaces with no named owner on one or both sides, or no verification activity in the schedule.

Where it must not go. Ruling on a cross-match exception. WE 4.2.2's arithmetic exists precisely because two thirds of the raw exceptions were legitimate, and telling a granularity artefact from an omission requires knowing what the work actually produces — the ruling belongs to the owner of the item and must be recorded with their name. Deciding the architecture. The mesh-versus-layer choice depends on availability requirements, throughput, organisational boundaries, contractual structure and the expected growth in components — several of which are judgements about the future that no model has grounds for, and all of which belong to an accountable architect and the governance body.

Verification, concretely. Reproduce the interface arithmetic by hand — it is one formula — and state the assumed unit cost with its basis. Confirm each AI-derived interface against the source design before it enters the register. And when a model reports a WBS as compliant, spot-check at least the largest elements, because a hundred-per-cent check is only as good as the estimate values it was given.

KEY TERMS — KA 4.2

Term	Meaning
WBS	Deliverable-oriented hierarchical decomposition of the total scope of work.
Hundred-per-cent rule	The children of any element sum to exactly that element — an arithmetic invariant, and checkable.
Work package	The lowest WBS level: one owner, a verifiable outcome, an endorsed estimate, an assessable duration.
Control account	The level at which performance is measured and reported (Domain 7's earned value attaches here).
Product breakdown structure	Decomposition of the thing delivered; drives configuration management and acceptance.
Product-to-work cross-match	The traceability test that every product has work and every work package a product — a candidate list, not a finding.
Orphan work package	Budgeted work producing no product anyone asked for — waste that still reports 100 % complete.
Legitimately product-free work	Integration, testing, migration, training delivery, commissioning: real work with no product, and the class a product-derived WBS omits by construction.
Interface	A defined relationship across a boundary at which something must be agreed and verified.
Mesh vs layered	`n(n−1)/2` possible interfaces against `n` — the architecture choice priced in WE 4.2.3.
Interface agreement	The parties, content, format, direction, timing, exceptions, verifier and version of one interface.

SAMPLE MCQS — KA 4.2

MCQ 4.2-A `[4.2.3 · Application]` A programme integrates 12 components. The number of possible point-to-point interfaces is: - A. 12 - B. 66 - C. 132 - D. 144

Rationale: `n(n−1)/2 = 12 × 11/2 = 66` (4.2.3). A is the layered count; C counts each pair twice; D is `n²`.

MCQ 4.2-B `[4.2.3 · Evaluation]` In WE 4.2.3, the strongest argument for the integration layer is: - A. it saves 652,000 against the point-to-point design - B. adding one further component costs 216,000 on a mesh and 18,000 layered — and programmes acquire components - C. it reduces the interface count from 66 to 12 - D. point-to-point interfaces are technically inferior

Rationale: The marginal cost of growth is decisive because it compounds, whereas the one-off saving is a single number on today's component count (4.2.3). D is not established and would be an assertion.

MCQ 4.2-C `[4.2.1 · Analysis]` A WBS's five level-2 elements sum to 2,332,000 against an approved 2,400,000. The correct reading is that: - A. the project has 68,000 of contingency - B. the decomposition is incomplete, and the 68,000 will be consumed by whatever arrives first - C. the baseline should be reduced to 2,332,000 - D. the hundred-per-cent rule permits a 3 % tolerance

Rationale: Unallocated budget is a symptom of incomplete decomposition, not contingency (4.2.1); the rule admits no tolerance, and reducing the baseline would lock in the omission.

MCQ 4.2-D `[4.2.2 · Comprehension]` Configuration management applies naturally to a product breakdown rather than a WBS because: - A. products are more important than work - B. configuration items are products, and activities do not have versions - C. a WBS changes more often - D. product breakdowns are more detailed

Rationale: Versioning is a property of things, not of activities (4.2.2); applying configuration control to activities is one of the two standard confusions.

MCQ 4.2-E `[4.2.2 · Analysis]` A product-to-work cross-match returns 21 raw exceptions, of which classification shows 14 to be granularity artefacts or legitimately product-free work. The correct reading is that: - A. the matrix has 21 defects requiring correction - B. the finding is the 7-item residue after each exception has been ruled on by the owner of the item - C. a 66.67 % false-positive rate makes the check unreliable and it should be dropped - D. the false positives indicate the WBS is wrong at 14 places

Rationale: The cross-match produces a candidate list; the ruling produces the finding (4.2.2). A mistakes the candidate list for a defect report; C discards a check for doing exactly what a candidate-generation step is meant to do; D miscounts artefacts of matrix granularity as WBS defects.

MCQ 4.2-F `[4.2.2 · Application]` In WE 4.2.2 the classified findings are 214,000 of products with no work and 26,000 of work with no product. Removing the orphan work packages from the WBS: - A. reduces the unallocated residue from 68,000 to 42,000 - B. raises the unallocated residue from 68,000 to 94,000 - C. leaves the residue unchanged, since the two findings offset - D. reduces the approved baseline by 26,000

Rationale: Allocated cost falls to 2,306,000, so the gap against the approved 2,400,000 widens to 94,000 (4.2.2). A subtracts on the wrong side; C assumes an offset between a finding outside the baseline and one inside it; D confuses the estimate with the authorisation.

SELF-CHECK — KA 4.2

1. State the hundred-per-cent rule and why it is checkable. — Children sum exactly to their parent; it is arithmetic, so it can be audited rather than reviewed.
2. Why are interfaces systematically under-planned? — Components grow linearly and interfaces combinatorially, so a plan written for an earlier component count is always an underestimate.
3. What two disciplines matter more than the interface document? — Exactly one named owner on each side, and verification as a scheduled activity with duration and resource.

KNOWLEDGE AREA 4.3

Integrated baselines

Topics: 4.3.1 scope–schedule–cost integration · 4.3.2 configuration management · 4.3.3 baseline maintenance.

4.3.1 Scope, schedule and cost as one baseline

The principle. The performance measurement baseline is a **single three-dimensional statement**: the authorised scope, arranged in time, with cost attached. It is recorded in three documents for convenience, and treating those documents as three baselines is the origin of most integration failure.

The integration invariants — each testable, and each a real defect when it fails:

- Every scope element appears in the schedule; every schedule activity traces to a scope element.
- The cost baseline sums to the resourced schedule at the resource plan's rates (Domain 7).
- The **time-phased** cost baseline (`PV`, Domain 7's planned value) follows the schedule's dates — so a schedule change is a `PV` change, always, and a baseline whose `PV` curve did not move when the schedule did is not integrated.
- Contingency and management reserve are held explicitly, at a stated level, and are not distributed invisibly into estimates (Domain 8).
- Every deliverable has acceptance criteria; every acceptance activity has duration and resource.

Why the third invariant is the one that fails. It requires the cost and schedule tools to be connected, or a person to maintain the connection, and it is the first thing abandoned under pressure. The consequence is precise and severe: **earned value becomes meaningless.** Domain 7's `SPI` and `CPI` compare achievement against a baseline; if the baseline's time-phasing no longer matches the schedule the project is executing, the indices measure the gap between two documents rather than the state of the work — and they will keep producing confident numbers while doing it.

By how much, and in which direction, is computable from the same three figures the report already prints.

WORKED EXAMPLE

Worked example 4.3.1 — what a stale time-phased baseline does to Auriga's indices, and what it leaves exactly alone.

1. **Setup.** Project Auriga, Part Two's single-project thread, carries **BAC USD 4,000,000** over 25 weeks. At week 13 the reported position is **PV 2,080,000**, **EV 1,920,000**, **AC 2,120,000**, giving **CPI 0.91** and **SPI 0.92** (Domain 7, KA 7.3, whose definitions are used unchanged). At week 8 a change was approved that re-sequenced downstream work; its cost effect was posted but the `PV` curve was never re-phased, so the reported PV is still 2,080,000. Two variants are examined, each moving **USD 240,000** of planned value across the week-13 status date: **(a)** the approved change **deferred** that value beyond week 13; **(b)** it **pulled it forward** inside week 13.
2. **Formula.** `SPI = EV/PV`; `SV = EV − PV`; `CPI = EV/AC`; the composite forecast `EAC = AC + (BAC − EV)/(CPI × SPI)`, alongside the two CPI-only forms `BAC + AC − EV` and `BAC/CPI`.
3. **Substitution.** Reported `1,920,000/2,080,000`. Variant (a) `1,920,000/1,840,000`; variant (b) `1,920,000/2,320,000`. `CPI = 1,920,000/2,120,000` in all three columns.
4. **Result.**

Measure	Reported (stale PV)	(a) deferral — true PV 1,840,000	(b) acceleration — true PV 2,320,000
`SPI`	0.9231	**1.0435**	**0.8276**
`SV` (USD)	(160,000)	**80,000**	**(400,000)**
`CPI`	0.9057	0.9057	0.9057
`EAC = BAC + AC − EV`	4,200,000	4,200,000	4,200,000
`EAC = BAC/CPI`	4,416,667	4,416,667	4,416,667
composite `EAC`	4,608,056	**4,320,972**	**4,895,139**

The same EV, the same AC and the same authorised scope produce an `SPI` anywhere between **0.8276** and **1.0435** — a spread of **0.2159** — according only to whether the PV curve was maintained. `CPI` and both CPI-only forecasts are **identical in all three columns**. 5. **Interpretation.** The identity that falls out of the table is the reason this defect is so durable: **a stale PV curve corrupts every schedule measure and leaves every cost-only measure exactly correct.** So the cost report reconciles to the ledger, the CPI-based forecast is defensible line by line, the finance function signs it — and the schedule indices printed on the facing page are wrong by up to a fifth of an index point. A reviewer who tests only the arithmetic inside the report will find nothing wrong with it, because nothing inside it is wrong. The defect is in the relationship between the report and the schedule, which is the one thing the report cannot show.

The composite `EAC` is where the two halves meet, and it behaves in a way worth remembering because it makes the exposure quantifiable. Written as `AC + (BAC − EV) × PV/(CPI × EV)`, the composite is **linear in PV**, so its sensitivity is constant: `(BAC − EV)/(CPI × EV) =` **1.1962** here, meaning **each USD 1 of mis-phased planned value moves the forecast by USD 1.20**. The 240,000 of mis-phasing therefore moves the composite forecast by **USD 287,083 — 7.18 %** of BAC — and, because the relationship is linear, by exactly that amount in either direction. The error cannot be defended as conservative, which is the usual defence offered.

The two directions have opposite management consequences, and the flattering one is the more common because deferral is the more common change. Variant (a) reports 0.92 on a project actually running at **1.0435** against the authorised schedule: a recovery plan will be launched, overtime authorised and float consumed to fix a problem the project does not have — real cost incurred to correct a reporting defect, and the composite forecast overstated by 287,083 into the bargain. Variant (b) reports the same 0.92 on a project running at **0.8276**: true schedule variance is (400,000) rather than (160,000), the composite forecast is understated by 287,083, and both facts surface at the next milestone rather than in the report designed to surface them. Neither variant contains a false statement and neither is detectable from the indices themselves.

What makes it detectable is one subtraction, and it is the check KA 4.3.3 formalises: does the sum of the approved changes' schedule effects equal the movement in the PV curve? Here it plainly does not — the curve did not move at all — and the finding is available in minutes to anyone who asks the question. That is why the invariant is better stated as a **sequence obligation** than as good practice: **the PV curve is re-phased in the same transaction that approves the change, or it will not be re-phased at all.** Anything deferred to a later housekeeping pass competes with delivery and loses, for the same structural reason KA 4.4.2 gives for impact assessment.

One caution against over-reading the 0.2159 spread. It is the consequence of a single 240,000 re-sequencing on a project whose phasing is otherwise sound. A programme with a year of unmaintained phasing does not have a slightly wrong `SPI`; it has an `SPI` whose relationship to the work is unknown, and no sensitivity calculation recovers it. The honest treatment in that case is to **suspend the index and say so in the report**, restating it only when the baseline has been re-phased against the current schedule. Continuing to publish a schedule index with a footnote about baseline quality is the specific failure to avoid: readers take the number and leave the footnote.

The two-dimensional trap. The commonest expression of a non-integrated baseline is a change approved on cost alone. A change is assessed for its cost, approved, and the schedule is not re-baselined — so the project is delivered late against a baseline that never acknowledged the time, and the lateness is attributed to execution. KA 4.4 prices the general case.

4.3.2 Configuration management

Definition. Configuration management is the discipline of **knowing, at any moment, exactly which version of every controlled item is the approved one**, and how it came to be. Its four functions: identification (what is controlled, and its identifier), control (how a version changes and on whose authority), status accounting (what the current version of everything is), and audit (verifying that the actual state matches the recorded state).

Why it belongs in this domain rather than in quality. Integration depends on it absolutely. When two components are integrated, the question is not whether each works but whether **these versions** work together — and an interface verified against version 3 of a specification tells you nothing about version 5. Every interface agreement is therefore version-bound (KA 4.2.3), and configuration management is what makes the binding meaningful.

WORKED EXAMPLE

Worked example 4.3.2 — auditing Meridian's configuration register.

1. **Setup.** A configuration audit covers Meridian's **340** controlled items. It finds **296** with a single identified current version and a complete change history; **28** with no version reference at all; **11** with two items marked current; and **5** whose recorded current version does not match the version actually deployed.
2. **Formula.** Defect rate = non-conforming items ÷ total. Classify by consequence.
3. **Substitution.** `(28 + 11 + 5)/340`.
4. **Result. 44** non-conforming items — a **12.94 %** defect rate — of which 28 are unidentified, 11 ambiguous and **5 actively wrong**.
5. **Interpretation.** The three classes are not equally serious, and totalling them is the mistake a status report usually makes. The **28 unidentified** items cannot be integrated with confidence because no interface agreement can bind to them; the cost is rework at integration testing. The **11 ambiguous** items will each cause a decision by whoever picks first, and the wrong pick is discovered downstream. The **5 wrong** items are the serious finding: the register says something untrue, so verification performed against the register has verified nothing — and this is the class that produces the failure in which every component passed its own tests and the system did not work. A 12.94 % headline rate understates the position, because those five items alone can fail an integration.

The pairwise consequence, which is where the register is actually used. An item-level defect rate answers a question nobody asks. What integration needs to know is whether **an interface** can be verified, and an interface is verifiable only if both of its sides are on their recorded version. That turns an item-level rate into a relationship-level rate through an exponent, and the exponent is where the surprises are.

WORKED EXAMPLE

Worked example 4.3.2b — why an 87 % register gives a 76 % integration, and what the audit tolerance therefore has to be.

1. **Setup.** WE 4.3.2's audit found **296** of **340** controlled items conforming — **87.06 %**. Meridian has **31** genuinely required interfaces (Case study A's count), each side of each interface bound to a controlled item at a stated version. Treat all 31 as two-sided, which is the optimistic case; the three-party form — programme, national reporting gateway and identity provider — is computed alongside for comparison.
2. **Formula.** For a relationship spanning `k` controlled items whose individual conformance is `p`, and whose defects are independent, the probability that the whole relationship is verifiable is `p^k`. Expected at-risk relationships = `count × (1 − p^k)`. Inverting for design: the item conformance required to achieve a relationship-conformance target `q` across `k` items is `p = q^(1/k)`.
3. **Substitution.** `p = 296/340`. Two-sided `p² = 0.870588²`; three-party `p³`. At-risk `31 × (1 − p²)`. For an expected at-risk count of one interface in 31: `p = (30/31)^(1/2)`.
4. **Result.** Item conformance **87.06 %** yields two-sided interface conformance **75.79 %** and three-party conformance **65.98 %**. Of 31 required interfaces treated as two-sided, **7.50** are expected to be at risk — more where any of them is three-party, since `1 − p³ =` **34.02 %** against `1 − p² =` 24.21 %. Interface-level failure (**24.21 %**) is **1.87 times** item-level failure (12.94 %). To hold the expected at-risk count at **one** interface, item conformance must reach **98.37 %** — at most **5** non-conforming items out of 340. The audit found **44**.
5. **Interpretation.** The exponent is the whole lesson and it is routinely missed: **integration quality is item quality raised to the number of parties**, so a register that looks tolerable where it is measured is materially worse where it is used. 87 % is the kind of number that earns a green status. 75.79 % is not, and 65.98 % on a three-party interface is a different conversation with a different audience. The direction never reverses — the assembled relationship is always worse than its parts — which is why an item-level dashboard over-reports integration readiness systematically rather than occasionally.

The inversion is the practically useful half, because it converts an integration target into an audit tolerance instead of leaving the tolerance to be chosen by whoever writes the procedure. Meridian's audit had **no stated tolerance at all**; had it wanted 95 % of interfaces verifiable it needed `0.95^(1/2) =` **97.47 %** item conformance, which is at most **8** non-conforming items. State the rule in that direction: **the tolerance on the register is derived from the target on the interfaces, and the derivation runs backwards through the exponent.** A configuration tolerance set directly at item level — "under 10 % non-conforming is acceptable" — is a number with no argument behind it, and on a two-sided architecture a 10 % item tolerance silently permits a **19 %** interface failure rate that nobody has agreed to.

The audit interval falls out of the same arithmetic, and the answer is instructive because it is impossible. WE 4.3.2's 44 non-conforming items accumulated over the **6 months** since the previous audit — **7.33** items a month. To stay inside even the loose 5-item tolerance the audit would have to run every `5/7.33 =` **0.68 months**, about **3 weeks**, which no manual configuration audit sustains. That is not an argument for a heroic audit schedule; it is the diagnostic. **When the derived audit interval is shorter than the audit can feasibly run, the finding is about identification discipline, not about audit frequency.** The register is being written badly and has to be fixed at the point of writing — which is the automation case in the AI note below, and the reason software delivery largely solved this problem through version control while manual registers largely have not (see Industry variations).

Two cautions on the model, because it is a bound rather than a measurement. **Independence is a generous assumption**: register defects cluster by team, by system and by the week a migration happened, so where both sides of an interface are maintained by the same people with the same habit the real pairwise conformance is worse than `p^k`. Use `p^k` as an optimistic bound and label it as one. And `p` must be the conformance of the items **actually bound to interface agreements**, not of the whole register: the 340 items include many that no agreement references, and if the non-conforming 44 are concentrated among the referenced ones the interface exposure is worse again. A reviewer's first question is therefore not the headline defect rate but which items the interface register binds to, and what their conformance is measured separately.

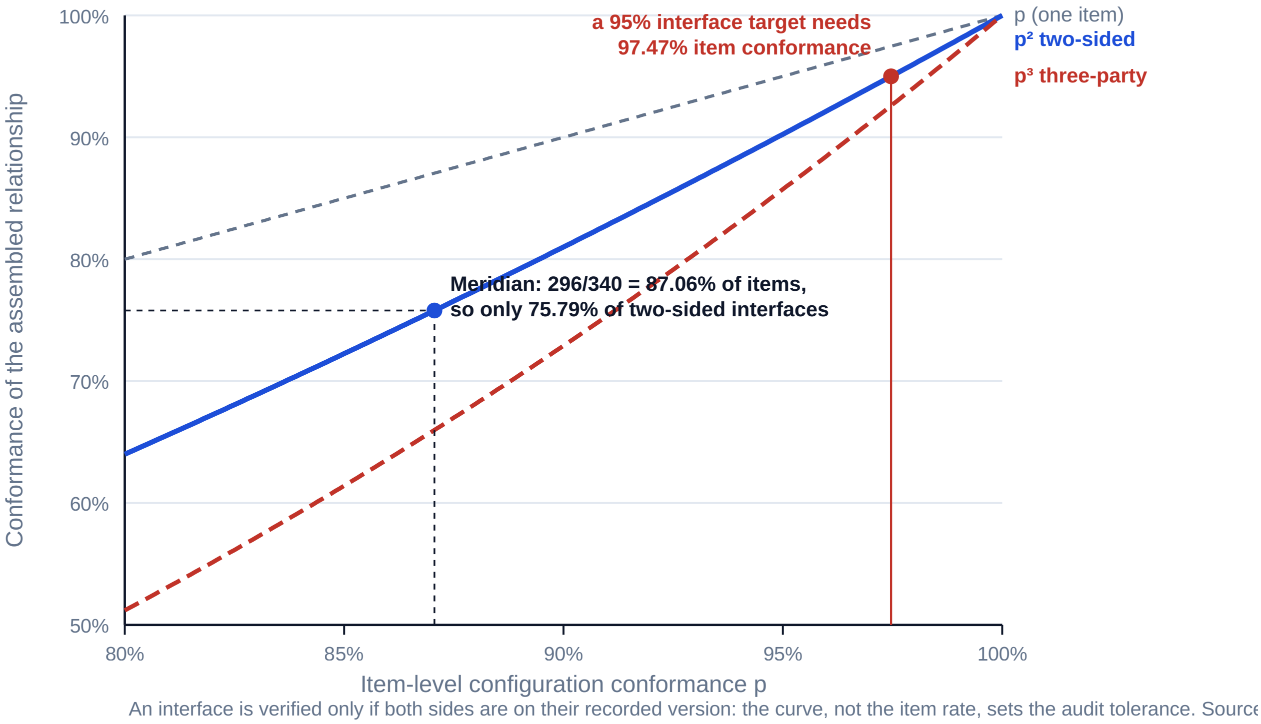


Fig 4.3.1 — Pairwise integrity is the power of item integrity

4.3.3 Baseline maintenance

The obligation. A baseline is not a historical record; it is the current authorised statement of the project, and maintaining it means that every change is applied **with traceability to the authority that approved it**, while the original remains reconstructable.

What must be preserved. The original baseline, unaltered; every approved change with its reference, date, authority and effect on scope, schedule and cost; the current baseline as the sum of the original and the approved changes; and the ability to reproduce any intermediate state. That last requirement is what makes variance analysis honest a year later — and it is the reason a baseline is maintained by accumulation rather than by replacement.

Re-baselining. Occasionally a baseline becomes so distant from reality that variance against it conveys nothing, and re-baselining is legitimate. It is also the most abused instrument in project control, because it makes an adverse variance disappear without anything having improved. Three disciplines keep it honest: re-baselining requires **the authority that approved the original**, not the project's own; the **reason is recorded** and is a substantive change in circumstance rather than accumulated variance; and the **original remains visible in reporting**, so that performance against the original commitment can still be stated. A project on its third baseline with no visible original has lost the ability to answer the only question that matters to the organisation that funded it.

Drift: the failure mode with no decision in it.

WORKED EXAMPLE

Worked example 4.3.3 — how Meridian's baseline moved 12.1 % without a decision.

1. **Setup.** Over year one, Meridian approved **34** changes, each below the project leader's then **10,000** authority, at an average direct cost of **6,800**. Of those, **14** carried critical-path impact averaging **0.3 weeks** each; cost of delay **14,280** per week; baseline **2,400,000**.
2. **Formula.** Direct drift = count × average cost. Schedule drift = affected count × average weeks × cost of delay. Express as a share of baseline, and per change.
3. **Substitution.** `34 × 6,800`; `14 × 0.3 × 14,280`.
4. **Result.** Direct **USD 231,200**; schedule impact `4.2 weeks =` **USD 59,976**; total **USD 291,176 — 12.1 %** of the baseline. Each individual change was **0.28 %** of it.
5. **Interpretation.** There was no point at which anyone decided to spend 291,176 or to accept 4.2 weeks of delay, and no individual approval was wrong on its own terms. This is Domain 3's Case study B mechanism with a number attached, and it is why a delegation schedule needs a **cumulative test**. But the test's parameters have to be derived rather than chosen, and here the arithmetic is unforgiving: a rule of "related changes aggregating above 100,000 in a rolling 90 days requires steering authority" would **not** have caught this, because 34 changes a year is about `34/4 = 8.5` per quarter, or **57,800** — comfortably under 100,000. Catching it needs either a threshold below **57,800** on a 90-day window or the same 100,000 threshold on a **180-day** window (`17 × 6,800 =` **115,600**, which trips it). The professional point generalises past this example: **a cumulative test set at a round number without reference to the observed change rate provides the appearance of a control and none of the function** — and the observed change rate is available from the change log of any project that has run for a quarter.

The check that finds what the cumulative test cannot. Drift of the kind above is entirely inside the change log, so summing the log detects it. The complementary failure — baseline movement that never reached the log — is invisible to any cumulative test, because cumulative tests read the log. It is caught instead by the reconciliation that maintenance by accumulation makes possible: original plus the sum of approved changes equals the current baseline, exactly. It is one addition and one subtraction, and it is habitually performed wrongly in the same way.

WORKED EXAMPLE

Worked example 4.3.3b — Meridian's reconciliation, and the residual that netting hid.

1. **Setup.** Meridian's original approved baseline is **USD 2,400,000**. The year-one change log holds the **34** small changes above (**231,200** in total) and **3** steering-approved changes at **86,000**, **142,000** and **57,500**. The cost tool reports a current baseline of **USD 2,948,900**.
2. **Formula.** Original + Σ approved changes = current baseline. Residual = current − expected. Then **decompose the residual by class before netting**: approved changes with no recorded baseline effect (which make the tool lower than the log implies) against baseline movements with no change reference (which make it higher).
3. **Substitution.** Approved total `231,200 + 86,000 + 142,000 + 57,500`. Expected current `2,400,000 + 516,700`. Residual `2,948,900 − 2,916,700`.
4. **Result.** Approved changes total **USD 516,700**; expected current baseline **USD 2,916,700**; residual **USD 32,200**, or **1.34 %** of the original. Investigation decomposes it: **2** approved changes worth **44,900** (18,400 and 26,500) carry no baseline effect, and baseline movements totalling **77,100** carry no change reference — `77,100 − 44,900 = 32,200`, which closes. The **gross** error is **USD 122,000**: **23.61 %** of the approved changes and **5.08 %** of the original baseline. Netting concealed **USD 89,800** of it.
5. **Interpretation.** The arithmetic that matters is that **the two classes have opposite signs, so netting them destroys the finding.** A residual of 32,200 on a 2.4 m baseline is 1.34 % and will be written off as timing or rounding by any reviewer working at that resolution, while the actual defect is **3.79 times** larger. Hence the single most useful sentence in this KA: **reconcile gross, never net.** A reconciliation that balances to zero is not evidence of integrity; it is evidence that the errors happened to be of similar size. The concealment is not random either — it is exactly twice the smaller class, `2 × 44,900 = 89,800`, which is a useful thing to be able to say out loud when someone proposes to report the net.

The two classes are not equally serious and they have different remedies. **44,900 of approved-but-unrecorded changes** means work has authority and no budget line: it will be delivered and charged, and the variance will surface as an overrun against a baseline that never received the change. The remedy is bookkeeping — post the effects; the authority is already on file. **77,100 of movement with no change reference** is the governance finding: budget exists that no recorded decision created. Its remedy is not a posting but an investigation, because there are two possible answers and they are very different. Either a decision was taken and not recorded — a Domain 3, KA 3.3.4 failure, and one of the three things KA 4.1.3 places outside tailoring — or no decision was taken at all, and someone has moved a baseline without authority. A reconciliation report that states the residual and not its classes has withheld precisely the information that distinguishes those two.

Note the relationship to drift, because the two controls are complementary rather than overlapping. The 291,176 of drift above was **entirely inside the change log** — every dollar authorised, recorded and traceable, invisible only because nobody summed the column. The 77,100 here is **outside the log altogether**, so no cumulative test at any threshold or window would ever see it. **Drift controls and reconciliation controls catch disjoint failures**, and a programme with an excellently derived cumulative test and no reconciliation has covered one of the two and will believe it has covered both.

Cost and cadence are worth stating because they are the usual objection. The reconciliation is one addition and one subtraction over three figures a programme already holds — call it an hour a month, **USD 1,320** a year at the blended rate, against 122,000 of error found in year one. Run it monthly rather than at gates, and for an arithmetic reason rather than a diligence one: the check does not get harder with delay, but the decomposition does. Finding which movements are unreferenced is a search over one month's transactions if run monthly and over a year's if run annually, and it is the decomposition that carries all the value.

What a reviewer should test, in order: that the three figures are quoted from three systems of record rather than from one report that derives two of them; that the residual is decomposed before it is characterised; and that a **zero** residual has been tested rather than assumed — because the commonest way to produce one is to derive the current baseline from the change log, which makes the identity true by construction and the control worthless. A reconciliation that has never produced a non-zero residual is not a clean programme; it is an unexamined formula.

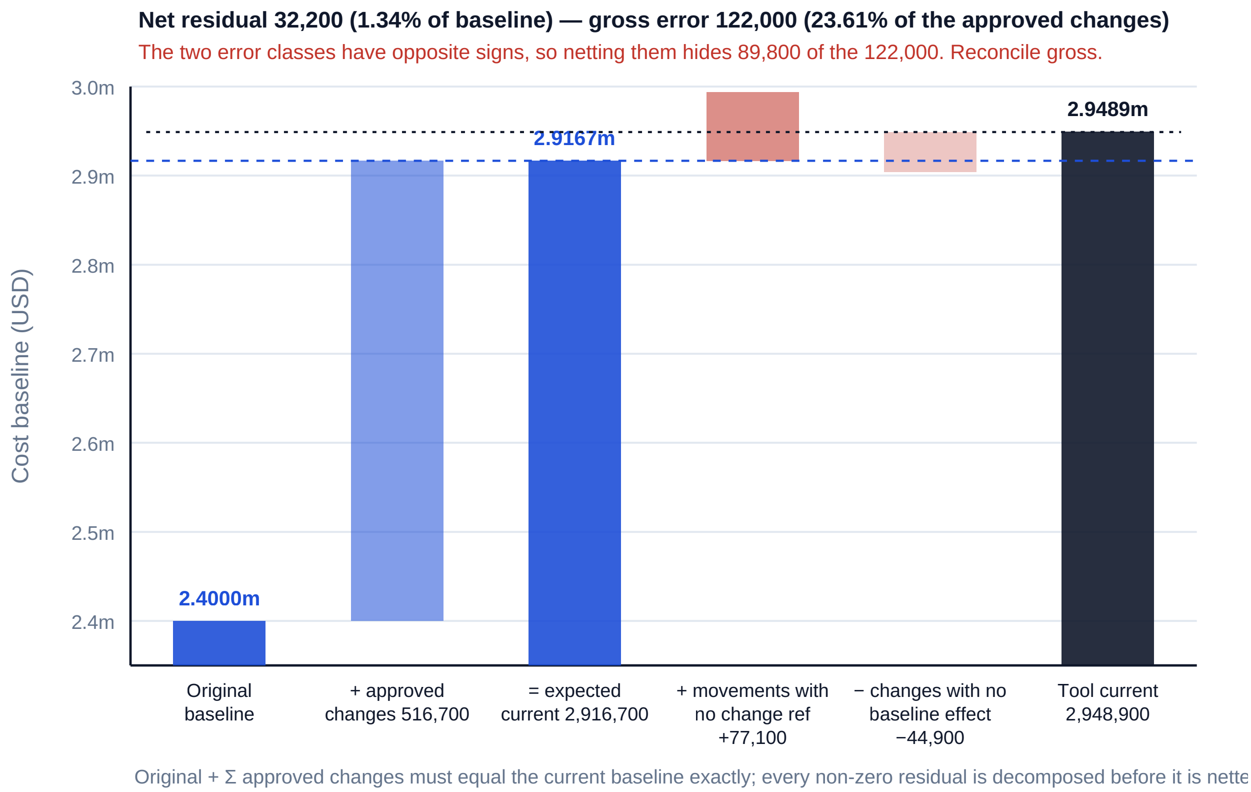


Fig 4.3.2 — Reconcile gross, never net

AI in this KA

Where it earns its place. Continuous baseline-integrity checking: does the time-phased cost baseline still match the schedule's dates; do the change log's entries sum to the difference between the original and current baselines; are there approved changes with no baseline effect recorded, or baseline movements with no change reference? All four are reconciliations with definite answers, and the fourth in particular is how drift is detected early, and the second is the reconciliation of WE 4.3.3b, which must be decomposed by class rather than netted. Configuration status accounting — comparing the register against a deployed inventory and reporting the discrepancy classes of WE 4.3.2. Joining the configuration register to the interface register to compute the pairwise exposure of WE 4.3.2b, which is the only way to know which interfaces the item defects actually threaten. Trend analysis on the change log to derive the observed change rate a cumulative test should be set from, and the register's defect accumulation rate an audit interval should be set from.

Where it must not go. Authorising a baseline change or a re-baseline. Reconciling a discrepancy by adjusting the register to match reality, which destroys the evidence that the discrepancy existed —

the correction must be a recorded, authorised change, and this is a real and tempting automation failure rather than a hypothetical one.

Verification, concretely. Every reported discrepancy is confirmed against both sources before it is treated as a finding; the change-log-sums-to-baseline-delta reconciliation is reproduced by hand at each reporting period, because it is a subtraction and it is the single most informative check in this KA; and no register is amended by a tool without an authorised change reference.

KEY TERMS — KA 4.3

Term	Meaning
Performance measurement baseline	The authorised scope, arranged in time, with cost attached — one statement in three documents.
Time-phased cost baseline	The `PV` curve; it must move whenever the schedule does.
Configuration management	Knowing which version of every controlled item is approved, and how it came to be.
Status accounting	The record of the current version of every controlled item.
Configuration audit	Verification that the actual state matches the recorded state.
Pairwise configuration integrity	`p^k` — the conformance of a relationship spanning `k` controlled items, always worse than the item rate `p`.
Derived audit tolerance	The item-level tolerance obtained by inverting `p^k` from an interface-level target: `p = q^(1/k)`.
Original-to-current reconciliation	Original baseline + Σ approved changes = current baseline, exactly, every reporting period.
Gross residual	The sum of the reconciliation's error classes before netting — the figure that carries the finding.
Baseline maintenance	Applying changes by accumulation, with traceability to authority, preserving the original.
Re-baselining	Replacing a baseline that no longer conveys information — legitimate, and the most abused instrument in project control.
Baseline drift	Cumulative movement through individually authorised small changes, with no decision on the total.
Cumulative test	A threshold on aggregated related changes over a stated period, set from the observed change rate.

SAMPLE MCQS — KA 4.3

MCQ 4.3-A `[4.3.1 · Analysis]` A schedule change is approved and the time-phased cost baseline is not updated. The most serious consequence is that: - A. the cost baseline is slightly inaccurate - B. earned value indices now measure the gap between two documents rather than the state of the work
- C. the schedule must be re-baselined - D. contingency is understated

Rationale: `SPI` and `CPI` compare achievement against the time-phased baseline; if it no longer matches the executing schedule they produce confident numbers about nothing (4.3.1).

MCQ 4.3-B `[4.3.2 · Evaluation]` In a 340-item configuration audit, 28 items have no version reference, 11 have two current versions and 5 have a recorded version that differs from what is deployed. Which is the most serious class, and why? - A. the 28, because they are the most numerous

- B. the 11, because ambiguity blocks decisions - C. the 5, because verification against the register has verified nothing - D. all three are equally serious at a 12.94 % defect rate

Rationale: A register that states something untrue invalidates any verification performed against it, which is the class that produces "every component passed and the system failed" (4.3.2).

MCQ 4.3-C `[4.3.3 · Application]` 34 changes averaging 6,800 direct cost were approved over a year, 14 of them carrying 0.3 weeks of critical-path impact at a delay cost of 14,280 per week. Total baseline drift is closest to: - A. USD 231,200 - B. USD 291,176 - C. USD 59,976 - D. USD 376,856

Rationale: `34 × 6,800 = 231,200` direct plus `14 × 0.3 × 14,280 = 59,976` schedule impact (4.3.3). A omits the schedule impact; C is the schedule impact alone; D applies the schedule impact to all 34 changes rather than the 14 that carried it.

MCQ 4.3-D `[4.3.3 · Evaluation]` For the drift above, a cumulative test of "related changes above 100,000 in a rolling 90 days" would: - A. have caught it, since total drift exceeds 100,000 - B. not have caught it, since a quarter's changes aggregate to about 57,800 - C. have caught it only if the threshold were raised - D. be inapplicable to changes below the delegation threshold

Rationale: The 90-day aggregate is `(34/4) × 6,800 ≈ 57,800`, below the threshold (4.3.3) — a cumulative test set at a round number without reference to the observed change rate has the appearance of a control and none of the function.

MCQ 4.3-E `[4.3.3 · Comprehension]` The discipline that most protects re-baselining from abuse is: - A. re-baselining no more than once a year - B. keeping the original baseline visible in reporting alongside the current one - C. having the project leader approve it - D. recalculating contingency at each re-baseline

Rationale: Visibility of the original preserves the ability to state performance against the original commitment (4.3.3). C is precisely the wrong authority.

MCQ 4.3-F `[4.3.2 · Application]` A configuration register holds 340 items of which 296 conform. Assuming independent defects, the expected share of **two-sided interfaces** whose verification cannot be relied on is closest to: - A. 12.94 % - B. 24.21 % - C. 25.88 % - D. 75.79 %

Rationale: `1 − (296/340)² = 1 − 0.7579 = 0.2421` (4.3.2b). A applies the item failure rate directly to interfaces; C doubles the item rate, which is the linear approximation and overstates slightly; D quotes the conformance rather than the failure.

MCQ 4.3-G `[4.3.3 · Analysis]` A reconciliation shows a residual of 32,200, which decomposes into 77,100 of baseline movement with no change reference and 44,900 of approved changes with no baseline effect. The correct reading is that: - A. the residual is 1.34 % of the baseline and immaterial - B. the gross error is 122,000 and must be reported as such, because the two classes have opposite signs - C. the cost tool is correct and the change log needs updating - D. netting is appropriate because the classes offset within the same baseline

Rationale: Opposite-signed classes make the net understate the gross by twice the smaller class (4.3.3b), so 89,800 of the 122,000 is concealed. A and D both read the net as the finding — A by applying a materiality test to it, D by asserting the offset is legitimate. C prejudges which record is wrong when the two classes have opposite remedies.

MCQ 4.3-H `[4.3.1 · Analysis]` A schedule change is approved and the `PV` curve is never re-phased. Which reported measure remains exactly correct? - A. `SPI` - B. `SV` - C. `CPI` - D. the composite `EAC = AC + (BAC − EV)/(CPI × SPI)`

Rationale: `CPI = EV/AC` contains no PV term, so it and the CPI-only forecasts are unaffected, which is why the defect passes a cost review (4.3.1). A and B are computed against PV directly; D is linear in PV and moves by USD 1.20 per USD 1 mis-phased on Auriga's figures.

SELF-CHECK — KA 4.3

1. Why is a schedule change always a cost-baseline change? — Because the cost baseline is time-phased; if its curve does not move, earned value stops measuring the work.
2. Which earned-value measures does a stale *PV* curve leave correct, and why does that matter? — `CPI` and both CPI-only forecasts, because they contain no PV term; it matters because the cost review therefore passes and the defect survives.
3. Which configuration-audit finding class is most serious, and why? — Recorded versions that differ from deployed ones, because verification against the register then proves nothing.
4. How is an item-level configuration tolerance derived? — By inverting `p^k` from the interface-level target: `p = q^(1/k)`. Chosen directly at item level it has no argument behind it.
5. Why must a reconciliation residual be reported gross? — Because opposite-signed error classes make the net understate the gross by twice the smaller class, and a zero residual is not evidence of integrity.
6. How should a cumulative-change threshold be set? — From the observed change rate in the change log, not from a round number.

KNOWLEDGE AREA 4.4

Integrated change control

Topics: 4.4.1 change flow · 4.4.2 impact assessment · 4.4.3 the change board and the decision log.

4.4.1 The change flow

Why "integrated". The word is the whole point: a change to scope is a change to schedule and to cost, and a control process that assesses one dimension has not controlled the change. The flow that does:

1. **Raise** — anyone may raise; the request states the change and its reason, and is logged on receipt with a reference, so that a rejected or withdrawn request still leaves a trace.
2. **Screen** — is it a change at all? Three outcomes are commonly confused and should be separated explicitly: a **change** (the baseline moves), a **clarification** (the baseline already covered it and someone misread it), and a **defect** (the work does not meet the baseline and must be corrected at no baseline movement). Misclassifying a defect as a change is how a supplier is paid twice; misclassifying a change as a clarification is how scope grows silently.
3. **Assess impact** — across all dimensions, per KA 4.4.2. This is the step that is under-resourced, and the reason is that assessment costs money before any decision has been taken to spend money.
4. **Decide** — at the authority the assessed impact requires, not the quoted cost (4.4.2's central point), with the decision recorded per Domain 3, KA 3.3.4.
5. **Implement and baseline** — update scope, schedule and cost together, with the change reference recorded against each, so that KA 4.3's reconciliation works.
6. **Verify and close** — confirm the change was implemented as approved, including its interface and documentation consequences.

Of the six steps, screening is the only one whose output is a **classification** rather than a number or a decision, and classifications are the cheapest thing in the flow to get right and the most expensive to reverse, because by the time a misclassification is discovered money has moved — in one direction or the other, and the two directions do not cost the same.

WORKED EXAMPLE

Worked example 4.4.1 — what Meridian's screening errors cost, in both directions.

1. **Setup.** Meridian logged **76** requests in year one. Screening ruled **51** changes, **16** clarifications and **9** defects; of the 51 changes, **37** were subsequently approved — the 34 below the leader's threshold plus the 3 taken to steering — and **14** were rejected or withdrawn. An assurance re-screen against the baseline found **8** misclassifications: **5** of the approved "changes" were **defects** — work already owed under the baseline, at the change log's average approved value of **6,800** — and **3** of the 16 "clarifications" were **changes**, carrying direct work of **9,200**, **14,600** and **5,400**, one of them with **0.5 weeks** of critical-path impact (cost of delay **14,280** per week).
2. **Formula.** Error rate = misclassifications ÷ requests. Defect-as-change cost = count × average approved value (the sum paid a second time for an existing obligation). Change-as-clarification cost = Σ direct + schedule impact (unbudgeted, unlogged, and outside the baseline). Then compare the **net** effect on the recorded change total with the **gross** error.
3. **Substitution.** `8/76`. `5 × 6,800`. `9,200 + 14,600 + 5,400 + 0.5 × 14,280`.
4. **Result.** Screening error rate **10.53 %**. Defects paid as changes **USD 34,000** — which is **14.71 %** of the 231,200 of direct drift computed in WE 4.3.3, so that figure includes that much work which was never a change at all. Changes taken as clarifications **USD 36,340**, none of it in any baseline or log. Total misclassification cost **USD 70,340**, or **USD 8,792.50** per error. The **net** effect on the recorded change total is `36,340 − 34,000 =` **USD 2,340** — **0.10 %** of the baseline — giving a corrected drift of **293,516**, or **12.23 %**, against WE 4.3.3's 291,176.
5. **Interpretation.** The two errors point in opposite directions and — for the second time in this domain, from wholly independent data — **netting them hides both.** A corrected drift figure that moves by 2,340 is inside anyone's materiality threshold while concealing 70,340 of gross misclassification, and the concealment is again exactly twice the smaller class: `2 × 34,000 =` **68,000** of the 70,340, where WE 4.3.3b concealed `2 × 44,900 =` 89,800 of 122,000. That recurrence is a property rather than a coincidence, and it is worth carrying out of the domain as one: **wherever a reconciliation has two error classes of opposite sign, the net understates the gross by twice the smaller class.** Report both figures or report neither.

The two errors also differ in **how they are discovered**, and that asymmetry decides where the control belongs. Paying twice for an existing obligation shows up in the cost report: money leaves, a variance appears, someone asks. Silent scope growth shows up nowhere — no change record, no baseline movement, no variance until the work is executed, at which point it is indistinguishable from ordinary productivity. So the direction that costs slightly more on this year's arithmetic (36,340 against 34,000) is also the direction with no downstream detector, and it is the one that needs the tighter control. A screening process designed only to stop overpayment has been designed against the visible half.

Whether the 34,000 can actually be recovered is a contractual question, not a project one. It turns on the terms, on any notice or claim period, on whether the work was accepted and paid without reservation, and on the governing jurisdiction. **Take contractual advice before treating a misclassification finding as recoverable income**: the arithmetic establishes that value was transferred, not that it can be reclaimed, and a leader who reports the second when they have only established the first will be corrected expensively.

The remedy is cheap and specific, and it is a change of form rather than of effort. Screening becomes a **recorded ruling** instead of an administrative sort: the baseline clause or requirement relied on is cited, a named technical authority signs, and the three outcomes are separate fields rather than a checkbox. Costed at half an hour of a technical authority and half an hour of the change manager per request — **USD 110** at the blended rate, **USD 8,360** across 76 requests — against 70,340 of error, a ratio of **8.41**. The ratio is not the point and will vary; the point is that the ruling is the only artefact in the flow that makes a classification **contestable later**, and an uncontestable classification is one nobody will revisit.

One caution about the base rate, because a 10.53 % error rate invites the wrong conclusion. It is not in itself evidence of poor screening: some proportion of requests genuinely sits on the change/defect boundary and will be ruled differently by two competent reviewers, and the honest comparison is against that irreducible rate rather than against zero. What a reviewer should test is whether the disputed rulings were disputed **on the record**. A screening process that has reversed no ruling in a year is not accurate; it is unexamined.

The discipline that holds it together. No work begins on an unapproved change. That is easy to write and hard to hold, because the pressure is always to start — and the specific damage is not the work itself but that it removes the option to say no, converting a decision into a ratification. Where genuine urgency exists, the answer is an **emergency change route** with a named authority, a short deadline and mandatory retrospective ratification, which is the same instrument as Domain 3's out-of-cycle mechanism. An emergency route that exists is used and recorded; one that does not exist is used and not recorded.

And the route's usage rate is a diagnostic, not a discipline problem. Meridian's 76 requests included **5** routed as emergencies — **6.58 %**. A rising share is almost never indiscipline; it is the normal route's latency being priced by the people who have to live with it, and the correct response is to reduce that latency — the paper deadline before the meeting interval, per Domain 3's lever asymmetry — rather than to tighten the emergency criteria, which merely moves the same work into unrecorded starts and removes the only trace it was leaving. So set a **review trigger** on the share rather than a target for it: when the share rises, examine the normal route, not the requesters.

4.4.2 Impact assessment — what a change actually costs

The central claim of this KA. The figure on a change request is the direct cost of doing the new work, and it is usually the smallest component of the change's true cost. Everything else — schedule consequence, rework of completed work, interface re-verification, regression testing, documentation and training — is real, incurred, and absent from the form.

A complete assessment covers: **direct cost** of the new work; **schedule** impact, on the critical path or on float, priced at the cost of delay; **rework** of work already completed to the old specification; **interface** re-verification for every affected interface (KA 4.2.3); **regression testing** of what was already proven; **documentation, training and communication** updates; **risk** profile change (Domain 8); and **benefit** change, since a change that reduces benefit has a cost that appears nowhere in a cost assessment (Domain 2).

WORKED EXAMPLE

Worked example 4.4.2 — the change that cost 3.29 times its quoted price.

1. **Setup.** Meridian receives a change request to add a national-reporting field set. The quoted direct build cost is **USD 40,000**. Assessment establishes: **2 weeks** of critical-path impact (cost of delay **14,280** per week); **22,000** of rework to records-application screens already completed; **3** affected interfaces requiring re-verification at **6,000** each; **14,000** of regression testing; and **9,000** of documentation and clinician-training updates.
2. **Formula.** True cost = direct + (schedule weeks × cost of delay) + rework + (interfaces × re-verification) + regression + documentation.
3. **Substitution.** `40,000 + 2 × 14,280 + 22,000 + 3 × 6,000 + 14,000 + 9,000`.
4. **Result.** `40,000 + 28,560 + 22,000 + 18,000 + 14,000 + 9,000 =` **USD 131,560 — 3.29 times** the quoted figure. The quoted cost is **30.4 %** of the true cost.
5. **Interpretation.** Two consequences follow, and the second is the more important because it is structural. First, a change board deciding on the quoted 40,000 is deciding on 30 % of the information, and it will approve changes it would have refused — not through any failure of judgement but because the number in front of it was the wrong number. Second, and this connects directly to Domain 3: **a delegation threshold applied to quoted direct cost is not a control.** Meridian's threshold is 25,000, so this change escalates. But a change quoted at **22,000** with the same two weeks of critical-path impact carries a true cost of at least `22,000 + 28,560 =` **USD 50,560** — twice the threshold — and is decided by the project leader alone, because the form asked for the direct cost. The remedy is a single sentence in the delegation schedule — the threshold reads on assessed total impact, not on quoted direct cost — and it is worth more than most process improvements a programme will make. Note, finally, what this arithmetic is not: it is not an argument against change. Changes are frequently worth 3.29 times their quoted cost, and Domain 2's business-case logic still decides. It is an argument for **assessing before deciding**.

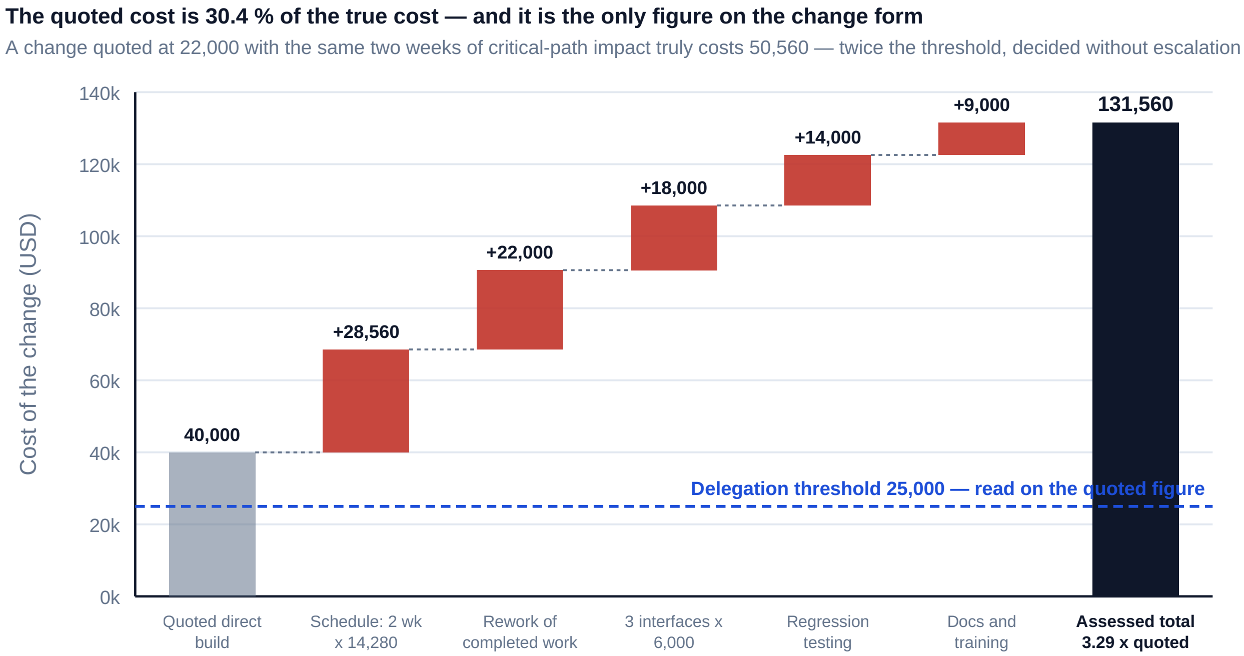


Fig 4.4.1 — What a change actually costs

Assessment discipline. Three rules make assessment reliable rather than performative. The assessment is done by the people who will do the work, not by the change board estimating on their behalf. It is **timeboxed and resourced** — an assessment budget exists, because otherwise assessment competes with delivery and loses, which is the actual reason changes are decided on quoted costs. And its **assumptions are stated**, particularly the schedule assumption, since whether two weeks lands on the critical path or on float is the difference between 28,560 and nothing, and it is a fact about the schedule rather than a matter of opinion (Domain 6's float).

4.4.3 The change board and the decision log

The change board. A body with authority to approve baseline changes within stated limits and to escalate beyond them. Its membership must include the authority to commit cost, the authority to commit schedule, and the technical authority to judge feasibility — and the last of those is the one usually missing, which is how changes are approved that cannot be implemented as described.

Cadence and latency. Domain 3's arithmetic applies unchanged: a change board meeting fortnightly with a one-week paper deadline imposes `E[wait] = 2/2 + 1 =` **2 weeks** on every escalated change. Where change volume is high, that latency is a delivery cost and is reduced the same way — a shorter paper deadline first, then a written-resolution route for changes below a stated impact.

The decision log, and the entry that must exist. Every change decision — approved, rejected, deferred or withdrawn — generates an entry: reference, date, decision-maker by name, the decision, the assessed impact across all dimensions, the basis, and the baseline effect. **Rejections matter as much as approvals**, and are the entries most often missing: a rejected change that leaves no trace returns in three months as a new request, is assessed again at full cost, and may be approved by a different body in ignorance of the first decision. The log is what prevents that, and it is the same log as Domain 3's, not a second one — one register of record for every decision that moves the baseline, whoever took it and at whatever value, which was Case study B's correction.

WORKED EXAMPLE

Worked example 4.4.3 — pricing the rejection entries Meridian did not write.

1. **Setup.** Of the **51** requests WE 4.4.1 ruled as changes, **14** were rejected or withdrawn. The decision log records **5** of them with a rationale; **9** left no trace. Within the same year, **4** of those 9 returned as fresh requests. A full impact assessment consumes **5 person-days** at the blended **USD 880 — USD 4,400**. One of the four returning requests was approved by the programme board in ignorance of the earlier refusal, at an assessed impact of **USD 63,000**. Writing a rejection entry — reference, date, decision-maker, reason, and the assessment already performed — takes about a quarter of an hour, **USD 27.50** at the same rate.
2. **Formula.** Duplicated assessment = recurrences × assessment cost. Total loss = duplicated assessment + the assessed impact of any decision reversed in ignorance. Compare with the cost of recording every rejection, and compare the **mean** loss per recurrence with the **modal** one.
3. **Substitution.** `4 × 4,400`; `+ 63,000`; against `14 × 27.50`.
4. **Result.** Rejections recorded **35.71 %**; recurrence among the untraced **44.44 %**. Duplicated assessment **USD 17,600**; one decision reversed in ignorance **USD 63,000**; total **USD 80,600**, or **USD 20,150** per recurrence. Recording all 14 rejections would have cost **USD 385**.
5. **Interpretation.** The distribution matters more than the total, and a leader should present both figures or neither. Three of the four recurrences cost only the wasted re-assessment — **USD 4,400** each, irritating and survivable. The fourth cost **63,000**, so the mean of 20,150 is **4.58 times** the modal case. Strip the reversed decision and the year's loss is 17,600 against 385 of recording cost, a ratio of **45.71**; include it and the ratio is **209.35**. Quote only the second and a competent reviewer will discount the whole argument as anecdote; quote only the first and you have understated the exposure by the one case that mattered. **The honest statement is that the modal cost of an untraced rejection is a wasted assessment and its tail cost is a reversed decision, and the control is bought for the tail.**

The tail case is also the only part the log uniquely prevents, and the mechanism is worth being precise about. Re-assessment waste can be attacked other ways — duplicate detection on request text, the third item in the AI note below, catches much of it. A **decision reversed in ignorance** cannot be caught that way: the second body was not careless, it simply had no record that the question had been asked and answered. That makes the rejection entry not an administrative courtesy but the instrument that makes a "no" **durable** — and a governance structure in which "no" does not persist is one where any sufficiently persistent requester eventually obtains a yes, from whichever authority happens to be sitting, at no point through anyone's misconduct.

Note the interaction with KA 4.4.2's assessment budget, because it converts this into an argument the budget holder cares about. Assessment is chronically under-resourced precisely because it precedes the decision to spend, and **17,600 of the year's assessment budget was spent twice on questions already answered** — four assessments of capacity that the same money could have applied to new requests. The rejection entry is therefore the cheapest available increase in assessment capacity, and it is worth putting to the budget holder in exactly those terms rather than as a compliance point.

The caution is about what the entry must contain. A rejection entry must carry the **assessment**, not merely the outcome: an entry reading "rejected — not affordable" saves nothing, because a returning request still has to be assessed from scratch to establish what it now costs. The entry that saves the

4,400 records the assessed impact, its basis and its date, so that a recurrence becomes a question about what has changed since — usually answerable in an hour, and occasionally answerable with a legitimate "the circumstances have changed and the answer is now yes", which is the outcome the log exists to make possible rather than to prevent.

AI in this KA

Where it earns its place. Assembling the components of an impact assessment — identifying which interfaces an affected component touches (from the interface register), which completed work packages are implicated, which test suites regress, which documents and training materials cite the changed behaviour. This is traversal of structured relationships, it is where human assessment is systematically incomplete, and it is checkable. Screening a request against the baseline to propose whether it is a change, a clarification or a defect — a proposal, for human ruling. Detecting duplicate or previously rejected requests in the log, which is exactly the failure the previous paragraph describes. Computing the assessed total against the delegation schedule and flagging where quoted and assessed costs fall on opposite sides of a threshold.

Where it must not go. Approving or rejecting a change. Estimating the direct cost or the schedule impact, which belongs to the people who will do the work and to the schedule respectively — a model asked for a schedule impact will produce a plausible number with no critical-path basis, and that number is worth 28,560 in the example above. And no classification of a defect as a change without human ruling, since that determination has commercial consequences.

Verification, concretely. Every AI-identified affected interface, work package and test suite is confirmed by the accountable owner before it enters the assessment; the schedule impact is taken from the schedule, by the planner, with the float position stated; the arithmetic of the total is reproduced by hand; and the decision record names the human decision-maker, with the AI's contribution recorded as assessment input rather than as authorship.

KEY TERMS — KA 4.4

Term	Meaning
Integrated change control	Assessing and authorising a change across scope, schedule and cost together.
Change vs clarification vs defect	Baseline moves · baseline already covered it · work fails to meet the baseline.
Screening ruling	The recorded classification: the baseline clause relied on, the outcome, and a named technical authority — the only artefact that makes a classification contestable later.
Emergency-route share	Emergency changes ÷ total requests — a latency diagnostic, so set a review trigger on it rather than a target.
Assessed total impact	Direct + schedule + rework + interface re-verification + regression + documentation + risk + benefit.
Quoted direct cost	The figure on the form — typically a minority of the true cost, and the wrong basis for a threshold.
Emergency change route	A named authority, a short deadline and mandatory retrospective ratification.
Change board	The body authorised to approve baseline changes within limits, including a technical authority.
Rejection entry	The log record of a change not approved — the entry most often missing and the reason requests recur.

SAMPLE MCQS — KA 4.4

MCQ 4.4-A `[4.4.2 · Application]` A change is quoted at 40,000 direct. Assessment adds 2 weeks of critical-path impact at 14,280 per week, 22,000 of rework, 3 interfaces at 6,000 each, 14,000 of regression testing and 9,000 of documentation. The assessed total is: - A. USD 40,000 - B. USD 103,000 - C. USD 131,560 - D. USD 109,560

Rationale: `40,000 + 28,560 + 22,000 + 18,000 + 14,000 + 9,000 = 131,560` (4.4.2). B omits the schedule impact; D omits the rework.

MCQ 4.4-B `[4.4.2 · Evaluation]` A programme's delegation threshold is 25,000 on quoted direct cost. A change quoted at 22,000 carries 2 weeks of critical-path impact at 14,280 per week. The structural defect is that: - A. the threshold is too low - B. a change with an assessed impact of at least 50,560 is decided without escalation, because the threshold reads on the quoted figure - C. the change should be split into smaller changes - D. the cost of delay should be excluded from change assessment

Rationale: The threshold's basis is the defect, not its level (4.4.2); the remedy is that it reads on assessed total impact.

MCQ 4.4-C `[4.4.1 · Analysis]` Classifying a defect as a change results in: - A. the baseline correctly reflecting the work - B. the supplier being paid twice for the same obligation - C. faster resolution at no cost - D. an unnecessary escalation

Rationale: A defect is work already owed under the baseline; treating it as a change adds budget for it (4.4.1). The mirror error — a change classified as a clarification — grows scope silently.

MCQ 4.4-D [4.4.3 · Comprehension] The change-log entry type most often missing and most consequential is: - A. the approval entry - B. the rejection entry - C. the implementation confirmation - D. the assessed cost

Rationale: An untraced rejection returns as a new request, is re-assessed at full cost, and may be approved by a different body in ignorance of the first decision (4.4.3).

MCQ 4.4-E [4.4.2 · Analysis] Why is impact assessment systematically under-resourced? - A. assessors lack the skill - B. it costs money before any decision has been taken to spend money, so it competes with delivery and loses - C. change boards prefer quoted figures - D. assessment adds no value to rejected changes

Rationale: The structural cause is that assessment is unfunded work preceding a decision (4.4.1, 4.4.2); the remedy is an explicit assessment budget.

MCQ 4.4-F [4.4.1 · Application] A re-screen finds 5 defects paid as changes at an average 6,800, and 3 changes ruled as clarifications carrying 29,200 of direct work plus 0.5 weeks of critical-path impact at 14,280 per week. The gross misclassification cost is: - A. USD 34,000 - B. USD 36,340 - C. USD 70,340 - D. USD 2,340

Rationale: `34,000 + 36,340 = 70,340` (4.4.1). A and B are the two classes taken alone; D is the net, which understates the gross by twice the smaller class and is the specific error the example exists to name.

MCQ 4.4-G [4.4.3 · Evaluation] Four untraced rejections recurred: three cost only a duplicated 4,400 assessment, and one was approved by another body at an assessed impact of 63,000. The defensible way to present the case for recording rejections is to: - A. quote the 209.35 ratio of total loss to recording cost - B. quote the 45.71 ratio excluding the reversed decision - C. quote both, stating that the modal cost is a wasted assessment and the tail cost is a reversed decision - D. quote the mean loss of 20,150 per recurrence as typical

Rationale: The mean is 4.58 times the modal case, so either ratio alone misleads — one as anecdote, the other by omitting the case that mattered (4.4.3). D presents a tail-dominated mean as typical.

SELF-CHECK — KA 4.4

1. Name the three screening outcomes and the cost of confusing two of them. — Change, clarification, defect; a defect treated as a change pays twice, a change treated as a clarification grows scope silently.
2. Which screening error needs the tighter control, and why? — A change ruled as a clarification, because overpayment surfaces in the cost report while silent scope growth has no downstream detector.
3. Why is a threshold on quoted direct cost not a control? — Because the quoted figure is typically a minority of the assessed impact, so changes above the real threshold are decided below it.
4. What must a rejection entry contain to be worth writing? — The assessed impact, its basis and its date, so that a recurrence is a question about what has changed since.
5. What makes a schedule impact assessment a fact rather than an opinion? — Whether the impact lands on the critical path or on float is a property of the schedule, and the float position must be stated.

Advanced topics — Domain 4

4.A.1 Integration across organisational boundaries

Where delivery spans organisations — client and supplier, joint venture partners, a programme and its constituent projects — integration acquires a failure mode that no internal design has: **each party integrates to its own baseline.** Version 4 of an interface specification on one side and version 5 on the other is not an argument about competence; it is the predictable consequence of two configuration registers with no reconciliation, and it is discovered at integration testing where it is most expensive.

Four provisions address it, and they belong in the contract rather than a plan (Domain 7; PFL-AI Domain 11): a **single interface register** identified as the register of record, with named owners on both sides of every interface; a **joint change process** for anything crossing the boundary, with one decision log rather than two reconciled ones; **matched configuration identification**, so that both parties refer to the same item by the same identifier at the same version; and **joint verification** of interfaces at a scheduled point, with resource committed on both sides. The recurring commercial mistake is to treat integration as each party's own responsibility up to its own boundary, which leaves the join itself unowned — and the join is where the failure happens.

4.A.2 Architectural decisions as governance decisions

Some integration choices are irreversible in practice long before they are irreversible in principle: the integration pattern of KA 4.2.3, the data model, the identity provider, a platform dependency. They share three properties that make them governance matters rather than technical ones — the cost of reversal rises steeply with time, the consequences fall outside the project's boundary and often outside its life, and they are typically taken by people whose delegated authority is defined in money, of which these decisions frequently involve very little.

That last point is the whole issue, and it is Domain 3's delegation schedule read on the wrong dimension: a decision costing 40,000 to take and 2,000,000 to reverse is not a 40,000 decision, and a schedule reading only on value cannot see it. The countermeasures are specific: an **architectural decision record** for each such choice, with its alternatives, rationale and reversal cost stated; those records reviewed by the governance body on the **reversibility** dimension rather than the value one; and a standing question in gate criteria — which decisions taken since the last gate are expensive to reverse, and who took them? Case study B of Domain 3 is exactly this failure with a control attached rather than an architecture attached.

4.A.3 The reviewer's integration eye

Invariants to test on any delivery architecture, each cheap and each diagnostic:

The plan set has been **read against itself before baseline approval**, with the findings priced, and the checks that failed are the ones spanning two owners. The **rolling-wave horizon is at least as long as the longest commitment lead time** inside it, or the exception is labelled a partial horizon. The WBS satisfies the **hundred-per-cent rule at every level**, and the value most likely to be missing is enabling change. Every product traces to work and every work package to a product — and each cross-match exception carries a **recorded ruling with a named ruler**, since two thirds of them will be legitimate and an unruled exception list is indistinguishable from an unrun check. The **interface register is built from need**, has a named owner on each side of every entry, and every entry has a **scheduled** verification activity with resource. The **time-phased cost baseline matches the schedule's current dates** — the single most informative check in the domain, the first thing abandoned under pressure, and the one whose failure leaves every cost measure correct. The change log's approved changes **sum to the difference** between the original and current baselines, and any residual is **decomposed before it is netted**; there are no baseline movements without a change reference and no approved changes without a baseline effect. Screening outcomes are **recorded rulings** citing the baseline provision relied on. Rejections appear in the log, carrying their assessment rather than only their outcome. The delegation threshold reads on **assessed total impact**. A **cumulative test** exists and its threshold and period are derived from the observed change rate — and it is present as well as the reconciliation above, the two being disjoint controls (4.3.3b). Configuration status accounting has been audited against deployment within a stated period, and the item-level tolerance is **derived from an interface-level target** rather than chosen. Re-baselines carry the original's authority, a substantive reason, and the original still visible in reporting. And every architectural decision with a reversal cost materially above its decision cost has a record and a named decision-maker.

Industry variations — Domain 4

- **Construction and infrastructure.** Interfaces are physical and contractual simultaneously, and interface management is a named discipline with its own register and manager; configuration management extends to as-built records, whose divergence from design is the standard handover dispute (Domain 16).
- **Software and digital.** Configuration management is largely automated through version control and build pipelines, which solves identification and control while leaving the interface agreement and its ownership as human work — and integration failure migrates there accordingly. The pairwise arithmetic of WE 4.3.2b explains why the automation is worth so much: it lifts item conformance to where the exponent stops biting, so the interface-level rate follows without anyone having to compute it.
- **Regulated manufacturing (pharmaceutical, medical devices, aerospace).** Configuration and change control are regulatory obligations with validated processes; change assessment must include re-validation cost, which frequently exceeds every other component and reverses the ranking of WE 4.4.2.
- **Public-sector programmes.** Baselines are published commitments, so re-baselining is externally visible and politically costly, which makes drift the preferred failure mode — and therefore makes the cumulative test of KA 4.3.3 the highest-value control available.
- **Multi-party ventures.** The reconciliation provisions of 4.A.1 are the domain's principal content; a single interface register named in the agreement is worth more than any internal process either party runs. And the exponent `k` in `p^k` is the number of parties, so a three-party interface is materially harder to keep verifiable than a two-party one at the same register quality — 65.98 % against 75.79 % on Meridian's figures — which is a reason to reduce the number of parties to a join before it is a reason to audit harder.
- **Healthcare.** Clinical safety cases are configuration-controlled artefacts bound to specific system versions, so a version change is a safety-case change — which makes Meridian's 5 mis-recorded configuration items a clinical-governance finding, not merely an administrative one.

Case study — Domain 4: the interface nobody owned (health, Meridian)

Situation. Meridian's integration testing began four weeks late and took eleven weeks against a planned five. The programme reported "integration complexity" and requested a schedule change. The assurance review that followed found something more specific.

What the review found. Of the **66** possible interfaces among the 12 components, **31** were genuinely required and had been specified. Of those 31, **9** had no named owner on one side — in every case the side outside the programme's own team: the national reporting gateway, the identity provider, the legacy records supplier. Those nine accounted for **all** of the four-week start delay and **four of the six weeks** of overrun, because each had to be renegotiated at the point of verification with someone who had never agreed to it. Separately, **5** configuration items were verified against register entries that did not match what was deployed (WE 4.3.2's most serious class), which produced two defects that each passed component testing.

What the arithmetic had already said. The programme had planned interface work at 12 interfaces — the layered count — while operating a partial mesh of 31. The plan was not optimistic; it was counting the wrong thing. At 18,000 an interface, 31 interfaces is **558,000** against a planned **216,000**, and the 342,000 difference had been absorbed as unplanned effort and schedule.

What the unowned interfaces cost, per interface. Eight of the elapsed weeks lost — the four-week late start and four of the six weeks of overrun — were attributed to the nine unowned interfaces, which were **29.03 %** of the 31 required. At the programme's cost of delay that is `8 × 14,280 =` **USD 114,240**, or **USD 12,693** per unowned interface, and none of it bought anything: the nine agreements had to be negotiated either way, and their negotiation effort already sat inside the 18,000 unit cost. What the programme paid for was negotiating them at the moment of verification rather than at the moment of specification. The register field that would have prevented it — owner on side B, named — costs nothing to add and nothing to maintain, which makes this the cheapest control in the domain and the one most often absent. The five mis-recorded configuration items compound it in exactly the way WE 4.3.2b predicts: with item conformance at 87.06 %, roughly **7.50** of the 31 interfaces were expected to be unverifiable on register grounds alone, and the two defects that passed component testing came out of that population rather than out of bad luck.

What changed. An interface register built from **need** rather than from either formula, with a named owner on **each** side of every entry and a scheduled verification activity for each — which made the nine unowned interfaces visible as nine missing agreements rather than as future complexity. Verification activities entered the schedule with duration and resource, which is what Domain 6 needed in order to show the float they consumed. And the configuration register was audited against deployment before integration rather than after.

What the domain teaches here. An unowned interface is not a risk; it is a scheduled failure with an unknown date. And the interface arithmetic is only useful applied to the required count: 12 was the architecture's promise, 66 was the theoretical maximum, and 31 was the number that had to be managed. Planning to the promise is the error.

Case study B — Domain 4: the baseline that could no longer answer the question (financial services)

Situation. A payments programme (the same organisation as Domain 3's Case study B, two years later) reached its third baseline. Reported performance was `CPI` 0.99 and `SPI` 1.00 — on baseline three. An audit asked a simpler question: what is performance against the **original approved commitment**? Nobody could produce the answer within a fortnight.

Why not. Each re-baseline had been implemented by **replacement** rather than accumulation. The new baseline was loaded, the old one archived as a file, and the change log referenced changes to "the baseline" without stating which. Intermediate states were not reconstructable, so the original-to-current reconciliation — approved changes summing to the difference between original and current — could not be performed at all. Two of the three re-baselines had also been approved by the programme board rather than by the investment committee that approved the original, and neither recorded a substantive change in circumstance; both cited accumulated variance, which is the reason that does not qualify.

How it resolved. The reconstruction took six weeks of a two-person effort against the change log, the archived files and the finance ledger, and established performance against the original as `CPI` **0.87** — a **14.9 %** cost overrun that three successive baselines had made invisible without ever stating an untruth. Corrections: baselines thereafter maintained by **accumulation** with every change referenced to its authority; the original baseline reported alongside the current one permanently; re-baselining reserved to the authority that approved the original, with a recorded substantive reason; and the original-to-current reconciliation performed and published every reporting period.

What the domain teaches here. A baseline exists to answer one question — how is this project performing against what was approved? — and a baseline maintained by replacement cannot answer it. Neither `CPI` 0.99 nor `CPI` 0.87 was a false statement; the first was true of a document and the second of the commitment, and only one of them was what the organisation had asked.

Executive perspective — Domain 4

What a programme director cannot delegate in this domain:

- **The authority statement in your own charter.** If you cannot state what you may decide alone without reading the document, you do not have it (4.1.1).
- **The interface count and its ownership.** Not the theoretical count and not the architecture's promise — the required count, with a named owner on each side and a scheduled verification for every entry. Unowned interfaces are scheduled failures with unknown dates (4.2.3, Case study A).
- **That the plan set has been read against itself before you sign the baseline.** Meridian's audit found 306,992 of unauthorised cost and schedule in three days for 5,280 — more than a year of authorised drift, before any work had started (4.1.2).
- **That the time-phased cost baseline still matches the schedule.** This is one check, it is the first thing abandoned under pressure, and without it earned value measures nothing (4.3.1). Be aware that its failure leaves `CPI` and the CPI-only forecasts exactly correct, so a clean cost review is not evidence that it has been done.
- **Reconciling gross, never net.** A residual of 32,200 concealed a gross error of 122,000, and the two classes had opposite remedies — one a posting, one an investigation (4.3.3b).
- **The tolerance on your configuration register, derived rather than chosen.** 87 % of items conforming is 76 % of two-sided interfaces verifiable; a 95 % interface target requires 97.47 % of items (4.3.2b).
- **The basis of your change threshold.** Assessed total impact, never quoted direct cost — a one-sentence change worth more than most process improvements (4.4.2).

- **A cumulative test derived from your own change log.** Meridian's baseline moved 12.1 % with no decision, and a 100,000/90-day rule would not have caught it (4.3.3).
- **Reversal cost as a governance dimension.** A decision costing 40,000 to take and 2,000,000 to reverse is not a 40,000 decision, and a value-only delegation schedule cannot see it (4.A.2).
- **Performance against the original baseline.** Keep it visible and reconcilable. It is the only question the funding organisation actually asked (4.3.3, Case study B).

Calculation exercises — Domain 4

Exercise 4.1 A programme integrates 9 components. One interface costs 24,000 to specify, build, test and document; an integration layer would cost 200,000. Compute the mesh and layered interface counts and costs, the saving, the breakeven layer cost, and the marginal cost of a tenth component under each architecture. Solution. Mesh `9 × 8/2 =` **36** interfaces at 24,000 = **USD 864,000**. Layered **9** interfaces = 216,000 plus 200,000 = **USD 416,000**. Saving **USD 448,000**. The layer is worth building while it costs below `864,000 − 216,000 =` **USD 648,000**. A tenth component adds `9 × 24,000 =` **216,000** on a mesh and **24,000** layered. Common error: counting `n²` or `n(n−1)`, which double- counts each pair.

Exercise 4.2 A parent WBS element is approved at 6,500,000. Its five children are estimated at 1,900,000, 1,250,000, 880,000, 1,640,000 and 410,000. A review then identifies an omitted element — commissioning and handover — estimated at 690,000. Compute the unallocated amount and its share of the parent, then the shortfall once the omission is restored. Solution. Children sum to **6,080,000**, leaving **420,000** unallocated — **6.46 %** of the parent. Restoring the omitted 690,000 gives an honest total of **6,770,000**, which exceeds the approved figure by **270,000**, or **4.15 %**. Common error: reading the 420,000 as contingency, which locks in the omission and leaves nothing to absorb it.

Exercise 4.3 A change is quoted at 75,000 direct. Assessment finds 3 weeks of critical-path impact at a cost of delay of 9,500 per week, 34,000 of rework, 4 affected interfaces at 7,500 each, 19,000 of regression testing and 11,000 of documentation. Compute the assessed total, the ratio to the quoted figure, and the quoted figure as a share of the true cost. Solution. `75,000 + 28,500 + 34,000 + 30,000 + 19,000 + 11,000 =` **USD 197,500 — 2.633 times** the quoted figure, of which the quote is **38.0 %**. Common error: omitting the schedule impact, which requires the float position from the schedule and is therefore the component most often left out on the grounds that it is "not yet known".

Exercise 4.4 Over a year a project approved 48 changes averaging 5,200 direct cost, 18 of which carried 0.25 weeks of critical-path impact each. Cost of delay is 9,500 per week; the baseline is 3,200,000. Compute total drift, its share of the baseline, each change's individual share, and the 90-day cumulative threshold that would have caught it. Solution. Direct `48 × 5,200 =` **249,600**; schedule `18 × 0.25 = 4.5` weeks at 9,500 = **42,750**; total **USD 292,350 — 9.14 %** of the baseline, while each change was **0.16 %** of it. A 90-day window holds about `48/4 = 12` changes, aggregating to **62,400**, so a cumulative threshold at or below **62,400** on a 90-day window would trip; a 100,000 threshold would not. Common error: setting the cumulative threshold at a round number without reference to the observed change rate, which produces a control with the appearance of function and none of it.

Exercise 4.5 A configuration register holds 500 controlled items, of which 468 conform. The programme has 44 required interfaces, 6 of them three-party and the remainder two-sided. Assuming independent defects, compute item conformance, two-party and three-party relationship conformance, the expected number of at-risk interfaces, and the item conformance that a 98 % two-party target would require. Solution. Item conformance `468/500 =` **93.60 %**. Two-party 0.936^2 = **87.6096 %**; three-party 0.936^3 = **82.0026 %**. Expected at-risk: two-party `38 × (1 − 0.876096) =` **4.7084**, three-party `6 × (1 − 0.820026) =` **1.0798**, total **5.7882** interfaces. For a 98 % two-party target, `p = 0.98^(1/2) =` **98.99 %**, i.e. at most **5** non-conforming items against the 32 currently non-conforming. Common error: applying the item failure rate straight to interfaces — `44 × 0.064 =` **2.8160** — which understates the exposure by **51.35 %** because it ignores the exponent.

Exercise 4.6 A programme's original approved baseline is 8,600,000. Its change log holds 27 approved changes totalling 742,300. The cost tool reports a current baseline of 9,318,600. Investigation finds 61,400 of approved changes with no recorded baseline effect and 37,700 of baseline movement with no change reference. Compute the expected current baseline, the net residual, the gross error, and the ratio between them. Solution. Expected current `8,600,000 + 742,300 =` **9,342,300**. Net residual `9,318,600 − 9,342,300 =` **(23,700)** — only **0.2756 %** of the original. Check: `37,700 − 61,400 = (23,700)` ✓. Gross error `61,400 + 37,700 =` **99,100** — **13.35 %** of the approved changes and **4.18 times** the net. Common error: treating a residual of a quarter of one per cent as rounding and closing the reconciliation, which discards a finding four times its size; note also that the residual here is negative, which does not make it benign — it means the log is ahead of the tool rather than behind it, and the two classes still require different remedies.

Exercise 4.7 Of 84 change requests, screening ruled 46 changes, 24 clarifications and 14 defects. A re-screen finds that 7 of the "changes" were defects, at an average approved value of 9,400, and that 4 of the "clarifications" were changes, with direct work of 12,000, 8,500, 6,200 and 15,300, one carrying 0.75 weeks of critical-path impact at a cost of delay of 9,500 per week. Compute the error rate, both error classes, the gross total, the cost per error, and the net effect on the recorded change total. Solution. Errors **11** of 84 — **13.10 %**. Defects paid as changes `7 × 9,400 =` **65,800**. Changes taken as clarifications `42,000 + 0.75 × 9,500 =` **49,125**. Gross total **USD 114,925**, or **USD 10,447.73** per error. Net effect on the recorded change total `49,125 − 65,800 =` **(16,675)** — **6.8921 times** smaller than the gross, and understating it by twice the smaller class (`2 × 49,125 = 98,250`). Common error: reporting the net, which here even has the wrong sign for the narrative — the log overstates the change total while scope has silently grown.

Exercise 4.8 At a status date a project reports PV 6,300,000, EV 5,900,000 and AC 6,150,000 against a BAC of 12,000,000. A change approved two months earlier deferred 450,000 of planned value beyond the status date and the `PV` curve was never re-phased. Compute reported and true `SPI` and `SV`, `CPI`, and the composite `EAC` on each basis. Solution. Reported `SPI = 5,900,000/6,300,000 =` **0.9365**, `SV =` **(400,000)**. True PV `6,300,000 − 450,000 = 5,850,000`, so true `SPI =` **1.0085** and `SV =` **50,000** — the project is marginally ahead of the authorised schedule while reporting 0.94. `CPI = 5,900,000/6,150,000 =` **0.9593** on both bases, and the CPI-only forecasts are unchanged at `BAC + AC − EV =` **12,250,000** and `BAC/CPI =` **12,508,475**. Composite `EAC = AC + (BAC − EV)/(CPI × SPI)`: reported **USD 12,939,558**, true **USD 12,454,589** — an overstatement of **USD 484,968**, which is exactly the sensitivity `(BAC − EV)/(CPI × EV) =` **1.0777** multiplied by the 450,000 of mis-phasing. Common error: concluding that a stale baseline always flatters. Deferral makes performance look worse than it is, and the response — a recovery plan on a project that does not need one — costs real money to correct a reporting defect.

Practitioner's toolkit — Domain 4

Adoption-ready artefacts; adapt headings to your organisation, then keep them stable.

Toolkit 4.T.1 — Charter on one page

Fields, in this order, and nothing else: purpose and measurable objectives · named sponsor · named project leader · **authority bounds** (spend, scope, resource, commitment — each a number or a sentence, not a paragraph) · scope boundary **including exclusions** · key deliverables and acceptance basis · fixed constraints, distinguished from planned dates · budget authority · material risks and assumptions · governance bodies and thresholds by reference · approval signature and date. If it exceeds two pages, the plan has begun; move the excess.

Toolkit 4.T.2 — Interface register

One row per **required** interface (built from need, not from `n(n−1)/2`): reference · the two components or parties · **owner on side A** and **owner on side B**, both named · content exchanged · format and protocol · direction · timing and frequency · error and exception behaviour · agreed specification **version** · verification activity, with its schedule reference, duration and resource · verification status and date. The two owner columns and the schedule reference are the fields that carry the value; an interface register without them is an inventory. Report monthly: entries with a missing owner, entries with no scheduled verification, and entries verified against a superseded version. Add one derived line to the same report — **expected at-risk entries**, `count × (1 − p^k)` from the configuration register's current item conformance `p` and each entry's party count `k` — which converts the configuration audit into a number the integration plan can act on (WE 4.3.2b).

Toolkit 4.T.3 — Integrated change request and baseline reconciliation

Request side. Reference · date raised · raiser · description and reason · **screening outcome** (change / clarification / defect, with the ruling recorded) · then the assessment, one line each: direct cost · schedule impact **with the float position stated** · rework · interfaces affected and re-verification cost · regression testing · documentation, training and communication · risk profile change · benefit change. Then: **assessed total impact** · authority required **on the assessed total** · decision, decision-maker by name and date · baseline effect on scope, schedule and cost · implementation and verification confirmation.

Reconciliation side, run every reporting period and published: original baseline · plus the sum of approved changes · equals current baseline (**the three must reconcile exactly**, each figure quoted from its own system of record rather than derived from another) · approved changes with no recorded baseline effect · baseline movements with no change reference · **the gross residual, stated before any netting, with each class shown separately** · rolling-window cumulative total against the derived threshold · movement in the time-phased `PV` curve reconciled to the approved changes' schedule effects · and current performance reported against **both** the current and the original baseline.

Exam preparation — Domain 4

What is assessed. Charter content and the primacy of the authority statement; the plan of plans and its consistency checks; tailoring as a recorded decision and its limits; WBS construction and the hundred-per-cent rule; work packages and control accounts; product versus work breakdown and where each belongs; **interface counting and integration architecture economics**; interface agreements and ownership; the integrated three-dimensional baseline and its invariants; configuration management's four functions and the audit finding classes; baseline maintenance by accumulation, and re-baselining discipline; **baseline drift and cumulative tests**; the change flow and the change/clarification/ defect distinction; **assessed impact versus quoted cost**; the change board and the decision log including rejections.

The calculations to be able to do under time pressure. Mesh and layered interface counts ($n(n-1)/2$ and n), their costs, the breakeven layer cost, and the marginal cost of one more component under each. The hundred-per-cent test and the shortfall once an omission is restored. Product and work-package coverage from a cross-match, and the classified residue after ruling. Plan-set audit exposure, converting unfunded work at a person-day rate and unscheduled elapsed time at the cost of delay, counting only the critical-path share. SPI, SV, CPI and the three EAC forms on both a stale and a re-phased PV, plus the composite forecast's sensitivity $(BAC - EV)/(CPI \times EV)$ per unit of mis-phased planned value. Pairwise configuration integrity p^k, expected at-risk relationships, and the inversion $p = q^{(1/k)}$ that turns an interface target into an item tolerance. The original-to-current reconciliation, its residual, and the decomposition of that residual into its two opposite-signed classes. Assessed total impact from its components, the ratio to the quoted figure, and the quoted share. Baseline drift from a change count, average cost, affected count, weeks and cost of delay — and the cumulative-window threshold that would catch it. Configuration defect rates by class.

The traps. Counting n^2 or $n(n-1)$ for interfaces · planning interface work at the architecture's promised count rather than the required count (Case study A) · reading unallocated budget as contingency (Exercise 4.2) · acting on raw cross-match exception counts without ruling on each one (WE 4.2.2) · assuming a stale PV curve always flatters, when deferral makes performance look worse (Exercise 4.8) · applying an item-level configuration rate to interfaces instead of p^k (Exercise 4.5) · choosing a configuration tolerance at item level rather than deriving it from an interface target · **netting a reconciliation residual instead of decomposing it** (Exercises 4.6, 4.7, WE 4.3.3b, WE 4.4.1) · omitting schedule impact from a change assessment because the float position "is not yet known" (Exercise 4.3) · applying a delegation threshold to quoted direct cost (4.4.2) · setting a cumulative threshold at a round number rather than from the observed change rate (Exercise 4.4) · treating a defect as a change, or a change as a clarification · treating a rising emergency-route share as indiscipline rather than as a latency symptom · leaving rejections out of the decision log, or logging an outcome without the assessment · re-baselining by replacement (Case study B) · applying configuration control to activities · setting a rolling-wave horizon shorter than the longest commitment lead time inside it.

How the domain connects. Domain 1 supplies the accountability principle and the cost of delay every calculation here is priced at. Domain 2 supplies the business case whose benefits the baseline must still deliver, and the enabling-change column that WE 4.2.1 finds missing from the WBS. Domain 3 supplies the delegation schedule this domain corrects the basis of, the latency arithmetic the change board inherits, and the decision log the change log is part of. Part Two then works the baseline: Domain 6 schedules the WBS and supplies the float that change assessment needs, Domain 7 measures against the time-phased baseline this domain insists on maintaining, Domain 8 attaches uncertainty to it, and Domain 16 hands over the configuration record. PFL-AI Domain 11 handles the contractual face of cross-boundary integration.

Domain 4 summary

Domains 1 to 3 produced accountability, a justified choice and a working decision structure, and none of them produced a project. Integration does — and integration fails in a characteristic way: the parts are managed and the joins are not.

The charter authorises and bounds; its irreplaceable content is the statement of what the leader may decide alone, and a charter that has become a plan has lost it. The plan of plans is integrated by a short list of consistency checks that only appear when the plans are read against one another — including the one that makes a programme report last month's position every month for its whole life. Those checks are arithmetic, and running them on Meridian's eleven subsidiary plans found **USD 306,992** of unfunded cost and unscheduled time — **12.79 %** of the baseline, more than a full year of authorised drift — for an audit costing **USD 5,280**, three days before any work began. The checks that failed all spanned two owners; the one that passed had a single owner on both sides, which is the general shape of the finding. And the same audit exposed a horizon two weeks shorter than the programme's longest procurement lead time — a plan that could not support its own commitments, which no diligence inside the wave could have fixed.

The WBS obeys the hundred-per-cent rule, which is arithmetic and therefore auditable: Meridian's five level-2 elements sum to **2,332,000** against an approved **2,400,000**, and the missing element is **clinician training and enabling change** at **214,000** — the same column Domain 2's benefits map omitted, leaving an honest baseline **6.1 %** above the approved figure and an NPV of **+1,186,898** that would still have been approved. The omission bought nothing.

Interfaces are where integration effort lives, and they grow combinatorially while components grow linearly: Meridian's 12 components admit **66** point-to-point interfaces costing **USD 1,188,000**, against **12** plus a layer at **USD 536,000** — a **54.9 %** saving, worth building while the layer costs below **972,000**, and decisive on the margin, where a thirteenth component costs **216,000** meshed and **18,000** layered. Case study A supplies the professional correction: the number to plan against is neither 66 nor 12 but the **31** genuinely required, of which **9** had no owner on the far side and accounted for the entire integration overrun.

Products and work verify each other, and the verification is arithmetic too — but only after each exception has been ruled on. Meridian's 148 products against 172 work packages produced **21** raw exceptions of which **14** were legitimate, a **66.67 %** false-positive rate, leaving **214,000** of omitted enabling change and **26,000** of work producing nothing: **USD 240,000**, exactly **10.00 %** of the baseline. Removing the orphans widened the unallocated gap from 68,000 to **94,000**, which is the counter-intuitive and correct direction.

The baseline is one three-dimensional statement, and its most informative invariant — that the time-phased cost baseline still matches the schedule — is the first abandoned under pressure, after which earned value measures the distance between two documents. How wrong is computable, and the shape of the answer explains the defect's durability: on Auriga's week-13 position, 240,000 of unre-phased planned value puts `SPI` anywhere between **0.8276** and **1.0435** against a reported **0.9231**, moves the composite `EAC` by **USD 287,083** in either direction at **USD 1.20** per USD of mis-phasing — and leaves `CPI` at **0.9057** and both CPI-only forecasts exactly correct. **A stale `PV` curve corrupts every schedule measure and leaves every cost-only measure right**, so the cost review passes and the defect survives.

Configuration management makes version-bound interface verification meaningful, and its audit classes are not equally serious: five items whose recorded version differs from what is deployed will fail an integration that a 12.94 % headline rate does not predict. Nor is the item rate the rate that matters. Integration quality is item quality raised to the number of parties, so Meridian's **87.06 %** of items gives **75.79 %** of two-sided interfaces and **65.98 %** of three-party ones — **7.50** of 31 interfaces expected to be unverifiable — and a 95 % interface target therefore requires **97.47 %** item conformance, at most **8** non-conforming items rather than the 44 found. **The tolerance on the register is derived from the target on the interfaces, backwards through the exponent**, and where the derived audit interval comes out at three weeks the finding is about how the register is written, not about how often it is audited.

Baselines are maintained by accumulation, never replacement, because the only question the funding organisation asked is performance against the original — the question Case study B's third baseline could not answer, and to which the answer turned out to be `CPI` **0.87** rather than 0.99. Accumulation is what makes the domain's cheapest control possible: original plus approved changes equals current, exactly. Meridian's reconciliation showed a residual of **32,200** — 1.34 %, dismissible — concealing a gross error of **USD 122,000**, since **77,100** of unreferenced movement and **44,900** of unposted approvals have opposite signs and hid **89,800** between them. **Reconcile gross, never net.** And the two controls are disjoint: summing the log catches drift inside it, the reconciliation catches what never entered it.

And change is integrated or it is not controlled. It begins with a classification, and Meridian got **10.53 %** of them wrong in both directions at once: **USD 34,000** of defects paid as changes — 14.71 % of the year's direct drift, work that was never a change at all — and **USD 36,340** of changes ruled as clarifications, unbudgeted and unlogged. Gross **USD 70,340**; net **USD 2,340**. The same identity as the reconciliation, from unrelated data: **the net understates the gross by twice the smaller class.** The two errors also differ in detectability, and the one with no downstream detector is silent scope growth, which is where the tighter control belongs. Rejections are the other cheap entry nobody writes: **9** of Meridian's 14 left no trace, **4** returned, three costing a duplicated **USD 4,400** assessment and one costing **USD 63,000** because a different body approved what had already been refused — a total of **USD 80,600** against **USD 385** of writing, with a mean **4.58 times** the modal case, which is why the control is bought for the tail and both figures must be quoted.

Meridian's baseline moved **12.1 %** through 34 individually authorised changes averaging **0.28 %** each, with no decision anywhere on the total, and a 100,000-in-90-days cumulative rule would not have caught it because a quarter's changes aggregate to **57,800** — so a cumulative test must be derived from the observed change rate, not chosen for roundness. A change quoted at **40,000** truly cost **131,560**, **3.29 times** the quoted figure, of which the quote was **30.4 %**; and therefore the domain's single most valuable sentence, worth more than most process improvements a programme will make: **the delegation threshold reads on assessed total impact, not on quoted direct cost.**

02

PART TWO

Delivering the work

Domains 5–10 — scope, planning and flow, cost and commercial, risk and resilience, quality and assurance, and procurement: the machinery of delivery.

05

DOMAIN 5

Scope, Requirements and Value Definition

Group: Delivering the work (Domain 1 of 6 in Part Two — the part's opening domain). **Target:** ~70 pages. **Binds to:** the PCI Book Pattern Specification and the shared registries (`docs/books/registries/`). This domain opens Part Two by settling what is to be delivered and which parts of it are worth delivering first, before Domain 6 begins to schedule anything. It closes the **Meridian Care Records** thread's definition work; **Project Auriga** takes over as the single-project illustration from Domain 6. British English; USD (+SAR where useful, indicative `USD 1 ≈ SAR 3.75`).

Registry note. This domain uses the registered cost of delay and baseline drift constructs and the annuity factor `AF(r, n)`, and submits four candidate rows to the shared formula registry with the draft (traceability defect rate by class; the requirement-defect correction ladder and its breakeven elicitation spend; value per unit of constrained effort; requirement-count reconciliation), per Editorial Charter §6.

Why this domain exists

Part One produced a project. Domain 1 established what the leader is answerable for, Domain 2 whether the work was worth choosing and whether the promise was honest, Domain 3 who may decide and how quickly, and Domain 4 how the parts fit and how a change moves all three baseline dimensions. All four assumed something they never established: that the **content** of the scope was known.

It usually is not, and the gap is not the one practitioners expect. The familiar complaint is that requirements were incomplete. The more expensive and far commoner condition is that they were **complete, agreed, delivered, verified — and did not produce the value the business case promised.** Domain 4's WBS audit found a missing element worth USD 214,000, and what it found missing was clinician training and enabling change: not a forgotten cost line but a **forgotten requirement** — that the receiving organisation be able to use what it was given. The omission did not originate in a spreadsheet. It originated in a room where nobody asked the clinicians anything.

The domain's central claim follows: **scope is not a list of things to build; it is a chain of justified links from a benefit to an accepted deliverable, and the chain is only as good as its weakest link — which is almost never the build.** Four consequences organise the chapter. An unwritten boundary is decided later by whoever is most insistent: KA 5.1 prices one undrawn boundary at USD 143,000, then shows the correct answer was to include the disputed sites — a decision the boundary statement would have surfaced and its absence prevented. Requirement defects are not equally expensive; they cost four times more at each stage they survive, and KA 5.2 shows a USD 84,000 elicitation programme returning USD 610,800, with **a single avoided live-service defect paying for the whole of it**. Requirements do not carry equal value per unit of effort, and where capacity binds KA 5.3 shows the intuitive ranking method losing USD 44,000 a year — USD 262,737 of present value — to simple enumeration. And scope grows by accumulation with nobody approving anything, a different failure from Domain 4's authorised drift needing a different detector: KA 5.4 finds USD 220,920 of movement that **left no trace in the change log at all**, and the only control that catches it is a count.

The through-line: **the cheapest place to be wrong about scope is at the start, and the arithmetic of how much cheaper is the most under-used business case in project management.**

Learning objectives. After this domain a candidate can: write a scope statement that is decidable rather than descriptive, and distinguish deliverables from outcomes and from activity; assemble a scope baseline and state which artefacts constitute it; write exclusions and assumptions that settle boundaries in advance, and **price an undrawn boundary on both its cost and its forgone benefit, including the payback period**; select elicitation techniques by the kind of requirement being sought and identify the stakeholder groups whose absence causes systematic requirement loss; **compute the correction cost of a requirement defect by the stage at which it is found, the saving from moving detection earlier, the return on an elicitation investment, and the schedule delay at which that investment stops paying**; specify a requirement so that it is unambiguous, testable, singular and traceable; **audit a traceability matrix forward and backward, report the defect rate by class rather than as a total, and say which class invalidates the record**; attribute benefit to requirement bundles and re-test a "must have" classification; **rank requirements by value per unit of constrained effort, and demonstrate where greedy ranking loses to enumeration and by how much**; distinguish controlled change from uncontrolled creep, **quantify creep by count reconciliation and price it at the cost of delay**; write acceptance criteria that are testable and **compute the breakeven dispute probability that justifies writing them**; distinguish verification from validation and price a conditional acceptance against the correction ladder; and govern AI-assisted requirements work without letting a model author, prioritise or accept scope.

The master programme. Meridian Care Records continues from Part One: the clinical-records rollout to **40 clinics**, approved cost **USD 2,400,000**, full-potential benefit **USD 979,200** a year, realistic benefit **USD 685,440** a year at 70 % adoption, cost of delay **USD 14,280 per week** (Domain 1), the ramped business case with NPV **+USD 1,332,898** (Domain 2, WE 2.2.2), steering latency `E[wait] = M/2 + L =` **4.0 weeks** (Domain 3), and the integrated baseline, interface architecture and authorised drift of **USD 291,176** (Domain 4). This domain works its requirements baseline of **480** approved requirements, its **425**-element design register, its Release 1 capacity of **20 development-weeks**, and the **USD 520,000** clinic-rollout element from Domain 4's WBS, whose per-clinic figure of **USD 13,000** prices the boundary question KA 5.1 opens with. From Domain 6, **Project Auriga** — the 25-week control-systems upgrade, BAC **USD 4,000,000** — takes over as the single-project illustration.

KNOWLEDGE AREA 5.1

Scope definition and the scope baseline

Topics: 5.1.1 the scope statement · 5.1.2 the scope baseline · 5.1.3 exclusions, assumptions and the boundary.

5.1.1 The scope statement

Definition. The scope statement is the authorised description of **what the project will deliver and what it will not**, expressed in terms specific enough that a competent third party could judge whether a given item falls inside it. That last clause is the whole test, and it is the one most scope statements fail.

Three registers, and only one of them is scope. Practitioners habitually conflate them, and the conflation is where value definition goes wrong:

Register	What it states	Example (Meridian)
Outcome	The change in the world the organisation is paying for	Clinicians retrieve a complete patient history in under 20 seconds at any of 40 clinics
Deliverable	The thing that will be handed over and accepted	A records application, 31 verified interfaces, migrated data, a trained user population
Activity	The work that produces the deliverable	Build, configure, test, migrate, train, cut over

Scope is the **deliverable** register, bounded by the outcome register and decomposed into the activity register (Domain 4's WBS). A scope statement written in the activity register — "implement a records system" — cannot be tested for completeness, because activities have no boundary of their own. A scope statement written purely in the outcome register cannot be accepted, because an outcome depends on behaviour the project does not control. The discipline is to write the deliverables, state the outcome each serves, and keep the two columns visibly distinct.

What makes a scope statement decidable. Four properties, each of which converts an argument into a lookup: **enumerated** deliverables (a countable list, not a category); **quantified** extent (40 clinics, not "the region"); **named** interfaces and dependencies, by reference to Domain 4's register; and **explicit exclusions**, which are the subject of 5.1.3 and the half most often omitted. A scope statement with all four is short. Length in a scope statement is almost always a symptom of the activity register having crept in.

5.1.2 The scope baseline

What it is. The scope baseline is the **approved** scope statement together with the WBS and the WBS dictionary — three artefacts, one authorised statement, and one of the three dimensions of the performance measurement baseline Domain 4, KA 4.3.1 insists is single. It changes only through integrated change control (Domain 4, KA 4.4), and every change to it is a change to schedule and cost by construction.

The WBS dictionary is where scope actually lives. The scope statement gives the boundary and the WBS gives the structure; the dictionary gives, per work package, the description of work, the deliverable, the acceptance basis, the responsible owner, the estimate and the dependencies. It is the document that answers "is this in scope?" at the level at which the question is actually asked, and it is the document most often left as a set of names. A WBS whose dictionary is a list of element titles has a structure and no content, and the deficit becomes visible at acceptance, when the acceptance basis has to be invented by whoever is in the room.

The four scope-baseline defects, in increasing order of cost to repair: unbounded — a category rather than a count, so extent is settled later by negotiation (5.1.3 prices one); undocumented at dictionary level, so acceptance criteria are improvised (KA 5.4.2 prices it); untraceable to benefit — deliverables no benefit requires, benefits no deliverable serves (KA 5.3.1); and unversioned, so nobody can say what was agreed, which is Domain 4's configuration failure arriving through the scope door.

5.1.3 Exclusions, assumptions and the boundary

Why exclusions are the highest-return sentences in the document. An exclusion costs a line to write and settles a dispute that would otherwise be resolved under time pressure, by whoever is most insistent, in month fourteen. It is Domain 3's decidability test applied to content rather than to authority: if this question arises, who answers it, and from what? Where the answer is "from the scope statement", the boundary holds. Where it is "from a conversation", the boundary is wherever the conversation ends.

Assumptions are exclusions with a deadline. An assumption states something the project has taken as true without controlling it — that the identity provider will be available, that clinical data quality in the legacy system meets a stated standard, that a named team is released on a stated date. Each carries an owner and a date by which it becomes a fact or a risk (Domain 8), because an assumption with neither is a hope in a register.

WORKED EXAMPLE

Worked example 5.1.3 — the boundary nobody drew, and the answer that was not "no".

1. **Setup.** Meridian's approved scope statement says the records application will be rolled out to "all clinics in the region". The approved count in the business case is **40** clinics. The regional service register, however, also lists **6** branch surgeries, **3** out-of-hours hubs and **2** prison health units — **11** further sites, each of which has been told by its own management that it is included. From Domain 4's WBS the clinic-rollout element is **USD 520,000** across the 40 clinics. Full-potential benefit is **USD 979,200** a year across the 40 clinics; realistic adoption is **70 %** (Domain 1).
2. **Formula.** Per-site rollout cost = rollout element ÷ approved site count. Boundary exposure = additional sites × per-site cost. Per-site benefit at realistic adoption = (full potential ÷ site count) × adoption. Payback period = exposure ÷ additional annual benefit.
3. **Substitution.** `520,000 / 40`; `11 × 13,000`; `(979,200 / 40) × 0.70`; `11 × 17,136`; `143,000 / 188,496`.
4. **Result.** Per-site rollout cost **USD 13,000**. The undrawn boundary carries an exposure of **USD 143,000 — 5.96 %** of the approved baseline. The same 11 sites would generate `11 × 17,136 =` **USD 188,496** a year at realistic adoption, against a full-potential per-site figure of **USD 24,480**. Payback on including them is **0.76 years — about 9.1 months**.
5. **Interpretation.** The instinct on seeing a 143,000 boundary exposure is to write the exclusion, and that instinct is wrong here, which is exactly why the example is worth working. Eleven sites costing 143,000 once and returning 188,496 a year pay for themselves in **nine months**; excluding them destroys value, and including them by accident in month fourteen destroys almost as much, because by then the rollout waves are scheduled, the training capacity is committed and the 11 sites arrive as a disruption rather than as a plan. The value of the boundary statement is therefore not that it says no. **It is that it forces the question to be asked while the answer is still cheap to act on** — which is the same argument Domain 2 made about options and the same one Domain 4 made about assessing before deciding. Three cautions before this arithmetic is reused. The 13,000 per-site figure is the rollout element only: it excludes licence cost, clinical safety assessment and the training element Domain 4 found missing at USD 214,000, so the true marginal cost per site is higher and the payback longer — a leader quoting nine months must say which cost basis it rests on. The 17,136 per-site benefit assumes the new sites adopt at the same 70 % as the original 40, which is a genuine assumption and not a fact: prison health units in particular have a different clinical workflow, and a lower adoption assumption for them would be defensible. And the decision is a **change** under Domain 4, KA 4.4 whichever way it goes — an 11-site extension is not a clarification, and recording it as one is precisely the misclassification KA 5.4.1 quantifies.

AI in this KA

Where it earns its place. Reporting every unbounded extent in a scope statement — every category noun without a count, every "all", "as required" and "including but not limited to". It is mechanical, exhaustive, and exactly the reading humans do not perform on a document they have already approved. Cross-checking the deliverable list against an organisational asset or site register to surface candidate boundary questions, which is how Meridian's 11 sites would have appeared in week three rather than month fourteen. Drafting WBS dictionary entries from an approved structure and existing estimates, for owner completion. Checking that every assumption carries an owner and a resolution date.

Where it must not go. Deciding a boundary. The 11-site question turns on clinical workflow, regional politics, adoption expectations and a benefit judgement, and it is a change decision belonging to a named authority. Nor should a model author the exclusion list, because an exclusion is a commitment about what will not be delivered and someone must be accountable for having made it.

Verification, concretely. Every flagged extent is confirmed against the source document, because the flag rate is high and the false-positive rate is not zero. Every candidate boundary question is put to the sponsor as a question, never resolved as an inference. The per-site arithmetic is reproduced by hand — one division — with its cost basis stated. And any AI-drafted dictionary entry is endorsed by the work-package owner before it enters the baseline, since the dictionary is the acceptance basis.

KEY TERMS — KA 5.1

Term	Meaning
Scope statement	The authorised description of what will and will not be delivered, specific enough to be tested against.
Outcome / deliverable / activity registers	The three levels routinely conflated; scope is the deliverable register, bounded by outcomes and decomposed into activities.
Decidable scope	Enumerated deliverables, quantified extent, named interfaces, explicit exclusions.
Scope baseline	The approved scope statement + WBS + WBS dictionary — one of Domain 4's three baseline dimensions.
WBS dictionary	Per work package: work, deliverable, acceptance basis, owner, estimate, dependencies — where scope actually lives.
Exclusion	A written statement of what is outside the boundary; the highest-return line in the document.
Assumption	Something taken as true without being controlled, carrying an owner and a resolution date.
Boundary exposure	The priced consequence of an unstated extent: additional units × unit cost.

SAMPLE MCQS — KA 5.1

MCQ 5.1-A `[5.1.1 · Comprehension]` A scope statement reads "implement a regional records system". Its principal defect is that: - A. it is too short - B. it is written in the activity register, so it has no boundary that can be tested - C. it does not name the sponsor - D. it omits the schedule

Rationale: Activities have no boundary of their own, so completeness cannot be assessed (5.1.1). Length is not the defect — decidability is; C and D belong to the charter and the schedule.

MCQ 5.1-B `[5.1.3 · Application]` A rollout element of USD 520,000 covers 40 approved sites. Eleven further sites believe they are included. The boundary exposure is: - A. USD 13,000 - B. USD 143,000 - C. USD 60,000 - D. USD 188,496

Rationale: `520,000/40 = 13,000` per site, `× 11 = 143,000` (5.1.3). A is the cost of one site, not of eleven; C divides the whole 2,400,000 baseline by 40 rather than the rollout element, giving a per-site 60,000 — the wrong cost base, and a per-site figure rather than an exposure; D is the annual benefit of the 11 sites, not their cost.

MCQ 5.1-C `[5.1.3 · Evaluation]` The 11 sites would cost USD 143,000 to include and generate USD 188,496 a year at realistic adoption. The correct professional conclusion is that: - A. they should be excluded, because the exposure exceeds the tolerance - B. the boundary statement's value is that it forces the decision while it is still cheap to act on — and here the decision is probably to include them - C. they are in scope already, since the statement says "all clinics in the region" - D. the exposure should be added to contingency

Rationale: A nine-month payback makes exclusion value-destroying (5.1.3); the boundary statement's function is to surface the question in time, not to say no. C is the reading that causes the dispute; D funds an unmade decision.

MCQ 5.1-D `[5.1.2 · Analysis]` A WBS dictionary consists only of element titles. The defect becomes visible: - A. immediately, at baseline approval - B. at acceptance, when the acceptance basis has to be invented by whoever is present - C. during estimating - D. only if the scope changes

Rationale: The dictionary carries the acceptance basis; its absence is invisible until acceptance is attempted (5.1.2). That is what makes it an expensive defect rather than a documentation nicety.

SELF-CHECK — KA 5.1

1. Which register is scope, and what happens when the other two are used instead? — The deliverable register; the activity register removes any testable boundary, and the outcome register cannot be accepted because it depends on behaviour the project does not control.
2. Why is an exclusion the highest-return line in a scope statement? — It costs a line and settles a dispute that would otherwise be resolved late, under pressure, by whoever is most insistent.
3. What is the point of pricing a boundary if the answer turns out to be "include them"? — The point is that the question gets asked while the answer is still cheap to act on; a nine-month payback accepted in week three is a plan, and the same decision in month fourteen is a disruption.

KNOWLEDGE AREA 5.2

Requirements elicitation, analysis and traceability

Topics: 5.2.1 elicitation and its economics · 5.2.2 analysis and specification quality · 5.2.3 traceability.

5.2.1 Elicitation and its economics

Definition. Elicitation is the deliberate discovery of what stakeholders need, as distinct from collection of what they ask for. The distinction is the discipline: a stated request is a proposed solution to an unstated need, and a requirements process that records requests has captured other people's designs without capturing the problem.

Techniques, matched to what is being sought. Each technique is good at one thing and blind to another, which is why a single-technique programme fails predictably:

Technique	Good at	Blind to
Structured interview	Depth, rationale, the why behind a request	Group disagreement; what the individual does not know they do
Facilitated workshop	Conflict surfacing, shared prioritisation	The quiet stakeholder and the absent one
Observation of work as done	The gap between documented and actual process	Rare cases, exceptions, and the volume-weighted picture
Document and data analysis	Volumes, exception rates, statutory obligations	Intent, and anything not already recorded
Prototype or walkthrough	Tacit needs made visible by reaction	Non-functional requirements — load, availability, security
Existing-system defect and support logs	What actually fails today	What has never been attempted

The stakeholder groups whose absence causes systematic loss. Requirement loss is not random; it clusters on groups that are structurally easy to omit — the people who will operate the service after handover, the people who will support it, the external users who never attend internal meetings, and whoever owns the **enabling change** that converts an output into an outcome. Domain 4 found the last of these missing as a USD 214,000 WBS element; Case study B of this domain finds the third of them missing at a cost of most of a programme's benefit.

The economics, which is the part usually asserted rather than computed. It is a commonplace that finding a requirement defect early is cheaper. The professional question is how much cheaper, and therefore how much elicitation is worth buying — and that is arithmetic.

WORKED EXAMPLE

Worked example 5.2.1 — the correction ladder, and what elicitation is worth.

1. **Setup.** Meridian's own change and defect records give the average cost of correcting one requirement defect by the stage at which it is found: **USD 400** at definition, **1,600** at design, **6,400** at build, **25,600** at test and integration, **102,400** in live service — a ladder rising by a factor of **four** at each stage. Projects of this type in the organisation surface about **96** requirement defects across a 480-requirement baseline (**0.2** per requirement), on this profile: **24** at definition, **22** at design, **26** at build, **18** at test, **6** in live service. A structured elicitation programme — clinician workshops across the 40 clinics, observation of records use as actually performed, prototype walkthroughs — costs **USD 84,000** (**USD 175** per requirement) and is expected to move **13** build-stage, **9** test-stage and **3** live-service defects into the definition stage.
2. **Formula.** Expected correction cost = Σ (defects at stage × cost at stage). Saving from earlier detection = Σ (defects moved × [cost at old stage − cost at definition]) — the **difference**, not the old cost, because the defect still costs 400 to correct at definition. Return = saving ÷ investment. Breakeven delay = (saving − investment) ÷ cost of delay.
3. **Substitution.** `24 × 400 + 22 × 1,600 + 26 × 6,400 + 18 × 25,600 + 6 × 102,400`; then `13 × 6,000 + 9 × 25,200 + 3 × 102,000`; then `610,800 / 84,000`; then `(610,800 − 84,000) / 14,280`.
4. **Result.** The unimproved expected correction cost is **USD 1,286,400** — an average of **USD 13,400** per defect. The elicitation programme saves `78,000 + 226,800 + 306,000 =` **USD 610,800**, for a net of **USD 526,800** and a return of **7.27 times** the investment. The expected correction cost falls to **USD 675,600**, or **USD 7,037.50** per defect. And the investment breaks even at a definition-phase extension of **36.89 weeks**.
5. **Interpretation.** Four things here matter more than the headline return. First, the single most useful sentence a leader can take from it: **moving one defect out of live service saves USD 102,000, which by itself pays for the entire USD 84,000 programme.** Nothing about a 7.27-times return is needed to authorise the spend; one defect is. Second, the breakeven delay answers the objection this investment always attracts — that elicitation delays the start. Elicitation would have to extend the definition phase by **more than 36.89 weeks** before it destroyed value, and a realistic four-week extension costs `4 × 14,280 =` **USD 57,120**, leaving **USD 469,680** net. The objection is real, and the breakeven is **more than nine times** the extension anyone is proposing — an order of magnitude short of decisive; a leader who cannot say that has to argue about it instead. Third, the saving is computed on the **difference** between stages, never on the later cost alone: the gross figures give `13 × 6,400 + 9 × 25,600 + 3 × 102,400 =` **620,800**, overstating the saving by 10,000 by forgetting that the defect still has to be fixed. Fourth, two real limits rather than hedges. The ladder is **this organisation's observed ladder**, not a law of nature — technology, regulatory regime and release cadence all change the ratio, and a programme that has not measured its own should say it is using an assumed one. And the arithmetic assumes the moved defects were **detectable at definition**; emergent integration behaviour, load characteristics that appear only at volume and clinical workflow effects that no prototype predicts are genuinely not, which is why the example moves half of the build and test defects rather than all of them. The honest claim is a large return on a subset, not the elimination of late defects.

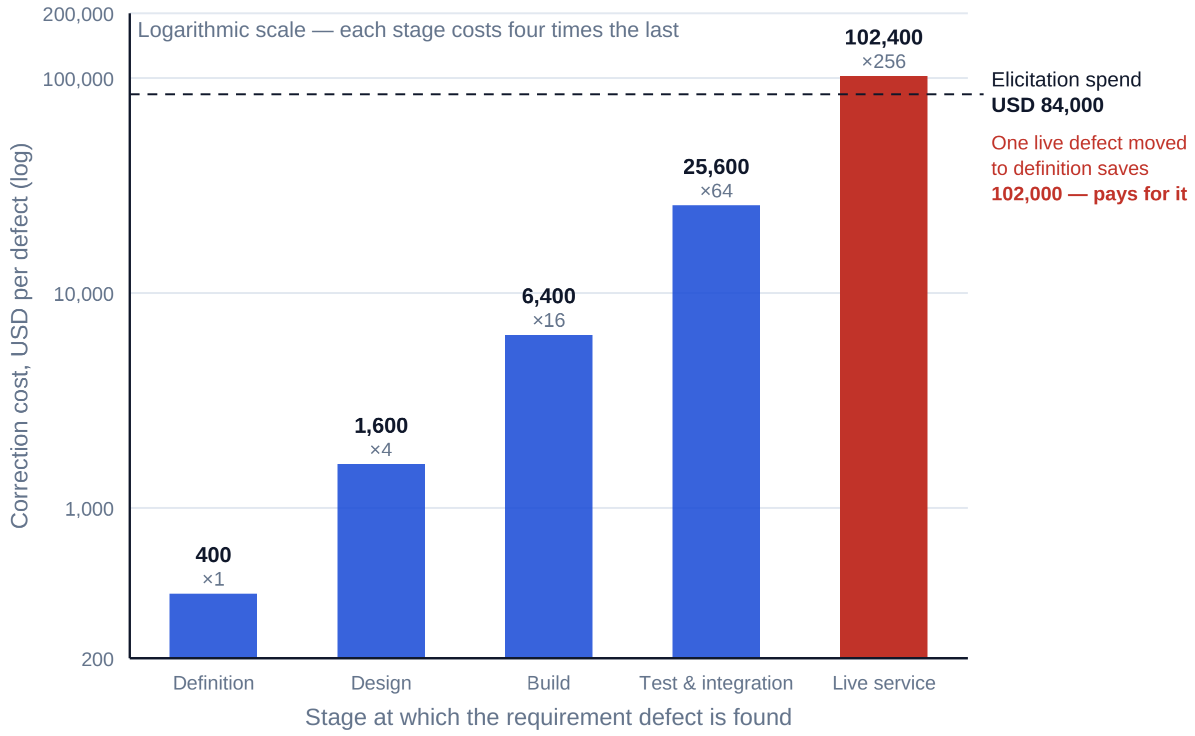


Fig 5.2.1 — What a requirement defect costs by the stage it is found

5.2.2 Analysis and specification quality

From elicited need to specified requirement. Analysis is the work of turning what stakeholders said into statements that can be designed against, built, tested and accepted. Six properties define a usable requirement, and each has a specific failure:

- **Necessary** — it traces to a benefit or an obligation. A requirement traceable to neither is someone's preference, and it will be built.
- **Unambiguous** — one reading only. The standard offenders are comparatives with no referent ("faster", "user-friendly"), passive constructions that hide the actor, and the conjunction "and/or".
- **Singular** — one requirement per statement. A compound requirement is accepted in part and disputed in part, and its status is permanently "in progress".
- **Testable** — a stated condition and a stated observable result. Untestable requirements are the subject of KA 5.4.2's arithmetic.
- **Feasible** — achievable within the architecture, the budget and the law, which requires a technical reader in the analysis loop.
- **Traceable** — identified, and linked forwards and backwards (5.2.3).

Requirement classes, and why they must be separated. Functional (what the system does), non-functional (how well — performance, availability, security, accessibility, usability), constraint (an imposed platform, standard or deadline), data (content, quality, retention), transition (migration, cutover, training, decommissioning) and regulatory (not negotiable and not prioritisable). They fail in different ways, which is why 5.2.3 reports them separately; and the two that go missing most reliably are **non-functional** requirements — because no user asks for availability — and **transition** requirements, which are Domain 4's missing 214,000 in requirement form.

The two analysis pathologies. Solution capture: recording the stakeholder's proposed solution as the requirement, which locks the design before the problem is understood and forfeits every cheaper option; the test is whether the statement expresses a need several designs could satisfy. Requirement inflation: elaborating each need into derived statements until the count impresses and the meaning disperses. A 480-requirement baseline for a records rollout is traceable; the same content expressed as 3,000 is traceable by nobody, and the count itself then obstructs prioritisation (KA 5.3).

5.2.3 Traceability

Definition. Traceability is the maintained, bidirectional linkage from **benefit → requirement → design element → test case → accepted deliverable**, such that any link can be followed in either direction. Its purpose is not documentation. It is to make four questions answerable by lookup rather than by opinion: what will happen to the benefit if this requirement is dropped; what must be re-tested if this design element changes; is there anything being built that nobody asked for; and can this deliverable be accepted.

Forward and backward are different tests. Forward traceability asks whether every requirement reaches a design element, a test case and an accepted deliverable — it detects **unmet scope**. Backward traceability asks whether every design element and every test case traces to an approved requirement — it detects **unrequested scope**, which is the mechanism of gold-plating and one of the routes by which creep enters (KA 5.4.1). Programmes overwhelmingly run the forward test only, and the reverse test is the cheaper of the two.

WORKED EXAMPLE

Worked example 5.2.3 — auditing Meridian's traceability matrix.

1. **Setup.** At entry to integration testing, Meridian's traceability audit covers the **480** approved requirements and the **425**-element design register. Forward: **416** requirements trace to a design element, a test case and a deliverable; **34** have no design element at all; **21** have a design element but no test case; **9** are recorded as accepted with no evidence of any verification having been performed. Backward: of the 425 design elements, **17** trace to no approved requirement.
2. **Formula.** Forward defect rate = non-conforming requirements ÷ approved requirements, reported **by class**. Reverse defect rate = orphaned design elements ÷ design register — a **different denominator**, and therefore not combinable with the forward rate.
3. **Substitution.** `(34 + 21 + 9) / 480`; and separately `34/480`, `21/480`, `9/480`; then `17/425`.
4. **Result. 64** forward non-conformances — a **13.33 %** forward defect rate — comprising **7.08 %** with no design, **4.38 %** with no test and **1.88 %** accepted without verification. **86.67 %** of requirements are fully traced. Separately, **17** orphaned design elements — a **4.00 %** reverse defect rate.
5. **Interpretation.** The classes are not totalled, for the reason Domain 4, KA 4.3.2 refused to total its configuration-audit classes: **they fail differently, and the total is the least informative number available.** The **34 with no design** are unmet scope that is still cheap — they surface at design review at 1,600 each on the ladder of 5.2.1, and the honest reading is that baseline and design are 34 requirements out of step. The **21 with no test case** will be delivered, accepted on assertion and found defective in live service at **102,400** each. The **9 accepted without verification evidence** are the serious finding, for a reason that has nothing to do with their number: the acceptance record states something untrue, so **every decision that relied on it — including the release decision — rested on nothing**, the same structural failure as Domain 4's five mis-recorded configuration items. A programme reporting "13.33 % traceability defects" has buried that. Two further observations. The rounded class rates sum to 13.34 % against a combined 13.33 % — display rounding, and one more reason not to present a total. And the **17 orphans** are design and build effort spent on work nobody approved, worth `17 × 6,400 =` **USD 108,800** at build-stage correction cost if they must be removed — invisible to the forward test most programmes run alone, and the reverse test costs one query.

AI in this KA

Where it earns its place. Traceability auditing is close to ideal AI work: exhaustive matrix traversal with definite answers, done badly and partially by humans, every finding checkable against a source — both directions, by class, with the orphan list produced as a matter of course. Ambiguity detection in requirement text — comparatives without referents, passive constructions hiding the actor, statements joined by "and", "and/or" and "as appropriate" — is linguistic pattern matching with a high true-positive rate, and is the highest-value AI application in the domain. Clustering interview notes and support-log entries into candidate themes, and flagging near-duplicates in a large baseline. Proposing candidate test conditions for analyst and tester ruling. And checking that every requirement carries a class and that the non-functional and transition classes exist at all — the absence of a whole class is a finding a model notices and a human does not.

Where it must not go. Authoring requirements. A requirement is a commitment made on behalf of stakeholders who exist, and a model generating plausible requirements from a project description produces exactly the failure this KA exists to prevent: a baseline that reads well and represents nobody. Ruling that a requirement is unnecessary or duplicated. Closing a traceability gap by creating the missing link, which converts a finding into a fabrication and is the most tempting automation failure here. And no AI output may serve as verification evidence for the nine requirements of WE 5.2.3 — that is the class the audit exists to catch.

Verification, concretely. Every reported gap is confirmed against both artefacts before it becomes a finding, and the orphan list is confirmed with the design owner, who can usually name the conversation the element came from. Ambiguity flags are triaged by the analyst, since some flagged constructions are correct in context. Clustered themes go back to the stakeholders who produced them, never straight into the baseline. The ladder arithmetic is reproduced by hand and its provenance stated as observed or assumed. And every AI contribution is recorded as an input, with the human author named.

KEY TERMS — KA 5.2

Term	Meaning
Elicitation	Deliberate discovery of need, as distinct from collection of requests.
Solution capture	Recording a stakeholder's proposed solution as the requirement, locking the design before the problem is understood.
Correction ladder	The cost of correcting one requirement defect by the stage found; Meridian's observed ladder rises ×4 per stage.
Breakeven elicitation spend	The investment at which the saving from earlier detection is exactly recovered; expressed either as a sum or as a tolerable delay.
Requirement classes	Functional · non-functional · constraint · data · transition · regulatory — reported separately because they fail differently.
Requirement inflation	Elaborating need into a count that impresses and disperses meaning, defeating traceability and prioritisation.
Forward traceability	Requirement → design → test → accepted deliverable; detects unmet scope.
Backward traceability	Design or test → approved requirement; detects unrequested scope, and costs one query.
Orphan	A design element or test case tracing to no approved requirement.

SAMPLE MCQS — KA 5.2

MCQ 5.2-A `[5.2.1 · Application]` On a ladder of 400 / 1,600 / 6,400 / 25,600 / 102,400 per defect by stage, an elicitation programme moves 13 build-stage, 9 test-stage and 3 live-service defects to the definition stage. The saving is: - A. USD 620,800 - B. USD 610,800 - C. USD 526,800 - D. USD 1,286,400

Rationale: The saving is the **difference** at each stage: `13 × 6,000 + 9 × 25,200 + 3 × 102,000 = 610,800` (5.2.1). A uses the gross later-stage costs and forgets the defect still costs 400 to fix at definition; C is the saving net of the 84,000 investment; D is the unimproved total correction cost.

MCQ 5.2-B `[5.2.1 · Evaluation]` The strongest single argument for the USD 84,000 elicitation programme is that: - A. it returns 7.27 times its cost - B. one defect moved out of live service saves USD 102,000 and pays for the whole programme - C. average correction cost falls from 13,400 to 7,037.50 per defect - D. early detection is recognised good practice

Rationale: The single-defect argument is decisive because it needs no assumption about how many defects move (5.2.1); A and C are true but depend on the whole estimated shift, and D is an appeal to practice rather than a computation.

MCQ 5.2-C `[5.2.3 · Analysis]` In a 480-requirement audit, 34 have no design element, 21 no test case, and 9 are recorded as accepted with no verification evidence. Which finding most undermines the **release decision itself**, and why? - A. the 34, because unmet scope is the largest group - B. the 21, because untested requirements reach live service - C. the 9, because the acceptance record states something untrue, so every decision that relied on it rested on nothing - D. all three equally, at a combined 13.33 % defect rate

Rationale: A false record invalidates the decisions taken on it, including the release decision (5.2.3) — the same structural failure as Domain 4's mis-recorded configuration items. A ranks by group size and B by future correction cost; both are real findings and both leave the release record truthful, which is why neither is the answer to this question. D is the reading the class breakdown exists to prevent.

MCQ 5.2-D `[5.2.3 · Comprehension]` Seventeen design elements trace to no approved requirement out of a 425-element register. This is: - A. a forward traceability defect of 3.54 % - B. a reverse traceability defect of 4.00 %, on a different denominator from the forward rate - C. immaterial, since the requirements are all still met - D. evidence that 17 requirements were lost

Rationale: `17/425 = 4.00 %`, and the design register is not the requirements baseline, so the two rates are not combinable (5.2.3). A puts the 17 over the 480-requirement baseline (`17/480 = 3.54 %`) and then mislabels a reverse finding as a forward one — the wrong denominator and the wrong test. C ignores unrequested effort worth 108,800 at build-stage cost. D inverts the finding: an orphan is work nobody asked for, not a requirement that went missing.

MCQ 5.2-E `[5.2.2 · Evaluation]` A requirement reads "the system shall provide faster record retrieval and improved reporting". Its defects are: - A. it is untestable only - B. it is ambiguous ("faster", "improved" — no referent) and compound (two requirements in one) - C. it is infeasible - D. it is untraceable

Rationale: Comparatives without a referent make it ambiguous and therefore untestable, and the conjunction makes it two requirements that will be accepted in part and disputed in part (5.2.2).

SELF-CHECK — KA 5.2

1. Why must the elicitation saving be computed on the difference between stages? — Because the defect still costs 400 to correct at definition; using gross later-stage costs overstates the saving, here by 10,000.
2. Which two requirement classes go missing most reliably, and why? — Non-functional, because no user asks for availability; and transition, because migration, cutover and training belong to somebody else.
3. What does backward traceability detect that forward traceability cannot? — Unrequested scope: 17 design elements nobody approved, worth 108,800 at build-stage correction cost, invisible to the forward test most programmes run.

KNOWLEDGE AREA 5.3

Value definition and prioritisation

Topics: 5.3.1 attributing value to requirements · 5.3.2 prioritisation methods and their failure modes · 5.3.3 capacity-constrained selection.

5.3.1 Attributing value to requirements

The link that closes the chain. Domain 2 built the benefits map from strategic driver to benefit to measure and owner. This KA connects the other end: from benefit to the **requirement bundle** that delivers it. Without that link, prioritisation has nothing to prioritise on and the benefits register has nothing to hold anyone to.

Bundles, not individual requirements. Value is rarely attributable to a single requirement, because benefits arise from usable capability and a capability needs several requirements to exist at all. The working unit is therefore the **requirement bundle**: the smallest set of requirements that together delivers a measurable benefit. Attributing value at the individual requirement level produces spuriously precise numbers and, worse, invites the delivery of half a capability — which returns nothing, at full cost.

Two tests that make attribution honest. The sum test: attributed benefit across all bundles must not exceed the business case's benefit, and where it falls short the balance must be explained rather than distributed. Meridian's five Release 1 candidate bundles carry **USD 957,000** of the **USD 979,200** full-potential annual benefit, and the remaining **USD 22,200** sits with minor requirements that support no bundle on their own — a statement, not a rounding. The counterfactual test: for each bundle, what happens to the benefit if it is not delivered? A bundle whose absence costs nothing has no value attributed to it, however popular it is, and a bundle whose absence collapses several other bundles' benefits is a dependency rather than a candidate.

Where attribution is legitimately impossible. Regulatory requirements, safety requirements and requirements imposed by contract are not prioritisable and should be removed from the value ranking entirely, held instead as constraints on capacity. Attempting to score them produces the absurdity of a statutory obligation ranking eleventh, and the process then has to override itself, which discredits the whole ranking.

5.3.2 Prioritisation methods and their failure modes

The methods, and what each is actually for. Ordered ranking (a strict sequence) is honest and does not scale past a few dozen items. Category schemes (must / should / could / will not, or an equivalent) scale and are the standard instrument in adaptive delivery (Domain 13), but they degrade in a specific way described below. Weighted scoring against explicit criteria is the Domain 2, KA 2.2.3 instrument applied at requirement level: it forces criteria into the open and remains an ordinal judgement, so its arithmetic must not be over-read. Value per unit of effort is the method that actually engages the constraint, and 5.3.3 shows both its power and its limit.

Must-have inflation, quantified. Category schemes fail because the categorisation is done by the requirement's originator, and every originator's requirement is a must-have. Of Meridian's **480** requirements, **374 — 77.92 %** — were marked "must have" on first pass. A structured re-test, in which each originator was asked to **name the consequence of releasing without it**, left **148 — 30.83 %** — standing as genuine must-haves. The discretionary pool that prioritisation can actually act on therefore grows from **106** requirements (**22.08 %**) to **332** (**69.17 %**), a **3.13-fold** increase in the scope available to trade. That number is the whole argument for the re-test: at 77.92 % must-have the classification carries almost no information and prioritisation is theatre, because four requirements in five are outside its reach.

Two further failure modes worth naming. Prioritising by cost rather than by value per unit of cost, which systematically favours cheap low-value work and is how a release ships twelve small conveniences and no capability. And prioritising without the dependency map, which produces sequences that cannot be built — the subject of the caution in 5.3.3 and of Domain 6's logic networks.

5.3.3 Capacity-constrained selection

The constraint decides the method. Domain 2, KA 2.2.3 established the principle at portfolio level and it holds unchanged here: **greedy ranking by value per unit of the scarce resource is a heuristic, not an optimum, and lumpy candidates require enumerating the feasible sets.** What follows is that result worked at requirement level, where the effect is larger because requirement bundles are lumpier than projects and the capacity window is shorter.

WORKED EXAMPLE

Worked example 5.3.3 — where the ranking loses to the enumeration.

1. **Setup.** Meridian's Release 1 has **20 development-weeks** of capacity before the first clinic wave must begin — a hard constraint set by the clinical training calendar, not by money. Five candidate requirement bundles, with their attributed annual benefit and their effort:

Bundle	Requirement bundle	Effort (dev-weeks)	Annual benefit (USD)	Value per week
A	Unified patient record and clinical search	13	299,000	**23,000**
B	e-Prescribing with decision support	10	228,000	**22,800**
C	Appointments, referrals and clinic workflow	10	225,000	**22,500**
D	Automated national reporting	5	110,000	**22,000**
E	Analytics dashboards	5	95,000	**19,000**

Total effort **43** weeks against 20 available; total attributed benefit **957,000** of the 979,200 full potential. 2. **Formula.** Greedy: rank by value per week, take each bundle that fits the remaining capacity. Enumeration: evaluate every subset whose total effort ≤ capacity and take the highest total value. Then value the difference: `annual difference × AF(r, n)`, with `AF(0.07, 8) = 5.971299` from Domain 2. 3. **Substitution.** Greedy takes **A** (23,000/week, 13 weeks, 7 remaining), skips B and C (10 weeks each, do not fit), takes **D** (5 weeks, 2 remaining), and cannot fit E. Enumeration compares every feasible subset, of which the material ones are `A+D = 18 weeks / 409,000`, `A+E = 18 / 394,000`, `B+D+E = 20 / 433,000`, `C+D+E = 20 / 430,000` and `B+C = 20 / 453,000`. 4. **Result.** Greedy selects **A + D**: **USD 409,000** a year, using **18 of 20** weeks — **90.0 %** capacity utilisation, with **2 weeks idle and no bundle small enough to use them**. Enumeration selects **B + C**: **USD 453,000** a year, using **20 of 20** weeks — **100 %** utilisation. The difference is **USD 44,000** a year, **10.76 %** more benefit, worth `44,000 × 5.971299 =` **USD 262,737** of present value over the eight-year appraisal period at 7 %. 5. **Interpretation.** The mechanism generalises to every constrained selection a leader will make, so it is worth understanding rather than memorising. Greedy fails **because the highest-ratio bundle is lumpy**: A leads on value per week by only 200 — 23,000 against B's 22,800, a **0.88 %** advantage —

and taking it strands two weeks that nothing can use. A 0.88 % ratio advantage bought a 10.76 % value loss, and the closer the ratios, the likelier that is. The rule follows directly: **where the constraint binds and the candidate set is small enough to enumerate, enumerate; otherwise rank greedily and then test whether the residual capacity can be filled, because stranded capacity is the signature of the failure.** Three cautions, the first severe. The enumeration assumes the bundles are **independent**; if C's clinic-workflow requirements depend on A's unified record then `B+C` is infeasible, so the dependency map (Domain 6, KA 6.1) is a **precondition** for this arithmetic, not a refinement of it, and without it the enumeration yields a confident answer that cannot be built. Second, a 44,000 difference between figures of 409,000 and 453,000 sits well inside the uncertainty of any benefit estimate, so the defensible claim is not "B+C is worth 44,000 more" but "the ranking method systematically strands capacity and the enumeration does not" — the structural argument survives uncertainty that the point figure does not. Third, deferring A has a cost the table cannot show: the unified record is the bundle clinicians associate with the programme's purpose, and adoption is behavioural (Domain 2's 70 %). A sponsor may defensibly overrule the arithmetic on adoption grounds, and should then record that they are buying adoption confidence for 44,000 a year — which is a decision, whereas an unexamined greedy ranking is not.

Fig 5.3.1 — Greedy ranking strands capacity; enumeration does not

What no prioritisation method does. It does not decide. Ranking and enumeration order candidates against one stated constraint and one stated value measure; they cannot see option value, sequencing dependencies, adoption behaviour, political commitment or the strategic cost of not doing something — the same limits Domain 2, KA 2.2.3 recorded at portfolio level. A release board that treats an enumeration output as its decision has automated its own accountability, which is Domain 1's principle failing quietly.

AI in this KA

Where it earns its place. Enumerating feasible subsets under a stated constraint is exactly the work this KA shows humans getting wrong, and it is deterministic — correct enumeration removes the greedy failure entirely and the answer is verifiable by re-computation. Constructing the dependency graph among bundles from the traceability matrix and design register, and reporting which enumerated sets it makes infeasible. Running the sum test against the business case and reporting the shortfall. Running the must-have re-test as a first pass by flagging every "must have" whose stated consequence of omission is absent or circular — how a 77.92 % figure becomes informative. And sensitivity analysis: how far the benefit estimates must move before the selection changes, which is more decision-relevant than the point answer.

Where it must not go. Attributing benefit. A benefit figure is a claim about the organisation's future behaviour, owned by a named benefit owner (Domain 2), and a model producing plausible attributions manufactures the precision the worked example warns is absent. Ruling on must-have status, which is a stakeholder consequence judgement. Selecting release content, which is a governance decision with adoption, political and sequencing dimensions no ranking captures. And no model re-weights the value measure, because whoever controls the weights controls the outcome.

Verification, concretely. Both the greedy and the enumerated answer are computed and reported, because the comparison is the finding — a single reported optimum hides whether the constraint bound at all. Every enumerated set is checked against the dependency graph, itself confirmed by the technical owner. The value-per-week arithmetic is reproduced by hand for the leading candidates. Attributed benefits are confirmed with their owners and their ranges stated. And the discounting reuses Domain 2's registered `AF(r, n)`, cited rather than re-derived.

KEY TERMS — KA 5.3

Term	Meaning
Requirement bundle	The smallest set of requirements that together delivers a measurable benefit — the unit of value attribution.
Sum test	Attributed benefit across bundles must not exceed the business case, and any shortfall is explained, not distributed.
Counterfactual test	What happens to the benefit if this bundle is not delivered? A zero answer means zero attributed value.
Must-have inflation	Originator-assigned categories collapsing into near-universal "must have", destroying the scheme's information content.
Value per unit of constrained effort	Attributed benefit ÷ effort in the scarce resource — the ranking that engages the constraint.
Greedy ranking	Taking candidates in ratio order while they fit; a heuristic that strands capacity when candidates are lumpy.
Enumeration	Evaluating every feasible subset under the constraint; the optimum where the candidate set is small enough.
Stranded capacity	Unused constrained resource that no remaining candidate is small enough to use — the signature of greedy failure.

SAMPLE MCQS — KA 5.3

MCQ 5.3-A `[5.3.3 · Application]` With 20 development-weeks available and bundles A (13 wk, 299,000), B (10 wk, 228,000), C (10 wk, 225,000), D (5 wk, 110,000) and E (5 wk, 95,000), greedy ranking by value per week selects: - A. A + D, worth 409,000 a year - B. B + C, worth 453,000 a year - C. A + B, worth 527,000 a year - D. B + D + E, worth 433,000 a year

Rationale: Greedy takes A first (23,000/week), then D, stranding 2 weeks — 409,000 (5.3.3). B is the enumerated optimum, not the greedy result; C exceeds capacity at 23 weeks; D is a feasible set greedy never reaches.

MCQ 5.3-B `[5.3.3 · Analysis]` In that selection, greedy loses to enumeration because: - A. value per week is the wrong measure - B. the highest-ratio bundle is lumpy and strands 2 of 20 weeks that nothing can fill - C. bundle A's benefit is overstated - D. the capacity constraint is not binding

Rationale: A's ratio advantage is under 1 % and its size wastes 10 % of capacity (5.3.3). The measure is not wrong; it is a heuristic, and the constraint is precisely what makes it fail.

MCQ 5.3-C `[5.3.3 · Evaluation]` Before the enumeration above can be relied upon, the indispensable additional input is: - A. a probability distribution on each benefit estimate - B. the dependency map among bundles, since an infeasible set must not be selected - C. the sponsor's ranking preference - D. a re-test of the must-have classification

Rationale: If C depends on A, `B+C` cannot be built and the answer is confidently wrong (5.3.3); dependency feasibility is a precondition, not a refinement. A and D are valuable and not indispensable.

MCQ 5.3-D `[5.3.2 · Analysis]` 374 of 480 requirements are marked "must have". The consequence for prioritisation is that: - A. the project is unusually well specified - B. the discretionary pool is 106 requirements — 22.08 % — so prioritisation can act on a fifth of the scope, and the classification carries almost no information - C. the must-haves should be delivered first in ranked order - D. capacity must be increased

Rationale: At 77.92 % must-have the category no longer discriminates; the re-test that asks for the consequence of omission raises the actionable pool to 332, a 3.13-fold increase (5.3.2).

MCQ 5.3-E `[5.3.1 · Comprehension]` Regulatory and safety requirements should be: - A. scored with the highest weight in the value model - B. removed from the value ranking and held as constraints on capacity - C. prioritised last, since they generate no benefit - D. attributed a nominal benefit for completeness

Rationale: They are not prioritisable; scoring them produces a statutory obligation ranking eleventh and forces the process to override itself, discrediting the ranking (5.3.1).

SELF-CHECK — KA 5.3

1. Why is value attributed to bundles rather than to individual requirements? — Because benefit arises from usable capability; attributing at requirement level manufactures precision and invites delivery of half a capability, which returns nothing at full cost.
2. What is the signature of a greedy prioritisation failure? — Stranded capacity: 2 of 20 weeks idle with no candidate small enough to fill them, costing 44,000 a year and 262,737 of present value.
3. What must be true before an enumeration is trustworthy? — The sets enumerated must be feasible against the dependency map; otherwise the optimum cannot be built.

KNOWLEDGE AREA 5.4

Scope change, creep and verification/acceptance

Topics: 5.4.1 change versus creep · 5.4.2 acceptance criteria and testability · 5.4.3 verification, validation and acceptance.

5.4.1 Change versus creep

The distinction, stated precisely. A **change** is a movement of the scope baseline that has been raised, screened, assessed, decided by an authority and recorded — Domain 4's flow, and Domain 4 quantified what its accumulation costs. **Creep** is a movement of the delivered content with **no corresponding baseline movement and no record**. The two are not degrees of the same thing. Change is a control operating; creep is the absence of one, and it is invisible to every instrument built to monitor change — including Domain 4's cumulative test, which reads a change log that by definition does not contain it.

The three routes creep takes, all of them familiar and none of them dishonest at the point of entry. Misclassification as clarification: the requester says the baseline already covered it, the screening step agrees without testing the claim against the requirement text, and scope grows with a paper trail that records no growth — the mirror error Domain 4, KA 4.4.1 named. Acceptance-criteria expansion: the requirement text is unchanged and its acceptance criterion grows, which is a scope change wearing a testing costume and is the commonest route where criteria are weak (5.4.2). Undocumented addition: someone with access and good intentions adds capability — a clinician asks a developer directly, or a developer improves something while in the code. This last is "gold-plating", and 5.2.3's reverse traceability test is the only cheap way to see it.

WORKED EXAMPLE

Worked example 5.4.1 — the movement that left no trace in the change log.

1. **Setup.** Meridian's requirements baseline at design freeze was **480** approved requirements. Through the following 13 months, integrated change control approved **12** requirement additions. At entry to acceptance testing the traceability matrix carries **531** requirements. The average direct build cost of an uncontrolled requirement is **USD 4,200**, and **16** of them consumed critical-path time averaging **0.25 weeks** each. Cost of delay **USD 14,280** per week; baseline **USD 2,400,000**. Domain 4, KA 4.3.3 separately established **USD 291,176** of authorised baseline drift over the programme's first year, so the two figures set beside each other are Meridian's cumulative baseline movement to date rather than two readings of one identical window.
2. **Formula. Requirement-count reconciliation:** baseline count + approved additions = expected traced count; actual − expected = unexplained requirements. Creep cost = (unexplained × average direct cost) + (affected count × average weeks × cost of delay). Then combine with the authorised drift and compute what share of total movement the change log captured.
3. **Substitution.** `480 + 12 = 492`; `531 − 492`; `39 × 4,200`; `16 × 0.25 × 14,280`; `291,176 + 220,920`; `291,176 / 512,096`.
4. **Result. 39** requirements are present in the delivered content and absent from every record — about **3 a month**, and **3.25 times** the number that came through change control. They cost **USD 163,800** direct plus `4.0` weeks of critical path at **USD 57,120**, a total of **USD 220,920 — 9.20 %** of the baseline, at **USD 5,664.62** each. Combined with Domain 4's authorised drift, Meridian's total baseline movement is **USD 512,096**, or **21.34 %** of the approved baseline, of which the change log accounts for only **56.86 %**.
5. **Interpretation.** The decisive number is **56.86 %**, and it should change how a leader reads a change report. Meridian had a working change process, a decision log and — after Domain 4 — a cumulative test derived from its own change rate, and all of that machinery was monitoring a little over half the movement it existed to control. The other 43 % was not hidden; it was simply never a change, because nobody raised it. Each crept requirement carries **USD 4,200** of direct cost — **0.175 %** of the baseline, the basis on which a delegation threshold reads a single item — so no threshold in Domain 3's delegation schedule can see one, and no cumulative test can aggregate what its log does not contain. Which gives the instrument, and it is almost embarrassingly cheap: **count the requirements.** Baseline count plus approved additions must equal the traced count, every reporting period — one addition and one subtraction against two registers that already exist, and the only control in this book that detects creep. Two further readings. Against Domain 2's approved NPV of **+USD 1,332,898**, total scope movement of 512,096 consumed **38.42 %** of the programme's whole justification, leaving **USD 820,802** — still positive, which is exactly why nobody stopped: creep is survivable right up until it is not. And a caution on the arithmetic: 4,200 is an average over requirements of very different size, and the 16-of-39 critical-path attribution comes from the schedule rather than the change log, carrying the schedule's own uncertainty (Domain 6). Report the 220,920 as an estimate with a stated basis and the **39** as a fact — because the count is a fact, and the count is what produces the action.

5.4.2 Acceptance criteria and testability

Definition. An acceptance criterion states the **condition** under which the deliverable will be judged to meet the requirement, in terms of an observable and measurable result. It is written when the requirement is written, by the requirement's owner with the tester present, because a criterion written later is written by whoever needs the deliverable accepted.

The three states a criterion can be in, and the middle one is the dangerous one:

State	Appearance	Consequence
Measurable	States a condition, an action and an observable result with a threshold	Acceptance is a test
Restated	Reads like a criterion, restates the requirement in other words ("the system shall retrieve records correctly")	Acceptance is an argument — and the gap is invisible until it happens
Absent	No criterion at all	Acceptance is an argument — and the gap is visible now

WORKED EXAMPLE

Worked example 5.4.2 — what it is worth to write a testable criterion.

1. **Setup.** Of Meridian's **480** requirements, **382** carry a measurable acceptance criterion, **71** carry a criterion that restates the requirement, and **27** carry none. Writing a measurable criterion, with a reviewer, costs **USD 320** of analyst and tester time. The organisation's history is that a deficient criterion produces a formal acceptance dispute in about **1 case in 4**; a dispute costs about **USD 9,600** in rework, re-test and clinical re-review; and about **1 dispute in 3** delays a clinic go-live by **0.5 weeks** at the cost of delay of **USD 14,280** per week.
2. **Formula.** Remediation cost = deficient count × cost per criterion. Expected cost per dispute = rework + (share delaying × weeks × cost of delay). Expected dispute cost = deficient count × P(dispute) × cost per dispute. Breakeven probability = remediation cost ÷ (deficient count × cost per dispute).
3. **Substitution.** `(71 + 27) × 320`; `9,600 + (1/3 × 0.5 × 14,280)`; `98 × 0.25 × 11,980`; `31,360 / (98 × 11,980)`.
4. **Result.** **98** requirements are deficient — **20.42 %** of the baseline, comprising **14.79 %** restated and **5.62 %** absent (the two class rates rounding to 20.41 against the combined 20.42 — display rounding, as in 5.2.3, and one more reason the classes and not the total are the report). Remediating all of them costs **USD 31,360**. Expected cost per dispute is **USD 11,980** (9,600 of rework plus **USD 2,380** of expected delay). Expected disputes are **24.5**, at an expected cost of **USD 293,510**. Remediation therefore nets **USD 262,150**, a return of **9.36 times**. And the **breakeven dispute probability is 2.67 %** against an observed **25 %**.
5. **Interpretation.** The breakeven settles the argument this work always loses. Writing testable acceptance criteria pays for itself if deficient criteria cause disputes more than **2.67 %** of the time; the observed rate is nine times that, and an organisation would have to believe its acceptance process almost frictionless before declining to spend 320 a requirement. Note next which class is worse, because the count says the opposite of the truth. The **27 absent** criteria are a visible gap: they appear on every completeness report and someone raises them. The **71 restated** criteria look complete, pass every completeness check, and are discovered at acceptance — deliverable built, clinic scheduled, supplier invoicing — which is Domain 4's "recorded state differs from actual state" failure moved from configuration to acceptance, and the class WE 5.2.3's nine unverified acceptances came from. **A restated criterion is worse than a missing one, and it is commoner.** Three cautions. The 24.5 is an expectation, not a forecast of 24 or 25 events, and must not be reported as a count of disputes that will happen. The rates and costs are the organisation's own history; a programme without that history should say it is using assumed rates and test the breakeven's sensitivity rather than the point result — and here the breakeven sits so far below the observed rate that the conclusion survives any plausible error in either. And remediation is not free of schedule: 98 criteria at four a day is about five weeks of one analyst's time, which has to be planned rather than absorbed, or it will not happen.

5.4.3 Verification, validation and acceptance

The distinction that Case study B turns on. Verification asks did we build what we specified? **Validation** asks does what we built produce the outcome we needed? They are different questions with different evidence and, critically, different failure signatures: a project can pass verification completely and fail validation completely, and it will report itself as a success throughout, because verification is the test it has instrumented. Validation requires the receiving organisation and real use, which is why it belongs to handover and benefits realisation (Domain 16) as well as here — and why Domain 1 insisted the leader is answerable for the outcome and not only the output.

Acceptance is the authorised act of taking a deliverable as meeting its criteria. It requires a named acceptor with authority, evidence against each criterion, and a recorded decision. Two provisions make it real rather than ceremonial: the acceptor is not the producer, and the evidence is retained — because the nine requirements of WE 5.2.3 were "accepted" and the record could not say by whom, on what evidence, or against what criterion.

Conditional acceptance, and its price. Deliverables are routinely accepted with open items, and this is often correct: waiting for perfection forgoes benefit at Meridian's 14,280 a week. But deferring an item moves it up the correction ladder of 5.2.1, and that movement is computable.

WORKED EXAMPLE

Worked example 5.4.3 — deferring six severity-one items.

1. **Setup.** At Meridian's Release 1 acceptance there are **41** open items, of which **6** are severity one — a defect against a requirement that materially affects clinical use. Fixing them before acceptance costs **USD 6,400** each on the correction ladder (build stage) and delays go-live by **1.5 weeks**. Deferring them to live service costs **USD 102,400** each. Cost of delay **USD 14,280** per week.
2. **Formula.** Fix-first cost = items × build-stage cost + delay weeks × cost of delay. Defer cost = items × live-service cost. Breakeven go-live delay = (defer cost − items × build-stage cost) ÷ cost of delay.
3. **Substitution.** `6 × 6,400 + 1.5 × 14,280`; `6 × 102,400`; `(614,400 − 38,400) / 14,280`.
4. **Result.** Fixing first costs **USD 59,820** (38,400 of correction plus **USD 21,420** of delay). Deferring costs **USD 614,400 — 10.27 times** as much, a difference of **USD 554,580**. Go-live would have to be delayed by **more than 40.34 weeks** before deferral became the cheaper option.
5. **Interpretation.** The 40.34-week breakeven makes this decision arithmetic rather than temperament, which is why it is worth computing in the room: the instinct at acceptance is that a 1.5-week delay is unaffordable, and the tolerable delay is about twenty-seven times that. But the result holds **per severity class, not per item**, and generalising it is the error to avoid. The **35** severity-two and severity-three items may be entirely correct to defer, because the live-service figure on the ladder prices correcting a requirement defect in a running clinical service — regression, re-release, re-training, clinical re-approval — and a cosmetic or low-frequency item does not incur most of that. A programme that applies the top of the ladder to all 41 items will refuse to go live at all, forgoing 14,280 a week for items worth a fraction of it. Two further points. The go-live delay already prices forgone benefit through the registered cost of delay (Domain 1), so a separate benefit-loss term would double-count. And the arithmetic assumes the six items can be fixed in 1.5 weeks; if they cannot, the question is not fix-versus-defer but what Release 1 contains — which belongs to KA 5.3, not to an acceptance meeting under time pressure.

AI in this KA

Where it earns its place. The requirement-count reconciliation of WE 5.4.1, run every period and producing the **list** of unexplained requirements rather than only the number. This is the domain's highest-value automation: the check is trivially cheap, nobody runs it, and it catches the 43 % of movement no change-log instrument can see. Classifying acceptance criteria as measurable, restated or absent — a linguistic test with a definite answer, and how 71 invisible criteria become visible before acceptance rather than during it. Detecting acceptance-criteria expansion by diffing criterion text against its baselined version, which is the creep route that leaves the requirement text untouched. Assembling the evidence pack per criterion and listing criteria with none. And applying the correction ladder by severity class to an open-item list to produce the fix-versus-defer comparison for decision.

Where it must not go. Accepting a deliverable — an authorised act by a named human who is not the producer. Ruling that a request is a clarification rather than a change, the misclassification route one of creep depends on, and a determination with commercial consequences. Deciding which open items may be deferred, which here is a clinical, safety and commercial judgement. And no AI-generated text may serve as acceptance evidence: a model asserting that a criterion is met is not evidence that it is, and treating it as such recreates the defect WE 5.2.3 found nine instances of.

Verification, concretely. Every unexplained requirement is investigated by a human before it is reported as creep, because some prove to be recorded changes with a broken reference — itself a finding, of a different kind. Criterion classifications are sampled and confirmed by the tester, particularly the restated class, since that judgement is the one with money attached. The fix-versus-defer arithmetic is reproduced by hand with the severity class stated for each item, because applying the live-service figure to a cosmetic defect is the error a tool makes silently. And every acceptance decision names its human acceptor, with any AI contribution recorded as analysis input.

KEY TERMS — KA 5.4

Term	Meaning
Change	An assessed, authorised, recorded movement of the scope baseline (Domain 4, KA 4.4).
Scope creep	Movement of delivered content with no baseline movement and no record; invisible to every change-log instrument.
Requirement-count reconciliation	Baseline count + approved additions = traced count; the only cheap control that detects creep.
Acceptance-criteria expansion	Creep entering through a growing criterion while the requirement text is unchanged.
Restated criterion	A criterion that repeats the requirement instead of stating a measurable condition — worse than an absent one, and commoner.
Breakeven dispute probability	The dispute rate at which writing testable criteria exactly pays for itself; **2.67 %** at Meridian against an observed 25 %.
Verification	Did we build what we specified?
Validation	Does what we built produce the outcome we needed? A project can pass the first completely and fail the second completely.
Conditional acceptance	Acceptance with open items; priced by moving each item up the correction ladder **by severity class**.

SAMPLE MCQS — KA 5.4

MCQ 5.4-A `[5.4.1 · Application]` A baseline of 480 requirements received 12 approved additions; the traceability matrix at acceptance-test entry carries 531. Uncontrolled requirements average 4,200 direct, and 16 of them consumed 0.25 weeks of critical path each at a delay cost of 14,280 per week. Total creep cost is: - A. USD 163,800 - B. USD 220,920 - C. USD 214,200 - D. USD 57,120

Rationale: `531 − (480 + 12) = 39` unexplained; `39 × 4,200 = 163,800` plus `16 × 0.25 × 14,280 = 57,120` gives 220,920 (5.4.1). A omits the schedule impact; C prices all 51 additions including the 12 that were properly approved; D is the schedule impact alone.

MCQ 5.4-B `[5.4.1 · Evaluation]` Domain 4 established USD 291,176 of authorised baseline drift over the programme's first year, giving cumulative movement of USD 512,096 against a USD 2,400,000 baseline. The most important consequence of the creep figure is that: - A. total movement is 21.34 % of the baseline - B. the change log captures only 56.86 % of the movement it exists to control, and no cumulative test on that log can see the rest - C. the average crept requirement cost 5,664.62 - D. the delegation threshold should be lowered below 4,200

Rationale: Creep leaves no change record, so change-log instruments — including Domain 4's cumulative test — monitor a little over half the movement (5.4.1); the remedy is the count reconciliation. A and C are true and subordinate; D cannot work, since no change is ever raised.

MCQ 5.4-C `[5.4.2 · Application]` Remediating 98 deficient criteria costs 320 each. A deficient criterion causes a dispute 25 % of the time, at an expected cost of 11,980 per dispute. The breakeven dispute probability is: - A. 25.00 % - B. 2.67 % - C. 9.36 % - D. 20.42 %

Rationale: `31,360 / (98 × 11,980) = 2.67 %` (5.4.2). A is the observed rate; C is the return multiple; D is the share of the baseline that is deficient.

MCQ 5.4-D `[5.4.3 · Evaluation]` Six severity-one items cost 6,400 each to fix before acceptance, with a 1.5-week go-live delay at 14,280 per week, or 102,400 each in live service. The result and its limit: - A. defer; the go-live delay is unaffordable - B. fix first — it costs 59,820 against 614,400, and the breakeven delay is 40.34 weeks — but the ladder must be applied by severity class, not to all 41 open items - C. fix first, and apply the same reasoning to all 41 open items - D. the comparison cannot be made without the benefit forgone during the delay

Rationale: Fix-first is cheaper by 554,580 and would remain so up to a 40.34-week delay (5.4.3); applying the live-service figure to minor items would prevent go-live altogether. D double-counts — the cost of delay already prices forgone benefit (Domain 1).

MCQ 5.4-E `[5.4.3 · Comprehension]` A programme reports 100 % of requirements verified and its benefits are 34 % of forecast. This is: - A. a verification failure - B. a validation failure — the specified thing was built, and it did not produce the outcome - C. a benefits-measurement error - D. evidence of scope creep

Rationale: Verification asks whether the specified thing was built and validation whether it produces the outcome; the two fail independently (5.4.3), which is Case study B.

SELF-CHECK — KA 5.4

1. Why can no cumulative change test detect creep? — Because creep generates no change record; the test reads a log that by definition does not contain it. Only a count reconciliation catches it.
2. Which acceptance-criterion class is most dangerous, and why? — The restated criterion: it passes completeness checks and is discovered at acceptance, with the deliverable built and the supplier invoicing.
3. What does the 40.34-week breakeven in WE 5.4.3 tell an acceptance meeting? — That the instinct "we cannot afford 1.5 weeks" is wrong by a factor of about twenty-seven for severity-one items — and that the same arithmetic does not license fixing everything.

Advanced topics — Domain 5

5.A.1 Scope in adaptive and hybrid delivery

The obvious objection to this domain is that adaptive delivery deliberately does not fix scope, so a scope baseline is the wrong instrument. The objection is half right and the conclusion is wrong. What adaptive delivery declines to fix is the **item list**; what it must still fix — more strictly than predictive delivery — is the **capacity, the cadence and the value envelope**. A backlog is not an absent baseline but a baseline on different variables, and Domain 13 works the mechanics.

Three of this domain's controls become more important, and they are the three adaptive programmes most often drop as belonging to a predictive method. **Value attribution** (KA 5.3.1), because iteration decisions are prioritisation decisions taken every few weeks and an unattributed backlog is prioritised on advocacy. **Acceptance criteria** (KA 5.4.2), because the definition of done is the only acceptance instrument available and a restated criterion fails inside a sprint exactly as it fails at a formal gate, only faster. And **backward traceability** (KA 5.2.3), because the shorter the path from idea to build, the easier it is for an item nobody approved to be delivered.

What genuinely changes is the creep instrument: a requirement-count reconciliation is meaningless where the count is supposed to move. The equivalent is a **value-envelope reconciliation** — attributed benefit of delivered items against the business case's benefit commitment, period by period. It detects the failure adaptive programmes actually suffer, which is not item growth but a backlog that has drifted away from the benefits it was funded to produce: delivering steadily, validating badly. Hybrid delivery needs both, applied to the parts that are each kind, with a named boundary between them — a hybrid programme running one control regime has silently chosen one method.

5.A.2 Requirements across a contractual boundary

Where the specification is written by one party and built by another, requirement quality becomes an allocation of risk — and the allocation is usually made accidentally, by the choice of specification style. An **output specification** states what the supplier must deliver, and the buyer retains the risk that delivering it produces no outcome: verification passes, validation fails, and there is no claim because the supplier built what was specified. An **outcome specification** transfers more of that risk to a supplier who will price it and who cannot control the buyer's own enabling change, so an outcome specification carrying no buyer obligations is either unpriceable or priced with a large margin. Neither is right in general; what is always wrong is not knowing which one has been signed.

Four provisions follow, and they belong in the contract rather than a plan (Domain 10; PFL-AI Domain 11). Requirement defects and their remedy are **allocated explicitly** — who bears the cost when a requirement turns out to be wrong rather than badly built, because on the ladder of 5.2.1 that difference is 6,400 against 102,400 and it will be argued. Acceptance criteria are **agreed and baselined before signature**, since a criterion negotiated after award is negotiated by the party who wants payment. **Traceability is a deliverable**, with a stated format and an audit right, or the buyer cannot run either test in KA 5.2.3 against work it did not do. And the buyer's own obligations — data, access, decisions within stated times, where Domain 3's `E[wait]` becomes a contractual exposure — are stated with the specificity demanded of the supplier, because an unstated buyer obligation is a common root of a claim.

Two boundaries on this section. Whether a given specification style, remedy or acceptance provision has the effect described depends on the contract's governing law and on the form of agreement used, and those differ materially between jurisdictions and between standard forms; nothing here states the law of any jurisdiction and none of it is legal advice. Drafting, risk allocation and any remedy question go to qualified counsel (Domain 10). What is portable is the method: decide which specification style you have signed, allocate requirement-defect risk deliberately rather than by default, baseline the acceptance criteria before signature, make traceability a deliverable, and state the buyer's own obligations as precisely as the supplier's.

5.A.3 The reviewer's scope eye

Invariants to test on any scope and requirements position, each cheap and each diagnostic.

The scope statement's extents are **counted, not categorised**, and every category noun has a number against it. The exclusion list exists and is **non-empty** — an empty exclusion list means the boundary questions have not been asked, not that there are none. Every assumption has an owner and a resolution date. The **WBS dictionary has content**, not only titles, and carries the acceptance basis. Every requirement carries a **class**, and the non-functional and transition classes are **present at all**; their total absence is the single fastest defect to spot and among the most expensive. Every requirement is **singular and testable**, and the acceptance criteria have been classified as measurable, restated or absent — with the restated count reported, because it is the one nobody has. **Traceability is audited in both directions**, reported **by class and never totalled**, with the orphan list produced; a programme that has never run the reverse test does not know what it is building. Every requirement bundle has an **attributed benefit with a named owner**, the attribution **passes the sum test** against the business case, and regulatory requirements are **held outside the value ranking** as constraints. Where a capacity constraint binds, both the greedy and the enumerated selections are computed, the **dependency map** has been used to exclude infeasible sets, and any **stranded capacity** is reported as a finding. The **requirement-count reconciliation** is run every reporting period — baseline plus approved additions against the traced count — and its unexplained list is investigated, not just counted. Criterion text is **diffed against its baselined version**, because that is where creep hides. And on any conditional acceptance, the correction ladder has been applied **by severity class** with the fix-versus-defer breakeven stated, and the acceptor is not the producer.

Industry variations — Domain 5

- **Construction and infrastructure.** Scope is employer's requirements plus a design-responsibility allocation, and the highest-value line states **who owns the design risk for each element**: a defect in a specification the contractor was told to design to is a claim, so the ladder of 5.2.1 becomes a commercial argument rather than an internal cost. Physical scope also carries an irreversibility the software cases lack — a wall in the wrong place is not a build-stage defect.
- **Software and digital.** The backlog carries the scope and the definition of done carries the criteria, so 5.A.1's transfers apply and the top of the correction ladder genuinely compresses under continuous deployment, which is the honest argument for it. What does not compress is the cost of a requirement being wrong: a fast cadence ships the wrong thing sooner.
- **Regulated manufacturing (pharmaceutical, medical devices, aerospace).** Bidirectional traceability from requirement through design and verification to release evidence is a regulatory obligation with an inspection consequence, so the nine unverified acceptances of WE 5.2.3 are a finding against the quality system rather than a project defect. Requirements are frozen at defined points, making late elicitation expensive by design — which raises the KA 5.2.1 return well above Meridian's.
- **Public-sector programmes.** Requirements originate partly in policy and statute, so a material share of the baseline is non-prioritisable (5.3.1) and the discretionary pool is smaller than the

count suggests. Consultation obligations make elicitation a legal process with a published record; the systematic omission is the **external user**, who attends no internal meeting — Case study B.

- **Defence and complex systems.** Requirements are verified against a formal cross-reference of requirement to verification method and evidence, over lifecycles long enough that the author has left. Traceability is the institutional memory, so a 4 % orphan rate is a configuration problem rather than an untidiness.
- **Healthcare.** Clinical safety requirements are bound to a safety case and to specific system versions (Domain 4's configuration point), so acceptance-criteria expansion is a safety-case change and the restated-criterion class of 5.4.2 carries clinical rather than merely commercial consequence — which makes Meridian's 71 restated criteria a clinical-governance finding.

Case study — Domain 5: the requirement nobody could test (health, Meridian)

Situation. Meridian's Release 1 acceptance ran eight weeks against a planned three. The programme reported "acceptance complexity" and requested a schedule change; the supplier reported that all delivered functionality met the specification. Both statements were true, and the assurance review found the mechanism.

What the review found. Of the 480 approved requirements, **98 — 20.42 %** — carried acceptance criteria that could not settle a dispute: **71** restated the requirement and **27** had none. Twenty-two of the 98 reached formal dispute during acceptance, against the **24.5** the organisation's own dispute rate predicted, and six of those delayed a clinic go-live. The traceability audit found the corollaries — **34** requirements with no design element, **21** with no test case and **9** recorded as accepted with no verification evidence, meaning the release recommendation had been assembled partly from a record that stated something untrue. The reverse test, run for the first time, found **17** of the 425 design elements tracing to no approved requirement.

What the arithmetic had already said, and nobody had computed. Remediating all 98 criteria would have cost `98 × 320 =` **USD 31,360** against an expected dispute cost of **USD 293,510** — a **9.36-times** return on a **breakeven dispute probability of 2.67 %** against an observed 25 %. The 71 restated criteria had passed every completeness check the programme ran, which is why the deficiency reached acceptance intact: the programme measured whether criteria **existed**, never whether they **said anything**.

How it resolved. Acceptance was suspended for four weeks — `4 × 14,280 =` **USD 57,120** — while criteria were rewritten for the 41 requirements still in dispute or awaiting test, at `41 × 320 =` **USD 13,120**. Six severity-one open items were fixed rather than deferred, at **USD 59,820** against a deferral cost of **USD 614,400** (WE 5.4.3). The nine unverified acceptances were reopened and verified. Of the 17 orphans, 11 were retained as retrospectively approved changes and 6 removed. And the requirement-count reconciliation was run for the first time, exposing the **39** crept requirements and **USD 220,920** of unrecorded movement that WE 5.4.1 quantifies.

What the domain teaches here. A criterion that exists and says nothing is worse than one that is missing, because the missing one appears on a report and the empty one does not — the programme's own completeness metric was the instrument that concealed it. Finding out at acceptance cost **57,120** of suspension plus 22 disputes; preventing it would have cost **31,360**, decided a year earlier by an analyst nobody would have escalated to. The second lesson is procedural: three controls — criterion classification, reverse traceability, count reconciliation — had never been run once, and each cost less than a day.

Case study B — Domain 5: one hundred per cent complete and thirty-four per cent useful (public sector)

Situation. A national licensing authority replaced its permit-application platform. At closure the programme reported **612** requirements delivered, **100 %** verified, on budget and two weeks early. Fourteen months later the benefits review found realised annual benefit of **USD 391,000** against a forecast **USD 1,150,000 — 34.0 %**. Nothing in the delivery record was untrue.

Why. Requirements had been elicited thoroughly, over four months, from a working group of **9** internal officers. External applicants — about **4,200** a year, the people whose behaviour the entire benefit case depended on — were not consulted: they were not employees and there was no mechanism to convene them. Of the 612 requirements, **47 — 7.68 %** — addressed the applicant's own journey; the rest described the officers' work. The platform therefore automated the authority's processing beautifully and left applicants submitting on paper, so the digital-channel adoption on which **78 %** of the forecast benefit rested barely arrived. The decomposition is worth stating, because it is why the shortfall was 66 points and not 78: **USD 897,000** of the forecast depended on digital adoption and **USD 253,000** did not, so a realised 391,000 implies that — taking the non-digital benefit as landing in full — only **USD 138,000** of the digital-dependent benefit, **15.4 %** of it, was ever delivered. Verification was complete because the specification was internally coherent; validation had never been attempted, because nobody had defined what a validated outcome looked like from outside the organisation.

How it resolved. Re-elicitation with applicant panels and observation of real submission attempts produced **63** further requirements, of which **41** were implemented as changes costing **USD 285,000**. Realised benefit reached **USD 816,500 — 71.0 %** of forecast, an uplift of **USD 425,500** a year, a payback of **0.67 years — about 8.0 months**. The authority's standing corrections: every benefit must name the population whose behaviour produces it, and that population must appear in the elicitation record; and closure requires a **validation** statement, not only a verification one.

What the domain teaches here. A hundred per cent verification rate against a specification nobody outside the organisation contributed to measures internal consistency, and it is routinely reported as success. The requirement worth 425,500 a year was available for the price of a conversation before the specification was frozen and cost 285,000 to add afterwards — entirely consistent with the ladder of KA 5.2.1. And the warning generalises past this sector: **the stakeholder group that produces the benefit is frequently the one with no seat in the requirements process**, because elicitation follows the organisational chart and benefits do not.

Executive perspective — Domain 5

What a programme director cannot delegate in this domain:

- **The boundary, stated as counts.** Every extent in your scope statement carries a number, and the exclusion list is non-empty. Meridian's undrawn boundary was worth **143,000** — and the right answer was to include the eleven sites, at a nine-month payback, which only the written boundary would have surfaced in time (5.1.3).
- **Who was in the room.** Requirement loss is not random: it clusters on operators, supporters, external users and whoever owns the enabling change. A licensing platform verified at **100 %** delivered **34 %** of its benefit because **4,200** applicants a year had no seat (Case study B).

- **The correction ladder, and that you have bought elicitation against it.** One defect moved out of live service saves **102,000** and pays for an **84,000** programme; the delay objection breaks even at **36.89 weeks** (5.2.1).
- **That the reverse traceability test has been run.** It costs one query, it is the only cheap detector of work nobody asked for, and Meridian's first run found **17** orphans worth **108,800** (5.2.3).
- **That capacity is not stranded.** Where a constraint binds, require both the ranked and the enumerated answer. The intuitive method left **2 of 20** weeks idle and lost **44,000** a year — **262,737** of present value (5.3.3).
- **The count reconciliation, monthly.** Baseline requirements plus approved additions against the traced count. Meridian's change log was monitoring **56.86 %** of the movement it existed to control; the missing **39** requirements cost **220,920** and no threshold could ever have seen one (5.4.1).
- **A validation statement at closure, not only a verification one.** They fail independently, and only one of them is the question the organisation asked (5.4.3).

Calculation exercises — Domain 5

Exercise 5.1 A traceability audit covers 640 approved requirements and a 578-element design register. It finds 51 requirements with no design element, 38 with no test case, and 12 recorded as accepted with no verification evidence; 23 design elements trace to no approved requirement. Compute the forward defect rate overall and by class, and the reverse defect rate. State which class is most serious and why. Solution. Forward non-conformances `51 + 38 + 12 =` **101**, a rate of `101/640 =` **15.78 %**, comprising **7.97 %** with no design, **5.94 %** with no test and **1.88 %** accepted without evidence. Reverse rate `23/578 =` **3.98 %**. The 12 are most serious: the acceptance record states something untrue, so every decision taken on it — including release — rested on nothing. Common error: combining the forward and reverse rates, or presenting the forward classes as a single 15.78 % figure. The denominators differ (640 requirements against 578 design elements) and the classes fail differently, so neither total is informative.

Exercise 5.2 An organisation's observed correction ladder is 500 / 2,000 / 8,000 / 32,000 / 128,000 per requirement defect by stage (definition, design, build, test, live). A review costing USD 62,000 is expected to move 11 build-stage, 8 test-stage and 2 live-service defects to the definition stage. Cost of delay is USD 11,500 per week. Compute the saving, the net, the return, and the definition-phase extension at which the investment stops paying. Solution. Saving `11 × 7,500 + 8 × 31,500 + 2 × 127,500 = 82,500 + 252,000 + 255,000 =` **USD 589,500**. Net **USD 527,500**; return **9.51 times**. Breakeven extension `527,500 / 11,500 =` **45.87 weeks**. Common error: computing the saving on the gross later-stage costs (`11 × 8,000 + 8 × 32,000 + 2 × 128,000 =` 600,000), which forgets that each defect still costs 500 to correct at definition and overstates the saving by 10,500.

Exercise 5.3 A release has 24 development-weeks of capacity. Four candidate requirement bundles: P (16 weeks, USD 496,000 a year), Q (12 weeks, 366,000), R (12 weeks, 360,000), S (6 weeks, 168,000). Compute the greedy selection by value per week and the enumerated optimum, their capacity utilisation, the annual difference, and its present value over 6 years at 8 % (`AF(0.08, 6) = 4.622880`). Solution. Ratios: P **31,000**/week, Q **30,500**, R **30,000**, S **28,000**. Greedy takes P

(16 weeks), cannot fit Q or R, takes S (6 weeks) — **P + S = USD 664,000** using **22 of 24** weeks (**91.7 %**), with 2 weeks stranded. Enumeration finds **Q + R = USD 726,000** at **24 of 24** weeks (**100 %**). Difference **USD 62,000** a year — **9.34 %** — worth `62,000 × 4.622880 =` **USD 286,619** of present value. Common error: ranking by value per week and stopping. The greedy answer is defensible on the ratio and wrong on the outcome; the diagnostic is the stranded capacity, and the check is one enumeration over four candidates. A second error is enumerating without a dependency map: if R requires P, `Q + R` is not a feasible set.

Exercise 5.4 At design freeze a project's requirements baseline held 560 requirements. Integrated change control subsequently approved 19 additions. At acceptance-test entry the traceability matrix carries 638. Uncontrolled requirements cost an average of USD 3,800 direct, and 21 of them consumed 0.2 weeks of critical path each. Cost of delay is USD 11,500 per week; the baseline is USD 3,100,000. Compute the unexplained count, the direct and schedule cost, the total, its share of the baseline, and each crept requirement's individual share. Solution. Expected traced count `560 + 19 =` **579**; actual 638, so **59** requirements are unexplained. Direct `59 × 3,800 =` **USD 224,200**; schedule `21 × 0.2 = 4.2` weeks at 11,500 = **USD 48,300**; total **USD 272,500** — **8.79 %** of the baseline, while each crept requirement is **0.12 %** of it. Common error: reconciling cost and never counting requirements. The change log can balance to the penny while 59 requirements enter without ever appearing in it; only the count reconciliation detects them, and no delegation threshold can, because 0.12 % of a baseline is below every threshold anyone sets.

Exercise 5.5 Of 720 requirements, 126 carry acceptance criteria that restate the requirement and 41 carry none. Writing a measurable criterion costs USD 280. A deficient criterion produces a dispute 20 % of the time; a dispute costs USD 8,400 in rework, and 1 dispute in 4 delays delivery by 0.5 weeks at a cost of delay of USD 11,500 per week. Compute the remediation cost, the expected cost per dispute, the expected dispute cost, the net, and the breakeven dispute probability. Solution. Deficient `126 + 41 =` **167**; remediation `167 × 280 =` **USD 46,760**. Expected delay per dispute `0.25 × 0.5 × 11,500 =` **USD 1,437.50**, so cost per dispute **USD 9,837.50**. Expected disputes `167 × 0.20 =` **33.4**, expected cost **USD 328,572.50**. Net **USD 281,812.50**. Breakeven probability `46,760 / (167 × 9,837.50) =` **2.85 %**. Common error: two, and both understate the case. Comparing the remediation cost against the rework cost alone omits the delay component (1,437.50 of the 9,837.50). And applying the 20 % dispute rate to all 720 requirements rather than to the 167 deficient ones inflates the expected cost by more than four times — an error that happens to favour the right decision, which is why it survives.

Practitioner's toolkit — Domain 5

Adoption-ready artefacts; adapt headings to your organisation, then keep them stable.

Toolkit 5.T.1 — Requirements traceability matrix

One row per requirement: reference · statement · **class** (functional / non-functional / constraint / data / transition / regulatory) · source, naming the **stakeholder or obligation**, not the document · benefit served and the **bundle** it belongs to (Toolkit 5.T.2) · design element · test case · deliverable · **acceptance criterion classification** (measurable / restated / absent) · verification evidence reference · acceptance decision, acceptor by name and date · baseline version approved at, and the change reference if added later.

Four reports every period carry the whole value of the matrix: rows missing a design, test or deliverable link, **by class and never totalled**; rows accepted with no verification evidence — the class that invalidates the record; **orphans**, from the reverse query over the design and test registers; and criteria classified restated or absent, with the restated count shown separately because it is the number nobody has. A matrix maintained and never queried is a cost with no control attached.

Toolkit 5.T.2 — Value attribution and constrained selection sheet

Attribution side. One row per **requirement bundle**: reference · requirements included · benefit delivered, with its **named owner** (Domain 2's benefits register) · attributed annual value with its range · effort in the **binding constrained resource**, named explicitly (usually a team, not money) · dependencies on other bundles · **value per unit of constrained effort** · and a flag for regulatory or safety bundles, removed from the ranking and held as constraints. Foot the attribution column and run the **sum test** against the business case, stating any shortfall rather than distributing it.

Selection side. State the constraint and its quantity. Compute and record **both** the greedy result and the enumerated optimum over feasible sets, with the dependency map applied; report capacity utilisation for each and flag any **stranded capacity** as a finding. Where the two differ, state the annual difference and its present value at the appraisal rate. Then record the decision, its maker by name, and — where it departs from the enumeration — what is being bought and at what annual cost, because overruled arithmetic is a decision and an unexamined ranking is not.

Toolkit 5.T.3 — Scope reconciliation and acceptance-criteria pattern set

Reconciliation side, run every reporting period and published on one page: baselined requirement count · plus approved additions (with change references) · minus approved deletions · **equals expected traced count** · against the **actual traced count** · equals **unexplained requirements**, listed individually and investigated, not merely counted. Then: criterion text **diffed against its baselined version**, with every expansion listed; the priced creep total, direct and schedule, at the cost of delay; and the cumulative total set beside Domain 4's authorised drift, so the report states **what share of total scope movement the change log actually captured**.

Acceptance-criterion patterns, four forms that make a criterion testable. **Threshold:** given [state], when [action], the result is [observable] within [limit] — for performance, capacity and time. **Enumeration:** the deliverable handles each of [n] listed cases, each individually verifiable — for functional coverage, and it forces the case list into the open. **Negative:** given [invalid input or failure condition], the result is [defined behaviour] — the pattern that catches the exception handling nobody specified. **Evidence:** acceptance requires [named artefact] reviewed by [named role] — for requirements whose satisfaction is a judgement (clinical safety, accessibility, regulatory conformance), which is how a judgement becomes auditable rather than an opinion. Prohibit four constructions outright: "correctly", "appropriately", "as required", and any comparative without a referent.

Exam preparation — Domain 5

What is assessed. The three registers and why scope is the deliverable register; decidable scope; the scope baseline's three artefacts and the primacy of the WBS dictionary; **pricing an undrawn boundary on both cost and forgone benefit**; elicitation technique selection and the stakeholder groups that cause systematic loss; **the correction ladder, the saving computed on stage differences, the return on elicitation and the breakeven delay**; requirement quality properties and the six requirement classes; solution capture and requirement inflation; **traceability in both directions, defect rates by class on their own denominators, and which class invalidates the record**; value attribution to bundles with the sum and counterfactual tests; **must-have inflation quantified**; **value per unit of constrained effort, greedy against enumeration, stranded capacity and the dependency precondition**; change against creep and the **count reconciliation**; acceptance-criterion states and the **breakeven dispute probability**; verification against validation; and **conditional acceptance priced by severity class**.

The calculations to be able to do under time pressure. Per-unit boundary exposure and its payback from forgone benefit. Expected correction cost from a stage profile and a ladder; the saving from moving `k` defects between stages, **always on the difference**; the return on an elicitation spend and the breakeven delay. Forward defect rate overall and by class, and the reverse rate on the design-register denominator. Value per unit of constrained effort; the greedy selection; the enumerated optimum over a small candidate set; capacity utilisation; and the annual difference converted to present value with `AF(r, n)`. The requirement-count reconciliation, and creep priced as direct plus schedule at the cost of delay, with its share of baseline and of total movement. Remediation cost, expected cost per dispute including its delay component, and the breakeven dispute probability. Fix-versus-defer against the ladder, and the breakeven go-live delay.

The traps. Scope written in the activity register, so no boundary can be tested (5.1.1) · reading a boundary statement as a way of saying no when the priced answer is to include (5.1.3) · computing an elicitation saving on gross later-stage costs rather than stage differences (5.2.1, Exercise 5.2) · assuming every late defect was detectable at definition (5.2.1) · totalling traceability classes, or combining forward and reverse rates on different denominators (5.2.3, Exercise 5.1) · running only the forward test and never seeing the orphans (5.2.3) · attributing value to individual requirements rather than bundles, and delivering half a capability (5.3.1) · scoring regulatory requirements in a value ranking (5.3.1) · accepting a 78 % must-have classification as information (5.3.2) · ranking greedily and stopping, ignoring stranded capacity (5.3.3, Exercise 5.3) · enumerating without a dependency map (5.3.3) · monitoring change and never counting requirements, so 43 % of movement stays invisible (5.4.1, Exercise 5.4) · measuring whether acceptance criteria exist rather than whether they say anything (5.4.2) · applying the dispute probability to all requirements rather than the deficient ones (Exercise 5.5) · applying the top of the ladder to every open item and so refusing to go live (5.4.3) · and reporting verification as though it answered the validation question (5.4.3, Case study B).

How the domain connects. Domain 1 supplies accountability for outcome rather than output and the cost of delay every calculation is priced at. Domain 2 supplies the benefits map requirements attach to, the ramped adoption profile the attributions inherit, the `AF(r, n)` used to discount a prioritisation difference, and the capacity-constrained selection principle KA 5.3.3 works at requirement level. Domain 3 supplies the decidability test the boundary discipline applies to content, and the thresholds that cannot see a 0.175 % crept requirement. Domain 4 supplies the WBS the scope baseline includes, the change flow that distinguishes change from creep, the authorised drift the creep figure is set beside, and the "record states something untrue" failure class recurring here as unverified acceptance and as the restated criterion. Forward: Domain 6 schedules these bundles and supplies the dependency map the enumeration requires; Domain 7 costs them; Domain 8 attaches uncertainty to the attributions; Domain 9 owns verification and acceptance in depth; Domain 10 carries the contractual face of specification risk (5.A.2); Domain 13 works the adaptive equivalents of 5.A.1; and Domain 16 answers the validation question at handover.

Domain 5 summary

Part One produced a project and assumed the content of its scope was known. It was not, and the expensive failure is not incompleteness but a scope that is complete, agreed, delivered, verified and valueless — the condition Case study B reports as **612 requirements, 100 % verified, 34.0 %** of forecast benefit, because the **4,200** external applicants whose behaviour produced the benefit had no seat in a requirements process built from the organisational chart. Only **47** of the 612 requirements — **7.68 %** — addressed them. The correction cost **USD 285,000** and returned **USD 425,500** a year, a payback of about **8.0 months**, for a conversation that had been available free eighteen months earlier.

Scope is the deliverable register, bounded by outcomes and decomposed into activities, and it is decidable only when its extents are counted. Meridian's boundary — "all clinics in the region" against an approved 40 — carried an exposure of **USD 143,000** at **USD 13,000** a site; and the professional result is the inversion, because those 11 sites would return **USD 188,496** a year at realistic adoption, a payback of about **9.1 months**. The boundary statement's value is not that it refuses; it is that it forces the question while the answer is still cheap to act on.

Requirement defects cost four times more at every stage they survive: **400 · 1,600 · 6,400 · 25,600 · 102,400**. On Meridian's detection profile the unimproved correction bill is **USD 1,286,400 — USD 13,400** a defect — and an **USD 84,000** elicitation programme returns **USD 610,800**, **7.27 times** its cost, computed on stage differences rather than gross later-stage costs, which would have overstated it by 10,000. Two sentences carry the whole argument: **one defect moved out of live service saves USD 102,000 and pays for the entire programme**, and the delay objection does not break even until the definition phase is extended by **36.89 weeks**. Traceability then audits the chain in both directions and reports **by class, never as a total**: **13.33 %** forward non-conformance across **7.08 %** with no design, **4.38 %** with no test and **1.88 %** accepted with no verification evidence — that last class being the one that makes every decision taken on the record worthless — plus a **4.00 %** reverse rate, **17** orphans worth **USD 108,800**, invisible to the forward test almost everyone runs alone.

Value is attributed to bundles, not requirements, and prioritisation only bites once the categories mean something: **77.92 %** of Meridian's requirements were marked must-have, and asking each originator to name the consequence of omission left **30.83 %**, raising the tradeable pool **3.13-fold**. Against a hard **20 development-week** constraint, greedy ranking by value per week takes the highest-ratio bundle, strands **2 of 20** weeks and returns **USD 409,000** a year; enumeration returns **USD 453,000 — 10.76 %** more, **USD 262,737** of present value at 7 % over eight years — and the ratio advantage that caused the loss was **under 1 %**. Stranded capacity is the signature; a dependency map is the precondition; and no ranking is a decision.

Finally, scope moves without anyone approving anything. Meridian's baseline of **480** requirements plus **12** approved additions should have traced **492**; it traced **531**, so **39** requirements — about three a month, **3.25 times** the approved additions — entered with no record at all, costing **USD 163,800** direct and **4.0** weeks of critical path, **USD 220,920** in total, **9.20 %** of the baseline, at **USD 4,200** of direct cost each — **0.175 %** of baseline, below every threshold any delegation schedule will ever set. Set beside Domain 4's **USD 291,176** of authorised drift, total movement is **USD 512,096 — 21.34 %** of baseline and **38.42 %** of the approved NPV — of which the change log captured **56.86 %**. No cumulative test on a change log can find the rest; **only counting the requirements can**, and it is one addition and one subtraction a month. On the way out, two more numbers worth keeping: writing testable acceptance criteria pays for itself above a **2.67 %** dispute rate against an observed **25 %**, and the **71** criteria that restate their requirement are more dangerous than the **27** that are missing, because completeness reports cannot see them. And fixing six severity-one defects before acceptance costs **USD 59,820** against **USD 614,400** deferred — a breakeven go-live delay of **40.34 weeks**, applied by severity class and not to all 41 open items.

06

DOMAIN 6

Planning, Scheduling and Delivery Flow (quantitative flagship)

Group: Delivering the work (Domain 6 of 6 in Part Two). **Target:** ~76 pages. **Binds to:** the PCI Book Pattern Specification and the shared registries (`docs/books/registries/`). This domain is the home of the schedule symbols — `ES`, `EF`, `LS`, `LF`, `TF` (total float), `FF` (free float) — restated by every domain that touches time. British English; USD (+SAR where useful, indicative `USD 1 ≈ SAR 3.75`).

Why this domain exists

A project leader's authority rests on one repeated act: making a credible statement about the future — when the work will finish, what could move that date, and what it would cost to move it back. This domain builds the machinery behind that credibility. It starts where planning actually starts — with levels of plan and the logic that binds work together (KA 6.1); builds the critical path method in full, forward and backward passes, and both kinds of float (KA 6.2); adds the realities the textbook network ignores — resources, constraints, milestones and rolling-wave elaboration (KA 6.3); and finishes with delivery flow across predictive, agile and hybrid worlds: recovery, compression economics, and forecasting under uncertainty (KA 6.4). Cost joins schedule in Domain 7 (earned value); risk quantification deepens in Domain 8. A leader who cannot read a network diagram is hostage to whoever can; this domain removes the hostage-taking.

Learning objectives. After this domain a candidate can: choose the right planning level for an audience and a decision; build a logic network with correct dependency types; **model an approval as a dated event against a governance calendar and compute what mis-modelling it costs**; run forward and backward passes and derive total and free float; identify the critical path and near-critical paths, **enumerate every path with its float, choose a near-critical band and price the monitoring decision**; explain why float is not spare time and **why activity floats must never be summed**; plan resources against the schedule, **produce a resource-feasible schedule under a hard cap, distinguish the critical chain from the critical path and price the peak that binds**; apply rolling-wave elaboration honestly and **price the commitment exposure a ranged planning package carries**; **build a compression menu with cost slopes, locate the least-cost duration and state the range of week-values over which it holds**; price fast-tracking as a probabilistic decision with a **breakeven rework probability**; distinguish predictive, agile and hybrid scheduling, **translate a throughput forecast into a required rate through the backward pass**, and govern each; **size and manage a schedule buffer from aggregated safety, and forecast from buffer consumption**; **convert a repetitive-delivery date into a takt, a crew count and a benefit integral**; and use three-point estimates and scenario analysis to forecast completion with stated uncertainty — with any AI-generated schedule subject to the family's verification rule.

The master worked project. One project runs through this domain and returns in Domains 7 and 8: **Project Auriga**, a control-systems upgrade for a regional utility, planned in weeks:

ID	Activity	Duration (wk)	Predecessors
A	Mobilise	2	—
B	Detailed design	6	A
C	Procure control hardware	8	B
D	Civil and cabling works	7	B
E	Installation	5	C, D
F	Testing and commissioning	4	E
G	Operator training and handover	2	D

Every calculation below uses this network. Auriga's `BAC` is **USD 4,000,000** and the value of a week — the client's early-completion bonus, and equally the cost of a week's delay — is **USD 45,000**, the figure Domains 7, 8 and 9 use unchanged. An engineer-week costs **USD 5,225** (Domain 7, KA 7.4.1).

The second scale. Scheduling questions change shape between one project and a rolling programme, so this domain also works the family's programme case, **Meridian Care Records** — the clinical-records rollout to **40 clinics**, approved cost **USD 2,400,000**, benefit **USD 979,200** a year at full potential and **USD 685,440** at the realistic 70 % adoption, with a **cost of delay of USD 14,280 per week** derived in Domain 1 and a steering committee whose expected decision latency Domain 3 computes as `E[wait] = M/2 + L =` **4 weeks** from a four-weekly meeting and a two-week paper deadline. Meridian supplies what Auriga cannot: a governance calendar the network has to obey (KA 6.1.2), a ranged planning package (KA 6.3.3) and forty near-identical units of repetitive work whose schedule is a rate rather than a network (6.A.5). A deployment specialist-week costs **USD 4,200** (Domain 1, KA 1.3.3b).

KNOWLEDGE AREA 6.1

Planning levels and logic networks

Topics: 6.1.1 levels of plan · 6.1.2 dependencies and logic · 6.1.3 network rules and quality.

6.1.1 Levels of plan

The principle. One schedule cannot serve a board, a site supervisor and a lender at once. Mature delivery organisations maintain a **hierarchy of schedules**, each derived from the one below, each honest with the one above:

Level	Name	Audience & decision	Typical granularity
L1	Executive summary	Board, sponsor — go/no-go, milestones	10–30 bars
L2	Management schedule	Steering, PMO — stage gates, interfaces	50–300 activities
L3	Control schedule	Project team — the CPM network; progress and float live here	300–5,000 activities
L4	Execution/lookahead	Site and squads — 2–6 week detail	daily/shift detail

The **control schedule (L3)** is where this domain's machinery operates: it carries the logic, the floats and the critical path, and it is the version under change control (Domain 4, KA 4.3). An L1 milestone that cannot be traced to L3 logic is decoration, and steering committees should treat it as such (Domain 3, KA 3.2).

Fig 6.1.1 — The schedule hierarchy: one logic, four honest views

6.1.2 Dependencies and logic

Definitions. Activities link through four dependency types — finish-to-start (**FS**, the default: design finishes, then procurement starts), start-to-start (**SS**: cabling can start once civils have started, often with a lag), finish-to-finish (**FF**: commissioning documentation finishes when commissioning finishes), and the rare start-to-finish (**SF**). A **lag** delays a successor (`FS+2` = start two weeks after the predecessor finishes); a **lead** (negative lag) overlaps it. Logic expresses physics and choice — what must follow, and what the team has chosen to sequence — and the two should be distinguishable in the schedule's notes, because only chosen logic can be re-chosen when recovery demands it (KA 6.4).

Common pitfall — the lag that hides work. A 3-week lag labelled "curing" is physics; a 3-week lag labelled nothing is often an activity somebody didn't model — unresourced, uncosted, invisible to progress measurement. Audit rule: every lag has a stated reason, and long lags (> 10 % of project duration) are activities in disguise.

WORKED EXAMPLE

Worked example 6.1.2 — re-choosing a dependency.

1. **Setup.** Survey work P (4 weeks) precedes report drafting Q (6 weeks), currently linked `FS+2` (two weeks of data cleaning after all surveying ends). The team proposes drafting in parallel as survey sections complete: `SS+1`. Compute Q's dates both ways (P starts week 0).
2. **Formula.** FS+lag: `ES(Q) = EF(P) + lag`. SS+lag: `ES(Q) = ES(P) + lag`.
3. **Substitution.** FS+2: `ES(Q) = 4 + 2 = 6`, `EF(Q) = 12`. SS+1: `ES(Q) = 0 + 1 = 1`, `EF(Q) = 7`.
4. **Result.** The chain finishes week **12** under FS+2 and week **7** under SS+1 — five weeks earlier from one logic choice.
5. **Interpretation.** Nothing was crashed and nobody works faster: the saving comes from overlapping, and its price is rework risk if early sections change after later surveying. This is why chosen logic must be visibly chosen (not fossilised as physics) — it is the cheapest recovery currency the project owns (KA 6.4.2), and spending it is a risk decision, not a scheduling trick.

The risk is priceable, and the breakeven is reassuring. Suppose the worst credible consequence is that late survey sections invalidate the early drafting and Q must be rewritten in full — six weeks lost — with probability `p`. The expected saving is `5 − 6p` weeks, so the overlap stops paying at `p = 5/6 =` **83.33 %**. At a more plausible `p` = 30 % the expected saving is `5 − 1.8 =` **3.20 weeks**. Re-choosing a dependency is therefore an unusually favourable trade: the saving is banked immediately and in full, while the loss is contingent — which is exactly the asymmetry KA 6.4.2 exploits again for fast-tracking, and exactly the asymmetry that makes overlap the first move in any recovery.

Two things break that comfortable arithmetic, and a reviewer should ask about both. The first is **detectability**. The model above assumes the rework is discovered while the report is still in draft. If the invalidated early sections reach a published report unnoticed, the consequence is not six schedule weeks but an escaped defect in a delivered product, priced by Domain 9's containment economics on a ladder an order of magnitude steeper. Overlap is safe in proportion to the strength of the check that sits between the overlapping activities. The second is that `p` is not a constant but a function of how much overlap is taken: `SS+1` drafts on one week of survey data, `SS+3` on three. Where a team can state the relationship it should optimise the overlap; where it cannot — the usual case — it should take the overlap that a downstream check can catch and say so, rather than pretending a single `p` describes every possible degree of overlap.

WORKED EXAMPLE

Worked example 6.1.2b — the approval that was drawn as a lag.

1. **Setup.** Meridian's wave-2 clinic fit-out design must be approved by the steering committee before the estate contractor mobilises. The planner has drawn this as `FS+3` on the design → mobilise link, labelled "approval". The committee meets every **M = 4** weeks — programme weeks 20, 24 and 28 — and papers are due **L = 2** weeks before each meeting (weeks 18, 22, 26). The design's forecast finish is **week 23**. The path carrying the approval has **2 weeks** of total float. Cost of delay **USD 14,280** per week (Domain 1, KA 1.3.3).
2. **Formula.** A submission finishing at `t` reaches the first meeting `m` for which `m − t ≥ L`; the wait is `m − t`. Its range across arrival times is `L` (best) to `M + L` (worst), and its mean for a decision arriving at a uniformly random point is `E[wait] = M/2 + L` (derived in Domain 3, KA 3.2.3 — not re-derived here).
3. **Substitution.** Finish week 23: the week-24 meeting closed its papers at week 22, so the first reachable meeting is **week 28** and the wait is `28 − 23 = 5` weeks. Finish week 22 instead: papers for the week-24 meeting are still open, so the wait is `24 − 22 = 2` weeks.
4. **Result.** The scheduled wait is **5 weeks**, not the 3 weeks drawn and not the 4-week expectation, so the approval lands in **week 28**. The 2-week understatement **exactly consumes the path's float**: `TF` falls from 2 to **0**. Pulling the design finish one week earlier, to week 22, cuts the wait to **2 weeks** and the approval to **week 24** — so the successor starts **4 weeks** earlier for **one week** of design work. Note the asymmetry: the wait shortens by 3 weeks (5 → 2) while the approval date moves by 4 (28 → 24), because the design's own week is saved as well. Four weeks are worth `4 × 14,280 =` **USD 57,120**, or **USD 45,120** net of a USD 12,000 compression cost.
5. **Interpretation. A mis-modelled lag makes a path critical while every duration is still "as planned".** No activity slipped, no estimate changed, nobody was late — and the path went from two weeks of protection to none, purely because a three-week label stood in for a five-week event. This is the pitfall above made arithmetical: the schedule was not optimistic, it was silent, and silence is not conservative. The audit habit follows directly — for every lag, ask what event it represents, then ask what that event's own logic is.

Once the network exists, the wait is not a random variable — it is a choice. Domain 3's `E[wait] = M/2 + L =` **4 weeks** is the correct planning figure for a decision arriving at an unknown time, which is what an escalation is. A planned approval is not that: its arrival date is a date the planner controls, so the honest model is deterministic against the meeting calendar, and the wait ranges from **2 to 6 weeks** depending on where the finish lands. Substituting the expectation for a planned submission is a category error in both directions — it flatters a submission that will miss its window and penalises one that will make it.

The highest-leverage week in a governance-bound schedule is the one immediately before a paper deadline. One week of compression bought four weeks of schedule here: a **4:1** return, and it arises entirely from the discreteness of the calendar. Compare it with Domain 3's own lever — shortening the paper lead time `L` by one week saves one week of expected wait, worth **USD 14,280**. The scheduling lever is four times the governance lever on this submission, costs nothing to exercise, and needs nobody's permission. It is also the more fragile: it works only if the submission actually lands before the deadline, so the planner who claims it must also protect it, which means treating the paper deadline as a milestone with float, not as an administrative detail.

The cautions. The 4:1 leverage is a property of this arrival date, not of the calendar — a design finishing at week 21 gains nothing from compression because it already makes the week-24 meeting, and a leader who reads "compress the design" as a general rule will buy weeks that convert into nothing. The gain is realised only if the approval sits on a binding path; on a path with four weeks of float the whole analysis is worth zero and the compression money should be spent elsewhere. And the wait is the committee's latency only — a deferral, a request for more information or a conditional approval restarts the cycle, which is why the register in toolkit 6.T.3 records approval events with their meeting dates rather than with nominal durations.

6.1.3 Network rules and quality

A network the passes can trust obeys checkable rules: every activity except the first has a predecessor and except the last a successor (**no dangles**); no loops; durations estimated by the owning discipline, not the scheduler; no date constraints doing logic's job (a "must start on" pin that overrides logic converts the schedule from a model into a poster); and logic density in a healthy range — too few links and float is fiction, too many and nothing can move. These rules are the schedule-quality gate a leader can run without scheduling software: ask for the dangle count, the constraint count and the reasons for both.

Two of those rules are countable, and counting them is the whole skill. Logic density is links ÷ activities. Auriga carries **7** dependency links across **7** activities — a density of **1.00** — and the number a reviewer compares it against is structural rather than conventional: binding `n` activities into one connected network takes at least `n − 1` links, so the floor here is **6** and Auriga sits exactly one link above it. That one surplus link is the second predecessor of E, which is to say **the surplus over `n − 1` is the count of genuine convergences in the network** — a directly meaningful figure, and the reason a density materially below 1 on a large network is proof of dangles rather than evidence of elegant simplicity. At the other extreme a density far above 2 usually means the same constraint has been expressed several times, which makes float unresponsive because every activity is held by a redundant link. Float distribution is the share of activities at or near zero float. Auriga shows `5/7 =` **71.4 %** at zero float, which on a real network would be a warning that the logic is over-tight or the constraints are doing the work.

Here the professional caution matters more than the metric: **Auriga is a seven-activity teaching network, and neither number means anything at that size.** Both measures are properties of a population of activities, and both take their meaning from an organisation's own calibrated history — a construction contractor's healthy density is not a software programme's. The transferable habit is therefore the arithmetic and the direction of the inference, not a benchmark: compute density and the zero-float share on your own portfolio of accepted schedules, and treat a new schedule that sits far outside your own distribution as a question to be answered rather than a defect to be asserted. A reviewer quoting a universal threshold for either is quoting something nobody derived.

AI in this KA

Scheduling tools now draft networks from scope statements and historical libraries. Useful — the draft logic is often 80 % right — and dangerous in a specific way: generated logic looks authoritative while encoding assumptions nobody made. The governed workflow: AI proposes the network; the discipline leads walk every link of their scope ("does B truly need all of A finished?"); the scheduler runs the quality rules above; and the leader signs the logic as a decision record (Domain 3, KA 3.3.4). **AI proposes; the professional verifies, decides and remains accountable.**

KEY TERMS — KA 6.1

Term	Meaning
Schedule hierarchy (L1–L4)	Nested plans for different audiences, derived from one logic.
Control schedule	The L3 CPM network under change control; home of float and progress.
FS / SS / FF / SF	The four dependency types.
Lag / lead	Imposed delay / overlap on a dependency.
Dangle	An activity missing a predecessor or successor; a network defect.
Date constraint	A pinned date overriding logic; each one weakens the model.
Logic density	Links ÷ activities; the surplus over `n − 1` counts the network's convergences.
Governance calendar	The meeting and paper-deadline dates an approval event must be scheduled against, rather than approximated by a lag.

SAMPLE MCQS — KA 6.1

MCQ 6.1-A `[6.1.2 · Application]` Cabling may begin one week after civil works begin. The correct dependency is: - A. FS+1 - B. SS+1 - C. FF+1 - D. FS−1

Rationale: The condition binds the two starts with a one-week lag: start-to-start plus one. A would wait for all civils to finish; C binds the finishes; D (a lead on FS) overlaps the finish, which is not what was stated.

MCQ 6.1-B `[6.1.3 · Analysis]` A schedule shows 14 "must finish on" constraints, all on milestones reported to the board. The most accurate reading is: - A. the schedule is well-governed, because board dates are protected - B. the milestones will be met, because the tool will honour the constraints - C. the schedule may no longer model reality — pinned dates can mask logic-driven slippage that float analysis would otherwise reveal - D. constraints are neutral scheduling hygiene with no analytical effect

Rationale: Pins override logic: the network can be slipping while pinned milestones hold still, hiding negative float until it is unrecoverable. A and B mistake suppression for control; D is false — every pin degrades the model's predictive value.

MCQ 6.1-E `[6.1.1 · Recall]` The schedule level that carries the logic, the floats and the critical path — and sits under formal change control — is: - A. L1, because the board owns the schedule - B. L2, because the PMO maintains it - C. L3, the control schedule - D. L4, because it has the most detail

Rationale: The L3 control schedule is the analytical model; L1/L2 summarise it and L4 elaborates near-term execution from it. Detail (D) is not the same as control — L4 churns weekly by design and is never the baselined network.

MCQ 6.1-C `[6.1.2 · Application]` Survey P (4 wk) feeds report Q (6 wk). Under `FS+2` the chain completes week 12. Re-linked `SS+1`, it completes week: - A. 7 - B. 11 - C. 12 - D. 5

Rationale: `ES(Q) = ES(P) + 1 = 1`, `EF(Q) = 7`. B subtracts one lag week from 12; C assumes logic changes nothing; D forgets Q's own duration must still run.

MCQ 6.1-D `[6.1.3 · Recall]` A "dangle" in a schedule network is: - A. an activity with negative float - B. an activity missing a predecessor or successor link - C. a milestone with zero duration - D. a lag longer than its predecessor

Rationale: Dangles are unbound activity ends — the passes cannot constrain them, so their dates and float are unreliable. A describes constraint-driven lateness; C describes every milestone; D is unusual but legal when justified.

MCQ 6.1-F `[6.1.2 · Analysis]` A design finishing in week 23 needs approval from a committee meeting in weeks 20, 24 and 28 whose papers are due two weeks before each meeting. The planner has drawn the approval as a three-week lag and the path carries two weeks of total float. The correct reading is: - A. the lag is conservative, since the average wait for this committee is four weeks - B. the approval will actually take five weeks, the two-week understatement consumes the path's float, and the path is critical with no duration having changed - C. the wait is four weeks, so one week of float remains - D. the lag is irrelevant because approvals are not project work

Rationale: Week 23 misses the week-24 meeting (papers closed at week 22), so the first reachable meeting is week 28 and the wait is 5 weeks against 3 drawn (6.1.2b). A calls a three-week lag conservative against a five-week event; C substitutes Domain 3's expectation for a planned submission whose date is known, which is the category error 6.1.2b names; D is the lag-that-hides-work pitfall stated as a principle.

MCQ 6.1-G `[6.1.3 · Analysis]` A 400-activity network reports 310 dependency links. Before looking at anything else, a reviewer can conclude that: - A. the logic is admirably lean - B. the network cannot be fully connected, so it contains dangles - C. the network is over-linked and float will be unresponsive - D. nothing at all — link counts carry no information

Rationale: Connecting 400 activities needs at least 399 links, so 310 proves unbound activities exist (6.1.3). A mistakes a structural impossibility for economy; C is the opposite defect, which appears at densities far above 2; D denies a countable check that costs one division.

■ SELF-CHECK — KA 6.1

1. Why must chosen logic be distinguishable from physical logic? — Because recovery (KA 6.4) works by re-choosing choices; physics cannot be renegotiated.
2. What three numbers give a fast schedule-quality read? — Dangle count, constraint count, unexplained-lag count — each should be at or near zero, with reasons for the rest.
3. When is Domain 3's *`E[wait] = M/2 + L`* the wrong figure for a schedule? — Whenever the submission date is planned rather than random: the wait is then a deterministic function of the finish date against the meeting calendar, ranging from `L` to `M + L` (2 to 6 weeks on Meridian).

4. What does a logic density of exactly 1.00 on a seven-activity network tell you? — That the network has one link more than the six needed to connect it, so it contains exactly one convergence — and that the metric is meaningless at this size.

KNOWLEDGE AREA 6.2

The critical path and float

Topics: 6.2.1 the forward and backward passes · 6.2.2 total and free float · 6.2.3 reading the critical path.

6.2.1 The forward and backward passes

Definitions. For each activity: `ES`/`EF` — earliest start/finish (forward pass, left to right: `ES` = latest `EF` of predecessors; `EF = ES + duration`); `LS`/`LF` — latest start/finish that do not delay the project (backward pass, right to left: `LF` = earliest `LS` of successors; `LS = LF − duration`). The project duration is the largest `EF` in the network.

Worked example 6.2.1 — Auriga, both passes. The passes are inherently tabular; the labelled table replaces the five-step skeleton, and the interpretation follows (pattern rule).

Act	Dur	ES	EF	LS	LF	TF	FF
A	2	0	2	0	2	0	0
B	6	2	8	2	8	0	0
C	8	8	16	8	16	0	0
D	7	8	15	9	16	1	0
E	5	16	21	16	21	0	0
F	4	21	25	21	25	0	0
G	2	15	17	23	25	8	8

Interpretation. Project duration **25 weeks**; the **critical path is A–B–C–E–F** (2+6+8+5+4 = 25), every activity on it with zero float. D can slip one week without moving the end date; G can slip eight. The passes turn a drawing into an instrument: every schedule statement a leader makes in this domain — "we finish in week 25", "D matters more than G", "procurement drives the job" — is read directly off this table.

Three checks confirm the table before anything is read off it, and they are the same three a reviewer runs on any pass, human or machine. **The critical path's durations sum exactly to the project duration:** `2 + 6 + 8 + 5 + 4 = 25`. **Every critical activity shows `ES = LS` and `EF = LF`:** A, B, C, E and F all do; D and G do not, and their gaps are precisely their floats. **Every activity's `TF` equals both `LS − ES` and `LF − EF`:** for G, `23 − 15 = 8` and `25 − 17 = 8`. A pass failing any of the three is arithmetically broken, and the failure is visible in seconds without re-running anything.

What the table also shows, and what the eye misses, is that **the second path is one week from binding**. A–B–D–E–F runs 24 weeks — 4.00 % below the project duration — so the network's whole protection against a slip in civil works is a single week. That is the observation KA 6.2.3 turns into a monitoring decision, and it is why float is the number to report rather than the critical path's membership list.

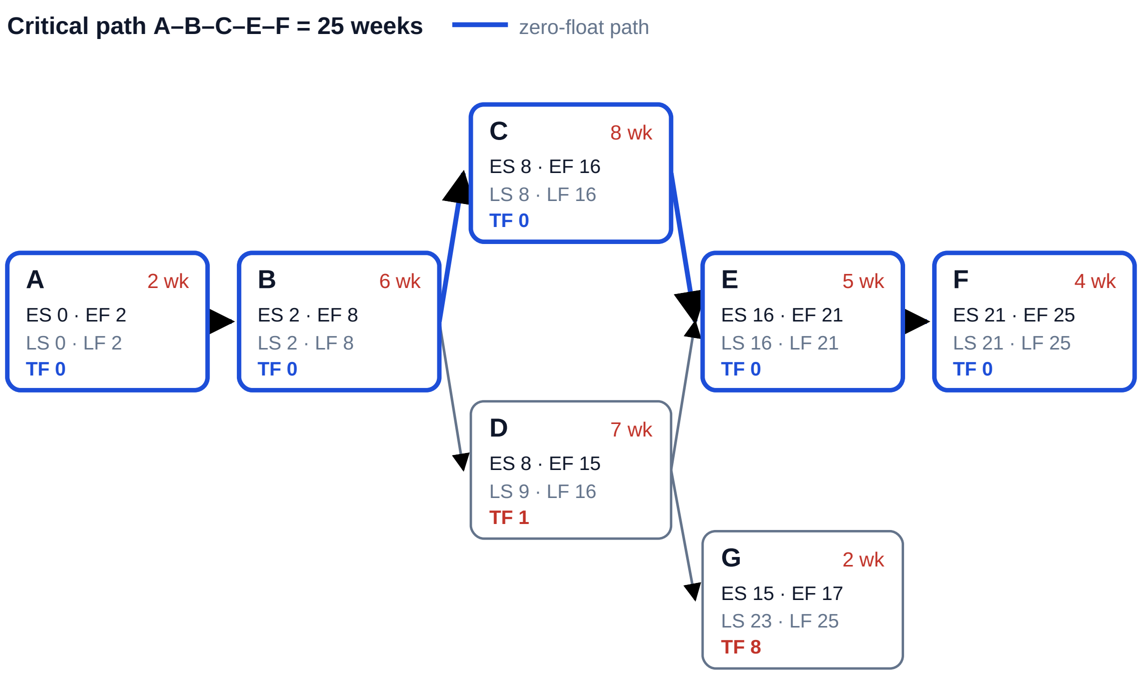


Fig 6.2.1 — The Auriga network with both passes

6.2.2 Total and free float

Definitions.

```
TF = LS − ES           — how far an activity can slip without delaying the project
FF = min(ES of successors) − EF    — how far it can slip without delaying ANY successor
```

`FF ≤ TF` always. The Auriga table carries the domain's subtlest lesson: **D has `TF` = 1 but `FF` = 0** — D can slip a week without moving the end date, but the moment it slips at all, G's earliest start moves. Total float is a project-level commodity; free float is a courtesy to your successors. A leader who grants a subcontractor "the float" without saying which kind has just given away someone else's schedule.

Float is not spare time. Float belongs to the path, not the activity: consume D's week and every activity sharing that path inherits zero. Contract regimes differ on who owns float (Domain 10, KA 10.3, treats float ownership clauses); the delivery answer is simpler — float is a risk buffer under the leader's governance, spent deliberately or not at all.

WORKED EXAMPLE

Worked example 6.2.2 — why activity floats must never be added.

1. **Setup.** A float report for Auriga lists two activities with float: D ($TF = 1$) and G ($TF = 8$). A subcontract manager reads the report as "the network carries nine weeks of slack" and proposes to spend it. How much cumulative slip can the network actually absorb on that chain without moving week 25, and what does the difference cost?
2. **Formula.** Float is a property of a **path**, not an activity. For a chain of activities whose only route to the end is one path of length Lp, the absorbable slip is the path float $PD - Lp$, shared among every activity on it. Where an activity also lies on a tighter path, its own slip is additionally capped by that path's float.
3. **Substitution.** Path A–B–D–G runs $2 + 6 + 7 + 2 = 17$ weeks against $PD = 25$, so its path float is **8**. Let D slip d and G slip g: the path finishes at $17 + d + g$, so $d + g \le 8$. D also sits on A–B–D–E–F (24 weeks), which caps $d \le 1$.
4. **Result.** The maximum cumulative slip the chain can absorb is **8 weeks** — for example $d = 1$ and $g = 7$, or $d = 0$ and $g = 8$ — against the **9** weeks the report's arithmetic implies. The report overstates the network's absorption capacity by **one week**, which at Auriga's USD 45,000 a week is **USD 45,000** of false comfort.
5. **Interpretation. The overstatement factor is the number of activities on the path.** For n activities lying on a single non-critical path with path float f, every one of them reports $TF = f$, so the float register sums to $n \cdot f$ while the path can absorb only f. A ten-activity chain with five weeks of path float shows **50 weeks** of "float" and holds **five**. Summing an activity float column is therefore not an approximation but a category error, and it is a mistake that scales with the size of the schedule — which is why it is commonest on exactly the large networks where it does most damage.

The practical consequence of float being shared is that two parties can each spend "their" float in good faith and jointly overrun, which is why float is a governed commodity (the float policy of the Executive perspective) rather than an entitlement recorded against each activity.

The reporting rule that follows is specific. A float report is per **path**, not per activity: list each path with its length, its float and the activities on it, and state the float remaining after what has already been spent. Auriga's report has three lines, not seven. A report with one line per activity cannot be read correctly even by someone who knows all this, because the shared structure has been discarded before the reader sees it.

The caution against over-applying it. Free float is additive in one narrow sense — an activity's FF is a genuinely local quantity, spendable without consulting anyone downstream — and this is exactly why FF is the right float to grant to a subcontractor and TF the wrong one. But FF is usually zero (D's is), so the honest answer to "how much float may we have?" is most often "none that is yours alone; here is what the path holds and who else is drawing on it". A counsel pointer belongs here, because float ownership is enforceability-sensitive: whether float is the contractor's, the employer's or the project's, and what a grant of it concedes about later delay claims, is a matter of the contract and the jurisdiction, varies between regimes and delay-analysis protocols, and should be taken from qualified legal advice (Domain 10, KA 10.3) rather than from a scheduling convention. The arithmetic above is contract-neutral; the answer given to the counterparty is not.

Common pitfall — managing only the critical path. D sits one week from critical. A team watching only A–B–C–E–F will discover D's importance the week it becomes too late to matter. Near-critical analysis (all paths within, say, 10 % of project duration) is standing practice; KA 6.4's scenario work builds on it, and KA 6.2.3 prices it.

6.2.3 Reading the critical path

The critical path is the longest path and therefore the shortest possible duration — both statements are the same fact, and fluency means holding both. Three leader-grade readings of the Auriga table: **(1)** procurement (C) is the single most schedule-driving activity — an expediting conversation is worth more than any amount of site pressure; **(2)** the C/D pair converge on E, so E's start is hostage to the later of two chains — convergence points are where slippage compounds and where progress review should focus (Domain 8, KA 8.1); **(3)** G's eight weeks of float make it the natural donor of resources in any recovery (KA 6.4).

WORKED EXAMPLE

Worked example 6.2.3 — choosing the near-critical band, and paying for it.

1. **Setup.** Auriga's team must decide what to monitor weekly. Enumerate every path, choose a near-critical band, and price the decision. Extending the weekly review to a second path costs about **0.5 engineer-weeks** per cycle for the **12** remaining cycles at **USD 5,225** an engineer-week (Domain 7, KA 7.4.1). A week of lateness costs **USD 45,000**.
2. **Formula.** Path float = `PD − path length`. A path is near-critical if its float ≤ the chosen band, conventionally a percentage of `PD`. Monitoring pays if `cost ≤ weeks of lateness avoided × cost of delay`.
3. **Substitution.** Paths: A–B–C–E–F `= 2+6+8+5+4 = 25`; A–B–D–E–F `= 2+6+7+5+4 = 24`; A–B–D–G `= 2+6+7+2 = 17`. Band at 10 %: `0.10 × 25 = 2.5` weeks. Monitoring cost `0.5 × 12 × 5,225`.
4. **Result.**

Path	Length (wk)	Float (wk)	Float as % of PD	Inside a 10 % band?
A–B–C–E–F	25	0	0.00 %	yes — critical
A–B–D–E–F	24	1	4.00 %	**yes**
A–B–D–G	17	8	32.00 %	no

Two of three paths fall inside the band, and because they share A, B, E and F they cover **6 of 7** activities — **85.71 %** of the network. Monitoring cost `0.5 × 12 × 5,225 =` **USD 31,350**, recovered by avoiding `31,350 / 45,000 =` **0.6967 weeks** of lateness. 5. **Interpretation. The band captures almost everything, and that is the point.** On a well-formed network the near-critical set is not a short annexe to the critical path but most of the schedule — here 85.71 % of activities. "Manage the critical path" is therefore not a management system; it is a reporting convention that survives because it fits on a slide. What the leader actually needs is the float column sorted ascending, with a line drawn where the band falls.

The band must be wider than the largest float you are unwilling to lose. A–B–D–E–F's float is 1 week, which is **4.00 %** of 25 — so any band of 4 % or more captures it and any band below 4 % does not. A team that had adopted a 2 % band (0.5 weeks) would monitor the critical path alone and would be blind to the path that is one week from binding. The percentage is not a convention to be inherited; it is a decision made by asking which paths would I want to hear about? and then setting the band to include them. Quoting a band without saying what it captures is the same defect as quoting a confidence level without a date (Domain 8, KA 8.A.2).

The monitoring economics are not close. Watching a second path to handover costs USD 31,350 and pays for itself if it prevents **0.70 weeks** of lateness once. If D slips two weeks undetected, E cannot start until week 17 and F finishes week 26 — a full week late, **USD 45,000**, or **1.44 times**

the entire cost of monitoring. Near-critical analysis is one of the few schedule controls whose case is so lopsided that it should not need making; it is omitted for want of the arithmetic rather than for want of the money.

The cautions. Path enumeration is trivial on seven activities and combinatorially hopeless on five thousand, where the practical instrument is the float histogram — count activities by float band and read the shape — rather than a path list. Path float is computed against the current network, so it changes with every crash and every slip (KA 6.4.2's path migration); a band chosen once and never revisited will be monitoring last quarter's near-critical set. And float measured against logic is not float against **capacity**: KA 6.3.1c shows an activity with eight weeks of logical float extending the project once a resource cap binds, so a watch-list drawn only from the float column can still miss what will actually make the project late.

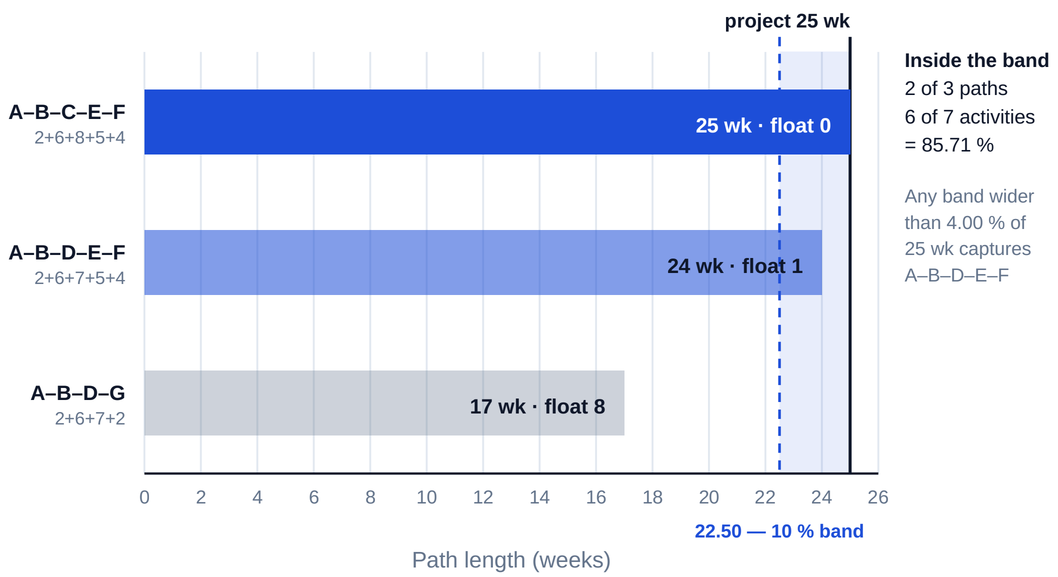


Fig 6.2.2 — Auriga's three paths, their floats, and the near-critical band

AI in this KA

CPM arithmetic is deterministic — ideal for machine computation and instant human verification. The two checks that catch nearly every pass error: the critical path's durations must **sum exactly** to the project duration, and every critical activity must show `TF` = 0 with `ES = LS`. An AI-produced schedule that fails either is broken; one that passes both may still have wrong logic — which is why KA 6.1's link-walking discipline precedes trust in any pass, human or machine.

KEY TERMS — KA 6.2

Term	Meaning
Forward / backward pass	Left-to-right ES/EF computation; right-to-left LS/LF computation.
ES EF LS LF	Earliest/latest start and finish per activity.
Total float TF	LS − ES; slip available without delaying the project.
Free float FF	Slip available without delaying any successor; FF ≤ TF.
Critical path	The zero-float longest path; the project's shortest possible duration.
Convergence point	A node where paths merge; slippage compounds there.
Path float	PD − path length; the slip a whole chain can absorb, shared by every activity on it.
Near-critical band	The float threshold below which a path joins the monitored set; a decision, not a convention.

SAMPLE MCQS — KA 6.2

MCQ 6.2-A [6.2.1 · Application] In the Auriga network, activity D (duration 7, predecessor B finishing week 8, successors E and G) has LS = 9 and ES = 8. Its total float is: - A. 0 - B. 1 - C. 8 - D. 7

Rationale: TF = LS − ES = 9 − 8 = 1. A confuses D with the critical activities; C is G's float; D is the activity's duration, not its float.

MCQ 6.2-B [6.2.2 · Analysis] An activity shows TF = 1 and FF = 0. Delaying it by one week will: - A. delay the project by one week - B. delay nothing, because total float absorbs the slip - C. delay at least one successor's earliest start while leaving the project end date unchanged - D. be impossible — free float can never be below total float

Rationale: Positive TF protects the end date (not A); zero FF means some successor moves immediately (not B). D inverts the invariant — FF ≤ TF always.

MCQ 6.2-C [6.2.3 · Application] Auriga's procurement activity C is crashed from 8 weeks to 6. The new project duration is: - A. 23 weeks - B. 24 weeks - C. 25 weeks - D. 21 weeks

Rationale: After one week of crashing, path B–D–E (8+7+5+4 via D) becomes co-critical at 24; the second crashed week buys nothing — duration stays 24. A assumes both weeks convert to project weeks; C forgets the crash entirely; D subtracts from the wrong baseline. (The full economics: KA 6.4.)

MCQ 6.2-F [6.2.1 · Application] Auriga's procurement C slips from 8 weeks to 9 (all else unchanged). The new project duration is: - A. 25 weeks - B. 26 weeks - C. 27 weeks - D. 24 weeks

Rationale: C was critical with zero float, so its extra week passes straight through: E runs 17–22, F finishes week 26. A assumes float absorbed a critical activity's slip; C adds the week twice; D subtracts it.

MCQ 6.2-D [6.2.1 · Application] Auriga's training activity G runs ES 15–EF 17 with LF = 25. Its total float is: - A. 8 - B. 10 - C. 2 - D. 0

Rationale: TF = LF − EF = 25 − 17 = 8 (equivalently LS − ES = 23 − 15). B ignores G's duration (25 − 15); C is G's duration; D confuses G with the critical path.

MCQ 6.2-E [6.2.3 · Analysis] A programme's binding path shows TF = −3 after a constraint-honest pass. The professional reading is: - A. the software has malfunctioned — float cannot be negative - B. the plan, as constrained, is already three weeks late; the number sizes the recovery problem and must be reported now - C. delete the constraints so the float returns to zero - D. the project will finish three weeks early

Rationale: Negative float is the gap between what logic needs and what constraints allow — information, not error (A) and not good news (D). C is the pin-and-squeeze suppression this domain's Case B dissects; it hides the problem until it is unrecoverable.

MCQ 6.2-G [6.2.2 · Analysis] A float report for Auriga lists D at TF = 1 and G at TF = 8. The cumulative slip the D–G chain can absorb without moving week 25 is: - A. 9 weeks — the sum of the two floats - B. 8 weeks, because both activities draw on the same path float of 8 - C. 1 week, the smaller of the two floats - D. 7 weeks, since D must retain a week

Rationale: Path A–B–D–G runs 17 weeks against a 25-week project, so $d + g \le 8$ (6.2.2). A sums a shared quantity, the error that scales with schedule size — a ten-activity chain with five weeks of path float reports fifty. C takes the minimum instead of the shared total; D invents a reservation rule that no pass produces.

MCQ 6.2-H [6.2.3 · Evaluation] A team adopts a near-critical band of 2 % of project duration on Auriga. The consequence is: - A. none — 2 % is a standard threshold - B. path A–B–D–E–F, whose float is 4.00 % of the 25-week duration, falls outside the band and goes unmonitored - C. all three paths are monitored, since 2 % is stricter - D. the band is invalid because bands must be at least 10 %

Rationale: The band is 0.5 weeks and A–B–D–E–F's float is 1 week, so the second path — one week from binding — is excluded (6.2.3). A appeals to a convention nobody derived; C inverts what "stricter" does to a threshold; D invents a rule, when the correct test is whether the band captures the paths the leader wants to hear about.

SELF-CHECK — KA 6.2

1. State both definitions of the critical path and why they are equivalent. — Longest path through the network; shortest possible project duration — the longest chain of unavoidable work is precisely what bounds how soon the whole can finish.
2. Why is $FF \le TF$ always? — Delaying any successor is one of the ways of delaying the project; the constraint set defining FF is stricter.
3. Auriga slips: D takes 10 weeks, not 7. New duration? — 27 weeks: D finishes week 18, E waits for it (18 > 16), E–F add 9 → the critical path re-routes through D (KA 6.4 works the recovery).
4. Why can a float column never be summed? — Because every activity on a path reports that path's float: n activities on a path with float f report $n \cdot f$ and the path holds f. Auriga's report implies 9 weeks and the chain absorbs 8.
5. Three checks that confirm a pass in seconds? — Critical durations sum to the project duration; critical activities show $ES = LS$ and $EF = LF$; every TF equals both $LS - ES$ and $LF - EF$.

KNOWLEDGE AREA 6.3

Resources, constraints, milestones and rolling wave

Topics: 6.3.1 resource planning and levelling · 6.3.2 constraints and milestones · 6.3.3 rolling-wave planning.

6.3.1 Resource planning and levelling

The principle. The passes of KA 6.2 assume infinite resources; reality staffs the network. Loading resources onto the schedule produces a **histogram** — demand per period — and the histogram almost always spikes. Two responses: **resource smoothing** uses float to flatten demand without moving the end date (slip D and G inside their float to pull crews off the peak); **resource levelling** accepts a later end date if the resource cap is hard. The order of operations is a leadership decision in disguise: smoothing spends float — the project's risk buffer — to save money, and someone accountable should price that trade explicitly, not discover it in a progress meeting.

WORKED EXAMPLE

Worked example 6.3.1 — smoothing Auriga's field crews.

1. **Setup.** Weeks 9–15: civil works D needs 3 crews; early installation staging inside C needs 2; the site cap is 4 crews. Overlap weeks 9–15 demand 5 — one over cap.
2. **Formula.** Smoothing test: can a demanding activity slip within `TF` (and, if successors matter, `FF`) to clear the peak?
3. **Substitution.** D has `TF` = 1, `FF` = 0: slipping D one week clears one peak week but consumes D's entire float and immediately moves G (`FF` = 0). The staging work inside C has discretionary timing across weeks 9–16.
4. **Result.** Re-sequence the staging within C to weeks 14–16 where D's demand tails off: peak demand falls to 4, the cap holds, the end date holds, and **no float is spent**.
5. **Interpretation.** The cheapest capacity is re-sequencing; the second cheapest is float; the expensive ones are overtime, second shifts and extension. A leader asks for the histogram with the options priced, not just the spike.

WORKED EXAMPLE

Worked example 6.3.1b — when the cap is hard: levelling, priced.

1. **Setup.** Now suppose the site cap is **3 crews** (a permit condition, not a preference). D needs all 3 for weeks 9–15; the 4-week staging package (2 crews) must finish before E can start (week 16). Delay costs USD 45,000 per week (the client's bonus forgone); a second-shift waiver for the staging crew costs USD 20,000 per week. What does the leader do?
2. **Formula.** Compare total cost of each feasible plan: pure levelling (extend) vs paid capacity (second shift) vs any float-funded re-sequence.
3. **Substitution.** Levelling: staging cannot start until D releases crews (week 16); it runs weeks 16–19, E starts week 20 — project 25 → 29, cost `4 × 45,000 = 180,000`. Second shift: staging runs weeks 12–15 off-shift, E starts on time — cost `4 × 20,000 = 80,000`. No float-funded option exists: D's `TF` = 1 cannot host a 4-week move.
4. **Result. Buy the second shift: USD 80,000** against USD 180,000 of extension — and the end date holds.
5. **Interpretation.** A hard cap converts a scheduling problem into a procurement problem. The histogram's job is to surface that conversion early enough for the cheap option to exist — second shifts, pre-assembly, off-site staging all have lead times of their own. Levelling that silently extends the project is a decision someone should have been asked to make.

The breakeven is wide, and saying so is what makes the recommendation robust. The shift wins for any premium below **USD 45,000** a week — the cost of delay itself — so at USD 20,000 the shift is bought at **44.4 %** of the price at which the leader would become indifferent, and the quoted premium could rise **2.25 times** before extension became the better plan. That is the sentence for the steering paper, because it answers the only question worth asking about a quoted price: how wrong can it be before the decision changes? The corollary is equally useful in the other direction — where the cost of delay is small, levelling into an extension is the correct answer and paying a premium to hold a date is the error.

The comparison is only valid because both plans deliver the same scope. Levelling to week 29 and second-shifting to week 25 produce the same works; if the second shift also carried a quality consequence — night-shift error rates are not day-shift error rates — the comparison would need Domain 9's containment economics on the other side of the ledger, and USD 80,000 would not be the whole price. A reviewer's question here is always what else changes when the shift pattern changes?

Two cautions on the arithmetic. The USD 45,000 assumes the extension actually costs four weeks of delay cost, which holds only because E is on the critical path with zero float — the same four-week levelling on a path with four weeks of float would cost nothing at all, and the histogram cannot tell you which case you are in. And the permit condition that fixes the cap at three crews is a genuine external constraint, not a preference: the option of simply exceeding it does not exist, and a plan that quietly assumes a waiver will be granted is a plan with an unpriced risk in it (Domain 8's register, not the schedule's float).

WORKED EXAMPLE

Worked example 6.3.1c — the resource-feasible schedule, and why the critical path is not the critical chain.

1. **Setup.** Both examples above levelled one activity against a cap. This one levels the **whole network** against a hard cap, which is the problem a planner actually faces. Auriga's specialist **engineering pool** — a different resource from the field crews above, and levelled separately, which is itself part of the difficulty — is capped at **6 engineers**. Each activity's engineer demand while it runs:

Activity	A	B	C	D	E	F	G
Duration (wk)	2	6	8	7	5	4	2
Engineers	2	4	1	3	5	6	2
TF (KA 6.2.1)	0	0	0	1	0	0	8

Activities may not be split. What is the shortest resource-feasible duration, what does the standard priority rule give, and what should the leader buy? 2. **Formula.** Load the early-start schedule and read the histogram: demand in week `t = Σ engineers of activities running in t`. Where demand exceeds the cap, defer activities by priority — conventionally **minimum total float first** — and re-check. Total demand `= Σ (duration × engineers)`; average utilisation `= total demand ÷ (cap × duration)`. 3. **Substitution.** Total demand `2×2 + 6×4 + 8×1 + 7×3 + 5×5 + 4×6 + 2×2 = 4+24+8+21+25+24+4 = 110` engineer-weeks against `6 × 25 = 150` of capacity. Early-start profile: weeks 0–2 A alone (2); 2–8 B (4); 8–15 C+D (`1+3 = 4`); 15–16 C+G (`1+2 = 3`); **16–17 E+G (`5+2 = 7`)**; 17–21 E (5); 21–25 F (6). 4. **Result.** Average utilisation `110/150 = `**73.33 %**, and the cap is still breached — by **one engineer, in the single week 16–17**, where E (5) overlaps G (2). Three plans follow.

Plan	What it does	Duration	Cost consequence vs the 25-week plan
Minimum-float priority rule	G (TF 8) yields to E (TF 0); G then finds no 2-engineer window until F ends	**27 wk**	2 weeks of delay = **USD 90,000**
Best resource-feasible schedule	G runs weeks 15–17; **E is deferred one week** to 17–22; F 22–26	**26 wk**	1 week of delay = **USD 45,000**
Buy the peak out	Hire **one** engineer for the single week 16–17; G 15–17, E 16–21, F 21–25	**25 wk**	one engineer-week purchased = **USD 5,225**

1. **Interpretation. Float against logic is not float against capacity.** G carries **eight weeks** of total float and is nevertheless the activity that makes the project late: under the cap it has nowhere to go, because every week between its earliest start and the project end is already spending five or six of the six available engineers. The set of activities that determines a resource-constrained duration is the **critical chain** — the binding sequence once both logic and capacity are honoured — and it need not be the critical path. Any statement of the form "G doesn't matter, it has eight weeks of float" is a statement about the logic network alone, and the logic network is not the project.

The standard priority rule loses money. Protecting the zero-float activity is the instinctive and usually correct move, and here it produces **27 weeks** where deliberately slipping E by one week produces **26** — a difference of **USD 45,000** left on the table by following the rule. The reason is structural: deferring G instead pushes it past a **nine-week** block (E's five weeks then F's four) in which no capacity is ever free, so the rule's refusal to defer the critical activity by one week costs the project two. Whether protecting the critical path is right depends on what lies behind it, which no priority rule inspects. Resource-constrained scheduling is a combinatorial optimisation, not a rule application; priority rules are **heuristics**, they are known to be beatable, and a planner who presents a levelled schedule should say which rule produced it and whether anything better was sought. This is precisely where an optimiser earns its place (see AI in this KA), and it is the honest reason to use one: not speed, but the USD 45,000.

Averages cannot answer capacity questions. The pool is **73.33 % utilised on average** across a schedule it cannot execute, and in the feasible 26-week plan it is only **70.51 %** utilised. Any resourcing conversation conducted in averages — "we have plenty of engineering capacity, we're only three-quarters loaded" — is a category error, because feasibility is a property of the **peak** and the peak is a property of the logic. The histogram is not a presentational nicety; it is the only instrument in the domain that answers the question the average appears to answer.

The whole overrun traces to one engineer-week, and that ratio is the lesson. The breach is one engineer for one week. Buying it costs **USD 5,225** and saves **USD 45,000** — a return of **8.61 times** — because a single week of excess demand at the wrong point in the network converts into a full project week. The general habit: price the excess area of the histogram in resource-weeks, compare it with the cost of delay, and expect the ratio to be startling. The breakeven price for that engineer-week is **USD 45,000**, or **8.61 times** the internal rate, which is why hiring in at a premium is so often right and so rarely proposed.

And optimise before you buy, because a bad baseline inflates what capacity looks worth. Measured against the priority rule's 27 weeks, that one engineer-week appears to save **two** weeks and to be worth up to **USD 90,000 — 17.22 times** the internal rate. Measured against the correct 26-week baseline it saves one and is worth up to **USD 45,000**. A leader negotiating a hire-in rate against the un-optimised schedule would rationally pay **twice** the defensible price, which is a general property of buying capacity against a heuristic baseline rather than a feature of this network.

Three cautions. The no-splitting rule is doing real work here: allowing G to run as two separate single weeks changes the answer, and whether splitting is permissible is a physical and contractual question, not a software setting. The example levels one resource; real projects level several simultaneously, which is why the problem is hard rather than tedious, and why a plan feasible on engineers can be infeasible on cranes. And the 6-engineer cap must be real — a cap that is actually a budget line is negotiable, and levelling against a negotiable cap manufactures a delay that a conversation would have removed.

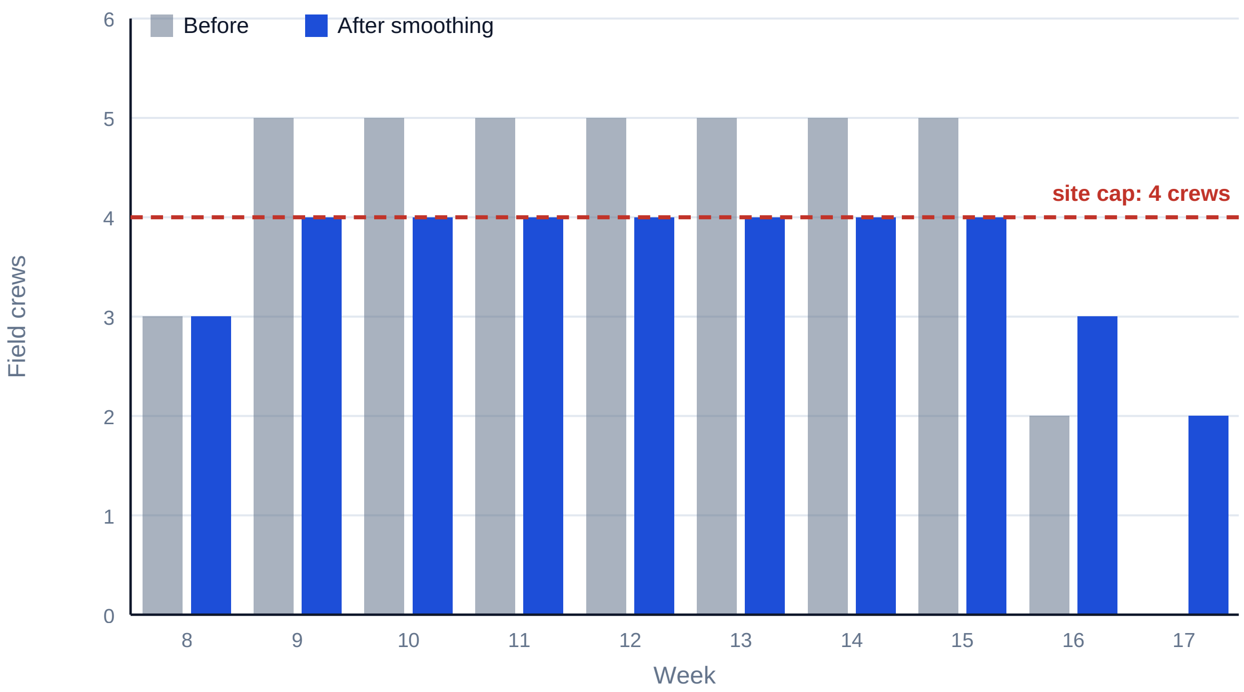


Fig 6.3.1 — Auriga field-crew histogram, before and after smoothing

6.3.2 Constraints and milestones

Constraints are externally imposed dates — an outage window, a regulatory deadline, a seasonal limit. Modelled honestly they are logic (an outage window is a predecessor with a date); modelled lazily they are pins that falsify float (KA 6.1.3). **Milestones** are zero-duration events that carry meaning: contractual obligations (Domain 10), gate decisions (Domain 3), integration points with other projects (Domain 15). A milestone's date is an output of the network, never an input — the moment a milestone is typed rather than computed, the schedule has stopped forecasting and started decorating.

Milestone hygiene: each has an owner, an unambiguous done-definition (Domain 5's acceptance criteria), traceable L3 logic, and — for contractual milestones — a stated float position reported honestly to the counterparty (Domain 11's reporting ethics).

6.3.3 Rolling-wave planning

The principle. Detail decays with distance: estimating month 18's tasks at daily granularity today manufactures precision, not knowledge. **Rolling wave** plans near-term work at execution detail (L4) and far work at planning packages (L3 summary), elaborating each wave as it approaches under change control.

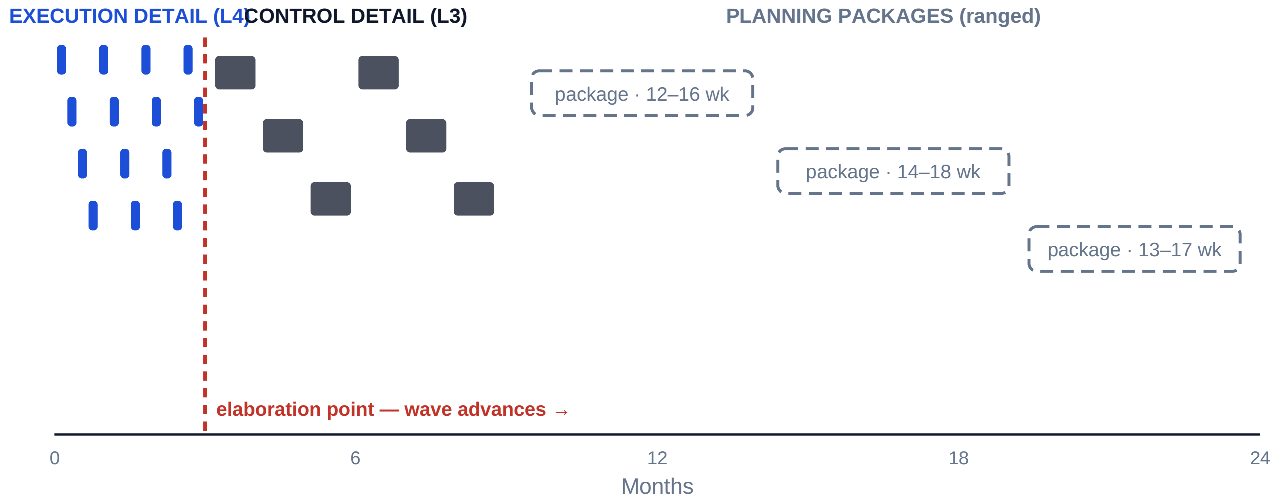


Fig 6.3.2 — The rolling-wave horizon

WORKED EXAMPLE

Worked example 6.3.3 — what a ranged planning package may be committed to.

1. **Setup.** Meridian's wave-3 clinic tranche is a far-wave planning package with a ranged duration of **14–18 weeks**, starting at programme **week 40**. The sponsor asks for a go-live date for the tranche. Elaborating the package — a workshop with the estate, clinical and training leads, **3 planner-weeks** at **USD 4,200** — would narrow the range to **15.5–16.5 weeks**. Cost of delay **USD 14,280** per week. What date may be committed, and is the elaboration worth buying first?
2. **Formula.** Milestone range = start + duration range. Commitment exposure = range width × cost of delay. Elaboration is worth buying if the exposure it removes exceeds its cost — noting that elaboration changes the precision of the promise, not the duration of the work.
3. **Substitution.** Milestone range $40 + 14 =$ week 54 to $40 + 18 =$ week 58; mid-point week 56. Exposure before $= 4 \times 14{,}280$; after $= 1 \times 14{,}280$. Elaboration cost $= 3 \times 4{,}200$.
4. **Result.** The tranche completes somewhere in **weeks 54–58**, mid-point **56**. Commitment exposure is **USD 57,120** before elaboration and **USD 14,280** after — a reduction of **USD 42,840** for **USD 12,600** of planning effort, a ratio of **3.40**. Committing to the mid-point and missing by two weeks costs **USD 28,560**.
5. **Interpretation. The mid-point is not a commitment, it is a coin toss.** Week 56 is the centre of a range, so a commitment to it is a commitment with roughly even odds — and the appearance of a firm date is what makes it dangerous, because nobody downstream will treat "week 56" as a 50 % statement. Exactly three honest options exist: commit to the upper bound (week 58, and hold the two weeks internally as the buffer they are), commit to the mid-point with the range stated beside it, or decline to commit until the package is elaborated. Which of the three is chosen is a governance decision (Domain 3), and Domain 8's methods set the confidence at which a buffered date should sit. What is not available is the fourth option everyone reaches for: quoting the mid-point as though it were the upper bound.

Elaboration is a purchasable reduction in commitment error, and it can be priced. USD 12,600 buys a **3.40-fold** return in exposure terms, and the arithmetic is available before the workshop is convened, which is what turns "we should elaborate wave 3" from an aspiration into a funded activity with an owner and a date. The general rule: elaborate when the exposure removed exceeds the elaboration cost, and elaborate before committing rather than after — the reverse order is how a programme acquires a public date it then has to defend.

Ranges must not be rolled up by adding mid-points. Three consecutive tranches each ranged 14–18 weeks do not make a 48-week programme with a 12-week range; the aggregate spread is narrower than the sum of the spreads where the tranches are independent, and wider where they share crews, estate or approvals. Domain 8, KA 8.2.4 supplies the aggregation, and this is the single most common quantitative error in rolling-wave reporting.

The warning. Elaboration does not shorten the work, and a leader who reads the USD 42,840 as a saving has misread it: nothing was saved, a promise merely became more accurate. Nor does narrowing the range make the mid-point right — the elaborated range 15.5–16.5 could sit anywhere inside the original 14–18, so the elaborated date may be later than the date the sponsor was informally given. That is the moment the honesty of a rolling-wave regime is actually tested, and it is why the elaboration event needs a re-baselining rule agreed in advance (Domain 4, KA 4.3) rather than negotiated in the meeting where the number appears.

AI in this KA

Resource optimisation is a genuine AI strength — levelling across thousands of activities and dozens of calendars is combinatorial work machines do better than planners. Governance holds the same shape: the optimiser proposes a levelled plan; the planner verifies the constraint set was real (crew interchangeability, shift rules, site logistics); the leader owns the trade the optimiser cannot see — whether spending float, money or scope is the right currency this month. An optimiser given a wrong constraint produces a beautifully efficient fiction.

Why the case is unusually strong here, and what still has to be checked. KA 6.3.1c is the argument in miniature: the priority rule every planner is taught produced 27 weeks where 26 was available, and USD 45,000 turned on a search nobody had time to run by hand. Resource-constrained scheduling is genuinely hard — the search space grows combinatorially with activities, resources and calendars — so this is one of the few places in delivery where a machine does not merely accelerate a human process but reaches answers humans systematically do not. The verification duties are correspondingly specific. **Check feasibility, not optimality:** re-derive the resource histogram from the proposed schedule and confirm no period exceeds any cap, which is one addition per period and catches the commonest failure — an optimiser that has quietly relaxed a constraint it could not satisfy. **Check that the objective was the right one:** minimising duration, minimising peak demand and minimising cost give three different schedules, and a tool asked for the wrong one returns an excellent answer to a question nobody had. **Check what was allowed to move:** splitting, overtime, calendar exceptions and crew substitution are assumptions with contractual and physical consequences, and an optimiser silently permitted to split an activity has produced a plan the site cannot run. And **compare against a stated baseline** — the priority-rule schedule — so that the improvement is a number rather than a claim.

KEY TERMS — KA 6.3

Term	Meaning
Resource histogram	Demand per period from the loaded schedule.
Smoothing / levelling	Flattening demand within float / accepting a later date under a hard cap.
Constraint	An externally imposed date; modelled as logic, not as a pin.
Milestone	Zero-duration event with an owner and a done-definition; its date is computed.
Rolling wave	Near-term detail, far-term packages, elaborated under change control.
Planning package	A far-wave summary activity with ranged duration awaiting elaboration.
Resource-feasible schedule	A schedule in which no period's demand exceeds any resource cap.
Critical chain	The binding sequence once logic and capacity are honoured; need not be the critical path.
Priority rule	The heuristic (commonly minimum total float first) used to break resource conflicts; beatable, and known to be.
Commitment exposure	Range width × cost of delay — what a date quoted from a ranged package risks.

SAMPLE MCQS — KA 6.3

MCQ 6.3-A `[6.3.1 · Application]` A planner clears a resource peak by delaying an activity with `TF` = 1, `FF` = 0 by one week. The immediate consequences are: - A. no schedule effect of any kind - B. project delay of one week - C. the end date holds, the activity's path float is exhausted, and at least one successor's earliest start moves - D. the resource peak worsens

Rationale: One week inside `TF` protects the end date (not B), but zero `FF` moves a successor and the path's buffer is now spent (not A) — the smoothing was legal but not free.

MCQ 6.3-B `[6.3.3 · Analysis]` A 30-month programme shows daily-level tasks throughout, including month 30. The strongest inference is: - A. the planning team is unusually diligent - B. far-horizon detail is manufactured precision that will churn on every update, obscuring real variance - C. the schedule will need no re-baselining - D. rolling-wave planning has been correctly applied

Rationale: Detail beyond the knowable horizon creates update churn and false confidence — the opposite of diligence in effect (A) and the opposite of rolling wave (D); constant churn makes re-baselining more likely, not less (C).

MCQ 6.3-E `[6.3.1 · Application]` D needs 3 crews in weeks 9–15 and concurrent staging needs 2; the site cap is 4. The excess demand the histogram must clear is: - A. 1 crew for 7 weeks - B. 5 crews for 7 weeks - C. 2 crews for 7 weeks - D. nothing — 3 + 2 = 5 is within a 4-crew cap across two activities

Rationale: Peak demand 5 against cap 4 leaves one excess crew-week in each of the seven overlap weeks — the precise quantity smoothing must relocate. B is total demand, not excess; C is the staging demand itself; D misreads the cap as per-activity when it is per-site.

MCQ 6.3-C `[6.3.1 · Application]` A hard 3-crew cap forces either a 4-week project extension (delay cost USD 45,000/week) or a second-shift waiver at USD 20,000/week for the same 4 weeks (end date held). The value-maximising choice and its saving are: - A. extend: saves USD 100,000 - B. second shift: saves USD 100,000 - C. second shift: saves USD 25,000 - D. they cost the same

Rationale: Extension costs `4 × 45,000 = 180,000`; the shift premium `4 × 20,000 = 80,000` — choosing the shift saves USD 100,000 and keeps the date. A picks the dearer plan; C compares one week only; D ignores the 65,000-per-week difference... which compounds four times.

MCQ 6.3-D `[6.3.3 · Recall]` A legitimate planning package in a rolling-wave schedule must carry: - A. daily-level task detail for its whole span - B. a ranged duration consistent with its estimate class and a dated elaboration event under change control - C. zero float, to keep pressure on the team - D. a pinned finish date agreed with the sponsor

Rationale: Far-wave honesty is ranged duration plus a governed elaboration point. A is the manufactured precision rolling wave exists to avoid; C fabricates criticality; D is the typed milestone anti-pattern (Case study B).

MCQ 6.3-F `[6.3.1 · Analysis]` Auriga's engineering pool is capped at 6. Activity G needs 2 engineers for 2 weeks and carries 8 weeks of total float, but every week from its earliest start to the project end already commits 5 or 6 engineers. The correct conclusion is: - A. G is irrelevant to the end date, because it has eight weeks of float - B. G determines the resource-constrained duration despite its logical float, because float against logic is not float against capacity - C. the cap must be wrong, since average utilisation is only 73.33 % - D. G should be deleted from the network and tracked separately

Rationale: Under the cap G has nowhere to go, so it joins the critical chain even though its total float is 8 (6.3.1c). A applies a logic-network statement to a capacity problem; C reasons from an average when feasibility is a property of the peak; D removes the activity that is about to make the project late from the model that would show it.

MCQ 6.3-G `[6.3.1 · Evaluation]` A levelled Auriga schedule produced by the minimum-total-float priority rule runs 27 weeks; a search finds a feasible 26-week schedule that deliberately defers the critical activity E by one week. The professional reading is: - A. the 26-week schedule is invalid, because critical activities must never be deferred - B. the priority rule is a heuristic and here costs USD 45,000; a levelled schedule should state the rule used and whether better was sought - C. both schedules are equally good, since both respect the cap - D. the rule is wrong and should never be used

Rationale: Deferring E by one week releases G two weeks earlier, so the "protect the critical path" instinct loses a week worth 45,000 (6.3.1c). A elevates a heuristic to a constraint; C ignores a one-week difference in duration; D over-corrects — the rule is a sound default whose output needs a stated baseline, not abolition.

SELF-CHECK — KA 6.3

1. Why is smoothing "spending the risk buffer"? — It consumes float, and float is the project's absorption capacity for the unknown (KA 6.2.2).
2. What makes a milestone honest? — Computed date, named owner, unambiguous done-definition, traceable logic.
3. What must a planning package carry to be legitimate? — A ranged duration consistent with its estimate class and a dated elaboration event under change control.
4. Why can average utilisation never establish that a resource plan is feasible? — Because feasibility is a property of the peak: Auriga breaches a 6-engineer cap while only 73.33 % loaded on average.
5. What is the difference between the critical path and the critical chain? — The critical path binds on logic alone; the critical chain binds once capacity is honoured too, and on Auriga it includes G, an activity with eight weeks of logical float.

KNOWLEDGE AREA 6.4

Delivery flow: predictive, agile and hybrid; recovery and forecasting

Topics: 6.4.1 flow across delivery modes · 6.4.2 schedule compression and recovery · 6.4.3 forecasting and scenario analysis.

6.4.1 Flow across delivery modes

Predictive scheduling (this domain so far) plans the whole network and measures variance against baseline. **Agile** delivery (Domain 13 in full) fixes cadence and capacity and lets scope flow: the schedule questions become throughput (items per iteration), cycle time and forecast by velocity. **Hybrid** programmes — the commonest reality — run engineered streams predictively and product streams by cadence, joined at **integration milestones** that appear in the CPM network as fixed-capacity deliveries. The leader's job is translation: a velocity forecast ("the reporting module completes in 4–6 sprints") enters the network as a ranged duration on the integration activity, and the network returns what the programme needs from the agile stream — the latest acceptable delivery, read from LS. Neither mode is senior; the network prices time, the cadence protects focus, and governance (Domain 3, KA 3.1.3) holds both to evidence.

WORKED EXAMPLE

Worked example 6.4.1 — the network's answer to an agile stream is a rate, not a date.

1. **Setup.** Auriga's operator-interface configuration is delivered by a cadence team working from a **72-item** backlog. Over its last eight iterations the team's throughput has run between **3.6** and **4.5** items a week, averaging **4.0**. The package must be complete before testing and commissioning **F** begins, and F's backward pass gives `LS(F)` = **21** (KA 6.2.1). The stream cannot start until the design freeze at **week 5**. Domain 13 supplies the throughput forecasting method; this example does the translation.
2. **Formula.** Forecast duration = backlog ÷ throughput, computed across the observed throughput range to give a ranged duration. Available window = `LS` of the successor − stream start. **Required throughput = backlog ÷ available window.** Deliverable scope at a given throughput = throughput × window.
3. **Substitution.** Durations: `72/4.5 = 16.0`, `72/4.0 = 18.0`, `72/3.6 = 20.0` weeks. Window `= 21 − 5 = 16` weeks. Required throughput `= 72/16`. At the mean rate the window delivers `4.0 × 16 = 64` items.
4. **Result.** The team's forecast duration is **16–20 weeks**, mean **18**. The window is **16 weeks**, so the required throughput is **4.5 items a week** — the team's **best observed iteration, sustained without variation for sixteen weeks**. At its mean rate the stream delivers **64** of 72 items, leaving **8 items — 11.11 % of the package — undelivered**, and finishing the full package two weeks late costs `2 × 45,000 =` **USD 90,000**. Starting at week 3 instead widens the window to 18 weeks and drops the required rate to **4.0** (the mean, and therefore roughly even odds); starting at week 1 widens it to 20 and drops the requirement to **3.6** — the worst rate the team has recorded.
5. **Interpretation. The backward pass does not ask the team when it will finish; it tells the team what rate it must hold.** "Deliver by week 21" is unactionable inside a cadence system and invites the team to agree to something it cannot forecast. "Sustain 4.5 items a week from week 5" is a statement the team can test against its own record — and in this case reject, which is the whole value of putting it that way. The conversion runs in both directions: the team's ranged forecast enters the network as a ranged duration on the integration activity, and the network returns a required rate read off `LS`. A hybrid programme that cannot perform this conversion has two plans and no plan.

A requirement equal to the best-ever observation is a plan with no margin, and saying so is the professional act. 4.5 is not impossible — it happened — but a rate achieved once is not a rate sustained for eight iterations, and the honest statement to the steering committee is that the current plan requires the team's historical maximum with zero allowance for variation. Domain 13, KA 13.2.4 sets out how to express such a forecast as a range with a stated meaning rather than as a single number; what belongs here is that the network is what turned an ambitious plan into a checkable claim.

Of the three levers in the arithmetic — the start date, the rate and the scope — only the rate belongs to the team. Moving the start and deferring items are the leader's to pull, which is why "the team must go faster" is at once the weakest available response and the most frequently chosen. It is also the one lever Domain 13 shows cannot be pulled by starting more work in parallel.

The cautions. Backlog items are being treated as interchangeable units of work, which is acceptable for forecasting a large package and wrong for a small one — 72 items averages out, 7 does not. `LS(F)` = 21 is read off the current network, so a crash or a slip anywhere on the critical path moves the required rate without the team touching anything, and the required rate must therefore be re-derived every cycle rather than fixed at planning. And the 16-week window assumes the integration itself is instantaneous: where the package needs a hardening or acceptance period before F can use it, that period is an activity in the network with its own duration, not a rounding allowance.

6.4.2 Schedule compression and recovery

The two levers. Crashing buys duration with money (more resources, overtime, expediting); **fast-tracking** buys it with risk (overlapping activities that logic preferred in sequence). Both obey the same law: work the critical path, and re-run the passes after every move, because **the critical path migrates**.

WORKED EXAMPLE

Worked example 6.4.2 — the crash that stopped paying.

1. **Setup.** Auriga's client offers a bonus of USD 45,000 (≈ SAR 168,750) per week saved. Expediting C costs USD 30,000 per week, up to two weeks. Should the leader buy one week, two, or none?
2. **Formula.** Crash only while (weeks actually saved × value per week) > crash cost — checked against the re-run passes, not the original network.
3. **Substitution.** Crash C by 1 (8→7): duration 25→24; net gain `45,000 − 30,000 = +15,000`. Crash C by 2 (8→6): path B–D–E–F (2+6+7+5+4) is now co-critical at 24 — duration stays 24; second week's net `0 − 30,000 = −30,000`.
4. **Result. Crash one week only**: +USD 15,000. The second week is pure loss.
5. **Interpretation.** Compression is subject to sharply diminishing returns because every crash promotes the next-longest path. The disciplined sequence: identify the binding path → price the cheapest week on it → re-run the passes → repeat until the next week costs more than it is worth. Leaders who mandate "crash procurement by two weeks" from the original network pay for weeks that no longer exist.

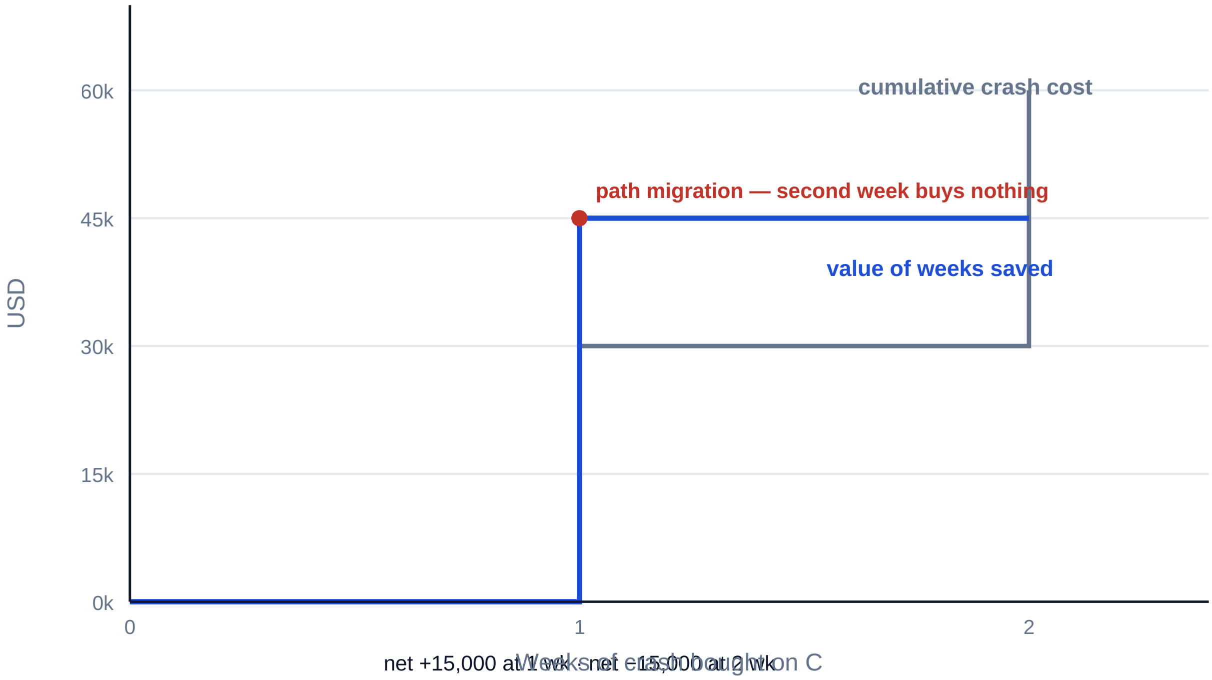


Fig 6.4.1 — The economics of crashing Auriga's procurement

WORKED EXAMPLE

Worked example 6.4.2b — the whole compression menu, and the least-cost duration.

1. **Setup.** The example above priced **one** lever. A leader planning a compression programme prices the **menu**. Auriga's available buys, each quoted by the owning discipline as a cost per week with a technical limit:

Activity	On paths	Cost slope (USD/wk)	Max weeks
C Procure hardware	A–B–C–E–F	30,000	2
B Detailed design	all three	40,000	2
D Civil and cabling	A–B–D–E–F, A–B–D–G	35,000	1
E Installation	both long paths	55,000	1
F Testing and commissioning	both long paths	65,000	1

A week is worth **USD 45,000**. What is the least-cost project duration? 2. **Formula.** For each target duration, find the **cheapest set of crashes that shortens every path** to that duration — not the cheapest activity, which is the standard error. Then net saving = weeks saved × 45,000 − cumulative crash cost, and the optimum is the duration maximising it. Equivalently: buy the next week while its **marginal** cost is below 45,000, and stop at the first week where it is not. 3. **Substitution.** 25 → 24 needs only A–B–C–E–F shortened: crash C ×1, **30,000**. 24 → 23 needs both long paths shortened (they are co-critical at 24), so the buy must be on a shared activity: B at **40,000** beats C ×1 + D ×1 at `30,000 + 35,000 = 65,000`. 23 → 22: B's second week, **40,000**. 22 → 21: B is exhausted, so E at **55,000** (again beating C + D at 65,000). 21 → 20: F at **65,000**, tying with C + D. 20 → 19: C ×1 + D ×1, **65,000**, exhausting the menu. 4. **Result.**

Duration (wk)	Cheapest plan	Marginal cost	Cumulative crash cost	Weeks saved × 45,000	Net saving
25	—	—	0	0	0
24	C ×1	30,000	30,000	45,000	+15,000
23	+ B ×1	40,000	70,000	90,000	+20,000
22	**+ B ×2**	**40,000**	**110,000**	**135,000**	**+25,000**
21	+ E ×1	55,000	165,000	180,000	+15,000
20	+ F ×1	65,000	230,000	225,000	(5,000)
19	+ C ×2, D ×1	65,000	295,000	270,000	(25,000)

The least-cost duration is **22 weeks**, three weeks compressed, for **USD 110,000** of crash spend and a net **USD 25,000** better than the 25-week plan. The shortest technically feasible duration is **19 weeks**, which destroys **USD 25,000**. 5. **Interpretation. The optimum is interior, and it is set by the marginal week.** The cost slopes rise — 30,000, 40,000, 40,000, 55,000, 65,000, 65,000 — because each crash promotes the next-longest path and forces the buy onto a shared, dearer activity. Compression is therefore an optimisation with an interior solution, and the objective is not "as fast as possible" but "as fast as pays". The same shape appears in Domain 9's cost-of-quality minimum for the same underlying reason: the last increments of any good are the expensive ones.

Averages mislead here and marginal figures do not. At 22 weeks the average crash cost is `110,000/3 =` **USD 36,666.67** a week, comfortably below the 45,000 a week is worth — and a manager reasoning from that average will buy the fourth week too, at a marginal 55,000, and lose **USD 10,000**. The rule is always marginal, never average, and the two diverge precisely because the slope rises.

The range of optimality is what makes the plan defensible. Since only the marginal cost matters, the optimal duration as a function of the value of a week `v` is a step function:

Value of a week `v`	Least-cost duration
below 30,000	25 weeks — compress nothing
30,000–40,000	24 weeks
40,000–55,000	**22 weeks**
55,000–65,000	21 weeks
above 65,000	19 weeks

Auriga's 45,000 sits **12.50 %** above the lower bound of its band and **18.18 %** below the upper, so the three-week plan survives a substantial error in the cost of delay — which is the sentence to put in the steering paper. Note also what the table says about other projects: the identical network with a week worth 25,000 should be compressed not at all, and with a week worth 80,000 should be compressed to its technical limit. **Compression policy is not a property of a schedule; it is a property of a schedule and a cost of delay together.** Where the project also carries a time-related indirect cost — site establishment, supervision, financing — that cost is added to the value of a week: an extra USD 18,000 a week takes `v` to 63,000 and moves the optimum from 22 weeks to **21**.

The optimum is a property of the option set, not of the network, and option sets decay. KA 6.4.2 concluded "crash one week only" and was right, because only C's expediting was on offer. With the full menu the answer is three weeks. The same effect runs the other way as delivery proceeds: 6.A.4 recovers two weeks late in the project, when B is complete and only C and E remain buyable, for **USD 85,000** — where the same two weeks bought from this menu at planning time cost **USD 70,000**. The **USD 15,000** difference is the price of the options that expired, and it is the quantitative case for pricing the compression menu at baseline rather than in the crisis. A compression menu is a perishable asset and should be reviewed on the same cycle as the risk register.

The cautions. Cost slopes are treated as linear within each activity's limit, which is a simplification: the second crashed week of an activity is often dearer than the first, and where a discipline can say so the table should carry two slopes rather than one. The limits are technical claims and belong to the disciplines that made them — a "maximum two weeks" that is really a preference will hide a cheap week. Every crash is a risk decision as well as a cost one: more people on the same work raises coordination load (Domain 12) and error rates (Domain 9), neither of which appears in a cost slope. And the arithmetic assumes the crashes are independent; two crashes competing for the same scarce specialist are not, which is a resource-feasibility question (KA 6.3.1c) to be re-run after the plan is chosen, not before.

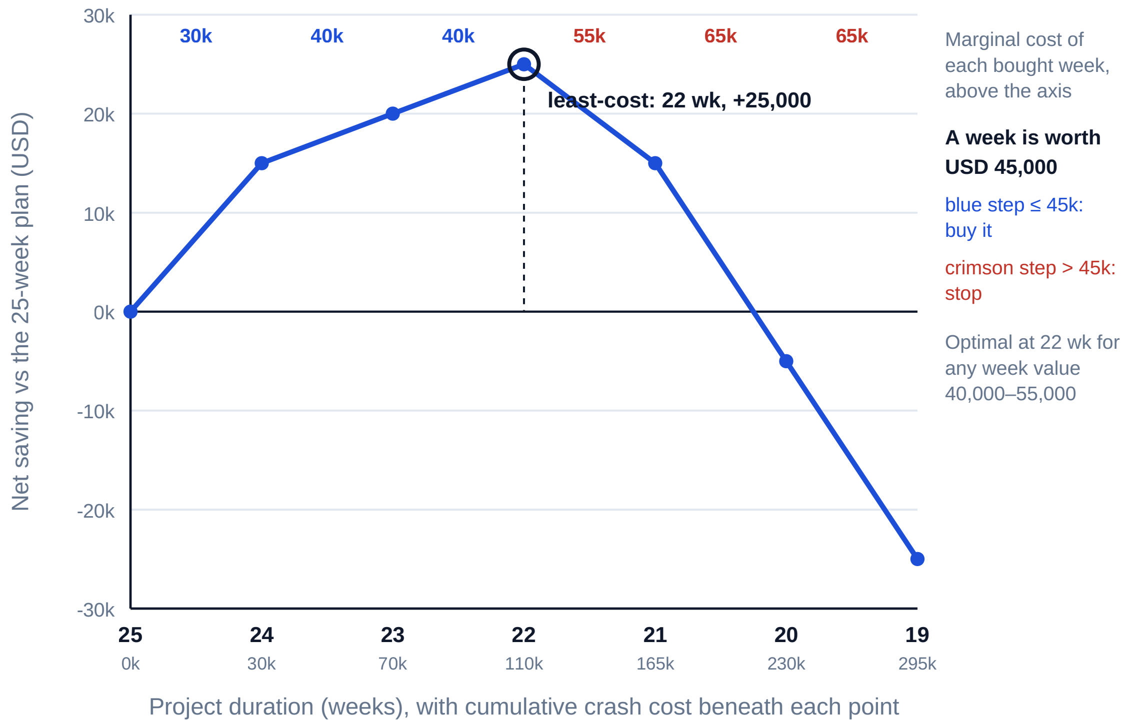


Fig 6.4.2 — The compression menu and the least-cost duration

WORKED EXAMPLE

Worked example 6.4.2c — fast-tracking priced as the probabilistic decision it is.

1. **Setup.** The alternative to buying weeks with money is buying them with overlap. Auriga can begin testing and commissioning **F** when installation **E** is 60 % complete — commissioning the completed sections while the last are installed — an overlap of **2 weeks** (40 % of E's five), taking the project from 25 weeks to **23**. The risk is specific: if the last-installed sections change the commissioning basis, the affected sections must be re-commissioned, costing **USD 140,000** and **1.5 weeks**. The commissioning lead assesses the probability at **25 %**. A week is worth **USD 45,000**. The crash alternative from the menu above buys the same two weeks for a certain **USD 70,000**.
2. **Formula.** Value each outcome against the do-nothing baseline: good outcome = weeks saved × cost of delay; bad outcome = weeks saved after rework × cost of delay − rework cost. Then `EV = good − p × (good − bad)`, and the breakeven probability is `p* = good ÷ (good − bad)`. Against a certain alternative worth `K`, the indifference probability is `p** = (good − K) ÷ (good − bad)`.
3. **Substitution.** Good: `2 × 45,000 = 90,000`. Bad: duration `23 + 1.5 = 24.5` weeks, so `0.5 × 45,000 − 140,000 = −117,500`. Swing `90,000 − (−117,500) = 207,500`. `p* = 90,000/207,500`; `p** = (90,000 − 20,000)/207,500` where the crash's net is `90,000 − 70,000 = 20,000`. `EV` at `p` = 0.25: `90,000 − 0.25 × 207,500`.
4. **Result.** The fast-track is worth **+USD 90,000** if the overlap holds and **−USD 117,500** if it does not — a swing of **USD 207,500**. It beats doing nothing while `p` < **43.37 %** and beats crashing while `p` < **33.73 %**. At the assessed 25 % its expected value is **+USD 38,125**, which is **USD 18,125** better than the crash plan's certain **+USD 20,000**.
5. **Interpretation. Expected value ranks the two options; variance is why a leader may still choose the other one.** The fast-track wins on expectation by 18,125 and carries a **USD 137,500** worse outcome in the quarter of futures where the overlap fails — the gap between a certain +20,000 and a possible −117,500. Nothing in the arithmetic settles which to take: that is a risk-appetite decision belonging to whoever carries the consequence, and Domain 8, KA 8.4.1 sets out when an expected-value comparison is the wrong test altogether. What the arithmetic does settle is that the decision must be made rather than drifted into, and by someone with the authority to accept the downside.

Two thresholds, not one, and their ratio is the useful part. `p*` = 43.37 % answers "is overlapping better than accepting the date?"; `p**` = 33.73 % answers "is it better than paying cash?". Their ratio is `70,000/90,000 =` **0.7778** — the crash option removes 22.22 % of the probability headroom the fast-track had against doing nothing. Whenever a certain alternative exists, the relevant threshold is the lower one, and a leader who computes only `p*` will overlap in cases where writing a cheque was better.

Both remedies can be mixed, and the mixture is usually right. One week of overlap plus one crashed week costs 30,000 with a smaller exposure than two weeks of overlap; the reason to prefer a hedge is that `p` is a **function of the overlap taken**, rising as the overlap deepens, and almost no team can state that function. Where `p(w)` is unknown — the usual case — the defensible practice is to take the smallest overlap that achieves the needed weeks, buy the remainder, and state both thresholds for the overlap actually taken.

The cautions. The outcome is modelled as binary when reality is a distribution of partial rework, so the two thresholds are guides rather than boundaries; the honest reading of `p*` = 43.37 % is "this has room to be wrong", not "43.37 % is the answer". The 25 % is an expert judgement, and it should be recorded in Domain 8's register with its owner and its basis rather than embedded in a schedule. Overlap also consumes coordination and supervision that the cost slope of a crash does not — two disciplines working the same plant at once — so the fast-track is cheaper on paper than in the field. And note that even the bad outcome finishes at **24.5 weeks**, still inside the original date: fast-tracking that fails is usually not a schedule disaster but a cost one, which is exactly why the decision belongs to whoever owns the cost.

Recovery. When the network slips (self-check 6.2.3: D at 10 weeks → 27), recovery options rank by cost-of-time: re-sequencing and logic re-choice (cheapest — KA 6.1's chosen logic); float harvesting from non-critical paths (G's 8 weeks fund nothing on the new critical path, but its resources might); crashing the new critical path (now through D); fast-tracking E against D's tail with an explicit rework risk (Domain 8's risk register prices it); and scope or acceptance re-negotiation (Domain 5) as the honest last resort. A **recovery plan** states the target date, the moves, their costs and risks, and the decision authority spending them — the template is toolkit 6.T.2.

6.4.3 Forecasting and scenario analysis

Three-point estimates. Uncertain durations are ranges, not points. The PERT expected duration and spread for an activity estimated optimistic `o`, most-likely `m`, pessimistic `p`:

```
tₑ = (o + 4m + p) / 6          σ = (p − o) / 6
```

WORKED EXAMPLE

Worked example 6.4.3 — Auriga's installation risk.

1. **Setup.** Installation E is estimated `o` = 4, `m` = 5, `p` = 12 weeks (the long tail: a legacy-system integration problem discovered late).
2. **Formula.** $t_e = (o + 4m + p)/6$; $\sigma = (p - o)/6$.
3. **Substitution.** $t_e = (4 + 20 + 12)/6 = 36/6$; $\sigma = (12 - 4)/6 = 8/6$.
4. **Result.** t_e = **6.0 weeks** (not 5); σ = **1.33 weeks**.
5. **Interpretation.** The deterministic network used 5 weeks; the risk-weighted expectation is 6 — the skewed tail alone adds a week to the honest forecast, pushing expected completion toward 26 weeks before any risk event "happens". Deterministic dates are the mode, not the mean; boards deserve the mean and the spread. Full quantitative schedule risk analysis — correlating activities, simulating paths, criticality indices — is Domain 8 (KA 8.2), built on exactly this input.

The bias has a closed form, and it is worth memorising. Subtracting `m` from t_e gives

$$t_e - m = (o - 2m + p) / 6$$

— the deterministic bias is one sixth of the estimate's **skew**, $(p - m) - (m - o)$. On E: $(4 - 10 + 12)/6 =$ **1.0000 week**. On the o = 6, m = 8, p = 16 estimate of Exercise 6.4: $(6 - 16 + 16)/6 =$ **1.0000 week** again, despite quite different numbers. Three consequences. A **symmetric** estimate has zero bias, so the deterministic plan is unbiased and PERT adds nothing — which is why applying it indiscriminately wastes effort. The bias depends only on how lop-sided the range is, not on how wide it is: a wide symmetric estimate (5, 9, 13) has a large σ of 1.3333 and no bias at all, so σ and bias answer different questions and neither substitutes for the other. And the bias is **positive whenever the tail is on the pessimistic side**, which is nearly always in delivery, because durations are bounded below by physics and unbounded above by circumstance. Right-skew is the normal condition, so the deterministic plan is normally optimistic — not because planners are optimists but because the mode of a right-skewed distribution is below its mean.

What this does and does not license. It does not license adding a week to every activity: doing that on all seven of Auriga's activities would inflate the plan by far more than the network's expected completion moves, because only the activities on the binding path contribute, and because path-level aggregation is not the sum of activity-level expectations. That aggregation — variances adding, merge bias at convergence points, criticality indices — is Domain 8, KA 8.A.2, worked on this same network, and it finds that E's start is only **13.16 %** likely in week 16. Nor does it license treating `σ = (p − o)/6` as a derived quantity: it is a convention that assumes the range spans roughly six standard deviations, it understates spread on strongly skewed estimates, and probabilities computed from it should be read to the nearest percentage point at best.

The reviewer's question is where the three points came from. `o`, `m` and `p` elicited from the same person in the same breath are usually one number with decorations: the range is anchored on the most likely value and adjusted by a habitual percentage, which produces symmetric estimates and therefore zero bias by construction. The tail that matters — the legacy-integration problem in E's `p` = 12 — comes from asking a different question ("what would have to go wrong, and how long would that take?"), which is a risk-identification act (Domain 8, KA 8.1) rather than an estimating one. An estimate whose `p` cannot be traced to a named scenario is not a three-point estimate.

Scenario analysis. Between the single date and the full simulation sits the leader's workhorse: three coherent scenarios (base / threat / opportunity), each a complete re-run of the passes under stated assumptions, each with owners for the assumptions that differ. Scenarios are cheap, auditable, and — because each is a real network — they expose path migration that a percentage-confidence number hides.

The earned-schedule bridge. Once cost joins schedule (Domain 7), progress data yields a time-based forecast: **earned schedule** `ES` asks when the value now earned was planned to have been earned, and `SPI(t) = ES / AT` (actual time) measures schedule efficiency in time units. If Auriga's week-22 status shows work that the baseline expected by week 20, `SPI(t) = 20/22 = 0.91` — the programme is running at 91 % of planned tempo, a signal the currency-based `SPI` famously loses late in a project. On Auriga's own week-13 position the value earned was planned to have been earned by week 12, so `ES` = **12.0000** and `SPI(t) = 12/13 =` **0.9231** — identical to the currency-based `SPI` of `1,920,000/2,080,000`, because the planned-value curve is locally linear there. Domain 7 (KA 7.3 and 7.A.1) builds the full machinery, including the point at which the two measures separate; it is flagged here because schedule forecasting belongs to whoever owns the network, not only to the cost engineer.

AI in this KA

Machine forecasting earns trust in exactly one way: calibration. A model that ingests progress data and produces completion forecasts must show its record — forecast vs actual across past periods — before its numbers enter a board pack (Domain 14, KA 14.3's verification workflow). The leader's questions are the auditor's: what data trained it, what does it do when the data thins, who re-ran its arithmetic, and what would make it wrong? Fluent confidence without a calibration record is the machine version of the site manager who is "sure it'll be fine".

KEY TERMS — KA 6.4

Term	Meaning
Crashing / fast-tracking	Buying duration with money / with overlap risk.
Path migration	The critical path moving as durations change; why passes are re-run.
Recovery plan	Target date, ranked moves, costs, risks, decision authority.
Three-point / PERT estimate	`tₑ = (o+4m+p)/6`, `σ = (p−o)/6`.
Scenario analysis	Complete network re-runs under coherent stated assumptions.
Integration milestone	The CPM node where an agile stream's delivery enters the network.
Cost slope	An activity's crash cost per week, with a technical limit on weeks available.
Least-cost duration	The duration maximising net saving; found where the marginal week's cost first exceeds the value of a week.
Required throughput	Backlog ÷ (successor's `LS` − stream start): what the network asks a cadence team to sustain.
Breakeven rework probability	`p* = good outcome ÷ (good − bad)`; the probability at which an overlap stops paying.
Deterministic bias	`tₑ − m = (o − 2m + p)/6` — one sixth of the estimate's skew.

SAMPLE MCQS — KA 6.4

MCQ 6.4-A `[6.4.2 · Application]` Saving a week is worth USD 45,000; crashing the critical activity costs USD 30,000 per week for up to two weeks; after one week of crashing, a parallel path becomes co-critical. The value-maximising decision is: - A. crash two weeks (net +USD 30,000) - B. crash one week (net +USD 15,000) - C. crash nothing (avoid all cost) - D. crash two weeks and fast-track the parallel path at no cost

Rationale: Week one converts to a project week: +15,000. Week two is absorbed by the co-critical path: −30,000. A prices weeks that don't materialise; C leaves +15,000 unclaimed; D invents a free fast-track — overlap always carries rework risk.

MCQ 6.4-B `[6.4.3 · Application]` An activity is estimated o = 4, m = 5, p = 12 weeks. Its PERT expected duration is: - A. 5.0 weeks - B. 7.0 weeks - C. 6.0 weeks - D. 6.5 weeks

Rationale: `(4 + 4×5 + 12)/6 = 36/6 = 6.0`. A is the most-likely value (the mode); B is the unweighted mean of the three points; D miscounts the weighting.

MCQ 6.4-C `[6.4.1 · Analysis]` A hybrid programme needs a software module from an agile team for an integration milestone. The schedule-sound way to join the two worlds is: - A. impose the CPM date on the team as a sprint deadline - B. enter the team's velocity-based forecast as a ranged duration and read the latest acceptable delivery from the backward pass - C. exclude the module from the network since agile work cannot be scheduled - D. convert the agile team to predictive planning for the integration period

Rationale: B translates in both directions — evidence-based forecast in, latest-start requirement out. A dictates without evidence; C leaves the network blind at a convergence point; D destroys the team's delivery system to decorate the network.

MCQ 6.4-F `[6.4.3 · Application]` An activity is estimated optimistic 3, most-likely 4, pessimistic 8 weeks. Its PERT expected duration and standard deviation are: - A. t_e = 4.5, σ = 0.83 - B. t_e = 4.0, σ = 0.83 - C. t_e = 5.0, σ = 1.67 - D. t_e = 4.5, σ = 5.0

Rationale: `tₑ = (3 + 16 + 8)/6 = 4.5`; `σ = (8 − 3)/6 = 0.83`. B reports the mode as the mean; C is the unweighted three-point average with a doubled spread; D confuses σ with the pessimistic-minus-optimistic range.

MCQ 6.4-D `[6.4.3 · Application]` At week 22, a programme has earned the value its baseline planned to earn by week 20. Its time-based schedule performance index `SPI(t)` is: - A. 1.10 - B. 0.91 - C. 0.80 - D. 20.0

Rationale: `SPI(t) = ES/AT = 20/22 = 0.91` — the programme delivers at 91 % of planned tempo. A inverts the ratio; C subtracts the two weeks from the wrong base; D reports earned schedule itself, not the index.

MCQ 6.4-E `[6.4.3 · Analysis]` Compared with quoting "78 % confidence of the date", giving the board three fully re-run schedule scenarios is stronger because: - A. three numbers always beat one - B. each scenario is an auditable network with named assumptions, showing how the date moves and where the path migrates - C. percentages are always statistically invalid - D. scenarios eliminate the need for risk analysis

Rationale: The scenario's power is auditability and mechanism — assumptions with owners, visible path migration. A is numerology; C overclaims (a calibrated percentage is legitimate, Domain 8); D reverses the relationship — scenarios are inputs to risk analysis, not substitutes.

MCQ 6.4-G `[6.4.2 · Evaluation]` A compression menu offers successive weeks at marginal costs of 30,000, 40,000, 40,000, 55,000, 65,000 and 65,000. A week is worth 45,000. The least-cost plan buys: - A. one week, because only the first step is clearly cheap - B. three weeks, stopping at the first marginal cost above 45,000 - C. six weeks, because the average cost of 49,167 is close to 45,000 - D. four weeks, because the average cost of the first four (41,250) is below 45,000

Rationale: Buy while the marginal step is at or below 45,000 — 30,000, 40,000, 40,000 — and stop at 55,000, giving 22 weeks and a net +25,000 (6.4.2b). C and D both reason from averages, the standard error: the fourth week costs 55,000 whatever the average of the first four is, and buying it loses 10,000. A stops at the single-lever answer of 6.4.2, which was correct only for a menu of one.

MCQ 6.4-H `[6.4.2 · Application]` An overlap saves 2 weeks worth 45,000 each; if it fails it costs 140,000 and gives back 1.5 weeks, and a certain crash plan would buy the same 2 weeks for 70,000. The probability of failure above which the overlap is worse than crashing is: - A. 43.37 % - B. 33.73 % - C. 50.00 % - D. 25.00 %

Rationale: The swing is `90,000 − (−117,500) = 207,500` and the crash's net is 20,000, so `p** = 70,000/207,500 = 33.73 %` (6.4.2c). A is the threshold against doing nothing, which is the wrong comparator once a certain alternative exists; C treats the choice as a symmetric coin toss — the answer if the two outcomes were equal in size, which they are not; D is the assessed probability, the input rather than the threshold.

MCQ 6.4-I `[6.4.1 · Application]` A cadence team must clear a 72-item package between week 5 and its successor's `LS` of week 21. Its observed throughput has ranged 3.6–4.5 items a week, averaging 4.0. The network's requirement is: - A. delivery by week 21, which the team should commit to - B. 4.5 items a week — the team's best observed rate, sustained for the full 16 weeks - C. 4.0 items a week, since the mean is the fair planning figure - D. 3.6 items a week, since planning should use the worst observation

Rationale: `72 ÷ (21 − 5) = 4.5` items a week (6.4.1). C and D quote the team's own forecast rates rather than deriving the requirement from the window — the mean fits an 18-week window and the worst rate a 20-week one, neither of which exists here. A restates the date instead of translating it, which is what leaves the requirement untested.

SELF-CHECK — KA 6.4

1. Why does compression show diminishing returns? — Each crash promotes the next-longest path; saved weeks stop converting to project weeks (path migration).
2. What makes a scenario more useful to a board than a confidence percentage? — It is a complete, auditable network with named assumptions — it shows how the date moves, not just that it might.
3. When may a machine forecast enter a board pack? — With a calibration record, verified arithmetic, stated data lineage and a named human owner.
4. Why is the average crash cost the wrong figure? — Because the decision is marginal: Auriga's average at 22 weeks is 36,666.67 against a week worth 45,000, yet the next week costs 55,000 and buying it loses 10,000.
5. What does the network give a cadence team, and in what units? — A required throughput — backlog ÷ (successor's `LS` − start) — which on Auriga's interface package is 4.5 items a week, the team's best observed rate.
6. When does a three-point estimate add nothing? — When it is symmetric: the bias is $(o - 2m + p)/6$, which is zero, so $t_e = m$ and only the spread is new information.

Advanced topics — Domain 6

6.A.1 The drum: constraint-aware flow

Where one resource is the system's constraint (a single commissioning team, one test rig), throughput scheduling subordinates everything to that drum: buffer it, never starve it, and let non-constraint efficiency go. This reframes utilisation worship — a 100 %-busy non-constraint builds inventory, not progress. In schedule terms: protect the constraint's feed path with deliberate float placement rather than spreading buffer evenly.

6.A.2 Schedule baselines and the update cycle

The control schedule lives on a cycle: status (actuals and remaining durations in), re-run passes, variance vs baseline, forecast, and — rarely, formally — re-baseline under Domain 4's change control. Two integrity rules carried from the family's controls tradition: never edit history (actualised dates are records), and never re-baseline to hide variance — a re-baseline is a governance decision with an audit trail, not a cosmetic one.

6.A.3 PDM edge cases: float under SS and FF links

Precedence-diagram (PDM) links change how float behaves, and two edge cases bite reviewers. **(1) The SS-linked short successor.** P (10 weeks) `SS+2` Q (3 weeks): Q can run weeks 2–5, finishing seven weeks before its predecessor — legal, and correct if the dependency truly binds only the starts; but if Q also needs P's output to finish, the network is missing an FF link, and Q's float is fiction. Paired `SS+FF` links express "starts together, finishes together" honestly. **(2) Duration-driven float.** Under an FF link the successor's float depends on its own duration: lengthen Q and its `LS` moves earlier — planners who "add duration for safety" on FF-linked activities silently destroy their float. The audit habit: on every SS/FF link, ask which end of each activity the logic genuinely binds, and whether the unbound end needs its own link. A network is only as honest as its least-considered link type.

6.A.4 Negative float and the honest schedule

A constraint-bound network can compute `TF < 0`: the plan, as constrained, is late already. Negative float is information — the size of the recovery problem — and suppressing it (deleting constraints, shortening durations by fiat) is the scheduling equivalent of cooking the books. The professional response mirrors KA 6.4.2: quantify it, trace the binding path, price the recovery options, and escalate the decision to whoever owns the trade (Domain 3's escalation design).

WORKED EXAMPLE

Worked example 6.A.4 — buying back two weeks of negative float.

1. **Setup.** A newly imposed outage constraint requires Auriga to finish by **week 23**; logic says 25 — so `TF` = −2 on A–B–C–E–F. Available buys: crash C at USD 30,000/week (max 2); crash E at USD 55,000/week (max 1). Find the cheapest feasible recovery to week 23.
2. **Formula.** Recover week by week on the currently binding path(s), re-running the passes after each buy (path migration, 6.4.2).
3. **Substitution.** Week one: crash C by 1 → duration 24, but B–D–E–F is now co-critical at 24. Week two must shorten both paths: the only shared activity available is E → crash E by 1 (USD 55,000) → duration 23. (C's second week alone would leave B–D–E–F at 24 — spent money, no schedule.)
4. **Result.** Feasible recovery: **crash C ×1 + crash E ×1 = USD 85,000**, finishing week 23.
5. **Interpretation.** Negative float is bought back on the binding path as it migrates — the second-cheapest week on the original critical path (C's second week, USD 30,000) is worthless, while the dearest activity (E, on both paths) is the only week-two purchase that works. This is why recovery plans list options in re-run order, not in unit-cost order (toolkit 6.T.2).

Note also what the restricted option set costs. Only C and E are buyable here because design B is complete and commissioning F cannot be accelerated at this stage; the same two weeks bought from the full planning-time menu of 6.4.2b — where B was still open at a 40,000 slope — cost **USD 70,000** against this **USD 85,000**. The **USD 15,000** gap is the price of options that expired, and it is the argument for pricing the compression menu at baseline and reviewing it on the risk register's cycle rather than assembling it in the crisis.

6.A.5 Repetitive delivery: takt, crews and the benefit integral

A rollout is not a network. Meridian installs the same clinic forty times, and a forty-times-repeated activity-on-node diagram teaches nothing: the schedule's content is a **rate**. **Takt** is the interval between successive completions — window ÷ units — and it is the quantity that converts a programme date into a crew count, because the rate a crew can sustain is a productivity figure the delivery team already owns. The arithmetic is three divisions, and the reason to do it is that the value of a rate behaves quite differently from the value of a date.

WORKED EXAMPLE

Worked example 6.A.5 — Meridian's rollout rate, and what accelerating it is worth.

1. **Setup.** Meridian's **40** clinics are installed by a crew of **6** deployment specialists, each completing **0.1** clinics a week (Domain 1, KA 1.3.3b), so **0.6** a week in total. The estate and training window requires all 40 live within **50 weeks** of rollout start. Specialists cost **USD 4,200** a week; per-clinic work content is **10** specialist-weeks. Adding specialists incurs Domain 1's ramp: **4 weeks** at **25 %** productivity, each newcomer absorbing **50 %** of an existing specialist's time. Benefit accrues at `6 hours × USD 85 =` USD 510 a week per adopting clinic and adoption runs at **70 %**, so **USD 357** a week per installed clinic. What crew does the window need, and does it pay?
2. **Formula.** Required takt = window ÷ units; required rate = units ÷ window; crew = required rate ÷ per-head rate. With reinforcement: ramp rate = base − newcomers × supervision × per-head + newcomers × ramp productivity × per-head; post-ramp rate = base + newcomers × per-head; duration = ramp + (units − ramp × ramp rate) ÷ post-ramp rate. For a linear rollout over a fixed horizon, the **unit-weeks** gained by finishing `ΔT` sooner = `(units ÷ 2) × ΔT`.
3. **Substitution.** Required takt `50/40 = 1.25` weeks; required rate `40/50 = 0.8` a week; crew `0.8/0.1 = 8`. With 2 newcomers: ramp rate `0.6 − 2 × 0.5 × 0.1 + 2 × 0.25 × 0.1 = 0.55`; post-ramp `0.6 + 0.2 = 0.8`; duration `4 + (40 − 4 × 0.55)/0.8`. Baseline `40/0.6`. Clinic-weeks gained `20 × ΔT`.
4. **Result.** The 50-week window needs a takt of **1.25 weeks** and therefore **8** specialists. With the ramp, the eight-person crew finishes in **51.2500 weeks** — an actual takt of **1.2813 weeks** — against **66.6667 weeks** for the crew of six, a saving of **15.4167 weeks**. Effort is nearly invariant: `6 × 66.6667 =` **400.0** specialist-weeks against `8 × 51.25 =` **410.0**, so the reinforcement costs **10** extra specialist-weeks, or **USD 42,000**. The acceleration yields `20 × 15.4167 =` **308.3333** clinic-weeks at USD 357 — **USD 110,075** — for a net **+USD 68,075**.
5. **Interpretation. Effort is invariant to the rate; only the benefit integral moves.** Forty clinics at 10 specialist-weeks each is **400 specialist-weeks** whether delivered by six people over 67 weeks or eight over 50. That is the structural reason repetitive-work acceleration is so often nearly free and so rarely attempted: the labour bill is set by the work content, not by the rate, and the only incremental cost is the transient of changing the rate — here 10 specialist-weeks of ramp. On a network, buying weeks costs a cost slope every time; on a rollout it costs a one-off. The two are different economies and must not be priced with the same instinct.

Accelerating a linear rollout is worth exactly half the fully-adopted cost of delay per week saved. The clinic-weeks gained are `(N/2) × ΔT`, so the value per week saved is `(40/2) × 357 =` **USD 7,140** — precisely half of Meridian's **USD 14,280**, and the identity holds for any linear ramp because the average number of live units during the ramp is half the final number. This matters because the temptation is to value rollout acceleration at the steady-state cost of delay, which **doubles** it. The reconciling check is worth noting: `40 × 357 =` **USD 14,280** exactly, Domain 1's figure, because 70 % of 40 clinics is the 28 adopting clinics it was derived from.

Where benefit only switches on at completion the full figure applies — and here the decision is robust to which regime holds. If the shared records system delivers nothing until coverage is complete, the 15.4167 weeks are worth `15.4167 × 14,280 =` **USD 220,150**, exactly twice the ramp-integral figure, and the net becomes **+USD 178,150**. Both readings say the same thing: buy the two specialists. That is the professionally important observation — a factor-of-two ambiguity in the valuation basis need not be resolved when the decision is unchanged either way, and the leader who notices this stops arguing about the benefits model and gets on with the rollout. Where the decision is sensitive to the factor of two, it is a benefits-realisation question (Domain 16) and must be settled by whoever owns the benefit, not by the planner.

The threshold is a volume, and it explains why Domain 1 reached the opposite verdict. Domain 1, KA 1.3.3b added three specialists to the last eight clinics and destroyed **USD 23,333.33**. The reason is now visible: the extra specialist-weeks a reinforcement consumes are **constant** — 10 for two newcomers, 15 for three, independent of how much work remains — while the weeks saved grow linearly with the volume remaining, `0.4166667N − 1.25` for two newcomers. On the full cost-of-delay basis two newcomers break even at `N =` **10.0588** clinics and on the ramp-integral basis at `N =` **17.1176**. Domain 1's eight clinics sit below both; forty sits far above. **Reinforcement pays above a computable remaining volume, so the same arithmetic gives opposite answers at the two ends of one rollout** — which is why the decision is re-taken as the programme runs down rather than settled once.

And the rate is capped by the drum, not by the crew. Suppose one data-migration specialist can handle only **0.7** clinics a week. The achievable rate is then `min(0.8, 0.7) =` **0.7**, the rollout takes `40/0.7 =` **57.1429 weeks**, and the eighth specialist buys nothing: the **5.8929 weeks** the drum costs, worth `20 × 5.8929 × 357 =` **USD 42,075**, can be recovered only by relieving the drum (6.A.1). This is the most expensive error available in repetitive scheduling — sizing the crew from the takt while a subordinate step sets the actual rate — and the only defence is to compute the sustainable rate of **every** step, not of the one whose people are most numerous.

Two further cautions. The 0.1-clinics-per-specialist-week rate is locally calibrated and clinics are not identical, so a takt computed from an average unit will be missed by the difficult units; the honest plan carries a unit-difficulty distribution and a recovery rule rather than a single takt. And a takt is a commitment to a cadence, which requires estate readiness, trained staff and approvals to arrive at the same cadence — which is where rollouts actually fail, the crew rarely being the binding constraint by the third month.

6.A.6 Buffer management: aggregating safety and reading its consumption

Every duration in the Auriga network contains protection. An eight-week procurement estimate is not anyone's honest expectation of eight weeks; it is an expectation with a margin folded in, sized so that the estimator is comfortable committing to it. That is rational behaviour with an expensive property: **safety held inside each activity cannot be shared, while safety held at the end of a chain can be.** A week of protection inside C is available only to C, and if C finishes early the protection is usually consumed anyway, since there is no reward for finishing early when the successor is not ready. Aggregating the protection into an explicit **buffer** converts private, unusable margin into a shared, managed and visible reserve. Buffer sizing and buffer-consumption control belong to the **critical-chain** tradition in schedule management, whose vocabulary KA 6.3.1c already uses; the arithmetic below is worked from first principles on Auriga, and the conventions are named as conventions rather than as results.

WORKED EXAMPLE

Worked example 6.A.6 — sizing and then managing Auriga's project buffer.

1. **Setup.** Auriga's disciplines are asked for two figures per critical-path activity: the duration in the network, and the duration they would give with an even chance of meeting it.

Activity	Network duration	50 % duration	Embedded safety
A Mobilise	2	1.5	0.5
B Detailed design	6	5	1.0
C Procure control hardware	8	6.5	1.5
E Installation	5	4	1.0
F Testing and commissioning	4	3	1.0
Chain	**25**	**20**	**5.0**

Size a project buffer, then manage against it. A week is worth **USD 45,000**. 2. **Formula.** Chain `= Σ` 50 % durations. Two sizing conventions: **half the aggregated safety**, $\Sigma s_i / 2$, or **root-sum-square**, $\sqrt{\Sigma s_i^2}$ — the same aggregation Domain 8 uses when variances add (KA 8.A.2). Committed duration = chain + buffer. In control: consumption ratio = (buffer consumed ÷ buffer) ÷ (chain complete ÷ chain); projected final consumption = buffer × consumption ratio. 3. **Substitution.** Chain `1.5 + 5 + 6.5 + 4 + 3 = 20`. Safety `0.5 + 1 + 1.5 + 1 + 1 = 5.0`. Half-safety buffer `5.0/2`. Root-sum-square `√(0.25 + 1 + 2.25 + 1 + 1) = √5.5`. At a status date with **45 %** of the chain complete and **30 %** of the buffer consumed: ratio `0.30/0.45`. 4. **Result.** The half-safety method commits to `20 + 2.5 =` **22.5000 weeks**, **2.5 weeks** inside the 25-week network and worth **USD 112,500**. The root-sum-square buffer is **2.3452 weeks**, a commitment of **22.3452 weeks**, **2.6548 weeks** inside the network and worth **USD 119,465.65**. In control, the consumption ratio is **0.6667**: projected final buffer use `2.5 × 0.6667 =` **1.6667 weeks**, a forecast chain completion at **21.6667 weeks** with **0.8333 weeks** of buffer unused — **USD 37,500** still in hand, and **3.3333 weeks** inside the original network date. 5. **Interpretation. Aggregation is where the weeks come from, and it is arithmetic rather than pressure.** Nothing was crashed, no estimate was challenged, nobody was told to work faster: the plan is 2.5 weeks shorter because five weeks of private protection became 2.5 weeks of shared protection, and shared protection covers more contingencies per week held. The two conventions differ by only **0.1548 weeks** here, and the root-sum-square form is the more defensible because it is the variance-addition result Domain 8 derives — which means it holds under **independence**, so where the activities share a cause (one subcontractor, one site condition, one approval body) the correct buffer is larger and approaches the simple sum of 5.0. A root-sum-square buffer across correlated activities is the schedule version of the aggregation error Domain 8, KA 8.2.4 prices for contingency.

Buffer consumption forecasts earlier than variance against baseline. A ratio below 1 means protection is being consumed more slowly than the chain is being completed; above 1, faster. Auriga's 0.6667 projects a finish with a third of the buffer unspent — information available at 45 % completion, from two numbers, without re-running a pass. It is the schedule counterpart of Domain 7's `EAC` family: an index observed to date, projected over what remains. The control regime should state thresholds in advance — below **0.5**, no action; **0.5 to 1.0**, prepare a recovery plan; above **1.0**, execute it — set and published before the first status date rather than derived from the first uncomfortable reading.

The linear projection is only as good as the resemblance between what is done and what remains. A chain whose risky activities sit at the end shows a comfortable ratio for months and consumes its whole buffer in the final quarter. That is the standard misuse, and it is detectable, because the same three-point estimates that sized the buffer say which activities carry the spread: where the remaining work is riskier than the completed work, weight the projection by remaining variance rather than remaining duration — and say that you have.

The method fails completely if the 50 % durations are not 50 % durations. Halving network durations by fiat produces the same table with none of the meaning: the buffer then protects estimates nobody believes, it is consumed in the first third of the chain, and the team correctly concludes that the plan was a compression exercise wearing a technique's clothing. Eliciting an even-chance duration requires that missing it half the time carries no penalty, which is a leadership commitment (Domain 12) rather than a scheduling one. Where that commitment cannot honestly be made, the aggregation should not be attempted and the schedule should run with visible float and a float policy instead.

And the buffer is a governed reserve, not a hiding place. It appears in the schedule as an explicit activity with an owner, its consumption is reported every cycle in the same pack as float, and it is never silently topped up — that is 6.A.2's re-baselining integrity rule applied to a buffer. A buffer nobody reports is indistinguishable from padding and will be removed by the first cost review that finds it.

Industry variations — Domain 6

The passes are universal; what counts as a schedule, and what discipline it needs, is sectoral:

- **Construction and EPC.** Deep L3/L4 hierarchies, thousand-activity networks, contractual float-ownership clauses (Domain 10) and delay-analysis protocols make float a legal quantity — the honesty settings of the Executive perspective are contract compliance here, not just culture.
- **Technology and product.** Hybrid is the default (KA 6.4.1): cadence-based streams joined to a thin predictive network at integration milestones; the commonest defect is a network so sparse that convergence points are invisible until they slip.
- **Shutdowns and turnarounds.** The network runs in hours; near-critical analysis widens to every path within minutes of critical, and the drum logic of 6.A.1 governs a single shared crane or permit desk. Compression economics (6.4.2) turn over in shifts, not weeks.
- **Pharmaceutical and regulated development.** Milestones are regulatory events with submission-window physics — genuine date constraints (6.3.2), modelled as logic, with the pin-and-squeeze temptation at its most dangerous because windows are unforgiving.
- **Rollouts and repetitive delivery** — retail estates, network upgrades, clinic and branch programmes. The schedule is a **rate**, not a network: takt, crew count and the sustainable rate of every step, with the drum found before the crew is sized, and acceleration priced as a benefit integral rather than a cost slope (6.A.5). The binding constraint is almost never the installation crew; it is the upstream supply of site readiness, trained staff and approvals arriving at the same cadence.
- **Public programmes.** Announced dates precede networks (Case study B); the leader's protection — milestones only after the backward pass — is hardest exactly where it matters most, and negative-float honesty (6.A.4) is the difference between an early re-plan and a public failure.

Case study — Domain 6: recovering Auriga (utilities / technology)

Situation. End of week 13. Civil and cabling works D — planned at 7 weeks from week 8 — hits contaminated-ground remediation; the discipline lead's honest re-estimate takes D to 10 weeks (finish week 18). The client's outage window for final commissioning opens at week 25 and closing the window costs USD 45,000 per week of delay. The team re-runs the passes.

Analysis. With D = 10: E starts week 18 (D now later than C's week 16), F finishes week 27 — **two weeks into the penalty window**, and the critical path has migrated to **A–B–D–E–F**. Procurement C, critical all project, now carries TF = 2. G still holds float.

The recovery decision. Options priced by the team:

Option	Weeks recovered	Cost	Net vs doing nothing
(1) Crash the old critical path (C)	0	30,000/wk	worthless — C is no longer binding
(2) Second civil crew on the remediation	1	35,000	+10,000
(3) Fast-track E against D's tail	1	12,000 expected (20 % × 60,000)	+33,000
(4) Do nothing	0	90,000 penalty exposure	baseline
(2) + (3) taken together	**2**	**47,000 expected**	**+43,000**

The leader takes (2) + (3): `2 × 45,000 = 90,000` of penalty avoided for **USD 47,000** of expected cost, a net **USD 43,000**, and week 25 with the window intact.

What the arithmetic settles, and what it does not. Three figures decide the case and are worth extracting, because each is a general instrument rather than a feature of this story.

The **ranking is by net, not by cost**. Option (3) is the cheapest and option (2) the dearer, yet both are bought because each recovers a week worth 45,000 and both clear that bar; a leader who stops at the cheapest option buys one week and pays a 45,000 penalty to save 35,000. The stopping rule is 6.4.2b's: buy while the marginal week costs less than a week is worth.

The **fast-track's breakeven probability** is `45,000/60,000 =` **75.00 %** — the rework would have to be three times likelier than the assessed 20 % before the overlap stopped paying. That is the sentence that makes the recommendation defensible to an assurance reviewer, and it is a stronger statement than the expected cost, because the assessed probability is the softest number in the analysis. Note also what the full treatment of 6.4.2c would add: the assessed 20 % gives an expected cost of 12,000, but the downside is a certain 60,000 against a certain 45,000 penalty avoided, so the bad outcome leaves the project 15,000 worse off than doing nothing — a small enough exposure that the expected-value test is the right test here, which is not always so.

The **new float table is the deliverable, not the new date.** With D at 10 weeks, procurement C — critical for the whole project until this week — carries TF = **2**, and the critical path is A–B–D–E–F. Every subsequent expediting conversation, every progress question and every recovery option must be aimed at the new binding path; the float table is what redirects the team, and issuing it is a separate act from announcing the recovered date. Teams that announce the date and not the table spend the following month managing the path that used to matter.

What the domain teaches here. Re-run the passes before spending a cent (the instinctive "crash procurement" would have bought nothing); price recovery in expected cost including risk, and state the breakeven for the probability you are least sure of; harvest overlap where the physics genuinely allows it; and report the new float positions to the client honestly — the schedule survived because the remediation estimate was honest a full five weeks before the window (Domain 11's reporting culture, applied to time).

Case study B — Domain 6: the milestone that was typed, not computed (public programme)

Situation. A government services programme publishes a go-live milestone eighteen months out — programme **week 78** — set in a ministerial announcement before the L3 schedule existed. When the network is finally built, the test environment cannot exist before **week 69** and user-acceptance testing needs **12 weeks**, so honest logic finishes at week 81: the backward pass returns `TF` = **−3** from day one. To show the pinned date, the plan carries a **9-week** UAT window — **75.00 %** of the tested requirement — and calls it a plan.

What happened, in numbers. For six monthly reporting cycles — **24 weeks** — the programme's L1 view showed the milestone green. The date was pinned, so each upstream slip was absorbed by silently shortening the testing window: by cycle seven the environment date had moved to **week 73** and the window to **5 weeks**, or **41.67 %** of the 12 weeks the test plan required. Two consequences were accumulating while the reporting said nothing.

The recovery problem grew by a factor of 2.33. At cycle one the gap between honest logic and the pinned date was 3 weeks; at cycle seven it was `73 + 12 − 78 =` **7 weeks**. Negative float does not sit still while it is being concealed — the upstream slips that the squeeze absorbed were real, so the pin converted a three-week problem into a seven-week one and spent 24 weeks of available runway doing it.

And the concealment moved the risk from schedule to product. If acceptance-test detection is taken as roughly proportional to the share of the test plan actually executed — a stated simplification, and exactly the kind of asserted detection rate Domain 9, KA 9.2.2 warns must be measured rather than assumed — then a regime designed to detect **75 %** of defects reaching it detects `0.75 × 5/12 =` **31.25 %** when only five twelfths of the plan is run. The escape fraction rises from **0.25** to **0.6875**: **2.75 times** as many defects reach live service. The squeeze did not remove the programme's problem; it converted a visible schedule variance into an invisible quality one, priced not by the schedule but by Domain 9's external-failure economics and paid by users.

In cycle seven an assurance review (Domain 3, KA 3.3) ran the passes without the pin, surfaced the negative float, and forced the choice the pin had deferred: descope the first release (Domain 5), or move the date. The minister moved the date. What that cost politically is not quantified here — it cannot honestly be — but the two figures above can be, and they are the ones that would have made the case at cycle one.

What the domain teaches here. A typed milestone is a promise without a plan. Negative float is not embarrassment to be formatted away; it is the earliest, cheapest warning the programme will ever get (6.A.4), and its value decays because the problem it names grows. The tell was available every cycle and needed no assurance review to find: **a test window shorter than the test plan is a negative-float report written in a different notation**, and any reviewer who compares the two numbers finds it in one line. And the leader's protection is procedural, not heroic: milestones enter public commitments only after the backward pass says they exist.

Executive perspective — Domain 6

What a project leader cannot delegate in this domain:

- **The logic sign-off.** Discipline leads own their links; the leader owns the assembled claim that this network is how the project actually works.

- **Float policy.** Who may spend float, in what increments, reported how — decided before the first progress update, not during the first crisis.
- **The honesty settings.** No pins doing logic's work; negative float reported the cycle it appears; re-baselining only through governance. Schedules fail morally before they fail mathematically.
- **The feasibility question.** A schedule is approved as a logic network and executed with people. The leader asks one question before signing: has this been levelled, against which caps, and what was the peak? Auriga's plan was infeasible on engineering capacity while only 73.33 % loaded on average (6.3.1c), and no amount of logical float protected it. A baseline signed without a resource histogram is a baseline signed without knowing whether it can be run.
- **The compression chequebook.** Crash and fast-track decisions are investments under uncertainty — the leader signs the expected-value arithmetic (6.4.2) and the risk acceptance (Domain 8), never a bare instruction to "go faster". The chequebook has a price list: the compression menu with its cost slopes, priced at baseline while every option is still open, because options expire and the same two weeks cost USD 70,000 at planning and USD 85,000 in the crisis (6.A.4).
- **The translation duty.** Boards hear dates; teams live in passes, floats and scenarios. The leader is the honest converter between the two languages — both directions.

Calculation exercises — Domain 6

Exercise 6.1 Using the Auriga network, compute ES/EF for every activity and the project duration, showing the convergence at E. Solution. A 0–2 · B 2–8 · C 8–16 · D 8–15 · E max(16,15)=16–21 · F 21–25 · G 15–17. Duration **25 weeks**; E's start is set by C (16), not D (15) — the convergence rule takes the later predecessor. Common error: taking the earlier predecessor at a merge (giving E 15–20 and a false 24-week project).

Exercise 6.2 Complete the backward pass and state TF and FF for D and G. Solution. From LF = 25: F 21/25 · E 16/21 · C 8/16 · G 23/25 · D LF = min(E LS 16, G LS 23) = 16, LS = 9. **D: TF = 1, FF = min(16,15) − 15 = 0. G: TF = 8, FF = 25 − 17 = 8.** Common error: computing D's FF against only E (giving 1) — free float tests every successor.

Exercise 6.3 The client offers USD 45,000 per week saved; crashing C costs USD 30,000/week (max 2 weeks); crashing E costs USD 55,000/week (max 1 week). Find the value-maximising plan. Solution. Crash C by 1: 25→24, +15,000. Second week of C: co-critical B–D–E–F holds 24 — −30,000, reject. Crash E by 1 (E is on both paths): 24→23, `45,000 − 55,000 = −10,000`, reject. **Plan: crash C by one week; net +USD 15,000.** Common error: crashing E first because it is "on more paths" — path count doesn't change its negative unit economics. Note the option set: with only C and E on offer the answer is one week, and Exercise 6.8 shows the same network reaching **22 weeks** once design B and commissioning F are also buyable — the optimum is a property of the menu, not of the network.

Exercise 6.4 Activity durations: o = 6, m = 8, p = 16 weeks. Compute t_e and σ, and state the deterministic bias. Solution. $t_e = (6 + 32 + 16)/6 =$ **9.0 weeks**; $\sigma = (16 - 6)/6 =$ **1.67 weeks**. The deterministic plan at m = 8 understates the expectation by a full week — right-skewed tails always pull the mean above the mode.

Exercise 6.5 A programme's network shows `TF` = −2 on the binding path. List, in cost order, the recovery families and the governance step each requires. Solution. (1) Logic re-choice/re-sequencing — planner's proposal, leader's sign-off; (2) float and resource harvesting from non-binding paths — float-policy authority; (3) crashing the binding path — expected-value case, budget authority; (4) fast-tracking — risk acceptance recorded in the register; (5) scope/acceptance renegotiation — sponsor and change control. Reporting the −2 honestly precedes all five (6.A.4).

Exercise 6.6 Enumerate Auriga's paths with their floats. Then state the cumulative slip the D–G chain can absorb without moving week 25, and the error a float report makes by summing D's and G's total floats. Solution. A–B–C–E–F **25**, float **0**; A–B–D–E–F **24**, float **1**; A–B–D–G **17**, float **8**. On the D–G chain `d + g ≤ 8` with `d ≤ 1` from the tighter path, so the absorbable cumulative slip is **8 weeks**, not the **9** implied by `1 + 8` — an overstatement of one week, or **USD 45,000** of false comfort at Auriga's cost of delay. Common error: adding an activity float column. The general form is that `n` activities on a path with float `f` report `n · f` while the path holds `f`, so the overstatement factor is the number of activities on the path.

Exercise 6.7 Auriga's engineering pool is capped at **6**. Engineer demands: A 2, B 4, C 1, D 3, E 5, F 6, G 2. Compute total demand, average utilisation against the 25-week plan, and find the week in which the early-start schedule breaches the cap. Then state the cheapest fix and its return. Solution. Total demand `4 + 24 + 8 + 21 + 25 + 24 + 4 =` **110** engineer-weeks against `6 × 25 =` **150** of capacity — **73.33 %** average utilisation. The early-start profile is 2, 4, 4 (C+D), 3 (C+G), then **7 in week 16–17** where E (5) overlaps G (2) — a breach of **one engineer for one week**. Hiring one engineer for that week costs **USD 5,225** and preserves the 25-week duration worth **USD 45,000** — a return of **8.61 times**. Common error: concluding from the 73.33 % utilisation that capacity is adequate; feasibility is a property of the peak, not the average.

Exercise 6.8 From the compression menu — C 30,000/wk (max 2), B 40,000/wk (max 2), D 35,000/wk (max 1), E 55,000/wk (max 1), F 65,000/wk (max 1) — find the least-cost duration when a week is worth 45,000, and state the range of week-values over which that answer holds. Solution. Marginal costs of successive weeks: **30,000** (C ×1), **40,000** (B ×1, the first buy that must shorten both long paths), **40,000** (B ×2), **55,000** (E), **65,000** (F), **65,000** (C ×2 + D). Buy while the step is at or below 45,000: three weeks, cumulative **USD 110,000**, **22 weeks**, net **+USD 25,000**. The answer holds for any week-value between **40,000 and 55,000**; below 40,000 the optimum is 24 weeks and above 55,000 it is 21. Common error: comparing the average crash cost at 22 weeks (**USD 36,666.67**) with 45,000 and buying a fourth week that costs 55,000 — a **USD 10,000** loss.

Exercise 6.9 An overlap saves 2 weeks at 45,000 each; if the overlap fails it costs 140,000 and returns 1.5 weeks of the saving. A certain crash plan buys the same 2 weeks for 70,000. Compute both breakeven probabilities and the expected value at an assessed 25 %. Solution. Good outcome **+90,000**; bad outcome `0.5 × 45,000 − 140,000 =` **−117,500**; swing **207,500**. Against doing nothing, `p* = 90,000/207,500 =` **43.37 %**. Against the crash plan's certain net of `90,000 − 70,000 = 20,000`, `p** = 70,000/207,500 =` **33.73 %**. At `p` = 0.25 the expected value is `90,000 − 0.25 × 207,500 =` **+38,125**, beating the crash plan by **18,125** while carrying a **137,500** worse downside. Common error: computing only `p*` and overlapping in cases where a certain alternative was better — once a paid alternative exists, `p**` is the relevant threshold.

Exercise 6.10 A critical chain's 50 % durations are 1.5, 5, 6.5, 4 and 3 weeks against network durations of 2, 6, 8, 5 and 4. Size the buffer both conventional ways, then forecast completion from a status showing 45 % of the chain complete and 30 % of the buffer consumed. Solution. Chain **20.0** weeks; embedded safety 0.5, 1, 1.5, 1, 1 = **5.0**. Half-safety buffer **2.5** → commitment **22.5000** weeks; root-sum-square `√5.5 =` **2.3452** → **22.3452** weeks, **2.6548** weeks inside the 25-week network, worth **USD 119,465.65**. Consumption ratio `0.30/0.45 =` **0.6667**; projected buffer use `2.5 × 0.6667 =` **1.6667** weeks; forecast chain completion **21.6667** weeks with **0.8333** weeks of buffer unused. Common error: applying root-sum-square to correlated activities, which understates the buffer — under shared causes the correct figure approaches the simple sum of 5.0 (Domain 8, KA 8.2.4).

Exercise 6.11 Meridian must install 40 clinics in 50 weeks. A specialist completes 0.1 clinics a week. Compute the required takt, the crew, and the value of the acceleration from the existing crew of six on both valuation bases. Solution. Takt `50/40 =` **1.25 weeks**; required rate **0.8** clinics a week; crew `0.8/0.1 =` **8**. With Domain 1's ramp (4 weeks, 25 % productivity, 50 % supervision) two newcomers give a ramp rate of **0.55** and a post-ramp rate of **0.8**, so duration is `4 + (40 − 2.2)/0.8 =` **51.2500** weeks against `40/0.6 =` **66.6667** — **15.4167** weeks saved for **10** extra specialist-weeks, **USD 42,000**. On the ramp-integral basis the gain is `20 × 15.4167 × 357 =` **USD 110,075** (net **+68,075**); on the switch-on-at-completion basis `15.4167 × 14,280 =` **USD 220,150** (net **+178,150**). Common error: valuing rollout acceleration at the fully-adopted cost of delay when benefit accrues per clinic — that doubles it, because the value per week of a linear ramp is `(N/2) ×` per-unit benefit, exactly half the steady-state figure.

Practitioner's toolkit — Domain 6

Adoption-ready artefacts; adapt headings to your organisation, then keep them stable.

Toolkit 6.T.1 — Schedule quality gate (run before trusting any network)

- [ ] Zero dangles; zero loops; every lag has a stated reason.
- [ ] Date constraints listed, each justified as genuine external physics — none doing logic's job.
- [ ] Durations owned by the executing discipline, estimate class stated (far waves: ranged).
- [ ] Passes re-run after every change; critical-path durations sum exactly to project duration.
- [ ] Near-critical paths listed with their floats, and the band stated with what it captures.
- [ ] Float report is **per path**, not per activity; no summed float column anywhere.
- [ ] Float report distinguishes `TF` and `FF`; float policy names who may spend it.
- [ ] Milestones computed, owned, done-defined; no typed dates.
- [ ] Every approval modelled against the governance calendar, not approximated by a lag.
- [ ] Resource histogram attached, caps named as real or negotiable, peak stated with the excess area in resource-weeks; levelling rule and any better schedule sought both recorded.
- [ ] Where a buffer is used: 50 % durations genuinely elicited, sizing convention stated, correlation considered, buffer visible with an owner and reported with float.
- [ ] AI-drafted logic, levelling or forecasts marked, verified against a stated baseline, and owned by a named human.

Toolkit 6.T.4 — Compression menu (priced at baseline, reviewed on the risk cycle)

Per candidate activity: current duration · which paths it lies on · cost slope per week · technical maximum weeks · owning discipline and the basis of the quote · risk consequence of taking it (coordination, quality, rework) · expiry — the date after which the option no longer exists. Then, on one line: the value of a week (cost of delay plus any time-related indirect cost), the resulting least-cost duration, and the range of week-values over which that duration stays optimal. Re-priced whenever the network changes, because the cheapest week is a property of the current binding paths.

Toolkit 6.T.2 — Recovery plan (one page)

Slip statement (activity, cause, size, date detected) · re-run pass results (new duration, new critical path, float table) · options priced in expected cost (re-sequence / harvest / crash / fast-track / renegotiate), each with risk and owner · selected plan and its decision authority · revised commitments and the stakeholder notice list · review date.

Toolkit 6.T.3 — Milestone register

Per milestone: computed date and current `TF` · owner · done-definition (acceptance reference) · contractual status and counterparty · L3 logic trace · reporting treatment (internal / contractual / public) · change history.

Exam preparation — Domain 6

The calculation traps. Taking the earlier predecessor at a convergence (Exercise 6.1) · computing `FF` against one successor instead of all (Exercise 6.2) · quoting `TF` after it has been spent · **summing an activity float column, which overstates absorption by the number of activities on the path (6.2.2)** · crashing from the original network instead of re-running passes (path migration, 6.4.2) · **comparing the average crash cost with the value of a week instead of the marginal one (6.4.2b)** · **inferring resource feasibility from average utilisation when it is set by the peak (6.3.1c)** · **valuing a rollout acceleration at the fully-adopted cost of delay when benefit accrues per unit, which doubles it (6.A.5)** · **rolling up ranged planning packages by adding mid-points (6.3.3)** · **substituting `E[wait] = M/2 + L` for a planned approval whose submission date is known (6.1.2b)** · treating the PERT mode as the mean, and forgetting that a symmetric estimate has no bias at all (6.4.3) · reading a pinned milestone as a forecast · confusing project-level float with activity-level slack in reports.

Reflection questions. 1. Your programme's board pack shows every milestone green. What three questions establish whether that is information or formatting? (Computed or typed? Passes re-run this cycle? Where is the float table?) 2. A subcontractor asks for "the float" on their package. What must you settle before answering? (Which float — *`TF`* or *`FF`*; the float policy; contractual float ownership — Domain 10; and whether anyone else is already drawing on the same path float — 6.2.2.) 3. When did you last see money spent on compression that the passes would have shown was worthless — and which governance step was missing? (6.4.2; toolkit 6.T.2.) 4. Has your last approved baseline been levelled against a real resource cap, and do you know its peak? (6.3.1c; toolkit 6.T.1. If the answer is an average utilisation figure, it is not an answer.) 5. Which of your approvals is currently modelled as a lag rather than as a date against a meeting calendar, and how much float is that costing you? (6.1.2b.)

Domain 6 summary

This domain turned "when will it finish?" from an opinion into an instrument. Planning levels give each audience a true view of one logic; dependency discipline makes the network a model rather than a drawing, and an approval modelled against its governance calendar rather than as a three-week label was worth four weeks of Meridian's schedule from one week of design work. The forward and backward passes yield the dates, the two floats and the critical path — with the master project showing why free float can vanish while total float survives, why convergence points compound slippage, why an activity float column can never be summed (nine weeks reported, eight available), and why a near-critical band of 10 % captures 85.71 % of the network rather than a footnote to it. Resources, constraints, milestones and rolling waves connect the network to the real world's crews, windows and horizons — and connect it hard: a plan only 73.33 % loaded on average was infeasible on its peak, an activity with eight weeks of logical float set the resource-constrained duration, and one engineer-week of excess demand was worth USD 45,000. Delivery flow closes the loop: a compression menu with rising cost slopes whose least-cost duration is 22 weeks and holds for any week worth 40,000–55,000, fast-tracking priced with two breakeven probabilities rather than a hope, recovery ranked from re-choice to renegotiation, buffers sized by aggregating safety and managed by their consumption ratio, rollouts scheduled as a takt and valued as an integral, and agile streams joined to the network through a required rate read off the backward pass. The leadership spine throughout: schedules fail morally before they fail mathematically — computed dates, visible float, reported negative float, resource feasibility tested before signature, and machine output verified against a stated baseline by named humans are what make a programme's word worth something.

07

DOMAIN 7

Cost, Resources and Commercial Awareness (quantitative flagship)

Group: Delivering the work (Domain 7 of 6 in Part Two — the cost counterpart to Domain 6's schedule). **Target:** ~78 pages. **Binds to:** the PCI Book Pattern Specification and the shared registries (`docs/books/registries/`). This domain is this book's home of the earned-value symbols — `PV`, `EV`, `AC`, `BAC`, `CV`, `SV`, `CPI`, `SPI`, `EAC`, `ETC`, `VAC`, `TCPI` — carried unchanged from the PCI family master table. Note the family notation-clash rule: `PV` here is **Planned Value**; present value is written `PV(x)` or in words (PFL-AI, Domain 3). British English; USD (+SAR where useful, indicative `USD 1 ≈ SAR 3.75`).

Why this domain exists

A leader who can defend a date but not a number is half-equipped. Domain 6 built the schedule; this domain builds the money that pays for it and the commercial arrangements that bind other people to deliver it. It starts with estimating and budgeting — how a credible number is constructed and what its accuracy claim actually means (KA 7.1); establishes the cost baseline and the forecasting that keeps it honest (KA 7.2); builds **earned value** in full, because it is the only technique that answers cost and schedule performance in one integrated language (KA 7.3); and closes with the commercial awareness a leader cannot delegate — resource economics, procurement strategy, contract models and cash flow (KA 7.4). Risk quantification deepens in Domain 8; contracts and supply networks get their own treatment in Domain 10. What belongs here is the leader's own numeracy: enough to know when a forecast is arithmetic and when it is hope.

Learning objectives. After this domain a candidate can: select an estimating method appropriate to the available definition and state its accuracy class; **read an accuracy class as a distribution and say what percentile the approved number sits at**; build a three-point cost estimate and distinguish a tail that is a spread from a tail that is a scenario; assemble a cost baseline separating contingency from management reserve, **and trend contingency draw against progress to a required draw efficiency**; measure `AC` so that it covers the same work as `EV`, **and compute what omitting accruals does to every index and forecast built on it**; distinguish commitments from actuals and **compute the uncommitted exposure of a forecast**; compute and interpret the full earned-value set (`CV`, `SV`, `CPI`, `SPI`); **show how much of a reported index is an artefact of the earning rule chosen, including the level-of-effort distortion identity**; forecast with each `EAC` method and choose the one matching the variance's cause; **state where each forecast's funding runs out if efficiency persists**; compute `VAC` and `TCPI`, **derive the identity family linking `TCPI` to every `EAC`, and convert a recovery-to-budget promise into the descope or the efficiency gain it actually requires**; read resource economics through blended rates, **burdened cost per productive hour and the priced cost of a crew-week**; distinguish the main contract models by who carries cost risk and **compute the hours breakeven between time-and-materials and a fixed price**; compute a cost-incentive fee and the point of total assumption, **and show how the**

share ratio and the ceiling move it; explain why cash flow and profit differ on a project **and compute the working capital a project absorbs**; **price the governance latency on a reserve release**; and govern AI-produced cost forecasts under the family verification rule.

The master worked project. Project Auriga continues from Domain 6 — the 25-week control-systems upgrade for a regional utility — now with money attached. Its approved cost baseline is `BAC` **= USD 4,000,000**, assembled from **USD 3,640,000** of control-account budgets plus **USD 360,000** of contingency, with **USD 240,000** of management reserve outside it for a total funding requirement of **USD 4,240,000** (KA 7.1.3). The delivering organisation's fixed price to the utility is **USD 4,400,000**, so the bid margin is **USD 400,000** — 9.09 % of price (KA 7.4.4). At the **data date, end of week 13**, the baseline says `PV` **= USD 2,080,000**; measurement gives `EV` **= USD 1,920,000** and `AC` **= USD 2,120,000**. Every calculation in KA 7.2–7.3 uses those four numbers, and Auriga's engineering blended rate of **USD 130.625 per hour** — an engineer-week of **USD 5,225**, the figure Domain 9 costs rework with — carries the resource arithmetic of KA 7.4. Domain 6's **cost of delay of USD 45,000 per week** is the price of time throughout. **Meridian Care Records** — the 40-clinic programme of Domains 1, 2, 15 and 16, approved cost **USD 2,400,000**, cost of delay **USD 14,280 per week** at its planned 70 % adoption — returns in Case study C, because reserve architecture and the latency of the authority that releases it are programme-scale cost questions a single project cannot show.

KNOWLEDGE AREA 7.1

Estimating and budgeting

Topics: 7.1.1 estimating methods and accuracy classes · 7.1.2 three-point estimating · 7.1.3 from estimate to budget.

7.1.1 Estimating methods and accuracy classes

The principle. An estimate's accuracy is governed by how well the work is defined, not by how much effort went into the arithmetic. The three standard methods trade definition against speed:

Method	How it works	Needs	Typical use
Analogous (top-down)	Scale a comparable past project	A true analogue and a scaling basis	Concept, screening (Domain 2)
Parametric	Cost = rate × parameter (per metre, per point, per kW)	A calibrated rate from real history	Early design, repetitive work
Bottom-up	Estimate each work package, then sum	A WBS at package level (Domain 4)	Baseline setting, control

Accuracy classes. Mature cost practice publishes an estimate's class alongside its number — a range that narrows as definition matures (the AACE Total Cost Management framework's class-5-to-class-1 progression is the reference treatment, described here in this book's own words). The professional discipline is simple and widely broken: **an estimate is never a single number without a range and a class**. "USD 4 million" is not an estimate; "USD 4 million, −15 %/+30 %, at a class consistent with 30 % design completion" is.

That second formulation contains far more than a caveat, and almost nobody reads what it says.

WORKED EXAMPLE

Worked example 7.1.1 — what Auriga's accuracy class actually claims.

1. **Setup.** Auriga's sanction estimate is **USD 4,000,000** with a declared class range of **−15 %/+30 %** at 30 % design completion. The programme board approves that number as the baseline. Read the range as a **triangular distribution** with the point estimate as its most likely value — the least assumption-heavy reading of a three-point statement — and ask three questions: how wide is the claim, where does the approved number sit inside it, and what would funding at a stated confidence have cost?
2. **Formula.** Bounds $lo = P(1 - d^-)$, $hi = P(1 + d^+)$. For a triangular distribution on (a, m, b): mean $= (a + m + b)/3$; the cumulative probability at the mode is $(m - a)/(b - a)$; median $= b - \sqrt{(b - a)(b - m)/2}$; the p-quantile above the mode is $b - \sqrt{(1 - p)(b - a)(b - m)}$; standard deviation $\sigma = \sqrt{(a^2 + m^2 + b^2 - ab - am - bm)/18}$. For comparison, the PERT weighting of the same three points is $(a + 4m + b)/6$.
3. **Substitution.** $lo = 4{,}000{,}000 \times 0.85$; $hi = 4{,}000{,}000 \times 1.30$; mean $= (3{,}400{,}000 + 4{,}000{,}000 + 5{,}200{,}000)/3$; $F(mode) = 600{,}000/1{,}800{,}000$; P80 $= 5{,}200{,}000 - \sqrt{0.20 \times 1{,}800{,}000 \times 1{,}200{,}000}$; median $= 5{,}200{,}000 - \sqrt{1{,}800{,}000 \times 1{,}200{,}000/2}$.
4. **Result.** Range **USD 3,400,000 to USD 5,200,000** — a band of **USD 1,800,000**, or **45.0 %** of the point estimate. σ = **USD 374,166**. The mean is **USD 4,200,000**; the median **USD 4,160,770**; the P80 **USD 4,542,733**. The approved USD 4,000,000 sits at the **33.33rd percentile** of the estimate's own declared range. The PERT weighting of the same three points gives **USD 4,100,000**.
5. **Interpretation.** The arithmetic turns a routine caveat into four statements a board can be held to, and the first one is uncomfortable.

Auriga was funded at a number it had a two-in-three chance of exceeding. Not because anybody was careless, but because an asymmetric range with the point estimate at its mode puts only $(m - a)/(b - a)$ of the distribution at or below that point — here exactly one third. The general result is worth carrying: **whenever the upside tail is longer than the downside tail, the point estimate is below the mean and below the median**, so approving the point estimate is approving a sub-50 % confidence position. It is a legitimate thing to do; it is not a legitimate thing to do unknowingly, and the sentence that makes it knowing is "this baseline is a P33".

The mean of the declared range is USD 4,200,000, which is exactly the week-13 EAC that Auriga's own data produces under method (a) (KA 7.3.3). That coincidence is worth pausing on, because it is the shape of most cost surprises: the eventual outturn was inside the estimate's stated accuracy from the first day. What went wrong was not the estimate but the **funding decision taken against it** — approving the mode and reporting the variance against the mode. A reviewer's question follows directly: is the current forecast outside the original class range, or merely above the point? Outside the range is an estimating failure; above the point but inside the range is a funding-level decision coming home.

Confidence has a price and it is computable. Funding at the P80 rather than the point requires **USD 542,733** more; funding at the mean requires **USD 200,000** more. Auriga's baseline in fact carries **USD 360,000** of contingency inside the 4,000,000 (KA 7.1.3) — which is a different structure again, because that contingency was sized bottom-up against identified risks rather than read off a distribution, and the two figures answer different questions. Where both are available they should be reconciled explicitly: a risk-based contingency far below the distribution-based number means either the register is incomplete (Domain 8, KA 8.2.4 aggregates it properly) or the range was drawn too wide.

Two cautions, both of which a reviewer should raise. The distribution shape is an assumption, not a fact: triangular gives a mean 200,000 above the point, the PERT weighting 100,000 — so the direction is robust and the magnitude is not, and no more than "the mean is 100,000 to 200,000 above the point, 2.5 % to 5.0 % of the estimate" should be claimed. And a class range is a statement about **definition maturity**, not a probability distribution anybody measured; treating it as one is a modelling convenience that must be labelled. What it is not is decoration. The honest use is the one above: convert the range into the percentile the approved number occupies, and say so out loud.

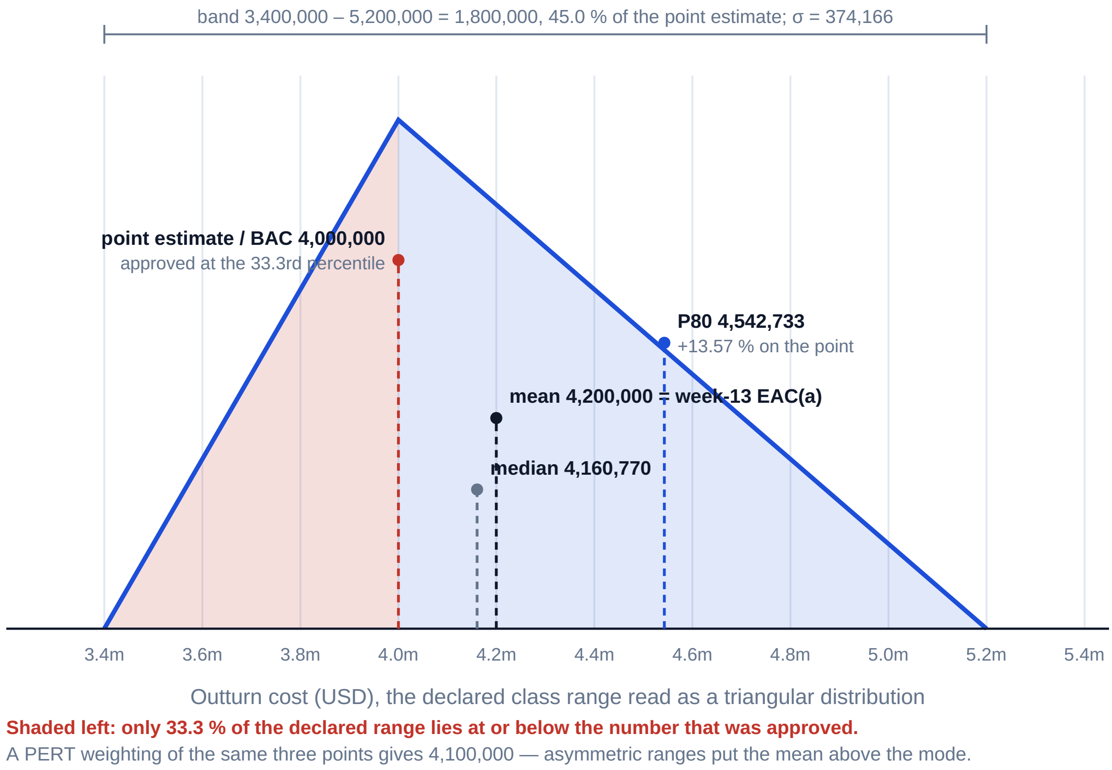


Fig 7.1.1 — What an accuracy class actually says

Common pitfall — precision mistaken for accuracy. A bottom-up estimate summing 400 packages to `USD 4,183,662` looks authoritative and is no more accurate than its worst assumption. Leaders should be more suspicious of an over-precise number than a rounded one.

7.1.2 Three-point estimating

Cost, like duration (Domain 6, KA 6.4.3), is a distribution. The same PERT weighting applies:

```
Cₑ = (o + 4m + p) / 6          σ = (p − o) / 6
```

WORKED EXAMPLE

Worked example 7.1.2 — Auriga's control-hardware package.

1. **Setup.** Procurement of control hardware is estimated **optimistic USD 680,000**, **most-likely USD 750,000**, **pessimistic USD 1,000,000** (the tail: a single-source controller with volatile lead-time pricing).
2. **Formula.** $C_e = (o + 4m + p)/6$; $\sigma = (p - o)/6$.
3. **Substitution.** $C_e = (680{,}000 + 3{,}000{,}000 + 1{,}000{,}000)/6 = 4{,}680{,}000/6$; $\sigma = 320{,}000/6$.
4. **Result.** C_e = **USD 780,000**; σ ≈ **USD 53,333**.
5. **Interpretation.** The most-likely figure is 750,000, but the expected cost is 780,000 — the right-skewed tail adds USD 30,000 before anything goes wrong. Budgeting at the mode systematically under-funds a portfolio of such packages; the difference between mode and mean, summed across a project, is a large part of what contingency exists to cover (7.1.3, and Domain 8's quantification).

Budget the mean, hold the tail centrally. Auriga's control account CA-20 is budgeted at **USD 780,000** — the C_e, not the mode (7.1.3's table). Budgeting the mode would have loaded **USD 30,000** of arithmetically certain under-funding into the baseline of this one package, and a project of forty such packages inherits forty such gaps that no manager can be held accountable for. The professionally right structure is the one used here: **each control account budgeted at its expected value, the dispersion held once, centrally, as contingency.**

The mode-to-mean gap is 0.5625 σ, and that ratio is the tell. It is 4.00 % of the mode here — small enough to be dismissed in a meeting and large enough to matter forty times over.

The pessimistic value is 4.125 σ above the mean, and that is a warning, not a spread. The optimistic value sits 1.875 σ below it. A tail four standard deviations out is not the upper end of a continuum of outcomes; it is a **discrete event** — the single-source controller not being available at the quoted price — which has its own probability and its own response. The professional treatment is to carry it in the risk register with that probability and cost (Domain 8, KA 8.2.2), not to smear it into a σ that then implies a symmetric spread nobody believes. Three-point estimating is the wrong instrument for a bimodal outcome, and asymmetry of this severity is the diagnostic.

Sizing from σ needs its approximation stated. Treating C_e and σ as a normal approximation, a P80 for this package is $780{,}000 + 0.8416 \times 53{,}333 =$ **USD 824,885**, so **USD 44,885** of contingency above the mean, or USD 74,885 above the mode. That is a defensible working figure and it is an approximation twice over — the PERT σ is itself a rule of thumb, and a distribution with a 4 σ tail is not normal. Quote it as an order of magnitude, aggregate properly at project level (Domain 8, KA 8.2.4, where correlation between packages does the real work), and never present a package-level P80 as though it were measured.

7.1.3 From estimate to budget

The structure. An estimate becomes a controllable budget through a deliberate hierarchy:

```
work-package estimates
  → control-account budgets          (WBS × organisation — the management-control points)
  → + contingency reserve            → the COST BASELINE (BAC): the PM's authority
  → + management reserve             → the total project funding requirement
```

Two rules carry the discipline. **Contingency is inside the baseline**, sized against identified risks (Domain 8) and spent by the project manager under a stated protocol. **Management reserve is outside the baseline**, held for unknown-unknowns, and released only by the sponsor or change authority (Domain 3's decision rights) — releasing it changes the baseline through change control (Domain 4, KA 4.4). Blurring the two is how projects appear to be "on budget" while consuming their own funding runway.

Time-phasing. The baseline is spread across the schedule to produce the **cumulative cost curve** (the S-curve) — and that curve is Planned Value (KA 7.3). A budget with no phasing cannot be measured against; the schedule (Domain 6) is therefore a precondition of cost control, not a parallel activity.

WORKED EXAMPLE

Worked example 7.1.3 — assembling Auriga's baseline, and reading the contingency trend.

1. **Setup.** Auriga's six control accounts, each budgeted at its expected value (7.1.2's discipline), with contingency sized bottom-up against the identified risks of Domain 8's register and management reserve set by the sponsor:

Control account	Budget (USD)
CA-10 Project management and engineering	620,000
CA-20 Control hardware procurement	780,000
CA-30 Civil and enabling works	540,000
CA-40 Installation	1,020,000
CA-50 Testing and commissioning	480,000
CA-60 Training and handover	200,000
Control-account total	**3,640,000**

Contingency is **USD 360,000**; management reserve is **USD 240,000**. At the week-13 data date the project is **48.00 %** complete by `EV/BAC` (KA 7.3.2) and has drawn **USD 190,000** of contingency — USD 150,000 for the ground-remediation risk and USD 40,000 for control-hardware price escalation. 2. **Formula.** `BAC` = control accounts + contingency. Total funding requirement = `BAC` + management reserve. Contingency draw index = (draw ÷ contingency) ÷ percent complete. Draw per point of progress = draw ÷ percent complete. Projected total draw = that rate × 100. Required draw efficiency for the remainder = remaining contingency ÷ (demonstrated rate × remaining points of progress). 3. **Substitution.** `BAC = 3,640,000 + 360,000`; funding `= 4,000,000 + 240,000`. Index `= (190,000/360,000)/0.48 = 0.5278/0.48`. Rate `= 190,000/48.00`. Required efficiency `= 170,000/(3,958.33 × 52.00)`. 4. **Result.** `BAC` = **USD 4,000,000**; total funding requirement **USD 4,240,000**. Contingency is **9.89 %** of the control-account total and **9.00 %** of `BAC`; management reserve is **6.00 %** of `BAC`. Contingency is **52.78 %** drawn at **48.00 %** complete — a draw index of **1.0995**. At the demonstrated rate of **USD 3,958.33 per point of progress** the full job needs **USD 395,833**, a shortfall of **USD 35,833** (9.95 % of the contingency, and 14.93 % of the management reserve). The remaining USD 170,000 must cover 52.00 points, i.e. USD 3,269.23 per point — a **required draw efficiency of 0.8259**, meaning the remaining work must consume contingency **17.41 % more slowly** than the first half did. 5. **Interpretation.** The third of the readings below is a reviewer's invariant most cost reports would fail.

Contingency has a `TCPI` of its own, and it should be reported. The structure is identical to KA 7.3.4's: a demonstrated rate, a remaining allowance, and the efficiency the remainder must achieve. Reporting "contingency 52.8 % drawn" invites a shrug; reporting "the remaining work must draw contingency 17.41 % more slowly than the work to date, and there is no stated reason why it would" invites a decision. The threshold to watch is the **draw index**: above 1.00 the reserve is depleting faster than the work is being done, and the projected shortfall — here USD 35,833 — is the number that eventually becomes a management-reserve request. It is visible now, at 48 % complete, which is the whole point of trending it.

The two reserves are not interchangeable and the arithmetic shows why. The 35,833 projected shortfall is 14.93 % of the management reserve, so the reserve can absorb it — but absorbing it is a **sponsor** act that changes the baseline through change control (Domain 4, KA 4.4), not a project-level top-up. A project that quietly funds a contingency overrun from management reserve has converted an early warning into an invisible one, and MCQ 7.1-B's erosion pattern is what that looks like three periods later.

The draw must reconcile to the cost variance, and here it does not. `CV` at week 13 is **(USD 200,000)** while contingency drawn is **USD 190,000** — a gap of **USD 10,000**. That gap is not a rounding artefact; it means either a risk was funded without a draw being recorded, or USD 10,000 of overrun has no identified cause. Both are findings. The invariant is worth carrying into every review: **identified-risk draws plus unattributed variance must equal the cost variance**, and the unattributed component is the one to interrogate, because a variance with no cause cannot be forecast (KA 7.2.2) and therefore cannot be recovered.

The caution: none of this arithmetic is valid if contingency was sized as a percentage rather than against a register. A flat "10 % contingency" has no risks behind it, so a draw against it cannot be reconciled to anything, and the trend above degenerates into an observation about spending speed. Domain 8, KA 8.2.4 is where the number comes from; this KA is where it is governed.

AI in this KA

Estimating is where AI assistance is most seductive and most in need of provenance. A model can produce a plausible parametric rate instantly, and it cannot tell you whether that rate came from comparable work, a different market, or nowhere at all. The governed workflow: **AI proposes** a method, a rate and a range; the estimator supplies the calibration evidence (which projects, which years, which escalation basis — the source discipline of the shared registry); the range and class are stated; and a named human owns the number. **AI proposes; the professional verifies, decides and remains accountable.** An estimate whose basis cannot be produced on request is not an estimate.

Two placements are worth being exact about, because 7.1.1 and 7.1.3 create work that machines do well and work they must not touch. Converting a declared class range into percentiles, means and a funding-confidence comparison — and repeating it across every package and every candidate distribution shape — is mechanical, exactly checkable and almost never done by hand; that is a good use. **Choosing the shape is not**, because the shape encodes a belief about how the tail behaves, and 7.1.2's four-sigma pessimistic value shows what happens when the wrong shape is applied to a discrete event: a scenario is quietly turned into a spread. Likewise, computing the contingency draw index and the required draw efficiency each period is routine; deciding whether the draw pattern means the register was incomplete or the work is simply front-loaded on its risks is a judgement about the project, and belongs to whoever will have to ask for the management reserve.

KEY TERMS — KA 7.1

Term	Meaning
Analogous / parametric / bottom-up	Scale an analogue · rate × parameter · sum the packages.
Accuracy class	The definition-linked range accompanying an estimate; never optional.
Control account	The WBS × organisation point where scope, budget and actuals integrate.
Contingency reserve	Inside the baseline; funds identified risks; PM-controlled.
Management reserve	Outside the baseline; unknown-unknowns; sponsor-controlled.
`BAC`	Budget at Completion — the time-phased cost baseline's total.
Total funding requirement	`BAC` + management reserve — the cash the sponsor must have available; Auriga's is USD 4,240,000.
Approval percentile	The position of the approved number inside its own declared range; Auriga's baseline is a P33.
Contingency draw index	(Contingency drawn ÷ contingency) ÷ percent complete; above 1.00 the reserve is depleting faster than the work is being done.
Required draw efficiency	Remaining contingency ÷ (demonstrated draw rate × remaining progress) — the `TCPI` of the reserve.

SAMPLE MCQS — KA 7.1

MCQ 7.1-A `[7.1.2 · Application]` A package is estimated o = 680,000, m = 750,000, p = 1,000,000. Its PERT expected cost is: - A. USD 750,000 - B. USD 780,000 - C. USD 810,000 - D. USD 840,000

Rationale: `(680,000 + 4 × 750,000 + 1,000,000)/6 = 780,000`. A is the mode; C is the unweighted three-point mean; D over-weights the pessimistic value.

MCQ 7.1-B `[7.1.3 · Analysis]` A project reports "on budget" while having consumed 60 % of its management reserve at 40 % complete. The correct reading is: - A. genuinely on budget — management reserve exists to be spent - B. the baseline is intact but the project's total funding is eroding faster than progress; the trend belongs in the next report to the sponsor - C. a baseline breach requiring immediate re-baselining - D. an accounting error, since management reserve is inside the baseline

Rationale: Management reserve sits outside the baseline (so D is wrong and A is technically true but misleading), and its release is a sponsor-level signal, not a project-level convenience. It is not yet a breach (C) — it is the early warning that precedes one.

MCQ 7.1-C `[7.1.1 · Recall]` Which statement about a bottom-up estimate summing to USD 4,183,662 is soundest? - A. its precision indicates high accuracy - B. its accuracy is bounded by its assumptions and definition maturity, and it must still carry a range and class - C. rounding it would reduce its accuracy - D. bottom-up estimates do not need accuracy classes

Rationale: Precision (digits) and accuracy (closeness to outturn) are independent; definition maturity governs the latter. Rounding changes no information (C), and every estimate carries a class (D).

MCQ 7.1-D `[7.1.1 · Analysis]` A point estimate of USD 4,000,000 carries a declared class range of −15 %/+30 %. Read as a triangular distribution, its expected cost is: - A. USD 4,000,000 - B. USD 4,100,000 - C. USD 4,200,000 - D. USD 4,300,000

Rationale: `(3,400,000 + 4,000,000 + 5,200,000)/3 = 4,200,000` (7.1.1). A treats the point estimate as the mean, which only holds for a symmetric range; B applies the PERT weighting, which weights the mode four times and is a different shape from the one asked for; D averages the two range endpoints and ignores the mode entirely.

MCQ 7.1-E `[7.1.3 · Application]` Control accounts total 3,640,000; contingency is 360,000; management reserve is 240,000. `BAC` and the total funding requirement are: - A. 3,640,000 and 4,000,000 - B. 4,000,000 and 4,240,000 - C. 4,240,000 and 4,240,000 - D. 4,000,000 and 4,000,000

Rationale: Contingency is inside the baseline and management reserve outside it (7.1.3). A excludes contingency from `BAC`, which would leave the project's own risk funding outside its authority; C puts management reserve inside the baseline, the error MCQ 7.1-B's misreading rests on; D omits the reserve from the funding requirement, so the sponsor holds no cash for it.

MCQ 7.1-F `[7.1.3 · Analysis]` Contingency of 360,000 is 190,000 drawn at 48.00 % complete. The draw efficiency the remaining work must achieve is: - A. 0.4722 - B. 0.8259 - C. 0.9095 - D. 1.0995

Rationale: `170,000/(3,958.33 × 52.00) = 0.8259` — the remainder must draw 17.41 % more slowly than demonstrated (7.1.3). A is the remaining balance as a share of the total, which is a level not an index; C inverts the two shares; D is the draw index itself — the rate at which contingency is being consumed relative to progress, not the efficiency required to recover.

SELF-CHECK — KA 7.1

1. Where do contingency and management reserve sit, and who spends each? — Contingency inside the baseline, PM-controlled; management reserve outside it, sponsor-controlled.
2. Why is the time-phased baseline a precondition for cost control? — Because the phased cumulative curve is Planned Value; without it there is nothing to measure performance against.
3. What two things must accompany every estimate? — A range and an accuracy class tied to definition maturity.
4. What percentile of its own declared range was Auriga's baseline approved at, and why? — The 33.33rd: an asymmetric −15 %/+30 % range puts only `(m − a)/(b − a)` of the distribution at or below the mode, so the approved number had a two-in-three chance of being exceeded.
5. What does a contingency draw index above 1.00 mean, and what number does it eventually produce? — The reserve is depleting faster than the work is being done; on Auriga it projects a USD 35,833 shortfall, which becomes a management-reserve request unless the draw rate changes.
6. What must the contingency draw reconcile to? — The cost variance. Auriga's USD 190,000 of draws against a `CV` of (USD 200,000) leaves USD 10,000 of variance with no identified cause, which is a finding because an unexplained variance cannot be forecast.

KNOWLEDGE AREA 7.2

The cost baseline, actuals and forecasting

Topics: 7.2.1 measuring actual cost · 7.2.2 the forecasting question · 7.2.3 baseline integrity.

7.2.1 Measuring actual cost

The principle. AC must cover the same work as EV, in the same period, or every index built from them is fiction. Three mechanics decide whether it does:

- **Accruals.** Work received but not yet invoiced belongs in this period's AC. Omit accruals and cost performance looks excellent until the invoices land — the commonest cause of a "sudden" overrun that was months old.
- **Commitment vs actual.** A purchase order is a commitment, not a cost. Both matter: AC drives performance, commitments drive the funding forecast. Confusing them double-counts or hides money.
- **Open-commitment hygiene.** Stale purchase orders left open inflate the forecast; cleansing them is a standing month-end task.

Each is usually stated and left; the size of what they do to a report is not intuitive.

WORKED EXAMPLE

Worked example 7.2.1 — the accrual that made Auriga look perfect.

1. **Setup.** At week 13 Auriga has genuinely incurred **USD 2,120,000** against EV of **USD 1,920,000**. The ledger, however, shows only **USD 1,920,000** of invoiced cost: the installation subcontractor's work for weeks 11–13 has been received and accepted but not yet invoiced, and **USD 200,000** of it has not been accrued. Weeks 14–17 then run at a true EV of **USD 480,000** for a true cost of **USD 520,000**, and the accrual backlog is invoiced in that period. Compare what the report says with what is true, in both periods and cumulatively.
2. **Formula.** `CPI = EV/AC`; `CV = EV − AC`; `EAC = BAC/CPI`; `TCPI = (BAC − EV)/(BAC − AC)`. Period index = period EV ÷ period AC. Cumulative index = cumulative EV ÷ cumulative AC.
3. **Substitution.** Reported `CPI = 1,920,000/1,920,000`; true `CPI = 1,920,000/2,120,000`. Reported `EAC(b) = 4,000,000/1.0000`; true `= 4,000,000/0.905660`. Period reported `= 480,000/(520,000 + 200,000)`; period true `= 480,000/520,000`. Cumulative after week 17 `= 2,400,000/2,640,000`.
4. **Result.**

At week 13	Reported	True
AC	1,920,000	2,120,000
CV	0	(200,000)
CPI	**1.0000**	**0.9057**
EAC(b)	4,000,000	**4,416,667**
TCPI to BAC	1.0000	**1.1064**

The single omission understates the forecast by **USD 416,667 — 10.42 % of** BAC. In the catch-up period the reported AC is `520,000 + 200,000 =` **USD 720,000**, so the reported period CPI is **0.6667** against a true period CPI of **0.9231** — an error of **0.2564**, and a reported figure implying the period's work cost **50.0 %** more than budget. Cumulatively after week 17 both versions converge on EV 2,400,000 against AC 2,640,000, CPI **0.9091**. 5. **Interpretation.** The invariant is the useful part: **an accrual omission is a timing error, so it cancels cumulatively and misstates every period in between.**

The direction of the distortion reverses, and the second half is the one that gets investigated. Weeks 1–13 report perfection; week 17 reports a collapse. The true story is that performance improved — 0.9057 to 0.9231 — in exactly the period the report shows falling apart. Anyone acting on the period reading will investigate the wrong thing, and the operational explanations offered in that meeting will be inventions, because there is nothing operational to explain.

The plausibility test is the detection mechanism. A single-period `CPI` of 0.6667 implies costs 50 % above budget for four weeks. Ask what physically happened to cause that; if the answer is thin, the cause is measurement, not production, and the ledger is where to look. This is MCQ 7.2-A's step change with the arithmetic attached: **a step in `CPI` with no operational counterpart is a measurement event.**

The forecast error is the real damage, not the index. Reported `TCPI` to `BAC` was 1.0000 — "no recovery required" — while the true figure was 1.1064. A board told the project needs par performance to finish at budget, when it in fact needs an 11 % efficiency gain, has been given the opposite of a warning. And the reported `EAC(b)` of exactly `BAC` is the tell: **when `CPI` reads exactly 1.0000, check the accruals before celebrating**, because a real project rarely lands on the integer.

The cure is a control, not diligence. `AC` must never be below invoices plus receipted-but-uninvoiced value, the reconciliation must be a signed month-end step, and the accrual basis must be stated in the report. The caution runs the other way too: an over-accrual flatters the following period exactly as symmetrically, so a project whose `CPI` improves in every period after a bad month should be asked whether the bad month was over-accrued.

WORKED EXAMPLE

Worked example 7.2.1b — commitments, coverage and the exposure nobody reports.

1. **Setup.** Auriga at week 13: AC **USD 2,120,000**; open purchase orders and subcontract commitments for work not yet received total **USD 1,180,000**. Of those open commitments, **USD 120,000** relates to scope already delivered and invoiced — stale lines nobody has closed. The forecast in use is method (b), EAC **USD 4,416,667** (KA 7.3.3). Twelve weeks remain.
2. **Formula.** Uncommitted balance of the baseline = BAC − AC − open commitments. Commitment coverage = (AC + open commitments) ÷ EAC. Uncommitted balance of the forecast = EAC − AC − open commitments. Escalation exposure = uncommitted forecast balance × price movement.
3. **Substitution.** Baseline uncommitted `= 4,000,000 − 2,120,000 − 1,180,000`. Coverage `= 3,300,000/4,416,667`; after cleansing `= 3,180,000/4,416,667`. Forecast uncommitted `= 4,416,667 − 3,300,000`.
4. **Result.** The baseline leaves **USD 700,000** uncommitted; the forecast leaves **USD 1,116,667**. Commitment coverage is **74.72 %** of the method-(b) forecast (78.57 % of the method-(a) forecast). Cleansing the USD 120,000 of stale lines moves coverage to exactly **72.00 %** and raises the baseline's uncommitted balance to **USD 820,000** — while changing AC, CPI, CV and every EAC by nothing at all. The USD 1,116,667 still to be committed over twelve weeks is **USD 93,056 per week** of new commitment, and a 5 % adverse price movement on it is **USD 55,833 — 27.92 %** of the entire cost variance the week-13 review is about to discuss.
5. **Interpretation.** The identity is the thing to remember, and it is exact: **USD 1,116,667 − USD 700,000 = USD 416,667 = |VAC(b)|.** The gap between what the baseline leaves to be bought and what the forecast says will actually have to be bought is the forecast variance at completion, seen from the funding side rather than the performance side. Two people arguing about whether the overrun is real are arguing about the same number twice, and the funding view is usually the one a sponsor accepts, because it is a statement about money not yet spent rather than a criticism of money already spent.

Commitment coverage answers a question CPI cannot: how much of the forecast is still exposed to prices? At 74.72 % coverage, a quarter of the outturn is unfixed. That is the number to report beside CPI when a market is moving, and its sensitivity — 5 % on the remainder being more than a quarter of the variance to date — is what makes escalation an active commercial matter rather than an estimating footnote. Long-lead items are where coverage is bought (Domain 10, KA 10.1.3 prices the lead time itself).

Cleansing commitments makes the picture look worse and be more true, which is why it does not happen. The stale USD 120,000 was silently claiming that scope was already secured. Removing it moves coverage down by 2.72 points and leaves an extra USD 120,000 visibly still to buy. Nothing in any performance index rewards the exercise; the whole return is a funding forecast that is not lying. That is precisely why it belongs in a standing month-end checklist with a named owner (Toolkit 7.T.1) rather than in anyone's judgement.

The caution: commitment coverage is not a measure of performance and must never be read as one. A project can be 100 % committed and badly over budget — indeed committing early at bad prices is one way to get there. Coverage bounds uncertainty in the forecast, not its level.

7.2.2 The forecasting question

A forecast answers one question — what will this cost when it is done? — and the honest answer depends entirely on **why** the current variance exists. That judgment is the leader's; the arithmetic is KA 7.3.3's EAC family. The rule to internalise now: **a forecast is a statement about the remaining work, not an extrapolation ritual.** A variance caused by a closed, non-recurring event says nothing about what is left; a variance caused by a systemic productivity shortfall says everything.

7.2.3 Baseline integrity

The baseline's authority rests on its stability. Four standing rules: budgets of completed or open work packages are never retrospectively adjusted (except to correct an authorised error); re-baselining happens only through change control, with an audit trail (Domain 4, KA 4.3); transfers between control accounts are logged, not silent; and the baseline never absorbs a variance by moving budget to where the money went. A baseline edited to match reality has stopped measuring anything — the cost analogue of Domain 6's pinned milestone.

KEY TERMS — KA 7.2

Term	Meaning
AC	Actual Cost of the work performed, including period accruals.
Accrual	Cost of work received but not yet invoiced, recognised in the period.
Commitment	A contractual obligation (e.g. a PO); a funding fact, not yet a cost.
Commitment coverage	(AC + open commitments) ÷ EAC — the share of the forecast whose price is already fixed; 74.72 % on Auriga.
Uncommitted forecast balance	EAC − AC − open commitments; its excess over the baseline's uncommitted balance equals |VAC|.
Re-baselining	Domain 4's instrument (KA 4.3.3) seen from earned value: it resets the PV curve every index is measured against, which is exactly why it cannot be used to retire an adverse variance.

SAMPLE MCQS — KA 7.2

MCQ 7.2-A [7.2.1 · Analysis] A project's CPI has read 1.02 for four months; then one month it drops to 0.91 with no change in productivity. The likeliest explanation is: - A. genuine sudden inefficiency - B. accruals were not being recognised, so earlier AC understated cost and this month absorbed the catch-up - C. the baseline was too generous - D. EV was over-claimed this month

Rationale: A step change in CPI without an operational change points at measurement, and missing accruals are the classic cause — earlier periods flattered, one period punished. D would raise, not lower, the earlier readings' credibility, and C would show as a stable, not stepped, pattern.

MCQ 7.2-B [7.2.3 · Recall] Which action preserves baseline integrity? - A. moving budget from an underspent control account to cover an overspend, without record - B. re-baselining through change control with an audit trail when scope genuinely changes - C. adjusting a completed package's budget to match its actual cost - D. reducing remaining budgets so the total still equals BAC

Rationale: Only governed change preserves the measurement. A is a silent transfer, C is retrospective adjustment, D is the classic "make the numbers add up" manoeuvre — each destroys comparability.

MCQ 7.2-C `[7.2.1 · Application]` `EV` is 1,920,000; invoiced cost is 1,920,000; a further 200,000 of work has been received and accepted but neither invoiced nor accrued. Reported and true `CPI` are: - A. 1.00 and 0.91 - B. 0.91 and 1.00 - C. 1.00 and 1.00 - D. 0.91 and 0.91

Rationale: The report divides `EV` by the understated `AC` of 1,920,000 (1.00); the truth divides by 2,120,000 (0.9057, shown as 0.91) — 7.2.1. B reverses the two; C treats the receipted work as a commitment rather than a cost, which is the substantive error the accrual rule exists to prevent; D assumes the accrual was recognised, in which case there would be nothing to detect.

MCQ 7.2-D `[7.2.1 · Analysis]` In the month a 200,000 accrual backlog is finally invoiced, the reported period `CPI` is 0.67 while true period performance was 0.92. The soundest reading is: - A. productivity collapsed by a third in that period - B. the period absorbed a timing correction: the cumulative index is now right and the period reading is not - C. `EV` was over-claimed in the period - D. the baseline for that period was too generous

Rationale: Accrual omissions cancel cumulatively and misstate the periods in between (7.2.1), so the cumulative `CPI` of 0.9091 is now correct while the period figure is an artefact. A takes the artefact at face value and would launch an investigation into production that has nothing to find; C would depress the earlier periods too, not this one alone; D would show as a stable bias rather than a single-period step.

MCQ 7.2-E `[7.2.1 · Application]` `AC` is 2,120,000, open commitments are 1,180,000 and the `EAC` in use is 4,416,667. The balance of the forecast still to be committed is: - A. USD 700,000 - B. USD 1,116,667 - C. USD 1,316,667 - D. USD 3,236,667

Rationale: `4,416,667 − 2,120,000 − 1,180,000 = 1,116,667` (7.2.1b). A measures against `BAC` rather than the forecast, and the 416,667 difference between A and B is exactly |`VAC`(b)|; C uses `EV` in place of `AC`, double-counting the cost variance; D omits `AC` altogether.

SELF-CHECK — KA 7.2

1. Why must *`AC`* include accruals? — So it covers the same work as `EV` in the same period; otherwise every index is misstated.
2. Is a purchase order a cost? — No: a commitment. It informs the funding forecast, not performance.
3. What single question does a forecast answer? — What the remaining work will cost, given why the variance exists.
4. State the accrual invariant and what it implies for period reporting. — An accrual omission is a timing error: it cancels cumulatively and misstates every period in between, so a period index with no operational counterpart is a measurement event, not a performance event.
5. What does commitment coverage tell you that *`CPI`* cannot? — How much of the forecast is still exposed to price movement: at Auriga's 74.72 %, a 5 % movement on the uncommitted remainder is USD 55,833, or 27.92 % of the cost variance to date.
6. Why does cleansing stale commitments never improve a report? — Because it changes no performance index and makes the funding forecast look worse while making it true, which is why it needs to be a checklist step with an owner rather than a matter of judgement.

KNOWLEDGE AREA 7.3

Earned value: measurement, variances and forecasting

Topics: 7.3.1 the three measures · 7.3.2 variances and indices · 7.3.3 the *EAC* family · 7.3.4 *VAC* and *TCPI*.

7.3.1 The three measures

Definitions. All three are expressed in the same budget currency so they are directly comparable:

- **Planned Value (PV)** — the budgeted cost of the work scheduled by the data date: the time-phased baseline of KA 7.1.3.
- **Earned Value (EV)** — the budgeted cost of the work actually performed: physical progress **valued at the budget rate**, never at what it cost.
- **Actual Cost (AC)** — the cost actually incurred for that work, accruals included (KA 7.2.1).

The single most important conceptual point: EV is measured at budget. That is what lets EV be compared with PV (both at budget → schedule progress) and with AC (budget vs actual for the same work → cost efficiency). Confusing "value earned" with "cost incurred" collapses the method.

Earning rules. How EV is claimed decides how much it can be gamed: **0/100** (nothing until complete — objective, good for short packages); **50/50**; **percent complete** (needs an objective basis); **units completed** (best where output is countable); **weighted milestones**; and **level of effort**, where EV is set equal to PV by the calendar, so it can never show a schedule variance. Level of effort dilutes whatever it is mixed into — formal practice segregates it and caps its share of a control account.

How much difference the choice makes is the question that decides whether earning rules are a technicality or a governance matter.

WORKED EXAMPLE

Worked example 7.3.1 — one physical state, three earned values.

1. **Setup.** A **USD 400,000** group of ten equal work packages inside Auriga's installation control account CA-40. At the data date: **4** packages complete, **3** in progress at an objectively measured **30 %** physical completion, **3** not started. The baseline expected **6** complete by now, so PV = **USD 240,000**. Actual cost for the group is **USD 210,000**. Compute EV, SPI and CPI under 0/100, 50/50 and objective percent-complete earning.
2. **Formula.** 0/100: EV = complete packages × package budget. 50/50: complete × budget + in-progress × budget/2. Percent complete: complete × budget + Σ(in-progress physical % × budget). Then SPI = EV/PV, CPI = EV/AC.
3. **Substitution.** 0/100 = 4 × 40,000. 50/50 = 160,000 + 3 × 20,000. Percent complete = 160,000 + 3 × 12,000.
4. **Result.**

Earning rule	EV	SPI	CPI	Reads as
0/100	160,000	**0.6667**	**0.7619**	badly late, badly over
50/50	220,000	**0.9167**	**1.0476**	slightly late, **under budget**
Percent complete (objective, 30 %)	196,000	**0.8167**	**0.9333**	late and over

The spread in `EV` is **USD 60,000 — 15.00 %** of the group's budget — and the `SPI` spread is **0.25**, on a physical state that did not change between rows. 5. **Interpretation.** The decisive observation is in the third column of the middle row. **The earning rule alone flips the account from over-running (`CPI` 0.9333) to under-running (`CPI` 1.0476).** Nothing was misreported, nothing was manipulated, and the two readings support opposite management decisions. That is why the earning rule is a control, not a preference.

The direction of each rule's bias is predictable, so a reviewer can compute it. 50/50 overstates by **USD 24,000** — 12.24 % of the objective figure — because the in-progress work is only 30 % done; 0/100 understates by **USD 36,000**, or 18.37 %. The breakeven is exact and worth carrying: **50/50 is unbiased precisely when in-progress work averages 50 % physical completion**, and at genuine 50 % it and objective percent-complete agree at USD 220,000 — while 0/100 still reads 160,000, because it credits nothing until a package closes. So 50/50 flatters a control account whose in-progress work is young — which is the normal condition of an account that has just started, and exactly when a project most wants reassurance.

Which rule to choose follows from package duration, not from taste. 0/100 is objective and cheap and its understatement is harmless where packages are short relative to the reporting period — a package that starts and finishes inside a month never sits half-earned at a data date. For long packages 0/100 produces a sawtooth that hides real progress, and objective percent complete earns its administrative cost. 50/50 is a convenience that should be reserved for packages short enough that the error cannot matter, and its use on long packages is the single commonest source of an over-stated `EV`.

The rule must be fixed before the period it measures. A rule chosen after the data date is not a measurement convention but a selection among answers, and the 15 % spread above is the size of the discretion involved. This is the cost analogue of Domain 6's pinned milestone: the defect is not arithmetic but the loss of a fixed reference. State the rule per package in the plan, disclose any change, and — the reviewer's test — recompute one account under a different rule to see how much of the reported position is the convention rather than the work.

The level-of-effort distortion, computed. Because level of effort sets `EV ≡ PV`, a control account that mixes it with discrete work reports a blend, and the blend has a closed form. With `s` the level-of-effort share of the account's budget and `SPI_d` the discrete work's own schedule performance index:

```
reported SPI = s × 1 + (1 − s) × SPI_d = 1 − (1 − s)(1 − SPI_d)
distortion   = reported SPI − SPI_d     = s × (1 − SPI_d)
```

An account 70 % level of effort whose discrete work is running at `SPI_d` **0.60** reports **0.88** — a distortion of **0.28**, which is to say the report conceals nearly all of the problem. The same account under a **20 %** level-of-effort cap would report **0.68**, a distortion of **0.08**. And the share required to make a discrete `SPI` of 0.60 report as 0.95 or better is `s ≥ 0.875` — so **87.5 % level of effort makes a two-fifths schedule shortfall disappear into a figure a steering committee would wave through.** The distortion is linear in the level-of-effort share, which is precisely why the countermeasure is a cap expressed as a share of budget, and why the share must be disclosed per control account rather than for the project as a whole.

7.3.2 Variances and indices

```
CV = EV − AC        SV = EV − PV
CPI = EV / AC       SPI = EV / PV
```

WORKED EXAMPLE

Worked example 7.3.2 — Auriga at week 13.

1. **Setup.** `BAC` = 4,000,000; at the data date `PV` = 2,080,000, `EV` = 1,920,000, `AC` = 2,120,000.
2. **Formula.** As above; variances in currency, indices as ratios (two decimals).
3. **Substitution.** `CV = 1,920,000 − 2,120,000`; `SV = 1,920,000 − 2,080,000`; `CPI = 1,920,000/2,120,000`; `SPI = 1,920,000/2,080,000`.
4. **Result.** `CV` = **(USD 200,000)** · `SV` = **(USD 160,000)** · `CPI` = **0.91** · `SPI` = **0.92**. Auriga is **48.0 % complete** (`EV/BAC`) having spent **53.0 %** of budget (`AC/BAC`).
5. **Interpretation.** Behind and over: for every dollar spent the project is producing 91 cents of budgeted work, and it has delivered 92 % of what it planned to by now. The percent-complete versus percent-spent pair (48 vs 53) is the same story in the form a board absorbs fastest. Note this is the cost view of the very slippage Domain 6's case study recovered on the schedule side — the two domains are describing one project.

The percent-complete/percent-spent gap is the cost variance, exactly. `%spent − %complete = (AC − EV)/BAC`, so the 5.00-point gap between 53.00 % and 48.00 % **is** the USD 200,000, and the identity holds always. That makes the two percentages a legitimate board summary rather than a simplification — nothing is lost in the translation, and a report where the gap and the `CV` disagree has an arithmetic error in it.

Express the overrun as a rate, because a rate can be projected and a total cannot. The work performed so far cost **10.42 %** more than it was budgeted to: `1/CPI − 1 = 1/0.905660 − 1`. That is the same 10.42 % that appears again as the ratio between removed scope and saved cash in KA 7.3.4, and it is the number that scales — USD 200,000 is a fact about the past, 10.42 % is a claim about the production process.

Express the schedule variance in time, because currency conceals the size of it. Auriga's `SV` of (USD 160,000) is exactly one week of planned value at this point on the curve, where the baseline is rising at USD 160,000 per week. So `SV` (160,000) means **one week behind**, which is both more useful and more challengeable than a currency figure, and it is the bridge to earned schedule in 7.A. 1. The general caution: `SV` in currency is not comparable between projects or even between phases of one project, because it is scaled by the local slope of the `PV` curve.

Both indices are cumulative, so they lag, and neither says whether things are improving. A cumulative `CPI` of 0.91 at 48 % complete cannot distinguish a project that has stabilised from one still deteriorating; only the period index can, and 7.2.1 shows how easily the period index is corrupted. The reviewer's pairing is therefore cumulative index for level and period index for direction, with the accrual reconciliation done before either is believed. And the whole set rests on the earning rules of 7.3.1: on the arithmetic there, a 15 % `EV` swing was available from the convention alone, which is larger than the variance being discussed here.

Fig 7.3.1 — Auriga's earned-value S-curves at week 13

7.3.3 The EAC family — forecasting

```
ETC = the remaining work's forecast cost          EAC = AC + ETC
(a) EAC = AC + (BAC − EV)                        remaining work at the BUDGETED rate
(b) EAC = BAC / CPI                              current cost efficiency PERSISTS
(c) EAC = AC + (BAC − EV) / (CPI × SPI)          cost AND schedule pressure compound
(d) EAC = AC + bottom-up ETC                     re-estimate the remainder from scratch
```

WORKED EXAMPLE

Worked example 7.3.3 — the same project, four futures.

1. **Setup.** Auriga's week-13 figures, forecast to completion by each method.
2. **Formula.** (a)–(c) above; `CPI` = 0.905660, `SPI` = 0.923077 at full precision.
3. **Substitution.** (a) `2,120,000 + (4,000,000 − 1,920,000)`; (b) `4,000,000 / 0.905660`; (c) `2,120,000 + 2,080,000/(0.905660 × 0.923077)`.
4. **Result.** (a) **USD 4,200,000** · (b) **USD 4,416,667** · (c) **USD 4,608,056**.
5. **Interpretation.** The same four measured numbers support forecasts spanning **USD 408,056** — over 10 % of `BAC`. The spread is not uncertainty in the arithmetic; it is the assumption about the remaining work, made explicit. Choosing among them is the leader's judgment, and it must be stated: if Auriga's overrun was the contaminated-ground remediation of Domain 6 — a discrete, now-closed event — method (a) is right and the others over-forecast. If it reflects a systemic productivity shortfall that will persist, (b). If schedule pressure is now driving overtime and expediting, (c). A forecast presented without its assumption named is not a forecast; it is a number.

Scale the spread against the remaining work, not against `BAC`. USD 408,056 is 10.20 % of `BAC` and **19.62 %** of the USD 2,080,000 still to be earned. The second figure is the honest one, because the disagreement is entirely about the remainder — the methods agree exactly about the past. A disagreement of a fifth about the work still to do is a large disagreement, and describing it as "about ten per cent" understates it by half.

Every method is an implied efficiency for the remaining work, and stating it that way ends most arguments. Method (a) assumes the remainder runs at **1.0000** — budgeted rate, i.e. a 10.42 % improvement on demonstrated performance. Method (b) assumes **0.9057**, today's `CPI` exactly. Method (c) assumes **0.8360**, the product `CPI × SPI`. So the choice is not between three formulae but between three claims about productivity, and each is falsifiable. The gap between (b) and (c) — **USD 191,389** — is the price the forecast puts on schedule pressure alone; asking "who is claiming that recovering the date will cost a further 16.4 % of efficiency on top of the 9.4 % already lost, and through what mechanism?" is a better question than comparing formula names.

Method (c) is not more conservative, it is a different hypothesis. `CPI × SPI` compounds two indices that are not independent and were never designed to be multiplied, and it is defensible only where a specific, describable mechanism links lateness to cost — overtime, expediting premiums, out-of-sequence work, extended site establishment. Where that mechanism exists, (c) is right and (b) under-forecasts. Where it does not, (c) is pessimism with an equation on it, and it does the same damage as optimism: it destroys the credibility of the next forecast.

Method (d) is the only one whose basis can be checked, and the only one that can be lower than (a). A bottom-up re-estimate of the remainder can legitimately land below USD 4,200,000 — because the remaining scope may be less exposed than the scope that overran — and no index-extrapolation method can ever produce that. Its cost is real (it takes a team a week or more, and it consumes the same engineers who are delivering) and it is the only method that survives cross-examination at a funding gate. The professional practice is therefore: report the index methods weekly as a trend, and commission (d) whenever the decision at stake is money.

WORKED EXAMPLE

Worked example 7.3.3b — where each forecast's funding actually runs out.

1. **Setup.** The board must authorise funding, and it will authorise the `EAC` it is given. Assume the demonstrated `CPI` of **0.905660** persists. For each forecast, how much budgeted work does the money it authorises actually buy — and at what point does the project stop?
2. **Formula.** Funded `ETC` = `EAC` − `AC`. Budgeted work that `ETC` buys at efficiency `e` = `ETC × e`. `EV` at exhaustion = `EV` + that amount; unfunded budgeted work = `BAC` − `EV` at exhaustion. Weeks short = unfunded work ÷ the baseline's average planned rate over the remaining weeks.
3. **Substitution.** Method (a): `ETC = 4,200,000 − 2,120,000 = 2,080,000`; buys `2,080,000 × 0.905660`. Remaining planned rate `= 2,080,000/12` over weeks 14–25.
4. **Result.**

Forecast	Funded ETC	Budgeted work it buys at CPI 0.9057	EV reached	Position
(a) 4,200,000	2,080,000	1,883,774	3,803,774	**95.09 % complete — money gone**
(b) 4,416,667	2,296,667	2,080,000	4,000,000	**exactly complete**
(c) 4,608,056	2,488,056	2,253,333	4,173,333	over-funded by 173,333 of budget-equivalent

Method (a) leaves **USD 196,226** of budgeted work unfunded — **4.91 %** of `BAC`, and at the baseline's average remaining rate of USD 173,333 per week, **1.13 weeks** of work. 5. **Interpretation.** The result is a sentence a sponsor cannot misunderstand: **on its own demonstrated performance, the method-(a) forecast funds Auriga to about one and a bit weeks short of the finish line.** That is a far more useful statement than "the forecast may be USD 216,667 light", because it names when the problem arrives and what it will feel like — a second funding request at 95 % complete, which is the worst moment in a project's life to ask for money, since every alternative to paying has already expired. Method (b)'s exactness is not a coincidence but the definition of the method: `BAC/CPI` is precisely the funding that completes the scope at today's efficiency, which is the same identity KA 7.3.4 finds from the `TCPI` side.

The asymmetry of forecast error is the governance point. Over-forecasting costs the organisation the opportunity value of committed but unspent funds — real, and recoverable at the next portfolio review (Domain 15). Under-forecasting costs a mid-project funding crisis at the moment of maximum sunk cost and minimum optionality, which is where escalation of commitment (Domain 2, KA 2.4.2) does its worst work. **The two errors are not symmetric, so a forecast should not be chosen as though they were**, and a leader who genuinely cannot distinguish between causes should say so and fund nearer (b) than (a) while commissioning method (d).

Method (c)'s "over-funding" is only over-funding if efficiency does not deteriorate. Its USD 2,488,056 buys USD 2,253,333 of budgeted work at `CPI` alone — an apparent surplus of USD 173,333 — but at the compounded rate of 0.8360 it completes the scope exactly, to the dollar. That is the test of whether (c) is right: it is the correct number if and only if the compounded efficiency materialises. Presenting (c) without that conditional is how a project acquires a reputation for hoarding contingency, and presenting (a) without the conditional above is how it acquires a reputation for surprises.

7.3.4 `VAC` and `TCPI`

```
VAC = BAC − EAC                        the forecast overrun/underrun
TCPI = (BAC − EV) / (BAC − AC)         efficiency the REMAINING work must achieve to hit BAC
TCPI = (BAC − EV) / (EAC − AC)         …to hit a revised EAC
```

WORKED EXAMPLE

Worked example 7.3.4 — what recovery would actually require.

1. **Setup.** Auriga's week-13 figures; test recovery to `BAC`, and to the method-(b) forecast.
2. **Formula.** `VAC = BAC − EAC`; `TCPI` as above.
3. **Substitution.** `VAC(a) = 4,000,000 − 4,200,000`; `VAC(b) = 4,000,000 − 4,416,667`. `TCPI(BAC) = 2,080,000 / (4,000,000 − 2,120,000) = 2,080,000/1,880,000`. `TCPI(EAC b) = 2,080,000 / (4,416,667 − 2,120,000)`.
4. **Result.** `VAC(a)` = **(USD 200,000)** · `VAC(b)` = **(USD 416,667)**. `TCPI` to `BAC` = **1.11**; `TCPI` to the (b) forecast = **0.91**.
5. **Interpretation.** To finish at budget, the remaining 52 % of work must run at **1.11** — an 11 % efficiency gain by a team currently achieving 0.91. That is a 22-point swing, and the honest question in the room is what specifically would deliver it. Meanwhile `TCPI` to the (b) forecast is 0.91 — exactly today's `CPI`, which is the arithmetic identity worth internalising: **forecasting with `BAC/CPI` is precisely the assumption that nothing changes.** `TCPI` is the reality check on every recovery promise: a required index far above demonstrated performance is a plan that needs a mechanism, not encouragement.

State the swing as a ratio, not a difference. "1.11 against 0.91" understates the demand, because the two are a ratio: `1.106383/0.905660 =` **1.2216**, so the remaining work must run **22.16 %** more efficiently than everything done so far. A 22 % productivity improvement, mid-project, on the same scope with the same team and the same suppliers, is not a management intention; it is an engineering change, a scope change or a commercial change, and the recovery plan should name which.

The identity generalises, and once it is seen the `EAC` family documents itself. `TCPI` computed to any `EAC` equals the efficiency that `EAC` assumes for the remaining work:

Target	`TCPI` = (`BAC` − `EV`) ÷ (target − `AC`)	Equals
`BAC` 4,000,000	2,080,000 / 1,880,000 = **1.1064**	the recovery demand
`EAC`(a) 4,200,000	2,080,000 / 2,080,000 = **1.0000**	budgeted rate — method (a)'s assumption
`EAC`(b) 4,416,667	2,080,000 / 2,296,667 = **0.9057**	`CPI` — method (b)'s assumption
`EAC`(c) 4,608,056	2,080,000 / 2,488,056 = **0.8360**	`CPI × SPI` — method (c)'s assumption

Choosing an `EAC` is choosing a `TCPI`, so the two should always be reported as a pair; and any `EAC` whose `TCPI` differs from the efficiency its author claims to be assuming contains an arithmetic error. That single check catches more defective cost reports than any other in this domain, and it takes one division.

The caution about `TCPI` runs the other way as well. A `TCPI` **below** demonstrated `CPI` is not good news — it means the target has already been relaxed to something easier than current performance, which is what re-baselining to the forecast does, and it is how a project comes to report favourable indices against a target nobody would have approved at sanction (7.2.3).

Worked example 7.3.4b — what `TCPI` 1.11 costs in scope, not in exhortation.

1. **Setup.** The board will not authorise more than the original **USD 4,000,000** of cost. Auriga's team is achieving `CPI` **0.905660** and can offer no mechanism to improve it. The only remaining lever is

scope: how much budgeted work must come out for the recovery to be arithmetic rather than aspiration?

2. **Formula.** With `D` the budgeted value of scope removed and the cost target `T`: `TCPI = (BAC − D − EV)/(T − AC)`. Setting `TCPI = CPI` and solving, `D = (BAC − EV) − CPI × (T − AC)`.
3. **Substitution.** `D = 2,080,000 − 0.905660 × (4,000,000 − 2,120,000) = 2,080,000 − 1,702,641`.
4. **Result.** `D` = **USD 377,358** — exactly `20,000,000/53` — which is **9.43 %** of `BAC` and **18.14 %** of the work remaining. For comparison, completing the full scope at the demonstrated efficiency needs **USD 416,667** of additional funding (that is `EAC`(b) − `BAC`, i.e. |`VAC`(b)|).
5. **Interpretation.** The identity between those two figures is the one to carry out of this domain: `416,667/377,358 =` **1.104167 = 1/**`CPI`. **At a** `CPI` **of 0.9057, one dollar of budgeted scope removed saves USD 1.10 of forecast cash** — because the scope you do not do is scope you do not do inefficiently. That converts a vague recovery conversation into a single exchange rate the sponsor can act on, and it explains why descoping is a more powerful lever than its size suggests: the leverage is exactly `1/CPI`, and it grows as performance worsens.

The number reframes the meeting. "We need a 22 % efficiency improvement" invites optimism. "Recovery to the approved 4,000,000 requires USD 377,358 of budgeted scope to come out, and here are the three candidate packages" invites a decision, with a named owner, at the authority that owns scope. `TCPI` is what makes the second sentence available.

Three cautions, and the first is the one that gets skipped. Removing 9.43 % of budgeted scope does not remove 9.43 % of the benefit, and it may remove far more: on a control-systems upgrade the cheap-to-build packages are frequently the ones the operating case depends on. Every descope must be priced against benefit (Domain 2, KA 2.3) and tested against acceptance criteria (Domain 5, KA 5.4.2) before its cost saving is booked. Second, a "descope" that is really a **deferral** does not save the money at all — it moves it to a later period and usually a higher price, which makes it a portfolio decision (Domain 15) rather than a project one, and the honest report says so. Third, the arithmetic assumes the removed scope carries its budgeted cost at the same demonstrated efficiency; removing scope from a package that was running well saves less than the formula says, and the sequence for choosing candidates therefore starts from the accounts with the worst performance, not the ones easiest to argue for.

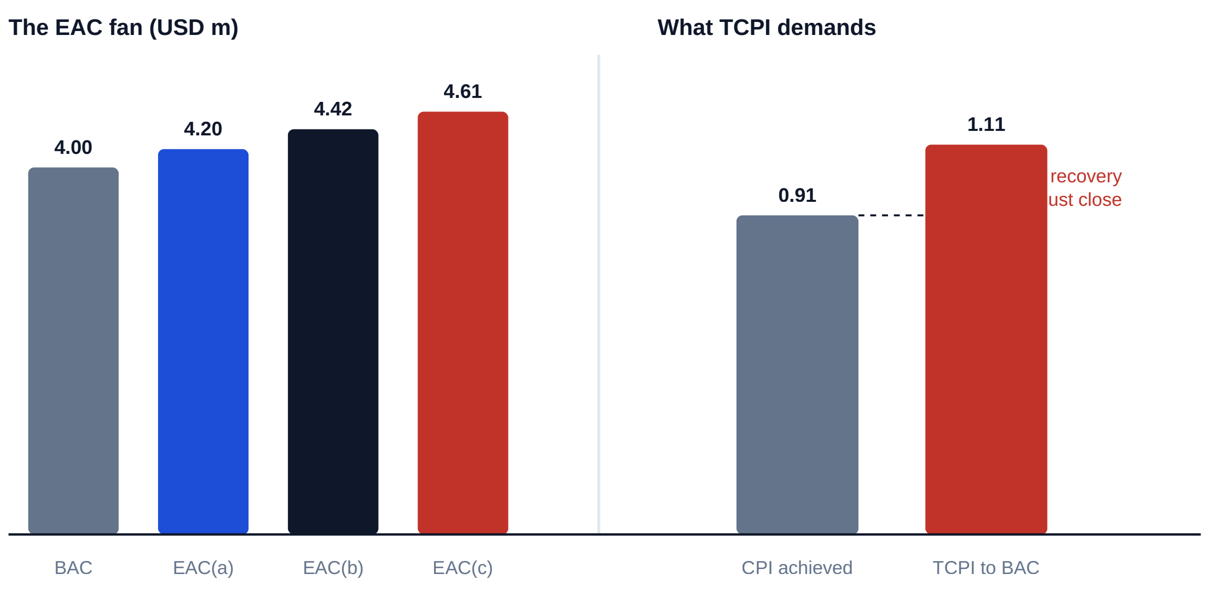


Fig 7.3.2 — The `EAC` fan and what `TCPI` demands

AI in this KA

Earned value is arithmetic on four numbers, so machine computation is safe and instant verification is cheap — which makes unverified AI output indefensible here rather than merely risky. The deterministic checks: `EV/BAC` must equal the claimed percent complete; `CV = EV − AC` and `CPI = EV/AC` recomputed independently; `TCPI` to `BAC/CPI` must equal `CPI` (7.3.4's identity); `TCPI` to each of the other `EAC`s must equal that method's own implied efficiency — 1.0000, `CPI`, `CPI × SPI` — which is a four-division audit of the whole forecast set; the percent-spent-minus-percent-complete gap must reproduce `CV/BAC`; and every `EAC` must state its assumption. A model asked to "check the earned-value numbers" will confirm arithmetic and miss the two things that actually go wrong: an earning rule applied inconsistently between periods, and a level-of-effort share that makes the reported `SPI` meaningless (7.3.1). Both are questions about convention rather than computation, so both stay with the professional. Where AI forecasts from trend data, the calibration rule of Domain 6 (KA 6.4) applies unchanged: show the record of past forecasts against outturn before the number enters a board pack, and name the human who owns it.

KEY TERMS — KA 7.3

Term	Meaning
`PV EV AC`	Budgeted cost of work scheduled · performed · the cost actually incurred.
`CV SV`	`EV − AC` · `EV − PV`, in currency.
`CPI SPI`	`EV/AC` · `EV/PV`, as ratios.
`EAC ETC`	Estimate at / to complete; `EAC = AC + ETC`.
`VAC`	`BAC − EAC` — forecast variance at completion.
`TCPI`	Efficiency the remaining work must achieve to hit a stated target.
Level of effort	Earning by calendar (`EV ≡ PV`); can never show schedule variance.
Earning rule	The convention by which physical progress becomes `EV`; fixed before the period it measures. On Auriga's ten-package group the choice moved `EV` by 15.00 % of budget.
Level-of-effort distortion	`s × (1 − SPI_d)` — the amount a level-of-effort share `s` adds to a reported `SPI`; 0.28 at a 70 % share with discrete `SPI` 0.60.
Implied `ETC` efficiency	The efficiency an `EAC` assumes for the remaining work; equal to the `TCPI` computed to that `EAC`.
Descope exchange rate	`1/CPI` — the forecast cash saved per unit of budgeted scope removed; USD 1.10 per dollar on Auriga.

SAMPLE MCQS — KA 7.3

MCQ 7.3-A `[7.3.2 · Application]` `BAC` 4,000,000; `PV` 2,080,000; `EV` 1,920,000; `AC` 2,120,000. `CPI` and `SPI` are: - A. 0.91 and 0.92 - B. 1.10 and 1.08 - C. 0.92 and 0.91 - D. 0.91 and 1.02

Rationale: `CPI = 1,920,000/2,120,000 = 0.91`; `SPI = 1,920,000/2,080,000 = 0.92`. B inverts both ratios; C swaps them (dividing `EV` by the wrong denominator); D miscomputes `SPI` against `BAC`-derived progress.

MCQ 7.3-B `[7.3.3 · Analysis]` The overrun was caused by a one-off ground-remediation event, now closed, and the remaining work is expected to run to budget. The appropriate `EAC` is: - A. `AC + (BAC − EV)` = 4,200,000 - B. `BAC/CPI` = 4,416,667 - C. `AC + (BAC − EV)/(CPI × SPI)` = 4,608,056 - D. `BAC` = 4,000,000

Rationale: A discrete, closed cause makes the variance **atypical**, so remaining work is forecast at the budgeted rate. B assumes the inefficiency persists and C that it compounds with schedule pressure — both contradict the stated cause. D ignores money already spent above budget.

MCQ 7.3-C `[7.3.4 · Application]` With `BAC` 4,000,000, `EV` 1,920,000 and `AC` 2,120,000, the `TCPI` required to complete at `BAC` is: - A. 0.91 - B. 1.00 - C. 1.11 - D. 1.08

Rationale: `(4,000,000 − 1,920,000)/(4,000,000 − 2,120,000) = 2,080,000/1,880,000 = 1.11`. A is the demonstrated `CPI`; B assumes recovery needs only par performance; D uses `PV` in place of `AC` in the denominator — `2,080,000/1,920,000 = 1.08` — and so makes recovery look nearly free (Exercise 7.3).

MCQ 7.3-D `[7.3.1 · Analysis]` A control account is 70 % level-of-effort by budget. Its reported `SPI` of 1.00 most likely means: - A. the account is exactly on schedule - B. little about schedule: level of effort earns by the calendar, so `EV ≡ PV` for most of the account regardless of progress - C. the discrete work is ahead, offsetting a delay - D. the earning rules were misapplied

Rationale: LOE sets `EV` equal to `PV` by construction, so a heavily-LOE account reads 1.00 whatever happens — which is why practice segregates and caps it. C invents an offset the data cannot show; the rules may have been applied entirely correctly (D), and that is the problem.

MCQ 7.3-E `[7.3.1 · Application]` Ten equal work packages of 40,000: four complete, three at an objectively measured 30 % physical completion, three not started. `EV` under 50/50 and under objective percent-complete earning is: - A. 160,000 and 196,000 - B. 220,000 and 196,000 - C. 220,000 and 220,000 - D. 196,000 and 220,000

Rationale: 50/50 credits half of each in-progress package (`160,000 + 60,000`); percent complete credits the measured 30 % (`160,000 + 36,000`) — 7.3.1. A gives the 0/100 figure for the first rule; C assumes the in-progress work is genuinely 50 % done, which is the only condition under which 50/50 is unbiased; D transposes the two.

MCQ 7.3-F `[7.3.1 · Analysis]` A control account is 70 % level of effort by budget and its discrete work is running at `SPI` 0.60. The account's reported `SPI` is: - A. 0.60 - B. 0.88 - C. 1.00 - D. 0.42

Rationale: `1 − (1 − 0.70)(1 − 0.60) = 0.88`, a distortion of `0.70 × 0.40 = 0.28` (7.3.1). A ignores that the level-of-effort portion reads 1.00 by construction; C assumes it swamps the account entirely; D multiplies the share by the discrete index instead of blending them.

MCQ 7.3-G `[7.3.4 · Analysis]` For which target is `TCPI` exactly 1.00 on Auriga's week-13 figures? - A. `BAC` = 4,000,000 - B. `AC + (BAC − EV)` = 4,200,000 - C. `BAC/CPI` = 4,416,667 - D. `AC + (BAC − EV)/(CPI × SPI)` = 4,608,056

Rationale: `TCPI` to any `EAC` equals the efficiency that `EAC` assumes, and method (a) assumes the budgeted rate (7.3.4). A gives 1.1064, the recovery demand; C gives `CPI` 0.9057; D gives `CPI × SPI` 0.8360 — each the assumption of its own method.

MCQ 7.3-H `[7.3.3 · Evaluation]` The board authorises the method-(a) forecast of 4,200,000 and the demonstrated `CPI` of 0.9057 persists. The funding is exhausted at: - A. 100.00 % complete - B. 95.09 % complete - C. 90.57 % complete - D. 52.00 % complete

Rationale: The funded `ETC` of 2,080,000 buys `2,080,000 × 0.905660 = 1,883,774` of budgeted work, taking `EV` to 3,803,774 (7.3.3b). A is the position only if the remainder runs at the budgeted rate, which is the assumption being tested; C applies `CPI` to the whole of `BAC` rather than to the funded remainder; D is the share of work remaining, not a completion point.

SELF-CHECK — KA 7.3

1. Why is *EV* measured at budget, not at cost? — So it is comparable with PV (schedule) and with AC (efficiency); at cost it would collapse into AC.
2. What does *TCPI* = 1.11 against *CPI* = 0.91 tell a leader? — Recovery to budget requires a 22-point efficiency swing; the plan needs a named mechanism, not optimism.
3. State the identity linking *TCPI* and *BAC/CPI*. — TCPI to an EAC of BAC/CPI equals the current CPI: that forecast is the assumption that nothing changes.
4. Generalise that identity. — TCPI to any EAC equals the efficiency that EAC assumes for the remaining work: 1.0000 for method (a), CPI for (b), CPI × SPI for (c). Choosing an EAC is choosing a TCPI, so report them as a pair.
5. How much scope must come out for Auriga to finish at 4,000,000 without improving? — USD 377,358 of budgeted work, 9.43 % of BAC; and because a dollar of budget removed saves 1/CPI = USD 1.10 of cash, that is equivalent to the USD 416,667 of extra funding method (b) requires.
6. When is 50/50 earning unbiased, and when does it flatter? — Unbiased when in-progress work averages exactly 50 % physical completion; it flatters whenever that average is lower, which is the normal state of a recently started account.

KNOWLEDGE AREA 7.4

Resource economics, procurement strategy and cash

Topics: 7.4.1 resource economics and blended rates · 7.4.2 contract models and cost risk · 7.4.3 incentive fees and the point of total assumption · 7.4.4 cash flow versus profit.

7.4.1 Resource economics and blended rates

Most project cost is people. A leader who reasons in headcount rather than **cost per unit of capability** will mis-price every option in Domain 6's compression decisions.

WORKED EXAMPLE

Worked example 7.4.1 — Auriga's engineering blended rate.

1. **Setup.** The engineering pool is 40 technicians at USD 95/hour, 25 engineers at USD 140/hour, 15 senior specialists at USD 210/hour.
2. **Formula.** Blended rate = Σ(count × rate) / Σ(count).
3. **Substitution.** `(40×95 + 25×140 + 15×210) / 80 = (3,800 + 3,500 + 3,150)/80 = 10,450/80`.
4. **Result. USD 130.63 per hour** (130.625 exactly; ≈ SAR 490 indicatively).
5. **Interpretation.** The blended rate makes options commensurable: a week of schedule compression staffed from this pool costs a computable amount, and shifting the mix changes the price as much as changing the headcount — adding five specialists moves the blend more than adding five technicians. This is the number that turns Domain 6's "crash it" instruction into an expected-value decision.

The mix moves the price more than the size does, and the arithmetic ranks the moves. Re-grading five technicians into specialists — same 80 people — takes the blend to **USD 137.8125**, **+5.50 %**. Adding five specialists to make 85 gives USD 135.2941, **+3.57 %**. Adding five technicians gives USD 128.5294, **−1.60 %**. So the ordering is re-grading, then growing at the top, then growing at the bottom, and a leader who negotiates headcount while the mix moves underneath them has not negotiated the cost at all. The practical control is to fix the mix in the plan, not just the number, and to report it.

The engineer-week that follows is the unit the rest of the book uses. `130.625 × 40 =` **USD 5,225** per engineer-week (≈ SAR 19,594 indicatively), which is the figure Domain 9, KA 9.2.3 uses to cost rework capacity and Domain 8 uses in response pricing. Registering the rate once and reusing it is what makes those numbers comparable across domains.

The blend is only valid for work drawn in the pool's proportions, and the error when it is not is large. A task performed entirely by senior specialists costs USD 210 per hour — **60.77 %** above the blend, an under-pricing of USD 79.375 an hour if the blend is used. That is the commonest misuse of a blended rate: applying it to a small, senior-heavy activity such as commissioning support or an interface investigation, where it can understate cost by more than half again. The discipline is to blend within a grade-homogeneous scope and to price senior-only work at senior rates.

WORKED EXAMPLE

Worked example 7.4.1b — what an hour of capacity actually costs.

1. **Setup.** The USD 95 per hour in the estimate above is a charge-out rate. Auriga's finance team builds the underlying cost of a technician hour: base pay **USD 55.00** per paid hour; statutory and employer on-costs **32 %**; **15 %** of paid hours are non-productive (leave, training, standby, weather, travel); recoverable overhead — site establishment, supervision, tools, information systems — **18 %**. What does one productive hour cost, and what does the estimate's rate actually recover?
2. **Formula.** Paid-hour cost `= base × (1 + on-costs)`. Cost per productive hour `= paid-hour cost ÷ (1 − non-productive share) × (1 + overhead)`. The paid-to-productive multiplier is `1/(1 − n)`.
3. **Substitution.** `55.00 × 1.32 = 72.60`; `72.60/0.85 = 85.4118`; `85.4118 × 1.18`.
4. **Result.** Paid-hour cost **USD 72.60**; per productive hour before overhead **USD 85.4118**; **full cost per productive hour USD 100.7859**. The paid-to-productive multiplier is **1.17647**. Against the USD 95.00 charge-out rate the estimate **under-recovers by USD 5.7859 an hour — 5.74 % of cost.**
5. **Interpretation.** The first duty is to know which rate a number is. **A rate quoted per paid hour understates the cost of a productive hour by `1/(1 − n)` — 17.65 % here — and the two are confused constantly**, because both are called "the labour rate". An estimate built on paid-hour rates against a schedule built on productive hours is under-funded by that multiplier before any risk materialises, and the discrepancy is invisible because both documents look right.

Utilisation is the most leveraged number on the sheet and nobody owns it. Cost per productive hour scales as `1/utilisation`, so losing five points from 85 % to 80 % raises it to **USD 107.0850 — +6.25 %**, exactly the ratio `0.85/0.80`. On Auriga, where labour is 62 % of `BAC` — **USD 2,480,000** — that is **USD 155,000**, which is **77.50 %** of the entire USD 200,000 cost variance the week-13 board meeting exists to discuss. **Five points of utilisation is three-quarters of the variance everyone is arguing about**, and it is lost in ways nobody records: waiting for access, waiting for a permit, waiting for a decision (Domain 3's governance latency, priced in Case study C), re-mobilising after a stand-down.

The 5.74 % under-recovery is a finding, not an error. It may be deliberate — a rate agreed commercially below full cost to win work, with the shortfall funded from elsewhere — or it may be an unnoticed drift as on-costs rose. Which of those it is matters, and the only way to know is to maintain the build-up and re-derive it annually. A charge-out rate carried unchanged for three years is almost certainly recovering less than it did.

A jurisdiction caution that must not be skipped. The composition of on-costs — statutory contributions, leave entitlements, end-of-service or severance provisions, insurance — and their accounting and tax treatment differ substantially between jurisdictions, and so does what may be recovered as overhead in a reimbursable contract. The 32 % and 18 % here are illustrative figures for a fictitious project. Build the rate with your own payroll and finance functions, and where a contract permits cost recovery, confirm the allowable basis with qualified advisers before relying on it.

WORKED EXAMPLE

Worked example 7.4.1c — the price of a crew-week, four ways.

1. **Setup.** Domain 6, KA 6.3.1 establishes the ladder of capacity options — re-sequence, spend float, overtime, second shift, extend — and ranks it by cost without pricing it. Price it. An Auriga field crew is **4 technicians × 40 hours** at the USD 95 charge-out rate. The options for adding one crew-week of capacity: re-sequence within float; overtime at a **1.5×** premium; an agency crew at **USD 128** per hour achieving **85 %** of in-house productivity; or accept the delay, on a zero-float activity, at Domain 6's cost of delay of **USD 45,000 per week**.
2. **Formula.** Straight-time crew-week `= 4 × 40 × rate`. Overtime crew-week `= straight × premium`. Agency cost per productive crew-week `= 4 × 40 × agency rate ÷ relative productivity`. Delay cost per week is given. Breakeven relative productivity between agency and overtime: `agency nominal ÷ overtime cost`.
3. **Substitution.** Straight `= 4 × 40 × 95 = 15,200`. Overtime `= 15,200 × 1.5`. Agency `= 20,480/0.85`. Breakeven `= 20,480/22,800`, equivalently `128/142.50`.
4. **Result.**

Option	Cost per crew-week of capacity	× straight time
Re-sequence within float	**0** cash (float consumed)	—
Straight-time crew (if available)	**15,200**	1.0000
Overtime at 1.5×	**22,800**	1.5000
Agency crew, 85 % productivity	**24,094**	1.5851
Accept the delay (zero float)	**45,000**	2.9605

The agency crew beats overtime only if its relative productivity is at least **89.82 %**; symmetrically, overtime beats the agency crew only while sustained-overtime productivity holds above **94.63 %**. The dearest capacity option is **53.54 %** of the delay cost, and USD 45,000 buys **2.96** straight-time crew-weeks, **1.97** overtime crew-weeks or **1.87** agency crew-weeks. 5. **Interpretation.** The ladder Domain 6 asserts is confirmed and — more usefully — the gaps between its rungs are now numbers.

Any of these beats the delay, so the decision is which to buy, not whether. On a zero-float activity every purchased option costs between **33.78 %** and **53.54 %** of doing nothing. That is the general shape once a cost of delay has been computed, and it is why computing one changes behaviour: without USD 45,000 on the table, "we cannot afford overtime" sounds prudent.

The two breakevens are the real content, because they are the arguments that actually happen. Nobody disputes that an agency crew costs more per hour; the dispute is about how productive it will be, and 89.82 % is the number that dispute is about. Likewise the case against sustained overtime is not that it costs 1.5× but that its productivity decays: **the moment sustained overtime productivity falls below 94.63 %, the agency crew is the cheaper purchase** — at 90 % productivity overtime effectively costs USD 25,333 per crew-week and has lost. Both breakevens are checkable against the project's own timesheet and output records, which converts a preference into a measurement.

Capacity only buys time where the activity is resource-limited. Adding a crew to an activity constrained by logic, curing time, a possession window or a single test facility buys nothing at all (Domain 6, KA 6.2 and 6.3.2), and the four prices above are then all wasted. Establish that the constraint is capacity before pricing capacity.

What the table excludes should be stated when it is used. Onboarding, induction and supervision load for new crews; the lead time each option needs before it can start (Domain 6's point, and often decisive); the quality consequence of fatigue or unfamiliarity, which Domain 9 prices as escaped defects at USD 12,000 each; and the float that the free option consumes, which is the project's risk buffer and has a value Domain 8 can quantify. **"Free" is the option whose price is hardest to see**, and a leader who spends float without saying so has bought capacity on credit.

7.4.2 Contract models and cost risk

Every contract model is an answer to one question: **who carries cost risk?**

Model	Buyer pays	Cost risk sits with	Suits
Firm fixed price (FFP)	An agreed price, whatever it costs	**Seller**	Well-defined scope; priced-in risk premium
Fixed price incentive (FPIF)	Target cost/fee with a share ratio, capped by a ceiling	**Shared, then seller above the ceiling**	Definable scope with real uncertainty
Cost plus fixed fee (CPFF)	Allowable cost + a fixed fee	**Buyer**	Development, unclear scope
Cost plus incentive fee (CPIF)	Allowable cost + a fee varying with performance	**Shared**	Scope where effort is uncertain but effort quality matters
Time and materials (T&M)	Rates × quantities	**Buyer**	Staff augmentation, short or open-ended work

The leader's discipline: **risk transferred is risk priced.** An FFP contract for ill-defined scope does not remove the risk — it converts it into a premium plus a claims exposure when the scope moves (Domain 10, KA 10.4). Conversely a cost-plus contract on well-defined work pays for uncertainty that no longer exists. Domain 10, KA 10.3.2 works the full outturn arithmetic — what each model pays the two parties at the same actual cost, and why an expected-value comparison of fixed price against cost-plus buys variance rather than money. What belongs here is the calculation a delivery leader does before that conversation starts: the point at which the simplest two options cross.

WORKED EXAMPLE

Worked example 7.4.2 — fixed price or time and materials: where they cross.

1. **Setup.** A defined configuration-and-testing package on Auriga. A supplier quotes a firm fixed price of **USD 480,000**. The alternative is time and materials from the same engineering pool at the blended **USD 130.625** per hour (KA 7.4.1). The team's own three-point estimate of the hours is **optimistic 2,900, most likely 3,200, pessimistic 4,100**.
2. **Formula.** Breakeven hours `= fixed price ÷ hourly rate`. T&M cost `= hours × rate`. Premium `= fixed price − T&M cost at the estimate`. PERT expected hours `= (o + 4m + p)/6`; triangular mean `= (o + m + p)/3`. For a triangular distribution, the probability of exceeding a value `x` above the mode is `(b − x)²/((b − a)(b − m))`.
3. **Substitution.** Breakeven `= 480,000/130.625`. T&M at the mode `= 3,200 × 130.625`. PERT hours `= (2,900 + 12,800 + 4,100)/6`. Exceedance `= (4,100 − 3,674.6411)²/(1,200 × 900)`.
4. **Result.** Breakeven **3,674.64 hours**. At the most likely 3,200 hours, T&M costs **USD 418,000**, so the fixed price carries a premium of **USD 62,000 — 14.83 %** over the estimate (12.92 % of the fixed price itself). At the pessimistic 4,100 hours T&M costs **USD 535,563**, USD 55,563 worse than the fixed price; at the optimistic 2,900 it costs **USD 378,813**, USD 101,188 better. PERT expected hours are **3,300** — T&M **USD 431,063**, a premium of **11.35 %**; the triangular mean is **3,400 hours** — T&M **USD 444,125**, a premium of **8.08 %**. The probability that the hours exceed the breakeven is **16.75 %**.
5. **Interpretation.** The identity is exact and worth keeping: **the fixed-price premium expressed as a percentage of the estimate is the hours overrun at which the two options break even.** 14.83 % premium, 14.83 % overrun tolerance. So the question at the table is never "is USD 62,000 a lot?" but "how confident are we that this package will not run more than 14.83 % over its estimated hours?" — a question the estimating team can actually answer, and one whose answer here is not very, since a 14.83 % overrun on 3,200 hours is well inside the declared range.

State the premium against expectation, not against the mode. Against expected hours the premium is **8.08 % to 11.35 %** depending on the distribution shape, not 14.83 %. Quoting the larger figure makes the fixed price look worse than it is, and quoting no figure at all is how fixed prices get accepted or rejected on temperament. What the buyer receives for that USD 35,875–48,938 is the removal of a 16.75 % chance of paying more than 480,000 and of a tail that reaches USD 535,563 — the variance-for-money trade Domain 10, KA 10.3.2 quantifies properly.

The distributions are not the same distribution, and this is the error the arithmetic hides. Under time and materials nobody is paid to be efficient: the supplier's revenue rises with the hours, so the hours estimate that was drawn for an in-house or fixed-price case does not transfer. The correct comparison prices T&M against a shifted distribution, and since the size of the shift is unknowable in advance, the practical countermeasures are structural — an hours cap or not-to-exceed envelope, a fixed rate schedule, an agreed productivity or output measure, and approval gates at stated cumulative hours. **A T&M arrangement without a cap is not a contract model, it is an open account.**

And the fixed price only covers the scope specified. The 14.83 % tolerance is worthless if the scope is 30 % defined, because the variations will arrive priced without competition (MCQ 7.4-C, and Domain 10, KA 10.4.1 on entitlement). The order of operations is therefore fixed: define the scope, then choose the model, then compute the breakeven — never the reverse.

7.4.3 Incentive fees and the point of total assumption

The mechanics. Under an incentive contract, buyer and seller agree a **target cost**, a **target fee** and a **share ratio** (buyer/seller) for over- and under-runs, with a **ceiling price** capping the buyer's exposure.

WORKED EXAMPLE

Worked example 7.4.3 — the incentive that stops incentivising.

1. **Setup.** Auriga's installation subcontract: target cost **USD 2,000,000**, target fee **USD 150,000**, share ratio **70/30** (buyer 70 %, seller 30 %), ceiling price **USD 2,450,000**. The seller finishes at an actual cost of **USD 2,300,000**.
2. **Formula.** Fee = target fee − (overrun × seller share). Buyer pays = actual cost + fee (subject to the ceiling). Point of total assumption `PTA = target cost + (ceiling − target price)/buyer share`, where target price = target cost + target fee.
3. **Substitution.** Overrun `2,300,000 − 2,000,000 = 300,000`; fee `150,000 − 300,000 × 0.30 = 150,000 − 90,000`. Buyer pays `2,300,000 + 60,000`. Target price `2,150,000`; `PTA = 2,000,000 + (2,450,000 − 2,150,000)/0.70`.
4. **Result.** Fee **USD 60,000** (down from 150,000); buyer pays **USD 2,360,000**. `PTA` **= USD 2,428,571.43** — and at that cost the buyer pays exactly the USD 2,450,000 ceiling.
5. **Interpretation.** The share ratio does its job up to the `PTA`: both parties lose money on overrun, so both want efficiency. **Above the `PTA` the seller bears 100 %** of further cost — which is the moment the incentive inverts: a seller heading past it has no financial reason to spend more on your project and every reason to argue that the extra cost is your scope change. Knowing where the `PTA` sits is therefore a **delivery** insight, not an accounting one: it predicts when a commercial relationship will turn adversarial (Domain 10, KA 10.4; Domain 11's negotiation).

Verify the `PTA` by walking it, because the check is one line and it catches sign errors. At a cost of 2,428,571.43 the overrun is 428,571.43, so the fee is `150,000 − 0.30 × 428,571.43 =` **USD 21,428.57**, and the buyer pays `2,428,571.43 + 21,428.57 =` **USD 2,450,000** — the ceiling, exactly. Any `PTA` that fails to reproduce the ceiling this way has been computed with the wrong share.

The fee is not yet exhausted at the `PTA`, and that boundary has a rule. The seller's fee reaches zero only at `2,000,000 + 150,000/0.30 =` **USD 2,500,000**, above the `PTA`. So on this structure the ceiling binds first, which is what makes the `PTA` the meaningful inflection. The general condition is worth carrying: **the ceiling binds before the fee is exhausted only while the ceiling is below** `target price + target fee × buyer share ÷ seller share` — here `2,150,000 + 150,000 × (0.70/0.30) =` **USD 2,500,000**. At a ceiling of exactly 2,500,000 the `PTA` and the zero-fee cost coincide at 2,500,000; above it the `PTA` is a nominal figure, because the seller has already lost its whole fee before the buyer's cap engages. A leader shown a `PTA` should therefore also be shown the zero-fee cost, and told which comes first.

The share ratio moves the `PTA`, and it moves it the way nobody expects.

Buyer / seller share	PTA	Fee at the PTA	Comes before the zero-fee cost?
50 / 50	2,600,000	(150,000)	No — zero-fee cost is 2,300,000
60 / 40	2,500,000	(50,000)	No — zero-fee cost is 2,375,000
66.67 / 33.33	**2,450,000**	**0**	Exactly coincides, at the ceiling
70 / 30 (Auriga)	**2,428,571**	21,429	Yes — zero-fee cost is 2,500,000
80 / 20	2,375,000	75,000	Yes
90 / 10	2,333,333	116,667	Yes

The more of the overrun the buyer absorbs, the earlier the inflection arrives — moving from 70/30 to a superficially more generous 90/10 brings the `PTA` forward by **USD 95,238**, because the buyer's fixed headroom to the ceiling is consumed faster. A buyer who concedes share in negotiation to appear reasonable has bought an earlier adversarial turn, and should know it. Note too the boundary in the third row: on this structure a buyer share below **66.67 %** means the ceiling never binds before the fee is gone.

Sensitivity to the ceiling is `1/buyer share`, so the ceiling is the stronger lever. Each dollar added to the ceiling moves the `PTA` by **1.4286** dollars at a 70 % buyer share. And the figure to monitor monthly is the headroom: with the seller's cost forecast at 2,300,000, the distance to the `PTA` is **USD 128,571 — 5.59 %** of that forecast. **A single 6 % cost movement puts this relationship past its inflection**, which is why toolkit 7.T.3 asks for the cost trend against the `PTA` at every commercial checkpoint rather than the `PTA` alone. Domain 10, KA 10.3.2 reads the same figure as a statement about the seller's marginal dollar.

A counsel pointer, because this is an enforceability-sensitive area. Ceilings, share ratios, fee adjustment mechanics and their interaction with variation, notice and claims provisions vary by contract form and by jurisdiction, and so does the treatment of a seller trading past a ceiling. The arithmetic here is contract-neutral; the drafting is not. Take qualified legal advice on the instrument before relying on any of it commercially (Domain 10, KA 10.4).

7.4.4 Cash flow versus profit

A project can be profitable and still fail for lack of cash — the same truth PFL-AI establishes for financings (its Domain 1). For a delivery leader the mechanism is payment terms: cost is incurred as work happens; cash arrives when invoices are approved and paid. On Auriga, with `AC` = USD 2,120,000 at week 13 and 60-day terms, roughly **USD 742,000** of incurred cost (about 35 %) is still unpaid — an exposure carried by whoever is funding the work. The leader's obligations are practical: know the terms in both directions (client and subcontractors), front-load nothing you cannot fund, and never let a retention or milestone-payment structure be agreed without someone computing its cash profile. Where the project sits inside a financed asset, this is precisely the CFADS conversation PFL-AI Domain 10 has with lenders.

That last obligation is the one routinely left undone, so here is the computation.

WORKED EXAMPLE

Worked example 7.4.4 — Auriga's cash, its profit, and the gap between them.

1. **Setup.** Auriga is delivered under a fixed price to the utility of **USD 4,400,000** against a `BAC` of 4,000,000 — a bid margin of USD 400,000. Work is valued monthly (weeks 4, 8, 12 ...) at the contract rate, **5 % retention** is withheld from each payment, and the client's terms are **60 days** from invoice. Half the retention is released at practical completion and half after the defects period. At week 13: `EV` **1,920,000**, of which **1,760,000** had been earned by the week-12 valuation, **880,000** by the week-8 valuation and **320,000** by the week-4 valuation; `AC` **2,120,000**, of which **USD 742,000** sits in supplier invoices within terms and accruals not yet paid. By week 13 only the week-4 invoice has been paid. What is the position?
2. **Formula.** Revenue recognised `= (price/BAC) × EV`. Margin to date `=` revenue `−` `AC`. Cash paid out `=` `AC` `−` payables. Cash received `=` net-of-retention value of invoices whose terms have expired. Funding absorbed `=` cash paid out `−` cash received. Days of spend `=` funding absorbed `÷` (`AC` ÷ elapsed days). Client exposure `=` receivables `+` retention held `+` unbilled work in progress.
3. **Substitution.** Price factor `= 4,400,000/4,000,000 = 1.10`. Revenue `= 1.10 × 1,920,000`. Certified to week 12 `= 1.10 × 1,760,000 = 1,936,000`, retention `= 96,800`. Week-4 payment `= 1.10 × 320,000 × 0.95`. Cash out `= 2,120,000 − 742,000`. Days of spend `= 1,043,600 ÷ (2,120,000/91)`.
4. **Result.**

Position at week 13	USD
Cost incurred (`AC`)	2,120,000
less payables (35.00 % of `AC`)	(742,000)
Cash paid out	**1,378,000**
less cash received (the week-4 invoice, net of retention)	(334,400)
Funding absorbed	**1,043,600**
Revenue recognised at 1.10 × `EV`	2,112,000
Margin to date	**(8,000)**
Receivables outstanding	1,504,800
Retention held	96,800
Unbilled work in progress (week 13, at 1.10)	176,000
Total client exposure	**1,777,600**

The funding absorbed is **26.09 %** of `BAC` and **44.80 days** of spend at the current rate of USD 23,297 a day. 5. **Interpretation.** Set the two headline numbers side by side. **The cost report's headline is a variance of USD 200,000. The bank account's headline is USD 1,043,600 — 5.22 times** larger, payable now, and absent from every document in the week-13 pack.

The working-capital identity ties the two statements together and should be used as a check. Funding absorbed `=` client exposure `−` payables `+` loss to date: `1,777,600 − 742,000 + 8,000 =` **1,043,600**, exactly. So cash is not a separate subject from performance; it is performance plus the timing of two sets of payment terms. A cash forecast that cannot be reconciled to the cost report through that identity has an error in one of them.

Express the requirement in days of spend, because that is the number that transfers. USD 1,043,600 means nothing across a portfolio; **44.80 days of spend** is immediately comparable between projects of different sizes and immediately actionable by a treasurer. A project running at 45 days of spend has a month and a half of its own cost outstanding at all times, and every week of client payment delay adds directly to it.

Payment terms are the largest single lever, and they are agreed by people who never see this table. Moving the client's terms from 60 days to 30 brings the week-8 invoice inside the data date — a further **USD 585,200** received — and cuts the funding absorbed to **USD 458,400**, a **56.08 %** reduction. Nothing about the work changes. That is the whole of the case for treating payment terms as a delivery parameter rather than a procurement formality, and for computing the profile before signature: after signature the same improvement has to be bought.

Retention is margin that is not cash, and the proportion is startling. Retention at completion will be `5 % × 4,400,000 =` **USD 220,000 — 55.00 %** of the entire bid margin — with **USD 110,000 (27.50 %** of margin) held beyond practical completion into the defects period. A project can therefore be complete, profitable and still waiting on more than a quarter of its profit, which is why release criteria and their evidence belong in the closeout plan (Domain 16) rather than being discovered afterwards.

The forecast method decides whether this contract makes money at all. At the fixed price of 4,400,000: method (a)'s `EAC` of 4,200,000 leaves a margin of **USD 200,000 (+4.55 %** of price); method (b)'s 4,416,667 leaves **(USD 16,667) (−0.38 %)**; method (c)'s 4,608,056 leaves **(USD 208,056) (−4.73 %)**. So KA 7.3.3's choice of assumption is not a reporting preference — it is the difference between a profitable contract and a loss-making one, and preserving the bid margin of 400,000 requires an `EAC` of exactly `BAC`, which is `TCPI` 1.11 (KA 7.3.4). **The `TCPI` conversation and the margin conversation are the same conversation**, and a commercial manager and a project controller who do not know that will present two irreconcilable papers to the same board.

An accounting caution. How revenue is recognised on a contract, and how retentions, accruals and unbilled work in progress are presented, are matters for the applicable financial reporting framework and the entity's own accounting policy, and they vary by jurisdiction. The 1.10 × `EV` basis used above is a transparent illustrative convention for a fictitious project, not a statement of how any entity must report. Agree the basis with the finance function before any figure of this kind leaves the project.

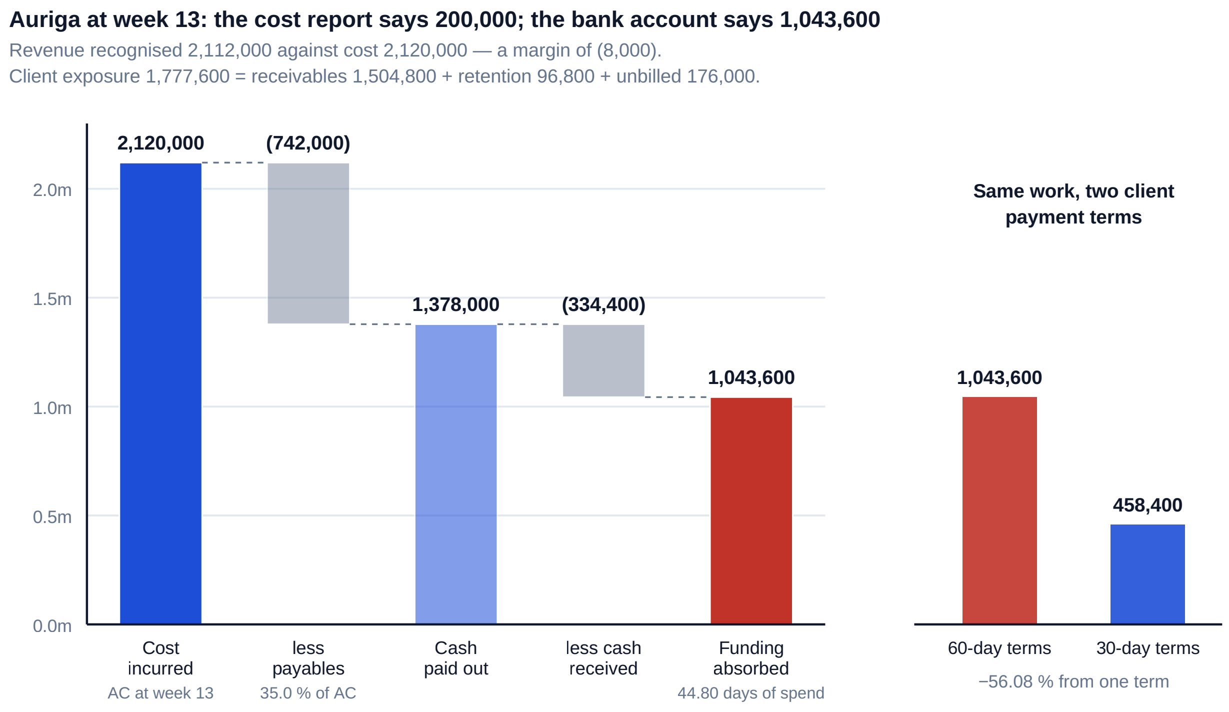


Fig 7.4.1 — Auriga at week 13: the cost report says 200,000; the bank account says 1,043,600

AI in this KA

Contract analytics — extracting terms, comparing models, flagging inconsistent clauses — is real and useful AI assistance, and it is also decision support rather than legal or commercial advice. Two boundaries: an AI-produced reading of a clause is verified against the clause itself before anyone relies on it (the document-against-summary check), and commercial and legal positions go to qualified counsel (Domain 10). Fee arithmetic and PTA computations are deterministic — they get the same golden-answer treatment as the earned-value set.

Three further placements are worth being specific about, because the temptation in this KA is to delegate the wrong half. Assembling a **rate build-up** from payroll and overhead data across many grades and cost centres — the mechanical part of 7.4.1b — is a strong application, and the resulting cost per productive hour is exactly checkable. Sweeping a portfolio of contracts for payment terms, retention percentages and release criteria and producing the aggregate cash profile of 7.4.4 is work most organisations do not do at all because it is tedious across dozens of documents; a model does it quickly and every extracted term is verifiable against its clause. What must **not** be delegated is the **utilisation** assumption and the **relative productivity** figure of 7.4.1c: both are judgements about a specific workforce in a specific setting, both drive their answers almost entirely, and a model asked for either will supply a confident, well-formatted number with no provenance — the failure mode Domain 9, KA 9.1 describes for detection rates. Where the evidence does not exist, the correct output is the breakeven — 89.82 %, 94.63 % — plus the statement that nobody has measured which side of it the project sits on.

KEY TERMS — KA 7.4

Term	Meaning
Blended rate	Weighted average cost per hour of a mixed resource pool.
FFP / FPIF / CPFF / CPIF / T&M	Contract models ordered by who carries cost risk.
Share ratio	The agreed buyer/seller split of over- and under-run.
Ceiling price	The cap on the buyer's total exposure under an incentive contract.
Point of total assumption (`PTA`)	The cost above which the seller bears 100 % of further overrun.
Zero-fee cost	`target cost + target fee ÷ seller share` — where the incentive fee is exhausted; on Auriga USD 2,500,000, above the `PTA`, which is why the ceiling binds first.
`PTA` headroom	The distance from the seller's current cost forecast to the `PTA`; 5.59 % on Auriga, and the figure to trend monthly.
Cost per productive hour	Burdened pay ÷ (1 − non-productive share) × (1 + overhead); USD 100.79 against a USD 95.00 charge-out rate.
Paid-to-productive multiplier	`1/(1 − n)` — 1.17647 at a 15 % non-productive share; the commonest silent under-funding in a labour estimate.
Cost per crew-week of capacity	The priced rungs of Domain 6's capacity ladder, from 15,200 at straight time to 45,000 for accepting the delay (7.4.1c).
T&M/fixed-price breakeven	`fixed price ÷ hourly rate`; the premium as a percentage of the estimate equals the hours overrun at which the two options cross.
Retention	A withheld percentage of payment, released on completion criteria; Auriga's 5 % is 55.00 % of the bid margin at completion.
Funding absorbed	Cash paid out − cash received; equals client exposure − payables + loss to date. Best stated in days of spend (44.80 on Auriga).

SAMPLE MCQS — KA 7.4

MCQ 7.4-A `[7.4.3 · Application]` Target cost 2,000,000; target fee 150,000; share 70/30; actual cost 2,300,000. The seller's fee is: - A. USD 150,000 - B. USD 60,000 - C. USD 90,000 - D. USD 45,000

Rationale: The seller absorbs 30 % of the 300,000 overrun: `150,000 − 90,000 = 60,000`. A ignores the incentive; C states the fee reduction rather than the fee; D applies the buyer's share to the fee.

MCQ 7.4-B `[7.4.3 · Analysis]` Target cost 2,000,000, target fee 150,000, ceiling 2,450,000, buyer share 70 %. The `PTA` is 2,428,571, and its delivery significance is that above it: - A. the contract becomes void - B. the buyer absorbs all further cost - C. the seller bears 100 % of further cost, so the incentive inverts and cost growth becomes a scope-change argument - D. the fee becomes negative but risk-sharing continues unchanged

Rationale: Beyond the `PTA` the ceiling binds the buyer, so every further dollar is the seller's — which predictably redirects the seller's effort from efficiency to entitlement. B reverses the exposure; A is fiction; D misses that sharing has stopped.

MCQ 7.4-C `[7.4.2 · Analysis]` A leader lets an FFP contract for scope that is only 30 % defined. The most likely outcome is: - A. cost risk is genuinely eliminated - B. a priced-in risk premium plus a claims-and-variations exposure as the scope is defined - C. the seller absorbs all scope growth at no cost to the buyer - D. the contract converts automatically to cost-plus

Rationale: Fixed price transfers risk at a price and only for the scope actually specified; undefined scope returns as variations. A and C mistake the contractual form for the underlying uncertainty; D invents a mechanism.

MCQ 7.4-D `[7.4.1 · Application]` A pool of 40 at USD 95/h, 25 at USD 140/h and 15 at USD 210/h has a blended rate of: - A. USD 148.33 - B. USD 130.63 - C. USD 112.31 - D. USD 140.00

Rationale: `10,450/80 = 130.63`. A averages the three rates unweighted (`445/3`); C blends only the two cheapest grades, dropping the specialists (`7,300/65`); D takes the middle rate as representative.

MCQ 7.4-E `[7.4.1 · Application]` Base pay USD 55.00 per paid hour; on-costs 32 %; 15 % of paid hours are non-productive; recoverable overhead 18 %. The full cost of one productive hour is: - A. USD 85.41 - B. USD 100.79 - C. USD 98.52 - D. USD 72.60

Rationale: `55.00 × 1.32 ÷ 0.85 × 1.18 = 100.7859` (7.4.1b). A stops before overhead; C adds 15 % rather than dividing by 0.85, understating by the difference between an uplift and a divisor; D is the paid-hour cost, which is what most rate cards actually contain.

MCQ 7.4-F `[7.4.1 · Evaluation]` An agency crew charges USD 128 per hour against an in-house USD 95 with overtime at 1.5×. The agency crew is the cheaper purchase per productive crew-week only if its relative productivity is at least: - A. 74.22 % - B. 89.82 % - C. 100.00 % - D. 66.67 %

Rationale: `128/142.50 = 0.8982` — the agency rate against the overtime rate (7.4.1c). A compares the agency rate with straight time and so ignores the overtime premium being avoided; C assumes any productivity shortfall disqualifies the option, which the arithmetic refutes; D inverts the 1.5 premium.

MCQ 7.4-G `[7.4.2 · Application]` A firm fixed price of USD 480,000 is offered against time and materials at USD 130.625 per hour, with the package estimated at 3,200 hours. The breakeven hours and the premium over the estimate are: - A. 3,674.64 hours and 14.83 % - B. 3,674.64 hours and 12.92 % - C. 3,428.57 hours and 14.83 % - D. 3,200 hours and nil

Rationale: `480,000/130.625 = 3,674.64`; the premium is `62,000/418,000 = 14.83 %`, which equals the hours overrun at breakeven (7.4.2). B expresses the premium against the fixed price rather than the estimate, understating the tolerance the buyer is purchasing; C uses the mid-grade rate of 140 instead of the blend; D assumes the estimate is the crossing point.

MCQ 7.4-H `[7.4.4 · Application]` The contract price is USD 4,400,000 against a `BAC` of 4,000,000. At week 13 `EV` is 1,920,000 and `AC` is 2,120,000. The margin recognised to date is: - A. USD 400,000 - B. (USD 8,000) - C. USD 200,000 - D. (USD 200,000)

Rationale: Revenue at `1.10 × 1,920,000 = 2,112,000` against cost of 2,120,000 (7.4.4). A is the bid margin, which assumes nothing has gone wrong; C and D read the cost variance as the margin, which omits the 10 % mark-up earned on the work performed and so mis-states the position in both directions.

■ SELF-CHECK — KA 7.4

1. State the one question every contract model answers. — Who carries cost risk.
2. Why is the *PTA* a delivery concern, not just a commercial one? — Above it the seller bears all further cost, so behaviour shifts from efficiency to entitlement.
3. How can a profitable project run out of cash? — Cost is incurred as work happens; cash arrives on payment terms — the gap must be funded.

4. Which way does the *`PTA`* move when the buyer takes a larger share of the overrun, and why? — Earlier: fixed headroom to the ceiling is consumed faster, so 70/30 to 90/10 brings it forward by USD 95,238. Below a 66.67 % buyer share this structure's ceiling never binds before the fee is exhausted.
5. What is the difference between a paid-hour rate and a productive-hour rate? — The factor `1/(1 − n)` — 17.65 % at Auriga's 15 % non-productive share — and confusing them under-funds a labour estimate before any risk materialises.
6. State Auriga's capacity ladder in money. — Per crew-week: re-sequencing free but paid in float, straight time 15,200, overtime 22,800, agency at 85 % productivity 24,094, accepting the delay 45,000 — so on a zero-float activity every purchased option beats doing nothing.
7. What does the funding absorbed at week 13 equal, and how is it best expressed? — USD 1,043,600 — client exposure 1,777,600 less payables 742,000 plus the 8,000 loss — best stated as **44.80 days of spend**, and halved to 458,400 by 30-day rather than 60-day client terms.

Advanced topics — Domain 7

7.A.1 Earned schedule — closing `SPI`'s late-project blind spot

`SPI` converges on 1.00 as a project finishes, whatever its lateness, because `EV` and `PV` both approach `BAC`. **Earned schedule** restates progress in time: `ES` is the date at which the value now earned was planned to have been earned, and `SPI(t) = ES / AT`. This is the bridge flagged in Domain 6 (KA 6.4.3), now with its cost-side companions — and the size of the blind spot it closes is worth computing rather than asserting.

WORKED EXAMPLE

Worked example 7.A.1 — the same project, read twice, at week 13 and at week 27.

1. **Setup.** Auriga's baseline `PV` curve reaches **1,920,000 at week 12** and **2,080,000 at week 13** — a local slope of USD 160,000 a week. At week 13 the project has earned exactly **1,920,000**. Now project forward: suppose the work is not complete at week 25 and at **week 27** `EV` stands at **3,920,000**, against a baseline whose `PV` was **3,880,000 at week 24** and reaches `BAC` **4,000,000 at week 25** (after which `PV` stays at `BAC`, having nothing left to plan). Planned duration `PD` = **25 weeks**. Read the schedule position both ways at both dates.
2. **Formula.** `ES` = the baseline time at which the `EV` now earned was planned to be earned, interpolated between the bracketing periods: `ES = t + (EV − PV_t)/(PV_{t+1} − PV_t)`. `SPI(t) = ES/AT`, with `AT` the actual time elapsed. Time forecast `IEAC(t) = PD/SPI(t)`. For comparison `SPI = EV/PV`.
3. **Substitution.** Week 13: `ES = 12` exactly, so `SPI(t) = 12/13`; `SPI = 1,920,000/2,080,000`. Week 27: `ES = 24 + (3,920,000 − 3,880,000)/(4,000,000 − 3,880,000) = 24 + 1/3`; `SPI(t) = 24.3333/27`; `SPI = 3,920,000/4,000,000`; `IEAC(t) = 25/0.901235`.
4. **Result.**

	SPI	ES	SPI(t)	Time forecast
Week 13	**0.9231**	12.0000	**0.9231**	27.0833 weeks — 2.0833 late
Week 27	**0.9800**	24.3333	**0.9012**	27.7397 weeks — 2.7397 late

At week 13 the two indices agree exactly. At week 27 they differ by **0.0788**: `SPI` reports the project **2 %** behind while it is in fact **2.74 weeks — 10.96 %** — late. 5. **Interpretation.** The week-13 agreement is not a coincidence and it is the more instructive half. **`SPI` and `SPI(t)` coincide wherever the `PV` curve is locally linear**, because `SV/slope` is then exactly the number of weeks of lateness: Auriga's `SV` of (160,000) is one week of planned value at 160,000 a week, so `ES` is 12 and `SPI(t)` is 12/13 — the same 0.9231 that `EV/PV` gives. That is why `SPI` is perfectly serviceable mid-project on a smooth S-curve, and it is worth saying so rather than presenting earned schedule as a correction to a broken measure.

The divergence appears where the curve flattens, which is exactly where the decisions get expensive. Past the planned finish, `PV` is pinned at `BAC` and has nothing left to compare against, so `SPI` is arithmetically compelled towards 1.00 and reports 0.98 for a project nearly three weeks late. `SPI(t)` keeps working because its denominator is time, which does not run out. The rule for practice: **quote `SPI` while the `PV` curve is still rising steeply, and `SPI(t)` from the point at which it flattens** — and never quote `SPI` in the last 15 % of a project without the time-based figure beside it (7.A.2's limitation list; the exam-preparation trap is the same one).

The time forecast is what makes the index worth computing, because it converts into money at a known rate. `PD/SPI(t)` gives 27.08 weeks at the data date — 2.0833 weeks late — which at Domain 6's cost of delay of USD 45,000 per week is **USD 93,750** of delay cost, a figure directly comparable with the compression options Domain 6 prices and with the recovery spend the week-13 case study argues about. Note the consistency check: at week 27 the remaining baseline work is `25 − 24.3333 =` 0.6667 weeks, which at the demonstrated time-efficiency of 0.9012 takes 0.7397 weeks — completion at 27.7397, exactly the `IEAC(t)`. Any earned-schedule presentation should reproduce that agreement.

Two cautions, both about the baseline rather than the technique. `ES` is read off the *`PV`* curve, so it inherits every defect of that curve — a curve phased by straight-line spread rather than by the schedule gives a confidently wrong `ES`, and the interpolation above assumes the curve is meaningful between period points. And re-baselining resets `ES` exactly as it resets `SPI` (7.2.3): a project that re-phases its `PV` curve to match its actual progress will report `SPI(t)` of 1.00 while remaining as late as it was the day before, which is why the integrity rules of KA 7.2.3 are a precondition for every index in this domain and not only for the cost ones.

7.A.2 EVM's limitations, stated plainly

Earned value measures conformance to plan, not value to the customer: a project can score 1.00 on both indices while building the wrong thing (Domain 5's acceptance criteria are the defence). It says nothing about quality (Domain 9). It is only as honest as its earning rules and its accruals. And it is blind to risk that has not yet materialised — Domain 8's contingency analysis, not `CPI`, tells you whether what remains is adequately funded. EVM is a measurement system, not a management philosophy; leaders who treat the indices as targets get optimised indices.

7.A.3 The reviewer's cost eye

Invariants a reviewer runs before trusting a cost report: `EV/BAC` equals the claimed percent complete; `CV = EV − AC` and `SV = EV − PV` reconcile to the stated indices; `EAC` = `AC` + a stated `ETC`, with the assumption named; `TCPI` to `BAC/CPI` equals `CPI`; sum of control-account budgets + contingency equals `BAC`; management reserve sits outside that total; no completed package's budget has moved since last period; accruals are present in the current period; and level-of-effort share is disclosed per control account. Any violation is a defect somewhere — find it before the board builds on it.

Seven more, each of which this domain has now derived, and each a single line of arithmetic:

- **The percent-spent minus percent-complete gap must equal `CV/BAC`.** Auriga: 53.00 % − 48.00 % = 5.00 %, and `200,000/4,000,000 = 5.00 %` (7.3.2).
- `TCPI` **to every reported** `EAC` **must equal that method's own implied efficiency** — 1.0000 for method (a), `CPI` for (b), `CPI × SPI` for (c). A mismatch means the forecast and its stated assumption are not the same document (7.3.4).
- **Contingency drawn plus unattributed variance must equal** `CV`**.** Auriga's USD 190,000 of draws against a `CV` of 200,000 leaves USD 10,000 with no named cause (7.1.3).
- `AC` **must not be below invoices plus receipted-but-uninvoiced value**, and a cumulative `CPI` of exactly 1.0000 is a prompt to check that before it is a cause for satisfaction (7.2.1).
- **Commitment coverage must be stated wherever prices are moving**, and the excess of the forecast's uncommitted balance over the baseline's must equal |`VAC`| (7.2.1b).
- **The earning rule per package must be the same rule as last period**, and one account should be recomputed under an alternative rule to size the discretion involved — 15 % of budget on Auriga's ten-package group (7.3.1).
- `SPI` **must not be quoted alone in the last 15 % of a project**, where it is compelled towards 1.00; the time-based figure belongs beside it (7.A.1).

And two outside the earned-value set that decide whether the report describes a viable project at all: the **funding absorbed**, stated in days of spend (44.80 on Auriga), and the `PTA` **headroom** on every incentive package (5.59 %). Neither appears in a conventional cost report, and both change decisions.

Industry variations — Domain 7

Each variation below changes a specific number in this domain, not merely its emphasis.

- **Construction and EPC.** Full EVM against a resource-loaded baseline, with `EV` claimed from measured-work valuations — so the earning rule is effectively units-completed and the earning-rule discretion of 7.3.1 is largely designed out, at the cost of a monthly measurement effort. Retention is the binding cash term: on Auriga's structure a 5 % retention is **USD 220,000**, or **55.00 %** of the bid margin, and half of it is held past completion. Variation and claims machinery under the standard international forms makes the `PTA` conversation routine rather than exceptional, so the headroom figure of 7.4.3 belongs on the monthly commercial report.
- **Government and defence programmes.** Formal earned-value management-system compliance with surveillance, control-account discipline and — the operative difference — a **hard cap on the level-of-effort share**, because 7.3.1's distortion identity `s × (1 − SPI_d)` is exactly what a surveillance regime exists to bound: a 20 % cap holds the distortion to 0.08 where a 70 % share allows 0.28. Forecasting is auditable rather than discretionary, which in practice means method (d) is commissioned at defined points rather than when someone asks for it.
- **Technology and product delivery.** Cost is overwhelmingly people, so the numbers that move the outturn are the blended rate, the mix and **utilisation**: 7.4.1b's result — five points of utilisation being 6.25 % of the labour bill, USD 155,000 on Auriga's labour content — is the whole game, and materials barely register. Where cadence replaces a fixed baseline, throughput and cost-per-increment stand in for `CPI` (Domain 13), and hybrid programmes report both; the commercial default is time and materials, which makes 7.4.2's cap-and-rate-schedule discipline the difference between a contract and an open account.

- **Energy and resources.** Long-lead equipment and commodity exposure move the centre of gravity to **commitment coverage** and escalation: 7.2.1b's arithmetic — a 5 % movement on the uncommitted remainder being 27.92 % of the cost variance to date — is why the coverage figure is reported monthly here and rarely elsewhere, and why early commitment is bought deliberately at a price (Domain 10, KA 10.1.3; PFL-AI Domain 3 for escalation and currency machinery). Contingency is sized quantitatively as a matter of course, so the draw-index trend of 7.1.3 has a register behind it.
- **Public services and transformation.** Benefits, not cost variance, are the accountability currency (Domain 16), and the cost-side consequence is the one Case study C computes: contingency gets sized against **delivery** risk while the exposure sits on the **benefit** side, where Meridian's unreserved annual gap of USD 293,760 was **2.4480 times** its entire contingency. The leader's task is keeping cost reporting honest while the value case is what gets debated — and extending reserve thinking to the risks that actually decide whether the money was worth spending.

Case study — Domain 7: the forecast the board actually needed (utilities)

Situation. Auriga's week-13 review. The cost report shows `CPI` 0.91, `SPI` 0.92, `CV` (200,000), `SV` (160,000). The programme director's paper proposes reporting `EAC` **USD 4,200,000** — method (a) — on the grounds that the overrun was the contaminated-ground remediation, a discrete event now closed and remediated (Domain 6's case study).

The challenge. The assurance reviewer asks three questions. Is the cause genuinely closed? Yes for the remediation — but the recovery bought a second civil crew and a fast-track of installation, both of which continue for six more weeks, and neither is in method (a)'s "remaining work at budgeted rate" assumption. What does recovery to *`BAC`* require? `TCPI` 1.11 against a demonstrated 0.91. What is the mechanism? There isn't one — the recovery spend increases cost to protect the date.

The outcome. The board is given a **range with named assumptions**: 4,200,000 if the closed event is the whole story; 4,416,667 if current efficiency persists; and a bottom-up (method d) re-estimate of the remaining 52 %, commissioned that week, as the number they will actually manage to. Contingency draw is authorised against the identified risk; management reserve is not touched. The minute records the `TCPI` gap and the explicit finding that **no recovery-to-budget plan exists** — so nobody later claims one was promised.

What the domain teaches here. A single-number forecast hides the only thing that matters: the assumption. `TCPI` converts optimism into an arithmetic claim someone has to defend, and the honest report is the one that survives the question "what would have to be true?"

Case study B — Domain 7: past the point of total assumption (technology)

Situation. A systems-integration subcontract ran on an incentive structure: target cost USD 2,000,000, target fee 150,000, 70/30 share, ceiling 2,450,000 — so `PTA` USD 2,428,571. By month eight the supplier's internal cost was tracking to 2,600,000, well past the `PTA`.

What happened. The supplier's behaviour changed abruptly and, in hindsight, rationally: fresh change requests on work previously treated as in-scope, slower responses on defect fixes that earned nothing, and a claim that the buyer's late environment provision had caused the growth. The buyer's team read it as bad faith. It was arithmetic: past the PTA every additional engineer-hour came out of the supplier's own margin, and the only route back to profit was re-characterising cost as buyer-caused scope.

The outcome. The parties re-set commercially — a re-baselined target cost recognising the genuine environment delay (documented, and partly the buyer's), a revised ceiling, and a time-and-materials envelope for the disputed remainder. Delivery recovered within two months of the reset. The retrospective's finding: the buyer had never computed the PTA, so a foreseeable inflection was experienced as a betrayal.

What the domain teaches here. Commercial structures create behaviour. Computing the PTA at signature — and watching the supplier's cost trend against it — turns an adversarial surprise into a managed conversation held early, while both parties still have options (Domain 10's supplier governance; Domain 11's negotiation).

Case study C — Domain 7: the reserve that could not be reached in time (public health)

Situation. Meridian Care Records — the 40-clinic shared clinical-records programme of Domains 1, 2, 15 and 16 — is at month 14. Its approved cost of **USD 2,400,000** is structured as **USD 2,160,000** of control-account budgets plus **USD 120,000** of contingency inside a baseline of **USD 2,280,000**, with **USD 120,000** of management reserve outside it. Contingency is **5.56 %** of the control-account total; management reserve is **5.26 %** of the baseline. Installation is running well — this is the programme that will deliver 40 of 40 clinics — and **USD 74,000** of contingency has been drawn against installation risks, leaving **USD 46,000**.

Adoption is not running well. Sixteen clinics are using the system where the case assumed twenty-eight (Domain 1, KA 1.3.2). The clinical lead brings a costed **adoption-support package** — floor-walking, workflow coaching and local super-users in the twelve clinics not yet using the system — at **USD 168,000**, expected to move adoption from 40 % to the planned 70 %.

The arithmetic the programme had not done. Twelve clinics at Domain 1's benefit rate of USD 510 a week each are worth **USD 6,120 a week**, or **USD 293,760** a year over a 48-week operating year. So the package pays for itself in **27.4510 weeks** of operation. That is not the problem.

The problem is funding it. Remaining contingency is USD 46,000, so **USD 122,000** must come from elsewhere — and USD 122,000 exceeds the entire USD 120,000 management reserve by USD 2,000. This is therefore not a reserve draw at all but a request for additional funding, decided by the programme board through change control (Domain 4, KA 4.4). Meridian's board sits on a four-week cycle with a two-week paper deadline, so Domain 3's governance latency applies unchanged: `E[wait] = M/2 + L = 4/2 + 2 =` **4 weeks** (Domain 3, KA 3.2.3).

Which cost of delay applies — and the error that is easy to make here. Meridian's registered cost of delay is **USD 14,280 a week**, and using it would price the wait at 4 × 14,280 = USD 57,120. That would be wrong by a factor of **2.3333**. The registered figure is the benefit of the whole 28-clinic adoption the case assumed; this decision unlocks only the **12** additional clinics, worth **USD 6,120 a week**. **The cost of delay applicable to a decision is the benefit that decision unlocks, not the programme's headline figure**, and applying a portfolio-level rate to a package-level choice inflates every business case built on it. So the four-week wait costs `4 × 6,120 =` **USD 24,480**.

The outcome. The board approved the package at its next meeting, and the four-week wait cost **14.57 %** of the package's own price — a whole month of benefit from twelve clinics, bought by nobody and noticed by no one. The retrospective made three findings, each a number rather than a sentiment.

The reserve was pointed entirely at the half of the programme that was succeeding. All USD 120,000 of contingency was sized against installation risks. The benefit-side exposure — the gap between 40 % and 70 % adoption — is worth **USD 293,760** a year, **2.4480 times** the whole contingency, and carried no reserve at all. Contingency sized only against delivery risk on a benefit-driven programme funds the risks that will not decide the outcome.

Decision-rights design is a cost parameter. Had the sponsor pre-authorised a delegated benefit-protection band of USD 150,000 to the programme director, decided in the director's own weekly cycle with no paper lead time — `E[wait] = 0.5` weeks — the latency cost would have been **USD 3,060** instead of USD 24,480, a saving of **USD 21,420** on this single decision. The delegation is free to create and costs nothing when unused; one decision pays for it many times over. Note the sizing constraint: the decision here is the **USD 122,000** of additional funding, so any band below that figure would have accelerated nothing — a delegation set below the size of the decisions it exists to accelerate is decorative, which is Domain 3, KA 3.2.3's threshold-design point measured in money. The USD 150,000 band above clears it with room for the next one.

The package was late because it was unbudgeted, not because it was unaffordable. At a 27.4510-week payback it would have been approved at sanction had anyone computed it. What the funding structure lacked was not money but a line — enabling change was outside the programme's cost baseline entirely, which is Domain 2, KA 2.3's enabling-change point seen from the cost side.

What the domain teaches here. Reserve architecture is not bookkeeping. Which risks a reserve is sized against decides what the programme can respond to; where the release authority sits decides how fast; and both are priced by the same cost of delay that prices everything else. **A reserve that cannot be reached inside the window in which the decision matters is not a reserve; it is a provision.** The instruments are Domain 3's delegation thresholds and Domain 8's aggregation — this domain's contribution is insisting that the latency be costed and that contingency be sized against the risks that decide the benefit, not only those that decide the delivery.

Executive perspective — Domain 7

What a project leader cannot delegate in this domain:

- **The forecast's assumption.** Analysts compute `EAC`; the leader owns which method and why, in one sentence a board can challenge. Reporting a number without its assumption is the domain's cardinal sin.
- **The `TCPI` reality test.** Before endorsing any recovery-to-budget plan: what index does the remaining work require, what is being demonstrated, and what specific mechanism closes the gap?
- **Reserve discipline.** Contingency and management reserve kept distinct, spent under stated authority, with consumption trended against progress — the erosion in MCQ 7.1-B is a leadership signal, not a bookkeeping detail.
- **Measurement integrity.** Earning rules fixed in advance, accruals present, level-of-effort share disclosed, no retrospective budget edits. Cost systems fail morally before they fail arithmetically — the same sentence as Domain 6, and the same reason.
- **The commercial shape.** Which model, why, and where the `PTA` sits. A leader who cannot say who carries cost risk on their largest package is not yet in control of it.

- **The confidence the baseline was approved at.** Not the number — the percentile. Auriga's 4,000,000 was a P33 of its own declared range, and a leader who does not know that figure for their own baseline is managing to a target whose difficulty nobody has stated.
- **The cash position, in days of spend.** USD 1,043,600 — 44.80 days — against a cost variance of 200,000. The variance is the subject of the meeting and the cash is the thing that can stop the project, and it is the leader who has to insist both appear in the same pack. Payment terms and retention structures are delivery parameters, priced before signature, not procurement detail.
- **Whether the reserve can be reached in time.** Case study C's finding generalises: a reserve behind a four-week decision cycle cost USD 24,480 to open on a decision worth USD 6,120 a week, and no amount of the reserve being adequate compensates for its being slow. Delegated bands for benefit-protecting spend are free to create, must be set above the size of the decisions they exist to accelerate, and are a leadership decision rather than an administrative one — as is insisting that a cost of delay be the benefit this decision unlocks and not the programme's headline rate.

Calculation exercises — Domain 7

Exercise 7.1 `BAC` 4,000,000; `PV` 2,080,000; `EV` 1,920,000; `AC` 2,120,000. Compute `CV`, `SV`, `CPI`, `SPI`, percent complete and percent spent. Solution. `CV` (200,000); `SV` (160,000); `CPI` 0.91; `SPI` 0.92; complete `1,920,000/4,000,000 =` **48.0 %**; spent `2,120,000/4,000,000 =` **53.0 %**. Common error: computing percent complete from `AC/BAC` — that is percent spent, and reporting it as progress overstates the project.

Exercise 7.2 Same data. Compute `EAC` by methods (a), (b) and (c), and the corresponding `VAC`. Solution. (a) `2,120,000 + 2,080,000 =` **4,200,000**, `VAC` **(200,000)**. (b) `4,000,000/0.905660 =` **4,416,667**, `VAC` **(416,667)**. (c) `2,120,000 + 2,080,000/(0.905660 × 0.923077) =` **4,608,056**, `VAC` **(608,056)**. Common error: rounding `CPI` to 0.91 before dividing — `4,000,000/0.91 = 4,395,604`, USD 21,062 adrift. Indices are display; arithmetic is full precision.

Exercise 7.3 Same data. Compute `TCPI` to `BAC` and to the method-(b) `EAC`, and interpret. Solution. To `BAC`: `2,080,000/1,880,000 =` **1.11**. To (b): `2,080,000/(4,416,667 − 2,120,000) = 2,080,000/2,296,667 =` **0.91**. The second equals the current `CPI` — the identity of 7.3.4: `BAC/CPI` forecasts are "nothing changes". Common error: using `PV` rather than `AC` in the denominator (giving 1.08) and concluding recovery is nearly free.

Exercise 7.4 Incentive subcontract: target cost 2,400,000; target fee 180,000; share 80/20; ceiling 2,900,000. The seller finishes at 2,650,000. Find the fee, what the buyer pays, and the `PTA`. Solution. Overrun 250,000; fee `180,000 − 250,000 × 0.20 =` **130,000**; buyer pays `2,650,000 + 130,000 =` **2,780,000** (below the 2,900,000 ceiling). Target price 2,580,000; `PTA = 2,400,000 + (2,900,000 − 2,580,000)/0.80 = 2,400,000 + 400,000 =` **2,800,000**. Common error: applying the buyer's 80 % share to the fee reduction.

Exercise 7.5 A pool of 30 at USD 105/h, 20 at USD 150/h and 10 at USD 220/h. Compute the blended rate, then the cost of adding four weeks of a 5-person crew at 40 h/week. Solution. `(30×105 + 20×150 + 10×220)/60 = (3,150 + 3,000 + 2,200)/60 = 8,350/60 =` **USD 139.17/h**. Crew cost `5 × 40 × 4 × 139.17 =` **USD 111,333**. Common error: using the blended rate for a crew drawn entirely from one grade — the blend applies to a representative mix, not to any arbitrary subset.

Exercise 7.6 A point estimate of USD 4,000,000 carries a declared class range of −15 %/+30 %. Treating it as a triangular distribution, find the bounds, the mean, the median, the P80, and the percentile at which the point estimate sits. Then state the PERT-weighted mean of the same three points. Solution. Bounds **3,400,000** and **5,200,000** — a band of 1,800,000, **45.0 %** of the point. Mean `(3,400,000 + 4,000,000 + 5,200,000)/3 =` **4,200,000**. Median `5,200,000 − √(1,800,000 × 1,200,000/2) =` **4,160,770**. P80 `5,200,000 − √(0.20 × 1,800,000 × 1,200,000) =` **4,542,733**. Percentile at the point `= (4,000,000 − 3,400,000)/1,800,000 =` **33.33 %**. PERT `(3,400,000 + 16,000,000 + 5,200,000)/6 =` **4,100,000**. Common error: reporting the point estimate as the expected cost — with an asymmetric range the mean always exceeds the mode, so the approved figure had a **66.67 %** chance of being exceeded.

Exercise 7.7 Control accounts total 3,640,000; contingency 360,000; management reserve 240,000. At 48.00 % complete, 190,000 of contingency has been drawn while `CV` is (200,000). Compute `BAC`, the total funding requirement, the draw index, the projected total draw and the draw efficiency the remainder must achieve — then state the reconciliation finding. Solution. `BAC` **4,000,000**; funding requirement **4,240,000**. Draw index `(190,000/360,000)/0.48 =` **1.0995**. Rate `190,000/48.00 =` **3,958.33 per point**, so projected total draw **395,833** and a shortfall of **35,833** (14.93 % of the management reserve). Required efficiency `170,000/(3,958.33 × 52.00) =` **0.8259** — the remainder must draw **17.41 %** more slowly. Finding: draws of 190,000 against a `CV` of 200,000 leave **10,000** of variance with no identified cause. Common error: reading the remaining balance (170,000/360,000 = 0.4722) as the required efficiency — that is a level, not an index, and it says nothing about the rate.

Exercise 7.8 `EV` 1,920,000; invoiced cost 1,920,000; a further 200,000 received and accepted but neither invoiced nor accrued. The next period earns 480,000 for a true cost of 520,000 and the backlog is invoiced in it. Compute reported and true `CPI` and `EAC(b)` at the data date, the reported and true period `CPI`, and the cumulative `CPI` afterwards. Solution. Reported `CPI` **1.0000**, `EAC(b)` **4,000,000**; true `CPI` **0.9057**, `EAC(b)` **4,416,667** — a forecast understatement of **416,667**, 10.42 % of `BAC`. Period reported `480,000/720,000 =` **0.6667** against a true **0.9231**, an error of **0.2564**. Cumulative afterwards `2,400,000/2,640,000 =` **0.9091** either way. Common error: treating the 0.6667 as a production collapse — it implies costs 50.0 % above budget for the period, and the true movement was an improvement from 0.9057 to 0.9231.

Exercise 7.9 Ten equal packages of 40,000 in one account: four complete, three at 30 % measured physical completion, three not started. `PV` 240,000; `AC` 210,000. Compute `EV`, `SPI` and `CPI` under 0/100, 50/50 and objective percent-complete earning. Then compute the reported `SPI` of a separate account that is 70 % level of effort whose discrete work runs at `SPI` 0.60. Solution. 0/100 `EV` **160,000** (`SPI` 0.6667, `CPI` 0.7619); 50/50 **220,000** (0.9167, **1.0476**); percent complete **196,000** (0.8167, 0.9333) — an `EV` spread of **60,000**, 15.00 % of budget, on one physical state. The level-of-effort account reports `1 − (1 − 0.70)(1 − 0.60) =` **0.88**, a distortion of **0.28**. Common error: reading the 50/50 `CPI` of 1.0476 as good news — the rule, not the work, produced it, and the same account read objectively is over-running at 0.9333.

Exercise 7.10 Auriga's week-13 data. Compute `TCPI` to `BAC` and to each of the three `EAC`s, and state what each equals. Solution. To `BAC 2,080,000/1,880,000 =` **1.1064** (the recovery demand). To `EAC(a) 2,080,000/2,080,000 =` **1.0000** = the budgeted rate. To `EAC(b) 2,080,000/2,296,667 =` **0.9057** = `CPI`. To `EAC(c) 2,080,000/2,488,056 =` **0.8360** = `CPI × SPI`. `TCPI` **to any** `EAC` **equals the efficiency that** `EAC` **assumes**, so choosing a forecast is choosing a `TCPI`. Common error: believing `TCPI` measures something independent of the forecast — it does not, and a `TCPI` that disagrees with the author's stated assumption is an arithmetic error in the report.

Exercise 7.11 The board will fund no more than the approved 4,000,000 and no efficiency improvement is available. How much budgeted scope must be removed, and what is the exchange rate between scope removed and cash saved? Solution. `D = (BAC − EV) − CPI × (BAC − AC) = 2,080,000 − 0.905660 × 1,880,000 =` **377,358** (exactly 20,000,000/53) — **9.43 %** of `BAC` and **18.14 %** of the remaining work. Completing the full scope instead needs **416,667** of extra funding, and `416,667/377,358 =` **1.104167 = 1/CPI**: **one dollar of budgeted scope removed saves USD 1.10 of forecast cash.** Common error: booking the saving without pricing the benefit lost — 9.43 % of budget is not 9.43 % of benefit, and on an integrated system it is frequently much more.

Exercise 7.12 A crew is 4 technicians × 40 h at USD 95/h. Price one crew-week of extra capacity by overtime at 1.5×, by an agency crew at USD 128/h achieving 85 % relative productivity, and by accepting a week's delay at USD 45,000. Then find the two productivity breakevens. Solution. Straight time **15,200**; overtime **22,800** (1.50×); agency **24,094** per productive crew-week (1.5851×); delay **45,000** (2.9605×). The agency crew beats overtime at a relative productivity of `128/142.50 =` **89.82 %** or better; overtime beats the agency crew only while sustained-overtime productivity holds above `22,800/24,094 =` **94.63 %**. Common error: comparing the agency's USD 128 with the in-house USD 95 and rejecting it — the relevant comparison is with the USD 142.50 overtime rate it displaces, and the 24,094 is still 53.54 % of the cost of doing nothing.

Exercise 7.13 A firm fixed price of USD 480,000 against time and materials at USD 130.625/h, with hours estimated optimistic 2,900 / most likely 3,200 / pessimistic 4,100. Find the breakeven hours, the premium over the most-likely estimate, the T&M cost at each of the three points, and the probability (triangular) that the hours exceed the breakeven. Solution. Breakeven `480,000/130.625 =` **3,674.64 h**. T&M at 3,200 h **418,000**, so a premium of **62,000 = 14.83 %** — identical to the hours overrun at breakeven. At 2,900 h **378,813**; at 4,100 h **535,563**, which is 55,563 worse than the fixed price. Exceedance `(4,100 − 3,674.6411)²/(1,200 × 900) =` **16.75 %**. Common error: expressing the premium against the fixed price (12.92 %) rather than against the estimate — that understates the overrun tolerance the buyer is actually purchasing.

Exercise 7.14 Contract price 4,400,000 against `BAC` 4,000,000. At week 13 `EV` 1,920,000, `AC` 2,120,000, payables 742,000, cash received 334,400. Compute the margin to date, the funding absorbed, the funding in days of spend, and the margin at completion under each of the three `EAC`s. Solution. Revenue `1.10 × 1,920,000 =` 2,112,000, so margin to date **(8,000)**. Cash out `2,120,000 − 742,000 =` 1,378,000; funding absorbed `1,378,000 − 334,400 =` **1,043,600** = 26.09 % of `BAC`. Daily spend `2,120,000/91 =` 23,297, so **44.80 days of spend**. Margin at completion: method (a) **+200,000** (+4.55 % of price); (b) **(16,667)** (−0.38 %); (c) **(208,056)** (−4.73 %). Common error: assuming the bid margin of 400,000 survives — it requires an `EAC` of exactly `BAC`, which is `TCPI` 1.11, so the margin conversation and the recovery conversation are one conversation.

Exercise 7.15 Target cost 2,000,000; target fee 150,000; ceiling 2,450,000. Compute the `PTA` at buyer shares of 50 %, 70 % and 90 %, the fee at the `PTA` in each case, and the zero-fee cost. State which of the `PTA` and the zero-fee cost binds first at 70/30. Solution. `PTA = 2,000,000 + 300,000/s`: at 50 % **2,600,000**, at 70 % **2,428,571**, at 90 % **2,333,333**. Fee at the `PTA = 150,000 − (PTA − 2,000,000)(1 − s)`: **(150,000)**, **21,429** and **116,667**. Zero-fee cost `= 2,000,000 + 150,000/(1 − s)`: 2,300,000, **2,500,000**, 3,500,000. At 70/30 the `PTA` (2,428,571) arrives **before** the fee is exhausted (2,500,000), so the ceiling binds first and the `PTA` is the real inflection; at 50/50 it does not, and the

> `PTA` there is nominal. Common error: assuming a more generous buyer share pushes the `PTA` further away — it brings it **closer**, by 95,238 between 70/30 and 90/10, because fixed headroom to the ceiling is consumed faster.

> **Exercise 7.16** Auriga's baseline reaches `PV` 1,920,000 at week 12 and 2,080,000 at week 13, and `EV` at week 13 is 1,920,000. Compute `ES`, `SPI(t)` and the time forecast. Then, at week 27 with `EV` 3,920,000 against baseline `PV` of 3,880,000 at week 24 and `BAC` at week 25, compute `SPI`, `ES`, `SPI(t)` and the time forecast, and explain the divergence. Solution. Week 13: `ES` **12.0000**, `SPI(t) 12/13 =` **0.9231** — equal to `SPI`, because the `PV` curve is locally linear and the `SV` of (160,000) is exactly one week of planned value. Forecast `25/0.923077 =` **27.0833 weeks**, 2.0833 late, which at USD 45,000 per week is **93,750** of delay cost. Week 27: `ES = 24 + 40,000/120,000 =` **24.3333**, `SPI(t)` **0.9012**, forecast `25/0.901235 =` **27.7397 weeks** — 2.7397 weeks or **10.96 %** late — while `SPI` reads **0.98**. Common error: quoting `SPI` late in a project: past the planned finish `PV` is pinned at `BAC`, so `SPI` is arithmetically driven to 1.00 and reported 2 % lateness for an 11 % overrun.

Practitioner's toolkit — Domain 7

Adoption-ready artefacts; adapt headings to your organisation, then keep them stable.

Toolkit 7.T.1 — Cost-report integrity checklist (run before publishing the month)

- [ ] Earning rules fixed in advance per package and unchanged this period; any change disclosed.
- [ ] `EV` claims sampled against physical evidence, not accepted on assertion.
- [ ] Accruals recognised for work received but not invoiced; open commitments cleansed.
- [ ] `PV` read from the current controlled baseline, phased over the approved schedule.
- [ ] No retro-fitted budgets; transfers between control accounts logged.
- [ ] `EV/BAC` reconciles to the reported percent complete; indices recompute from the four numbers.
- [ ] Every `EAC` states its method **and its assumption**; `TCPI` to `BAC` reported beside it.
- [ ] Contingency and management reserve shown separately, with consumption vs progress.
- [ ] Level-of-effort share disclosed per control account.
- [ ] AI-produced forecasts marked, calibration record attached, human owner named.

Toolkit 7.T.2 — Estimate basis sheet (one per estimate)

Method (analogous/parametric/bottom-up) · definition maturity and accuracy class · range (−x %/+y %) · rate sources with dates and escalation basis · quantities and their source · exclusions stated explicitly · risks feeding contingency (link to the register) · reviewer and date. An estimate whose basis sheet cannot be produced is not releasable.

Toolkit 7.T.3 — Commercial one-pager (per major package)

Model (FFP/FPIF/CPFF/CPIF/T&M) and why · who carries cost risk · target cost, fee, share ratio, ceiling · `PTA`**, the zero-fee cost, which binds first, and the current cost trend against the** `PTA` **expressed as headroom in per cent** · for T&M, the hours cap, the rate schedule and the breakeven against any fixed-price alternative · payment terms both directions and the cash profile · retention and release criteria · variation and claims route · escalation contacts. Reviewed at every commercial checkpoint, not filed at signature.

Toolkit 7.T.4 — Cash and commitment page (one per period, beside the cost report)

A single page, six lines, none of which appears in a conventional cost report and each of which has changed a decision on the projects in this domain:

- **Cash paid out** = `AC` − payables, with the payables figure and its share of `AC`.
- **Cash received**, and the date of the oldest unpaid certified invoice.
- **Funding absorbed** = the difference — reported both in currency and in **days of spend**, so it is comparable across the portfolio.
- **Client exposure** = receivables + retention held + unbilled work in progress, with the reconciliation check funding absorbed = client exposure − payables + loss to date.
- **Retention**: held now, due at completion, held beyond completion — and each as a percentage of the contract margin.
- **Commitment coverage** = (`AC` + open commitments) ÷ `EAC`, with the uncommitted forecast balance, the weekly rate at which it must now be committed, and the exposure to a stated price movement. Stale open commitments cleansed this period are stated explicitly, since cleansing them worsens the page and improves its truth.

Exam preparation — Domain 7

The calculation traps. Percent spent reported as percent complete (Exercise 7.1) · rounding `CPI` before dividing (Exercise 7.2) · `PV` instead of `AC` in the `TCPI` denominator (Exercise 7.3) · applying the wrong party's share to an incentive fee (Exercise 7.4) · `EV` valued at actual cost rather than budget · reading a level-of-effort-heavy `SPI` as schedule performance (MCQ 7.3-D) · quoting `SPI` late in a project where it must converge on 1.00 (7.A.1) · confusing commitments with actuals · treating contingency and management reserve as one pot · reporting a point estimate as though it were the expected cost of an asymmetric range (Exercise 7.6) · reading a remaining reserve balance as the reserve's required efficiency (Exercise 7.7) · taking a single-period index at face value in the month an accrual backlog clears (Exercise 7.8) · using 50/50 earning on long packages and reading the resulting favourable `CPI` as performance (Exercise 7.9) · assuming `TCPI` says something independent of the `EAC` it is computed to (Exercise 7.10) · booking a descope saving without pricing the benefit removed (Exercise 7.11) · comparing an agency hourly rate with straight time rather than with the overtime rate it displaces (Exercise 7.12) · expressing a fixed-price premium against the price rather than the estimate (Exercise 7.13) · assuming a bid margin survives a forecast overrun (Exercise 7.14) · assuming a more generous share ratio pushes the `PTA` further away (Exercise 7.15) · adding a non-productive time allowance as an uplift rather than dividing by utilisation (MCQ 7.4-E).

The identities worth memorising. `%spent − %complete = CV/BAC` · `1/CPI − 1` is the rate at which the work performed exceeded its budget · `TCPI` to any `EAC` equals that `EAC`'s implied efficiency (1.0000 / `CPI` / `CPI × SPI`) · a dollar of budgeted scope removed saves `1/CPI` dollars of forecast cash · reported `SPI` = `1 − (1 − s)(1 − SPI_d)`, so level-of-effort distortion is `s × (1 − SPI_d)` · paid-hour to productive-hour cost is `1/(1 − n)`, and cost per productive hour scales as `1/utilisation` · a fixed-price premium in per cent equals the hours overrun at which T&M breaks even · funding absorbed = client exposure − payables + loss to date · `E[wait] = M/2 + L` prices the latency of the authority that releases a reserve.

Reflection questions. 1. Your cost report shows one `EAC`. What must accompany it before you will sign it? (The method and the assumption about remaining work; *`TCPI`* to *`BAC`* beside it.) 2. On your largest package: who carries cost risk, where is the `PTA`, and how close is the supplier's cost trend to it? (7.4.2–7.4.3; toolkit 7.T.3.) 3. Which invariant in 7.A.3 would have caught the last cost surprise you experienced — and why wasn't it running? 4. What percentile of its own declared range was your current baseline approved at, and who in the governance chain knows that number? (7.1.1.) 5. How many days of spend is your project absorbing, and what would 30-day rather than 60-day client terms be worth? (7.4.4; toolkit 7.T.4.) 6. If your board refused any further funding tomorrow, how much budgeted scope would have to come out — and which packages, priced against benefit? (7.3.4b.) 7. Where does the authority to release your largest reserve sit, what is `E[wait]` for it, and what does that wait cost per week? (Case study C.)

Domain 7 summary

Cost leadership begins with an honest number, and honesty here is arithmetic rather than character: a method matched to definition maturity, a range and class always stated — and read, since Auriga's approved USD 4,000,000 sat at the 33.33rd percentile of its own declared range, whose mean of 4,200,000 is the very forecast week 13 produced. Three-point thinking where tails are real, with a 4.125 σ tail recognised as a scenario for Domain 8 rather than a spread; a budget built through control accounts into a time-phased baseline with contingency inside it (USD 360,000) and management reserve outside (USD 240,000) for a funding requirement of USD 4,240,000; and the contingency trend read as an index of its own, where a draw of 52.78 % at 48.00 % complete demands that the remainder run 17.41 % more frugally than the first half.

Measurement then has to be earned — `AC` with accruals, `EV` at budget under earning rules fixed in advance — and both halves of that sentence carry a number. One unrecognised USD 200,000 accrual makes a project reporting `CPI` 0.91 report 1.00 and understates its forecast by USD 416,667, then punishes the innocent period in which the backlog clears. One choice of earning rule moves `EV` by 15.00 % of a control account's budget and flips it from over-running to under-running. Everything downstream is arithmetic on numbers that fragile: `CV`, `SV`, `CPI`, `SPI` at Auriga's week 13 (0.91 and 0.92; 48 % complete against 53 % spent, whose 5-point gap is the USD 200,000); the `EAC` family spanning USD 408,056 — a fifth of the remaining work — because each encodes a different implied efficiency, and funding the cheapest of them buys the project only to 95.09 % complete; and `VAC` and `TCPI` turning recovery talk into a defensible claim, since `TCPI` to any `EAC` is exactly the efficiency that forecast assumes, and recovery to budget here means either 22.16 % more productivity or USD 377,358 of scope out, at an exchange rate of `1/CPI` — USD 1.10 of cash saved per dollar of budget removed.

The commercial half is the same discipline pointed outward. Blended rates make options commensurable and the mix moves the price more than the headcount does; behind the rate sits a cost per productive hour of USD 100.79 where five points of utilisation are worth USD 155,000 — three-quarters of the variance being debated; Domain 6's capacity ladder prices out at 15,200 straight, 22,800 overtime, 24,094 agency and 45,000 to do nothing, with the arguments living at breakevens of 89.82 % and 94.63 %. Contract models allocate cost risk at a price whose crossing point is computable — a 14.83 % fixed-price premium is exactly a 14.83 % tolerance on hours — and the point of total assumption predicts when a supplier's incentives invert, moving closer as the buyer concedes share and becoming nominal once the ceiling passes the zero-fee cost. Beneath all of it runs cash: a project whose reported problem is USD 200,000 is absorbing USD 1,043,600 — 44.80 days of spend — half of which one payment term would return, holding retention worth 55.00 % of its margin, and whose profitability is decided by which `EAC` is believed. At programme scale Meridian adds the last lesson: a reserve sized entirely against delivery risk while the exposure sat on the benefit side, behind an authority whose four-week wait cost USD 24,480 — 14.57 % of the decision it was blocking, and a reminder that the cost of delay applicable to a decision is the benefit that decision unlocks, not the programme's headline rate.

Throughout, the leadership rule holds — the number is only as good as the assumption named beside it, and machine-produced forecasts reach a board only with a calibration record and a human owner. Domain 8 quantifies the risk this domain reserves for; Domain 9 costs the rework this domain's engineer-week prices; Domain 10 takes the commercial relationships into their own depth.

08

DOMAIN 8

Risk, Uncertainty and Resilience (quantitative)

Group: Delivering the work (Part Two). **Target:** ~74 pages. **Binds to:** the PCI Book Pattern Specification and the shared registries (`docs/books/registries/`). This domain is the home of `EMV` and supplies the quantification behind the contingency that Domain 7 (KA 7.1.3) reserves and the schedule ranges Domain 6 (KA 6.4.3) forecasts with. It supplies Domain 9 with the tolerance-and-confidence language its acceptance decisions use, Domain 10 with the counterparty arithmetic behind risk transfer, and Domain 15 with the programme-level aggregation its portfolios depend on. British English; USD (+SAR where useful, indicative `USD 1 ≈ SAR 3.75`).

Why this domain exists

Domains 6 and 7 built plans and budgets; both left an obligation outstanding. Domain 6 forecast with ranges but did not say where the ranges come from. Domain 7 put a contingency reserve inside the baseline and said it is "sized against identified risks" without showing how. This domain closes both gaps and adds the part neither covers: what a leader does when the unexpected happens anyway. It builds risk identification honestly, including opportunities (KA 8.1); works the analysis from qualitative screening through `EMV`, decision trees and aggregated contingency (KA 8.2); sizes and governs reserves (KA 8.3); and turns to resilience, crisis leadership, cognitive bias and AI-enabled risk sensing (KA 8.4). The discipline throughout is that **risk management is a decision-support activity, not a documentation activity** — a register that changes no decision has cost money and bought nothing.

Beneath that lies the domain's central quantitative claim, and it is the one most often missed: **aggregate risk is not the sum of its parts, and the term that decides the difference is variance.** Every number a leader is actually asked for — a contingency, a confidence level, a buffer, a date the board can commit to — is a statement about the spread of an aggregate distribution, not about its average. Expected monetary value, which is where most training stops, fixes only the average; it is silent on the shape, and the shape is what breaks projects. Three consequences run through the rest of this domain. Two registers with identical total `EMV` can need reserves that differ by a third, because one concentrates its exposure in a few large, unlikely events and the other spreads it. Two responses of identical cost that remove identical `EMV` can differ by a third in the reserve they release, because one attacks a probability and the other an impact. And a reserve labelled P80 can be a P73 reserve the moment a shared driver is admitted — the label failing before the money does. A leader who can compute those three things is doing risk management; a leader who can only multiply probability by impact is doing arithmetic.

Learning objectives. After this domain a candidate can: distinguish risk from uncertainty and issue; write a risk statement that supports a decision; identify opportunities as rigorously as threats;

estimate how complete a register is from the overlap between two independent identification methods, and use the answer to size resilience rather than to reassure; run qualitative screening, **compute the span of expected values a single matrix cell can conceal and derive the band ratio a defensible scale needs**; compute EMV for individual risks and interpret the total, including the sensitivity of its ranking to the probabilities assumed; build and solve a decision tree, including the value of information; **value imperfect information, show that the value depends on how the sample is designed as much as on what it costs, and compute the prior probability below which buying it destroys value**; aggregate risks to a mean and a confidence level rather than summing point estimates, **and check the normal approximation against the exact enumerated distribution**; **quantify what one shared driver does to an aggregate, and state the confidence an independence-based reserve actually buys**; **rank responses by the confidence-level reduction they buy rather than by EMV reduction, and explain from the algebra why an impact lever removes between 1.5 and 3 times the variance of a probability lever at equal EMV cost**; convert a risk-appetite statement into an operable tolerance, a confidence level and an escalation threshold; size a contingency reserve defensibly, distinguish it from management reserve, **and test a part-consumed reserve against the register still open**; select and cost responses **across the response families with their secondary risks priced, and show where omitting the secondary risks selects the wrong response**; explain resilience as distinct from prediction and **price a resilience measure on both its expected value and its tail**; recognise the biases that corrupt estimates and reviews, **and reconcile a bottom-up register against a reference class of comparable outturns**; lead in a crisis; **compute merge bias at a convergence point, correct it for shared predecessors, and state the date range over which it stops mattering**; and govern AI-produced risk analysis, **including deriving the base rate at which a sensitive monitor beats a specific one**.

The master worked project — and its programme. Project Auriga continues from Domains 6 and 7 — the 25-week, BAC **USD 4,000,000** control-systems upgrade. Its risk register, used throughout KA 8.2–8.3, carries the ground-conditions risk that actually materialised in Domain 6's case study, so the reader sees the same event before and after the fact. Its five-risk register has a total EMV of **USD 278,000**, an independence-based P80 of **USD 490,624**, and — once one shared subcontractor is admitted — a σ of **USD 340,339** that turns that same reserve into a **73.4 %** reserve.

Single-project arithmetic is not, however, where most contingency is lost, so this domain also works **Meridian Care Records** at programme scale, the fictional public-health programme of Domains 1, 2, 5, 15 and 16: **40 clinics**, approved cost **USD 2,400,000**, a cost of delay of **USD 14,280 per week**, and a board risk appetite that this domain converts into a number. Its five quantified risks total **USD 119,000** of EMV against a 5 % tolerance of **USD 120,000** — which is the whole problem in one comparison, and the reason KA 8.3 exists.

KNOWLEDGE AREA 8.1

Threats, opportunities and identification

Topics: 8.1.1 risk, uncertainty and issues · 8.1.2 writing a usable risk statement · 8.1.3 identification methods and their blind spots.

8.1.1 Risk, uncertainty and issues

Definitions. A **risk** is an uncertain event that, if it occurs, affects an objective — threats reduce achievement, **opportunities** improve it. **Uncertainty** in the broader sense covers what cannot be enumerated with probabilities at all — the conditions where scenario thinking and resilience substitute for calculation. An **issue** is a risk that has occurred; it is managed, not analysed. Keeping these apart matters practically: registers clogged with issues stop being forward-looking, and treating deep uncertainty as if it were quantifiable produces false precision (KA 8.4.1).

Risk appetite and thresholds convert judgment into something operable: how much exposure the organisation accepts, expressed as thresholds that trigger escalation or response — connecting this domain to Domain 3's decision rights. Appetite that is never expressed numerically cannot be applied consistently by different people, which is the usual reason two projects in one portfolio treat the same exposure differently. The conversion is arithmetic and it has a result worth anticipating: a well-formed appetite statement **determines the confidence level at which contingency is held**, so P80 is not a convention to inherit but a consequence to derive. Worked example 8.3.2 does that conversion on Meridian, because the reserve is where the number lands.

8.1.2 Writing a usable risk statement

Most registers fail at the sentence level. A usable statement has three parts — **cause → event → consequence** — and names the affected objective:

> Because the controller is single-sourced with a volatile lead time (**cause**), it may be that delivery slips beyond the installation window (**event**), resulting in a delay to commissioning and additional preservation cost (**consequence**).

Compare "supplier risk", which supports no decision: it identifies no cause to attack, no event to monitor and no consequence to size. The three-part form is not a formatting preference — each part maps to a different response type (attack the cause, monitor the event, mitigate the consequence), which is why KA 8.3's response selection depends on it.

8.1.3 Identification methods and their blind spots

Methods: structured workshops, checklists and prompt lists from prior projects, assumption analysis (every assumption is a risk in disguise — Domain 2's assumption register), interviews for what people will not say in a group, and lessons from comparable work (Domain 9's lessons-learned). Each has a blind spot, and knowing them is the professional part:

- **Workshops** surface what the group already shares and suppress what a junior member suspects — so run them with an explicit invitation to dissent and a route for private input.
- **Checklists** find known risks and, by their comfort, discourage looking for novel ones.
- **Assumption analysis** is the highest-yield method and the least used, because it requires admitting how much of the plan is assumed.
- **Historical data** encodes the past's risks, not this project's novelties.

Opportunities are systematically under-identified — most registers are 90 % threats — because the process is framed defensively and because nobody is rewarded for one. A leader who wants them must ask for them separately and size them the same way (Auriga's `R5` below is one).

How complete is the register? Every method above leaves something out, and the resilience argument of KA 8.4.1 rests on the claim that the register is always incomplete. That claim can be measured rather than asserted, and the measurement costs nothing beyond running two identification methods separately and comparing their lists. If method 1 finds n_1 risks, method 2 finds n_2, and m risks appear on both, then — treating the two methods as independent samples from the same underlying population of risks — the population is estimated by the standard capture–recapture estimator:

```
N̂ = (n₁ × n₂) / m            (Lincoln–Petersen)
N̂c = ((n₁+1)(n₂+1))/(m+1) − 1  (Chapman, less biased at small m)
```

The logic is the same as estimating the size of a fish population by tagging and re-sampling: the smaller the overlap between two independent samples, the larger the population they were drawn from. Applied to a risk register it converts an uncomfortable feeling into a number a leader can act on.

WORKED EXAMPLE

Worked example 8.1.3 — how much of Meridian's register is missing?

1. **Setup.** Meridian's identification ran two methods separately and deliberately. A structured workshop with clinical, technical and operational representatives produced **34** distinct risks. Independently, and without seeing the workshop output, an assumption-analysis review of the programme plan (against Domain 2's assumption register) produced **22**. Comparing the lists, **14** risks appear on both. Each register entry is written in the three-part form of 8.1.2, so "the same risk" means the same cause-and-event pair, not similar wording. The assumption review cost **USD 12,000** of reviewer time.
2. **Formula.** $\hat{N} = n_1 n_2/m$; distinct identified = $n_1 + n_2 - m$; estimated unidentified = $\hat{N} - (n_1 + n_2 - m)$; coverage = distinct ÷ $\hat{N}$. Cross-check with Chapman's $\hat{N}c = ((n_1+1)(n_2+1))/(m+1) - 1$.
3. **Substitution.** $\hat{N} = (34 \times 22)/14 = 748/14$; distinct $= 34 + 22 - 14$; $\hat{N}c = (35 \times 23)/15 - 1 = 805/15 - 1$.
4. **Result.** Estimated population **53.4286** risks (Chapman: **52.6667**). Distinct identified **42**. Estimated unidentified **11.4286** risks — **21.39 %** of the estimated population, so coverage is **78.61 %** (Chapman: 10.6667 missing, **79.75 %** coverage). The workshop on its own covered **63.64 %** of the estimated population. The assumption review added **8** risks the workshop had missed, at **USD 1,500** each.
5. **Interpretation.** Four things follow, and the fourth is the one that changes a decision.

The workshop is not "the risk identification". It found under two-thirds of the estimated population, and it is routinely the only method run. The eight risks the assumption review added are, by construction, exactly the ones the workshop's shared frame of reference could not see — the failure mode named at the top of this topic, now with a size attached. At USD 1,500 per new risk the review pays for itself if the average newly found risk carries more than USD 1,500 of avoidable exposure, which on a USD 2,400,000 programme is close to a certainty and is why the second method is not optional.

The bias runs one way, so the estimate is a floor. Capture–recapture assumes the two methods are independent and that every risk is equally detectable. Neither holds exactly. Both methods were run by people inside the same programme, so they share some blind spots; shared blind spots raise m, and a higher m lowers $\hat{N}$. Unequal detectability — obvious risks found by both methods, subtle ones by neither — pushes the same way. So **11.43 is a lower bound on what is missing**, and a leader should present it as "at least eleven risks we have not written down", never as a precise count. The Chapman cross-check moving the estimate by less than one risk is evidence that the arithmetic is stable, not that the assumptions are true.

The estimate does not license adding `EMV` for imaginary risks. The unidentified 11.43 have no cause, no event, no consequence and no owner; they cannot be quantified, responded to or escalated. What they justify is **management reserve** (8.3.2) and **resilience** (8.4.1) — money and capacity held against exposure that has no line item. That is precisely the distinction between the two reserve types, now with a quantitative basis rather than a shrug, and it is why an organisation that folds management reserve into contingency has no way to fund what this arithmetic finds.

And the number is a decision rule, not a report line. Meridian's programme office set a coverage threshold: below 85 %, commission a third independent method before the tranche gate. At 78.61 % the threshold fired, and a supplier-side review was run. The professional caution is the mirror image: **coverage above the threshold is not evidence that identification is finished** — it is evidence that two methods agree, which on a novel programme is as consistent with a shared blind spot as with completeness. Deep uncertainty (8.1.1) is invisible to this arithmetic by definition, because it cannot be enumerated at all.

AI in this KA

Generating candidate risk lists is a real AI strength: broad, fast, and free of the group's shared blind spot — genuinely useful against the workshop failure mode above. It is equally a machine for **plausible generic risk**, producing items that fit any project and support no decision. The governed workflow: use AI to widen identification (especially by asking it to challenge the plan's assumptions), then require every surviving item to pass the three-part statement test of 8.1.2 with a named owner. **AI proposes; the professional verifies, decides and remains accountable** — and here the verification is specifically that the risk is this project's, not a template's.

KEY TERMS — KA 8.1

Term	Meaning
Risk / opportunity	Uncertain event affecting an objective, adversely / favourably.
Uncertainty	Conditions that cannot be meaningfully enumerated with probabilities.
Issue	A risk that has occurred; managed, not analysed.
Risk appetite / threshold	Accepted exposure, expressed operably enough to apply consistently.
Cause → event → consequence	The three-part risk statement; each part maps to a response type.
Capture–recapture estimate	Register completeness inferred from the overlap of two independent identification methods: $\hat{N} = n_1 n_2 / m$.
Coverage	Distinct risks identified as a share of the estimated population; an optimistic figure, since shared blind spots deflate the population and so inflate it.

SAMPLE MCQS — KA 8.1

MCQ 8.1-A `[8.1.2 · Analysis]` Which register entry best supports a decision? - A. "Supplier risk — high" - B. "Because the controller is single-sourced with volatile lead times, delivery may slip beyond the installation window, delaying commissioning and adding preservation cost" - C. "Delay to commissioning — probability 35 %" - D. "Supplier may cause problems — owner: procurement"

Rationale: Only B names a cause to attack, an event to monitor and a consequence to size (8.1.2). C states a consequence with a probability but no cause, so no response can be designed; A and D support nothing at all.

MCQ 8.1-B `[8.1.1 · Recall]` The ground contamination has been discovered and remediation is under way. In the register this is: - A. a risk with probability 1.0 - B. an issue — it has occurred, and is managed rather than analysed - C. an opportunity, since remediation was funded - D. removed entirely, with no further record

Rationale: Occurred risks become issues (8.1.1). A is the common fudge that clogs registers; D loses the audit trail and the lesson; C is nonsense.

MCQ 8.1-C `[8.1.3 · Analysis]` A register of 60 items contains 3 opportunities. The soundest inference is: - A. this project genuinely has few opportunities - B. the identification process is framed defensively; opportunities must be asked for separately and sized the same way - C. opportunities do not belong in a risk register - D. the register is too long and should be cut

Rationale: A 95 % threat ratio reflects process framing rather than reality (8.1.3). C contradicts the definition of risk; D addresses a different problem.

MCQ 8.1-D `[8.1.3 · Application]` A workshop finds 34 risks and an independent assumption review finds 22, with 14 common to both. The estimated risk population and the estimated number still unidentified are: - A. 56 and 14 - B. 53.43 and 11.43 - C. 42 and nil — the two methods between them found everything - D. 78.61 and 36.61

Rationale: $\hat{N} = (34 \times 22)/14 = 53.4286$, and 42 distinct entries were identified, leaving **11.4286** (8.1.3). A adds the two lists without removing the 14 duplicates and then treats the overlap as the missing count; C assumes two methods exhaust the population, which is the assumption the estimator exists to test; D takes the coverage percentage (78.61) as a population count and subtracts the 42 from it.

MCQ 8.1-E `[8.1.3 · Analysis]` Two identification methods overlap heavily, giving a high coverage estimate. The soundest professional reading is: - A. identification is complete and can be closed - B. a large overlap deflates the population estimate, so high coverage is as consistent with a shared blind spot as with completeness - C. one of the two methods was run incorrectly - D. the estimate is invalid whenever the overlap exceeds half of either list

Rationale: $\hat{N} = n_1 n_2 / m$ falls as m rises, so methods sharing a frame of reference produce reassuring coverage; the estimate is a floor on what is missing (8.1.3). A is the error the arithmetic is meant to prevent; C infers a process fault from a statistic that has other explanations; D invents a threshold the estimator does not have.

SELF-CHECK — KA 8.1

1. State the three parts of a usable risk statement. — Cause, event, consequence, against a named objective.
2. Why is assumption analysis the highest-yield identification method? — Every assumption is a risk in disguise, and plans rest on more of them than teams admit.
3. What is the register consequence of treating an occurred risk as a 100 % risk? — It clogs the forward-looking register; occurred risks are issues.
4. What does a capture–recapture coverage estimate justify, and what does it not? — It justifies management reserve and resilience sized against unidentified exposure; it does not justify adding `EMV` for risks that have no cause, event, consequence or owner.

5. Which way does the bias run in a register-completeness estimate? — Shared blind spots raise the overlap and lower the population estimate, so the count of missing risks is a floor.

KNOWLEDGE AREA 8.2

Analysis: from screening to quantification

Topics: 8.2.1 qualitative screening and its limits · 8.2.2 expected monetary value · 8.2.3 decision trees and the value of information · 8.2.4 aggregating to a contingency.

8.2.1 Qualitative screening and its limits

Probability-and-impact scoring on a matrix is a **screening** tool: cheap, fast, and adequate for ranking a long list into an attention order. Its limits are structural and must be stated wherever it is used. Ordinal scales cannot be arithmetic — a "4" is not twice a "2", so scores must not be multiplied and summed as if they were money. Band boundaries create false distinctions (a risk at the top of "medium" outranks one at the bottom of "high" in reality but not on the matrix). And the matrix is blind to **correlation**: five risks driven by one supplier are one risk wearing five costumes.

The professional practice: screen qualitatively, then **quantify what matters** — the items that could breach a threshold, plus anything a decision now depends on.

How coarse a screen is it? That is answerable, and the answer is the argument for the quantification threshold rather than a matter of taste. A matrix cell is a rectangle in probability-by-impact space, so the range of expected values it contains follows directly from the widths of its two bands.

WORKED EXAMPLE

Worked example 8.2.1 — what one matrix cell conceals.

1. **Setup.** A delivery organisation screens on a 3 × 3 matrix. Probability bands: **Low** 0.01–0.10, **Medium** 0.10–0.35, **High** 0.35–0.70. Cost-impact bands: **Low** USD 10,000–60,000, **Medium** USD 60,000–250,000, **High** USD 250,000–800,000. Two risks are both scored **Medium probability / Medium impact** and therefore receive the same attention, the same review cadence and the same response budget. How different can they actually be?
2. **Formula.** Within one cell, `EMV` ranges from `p_min × I_min` to `p_max × I_max`, so the span factor is `(p_max/p_min) × (I_max/I_min)` — the product of the two band ratios. Compare that against the step between adjacent cells, taken at band midpoints.
3. **Substitution.** Cell floor `0.10 × 60,000`; cell ceiling `0.35 × 250,000`. Span factor `(0.35/0.10) × (250,000/60,000) = 3.5 × 4.1667`. Midpoint of Medium/Medium `0.225 × 155,000`; of High/High `0.525 × 525,000`.
4. **Result.** The Medium/Medium cell contains expected values from **USD 6,000** to **USD 87,500** — a span factor of **14.5833**, exactly `3.5 × 4.1667`. The step from the Medium/Medium midpoint (**USD 34,875**) to the High/High midpoint (**USD 275,625**) is a factor of **7.9032**. So the range inside one cell is **1.8452 times** the whole step from that cell to the one a full band higher on **both** axes.
5. **Interpretation.** The identity is the useful part, and it converts a familiar complaint into a design specification.

Within-cell span = probability band ratio × impact band ratio. Nothing about the shading, the wording of the bands or the number of colours changes it; only the ratios do. A matrix whose bands span factors of 3.5 and 4.17 cannot resolve two risks that differ by a factor of fourteen, which means **its internal resolution is coarser than its own band structure implies**. Two risks in the same cell here can differ by more than the distance the matrix draws between "medium" and "high". That is why ordinal scores must not be multiplied and summed (the point above), and it is also why "we manage all amber items the same way" is a funding error rather than a simplification: at the extremes of this cell, a response budget set by cell allocates the same money to an exposure of 6,000 and one of 87,500.

The design rule falls straight out. To hold the within-cell span below a factor of 4, each axis needs a band ratio below 2. Covering probabilities from 0.01 to 0.70 — a range of **70** — in bands of ratio 2 requires **7** bands ($\log_2 70 = 6.1293$); covering impacts from 10,000 to 800,000 — a range of **80** — requires **7** as well ($\log_2 80 = 6.3219$). A 7 × 7 matrix on ratio-constant bands is a defensible screen; a 3 × 3 matrix on the same total range is not, and no amount of careful colouring repairs it. Note what this also says about **linear** bands, which are commoner and worse: equal-width bands have enormous ratios at the bottom of the scale (a band from 0.01 to 0.10 is a ratio of 10) and negligible ones at the top, so a linear matrix is least reliable exactly where low-probability high-impact risks live — the class 8.3.1 says must be handled structurally.

What screening is still for. The conclusion is not that matrices are useless; it is that they are a triage instrument whose output is an attention order, and that they must be paired with a stated quantification threshold. Meridian's threshold was set at the Medium/Medium floor: any risk whose assessed impact reaches USD 60,000 is quantified individually, **plus every identified opportunity whatever its size**, because the under-identification of 8.1.3 means an impact threshold applied symmetrically would remove the few opportunities a register contains. That took its 42-item register down to the five entries 8.2.2b aggregates — four threats above the floor and one opportunity of USD 40,000 below it. The reviewer's test is simply whether that threshold is written down: a screen with no threshold is a screen that has replaced quantification instead of ordering it. One caution. Recalibrating the bands to reduce span makes the matrix harder to complete quickly, which is the property that made it useful — so the trade-off should be made deliberately at the organisational level, not per project. (Nothing above touches the correlation blindness named at the top of this topic; band width cannot repair it.)

8.2.2 Expected monetary value

```
EMV = probability × impact          (per risk; threats positive, opportunities negative cost)
```

WORKED EXAMPLE

Worked example 8.2.2 — Auriga's risk register, quantified.

1. **Setup.** Four threats and one opportunity, each with an assessed probability and cost impact (`BAC` USD 4,000,000).
2. **Formula.** `EMV = p × impact`; total exposure = Σ `EMV`, opportunities carried negative.
3. **Substitution and result.**

ID	Risk	p	Impact (USD)	EMV (USD)
R1	Controller lead-time slip	0.35	240,000	84,000
R2	Ground conditions worse than surveyed	0.50	180,000	90,000
R3	Integration rework	0.25	320,000	80,000
R4	Permit delay	0.15	400,000	60,000
R5	Early-delivery rebate (opportunity)	0.30	(120,000)	**(36,000)**
	Total expected exposure			**278,000**

1. **Interpretation.** Total `EMV` is **USD 278,000 — 6.95 %** of `BAC`, and USD 314,000 if the opportunity is ignored, an overstatement of **12.95 %**, which is why opportunities must be in the same arithmetic rather than mentioned in prose. Five things the table teaches, in the order a reviewer should test them.

`EMV` **is an average of outcomes that will not happen.** No single risk will cost 84,000 — R1 costs 240,000 or nothing. So `EMV` is the right basis for funding a portfolio of risks and the wrong basis for deciding whether one specific risk is survivable. The strongest form of the point is arithmetic: the probability that all four threats occur together, producing the 1,140,000 worst case, is `0.35 × 0.50 × 0.25 × 0.15 =` **0.65625 %**, while the probability that the register's net cost is **zero or better** — no threat occurs, or only the opportunity does — is **20.72 %** (enumerated in 8.2.4). One future in five costs nothing; the register's central estimate describes neither of those futures.

Ranking by `EMV` reorders the register. R4 has the largest impact (400,000) but the smallest threat `EMV` (60,000), while R3's lower impact carries more expected cost. A leader managing by impact alone would attend to them in the wrong order — though see 8.3.1, because impact still governs survivability.

That ranking is fragile, and the fragility is computable. R4's `EMV` overtakes R1's as soon as its probability reaches `84,000/400,000 =` **0.21**, and tops the whole register at `90,000/400,000 =` **0.225**. So a six-point revision in one probability — well inside the precision of any real assessment — reverses the top of the order. Likewise a plausible ±0.05 error on R2's probability moves its `EMV` by ±USD 9,000 and the total by **±3.24 %**. The professional consequence is a presentation rule: quote total exposure to the nearest thousand at most, and where a response decision turns on a rank, state the probability at which the rank flips rather than defending the rank itself.

The total is a mean, and means are additive whatever else is true. Σ`EMV` is the mean of the aggregate distribution under any dependence structure between the risks — correlation moves the spread and not the average (8.2.4b proves it on this register, and the AI note at the end of this KA turns it into a hand-check).

And the sign convention has to be enforced, not assumed. R5's `EMV` is carried as (36,000) — a reduction in expected cost — because it is an opportunity against the same objective (cost) as the four threats. Mixing objectives inside one total is the commoner and more damaging version of the same error, and 8.2.2b takes it up.

WORKED EXAMPLE

Worked example 8.2.2b — Meridian's programme register, and the objective a total belongs to.

1. **Setup.** Meridian's 42-item register (8.1.3) has five entries above the quantification threshold of 8.2.1. All five are assessed against a **single objective — cost against the approved USD 2,400,000** — and the programme separately holds benefit risks, schedule risks and reputational risks in their own aggregations.

ID	Risk (cause → event → consequence, abbreviated)	p	Impact (USD)	EMV (USD)
M1	Legacy extract quality → second data run needed at clinics	0.40	90,000	36,000
M2	Clinician availability → extended parallel running at the 8 largest clinics	0.35	120,000	42,000
M3	National records-format revision → interface rebuild	0.20	150,000	30,000
M4	Trainer attrition → contracted backfill for the training programme	0.30	70,000	21,000
M5	Shared-licence rebate on completing the rollout early (opportunity)	0.25	(40,000)	**(10,000)**
	Total expected cost exposure			**119,000**

1. **Formula.** `EMV = p × impact`, summed within one objective. Express the total against the objective's tolerance, not against the whole approved cost, because the tolerance is what governance actually controls (8.3.2).
2. **Substitution.** `0.40 × 90,000`, `0.35 × 120,000`, `0.20 × 150,000`, `0.30 × 70,000`, `0.25 × (40,000)`. Tolerance = `5 % × 2,400,000`.
3. **Result.** Total expected cost exposure **USD 119,000**, which is **4.96 %** of the approved USD 2,400,000 — and **99.17 %** of the board's 5 % tolerance of **USD 120,000**.
4. **Interpretation.** The comparison in step 4 is the whole finding, and it is invisible if the total is expressed as a percentage of approved cost.

The mean alone consumes 99.17 % of the tolerance. Because a mean is roughly the 50th percentile of an aggregate, that means Meridian has approximately a **coin-flip** chance of breaching its stated tolerance before any confidence level is discussed. No contingency figure can repair this, because contingency covers the risks; the tolerance is the space the risks have to fit inside. Only three things can: reduce the exposure by response (KA 8.3), widen the tolerance through governance, or reduce scope. Presenting a P80 contingency without saying this would be technically correct and professionally negligent, and 8.3.2 completes the arithmetic.

Why one objective per total, always. M2's consequence is cost — paying for parallel running. Had it been written as "adoption below plan", its consequence would be benefit, measured against Meridian's USD 685,440 a year (Domain 1, KA 1.3.2) and against a different tolerance owned by a different body. Summing a cost impact and a benefit impact produces a number that is compared with nothing, cannot be funded from any reserve, and is the commonest reason a programme register's "total exposure" is quietly ignored by the board that receives it. The register may hold every objective; a **total** may hold only one. Where a single event hits two objectives — a delay that costs money and defers benefit at USD 14,280 a week — it appears in both aggregations with the relevant consequence, and the double appearance is disclosed rather than netted.

Scale changes which comparison matters, not the arithmetic. Auriga's 278,000 against a 4,000,000 BAC reads as comfortable at 6.95 %; Meridian's 119,000 against a 2,400,000 approved cost reads as comfortable at 4.96 % and is not, because the governing constraint is a 120,000 tolerance rather than the budget. The reviewer's question is therefore never "what percentage of the budget is the exposure?" but **"what is the exposure as a share of the room we have?"** — and if nobody can state the room, that is the finding.

8.2.3 Decision trees and the value of information

A decision tree makes a choice under uncertainty explicit: decision nodes (what we control), chance nodes (what we do not), and outcomes valued and rolled back to the present decision.

WORKED EXAMPLE

Worked example 8.2.3 — should Auriga survey before excavating?

1. **Setup.** Excavation may hit contamination: probability **0.40**, cost if hit **USD 300,000**. Option A: proceed directly. Option B: commission a **USD 25,000** ground survey first, which reliably reveals the condition; if contamination is present, a planned mitigation costs **USD 90,000** instead of the reactive 300,000.
2. **Formula.** Roll back each branch: EV(cost) = Σ probability × outcome cost. Value of information = EV(cost without) − EV(cost with).
3. **Substitution.** A: `0.40 × 300,000`. B: `25,000 + 0.40 × 90,000 = 25,000 + 36,000`.
4. **Result.** A **USD 120,000** expected cost; B **USD 61,000**. **Value of the survey = USD 59,000**, against its USD 25,000 price.
5. **Interpretation.** The survey is worth buying, and the arithmetic shows why: it converts a reactive 300,000 into a planned 90,000 in the 40 % of futures where the problem exists. This is the general shape of **buying information** — worth it when it changes what you would do, worth nothing when it does not, however reassuring it feels. **Information is valuable only in proportion to the better action it enables.** (Domain 6's case study is the counterfactual: Auriga did not survey, met the condition reactively, and paid the recovery this tree prices.)

Three breakevens make the conclusion defensible rather than merely correct, and each is one division.

How much could the survey have cost? Branch B beats branch A while the survey price stays below `120,000 − 0.40 × 90,000 =` **USD 84,000 — 3.36 times** the actual price. That is the sentence for a procurement conversation: a survey quoted at three times the budget figure would still have been the right purchase, so the decision was never close and the cost-reduction pass that cut it (see the case study) was not making a marginal call.

At what prior does information stop paying? The survey pays while `25,000 < p × (300,000 − 90,000)`, i.e. `p > 25,000/210,000 =` **11.90 %**. Below a roughly one-in-eight belief in contamination, the survey destroys value. This is the number that governs whether the same tree is worth building on the next site, and it explains why an apparently identical decision can go the other way on ground everyone considers clean.

And what would make the informed action worthless? If mitigation once forewarned still cost `M`, branch B is `25,000 + 0.40M`, which reaches branch A's 120,000 at `M =` **USD 237,500**. So the entire value of knowing rests on a planned response costing less than 237,500; at the 250,000 the text above uses as an illustration, B's EV is `25,000 + 100,000 = 125,000` and the survey destroys 5,000 of value. **Information's value is bounded by the response it enables**, which is why "we should get more data" is not a proposal until the responder and the response are named.

One further property is worth naming because 8.2.3b relaxes it. This survey is assumed to **reveal the condition reliably** — a perfect signal — so its gross value equals the value of perfect information, `120,000 − 36,000 =` **USD 84,000**, and it captures 100 % of what could possibly be known. Real signals are imperfect, and their value falls in a way that depends less on their price than on how they are designed.

WORKED EXAMPLE

Worked example 8.2.3b — Meridian's pilot, and the value the sample design decides.

1. **Setup.** Meridian's migration approach may be unfit for clinics running the **legacy patient index**; **12 of the 40** clinics run it, and which twelve is known from the asset data. Whether the approach fails on that configuration is not known: the programme assesses the probability at **0.30**. Discovered only at full rollout, the failure costs **USD 480,000** of rework plus **6 weeks** of programme delay at the cost of delay of **USD 14,280 per week**. Discovered in a pilot, the approach is redesigned for **USD 130,000** plus **2 weeks** of delay. A two-clinic pilot costs **USD 45,000**. Three options: no pilot; a pilot at two clinics chosen at random from the 40; a pilot at two clinics chosen so that at least one runs the legacy index.
2. **Formula.** Late total = rework + 6 × cost of delay. Early total = redesign + 2 × cost of delay. For a random pair, the flaw is detected only if at least one pilot clinic runs the legacy index: `P(detect) = 1 − C(28,2)/C(40,2)`. EV(cost) = pilot price + `p ×` [detect × early + (1 − detect) × late]. Breakeven prior: pilot price ÷ (late − expected cost given the pilot).
3. **Substitution.** Late `480,000 + 6 × 14,280`; early `130,000 + 2 × 14,280`. Random-pair detection `1 − (28 × 27)/(40 × 39) = 1 − 756/1,560`. Designed pair detection `= 1`.
4. **Result.** Late total **USD 565,680**; early total **USD 158,560**.

Option	Detection given a flaw	Expected cost (USD)	Value vs no pilot
No pilot	—	`0.30 × 565,680 =` **169,704.00**	—
Random two clinics	**51.5385 %**	`45,000 + 0.30 × 355,856.62 =` **151,756.98**	**17,947.02**
Designed two clinics	**100 %**	`45,000 + 0.30 × 158,560 =` **92,568.00**	**77,136.00**

The **design choice alone is worth USD 59,188.98** and costs nothing: both pilots run two clinics at the same USD 45,000. Perfect free information would cost `0.30 × 130,000 =` **USD 39,000**, so the value of perfect information is **USD 130,704** and the designed pilot captures **59.02 %** of it; the random pilot captures **13.73 %**. 5. **Interpretation.** The finding is not that pilots are valuable — everyone believes that — but that **the value of an imperfect signal is set by its design, and a badly designed pilot is roughly a quarter as valuable as a well designed one at identical cost.**

Where the loss comes from. A random pair of clinics misses every legacy-index site nearly half the time (**48.4615 %**), and in those futures the programme pays the full 565,680 plus the 45,000 it spent finding nothing. A pilot chosen to include the configuration under suspicion cannot miss. Nothing about the money changed; only the question the pilot was capable of answering. This is the same lesson Domain 9 (KA 9.3.2) states for acceptance sampling — a convenience sample of the accessible items establishes nothing — arriving here through the value of information rather than through a confidence bound, and it is worth recognising as one idea in two instruments.

The breakevens diverge sharply, which is the decision rule. The designed pilot pays while the prior exceeds `45,000/(565,680 − 158,560) =` **11.0533 %**; the random pilot needs `45,000/209,823.38 =` **21.4466 %** — a prior **1.9403 times** higher. So there is a whole band of beliefs, from 11.05 % to 21.45 %, in which **the right decision is to pilot and the wrong decision is to pilot badly**: a programme that runs the random pilot in that band destroys value while appearing prudent. The professional habit that follows is to state, of any proposed investigation, which uncertainty it can resolve and with what probability — not merely what it costs.

Two cautions and one governance point. The arithmetic treats the flaw as a single binary condition; if the approach could fail partially, or fail on a configuration nobody has thought of, the pilot's real detection probability is lower than 1 even when designed, and the honest figure is a range. The 2-week and 6-week delays are themselves estimates, and because the cost of delay is a rate, an error in the weeks propagates linearly into the answer: one week either way on the late case moves the no-pilot option by `0.30 × 14,280 =` USD 4,284. And deliberately selecting the pilot sites has a stakeholder consequence that no expected value captures — the clinics chosen are those most likely to have a bad experience, which is a Domain 11 conversation that has to be held before the arithmetic is acted on, not after.

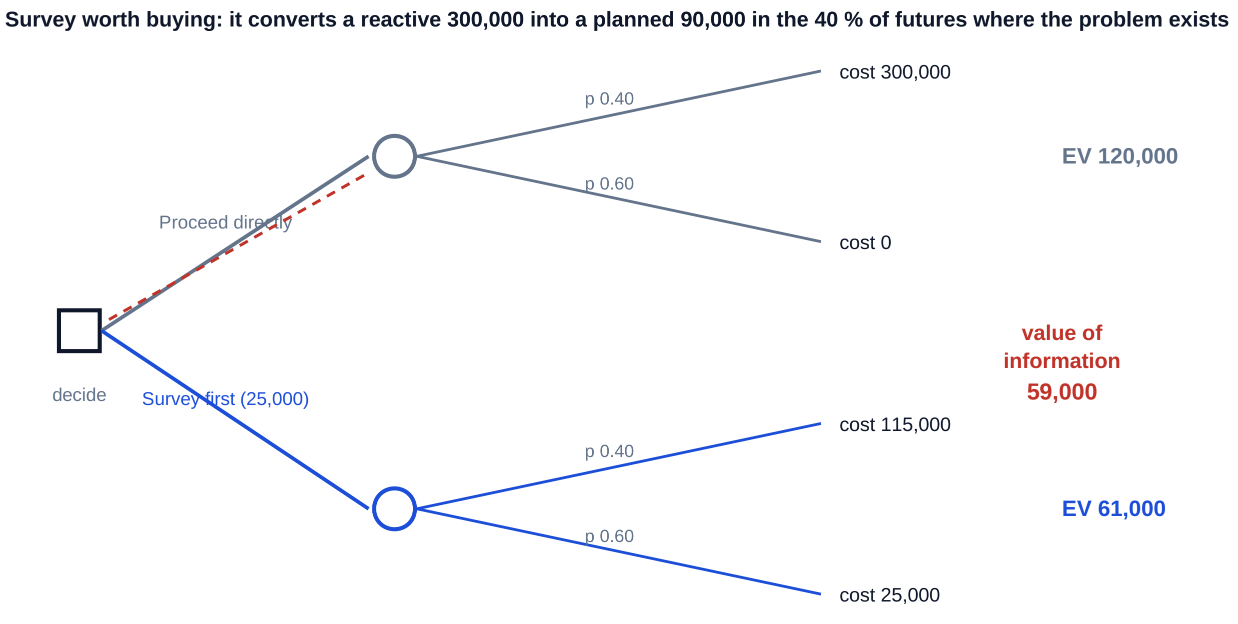


Fig 8.2.1 — The survey decision tree

8.2.4 Aggregating to a contingency

The error to avoid. Summing worst cases produces a number nobody will fund (240,000 + 180,000 + 320,000 + 400,000 = 1,140,000, or 28.5 % of `BAC`); summing `EMV`s alone (278,000) funds the average, which by construction is exceeded about half the time. Neither is a contingency. What a leader needs is a **confidence level**: an amount that covers the aggregate outcome with stated probability.

WORKED EXAMPLE

Worked example 8.2.4 — Auriga's contingency at P80.

1. **Setup.** The five register risks of 8.2.2, treated as independent (an assumption examined below).
2. **Formula.** For independent risks: mean = Σ `p × impact`; variance = Σ `p(1−p) × impact²`; standard deviation = √variance. A P80 amount ≈ mean + 0.8416 × σ (the 80th percentile of a normal approximation).
3. **Substitution.** Mean **278,000** (from 8.2.2). Variance `0.35×0.65×240,000² + 0.50×0.50×180,000² + 0.25×0.75×320,000² + 0.15×0.85×400,000² + 0.30×0.70×120,000²` = **63,828,000,000**; σ = **252,642**. P80 = `278,000 + 0.8416 × 252,642`.
4. **Result.** σ **USD 252,642**; **P80 ≈ USD 490,624**. Against the worst-case sum of 1,140,000 and the `EMV` sum of 278,000 — and against the "10 % of `BAC`" rule of thumb, which would give 400,000 with no reasoning attached.
5. **Interpretation.** The P80 figure is defensible in a way the other three are not: it states what confidence it buys. Five things belong beside it, and the first two are checks a reviewer can run in five minutes.

What confidence does the rule of thumb actually buy? The "10 % of `BAC`" convention gives 400,000, which on this distribution sits at `z = (400,000 − 278,000)/252,642 =` **0.4829**, i.e. the **68.54th** percentile. So the rule of thumb is not "conservative" or "prudent"; it is a **P68.5** reserve, and it happens to be one on this register only by coincidence — on a register with the same mean and half the σ it would be a **P83** reserve. The convention's real defect is not that it is wrong but that it **conceals which question it is answering**, and the two-line calculation above converts it into a number that can be argued with.

Check the approximation against the exact distribution, because with five risks you can. Five independent binary risks have `2⁵ = 32` outcomes; enumerating them gives the aggregate exactly with no normal assumption at all. Doing so shows that **USD 490,624 in fact covers 78.60 %** of futures, not 80.00 % — the approximation overstates the confidence by **1.40 percentage points**, which is small enough to use and large enough to disclose. The exact distribution also shows why a percentile is a slippery object on a lumpy aggregate: the outcomes cluster at 32 discrete levels, and the smallest level whose cumulative probability reaches 0.80 is **USD 500,000**, at which point the cumulative probability has already jumped to **83.43 %**. There is no outcome at all whose probability of non-exceedance is exactly 80 %. The professional habit: quote the confidence level as an approximation with its method named, and never defend the fourth digit of a percentile on a register of five risks. (The rule of thumb fares similarly — 400,000 covers **68.58 %** exactly against the approximation's 68.54 %.)

The price of confidence is computable, and it is the conversation governance should be having. The same register gives P50 = the mean = **USD 278,000** (6.95 % of `BAC`), P80 = **USD 490,624** (**12.27 %**), and P90 = `278,000 + 1.2816 × 252,642 =` **USD 601,786** (**15.04 %**). Moving from P80 to P90 therefore costs **USD 111,163**, or about **USD 11,116 per percentage point** of confidence. That is the number a sponsor needs in order to choose, and it is why "the confidence level is a policy choice" is a statement about a priced trade rather than a disclaimer.

Independence is an assumption, usually optimistic — if R1 and R3 share a supplier they are correlated, the variance is larger, and the true P80 is higher; correlation is what turns a bad month into a crisis. Worked example 8.2.4b prices it on this register rather than leaving it as a warning.

And the normal approximation is a convenience for a handful of Bernoulli risks; proper practice runs a Monte Carlo simulation over the register (and over the schedule, extending Domain 6's three-point durations) to produce a distribution rather than a formula. Note what the 32-outcome enumeration establishes about that: the approximation's error here is a point and a half, so simulation on a five-risk register buys presentation rather than accuracy. Its value appears when the register is large, when impacts are themselves distributions rather than points, or when correlations must be modelled — and the enumeration above is the check that a simulation is doing what its owner thinks, because both must return the same mean of 278,000.

WORKED EXAMPLE

Worked example 8.2.4b — what one shared subcontractor does to Auriga's reserve.

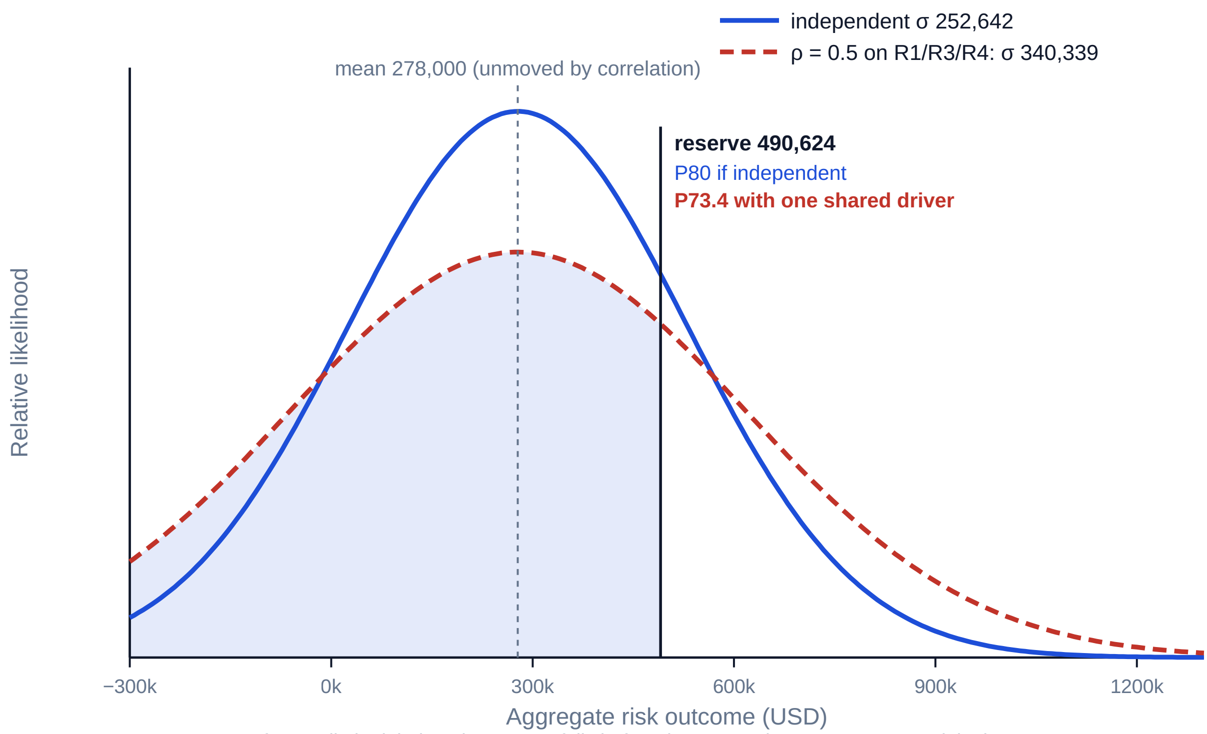
1. **Setup.** Auriga's register is re-examined and one fact emerges from the procurement schedule: **R1** (controller lead time), **R3** (integration rework) and **R4** (permit delay, whose application package the same firm prepares) all depend on the same specialist subcontractor. The risk team assesses a pairwise correlation of **ρ = 0.5** among the three — a judgment, and one that must be recorded as such. R2 and R5 remain independent of everything.
2. **Formula.** For correlated risks, $\mathrm{Var}(\Sigma X) = \Sigma\ \mathrm{Var}(X_i) + 2\ \Sigma_{i<j}\ \rho_{ij}\ \sigma_i\ \sigma_j$. The mean is unchanged: $E(\Sigma X) = \Sigma\ E(X_i)$ whatever the dependence. Per-risk $\sigma_i = \sqrt{(p_i(1-p_i))} \times \mathrm{impact}_i$.
3. **Substitution.** Per-risk σ: R1 **114,472.70**, R2 **90,000.00**, R3 **138,564.06**, R4 **142,828.57**, R5 **54,990.91**. Correlation term $2 \times 0.5 \times (\sigma_1\sigma_3 + \sigma_1\sigma_4 + \sigma_3\sigma_4)$ = 15,861,803,176.18 + 16,349,972,477.04 + 19,790,907,002.96.
4. **Result.** Added variance **52,002,682,656**, taking total variance from 63,828,000,000 to **115,830,682,656** and σ from 252,642 to **USD 340,339.07** — a factor of **1.3471**. The correlated P80 is 278,000 + 0.8416 × 340,339.07 = **USD 564,429.36**, **USD 73,805.82** (or **15.04 %**) above the independence figure, and **14.11 %** of BAC against 12.27 %. And the reserve already approved at 490,624 now covers $z = 0.6247$ of the correlated distribution — a **73.39 %** reserve.
5. **Interpretation.** Three results, and the third is the one to take to a board.

The mean did not move. Total EMV is still exactly 278,000, because correlation redistributes probability without changing an average. This is why a register can pass every EMV check, reconcile perfectly against a simulation's mean, and still be under-reserved: **the error lives entirely in the second moment, where no EMV review looks.** It is also the reason ΣEMV remains a valid hand-check on a simulation even when correlation is modelled.

The correlation term is large because it scales with σ products, not with EMVs. Three risks — 60 % of the register by count — added 81.5 % as much variance again as all five risks contributed independently. The general form of that, and why concentration hurts faster than counting suggests, is 8.A.1's. A programme with twenty risks on one scarce resource is not twenty risks.

And the label fails before the money does. Nobody wrote down a wrong reserve; they wrote down a reserve that was P80 under an assumption, and the assumption was never tested. The board approved "an 80 % confidence contingency" and holds a **73.4 %** one — a 6.6-point shortfall in confidence that appears nowhere in the financial statements and can only be found by asking which risks share a driver. Restoring 80 % costs the **USD 73,806** computed above; that is the priced version of 8.A.1's warning and the exact question the Executive perspective says a leader must not delegate.

One caution on the arithmetic itself. **ρ = 0.5 is a judgment, and the answer is sensitive to it**: the correlation term scales linearly in ρ, so halving the assumed correlation to 0.25 halves the added variance and the honest presentation is a range with the assumption named, not a single figure. Where the shared driver is identifiable at all, 8.A.1 argues for a different model altogether, and Case study B computes what that model does to the answer.



Same mean, fatter tail: the label on the reserve fails before the money does. Source: PCI original.

Fig 8.2.2 — Independence is worth 6

WORKED EXAMPLE

Worked example 8.2.4c — the response that buys variance, on Meridian's register.

1. **Setup.** Meridian's five quantified risks (8.2.2b) give a mean of **USD 119,000**, a variance of **10,149,000,000** and σ of **USD 100,742.25**, so the P90 the board's appetite implies (8.3.2) is `119,000 + 1.2816 × 100,742.25 =` **USD 248,111.26**. Two responses are proposed, each costing **USD 25,000**. **Response A** halves the probability of M2 from 0.35 to 0.175 by seconding clinical backfill so that parallel running is less likely to extend. **Response B** cuts the impact of M3 from USD 150,000 to USD 60,000 by building the interface behind an abstraction layer, so a records-format revision is a configuration change rather than a rebuild; the probability stays 0.20 because the national timetable is outside the programme's control.
2. **Formula.** `EMV` contribution `= p × I`; variance contribution `= p(1−p)I²`; recompute mean, variance, σ and P90 after each response and compare the **P90 reduction per USD spent**.
3. **Substitution.** A: `EMV` 42,000 → `0.175 × 120,000 = 21,000`; variance 3,276,000,000 → `0.175 × 0.825 × 120,000² = 2,079,000,000`. B: `EMV` 30,000 → `0.20 × 60,000 = 12,000`; variance 3,600,000,000 → `0.20 × 0.80 × 60,000² = 576,000,000`.
4. **Result.**

	EMV removed	Variance removed	New mean	New σ	New P90	P90 reduction
A halve M2's probability	**21,000**	1,197,000,000	98,000	94,615.01	**219,258.60**	**28,852.67**
B cut M3's impact	18,000	**3,024,000,000**	101,000	84,409.72	**209,179.49**	**38,931.77**

A removes **USD 3,000 more** EMV; B removes **USD 10,079.11 more of the P90 requirement** — a reduction **34.93 %** larger, for the same USD 25,000. Per dollar spent, A returns **1.1541** of P90 reduction and B returns **1.5573**. 5. **Interpretation.** The two rankings disagree, and which one is right depends on what the organisation is short of.

If the binding constraint is the reserve, buy variance; if it is the expected cost, buy the mean. Meridian's constraint is the tolerance (8.2.2b), so B is the correct purchase even though it looks worse on the metric most registers publish. A programme whose problem is a thin margin rather than a thin contingency would choose A. The professional error is not choosing wrongly but **choosing without stating which constraint is binding** — at which point the EMV column decides by default, and the default is wrong whenever a confidence level is the thing under pressure.

The algebra says why, and it generalises. EMV is linear in both p and I, so halving either halves the EMV identically. Variance goes as $p(1-p)I^2$, which is **quadratic in impact and non-monotonic in probability**. Halving an impact therefore removes exactly **75 %** of that risk's variance contribution, always. Halving a probability leaves $(1 - p/2)/(2(1 - p))$ of it — 0.5278 at p = 0.10, 0.6346 at p = 0.35, 0.75 at p = 0.50 — so it removes between 25 % and 50 %. Dividing gives the invariant: **for any risk with $p \le 0.5$, halving the impact removes between 1.5 and 3 times as much variance as halving the probability, at identical EMV cost** (1.5882 at p = 0.10, 1.7143 at 0.20, 2.0526 at 0.35, exactly 3 at 0.50). Impact reduction is the reserve lever; probability reduction is the margin lever. This is also the arithmetic behind the intuition that "cap the downside" beats "make it less likely" — and it is the reason **modularity and staged commitment (8.4.1) release contingency**, since both are impact levers.

Two cautions. Impact reduction is frequently harder to engineer than probability reduction and sometimes impossible — an abstraction layer exists to be built, but no design change makes a regulator less likely to publish. The invariant tells a leader where to look first, not what is available. And these responses were compared at equal cost precisely so that the levers could be isolated; in practice each option's own cost, secondary risks and schedule effect enter as well, which is the fuller comparison worked example 8.3.1 runs.

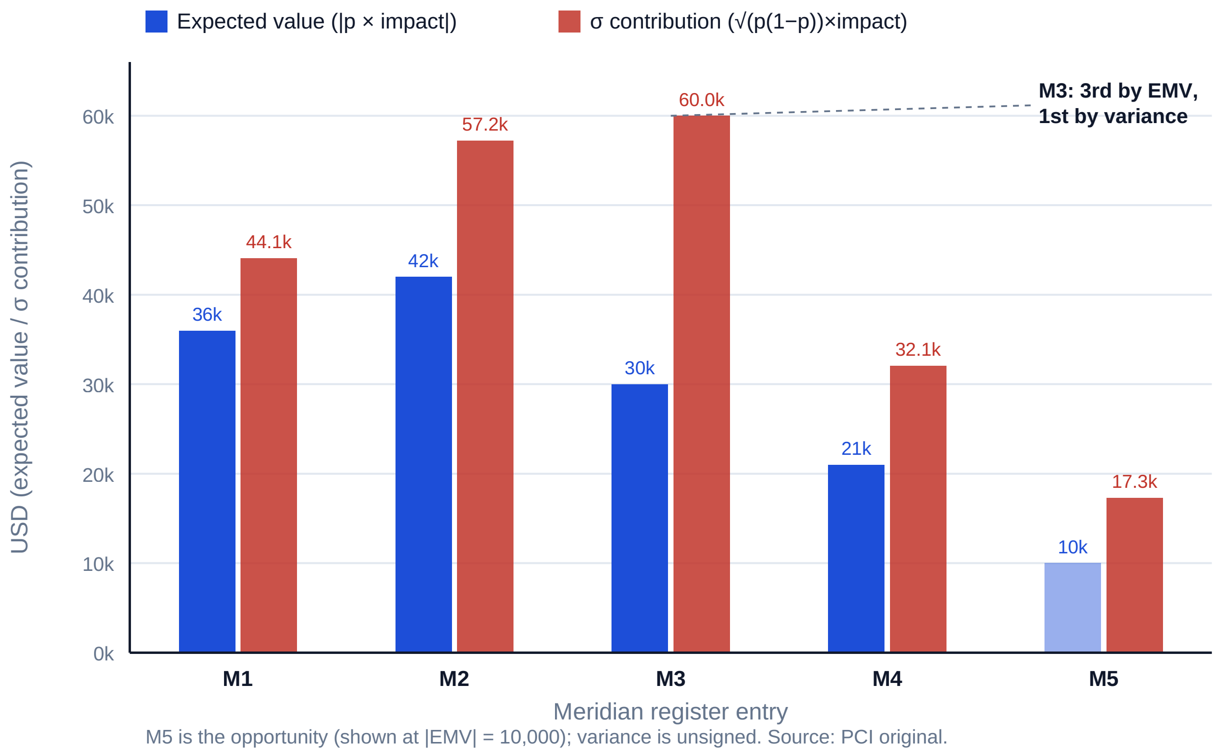


Fig 8.2.3 — Expected value and variance rank the same register differently

AI in this KA

Simulation, correlation analysis and scenario generation are legitimate machine work, and the outputs are unusually seductive because a distribution looks like evidence. Three checks before any simulated number reaches a board. **Interrogate the inputs**: a simulation is an amplifier of its assumed distributions and correlations, and those are judgments — garbage in, precisely-shaped garbage out. **Recompute a landmark by hand**: the mean should match Σ`EMV`; if it does not, the model is not doing what you think. **Ask what would change the decision**: a P80 quoted to the dollar with no sensitivity is decoration. The register's inputs remain human judgments with named owners, whatever produced the output.

KEY TERMS — KA 8.2

Term	Meaning
EMV	probability × impact; the average of outcomes that will not individually occur.
Decision tree	Explicit rollback of decisions and chance events to a present choice.
Value of information	The reduction in expected cost that knowing enables; zero if it changes no action.
P50 / P80	Confidence levels — amounts covering the aggregate outcome with stated probability.
Correlation	Shared drivers that raise aggregate variance; independence is usually optimistic.
Monte Carlo simulation	Repeated sampling to produce an outcome distribution rather than a point.
Within-cell span	The EMV range a single matrix cell contains: the product of its two band ratios.
Quantification threshold	The impact above which a screened risk is quantified individually; a screen without one has replaced quantification.
Variance contribution	$p(1-p)I^2$ for one risk; quadratic in impact, so it ranks a register differently from EMV.
Value of perfect information	The most any signal could be worth: expected cost without information less expected cost knowing the truth free.
Price of confidence	The cost per percentage point of moving a reserve between confidence levels.

SAMPLE MCQS — KA 8.2

MCQ 8.2-A `[8.2.2 · Application]` A risk has probability 0.15 and impact USD 400,000. Its EMV is: - A. USD 400,000 - B. USD 60,000 - C. USD 340,000 - D. USD 26,667

Rationale: `0.15 × 400,000 = 60,000`. A is the impact; C is impact less EMV; D divides instead of multiplying.

MCQ 8.2-B `[8.2.2 · Analysis]` R4 has the register's largest impact (400,000) but its smallest threat EMV (60,000). The correct managerial reading is: - A. R4 should be ignored, having the lowest EMV - B. EMV sets the funding priority, while impact still governs whether the event is survivable — both readings are needed - C. impact alone should drive priority - D. the assessment must be wrong, since impact and EMV disagree

Rationale: EMV is the right basis for funding a portfolio of risks; a single large impact may still be existential regardless of probability (8.2.2, 8.3.1). A and C each discard half the information; D misunderstands that the two measure different things.

MCQ 8.2-C `[8.2.3 · Application]` Proceeding directly has an expected cost of `0.40 × 300,000`. A USD 25,000 survey reduces the mitigated cost to USD 90,000. The value of the information is: - A. USD 25,000 - B. USD 59,000 - C. USD 84,000 - D. nil — the survey costs more than it saves

Rationale: `120,000 − (25,000 + 36,000) = 59,000`. A is the survey's price, not its value; C omits the survey cost from branch B; D reverses the conclusion.

MCQ 8.2-D `[8.2.4 · Analysis]` A register's EMV sum is 278,000 and its worst-case sum is 1,140,000. Setting contingency at 278,000 means: - A. an appropriately funded reserve - B. funding the average outcome, which by construction is exceeded roughly half the time - C. a conservative reserve, since not all risks will occur - D. the same as a P80 reserve

Rationale: The mean is the ~50th percentile of the aggregate (8.2.4), so it is exceeded about half the time — the reason a confidence level is chosen explicitly. C mistakes the mean for conservatism; D confuses two different statistics (490,624 here).

MCQ 8.2-E `[8.2.1 · Analysis]` Why must ordinal probability-impact scores not be multiplied and summed as money? - A. the matrix is only for threats - B. ordinal bands are ranks, not quantities — a "4" is not twice a "2", so the arithmetic is meaningless - C. multiplication requires more than five bands - D. scores may be summed provided they are weighted

Rationale: Ordinal scales support ordering, not arithmetic (8.2.1). Weighting (D) does not repair a scale that never carried magnitude.

MCQ 8.2-F `[8.2.1 · Application]` A matrix's medium probability band runs 0.10–0.35 and its medium impact band USD 60,000–250,000. The span of expected values inside that single cell is a factor of: - A. 3.5 - B. 14.58 - C. 4.17 - D. 2.5

Rationale: Within-cell span is the product of the two band ratios, `3.5 × 4.1667 = 14.5833` (8.2.1). A gives the probability ratio alone and C the impact ratio alone — each answers half the question; D subtracts one from the probability band ratio (3.5 − 1), the slip that reads a ratio as an increment.

MCQ 8.2-G `[8.2.4 · Analysis]` A register's contingency was set at a P80 of USD 490,624 on an independence assumption. Three of its risks are then found to share a subcontractor, raising σ from 252,642 to 340,339. The reserve is now: - A. still a P80 reserve, because the mean has not changed - B. a 73.4 % reserve — correlation widens the distribution without moving the mean, so the same amount covers fewer futures - C. inadequate by USD 73,806 of expected cost - D. unaffected, since correlation is a modelling choice rather than a fact

Rationale: `z = (490,624 − 278,000)/340,339 = 0.6247`, i.e. **73.4 %** (8.2.4b). A confuses the mean's invariance with the percentile's; C misreads the USD 73,806 — it is the extra reserve needed to restore 80 % confidence, not an increase in expected cost, which is unchanged at 278,000; D treats a shared supplier as an opinion.

MCQ 8.2-H `[8.2.3 · Analysis]` Twelve of 40 clinics run the configuration under suspicion. A two-clinic pilot chosen at random detects the problem with probability 0.5154; a two-clinic pilot chosen to include a suspect configuration detects it with certainty. Both cost USD 45,000. The right conclusion is: - A. the pilots are equivalent, since the cost and sample size are identical - B. the design choice is worth USD 59,189 at no additional cost, because the value of information depends on what the sample is capable of detecting - C. the random pilot is preferable because it is unbiased - D. neither pilot is worthwhile, since 45,000 exceeds the redesign saving

Rationale: Expected costs are 151,756.98 random against 92,568.00 designed (8.2.3b). A prices inputs rather than the information bought; C applies a sampling virtue that is irrelevant when a specific condition is being tested and Domain 9 (KA 9.3.2) warns about in its own instrument; D compares the pilot price with the wrong quantity — the comparison is against the 565,680 late case, and the designed pilot pays above an 11.05 % prior.

MCQ 8.2-I `[8.2.4 · Application]` Two responses cost the same and each removes the same proportion of a risk. Response A halves the probability; Response B halves the impact. Compared on the reserve they release: - A. they are equivalent, since both halve the `EMV` - B. B releases more, because variance goes as impact squared — halving an impact removes 75 % of that risk's variance while halving a probability removes between 25 % and 50 % - C. A releases more, because probability drives whether the event happens at all - D. neither releases reserve, since contingency is set from `EMV`

Rationale: EMV is linear in both, so A's premise is right and its conclusion wrong; variance is $p(1-p)I^2$ (8.2.4c). C reverses the result; D confuses a mean-based reserve with a confidence-based one, which is the error MCQ 8.2-D addresses.

SELF-CHECK — KA 8.2

1. Why is *EMV* wrong for judging whether one risk is survivable? — It averages outcomes that will not occur; the actual event is impact-or-nothing.
2. When is information worth nothing however reassuring? — When it would not change the action taken.
3. Which assumption in the P80 calculation is usually optimistic, and why does it matter? — Independence; correlated risks raise variance and the true confidence amount.
4. What determines the span of expected values inside one matrix cell? — The product of its probability and impact band ratios; nothing else.
5. What does correlation change and what does it leave alone? — It widens the distribution and raises every percentile; the mean is unchanged, which is why EMV reviews cannot detect it.
6. Which lever should a leader short of contingency pull, and why? — Impact, because variance is quadratic in impact: at equal EMV reduction it removes 1.5 to 3 times as much variance.
7. Why express total exposure against a tolerance rather than against the budget? — The tolerance is the room governance controls; Meridian's 119,000 is 4.96 % of approved cost and 99.17 % of the tolerance.

KNOWLEDGE AREA 8.3

Responses, reserves and governance

Topics: 8.3.1 selecting and costing responses · 8.3.2 reserves and their authority · 8.3.3 monitoring and the register that earns its keep.

8.3.1 Selecting and costing responses

The response families — **avoid** (remove the cause or the exposure), **reduce** (lower probability or impact), **transfer** (insurance, contract terms — a price, not a disappearance; Domain 7, KA 7.4.2), **accept** (with or without a fallback), and for opportunities **exploit, enhance, share, ignore**. Two rules make selection professional rather than reflexive.

A response is an investment. It costs money and time and reduces EMV; if the reduction is smaller than the cost, the response destroys value and acceptance is the correct answer. Auriga's fast-track decision from Domain 6 is exactly this arithmetic: one week of client bonus (USD 45,000) against an accepted rework risk of $0.20 \times 60{,}000 = 12{,}000$ — a net **+USD 33,000**, which is why the recovery plan took it.

Impact governs survivability even at low probability. A 3 % chance of an event the project cannot survive is not managed by its EMV; it is avoided, transferred, or the project's viability is reconsidered. EMV funds portfolios; existential risks are handled by structure, and confusing the two is how organisations optimise their way into single points of failure.

Secondary risks. Every response creates its own — fast-tracking creates rework risk, a transfer creates counterparty risk, mitigation creates delivery risk on the mitigation itself. A register whose responses have no secondary entries has not been thought through. That is usually asserted as a completeness rule; it is in fact a selection rule, because the secondary risks can reverse which response is cheapest.

WORKED EXAMPLE

Worked example 8.3.1 — four responses to Auriga's R1, priced against each other.

1. **Setup.** R1 — controller lead-time slip — carries p 0.35 and an impact of **USD 240,000**, so its EMV is **USD 84,000** and its impact is **6.00 %** of BAC (survivable, so the EMV logic below is legitimate; see the caution in step 5). Four options are costed:
2. **Accept**, with the slip absorbed by contingency.
3. **Reduce** — dual-qualify a second controller supplier for **USD 22,000**, cutting p to 0.15. Secondary risk: the alternate's first article fails inspection, p 0.10, impact USD 40,000.
4. **Transfer** — a delivery-guarantee term in the supply contract priced at **USD 18,000** that recovers **60 %** of the impact. Secondary risk: the supplier is a single-site specialist whose ability to pay is impaired by the very disruption that triggers the claim, assessed at **0.25** conditional on the event.
5. **Avoid** — redesign to a commodity controller for **USD 95,000**, removing R1 entirely. Secondary risk: the redesign introduces integration rework, p 0.20, impact USD 120,000.
6. **Formula.** Total expected cost of an option = response cost + residual EMV + Σ secondary EMV. Choose the minimum. For the transfer, residual EMV = $p \times \text{impact} \times (1 - \text{recovery})$ and the counterparty secondary = $p \times P(\text{cannot pay}) \times \text{recovered amount}$.
7. **Substitution.** Reduce $22{,}000 + 0.15 \times 240{,}000 + 0.10 \times 40{,}000$. Transfer $18{,}000 + 0.35 \times 240{,}000 \times 0.40 + 0.35 \times 0.25 \times (0.60 \times 240{,}000)$. Avoid $95{,}000 + 0 + 0.20 \times 120{,}000$.
8. **Result.**

Option	Response cost	Residual EMV	Secondary EMV	Total	Total ignoring secondaries
Accept	0	84,000	0	**84,000**	84,000
Reduce	22,000	36,000	4,000	**62,000**	58,000
Transfer	18,000	33,600	12,600	**64,200**	51,600
Avoid	95,000	0	24,000	**119,000**	95,000

With secondary risks priced, **reduce wins at USD 62,000**. Ignoring them, transfer appears best at 51,600 — better than reduce by 6,400 — when in truth it is worse by 2,200: a **swing of USD 8,600** in the comparison, and a change of decision. 5. **Interpretation.** Four conclusions, and the first is the reason this example exists.

Omitting secondary risks does not make a response look slightly better; it changes which response is chosen. The transfer's secondary is large precisely because it is correlated with the event it covers — the supplier's insolvency and the delivery failure have the same cause. That is the general case rather than an exotic one, and it is why "transfer is a price, not a disappearance" (Domain 7, KA 7.4.2; Domain 10, KA 10.3) needs a number beside it. The breakeven is checkable: the transfer ties the reduce option when the counterparty's probability of being unable to pay reaches $(62{,}000 - 18{,}000 - 33{,}600)/(0.35 \times 144{,}000) =$ **20.63 %**. Below that, transfer; above it, dual-qualify. A leader who cannot form a view on that one number has no basis for choosing between the two, and asking for it is a better use of a review than re-reading the register. One pointer, because this is an enforceability-sensitive area: whether a guarantee or indemnity is enforceable, what it actually reaches, and where the claim ranks if the counterparty fails are legal questions that turn on the wording and on the jurisdiction. Take the 60 % recovery this table multiplies from qualified counsel (Domain 10, KA 10.3), not from the register.

Avoidance is not the safe answer. It is the most expensive option here by a wide margin — **USD 35,000 worse than doing nothing** — and it does not even eliminate exposure, because the redesign carries 24,000 of secondary `EMV` of its own. Avoidance moved the risk from procurement to engineering and paid 95,000 for the move. The instinct that the strongest-sounding response family is the most responsible one is the commonest failure in response selection, and the arithmetic is the only reliable corrective.

Acceptance is a priced position, not a default. Reduce creates **USD 22,000** of value against accept — a **26.19 %** improvement on the 84,000 — so acceptance is wrong here. But it becomes right the moment the reduce option's cost rises above 44,000 (`84,000 − 36,000 − 4,000`), which a single change in the alternate supplier's qualification scope could do. Acceptance and response are the two ends of one calculation, and the register should record the cost at which the answer flips, not just the answer.

Two boundaries on all of it. First, this comparison is on **expected cost only**. Worked example 8.2.4c's result applies here too: the reduce option is a probability lever and the transfer is effectively an impact lever, so if the binding constraint were the confidence level rather than the expected cost, the ranking would need recomputing on variance — dual-qualifying leaves `0.15 × 0.85 × 240,000² = 7,344,000,000` of variance where the transfer leaves `0.35 × 0.65 × 96,000² = 2,096,640,000`. Second, and overriding: **if R1's impact were existential rather than 6 % of `BAC`, the avoid option's 119,000 would be cheap** and the whole table would be the wrong instrument. That is the next rule, and it is not a refinement of this one but a limit on it.

8.3.2 Reserves and their authority

Domain 7 (KA 7.1.3) established the structure; this domain supplies the sizing and the governance:

Reserve	Covers	Sized by	Spent by
Contingency	Identified risks in the register	Aggregation to a stated confidence (8.2.4)	Project manager, under a published protocol
Management reserve	Unknown-unknowns and scope change	Judgment and organisational policy	Sponsor / change authority, via change control

Three governance rules. **The draw protocol is published in advance** — which risk, what evidence, what approval — because a reserve released ad hoc is indistinguishable from an overrun. **Consumption is trended against risk retirement**, not against time: burning 60 % of contingency while 20 % of the register has been retired is the signal MCQ 7.1-B describes. And **contingency released by retired risks is returned, not reallocated** to convenient overspends; otherwise the reserve silently becomes a slush fund and the next real risk is unfunded.

Two questions remain, and both are arithmetic rather than procedural: at what confidence should the reserve be held, and is the reserve still adequate now that some of it has been spent. The first is answered from the appetite, not from convention.

WORKED EXAMPLE

Worked example 8.3.2 — deriving Meridian's confidence level from its appetite.

1. **Setup.** Meridian's board has expressed its appetite in one sentence: it will accept **no more than a 10 % chance** that the programme's outturn cost exceeds the approved **USD 2,400,000** by more than **5 %**. The quantified register (8.2.2b) has a mean of **USD 119,000**, σ of **USD 100,742.25**. No confidence level has yet been chosen for contingency.
2. **Formula.** Tolerance `= 5 % × approved cost`. "No more than a 10 % chance of exceeding" is the definition of a **P90** requirement, so the appetite requires `mean + 1.2816 σ ≤ tolerance`. Rearranged, the maximum σ the appetite permits at a given mean is `σ_max = (tolerance − mean)/1.2816`.
3. **Substitution.** Tolerance `0.05 × 2,400,000`. Required P90 `119,000 + 1.2816 × 100,742.25`. Permitted σ `(120,000 − 119,000)/1.2816`.
4. **Result.** Tolerance **USD 120,000**. The register's P90 is **USD 248,111.26 — 2.0676 times** the tolerance, and **10.34 %** of approved cost against an appetite of 5 %. The σ the appetite permits at the current mean is **USD 780.27**, against an actual σ of 100,742.25: the appetite is not merely breached but **unreachable by any contingency decision**.
5. **Interpretation.** Three findings, each of which is a different conversation.

The confidence level was never a choice. The board said "10 % chance of exceeding", which is P90; nobody needed to prefer P80 or P90, and a programme that holds contingency at P80 while its board has stated a 10 % tolerance is inconsistent with its own governance. This is the general result: **a well-formed appetite statement determines the confidence level.** The corollary is more useful in practice — where a project has inherited "P80 by convention", the honest reconstruction is to ask what appetite statement P80 implies (a 20 % acceptance of exceedance) and put that sentence in front of the sponsor. Most sponsors have never seen it written down, and some do not agree with it.

The appetite constrains σ, not just the mean, and that is where it becomes unreachable. With the mean at 119,000 the tolerance leaves USD 1,000 of room, so the permitted σ is 780 — a register with essentially no uncertainty at all. No reserve fixes this, because contingency is money held inside the tolerance; the tolerance is the room. The three real options are the ones 8.2.2b named, and the arithmetic now sizes them: reduce the exposure (responses, per 8.2.4c and 8.3.1), widen the tolerance (a governance decision the board must take knowingly), or reduce scope. The professional obligation is to present the choice, not to select the confidence level that makes the paper pass.

And the gap is the number that makes the case. Restoring the appetite requires taking the P90 from 248,111 to 120,000 — a reduction of **USD 128,111** — which on the evidence of 8.2.4c is bought mainly by impact levers. Response B there removed USD 38,932 of P90 for USD 25,000, so the order of magnitude is three to four such responses, and that is a fundable programme of work rather than an argument. Two cautions. The 1.2816 factor is the normal approximation again, and on a five-risk register it carries the sort of error 8.2.4 measured, so the target should be expressed as a range. And an appetite expressed against cost says nothing about schedule, benefit or reputation; each objective needs its own statement, and a board that has expressed only one has expressed less than it thinks (Domain 15, KA 15.4).

WORKED EXAMPLE

Worked example 8.3.2b — is Auriga's reserve still adequate at week 13?

1. **Setup.** Auriga's contingency was set at the P80 of **USD 490,624** (8.2.4). At the week-13 review — `PV` 2,080,000, `EV` 1,920,000, `AC` 2,120,000, `CPI` 0.91, `SPI` 0.92 (Domain 7, KA 7.3) — three things have happened. **R2** (ground conditions) has **occurred** and is now an issue. **USD 232,000** of contingency has been drawn under the published protocol: **USD 195,000** against R2, whose outturn exceeded its 180,000 assessment, and **USD 37,000** between two register entries that sat below the quantification threshold of 8.2.1 and occurred. **R4** (permit delay) has been **retired**: the consenting window closed without incident. **R1**, **R3** and **R5** remain open at unchanged assessments. Is the remaining reserve still a P80 reserve?
2. **Formula.** Requirement = P80 of the **open** register (mean + 0.8416 σ, recomputed from the open entries only). Remaining reserve = original reserve − drawn. Adequacy ratio = remaining ÷ requirement. Implied confidence = `Φ((remaining − open mean)/open σ)`. Release from a retired risk = original P80 − P80 recomputed without it.
3. **Substitution.** Open register: R1 `0.35 × 240,000`, R3 `0.25 × 320,000`, R5 `0.30 × (120,000)`; variance `13,104,000,000 + 19,200,000,000 + 3,024,000,000`. Remaining `490,624 − 232,000`.
4. **Result.** Open mean **USD 128,000**, open variance **35,328,000,000**, open σ **USD 187,957.44**, so the P80 **requirement is USD 286,184.98**. Remaining reserve **USD 258,623.54 — 47.29 %** of the reserve has been drawn. Adequacy ratio **0.9037**, a **shortfall of USD 27,561.44**, and the remaining reserve in fact covers `z = 0.6950` of the open distribution: a **75.65 %** reserve, not an 80 % one.

Separately, retiring R4 releases more than its `EMV`. The register without R4 has mean **218,000**, σ **208,393.86** and P80 **USD 393,384.27**, so R4's retirement releases `490,624 − 393,384 =` **USD 97,239.27** against an `EMV` of only 60,000 — a ratio of **1.6207**. 5. **Interpretation.** Two results, and together they replace the usual reserve test with a better one.

The naive test passes and the real test fails. On the conventional comparison, 47.29 % of the reserve is drawn while R4's retirement alone accounts for **21.58 %** of the original `EMV` and R2's occurrence for a further 32.37 % — drawdown and register progress look broadly in step, and a status report saying so would not be challenged. Recomputing the requirement from the open register shows the reserve is **90.37 %** of what its own policy requires and is buying **75.65 %** confidence rather than 80 %. The discipline that follows: **a part-consumed reserve is tested against the register still open, at the stated confidence, not against elapsed time or against the original register.** The number to report each period is the adequacy ratio, and the trigger — a ratio below 1.00 — is a request to top up from management reserve or to reduce exposure, taken deliberately while USD 27,561 is a small decision.

Retiring a low-probability, high-impact risk releases far more than its `EMV`. R4 carried 32.0 % of the original variance on 21.6 % of the `EMV`, because variance is quadratic in impact (8.2.4c). A leader who returns only the retired risk's `EMV` of 60,000 leaves **USD 37,239** sitting in the reserve doing nothing — money that the "return, do not reallocate" rule above was never meant to trap. The corollary is uncomfortable and worth stating: **an unretired risk whose window has quietly passed is expensive**, because it holds variance-weighted reserve against an exposure that no longer exists. Register hygiene is a funding activity, which is not how it is usually presented.

Two cautions. Recomputing the requirement each period tempts a project into re-deriving its contingency downwards whenever the arithmetic is convenient; the protocol should fix the confidence level and the method in advance, so that only the register changes. And the whole calculation still assumes independence among R1, R3 and R5 — on 8.2.4b's finding, the true requirement is higher and the adequacy ratio worse, which is the first question a reviewer should ask of this table rather than the last.

8.3.3 Monitoring and the register that earns its keep

A live register has movement: probabilities revised as evidence arrives, risks retired when their window closes, new entries as the project learns, owners who report rather than merely appear. The tests of whether it is working are behavioural, not documentary — **has a decision changed because of it this month?**, **are the top items the ones the team actually worries about?**, and **did anything that hurt us appear in it beforehand?** A register that answers no to all three is theatre, and the honest response is to fix the process rather than reformat the document. **Early-warning indicators** are the register's operational edge: leading measures tied to specific risks (supplier confirmations slipping, defect discovery rate rising, permit queue lengthening) that fire before the event, which is what makes a response affordable (Domain 6's lead-time point).

KEY TERMS — KA 8.3

Term	Meaning
Avoid / reduce / transfer / accept	The threat-response families; transfer is a price, not a disappearance.
Secondary risk	Risk created by a response; a register without them is incomplete.
Draw protocol	The published rules for releasing contingency.
Risk retirement	Closing a risk whose window has passed; frees contingency for return — often far more than the risk's `EMV`.
Early-warning indicator	A leading measure tied to a risk, firing before the event.
Tolerance	The amount by which an objective may be exceeded before appetite is breached; the room inside which contingency sits.
Adequacy ratio	Remaining reserve ÷ the confidence-level requirement of the register still open; below 1.00 is a governance trigger.
Counterparty risk (priced)	The secondary `EMV` of a transfer: `p × P(cannot pay) × amount recovered`, usually correlated with the event transferred.

SAMPLE MCQS — KA 8.3

MCQ 8.3-A `[8.3.1 · Application]` A mitigation costs USD 50,000 and reduces a risk's `EMV` from 84,000 to 20,000. The decision and its basis are: - A. reject — 50,000 is a large outlay - B. accept — the `EMV` reduction of 64,000 exceeds the 50,000 cost - C. accept — any reduction in `EMV` justifies a response - D. indifferent — cost and benefit are equal

Rationale: Responses are investments: `84,000 − 20,000 = 64,000` of reduction for 50,000 of cost is value-creating. A prices the outlay without the benefit; C would justify unlimited spend; D miscomputes.

MCQ 8.3-B `[8.3.1 · Analysis]` A risk has probability 0.03 and an impact the project could not survive. The correct treatment is: - A. accept it — the `EMV` is small - B. treat it structurally: avoid, transfer, or reconsider viability, because `EMV` funds portfolios while survivability is governed by impact - C. fund its `EMV` in contingency and monitor - D. exclude it from the register as improbable

Rationale: Existential exposure is not an averaging problem (8.3.1). A and C both apply portfolio logic to a single point of failure; D removes the entry that most needs governance attention.

MCQ 8.3-C [8.3.2 · Analysis] Contingency freed by a retired risk is used to cover an unrelated overspend. This is: - A. efficient reserve management - B. a governance failure: the reserve silently becomes a slush fund and the next genuine risk is unfunded - C. acceptable if the total baseline is unchanged - D. required, since contingency is inside the baseline

Rationale: Contingency is tied to identified risks; reallocating it to overspends destroys the link between reserve and exposure (8.3.2). C is the reasoning that makes the failure invisible.

MCQ 8.3-D [8.3.1 · Analysis] On one risk, transferring appears cheaper than reducing — USD 51,600 against 58,000 — until secondary risks are priced, after which reducing is cheaper at 62,000 against 64,200. The correct reading is: - A. the second calculation is double-counting, since the transfer already covers the risk - B. secondary risks are a selection rule, not only a completeness rule: here the transfer's counterparty exposure is correlated with the very event transferred, and pricing it changes the decision - C. the difference is immaterial at USD 2,200 and either option may be taken - D. reducing is always preferable to transferring

Rationale: The transfer's secondary EMV of 12,600 arises because the supplier's ability to pay is impaired by the disruption that triggers the claim (8.3.1). A misses that the counterparty may not perform; C ignores that the 2,200 is a reversal of a 6,400 apparent advantage and that the breakeven default probability, 20.63 %, is the thing to form a view on; D generalises one case into a rule.

MCQ 8.3-E [8.3.2 · Application] A risk with p 0.15 and impact USD 400,000 is retired when its window closes. Its EMV was USD 60,000, and recomputing the register's P80 without it falls by USD 97,239. The amount to return to the sponsor is: - A. USD 60,000 — the risk's EMV - B. USD 97,239, because the reserve was sized on a confidence level and the retired risk carried 31.96 % of the variance on 21.58 % of the EMV - C. nil, since contingency stays inside the baseline - D. USD 37,239 — the difference between the two figures

Rationale: A confidence-based reserve releases what recomputing it releases; variance is quadratic in impact, so a low-probability high-impact risk carries disproportionate reserve (8.3.2b). A returns the mean-based amount and leaves USD 37,239 idle; D returns only that idle remainder; C confuses where contingency sits with whether it may be released.

MCQ 8.3-F [8.3.2 · Analysis] A board states it will accept no more than a 10 % chance of exceeding approved cost by more than 5 %. The confidence level for contingency is therefore: - A. P80, the industry convention - B. P90, because "no more than a 10 % chance of exceeding" is the definition of the 90th percentile, and the tolerance is 5 % of approved cost - C. P95, to be prudent - D. undetermined — appetite statements are qualitative

Rationale: The appetite determines the confidence level (8.3.2); the leader's job is to derive it, not to select it. A inherits a convention that implies a 20 % acceptance of exceedance the board did not state; C substitutes caution for the board's decision; D is the position the arithmetic refutes.

SELF-CHECK — KA 8.3

1. When is acceptance the professionally correct response? — When the response's cost exceeds the EMV reduction it buys — and the risk is survivable.
2. What does transfer actually achieve? — It prices the risk to a counterparty and creates counterparty risk; it does not remove the exposure.
3. What should contingency consumption be trended against? — Risk retirement, not elapsed time.
4. State the adequacy test for a part-consumed reserve. — Remaining reserve divided by the stated confidence level's requirement recomputed on the register still open; below 1.00 is a trigger.
5. Where does an appetite statement's confidence level come from? — The statement itself: "no more than a 10 % chance of exceeding" is P90.

6. Why can avoidance be the most expensive response? — It is paid in full whether or not the event would have occurred, and it creates its own secondary risks — Auriga's avoid option costs USD 35,000 more than accepting.

KNOWLEDGE AREA 8.4

Resilience, bias and crisis leadership

Topics: 8.4.1 resilience versus prediction · 8.4.2 cognitive bias · 8.4.3 crisis leadership · 8.4.4 AI-enabled risk sensing.

8.4.1 Resilience versus prediction

Quantification (KA 8.2) handles risks you can name. **Resilience** is the capability to absorb what you cannot: it assumes the register is incomplete, because it always is. Its levers are structural rather than analytical — **buffers** (float placed deliberately, contingency, capacity headroom), **optionality** (staged commitments, alternative suppliers qualified in advance, reversible decisions — Domain 1, KA 1.3.3), **modularity** (a failure contained rather than propagated across interfaces), **redundancy** where failure is intolerable, and **fast detection** (early-warning indicators, short feedback loops), because response cost rises with delay.

The trade-off is real and should be stated rather than finessed: resilience costs efficiency. A project optimised to the last dollar and week has no capacity to absorb surprise, and one buffered everywhere is uncompetitive. The leadership judgment is where to be resilient — at the points where failure propagates or cannot be tolerated — not whether.

That judgment is usually left qualitative, and it need not be. A resilience measure has a price, a probability of being needed and a recovery time it shortens, which is enough to price it twice: once on expected value and once on the tail. The two answers frequently disagree, and knowing what to do when they do is the substance of this topic.

WORKED EXAMPLE

Worked example 8.4.1 — Meridian's standby trainer retainer, on the average and on the tail.

1. **Setup.** Meridian's rollout depends on a single training provider. If that provider's capacity fails — attrition, a competing contract — clinics cannot go live, and the programme's cost of delay is **USD 14,280 per week**. The probability of a capacity failure over the rollout is assessed at **0.25**. Recovering unaided means re-tendering: **7 weeks**. A **USD 18,000** annual retainer with a second, pre-qualified provider reduces recovery to **2 weeks**. The programme separately notes the tail: unaided, the 95th-percentile recovery is **12 weeks**, which would miss the winter demand window and defer the remaining **40 %** of the benefit by a full **13-week** quarter; with the retainer the 95th-percentile recovery is **3 weeks** and the window holds.
2. **Formula.** Expected delay cost = `P(failure) × recovery weeks × cost of delay`; add the retainer where taken. Breakeven probability = retainer ÷ (weeks saved × cost of delay). Tail cost = 95th-percentile weeks × cost of delay, plus any window-miss deferral.
3. **Substitution.** Without: `0.25 × 7 × 14,280`. With: `18,000 + 0.25 × 2 × 14,280`. Breakeven `18,000/(5 × 14,280)`. Tail deferral `0.40 × 14,280 × 13`.
4. **Result.** Expected cost **without** the retainer **USD 24,990.00**; **with** it **USD 25,140.00** — the retainer is a **net loss of USD 150.00**. It breaks even at a failure probability of `18,000/71,400 =` **25.2101 %**, against the assessed 25.00 %: it misses by **0.2101 of a percentage point**.

On the tail the picture inverts. Without the retainer the 95th-percentile outcome costs `12 × 14,280 =` **USD 171,360** of delay **plus USD 74,256** of deferred benefit — **USD 245,616**. With it, `18,000 + 3 × 14,280 =` **USD 60,840**. The retainer removes **USD 184,776** of tail exposure and reduces it by a factor of **4.0371**. 5. **Interpretation.** The two calculations disagree, and the professional content of this example is what a leader does about that.

When the breakeven sits inside the estimating error, expected value has stopped being the deciding test. The breakeven probability is 25.2101 % and the assessment is 25 %. Nobody can estimate a provider-capacity failure to a fifth of a percentage point; the difference is noise. The correct reading is not "reject, it loses USD 150" — that is false precision dressed as rigour — but **"on expected value this decision does not matter, so decide it on the tail and on survivability"**. That is exactly the boundary 8.3.1 draws between portfolio logic and structural logic, arriving here from the other direction: the arithmetic hands the decision back to judgment, and names why it is doing so.

On the tail it is not close. The retainer costs USD 18,000 and removes USD 184,776 of 95th-percentile exposure — a ratio of **10.27 to 1**. And the tail is not a smooth extension of the average, because a 12-week recovery crosses a **threshold**: it misses the winter window, at which point a further 74,256 of deferred benefit appears that no expected-value calculation containing only a weekly rate would ever show. **Resilience is bought where the consequence function has a step in it**, which is a sharper statement of "where failure cannot be tolerated" and a testable one: find the thresholds, then ask what recovery time clears them.

The efficiency cost is real and should be named, not hidden in the tail argument. USD 18,000 buys nothing at all in the 75 % of futures where the provider performs; that is what "resilience costs efficiency" means in cash, and it is the reason the lever has to be aimed rather than applied everywhere. The discriminator is the step: dependencies whose failure crosses a threshold get structural treatment, dependencies whose failure is priced linearly are accepted and funded through contingency. Two cautions. The 95th-percentile recovery times are themselves estimates with wider errors than the mean, so the tail argument should be presented as an order of magnitude — 10 to 1 — rather than as USD 184,776 to the dollar. And a retainer is itself a contract with a counterparty, so it carries the secondary risk 8.3.1 priced: a second provider that cannot mobilise in two weeks when called has sold an option that does not exercise, which is a procurement-terms question (Domain 10, KA 10.3) and not a risk-register one.

8.4.2 Cognitive bias

Estimates and reviews are produced by people, and the predictable distortions are worth naming because each has a countermeasure:

Bias	Effect	Countermeasure
Optimism bias / planning fallacy	Systematic underestimation of duration and cost	Reference-class forecasting against comparable completed projects
Anchoring	First number dominates subsequent estimates	Independent estimates before any figure is shared
Availability	Recent or vivid events overweighted	Structured checklists and historical base rates
Groupthink / social pressure	Dissent suppressed in workshops	Pre-mortem, private input channels, explicit dissent invitation
Sunk-cost / escalation of commitment	Continuing because of what is spent	Decision framed on remaining cost and benefit only (Domain 2's kill criteria)
Confirmation bias	Evidence sought that supports the plan	Assign someone to argue the opposite; test the disconfirming case

The **pre-mortem** deserves particular mention as the highest-yield, cheapest technique in this domain: before committing, ask the team to imagine the project has failed and to write down why. It licenses the dissent a status meeting suppresses, and it reliably surfaces risks the workshop of 8.1.3 missed.

Reference-class forecasting is the countermeasure with arithmetic behind it, and the arithmetic does something the table above cannot: it prices the risks nobody identified. The method is to stop asking "what will this project cost?" and instead ask "what did comparable completed projects cost against what they were approved for?" — then apply that distribution of outturn ratios to the current estimate. The word doing the work is **completed**: the reference class contains the overruns, the surprises and the unidentified risks of every project in it, which is precisely the material a bottom-up register cannot contain.

WORKED EXAMPLE

Worked example 8.4.2 — Meridian against its reference class.

1. **Setup.** Meridian's approved cost is **USD 2,400,000**, built bottom-up, with a quantified risk register whose P80 contingency requirement is **USD 203,784.67** (8.2.2b, at the 0.8416 factor). The sponsoring authority holds records of **12** comparable completed clinical-system rollouts. Each is expressed as an **outturn ratio** — final cost ÷ approved cost at the equivalent gate — and sorted: 0.95, 1.02, 1.05, 1.08, 1.12, 1.15, 1.18, 1.22, 1.28, 1.35, 1.48, 1.72.
2. **Formula.** Mean ratio = Σ ratios ÷ n. Median = average of the two central values at even n. The 80th percentile is read from the order statistics at position `(n + 1) × 0.80`, interpolating between neighbours. Reference-class forecast = approved cost × ratio; uplift = forecast − approved cost.
3. **Substitution.** Σ = 14.60, so mean `= 14.60/12`. Median `= (1.15 + 1.18)/2`. Order position `13 × 0.80 = 10.40`, so `P80 = 1.35 + 0.40 × (1.48 − 1.35)`.
4. **Result.**

Statistic	Ratio	Forecast (USD)	Uplift on 2,400,000	Uplift %
Best case in class	0.95	2,280,000	(120,000)	(5.00 %)
Median	1.1650	2,796,000	396,000	16.50 %
Mean	1.216667	**2,920,000**	520,000	21.67 %
P80	1.4020	**3,364,800**	**964,800**	**40.20 %**
Worst case in class	1.72	4,128,000	1,728,000	72.00 %

11 of the 12 projects (**91.67 %**) exceeded their approved cost. The reference-class P80 uplift of **USD 964,800** is **4.7344 times** the register-based P80 of 203,785 — a gap of **USD 761,015.33**. 5. **Interpretation.** The gap is the finding, and what to do with it is the professional skill.

Do not add the two numbers. They are two estimates of the same quantity by different methods, not two separate exposures, and summing them double-counts every risk the register identified. The reference class already contains ground-condition surprises, interface rebuilds and trainer attrition, because the twelve projects in it experienced them.

Use the reference class as a challenge on the total, and treat the gap as an estimate of what the register cannot see. Two causes account for it, and only one is measurable from the data here. The register-completeness arithmetic of 8.1.3 estimated **11.4286** unidentified risks; if each resembled the average quantified entry at `119,000/5 =` **USD 23,800** of `EMV`, they would add **USD 272,000.00 — 35.74 %** of the gap. The residual **USD 489,015.33** cannot be decomposed from the evidence available and must be presented as such: part of it is **optimism in the impacts already assessed** (8.4.2's subject), and part is **scope growth** rather than risk at all, because projects in a reference class typically had less well-defined scope at the equivalent gate and grew into their outturns (Domain 5). Presenting the residual as a single cause would be exactly the false precision this topic warns against.

What the numbers change. Not the approved cost, which is a governance artefact, but three things: the **management reserve**, which now has a defensible basis (8.1.3's unidentified risks and the residual gap, held outside the baseline under sponsor control); the **funding requirement** presented to the authority, which should carry the reference-class range rather than a single figure;

and the **tolerance conversation** of 8.3.2, since a 5 % tolerance against a class whose median outturn is 16.5 % is a statement about this programme being materially better managed than its peers — a claim that may be true and must then be evidenced, not assumed.

Three cautions on the method itself. The class must be genuinely comparable: twelve rollouts of clinical systems, not twelve public-sector programmes of any kind, and the leader should be able to say what makes them comparable and what makes each different. Twelve is a small class, so the P80 read from order statistics is coarse — the interpolation between the 10th and 11th values moves the answer by USD 312,000 across that single interval, which is the honest measure of its precision. And the class's own approved costs were set under whatever appraisal regime applied at the time; a class drawn from an era of weaker scope definition overstates the uplift a well-defined project needs, which is a reason to prefer recent comparators and to say so.

8.4.3 Crisis leadership

When a serious risk materialises, the leader's job changes shape. The sequence that works: **stabilise** (stop the harm; secure safety, then the situation), **establish facts** (separate what is known from assumed — crises run on rumour), **decide with a clock** (a good decision now beats an optimal one later, and say explicitly which decisions are reversible), **communicate early and honestly** (bad news does not improve with age; Domain 11's stakeholder work is now executing under compressed time), **protect the team** (crises consume people, and exhausted teams make the second mistake), and **capture the lesson while it is vivid** (Domain 9's lessons-learned). Domain 6's recovery machinery is the schedule expression of the same posture: re-run the passes, price the options, escalate the trade with named authority.

8.4.4 AI-enabled risk sensing

The genuine capability is **detection at scale**: anomaly detection across cost, schedule, quality and supplier data; pattern recognition against historical projects; and sentiment or volume signals in issue logs and correspondence that precede formal escalation. Used well, this shortens the detection half of 8.4.1's fast-detection lever — which is where its value actually lies.

The honest boundaries: models detect **patterns like the past**, so novel risk is invisible to them; they produce **false positives** that erode attention if unfiltered; and a **calibration record is mandatory** before their alerts influence decisions (Domain 6, KA 6.4; Domain 14). The failure mode to watch for is not a wrong alert but **displaced judgment** — a team that stops thinking about risk because a dashboard is watching. The register's owners remain human, the analysis stays challengeable, and the principle holds: AI proposes; the professional verifies, decides and remains accountable.

"Tune the thresholds" is the standard remedy for false positives, and it is incomplete advice because it does not say **which way**. The direction is decided by the base rate and by the ratio of what a caught problem saves to what an alert costs to investigate — both of which a programme can measure, and neither of which is a matter of opinion.

WORKED EXAMPLE

Worked example 8.4.4 — which way to tune Meridian's risk monitor.

1. **Setup.** Meridian runs an anomaly monitor over the weekly rollout data from its **40** clinics. In a given clinic-week a genuine emerging problem is present with probability **0.01** — the measured base rate over the first two quarters, i.e. **0.4** genuine problems a week across the estate. Caught early, a problem is remediated for USD 4,000; caught late it costs USD 26,000, so catching one **saves USD 22,000**. Investigating an alert costs **USD 900** — half a day of a delivery manager. The vendor offers two calibrations: **A, sensitive** (sensitivity 0.85, specificity 0.90) and **B, specific** (sensitivity 0.60, specificity 0.98).
2. **Formula.** With `N` clinic-weeks, base rate `b`, sensitivity `s` and specificity `k`: true positives `= Nbs`; false positives `= N(1−b)(1−k)`; alerts = their sum; precision = TP ÷ alerts. Net value = `TP × saving − alerts × investigation cost`. An alert is worth investigating while `precision × saving ≥ cost`, so the breakeven precision is `cost ÷ saving`. Setting the two configurations' net value equal gives the crossover base rate.
3. **Substitution.** A: `TP = 40 × 0.01 × 0.85`; `FP = 40 × 0.99 × 0.10`. B: `TP = 40 × 0.01 × 0.60`; `FP = 40 × 0.99 × 0.02`. Breakeven precision `900/22,000`. Net value as a function of `b`: A `= 748,000b − (30b + 4) × 900 = 721,000b − 3,600`; B `= 528,000b − (23.2b + 0.8) × 900 = 507,120b − 720`.
4. **Result.**

	TP/week	FP/week	Alerts	Precision	Missed/week	Net value/week
A sensitive, at `b` = 1.0 %	0.3400	3.9600	4.3000	**7.91 %**	0.0600	**USD 3,610.00**
B specific, at `b` = 1.0 %	0.2400	0.7920	1.0320	**23.26 %**	0.1600	**USD 4,351.20**
A sensitive, at `b` = 7.5 %	2.5500	3.7000	6.2500	40.80 %	0.4500	USD 50,475.00
B specific, at `b` = 7.5 %	1.8000	0.7400	2.5400	70.87 %	1.2000	USD 37,314.00

The breakeven precision is `900/22,000 =` **4.0909 %**. The crossover base rate is `(3,600 − 720)/(721,000 − 507,120) =` **1.3465 %**: below it the **specific** configuration wins, above it the **sensitive** one. Configuration A stops paying at all below a base rate of **0.4993 %**; B keeps paying down to **0.1420 %**. Over Meridian's 26-week rollout at its measured 1.0 % base rate, B returns **USD 113,131.20** against A's **USD 93,860.00** — a difference of **USD 19,271.20** — at the cost of **2.60** more missed problems across the rollout (4.16 against 1.56). 5. **Interpretation.** Three results, and they are not the ones the received wisdom predicts.

A monitor whose alerts are wrong 92.09 % of the time still pays. At configuration A's 7.91 % precision the monitor returns USD 3,610 a week, because precision only has to clear `cost ÷ saving = 4.0909 %` — and 7.91 % is **1.9328 times** that threshold. So "we get too many false alarms" is not, by itself, an economic argument for anything. It is an **attention** argument, and attention is the resource the calibration should be protecting; the arithmetic makes clear that the choice between A and B is about how a scarce delivery manager's time is spent, not about whether the monitor is worth running.

The direction of tuning depends on the base rate, and it flips. At Meridian's 1.0 % base rate the specific configuration is better by USD 741.20 a week; at a 7.5 % base rate the sensitive one is better by USD 13,161 a week. The crossover — **1.3465 %** — is uncomfortably close to Meridian's measured 1.0 %, which means the recommendation is **fragile** and must be presented as such: a modest deterioration in the estate would reverse it. Two governing quantities, one ratio, one crossover; that is the whole of it, and it is why calibration is a measurement duty rather than a preference. The corollary for practice is that the **base rate must be measured before the threshold is set**, and re-measured when the delivery environment changes — which is the calibration record the paragraph above requires, now with a stated purpose.

And the expected-value answer can be overruled, for a nameable reason. B wins at Meridian's base rate by missing more: **4.16** genuine problems across the rollout against A's 1.56. If a missed problem is a USD 22,000 remediation, that trade is already priced and B is right. If any single missed problem could be a clinical-safety event or an irreversible data loss, then 8.3.1's rule binds — **existential exposure is not managed by its expected value** — and the sensitive configuration is correct regardless of the USD 19,271 it forgoes. The professional obligation is to state which regime applies before quoting the arithmetic, and the failure to do so is how an optimisation becomes an incident. Two cautions on the model. Sensitivity and specificity are themselves estimates from a calibration sample and drift as the estate changes (Domain 14, 14.A.2 on revalidation). And the arithmetic assumes alerts are independent draws; a monitor that fires repeatedly on one underlying condition inflates the alert count without adding information, which is the correlation problem of 8.A.1 appearing in the detection layer.

KEY TERMS — KA 8.4

Term	Meaning
Resilience	Capability to absorb unidentified risk; buffers, optionality, modularity, redundancy, fast detection.
Reference-class forecasting	Estimating from comparable completed projects to counter optimism bias.
Pre-mortem	Imagining failure before committing, to license dissent and surface missed risk.
Escalation of commitment	Continuing because of sunk cost; countered by forward-looking framing.
Displaced judgment	Ceasing to think because a tool is watching.
Outturn ratio	Final cost ÷ approved cost for a comparable completed project; the unit of a reference class.
Consequence threshold	A point where the consequence function steps rather than scales; where resilience is bought.
Base rate	The prevalence of the condition a monitor looks for; with the saving-to-investigation ratio it decides how a monitor should be tuned.
Precision	True alerts ÷ all alerts; worth investigating while precision exceeds investigation cost ÷ saving.

SAMPLE MCQS — KA 8.4

MCQ 8.4-A `[8.4.2 · Application]` A programme's estimates have run 25 % low across four comparable past projects. The most effective countermeasure is: - A. instruct estimators to be more careful - B. reference-class forecasting — estimate from the distribution of comparable completed projects rather than from the plan - C. add a 25 % contingency and continue the same process - D. escalate to the sponsor for a larger budget

Rationale: Optimism bias is systematic, so the fix is methodological (8.4.2). A relies on exhortation against a structural effect; C treats the symptom while preserving the cause; D funds it without correcting it.

MCQ 8.4-B [8.4.1 · Analysis] A project has been optimised to eliminate all float and buffers. Its risk position is: - A. improved — waste has been removed - B. degraded — it now has no capacity to absorb the risks not in the register, and resilience is what covers those - C. unchanged, provided contingency is funded - D. improved, since efficiency reduces exposure time

Rationale: Resilience assumes register incompleteness (8.4.1); removing all absorption capacity maximises fragility. C confuses money with time and capacity — funded contingency cannot buy back a schedule with nowhere to move.

MCQ 8.4-C [8.4.4 · Analysis] An AI monitor has raised 40 alerts this month, 3 of which mattered. The correct response is: - A. disable the monitor - B. tune thresholds and require a calibration record, because unfiltered false positives erode the attention the tool exists to direct - C. investigate all 40 equally - D. accept the ratio as inherent to anomaly detection

Rationale: False positives consume the scarce resource (attention) that detection is meant to focus (8.4.4). A discards real capability, C guarantees the erosion, D abandons the calibration duty.

MCQ 8.4-D [8.4.3 · Recall] In the first hour of a crisis, the leader's priority order is: - A. communicate, then investigate, then stabilise - B. stabilise and secure safety, establish facts, then decide with a clock and communicate early - C. establish blame, then stabilise - D. wait for complete information before acting

Rationale: Stabilisation and factual grounding precede decisions, which are taken against a clock (8.4.3). D is the failure the sequence exists to prevent; C is never a first-hour activity.

MCQ 8.4-E [8.4.1 · Analysis] A resilience measure costs USD 18,000, is needed with probability 0.25 and saves five weeks of a USD 14,280-per-week delay. Its breakeven probability is 25.2101 %. The correct professional conclusion is: - A. reject it — the expected loss is USD 150 - B. the breakeven lies inside the estimating error of the probability, so expected value cannot decide it; decide on the tail and on survivability - C. accept it — a resilience measure is always worth buying - D. re-estimate the probability to three decimal places and re-run the comparison

Rationale: No one estimates a capacity failure to a fifth of a percentage point, and the tail comparison is decisive at roughly 10 to 1 (8.4.1). A treats a noise-level difference as a result; C abandons the arithmetic entirely; D is false precision — the input cannot support the digits.

MCQ 8.4-F [8.4.4 · Application] A monitor's alerts are correct 7.91 % of the time. Investigating one costs USD 900 and catching a genuine problem saves USD 22,000. Investigating every alert is: - A. wasteful — over 92 % of investigations find nothing - B. worthwhile: the breakeven precision is 900/22,000 = 4.09 %, so 7.91 % returns 1.93 times the investigation cost - C. worthwhile only if precision exceeds 50 % - D. impossible to assess without the monitor's sensitivity

Rationale: An alert is worth investigating while precision exceeds cost ÷ saving (8.4.4). A counts the failures rather than valuing the successes — the arithmetic's whole point; C invents a threshold; D confuses what is needed to compare two calibrations with what is needed to decide whether to investigate a given alert.

MCQ 8.4-G [8.4.2 · Analysis] A bottom-up register gives a P80 contingency of USD 203,785 while a reference class of twelve comparable completed projects gives a P80 uplift of USD 964,800. The professional treatment is: - A. add them, giving USD 1,168,585 of required funding - B. treat them as two estimates of one quantity: use the reference class to challenge the total, attribute part of the gap to unidentified risks and the rest to optimism and scope growth, and hold the difference as management reserve - C. discard the reference class, since the register is specific to this project - D. discard the register, since the reference class is evidence and the register is judgment

Rationale: The reference class already contains the register's risks, so adding double-counts (8.4.2). C loses the only measurement of what the register cannot see; D discards the only basis for response and ownership — the reference class names no risk, no owner and no action.

SELF-CHECK — KA 8.4

1. Why is resilience not a substitute for quantification, and vice versa? — Quantification handles named risks; resilience absorbs the unnamed. Each fails at the other's job.
2. What is a pre-mortem for? — To license dissent before commitment and surface risks a workshop suppresses.
3. What is AI risk sensing's structural blind spot? — Novel risk: it detects patterns like the past.
4. Where is resilience bought, expressed as a test? — Where the consequence function has a step in it: find the thresholds, then ask what recovery time clears them.
5. Why must a bottom-up contingency and a reference-class uplift never be added? — They estimate the same quantity; the reference class already contains the register's risks.
6. What two quantities decide whether a monitor should be tuned sensitive or specific? — The base rate and the ratio of the saving from a caught problem to the cost of investigating an alert.

Advanced topics — Domain 8

8.A.1 Correlation and why aggregate risk is worse than it looks

Independence made 8.2.4's arithmetic tractable and understated the answer; worked example 8.2.4b prices the understatement on Auriga's own register — σ rising by a factor of 1.3471 and a P80 reserve degrading to 73.4 % on one shared subcontractor. This topic takes the organisational form of the same error, which is larger and less visible.

The mechanism is worth stating once in its general form. When risks share a driver — one supplier, one technology, one regulator, one labour market — they move together, and the correlation term $2 \Sigma_{i<j} \rho_{ij} \sigma_i \sigma_j$ grows with the number of pairs, $k(k-1)/2$, while the independent term grows only with k. **Concentration therefore hurts quadratically**: doubling the number of risks on one driver roughly quadruples the variance it contributes. That is the arithmetic behind the observation that a bad month becomes a crisis, and it is why counting entries is not a measure of exposure.

Three practical implications follow. Identify **common drivers** explicitly and give each a column in the register, so that grouping by driver is a sort rather than a project. Model correlation in simulation rather than assuming it away, and disclose the coefficients as the judgments they are. And where a driver is identifiable, prefer to **model the driver itself as a single risk with a structural response** (8.3.1) over modelling its symptoms as correlated line items — a correlation coefficient is a poor description of a subcontractor that either performs or does not, and it understates the extreme tail where all the symptoms occur together.

The organisational version of this error is believing a portfolio of projects is diversified when every one depends on the same scarce engineering resource, and it is the subject of Case study B — where the arithmetic is done.

8.A.2 Schedule risk analysis and the merge-bias trap

Applying this domain's methods to Domain 6's network produces **quantitative schedule risk analysis**: three-point durations (KA 6.4.3) simulated across the logic to yield a completion distribution and, valuably, a **criticality index** — how often each activity lands on the critical path across simulations — which is more decision-useful than a single deterministic critical path. The trap it exposes is **merge bias**: where paths converge (Domain 6's node E), the merged event waits for the latest predecessor, so the probability of being on time is the product of the paths' probabilities, not the best of them. Two paths each 80 % likely to meet a date give the merge point 64 %. Deterministic CPM cannot see this, which is why convergence points are systematically optimistic and why Domain 6 flagged them for attention.

That much is standard. What is usually left out is that the simple product is itself wrong when the paths share predecessors — as converging paths almost always do — and that merge bias **fades as the date becomes less aggressive**, so quoting it without a date is meaningless. Both are worth working on the real network.

WORKED EXAMPLE

Worked example 8.A.2 — merge bias at Auriga's node E, three ways.

1. **Setup.** Domain 6's network reaches installation **E** through two paths: A–B–C (mobilise, design, procure hardware) and A–B–D (mobilise, design, civil and cabling). Deterministic durations put A–B–C at 16 weeks and A–B–D at 15, so $ES(E) = 16$ with one week of float on D. The estimating team supplies three-point durations:

Activity	o	m	p	$t_e = (o+4m+p)/6$	$\sigma = (p-o)/6$
A Mobilise	1.5	2	3.5	2.166667	0.333333
B Detailed design	5	6	10	6.500000	0.833333
C Procure control hardware	6	8	14	8.666667	1.333333
D Civil and cabling works	5	7	12	7.500000	1.166667

What is the probability that E can actually start in week 16? 2. **Formula.** Path $t_e = \Sigma$ activity t_e; path $\sigma = \sqrt{\Sigma \sigma_i^2}$ (variances add). Probability of meeting a date T on one path $= \Phi((T - t_e)/\sigma)$. For a merge of **independent** paths, multiply. For paths sharing predecessors, decompose: $start\_E = AB + \max(C, D)$, and condition on the shared part — $P = \int f_{AB}(t) \cdot \Phi((T-t-t_eC)/\sigma C) \cdot \Phi((T-t-t_eD)/\sigma D)\, dt$. 3. **Substitution.** A–B–C: $t_e = 2.166667 + 6.5 + 8.666667$, variance $0.111111 + 0.694444 + 1.777778$. A–B–D: $t_e = 2.166667 + 6.5 + 7.5$, variance $0.111111 + 0.694444 + 1.361111$. Shared A+B: $t_e = 8.666667$, $\sigma = \sqrt{0.805556}$. 4. **Result.** Path A–B–C t_e **17.3333** weeks, σ **1.6073**; path A–B–D t_e **16.1667**, σ **1.4720**; shared A+B σ **0.8975**.

Reading of the same network	P(E can start in week 16)
Deterministic CPM	100 % (a date, not a probability)
Critical path alone	**20.34 %**
Independent-path product (textbook merge bias)	**9.25 %**
Correct, sharing A and B	**13.16 %**

And the effect decays with the date:

Target for ES(E)	Dominant path alone	Correct merge	Reduction from the merge
Week 16	20.34 %	13.16 %	**35.31 %**
Week 17	41.79 %	34.33 %	17.85 %
Week 18	66.08 %	61.53 %	6.90 %
Week 18.686 (the dominant path's P80)	80.00 %	**77.61 %**	2.99 %
Week 20	95.15 %	94.80 %	**0.37 %**

The date at which E's start is 80 % likely is **week 18.8093** — a buffer of **2.8093 weeks** over the deterministic 16 — against **week 18.6860** if only the dominant path is considered. So ignoring the merge understates the buffer by **0.1233 weeks**, which at Auriga's cost of delay of USD 45,000 per week is **USD 5,546.37**. 5. **Interpretation.** Four conclusions, and the last two are the ones that stop merge bias being misapplied.

The deterministic date is not a forecast. Week 16 has a **13.16 %** chance of being met. Nothing is wrong with the network; the date is the mode of a distribution and is being read as a commitment (Domain 6, KA 6.4.3 makes the same point on a single activity). The instrument that fixes it is a buffer chosen at a stated confidence, exactly as contingency is (8.2.4) — the schedule version of the same discipline, and the reason a project with a P80 cost reserve and a P0 schedule date is internally inconsistent.

The textbook product overstates the bias. Multiplying 20.34 % by 45.49 % gives 9.25 %, but the two paths share A and B: a good mobilisation and design help both paths simultaneously, so the failures are correlated and the true figure is **13.16 %** — 42.2 % higher than the naive product. The ordering is an invariant worth remembering: **naive product ≤ correct merge ≤ dominant path alone.** A leader given a merge-bias number should ask which of the three it is; a schedule simulation that honours the logic computes the middle one automatically, while a spreadsheet multiplying path probabilities computes the lower bound and calls it the answer.

Merge bias is a property of a date, not of a network. At week 16 the merge removes 35.31 % of the probability; at week 20 it removes 0.37 %. The reason is structural: as the date becomes generous, the dominant path's probability approaches 1 and the secondary path's approaches 1 faster, so the product approaches the dominant path's own figure. **Convergence points are dangerous at aggressive dates and almost irrelevant at conservative ones** — which is why the buffer understatement above is only 0.12 weeks even though the week-16 distortion is severe. Quoting "merge bias adds X weeks" without the confidence level attached is therefore not a result.

And the practical consequence is where to look, not how much to add. The decision-useful output of this analysis is not the 2.81-week buffer but the **criticality index**: D reaches E only 1.17 weeks behind C on expectation with a σ of 1.47, so D lands on the critical path in a substantial minority of futures despite showing one week of float deterministically. That is what makes it worth monitoring, and it is the same finding Domain 6 reached by re-running its passes after a crash — here obtained without crashing anything. Two cautions. PERT's $\sigma = (p - o)/6$ is a convention rather than a derivation and understates spread on strongly skewed estimates, so the probabilities above should be read to the nearest point. And the arithmetic treats C and D as independent given A and B, which fails if the same site conditions or the same subcontractor drive both — in which case 8.2.4b's correlation treatment applies to the schedule exactly as it did to the cost aggregation.

8.A.3 The reviewer's risk eye

Invariants worth testing in an hour: every entry has cause, event and consequence and a named owner; no occurred risks parked as high-probability entries; opportunities present in credible proportion; ΣEMV reconciles to the register; contingency stated at a named confidence level with its independence assumption disclosed; contingency consumption trended against risk retirement, not time; responses costed against EMV reduction, with secondary risks recorded; existential risks handled structurally rather than by EMV; early-warning indicators defined for the top items; and at least one decision this month traceable to the register. Failure of the last one matters most: it means the process is documentation, not decision support.

Seven further tests, each of which this domain has now made arithmetic, and each of which takes one division:

- **Does every total belong to one objective?** A "total exposure" mixing cost and benefit impacts is compared with nothing and can be funded from nothing (8.2.2b).
- **Is total exposure stated against the tolerance, not the budget?** Meridian's 4.96 % of approved cost is 99.17 % of the room (8.2.2b).
- **Does the confidence level follow from an appetite statement, or from habit?** If the latter, write down the appetite that the inherited level implies and put it to the sponsor (8.3.2).
- **Is the reserve adequate against the register still open?** Remaining ÷ recomputed requirement; a ratio below 1.00 is a trigger, and Auriga's was 0.9037 while the naive drawdown test passed (8.3.2b).
- **Have retired risks released their variance share, not just their EMV?** R4's retirement released USD 97,239 against an EMV of 60,000 (8.3.2b).
- **Is there a quantification threshold, written down?** A screen with no threshold has replaced quantification rather than ordering it, and one matrix cell can conceal a 14.6-fold span of expected values (8.2.1).
- **Has register completeness been estimated rather than assumed?** Two independent identification methods and their overlap give a floor on what is missing, and that floor — not a percentage convention — is the basis for management reserve (8.1.3).

Industry variations — Domain 8

- **Construction and infrastructure.** Ground conditions, weather and permits dominate, and they are **shared drivers** rather than independent risks: one wet season moves every external activity together, so the correlation term of 8.A.1 is the dominant part of the variance and an independence-based reserve is furthest from the truth here. Quantitative schedule risk analysis is contractually expected on major programmes, which makes 8.A.2's distinction — naive product against correct merge — a commercial point and not an academic one. Risk allocation is largely executed through contract terms (Domain 10), so the counterparty arithmetic of 8.3.1 is the profession's central skill in this sector: a transfer to a thinly capitalised subcontractor is priced at its breakeven default probability or it is not priced at all.
- **Energy and resources.** Commodity price and currency exposure sit alongside delivery risk and are managed with financial instruments as well as project responses (PFL-AI Domains 3, 11). Two consequences for this domain's methods: price risk is **continuous** rather than binary, so the Bernoulli variance of 8.2.4 is the wrong model and simulation over a price distribution is the right one; and because hedges are impact levers rather than probability levers, they release reserve at the favourable end of 8.2.4c's 1.5-to-3 range, which is a large part of why they are bought.

- **Technology and digital.** Requirement volatility and integration risk dominate; resilience is bought through modularity, staged release and reversibility rather than buffers, so 8.4.1's levers apply in a different mix. The reason is 8.2.4c's identity: modularity and reversibility are **impact** levers, and impact levers are what release contingency. Reversibility also makes information cheap to buy, so the value-of-information arithmetic of 8.2.3 favours acting and observing over surveying and deciding — the opposite conclusion to the utilities case study, from the same tree with a different reactive cost.
- **Pharmaceutical and regulated development.** Low-probability/high-consequence events and regulatory outcomes make structural treatment (8.3.1) the norm; risk files are auditable artefacts, not internal management tools. The characteristic quantitative failure is presenting an EMV-optimal response for an exposure that a regulator treats as unacceptable at any probability — the same error 8.4.4 names in monitor tuning, where a missed detection cannot be traded against investigation cost. The correct output is the optimum among admissible options, which is a smaller set than the arithmetic alone would suggest.
- **Public programmes.** Optimism bias is explicitly recognised in appraisal guidance, and reference-class adjustment may be mandated; where such guidance applies, the arithmetic of 8.4.2 is a submission requirement rather than an option, and the professional skill is the reconciliation — never adding a mandated uplift to a bottom-up contingency. Political and reputational risk carries weight that no EMV captures well, and it typically has the **step consequence function** of 8.4.1: a threshold crossed rather than a cost incurred, which is why public programmes buy resilience that looks uneconomic on expected value.
- **Financial services and operational change.** Regulatory deadlines make the consequence function almost purely a step, so 8.4.1's threshold test governs and buffer economics dominate expected-value economics. Detection is heavily instrumented, which puts 8.4.4's crossover arithmetic at the centre — and it lands the other way. Base rates for genuine control failures are very low, which favours the specific configuration, but the cost of an escaped failure is very high, which favours the sensitive one, and the second effect wins because the crossover base rate falls roughly in proportion to the saving-to-investigation ratio. On Meridian's two calibrations, raising that ratio tenfold — an escaped problem costing USD 220,000 rather than 22,000 — moves the crossover from **1.3465 %** to **0.1313 %**, so the sensitive configuration becomes correct at base rates ten times lower. Same two quantities, opposite answer, and it is the ratio rather than the rarity that decides.

Case study — Domain 8: the survey Auriga did not commission (utilities)

Situation. At planning, Auriga's team considered a USD 25,000 ground survey ahead of the civil works. The register carried R2 — ground conditions worse than surveyed, p 0.50, impact USD 180,000 — and the survey was cut in a cost-reduction pass as "nice to have". The decision was not analysed; it was traded against a round-number savings target.

What happened. At week 13 the contamination was found. The reactive response cost more than the planned one would have — a second civil crew at USD 35,000, a fast-track carrying $0.20 \times 60{,}000 = 12{,}000$ of expected rework, and the exposure to a USD 45,000-per-week client penalty window that made the whole recovery urgent (Domain 6's case study; Domain 7's forecast consequence, EAC USD 4.2m against BAC 4.0m).

The analysis, done afterwards. On the decision-tree arithmetic of 8.2.3 — 0.40 probability at the time of the decision, USD 300,000 reactive versus USD 90,000 planned — the survey was worth **USD 59,000** against its USD 25,000 price. Cutting it saved 25,000 and cost the project substantially more, and the register had contained the information needed to see that. Two further figures from the same tree show how far from marginal the call was: the survey would still have paid at any price up to **USD 84,000 — 3.36 times** what it was quoted at — and it would only have stopped paying had the team's belief in contamination fallen below **11.90 %**, against the 0.40 the register itself carried. The cost-reduction pass therefore did not make a close judgment badly; it made a decision whose arithmetic was not close in either direction, without doing the arithmetic.

What the domain teaches here. Two things, both uncomfortable. **A cost-reduction pass that does not consult the register is a risk decision taken blind** — savings targets and risk exposure must be traded explicitly. And **the register was not the failure; its irrelevance to the decision was** (8.3.3's test: did a decision change because of it?). The corrective adopted afterwards was procedural: any proposed saving above a threshold must state which register entries it affects and by how much.

Case study B — Domain 8: the diversified portfolio that wasn't (technology programme)

Situation. A transformation programme's board reviewed 22 project-level risks spread across six workstreams and concluded exposure was well diversified. The register held **14** risks each assessed at `p` 0.25 with an impact of **USD 60,000**, and **8** each at `p` 0.20 with an impact of **USD 80,000** — a total `EMV` of **USD 338,000** in which no single item exceeded **4.73 %** of the aggregate. Contingency was set at **USD 380,000**: the mean plus a **12.43 %** margin, assuming independence. On that assumption the reserve covers **62.41 %** of futures; the independence-based P80 would have been **USD 449,784**, so even on its own terms the reserve was thin, and nobody had computed that either.

What happened. Fourteen of the 22 risks depended on the same six-person integration team. When two members left within a month, **11 of the 14** materialised or worsened together, along with 2 of the 8 independent risks: an outcome of `11 × 60,000 + 2 × 80,000 =` **USD 820,000**, or **USD 440,000** beyond the reserve. The programme's recovery required a re-baseline and a management-reserve release.

The arithmetic, done afterwards. The concentration was priced three ways, and the ladder is the lesson.

Model of the 14 shared risks	σ of the aggregate	P80	Confidence the 380,000 reserve buys	Where the 820,000 outcome sits
Independent (as assumed)	132,823	449,784	62.41 %	99.99th percentile — **1 in 7,026**
Pairwise ρ = 0.6	302,245	592,369	55.53 %	94.46th percentile — **1 in 18**
One driver, all 14 together	**374,823**	**653,451**	**54.46 %**	90.08th percentile — **1 in 10**

The mean is **USD 338,000** in all three columns: correlation moved nothing about the average and everything about the answer (8.2.4b). The correlation term at ρ = 0.6 is `2 × 0.6 × 91 pairs × 675,000,000 =` **USD 73,710,000,000** of variance — **4.18 times** the entire independent variance of the 22-risk register — because 14 risks generate 91 pairs and the correlation term grows with pairs while the independent term grows with entries (8.A.1).

What the board had actually been told. That the outcome was a one-in-seven-thousand event. It was a **one-in-ten** event, and the difference between those two statements is a single untested assumption. Note also that the ladder's third rung is the lowest confidence figure: modelling the integration team as one risk with an impact of `14 × 60,000 =` USD 840,000 produces the widest distribution of the three, because a perfectly shared driver is the limiting case of correlation. That is the honest model of a six-person team, and it makes the exposure legible in a way fourteen line items never did.

What was done differently. The register was restructured by **common driver** rather than by workstream, immediately revealing the concentration; the integration capability was treated as a single structural risk with a structural response (cross-training, a retained partner, staged scope) rather than as fourteen separate `EMV` line items; and contingency was re-derived by simulation with correlation modelled explicitly. The structural response is also the cheaper one on 8.2.4c's logic: cross-training and staged scope are **impact** levers on the driver, and impact levers release reserve at up to three times the rate of probability levers.

What the domain teaches here. Counting entries measures paperwork; counting **drivers** measures exposure. A register with many risks and few drivers is concentrated, and independence is the assumption most likely to be both convenient and wrong. The one-sentence test the Executive perspective demands — which of these risks share a driver? — would have moved this programme's stated confidence by eight percentage points and its P80 by **USD 203,667** before a single event occurred.

Executive perspective — Domain 8

What a project leader cannot delegate in this domain:

- **The confidence level, and where it comes from.** Whether contingency is set at P50 or P80 is a risk-appetite decision belonging to governance, and the leader ensures it is chosen, stated and understood — not inherited from a spreadsheet default. The stronger version of the duty is to derive it: an appetite statement of the form "no more than a 10 % chance of exceeding by more than 5 %" fixes the level at P90 and the tolerance at a number, and a leader who has never seen that sentence written down should write it and take it to the sponsor (8.3.2).
- **The independence question.** Asking, of any aggregate number, which risks share a driver. It is one sentence and it is where Case study B's programme lost control — worth 7.95 points of stated confidence and USD 203,667 of P80 before anything happened.
- **Which constraint the responses are buying against.** Expected cost and confidence level are different objectives, and the same USD 25,000 buys different amounts of each: on Meridian, an impact lever returned 34.93 % more P90 reduction than a probability lever that removed more `EMV` (8.2.4c). The leader states which constraint binds before the register's `EMV` column decides by default.
- **The room, not the budget.** Insisting that total exposure be presented against the objective's tolerance rather than as a percentage of cost — the comparison that showed Meridian's mean consuming 99.17 % of its 5 % tolerance, and that no percentage-of-budget statement would ever have surfaced.
- **The structural exposures.** Existential risks handled by structure, never by `EMV` — and named personally by the leader, because low probability makes them easy to leave in the register.
- **The register's relevance.** Insisting on the 8.3.3 test: which decision changed because of it this month? A register that changes nothing gets fixed or stopped.

- **The bias countermeasures.** Running the pre-mortem, commissioning independent estimates, framing continuation decisions on remaining cost and benefit — because the leader is the only person positioned to make dissent safe.
- **The AI boundary.** Detection is delegable; judgment is not, and a team that has stopped thinking because a dashboard is watching is the outcome to prevent.

Calculation exercises — Domain 8

Exercise 8.1 Risks: A p 0.40 impact 150,000; B p 0.20 impact 500,000; C p 0.60 impact 80,000; D (opportunity) p 0.25 impact (200,000). Compute each EMV and the total exposure. Solution. A **60,000**; B **100,000**; C **48,000**; D **(50,000)**. Total **USD 158,000**. Common error: omitting the opportunity, giving 208,000 and overstating required contingency by 31.6 %.

Exercise 8.2 Using Exercise 8.1's four risks and assuming independence, compute σ and a P80 contingency (mean + 0.8416 σ). Solution. Variances: $0.40 \times 0.60 \times 150{,}000^2 = 5.40e9$; $0.20 \times 0.80 \times 500{,}000^2 = 40.0e9$; $0.60 \times 0.40 \times 80{,}000^2 = 1.536e9$; $0.25 \times 0.75 \times 200{,}000^2 = 7.50e9$. Total **54.436e9**; σ = **USD 233,315**; P80 = $158{,}000 + 0.8416 \times 233{,}315 =$ **USD 354,358**. Common error: taking √(Σσ) rather than √(Σvariance).

Exercise 8.3 A risk (p 0.30, impact 400,000) can be mitigated for USD 70,000, reducing probability to 0.10. Is the mitigation worthwhile? Solution. EMV before $0.30 \times 400{,}000 =$ 120,000; after $0.10 \times 400{,}000 =$ 40,000; reduction **80,000** against a cost of 70,000 → **net +USD 10,000, accept** (marginally — and the leader should note that a 400,000 impact may warrant response on survivability grounds regardless). Common error: comparing the mitigation cost against the impact rather than the EMV reduction.

Exercise 8.4 Two independent paths each have a 0.80 probability of meeting a milestone date and converge on it. What is the probability the milestone is met, and what is the effect called? Solution. $0.80 \times 0.80 =$ **0.64**. **Merge bias** — the merged event waits for the later path, so convergence points are systematically optimistic in deterministic CPM (8.A.2). Common error: answering 0.80, the best of the two. Second common error, and the subtler one: multiplying when the two paths **share predecessors**, which correlates them and makes 0.64 a lower bound rather than the answer — on Auriga's node E the naive product gives 9.25 % where the correct figure is 13.16 %.

Exercise 8.5 Three risks: X p 0.30 impact 200,000; Y p 0.40 impact 150,000; Z p 0.20 impact 250,000. X and Z depend on the same supplier, with an assessed correlation of ρ = 0.6; Y is independent of both. Compute the P80 contingency assuming independence, the P80 with the correlation modelled, and the confidence the independence-based figure actually buys. Solution. EMVs **60,000**, **60,000**, **50,000**; mean **USD 170,000**. Variances $0.30 \times 0.70 \times 200{,}000^2 = 8.40e9$; $0.40 \times 0.60 \times 150{,}000^2 = 5.40e9$; $0.20 \times 0.80 \times 250{,}000^2 = 1.00e10$; total **23.80e9**, σ **USD 154,272.49**, so P80 = $170{,}000 + 0.8416 \times 154{,}272.49 =$ **USD 299,835.72**. Correlation term $2 \times 0.6 \times \sigma_X \sigma_Z = 1.2 \times 91{,}651.51 \times 100{,}000 =$ **10,998,181,667.89**, giving total variance **34,798,181,667.89**, σ **USD 186,542.71** and a corrected P80 of **USD 326,994.34** — **9.06 %** higher. The independence figure of 299,836 sits at $z = (299{,}836 - 170{,}000)/186{,}542.71 =$ 0.6960 of the correlated distribution, i.e. it is a **75.68 %** reserve, not an 80 % one. Common error: adding the two σ values (154,272 + a correlation adjustment) instead of adding variances, or — more damaging — reporting the reserve as P80 after discovering the shared supplier, which leaves the label attached to the wrong number.

Exercise 8.6 A risk carries `p` 0.20 and an impact of USD 300,000. Two responses cost the same: one halves the probability, the other halves the impact. Compare their effect on `EMV` and on variance, and state the general rule. Solution. Base `EMV` **60,000**; base variance `0.20 × 0.80 × 300,000² =` **14.40e9**. Halving the probability: `EMV` **30,000**, variance `0.10 × 0.90 × 300,000² =` **8.10e9**, so **6.30e9** removed. Halving the impact: `EMV` **30,000**, variance `0.20 × 0.80 × 150,000² =` **3.60e9**, so **10.80e9** removed — a factor of **1.714286** more, at identical `EMV` reduction. The rule: `EMV` is linear in both `p` and `I`, so either halving removes the same expected cost; variance is `p(1−p)I²`, so halving the impact always removes 75 % of a risk's variance while halving the probability removes between 25 % and 50 %, giving a ratio between **1.5** and **3** for any `p ≤ 0.5` (8.2.4c). Common error: concluding the two responses are equivalent because their `EMV` reduction is identical — true, and irrelevant whenever the binding constraint is a confidence level rather than an expected cost.

Exercise 8.7 A risk has `p` 0.25 and an impact of USD 400,000. Three responses are offered. **Reduce:** USD 28,000, cutting `p` to 0.10, with a secondary risk of `p` 0.15 and impact 40,000. **Transfer:** USD 26,000 premium recovering 70 % of the impact, with the counterparty unable to pay with probability 0.30 given the event. **Avoid:** USD 150,000, removing the risk, with a secondary risk of `p` 0.20 and impact 90,000. Rank the four options including acceptance, first ignoring the secondary risks and then including them, and compute the counterparty default probability at which the transfer ties the best alternative. Solution. Accept **USD 100,000**. Ignoring secondaries: transfer `26,000 + 0.25 × 400,000 × 0.30 =` **56,000**; reduce `28,000 + 0.10 × 400,000 =` **68,000**; avoid **150,000** — transfer appears best by 12,000. Including secondaries: reduce `68,000 + 0.15 × 40,000 =` **74,000**; transfer `56,000 + 0.25 × 0.30 × 280,000 =` **77,000**; avoid `150,000 + 0.20 × 90,000 =` **168,000** — so **reduce wins**, and the ranking reversed. Reduce creates **USD 26,000** of value against acceptance, a **26.00 %** improvement. The transfer ties reduce at `(74,000 − 26,000 − 30,000)/(0.25 × 280,000) =` **25.71 %** of counterparty default. Common error: omitting the secondary risks, which here selects the wrong response; and treating the transfer's counterparty exposure as independent of the event, when the disruption that triggers the claim is usually what impairs the counterparty.

Exercise 8.8 A risk workshop identifies 28 risks; an independent checklist-and-interview review identifies 19; 11 appear on both lists. Estimate the risk population, the number still unidentified and the coverage achieved, cross-check with the Chapman estimator, and state what the answer may and may not be used for. Solution. `N̂ = (28 × 19)/11 = 532/11 =` **48.3636**. Distinct identified `= 28 + 19 − 11 =` **36**, so **12.3636** risks are estimated unidentified — **25.56 %** of the population — and coverage is **74.44 %**. Chapman: `(29 × 20)/12 − 1 =` **47.3333**, giving **11.3333** missing — the two estimators differ by **1.0303**, which is the estimated missing count divided by `m + 1` (`12.3636/12`), the identity Toolkit 8.T.4 uses as its divergence test. The workshop alone covered **57.89 %**. The estimate justifies **management reserve and resilience** sized against unidentified exposure and a decision on whether to run a third identification method; it does **not** justify adding `EMV`, because the missing risks have no cause, event, consequence or owner (8.1.3). Common error: reading a high overlap as evidence of completeness — a large `m` deflates `N̂`, so shared blind spots produce reassuring coverage; and adding the estimated missing count to contingency, which funds risks that cannot be responded to or retired.

Exercise 8.9 A monitor watches 60 sites weekly. A genuine emerging problem is present in a site-week with probability 0.03. Catching one early saves USD 15,000; investigating an alert costs USD 750. Configuration A has sensitivity 0.90 and specificity 0.88; configuration B has sensitivity 0.65 and specificity 0.97. Compute each configuration's weekly net value and A's precision, the

> breakeven precision for investigating an alert, and the base rate at which the two configurations are equally good. Solution. At `b` = 0.03, A: true positives `60 × 0.03 × 0.90 =` **1.62**, false positives `60 × 0.97 × 0.12 =` **6.984**, alerts **8.604**, precision **18.83 %**; net `1.62 × 15,000 − 8.604 × 750 =` **USD 17,847.00**. B: true positives **1.17**, false positives `60 × 0.97 × 0.03 =` **1.746**, alerts **2.916**; net `1.17 × 15,000 − 2.916 × 750 =` **USD 15,363.00** — so **A wins by USD 2,484**. Breakeven precision `750/15,000 =` **5.00 %**, which A's 18.83 % clears by a factor of 3.77. As functions of the base rate, `Net_A = 774,900b − 5,400` and `Net_B = 557,100b − 1,350`; equating gives `b* = 4,050/217,800 =` **1.8595 %**. Below that the specific configuration wins; above it the sensitive one. A stops paying below **0.6969 %**, B below **0.2423 %**. Common error: choosing the configuration with the higher precision — B's precision is better at every base rate and its net value is worse above 1.86 %; and applying the crossover from one programme to another, since it moves with both the base rate and the saving-to-investigation ratio.

Practitioner's toolkit — Domain 8

Adoption-ready artefacts; adapt headings to your organisation, then keep them stable.

Toolkit 8.T.1 — Risk register columns that earn their keep

ID · **cause → event → consequence** statement · affected objective · **common driver** (8.A.1) · probability · impact (cost and time) · `EMV` · qualitative band (screening only) · response family and description · **response cost and expected `EMV` reduction** · **secondary risks created** · owner (a person) · early-warning indicator · review date · status (open/retired/**issue**) · retirement evidence. Omit any column and the register loses a specific capability; the two most commonly missing — common driver and response cost — are the two that make it decision-useful.

Toolkit 8.T.2 — Contingency derivation sheet (one per baseline)

Register version and date · **objective the total belongs to**, with the other objectives' aggregations cross-referenced · Σ`EMV` · **the objective's tolerance, and Σ`EMV` as a percentage of it** · variance and σ by risk, ranked by **variance contribution** as well as by `EMV` · **independence assumption stated and challenged** (which risks share drivers, with the assumed ρ and the P80 both ways) · method (formula approximation or simulation, with correlation treatment) · **the appetite statement the confidence level is derived from**, and who owns it · resulting contingency · **the price of confidence** (cost per percentage point between the adjacent levels) · management reserve separately, with its basis including the **register-completeness estimate** of 8.1.3 · draw protocol reference · and, from the second period onwards, the **adequacy ratio**: remaining reserve ÷ the requirement recomputed on the register still open. A contingency figure without this sheet is a number without a defence; a contingency figure without the last line is a number that was defensible once.

Toolkit 8.T.3 — Pre-mortem session guide

Before commitment, one hour: (1) state the plan and the commitment about to be made; (2) each participant writes privately for ten minutes — "it is twelve months on and this has failed; why?"; (3) round-robin with no challenge permitted during collection; (4) cluster into causes and identify **common drivers**; (5) convert each into a three-part risk statement with an owner; (6) identify the three cheapest early-warning indicators. Rule: the most senior person speaks last, always.

Toolkit 8.T.4 — Register-completeness check (one per gate)

A twenty-minute calculation that turns "the register is probably incomplete" into a funded position. (1) Run two identification methods **separately**, by different people, with no sight of each other's output — a workshop and an assumption-analysis review are the standard pair (8.1.3). (2) Reconcile the two lists on **cause-and-event identity**, not wording; record n_1, n_2 and the overlap m. (3) Compute $\hat{N} = n_1 n_2 / m$ and the Chapman cross-check $((n_1+1)(n_2+1))/(m+1) - 1$. The two differ by exactly the estimated missing count divided by $m + 1$, so the divergence needs no threshold of its own: read it as a fraction of the missing count, and where it approaches that count m is too small to support the estimate and a third method is the answer rather than a third decimal place. (4) Record distinct identified, estimated unidentified and coverage. (5) State the coverage threshold that triggers a further method, and whether it fired. (6) Carry the estimated unidentified count into the management-reserve basis on Toolkit 8.T.2 — **never** into contingency. Standing caution to print on the sheet: the missing count is a floor, not an estimate (8.1.3).

Toolkit 8.T.5 — Response option comparison (one per quantified risk)

One row per option — accept, avoid, reduce, transfer, and for opportunities exploit, enhance, share, ignore — with five columns that must all be filled before a choice is recorded: **response cost** · **residual** EMV (p and impact after the response, both restated) · **secondary risks created**, each with its own p and impact and its own owner · **total expected cost** · and **variance after**, $p(1-p)I^2$ on the residual, because a response is also a reserve decision (8.2.4c). Two lines beneath the table complete it: the **cost at which the ranking flips** to the next-best option, and — for any transfer — the **counterparty default probability at which the transfer ceases to be preferred**, noting whether that default is correlated with the event transferred. A comparison without the secondary-risk column selects the wrong option often enough to be a known defect (8.3.1); one without the flip point cannot be reviewed, because a reviewer cannot tell whether the choice was close.

Exam preparation — Domain 8

The traps. Multiplying and summing ordinal scores as if they were money (8.2.1) · omitting opportunities from total exposure (Exercise 8.1) · setting contingency at ΣEMV and calling it funded (MCQ 8.2-D) · √(Σσ) instead of √(Σvariance) (Exercise 8.2) · comparing response cost against impact rather than EMV reduction (Exercise 8.3) · managing an existential low-probability risk by its EMV (8.3.1) · assuming independence without testing common drivers (8.A.1) · answering the best path probability at a merge point (Exercise 8.4) · leaving occurred risks in the register as p = 1.0 · treating a transfer as removing exposure.

The traps this domain's later topics add. Adding a correlation adjustment to σ rather than to the variance, and — worse — leaving the "P80" label on a reserve after a shared driver has been found (Exercise 8.5, MCQ 8.2-G) · treating two responses as equivalent because their EMV reduction is equal, when one is an impact lever and the other a probability lever (Exercise 8.6) · omitting secondary risks from a response comparison, which reverses the ranking whenever a transfer's counterparty default is correlated with the event (Exercise 8.7, MCQ 8.3-D) · returning only a retired risk's EMV when the reserve was sized on a confidence level (MCQ 8.3-E) · testing a part-consumed reserve against elapsed time or the original register instead of the register still open (8.3.2b) · choosing a confidence level by convention when the appetite statement determines it (MCQ 8.3-F) · summing a bottom-up contingency and a reference-class uplift (MCQ 8.4-G) · reading a large overlap between two identification methods as evidence of completeness (MCQ 8.1-E) · comparing a monitor's precision against a target instead of against cost ÷ saving, or choosing the higher-precision calibration without computing net value (MCQ 8.4-F, Exercise 8.9) · multiplying path probabilities at a merge point when the paths share predecessors, and quoting a merge-bias figure without the date it applies to (8.A.2) · treating a breakeven that falls inside the estimating error as a result (MCQ 8.4-E) · and expressing total exposure as a percentage of budget when the governing constraint is a tolerance (8.2.2b).

The calculations to be able to do under time pressure. `EMV` and Σ`EMV` with opportunities signed correctly · variance $p(1-p)I^2$, σ = √(Σvariance), and a percentile as mean + zσ · the correlation term $2\Sigma\rho\sigma_i\sigma_j$, and inverting a reserve into the confidence it buys · decision-tree rollback and the value of information, with its breakeven price and breakeven prior · p^* for a response: the cost at which acceptance becomes correct · the adequacy ratio · an appetite statement converted into a tolerance and a confidence level · $\hat{N} = n_1 n_2 / m$ · PERT t_e and σ, path σ, and a merge probability both ways · and a monitor's precision against `cost ÷ saving`.

Reflection questions. 1. Take your project's contingency figure: what confidence level does it represent, who chose it, and what independence assumption sits underneath? 2. Group your register by common driver rather than by workstream. How concentrated is it actually? 3. Which decision changed last month because of your register — and if none, is the process decision support or documentation? 4. Write down the appetite statement your confidence level implies — "we accept an X % chance of exceeding by more than Y" — and consider whether your sponsor would sign it. 5. Take your last response decision. What were its secondary risks, and would pricing them have changed the choice? 6. Rank your register by variance contribution rather than by `EMV`. Which entry moves furthest, and is it being managed as though it matters that much?

Domain 8 summary

Risk management earns its place only when it changes decisions. That starts at the sentence level: cause → event → consequence, against a named objective, with opportunities identified as deliberately as threats because defensive framing otherwise buries them — and it can be tested for completeness rather than assumed complete, two independent methods and their overlap putting a floor under what is missing (Meridian's **11.43** unidentified risks, **78.61 %** coverage) and giving management reserve a basis at last. Analysis then progresses from qualitative screening — useful for ordering, invalid as arithmetic, and coarser than it looks, since one matrix cell conceals an `EMV` span equal to the product of its band ratios (**14.5833** here, **1.8452 times** the step between cells two bands apart) — to quantification: `EMV` for each risk (Auriga's register totalling **USD 278,000**, 6.95 % of `BAC`, and USD 314,000 if the opportunity is ignored, an overstatement of 12.95 %), with the total always belonging to **one objective** and always stated against that objective's **tolerance** rather than its budget, which is how Meridian's comfortable-looking 4.96 % of approved cost turns out to consume **99.17 %** of the room the board allowed.

Decision trees price choices and the **value of information** — the survey worth **USD 59,000** against its 25,000 price, still worth buying at any price up to **USD 84,000**, and worthless if it changed no action — and, once the signal is imperfect, the value depends on the **design** of the enquiry as much as its price: Meridian's designed two-clinic pilot was worth **USD 77,136** against the random pilot's **USD 17,947** at identical cost, and needed only an **11.05 %** prior to pay against the random pilot's 21.45 %. Aggregation then converts a register into a **stated confidence** — Auriga's P80 of **USD 490,624** from a mean of 278,000 and σ of 252,642, defensible precisely because it says what confidence it buys, unlike a worst-case sum, an `EMV` sum or a 10 % rule of thumb that turns out on enumeration to be a **P68.5** reserve. Two refinements matter more than the base calculation. **Correlation moves the spread and not the mean**: one shared subcontractor takes Auriga's σ to **USD 340,339** and makes that same 490,624 a **73.4 %** reserve, the label failing before the money does. And because variance goes as $p(1-p)I^2$, **impact levers release between 1.5 and 3 times the**

reserve of probability levers at equal `EMV` cost — which is why Meridian's second response bought **34.93 %** more P90 reduction for the same USD 25,000 while removing less `EMV`.

Responses are investments judged against the reduction they buy, priced across the families with their **secondary risks** included, because omitting them is not an incompleteness but a selection error: on Auriga's R1 the transfer looks best at 51,600 and is worst-but-one at 64,200 once a counterparty default correlated with the event is priced, and the reduce option wins at 62,000. Existential exposures are handled structurally rather than by averaging. Reserves are sized by aggregation at a confidence level **derived from the appetite statement** rather than inherited — "no more than a 10 % chance of exceeding by more than 5 %" is P90, and on Meridian it is unreachable by any contingency decision — separated by authority, and tested each period against the register still open, which showed Auriga's part-consumed reserve at an adequacy ratio of **0.9037** and a real confidence of **75.65 %** while the conventional drawdown test passed. Retirement releases variance, not `EMV`: R4's closure freed **USD 97,239** against an `EMV` of 60,000.

Beyond what can be named sits resilience — buffers, optionality, modularity, redundancy and fast detection, bought at an efficiency cost that should be chosen rather than stumbled into, and bought specifically where the consequence function has a **step** in it: Meridian's trainer retainer loses USD 150 on expected value, misses its breakeven by two-tenths of a percentage point, and removes **USD 184,776** of tail exposure, which is what decides it. Alongside sit the bias countermeasures (pre-mortem, independent estimates) and **reference-class forecasting**, whose uplift is never added to a bottom-up contingency but used to challenge it — Meridian's class of twelve giving a P80 uplift of **USD 964,800** against a register-based **USD 203,785**, a gap of which about a third is attributable to the unidentified risks the completeness estimate found and the rest to optimism and scope growth that must be named as undecomposable. AI-enabled detection is governed by the same arithmetic: an alert is worth investigating while precision exceeds `cost ÷ saving` — **4.09 %** on Meridian, cleared nearly twice over by a monitor wrong 92 % of the time — and the direction of tuning is decided by a crossover base rate (**1.3465 %**), not by preference. Applied to the schedule, the same tools give merge bias its honest form: Auriga's node E has a **13.16 %** chance of its deterministic date, not the 20.34 % of its critical path nor the 9.25 % of a naive product, and the effect fades from a 35.31 % distortion at week 16 to 0.37 % by week 20, so a merge-bias figure without a date is not a result. Crisis leadership closes the domain: stabilise, establish facts, decide against a clock, communicate early. Independence is the assumption most likely to be convenient and wrong: count drivers, not entries. Domain 9 turns to quality and the assurance that catches what risk analysis missed.

09

DOMAIN 9

Quality, Assurance and Continuous Improvement

Group: Delivering the work (Domain 9 of 6 in Part Two). **Target:** ~76 pages. **Binds to:** the PCI Book Pattern Specification and the shared registries (`docs/books/registries/`). This domain works **Project Auriga** — the 25-week control-systems upgrade carried through Domains 6, 7 and 8 — and supplies the cost-of-quality, containment, rework-capacity, acceptance-sampling and rolled-yield arithmetic that Domain 10 uses for supplier acceptance, Domain 14 for AI-output verification and Domain 16 for handover readiness. It builds on Domain 3's gate economics and Domain 5's requirement-defect ladder and does not repeat either. British English; USD (+SAR where useful, indicative `USD 1 ≈ SAR 3.75`).

Why this domain exists

Domain 5 defined what the project must produce and how acceptance would be tested. Domain 6 built the schedule, Domain 7 the money, Domain 8 the risk. Every one of them assumed something this domain has to supply: **that the work will be right, and that somebody has priced being wrong.**

Quality is the discipline most often reduced to a virtue. Organisations declare a commitment to "the highest quality", write a plan nobody reads, and discover in commissioning that quality is a budget line, a capacity constraint and a schedule driver — all three set months earlier by people who did not know they were setting them. The recurring failure is not indifference. It is the belief that quality is free at the top and infinitely expensive to give up, so the only two available positions are "we do not compromise on quality" and an unadmitted compromise. Neither is a management position, because neither can be costed.

The domain's central claim is that **quality is an optimisation with an interior solution, and the optimum is not zero defects.** Prevention, appraisal, internal failure and external failure move against each other; their sum has a minimum; that minimum sits at a positive escape rate; and where it sits is fixed by one ratio — what an escaped defect costs against what finding it costs. An organisation that cannot compute that ratio is not choosing its quality regime but inheriting one. KA 9.1 computes it. KA 9.2 shows where the money actually goes — into layers of containment whose economics differ by an order of magnitude — and why rework is a capacity problem before it is a cost problem, with a schedule consequence no cost-of-quality model contains. KA 9.3 turns to the decisions: accepting work, disposing of nonconformance, and the arithmetic of sampling, where the central and uncomfortable result is that **a passing sample routinely fails to exclude the defect rate at which its own plan is the wrong plan.** KA 9.4 closes on improvement and on the two data problems that now dominate delivery quality: the quality of the data a project runs on, and the quality of the output an AI system hands a professional who must sign for it.

Learning objectives. After this domain a candidate can: distinguish quality from grade and both from fitness for purpose, and write a quality objective that is measurable rather than aspirational; decompose the cost of quality into prevention, appraisal, internal failure and external failure and explain why internal failure cost rises before it falls; **compute total cost of quality across candidate regimes, locate the interior minimum and state the range of external-failure unit cost over which it holds**; explain why zero defects is not the objective and what would make it one; design a containment chain and compute its escape fraction as the product of layer escape rates; **compute the expected containment cost per introduced defect and use it as the breakeven price of prevention**; show arithmetically why equal money on prevention beats equal money on appraisal, and name the condition under which it does not; **convert a rework share of capacity into a duration multiplier and a delay cost and explain why the relationship is hyperbolic**; run a formal acceptance decision and select a nonconformance disposition with its authority; **compute what a clean acceptance sample does and does not establish, derive the breakeven defective fraction from the ratio of verification cost to failure cost, and test an acceptance plan for self-consistency**; conduct root-cause analysis to the level at which the cause can be removed and price removal against expected recurrence; **compute rolled throughput yield as the product of stage yields, target improvement at the binding step and derive the uniform step yield an end-to-end target requires**; assess data quality across dimensions and compute composite fitness; and specify a verification regime for AI-produced work whose sample size is derived from the consequence of an escaped error rather than from convenience.

The master worked project. Project Auriga continues from Domains 6, 7 and 8 — the 25-week control-systems upgrade for a regional utility, `BAC` **USD 4,000,000**, at week 13 showing `CPI` 0.91 and `SPI` 0.92 (Domain 7, KA 7.3), with a **cost of delay of USD 45,000 per week** and a critical path through installation and testing and commissioning at zero float (Domain 6). This domain attaches quality economics to it: a cost of quality of **USD 364,000** at the chosen regime (9.1 % of `BAC`), a containment chain that lets **8** of **80** introduced control-logic defects escape at **USD 12,000** each, a commissioning phase whose 30 % rework share turns a 4-week activity into **4.7619 weeks**, and an acceptance sample of 20 instrument loops that cannot exclude a defect rate of **13.91 %**. Auriga's engineer-week costs **USD 5,225** throughout, from Domain 7's blended engineering rate of USD 130.625 per hour over a 40-hour week.

KNOWLEDGE AREA 9.1

Quality planning

Topics: 9.1.1 quality, grade and fitness for purpose · 9.1.2 the cost of quality and its interior minimum · 9.1.3 quality objectives, metrics and tolerances.

9.1.1 Quality, grade and fitness for purpose

Definition. Quality is the degree to which delivered work conforms to its stated requirements. **Grade** is the level of capability, feature richness or material specification those requirements ask for. The two are independent, and confusing them is the commonest conceptual error in delivery: a low-grade product built exactly to a low-grade specification is high quality, and a high-grade product riddled with nonconformances is not. A leader may legitimately reduce grade — that is a scope decision under Domain 5, taken with the sponsor and recorded. A leader may never legitimately reduce quality, because quality is conformance to what was agreed, and reducing it is delivering something other than the thing promised while saying nothing.

Fitness for purpose is the third and hardest test: whether the conforming, correctly graded output serves the use it was built for. Domain 5's Case study B — a public-sector system one hundred per cent complete against specification and thirty-four per cent useful — is the canonical failure, and the reason quality management cannot stop at conformance. Conformance is verifiable against a document; fitness for purpose only against a user, an operating context and a benefit measure (Domain 2). A regime that tests only the first will pass work that fails the second with complete documentary integrity.

The reference points are worth naming precisely and describing in the book's own words. The **ISO 9000 family** supplies the vocabulary and the management-system requirements for quality (ISO 9001 being the certifiable requirements standard); it evidences that an organisation has defined and follows processes, not that any deliverable is correct. **Statistical process control** and the **Six Sigma** family of improvement methods supply measurement machinery for repetitive processes; projects borrow their arithmetic — yields, sampling, capability — with care, because a project is not a repeating process and its samples are small. Neither tradition answers the question this domain treats as primary: how much conformance is it worth buying?

Three distinctions to hold. Quality assurance is process-directed and preventive. Quality control is product-directed and detective. Quality improvement is directed at the causes of the answers the first two give. They have different owners, cadences and costs, and an organisation funding only the second will pay for the same defect indefinitely.

9.1.2 The cost of quality and its interior minimum

Definition. The **cost of quality** (`CoQ`) is the total cost incurred because the work might not conform, plus the total cost incurred because it did not. It has four categories, and every serious quality decision is a trade among them:

Category	What it buys	Auriga examples	Behaviour as quality rises
Prevention	Fewer defects introduced	Design standards, supplier qualification, training, model-based design review, configuration control	Rises
Appraisal	Defects found before the customer	Peer review, factory acceptance test, site acceptance test, inspection, audit	Rises
Internal failure	Nothing — it is loss	Rework, retest, scrap, document re-issue, disruption inside the project	Rises, then falls
External failure	Nothing — loss plus reputation	Field defects, warranty attendance, outage credits, client disruption, claims	Falls

The first two are the **cost of conformance**, the last two the **cost of nonconformance**. Two features of that table are routinely got wrong. Internal failure cost is **not monotone**: an organisation that improves testing before it improves engineering finds more defects internally and its internal failure cost rises — a movement almost always misread as deterioration, and the commonest reason quality programmes are abandoned in their first period. And external failure is the only category whose unit cost the project does not control, because it is set by the operating context: the same defect costs a trifle in a marketing website and a great deal in a live utility control system.

WORKED EXAMPLE

Worked example 9.1.2 — Auriga's cost of quality, and where its minimum actually is.

1. **Setup.** Auriga's control-logic and configuration scope can be delivered under five candidate regimes. For each, the estimating team has stated prevention spend, appraisal spend, the defects the regime would allow to be **introduced**, the fraction it would **detect** before handover, and the observed average internal correction cost per defect found (which rises as detection reaches deeper, later layers). An escaped defect in live operation costs Auriga's client **USD 12,000** on average — diagnosis, a site attendance inside an outage window, retest and the contractual service credit. `BAC` is **USD 4,000,000**.

Regime	Prevention	Appraisal	Introduced	Detected	Internal unit cost
R0 test at the end	12,000	48,000	160	75.00 %	1,000
R1 basic	40,000	60,000	120	85.00 %	1,300
R2 planned	96,000	64,000	80	90.00 %	1,500
R3 enhanced	170,000	80,000	64	93.75 %	1,650
R4 maximum	285,000	100,000	50	96.00 %	1,750

1. **Formula.** Per regime: internal failures = introduced × detected; escapes = introduced − internal failures; internal failure cost = internal failures × internal unit cost; external failure cost = escapes × external unit cost; `CoQ` = prevention + appraisal + internal failure + external failure. Choose the regime minimising `CoQ`.
2. **Substitution.** R2: internal failures `80 × 0.90 = 72`; escapes `8`; internal failure cost `72 × 1,500 = 108,000`; external failure cost `8 × 12,000 = 96,000`; `CoQ` = `96,000 + 64,000 + 108,000 + 96,000`.
3. **Result.**

Regime	Escapes	Prevention	Appraisal	Internal failure	External failure	Total CoQ	% of BAC
R0	40	12,000	48,000	120,000	480,000	**660,000**	16.5 %
R1	18	40,000	60,000	132,600	216,000	**448,600**	11.2 %
R2	**8**	96,000	64,000	108,000	96,000	**364,000**	**9.1 %**
R3	4	170,000	80,000	99,000	48,000	**397,000**	9.9 %
R4	2	285,000	100,000	84,000	24,000	**493,000**	12.3 %

The minimum is **USD 364,000** at regime **R2**, with **8 escaped defects** — not at the lowest attainable defect count. 5. **Interpretation.** Five things follow, and a leader who can state all five commands the quality budget rather than spectating.

The optimum is interior, and not zero. R4 costs **USD 129,000** more than R2 to remove six further escapes — **USD 21,500 per escape avoided** against damage of USD 12,000, or **1.79 times** the harm it prevents. The last defects are the expensive ones, so an organisation treating zero defects as the objective is committing to buy protection at roughly twice its value. The honest objective is not "no defects" but **"no defect whose prevention costs less than its consequence"** — a computable instruction, and the one this table executes.

The marginal test is the decision, not the total. R1 → R2 costs **USD 60,000** more in conformance and returns **USD 24,600** of internal and **USD 120,000** of external failure — a net **USD**

84,600 gain. R2 → R3 costs **USD 90,000** more and returns **USD 9,000** and **USD 48,000** — a net **USD 33,000** loss. The rule: spend the next increment while its conformance cost is below the failure cost it removes, and stop at the first increment where it is not.

The range of optimality is what makes the recommendation defensible. R2 stays optimal for any external-failure unit cost between **USD 3,540** (below which R1 wins) and **USD 20,250** (above which R3 wins) — a **5.72-fold** band. Auriga's assumed USD 12,000 sits **3.39 times** above the lower bound and at **59.3 %** of the upper. That is the sentence for a steering committee, because it answers the only question worth asking about an estimated input: how wrong can it be before the recommendation changes? In a safety-related or heavily regulated context the escaped-defect cost is a multiple of USD 12,000: above USD 20,250 the optimum moves to R3 and above **USD 60,000** to R4. Same arithmetic, different context, different regime — which is the quantitative statement of why quality regimes are not transferable between sectors.

Internal failure cost rises before it falls, and this must be said in advance. R0 → R1 raises it from 120,000 to **132,600** while total `CoQ` falls by **USD 211,400**: better detection finds defects that were escaping, and the cost moves from the external column to the internal one at a fifth of the unit price. A programme reported on internal failure cost alone will look like a failure in its first period, which is where such programmes are cancelled.

The cautions. The introduced-defect counts and detection rates are **estimates from comparable work** and must be labelled as such; the conclusion is robust to the external unit cost across a wide band but not to a detection rate that has been asserted rather than measured. The model prices only what is in it — it excludes the **schedule** consequence of rework (KA 9.2.3 computes that separately and it is material), the tail risk of a single catastrophic escape rather than an average one, and any licence or reputational consequence not expressible per defect. And the four categories must be **defined once and counted consistently**, because the easiest way to make a quality programme look successful is to reclassify internal failure as appraisal.

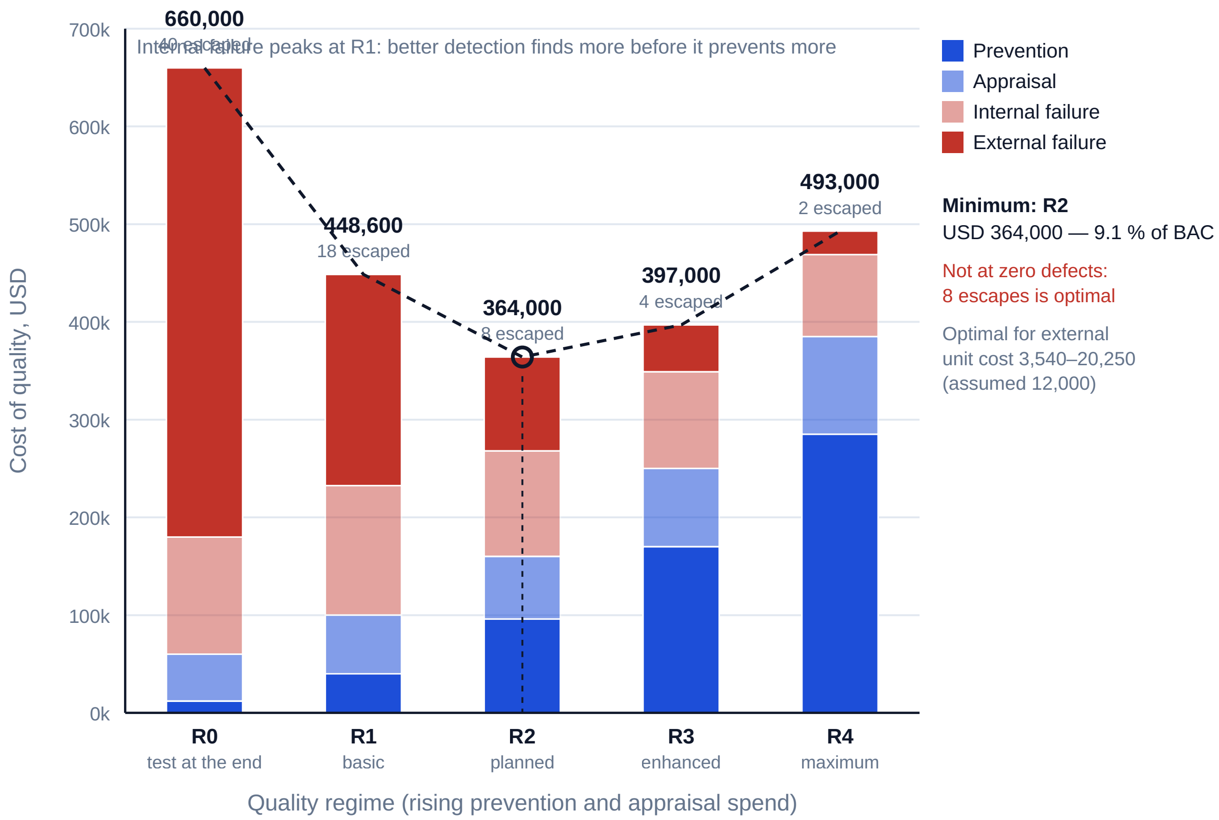


Fig 9.1.1 — Auriga's cost of quality and its interior minimum

9.1.3 Quality objectives, metrics and tolerances

The plan's job. A quality management plan is the record of five decisions, each with an owner and a number. What conformance means for each deliverable class — the acceptance criteria of Domain 5, KA 5.4.2, referenced not restated. What tolerance applies, because a specification without a tolerance is untestable and will be argued about at acceptance. Which containment layers exist, with their coverage and expected detection rates (KA 9.2). Who accepts, and on what authority (KA 9.3). And what it costs — the four-category budget of 9.1.2, reconcilable to Domain 7's cost baseline rather than a parallel document.

Metrics that survive contact with delivery. A quality metric earns its place if a named person changes a decision when it moves. Four do, on Auriga: **escape rate** (defects reaching the next stage or the customer — the primary outcome measure, and what 9.1.2 optimises); **first-time-right rate** per step and rolled end to end (KA 9.4.2 — the primary process measure); **rework share of capacity** (KA 9.2.3 — the primary schedule measure, and the one most often absent); and **cost of quality as a percentage of** `BAC` split four ways, of which Auriga's 9.1 % is the figure the sponsor should know.

Tolerance and the two errors. Every tolerance trades rejecting conforming work against accepting nonconforming work. It is therefore the same class of decision as Domain 3's delegation threshold — a certain, recurring, invisible cost against an uncertain, occasional, highly visible one — and organisations reliably optimise against the visible one. The countermeasure is the same: compute both, state the breakeven, and let the accountable person choose knowing the price.

AI in this KA

Where it earns its place. Assembling a four-category cost-of-quality picture from accounting data never coded for it — mapping rework time codes, inspection orders, warranty claims and service credits into the four categories — is a classification task over messy records and the single largest practical obstacle to doing cost-of-quality work at all. Testing acceptance criteria and tolerances for measurability and flagging those that cannot be tested. Sweeping a specification set for missing tolerances. Running the 9.1.2 model across dozens of regime and unit-cost combinations to produce the range-of-optimality band rather than a point estimate.

Where it must not go. Setting the external-failure unit cost, which is a judgement about consequence in an operating context and, where safety or licence is involved, not an expected-value judgement at all. Choosing the regime — a risk-appetite decision belonging to the sponsor and, on a safety-related system, to the accountable engineer. And supplying detection rates or introduced-defect counts from plausibility: a model asked for these returns confident, well-formatted numbers with no provenance which then drive a real budget. Where the history does not exist, the correct output is that it does not exist, plus the range over which the conclusion holds.

Verification, concretely. Reproduce the whole `CoQ` table by hand or in a spreadsheet whose formulae are visible — four multiplications and a sum per regime — and state every input's source beside it. Publish the marginal step analysis and the range of optimality, not the point minimum, because those tell a board whether the recommendation is robust or fragile. And where an AI system has classified historical costs into the four categories, sample-check the classification on a stated sample size: category misclassification makes the model wrong in a direction nobody notices.

KEY TERMS — KA 9.1

Term	Meaning
Quality	The degree to which delivered work conforms to its stated requirements.
Grade	The level of capability or specification the requirements ask for; independent of quality.
Fitness for purpose	Whether conforming, correctly graded output serves the use it was built for.
Cost of quality (`CoQ`)	Prevention + appraisal + internal failure + external failure.
Cost of conformance	Prevention + appraisal — money spent so defects do not occur or do not escape.
Cost of nonconformance	Internal + external failure — money lost because they did.
Escape	A defect reaching the next stage, or the customer, undetected.
External-failure unit cost	The average cost of one escaped defect in the operating context; set by context, not the project.
Range of optimality	The band of an uncertain input over which the recommended option stays recommended.
Tolerance	The range within which output is conforming; a specification without one is untestable.

SAMPLE MCQS — KA 9.1

MCQ 9.1-A `[9.1.2 · Application]` A regime allows 80 defects to be introduced and detects 90 % before handover; internal correction averages 1,500 and an escaped defect costs 12,000. Prevention is 96,000 and appraisal 64,000. Total cost of quality is: - A. USD 268,000 - B. USD 364,000 - C. USD 204,000 - D. USD 460,000

Rationale: `96,000 + 64,000 + (72 × 1,500) + (8 × 12,000) = 364,000` (9.1.2). A omits external failure; C counts nonconformance only and drops conformance; D applies the 12,000 external unit cost to all 80 introduced defects rather than to the 8 that escape.

MCQ 9.1-B `[9.1.2 · Evaluation]` Moving to the next-stricter regime costs 90,000 more in conformance and removes 9,000 of internal failure and 4 escaped defects. At what external-failure unit cost does the stricter regime become worth buying? - A. USD 12,000 - B. USD 22,500 - C. USD 20,250 - D. USD 81,000

Rationale: The step pays when `90,000 ≤ 9,000 + 4u`, so `u ≥ 81,000/4 = 20,250` (9.1.2). B divides the conformance cost by escapes avoided and forgets the internal-failure saving; D is the numerator, not the unit cost; A is the assumed unit cost, which is the thing being tested.

MCQ 9.1-C `[9.1.2 · Analysis]` An organisation strengthens testing and its internal failure cost rises in the first period while total cost of quality falls. The correct reading is that: - A. the testing programme is failing and should be reversed - B. defects have moved from the external column to the internal one at a much lower unit cost - C. internal and external failure have been misclassified - D. prevention spending was set too low

Rationale: Internal failure cost is not monotone in quality — better detection finds defects that were previously escaping (9.1.2). Reading the rise as deterioration is the standard reason such programmes are cancelled in their first period.

MCQ 9.1-D `[9.1.1 · Comprehension]` A supplier delivers a product built exactly to a deliberately basic specification. It is best described as: - A. low grade and low quality - B. low grade and high quality - C. high grade and low quality - D. not assessable until the client uses it

Rationale: Grade is the specified level; quality is conformance to it (9.1.1). D confuses quality with fitness for purpose, which is a third and separate test.

MCQ 9.1-E `[9.1.2 · Evaluation]` Removing the last 6 escaped defects costs 129,000 more in total cost of quality, and each escape would have cost 12,000. The strongest professional statement is that: - A. the spend is justified because defects should be eliminated - B. the spend buys protection at 21,500 per defect, 1.79 times the harm it prevents, so it is not justified on these figures - C. the spend is justified because 129,000 is small against a 4,000,000 budget - D. the comparison cannot be made without a Monte Carlo simulation

Rationale: `129,000/6 = 21,500` against a 12,000 consequence, a ratio of 1.79 (9.1.2). A is the zero-defects fallacy; C is affordability, not value; D over-reaches — the expected-value comparison is valid on stated averages, though a safety case would also require the tail to be examined.

SELF-CHECK — KA 9.1

1. Why is the cost-of-quality minimum not at zero defects? — Because the marginal conformance cost of removing the last defects exceeds their consequence; on Auriga the last six escapes cost 21,500 each to remove against 12,000 of harm.
2. What single input determines where the optimum sits, and who sets it? — The external-failure unit cost, set by the operating context and not by the project; Auriga's optimum holds for 3,540–20,250 and moves at 20,250 and again at 60,000.
3. Which cost category is non-monotone, and why does it matter? — Internal failure: it rises as detection improves before falling as introduction falls, and misreading that rise is why quality programmes get cancelled in their first period.

KNOWLEDGE AREA 9.2

Assurance and control

Topics: 9.2.1 assurance and control as different instruments · 9.2.2 containment layers and the economics of detection · 9.2.3 rework, capacity and the schedule consequence.

9.2.1 Assurance and control as different instruments

Quality assurance acts on the **process** and is preventive: it asks whether the way the work is being done is capable of producing conforming output, and its evidence is process evidence — competence records, controlled documents, a managed configuration, a followed method. Quality control acts on the **product** and is detective: it asks whether this output conforms, against its criteria. Both are the first line of the assurance model built in Domain 3, KA 3.3.2; the second and third lines form opinions about whether the first line works, and Domain 3's failure modes — duplication, gap, capture, and treating a favourable opinion as a transfer of accountability — apply here unchanged and are not restated.

The distinction is operationally load-bearing because control cannot make output conform; it can only report that it does not. Every unit of control spend therefore buys **information**, and information is worth what the decision it changes is worth. An inspection whose result cannot change anything — carried out after the only feasible correction window has closed, or reported to someone with no authority to act — is pure cost, and the commonest wasted quality spend in delivery is not too little inspection but inspection positioned too late. Domain 3, KA 3.3.1 computes the same result for a governance gate and establishes that gate value is destroyed by elapsed time and by weak detection; the containment arithmetic below is its product-level counterpart, and the two should be read together.

9.2.2 Containment layers and the economics of detection

Definition. A **containment layer** is a review, test or inspection with a stated coverage, a stated **detection rate** (the share of the defects reaching it that it finds) and a stated **unit correction cost**. A delivery process is a chain of such layers, and the property that matters is the **escape fraction**: the product of the layers' individual escape rates.

```
escape fraction = ∏ (1 − dᵢ)      over layers i, with dᵢ the detection rate of layer i
escapes         = defects introduced × escape fraction
```

The product form is the lesson. Layers are multiplicative, not additive, so three unimpressive layers can outperform one strong one — and, symmetrically, removing one layer does not degrade the result by its own detection rate but multiplies the escape fraction by $1/(1 - d_i)$, which for a 50 % layer is a doubling.

The correction-cost ladder. Correction cost rises with the stage at which a defect is found, because later stages have built more on top of it and involve more parties. Auriga's observed ladder for a control-logic or configuration defect:

Layer	What it is	Detection rate d_i	Correction cost per defect	Ratio to the first layer
L1	Design peer review	0.50	USD 800	×1.000
L2	Factory acceptance test	0.60	USD 2,000	×2.500
L3	Site acceptance test	0.50	USD 3,500	×4.375
—	Live operation (escape)	—	USD 12,000	×15.000

Domain 5, KA 5.2.1 establishes the same shape for requirement defects on a much steeper ladder — a factor of four per stage, from USD 400 at definition to USD 102,400 in live service — because a requirement defect propagates through design, build and test before it is visible. The two ladders are not interchangeable and must not be averaged: a single "cost per defect" figure that mixes them will mis-price every containment decision made from it.

WORKED EXAMPLE

Worked example 9.2.2 — Auriga's containment chain, and the price of one avoidable defect.

1. **Setup.** Under regime R2 (9.1.2), **80** control-logic defects are introduced. They pass through the three layers above, with detection rates 0.50, 0.60 and 0.50 and correction costs of 800, 2,000 and 3,500. An escape costs USD 12,000.
2. **Formula.** Escape fraction = $\prod(1 - d_i)$. Defects found at layer i = defects reaching it × d_i. Internal failure cost = `Σ (found at i × unit cost at i)`. Expected containment cost per **introduced** defect = (internal + external failure cost) ÷ defects introduced.
3. **Substitution.** `(1 − 0.50)(1 − 0.60)(1 − 0.50) = 0.50 × 0.40 × 0.50`. Flow: L1 finds `80 × 0.50 = 40`, leaving 40; L2 finds `40 × 0.60 = 24`, leaving 16; L3 finds `16 × 0.50 = 8`, leaving **8**. Internal cost `40 × 800 + 24 × 2,000 + 8 × 3,500`.
4. **Result.** Escape fraction **0.100** (detection **90.0 %**); escapes **8**. Internal failure cost `32,000 + 48,000 + 28,000 =` **USD 108,000** — an average of **USD 1,500** per defect found internally, reconciling to 9.1.2. External failure cost `8 × 12,000 =` **USD 96,000**. Total nonconformance **USD 204,000**, which over 80 introduced defects is an **expected containment cost of USD 2,550 per introduced defect**.
5. **Interpretation.** That last figure is the most useful number in the domain and is almost never computed. **USD 2,550 is what one avoidable Auriga defect costs, whoever eventually finds it** — the weighted average of being caught cheaply at peer review, expensively at site acceptance and catastrophically in service. It converts every prevention proposal into a one-line test: does this remove a defect for less than 2,550? It also prices the layers. Removing L3 raises the escape fraction to `0.50 × 0.40 =` 0.200 — escapes double from 8 to 16, adding `8 × 12,000 = 96,000` of external failure while saving only `8 × 3,500 = 28,000` of L3 correction, so **L3 must cost more than USD 68,000 to run** before its removal is arguable. That is a checkable claim rather than an opinion about whether site testing is worth it. Two cautions. The model assumes layers are **independent detectors**, and they are not: two layers staffed by the same team reading the same document with the same blind spot miss the same defects, so the real escape fraction is worse than the product — the countermeasure is to make consecutive layers methodologically different (a review, then an executed test, then a test in the target environment) rather than more of the same. And detection rates are estimates; where no history exists, state a range and compute the decision across it.

WORKED EXAMPLE

Worked example 9.2.2b — prevention or appraisal, on the same USD 40,000.

1. **Setup.** Auriga has USD 40,000 of uncommitted quality budget at the end of design and two proposals. **(a) Appraisal:** extend site acceptance test coverage, raising L3's detection rate from 0.50 to 0.75. **(b) Prevention:** a design-standards and supplier-qualification package cutting defects introduced from 80 to 60, detection rates unchanged. The baseline is the R2 chain: nonconformance USD 204,000.
2. **Formula.** Re-run the chain under each proposal; compare (added spend + internal + external failure) against the baseline 204,000.
3. **Substitution.** (a) 80 introduced; L1 finds 40, L2 finds 24, L3 finds `16 × 0.75 = 12`, escapes **4**; internal `32,000 + 48,000 + 12 × 3,500 = 122,000`; external `4 × 12,000 = 48,000`. (b) 60 introduced; L1 finds 30, L2 finds 18, L3 finds 6, escapes **6**; internal `24,000 + 36,000 + 21,000 = 81,000`; external `6 × 12,000 = 72,000`.
4. **Result.** **(a)** `40,000 + 122,000 + 48,000 =` **USD 210,000 — USD 6,000 worse** than doing nothing. **(b)** `40,000 + 81,000 + 72,000 =` **USD 193,000 — USD 11,000 better** than doing nothing, and **USD 17,000 better than (a)** on identical money.
5. **Interpretation.** Prevention wins **even though it removes fewer escapes**: (a) halves escapes from 8 to 4, (b) only cuts them from 8 to 6. The reason is the containment cost per introduced defect. Option (b) removes 20 defects before any layer has to pay to find them, worth `20 × 2,550 =` **USD 51,000** against USD 40,000 of spend — a net **USD 11,000**, exactly the result above and the identity worth remembering. Option (a) removes no defect; it **relocates** detection to a later and dearer layer, so its saving on external failure is partly consumed by internal failure rising from 108,000 to 122,000. Both rules are computable: **prevention pays whenever it removes a defect for less than the expected containment cost per introduced defect** — USD 2,550 here, so a 40,000 programme must avoid at least `40,000/2,550 =` **15.69, i.e. 16 defects** — and **appraisal pays only when the layer it strengthens is cheaper than the layer or escape it displaces work from.** Raising detection at the earliest layer would have been a different and better proposition.

Two observations belong in the record. Option (b) takes total `CoQ` to `160,000 + 40,000 + 81,000 + 72,000 =` **USD 353,000**, below the USD 364,000 that 9.1.2 called optimal. That is not a contradiction: 9.1.2 optimised along a ladder of **pre-costed regimes**, and prevention more efficient than the ladder assumed shifts the whole curve down — so a cost-of-quality optimum is conditional on the prevention technology available and should be recomputed when that changes. And option (a) reaches 4 escapes for a total `CoQ` of **USD 370,000** where regime R3 reaches 4 escapes at **USD 397,000** — two routes to the same escape count, USD 27,000 apart, because R3 got there partly by introducing fewer defects. **The escape count is not a sufficient description of a quality regime.**

9.2.3 Rework, capacity and the schedule consequence

The principle no cost-of-quality model contains. Rework is not only money; it is **capacity**. Correcting a defect consumes the same people, equipment and test facility that would otherwise produce new output, so a rework share of capacity is a direct reduction in throughput — and where the activity is on the critical path, the reduction converts straight into elapsed time. The relationship is not linear:

```
effective capacity  = nominal capacity × (1 − r)      r = rework share of capacity
duration            = first-pass work content ÷ effective capacity
duration multiplier = 1 / (1 − r)
```

`1/(1 − r)` is hyperbolic. The tenth percentage point of rework costs far less than the fortieth, which is why a team absorbs a modest rate almost invisibly and then loses control of a phase over what looks like a small further deterioration.

WORKED EXAMPLE

Worked example 9.2.3 — Auriga's commissioning, and what 30 % rework costs.

1. **Setup.** Auriga's activity **F, testing and commissioning**, is planned at **4 weeks** with **6** engineers — a nominal capacity of **24 engineer-weeks**. First-pass work content is **20 engineer-weeks**, so the plan carries a rework allowance of **4 engineer-weeks**, or **16.67 %** of capacity. F is on the critical path with zero float (Domain 6, KA 6.2.1); Auriga's cost of delay is **USD 45,000 per week**; an engineer-week costs **USD 5,225** (Domain 7, KA 7.4.1). At the two-week point, rework is running at **30 %** of capacity.
2. **Formula.** duration = content ÷ (crew × (1 − `r`)); overrun = duration − planned duration; delay cost = overrun × cost of delay; rework consumed = `r` × crew × duration; labour above allowance = (rework consumed − allowance) × engineer-week cost.
3. **Substitution.** `20 ÷ (6 × 0.70) = 20 ÷ 4.2`. Rework consumed `0.30 × 6 × 4.7619`.
4. **Result.**

Rework share r	Multiplier 1/(1−r)	Duration (weeks)	Overrun (weeks)	Delay cost (USD)
16.67 % (as planned)	1.2000	4.0000	0.0000	0
25 %	1.3333	4.4444	0.4444	20,000
30 % (actual)	**1.4286**	**4.7619**	**0.7619**	**34,286**
40 %	1.6667	5.5556	1.5556	70,000
50 %	2.0000	6.6667	2.6667	120,000

At 30 %: duration **4.7619 weeks**, overrun **0.7619 weeks**, delay cost **USD 34,286**. Rework consumes **8.5714 engineer-weeks** against a 4-engineer-week allowance — **4.5714 engineer-weeks** of unbudgeted labour at 5,225, or **USD 23,886**. Total consequence: **USD 58,171**. 5. **Interpretation. The cost is mostly schedule, and the schedule cost is invisible in the quality report.** Of the 58,171, **59 %** is delay and 41 % labour; a quality function reporting rework hours reports the smaller half. On a critical-path activity the rework share of capacity is a schedule metric that happens to be collected by the quality system, and it belongs in the same report as float.

The convexity is the management lesson. Moving from 30 % to 40 % costs a further **USD 35,714** of delay and 40 % to 50 % costs **USD 50,000**, where 10 % to 20 % would have cost only **USD 20,833**. Every additional point is dearer than the last, so intervene early — while the numbers still look tolerable — and trigger on the rate, not the accumulated slip.

The allowance is a decision and should be visible as one. A 16.67 % allowance is legitimate and honest, and it is also a forecast that can be wrong; a plan with **no** allowance is not a plan without rework but a plan whose first defect creates a slip. State the allowance, measure against it weekly, and treat the variance as a leading indicator — the rework share reveals itself two weeks before the milestone does.

The easy-to-miss point: the natural response — reduce test coverage to recover the schedule — is a decision to raise the escape fraction, and 9.2.2 prices it. Cutting L3's detection from 0.50 to 0.25 raises escapes from 8 to 12, adding `4 × 12,000 =` **USD 48,000** of external failure against `4 × 3,500 =` USD 14,000 of internal correction saved. Any schedule recovery must beat that net USD 34,000, computed, before it is taken.

AI in this KA

Where it earns its place. Detecting defects at scale: static analysis of control logic and code, cross-checking a configuration set against a design baseline, comparing an as-installed instrument schedule against the design register, sweeping test evidence for missing or contradictory results. These have checkable right answers, and a model doing them is functioning as a **containment layer** — and should be characterised as one, with a measured detection rate and a unit cost, so it can be placed in the chain of 9.2.2 rather than trusted qualitatively. Also: classifying a rework log by cause and computing the rework share of capacity per team and per week, which most organisations cannot report because the time data is not coded for it.

Where it must not go. It must not be the **only** layer for anything consequential, because 9.2.2's independence caution applies with unusual force: a model's misses are systematic, so two checks by the same model are one check, and a model that generated the artefact is not an independent reviewer of it. It must not issue a test result, sign an inspection record or grant a release — those are attributable acts under Domain 3, KA 3.A.2. And its detection rate must not be asserted: a claim that an automated check finds "most" defects is not a number, and a chain computed from asserted rates gives a false escape fraction that is then used to justify removing a human layer.

Verification, concretely. Measure the model's detection rate against a known, held-back defect set before relying on it, and re-measure when the model, the prompt or the artefact type changes — a detection rate is a property of a configuration, not of a tool. Record its false-positive rate too, because a layer raising many false findings consumes the engineering attention the next layer needs. Keep a human layer **methodologically different** from the automated one. And reproduce the arithmetic by hand: the escape fraction is a product of three numbers and the duration multiplier is one division.

KEY TERMS — KA 9.2

Term	Meaning
Quality assurance	Process-directed, preventive activity: is the way we work capable of producing conforming output?
Quality control	Product-directed, detective activity: does this output conform?
Containment layer	A review, test or inspection with a stated coverage, detection rate and unit correction cost.
Detection rate (d_i)	The share of defects reaching a layer that the layer finds.
Escape fraction	$\prod(1 - d_i)$ across the chain — the share of introduced defects reaching the customer.
Correction-cost ladder	The rising cost of correcting a defect by the layer at which it is found.
Expected containment cost per introduced defect	(Internal + external failure) ÷ defects introduced; the breakeven price of prevention.
Rework share of capacity (r)	The fraction of a team's available effort consumed correcting output.
Duration multiplier	$1/(1 - r)$ — the hyperbolic effect of rework on elapsed time.
Rework allowance	The capacity a plan reserves for rework; a forecast, and a decision.

SAMPLE MCQS — KA 9.2

MCQ 9.2-A [9.2.2 · Application] Three containment layers have detection rates 0.50, 0.60 and 0.50. Of 80 introduced defects, how many escape? - A. 8 - B. 3 - C. 12 - D. 27

Rationale: Escape fraction $0.50 \times 0.40 \times 0.50 = 0.10$, so $80 \times 0.10 = 8$ (9.2.2). B adds the detection rates to 1.60 and treats the chain as detecting everything with a remainder; C uses two layers only; D applies the average detection rate (0.5333) three times as if each layer saw all 80.

MCQ 9.2-B [9.2.2 · Analysis] On a chain where 80 defects are introduced, internal failure costs 108,000 and external failure 96,000. The expected containment cost per introduced defect is: - A. USD 1,500 - B. USD 2,550 - C. USD 12,000 - D. USD 2,000

Rationale: $(108{,}000 + 96{,}000)/80 = 2{,}550$ (9.2.2). A is the average cost per defect found internally, omitting escapes; C is the escape unit cost; D divides internal failure alone by 80 and rounds — all three understate the breakeven price of prevention.

MCQ 9.2-C [9.2.2 · Evaluation] USD 40,000 will either raise the last layer's detection rate from 0.50 to 0.75, or cut defects introduced from 80 to 60. On the figures of 9.2.2, the better choice and its reason are: - A. raising detection, because it halves escapes from 8 to 4 - B. cutting introduction, because it removes 20 defects at an expected containment cost of 2,550 each — USD 51,000 of value for USD 40,000 - C. either, since both cost the same - D. raising detection, because internal correction is cheaper than external failure

Rationale: Prevention returns 51,000 against 40,000 — a net 11,000 — and beats the appraisal option by 17,000 (9.2.2b). A counts escapes rather than cost and misses that raising late-layer detection moves work to a dearer layer, pushing internal failure from 108,000 to 122,000; D states a true premise that does not reach the conclusion.

MCQ 9.2-D [9.2.3 · Application] A 4-week activity with 6 engineers has 20 engineer-weeks of first-pass content. Rework runs at 30 % of capacity. The activity will take: - A. 4.00 weeks - B. 4.76 weeks - C. 5.20 weeks - D. 4.29 weeks

Rationale: `20 ÷ (6 × 0.70) = 4.7619` weeks (9.2.3). C adds 30 % to the 4-week plan, the linear error; D mishandles the allowance by scaling the 3.33 weeks of pure content; A ignores the overrun.

MCQ 9.2-E `[9.2.3 · Analysis]` Why does the marginal cost of rework rise as the rework share rises? - A. because rework is charged at a premium rate - B. because the duration multiplier `1/(1 − r)` is convex, so equal increments of `r` add increasing amounts of elapsed time - C. because defects found later cost more to fix - D. because float is consumed first and then lost

Rationale: The convexity is the mechanism: on Auriga's figures 10 %→20 % adds 0.4630 weeks while 40 %→50 % adds 1.1111 (9.2.3). C is the correction-cost ladder, a different effect (9.2.2); A and D are not general.

MCQ 9.2-F `[9.2.2 · Analysis]` An automated check and a manual review are placed as consecutive containment layers, both driven from the same design document by the same team. The escape fraction computed as the product of their detection rates will be: - A. correct - B. too optimistic, because the layers are not independent detectors - C. too pessimistic, because two layers always find more than one - D. correct only if their detection rates are equal

Rationale: The product form assumes independence; layers sharing a source, a method or a blind spot miss the same defects, so the true escape fraction is worse than the product (9.2.2). The countermeasure is to make consecutive layers methodologically different.

SELF-CHECK — KA 9.2

1. What does removing a 50 % containment layer do to the escape fraction? — Doubles it: the layer's escape rate of 0.50 leaves the product, multiplying escapes by `1/(1 − 0.50)`.
2. State the breakeven test for a prevention proposal. — It pays if it removes a defect for less than the expected containment cost per introduced defect — USD 2,550 on Auriga, so a USD 40,000 programme must avoid 16 defects.
3. Why is a rework share of capacity a schedule metric? — Because effective capacity is `nominal × (1 − r)` and duration is content ÷ effective capacity, so on a zero-float activity every point of rework is elapsed time: 30 % turned Auriga's 4-week commissioning into 4.7619 weeks and USD 34,286 of delay.

KNOWLEDGE AREA 9.3

Acceptance, nonconformance and root-cause analysis

Topics: 9.3.1 the acceptance decision and nonconformance disposition · 9.3.2 acceptance sampling and what a passing sample establishes · 9.3.3 root-cause analysis and recurrence.

9.3.1 The acceptance decision and nonconformance disposition

Definition. Acceptance is a formal decision by a named authority that a deliverable meets its acceptance criteria and may pass to the next stage, to the client, or into service. Domain 5, KA 5.4.3 established the criteria and the verification/validation distinction; what belongs here is the decision, its authority, and what happens when the answer is no.

Acceptance goes wrong in four recognisable ways, each with a countermeasure that costs nothing. **Acceptance by silence** — nobody rejects, so the work is deemed accepted, usually by the expiry of a contractual review period; the countermeasure is a positive acceptance record with a named signatory, since the absence of a rejection is not a decision. **Acceptance by exhaustion** — the deliverable is accepted at the fourth submission because the reviewers have no more time, not because it changed; the countermeasure is to record the open nonconformances at acceptance, converting exhaustion into a visible conditional pass. **Acceptance without authority** — the signatory is not entitled to bind the receiving organisation, which is Domain 3's single-A defect at deliverable level. And **acceptance against criteria written after the evidence**, where the criteria were adjusted to what the deliverable does, so the test is guaranteed to pass and evidences nothing.

Nonconformance disposition. When output does not conform, exactly five dispositions exist, each with a different authority and a different downstream obligation:

Disposition	What it means	Authority required	Downstream obligation
Rework	Bring it into full conformance	Delivery management	Retest to the same criteria
Repair	Make it fit for use without full conformance	Design authority	The as-built record must show the repair
Use as is (concession)	Accept the nonconformance as it stands	The party bearing the consequence — client or design authority, never the producer alone	Register it; assess cumulative effect
Regrade	Use it for a lesser purpose for which it conforms	Design authority plus the owner of that purpose	Traceability so it cannot migrate back
Reject / scrap	Do not use it	Delivery management	Root-cause analysis if value or recurrence warrants

Two disciplines make this table work rather than decorate a procedure. **The producer never grants its own concession** — a concession transfers a consequence, so its authority sits with whoever will carry it, and a process in which the delivery team accepts its own nonconformances has no acceptance function at all. And **concessions must be counted, not merely recorded**: individually minor concessions accumulate into a delivered product materially different from the specified one, which is Domain 3's Case study B at deliverable level. The cumulative-effect test — related concessions aggregating above a threshold within a period require the authority appropriate to the aggregate — is the countermeasure, and it is the instrument Domain 4, KA 4.4 applies to change.

9.3.2 Acceptance sampling and what a passing sample establishes

Verifying every item is often uneconomic, so acceptance is taken on a sample. The arithmetic is elementary, almost never done, and produces two results practitioners find genuinely surprising.

For a population with true defective fraction p, the probability that a sample of n contains **no** defective item is $(1 - p)^n$. Turning that round gives the **confidence bound**: if a sample of n is clean, the largest defective fraction consistent with that observation at confidence $1 - \alpha$ is

```
p_upper = 1 − α^(1/n)          (α = 0.05 for a 95 % upper bound)
```

And the economics of how much to verify reduce to a single ratio. With c the cost of verifying one item and u the cost of one escaped defective item, full verification of N items costs cN while verifying n and accepting the rest costs $cn + u \cdot p \cdot (N - n)$. Setting them equal, the $(N - n)$ terms cancel and

```
p* = c / u
```

Full verification pays exactly when the defective fraction exceeds the ratio of verification cost to failure cost — and that breakeven is independent of the sample size. It is the number to put on the table before anyone argues about sample sizes.

WORKED EXAMPLE

Worked example 9.3.2 — Auriga's instrument loops: what a clean sample of 20 proves.

1. **Setup.** Auriga's installation includes **120** instrument loops. Verifying one costs **USD 1,400** (a two-engineer point-to-point check and signal injection). A defective loop reaching service costs **USD 12,000**. The subcontractor proposes verifying **20** loops and accepting the batch if none is defective. The sample is drawn and is clean. What has been established?
2. **Formula.** `P(clean sample) = (1 − p)ⁿ`; 95 % bound `p_upper = 1 − 0.05^(1/n)`; breakeven `p* = c/u`; minimum self-consistent sample `n* = ln(0.05)/ln(1 − p*)`.
3. **Substitution.** `(1 − 0.05)²⁰`; `1 − 0.05^(1/20)`; `1,400/12,000`; `ln(0.05)/ln(1 − 0.116667)`.
4. **Result.** A population 5 % defective passes this plan **35.85 %** of the time (at 2 % defective, **66.76 %**; at 10 %, **12.16 %**). The 95 % upper bound after a clean sample of 20 is **13.91 %** — on 120 loops, up to **16.69**, i.e. as many as 16 defective loops. The breakeven defective fraction is `1,400/12,000 =` **11.67 %**. The smallest sample whose 95 % bound falls below that breakeven is **25** (the bound is 11.29 % at n = 25 and 11.73 % at n = 24), costing **USD 7,000** more than the proposed 20.
5. **Interpretation.** The decisive observation compares two of those numbers. **The plan's own confidence bound (13.91 %) is higher than the defective fraction at which its strategy is wrong (11.67 %).** A clean sample of 20 cannot exclude the very defect rate at which the correct decision would have been to verify all 120 loops. The plan is **not self-consistent**, and no amount of it passing changes that. Five more loops — USD 7,000 on a USD 4,000,000 project — repairs it, and that is the cheapest quality decision anywhere in this domain.

What the sample does establish is worth stating precisely. It establishes that the population is probably not grossly defective: rates above about 14 % are unlikely given a clean 20. It does **not** establish conformance, and it does not establish any tighter rate — bounding `p` at 5 % with 95 % confidence needs **59** items (`ln 0.05 / ln 0.95 = 58.404`, and the familiar rule-of-three approximation `3/p` gives 60 as a mental check).

The economics cut against instinct. At a true 5 % defective rate the sample-of-20 plan is the **cheapest** of the three options: `1,400 × 20 + 12,000 × 0.05 × 100 =` **USD 88,000**, against **USD 119,200** for a 59-loop plan and **USD 168,000** for verifying all 120. Sampling is not laziness and full inspection is not automatically responsible — below `p* = 11.67 %` it destroys value. But the plan must be able to see whether `p` is below the breakeven, and the 20-loop plan cannot. Hence the discipline: **choose `n` from the confidence bound the economics require, not from what feels proportionate.** Three cautions. The `(1 − p)ⁿ` form treats draws as independent, a good approximation when `n` is small relative to `N` and conservative here. Defects are often **clustered** — one mis-trained crew, one bad batch — so the sample must be random, and a convenience sample of the accessible loops establishes nothing. And zero-defect acceptance interacts badly with measurement error, so the disposition rules of 9.3.1 must cover a contested sample result before the sampling starts.

9.3.3 Root-cause analysis and recurrence

The purpose, stated to exclude what usually happens. Root-cause analysis exists to reach the level at which a cause can be **removed**, so the defect class does not recur. It is not an explanation, not an apology and not a search for a person. The standard techniques — iterative "why" questioning, cause-and-effect decomposition, fault-tree reasoning backwards from an undesired event, and change analysis comparing a working case with a failing one — share one failure mode: **stopping at the level at which somebody can be blamed**, which is always shallower than the level at which something can be fixed. "The engineer loaded the wrong parameter set" describes the event. "There was no controlled baseline against which a parameter set could be checked, and no process step that would have detected the mismatch" is a cause, because it names something a manager can change.

The three tests of a completed analysis. A root cause has been reached when removing it would have prevented this occurrence; removing it prevents the whole **class**, not just the instance; and the removal is within somebody's authority to enact. If the third test fails, the analysis has produced a finding for a higher level, and Domain 3's escalation machinery is how it travels.

Recurrence is the metric. The **recurrence rate** — repeat nonconformances as a share of all — measures whether the programme works; a stable or rising rate means causes are being described rather than removed. **Cause concentration** — the share attributable to the largest single cause — tells you where to spend, and is almost always higher than intuition suggests.

WORKED EXAMPLE

Worked example 9.3.3 — Auriga's dominant cause, priced.

1. **Setup.** Auriga's 8 escaped control-logic defects (9.2.2) trace to three root causes: **5** to an uncontrolled supplier firmware configuration baseline (two field devices shipped with a superseded parameter set and nothing in the process compared them to a controlled reference), **2** to an ambiguous interlock specification, **1** to a one-off wiring error. Removing the dominant cause — imposing configuration control on the supplier's firmware, re-baselining, and adding a baseline comparison to the factory acceptance test — costs **USD 18,000**. Over the remaining installation programme and the 12-month warranty period the same cause is expected to produce **5** further occurrences at **USD 12,000** each.
2. **Formula.** Cause concentration = defects from the largest cause ÷ total. Net value of removal = expected recurrences × unit consequence − removal cost. Breakeven recurrences = removal cost ÷ unit consequence.
3. **Substitution.** `5/8`; `5 × 12,000 − 18,000`; `18,000/12,000`.
4. **Result.** Cause concentration **62.5 %**. Expected avoided cost **USD 60,000** against a removal cost of **USD 18,000** — a net **USD 42,000** and a payback ratio of **3.33 times**. Removal breaks even at **1.5**, so **2** recurrences.
5. **Interpretation.** The payback ratio is not the interesting number; the **breakeven of 1.5 recurrences** is, because it changes what the analysis must prove. Nobody needs to defend a forecast of five recurrences; the investment is justified if the cause recurs **twice**, which for a cause that has already recurred five times is an observation rather than a forecast. That is the general shape of a root-cause business case, and it is why such cases are easier to make than they are made: arguing about the expected recurrence count is arguing about the wrong number.

Three professional points. **Cause concentration is the budget allocator**: with 62.5 % on one cause, a programme treating all three equally spends most of its effort on 37.5 % of the problem, and this concentration is the normal finding once causes are classified consistently rather than described individually. **The one-off must be tested, not assumed** — a single occurrence is what the fifth firmware defect also looked like at the time, so the honest test is whether a mechanism exists that would catch a recurrence. And **the removal must be verified as effective, on a date**: an action closed in a corrective-action log is not a cause removed, and the evidence is the absence of recurrence over a stated observation window, which is why the recurrence rate is reported and not merely the closure rate.

AI in this KA

Where it earns its place. Clustering a nonconformance log by cause — the task producing the concentration figure above, that humans do badly because recurrences are weeks apart and worded differently every time, and that has a checkable answer. Extracting a structured acceptance and concession register from correspondence and minutes, flagging deliverables accepted without a named signatory or with open nonconformances unrecorded. Computing sampling plans and confidence bounds across candidate n values, including the self-consistency test. Drafting the first pass of a root-cause narrative from evidence, for a human to interrogate and own.

Where it must not go. It must not accept a deliverable, grant a concession or close a corrective action — all three are attributable decisions with a consequence-bearing owner (9.3.1). It must not conclude a root-cause analysis: a plausible causal narrative is exactly what a language model produces most readily and what is least verifiable, and a fluent wrong cause is worse than no analysis because it terminates the enquiry. And it must not estimate the recurrence count, the defective fraction or the escaped-defect cost — the three inputs on which the calculation turns and for which a model has no evidence.

Verification, concretely. For a clustered log, re-read a stated sample of records and confirm the cause assignment, because the concentration figure that will drive the budget is only as good as the classification. For any proposed cause, apply the three tests of 9.3.3 explicitly and in writing, and reject a cause failing the second. Reproduce every sampling number by hand — the bound is one power, the breakeven one division. And put the self-consistency comparison in the acceptance paper, since it is the line that tells a reviewer whether the plan can support the decision it is being used to make.

KEY TERMS — KA 9.3

Term	Meaning
Acceptance	A formal decision by a named authority that a deliverable meets its criteria and may pass on.
Acceptance by silence	Deemed acceptance through the expiry of a review period; not a decision.
Nonconformance	Output that does not meet its specified requirement.
Concession (use as is)	Acceptance of a nonconformance as it stands, authorised by the party bearing the consequence.
Regrade	Use of nonconforming output for a lesser purpose for which it conforms, with traceability.
Cumulative-effect test	The rule that related concessions aggregating above a threshold require the authority appropriate to the aggregate.
Acceptance sampling	Accepting or rejecting a population on the evidence of a sample.
Confidence bound (`p_upper`)	`1 − α^(1/n)` — the largest defective fraction consistent with a clean sample of `n`.
Breakeven defective fraction (`p*`)	`c/u` — verification cost ÷ escaped-defect cost; above it, full verification pays. Independent of `n`.
Self-consistent sampling plan	One whose confidence bound lies below its own breakeven defective fraction.
Cause concentration	The share of nonconformances attributable to the single largest root cause.
Recurrence rate	Repeat nonconformances as a share of all; the test of whether causes are being removed.

SAMPLE MCQS — KA 9.3

MCQ 9.3-A `[9.3.2 · Application]` A sample of 20 items is drawn and none is defective. At 95 % confidence, the largest defective fraction consistent with that result is closest to: - A. 0 % - B. 5 % - C. 13.9 % - D. 2.5 %

Rationale: `1 − 0.05^(1/20) = 0.1391` (9.3.2). A treats a clean sample as proof of conformance; B is the fraction a 59-item sample would bound; D halves the 5 % significance level.

MCQ 9.3-B `[9.3.2 · Analysis]` Verifying one item costs 1,400; an escaped defective item costs 12,000. Above what defective fraction does verifying the whole population beat sampling? - A. 8.57 % - B. 11.67 % - C. it depends on the sample size - D. 1.4 %

Rationale: `p* = c/u = 1,400/12,000 = 11.67 %`, and the `(N − n)` terms cancel so the breakeven is independent of `n` (9.3.2). A inverts the ratio; C is the intuition the algebra refutes.

MCQ 9.3-C `[9.3.2 · Evaluation]` A 20-item zero-defect acceptance plan has a 95 % bound of 13.91 % and a breakeven defective fraction of 11.67 %. The most serious criticism of the plan is that: - A. 20 items is too few to be statistically valid - B. its confidence bound exceeds its own breakeven, so a clean sample cannot exclude the defect rate at which full verification would have been correct - C. it should use a one-defect acceptance number instead of zero - D. sampling is inappropriate for safety-related work

Rationale: Self-consistency is the precise defect, repaired by raising `n` to 25 (9.3.2). A is an unquantified assertion; C would weaken the plan further; D is a different argument these figures do not establish.

MCQ 9.3-D `[9.3.1 · Comprehension]` A delivery team decides to accept its own nonconforming output as fit for use without correction. The defect in that process is that: - A. rework is always preferable to concession - B. the authority for a concession belongs to the party that will bear the consequence, not the producer - C. concessions are never permissible - D. the disposition should have been regrade

Rationale: A concession transfers a consequence, so its authority sits with whoever carries it (9.3.1). A and C overstate — concession is a legitimate disposition; D presumes a lesser purpose exists.

MCQ 9.3-E `[9.3.3 · Application]` Removing a root cause costs 18,000; each recurrence costs 12,000. The strongest way to present the case is that removal: - A. has a payback ratio of 3.33 times on five expected recurrences - B. breaks even at 1.5 recurrences, so needs the cause to recur only twice - C. saves 60,000 - D. reduces the cause concentration from 62.5 %

Rationale: The breakeven recurrence count is the robust argument because it does not depend on forecasting the recurrence rate (9.3.3). A and C are true but rest on the five-recurrence estimate; D describes a measure, not a value.

MCQ 9.3-F `[9.3.3 · Analysis]` Which statement identifies a root cause rather than an event description? - A. the engineer loaded the wrong parameter set - B. the device was delivered with a superseded configuration - C. there was no controlled baseline against which a parameter set could be checked, and no process step that would have detected the mismatch - D. the site acceptance test did not detect the fault

Rationale: Only C names something within a manager's authority to change whose removal prevents the whole class (9.3.3). A stops where a person can be blamed; B and D describe the occurrence and a layer's miss.

SELF-CHECK — KA 9.3

1. What does a clean sample of 20 establish? — That the defective fraction is probably below 13.91 % at 95 % confidence; neither conformance nor any tighter rate, and bounding `p` at 5 % would need 59 items.
2. State the breakeven for full verification and what it depends on. — `p* = c/u` — 11.67 % on Auriga — and it is independent of the sample size.
3. Who may authorise a concession, and what must accompany it? — The party bearing the consequence, never the producer alone; plus a register entry and a cumulative-effect test, since individually minor concessions aggregate into a different product.

KNOWLEDGE AREA 9.4

Lessons learned, continuous improvement, data quality and AI-output quality

Topics: 9.4.1 lessons that change behaviour · 9.4.2 first-time-right and the compounding of stage yields · 9.4.3 data quality as a delivery constraint · 9.4.4 AI-output quality and the verification that earns its cost.

9.4.1 Lessons that change behaviour

Lessons-learned processes are near-universal and largely ineffective, for structural rather than cultural reasons. A lesson is captured at the end of a project, by the people leaving it, into a repository consulted by nobody, as a narrative rather than an instruction. Nothing in that chain contains a mechanism by which a future decision changes.

The four properties of a lesson that works, each a test the entry passes or fails. It is **specific to a decision** — it names the decision class it should alter, not a theme. It has an **owner in the standing process**, so the change lands in a template, checklist, gate criterion, estimating parameter or contract clause rather than in a document. It carries a **number** where one exists — Auriga's expected containment cost per introduced defect (USD 2,550), its rework allowance (16.67 %), the self-consistent sample size for a zero-defect plan — because a parameter propagates and a paragraph does not. And it is **captured when it is learned**, not at closeout, since late capture loses both fidelity and the chance to apply it on the same project.

The improvement loop, and its only honest measure. Continuous improvement is the cycle of measuring a process, changing it and re-measuring — the plan–do–check–act discipline from quality management, applied to delivery processes. Its integrity rests entirely on the last step, which is the one usually skipped: an improvement implemented but never re-measured is a change, not an improvement, and organisations accumulate large numbers of them. The measure that keeps the loop honest is the **recurrence rate** of 9.3.3, because it is the only figure that cannot be satisfied by activity.

9.4.2 First-time-right and the compounding of stage yields

Definition. The **first-time-right yield** of a step is the share of items passing it without rework. The **rolled throughput yield** (RTY) of a chain of sequential steps is the product:

$$RTY = \prod y_i \qquad y_i = \text{first-time-right yield of step } i$$

Sequential steps compound multiplicatively, which is why chains of individually respectable steps produce disreputable end-to-end results, and why step-level metrics can all look healthy while delivery does not.

WORKED EXAMPLE

Worked example 9.4.2 — Auriga's handover chain: six good steps, one coin flip.

1. **Setup.** Auriga's 60 handover packages (one per plant area) pass through six sequential steps with observed first-time-right yields: design package issue **0.95**, supplier document review **0.92**, factory acceptance **0.90**, site installation **0.94**, site acceptance test **0.88**, client handover documentation **0.85**. A first-pass failure at any step triggers a rework loop costing on average **USD 2,600**.
2. **Formula.** `RTY = ∏ yᵢ`. First-pass failures at step `i` = items reaching it first time × `(1 − yᵢ)`; total rework loops = `Σ` those failures. Uniform yield required for a target `RTY` over `k` steps = `RTY^(1/k)`.
3. **Substitution.** `0.95 × 0.92 × 0.90 × 0.94 × 0.88 × 0.85`. Flow from 60: failures `3.0000, 4.5600, 5.2440, 2.8318, 5.3237, 5.8561`.
4. **Result.** `RTY` = **55.3074 %**. The arithmetic mean of the six step yields is **90.6667 %** — a figure describing nothing anyone experiences. Of 60 packages, **33.18** pass the whole chain first time and **26.82** require at least one rework loop, costing **USD 69,720**.
5. **Interpretation.** The gap between 90.67 % and 55.31 % is the point, and it is the arithmetic behind a complaint heard in every delivery organisation: every function reports good numbers and the handover is still chaos. Both figures are correct; only one is relevant, because a package must survive all six steps and the customer experiences the product, not the mean.

Where to improve is determined by the product form, not by judgement. Raising the weakest step — handover documentation, 0.85 → 0.95 — multiplies `RTY` by `0.95/0.85` to **61.8142 %**, a gain of **6.507 percentage points**, cutting rework loops from 26.82 to 22.91 and saving **USD 10,151** on this chain alone. Raising the strongest step, 0.95 → 0.98, gives **57.0540 %** — a gain of **1.747 points**, or **3.73 times less**, for effort that is usually harder because the step is already good. The binding step is identifiable from the yields with no judgement at all.

The target arithmetic should be said out loud to sponsors. An end-to-end first-time-right target of **80 %** over six steps requires **96.35 %** at every single step (`0.80^(1/6)`); a 90 % target requires **98.26 %**. That is why "we are about 90 % right" organisations fail at handover, and why a target set without this calculation is an instruction nobody can execute.

Three cautions. The failure counts are **first-pass failures**, counted on the flow reaching each step first time; a package failing twice contributes two loops in reality, and the model states its convention rather than hiding it. `RTY` assumes steps are **sequential and each must be passed** — parallel or optional steps need a different structure, and a chain drawn wrongly gives a confidently wrong number. And a step yield is meaningful only where the step has a **defined pass criterion**: yields measured against an undefined standard record the reviewer's mood, so Domain 5, KA 5.4.2's testability requirement is the prerequisite for this whole measure.

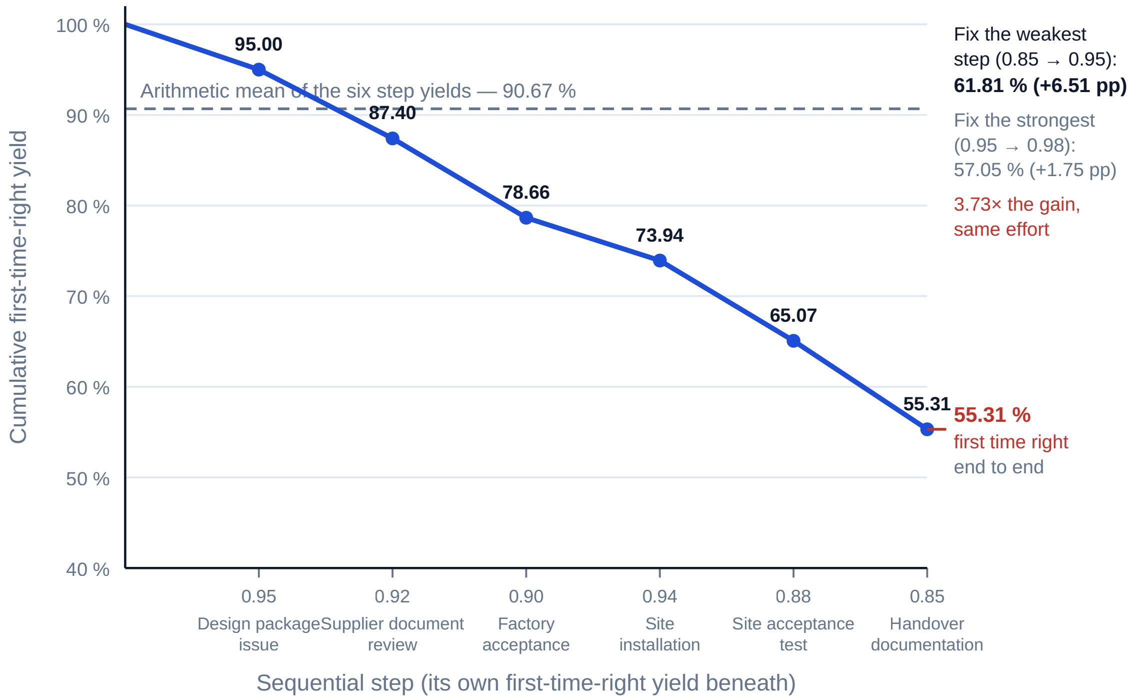


Fig 9.4.1 — Rolled throughput yield: six good steps, one coin flip

9.4.3 Data quality as a delivery constraint

A project's decisions are made on its data — the asset register, the requirements baseline, the cost ledger, the risk register, progress measurement — and data with defects produces decisions with defects, at a scale and speed no deliverable inspection catches. Data quality is also what determines whether the analytics, dashboards and AI systems of Domain 14 are usable at all, and the standard sequencing error in digital delivery is to buy the analytics before fixing the data.

The dimensions, each measurable as a pass rate over records: **completeness** (required fields populated), **accuracy** (values correspond to reality), **validity** (values conform to defined format, type and range), **consistency** (the same fact agrees across records and systems), **timeliness** (current enough for the decision), **uniqueness** (no unintended duplicates). International reference points may be cited by name — the ISO 8000 series on data quality and the ISO/IEC 25012 data quality model — and both are frameworks for defining and assessing dimensions rather than sources of a target number; the target is a project decision.

WORKED EXAMPLE

Worked example 9.4.3 — Auriga's asset register, and why 96 % is 78 %.

1. **Setup.** Auriga must hand over an asset register of **3,200** equipment records to the utility's maintenance system. Sampling gives per-dimension conformance of completeness **0.960**, accuracy **0.930**, validity **0.980**, consistency **0.910**, timeliness **0.990**, uniqueness **0.995**. A record is fit for use only if it satisfies **all six**.
2. **Formula.** Composite fitness = ∏ dimension conformance rates (treating dimensions as independent). Conforming records = composite × record count.
3. **Substitution.** `0.960 × 0.930 × 0.980 × 0.910 × 0.990 × 0.995`.
4. **Result.** Composite fitness **78.43 %** against an arithmetic mean of the six dimensions of **96.08 %**. Of 3,200 records, approximately **2,510** are fit for use and **690** are not.
5. **Interpretation.** This is 9.4.2's compounding applied to data, producing the same surprise for the same reason: a dashboard reporting six dimensions all above 91 % is reporting a register in which nearly a quarter of records fail somewhere. The remediation logic is also the same — attack the weakest dimension. Raising **consistency** from 0.910 to 0.980 lifts the composite to **84.46 %**, a gain of **6.03 points** and roughly **193** more usable records, for work that is usually reconciliation rather than re-survey. Three professional points. Independence across dimensions is an **assumption and usually optimistic**: records that are incomplete are often also inconsistent, so for correlated defects the true composite tends to be better than the naive product, and the figure should be sample-verified end to end rather than only dimension by dimension. The composite must be computed against **fitness for a named decision** — a register good enough for statutory reporting may be inadequate for condition-based maintenance, and a single "data quality percentage" divorced from a use is a number without a meaning. And data quality is an **acceptance criterion**, not a wish: the 690 non-conforming records are a nonconformance under 9.3.1, with a disposition, an owner and a date, or they will be discovered by the receiving organisation after Domain 16's transition, when the project team has dispersed.

9.4.4 AI-output quality and the verification that earns its cost

When an AI system produces work a professional must sign for — drafted commissioning procedures, generated test cases, extracted requirements, a summarised evidence pack — the professional's obligation is unchanged: they are accountable for the output (Domain 1; AI proposes; the professional verifies, decides and remains accountable). The practical question is therefore not whether to verify but **how much**, and that is a sampling question with an answer, which is where 9.3.2's arithmetic becomes the most useful tool in this domain.

Two properties of AI output make the answer differ from a supplier's. Errors are **systematic rather than random** — a model that misunderstands an instruction misunderstands it consistently, so defects cluster by type and stratified sampling across item types is essential. And output is **fluent regardless of correctness**, which removes the informal signal reviewers rely on with human work: an unsure human writes hesitantly, and a wrong model does not.

WORKED EXAMPLE

Worked example 9.4.4 — verifying 240 AI-drafted commissioning steps.

1. **Setup.** Auriga's commissioning team uses an AI assistant to draft **240** commissioning steps from the design documents. A reviewer checks a random **20** and finds no error. Reviewing one step costs **USD 40** of engineer time; an erroneous step reaching execution costs **USD 1,800** (an aborted test, a re-scheduled outage slot, a re-issued procedure). What verification is defensible?
2. **Formula.** As 9.3.2: 95 % bound `1 − 0.05^(1/n)`; breakeven error fraction `p* = c/u`; sample to bound the error rate at a target `p`: `n = ln(0.05)/ln(1 − p)`.
3. **Substitution.** `1 − 0.05^(1/20)`; `40/1,800`; `ln(0.05)/ln(0.98)`.
4. **Result.** The clean sample of 20 bounds the error rate at **13.91 %** — up to **33.39** of the 240 steps, i.e. as many as 33 erroneous steps. The breakeven error fraction is `40/1,800 =` **2.22 %**. Bounding the error rate at **2 %** would require **149** items — **62.1 %** of the population, at a review cost of **USD 5,960** against **USD 9,600** to review all 240. If the error rate really were at the bound, the 220 unreviewed steps would carry an expected consequence of `0.1391 × 220 × 1,800 =` **USD 55,087**.
5. **Interpretation.** The conclusion is uncomfortable and it is arithmetic, not conservatism. **Because the breakeven error fraction (2.22 %) is far below what a small sample can bound (13.91 %), and because the sample needed to get below the breakeven is 62 % of the population, the defensible regime here is full review** — USD 9,600 against a potential exposure of USD 55,087 on the unreviewed remainder. And that is precisely where the claimed productivity gain goes: **the saving from generating a deliverable is real only to the extent that the verification it requires is cheaper than producing it correctly in the first place.** A leader who accepts a time saving on generation without pricing the verification has not made a saving; they have moved a cost somewhere nobody is measuring it, and accepted an accountability whose evidence base is a sample of 20.

Two refinements make this practical rather than merely discouraging. The ratio `c/u` is a **design variable**: reducing `u` — by placing an independent, cheap containment layer after the AI-drafted step and before execution, as 9.2.2 would — raises the breakeven error fraction and can make sampling defensible where full review was not. The economics reward **building a containment chain around AI output**, not reviewing harder. And the argument is **per item class, not per tool**: a low-consequence output may be sampled lightly and legitimately, so a verification policy applying one sample size to everything is wrong in both directions at once. The governance instrument is a **verification standard proportional to consequence** — Domain 3, KA 3.A.2's requirement with the arithmetic attached — and Domain 14 develops the wider frame, including the standards that may be cited by name (ISO/IEC 42001 for AI management systems, ISO/IEC 23894 for AI risk management).

AI in this KA

Where it earns its place. Reading a large lessons repository and clustering entries into recurring themes with counts, which is what makes a repository usable. Computing rolled throughput yields and their sensitivity to each step, identifying the binding step and the uniform yield a target implies. Profiling a data set across the six dimensions of 9.4.3 — a strong application, because dimension profiling is mechanical, high-volume and exactly checkable — and proposing the remediation rules that would fix the largest number of records. Generating candidate improvement actions from a nonconformance pattern, for human triage.

Where it must not go. It must not decide which lessons enter the standing process — that is a change to how the organisation works and needs an owner who can be held to it. It must not be the sole judge of its own output quality: a model's self-assessed confidence is not a detection rate, and a second pass by the same model is not an independent layer (9.2.2). It must not remediate data silently — an inferred value that fills a completeness gap improves the metric and can degrade the register, the worst combination available. And it must not set the verification sample size for its own output, because that is the decision whose independence the whole argument of 9.4.4 rests on.

Verification, concretely. Characterise every AI containment layer with a **measured** detection rate against a held-back defect set, and re-measure on any change of model, prompt or artefact type. Compute the verification sample from the consequence ratio `c/u` and record that computation in the quality plan, so the sample size has a stated basis rather than being a habit. Where data has been machine-remediated, hold an untouched control sample and compare, since a plausible wrong fill is detectable only against ground truth. And reproduce the yields, composites and bounds by hand — every calculation in this KA is a product, a power or a division.

KEY TERMS — KA 9.4

Term	Meaning
First-time-right yield (`yᵢ`)	The share of items passing a step without rework.
Rolled throughput yield (`RTY`)	`∏ yᵢ` — the share of items passing every sequential step first time.
Binding step	The step whose yield improvement raises `RTY` most; identifiable from the yields alone.
Uniform yield requirement	`RTY^(1/k)` — the per-step yield an end-to-end target implies over `k` steps.
Data quality dimensions	Completeness, accuracy, validity, consistency, timeliness, uniqueness.
Composite fitness	The product of dimension conformance rates; the share of records fit for use.
Recurrence rate	Repeat nonconformances as a share of all; the honest measure of an improvement loop.
Verification standard proportional to consequence	A rule setting review depth from the cost of an escaped error, not from convenience.
Systematic error	An error repeating consistently across similar items — the characteristic shape of AI error, requiring stratified sampling.

SAMPLE MCQS — KA 9.4

MCQ 9.4-A `[9.4.2 · Application]` Six sequential steps have first-time-right yields 0.95, 0.92, 0.90, 0.94, 0.88 and 0.85. The rolled throughput yield is closest to: - A. 90.7 % - B. 55.3 % - C. 44.7 % - D. 85.0 %

Rationale: `RTY` is the product, 0.55307 (9.4.2). A is the arithmetic mean of the six yields, which describes nothing anyone experiences; C is the complement of the product; D is the worst single step.

MCQ 9.4-B `[9.4.2 · Evaluation]` On that chain, improvement effort is best directed at the step with yield 0.85 rather than the one with 0.95 because: - A. the 0.85 step is later in the chain - B. raising 0.85 to 0.95 multiplies `RTY` by 0.95/0.85 and gains 6.51 points, against 1.75 points for raising 0.95 to 0.98 - C. late steps are always cheaper to improve - D. the 0.95 step is already compliant

Rationale: The multiplicative form makes the gain proportional to the ratio of new to old yield, so the weakest step gives the largest lift — here 3.73 times as much (9.4.2). A confuses position with leverage; C is unsupported.

MCQ 9.4-C `[9.4.2 · Analysis]` An end-to-end first-time-right target of 80 % across six sequential steps requires a per-step yield of about: - A. 80.0 % - B. 96.4 % - C. 93.3 % - D. 88.9 %

Rationale: `0.80^(1/6) = 0.9635` (9.4.2). A applies the end-to-end figure per step; C divides 80 % by 6 and adds it back; D treats the losses as additive.

MCQ 9.4-D `[9.4.3 · Application]` Six data quality dimensions score 0.960, 0.930, 0.980, 0.910, 0.990 and 0.995, and a record is fit for use only if it satisfies all six. Composite fitness is closest to: - A. 96.1 % - B. 78.4 % - C. 91.0 % - D. 21.6 %

Rationale: The product is 0.7843 (9.4.3). A is the arithmetic mean; C is the weakest dimension; D is the complement of the product.

MCQ 9.4-E `[9.4.4 · Evaluation]` Reviewing one AI-drafted item costs 40; an escaped erroneous item costs 1,800; a clean sample of 20 has been taken from 240 items. The defensible conclusion is: - A. the sample is clean, so the output may be accepted - B. the breakeven error fraction is 2.22 % while a clean sample of 20 bounds the rate only at 13.91 %, and reaching the breakeven needs 149 items — so full review at 9,600 is indicated - C. the sample should be increased to 59 items, bounding the rate at 5 % - D. the output should not be used at all

Rationale: The bound must lie below the breakeven for the plan to support its own decision (9.4.4, applying 9.3.2). A accepts on a sample that cannot exclude 33 erroneous items; C bounds at 5 %, still more than double the breakeven; D is not supported — the arithmetic supports full review, not abandonment.

MCQ 9.4-F `[9.4.4 · Analysis]` Which measure most raises the breakeven error fraction and so makes sampling AI output defensible? - A. increasing the sample size - B. adding an independent, cheap containment layer after the AI step and before execution, which lowers the cost of an escaped error - C. asking the model to check its own output - D. improving the prompt

Rationale: The breakeven is `c/u`, so lowering `u` raises it (9.4.4 with 9.2.2). A changes the bound, not the breakeven; C is not an independent layer; D may lower the error rate but does not change the economics of verification.

SELF-CHECK — KA 9.4

1. Why does a chain of six steps averaging 90.67 % deliver 55.31 %? — Because sequential yields compound multiplicatively; the mean describes no item's experience, and every item must pass all six.
2. What per-step yield does an 80 % end-to-end target require over six steps? — 96.35 % (`0.80^(1/6)`), which is why targets set without this calculation cannot be executed.
3. How is the verification sample for an AI-produced deliverable set? — From the consequence ratio: the sample must bound the error rate below `p* = c/u`; on Auriga's commissioning steps that is 2.22 %, needing 149 of 240 items, so full review is the defensible regime.

Advanced topics — Domain 9

9.A.1 Quality across a contractual boundary

Where a supplier produces the work, every mechanism in this domain must survive a commercial boundary, and each acquires a failure mode a leader should assume present until disproved.

Containment layers become contractual artefacts. A layer exists only if the contract obliges the supplier to run it, entitles the client to witness or audit it, and defines what its evidence must show. Absent those, the client's chain is one layer shorter than the plan says and its escape fraction is worse by a factor of $1/(1 - d_i)$ for the layer that turns out not to be real. **Detection rates become commercial information**: a supplier has no incentive to disclose a weak internal detection rate, so the client's chain arithmetic runs on the supplier's optimism unless the client's own layer is independently staffed. **Nonconformance dispositions become negotiations** — a concession the client's design authority should grant is routinely proposed by the supplier alongside a schedule benefit, and 9.3.1's rule about who bears the consequence has to hold against commercial pressure. And **cost of quality relocates without falling**: a fixed-price contract moves prevention and appraisal into the supplier's cost base, where the client cannot see them and can only infer them from escapes — and the supplier's optimum, computed on their external-failure unit cost (a warranty obligation) rather than the client's (an operating consequence), is systematically lower. That divergence is the most important quality fact about fixed-price delivery, and the countermeasures are contractual: specified containment layers with witness rights, defect-liability terms that move u closer to the client's true figure, and acceptance criteria and sampling plans agreed before award rather than argued at delivery. Domain 10 handles the contracting machinery; the quality point is that a contract not specifying the containment chain has not specified the quality.

9.A.2 Governing quality in adaptive and hybrid delivery

Iterative delivery changes where quality is decided, not whether it is. **The definition of done replaces the inspection**: quality is built into the step rather than inspected at the end, so the definition of done is the containment layer and must be stated with the specificity of an acceptance criterion and measured with the discipline of a detection rate, or it is a slogan. **Technical debt is deferred internal failure, and it compounds**: work knowingly left non-conforming raises the introduced-defect count of every subsequent increment, so the honest treatment is a quantified backlog item that 9.2.2's arithmetic can see, not a cultural complaint. **Rework capacity is the flow constraint**: KA 9.2.3's multiplier $1/(1 - r)$ is exactly how velocity degrades — a 30 % rework share is a 1.4286-times duration multiplier on everything, presenting as a team that has become slower rather than as a quality problem — and the measure that surfaces it is the rework share of capacity per iteration, reported alongside throughput.

The hybrid case, which is most real programmes, adds one requirement: parts of the work under different methods have **different containment chains and different escape fractions**, and the escape fraction of a deliverable is the product across the chain it actually traverses, not an average of the two regimes. A component built iteratively with a strong definition of done and then installed under a sequential commissioning regime inherits both chains; a leader reasoning about the average will under-estimate escapes on the path that matters.

9.A.3 The reviewer's quality eye

Invariants to test on any quality regime, each cheap and each diagnostic.

The **four cost-of-quality categories are defined and counted consistently**, with internal failure not quietly reclassified as appraisal. There is a **computed total cost of quality** with a stated percentage of budget and a **marginal analysis**, not merely a chosen regime. The **external-failure unit cost is stated with its source**, and the recommendation carries a **range of optimality** rather than a point. The **escape fraction is computed as a product** of layer detection rates, and consecutive layers are **methodologically different** rather than repetitions. Every layer has a **measured** detection rate, a coverage statement and a unit correction cost — including automated and AI layers, which are characterised, not trusted. There is an **expected containment cost per introduced defect**, and every prevention proposal is tested against it. The plan states a **rework allowance** as a percentage of capacity, actual rework share is measured against it weekly, and the **schedule** consequence is priced at the cost of delay on any zero-float activity. Every acceptance is a **positive record with a named signatory** and a list of open nonconformances. Every concession is authorised by the **party bearing the consequence** and subject to a **cumulative-effect test**. Every sampling plan is **self-consistent** — its confidence bound below its own breakeven `c/u`. Root causes pass the **three tests**, and the **recurrence rate** is reported rather than the closure rate. `RTY` is computed as a product, improvement is directed at the **binding step**, and any end-to-end target carries the **uniform per-step yield** it implies. Data quality is expressed as a **composite against a named use**, not an average of dimensions. And AI-produced work carries a **verification sample derived from** `c/u`, recorded in the quality plan — the test that catches the commonest new defect in modern delivery, which is an accountability accepted on the evidence of a sample of twenty.

Industry variations — Domain 9

- **Utilities, energy and process industries.** External failure is dominated by outage windows and availability obligations, so `u` is high, `p* = c/u` is low, and full verification is far more often the economic answer than elsewhere — which is why Auriga's regime sits at the strict end. Commissioning is the concentrated rework risk and is on the critical path almost by construction.
- **Regulated life sciences and medical devices.** Quality is a licence condition, not an optimisation: certain containment layers and record requirements are mandatory whatever the arithmetic says, and the cost-of-quality model runs inside that constraint. The professional error is presenting an economic optimum that breaches a regulatory requirement; the correct output is the optimum among compliant options.
- **Software and digital services.** Correction is cheap and deployment reversible, so `u` is low and rapid detection with fast rollback frequently beats heavy prevention. The characteristic failure is applying that logic to the irreversible parts — data migrations, external interfaces, published records — where a rollback does not undo the consequence.
- **Construction and infrastructure.** Physical rework is expensive and often infeasible, so prevention and early appraisal dominate and the correction-cost ladder is steeper than Auriga's. Concession management is central because a physical nonconformance frequently cannot be reworked at any acceptable cost, which puts real pressure on 9.3.1's rule about who may grant one.
- **Aerospace, rail and safety-critical transport.** Escapes carry consequences not expressible as an average cost, so 9.1.2's expected-value logic is bounded by a safety case and traceability obligations; sampling plans are typically fixed by an approval regime rather than derived, and the leader's task is to know the arithmetic well enough to see when a mandated plan is weaker than the economics would require.

- **Public services and administrative programmes.** Fitness for purpose dominates conformance, because the deliverable is a service experienced by citizens who did not write the specification; and data quality (9.4.3) is frequently the binding constraint on the whole benefit, since the records were assembled for a different purpose over decades.

Case study — Domain 9: the fortnight in commissioning (utilities, Auriga)

Situation. Auriga entered testing and commissioning — activity F, 4 weeks, 6 engineers, on the critical path with zero float — after the week-13 review that had already shown `CPI` 0.91 and `SPI` 0.92 (Domain 7). Two weeks into F, the commissioning manager reported that the team was "working through a lot of retests" and asked for a decision by the end of the week. The programme had no rework metric, so nobody could say how much.

What the arithmetic showed. The project leader had the time coding re-cut and computed four numbers. Rework was consuming **30 %** of commissioning capacity against a plan that assumed **16.67 %** — 4 engineer-weeks of allowance in a 24-engineer-week phase. At 30 %, `20 ÷ (6 × 0.70)` gives **4.7619 weeks**, an overrun of **0.7619 weeks** worth **USD 34,286** at Auriga's cost of delay of 45,000 per week, plus **4.5714 engineer-weeks** of unbudgeted labour at USD 5,225 — **USD 23,886**. Total consequence: **USD 58,171**, of which 59 % was schedule and therefore invisible in every report the programme produced. The trend mattered more than the level: at 40 % the phase would run **5.5556 weeks** and the delay cost **USD 70,000** — a further **USD 35,714** for ten points, against **USD 20,833** for the ten points between 10 % and 20 %.

The proposal, and why it was rejected. The commissioning manager recommended halving site acceptance test coverage to recover the schedule. Priced against 9.2.2's chain, dropping L3's detection rate from 0.50 to 0.25 raises escapes from **8** to **12**, adding `4 × 12,000 =` **USD 48,000** of external failure against `4 × 3,500 =` USD 14,000 of internal correction saved — a net loss of **USD 34,000** before any schedule benefit. The recovery on offer could not close that gap, and the leader was able to say so in one sentence with a number attached, which is the difference between a governance conversation and an argument about commitment to quality.

What was done instead. Two decisions. **Buy capacity.** With 11.6 engineer-weeks of first-pass content remaining, adding two engineers on an off-shift basis takes the remaining duration from 2.7619 to **2.0714 weeks** — a total phase duration of **4.0714 weeks** and an overrun of only **0.0714 weeks**, or **USD 3,214** of delay instead of USD 34,286, a saving of **USD 31,071** for 4.1429 engineer-weeks of extra capacity costing USD 21,646 plus a 25 % off-shift premium of USD 5,412, or **USD 27,058** in total. Net **USD 4,013** — thin, worth doing, and worth recognising as thin. **Remove the cause.** Root-cause analysis showed **5 of 8** escapes — a cause concentration of **62.5 %** — traced to an uncontrolled supplier firmware configuration baseline. Removal cost **USD 18,000** against **5** expected recurrences at 12,000 — **USD 60,000** avoided, a net **USD 42,000**, breaking even at **1.5** recurrences.

What the domain teaches here. The most valuable calculation was the counterfactual. Had Auriga taken option (b) of 9.2.2b — USD 40,000 of prevention at the end of design, cutting introduced defects from 80 to 60 — site acceptance would have found 6 defects rather than 8, removing **1.5 engineer-weeks** of rework, shortening F to **4.5119 weeks** and saving **USD 11,250** of delay on top of the USD 11,000 of cost-of-quality saving the model had already shown. The prevention option was worth **USD 22,250**, not USD 11,000, and the missing half was schedule. **The cost-of-quality model systematically understates prevention, because it contains no clock.** The practical consequence is that any prevention proposal touching a critical-path activity should be priced twice — once in the CoQ model and once at the cost of delay. Note also which decisions were still available at week 15: buying capacity at a net USD 4,013, and removing a cause for the next occurrences. The decision worth USD 22,250 had expired at the end of design, and nobody had been asked to make it.

Case study B — Domain 9: the clean sample (rail signalling, transport)

Situation. A metropolitan signalling renewal programme accepted a supplier's batch of **480** axle-counter installations on the evidence of a **30-item** zero-defect sample. Verifying one installation cost **USD 900** — a possession slot, two technicians and a test set. A defective installation reaching service cost the programme an estimated **USD 21,000**: an emergency possession, retest, re-commissioning and the operator's performance regime. The sample was drawn at random, was clean, and the batch was accepted. Over the following eleven months, **21** of the 450 unsampled installations were found defective in service — an observed defective fraction of **4.667 %** — at a realised cost of **USD 441,000**.

What the arithmetic showed afterwards. The plan had been unsound before it was run, and the test that would have shown it takes two lines. The breakeven defective fraction was `p* = 900/21,000 =` **4.286 %**. The 95 % upper confidence bound after a clean sample of 30 is **9.50 %** — more than twice the breakeven. The plan could not exclude a defect rate at which verifying all 480 installations would have been correct; its bound exceeded its own breakeven by **5.22 percentage points**. The self-consistent sample size — the smallest `n` whose 95 % bound falls below 4.286 % — is **69** (the bound is 4.249 % at 69 and 4.310 % at 68), costing `69 × 900 =` **USD 62,100** against the **USD 27,000** actually spent: an extra **USD 35,100**. At the defect rate that in fact obtained, a 30-item plan passes the batch **23.84 %** of the time and a 69-item plan **3.70 %**, so the extra 39 items would have converted a one-in-four chance of missing the problem into a one-in-twenty-seven chance — an expected saving of `(0.2384 − 0.0370) × 441,000 =` **USD 88,838** for USD 35,100 of additional testing, a net **USD 53,738**.

How it resolved. The realised escape cost of USD 441,000 slightly exceeded the **USD 432,000** that verifying all 480 installations would have cost — a coincidence worth noticing, because it means even the most conservative option would have been only marginally justified after the event and was genuinely hard to justify before it. That is why the correction was not "inspect everything". Three changes went into the framework contract for the following batches. Acceptance plans acquired a **self-consistency test**: the plan's 95 % bound must lie below the breakeven `c/u`, with both figures stated in the acceptance paper. The escaped-defect unit cost was **defined contractually**, including the operator's performance regime, so `p*` could be computed rather than argued. And the supplier's own containment layers were **specified with witness rights**, on the reasoning of 9.A.1 — the client's escape fraction had been computed on the assumption that the supplier ran an internal check the contract had never required.

What the domain teaches here. A passing sample is not evidence of conformance; it is a bound, and the bound has to be compared with something. The something is `p* = c/u`, and the comparison is the whole of acceptance-sampling judgement: **a plan whose confidence bound exceeds its own breakeven defective fraction cannot support the decision it is being used to make, however cleanly it passes.** The remedy was cheap, available before any test was run, and needed no statistical sophistication beyond one power and one division.

Executive perspective — Domain 9

What a programme director cannot delegate in this domain:

- **The external-failure unit cost, and therefore the quality regime.** `u` sets where the cost-of-quality minimum sits and what `p* = c/u` is; it is a judgement about consequence in an operating context, and every quality decision below it inherits it (9.1.2, 9.3.2).
- **The decision that zero defects is not the objective — and the version that is.** "No defect whose prevention costs less than its consequence" is computable; a commitment to zero defects buys the last escapes at, on Auriga's figures, 1.79 times their harm (9.1.2).
- **The rework share of capacity as a schedule metric.** Require it weekly against a stated allowance on every critical-path activity, priced at the cost of delay. Its absence is why commissioning overruns arrive as surprises (9.2.3).
- **Who may grant a concession, and whether concessions are counted.** Never the producer alone, never without a cumulative-effect test; this is Domain 3's Case study B at deliverable level and it is how a delivered product ends up materially different from the specified one (9.3.1).
- **The self-consistency of every acceptance plan you rely on.** Two numbers — the 95 % bound and `c/u` — and the plan either supports its decision or does not. Ask for both in the acceptance paper (9.3.2, Case study B).
- **The verification standard for AI-produced work, derived rather than habitual.** The sample must bound the error rate below `c/u`; where it cannot, the productivity gain was never real and the accountability has been accepted on the evidence of a sample of twenty (9.4.4).

Calculation exercises — Domain 9

> **Exercise 9.1** A programme with a budget of USD 6,000,000 can adopt one of four quality regimes. An escaped defect costs **USD 9,000**.

Regime	Prevention	Appraisal	Introduced	Detected	Internal unit cost
Q1	15,000	45,000	150	80 %	1,000
Q2	55,000	65,000	110	90 %	1,200
Q3	120,000	85,000	80	95 %	1,400
Q4	220,000	110,000	50	96 %	1,500

Compute the total cost of quality for each, identify the optimum, and state the external-failure unit cost at which the optimum would move one regime stricter. Solution. Escapes 30, 11, 4, 2. Internal failure `120 × 1,000 = 120,000`; `99 × 1,200 = 118,800`; `76 × 1,400 = 106,400`; `48 × 1,500 = 72,000`. External failure 270,000; 99,000; 36,000; 18,000. Totals **USD 450,000** (7.50 % of budget), **USD 337,800** (5.63 %), **USD 347,400** (5.79 %), **USD 420,000** (7.00 %). The optimum is **Q2**. The Q2 → Q3 step costs `205,000 − 120,000 =` 85,000 more in conformance and returns 12,400 of internal failure and 7 escapes, so it pays once `u ≥ (85,000 − 12,400)/7 =` **USD 10,371.43**. The assumed 9,000 sits only **13.22 %** below that flip point, so the recommendation is **fragile** and the paper must say so. (For completeness, Q1 → Q2 pays above **USD 3,094.74** and Q3 → Q4 above **USD 45,300**.) Common error: choosing the regime with the fewest escapes, or reporting the minimum without the range of optimality — which converts a fragile recommendation into an apparently settled one.

Exercise 9.2 A process introduces **150** defects. Three containment layers have detection rates **0.40**, **0.50** and **0.60**, with correction costs of **USD 700**, **USD 2,200** and **USD 5,000**. An escaped defect costs **USD 16,000**. Compute the escape fraction, the escapes, the internal and external failure costs, and the expected containment cost per introduced defect. Then evaluate a prevention programme costing **USD 95,000** that would avoid **30** defects. Solution. Escape fraction $0.60 \times 0.50 \times 0.40 =$ **0.120** (detection 88.0 %). Flow: L1 finds $150 \times 0.40 = 60$ (90 remain), L2 finds $90 \times 0.50 = 45$ (45 remain), L3 finds $45 \times 0.60 = 27$, escapes **18**. Internal failure $60 \times 700 + 45 \times 2{,}200 + 27 \times 5{,}000 = 42{,}000 + 99{,}000 + 135{,}000 =$ **USD 276,000**. External failure $18 \times 16{,}000 =$ **USD 288,000**. Total nonconformance **USD 564,000**, so the expected containment cost per introduced defect is $564{,}000/150 =$ **USD 3,760**. The prevention programme returns $30 \times 3{,}760 =$ **USD 112,800** for USD 95,000 — a net **USD 17,800**; it breaks even at $95{,}000/3{,}760 =$ **25.27, i.e. 26 defects avoided**. Common error: computing the escape fraction as $1 -$ the sum of the detection rates, or applying each detection rate to the original 150 rather than to the defects reaching that layer.

Exercise 9.3 A test phase is planned at **6 weeks** with **8** engineers (48 engineer-weeks of capacity) against **40** engineer-weeks of first-pass work content. Cost of delay is **USD 30,000** per week and an engineer-week costs **USD 5,000**. Compute the planned rework allowance as a share of capacity; then the duration, overrun, delay cost and unbudgeted labour if rework runs at **35 %**; and the rework share at which the phase would take twice its planned duration. Solution. Allowance $48 - 40 = 8$ engineer-weeks = **16.67 %** of capacity (equivalently $1 - 40/48$). At $r = 0.35$: duration $40 \div (8 \times 0.65) =$ **7.6923 weeks**, overrun **1.6923 weeks**, delay cost **USD 50,769**. Rework consumed $0.35 \times 8 \times 7.6923 =$ **21.5385** engineer-weeks, **13.5385** above allowance, costing **USD 67,692**. Total consequence **USD 118,462**. For a 12-week duration, $40 \div (8(1 - r)) = 12$ gives $1 - r = 0.41667$, so $r =$ **58.33 %**. Common error: adding 35 % to the 6-week plan (giving 8.1 weeks) instead of dividing by $(1 - r)$, and forgetting that the plan already contained a 16.67 % allowance — so the variance is 18.33 points, not 35.

Exercise 9.4 A population of **300** welds is to be accepted. Verifying one weld costs **USD 260**; a defective weld reaching service costs **USD 6,500**. The proposed plan verifies **30** welds and accepts on zero defects. Compute (a) the probability the plan passes a population that is 4 % defective, (b) the 95 % upper confidence bound after a clean sample of 30, (c) the breakeven defective fraction, and (d) the smallest self-consistent sample size and its cost. Solution. (a) $0.96^{30} =$ **29.39 %**. (b) $1 - 0.05^{(1/30)} =$ **9.50 %** — on 300 welds, up to **28.51** defective. (c) $p^* = c/u = 260/6{,}500 =$ **4.00 %**. (d) $n = \ln(0.05)/\ln(0.96) = 73.385$, so **74** (the bound is 3.967 % at 74 and 4.021 % at 73), costing $74 \times 260 =$ **USD 19,240** against USD 7,800 for the 30-weld plan and **USD 78,000** for verifying all 300 — the self-consistent plan costs only **24.7 %** of full verification. Note that the proposed plan's bound (9.50 %) is more than twice its breakeven (4.00 %): it cannot support the decision it is being used to make. Common error: treating a clean sample as evidence of conformance, or comparing the bound with the target defect rate instead of with $p^* = c/u$ — the target is an aspiration, the breakeven is the economics.

Exercise 9.5 A document chain has five sequential steps with first-time-right yields **0.98**, **0.94**, **0.91**, **0.96** and **0.89**. **200** items enter and each first-pass failure costs **USD 1,900**. Compute the rolled throughput yield, the number of first-pass failures, the gain from raising the weakest step to 0.96 against raising the strongest to 0.99, and the uniform per-step yield a 90 % end-to-end target would require. Solution. $RTY = 0.98 \times 0.94 \times 0.91 \times 0.96 \times 0.89 =$ **71.6237 %** (the arithmetic mean of the five yields is 93.60 %, which describes nothing). First-pass failures, computed on the flow reaching each step: $4.0000 + 11.7600 + 16.5816 + 6.7063 + 17.7047 =$ **56.7527**, costing **USD**

> **107,830**; **143.2473** items pass clean. Raising the weakest step 0.89 → 0.96 multiplies `RTY` by `0.96/0.89` to **77.2570 %**, a gain of **5.633 points**, cutting failures to **45.4860** and saving **USD 21,407**. Raising the strongest 0.98 → 0.99 gives **72.3545 %**, a gain of **0.731 points — 7.71 times less**. A 90 % end-to-end target requires `0.90^(1/5) =` **97.91 %** at every step. Common error: averaging the yields rather than multiplying them, and — more damaging in practice — directing improvement at the step that is easiest to change rather than at the binding one, which here costs a factor of 7.71 in return.

Practitioner's toolkit — Domain 9

Adoption-ready artefacts; adapt headings to your organisation, then keep them stable.

Toolkit 9.T.1 — Cost-of-quality and containment sheet

One page per deliverable class, completed at planning and revisited at each gate. **Top block, the four categories:** prevention, appraisal, internal failure and external failure, each with a budget, an actual, and a definition of what is counted in it — the definitions matter more than the numbers, because inconsistent classification is the commonest way the model is made to lie. **Middle block, the containment chain:** one row per layer with its coverage, its measured detection rate and its source, its unit correction cost, and a note of whether the next layer is methodologically different; a computed escape fraction $\prod(1 - d_i)$; escapes; and the **expected containment cost per introduced defect**, which is what every prevention proposal is then tested against. **Bottom block, the decision:** total cost of quality with its percentage of budget, the marginal step analysis for one regime stricter and one looser, and the **range of optimality** on the external-failure unit cost. A sheet whose bottom block is blank has recorded a quality regime rather than chosen one.

Toolkit 9.T.2 — Acceptance and disposition record with the self-consistency test

Per deliverable: reference and version · acceptance criteria referenced to the requirements baseline · the verification evidence · **the sampling plan, if any, with `n`, the acceptance number, the 95 % bound `1 − 0.05^(1/n)`, the verification unit cost `c`, the escaped-defect unit cost `u`, the breakeven `p* = c/u`, and a pass/fail on the test that the bound lies below the breakeven** · the acceptance decision with a **named signatory and role** · the open nonconformances at acceptance, each with its disposition (rework / repair / concession / regrade / reject), the authority that granted it and the party bearing the consequence · and the running **cumulative concession total** against its threshold. Monthly integrity counts, each a number: deliverables accepted without a named signatory; concessions granted by the producer; sampling plans failing the self-consistency test; and cumulative concession value against threshold. Those four counts will find a quality function's defects before an auditor does.

Toolkit 9.T.3 — Improvement register: rework, yield and cause

Three linked sections, all numeric, reported at the same cadence as cost and schedule. **Rework and capacity:** per activity, the planned rework allowance as a share of capacity, the measured share this period, the implied duration multiplier `1/(1 − r)`, and — for any activity with zero or minimal float — the overrun priced at the cost of delay. **Yield:** per process chain, each step's first-time-right yield, the rolled throughput yield as their product, the identified binding step, and the uniform per-step yield the current end-to-end target implies. **Cause:** per nonconformance, the root cause tested against the three tests (prevents this occurrence, prevents the class, within someone's authority to remove), the removal cost, the breakeven recurrence count `removal cost ÷ unit consequence`, the owner in the **standing process** the change lands in (template, checklist, gate criterion, estimating parameter, contract clause) and the date effective; plus the two portfolio measures — **cause concentration** and **recurrence rate** — which together answer the only question that matters about an improvement programme: are causes being removed, or described?

Exam preparation — Domain 9

What is assessed. Quality, grade and fitness for purpose as three distinct tests; the four cost-of-quality categories and their opposing behaviour, including the non-monotonicity of internal failure; computing total cost of quality across regimes, locating the interior minimum and stating the range of optimality; why zero defects is not the objective; assurance versus control; containment layers, the multiplicative escape fraction and the independence caveat; the correction-cost ladder and its distinction from Domain 5's requirement ladder; the expected containment cost per introduced defect as the breakeven price of prevention; prevention against appraisal at equal spend; rework as a share of capacity, the duration multiplier and its convexity; acceptance decisions, their four failure modes and the five nonconformance dispositions with their authorities; acceptance sampling, the confidence bound, the breakeven defective fraction `c/u` and the self-consistency test; root-cause analysis, the three tests, cause concentration and breakeven recurrence; lessons that change behaviour; rolled throughput yield, the binding step and the uniform yield requirement; data quality dimensions and composite fitness; and the derivation of a verification regime for AI-produced work.

The calculations to be able to do under time pressure. Total cost of quality for a regime and the marginal step test between two regimes, including the breakeven external-failure unit cost. Escape fraction as $\prod(1 - d_i)$ and the defect flow layer by layer. Expected containment cost per introduced defect, and the breakeven number of defects a prevention spend must avoid. Duration from `content ÷ (crew × (1 − r))`, the overrun, and its cost at a cost of delay. $P(\text{clean sample}) = (1 - p)^n$; the 95 % bound `1 − 0.05^(1/n)`; the breakeven `c/u`; and the smallest `n` satisfying `1 − 0.05^(1/n) ≤ c/u`. Cause concentration and breakeven recurrence count. `RTY` as a product, the gain from a step improvement as a ratio of yields, and the uniform yield `RTY^(1/k)`. Composite data fitness as a product of dimension rates.

The traps. Treating zero defects as the objective and ignoring that the last escapes cost more than they save (9.1.2, MCQ 9.1-E) · reporting the cost-of-quality minimum without the range of optimality, which turns a fragile recommendation into a settled one (Exercise 9.1) · reading a rise in internal failure cost as deterioration when quality is improving (9.1.2, MCQ 9.1-C) · adding detection rates instead of multiplying escape rates (Exercise 9.2, MCQ 9.2-A) · applying each layer's detection rate to the original defect count rather than to the defects reaching it (Exercise 9.2) · treating two layers with a shared method or source as independent detectors (MCQ 9.2-F) · adding a rework percentage to a duration instead of dividing by `(1 − r)` (Exercise 9.3, MCQ 9.2-D) · forgetting that a plan already contains a rework allowance, so the variance is the difference and not the whole share (Exercise 9.3) · comparing escape counts rather than costs when choosing between prevention and appraisal (9.2.2b, MCQ 9.2-C) · treating a clean sample as evidence of conformance (9.3.2, MCQ 9.3-A) · believing the breakeven defective fraction depends on the sample size (MCQ 9.3-B) · comparing the confidence bound with a target rate instead of with `c/u` (Exercise 9.4) · letting the producer grant its own concession, or recording concessions without a cumulative test (9.3.1, MCQ 9.3-D) · stopping root-cause analysis where a person can be blamed (9.3.3, MCQ 9.3-F) · averaging stage yields instead of multiplying them (9.4.2, MCQ 9.4-A) · improving the strongest step rather than the binding one (Exercise 9.5, MCQ 9.4-B) · setting an end-to-end yield target without computing the per-step yield it requires (MCQ 9.4-C) · reporting a data quality average rather than a composite against a named use (9.4.3, MCQ 9.4-D) · and accepting AI-produced work on a sample whose confidence bound exceeds `c/u` (9.4.4, MCQ 9.4-E).

How the domain connects. Domain 3 supplies the assurance-line model and the gate economics whose product-level counterpart is the containment chain, and the escalation machinery that carries a root cause exceeding the project's authority. Domain 4's configuration management is the control whose absence produced this domain's dominant cause, and its cumulative test is the instrument reused for concessions. Domain 5 supplies acceptance criteria, testability and the requirement-defect ladder this domain deliberately does not duplicate. Domain 6 supplies the float that determines whether a rework overrun is a cost or a delay, and Domain 7 the blended rate and cost of delay that price both. Domain 8's risk register is where an escape probability belongs once quantified. Domain 10 carries quality across the contract boundary, where the supplier's optimum and the client's diverge. Domain 14 develops the AI governance frame that 9.4.4's sampling arithmetic serves. And Domain 16 receives the consequences: a handover chain's rolled yield and an asset register's composite fitness are readiness measures before they are quality measures.

Domain 9 summary

Quality is conformance to what was agreed; grade is what was agreed; fitness for purpose is whether the conforming, correctly graded thing works for its use. All three are testable, and only the first is testable against a document.

The domain's contribution is to make quality an **optimisation with an interior solution.** Auriga's cost of quality across five candidate regimes runs **660,000 · 448,600 · 364,000 · 397,000 · 493,000**, and the minimum — **USD 364,000**, **9.1 % of** `BAC` — sits at a regime that lets **8** of **80** introduced defects escape, not at the strictest one. Removing the last six escapes would cost **USD 129,000**, or **USD 21,500 each**, against a consequence of USD 12,000 — **1.79 times** the harm prevented. The honest objective is therefore not zero defects but no defect whose prevention costs less than its consequence, and the regime holds for any escaped-defect cost between **USD 3,540** and **USD 20,250**, which is what makes it defensible. Internal failure cost rises before it falls (132,600 at R1 against 120,000 at R0) because better detection finds what was escaping — the movement that gets quality programmes cancelled in their first period.

Where the money goes is a **chain of containment layers** whose escape fraction is the product of their escape rates: Auriga's `0.50 × 0.40 × 0.50 =` **0.100**, giving 8 escapes, **USD 108,000** of internal failure at an average of 1,500 and **USD 96,000** of external failure — **USD 2,550 of expected containment cost per introduced defect**, which is the breakeven price of prevention and the most useful single number in the domain. It settles the prevention-or-appraisal question arithmetically: on the same USD 40,000, cutting introduced defects from 80 to 60 is worth **USD 11,000** while raising the last layer's detection from 0.50 to 0.75 is worth **minus USD 6,000** — prevention wins by USD 17,000 despite removing fewer escapes, because appraisal relocates detection to a dearer layer while prevention removes the cost entirely. And rework is capacity before it is cost: a 30 % rework share against a 16.67 % allowance turned Auriga's 4-week commissioning into **4.7619 weeks**, costing **USD 34,286** of delay and **USD 23,886** of unbudgeted labour, with the multiplier `1/(1 − r)` convex enough that the next ten points cost **USD 35,714** and the ten after that USD 50,000.

Acceptance is a decision with a named signatory, and the five dispositions carry different authorities — a concession belongs to whoever bears the consequence, never to the producer, and must be counted cumulatively. Sampling yields the domain's sharpest result: a clean sample of 20 instrument loops bounds the defective fraction only at **13.91 %**, while the breakeven at which full verification pays is `p* = c/u = 1,400/12,000 =` **11.67 %**, independent of the sample size — so **the plan cannot exclude the defect rate at which its own strategy is wrong**, and five more loops (USD 7,000) repairs it. Case study B is the same defect realised: a 30-item plan with a 9.50 % bound against a 4.286 % breakeven passed a batch that was 4.667 % defective and cost **USD 441,000**, when **USD 35,100** of additional testing carried an expected saving of **USD 88,838**. Root causes are reached only when removal prevents the class and lies within somebody's authority; Auriga's dominant cause carried **62.5 %** of escapes and its removal broke even at **1.5 recurrences**.

Improvement compounds, and so does its absence. Six handover steps yielding 0.95, 0.92, 0.90, 0.94, 0.88 and 0.85 average **90.67 %** and deliver **55.31 %** — 26.82 of 60 packages needing rework at **USD 69,720** — and the binding step, not the weakest-looking process, is where effort belongs: fixing the 0.85 step gains **6.51 points** against **1.75** for fixing the 0.95 step, **3.73 times** the return. An 80 % end-to-end target requires **96.35 %** at every one of six steps. The same arithmetic governs data: six dimensions averaging **96.08 %** leave a composite fitness of **78.43 %**, about **690** of 3,200 asset records unfit for use. And it governs AI output, where the breakeven error fraction of **2.22 %** sits far below the **13.91 %** a clean sample of 20 can bound and the sample needed to reach it is **149 of 240 items** — so full review at **USD 9,600** is the defensible regime, and the productivity gain claimed for generation was never real until the verification was priced.

The through-line: **quality is a computable trade, its optimum is not zero, and every claim of conformance is a bound that must be compared with the ratio of what checking costs to what being wrong costs.** Compute that ratio, and quality stops being a virtue and becomes a decision.

10

DOMAIN 10

Procurement, Contracts and Supply Networks

Group: Delivering the work (Domain 10, the closing domain of Part Two). **Target:** ~74 pages. **Binds to:** the PCI Book Pattern Specification and the shared registries (`docs/books/registries/`). This domain completes the commercial thread Domain 7 opened: Domain 7 supplied the contract-model taxonomy and the **point of total assumption** (`PTA`, KA 7.4.3), and this domain takes them into the decisions that precede a contract and the behaviour that follows one. It continues **Project Auriga** — the 25-week control-systems upgrade for a regional utility, `BAC` **USD 4,000,000**, cost of delay **USD 45,000 per week** — from Domains 6, 7 and 8. British English; USD (+SAR where useful, indicative `USD 1 ≈ SAR 3.75`).

Registry note. This domain uses the registered cost of delay, `E[wait]`, `EMV` and `PTA` constructs unchanged, and submits five candidate rows to the shared formula registry: total cost of ownership and its breakeven volume `Q*`; the weighted evaluation score `S(w)` and its crossover weighting `w*`; the marginal-dollar allocation identity; expected cost of disruption with its breakeven disruption probability `p*`; and notice-bar exposure. Each carries a verified golden example below.

Why this domain exists

Domains 5 to 9 assumed that the organisation would have the capability to do the work. On most real projects it does not, and buys a substantial part of it. Auriga's own baseline says so: of its USD 4,000,000, the installation package alone is a **USD 2,000,000** subcontract, and the controllers, the civils and the commissioning support are all bought. More than half the project is delivered by people the project leader cannot instruct, on terms someone else negotiated, against a specification written before the design was finished.

That is the gap this domain closes, and the reason it sits last in Part Two rather than first. Every technique in the preceding five domains — the requirement, the critical path, the earned-value forecast, the risk response — becomes something the leader can only ask for once the boundary of the contract is crossed. The instrument that converts an intention into an obligation is the contract, and the instrument that decides who is obliged is the procurement process. Both are usually treated as functions belonging to somebody else, and both are among the least reversible decisions a project makes. A poorly chosen supplier can be managed; a poorly chosen contract structure cannot, because it is the thing you would have to use to manage it.

The domain's central claim is that **procurement decisions are delivery decisions with a computable price, and almost all of them are decided on the wrong number.** Make-or-buy is decided on unit price when the answer lies in total cost of ownership including transition and exit. Tenders are decided on a weighting that nobody has tested, when the weighting itself picks the winner. Contract types are chosen by habit, when their incentive arithmetic determines what the supplier will do at month eight. Sourcing is called resilient because there are two suppliers, when the exposure sits in a sub-tier neither of them will mention. In each case the right number exists, is small enough to compute in a meeting, and changes the decision.

Learning objectives. After this domain a candidate can: compute a procurement chain's total lead time including governance latency and treat it as a schedule predecessor; build a make-or-buy comparison on total cost of ownership including transition-in, operating and exit costs, compute the breakeven volume, and show how pricing the capability stand-up delay moves it; select a route to market and state what each route buys; construct a price/quality evaluation model, compute the weighting at which the winner changes, and state the professional obligation to publish the model before bids are opened; test a bid for abnormal pricing against a named benchmark; compute buyer outturn and supplier margin under fixed-price, cost-plus and target-cost structures at the same actual cost, state the marginal-dollar allocation each implies, and say where the PTA binds; demonstrate that an incentive structure only creates value if it changes behaviour, and compute the behaviour change the supplier needs to accept it; size a performance regime so that its cap deters rather than prices non-performance; assemble a claim's heads of cost and quantify what a notice provision puts at risk; price a dispute against its irrecoverable cost; assess an ethical-sourcing programme honestly, including where the expected-value case fails and obligation governs instead; compare single, dual-split and dual-qualified sourcing on expected cost, compute the breakeven disruption probability and the premium resilience is worth, and show why a shared sub-tier destroys the calculation; and govern AI-assisted contract and supplier analysis without letting it become commercial or legal advice.

The master project. Project Auriga continues. This domain buys its **84 substation controllers**, lets its **USD 2,000,000 installation subcontract** on the target-cost structure Domain 7 priced (target fee 150,000, share 70/30, ceiling 2,450,000, PTA **USD 2,428,571.43**), decides whether to build or buy its remote-terminal-unit configuration capability, evaluates three installation bids whose ranking depends entirely on a weighting, prices a variation claim at the blended engineering rate of **USD 130.625 per hour** established in KA 7.4.1, and tests three sourcing structures against the controller lead-time risk that Domain 8 carries as R1 (p 0.35, impact USD 240,000). Every figure in this domain is either computed here or cited from the domain that derived it.

KNOWLEDGE AREA 10.1

Make-or-buy and the procurement lifecycle

Topics: 10.1.1 what procurement is for · 10.1.2 make-or-buy on total cost of ownership · 10.1.3 the procurement lead time as a delivery constraint.

10.1.1 What procurement is for

Definition. Project procurement is the process by which a project **converts a need it cannot meet internally into an enforceable obligation held by another organisation**, and then manages that obligation to delivery and closure. The definition is deliberately narrow at one end and wide at the other: it excludes nothing about performance management, and it insists on enforceability, because a supplier relationship that rests only on goodwill has no mechanism for the week goodwill runs out.

The lifecycle, and what each stage decides. Six stages, and the value of naming them is that each one closes options the next cannot reopen.

1. **Define the need** — the specification, and the choice between specifying an output (what it must achieve) and an input (how it must be built). This choice, not the contract type, determines who is answerable for fitness for purpose, and it is made before anyone in procurement is involved.
2. **Decide make-or-buy** — treated in 10.1.2, and the only stage at which the answer "we should not buy this at all" is still available.
3. **Choose the route to market and the contract structure** — 10.2.1 and 10.3.1.
4. **Tender, evaluate and award** — KA 10.2. The stage with the most process and the least remaining freedom, which is why it absorbs attention disproportionate to its leverage.
5. **Administer the contract** — KA 10.3. Where nearly all of the value is won or lost, and where staffing is nearly always thinnest.
6. **Close out** — final account, retention release, warranties, knowledge and data transfer, and the lessons that inform the next specification. Routinely under-resourced because the project is over, which is precisely why final accounts settle badly.

The three questions a leader must answer before procurement starts, none of which a procurement function can answer for them. What must this supplier be accountable for, in words that will still mean the same thing in a dispute? What will I do if they fail — and is that response available under the structure I am about to sign? What information am I entitled to, on what cadence, and is it enough to run Domain 7's earned-value system across the boundary? A project that reaches award without those three answers has bought a supplier rather than a capability.

The boundary problem. Domain 3 established that governance must span the contract boundary and that decision rights are most often left undefined exactly there (KA 3.1.2). The specific asymmetry to watch is **threshold mismatch**: the client's change-approval threshold and the supplier's are set independently, by different organisations, and are almost never equal. Where the supplier may commit to a change at 25,000 and the client's project leader may approve one only at 10,000, every change between those figures arrives already committed on one side and unapproved on the other — a design failure at the moment the contract was signed, not a governance failure at the moment it appears.

10.1.2 Make-or-buy on total cost of ownership

The principle. A make-or-buy decision compares **total cost of ownership over the whole period the capability is needed, including the cost of getting into the arrangement and the cost of getting out of it.** Almost every make-or-buy decision that is later regretted was made on unit price, and unit price is the one component that systematically favours the option with the higher fixed cost — because it hides that cost.

```
TCO(option) = transition-in + (unit cost × volume) + transition-out
Breakeven volume  Q* = (F_make − F_buy) / (v_buy − v_make)
```

where `F` is the option's total fixed cost (transition-in plus exit) and `v` is its cost per unit of output. `Q*` exists only where the option with the higher fixed cost also has the lower unit cost — which is the normal case, and the reason the decision is a genuine choice rather than an arithmetic one.

The four costs the unit price hides. Transition-in — recruitment, training, tooling, test environments, licences, and the management attention consumed while none of it works yet. Exit — and this is the one most often scored at zero: redeployment or severance, decommissioning, data extraction in a usable form, knowledge transfer, re-qualification of whoever takes over, and the cost of running both arrangements in parallel during handover. Stand-up elapsed time, priced at the project's cost of delay wherever the capability is on the critical path. And residual management cost, which does not vanish when work is outsourced — it changes shape, from supervising people to administering a contract, and a buy case that assumes it falls to zero is understating by the cost of the contract manager it will need.

WORKED EXAMPLE

Worked example 10.1.2 — Auriga's remote-terminal-unit configuration: make or buy?

1. **Setup.** Auriga must configure and commission remote terminal units. Building the capability in-house costs **USD 420,000** to stand up (two recruits, training, a test rig, licences), **USD 3,600** per unit configured, and **USD 60,000** to wind down at the end (rig decommissioning, redeployment). Buying it from a specialist costs **USD 95,000** to transition in (specification, tender, mobilisation, interface definition), **USD 5,400** per unit, and **USD 145,000** to transition out (data extraction into a supportable format, knowledge transfer, re-qualification of the utility's own team at handover). Standing the in-house capability up takes **14 weeks** against the supplier's **5 weeks**, and the capability is on the critical path. Auriga requires **84 units**.
2. **Formula.** `TCO = F + vQ`, with `F` = transition-in + exit; breakeven `Q* = (F_make − F_buy)/(v_buy − v_make)`; and, where the stand-up is on the critical path, add (weeks of difference × cost of delay) to the make case.
3. **Substitution.** `F_make = 420,000 + 60,000 = 480,000`; `F_buy = 95,000 + 145,000 = 240,000`. `Q* = (480,000 − 240,000)/(5,400 − 3,600) = 240,000/1,800`. At `Q` = 84: make `480,000 + 3,600 × 84`; buy `240,000 + 5,400 × 84`. Delay term `(14 − 5) × 45,000 = 9 × 45,000`.
4. **Result.** Breakeven volume `Q*` **= 133.33 units**. At the required 84 units, make costs **USD 782,400** and buy costs **USD 693,600 — buy is USD 88,800 cheaper**, 11.35 % of the make case. With the 9-week stand-up delay priced at **USD 405,000**, the make case rises to **USD 1,187,400** and the breakeven volume moves to **358.33 units**.
5. **Interpretation.** On unit price alone, making looks decisively cheaper: 3,600 against 5,400 is a **33.33 %** saving per unit, and that is the number a make-or-buy paper usually leads with. It is the wrong test, and the arithmetic shows exactly why: the unit-price advantage of 1,800 has to recover a fixed-cost disadvantage of 240,000, which needs 133.33 units, and Auriga has 84. The decision is therefore **a bet on volume, and the honest question is not "is making cheaper?" but "do we believe the volume will exceed 134 units?"** That reframing is the deliverable. It also yields the sensitivity that matters: a second phase of 60 further units takes the total to 144 and reverses the answer — make USD 998,400 against buy USD 1,017,600, a **USD 19,200** advantage to making — so a decision to buy at 84 units should be written so that it can be revisited if the second phase is authorised, with the exit cost of 145,000 understood as the price of that reversibility rather than as an overhead. Two further cautions belong in the paper, and both are about what the number cannot carry. First, **the schedule term dominates everything else here**: pricing the 9-week difference at Auriga's cost of delay adds 405,000 and moves the breakeven from 133.33 to 358.33 units, a factor of **2.69** — so on a project with a critical-path constraint the make case is not marginal, it is unavailable, and no volume Auriga will ever see makes it competitive. Second, the calculation is silent on **capability strategy**. A utility that intends to own its control systems for twenty years may rationally build a capability that loses money on this project, and that is a legitimate decision — but it must be taken as a deliberate investment with the 88,800 (or 493,800 with delay) named as its price, not smuggled in behind a unit-price comparison. The failure mode is not choosing to make; it is choosing to make while believing it is cheaper.

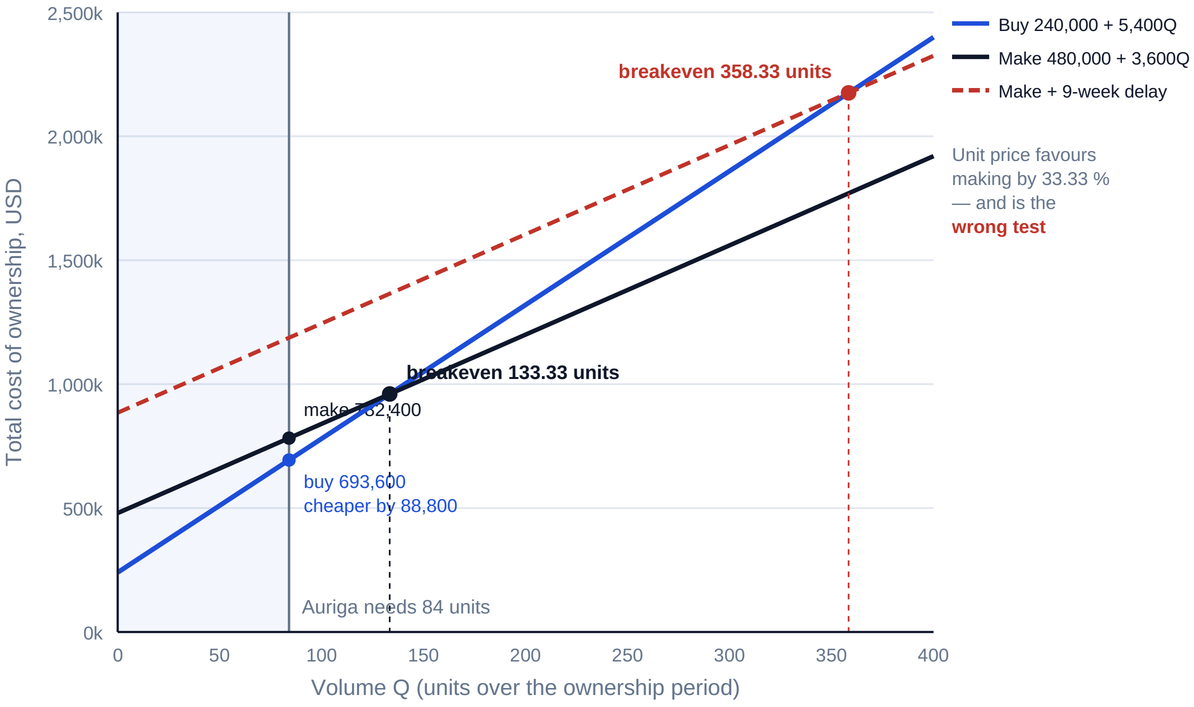


Fig 10.1.1 — Make-or-buy on total cost of ownership: the breakeven the unit price hides

10.1.3 The procurement lead time as a delivery constraint

The principle. A procurement process has a duration, that duration is a predecessor in the schedule, and it is systematically absent from plans because it is nobody's activity. Domain 6's critical path is computed over the work; the procurement chain that must complete before the work can start is computed, if at all, by a different function on a different document.

The chain, and its two kinds of time. Every procurement contains process time (specification, tender, evaluation, approval, execution) and physical time (manufacture, transport, acceptance). Practitioners assume the physical time dominates. It usually does not.

WORKED EXAMPLE

Worked example 10.1.3 — the controller procurement chain against a 25-week project.

1. **Setup.** Auriga's 84 substation controllers are a long-lead item. The chain: write and internally review the specification **3 weeks**; obtain authority to go to market — a decision for a body meeting every 4 weeks with a 1-week paper deadline; tender period **4 weeks**; evaluation and clarification **3 weeks**; award approval — the same body again; contract execution and mobilisation **2 weeks**; manufacture **11 weeks**; delivery, site acceptance and functional test **2 weeks**.
2. **Formula.** Total lead time = Σ process legs + Σ governance latency + Σ physical legs, with governance latency `E[wait] = M/2 + L` per approval (Domain 3, KA 3.2.3).
3. **Substitution.** Each approval: `E[wait] = 4/2 + 1 = 3.0` weeks, twice. Total `3 + 3 + 4 + 3 + 3 + 2 + 11 + 2`.
4. **Result. 31 weeks** of total procurement lead time, against a **25-week** project — an excess of **6 weeks**. Governance latency accounts for **6 weeks**, **19.35 %** of the chain, worth **USD 270,000** at Auriga's cost of delay. Process time totals **18 weeks** and physical time **13 weeks**, so **58.06 %** of the chain is administration.
5. **Interpretation.** The headline is uncomfortable and completely ordinary: **the procurement chain for a critical component is longer than the project that needs it.** Three consequences follow, and a leader who has done this arithmetic behaves differently at three specific moments. First, long-lead items must be committed **before the baseline is approved**, which means committing on an incomplete specification and accepting the variation exposure that follows — an explicit, priced trade, not an oversight, and the origin of Domain 8's R1 (controller lead-time slip, `p` 0.35, impact 240,000). Second, the largest single compressible block is not the manufacture but the **process**: 18 of 31 weeks, of which 6 are pure governance latency. Domain 3's result applies unchanged and is worth more here than anywhere — cutting the paper deadline from 1 week to nothing saves a full week per approval, so two approvals give back **2 weeks and USD 90,000** for the price of an administrative decision, while collapsing the two approvals into one delegated award authority gives back **3 weeks and USD 135,000**. Third, and least popular: the honest schedule shows procurement on the critical path from day one. A plan that starts at "contract awarded" is not optimistic, it is incomplete by 18 weeks, and the difference will be discovered at the worst possible moment — when the site team is standing ready and the controllers are in manufacture. The professional caution: the 11-week manufacture figure is a supplier's number, and it is the one figure in the chain that a supplier has an incentive to state optimistically at bid stage and revise after award. It should be tested against the supplier's actual recent performance, not against its quotation.

AI in this KA

Where it earns its place. Assembling total-cost-of-ownership comparisons from disparate sources — payroll, licence schedules, historical supplier invoices, exit provisions buried in incumbent contracts — is a data-collation task with a clear right answer, and the exit column is precisely the one humans leave blank because the evidence is scattered. Extracting stated lead times from a set of bids and comparing them with the same suppliers' delivery history. Modelling the breakeven volume across ranges of unit cost, transition cost and delay, which is deterministic and cheap to sweep. Reading an incumbent contract for its termination, transition-out, data-return and intellectual-property clauses and listing the exit obligations, which is where the exit cost actually comes from.

Where it must not go. It must not supply the estimates it is asked to compare. A model asked "what does it cost to stand up an RTU configuration team?" will produce a confident figure with no provenance, and that figure will then anchor a real decision — the same defect Domain 3 identified for critical-path shares. It must not decide capability strategy, which is a decision about what the organisation intends to be able to do, and belongs to an accountable executive. And its reading of an exit clause is a summary, not the clause: the obligation is what the contract says.

Verification, concretely. Every input to a make-or-buy paper carries a named source and a date. The breakeven volume is recomputed by hand — it is one subtraction and one division — and stated in the paper alongside the point estimate, because the breakeven is what tells the decision-maker whether the recommendation is robust. Where an AI tool has listed exit obligations, each one is confirmed against the clause before it enters the cost.

KEY TERMS — KA 10.1

Term	Meaning
Procurement	Converting a need the project cannot meet internally into an enforceable obligation held by another organisation, and managing it to closure.
Total cost of ownership (TCO)	Transition-in + operating cost over the period of need + transition-out; the only valid basis for make-or-buy.
Transition-in / transition-out	The cost of entering and of leaving an arrangement; the exit column is the one most often scored at zero.
Breakeven volume (`Q*`)	The volume at which two options cost the same: `(F_make − F_buy)/(v_buy − v_make)`.
Output vs input specification	Specifying what must be achieved against how it must be built; determines who is answerable for fitness for purpose.
Threshold mismatch	Client and supplier change-approval thresholds set independently, so changes between them arrive committed on one side and unapproved on the other.
Long-lead item	A component whose procurement chain exceeds the time available after baseline approval, and must therefore be committed earlier.
Procurement lead time	Process legs + governance latency + physical legs; a schedule predecessor, not an administrative overhead.

SAMPLE MCQS — KA 10.1

MCQ 10.1-A `[10.1.2 · Application]` In-house provision costs 420,000 to stand up, 3,600 per unit and 60,000 to exit; buying costs 95,000 to transition in, 5,400 per unit and 145,000 to exit. The breakeven volume is: - A. 66.67 units - B. 133.33 units - C. 180.56 units - D. 227.78 units

Rationale: `(480,000 − 240,000)/(5,400 − 3,600) = 240,000/1,800 = 133.33` (10.1.2). A divides the fixed-cost difference by the make unit cost instead of the unit-cost difference (`240,000/3,600 = 66.67`). C omits both exit costs (`325,000/1,800 = 180.56`). D assigns each option's exit cost to the other option (`410,000/1,800 = 227.78`).

MCQ 10.1-B `[10.1.2 · Analysis]` At 84 units the make option costs 782,400 and the buy option 693,600, yet the make option's unit cost is 33.33 % lower. The correct reading is that: - A. the unit costs must have been miscalculated - B. the fixed-cost difference of 240,000 has not been recovered at this volume, so the decision is a bet on volume exceeding 133.33 units - C. unit cost is the more reliable comparison because it excludes one-off items - D. the two options are equivalent because the difference is under 15 %

Rationale: The unit-price advantage must recover a fixed-cost disadvantage, and 84 units does not (10.1.2). C inverts the principle — excluding the one-off items is the error. D substitutes a tolerance for an answer.

MCQ 10.1-C `[10.1.2 · Evaluation]` Standing the in-house capability up takes 9 weeks longer than mobilising the supplier, and the capability is on the critical path at a cost of delay of 45,000 per week. The effect on the breakeven volume is to move it from 133.33 units to: - A. 133.33 units — elapsed time does not affect a cost comparison - B. 225.00 units - C. 358.33 units - D. 491.67 units

Rationale: `(480,000 + 405,000 − 240,000)/1,800 = 645,000/1,800 = 358.33` (10.1.2). A omits the delay entirely. B prices the delay alone and forgets the fixed-cost difference (`405,000/1,800 = 225.00`). D omits the buy option's fixed cost from the numerator (`885,000/1,800 = 491.67`).

MCQ 10.1-D `[10.1.3 · Application]` A procurement chain has 12 weeks of process legs, 13 weeks of manufacture and delivery, and two approvals by a body meeting every 4 weeks with a 1-week paper deadline. Total lead time is: - A. 25 weeks - B. 29 weeks - C. 31 weeks - D. 33 weeks

Rationale: Each approval adds `E[wait] = 4/2 + 1 = 3.0` weeks, so `12 + 13 + 6 = 31` (10.1.3, Domain 3 KA 3.2.3). A omits governance latency altogether; B counts only half of each interval and omits the paper deadlines; D adds the whole meeting interval twice.

MCQ 10.1-E `[10.1.1 · Comprehension]` Which stage of the procurement lifecycle is the last at which the option "we should not buy this at all" remains available? - A. define the need - B. decide make-or-buy - C. choose the route to market - D. tender and evaluate

Rationale: After the make-or-buy decision, subsequent stages choose how to buy, not whether (10.1.1). The stage with the most process — tender and evaluate — has the least remaining freedom.

SELF-CHECK — KA 10.1

1. Why does unit price systematically favour the wrong option in make-or-buy? — It excludes fixed transition-in and exit costs, which are exactly the costs the low-unit-cost option carries.
2. What single figure converts a make-or-buy recommendation into a testable proposition? — The breakeven volume: the decision becomes "do we believe volume will exceed `Q*`?"
3. Which part of a procurement chain is usually most compressible, and why is that surprising? — The process and governance legs — 18 of Auriga's 31 weeks — not the manufacture, which is where attention goes.

KNOWLEDGE AREA 10.2

Tendering and evaluation

Topics: 10.2.1 routes to market · 10.2.2 the evaluation model and the weighting that chooses the winner · 10.2.3 probity, and fixing the model before the bids are opened.

10.2.1 Routes to market

The choice. A route to market is a decision about **how much competition to buy and what to pay for it**, and competition is not free: it costs elapsed time, bid cost on both sides, and the goodwill of suppliers who lose repeatedly. The main routes, with what each actually purchases:

Route	What it buys	What it costs	Suits
Open competition	The widest price discovery and the strongest probity record	The longest chain (10.1.3), high evaluation effort, high aggregate bid cost	Novel requirements, unknown market, regulated buyers
Selective (pre-qualified) competition	Price discovery among capable bidders only	Pre-qualification effort; the risk of excluding a capable newcomer	Most substantial project packages
Framework call-off	Speed — the tender is already done	The framework's rates, which may have drifted from the market; and the aggregation illusion below	Repeat purchases of definable scope
Single-source negotiation	Speed and continuity of a known capability	The absence of price discovery; a documented justification is mandatory	Genuine sole source, proprietary interfaces, emergency
Two-stage	Early supplier involvement on buildability while price is settled later	Weak price tension at stage two, when the supplier is embedded	Complex or undefined scope needing supplier design input

Three cautions worth stating. Competition is not price tension. Three bidders who all believe one of them is favoured produce three prices and no competition; price tension requires that each bidder believes it can win, which is a function of how the model is published (10.2.3), not of the number of bidders. The aggregation illusion in frameworks: a call-off placed without any comparison inside the framework buys the framework's rate, not the market's, so framework rates need periodic benchmarking against a live test — a framework that has never been benchmarked is a single-source arrangement with paperwork. Two-stage arrangements fail predictably at stage two, when the supplier is the only party who understands the design it helped produce and the buyer's walk-away option has quietly disappeared; the countermeasure is to fix the stage-two commercial terms — fee percentage, overhead recovery, basis of preliminaries — at stage one, while that option still exists.

10.2.2 The evaluation model, and the weighting that chooses the winner

The principle. An evaluation model converts a set of bids into a ranking. It is a piece of arithmetic, it is designed by people, and **the design determines the outcome at least as much as the bids do.** A leader who does not understand that is not evaluating bids; they are ratifying whatever the model was set up to produce.

```
S_i(w) = w × P_i + (1 − w) × Q_i
```

where `S_i` is bidder `i`'s weighted score, `P_i` its normalised price score, `Q_i` its quality score, and `w` the weight given to price. Because each `S_i(w)` is linear in `w`, two bidders' scores cross at exactly one weighting:

```
w* = (Q_b − Q_a) / [(P_a − Q_a) − (P_b − Q_b)]
```

WORKED EXAMPLE

Worked example 10.2.2 — three bids, three winners.

1. **Setup.** Auriga's installation package attracts three compliant bids. Alpha USD 2,000,000, quality **62**/100; Beta USD 2,200,000, quality **78**; Gamma USD 2,480,000, quality **92**. Price is normalised on the ratio convention — the lowest price scores 100, others score `lowest ÷ own × 100`. The evaluation panel has not yet fixed the weighting.
2. **Formula.** `S_i(w) = w P_i + (1 − w) Q_i`; crossover `w* = (Q_b − Q_a)/[(P_a − Q_a) − (P_b − Q_b)]`.
3. **Substitution.** Price scores: Alpha `2,000,000/2,000,000 × 100 = 100.000000`; Beta `2,000,000/2,200,000 × 100 = 90.909091`; Gamma `2,000,000/2,480,000 × 100 = 80.645161`. At `w` = 0.70: Alpha `0.70 × 100 + 0.30 × 62`; Beta `0.70 × 90.909091 + 0.30 × 78`; Gamma `0.70 × 80.645161 + 0.30 × 92`. Then at `w` = 0.40, and the crossovers.
4. **Result.**

Bidder	Price (USD)	Quality	Price score	S at 70/30	S at 60/40	S at 40/60
Alpha	2,000,000	62	100.00	**88.60**	84.80	77.20
Beta	2,200,000	78	90.91	87.04	**85.75**	83.16
Gamma	2,480,000	92	80.65	84.05	85.19	**87.46**

Three different winners from one bid set. The crossovers are Beta/Gamma at a price weight of **57.70 %** and Alpha/Beta at **63.77 %**: Gamma wins below 57.70 %, Beta in the band between, and Alpha above 63.77 %. 5. **Interpretation.** The first and most important reading is procedural, not commercial: **with the bids in hand, a panel could choose a weighting to produce any of the three outcomes it liked.** That is why the weighting must be published before bids are opened (10.2.3), and it is not a formality — it is the only thing standing between an evaluation and a rationalisation. The second reading is about the model's fragility. Beta wins in a band **6.07 percentage points** wide, which is narrower than the uncertainty in the quality scores themselves; a panel whose consensus moved Beta's quality score by two points would relocate the boundary. A model whose output is knife-edge is telling the buyer something useful — that the three bids are, on this model, close to indistinguishable, and the decision should turn on something the model does not capture rather than on the third decimal place. The third reading is the commercial one, and it is the test that should discipline any quality weighting. Gamma's premium over Alpha is **USD 480,000 — 24.00 %**, or USD 16,000 per quality point. What does that buy? Suppose the panel's own risk assessment maps the quality scores to the probability of the integration rework that Domain 8 carries as R3 (impact USD 320,000): Alpha 0.30, Beta 0.18, Gamma 0.10, giving expected rework of **96,000**, **57,600** and **32,000**. Risk-adjusted cost is then Alpha **2,096,000**, Beta **2,257,600**, Gamma **2,512,000** — and Alpha is still cheapest by a wide margin, because Gamma's 480,000 premium buys only `96,000 − 32,000 =` **64,000** of avoided expected cost. The premium is **7.5 times** the risk it removes. Add a generous allowance for schedule as well — say Gamma's quality also avoids three weeks of integration delay at 45,000 a week, worth 135,000 — and the total avoided is 199,000, still **281,000** short. Expressed as a breakeven: Gamma's premium is worth paying only if its quality avoids more than 480,000 of expected cost, which at Auriga's cost of delay is **10.67 weeks** of delay. That is the professional obligation in a sentence: **a quality weighting is a statement about how much money quality is worth, and if the number it implies cannot be defended against the risks it is supposed to remove, the weighting is not a judgement, it is a preference.** Two cautions. Quality may buy things that are genuinely outside this arithmetic — safety performance, regulatory standing, workforce capability retention, the ability to be relied on in a crisis — and where it does, those must be **named and, so far as possible, priced**, not asserted as a reason the arithmetic does not apply. And the risk mapping above is the panel's own assessment; it must be recorded as an assumption with its basis, because the whole conclusion is proportional to it.

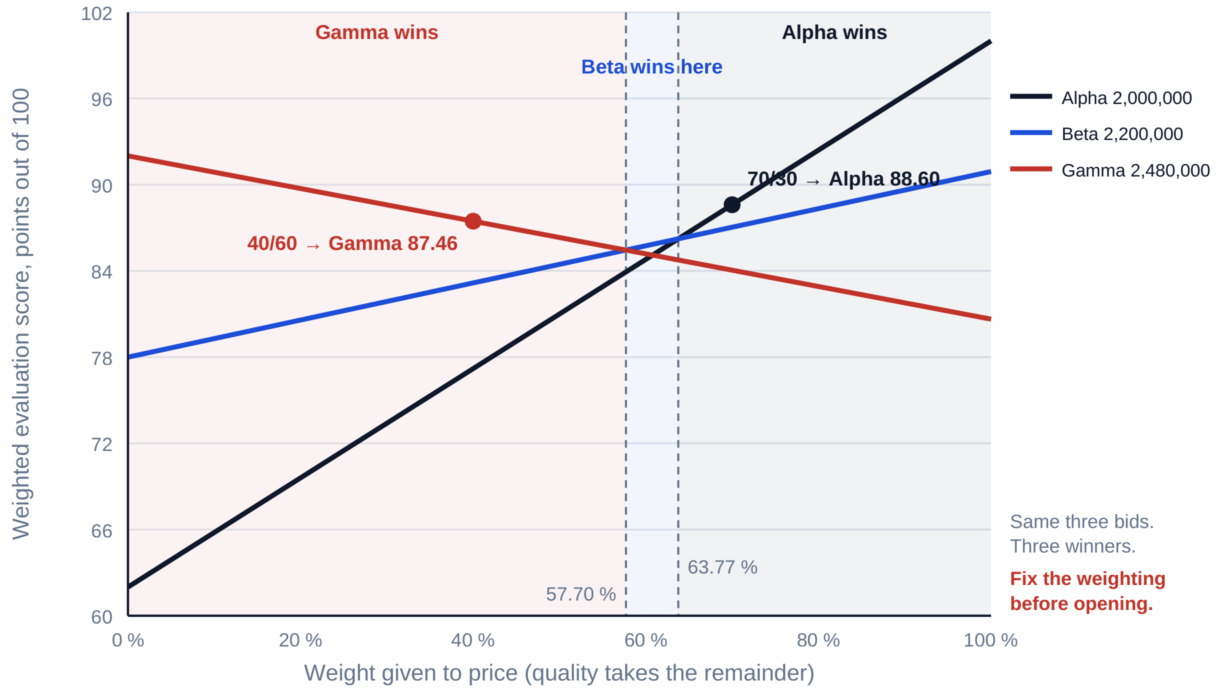


Fig 10.2.1 — The weighting chooses the winner

Normalisation is part of the model, and it moves the answer too. The ratio convention above is one of several. A linear convention — `100 × (1 − (bid − lowest)/lowest)` — gives Alpha 100, Beta **90.00** and Gamma **76.00**, because it penalises price differences more steeply. Under it the winner still changes with the weighting, but the boundaries move: Beta/Gamma crosses at **50.00 %** and Alpha/Beta at **61.54 %**. The Beta/Gamma boundary has shifted by **7.70 percentage points** purely because of a choice about arithmetic that most invitation documents do not state. The obligation is therefore wider than publishing the weights: **publish the normalisation formula, the scoring scale, the moderation process and the treatment of non-compliance**, because each of them is a lever on the outcome.

Testing a bid for abnormal pricing. A price far below the others may be an efficiency the buyer should welcome, or a bid that cannot be delivered and will return as claims (Domain 7, KA 7.4.2). The test is quantitative and must specify its benchmark, because the benchmark changes the answer. Against the bid spread, the mean of the three bids is **USD 2,226,666.67** and Alpha is **10.18 %** below it — enough to trigger a clarification duty under most published thresholds. Against the buyer's own estimate, Alpha at 2,000,000 sits exactly on Auriga's target cost, and is not low at all. Both readings are true, and the professional response is neither to disqualify nor to ignore: it is to seek a structured explanation of the price build-up, satisfy the panel that the scope and the risk are actually covered, and record the answer — because if the bid is genuinely under-priced, the buyer will meet the difference again as variations, and the record is what makes that argument winnable.

10.2.3 Probity, and fixing the model before the bids are opened

The obligation. The evaluation model — weights, normalisation, scoring scale, sub-criteria and their weights, the treatment of non-compliant bids — must be **fixed, documented and disclosed to bidders before bids are opened**, and not changed afterwards. This is a professional obligation independent of whatever a particular jurisdiction's procurement law requires, and 10.2.2 is the reason: the model selects the winner, so a model chosen after the prices are known is not an evaluation method but a means of choosing a supplier while appearing not to.

The mechanics that make it real, in the order they occur. Publish the model in the invitation, with sub-criteria weights, not merely the top-level split. Receive bids to a single controlled point with a recorded receipt time, and hold them unopened. **Separate the price and quality submissions** and score quality before any price is seen — this is the single most effective anchoring control available, and its absence is the commonest defect in otherwise diligent processes. Have each panel member score independently and record those scores before moderation, so that the moderation is a documented convergence rather than a consensus with no history. Record the reason for every score, especially the low ones, since that is what a debrief and any challenge will test. Manage conflicts of interest by declaration and exclusion, before the panel is constituted. And where a change to the model becomes genuinely unavoidable — a criterion turns out to be unassessable — **re-run the evaluation under the original model as well**, disclose both results, and have the decision taken by an authority senior to the panel, because the only defence against a post-hoc change is transparency about what it altered.

Debriefing, and why it is a commercial instrument. A specific, evidence-based debrief to unsuccessful bidders costs a few hours and buys three things: better bids next time, a market that continues to bid, and a documented rationale that has already survived contact with the party most motivated to attack it. A vague debrief buys a challenge.

AI in this KA

Where it earns its place. Compliance checking a bid against a long requirements schedule and listing omissions and qualifications — high-volume, low-judgement, and exactly where human panels tire and miss things. Extracting a comparable price build-up from bids submitted in different formats, so that like is compared with like. Sweeping the evaluation model across weightings and normalisation conventions to reveal how sensitive the outcome is — the arithmetic of 10.2.2 run over hundreds of combinations, which is deterministic, verifiable and genuinely informative if it is done before bids are opened, as model design. Drafting the clarification questions that a price build-up implies.

Where it must not go. It must not score quality. A quality score is an expert judgement for which a named human must be answerable, and an AI-generated score cannot be defended in a debrief or a challenge because no one can explain it in the terms the criterion was written in. It must not be used to sweep weightings **after** bids are opened — the same computation that is good model design before opening is the mechanism of a rigged evaluation after it, and the difference is entirely one of timing. And it must not decide non-compliance, which is a contractual judgement with legal consequences.

Verification, concretely. Any AI-produced compliance list is confirmed against the bid text before it affects a score, with the confirmation recorded against the bidder. Any AI-assembled price comparison is reconciled to the bidders' own totals to the cent. The evaluation arithmetic itself — weighted scores and crossovers — is reproduced by hand for the top two bidders, because it is four multiplications and the whole award rests on it. And the audit trail must show that any weighting sensitivity analysis was performed and locked **before** the receipt deadline, with its date.

KEY TERMS — KA 10.2

Term	Meaning
Route to market	The competitive structure chosen for a procurement; each route buys a different amount of price discovery at a different cost in time.
Aggregation illusion	Treating a framework call-off as competitively priced because the framework was competed; the rate may have drifted from the market.
Evaluation model	Weights, normalisation, scoring scale and moderation rules; a piece of arithmetic that materially selects the winner.
Price normalisation	The formula converting prices into scores; the ratio and linear conventions give different scores and different crossovers.
Crossover weighting (`w*`)	The price weight at which two bidders' weighted scores are equal.
Quality premium test	Comparing a higher-quality bid's price premium with the expected cost it demonstrably avoids.
Abnormally low tender	A bid materially below its benchmark; the benchmark must be stated, since the bid spread and the buyer's estimate can disagree.
Two-envelope evaluation	Scoring quality before price is revealed, to remove price anchoring from quality judgement.

SAMPLE MCQS — KA 10.2

MCQ 10.2-A `[10.2.2 · Application]` Bids: Alpha 2,000,000 quality 62; Beta 2,200,000 quality 78; Gamma 2,480,000 quality 92. Price is scored `lowest ÷ own × 100`. At a 70/30 price/quality weighting the winner and score are: - A. Gamma, 84.05 - B. Beta, 87.04 - C. Alpha, 88.60 - D. Alpha, 81.00

Rationale: `0.70 × 100 + 0.30 × 62 = 88.60`, against Beta 87.04 and Gamma 84.05 (10.2.2). A and B name the other bidders' correct scores but not the winner at this weighting; D is Alpha's score at a 50/50 weighting.

MCQ 10.2-B `[10.2.2 · Analysis]` For the same three bids, the price weight at which Beta overtakes Gamma is closest to: - A. 50.00 % - B. 57.70 % - C. 60.78 % - D. 63.77 %

Rationale: `w* = (92 − 78)/[(90.909091 − 78) − (80.645161 − 92)] = 14/24.263930 = 57.70 %` (10.2.2). A is the Beta/Gamma crossover under the linear normalisation convention, not the ratio convention. C is the Alpha/Gamma crossover, which lies inside Beta's winning band and is therefore not a boundary at all. D is the Alpha/Beta crossover.

MCQ 10.2-C `[10.2.2 · Evaluation]` Gamma's price premium over Alpha is 480,000. On the panel's own risk mapping, Gamma's higher quality reduces expected integration rework from 96,000 to 32,000. The strongest statement a leader can make about the premium is that it is: - A. justified, because Gamma scores 30 quality points higher - B. 7.5 times the expected cost it avoids, so it must be justified by something the risk assessment does not capture — named and priced - C. unjustifiable in all circumstances - D. justified, because 480,000 is only 24 % of Alpha's price

Rationale: `480,000/(96,000 − 32,000) = 7.5` (10.2.2). A and D restate inputs as conclusions. C overreaches: quality may buy safety, regulatory standing or capability the mapping omits — but those must be named, not assumed.

MCQ 10.2-D `[10.2.3 · Analysis]` A panel opens the priced envelopes, then confirms the price/quality weighting at 40/60. The defect is that: - A. 40/60 is too low a price weighting for a construction package - B. the weighting should have been an odd split to avoid ties - C. with the bids known, the weighting can be selected to produce any of three winners, so a model fixed after opening is not an evaluation method - D. the panel should have used the linear normalisation convention

Rationale: The bid set in 10.2.2 has three different winners across the weighting range, so post-hoc weighting selects the supplier (10.2.3). A and D are matters of judgement, not defects; B is invented.

MCQ 10.2-E `[10.2.2 · Analysis]` Alpha's 2,000,000 bid is 10.18 % below the mean of the three bids but exactly equal to the buyer's own target cost. The correct professional response is to: - A. disqualify Alpha as abnormally low - B. ignore the test, since the bid matches the buyer's estimate - C. seek a structured explanation of the price build-up, satisfy the panel that scope and risk are covered, and record the answer - D. average the two benchmarks and apply the threshold to the result

Rationale: The two benchmarks disagree, which is information rather than a tie to be broken; the response is clarification and a record, because an under-priced bid returns as variations (10.2.2, Domain 7 KA 7.4.2). D invents a procedure.

SELF-CHECK — KA 10.2

1. Why must the evaluation weighting be published before bids are opened? — Because the weighting selects the winner: one bid set can yield three different winners across the weighting range.
2. What does a very narrow winning band tell the buyer? — That the bids are close to indistinguishable on this model, so the decision should turn on something the model does not capture.
3. Name the control that removes price anchoring from quality scoring. — Two-envelope evaluation: score quality before any price is revealed.

KNOWLEDGE AREA 10.3

Contract strategy and supplier governance

Topics: 10.3.1 contract strategy as risk allocation · 10.3.2 incentive arithmetic: buyer and supplier outturn · 10.3.3 supplier governance and performance regimes.

10.3.1 Contract strategy as risk allocation

The starting point, cited not re-derived. Domain 7 KA 7.4.2 sets out the contract models — firm fixed price, fixed-price incentive, cost plus fixed fee, cost plus incentive fee, time and materials — ordered by who carries cost risk, together with the discipline that **risk transferred is risk priced**. That taxonomy is assumed here. What this KA adds is the arithmetic of what each model does to the two parties' money at the same actual cost, and what that predicts about behaviour.

The four questions a contract strategy must answer. Who carries which risk — and the test of a sound allocation is whether the party carrying a risk can actually **influence** it; a risk allocated to a party with no control over it is not managed, it is priced, and priced badly. What the payment mechanism rewards — the marginal-dollar question of 10.3.2. What happens when things go wrong — the change, notice, suspension, step-in and termination machinery of KA 10.4, which must exist before it is needed. What the buyer is entitled to know — access, records, audit and progress data granular enough to run Domain 7's earned-value system across the boundary, because a supplier reporting percentage complete on its own definition supplies comfort, not control.

Packaging, and the interface cost of splitting work. How work is divided between contracts is a contract-strategy decision with a computable cost, and Domain 4's result applies directly: `n` packages let by a buyer who integrates them itself create up to `n(n−1)/2` pairwise interfaces, all of which the buyer owns by default (Domain 4, KA 4.2.3). Five packages create ten interfaces; a single main-contract-plus-subcontracts structure creates five to one integrator. Splitting work buys price tension in each package and sells the integration risk to yourself, and the trade should be stated in those terms: **the price saving from splitting must exceed the cost of the interfaces it creates**, and on a technically integrated scope it frequently does not.

10.3.2 Incentive arithmetic: buyer and supplier outturn

The principle. A contract's payment mechanism determines, for every additional dollar of cost, who pays it. That allocation is what the supplier's management responds to at month eight, long after the strategy paper has been filed.

The marginal-dollar allocation identity. For any payment mechanism, at any cost level,

```
buyer's share of the next dollar of cost + supplier's share = 1
```

The identity is trivial and its consequences are not: **a contract cannot make a dollar of cost disappear; it can only decide who holds it.** Every claim that a structure "removes cost risk" is therefore a claim about allocation, and the allocation can be read off the mechanism:

Mechanism	Buyer's share of the marginal dollar	Supplier's share	What the supplier is rewarded for
Firm fixed price	0.00	1.00	Efficiency — and, unless specification and inspection are tight, for reducing quality and scope
Cost plus fixed fee	1.00	0.00	Nothing; cost recovery is certain and effort is not rewarded
Target cost, 70/30 share, **below** the PTA	0.70	0.30	Efficiency, at 30 cents in the dollar — both parties lose on overrun
Target cost, **above** the PTA	0.00	1.00	Entitlement, not efficiency: the marginal incentive is a fixed price the supplier never priced

The last row is Domain 7's point of total assumption read as a behavioural statement rather than a commercial one, and Auriga's installation subcontract puts it at **USD 2,428,571.43** (Domain 7, KA 7.4.3 — target cost 2,000,000, target fee 150,000, share 70/30, ceiling 2,450,000). Nothing about that figure is re-derived here; what matters here is that it is the cost level at which the supplier's marginal share jumps from 0.30 to 1.00, and therefore the level at which a cooperative relationship becomes an adversarial one for reasons that have nothing to do with the people involved.

WORKED EXAMPLE

Worked example 10.3.2 — the same scope, three mechanisms, one cost distribution.

1. **Setup.** Auriga's installation scope. The supplier's own assessment of its outturn cost, disclosed in negotiation: **1,850,000** with probability **0.20**, **2,000,000** with **0.40**, **2,300,000** with **0.30**, **2,600,000** with **0.10**. The supplier requires an expected margin of **USD 150,000**. Three candidate mechanisms: a firm fixed price; cost plus a fixed fee of 150,000; and the target-cost structure of Domain 7 (target cost 2,000,000, target fee 150,000, share 70/30, ceiling 2,450,000, `PTA` 2,428,571.43).
2. **Formula.** `E[cost] = Σ p × cost`. Fixed price (risk-neutral) `= E[cost] + required margin`. Cost-plus buyer outturn `= cost + fee`. Target-cost fee `= target fee − overrun × supplier share` (negative overrun increases the fee), buyer outturn `= cost + fee`, capped at the ceiling. Supplier margin = buyer outturn − cost.
3. **Substitution.** `E[cost] = 0.20 × 1,850,000 + 0.40 × 2,000,000 + 0.30 × 2,300,000 + 0.10 × 2,600,000 = 370,000 + 800,000 + 690,000 + 260,000`. Fixed price `= 2,120,000 + 150,000`. Target-cost fee at 1,850,000: `150,000 + 150,000 × 0.30 = 195,000`; at 2,300,000: `150,000 − 300,000 × 0.30 = 60,000`; at 2,600,000 the uncapped outturn 2,570,000 exceeds the ceiling, so the buyer pays 2,450,000 and the supplier absorbs the rest.
4. **Result.** `E[cost]` = **USD 2,120,000**; risk-neutral firm fixed price **USD 2,270,000**.

Actual cost	Firm fixed price: buyer / supplier margin	Cost plus fixed fee: buyer / margin	Target cost: buyer / margin
1,850,000	2,270,000 / **420,000**	2,000,000 / 150,000	2,045,000 / 195,000
2,000,000	2,270,000 / 270,000	2,150,000 / 150,000	2,150,000 / 150,000
2,300,000	2,270,000 / (30,000)	2,450,000 / 150,000	2,360,000 / 60,000
2,600,000	2,270,000 / **(330,000)**	2,750,000 / 150,000	2,450,000 / (150,000)
Expected	**2,270,000 / 150,000**	**2,270,000 / 150,000**	**2,222,000 / 102,000**

The two expected buyer outturns for fixed price and cost-plus are **identical at USD 2,270,000**. The target-cost structure gives the buyer **USD 2,222,000** and the supplier **USD 102,000** — the buyer's **48,000** saving is exactly the supplier's **48,000** shortfall. 5. **Interpretation.** Three results, each of which changes how a leader argues about contract type.

First: at risk-neutral pricing, transferring risk is free in expectation and buys only variance. The fixed price and the cost-plus arrangement cost the buyer the same 2,270,000 on average, because the fixed-price premium of `2,120,000 − 2,000,000 =` 120,000 over the efficient cost is the expected cost of the risk. What the buyer gains is certainty: the standard deviation of its outturn falls from **USD 230,434** under cost-plus to **zero** under fixed price. That is the honest case for fixed price — not that it is cheaper, but that it converts a distribution into a number — and it is worth paying a real premium for where the buyer's own funding or covenants cannot absorb the variance. A risk-averse supplier will charge one: a 25 % loading on the 120,000 risk element takes the price to **USD 2,300,000**, and that 30,000 is the visible, arguable price of certainty. The corollary is the part buyers forget: the transfer is only as good as the supplier's balance sheet. At the 2,600,000 outcome the fixed-price supplier loses **330,000**, and a supplier that cannot absorb that does not deliver a fixed price — it delivers an insolvency and a re-procurement, at which point the risk returns to the buyer with the delay added (KA 10.4.4).

Second: an incentive structure does not create value, it reallocates it — unless behaviour changes. As specified, the target-cost arrangement is simply 48,000 transferred from supplier to buyer, and a supplier that can do this arithmetic will respond by raising the target cost, raising the target fee, or declining. The reason is structural and worth stating precisely: **the target fee is the fee at the target, not the expected fee.** Any real probability of overrun makes the expected fee less than the target fee, so a supplier facing the distribution above expects 102,000 against a requirement of 150,000. Negotiations that stall over "the fee" are usually stalling over this, unnamed.

Third: the behaviour change required can be computed, and it is the number the negotiation is actually about. Suppose the incentive causes the supplier to shift probability towards the underrun outcome, the remaining mass staying in its original 3:1 ratio between the 2,300,000 and 2,600,000 branches. The supplier's expected margin is then `187,500q + 64,500` where `q` is the probability of the 1,850,000 outcome, so it reaches 150,000 at `q` **= 0.456** — the underrun probability must rise from 0.20 to 0.456, a factor of **2.28**. At that point expected cost falls from 2,120,000 to **USD 1,985,600**, so the incentive has created **USD 134,400** of genuine value; the buyer's expected outturn is **USD 2,135,600**, which is 134,400 below the fixed price; and the supplier is exactly indifferent. In other words, **at the point where the supplier will just accept the structure, the buyer captures the entire value created.** That is the real subject of the negotiation: not the fee in the abstract, but how the 134,400 is split, and a buyer who understands this can concede part of it knowing precisely what it is buying — a supplier with an actual reason to be efficient. The professional cautions are three. The distribution is the supplier's assessment and it has every incentive to shade it pessimistically, so it must be tested against comparable outturns rather than accepted. The behaviour shift is a hypothesis, not a fact; the structure should therefore carry measurable efficiency commitments rather than resting on the incentive alone. And none of this arithmetic survives contact with a badly defined scope: on undefined scope every mechanism converges on cost-plus with a dispute attached, whatever the contract says (Domain 7, KA 7.4.2).

10.3.3 Supplier governance and performance regimes

The principle. A contract is a document until someone administers it. Supplier governance is the machinery that turns obligations into observed performance, and its design errors are the same three every time: it measures what is easy rather than what matters, it meets on a cadence slower than the decisions it must take, and its sanctions are too small to change anyone's behaviour.

Governing across the boundary. Domain 3's structures apply, with three additions specific to a contract. Decision rights must be **mapped on both sides**, with the threshold mismatch of 10.1.1 resolved explicitly. The governance cadence must be set against the decision rate, not the reporting rhythm: a supplier review every eight weeks with a two-week paper deadline imposes `E[wait] = 8/2 + 2 =` **6 weeks** on every decision it owns (Domain 3, KA 3.2.3), which is longer than most delivery problems will politely wait, so an out-of-cycle route with a named decision-maker on each side is not a refinement but a requirement. And the buyer must specify the **data** it receives — cost and progress at a granularity that lets its own earned-value system run across the boundary — because a governance body reading the supplier's own summary is governing the summary.

Performance regimes: what a cap actually does. Service credits, liquidated sums for late delivery and KPI-linked fee adjustments all work by making failure expensive. Whether they work at all depends on a comparison almost nobody makes: **the cap against the supplier's cost of compliance.**

Auriga's five-year support contract runs at **USD 2,200,000** a year with service credits capped at **5 %** — **USD 110,000**. Meeting the availability target requires the supplier to hold a standby crew costing **USD 180,000** a year. The supplier's arithmetic is immediate: fail, pay the cap, and save **USD 70,000**. The regime has not deterred non-performance; it has **priced** it, and at a discount. Two different caps follow from two different purposes, and confusing them is the design error. For the credits to deter, the cap must exceed the cost of compliance — at least `180,000/2,200,000 =` **8.18 %** of contract value. For the credits to compensate, they must cover the buyer's own loss from unavailability, assessed at **USD 320,000** a year, which needs `320,000/2,200,000 =` **14.55 %**. At the 5 % cap actually agreed, the credits recover **34.38 %** of the buyer's loss and deter nothing. The professional discipline is to state which purpose the regime serves, compute the cap that purpose requires, and — where the required cap is unobtainable in negotiation — record that the residual exposure is the buyer's, so that it appears in the risk register at its real size rather than being assumed away because "there are service credits".

Relationship management, and why it is not softness. Suppliers allocate their best people by judgement, not by contract, and they allocate them away from clients who are slow to pay, slow to decide, late with access and inclined to treat every issue as a breach. The behaviours that keep a supplier's A-team on a project are concrete and cheap: decide inside the contractual period, pay to terms, give access and information when promised, escalate issues before they become entitlements, and keep the commercial conversation separate from the delivery one so that neither poisons the other. **A buyer who is difficult to work with pays for it in the next tender and in every discretionary decision the supplier takes in between** — a real cost that appears in no report.

AI in this KA

Where it earns its place. Contract analytics at scale: extracting obligations, notice periods, liability caps, change mechanisms and termination rights from a portfolio of agreements into a structured obligations register, and flagging inconsistencies between the main agreement and its schedules. Modelling the payment mechanism across cost outcomes — the whole of 10.3.2, swept over share ratios, ceilings and distributions — which is deterministic and tedious. Monitoring supplier performance data for trend breaks. Comparing a supplier's reported progress with its own historical reporting patterns to flag optimistic drift.

Where it must not go. It must not give legal or commercial advice, and it must not draft a clause that is signed without qualified review; the boundary Domain 7 states holds here in full. It must not set a liability cap, a share ratio or a target cost — those are risk-appetite decisions belonging to accountable people. It must not generate the supplier's cost distribution: a model asked for probabilities will supply them, and they will look exactly like data. And an AI reading of a clause is a summary; the obligation is the clause.

Verification, concretely. Every extracted obligation is cited to a clause number and spot-checked at a stated sample rate before the register is relied on. Fee, share and `PTA` arithmetic is reproduced by hand for at least three cost points, including one above the `PTA`, because that is where the mechanism changes shape. Any distribution used in an incentive model carries a named source, and the paper states the result at the pessimistic and optimistic ends, not only at the mean.

KEY TERMS — KA 10.3

Term	Meaning
Marginal-dollar allocation identity	For any payment mechanism, buyer share + supplier share of the next dollar of cost = 1; a contract allocates cost, it does not remove it.
Point of total assumption (PTA)	The cost above which the supplier bears 100 % of further overrun (Domain 7, KA 7.4.3); read here as the point at which behaviour turns from efficiency to entitlement.
Risk-neutral fixed price	Expected cost plus the required margin; at this price, transferring risk costs the buyer nothing in expectation and buys variance reduction only.
Risk loading	The premium a risk-averse supplier adds above the risk-neutral price; the arguable, visible price of certainty.
Target fee vs expected fee	The fee at the target cost, against the probability-weighted fee actually expected; any real overrun risk makes the second lower than the first.
Influence test	A risk should be allocated to the party that can influence it; a risk allocated elsewhere is priced, not managed.
Deterrence cap vs compensation cap	The service-credit cap needed to exceed the supplier's cost of compliance, against the cap needed to cover the buyer's loss; they differ.
Out-of-cycle route	Domain 3's mechanism (KA 3.3.3) applied across a contract boundary: named authorities on **both** sides, since a route that exists for only one party is not a route.

SAMPLE MCQS — KA 10.3

MCQ 10.3-A `[10.3.2 · Application]` A supplier's cost outcomes are 1,850,000 (0.20), 2,000,000 (0.40), 2,300,000 (0.30) and 2,600,000 (0.10), and it requires a 150,000 expected margin. The risk-neutral firm fixed price is: - A. USD 2,150,000 - B. USD 2,270,000 - C. USD 2,120,000 - D. USD 2,450,000

Rationale: `E[cost] = 2,120,000`, so the price is `2,120,000 + 150,000 = 2,270,000` (10.3.2). A adds the margin to the target cost rather than the expected cost — the commonest error, and it understates by 120,000. C is the expected cost with no margin. D is the ceiling of the target-cost alternative.

MCQ 10.3-B `[10.3.2 · Analysis]` Under a target-cost contract with a 70/30 share, above the point of total assumption the buyer's and supplier's shares of the next dollar of cost are: - A. 0.70 and 0.30, unchanged - B. 0.00 and 1.00 - C. 1.00 and 0.00 - D. 0.50 and 0.50

Rationale: Above the `PTA` the ceiling binds the buyer, so the supplier carries every further dollar (Domain 7 KA 7.4.3; 10.3.2). A misses that sharing has stopped; C reverses the exposure; D invents a split.

MCQ 10.3-C `[10.3.2 · Evaluation]` The expected buyer outturn is 2,270,000 under both firm fixed price and cost plus fixed fee, while the outturn standard deviation is 230,434 under cost-plus and zero under fixed price. The correct conclusion is that at risk-neutral pricing, fixed price: - A. is cheaper in expectation and therefore always preferable - B. costs the same in expectation and buys variance reduction, whose value depends on the buyer's ability to absorb variance — and on the supplier's solvency - C. is more expensive in expectation by the risk premium - D. eliminates the cost risk from the project

Rationale: The premium equals the expected cost of the risk, so nothing is gained in expectation except certainty (10.3.2). D contradicts the marginal-dollar identity — the risk is allocated, not removed, and returns if the supplier fails.

MCQ 10.3-D [10.3.2 · Analysis] As specified, the target-cost structure gives the buyer an expected outturn 48,000 below the fixed price and the supplier an expected margin 48,000 below its requirement. The most useful inference is that: - A. the target-cost structure is superior for the buyer and should be used - B. an incentive structure reallocates value unless it changes behaviour; the supplier will raise the target cost or fee, or decline - C. the supplier has mis-stated its cost distribution - D. the share ratio should be 50/50

Rationale: The buyer's gain is exactly the supplier's loss, so no value is created (10.3.2). A treats a transfer as an improvement and will not survive negotiation; C is possible but not inferable; D is an unmotivated fix.

MCQ 10.3-E [10.3.3 · Evaluation] Service credits are capped at 5 % of a 2,200,000 contract. Compliance costs the supplier 180,000 a year and non-performance costs the buyer 320,000 a year. The regime: - A. deters non-performance, because 110,000 is a material sum - B. prices non-performance at a 70,000 discount to compliance, and recovers only 34.38 % of the buyer's loss - C. is adequate because service credits are a secondary remedy - D. should be capped at 5 % of the buyer's loss instead

Rationale: `0.05 × 2,200,000 = 110,000 < 180,000`, so failing is the supplier's cheaper course; and `110,000/320,000 = 34.38 %` of the buyer's loss (10.3.3). Deterrence needs 8.18 %, compensation 14.55 %.

SELF-CHECK — KA 10.3

1. State the marginal-dollar allocation identity and its consequence. — Buyer share + supplier share of the next dollar of cost equals 1; a contract allocates cost, it cannot remove it.
2. Why does a supplier expect less than the target fee? — Because the target fee is the fee at the target cost; any real probability of overrun reduces the probability-weighted fee below it.
3. What two different caps can a service-credit regime need, and why? — A deterrence cap exceeding the supplier's cost of compliance (8.18 % on Auriga's support contract) and a compensation cap covering the buyer's loss (14.55 %); they serve different purposes and rarely coincide.

KNOWLEDGE AREA 10.4

Claims, change and disputes; ethical sourcing; supply resilience

Topics: 10.4.1 claims awareness and the anatomy of an entitlement · 10.4.2 dispute avoidance and the escalation ladder · 10.4.3 ethical and sustainable sourcing · 10.4.4 supply-chain resilience: single, dual-split and dual-qualified sourcing.

10.4.1 Claims awareness and the anatomy of an entitlement

The professional position. A project leader is not a lawyer and should not act as one. What a leader must be able to do is recognise a claim situation as it forms, preserve the record that will decide it, and quantify the exposure well enough to escalate it accurately. **Claims are decided on contemporaneous records, and the records are made — or not made — long before anyone uses the word claim.**

The four elements of a claim, all of which must be present. **Entitlement** — an identified contractual basis: an instruction, a variation, a relevant event, a breach. Without it the merits do not matter. **Causation** — a demonstrated link from that basis to the effect claimed, which is where most claims are actually won or lost, because concurrent causes and the claimant's own delays are pleaded against it. **Quantum** — the money, built up by head of cost and evidenced. **Notice** — given in the form and within the period the contract requires, which in many contracts is a precondition rather than a courtesy.

The heads of cost, and what each requires. Direct cost — additional labour, plant and materials, evidenced by records at the time. Prolongation — time-related site and management costs for the period of extended duration; requires a demonstrated critical-path effect (Domain 6), not merely a late finish. Disruption — lost productivity on work that was not itself changed, the hardest head to prove and usually established by a **measured mile**: comparing productivity in an unimpacted period with the impacted one. Finance cost on the funded difference, where the contract or the governing law allows it. Overhead and profit, at the contractual percentage. Acceleration and mitigation costs, where instructed or reasonably incurred.

WORKED EXAMPLE

Worked example 10.4.1 — a variation claim, and what a notice provision puts at risk.

1. **Setup.** Auriga's installation supplier is instructed to relocate two remote-terminal-unit cabinets. It claims: **240 hours** of additional labour at the blended engineering rate of **USD 130.625** per hour (Domain 7, KA 7.4.1); **USD 18,400** of materials; **1.5 weeks** of prolongation at **USD 14,000** per week of site establishment; and disruption on **1,204** planned hours of unaffected work whose productivity fell to a factor of **0.86** of the measured-mile baseline. Overhead and profit is **12 %** by the contract. The instruction issued on day 1; the supplier gave notice on **day 41**, against a contractual requirement of **28 days**, and under this contract the prolongation and disruption heads depend on valid notice.
2. **Formula.** Claim `= (direct + prolongation + disruption) × (1 + OH&P)`, with disruption `= (planned hours ÷ productivity factor − planned hours) × rate`. Notice-bar exposure `=` (heads dependent on notice) `× (1 + OH&P)`.
3. **Substitution.** Labour `240 × 130.625`; disruption hours `1,204/0.86 = 1,400`, so extra `1,400 − 1,204 = 196` hours at 130.625; prolongation `1.5 × 14,000`. Subtotal then `× 1.12`.
4. **Result.** Labour **USD 31,350.00**; materials 18,400; prolongation **USD 21,000**; disruption **USD 25,602.50**. Subtotal **USD 96,352.50**; overhead and profit **USD 11,562.30**; **total claim USD 107,914.80** (≈ SAR 404,681 indicatively). The heads exposed to the notice bar total **USD 52,194.80 — 48.37 %** of the claim — leaving **USD 55,720.00** unaffected.
5. **Interpretation.** Nearly half of a properly constructed claim is at risk from a diary entry nobody made, and that is the single most useful thing a delivery leader can know about claims: **notice provisions destroy more value than negotiation does, and they do it silently.** The lesson runs in both directions, which is what makes it a professional rather than a tactical point. As a seller, the discipline is a notice register — every instruction, every relevant event, logged on the day with its notice deadline computed, because the 28 days runs from the event and not from the moment the cost becomes obvious. As a buyer, resisting a genuine entitlement purely on a time bar buys a number and sells a relationship: the supplier's response is predictable and expensive — its A-team moves, its pricing of every subsequent variation stiffens, and its own notices thereafter arrive on day one for everything, generating administrative load out of all proportion to the 52,194.80. Three further cautions. **Whether a notice provision of this kind operates as a condition precedent varies by jurisdiction and by the law of the contract**; in some legal systems such bars are enforced strictly, in others they may be read down or fall foul of statutory controls, and nothing here is legal advice or a statement of any particular jurisdiction's law. The measured-mile figure is only as good as the comparison period — an unimpacted baseline drawn from a different crew, season or work type is not a measured mile, and this is the head most often successfully challenged. And the productivity arithmetic must be done the right way round: dividing by 0.86 (giving 196 extra hours) is correct, whereas multiplying by 0.14 gives 168.56 hours and understates the claim by 27.44 hours, or **USD 3,584.35** before overhead and profit.

10.4.2 Dispute avoidance and the escalation ladder

The principle. Disputes are expensive in a way that is systematically mis-estimated, because the irrecoverable cost of resolving one is compared with the amount in issue only after the decision to fight has already been taken.

The ladder, and what each rung costs. Direct negotiation between the people accountable for delivery; then structured negotiation between executives outside the project; then a neutral evaluation or mediation; then a determinative but interim process — adjudication, expert determination or a dispute board, depending on the contract and the jurisdiction; then arbitration or litigation. Cost and elapsed time rise by roughly an order of magnitude across the ladder, and **the relationship's usable life ends somewhere around the fourth rung**, which matters when the supplier still has work to do.

WORKED EXAMPLE

Worked example 10.4.2 — the arithmetic of settling.

1. **Setup.** A **USD 400,000** claim is in dispute at week 30 of Auriga. Settlement now at **55 %** — USD 220,000 — is achievable, with **USD 15,000** of internal and advisory cost and **3 weeks** to conclude. The alternative is arbitration: the buyer's advisers assess the outcomes as the claim largely upheld (400,000) with probability **0.55**, partly upheld (180,000) with **0.30**, and dismissed (nil) with **0.15**; the buyer's own **irrecoverable** costs are **USD 340,000** whatever the result, and the process takes **78 weeks**.
2. **Formula.** Expected award `= Σ p × award`. Expected total `=` expected award `+` irrecoverable cost. Compare with the negotiated total.
3. **Substitution.** `0.55 × 400,000 + 0.30 × 180,000 + 0.15 × 0 = 220,000 + 54,000 + 0`. Expected total `= 274,000 + 340,000`. Negotiated total `= 220,000 + 15,000`.
4. **Result.** Expected award **USD 274,000 — 68.50 %** of the claim, above the 55 % on offer. Expected total cost of arbitrating **USD 614,000** against **USD 235,000** for settling: a saving of **USD 379,000**, or **61.73 %**, and 75 weeks of elapsed time.
5. **Interpretation.** The decisive figure is not the 379,000 but the structural one behind it: **the irrecoverable cost of the formal route, USD 340,000, exceeds the entire negotiated settlement of 235,000.** On these numbers the buyer cannot come out ahead by arbitrating even if it wins outright, because a total victory saves the 235,000 it would have paid and spends 340,000 to do it — a net loss of 105,000 in the best case. That is the arithmetic that should be on the table before anyone instructs anybody, and it is very often not, because the comparison usually made is between the settlement and the expected award (274,000 against 220,000), which favours fighting and omits the cost of doing so. Three cautions keep this honest. Costs recovery, interest and the availability of interim determinative processes **vary substantially by jurisdiction and by forum**, and the 340,000 irrecoverable assumption must be replaced with advice specific to the contract's governing law — this is illustrative arithmetic, not a rule. Some disputes should be fought despite the arithmetic: where a point of principle will otherwise be conceded across a portfolio of contracts, where a regulator or auditor requires the matter tested, or where settling would create a precedent worth more than 379,000 to the other side. And the 78 weeks carries its own cost beyond money — the management attention consumed, and the supplier's behaviour on the remaining work while the matter is live.

The avoidance machinery, which is cheaper than any rung of the ladder. A single agreed set of records, maintained jointly, so that the facts are not themselves in dispute. Early-warning obligations running both ways with short periods, so that problems arrive as information rather than as entitlements. A change mechanism fast enough to be used — Domain 4's integrated change control across the boundary, with the threshold mismatch of 10.1.1 resolved. A named commercial lead on each side with authority to settle at a stated value, and an out-of-cycle route above it (Domain 3, KA 3.3.3). And a scheduled joint review of open commercial items, so that they are closed at fifty rather than argued at five hundred.

10.4.3 Ethical and sustainable sourcing

The obligation, stated carefully. Buyers increasingly carry duties in respect of conditions in their supply chains — labour standards and forced labour, health and safety, environmental impact, anti-bribery and sanctions compliance, and the accuracy of what they publish about all of it. Several jurisdictions have enacted human-rights, modern-slavery or supply-chain due-diligence legislation, and their scope, thresholds and reporting requirements **differ materially**; the applicable regime depends on where the buyer is established, where it sells, and the contract's governing law. Nothing in this section states the requirements of any particular jurisdiction, and none of it is legal advice. What is portable is the method: risk-tier the supply base, apply proportionate diligence, contract for the standard and for the right to verify it, and act on what verification finds.

What "contracting for it" means concretely. A stated standard the supplier must meet, not an aspiration. A right of audit — including of sub-tiers — with a duty to disclose them, which is also the control that makes 10.4.4's resilience analysis possible. A duty to notify adverse findings. A remediation-first response, because terminating a supplier where a labour abuse is found removes the buyer's leverage and frequently harms the workers concerned; termination belongs at the end of an escalation, not the start. And flow-down obligations, since the risk concentrates where the buyer's visibility ends.

The arithmetic, and its honest limits. Consider a proportionate programme across Auriga's supply base: **34** first-tier suppliers desktop-screened at **USD 1,200** each, and the **6** assessed as highest risk audited on site at **USD 14,000** each — a programme cost of **USD 124,800**. Against it: the assessed probability of a material breach in the supply chain over a three-year window is **0.08**, with a consequence — remediation, re-procurement, contractual and reputational cost — assessed at **USD 1,900,000**, giving an expected exposure of **USD 152,000** (`EMV`, Domain 8 KA 8.2.2). If the programme is **70 %** effective at detecting and preventing, it avoids `0.70 × 152,000 =` **USD 106,400** — which is **USD 18,400 less** than it costs.

That result is uncomfortable and it is stated deliberately, because pretending otherwise is worse. Three things follow. First, the breakevens are useful and should be in the paper: the programme pays at a detection effectiveness above **82.11 %**, or at a breach probability above **9.38 %** — both entirely plausible in higher-risk categories and geographies, which is precisely why the programme should be **risk-tiered rather than uniform**, concentrating spend where the probability is highest instead of screening everyone equally. Second, `EMV` is a poor instrument here for the reason Domain 8 gives: it is an average of outcomes that will not individually occur, and the consequence distribution for a serious supply-chain finding is heavily skewed — the 1,900,000 is a mean whose upper tail includes losing a licence, a listing or a market. A decision taken on the mean of a fat-tailed distribution is a decision taken on the wrong statistic. Third, and decisively: **where legal duties apply, this is not an expected-value decision at all.** Compliance is not optional because the arithmetic is marginal, and where the obligation is one of values rather than law, the organisation should say so plainly rather than construct a business case it does not believe. The arithmetic's proper job is to allocate the programme's effort, not to decide whether to have one.

10.4.4 Supply resilience: single, dual-split and dual-qualified sourcing

The principle. Resilience is bought, and the thing being bought is the ability to keep going when a supplier cannot deliver. The mistake made almost universally is to treat "two suppliers" as the product being purchased, when what actually reduces exposure is **a qualified, exercisable alternative** — and those are not the same purchase.

```
Expected cost of disruption = Σ P(state) × consequence(state)
Total expected cost = certain cost + expected cost of disruption
Breakeven disruption probability  p* = extra certain cost ÷ reduction in consequence
```

WORKED EXAMPLE

Worked example 10.4.4 — Auriga's 84 controllers: three sourcing structures.

1. **Setup.** Domain 8 carries controller lead-time slip as R1. Three structures are on the table. **Option 1, single source:** supplier A at **USD 9,600** per unit, one supplier qualified at **USD 40,000**. **Option 2, dual source with a 60/40 volume split:** A at 9,600 and B at **USD 10,600**, both qualified (**USD 80,000**), plus **USD 20,000** to hold two build configurations to a common specification. **Option 3, single award with a qualified alternate:** full volume to A at 9,600, both suppliers qualified (80,000), and **USD 25,000** a year to keep the alternate exercisable — specification maintenance and an annual sample build. Each supplier has an independent **0.18** probability of a disruption in the delivery window. Consequences: a single supplier disrupted with no qualified alternate requires a 14-week re-source — `14 × 45,000 = 630,000` of delay plus 120,000 of re-qualification and rework plus a 150,000 contractual step — **USD 900,000**; under the dual split, one supplier disrupted means a 3-week surge ramp on the other, `3 × 45,000 = 135,000` plus a 60,000 surge premium — **USD 195,000**; with a qualified alternate, activation takes 5 weeks — `5 × 45,000 = 225,000` plus 70,000 of switching and re-qualification — **USD 295,000**.
2. **Formula.** Certain cost `= units × unit price + qualification + holding cost`. Expected disruption `= Σ P(state) × consequence`, with `P(exactly one of two) = 2p(1 − p)` and `P(both) = p²` under independence. Breakeven `p* =` extra certain cost ÷ consequence reduction.
3. **Substitution.** Option 1: `84 × 9,600 + 40,000`, disruption `0.18 × 900,000`. Option 2: blended unit `0.60 × 9,600 + 0.40 × 10,600 = 10,000`, so `84 × 10,000 + 80,000 + 20,000`; disruption `2 × 0.18 × 0.82 × 195,000 + 0.18² × 900,000`. Option 3: `84 × 9,600 + 80,000 + 25,000`, disruption `0.18 × 295,000`.
4. **Result.**

	Certain cost (USD)	Expected disruption (USD)	Total expected (USD)
Option 1 — single source	846,400	162,000	**1,008,400**
Option 2 — dual source, 60/40 split	940,000	86,724	**1,026,724**
Option 3 — qualified alternate	911,400	53,100	**964,500**

Option 3 is best: **USD 43,900** better than single sourcing and **USD 62,224** better than the dual split. Its breakeven disruption probability against Option 1 is `65,000/(900,000 − 295,000) =` **10.74 %**; the dual split's, from `510,000p² − 510,000p + 93,600 = 0`, is **24.22 %**. The maximum premium worth paying for Option 3's resilience — the whole reduction in expected disruption cost — is `162,000 − 53,100 =` **USD 108,900**, or **12.87 %** of the single-source cost. 5. **Interpretation.** The result that carries the teaching is the ranking: **the resilience is in the qualified alternate, not in the split volume.** Splitting the order does two things the buyer rarely accounts for. It pays a price premium — the blended unit cost rises from 9,600 to 10,000, worth 33,600 on this volume — and, under independence, it **doubles the exposure surface**: the probability that something goes wrong rises from 0.18 to `1 − 0.82² =` 0.3276, and although each individual event is now far cheaper, the arithmetic has to earn back the premium before it helps. Dual qualification, by contrast, retains the full volume with the single supplier (and its discount) and buys only the option, at 65,000 of extra certain spend against a 108,900 reduction in expected exposure. That leads to the general design rule: **qualify two, award one, and keep the second exercisable** — with the emphasis on exercisable, because a second supplier that has not built the part for three years is a name on a list, not an alternative, and the 25,000 a year is what distinguishes them. The breakevens are the negotiating numbers: Option 3 pays whenever the disruption probability exceeds **10.74 %**, while the dual split needs more than **24.22 %** — so a colleague arguing for split sourcing on resilience grounds is implicitly asserting a disruption probability more than twice as high as the one that justifies the cheaper structure, and should be asked for it.

The caution that overturns everything above: correlation. All three calculations assume the two suppliers fail independently. Suppose instead that both draw a critical module from a single sub-tier source, so that the joint probability of disruption is **0.12** rather than 0.0324 (each supplier's marginal probability still 0.18, leaving `0.18 − 0.12 =` 0.06 of idiosyncratic risk each). Then `P(exactly one) = 2 × 0.06 =` 0.12, and the expected disruption costs become `0.12 × 195,000 + 0.12 × 900,000 =` **USD 131,400** for Option 2, total **1,071,400**, and `0.06 × 295,000 + 0.12 × 900,000 =` **USD 125,700** for Option 3, total **1,037,100**. Both are now **worse than single sourcing's 1,008,400**: the buyer has paid for resilience and bought exposure. The rule to take away is blunt — **two suppliers with one sub-tier is one supplier with two invoices** — and the correct response is not to choose between the options but to attack the sub-tier: require disclosure of the critical sub-tier sources as a contractual obligation (10.4.3), qualify a second module design, or hold buffer stock sized to the re-source interval. This is Domain 8's correlation result (KA 8.A.1) in a commercial setting, and it is the reason a supply network must be mapped rather than assumed.

AI in this KA

Where it earns its place. Building the notice register of 10.4.1 — extracting every instruction, relevant event and correspondence item from a project's document set, computing each notice deadline and flagging what is approaching or missed — is high-volume, deterministic work with an unambiguous right answer, and it is exactly the work that goes undone. Assembling a chronology from thousands of documents for a delay analysis, for expert verification. Screening a supply base against sanctions, adverse-media and insolvency signals, and flagging changes. Reading a set of supplier disclosures to build a sub-tier map and identify shared dependencies — the concentration that 10.4.4 shows destroys a resilience case. Modelling sourcing options and their breakevens, as computed above.

Where it must not go. It must not assess entitlement, causation or the strength of a claim; those are legal judgements on which qualified advice is required and on which a confident, unattributable opinion is actively dangerous. It must not generate a delay analysis presented as evidence — a chronology is an input to an expert's opinion, not the opinion. It must not decide a supplier's ethical standing from inference: a sanctions or adverse-media hit is a prompt to investigate, and treating it as a finding is both unfair and, in some jurisdictions, unlawful. And it must not supply the probabilities in 10.4.4; those come from delivery history and market intelligence, and a model asked for them will produce numbers indistinguishable from data.

Verification, concretely. Every notice deadline the tool computes is checked against the clause and the event date before anyone relies on it, and the notice register is reconciled to the correspondence log at a stated sample rate. Every chronology entry cited in a claim is verified against the source document. Every screening hit is human-reviewed before any action, with the review recorded. And the sourcing arithmetic is reproduced by hand at two probabilities — the point estimate and the breakeven — because the breakeven is the number that decides whether the recommendation survives an argument.

KEY TERMS — KA 10.4

Term	Meaning
Entitlement	The contractual basis for a claim; without it, merits are irrelevant.
Causation	The demonstrated link from the contractual basis to the effect claimed; where most claims are decided.
Quantum	The claim's money, built up by head of cost and evidenced contemporaneously.
Notice provision	The required form and period for notifying a claim; in some contracts and jurisdictions a condition precedent.
Prolongation	Time-related costs of extended duration; requires a demonstrated critical-path effect.
Disruption / measured mile	Lost productivity on unchanged work, established by comparing an unimpacted period's productivity with the impacted period's.
Notice-bar exposure	The value of the heads of claim that depend on valid notice, grossed up for overhead and profit.
Irrecoverable cost	The share of a formal dispute's cost that cannot be recovered whatever the outcome; the figure that usually decides whether fighting can pay.
Risk-tiered diligence	Concentrating supply-chain assurance effort where breach probability is highest, rather than screening uniformly.
Qualified alternate	A second supplier approved and kept exercisable, receiving no volume; the structure that actually buys resilience.
Sub-tier concentration	A shared dependency below the first tier that correlates suppliers' failures and destroys a dual-sourcing case.
Expected cost of disruption	Σ P(state) × consequence(state); compared against the certain cost of the structure that reduces it.

SAMPLE MCQS — KA 10.4

MCQ 10.4-A `[10.4.1 · Application]` Disruption is claimed on 1,204 planned hours whose productivity fell to 0.86 of the measured-mile baseline, at USD 130.625 per hour. The disruption cost is: - A. USD 22,018.15 - B. USD 25,602.50 - C. USD 29,770.35 - D. USD 157,272.50

Rationale: Hours required `= 1,204/0.86 = 1,400`, so extra hours `= 196` and cost `= 196 × 130.625 = 25,602.50` (10.4.1). A multiplies the planned hours by `(1 − 0.86)` instead of dividing by 0.86 — 168.56 hours rather than 196, understating by USD 3,584.35. C applies the correct division to 1,400 hours instead of 1,204 (227.91 extra hours). D prices all 1,204 hours rather than the extra ones.

MCQ 10.4-B `[10.4.1 · Analysis]` A USD 107,914.80 claim comprises labour and materials of 49,750 and prolongation and disruption of 46,602.50, each grossed up by 12 % overhead and profit. Notice was given on day 41 against a 28-day requirement, and the prolongation and disruption heads depend on notice. The amount at risk is: - A. USD 46,602.50 - B. USD 52,194.80 - C. USD 55,720.00 - D. USD 107,914.80

Rationale: `46,602.50 × 1.12 = 52,194.80`, 48.37 % of the claim (10.4.1). A omits the overhead and profit attaching to those heads; C is the surviving amount; D assumes the whole claim falls, which the direct heads do not.

MCQ 10.4-C `[10.4.2 · Evaluation]` A 400,000 claim can be settled for 220,000 plus 15,000 of costs. Arbitration has an expected award of 274,000 and irrecoverable costs of 340,000. The strongest argument for settling is that: - A. the expected award of 274,000 exceeds the 220,000 settlement - B. arbitration takes 78 weeks - C. the irrecoverable cost of 340,000 exceeds the entire negotiated settlement of 235,000, so arbitrating cannot pay even on a total win - D. settlements are always cheaper than awards

Rationale: A actually argues against settling and is the comparison usually made; it omits the cost of obtaining the award. B is real but secondary. C is decisive: a total victory saves 235,000 and spends 340,000 (10.4.2). D is an unsupported generalisation.

MCQ 10.4-D `[10.4.4 · Application]` Single sourcing costs 846,400 with a 0.18 probability of a 900,000 disruption. A qualified alternate adds 65,000 of certain cost and cuts the consequence to 295,000. The breakeven disruption probability is: - A. 7.22 % - B. 10.74 % - C. 18.00 % - D. 22.03 %

Rationale: `p* = 65,000/(900,000 − 295,000) = 65,000/605,000 = 10.74 %` (10.4.4). A divides by the 900,000 consequence rather than the reduction; C restates the assumed probability; D divides by 295,000.

MCQ 10.4-E `[10.4.4 · Analysis]` Two suppliers each have a 0.18 disruption probability, but both draw a critical module from one sub-tier source, so the joint probability is 0.12 rather than 0.0324. The effect on a dual-split sourcing case is that expected disruption cost: - A. falls, because two suppliers are still better than one - B. is unchanged, since the marginal probabilities are unchanged - C. rises from 86,724 to 131,400, making the dual split worse than single sourcing - D. rises, but the dual split remains the best option

Rationale: `P(exactly one) = 2(0.18 − 0.12) = 0.12`, so expected cost is `0.12 × 195,000 + 0.12 × 900,000 = 131,400`, taking the total to 1,071,400 against single sourcing's 1,008,400 (10.4.4). B mistakes marginals for the joint distribution, which is the error the whole example exists to expose.

MCQ 10.4-F `[10.4.3 · Evaluation]` A 124,800 supply-chain diligence programme avoids an expected 106,400 of exposure. The correct professional conclusion is that: - A. the programme should be cancelled, since it fails an expected-value test - B. the programme should be risk-tiered to raise effectiveness where probability is highest, the breakevens (82.11 % effectiveness, 9.38 % probability) stated, and any legal duty met regardless of the arithmetic - C. the consequence figure should be raised until the business case works - D. expected monetary value is the correct basis for this decision

Rationale: The arithmetic allocates effort; it does not decide whether a legal or values obligation applies, and the consequence distribution is fat-tailed so its mean is the wrong statistic (10.4.3, Domain 8 KA 8.2.2). C is the manipulation the honest presentation of breakevens is designed to prevent.

SELF-CHECK — KA 10.4

1. Name the four elements of a claim and the one on which most claims are decided. — Entitlement, causation, quantum and notice; causation decides most of them.
2. Which figure usually decides whether a formal dispute can pay, and why is it missed? — The irrecoverable cost, because the comparison usually made is between the settlement and the expected award, which omits the cost of obtaining it.
3. Why is dual-splitting an order a weaker resilience purchase than dual qualification? — It pays a price premium and, under independence, doubles the exposure surface, while the option value comes from having an exercisable alternate, which dual qualification buys without either cost.

Advanced topics — Domain 10

10.A.1 Supply-network mapping and the visibility that stops at tier one

Almost every organisation's supplier data stops at the parties it pays. Almost every serious supply disruption originates below that line. The consequence is a systematic error in the direction of optimism: exposure is assessed on the first tier, where it is diversified, and materialises in the lower tiers, where it is concentrated.

Mapping is therefore a deliberate exercise with a defined scope, and the scope should be set by criticality rather than by spend — the cheapest component on a critical path can stop a project that a major package cannot. For each critical item, the questions are: which sub-tier sources does each of my suppliers depend on; do any of my suppliers share one; what is the single-source geography, qualification or intellectual-property constraint that makes substitution slow; and what is the **re-source interval** — the elapsed time from a failure to an alternative in production, which is the figure that converts a supply risk into a schedule risk at the project's cost of delay. The contractual enabler is a disclosure obligation with an audit right (10.4.3), because a supplier will not volunteer a dependency that is also a commercial vulnerability.

Two structural insights are worth carrying. Domain 4's interface arithmetic applies to supply networks: `n` suppliers who must interoperate create up to `n(n−1)/2` pairwise relationships, and a buyer who splits packages to gain price tension acquires all of them (Domain 4, KA 4.2.3). And Domain 8's correlation result governs the aggregate: a register with many supplier risks and few underlying drivers is concentrated, however it is presented (Domain 8, KA 8.A.1). Restructuring a supplier risk register **by shared driver** rather than by supplier is a half-day exercise that regularly reveals a single point of failure behind a dozen entries — and, as 10.4.4 computes, a single shared sub-tier is enough to make a paid-for dual-sourcing arrangement worse than the single source it replaced.

10.A.2 Contracting for AI-delivered and outcome-based services

Where a supplier's deliverable is produced with, or consists of, an AI-enabled service, several contractual assumptions that normally go unexamined stop holding, and the gaps must be closed deliberately rather than left to a standard schedule.

The deliverable can change without a variation. A model updated by the supplier can alter output quality, latency and behaviour while the contract's description of the service remains satisfied. Insist on a defined baseline of behaviour with measurable acceptance criteria, notice of material model or version changes, and a right to re-test against the baseline after one — otherwise the buyer has bought a moving target and has no mechanism to notice it moving.

Data rights need four separate answers, not one. Who owns the input data; what the supplier may do with it, specifically including whether it may train models serving other customers; what happens to derived data, embeddings and model improvements at termination; and which sub-processors and jurisdictions are involved. "Confidentiality" answers none of these.

Performance obligations must be honest about probabilistic output. A service whose output is correct with some probability cannot be warranted as correct, and a supplier accepting such a warranty has either mispriced it or will resist it in practice. The workable structure specifies accuracy against a defined test set, a human-verification obligation on the buyer's side for consequential uses, and an allocation of responsibility for errors that reflects who was required to verify — the contractual expression of the family principle that **AI proposes; the professional verifies, decides and remains accountable**. If the contract does not say who the professional is, no one is.

Outcome-based structures inherit the same problem commercially. Paying for an outcome shifts the burden onto a definition that must now bear weight: what is measured, who measures it, what happens when the measure is disputed, and what proportion of the outcome the supplier actually controls. Where a supplier controls only part of it, an outcome-based payment is a lottery with a margin attached, and it will be priced as one.

10.A.3 The reviewer's procurement eye

Invariants to test on any procurement or contract arrangement, each cheap and each diagnostic.

Every make-or-buy decision states a **total cost of ownership including a non-zero exit cost** and a **breakeven volume**, and where the capability is on the critical path, the stand-up delay is priced at the cost of delay. The procurement chain for every long-lead item has a **computed total lead time** including governance latency, and is a predecessor in the schedule rather than an assumption behind it. The evaluation model — weights, sub-weights, normalisation formula, scoring scale, moderation and non-compliance treatment — was **fixed and disclosed before the receipt deadline**, with a dated record, and quality was scored before price was seen. Any quality premium has been tested against the expected cost it avoids, and the ratio is stated. Every bid materially below its benchmark has a recorded explanation, with the benchmark named. Every payment mechanism has its **marginal-dollar allocation** written down, and every target-cost contract has its PTA computed and known to the delivery team before mobilisation, not after. Every performance regime states whether its cap is set to deter or to compensate, and shows the arithmetic for whichever it claims. Every contract's notice periods are in a live **notice register** with computed deadlines. Every critical item has a mapped sub-tier and a stated **re-source interval**. And the resilience claim for every dual-sourced item is tested for **correlation**, because a resilience arrangement that has not been tested for a shared sub-tier is an assertion, and 10.4.4 shows what the assertion costs when it is wrong.

Industry variations — Domain 10

- **Public sector and regulated buyers.** Procurement route, timescales, publication, evaluation disclosure, standstill and challenge rights may be prescribed by law, and those rules — which differ by jurisdiction — override commercial preference. The practical effects are that the chain of 10.1.3 is longer and largely incompressible, that the model-fixing discipline of 10.2.3 is a legal requirement rather than good practice, and that the leader's remaining levers are the paper deadline, the delegation of award authority, and early market engagement before the formal process begins.
- **Construction and infrastructure.** Standard-form contracts dominate, with well-developed variation, extension-of-time, notice and dispute machinery, and interim determinative processes available in some jurisdictions. Notice discipline (10.4.1) carries more value here than anywhere else, retention and payment cycles drive cash (Domain 7, KA 7.4.4), and multi-party governance follows the contract structure rather than the organisation chart (Domain 3, KA 3.1.2).
- **Technology and software services.** Scope is defined late by nature, so fixed price on undefined scope is the characteristic error and the AI-service gaps of 10.A.2 are the characteristic gap. Exit cost is systematically understated because data extraction, re-implementation and licence

discontinuity are discovered only at termination — which is exactly why the make-or-buy exit column matters most in this sector.

- **Energy, utilities and process industries.** Long-lead, engineered-to-order equipment with limited qualified sources; the re-source interval of 10.A.1 is measured in quarters, so 10.4.4's qualified alternate and buffer strategies dominate, and the procurement chain frequently starts before the design is complete. Auriga is this shape.
- **Pharmaceutical and life sciences.** Supplier qualification is a regulated activity: changing a supplier or a manufacturing site may require re-validation and regulatory notification, which makes the alternate expensive to keep exercisable and the switching consequence far larger than a commercial calculation suggests. Dual qualification must be planned at development, not at supply.
- **Consumer manufacturing and retail.** High volumes make the unit-price term dominant, so the make-or-buy breakeven of 10.1.2 is often comfortably passed — but the ethical-sourcing exposure of 10.4.3 is at its most acute, sub-tiers are numerous and opaque, and reputational consequence distributions are the most heavily skewed of any sector.

Case study — Domain 10: the weighting confirmed after the envelopes (utilities, Auriga)

Situation. Auriga's installation package attracted three compliant bids: Alpha USD 2,000,000, Beta 2,200,000, Gamma 2,480,000. The invitation stated a price/quality weighting of **70/30**. The panel, under time pressure and conscious that the previous installation contractor had performed poorly, opened the priced envelopes alongside the technical submissions, scored quality at 62, 78 and 92, and then — recording it as a "confirmation" — applied a **40/60** weighting on the grounds that quality mattered more on this scope than the invitation had implied. Gamma won with **87.46** against Alpha's **77.20**, and the award recommendation went to the utility's procurement board.

What happened. Alpha requested a debrief, was told that quality had been weighted at 60 %, and challenged the award. The utility's internal audit function reconstructed the evaluation and reached three findings the panel had not anticipated. Under the **published** 70/30 model the same scores gave Alpha **88.60**, Beta 87.04 and Gamma 84.05 — **Alpha won**. The bid set had **three** possible winners across the weighting range, with boundaries at **57.70 %** and **63.77 %** price weight, so the choice of weighting had selected the supplier. And the panel's own risk mapping — integration rework probabilities of 0.30, 0.18 and 0.10 against a 320,000 impact — made Gamma's **USD 480,000** premium worth only `96,000 − 32,000 =` **USD 64,000** of avoided expected cost, a ratio of **7.5 to 1**, so the quality case the re-weighting was meant to express did not survive its own numbers.

How it resolved. The award was withdrawn and re-made under the published model. Alpha was appointed at 2,000,000 with two additions the audit finding made possible: a supplier-development condition funding 90,000 of integration-team secondment against the identified rework risk, and a target-cost structure with the `PTA` computed and shared with the delivery team before mobilisation (Domain 7, KA 7.4.3). The process cost **11 weeks** — 3 weeks of probity review, 3 weeks of re-evaluation, 3 weeks of re-approval latency at `E[wait] = 4/2 + 1`, and 2 weeks to execute — worth **USD 495,000** at Auriga's cost of delay, plus **USD 62,000** of external review: **USD 557,000** in total.

What the domain teaches here. The arithmetic is brutal and simple: **the probity failure cost USD 557,000 — more than the entire USD 480,000 price gap the re-weighting was arguing about, and 1.16 times it.** The panel was not corrupt and was not even wrong about quality mattering; it was wrong about when that judgement may be made. Had the same conviction been expressed in the invitation as a 40/60 weighting, published before bids were received, the award to Gamma would have been unimpeachable — and the bidders would have priced and pitched differently, which is the point of publishing it. Two further lessons. Scoring quality **after** seeing price is not a procedural nicety: the panel's quality scores were never tested for the anchoring the sequence introduced, and could not be defended in the debrief. And the 7.5-to-1 ratio is the discipline a quality weighting needs before it is set, not after it is challenged — a panel that had computed it during model design would have chosen 70/30 with conviction rather than 40/60 with hindsight.

Case study B — Domain 10: two suppliers, one foundry (rail rolling stock)

Situation. A rolling-stock refurbishment programme needed **1,200** traction converters. Wary of single-source exposure after an earlier disruption, the sourcing team dual-sourced on a **55/45** split: supplier V at **USD 14,800** per unit and supplier W at **USD 15,600**, a blended **USD 15,160** and a contract value of **USD 18,192,000**. Awarding the full volume to V would have secured a **3 %** volume discount — **USD 14,356** a unit, **USD 17,227,200** — so the programme paid **USD 964,800** for resilience. The board minute recorded the arrangement as "dual-sourced, exposure mitigated". Each supplier was assessed at a **0.15** probability of disruption in the delivery window, treated as independent, giving an assumed expected disruption cost of `0.2550 × 294,000 + 0.0225 × 1,297,000 =` **USD 104,152.50** — with a one-supplier disruption costing 2 weeks of fleet unavailability at USD 128,000 a week plus a 38,000 surge premium (**294,000**), and both suppliers disrupted costing 9 weeks plus 145,000 of re-engineering (**1,297,000**).

What happened. Both suppliers slipped in the same quarter. Their converters used the same power module, and that module had a single qualified foundry, which reallocated capacity. Neither supplier had disclosed the dependency and neither had been asked. Reconstructed with the true joint probability of **0.11**, the arrangement's expected disruption cost was `0.12 × 294,000 + 0.11 × 1,297,000 =` **USD 166,190**, not 104,152.50 — the independence assumption had understated exposure by **USD 62,037.50**, or **59.56 %** — and the realised outcome was the 9-week case.

How it resolved. A sub-tier mapping exercise to tier three found the single foundry behind both suppliers and two further shared dependencies. The programme then did the arithmetic it had never done: qualifying a **second power-module design** would cost **USD 380,000** — against the **USD 964,800** already spent on a volume split that had bought nothing, a ratio of **2.54 to 1**. The arrangement was restructured to a single award to V at the discounted 14,356, plus the 380,000 second module design, plus a contractual sub-tier disclosure and audit obligation. With the module risk reduced (joint probability 0.02, leaving V's idiosyncratic 0.04, so 0.06 in total) and the recovery shortened to 4 weeks plus 90,000, expected disruption fell to **USD 36,120** and the total position to **USD 17,643,320** against the dual-split arrangement's **USD 18,358,190** — better by **USD 714,870**, with the actual exposure reduced rather than relabelled.

What the domain teaches here. Two suppliers with one sub-tier is one supplier with two invoices, and the money spent on the split is not resilience, it is a price premium with a resilience narrative attached. Three transferable points. The independence assumption is the one most likely to be both convenient and wrong, and it understated this programme's exposure by nearly 60 % — Domain 8's correlation lesson (KA 8.A.1) arriving through a purchase order. Resilience spending must be directed at the **actual single point of failure**, which requires knowing where it is: 380,000 spent on the module would have delivered what 964,800 spent on the split did not. And a sub-tier disclosure obligation is not administrative box-ticking — it is the contractual instrument that makes the analysis possible at all, and it costs nothing to include at tender and cannot be obtained afterwards.

Executive perspective — Domain 10

What a programme director cannot delegate in this domain:

- **The make-or-buy basis, and its exit column.** Insist on total cost of ownership with a non-zero exit cost and a stated breakeven volume, and require the stand-up delay to be priced at the cost of delay wherever the capability is on the critical path. A unit-price comparison is not a make-or-buy paper (10.1.2).
- **The evaluation model, fixed and dated before the receipt deadline.** Weights, sub-weights, normalisation and the sequence in which quality and price are seen. One bid set can have three winners; whoever chooses the model after opening has chosen the supplier, and Case study A shows the cost of learning that afterwards (10.2.2, 10.2.3).
- **The marginal-dollar allocation of every material contract, and where its `PTA` sits.** You should be able to say, without notes, who pays the next dollar of cost under each of your major contracts, and at what cost level that changes (10.3.2; Domain 7, KA 7.4.3).
- **Whether your performance regimes deter or merely price non-performance.** Compare every cap with the supplier's cost of compliance and with your own loss. A 5 % cap against a 180,000 compliance cost is a discount voucher for failure (10.3.3).
- **The notice register.** Your suppliers keep one, or their advisers do. Nearly half of a properly constructed claim can turn on a diary entry, in both directions (10.4.1).
- **The correlation test on every resilience claim.** Ask, of every dual-sourced critical item, which sub-tier sources the two suppliers share — and treat "we don't know" as the answer that it is. Two suppliers with one sub-tier is one supplier (10.4.4, Case study B).

Calculation exercises — Domain 10

> **Exercise 10.1** In-house provision of a support capability costs 260,000 to stand up, 1,450 per unit and 40,000 to exit. Outsourcing costs 55,000 to transition in, 1,900 per unit and 85,000 to transition out. The requirement is 520 units. Compute the breakeven volume and the better option; then recompute with a 7-week capability stand-up delay on the critical path at a cost of delay of 18,000 per week. Solution. `F_make = 300,000`, `F_buy = 140,000`, so `Q* = 160,000/(1,900 − 1,450) = 160,000/450 =` **355.56 units**. At 520 units: make `300,000 + 754,000 =` **1,054,000**; buy `140,000 + 988,000 =` **1,128,000 — make is 74,000 cheaper**, because 520 exceeds the breakeven. Adding the delay: `7 × 18,000 =` **126,000**, so make becomes **1,180,000** and **buy is now 52,000 cheaper**; the breakeven moves to `(300,000 + 126,000 − 140,000)/450 =` **635.56 units**. Common error: omitting the exit costs from both sides, which gives a breakeven of `205,000/450 =` **455.56 units** — it happens to reach the right answer at 520 units, for the wrong reason, and would reach the wrong one at any volume between 356 and 456.

> **Exercise 10.2** Two bids: P at 1,600,000 with quality 70, and Q at 1,840,000 with quality 88. Price is scored `lowest ÷ own × 100`. Determine the winner at a 70/30 and a 50/50 price/quality weighting, compute the crossover weighting, and state the implied price per quality point. Solution. Price scores: P **100.00**, Q `1,600,000/1,840,000 × 100 =` **86.96**. At 70/30: P `70 + 21 =` **91.00**, Q `60.87 + 26.4 =` **87.27 — P wins**. At 50/50: P `50 + 35 =` **85.00**, Q `43.48 + 44 =` **87.48 — Q wins**. Crossover `w* = (88 − 70)/[(100 − 70) − (86.956522 − 88)] = 18/31.043478 =` **57.98 %**. Q's premium is `240,000`

— **15.00 %** — for 18 quality points, **13,333.33 per point**, which must be tested against the expected cost those points avoid. Common error: scoring price as a share of the highest bid or on a linear scale without saying so; the normalisation convention is part of the model and moves the crossover, as 10.2.2 shows by 7.70 percentage points on a three-bid set.

Exercise 10.3 A target-cost subcontract has a target cost of 1,500,000, a target fee of 120,000, a 60/40 share (buyer 60 %) and a ceiling of 1,800,000. The actual cost is 1,700,000. Compute the fee, buyer outturn and supplier margin, the `PTA`, and compare with a firm fixed price of 1,650,000 and with cost plus a fixed fee of 120,000. Solution. Overrun `1,700,000 − 1,500,000 = 200,000`; fee `120,000 − 200,000 × 0.40 =` **40,000**; buyer pays **1,740,000** (below the ceiling); supplier margin **40,000**. Target price `1,620,000`, so `PTA = 1,500,000 + (1,800,000 − 1,620,000)/0.60 = 1,500,000 + 300,000 =` **1,800,000** — here the `PTA` coincides with the ceiling, and at a cost of 1,800,000 the fee is exactly nil. Firm fixed price: buyer pays **1,650,000**, supplier margin **(50,000)**. Cost plus fixed fee: buyer pays **1,820,000**, margin **120,000**. The buyer's outturn spans **170,000** across the three mechanisms on the same actual cost, and the marginal-dollar allocation is 0/1.00 under fixed price, 1.00/0 under cost-plus, and 0.60/0.40 under the target cost below the `PTA`. Common error: computing the fee as `target fee − overrun × buyer share` (120,000 − 120,000 = 0), which reverses the share ratio; the supplier bears its own share of the overrun, not the buyer's.

Exercise 10.4 A single-source arrangement has a certain cost of 1,240,000 and a 0.22 probability of a disruption costing 640,000. Qualifying an alternate adds 58,000 of certain cost and cuts the consequence to 210,000. Compute both total expected costs, the breakeven disruption probability and the maximum premium the resilience is worth. Then recompute assuming a shared sub-tier makes the alternate unavailable in 0.14 of the 0.22. Solution. Single: `1,240,000 + 0.22 × 640,000 =` **1,380,800**. Alternate: `1,298,000 + 0.22 × 210,000 =` **1,344,200** — better by **36,600**. Breakeven `p* = 58,000/(640,000 − 210,000) = 58,000/430,000 =` **13.49 %**. Maximum premium worth paying `= 140,800 − 46,200 =` **94,600**. With the shared sub-tier: expected disruption `(0.22 − 0.14) × 210,000 + 0.14 × 640,000 = 16,800 + 89,600 =` **106,400**, total **1,404,400** — **worse than single sourcing**, and the correct response is to attack the sub-tier rather than to choose between the two structures. Common error: computing the breakeven by dividing the extra certain cost by the full consequence (58,000/640,000 = 9.06 %) rather than by the reduction in consequence, which overstates the case for the alternate.

Exercise 10.5 A variation claim comprises 320 hours of labour at 118.50 per hour, 12,600 of materials, 2 weeks of prolongation at 16,500 per week, and disruption on 900 planned hours whose productivity fell to 0.90 of the measured-mile baseline. Overhead and profit is 10 %. Notice was late, and the prolongation and disruption heads depend on valid notice. Compute the total claim, the amount at risk and the percentage lost. Solution. Labour `320 × 118.50 =` **37,920**; materials **12,600**; prolongation `2 × 16,500 =` **33,000**; disruption — hours required `900/0.90 = 1,000`, extra **100** hours, so `100 × 118.50 =` **11,850**. Subtotal **95,370**; with 10 % overhead and profit, **total claim 104,907.00**. At risk: `(33,000 + 11,850) × 1.10 =` **49,335.00**, which is **47.03 %** of the claim; surviving `(37,920 + 12,600) × 1.10 =` **55,572.00**. Common error: computing disruption as `900 × 0.10 × 118.50 = 10,665`, which multiplies by the productivity shortfall instead of dividing by the productivity factor and understates the head by 1,185 before overhead and profit.

Practitioner's toolkit — Domain 10

Adoption-ready artefacts; adapt the headings to your organisation, then keep them stable.

Toolkit 10.T.1 — Make-or-buy total-cost-of-ownership sheet

One page per decision, completed before any market engagement. Two columns, make and buy, and these rows: transition-in cost, itemised (recruitment, training, tooling, environments, licences, specification and tender effort, mobilisation); unit cost with the volume basis stated; residual management cost — **never zero on either side**; **exit cost**, itemised (redeployment or severance, decommissioning, data extraction in a supportable format, knowledge transfer, re-qualification, parallel running); stand-up elapsed time in weeks and whether the capability is on the critical path; and the delay term, priced at the project's cost of delay. Below the columns, three computed figures: the **breakeven volume** `Q*`, the outturn at the required volume, and the outturn at the plausible upper volume. The sheet's purpose is to make the decision a proposition about volume rather than a preference about sourcing; a paper that cannot fill in the exit row has not done the analysis, and a paper with a zero in it is asserting something the incumbent's contract will disprove.

Toolkit 10.T.2 — Evaluation model lock sheet

One page, completed, dated, signed by the accountable authority and **issued with the invitation**. Contents: the criteria and their weights; the sub-criteria and their weights; the **price normalisation formula written out in full**; the quality scoring scale with what each score means in words; the sequence of evaluation, stating explicitly that quality is scored before price is disclosed; the moderation process and how individual scores are recorded before it; the treatment of non-compliant, qualified and abnormally low bids, with the benchmark and threshold for the last of these named; the panel members and their conflict declarations; and — the row that turns the sheet into a control — the **pre-award sensitivity analysis**: the winner at three weightings and the crossover weightings between the leading bidders, computed before the receipt deadline and dated. Any change after issue is recorded on the sheet with its author, its reason and the result under **both** the original and the amended model.

Toolkit 10.T.3 — Supplier resilience and sub-tier register

One row per critical item, not per supplier, and criticality set by schedule effect rather than by spend. Columns: item; why it is critical (which path, which milestone); first-tier supplier(s) and volume allocation; **disclosed sub-tier sources for the critical component**, to the depth the disclosure obligation reaches; **shared dependencies flagged across rows** — this is the column the register exists for; qualification status of any alternate (qualified / qualified and exercised in the last 12 months / named only); **re-source interval** in weeks; consequence of disruption, priced at the cost of delay plus switching cost; assessed disruption probability with its source; expected cost of disruption; the annual cost of keeping the alternate exercisable; and the **breakeven disruption probability** at which that cost is justified. Reviewed quarterly, with two standing questions: has any shared dependency appeared since the last review, and has any alternate lapsed from exercisable to named only?

Exam preparation — Domain 10

What is assessed. The procurement lifecycle and what each stage forecloses; make-or-buy on total cost of ownership and its breakeven volume; the procurement chain as a schedule constraint including governance latency; routes to market; the evaluation model, price normalisation, the crossover weighting and the obligation to fix the model before opening; abnormally low tender testing against a named benchmark; contract strategy as risk allocation and the influence test; buyer and supplier outturn under fixed-price, cost-plus and target-cost mechanisms; the marginal-dollar allocation identity and the PTA as a behavioural boundary; why an incentive reallocates value unless behaviour changes; performance-regime caps for deterrence against compensation; the four elements of a claim, the heads of cost and notice-bar exposure; the dispute ladder and irrecoverable cost; risk-tiered ethical sourcing and the honest limits of its arithmetic; and single, dual-split and dual-qualified sourcing with correlation.

The calculations to be able to do under time pressure. `TCO = F + vQ` and `Q* = (F_make − F_buy)/(v_buy − v_make)`, with and without a delay term. A procurement chain's total lead time including `E[wait] = M/2 + L` per approval. Price normalisation on both conventions, `S_i(w) = wP_i + (1 − w)Q_i`, and the crossover `w*`. Target-cost fee, buyer outturn and supplier margin at a stated actual cost, plus the PTA. Expected cost, risk-neutral fixed price and expected supplier margin from a cost distribution. A service-credit cap against a compliance cost and against a buyer loss. Claim heads including measured-mile disruption, overhead and profit, and notice-bar exposure. Expected cost of disruption under independence and under a stated joint probability, with the breakeven probability `p*` and the maximum resilience premium.

The traps. Deciding make-or-buy on unit price, which systematically favours the high-fixed-cost option (10.1.2) · scoring exit cost at zero (Toolkit 10.T.1) · dividing the fixed-cost difference by a unit cost rather than by the unit-cost difference (MCQ 10.1-A) · omitting governance latency from a procurement chain (Exercise implied by 10.1.3; Domain 3 KA 3.2.3) · treating the evaluation weighting as a detail rather than as the thing that selects the winner (10.2.2, Case study A) · failing to state the normalisation convention, which moves a crossover by up to 7.70 percentage points (10.2.2) · testing an abnormally low bid against an unnamed benchmark (10.2.2) · adding the required margin to the target cost instead of the expected cost when pricing a fixed-price bid (MCQ 10.3-A) · computing an incentive fee with the buyer's share instead of the supplier's (Exercise 10.3) · assuming sharing continues above the PTA (MCQ 10.3-B; Domain 7 KA 7.4.3) · reading a buyer's expected saving under an incentive as value created when it is value transferred (10.3.2) · comparing a service-credit cap with nothing at all (10.3.3) · computing measured-mile disruption by multiplying by the productivity shortfall instead of dividing by the productivity factor (Exercise 10.5) · comparing a settlement with the expected award and omitting irrecoverable cost (10.4.2) · dividing an extra certain cost by the full consequence rather than by the consequence reduction when computing a breakeven probability (Exercise 10.4) · and assuming supplier independence, which understated Case study B's exposure by 59.56 %.

How the domain connects. Domain 3 supplies the governance latency that makes the procurement chain 31 weeks long and the escalation design that a dispute ladder rests on. Domain 4 supplies the interface arithmetic that prices a packaging decision and the change control that must run across the contract boundary. Domain 5's specification determines whether any contract mechanism can work — on undefined scope they all converge on cost-plus with a dispute attached. Domain 6 supplies the critical path that makes a stand-up delay or a re-source interval expensive, and the cost of delay of USD 45,000 a week at which everything here is priced. Domain 7 supplies the contract-model taxonomy, the blended rate, the cash-flow consequence of payment terms and the PTA — all cited here, none re-derived. Domain 8 supplies EMV, the correlation result that Case study B demonstrates commercially, and the R1 and R3 risks these sourcing and evaluation decisions act on. Domain 11 handles the negotiation behaviour that determines where inside a bargaining range these arithmetic results actually land, and Domain 13 the contracting problem for adaptive delivery, where scope is deliberately not fixed. PFL-AI Domain 11 treats the risk-allocation face of the same problem from the lender's side.

Domain 10 summary

Procurement decisions are delivery decisions, and almost all of them are taken on the wrong number.

Make-or-buy is a proposition about volume, not about unit price. Auriga's remote-terminal-unit capability looks 33.33 % cheaper to build per unit and is USD 88,800 more expensive to own at the 84 units required, because the unit-price advantage of 1,800 has to recover a fixed-cost disadvantage of 240,000 — a breakeven of **133.33 units**. Price the 9-week capability stand-up at the project's cost of delay and the breakeven moves to **358.33 units**, which is not a marginal case but an unavailable one. The exit column is where make-or-buy papers fail, and it is never zero. The procurement chain that delivers the controllers runs to **31 weeks against a 25-week project**, of which 18 weeks are process and 6 are pure governance latency worth **USD 270,000** — which is why long-lead items are committed before the baseline and why the compressible time is administrative, not industrial.

Tender evaluation is arithmetic that selects a supplier. One bid set — Alpha 2,000,000/62, Beta 2,200,000/78, Gamma 2,480,000/92 — produces **three different winners**: Gamma below a 57.70 % price weight, Beta between 57.70 % and 63.77 %, Alpha above. Changing only the normalisation convention moves the Beta/Gamma boundary by **7.70 percentage points**. That is the whole case for fixing and publishing the model, in full, before bids are opened, and for scoring quality before price is seen; Case study A paid **USD 557,000** to learn it, against the **USD 480,000** price gap the re-weighting was arguing about. The discipline that should precede any quality weighting is the premium test: Gamma's 480,000 premium bought **USD 64,000** of avoided expected rework, a ratio of **7.5 to 1**.

Contract strategy is the allocation of the marginal dollar, and the allocation always sums to one — a contract cannot make cost disappear. On the same cost distribution (expected cost USD 2,120,000), a risk-neutral firm fixed price of **USD 2,270,000** and cost plus a 150,000 fee cost the buyer the same in expectation; what fixed price buys is the collapse of a **USD 230,434** standard deviation to zero, and it is only as good as the supplier's balance sheet — at the 2,600,000 outcome that supplier loses 330,000. The target-cost structure as specified moves **USD 48,000** from supplier to buyer and creates nothing: the supplier's expected margin is 102,000 against a 150,000 requirement, because the target fee is the fee at the target and not the expected fee. Value appears only when behaviour changes — the underrun probability must rise from 0.20 to **0.456** for the supplier to accept the structure, at which point expected cost falls by **USD 134,400**, all of it accruing to the buyer, which is what the negotiation is really about. Above Auriga's PTA of **USD 2,428,571.43** (Domain 7, KA 7.4.3) the supplier's marginal share jumps from 0.30 to 1.00 and cooperation turns to entitlement for structural reasons. And a performance regime capped at 5 % of a 2,200,000 contract against a 180,000 compliance cost prices failure at a **USD 70,000 discount**: deterrence needs **8.18 %**, compensation **14.55 %**, and confusing the two is the design error.

Claims, disputes and resilience are where records earn their keep. A properly built USD 107,914.80 variation claim had **48.37 %** of its value — USD 52,194.80 — exposed to a notice provision, which is why a notice register is worth more than any negotiating skill, in both directions. In a 400,000 dispute settling at 235,000 against an expected arbitration cost of 614,000, the decisive figure is that the **USD 340,000 of irrecoverable cost exceeds the entire settlement**, so fighting cannot pay even on a total win. An ethical-sourcing programme costing 124,800 against 106,400 of avoided expected exposure fails an expected-value test at 70 % effectiveness — and the honest response is to risk-tier it, state the breakevens (**82.11 %** effectiveness, **9.38 %** probability), recognise the mean of a fat-tailed consequence as the wrong statistic, and meet any legal duty regardless. Resilience, finally, is bought in the wrong form almost universally: for Auriga's 84 controllers, single sourcing costs **1,008,400** in expectation, a 60/40 dual split **1,026,724**, and a **qualified alternate 964,500** — so qualify two, award one, keep the second exercisable, at a breakeven disruption probability of **10.74 %** against the split's **24.22 %**. Then test the assumption that matters: introduce a shared sub-tier at a joint probability of 0.12 and both resilience options become **worse than single sourcing** — which Case study B lived, at a cost of 964,800 spent on a split that bought nothing while 380,000 spent on the actual single point of failure would have.

The through-line: **every procurement decision has a number that decides it, the number is usually not the one on the front of the paper, and the breakeven is more useful than the point estimate.** Compute the breakeven volume, the crossover weighting, the marginal dollar, the deterrence cap and the breakeven disruption probability, and a procurement conversation stops being about preference and becomes about evidence.

03

PART THREE

Leading people and organisations

Domains 11–13 — stakeholders and influence, leadership and teams, and adaptive delivery: the work that is done through other people.

11

DOMAIN 11

Stakeholders, Communication and Influence

Group: Leading people and organisations (the first of the three domains in Part Three, Domains 11–13). **Target:** ~70 pages. **Binds to:** the PCI Book Pattern Specification and the shared registries (`docs/books/registries/`). This domain returns to the **Meridian Care Records** programme at programme scale, applies Domain 4's interface arithmetic to people, prices Domain 3's governance latency as information latency, and supplies the engagement-allocation and negotiation arithmetic used by Domains 12, 13, 15 and 16. British English; USD (+SAR where useful, indicative `USD 1 ≈ SAR 3.75`). Internal effort in this domain is valued at a stated blended rate of **USD 110 per hour**; a neutral facilitator at **USD 150 per hour**.

Why this domain exists

Part Two built the delivery machinery: scope, schedule, cost, risk, quality, supply. Every one of those disciplines assumed that the people whose agreement the project needs would give it, roughly when asked, on roughly the terms proposed. That assumption is the largest unpriced item in most delivery plans, and this domain prices it.

Two facts sit behind the domain's central claim. The first is Domain 1's arithmetic, which nobody disputes and few programmes act on: Meridian releases **USD 24,480 a year** for each clinic that actually uses the system, so **moving adoption creates more value than moving the installation date** — and adoption is a stakeholder outcome, not a delivery one. The second is that stakeholder work is nevertheless the first thing cut, because it has no unit of account. A schedule has weeks, a budget has dollars, and engagement has "relationships", which are not measurable and therefore not defensible when capacity is short. The domain's central claim is the correction: **engagement is a resource-allocation problem with a computable return, and communication is an architecture with a computable load.** Both can be designed, priced and reviewed like any other part of the delivery system, and where they genuinely cannot be — legitimacy, trust, the moral weight of a community's objection — the professional obligation is to say so plainly rather than to manufacture a metric that survives challenge better than the judgement it replaced.

The domain proceeds from the map to the mechanism. KA 11.1 treats the stakeholder set as a system rather than a list, assesses it on influence, interest, attitude and **consent risk**, and allocates a finite engagement capacity against those four things. KA 11.2 designs communication as an architecture: it applies Domain 4's interface result to people, computes the load a mesh imposes, and then computes something almost no reporting regime measures — **how old the number is when an executive decides on it**. KA 11.3 makes negotiation preparation quantitative through reservation values, the priced alternative and the zone of possible agreement, then shows that most of the value in a multi-issue negotiation is created by trading rather than claimed by conceding. KA 11.4 handles the stakeholders who cannot be managed — the public, communities, and audiences across languages and decision conventions — and the newest failure mode in the domain: communication that a machine wrote and nobody checked.

The through-line: **the stakeholder objection you have not paid for in engagement, you will pay for in delay, at the cost of delay, with interest.**

Learning objectives. After this domain a candidate can: distinguish a stakeholder system from a list and name the parties a register habitually omits; assess stakeholders on influence, interest, attitude and consent risk, and state what each dimension predicts that the others do not; **price a late objection at the cost of delay and compute the earlier engagement spend that would have avoided it, with the breakeven probability that justifies it**; **allocate a finite engagement capacity as a three-term design — value-led core, consent-risk floors, salience-proportional residual — and demonstrate what a salience-only allocation forgoes**; apply Domain 4's interface arithmetic to communication and state the maximum parties a mesh and a routed design each support within a stated capacity; **compute the age of the information an executive decides on from the reporting period, consolidation time and paper lead time, and price the reduction a shorter cut buys**; write an honest status and an escalation-grade message; **build a reservation value from a fully priced alternative, bound the zone of possible agreement, and compute what preparation is worth in transferred value**; compute the joint gain a multi-issue trade produces against a split-the-difference compromise; diagnose conflict by source rather than by personality; price a consultation against the consent risk it reduces while stating what the arithmetic cannot settle; communicate across decision conventions without assuming one is universal; and govern AI-assisted communication so that no external statement is issued unverified and no synthetic voice speaks for the programme.

The master programme. Meridian Care Records continues from Domains 1–4: the clinical-records rollout to **40 clinics**, approved cost **USD 2,400,000**, full-potential benefit **USD 979,200** a year, realistic benefit **USD 685,440** a year at **70 %** adoption, and a **cost of delay of USD 14,280 per week** (Domain 1, KA 1.3.2–1.3.3). Three inherited results do the work in this domain. Each adopting clinic is worth **USD 24,480** a year, so adoption is the largest lever available to the leader. Meridian's steering committee imposes a governance latency of `E[wait] = M/2 + L = 4/2 + 2 =` **4.0 weeks** (Domain 3, KA 3.2.3), which is what makes a late objection expensive. And one interface costs **USD 18,000** to specify, build, test and document (Domain 4, WE 4.2.3), which is what makes a late objection expensive twice. This domain adds the stakeholder register — **62** identified parties, of which **14** require an individually managed relationship — and spends the programme's **480 hours** of annual engagement capacity three different ways to show what the choice is worth.

KNOWLEDGE AREA 11.1

Stakeholder systems and engagement strategies

Topics: 11.1.1 the stakeholder system · 11.1.2 assessment — influence, interest, attitude and consent risk · 11.1.3 engagement strategy under a capacity constraint.

11.1.1 The stakeholder system

Definition. A **stakeholder** is any party whose decisions, consent, resources, behaviour or objection can materially affect the project's outputs, outcomes or legitimacy, or who is materially affected by them. A **stakeholder system** is that set of parties together with the relationships between them — who influences whom, who speaks for whom, and which pairs will align or divide under pressure.

The distinction is not pedantry. A list supports a communication schedule; only a system supports prediction, because the events that damage projects are relational — a supplier and a regulator reaching a shared view of a specification the client has not seen, a user group and a professional body forming a coalition, a funder deferring to an operational director whose interests diverge from its own. A leader working from a list is repeatedly surprised by conclusions formed in conversations they were not part of, and each surprise arrives late.

What registers habitually omit, consistently enough to be a checklist. Parties with no organisational representation — future users, patients, tenants, passengers, the public — whose interests are real, whose consent may be legally required, and who have no one on the distribution list. Internal parties whose work the project changes without commissioning them: finance, procurement, legal, information security, service desks, training functions. Parties visible only at handover, where Domain 16's readiness failures originate. The supplier's own subcontractors, whose interests are not the supplier's (Domain 10). Predecessors and successors — the sponsor of the previous attempt, the owner of the system being replaced, the programme inheriting the dependency. And the ones who left: registers record roles, but relationships attach to people, a point 11.A.2 makes arithmetic.

The four questions that make a register useful. Not "who are the stakeholders?" but: what does this party **decide, consent to, resource, or do** that the project needs? What does the project change **for them**, and is that a gain or a loss in their own terms? What is their **relationship to the other parties** — who do they follow, and who follows them? And what is the **cost if their agreement arrives late**? The last question converts a register from an administrative artefact into a planning input, because it produces a number, and 11.1.2 computes it.

Note the vocabulary. Stakeholders are not managed; work is managed, and stakeholders are engaged, consulted, negotiated with and occasionally opposed. "Stakeholder management" quietly licenses the posture — that the party's view is an obstacle to be handled — that guarantees the late objection this domain prices. An objection is information, sometimes about the stakeholder and sometimes about the project, and the leader's first task is to establish which.

11.1.2 Assessment — influence, interest, attitude and consent risk

The four dimensions, and what each predicts. Most assessment practice uses two; four are needed, because two of the failure modes are invisible in the first pair.

- **Influence** — the party's capacity to affect the project's decisions, resources or permissions. Predicts how much damage a disagreement does.
- **Interest** — how much the party cares, measured by attention rather than by stake. Predicts how quickly they will notice, and it is emphatically not a measure of how much is at stake for them: a party can have an enormous stake and no attention, and that combination is the most dangerous in the register.
- **Attitude** — supportive, neutral, sceptical or opposed, assessed on evidence rather than on courtesy. Predicts what an engagement contact will achieve. A supportive party needs information; a sceptical one needs argument; an opposed one needs negotiation or containment, and sending each of them the same monthly bulletin serves none of them.
- **Consent risk** — whether the party holds a veto, a statutory approval, a licence, a signature, a clinical or safety authority, or a practical ability to withhold cooperation. Predicts the consequence of being wrong about the other three, and it is a property of the party's position rather than of its disposition. A helpful regulator still holds the veto.

The high-influence, low-interest, high-consent-risk quadrant is where projects are damaged. Such a party is quiet for months, is therefore assessed as low priority by any allocation rule that reads interest, and then reads the specification properly at the point of approval. What happens next has a price.

WORKED EXAMPLE

Worked example 11.1.2 — the objection that arrived in week 34.

1. **Setup.** Meridian's national reporting body is assessed influence **5**, interest **2**, attitude neutral, consent risk **high** (it approves the statutory data extract). It received the monthly bulletin and nothing else. In **week 34** of the rollout it objects that the extract's record-level structure does not meet its submission requirement. Resolving the objection requires a specification change approved by the steering committee — Domain 3's `E[wait]` of **4.0 weeks** — followed by **5 weeks** of rework and re-verification, all on the critical path. The direct work is re-specification and re-verification of the reporting-gateway interface and the two interfaces that depend on it, at Domain 4's unit cost of **USD 18,000** each, plus **USD 42,000** of application rework. Cost of delay **USD 14,280 per week**. A technical pre-consultation at design stage would have taken **24 hours** of a data specialist's time and **6 hours** of the project leader's, at the blended internal rate of **USD 110 per hour**.
2. **Formula.** Assessed total = (governance latency + rework weeks) × cost of delay + interface re-verification + application rework. Avoidance cost = hours × rate. Breakeven probability = avoidance cost ÷ assessed total.
3. **Substitution.** Delay `(4.0 + 5) × 14,280`. Direct `3 × 18,000 + 42,000`. Avoidance `(24 + 6) × 110`. Breakeven `3,300 ÷ 224,520`.
4. **Result.** Delay **9.0 weeks**, costing **USD 128,520**; direct cost **USD 96,000**; **assessed total USD 224,520** (≈ SAR 841,950 indicatively). Avoidance cost **USD 3,300**. The consequence is **68.0 times** the avoidance cost, and the pre-consultation pays whenever the probability of the objection exceeds **1.47 %**.
5. **Interpretation.** The 68-fold ratio is arresting and it is not the useful number; **1.47 %** is. A leader arguing for engagement with a ratio is making a rhetorical case, and a sceptical sponsor will answer that the objection might not have happened. A leader arguing with the breakeven probability has made that answer impossible: the sponsor must now assert that the chance of a statutory approver objecting to a specification it has never been shown is below one and a half per cent, which nobody will say out loud. The asymmetry generalises far beyond this case, because engagement costs are hours and consequences are weeks of critical path — so the breakeven probability for engaging a high-consent-risk party is almost always in the low single figures. Three cautions belong with it. The arithmetic prices the avoidable objection; a pre-consultation that discovers a genuine incompatibility does not eliminate the cost, it moves it earlier, where it is cheaper (Domain 5's design-maturity argument, and Case study B's 7.45-fold escalation). The 5 weeks of rework is an estimate and the 4.0 weeks of governance latency is not — it is a computed property of Meridian's committee design, which means **a governance design and an engagement design are the same conversation**: at Domain 3's redesigned latency of 3.0 weeks the same objection costs 210,240 rather than 224,520. And the pre-consultation must be a technical engagement with the people who will apply the requirement, not a courtesy briefing of their director, because the director did not know either.

11.1.3 Engagement strategy under a capacity constraint

The constraint, stated honestly. Engagement capacity is finite and small. Meridian's is **40 hours a month** — the project leader's own engagement time plus two part-time engagement leads — giving

480 hours over the twelve months of rollout. Every engagement strategy is an allocation of those 480 hours whether or not anyone writes it down, and the strategies that are not written down allocate by whoever asks most persistently — a default that correlates with neither value nor risk.

The conventional allocation and its defect. Standard practice scores each party on influence and interest, multiplies to a **salience** score, and allocates attention in proportion. It is a reasonable first cut with one structural defect: it allocates by power and attention, and neither is the same as value at stake or consequence of refusal. So it over-serves the loud and the powerful and under-serves two groups at once — the users who determine whether any benefit is realised, and the quiet approver of 11.1.2.

The three-term design. The correction is to allocate in three terms, in this order:

1. A **value-led core** — the effort that demonstrably moves the largest benefit lever, sized from the benefit arithmetic rather than from the score.
2. **Consent-risk floors** — a minimum allocation to each party that can stop the work, set by what the relationship actually requires, not by its interest score.
3. A **salience-proportional residual** — what is left, distributed by influence × interest.

WORKED EXAMPLE

Worked example 11.1.3 — allocating Meridian's 480 engagement hours.

1. **Setup.** Seven groups, scored influence × interest: regional health authority $5 \times 3 = 15$; clinic directors $4 \times 5 = 20$; clinicians $2 \times 5 = 10$; practice administrators $2 \times 4 = 8$; national reporting body $5 \times 2 = 10$; patient representatives $3 \times 3 = 9$; records-system supplier $2 \times 4 = 8$. Total salience **80**. Pilot evidence supports a structured adoption intervention — a clinical-workflow session plus two follow-ups — costing **8 hours per clinic** and raising expected adoption by **12.5 percentage points**, from 70 % to 82.5 %. Each adopting clinic is worth **USD 24,480** a year (Domain 1). Consent-risk floors are set at **60 hours** for the regional health authority (which holds the funding), **40** for the national reporting body and **30** for the patient representative group.
2. **Formula.** Salience allocation = capacity × salience ÷ Σ salience. Three-term allocation = value core (8 h × clinics) + Σ floors + residual apportioned by salience. Benefit gain = uplift in adopting clinics × annual value per clinic. Value per engagement hour = gain ÷ core hours.
3. **Substitution.** Salience: $480 \div 80 =$ 6 hours per point. Core 8×40. Residual $480 - 320 - 130$, apportioned across salience $8 + 8 = 16$. Gain $40 \times 0.125 \times 24{,}480$.

Group	Salience	Salience-proportional	Three-term design	Basis
Regional health authority	15	90 h	**60 h**	consent floor
Clinic directors	20	120 h	**160 h**	value core
Clinicians (users)	10	60 h	**160 h**	value core
Practice administrators	8	48 h	**15 h**	residual
National reporting body	10	60 h	**40 h**	consent floor
Patient representatives	9	54 h	**30 h**	consent floor
Records-system supplier	8	48 h	**15 h**	residual
Total	**80**	**480 h**	**480 h**	

1. **Result.** The value core is **320 hours**; floors take **130**; the residual is **30**. The core delivers **5** additional adopting clinics and **USD 122,400** a year of recurring benefit, at **USD 382.50 of benefit per engagement hour** in year one alone. The salience allocation gives the two adoption-determining groups only **180 hours**, enough for **22** of the 40 clinics — an uplift of **6.875 percentage points**, **2.75** clinics and **USD 67,320** a year — so it forgoes **USD 55,080 a year, permanently**. The core's opportunity cost is `320 × 110 =` **USD 35,200** of existing capacity, against which the breakeven uplift is `35,200 ÷ 24,480 =` **1.4379** clinics, or **3.59 percentage points** of adoption.
2. **Interpretation.** The decisive figure is the **3.59-point breakeven**, because it converts an argument about the value of engagement into a question a clinical lead can answer: does a workflow session plus two follow-ups, run in every clinic, move programme adoption by more than three and a half points? Almost certainly — and if it does not, the intervention should be redesigned rather than defended. Note also what the reallocation costs: nothing. The 480 hours are already paid for, so the 55,080 forgone is not an underspend but a **misallocation of capacity that was going to be spent anyway**, which is why this is the cheapest large improvement in the domain. Two second-order results matter more than they look. First, raising adoption to 82.5 % raises the **cost of delay** itself, because the benefit rate rises with adopting clinics: 33 clinics at 510 a week is **USD 16,830 per week**, up **17.86 %**. So engagement makes every schedule lever more valuable — Domain 1's 8-week compression for 60,000, worth a net **+USD 54,240** at 70 % adoption, becomes worth `8 × 16,830 − 60,000 =` **+USD 74,640** at 82.5 %. **Sequence engagement before acceleration** is therefore an arithmetic ordering, not a preference. Second, the floors cannot be justified by this arithmetic and must not be: their return is the avoided 224,520 of 11.1.2, which is a risk return with a breakeven probability, not a benefit return with a rate. A design that computed only the value core would fund adoption brilliantly and be stopped in week 34.

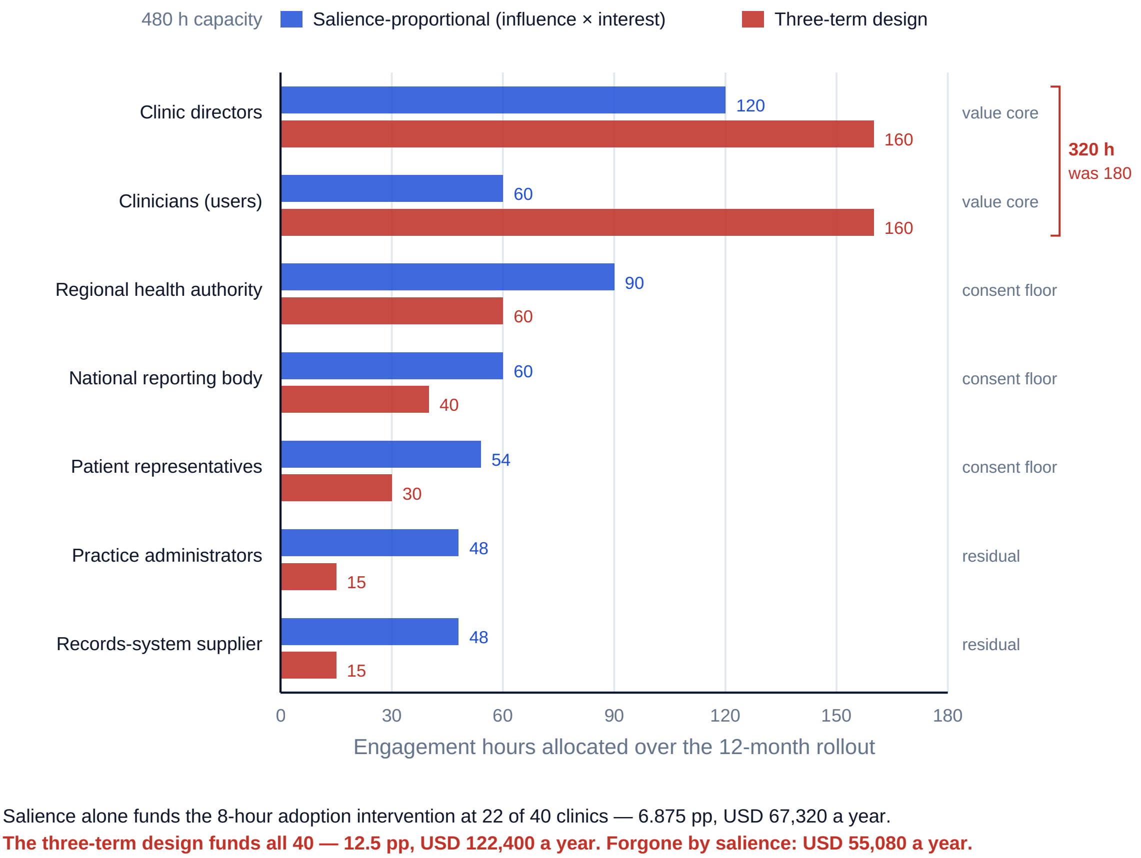


Fig 11.1.1 — Two allocations of the same 480 hours

From allocation to plan. An allocation is not yet a strategy. Each group needs a stated **objective** as a dated future state ("the reporting body has confirmed the extract specification in writing by week 12", not "maintain good relations"), a **named owner** at a level the party will engage with, a **method** matched to attitude, a **cadence**, and a **test of whether it worked**. Objectives written as states rather than activities are what make an engagement plan reviewable: "held four meetings" cannot fail, and therefore cannot inform.

AI in this KA

Where it earns its place. Building a first-pass register from documents the project already has — contract schedules, distribution lists, minutes, planning submissions, service-desk records — and listing parties named in them that the register omits: a set-difference task over unstructured text, the omission checklist of 11.1.1 mechanised, and one that routinely finds internal functions nobody thought to list. Classifying a large volume of consultation responses into themes with counts, so that a leader reads a structured summary of two thousand comments rather than a sample of twenty. Drafting a salience and consent-risk assessment for human challenge, useful precisely because a draft is easier to argue with than a blank page. Modelling allocations across capacity and floor assumptions, which is arithmetic.

Where it must not go. Assigning **attitude** or **influence**. A model reads text; influence is political and attitude is often deliberately unstated, so a fluent assessment of either will be confidently wrong in the direction the source documents are polite. Inferring a party's negotiating position from its published statements as though that settled it. And any "stakeholder sentiment score" entering a report as a governance input — a number with no measurement behind it displaces the judgement it was meant to support, permanently, because numbers survive committee meetings better than qualifications do.

Verification, concretely. Every register entry carries a named human owner who has confirmed the party's role, authority and consent risk against a source — the contract, the statutory instrument, the delegation schedule — not against the model's summary of it. Classified consultation responses are re-read on a stated sample (10 % is a defensible floor) with the disagreement rate recorded, and the classification is rejected above a pre-set tolerance. The allocation arithmetic is reproduced by hand: it is a division and two multiplications.

KEY TERMS — KA 11.1

Term	Meaning
Stakeholder	A party whose decisions, consent, resources, behaviour or objection can materially affect the project's outputs, outcomes or legitimacy, or who is materially affected by them.
Stakeholder system	The stakeholder set together with the relationships between its members — who influences whom, and which pairs align under pressure.
Influence	Capacity to affect the project's decisions, resources or permissions; predicts the damage a disagreement does.
Interest	Attention paid, not stake held; predicts how quickly a party notices.
Attitude	Supportive, neutral, sceptical or opposed, assessed on evidence; predicts what an engagement contact will achieve.
Consent risk	Whether the party holds a veto, approval, licence, signature or practical ability to withhold cooperation; a property of position, not disposition.
Salience	Influence × interest — a useful first cut and an inadequate allocation rule.
Value-led core	Engagement effort sized from the benefit arithmetic rather than from the salience score.
Consent-risk floor	A minimum engagement allocation to a party that can stop the work, justified by breakeven probability rather than by benefit rate.
Engagement objective	A stated future state of a relationship with a date and a test, never an activity count.

SAMPLE MCQS — KA 11.1

MCQ 11.1-A `[11.1.1 · Comprehension]` The practical difference between a stakeholder list and a stakeholder system is that only the system: - A. records contact details and communication preferences - B. captures the relationships between parties, and so supports prediction of positions formed in conversations the project is not part of - C. is approved by the sponsor - D. includes external as well as internal parties

Rationale: The system's added content is the relationships between members (11.1.1). A is administrative; C is governance; D is a completeness property a list can also have.

MCQ 11.1-B `[11.1.2 · Application]` An objection causes 4.0 weeks of governance latency and 5 weeks of rework on the critical path at a cost of delay of 14,280 per week, plus three interface re-verifications at 18,000 each and 42,000 of application rework. The assessed total is: - A. USD 96,000 - B. USD 128,520 - C. USD 167,400 - D. USD 224,520

Rationale: `9.0 × 14,280 + 54,000 + 42,000 = 224,520` (11.1.2). A is the direct cost alone; B is the delay alone; C omits the 4.0 weeks of governance latency and counts only the rework weeks.

MCQ 11.1-C `[11.1.2 · Evaluation]` A 30-hour pre-consultation costing 3,300 would have addressed the objection above. The most persuasive argument to a sceptical sponsor is that: - A. the consequence is 68.0 times the avoidance cost - B. the pre-consultation pays whenever the probability of the objection exceeds 1.47 % - C. engaging regulators early is good practice - D. 3,300 is immaterial against a 2,400,000 budget

Rationale: The breakeven probability forces the sceptic to assert something indefensible, whereas a ratio invites the answer "it might not have happened" (11.1.2). D is true and irrelevant, since immateriality is not a reason to spend.

MCQ 11.1-D `[11.1.3 · Analysis]` Allocating engagement capacity in proportion to influence × interest systematically under-serves: - A. the most powerful stakeholders - B. the parties with the highest interest - C. the user group that determines benefit realisation and the quiet party holding a veto - D. the project's own delivery team

Rationale: Salience measures power and attention, not value at stake or consequence of refusal (11.1.3) — which is why the design adds a value-led core and consent-risk floors.

MCQ 11.1-E `[11.1.3 · Application]` A 320-hour adoption core raises adoption by 12.5 percentage points across 40 clinics, each adopting clinic being worth 24,480 a year. The benefit per engagement hour in year one is: - A. USD 76.50 - B. USD 382.50 - C. USD 3,060.00 - D. USD 2,142.00

Rationale: `40 × 0.125 × 24,480 = 122,400`; `122,400 ÷ 320 = 382.50` (11.1.3). A divides one clinic's annual value by the 320 hours (`24,480 ÷ 320`); C divides the gain by the 40 clinics rather than by hours; D divides the whole 685,440 annual benefit by 320 hours instead of the uplift.

SELF-CHECK — KA 11.1

1. Name the assessment quadrant that damages projects, and why an allocation rule misses it. — High influence, low interest, high consent risk: any rule that reads interest scores it low, and it then reads the specification at the point of approval.
2. State the three terms of an engagement allocation, in order. — Value-led core sized from the benefit arithmetic; consent-risk floors for parties that can stop the work; salience-proportional residual.
3. Why does raising adoption make schedule compression more valuable? — The cost of delay is the benefit rate, and the benefit rate rises with adopting clinics: 14,280 a week at 70 % adoption, 16,830 at 82.5 %.

KNOWLEDGE AREA 11.2

Executive communication and reporting

Topics: 11.2.1 communication as a designed architecture · 11.2.2 the executive report and the age of its numbers · 11.2.3 honest status and the escalation-grade message.

11.2.1 Communication as a designed architecture

The claim. A communication plan is usually a table of meetings and distribution lists. It should be an **architecture**, because communication has the same combinatorial property as integration, and for the same reason: what costs money is not the parties, it is the **channels between them**.

Domain 4, KA 4.2.3 established the arithmetic and this domain does not re-derive it. For n parties each communicating directly with every other, the number of channels is $n(n - 1)/2$; routing every party through a single integrating point gives n. Applied to people it explains something practitioners experience and rarely diagnose: a programme that has added four stakeholder groups has not added four relationships but a great many, and the engagement plan written for the original count is still in use.

WORKED EXAMPLE

Worked example 11.2.1 — the communication load Meridian was carrying.

1. **Setup.** Meridian's register holds **62** parties, of which **14** require an individually managed relationship. In the design as found, each of those 14 dealt directly with each of the others and with the programme, and maintaining one such channel — preparation, contact, follow-up, recording — costs **1.5 hours per month**. The alternative routes every party through the programme's engagement function as the integrating point, at a hub overhead of **12 hours a month** (two stakeholder forums, all-in). Engagement capacity is **40 hours a month** (11.1.3).
2. **Formula.** Mesh channels `n(n − 1)/2`; routed channels `n` (Domain 4, KA 4.2.3). Load = channels × hours per channel (+ hub overhead). Sustainable party count = the largest `n` whose load fits the capacity.
3. **Substitution.** Mesh `14 × 13/2 = 91`, load `91 × 1.5`. Routed `14`, load `14 × 1.5 + 12`. Sustainable mesh count: largest `n` with `n(n − 1)/2 × 1.5 ≤ 40`.
4. **Result. 91** channels demanding **136.5 hours a month** — **341.25 %** of capacity — against **14** channels demanding **33.0 hours**, or **82.5 %**. An unrouted design sustains at most **7** parties (21 channels, 31.5 hours); the eighth takes it to 42.0 hours and over capacity. A routed design sustains **18** (39.0 hours). Routing saves **103.5 hours a month**, **1,242 hours a year**, worth **USD 136,620** at the blended rate.
5. **Interpretation.** The number to carry away is **seven**. An unrouted design supports seven actively managed parties within Meridian's capacity and Meridian has fourteen, so the design was never feasible — and infeasible communication designs do not fail by anyone reporting a shortfall. They fail silently: contacts are skipped, the skipped ones are the low-interest parties because they do not chase, and 11.1.2's objection is the result. The overload is therefore not a workload complaint but the **mechanism** by which the late objection is generated, which is the professional insight this arithmetic buys. Three cautions. Routing is not free of distortion: a hub is a single point through which every message is paraphrased, and the parties who must not be paraphrased — the sponsor, the statutory approver, the clinical authority — need bilateral channels regardless of load. Meridian's honest design is therefore **hybrid**: a deliberate three-party mesh among those three plus all 14 routed to the hub, giving `3 + 14 = 17` channels, **37.5 hours a month**, **93.75 %** of capacity. That is tight and it is the truth; a plan showing 82.5 % would understate the design actually chosen. Next, the 1.5-hour unit is an average over relationships of very different weight, and a leader quoting one figure for a regulator and a practice administrator should say so. And the hub overhead is real work that must be resourced as work: an engagement function that is nobody's named job is a mesh with a diagram on top of it.

11.2.2 The executive report and the age of its numbers

What an executive report is for. Not to describe the project: to enable a decision the reader has authority to take, or to confirm that none is required. Everything serving neither purpose consumes the scarcest resource in the governance system, the attention of people who can decide (Domain 3, Fault 3). The structural consequence is a fixed order, the reverse of how most reports are written: **the decision or confirmation first**, then the two or three facts that drive it, then the forecast with its assumptions, then exceptions, then the rest by reference. A report that reaches its decision on page four was written for the author's comfort.

The property nobody measures. A report's usefulness depends on more than its content: it depends on **how old the facts in it are when the decision is taken**. Three quantities determine that, and all three are known at design time. Let

```
P = the reporting period (e.g. 4 weeks for a monthly pack)
C = consolidation time — data cut-off to pack issue
L = the paper lead time — how far before the meeting submissions close (Domain 3)
```

Then, measuring back from the decision meeting:

```
Age of the newest fact   = C + L
Mean age of the facts    = C + L + P/2
Maximum blindness        = P + C + L
```

The third is the important one: **maximum blindness** is how long a problem arising just after a cut-off remains invisible to the decision body. It is the number that should be quoted when anyone proposes a longer reporting cycle, and it is almost never computed.

WORKED EXAMPLE

Worked example 11.2.2 — how old is the number Meridian's committee decides on?

1. **Setup.** Meridian reports monthly: `P` = **4 weeks**, data cut-off at period end, consolidation and internal review `C` = **1 week** from cut-off to pack issue, and the steering committee's paper lead time `L` = **2 weeks** (Domain 3, KA 3.2.3). The proposed redesign keeps the monthly committee but cuts data **weekly** into a standing dashboard — `P` = 1, `C` = 0.5, `L` = 1 — so the pack presents the latest weekly cut rather than a monthly consolidation. Cost of delay **USD 14,280 per week**.
2. **Formula.** Newest `= C + L`; mean `= C + L + P/2`; maximum blindness `= P + C + L`. Saving priced at the cost of delay.
3. **Substitution.** As found: `1 + 2`; `1 + 2 + 4/2`; `4 + 1 + 2`. Redesigned: `0.5 + 1`; `0.5 + 1 + 0.5`; `1 + 0.5 + 1`.
4. **Result.**

	Newest fact	Mean fact age	Maximum blindness
Monthly pack (`P` 4, `C` 1, `L` 2)	3.0 weeks	**5.0 weeks**	**7.0 weeks**
Weekly cut (`P` 1, `C` 0.5, `L` 1)	1.5 weeks	**2.0 weeks**	**2.5 weeks**
Weeks saved	1.5	3.0	4.5
Priced at 14,280/week	USD 21,420	**USD 42,840**	**USD 64,260**

1. **Interpretation.** Meridian's committee has been deciding on a picture whose average fact is **five weeks old**, and a problem arising in the first week of a period can run **seven weeks** before the body that could stop it sees it — at 14,280 a week, **USD 99,960** consumed before the decision system is aware of it. That reframes the usual argument: the objection to monthly reporting is not that it is "not current enough", which is taste, but that its **maximum blindness is 7.0 weeks and can be 2.5**, which is arithmetic. Note the ordering of the levers, mirroring Domain 3's result

about paper deadlines. Shortening the reporting period is the strongest lever on maximum blindness and costs almost nothing once the data layer exists; consolidation time is next and is usually pure process; the paper lead time is Domain 3's lever, shared with the governance design. The professional caution is that these savings are **conditional and not additive across the year** — they accrue only where a faster picture would have produced a faster decision and the delay was on the critical path. The honest claim is therefore a per-occurrence one, 42,840 on the mean and 64,260 in the worst case, with the number of qualifying occurrences stated as an assumption. Presenting it as an annual saving is the error that gets a good proposal rejected by a finance director who can see the double count.

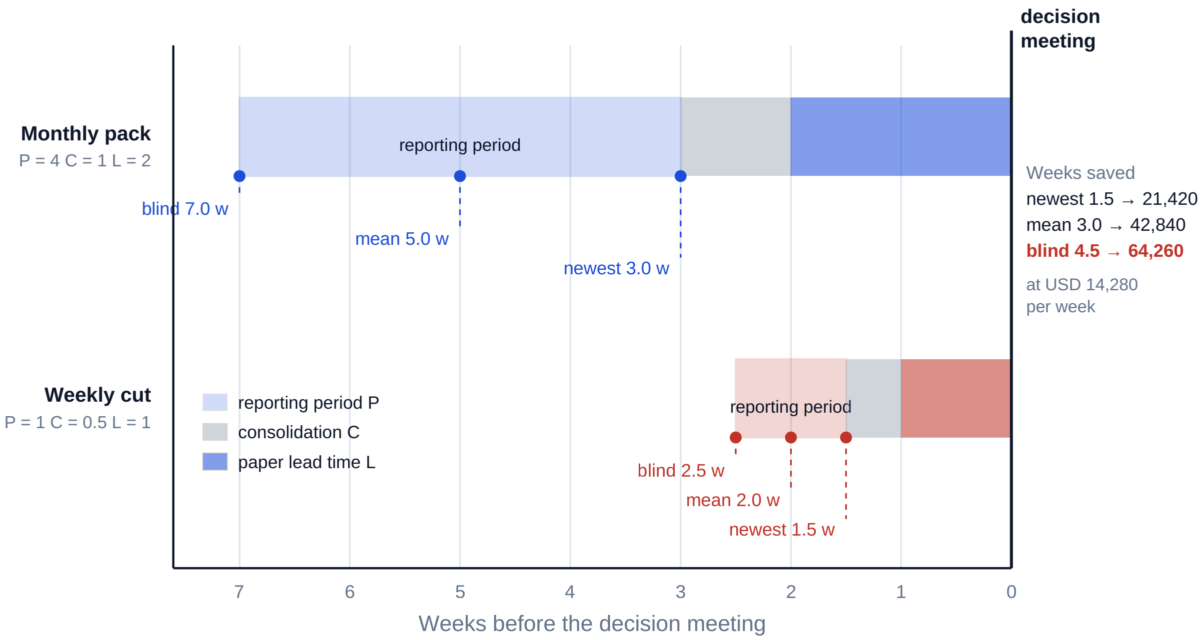


Fig 11.2.1 — The age of the number an executive decides on

11.2.3 Honest status and the escalation-grade message

The status problem. Traffic-light status is universal and structurally corrupt, for a reason worth stating precisely: the light is set by the person whose performance it evaluates, against undefined thresholds, for an audience that reacts to colour before content. The predictable result is **amber compression** — nearly everything is amber, red is reserved for the unrecoverable, and the status carries almost no information at the point it is most needed. The fix is not more honesty but **pre-agreed objective thresholds** set at baseline and applied mechanically: "red where forecast completion exceeds baseline by more than four weeks or forecast cost exceeds budget by more than 5 %" removes the author's discretion and with it the incentive that corrupted the signal, and it makes the status arguable on the facts, which is what a governance body needs.

What a red status must carry, and this is the difference between reporting a problem and escalating one. The **exception**, quantified against its threshold. The **cause**, distinguished from the symptom. The **impact** if nothing changes, priced. The **options**, each with cost, consequence and risk. The **recommendation**, made by name. The **decision required**, from whom, **by when**, with the consequence of missing that date. And the **date the leader knew**, which is the field that makes Domain 3's escalation lead time measurable.

Bad news. Three rules earn their place. Escalate the forecast, not the event — a leader who waits for a variance to occur has converted a decision into a report. Never let a governance body learn something material from a third party first: once it happens, the leader's reporting is discounted for the remainder of the project. And separate the news from the plea, because a message mixing an exception with a request for sympathy invites a response to the second. Domain 12 handles the personal dimension; the reporting obligation here is narrow and absolute — the material fact reaches the accountable decision-maker, in writing, on the date it is known.

The tailoring rule. Different audiences need different depth, never different facts. Tailoring selects what a reader needs to decide and expresses it in their vocabulary; it does not mean the board's version and the delivery team's version disagree. Where two versions of one status exist, the project has created a discovery problem for itself, and Domain 3's decision record is where the discrepancy will be found.

AI in this KA

Where it earns its place. Assembling a report to a fixed skeleton from the data layer — figures, variances, trends, exception lists — which removes the copy-and-paste error class that infects manually built packs. **Consistency checking**, the highest-value use in this KA and the least used: reading a drafted report against its own numbers and flagging where the narrative and the arithmetic disagree, where a status contradicts its stated threshold, or where a figure differs from the previous pack without explanation. Rewriting a technical explanation for a non-specialist reader, for human approval. Producing audience-specific depth variants from one factual base — the tailoring rule mechanised.

Where it must not go. Setting the status, which is an accountable judgement against a threshold. Writing the recommendation. And — the failure that is easy to walk into and hard to see afterwards — **smoothing**. A model asked to make a report "clearer", "more balanced" or "less alarming" will reliably produce a calmer document, because calm prose is what fluent business writing looks like in the material it learned from, and the qualifier carrying the warning is the first thing to go. Any instruction to soften a report is an instruction to alter its meaning, and must be treated as an editorial decision taken by a person, on the record.

Verification, concretely. The status and the recommendation are entered by the accountable human and excluded from any automated drafting step. Every figure traces to a query against the data layer, available on request rather than reconstructed after a challenge. A named person signs the pack as issued, and that signature means they have read it — which implies a report short enough to read. Every consistency flag is resolved and the resolution recorded, because an unresolved flag in an issued report is documented knowledge of an inconsistency, which is worse than no checker at all.

KEY TERMS — KA 11.2

Term	Meaning
Communication architecture	The designed set of channels between parties — mesh, routed or hybrid — with its load computed against capacity.
Channel load	Channels × hours per channel per period; compared with engagement capacity to give the sustainable party count.
Integrating point (hub)	The function through which routed communication passes; carries its own overhead and its own distortion risk.
Reporting period (`P`)	The interval of activity a report covers.
Consolidation time (`C`)	Elapsed time from data cut-off to pack issue.
Maximum blindness	`P + C + L` — how long a problem arising just after a cut-off stays invisible to the decision body.
Amber compression	The collapse of traffic-light status into a single uninformative colour, caused by author discretion over undefined thresholds.
Escalation-grade message	Exception, cause, impact priced, options, recommendation by name, decision required by whom and by when, and the date the leader knew.
Tailoring (of a message)	Varying depth and vocabulary by audience while holding the facts identical. **Context flag:** distinct from tailoring of method (Domain 4, KA 4.1.3), which is a recorded decision to adapt process; the shared word carries two different concepts and neither reading substitutes for the other.

SAMPLE MCQS — KA 11.2

MCQ 11.2-A `[11.2.1 · Application]` Fourteen parties each communicate directly with every other. The number of channels is: - A. 14 - B. 91 - C. 182 - D. 196

Rationale: `n(n − 1)/2 = 14 × 13/2 = 91` (11.2.1, citing Domain 4 KA 4.2.3). A is the routed count; C counts each pair twice; D is `n²`.

MCQ 11.2-B `[11.2.1 · Analysis]` At 1.5 hours per channel per month and 40 hours of engagement capacity, the largest number of parties an unrouted design sustains is: - A. 7 - B. 8 - C. 14 - D. 26

Rationale: Seven parties give 21 channels and 31.5 hours; eight give 28 channels and 42.0 hours, which exceeds capacity (11.2.1). C is the actual party count, which is why the design was infeasible; D divides capacity by the unit cost and ignores the combinatorial term.

MCQ 11.2-C `[11.2.2 · Application]` A monthly report has a 4-week period, 1 week of consolidation and a 2-week paper lead time. The mean age of the facts at the decision meeting is: - A. 2.0 weeks - B. 3.0 weeks - C. 5.0 weeks - D. 7.0 weeks

Rationale: `C + L + P/2 = 1 + 2 + 2 = 5.0` weeks (11.2.2). B is the newest fact only; D is maximum blindness; A counts only the period's half-life.

MCQ 11.2-D `[11.2.2 · Evaluation]` The strongest argument for cutting data weekly rather than monthly is that it: - A. makes the pack look more current to the committee - B. reduces maximum blindness from 7.0 to 2.5 weeks — 4.5 weeks, or USD 64,260 at the cost of delay - C. saves consolidation effort - D. reduces the paper lead time

Rationale: Maximum blindness is the decision-relevant quantity and the redesign is arguable arithmetically (11.2.2). A is presentational; C is false, since more frequent cuts add consolidation events; D is a separate governance lever (Domain 3).

MCQ 11.2-E `[11.2.3 · Analysis]` The root cause of amber compression is that: - A. project leaders are optimistic by disposition - B. the status is set by the person it evaluates, against undefined thresholds - C. boards react badly to red status - D. reporting cycles are too long

Rationale: The corruption is structural — discretion plus undefined thresholds — so the remedy is pre-agreed objective thresholds applied mechanically (11.2.3). A and C are contributing conditions, not the cause.

SELF-CHECK — KA 11.2

1. Why is an overloaded communication architecture a risk mechanism rather than a workload complaint? — Skipped contacts fall on the parties who do not chase, which are the low-interest, high-consent-risk parties whose late objection 11.1.2 prices.
2. State the three components of report age and which one to quote against a longer cycle. — `C + L` (newest), `C + L + P/2` (mean), `P + C + L` (maximum blindness); quote maximum blindness.
3. What must a red status carry beyond the exception? — Cause, priced impact, options, a named recommendation, the decision required from whom by when, and the date the leader knew.

KNOWLEDGE AREA 11.3

Negotiation and conflict

Topics: 11.3.1 preparation — reservation value, the priced alternative and the zone of possible agreement · 11.3.2 creating value before claiming it · 11.3.3 conflict — sources, modes and the resolution route.

11.3.1 Preparation — reservation value, the priced alternative and the zone of possible agreement

The three quantities. Negotiation in a delivery context is not mainly a matter of technique. It is almost entirely a matter of three numbers, two of which are knowable before the meeting.

- Your **best alternative** — what happens if no agreement is reached, **fully priced**, including delay at the cost of delay, internal effort, and the risk the alternative carries, valued as an expected monetary value (`EMV`, Domain 8).
- Your **reservation value** — the worst terms you would rationally accept, derived from the priced alternative. For a buyer it is a maximum; for a seller a minimum. A reservation value that is not derived from a priced alternative is a preference, and preferences move under pressure.
- The **zone of possible agreement (ZOPA)** — the interval between the two parties' reservation values. If it is empty there is no deal at any level of skill, and the professional act is to stop and change the alternatives rather than to keep meeting.

The single most consequential preparation failure is pricing the alternative on its **invoice** rather than its **consequences**. A competitive re-tender does not cost the difference in licence fees; it costs that plus the weeks it takes at the cost of delay, plus internal effort, plus risk. Omit those and the reservation value is set too low — and a buyer whose reservation value is too low walks away from deals it should accept, which is the opposite of the error everyone worries about.

WORKED EXAMPLE

Worked example 11.3.1 — the Meridian support-and-enhancement extension.

1. **Setup.** Meridian must extend the records-system supplier's support and enhancement contract across the rollout year. **Meridian's alternative** is to re-tender and transition to another supplier: licence and transition **USD 640,000**, **6 weeks** of transition delay at the cost of delay of **14,280** per week, **USD 74,320** of internal transition effort, and transition risk assessed at an `EMV` of **USD 45,000**. **The supplier's position**, as estimated by Meridian from the tender history and published rate cards: cost to serve **USD 520,000**, and a best alternative use of the same team worth **USD 95,000** of contribution. The supplier opens at **900,000**. Assume, as a stated benchmark, that a settlement splits the zone equally.
2. **Formula.** `RV(buyer)` = fully priced alternative. `RV(seller)` = cost to serve + forgone contribution. ZOPA width = `RV(buyer) − RV(seller)`. Midpoint settlement = mean of the two reservation values. Surplus = own reservation value less price (buyer) or price less own reservation value (seller). Share of ZOPA = surplus ÷ width.
3. **Substitution.** `RV(buyer) = 640,000 + 6 × 14,280 + 74,320 + 45,000`. `RV(seller) = 520,000 + 95,000`. Width `845,000 − 615,000`. Midpoint `(845,000 + 615,000)/2`.
4. **Result.** The alternative's delay component is **USD 85,680**, giving `RV(buyer)` = **USD 845,000**. `RV(seller)` = **USD 615,000**. The **ZOPA runs from 615,000 to 845,000**, a width of **USD 230,000**, and the supplier's 900,000 opening is **outside it** — there is no deal at that price. The midpoint settlement is **USD 730,000**, giving each side **USD 115,000** of surplus. Had Meridian not priced its alternative and instead anchored on the supplier's ask, settling at **780,000**, its surplus would be **USD 65,000 — 28.3 %** of the zone against the supplier's **71.7 %** — and **USD 50,000** of value would have transferred purely through the absence of preparation.
5. **Interpretation.** Fifty thousand dollars is the price of an unpriced alternative, available to whichever party has done the arithmetic — roughly fifteen times the 3,300 of 11.1.2, which is this domain's general pattern: preparation is always cheap relative to consequence. Now the subtler result. Suppose Meridian spends **USD 25,000** genuinely pre-qualifying a second supplier, cutting the transition delay from 6 weeks to **2** (saving 57,120) and the transition risk `EMV` from 45,000 to **12,000** (saving 33,000). Its alternative now costs **USD 754,880**, so its reservation value **falls** to that figure, the ZOPA narrows to **USD 139,880**, and the midpoint settlement falls to **USD 684,940 — USD 45,060** less cash, a net gain of **USD 20,060** after the spend, with **USD 33,000** less risk carried; the breakeven pre-qualification spend is therefore **USD 45,060**. And here is the trap: Meridian's surplus has fallen from 115,000 to **USD 69,940**, so a negotiator measured on surplus captured will report the improved position as a worse outcome. Surplus is measured against your own reservation value, which the investment deliberately moved. **The measures that govern are cash paid and risk carried, not surplus captured** — worth insisting on, because surplus-based incentives reward negotiators for keeping their alternatives bad. Three cautions. The counterparty's reservation value is an estimate from observable evidence and must be presented as a range with the assumptions that move it; a single confident figure invites a concession plan built on a guess. The equal-split benchmark is a benchmark, not a prediction — real splits turn on patience, alternatives, internal deadlines and who is measured on what. And a ZOPA computed on price alone is the smaller half of the problem, which is 11.3.2's subject.

11.3.2 Creating value before claiming it

The distinction. Claiming value moves a fixed quantity between the parties. Creating value increases the quantity available, and it is possible whenever the parties value the issues **differently** — which, across any real contract, they almost always do. A negotiation conducted on price alone forgoes all of it, and a negotiation conducted by conceding equally on every issue forgoes most of it.

WORKED EXAMPLE

Worked example 11.3.2 — the trade that created 82,000.

1. **Setup.** Beyond price, three issues are open in the Meridian extension. **A**: priority-incident response time cut from 4 hours to 2 — worth **USD 84,000** to Meridian, costing the supplier **USD 30,000**. **B**: two named engineers dedicated on site through the rollout — worth **USD 36,000** to Meridian, costing the supplier **USD 60,000**. **C**: extending the licence to 8 additional clinics in year 2 — worth **USD 40,000** to Meridian, costing the supplier **USD 12,000**. Compare conceding half of each issue with trading whole issues.
2. **Formula.** Joint gain on a set of issues = Σ(value to buyer) − Σ(cost to seller). Compromise: half of every issue. Trade: take the issues with positive joint value, drop the rest. Share the joint gain through price.
3. **Substitution.** Compromise `0.5 × (84,000 + 36,000 + 40,000) − 0.5 × (30,000 + 60,000 + 12,000)`. Trade `(84,000 + 40,000) − (30,000 + 12,000)`.
4. **Result.** The compromise gives Meridian **USD 80,000** of value at a cost to the supplier of **USD 51,000** — a joint gain of **USD 29,000**. Trading A and C in full and dropping B gives Meridian **USD 124,000** at a cost of **USD 42,000** — a joint gain of **USD 82,000**, which is **2.83 times** the compromise and **USD 53,000** better in absolute terms. Issue B has a joint value of `36,000 − 60,000 =` **−USD 24,000**: it destroys value, so any agreement containing it is jointly worse than the same agreement without it. Sharing the 82,000 equally means compensating the supplier its 42,000 of cost plus **USD 41,000**, so the price rises **USD 83,000** — from the 684,940 of 11.3.1 to **USD 767,940** — and each side is **USD 41,000** better off than under the compromise-free baseline.
5. **Interpretation.** The compromise looks fair, feels reasonable and destroys **53,000** of joint value, which is the most useful thing here: **"splitting the difference" is not a neutral procedure, it is an expensive one.** Two further results are worth carrying. Issue B is the one a buyer instinctively pushes hardest, because on-site presence feels like commitment, and it is the one issue that should be traded away — the arithmetic says so before any relationship is spent on it. A leader who cannot compute joint value will spend negotiating capital on B and concede on C, which is worth 3.33 times its cost to the other side. And the price rise of 83,000 is what makes the trade acceptable to the supplier: a buyer capturing the whole 82,000 by refusing to move on price has converted a value-creating trade into a claiming exercise and will be met accordingly at renewal (Domain 10 makes the same point about gainshare). The cautions are real. These valuations are the buyer's own estimates of both sides' numbers, and the trade only survives if both parties disclose relative priorities — which is not disclosing reservation values, and the distinction is the whole of negotiation ethics in practice: revealing that response time matters more to you than on-site presence is cooperative, revealing the maximum you would pay is simply losing. Where a relationship has no history of reciprocity, an early unilateral disclosure of priorities will be used to claim rather than to create, which is why value creation is easier in a second contract than in a first.

11.3.3 Conflict — sources, modes and the resolution route

Diagnose by source, not by personality. Conflict on projects is routinely attributed to individuals and is usually produced by structure. Five sources, each with a different remedy, and misdiagnosis wastes the intervention:

- **Structural** — two people have been given incompatible objectives, or a matrix has no precedence rule (Domain 3, KA 3.1.2). No amount of facilitation resolves this; only a decision about authority does.
- **Resource** — two workstreams need the same scarce person. Resolved by priority at the level that owns both, and prevented by Domain 6's resource planning.
- **Information** — the parties hold different facts, or the same facts at different ages (11.2.2). The cheapest conflict to resolve if diagnosed early and the most damaging if left, because it teaches each party that the other is unreliable.
- **Interest** — the parties want genuinely different outcomes. This is a negotiation (11.3.1–11.3.2), not a misunderstanding, and treating it as one is condescending as well as ineffective.
- **Interpersonal** — genuine relational breakdown. Real, less common than attributed, and Domain 12's territory.

Choosing a response. The five familiar modes — competing, accommodating, avoiding, compromising, collaborating — are situational rather than ranked, and the skill is selection. Collaboration is the highest-value mode and the most expensive in elapsed time, so it fits a large stake and a continuing relationship and not a decision needed on Friday. Competing is right where safety, legality or a non-negotiable requirement is at stake, and a leader who cannot compete on those will be rolled on them. Accommodating is right where the issue matters little to you and much to the other party — it is how reciprocity is built and the mode most under-used by anxious leaders. Avoiding is right only where events will genuinely resolve the issue, which is rarer than it is invoked. Compromising is the fast default whose cost 11.3.2 has just computed: acceptable for small single-issue matters, expensive everywhere else.

The resolution route, and its cost. Escalation is not a mode; it is what happens when the chosen mode has failed, and it has a price. Domain 3, KA 3.3.3 computed escalation latency and Domain 10, KA 10.4 the dispute ladder for contractual conflict; neither is re-derived here. The point specific to this KA is the **threshold**: an unresolved conflict holding a critical-path decision consumes the cost of delay every week, so the comparison is not "facilitation versus toughing it out" but the intervention's cost against the delay it stops, divided by the probability it works. Exercise 11.5 computes one such case, and the shape of the answer is that breakeven success probabilities for cheap facilitation sit in the low single figures — so the intervention is almost always worth attempting and almost never attempted early.

Two obligations bind a leader who is party to the conflict. Separate the position from the person in the written record as well as in the room, because the record is what the organisation reads later. And hand the resolution to someone who is not a party: a leader chairing a conflict they are in has converted a resolvable dispute into a legitimacy problem, and legitimacy is not recovered by being right.

AI in this KA

Where it earns its place. Preparation, which is where negotiations are decided. Building the reservation-value model of 11.3.1 from its components and testing which assumptions move it — a sensitivity task with a clear right answer. Enumerating trade combinations: with three issues there are few and with twelve there are thousands, and finding the sets with positive joint value is genuine combinatorial work humans do badly. Rehearsing counter-arguments, so that a leader has met the strongest version of the other side's case before the room. Summarising a long contract and correspondence history into a chronology with the source document cited for each entry.

Where it must not go. Estimating the counterparty's reservation value from plausibility: asked for that number a model will supply a confident one with no provenance, it will enter the concession plan as a fact, and real money will move against it. Conducting the negotiation, or drafting messages sent without a human deciding every commitment they contain — a sentence that concedes an issue concedes it. Generating a "fair" outcome, which is a judgement about legitimacy, not a computation. And any use in a live conflict between named individuals: an assessment of a colleague's motives generated by a tool and stored in a project record is an employment and dignity problem regardless of its accuracy.

Verification, concretely. Every input to a reservation value carries a source and a date, and the alternative's delay component is priced at the project's own cost of delay rather than a rate someone remembered. The counterparty's estimated reservation value is recorded as a range with the evidence for each bound, and the concession plan is tested at both. Nothing model-derived is stated to the counterparty as fact. And the ZOPA arithmetic is reproduced by hand before the meeting — two additions and a subtraction, and the only number in the room that must be exactly right.

KEY TERMS — KA 11.3

Term	Meaning
Best alternative	What happens absent agreement, priced fully — invoice cost plus delay at the cost of delay plus internal effort plus risk as an `EMV`.
Reservation value	The worst terms a party would rationally accept, derived from its priced alternative; a maximum for a buyer, a minimum for a seller.
Zone of possible agreement (ZOPA)	The interval between the two reservation values; empty means no deal at any level of skill.
Surplus	Own reservation value less the price agreed (buyer), or price less own reservation value (seller).
Joint gain	Σ(value to one party) − Σ(cost to the other) over the issues traded; negative for issues that destroy value.
Logrolling (trading whole issues)	Conceding entirely on issues valued asymmetrically, rather than partially on all of them.
Conflict source	Structural, resource, information, interest or interpersonal — each with a different remedy.
Resolution route	The chosen mode, its cost, and the escalation path with its latency if the mode fails.

SAMPLE MCQS — KA 11.3

MCQ 11.3-A `[11.3.1 · Application]` A supplier's cost to serve a contract is 520,000 and the best alternative use of the same team would earn 95,000 of contribution. Its reservation value is: - A. USD 425,000 - B. USD 520,000 - C. USD 615,000 - D. USD 95,000

Rationale: `520,000 + 95,000 = 615,000` (11.3.1). B omits the forgone contribution, which is the commonest error and would put the ZOPA floor 95,000 too low; A subtracts it.

MCQ 11.3-B `[11.3.1 · Analysis]` With a ZOPA of 615,000–845,000, a buyer that fails to price its alternative and settles at 780,000 rather than the 730,000 midpoint has: - A. captured 71.7 % of the zone - B. transferred USD 50,000 of value to the seller through the absence of preparation - C. made an error of USD 165,000 - D. exceeded its reservation value

Rationale: `780,000 − 730,000 = 50,000`; the buyer's share falls to 28.3 % (11.3.1). A is the seller's share; C is the seller's surplus; D is false, since 780,000 is inside the zone.

MCQ 11.3-C `[11.3.1 · Evaluation]` Pre-qualifying a second supplier for 25,000 cuts the buyer's reservation value from 845,000 to 754,880 and the midpoint settlement from 730,000 to 684,940. The correct reading is that the buyer is: - A. worse off, because its surplus falls from 115,000 to 69,940 - B. better off by USD 20,060 in cash after the spend, and carrying USD 33,000 less risk - C. unaffected, since both reservation values moved - D. better off by USD 45,060, ignoring the pre-qualification cost

Rationale: Surplus is measured against a reservation value the investment deliberately moved; cash paid and risk carried govern (11.3.1). A is the trap; D omits the 25,000.

MCQ 11.3-D `[11.3.2 · Application]` Three issues are worth 84,000, 36,000 and 40,000 to the buyer and cost the seller 30,000, 60,000 and 12,000. Conceding half of each yields a joint gain of 29,000. Trading the two value-creating issues in full yields: - A. USD 58,000 - B. USD 82,000 - C. USD 124,000 - D. USD 160,000

Rationale: `(84,000 + 40,000) − (30,000 + 12,000) = 82,000` (11.3.2). A trades all three issues, including the one whose joint value is −24,000, giving `160,000 − 102,000 = 58,000`; C is the buyer's value on the traded pair with no deduction of the seller's cost; D is the buyer's value across all three issues.

MCQ 11.3-E `[11.3.3 · Analysis]` Two workstream leads are in open conflict; each has been given an objective the other's success would prevent. Facilitation has failed twice. The diagnosis is: - A. interpersonal conflict requiring coaching - B. structural conflict requiring a decision about objectives and authority - C. information conflict requiring a single authoritative source - D. interest conflict requiring negotiation

Rationale: Incompatible assigned objectives are structural, and no facilitation resolves a structure (11.3.3) — the repeated failure of facilitation is itself the diagnostic.

SELF-CHECK — KA 11.3

1. What is the commonest error in setting a reservation value, and which way does it bite? — Pricing the alternative on its invoice rather than its consequences, which sets the reservation value too low, so a buyer walks away from deals it should accept.
2. Why is splitting the difference expensive? — It concedes partially on every issue, including those with negative joint value; trading whole asymmetrically valued issues produced 82,000 against the compromise's 29,000.
3. What may be disclosed in a value-creating negotiation, and what may not? — Relative priorities may; reservation values may not.

KNOWLEDGE AREA 11.4

Public and community stakeholders, cross-cultural communication and AI-generated communication risk

Topics: 11.4.1 public and community stakeholders and consent · 11.4.2 cross-cultural and multilingual communication · 11.4.3 misinformation and AI-generated communication risk.

11.4.1 Public and community stakeholders and consent

What is different about them. Four properties change the method. They are **unbounded** — there is no membership list, so representativeness is a judgement rather than a fact. They are **heterogeneous**, so a single "community view" is usually an artefact of who attended. Their engagement is often **statutory**, with prescribed forms, periods and grounds of objection that vary entirely by jurisdiction — and nothing here should be read as describing the requirements of any particular one. And their influence is **discontinuous**: low for long periods, then decisive through a planning objection, a legal challenge, a political intervention or simple non-cooperation.

Consultation, and the distinction that decides its value. Consultation before a decision is engagement: it can change the design, so participants have a reason to invest in it. Consultation after a decision is notification with a comment box, and it reliably produces objections rather than absorbing them, because obstruction is the only remaining way to influence the outcome. Case study B computes what that distinction was worth on one project — a factor of **3.90** on total cost — and Domain 5's design-maturity argument explains the mechanism: accommodating a concern costs what it costs to change the design at the maturity you have reached.

Pricing consent risk. Consent risk can be priced with Domain 8's `EMV`, provided the probabilities are presented as judgements and the breakeven is stated.

WORKED EXAMPLE

Worked example 11.4.1 — is Meridian's public consultation worth holding?

1. **Setup.** Meridian's records system creates a regional data-sharing capability with a legitimate public interest in it. Without early public consultation, the programme assesses the probability of a formal privacy complaint that suspends the data-sharing element at **0.35**; with a structured consultation costing **USD 48,000** it assesses **0.10**. A complaint would cost **12 weeks** of delay at the cost of delay of **14,280** per week plus **USD 60,000** of legal and response cost.
2. **Formula.** `EMV` = probability × consequence (Domain 8). Compare `EMV` without consultation against consultation cost + `EMV` with it. Breakeven consultation cost = the difference in the two `EMV` terms. Required probability reduction = consultation cost ÷ consequence.
3. **Substitution.** Consequence `12 × 14,280 + 60,000`. Without `0.35 × 231,360`. With `48,000 + 0.10 × 231,360`. Breakeven `80,976 − 23,136`. Required reduction `48,000 ÷ 231,360`.
4. **Result.** The consequence is **USD 231,360** (delay **171,360** plus response **60,000**). `EMV` without consultation **USD 80,976**; with consultation **USD 71,136**. The consultation is worth **USD 9,840**, it remains worth holding while it costs less than **USD 57,840**, and it must reduce the probability of complaint by at least **20.75 percentage points** — from 0.35 to **0.1425** — to break even.
5. **Interpretation.** The point estimate of 9,840 is the weakest number here and should never be the one presented: it is the difference of two products of judged probabilities, so it is fragile in both directions and a reviewer can argue it away in a sentence. The **20.75-point required reduction** is the number to present, because it asks a question consultation practitioners can answer from experience — does a structured, pre-decision consultation cut the likelihood of a formal complaint by more than a fifth? — and because it exposes how much of the case rests on judgement rather than concealing it. The breakeven cost of **57,840** answers a different and often more useful question: how much more consultation could be justified on risk grounds alone. Two limits must then be stated plainly, and stating them is part of the professional standard rather than a caveat on it. First, **the probabilities are not measurements.** They are structured judgements, to be recorded with their basis and their author; a programme reporting 0.35 without saying who judged it and on what has manufactured a number. Second, and more important: **consultation is not only a risk control.** A community's entitlement to be heard about a change to its own services does not depend on whether hearing it reduces an expected cost, and a programme that would abandon consultation because the arithmetic came out at −9,840 has misunderstood what the arithmetic is for. The `EMV` supports the case for consultation where it is resisted; it is not a licence to withdraw it where it is not. Legitimacy is not quantifiable, and this book will not pretend otherwise.

11.4.2 Cross-cultural and multilingual communication

What actually varies. Language is the visible difference and rarely the expensive one. Four others cost more, and each is observable rather than stereotypical — the discipline is to observe how a specific organisation behaves rather than to predict it from nationality, which is both unreliable and offensive.

- **Directness.** Whether disagreement is stated or signalled. Where it is signalled, a project that hears only stated objections has a silent risk register, and the counter-move is to seek disagreement in private and in writing rather than to ask for it in a meeting.

- **Decision convention.** Whether a meeting decides or ratifies a decision reached beforehand, and whether authority is individual or collective. A leader who brings a decision to a meeting in a consensus-forming organisation gets agreement in the room, no decision, and the discovery weeks later.
- **Attitude to hierarchy.** Whether bad news may travel directly to the accountable person or must pass through the chain — which determines the actual escalation latency, potentially several times the designed one of Domain 3, KA 3.3.3.
- **Time and commitment.** Whether a date is a commitment, an intention or an aspiration; and whether a written record is the agreement or a note about it.

Working method. Establish the convention explicitly and early — "a decision is recorded in the log and takes effect then; agreement in a meeting is not yet a decision" — substituting an agreed project convention for a clash of unexamined ones. Provide material in the language of the people who must act on it, not merely of those who sign; for Meridian that means clinical-workflow material in the language clinicians work in. Use **back-translation** for anything carrying a commitment, instruction or safety implication. Confirm understanding by asking the recipient to state the action, not to confirm receipt. And allow in the schedule for the elapsed time consensus formation actually takes: a plan assuming decisions are made in meetings, in an organisation where they are not, carries a systematic and invisible underestimate in every approval duration.

11.4.3 Misinformation and AI-generated communication risk

The exposure. A programme of any public significance operates in an information environment where claims about it circulate faster than it can respond — some mistaken, some deliberate, and increasingly some machine-generated. Three risks matter to a delivery leader, and none is a communications-department problem.

Misinformation about the project. Incorrect claims about cost, purpose, data handling or safety gather support because they arrive first and are more interesting than the truth. Corrections propagate less well than the original claim — a real and well-observed asymmetry, and one this book will not attach a multiplier to, because no honest general figure exists. The countermeasures are structural rather than rhetorical: publish the authoritative facts before the questions arrive, so a correction is a reference rather than a rebuttal; maintain a single named source of truth that journalists and community groups can check; respond to the substance and never to the motivation; and monitor deliberately, because the first a programme usually hears of a claim is when a governance member forwards it.

Impersonation and synthetic content. A programme's name, branding, spokespeople and letterhead can be reproduced convincingly at negligible cost, including synthetic voice and video of real individuals. Two controls carry most of the weight: a published, stable statement of the channels through which the programme communicates officially, so a recipient can verify; and an internal rule that no commitment is ever created by an inbound communication alone, whatever it appears to be — a fraud control as much as a communications one.

AI-drafted communication that nobody verified. The risk most within the leader's control and most often mishandled, because the failure does not look like a failure. A model drafting a stakeholder bulletin produces fluent, plausible text containing dates, commitments and attributions it inferred rather than knew. A bulletin stating that a clinic will go live in March, when no such commitment exists, has created an expectation, possibly a representation, and certainly a dispute — and it will be quoted back to the programme by the party it misled. **That is a governance defect, not a copy-editing one**: it is an unauthorised commitment, and it warrants the seriousness Domain 3 attaches to a decision taken without authority.

WORKED EXAMPLE

Worked example 11.4.3 — the verification step, priced.

1. **Setup.** Meridian issues **96** external communications a year: 48 weekly bulletins, 24 clinic-specific notices, 12 authority updates and 12 public updates. AI-assisted drafting saves **1.2 hours** each. A mandatory verification step — every factual claim, date and commitment checked against a source by a named person — adds **0.5 hours** each. Blended rate **USD 110** per hour. Without verification, the programme assesses that **4 %** of items would carry a material misstatement, each costing **USD 6,000** in correction, clarification and relationship repair.
2. **Formula.** Net hours = items × (drafting saving − verification time). Verification cost = verification hours × rate. Unverified exposure = items × error rate × cost per error. Breakeven error rate = verification cost ÷ (items × cost per error).
3. **Substitution.** Net `96 × (1.2 − 0.5)`. Verification `96 × 0.5 × 110`. Exposure `96 × 0.04 × 6,000`.
4. **Result.** Drafting saves **115.2 hours**; verification consumes **48.0**; the net saving is **67.2 hours**, worth **USD 7,392**. Verification costs **USD 5,280** and stands against an unverified exposure of **USD 23,040** across an expected **3.84** defective items — a return of **4.36 times**. The verification step pays at any error rate above **0.92 %**.
5. **Interpretation.** The result contradicts the argument usually made in both directions. AI-assisted drafting **does** save time — 67.2 hours a year net, **1.4 hours a week** across the 48-week operating year, returned to the work 11.1.3 showed is worth 382.50 an hour — and the verification step **is not** what destroys the saving: it consumes **41.7 %** of it and returns 4.36 times its cost. The decisive figure is the **0.92 % breakeven error rate**, because nobody will claim that unverified machine-drafted communications about a health programme are accurate more than 99.08 % of the time at the level of individual dates and commitments — so the verification step is not a judgement call. Three qualifications. The 4 % error rate and the 6,000 per-error cost are assumptions and should be replaced with measured values as soon as the programme has any; the verification log produces the error rate as a by-product, which is the cheapest measurement available and almost never kept. The exposure prices correction and relationship repair only — a misstatement about patient data would not be a 6,000 event, so the highest-consequence categories warrant a heavier check than 0.5 hours. And the saving is only real if drafting replaces work rather than generating more communication: 96 items a year is a designed number, and a programme issuing 150 because drafting became cheap has not saved 67 hours, it has spent 40 more and added 54 items of exposure.

AI in this KA

Where it earns its place. Drafting in multiple languages with **back-translation** as a check — a genuine capability improvement for programmes that previously could not afford multilingual material at all. Monitoring public and social channels at volume for claims about the programme, so the first a leader hears of a misstatement is not a governance member's forwarded message. Readability and plain-language testing, which is objective and tedious. Producing accessible formats — summaries, alternative text, simplified versions — from one approved factual base. And drafting the first version of the factual-claims register that 11.4.3's verification step needs.

Where it must not go. External publication without human verification, for the reasons priced above. Any **synthetic voice or likeness** of a real person, however convenient and however disclosed — a programme cannot credibly campaign against impersonation while practising it. Automated replies to community objections, which read exactly as they are and convert a substantive objection into a legitimacy grievance. Personalised persuasive targeting of identified individuals, a manipulation risk no delivery benefit justifies. And translation of safety-critical or legally operative text without qualified human review, where a plausible error is worse than an obvious one.

Verification, concretely. A **two-person rule** on every external communication — a drafter and a named verifier, recorded, the verifier accountable for every date, figure and commitment in the text. A **factual-claims register** listing each claim and its source document, retained with the issued item so that a challenge is answered from the record rather than from memory. Back-translation on anything carrying a commitment or an instruction. A published statement of where the programme uses AI in its communications, because asking a community to trust you while concealing how you speak is the wrong economy. And the verification log's error rate reported monthly, since it is the only measurement that says whether the controls work.

KEY TERMS — KA 11.4

Term	Meaning
Community stakeholder	An unbounded, heterogeneous group with a legitimate interest in a project's effects, often with statutory engagement rights that vary by jurisdiction.
Pre-decision consultation	Consultation capable of changing the design — engagement; as distinct from post-decision notification.
Consent risk pricing	`EMV` of a consent failure, presented with its breakeven cost and required probability reduction, never as a point estimate alone.
Decision convention	Whether a meeting decides or ratifies, and whether authority is individual or collective.
Back-translation	Independent re-translation to the source language as a check on anything carrying a commitment, instruction or safety implication.
Correction asymmetry	The observed tendency of corrections to propagate less well than the original claim; real, and not reduced here to a multiplier.
Synthetic content risk	Convincing reproduction of a programme's identity, spokespeople, voice or documents at negligible cost.
Factual-claims register	Each claim in an external communication with its source document, retained with the issued item.
Two-person rule	A named drafter and a named verifier for every external communication.

SAMPLE MCQS — KA 11.4

MCQ 11.4-A `[11.4.1 · Application]` A consent failure would cost 231,360. Its probability is 0.35 without consultation and 0.10 with a consultation costing 48,000. The consultation is worth: - A. USD 57,840 - B. USD 9,840 - C. USD 80,976 - D. USD 23,136

Rationale: `0.35 × 231,360 = 80,976` against `48,000 + 0.10 × 231,360 = 71,136`; the difference is 9,840 (11.4.1). A is the breakeven consultation cost; C and D are the two `EMV` terms.

MCQ 11.4-B `[11.4.1 · Evaluation]` For the same case, the figure a leader should present to a sceptical board is: - A. the expected saving of 9,840 - B. the required probability reduction of 20.75 percentage points - C. the consequence of 231,360 - D. the consultation cost of 48,000

Rationale: The point estimate is the difference of two judged probabilities and is fragile; the required reduction states exactly what must be believed and can be answered from experience (11.4.1).

MCQ 11.4-C `[11.4.2 · Analysis]` A programme translates all its material accurately but assumes that agreement reached in a meeting is a decision. In a consensus-forming counterpart organisation the predictable consequence is: - A. the translation will be misunderstood - B. agreement in the room, no decision, and the difference discovered weeks later - C. escalation will be faster than designed - D. the counterpart will refuse to meet

Rationale: Decision convention, not language, is the expensive variable (11.4.2); the plan's approval durations are systematically understated as a result.

MCQ 11.4-D `[11.4.3 · Application]` Ninety-six external communications a year, verification at 0.5 hours each and USD 110 an hour, against an assessed 4 % error rate at USD 6,000 per error. The return on the verification step is: - A. 2.18 times - B. 4.36 times - C. 0.23 times - D. 3.12 times

Rationale: Exposure `96 × 0.04 × 6,000 = 23,040` against verification `96 × 0.5 × 110 = 5,280`, a ratio of 4.36 (11.4.3). A uses 1.0 hour per item rather than 0.5 (`23,040 ÷ 10,560`); C inverts the ratio (`5,280 ÷ 23,040`); D uses the 7,392 net hourly saving as if it were the verification cost.

MCQ 11.4-E `[11.4.3 · Analysis]` An AI-drafted bulletin states a go-live month to which the programme has not committed. This is best treated as: - A. a copy-editing error, corrected in the next issue - B. a governance defect — an unauthorised commitment made outside the decision system - C. a supplier performance issue - D. an acceptable risk given the drafting time saved

Rationale: The defect is the creation of an expectation and possibly a representation without authority, which is Domain 3's category of a decision taken without authority (11.4.3).

SELF-CHECK — KA 11.4

1. What distinguishes consultation from notification, and why does it change the cost? — Consultation can change the design, so accommodating a concern costs what a change costs at that design maturity; after the decision, the only remaining influence is obstruction.
2. Name the cross-cultural variable that most distorts a schedule, and how. — Decision convention: where meetings ratify rather than decide, every approval duration in the plan is understated.
3. Why is an unverified AI-drafted commitment a governance matter? — It is an unauthorised commitment, created outside the decision system, and it will be quoted back to the programme by the party it misled.

Advanced topics — Domain 11

11.A.1 Influence without authority, and the arithmetic of a blocking minority

Project leaders routinely need agreement from people they cannot instruct, which is why influence is a professional competence rather than a personality trait. Four sources require no authority: **expertise** (being the person who knows, demonstrated rather than asserted); **information** (holding the integrated picture nobody else holds — a by-product of the leader's position, squandered by hoarding it); **reciprocity** (having accommodated others on issues that mattered to them and not to you — 11.3.3's under-used mode, and the reason accommodating is an investment rather than a weakness); and **coalition** (the support of parties the target respects). All four are built before they are needed, which is the practical difficulty: influence is a stock accumulated in quiet periods and spent in loud ones, and a leader who begins building it when the difficult decision arrives has begun too late.

Coalition work has an arithmetic worth doing, because voting rules make the target countable. In a nine-member body requiring a two-thirds majority, **6 of 9** votes carry a motion, so the smallest blocking coalition is `9 − 6 + 1 =` **4**. Those two numbers answer different questions and are habitually confused: a leader seeking approval must secure **six** supporters — not five, and not "a majority" — while a leader assessing risk must ask whether any **four** parties share an interest in refusal. Domain 3, KA 3.1.2 showed that unanimity is indistinguishable from an inability to decide; this is the same result at finer resolution, and the consequence is that an influence plan for a supermajority body targets a named list of six, with the four most likely refusers engaged first, rather than a general campaign of goodwill.

11.A.2 The stakeholder register as a decaying asset

A register is treated as a document and behaves like a perishable one, because the entries are roles and the relationships are with people. Meridian's register holds **62** entries; over twelve months **19** changed role holder — an annual turnover of **30.6 %**. The consequence is computable from the refresh cadence: with an annual refresh, entries are on average half a year old, so about **15.3 %** of the register is wrong at any moment; with a quarterly refresh, about **3.8 %**. Fifteen per cent of 62 means roughly nine relationships the programme believes it has and does not — and the nine are not distributed evenly, because turnover is highest in exactly the junior and operational roles that carry the day-to-day cooperation.

Three practices follow. Refresh on a **cadence tied to observed turnover** rather than annually by habit; the arithmetic above is the justification and it costs an hour a quarter. Record, for each key relationship, **who holds it on the programme side**, because the programme's own turnover does the same damage in the other direction and a departing engagement lead takes the relationship with them unless it is deliberately transferred. And treat a change of role holder as a **re-engagement trigger**, not an administrative update: a new incumbent has agreed nothing, has not seen the specification, and is the single most common source of the 11.1.2 objection arriving in week 34.

11.A.3 The reviewer's stakeholder eye

Invariants to test on any engagement and communication design, each cheap and each diagnostic.

Every register entry names a **decision, consent, resource or action** the project needs from that party, and the **cost if it arrives late**. Every party is scored on **four** dimensions, not two, with **consent risk** scored from position rather than disposition. The engagement allocation is **explicit and sums to a stated capacity**, with a value-led core sized from the benefit arithmetic, consent-risk floors and a residual — not a single proportional rule — and every floor is justified by a stated **breakeven probability**. The communication architecture's **channel load is computed** against capacity, and the design names which relationships are deliberately bilateral and why. Report `P`**,** `C` **and** `L` **are stated and maximum blindness computed**, and any proposal to lengthen the cycle quotes the new figure. Status thresholds are **objective and set at baseline**, and the report records **the date the leader knew**. Every negotiation of material value has a **written reservation value derived from a priced alternative** — including delay at the cost of delay and risk as an `EMV` — plus an estimated counterparty range with its evidence, and multi-issue negotiations carry an **issue-by-issue value and cost table** so that value-destroying issues are identified before capital is spent on them. Every external communication has a **named verifier** and a factual-claims register. The register's **refresh cadence is derived from observed turnover**. And the test that subsumes several others: for each of the last three stakeholder surprises, ask **which dimension of the assessment would have predicted it** — a surprise no dimension would have predicted indicates a missing dimension, and one all four would have predicted indicates an allocation that was never made.

Industry variations — Domain 11

- **Public sector and government.** Consultation is often statutory in form, period and grounds, so the engagement plan is partly a compliance instrument and the discretion sits in how early rather than whether; political stakeholders change on an electoral cycle unrelated to the project, making 11.A.2's re-engagement trigger a scheduled event rather than a contingency.
- **Healthcare.** Clinical authority is not delegable to a project body (Domain 3), so clinical stakeholders hold consent risk regardless of their interest score — which is why Meridian's design floors them; and the benefit is almost entirely adoption-determined, so 11.1.3's value core dominates.
- **Construction and infrastructure.** Community and landowner objection rights convert directly into programme delay, so consent risk is the dominant dimension; the interface-to-people mapping of 11.2.1 is at its most literal, with parties numerous, physically co-located and contractually separated.
- **Energy and resources.** Community consent is frequently the critical path rather than a risk to it (Case study B), long asset lives mean the engagement outlives the project team, and consultation held after a layout decision is the standard and expensive error.
- **Technology and product organisations.** Users are numerous, unrepresented and reachable directly, which makes 11.1.3's value core computable from telemetry rather than judgement; the characteristic failure is over-weighting the loud internal stakeholder whose interest score is high and whose consent risk is nil.
- **Financial services and regulated industries.** The regulator is the archetypal high-influence, low-interest, high-consent-risk party of 11.1.2, and engagement is bounded by rules about what may be discussed and when — so the floor is spent on formal, documented pre-submission engagement rather than relationship building, and the record of that engagement is itself a regulatory artefact.

Case study — Domain 11: the stakeholder who was only informed (health, Meridian)

Situation. At week 36 of Meridian's rollout, 31 of 40 clinics were installed and the programme was reported amber. Adoption among installed clinics was running at 62 %, below the 70 % the business case assumed. The national reporting body had objected in week 34 to the structure of the statutory data extract, and the objection was being handled as a technical issue by the integration team. The steering committee had been told none of this in those terms: the week-32 pack described adoption as "building" and the extract as "under discussion with the regulator".

What the arithmetic showed. The project leader computed four numbers before the next governance review. The **objection**: 4.0 weeks of governance latency plus 5 weeks of rework, three interface re-verifications at 18,000 and 42,000 of application rework — **USD 224,520**, against a technical pre-consultation at design stage costing **USD 3,300**, a breakeven probability of **1.47 %**. The **communication load**: 14 actively managed parties, unrouted, 91 channels at 1.5 hours a month — **136.5 hours** against a capacity of 40, or **341.25 %** — which explained without anyone needing to be blamed why the reporting body had received a bulletin and nothing else. The **report age**: `P` 4, `C` 1, `L` 2, a mean fact age of **5.0 weeks** and a maximum blindness of **7.0 weeks**, so the extract issue could have run seven weeks before the committee saw it, and had. And the **allocation**: the plan's 480 hours, distributed in proportion to influence × interest, gave the two adoption-determining groups **180 hours** — enough to run the 8-hour adoption intervention in 22 of 40 clinics, **USD 67,320** a year of the available **USD 122,400**, forgoing **USD 55,080** a year permanently.

How it resolved. Four changes, none requiring new money. The communication architecture became **hybrid**: all 14 parties routed through a named engagement function, with a deliberate three-party bilateral mesh for the sponsor, the clinical authority and the reporting body — 17 channels, **37.5 hours a month**, **93.75 %** of capacity, stated as tight in the plan rather than presented as comfortable. Data cut **weekly** into a standing dashboard with the monthly pack drawing the latest cut, taking maximum blindness from **7.0 to 2.5 weeks** and mean fact age from 5.0 to 2.0 — worth **USD 42,840** on the mean and **USD 64,260** in the worst case per qualifying occurrence. The 480 hours were **reallocated** on the three-term design: a 320-hour adoption core, 130 hours of consent-risk floors including 40 for the reporting body, and a 30-hour residual. And status thresholds were fixed at baseline, so "amber" acquired a definition and the week-32 pack's language became impossible to repeat.

The outcome, and the part that mattered. The extract objection took 9 weeks to clear as computed; nothing recovered that. Adoption among installed clinics reached 79 % by month 12 — short of the 82.5 % the intervention was designed for, and the shortfall traced to four clinics where the workflow sessions were run by the supplier rather than by clinical peers, a finding the programme would not have had without an allocation explicit enough to audit. The durable change was elsewhere. "We need more stakeholder engagement" had sat in the risk register for two quarters and produced nothing. "Our communication design requires 341 % of our engagement capacity, our committee decides on five-week-old facts, and our allocation forgoes 55,080 a year" was approved in one meeting.

What the domain teaches here. Engagement failures present as behaviour and are usually capacity and allocation failures with an arithmetic signature. The overloaded architecture is not a workload complaint; it is the mechanism that produced the week-34 objection, because the contacts an overloaded design drops are precisely the low-interest, high-consent-risk ones. And note which correction was largest for the least money: the 480 hours were already being spent.

Case study B — Domain 11: the consultation that was held twice (energy)

Situation. A regional utility developed a 90 MW onshore wind project with a new grid connection across farmland. Its financial model priced the cost of delay at **USD 62,000 per week** in forgone contracted revenue net of variable cost. The development team ran a full public consultation costing **USD 180,000** — after the turbine layout and connection corridor had been fixed, on the reasonable view that consulting on an incomplete design would confuse people.

What happened. The consultation produced 340 responses, the substantive ones concentrating on two turbine positions and one section of the corridor. Because the layout was fixed, the only available responses were to refuse the objection or to change the design at detailed-design maturity. Two objections proceeded formally. The developer ran a **second consultation** at **USD 240,000**, conceded a layout change costing **USD 410,000** at detailed design, and lost **22 weeks** of planning determination — **USD 1,364,000** at the project's cost of delay. Total: **USD 2,194,000**.

How it resolved, and what the alternative would have cost. The post-project review modelled the counterfactual with the same objections arising. Consultation at **options** stage, on three candidate layouts, would have cost **USD 260,000** — 80,000 more, because two rounds and three options are more work than one round and one answer. The same two turbine positions and the same corridor section would have been changed while the layout was still a set of options, at **USD 55,000** rather than 410,000 — a **7.45-fold** escalation avoided, and Domain 5's design-maturity argument in money. Determination would have run 4 weeks over plan rather than 22: **USD 248,000**. Total: **USD 563,000**. The early programme costs **USD 80,000** more in consultation and **USD 1,711,000** less in everything else — a net saving of **USD 1,631,000**, a total cost ratio of **3.90**, and **USD 21.39** avoided downstream for each additional dollar of early consultation.

What the domain teaches here. Consultation before a decision is engagement; after a decision it is notification, and it reliably produces objections because obstruction is the only influence left. The mechanism is not goodwill but **design maturity**: the identical concern cost 55,000 to accommodate at options stage and 410,000 at detailed design, and the 22 weeks of delay followed from having nothing to offer. Note what the developer's reasoning got right — consulting on an incomplete design is harder — and where it went wrong: it treated that difficulty as a reason to defer, when the arithmetic says the difficulty is the price of the option. And note the number without which none of this is arguable: without 62,000 a week, the 22 weeks are an inconvenience and the whole case dissolves into opinion.

Executive perspective — Domain 11

What a programme director cannot delegate in this domain:

- **The engagement allocation, in hours, summing to your stated capacity.** Not a communication plan — an allocation, with a value-led core sized from the benefit arithmetic and a breakeven stated for every consent-risk floor. Meridian's misallocation forgave 55,080 a year of capacity it was spending anyway (11.1.3).
- **The channel load of your own communication design.** Compute it. An architecture demanding 341 % of capacity is not a workload problem, it is the mechanism that generates your next late objection, and it fails silently on the parties who do not chase (11.2.1).
- **Maximum blindness.** `P + C + L`. Know the number, quote it whenever anyone proposes a longer cycle, and be able to say how old the average fact in your last pack was when the committee decided (11.2.2).
- **The status definition, set at baseline, and the date you knew.** Discretion over an undefined threshold is what corrupted the signal; a definition and a knew-date restore it, and the knew-date is what makes your escalation lead time measurable (11.2.3, Domain 3).
- **Your reservation value on every material negotiation, in writing, before the meeting.** Derived from a fully priced alternative including delay at your cost of delay and risk as an `EMV`. An unpriced alternative cost Meridian 50,000 on one contract; and never let anyone measure your negotiators on surplus captured, which rewards them for keeping the alternative bad (11.3.1).
- **That no external communication leaves the programme unverified.** A named verifier and a factual-claims register on every item. The verification step returns 4.36 times its cost and pays at any error rate above 0.92 % — and an AI-drafted commitment nobody checked is an unauthorised commitment, not a typographical one (11.4.3).

Calculation exercises — Domain 11

Exercise 11.1 A programme actively manages 11 parties. Each maintained channel costs 2.0 hours a month; engagement capacity is 45 hours a month; a routed design carries a hub overhead of 10 hours a month. Compute the mesh and routed loads and their utilisation, the largest number of parties each design sustains, and the annual hour saving from routing. Solution. Mesh channels `11 × 10/2 =` **55**, load `55 × 2 =` **110 hours a month**, or `110/45 =` **244.4 %** of capacity. Routed `11 × 2 + 10 =` **32 hours**, **71.11 %**. Largest sustainable unrouted count: 7 parties give 21 channels and **42 hours** (within 45); 8 give 28 channels and **56 hours** (over). So **7**. Routed: `2n + 10 ≤ 45` gives `n ≤ 17.5`, so **17** (44 hours). Annual saving `(110 − 32) × 12 =` **936 hours**. Common error: computing n^2 (121) or `n(n − 1)` (110 channels rather than 55) — the second is doubly seductive here because it coincides numerically with the correct hour figure.

Exercise 11.2 A programme has 600 engagement hours over 15 months and five stakeholder groups scored influence × interest: A `5 × 4`, B `3 × 5`, C `4 × 2`, D `2 × 4`, E `3 × 3`. Compute the salience-proportional allocation. Then rebuild it as a three-term design in which C (the statutory approver) takes a consent floor of 120 hours and D (the adopting user group across 50 sites) takes a value-led core of 260 hours, with the residual apportioned by salience among A, B and E. Each adopting site is worth USD 19,200 a year and the core is expected to raise adoption by 10 percentage points. Compute the annual gain, the benefit per engagement hour and the core's hours per site. Solution. Salience: A 20, B 15, C 8, D 8, E 9, total **60**; `600/60 =` **10 hours per point** → A **200**, B **150**, C **80**, D **80**, E **90**. Three-term: C **120**, D **260**, residual `600 − 120 − 260 =` **220** across salience `20 + 15 + 9 = 44` at 5 hours a point → A **100**, B **75**, E **45**; total 600. Gain `50 × 0.10 × 19,200 =` **USD 96,000** a year; per engagement hour `96,000/260 =` **USD 369.23**; core intensity `260/50 =` **5.20 hours per site**. Common error: allocating by salience alone, which gives the statutory approver 80 hours against a required floor of 120 and the adopting group 80 against a required 260 — under-serving simultaneously the party that can stop the work and the party that determines whether any benefit exists.

Exercise 11.3 A monthly report has a 4-week period, 1.5 weeks of consolidation and a 1-week paper lead time; cost of delay is 9,500 per week. Compute the newest fact age, mean fact age and maximum blindness, then the same three under a fortnightly cycle with 0.5 weeks of consolidation and a 0.5-week paper lead time, and price each saving. Solution. As found: newest `C + L = 1.5 + 1 =` **2.5 weeks**; mean `C + L + P/2 = 2.5 + 2 =` **4.5 weeks**; maximum blindness `P + C + L =` **6.5 weeks**. Redesigned: newest `0.5 + 0.5 =` **1.0**; mean `1.0 + 1.0 =` **2.0**; blindness `2 + 0.5 + 0.5 =` **3.0**. Savings **1.5**, **2.5** and **3.5 weeks**, priced at **USD 14,250**, **USD 23,750** and **USD 33,250**. Common error: reporting the newest fact age as the age of the picture, which understates the mean by `P/2` — 2.0 weeks here, or 19,000 at this cost of delay.

Exercise 11.4 A buyer's fully priced alternative to renewing a contract costs USD 1,200,000. The incumbent's cost to serve is USD 820,000 and its best alternative use of the same team would earn USD 130,000 of contribution. Compute both reservation values, the ZOPA and its width, the midpoint settlement and each side's surplus there. Then compute the buyer's surplus and share of the zone if it anchors on the incumbent's opening ask and settles at USD 1,090,000, and the value transferred relative to the midpoint. Solution. `RV(seller) = 820,000 + 130,000 =` **USD 950,000**; `RV(buyer) =` **USD 1,200,000**. ZOPA **950,000 – 1,200,000**, width **USD 250,000**. Midpoint **USD 1,075,000**, surplus **USD 125,000** each. At 1,090,000 the buyer's surplus is `1,200,000 − 1,090,000 =` **USD 110,000**, **44.0 %** of the zone against the seller's **56.0 %**, and **USD 15,000** has transferred relative to the midpoint. Common error: treating the seller's cost to serve as its reservation value,

> which puts the ZOPA floor at 820,000 — 130,000 too low — and leads a buyer to pursue prices the seller will never accept, or to conclude wrongly that a workable deal is unavailable.

> **Exercise 11.5** An unresolved priority conflict between two workstreams has held a critical-path decision for 3 weeks at a cost of delay of 14,280 a week, and escalation preparation has consumed 4 people × 5 hours at the blended rate of 110. A facilitated resolution session would take 6 hours of a neutral facilitator at 150 an hour plus 14 hours of participant time at 110. Compute the cost already incurred, the cost of the facilitated route, the ratio between them, and the minimum probability of success that would have justified the facilitation. Solution. Incurred: delay `3 × 14,280 =` **USD 42,840** plus preparation `20 × 110 =` **USD 2,200**, total **USD 45,040**. Facilitation: `6 × 150 + 14 × 110 = 900 + 1,540 =` **USD 2,440**. Ratio **18.46**. Breakeven success probability `2,440/45,040 =` **5.42 %**. Common error: comparing the facilitator's fee alone (900) with the delay and concluding the intervention is trivially cheap — participant time of 1,540 is the larger part of its cost and is the real reason facilitation is resisted; conversely, omitting the 2,200 already spent on escalation preparation understates what the conflict has cost by 5 % of the total.

Practitioner's toolkit — Domain 11

Adoption-ready artefacts; adapt headings to your organisation, then keep them stable.

Toolkit 11.T.1 — Stakeholder register and engagement allocation sheet

One row per party: reference · party and named individual · **what the project needs from them** (decide / consent / resource / do) · **cost if their agreement arrives late**, priced at the cost of delay · influence (1–5) · interest (1–5) · attitude, with the evidence · **consent risk** (none / withholds cooperation / holds approval or veto, with the instrument) · relationships (who they follow, who follows them) · engagement objective as a **dated future state** · named owner on the programme side · method · cadence · **allocated hours** · test of whether it worked · date last verified.

Beneath the rows, the allocation summary: stated capacity in hours; the **value-led core** with the benefit arithmetic that sizes it; each **consent-risk floor** with its breakeven probability; the **residual** and its apportionment; and a total that must equal capacity. Report monthly: parties with no named programme owner, floors underspent, objectives past their date, and entries unverified within the refresh cadence derived from observed turnover (11.A.2). An allocation that does not sum to a stated capacity is a wish list.

Toolkit 11.T.2 — Communication architecture and report-age sheet

Part one, the architecture. Actively managed parties `n` · channels under a mesh `n(n − 1)/2` and under routing `n` · hours per channel per period, with the basis · hub overhead · computed load and utilisation against stated capacity · the **sustainable party count** under each design · and an explicit list of **deliberately bilateral relationships** with the reason each cannot be intermediated. Part two, report age. For each recurring report: `P` · `C` · `L` · computed newest fact age, mean fact age and **maximum blindness** · the cost of delay · and the priced saving from each candidate reduction. The purpose is that both quantities are **visible at design time**; a communication plan that cannot fill this in is a distribution list.

Toolkit 11.T.3 — Negotiation preparation sheet

Our position. The alternative, itemised: invoice or transition cost · elapsed weeks × cost of delay · internal effort · risk as an EMV, with its basis · **total = our reservation value**. What would improve the alternative, at what cost, and the **breakeven spend** — the shift it produces in the reservation value. Their position, estimated. Cost to serve · forgone contribution · **estimated reservation value as a range**, with the evidence for each bound. The zone. ZOPA low and high, width, and the settlement at the equal-split benchmark. The issues. One row per non-price issue: value to us · cost to them (estimated) · **joint value** · trade or drop, with every negative-joint-value issue flagged as not to be pursued. Conduct. Concession sequence and what each concession buys · what we will disclose (relative priorities) and what we will not (reservation value) · who speaks · what requires escalation before agreement. Reviewed against outcome afterwards, because the estimate of the counterparty's reservation value is the input that most needs calibrating and is almost never checked against what happened.

Exam preparation — Domain 11

What is assessed. The stakeholder system and the parties registers omit; four-dimensional assessment and what each dimension predicts; **pricing a late objection and its breakeven probability**; **the three-term engagement allocation under a capacity constraint** and what a salience-only rule forgoes; communication as an architecture, with **channel load and sustainable party count**; **report age — newest, mean and maximum blindness**; honest status, objective thresholds and the escalation-grade message; **reservation values from priced alternatives, the ZOPA, surplus and share**; **joint gain from trading whole issues against splitting the difference**; conflict diagnosis by source; consultation before versus after a decision, and **consent risk priced as an EMV with its required probability reduction**; cross-cultural decision conventions; and the verification of AI-drafted communication.

The calculations to be able to do under time pressure. Assessed cost of an objection from governance latency plus rework at the cost of delay, plus interface and rework components, and its breakeven probability. A salience-proportional allocation, then a three-term allocation to the same capacity; benefit per engagement hour; the breakeven adoption uplift. Mesh and routed channel counts and loads, and the sustainable party count under a stated capacity. `C + L`, `C + L + P/2` and `P + C + L`, each saving priced. Reservation values from a fully priced alternative (including delay and EMV), ZOPA width, midpoint, surplus, share of zone and value transferred against a benchmark. Joint gain for a compromise and for a trade, and the price rise that shares it. EMV with and without a mitigation, the breakeven mitigation cost and the required probability reduction. Verification cost against exposure, and the breakeven error rate.

The traps. Computing n^2 or $n(n-1)$ for channels (Exercise 11.1) · quoting the newest fact age as the age of the picture, understating the mean by $P/2$ (Exercise 11.3) · omitting governance latency when pricing an objection (MCQ 11.1-B) · allocating capacity by influence × interest alone, which under-serves the benefit-determining group and the quiet approver simultaneously (11.1.3, Exercise 11.2) · treating a seller's cost to serve as its reservation value and putting the ZOPA floor too low (Exercise 11.4) · pricing an alternative on its invoice rather than its consequences, which sets a reservation value too low and forfeits deals that should be accepted (11.3.1) · reading a fall in surplus as a worse outcome when the reservation value was deliberately moved (MCQ 11.3-C) · splitting the difference across issues including those with negative joint value (11.3.2) · presenting an EMV point estimate rather than the required probability reduction (11.4.1) · annualising a report-age saving that accrues only per qualifying occurrence (11.2.2) · and treating an unverified AI-drafted commitment as a copy-editing error (11.4.3).

How the domain connects. Domain 1 supplies the cost of delay every calculation here is priced at, and the adoption arithmetic that makes engagement the largest available lever. Domain 3 supplies the governance latency `E[wait]` that makes a late objection expensive and the paper lead time `L` that enters report age. Domain 4 supplies the interface arithmetic `n(n − 1)/2` applied here to people, and the 18,000 interface unit cost. Domain 5 supplies the design-maturity result behind Case study B. Domain 6 consumes engagement durations as schedule input, since a consensus-forming approval is a longer predecessor than the plan assumes (11.4.2). Domain 8 supplies `EMV`, used for the priced alternative and for consent risk. Domain 10 supplies the procurement and dispute machinery this domain's negotiation behaviour operates inside, and defers negotiation itself to here. Domain 12 takes the interpersonal and team dimension of conflict and influence; Domain 14 the systematic treatment of AI governance whose communication face appears in 11.4.3; Domain 15 the same arithmetic at portfolio scale; and Domain 16 the adoption measurement this domain's value core exists to move.

Domain 11 summary

Stakeholder work has no natural unit of account, which is why it is cut first and why it is the largest unpriced item in most delivery plans. This domain gives it units. Meridian's national reporting body — influence 5, interest 2, consent risk high, and sent a bulletin — objected in week 34 at an assessed cost of **USD 224,520**: 9.0 weeks of critical path (4.0 of them Domain 3's governance latency) at 14,280 a week, plus three interface re-verifications and application rework. The pre-consultation that would have surfaced it cost **USD 3,300**, so engagement paid at any probability of objection above **1.47 %** — and that breakeven, not the 68-fold ratio, is what wins the argument, because a sponsor can dismiss a ratio and cannot assert one and a half per cent.

Engagement is an allocation of a stated capacity. Meridian's 480 hours, distributed in proportion to influence × interest, gave the two adoption-determining groups 180 hours and forgave **USD 55,080 a year, permanently**, of capacity that was being spent anyway; the three-term design — a **320-hour** value-led core, **130 hours** of consent-risk floors, a **30-hour** residual — delivers **5** more adopting clinics and **USD 122,400** a year at **USD 382.50 per engagement hour**, on a breakeven uplift of **3.59 percentage points**. It also raises the cost of delay to **USD 16,830** a week, which makes every schedule lever more valuable and settles the sequencing question: engagement before acceleration.

Communication is an architecture with a computable load. Fourteen actively managed parties admit **91** channels unrouted — **136.5 hours a month** against a capacity of 40, **341.25 %** — while routing gives 14 channels and 33.0 hours; an unrouted design sustains **7** parties and a routed one **18**, and Meridian's honest hybrid runs 17 channels at **37.5 hours**, **93.75 %** of capacity. Reporting has an age: `C + L` for the newest fact, `C + L + P/2` for the mean, `P + C + L` for **maximum blindness** — 3.0, 5.0 and **7.0 weeks** as found, 1.5, 2.0 and **2.5** after weekly cuts, saving up to **USD 64,260** per qualifying occurrence. Status is corrupted by discretion over undefined thresholds and repaired by thresholds set at baseline plus a recorded date the leader knew.

Negotiation is preparation. A reservation value derived from a fully priced alternative — 640,000 plus 6 weeks at 14,280 plus 74,320 of effort plus a 45,000 `EMV` — put Meridian's at **USD 845,000** against the supplier's **USD 615,000**, a **USD 230,000** zone whose midpoint is **730,000**; failing to price the alternative and settling at 780,000 would have transferred **USD 50,000** for nothing. Improving the alternative for 25,000 cut the settlement to **USD 684,940** and the risk carried by

33,000 while reducing surplus to 69,940 — the trap that reveals which measures govern. And most of the value was never in the price: trading two asymmetrically valued issues produced a joint gain of **USD 82,000** against the **USD 29,000** of a split-the-difference compromise, with issue B destroying **24,000** and therefore never worth pursuing.

Public stakeholders cannot be managed, only engaged, and consultation before a decision is engagement while consultation after it is notification: Case study B's project paid **USD 2,194,000** for the second where **USD 563,000** would have bought the first, a ratio of **3.90**, because the identical concern cost 55,000 at options stage and **410,000** at detailed design. Consent risk prices as an `EMV` — 80,976 against 71,136, worth 9,840, breaking even at a **20.75-point** reduction in probability — with the plain statement that legitimacy is not quantifiable and consultation is not only a risk control. And the newest failure mode is the cheapest to close: verifying AI-drafted external communication costs **USD 5,280** against an exposure of **USD 23,040**, returns **4.36 times**, pays at any error rate above **0.92 %**, and still leaves **67.2 hours** of net saving — while an unverified machine-written commitment is not a typographical error but a decision taken without authority.

The through-line: **the objection you have not paid for in engagement, you will pay for in delay, at the cost of delay** — and every quantity in that sentence is computable at design time, from numbers a programme already has.

12

DOMAIN 12

Leadership, Teams and Organizational Behaviour

Group: Leading people and organisations (Domain 12 of 3 in Part Three). **Target:** ~72 pages. **Binds to:** the PCI Book Pattern Specification and the shared registries (`docs/books/registries/`). Domain 3 established who may decide and at what price, Domain 4 how the parts connect; this domain establishes what it takes for a group of humans to work as one team inside those structures. It runs **Meridian Care Records** at programme scale and **Project Auriga** at single-project scale. British English; USD (+SAR where useful, indicative `USD 1 ≈ SAR 3.75`).

Registry note. This domain uses the registered cost of delay and the registered **interface count** `n(n − 1)/2` (PML-AI D4, KA 4.2.3) unchanged, applying the second to human communication paths, and cites Domain 3's `E[wait]` rather than re-deriving it. It submits four candidate registry rows: **coordination overhead share** `c·n(n − 1)/(2h)`; **net team capacity** `hn − c·n(n − 1)/2` with its maximising size; **coaching time per report** `(R − e(n − b))/n` with its sufficiency threshold; and the **three-component turnover cost**. Each carries a verified golden example below. Every parameter in them — the cost of a communication link, the shape of a ramp-up curve, the minutes a person needs — is a **locally calibrated planning figure, not a measured constant**, and the domain says so at each use.

Why this domain exists

Domains 1 to 11 built a delivery system: an accountable leader, an honest business case, decision rights with a computable latency, an integrated architecture, a scope, a schedule, a cost model, a risk register, a quality regime, a supply chain and a stakeholder strategy. Every one is executed by people who can leave, disengage, misunderstand each other, fear speaking up, or run out of the one resource nobody baselines — attention. **The system is only as good as the human capacity available to run it**, and that capacity is not a constant the leader inherits but a variable the leader designs.

This is the most abused domain in the discipline, and the abuse takes a specific form: leadership content asserting effects it cannot demonstrate. Confident claims about what motivates people, what a personality instrument predicts, what a culture programme returns circulate widely and are mostly unfalsifiable as stated. So this domain carries two obligations it does not carry elsewhere. The first is **honesty about the evidential status of every claim**: some of what follows is arithmetic, which is checkable; some is widely used vocabulary, useful without being predictive; some is professional judgement, labelled as such. The second is to find the parts of the domain that genuinely are quantitative, because they exist and are neglected — team size, turnover, span of control, delegation and time-zone overlap all have arithmetic, and each turns a conversation about commitment into a conversation about design, which is the only kind that gets a decision.

Hence the central claim: **the human capacity of a delivery organisation is a designed quantity with a computable cost, and most of what is described as a leadership or culture problem is**

a structural problem that arithmetic can locate. A leader who can compute the coordination overhead of a team, the coaching capacity of a span, the cost of a departure and the latency of a distributed handoff can change those things; a leader who cannot will be asked to exhort people instead, and exhortation moves none of them. The domain proceeds outward from the individual: KA 12.1 handles leadership itself and the leader's attention; KA 12.2 the group, its size arithmetic, its safety and its turnover; KA 12.3 what a leader does with individuals; KA 12.4 distance, culture, ethics and leading people whose work is increasingly mediated by AI.

Learning objectives. After this domain a candidate can: distinguish leadership from management by what each supplies; state what each leadership theory family is useful for and what it does not establish, and select behaviour from situation rather than preference; apply emotional intelligence as four practised operations rather than a trait claim; **audit a leader's week and compute the discretionary attention actually available**; explain the motivation frameworks with their evidential status stated, and design conditions rather than incentives alone; **compute communication paths from `n(n − 1)/2`, convert them into coordination overhead as a share of team capacity, identify the size at which a further member adds no net capacity, and price the capacity a structured design releases**; define psychological safety and specify observable signals without overclaiming its effects; **compute the full cost of one departure as recruitment plus ramp-up plus lost productivity, and state which components are real costs and which are transfers**; **compute coaching time per report against span of control and derive the maximum sustainable span from a stated sufficiency convention**; make delegation an economic decision with a breakeven, citing Domain 3's threshold arithmetic rather than repeating it; structure and time a difficult conversation and price its deferral; **compute working-hour overlap from a common time reference, price the round-trip latency of a zero-overlap pair, and design a shared and rotated overlap window**; lead multicultural teams without substituting stereotype for enquiry; hold the ethical obligations that cannot be delegated; and govern AI's use in people decisions, where the constraints are tighter than anywhere else in the book.

The master threads. Meridian Care Records supplies the programme case: the rollout to **40 clinics**, benefits **USD 685,440** a year at 70 % adoption, **cost of delay USD 14,280 per week** (Domain 1), and a steering committee whose latency `E[wait] = 4/2 + 2` is **4.0 weeks** (Domain 3, KA 3.2.3). This domain follows that team as it grows from 12 people to 40 and asks what the growth cost. Project Auriga supplies the project case: the 25-week control-systems upgrade, `BAC` **USD 4,000,000**, cost of delay **USD 45,000 per week**, a blended engineering rate of **USD 130.625 per hour** — an engineer-week of **USD 5,225** (Domain 7, KA 7.4.1) — and the **six-engineer** commissioning team of Domain 9. This domain audits that leader's week, prices a resignation from that team, and prices the decision to delegate.

KNOWLEDGE AREA 12.1

Leadership theories in practice and emotional intelligence

Topics: 12.1.1 what leadership supplies that management does not · 12.1.2 the theory families and what each is honestly good for · 12.1.3 emotional intelligence as an operational discipline · 12.1.4 the leader's attention as the binding constraint.

12.1.1 What leadership supplies that management does not

Definition. In a delivery context, **management** is the allocation and control of resources against a plan; **leadership** is the production of willing, aligned and informed effort under conditions the plan did not anticipate. These are different outputs, not a hierarchy of seniority, and a project needs both from the same person, often in the same hour.

The test is what fails when each is absent. Without management, work is enthusiastic and uncoordinated. Without leadership, work is coordinated and inert: people do what the plan says after the plan has stopped being true, escalate nothing, and wait. The second failure is far harder to detect, because **a project failing for want of leadership produces perfectly compliant status information until the moment it produces a disaster.**

Leadership supplies four things a plan cannot. **A reason the work matters that survives difficulty** — not a slogan but a specific account connected to Domain 1's benefits chain (KA 1.3.2); Meridian's is twenty-eight clinics of clinicians spending six hours a week on patients instead of paperwork, a fact with a number attached. **Interpretation under ambiguity**: when plan and situation diverge, someone must say what the team should do before governance can rule, and that act is done in public and is the main way a team learns what the leader values. **The conditions in which people tell the truth**, on which every control in this book depends, and which a team defeats once it learns that bad news is punished (KA 12.2.3). And **named, bounded discretion** — Domain 3's bounded envelope (KA 3.1.3) at individual scale, priced in KA 12.3.

Leadership does not substitute for authority (a leader without decision rights is eventually defeated by the delegation schedule), for capacity (no inspiration converts 40 hours into 60 sustainably), or for competence in the work — a leader who cannot follow the technical argument cannot tell a confident answer from a correct one, which is exactly the judgement Domain 9's assurance depends on.

12.1.2 The theory families and what each is honestly good for

Leadership theory is a century of overlapping vocabularies, and the professional skill is knowing what each family is for rather than which is true. **No claim is made here about effect sizes, and none should be made from this material.**

Trait approaches ask what leaders are like. Their contribution is vocabulary for individual differences; their limitation is that describing a difference is not explaining an outcome. The practical caution: personality instruments are widely used for team composition and selection, and their predictive claims are much weaker than their marketing — use them, if at all, as a **conversation aid inside a team**, never as a selection filter or a performance judgement. **Behavioural approaches** ask what leaders do, usually along attention to the task against attention to the relationship; this is useful because behaviour is observable, coachable and changeable, and it underpins the instruction most delivery leaders need — your team reads what you spend time on.

Situational and contingency approaches vary direction against support according to the competence and confidence of the person on the task. As predictive theory the evidence is mixed; as a **discipline for choosing behaviour deliberately** it is the most useful family here, because it forces the question a leader otherwise skips — what does this person, on this task, this week, need from me? A competent engineer on familiar work needs a boundary and to be left alone; the same engineer on unfamiliar work needs direction they will not ask for. Getting that backwards is the commonest leadership error in technical delivery, and it is a failure to diagnose, not a character flaw.

Transformational and transactional approaches distinguish leadership that changes what people want from leadership that trades reward for performance; both are needed, and transformational language is easily counterfeited — a team can tell within a fortnight whether the vision matches the decisions. **Servant and steward framings** name obligations that otherwise go unstated: removing impediments, absorbing external pressure rather than transmitting it, defending the team's ability to work honestly; their failure mode is being read as licence to avoid difficult decisions. **Distributed leadership** observes that leadership acts are performed by many people, not only the title-holder —

the family that matters most on a programme like Meridian, and one that connects to Domain 3: distributed leadership without distributed decision rights is a slogan, and distributed rights without the information to use them are abdication (KA 3.2.3).

These are lenses, not a menu. A defensible leader takes a few operational habits from the list: diagnose before behaving; watch what you spend time on because your team does; state boundaries explicitly; make the reason for the work truthful and specific; and treat obligations to the team as obligations rather than preferences.

12.1.3 Emotional intelligence as an operational discipline

Definition. Emotional intelligence, as used here, is the practised ability to notice emotional information — one's own and others' — and act on it deliberately rather than reactively. It is treated as **four operations that can be practised**, not a trait or a score. The popular literature makes strong claims about its predictive power; those claims are contested, and nothing here depends on them, because each operation is defensible as professional craft.

Self-awareness is knowing which situations reliably degrade your judgement. The delivery-specific ones recur: being late, being contradicted in front of your sponsor, and receiving bad news you suspect was withheld. A leader who knows the third makes them punitive can suppress the first reaction for an hour, and that hour is often the difference between a team that escalates next time and one that does not. **Self-regulation** is having a rule for the predictable bad moments, decided in advance; the most valuable in delivery is first reaction to bad news is questions only, no judgements, because it directly protects the information flow every control consumes. **Social awareness** is noticing who has stopped contributing, whose agreement was compliance, and which silence is consent rather than resignation — then **testing the reading rather than trusting it**, since confident misreading of others is common and is made worse by believing oneself emotionally intelligent. **Relationship management** is acting on the reading: the conversation held early (KA 12.3.3), the disagreement surfaced rather than absorbed, the repair after a bad exchange.

The concept is misused three ways: as a **label** applied to people, which converts a practicable skill into a fixed attribute; as a **substitute for candour**, where avoidance is called sensitivity; and as a **selection instrument** with claimed predictive validity — the same category error as the trait instruments above.

12.1.4 The leader's attention as the binding constraint

Every behaviour above consumes the same resource, and it is the only resource on a project never baselined, levelled or reported. A schedule assuming a leader will coach six people, hold two difficult conversations, prepare a steering paper, manage a supplier and absorb a client escalation in one week has made a resource assumption as material as any in Domain 6 — and, unlike Domain 6's, nobody has checked whether it is feasible. The check takes twenty minutes.

WORKED EXAMPLE

Worked example 12.1.4 — auditing the Auriga project leader's week.

1. **Setup.** Auriga's project leader works to a **45.0-hour** planning week (the project's stated assumption; it is not a claim about what is healthy or contractual, and where norms differ the number and every result below change). Four weeks of diary and interruption logging give the recurring commitments: governance and reporting **6.0 h**; client and regulator liaison **5.5 h**; supplier and commercial **4.0 h**; technical review and approvals **7.5 h**; team ceremonies **5.0 h**; unplanned interruption and escalation **9.0 h** (observed, not planned). The leader intends to reserve **1.5 h** for personal planning.
2. **Formula.** Discretionary attention = working week − Σ recurring commitments. People-work capacity = discretionary attention − personal reserve. Shares are taken against the working week.
3. **Substitution.** `6.0 + 5.5 + 4.0 + 7.5 + 5.0 + 9.0`; then `45.0 − 37.0`; then `8.0 − 1.5`.
4. **Result.** Recurring commitments **37.0 hours**, **82.2 %** of the week. Discretionary attention **8.0 hours**, **17.8 %**. Capacity for people work — coaching, difficult conversations, one-to-ones, recruitment, team design — **6.5 hours**, or **14.4 %** of the week.
5. **Interpretation.** Six and a half hours is the entire leadership budget of this project, and every commitment in KA 12.3 is paid from it. Three consequences follow, and the third changes behaviour. **The largest single line is not a decision anyone made**: 9.0 hours of interruption is **20.0 %** of the week, more than governance and technical review combined, and much of it is the absence of Domain 3's bounded envelope (KA 3.1.3), so decisions arrive that need not have. Recovering a third of it returns 3.0 hours and lifts people-work capacity from 6.5 to 9.5 — a **46.2 %** increase in the leadership budget from a change that removes nothing. **The 7.5-hour technical-review line is a trade-off, not waste**: it is what lets the leader tell a confident answer from a correct one, and cutting it to buy coaching time trades one competence for another. And **6.5 hours cannot be spread thinly across an arbitrary number of people** — the arithmetic KA 12.3.2 completes, and the reason span of control is a leadership question rather than an organisational-chart question. The professional caution: this audit is only worth doing on **observed** data. A leader who fills it in from intention will produce 15 discretionary hours and will not know why the coaching never happens.

AI in this KA

Where it earns its place. Classifying a quarter of a leader's meetings into the commitment categories above and computing the shares — high-volume, low-judgement work nobody does by hand. Rehearsing a difficult conversation by generating the likely responses and the leader's weakest answers, since the failure mode of these conversations is being unprepared for the second sentence. Drafting from the project record the factual chronology a performance conversation must rest on. Summarising team-survey free text into themes with the source comments retained, so the leader reads the actual words.

Where it must not go. Three prohibitions, firmer here than anywhere else in this book because the subject is people. **No inference about an individual's emotional state, engagement, personality or likelihood of leaving** from behavioural telemetry — keystrokes, message sentiment, camera analysis, attendance: such inference is unreliable, frequently unlawful depending on jurisdiction, and corrosive of the trust every control in this domain depends on, and a leader who deploys it loses the truthful upward information of 12.1.1 permanently and will not be told why. **No AI-generated assessment of a person** entering a performance, promotion or termination decision.

And **no AI-authored message purporting to be the leader's own voice** in a sensitive conversation: the counterfeit is detectable, and the damage when detected exceeds any time saved.

Verification, concretely. Any AI classification of the leader's diary is checked against the raw entries for one week in four, with categories defined by the leader, not the model. Any factual chronology is verified line by line against the source record, because a single wrong date destroys the conversation. Where AI is used for rehearsal, no personal data about the other party is entered — the rehearsal works on the situation, not the person.

KEY TERMS — KA 12.1

Term	Meaning
Leadership (delivery context)	The production of willing, aligned and informed effort under conditions the plan did not anticipate.
Management (delivery context)	The allocation and control of resources against a plan.
Situational diagnosis	Deciding what direction and support this person needs on this task before choosing a behaviour.
Emotional intelligence	Four practised operations — self-awareness, self-regulation, social awareness, relationship management — skills, not a trait score.
Discretionary attention	The leader's working week less recurring commitments; the resource all leadership acts are paid from.
People-work capacity	Discretionary attention less the personal planning reserve: 6.5 h/week on Auriga.
Interruption load	Observed unplanned escalation and interruption time; the largest and most designable line in the attention audit.

SAMPLE MCQS — KA 12.1

MCQ 12.1-A `[12.1.4 · Application]` A leader's 45.0-hour planning week contains 37.0 hours of recurring commitments. The share of the week available as discretionary attention, before any personal planning reserve, is: - A. 14.4 % - B. 17.8 % - C. 20.0 % - D. 82.2 %

Rationale: `(45.0 − 37.0)/45.0 = 17.8 %` (12.1.4). A is the people-work share after the 1.5-hour reserve — the answer to a different question; C is the interruption load's share of the week; D is the committed share.

MCQ 12.1-B `[12.1.1 · Analysis]` A project reports faithfully on plan for four months, then discloses a six-month overrun in one meeting. The failure this most strongly suggests is: - A. inadequate management, because variance control was absent - B. inadequate leadership, because the conditions for truthful upward information were absent - C. inadequate governance, because the committee met too rarely - D. inadequate scope definition

Rationale: Compliant reporting is evidence that management existed; what was missing was the condition under which people report what the reports do not ask for (12.1.1, and Domain 3's escalation-lead-time indicator at 3.3.3).

MCQ 12.1-C `[12.1.2 · Evaluation]` A delivery organisation proposes using a personality instrument to select project leaders. The soundest professional position is that the instrument: - A. should be used, because it is widely adopted - B. may be useful as a conversation aid inside a team, but its predictive claims are too weak to support a selection decision - C. should be used only for senior appointments - D. is invalid and has no legitimate use

Rationale: Describing an individual difference is not explaining an outcome, so the legitimate use is internal dialogue, not selection (12.1.2). D overstates in the other direction, which is also a failure of judgement.

MCQ 12.1-D `[12.1.2 · Application]` A competent engineer is put on genuinely unfamiliar work and does not ask for help. The situational diagnosis indicates that the leader should: - A. leave them alone, since they are competent - B. increase direction on this task while keeping the relationship supportive - C. reduce their scope permanently - D. treat the silence as confidence

Rationale: The diagnosis is per person **per task** (12.1.2); competence on familiar work does not transfer, and the absence of a request for help is not evidence of not needing it.

SELF-CHECK — KA 12.1

1. What does leadership supply that a plan cannot? — A truthful reason the work matters, interpretation under ambiguity, the conditions for truthful upward information, and named bounded discretion.
2. Why is the Auriga leader's interruption load the most important line in the attention audit? — It is the largest single line at 9.0 hours (20.0 % of the week), nobody decided it, and much of it is the absence of a bounded envelope, so it is the most designable.
3. State the honest limit of trait and personality approaches. — Describing an individual difference is not explaining an outcome; their predictive claims are too weak for selection or performance decisions.

KNOWLEDGE AREA 12.2

Motivation, team formation and psychological safety

Topics: 12.2.1 motivation without the folklore · 12.2.2 team formation and the arithmetic of team size · 12.2.3 psychological safety as a design variable · 12.2.4 retention and the cost of turnover.

12.2.1 Motivation without the folklore

Definition. Motivation, for a delivery leader's purposes, is the direction, intensity and persistence of effort a person applies to work. It is not a quantity a leader installs; it is a response to conditions, some of which the leader controls. The frameworks in general use are worth knowing with their evidential status attached, because they are quoted constantly and interrogated rarely.

Need hierarchies (the family associated with Maslow) propose that needs are ordered and that higher needs become salient once lower ones are met. The ordering has not held up as a predictive model; the framework is best used as **vocabulary** for noticing that a person whose employment feels insecure will not be reached by an appeal to professional growth. **Two-factor accounts** (associated with Herzberg) separate factors whose absence causes dissatisfaction — pay, conditions, supervision, security — from those whose presence produces satisfaction — achievement, recognition, responsibility, growth. The empirical separation is contested; the **operational insight is sound**: fixing a dissatisfier removes a complaint, it does not produce engagement. A team with a broken build pipeline and no career conversations will not be engaged by fixing the pipeline; it will stop complaining about the pipeline. This is the most frequently misapplied idea in the domain, because organisations reliably spend on dissatisfiers and count the spending as motivation work.

Self-determination framings make intrinsically motivated effort depend on autonomy, competence and relatedness. This family has the strongest current standing and is the most actionable in project work, because all three levers are the leader's: autonomy is Domain 3's bounded envelope at individual scale, competence is whether people get work they can succeed at with the support to do so, relatedness is whether they belong to a team rather than a resource pool. **Expectancy and goal accounts** make effort follow from believing that effort leads to performance, that performance leads to an outcome, and that the outcome is worth having — and hold that specific, difficult, accepted goals draw more effort than vague ones. This family explains most demotivation in troubled projects without reference to character: a team on a plan it privately believes impossible has a broken **effort → performance** link, and no incentive attached to the outcome can repair it. That is why an honest re-baseline is often the most motivating act available to a leader of a late project, and why "stretch" targets nobody believes demotivate.

For practice — professional judgement, not measured effect — design the **conditions** and be sceptical of designing the **incentives**: a credible plan, work people can succeed at, an explicit boundary of their own judgement, visible connection between their work and the outcome, specific and accepted goals, and recognition specific enough to be believed. Financial incentives have a real but narrow role and two known hazards: they attach effort to the measure rather than the objective, which is Domain 9's measurement problem in another form, and they are hard to withdraw without a cost exceeding their original benefit.

12.2.2 Team formation and the arithmetic of team size

Formation as a described sequence. The forming–storming–norming–performing vocabulary describes how a new team settles: politeness and dependence on the leader; conflict as real differences about approach and status surface; the settling of working norms; productive work. Two qualifications. It is a **description, not a mechanism**, and teams do not traverse it once — a change of membership, sponsor or circumstance returns a team to conflict. And the leader's task differs by stage: in forming supply structure and purpose; in storming make the conflict about the work rather than the people and do not suppress it, because a team that never storms has usually concluded that disagreement is unsafe; in norming make the emerging norms explicit so they can be chosen; in performing protect the team from disruption, which mainly means from your own organisation.

The part that is arithmetic. Domain 4 established that components grow linearly while interfaces grow combinatorially, and registered the count (KA 4.2.3):

```
Communication paths = n(n − 1)/2     — every person to every other
```

That is the same formula Domain 4 used to count 66 technical interfaces across 12 components; it counts 66 communication paths across a 12-person team, and the two are one problem seen from two sides. What turns the count into a management number is a **cost per path**. Let

```
c = hours of total team time consumed per active pairwise relationship per week
h = productive hours per person per week
n = team size
```

Then coordination overhead is `c · n(n − 1)/2` hours a week, and as a share of gross capacity `h · n`:

```
Coordination share = c(n − 1) / (2h)
```

With `c` = 0.5 hours and `h` = 40 hours the share collapses to the identity **(n − 1)/160** — linear in team size and independent of everything else. **The parameter `c` is a locally calibrated planning figure,**

not a constant: 0.5 hours of total team time per relationship per week is roughly 15 minutes each from two people, plausible for colleagues who genuinely need to stay aligned, and the figure this book uses illustratively. Calibrate it from your own data if you have any; the shape of the result, not the parameter, is the lesson.

WORKED EXAMPLE

Worked example 12.2.2 — what Meridian's team growth cost, and what structure recovers.

1. **Setup.** Meridian's programme team grew from **12** people to **40** as the rollout scaled from pilot to national, with no change in how it worked: one team in which anyone might coordinate with anyone. Take `c` = **0.5** hours per pairwise relationship per week and `h` = **40** productive hours. The alternative is **five pods of eight**, each owning a cluster of clinics end to end with one representative per pod in a weekly forum — the human form of Domain 4's layered architecture (KA 4.2.3). Compare all three states.
2. **Formula.** Paths `n(n − 1)/2`; overhead `c ×` paths; share `= c(n − 1)/(2h)`. For pods, paths `= (n/s) × s(s − 1)/2 + (n/s)(n/s − 1)/2` with pod size `s`, and share `= c ×` paths `/(h·n)`.
3. **Substitution.** At 12: `12 × 11/2 = 66`; overhead `0.5 × 66 = 33.0` h; share `33.0/(40 × 12)`. At 40: `40 × 39/2 = 780`; overhead `390.0` h; share `390.0/1,600`. Pods: `5 × (8 × 7/2) + (5 × 4/2) = 140 + 10 = 150`; overhead `75.0` h; share `75.0/1,600`.
4. **Result.** At 12 people: **66** paths, **33.0** hours a week, **6.875 %** of capacity — **0.825** of a full-time person. At 40: **780** paths, **390.0** hours, **24.375 %** — **9.75** full-time people. In five pods of eight: **150** paths, **75.0** hours, **4.6875 %** — **1.875** full-time people. The pod design releases **630** paths, **315.0** hours a week, or **7.875 full-time-equivalent people**, cutting coordination by **19.6875 percentage points** of capacity.
5. **Interpretation.** The programme tripled its headcount and, fully connected, spent nearly a quarter of the result on talking about the work. **It was invisible in every report**: no line in Meridian's cost model says "coordination", the hours were charged to the work packages people were coordinating about, and the effect presented as unexplained productivity decline — exactly how it presents in practice. **The leverage is structural, not personal**: the pod design reduces nobody's workload and adds nobody; it removes 630 relationships that were not needed by giving each pod an end-to-end scope and one named path outward, and 7.875 FTE is more capacity than most improvement programmes deliver in a year. **Structure barely degrades with scale while a mesh degrades quadratically**: the pod share moves only from 4.375 % at 8 people to 5.156 % at 88, while the mesh share passes **50.0 %** at **81**. That is also the exact size at which net capacity `hn − c·n(n − 1)/2` stops rising — **1,620 hours (40.5 FTE)** at both 80 and 81 people — so **the 81st member of a fully connected team adds nothing**, and members beyond it subtract. That is the arithmetic behind the familiar observation that adding people to a late project can make it later, with the qualification that this is a model of a fully connected team: the honest reading is not "teams cannot exceed 80" but "**a team not given structure has a hard capacity ceiling, and it is lower than anyone expects**". **The sensitivity decides how seriously to take any of it.** At `c` = 0.25 the 40-person mesh costs **12.1875 %** and the 50 % point moves to 161 people; at `c` = 0.8, **39.0 %** and 51; at `c` = 1.0, **48.75 %** and 41. So the conclusion — structure your team — is robust across the plausible range while the specific ceiling is not, and a leader who quotes 81 as a fact has misused the model. Quote your own team's share at your own calibrated `c`, as an estimate.

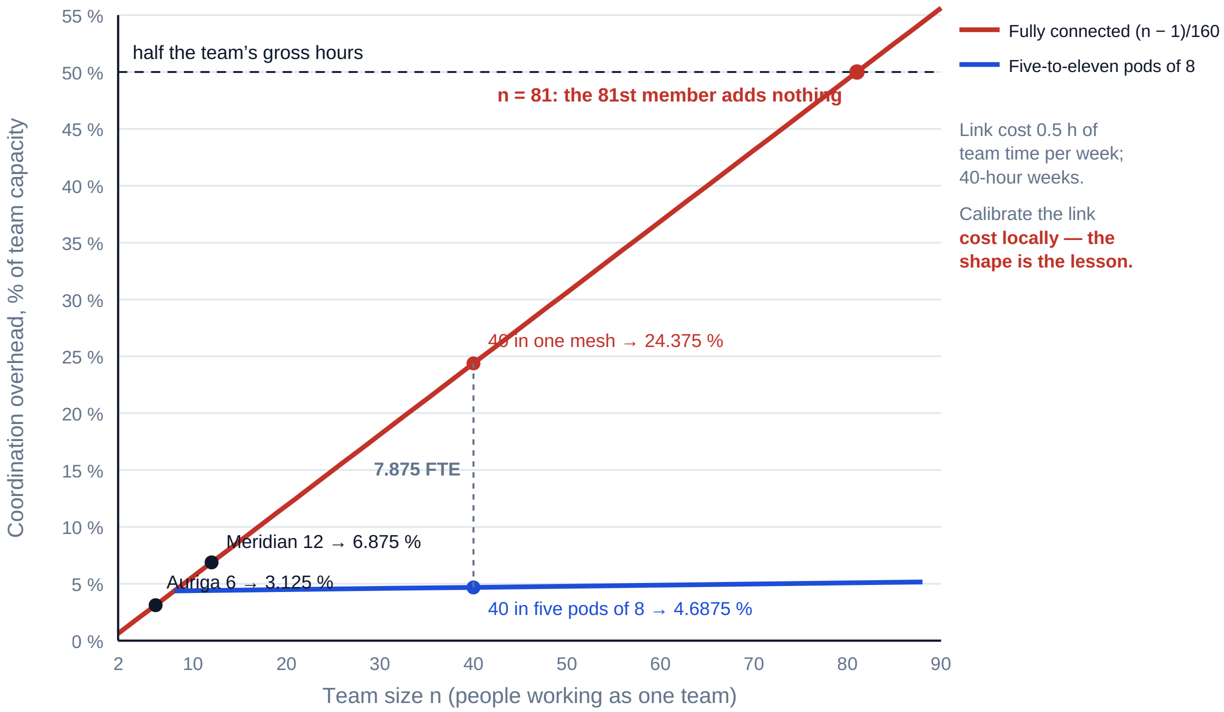


Fig 12.2.1 — The coordination ceiling, and what structure buys

At Auriga's scale. The six-engineer commissioning team of Domain 9 has **15** paths, **7.5** hours a week of coordination, **3.125 %** of capacity — **USD 979.69** a week at USD 130.625 an hour, or **USD 24,492.19** across the 25-week project. That is small, and it should be: small teams are cheap to coordinate, which is the actual reason to prefer them. The instructive number is marginal. A seventh member brings 40 hours and 6 new relationships costing 3.0 hours, so the net addition is **37.0 hours** — plainly worth having; the same calculation at the 81st member returns zero. **The marginal capacity of the nth member is** `h − c(n − 1)`, and a leader who knows that will never again argue about team size without a number.

12.2.3 Psychological safety as a design variable

Definition. Psychological safety is a shared belief that the team is safe for interpersonal risk-taking — that a person can ask a question, admit an error, disagree with a senior colleague or report bad news without penalty to their standing. The term comes from organisational-behaviour research on team learning; this book uses it in the registered sense (`registries/TERMINOLOGY.md`).

What it is not. Not comfort, not niceness, not the absence of disagreement — a team with high safety and low standards is pleasant and produces nothing, and that conflation is the main reason the concept is dismissed by experienced delivery people. The useful frame pairs safety with **accountability for standards**: safety governs whether people can speak, standards govern what they are held to. Low safety with high standards produces anxiety and concealment; high safety with low standards produces comfortable underperformance; the combination the concept exists to name is high safety with high standards.

On evidential status. This book does **not** claim a measured effect size for psychological safety on delivery outcomes, and a professional should be sceptical of anyone quoting one from a project context. What can be asserted is narrower and sufficient: **every control system in this book consumes voluntary upward information**, and a team that has learned that raising problems is penalised will not supply it. Domain 3's escalation-lead-time indicator, Domain 8's risk identification, Domain 9's nonconformance reporting and Domain 4's interface issues all fail silently in the same way. That is an argument from the architecture of the delivery system, not from a research finding, and it is strong enough.

Observable signals, because belief is not measurable and behaviour is. Presence: junior people ask questions in front of senior people; errors are reported by the person who made them, before they are found; bad news arrives with time to act on it; disagreement is expressed in the meeting rather than after it; someone says "I don't know". Absence: pre-meetings where the real conversation happens; problems that arrive already solved and slightly too late; unanimous agreement followed by inaction; and a rising share of issues discovered by assurance rather than reported by the team — a countable signal available from data Domain 9 already produces.

What a leader does, in order of effect and all of it cheap. **Respond well to the first piece of bad news** — the climate is set by a handful of visible early responses, which is 12.1.3's self-regulation rule doing real work. **Report your own errors** with the specificity you would ask of others. **Ask a question you do not know the answer to, in public. Separate the issue from the person** when escalating, which is Domain 3's escalation-culture provision written into governance. And **name the standard alongside the safety**, so nobody mistakes candour for tolerance of poor work.

12.2.4 Retention and the cost of turnover

Turnover is how every failure in this knowledge area becomes a cost, and it is almost never costed. Organisations know the recruitment fee, treat the vacancy as a saving, and never compute the rest. The build-up below is deliberately conservative, and its most important feature is what it **excludes**.

WORKED EXAMPLE

Worked example 12.2.4 — the full cost of one resignation from Auriga's team of six.

1. **Setup.** One of the six engineers on Auriga's commissioning team resigns, giving notice in week 14 of the 25-week project. The blended engineering rate is **USD 130.625** an hour, so an engineer-week is **USD 5,225** (Domain 7, KA 7.4.1). Recruitment: an agency fee of **USD 24,000** and **20 hours** of leader and team time in shortlisting and interviewing. The post is vacant **6 weeks**; the work is covered by overtime at a **50 % premium** on the affected hours. The replacement's effectiveness ramps linearly from **30 %** in week 1 to **100 %** in week 8, with **4 hours a week** of colleague mentoring across those 8 weeks. The leaver's last two weeks run at **60 %** effectiveness, and handover consumes **12 hours** across the team. Every one of those is a stated planning assumption for this project, not a measured constant; the ramp shape in particular is a professional judgement and the result is sensitive to it.
2. **Formula.** Total = recruitment + ramp-up + lost productivity, where recruitment = fee + (selection hours × rate); ramp-up = (weeks − Σ effectiveness) × engineer-week + (mentoring hours × rate); lost productivity = leaver disengagement + handover + the **incremental** cost of covering the vacancy.
3. **Substitution.** Recruitment `24,000 + 20 × 130.625`. Ramp effectiveness `0.30 + 0.40 + … + 1.00 = 5.20`, so lost effective weeks `8 − 5.20 = 2.80`; ramp-up `2.80 × 5,225 + 32 × 130.625`. Lost productivity: `0.8 × 5,225`, `12 × 130.625`, and `0.50 × 6 × 5,225`.
4. **Result.** Recruitment **USD 26,612.50**; ramp-up **USD 18,810.00** (14,630 of lost effectiveness plus 4,180 of mentoring); lost productivity **USD 21,422.50** (4,180 + 1,567.50 + 15,675). **Total USD 66,845.00 — 12.79 engineer-weeks**, **26.65 %** of the post's annual charge-out value of 250,800 over 48 weeks, and **1.671 %** of Auriga's `BAC` of 4,000,000. Two departures in a year cost **USD 133,690**.
5. **Interpretation.** Start with the exclusion, because it makes the number defensible. **The vacancy does not cost the salary; the salary is saved.** What it costs is the work not done and whatever is paid to get it done anyway — here a 50 % overtime premium of 15,675 on six engineer-weeks, not the 31,350 those weeks are worth. Counting the saved pay as a loss inflates the figure to 82,520, is the commonest error in turnover costing, and destroys the credibility of the calculation with any finance function that reads it. **The overtime decision is then not close.** Uncovered, six engineer-weeks missing from a six-person crew is exactly **1.0 week** of elapsed delay on a zero-float activity, costing **USD 45,000** at Auriga's cost of delay — **2.87 times** the premium. The breakeven premium is `45,000/(6 × 5,225) =` **143.54 %**: the project could pay two-and-a-half times basic rate and still beat the delay, worth knowing before the negotiation. **The number also sets the retention budget a leader may argue for — and only that.** At 66,845 per departure, an intervention costing 20,000 a year breaks even by preventing **0.2992** of a departure, roughly one every three and a third years: a low bar. But **the arithmetic tells you what a prevented departure is worth; it does not tell you that any intervention prevents one.** No claim is made here that a named practice reduces turnover by a stated amount, and a leader presenting such a claim without their own data is doing the thing this domain exists to stop. The defensible proposal is "this is the value at risk, here is what we propose to try, here is how we will know" — Domain 2's discipline applied to a people decision. **Finally, note where the cost sits**: recruitment is only **39.8 %** of the total and **60.2 %** is ramp-up and lost productivity, the components nobody records; the **28.1 %** that is ramp-up is incurred after the vacancy is filled and the problem considered closed, which is why a team that has replaced three people stays slower than its headcount suggests for a quarter after the last one started.

AI in this KA

Where it earns its place. Computing the coordination arithmetic across candidate structures — pod sizes and boundary options, each deterministic and checkable — and finding the design that minimises required paths given a stated dependency set, which is a genuine combinatorial problem. Extracting the actual communication graph from collaboration metadata **in aggregate** to test whether the assumed structure matches reality. Analysing free-text survey responses into themes with the source comments retained. Building the turnover model and its sensitivities.

Where it must not go. Individual-level inference from communication data. The aggregate question — are our pods real, or is everyone still talking to everyone? — is legitimate; the same data at individual level becomes surveillance, is unreliable as a measure of anything worth knowing, is restricted or unlawful in many jurisdictions, and converts truthful upward information into managed information permanently. There is no version of inferring who is disengaged that this book endorses, and none of predicting which individuals will resign and acting on it: the prediction is weak, the action is visible, and the effect on those not selected is corrosive.

Verification, concretely. Reproduce `n(n − 1)/2` and the coordination share by hand — two operations each. State `c` with its basis and present the result across at least three values, because the conclusion is robust and the ceiling is not. For any aggregate communication analysis, confirm with the teams that the graph matches how they actually work: a low observed link count can mean a well-structured team or a team that has stopped talking, and only the people in it can tell you which.

KEY TERMS — KA 12.2

Term	Meaning
Communication path	One pairwise relationship in a team; counted by `n(n − 1)/2` (Domain 4, KA 4.2.3).
Link cost `c`	Hours of total team time consumed per active pairwise relationship per week; a locally calibrated planning figure.
Coordination share	`c(n − 1)/(2h)` — coordination overhead as a fraction of team capacity; `(n − 1)/160` at `c` = 0.5, `h` = 40.
Net team capacity	`hn − c·n(n − 1)/2`; maximised at 80–81 people under the illustrative parameters.
Marginal capacity of the nth member	`h − c(n − 1)`; zero at `n` = 81 under the same parameters.
Pod (structured team)	A sub-team with end-to-end ownership and one named path outward — the human form of Domain 4's layered architecture.
Psychological safety	A shared belief that the team is safe for interpersonal risk-taking — candour without penalty.
Safety-standards pairing	Safety governs whether people can speak; standards govern what they are held to. Both are required.
Turnover cost	Recruitment + ramp-up + lost productivity, excluding the saved pay of a vacant post.
Ramp-up loss	Weeks of a replacement's tenure minus the sum of their weekly effectiveness, valued at a person-week.

SAMPLE MCQS — KA 12.2

MCQ 12.2-A `[12.2.2 · Application]` A 12-person team works as a single group. Each pairwise relationship consumes 0.5 hours of total team time per week; people are productive 40 hours a week.

Coordination overhead as a share of team capacity is: - A. 3.4375 % - B. 6.875 % - C. 13.75 % - D. 82.5 %

Rationale: Paths `12 × 11/2 = 66`; overhead `0.5 × 66 = 33.0` h; capacity `40 × 12 = 480` h; `33.0/480 = 6.875 %` (12.2.2). A counts only one side of each relationship, halving the cost; C omits the division by two in `n(n − 1)/2`, counting each pair twice; D divides the 33 coordination hours by a single 40-hour week, attributing all coordination to the leader.

MCQ 12.2-B `[12.2.2 · Analysis]` Under the same parameters, the team size at which one further member adds no net capacity at all is: - A. 41 - B. 81 - C. 160 - D. 161

Rationale: Marginal capacity is `h − c(n − 1) = 40 − 0.5(n − 1)`, zero at `n` = 81 (12.2.2). A is the answer if the link cost is doubled to 1.0 hours; C mistakes `n − 1` = 160 for `n`; D is the size at which coordination consumes the team's entire gross capacity, a different and later point.

MCQ 12.2-C `[12.2.4 · Evaluation]` A departure costs 26,612.50 in recruitment, 18,810.00 in ramp-up, and — in lost productivity — 4,180 of leaver disengagement, 1,567.50 of handover, and a 50 % overtime premium of 15,675 to cover a six-week vacancy worth 31,350 in engineer-weeks. The defensible total is: - A. USD 26,612.50 - B. USD 45,422.50 - C. USD 66,845.00 - D. USD 82,520.00

Rationale: `26,612.50 + 18,810.00 + 21,422.50 = 66,845.00` (12.2.4). A counts recruitment only; B omits lost productivity; D substitutes the vacant post's 31,350 of pay for the 15,675 premium actually incurred — the saved-salary double count.

MCQ 12.2-D `[12.2.2 · Evaluation]` Restructuring a 40-person team from one group into five pods of eight, each with a single representative in a coordination forum, changes coordination from 780 paths to 150. At 0.5 hours per path per week, the capacity released is: - A. 630 - B. 7.875 FTE - C. 9.75 FTE - D. 19.6875 FTE

Rationale: `0.5 × (780 − 150) = 315.0` hours a week `÷ 40 = 7.875` FTE (12.2.2). A is the reduction in paths; C is the mesh design's total coordination cost, not the saving; D is the saving in percentage points of capacity, mislabelled as FTE.

SELF-CHECK — KA 12.2

1. State the coordination-share identity and its one calibrated parameter. — Share `= c(n − 1)/(2h)`, which is `(n − 1)/160` at `c` = 0.5 hours per path per week and `h` = 40 hours; `c` is calibrated locally and the ceiling result is sensitive to it.
2. Why does a vacant post not cost its salary? — The salary is saved; the cost is the work not done and whatever is paid to get it done anyway — on Auriga a 50 % overtime premium of 15,675 rather than the 31,350 the vacant weeks are worth.
3. What claim does this book make about psychological safety, and what claim does it refuse? — It claims that every control in the book consumes voluntary upward information which low safety suppresses; it refuses to quote a measured effect size on delivery outcomes.

KNOWLEDGE AREA 12.3

Delegation, coaching and difficult conversations

Topics: 12.3.1 delegation as an economic decision · 12.3.2 span of control and coaching capacity · 12.3.3 the difficult conversation · 12.3.4 performance management honestly.

12.3.1 Delegation as an economic decision

Definition. Delegation is the transfer of authority to decide and act on a defined class of work, together with the information and criteria needed to do so, with the leader retaining accountability. The last two clauses separate it from its counterfeits: transferring the work without the authority is allocation, and transferring the authority without the information is abdication (Domain 3, KA 3.2.3). Delegation is habitually discussed as a development activity and decided as a matter of trust. It is also an economic decision with a computable breakeven, and the arithmetic reliably points the other way from instinct.

WORKED EXAMPLE

Worked example 12.3.1 — should Auriga's leader delegate the weekly interface coordination?

1. **Setup.** Weekly coordination of Auriga's technical interfaces (Domain 4's interface register) consumes **3.5 hours** of the project leader's week and will do so for the remaining **12 weeks**. The leader's management charge rate is **USD 185 per hour**; the senior engineer who would take the task is charged at **USD 140** (Domain 7's engineer grade). Delegated, the engineer needs **5.0 hours** a week for the first 4 weeks and **3.5 hours** thereafter, and the leader retains **1.0 hour** a week of oversight for 4 weeks and **0.5 hours** thereafter. The leader's people-work capacity is **6.5 hours** a week (KA 12.1.4).
2. **Formula.** Cost of each option = Σ(hours × rate). Released leader hours = status-quo leader hours − retained oversight. Breakeven value of a released leader hour = incremental direct cost ÷ released hours.
3. **Substitution.** Status quo `3.5 × 12 = 42.0` leader hours `× 185`. Delegated: engineer `5.0 × 4 + 3.5 × 8 = 48.0` h `× 140`; oversight `1.0 × 4 + 0.5 × 8 = 8.0` h `× 185`.
4. **Result.** Status quo **USD 7,770.00**. Delegated **USD 8,200.00** (6,720 + 1,480) — delegation is **USD 430.00** more expensive in direct labour. It releases **34.0** leader hours. The breakeven value of a released leader hour is `430.00/34.0 =` **USD 12.65**, or **6.836 %** of the leader's own charge rate.
5. **Interpretation.** The decision is not close, and the arithmetic shows why the instinctive framing gets it wrong. Priced as a labour comparison, delegation looks like a 430-dollar mistake; priced correctly it is the purchase of 34 leader hours for 430 dollars, and it pays if a leader hour is worth more than **USD 12.65** — under **7 %** of what the organisation charges for it. **The denominator makes it urgent**: the 3.5 hours the task consumes is **53.85 %** of the leader's entire discretionary people-work capacity of 6.5 hours, and the 34 released hours are **5.23 weeks** of that capacity. One recurring coordination task was consuming more than half of the only resource from which coaching, difficult conversations and team design are paid. **The risk side.** The objection is not cost but error: if the delegate mishandles one interface, Domain 4 prices re-verification at **USD 6,000** (KA 4.4.2). Value the 34 released hours at the leader's own rate — `34 × 185 =` **USD 6,290** — and net benefit is **USD 5,860**, so the delegation fails only if the probability of a 6,000-dollar incident exceeds `5,860/6,000 =` **97.67 %**. That is structurally Domain 3's threshold result (KA 3.2.3) and for the same reason, worth naming rather than re-deriving: **the cost of not delegating is certain, recurring and invisible, while the cost of a delegated error is uncertain, occasional and highly visible**, and organisations optimise against the visible cost. Two cautions. The 6,290 valuation holds **only if the released hours are actually spent on something worth 185 dollars an hour** — delegation that produces a busier leader rather than a coaching leader creates none of this value, which is why the released time should be reallocated at the moment of delegation, in writing. And the delegate must receive the information and criteria, not just the task: without them the 97.67 % breakeven is not the relevant number, because the error probability is no longer small.

12.3.2 Span of control and coaching capacity

Span of control is usually argued as a structural convention — six to eight, or whatever the organisation last used. The better framing is capacity: **a leader can supervise many people and can coach few, and the binding constraint is the second.** Let `R` be people-work capacity in hours a week, `n` the direct reports, `b` the team size at which `R` was measured, and `e` the additional fixed load each report above `b` adds to the leader's week — more one-to-ones, more interfaces held, more escalations absorbed. Then

```
Coaching time per report = (R − e(n − b)) / n
```

The second term is what most span discussions omit, and omitting it is not conservative — it overstates the sustainable span.

WORKED EXAMPLE

Worked example 12.3.2 — how many people can Auriga's leader actually lead?

1. **Setup.** From KA 12.1.4, `R` = **6.5** hours a week, measured at `b` = **6** direct reports. Each report above six adds `e` = **0.5** hours of fixed load — a planning judgement calibrated from the observed interruption log, not a constant. The project's stated sufficiency convention is that meaningful weekly coaching of a delivery professional requires at least **45 minutes** of undivided attention; this too is a deliberately adopted convention, not a measured threshold, and an organisation with different practices should set its own and re-run the arithmetic.
2. **Formula.** Per-report time `= (R − e(n − b))/n`. Maximum sustainable span: solve `(R − e(n − b))/n ≥ T`, giving `n ≤ (R + eb)/(T + e)`.
3. **Substitution.** `(6.5 − 0.5(n − 6))/n = (9.5 − 0.5n)/n`. Threshold `n ≤ (6.5 + 0.5 × 6)/(0.75 + 0.5) = 9.5/1.25`.
4. **Result.** Coaching time per report: **65.00** minutes at 6 reports, **51.43** at 7, **41.25** at 8, **33.33** at 9, **27.00** at 10, **21.82** at 11 and **17.50** at 12. The threshold is `n ≤` **7.6**, so the maximum sustainable span is **7 direct reports**. Coaching capacity reaches exactly **zero** at **19** reports.
5. **Interpretation.** Three results, in ascending order of usefulness. **The obvious one**: doubling the span roughly halves each person's time, and at 12 reports each receives 17.5 minutes a week, which is not coaching — it is a status check with a friendly tone. **The non-obvious one**: because the leader's own fixed load grows with the team, the decline is faster than `1/n`, and ignoring that changes the answer. On the naive curve `6.5/n` the test is met up to `n ≤` **8.667**, a maximum span of **8** (48.75 minutes at 8, 43.33 at 9) — so **omitting the leader's growing fixed load overstates the sustainable span by one report**, exactly the margin by which most spans are set too wide. **The decisive one**: the sustainable span is a function of `R`, and `R` is designable. Recovering the 3.0 hours of interruption identified in KA 12.1.4 raises `R` to 9.5 and moves the threshold to `n ≤ 12.5/1.25 =` **10** reports. **The way to widen a span is to raise the leader's people-work capacity, not to divide a fixed budget more thinly** — and an organisation that loads its leaders with governance and interruption has made a span decision without knowing it. Two cautions: this bounds coaching, not supervision — a leader can hold accountability for 30 people through team leads, which is what KA 12.2.2's pod design does, and the coaching arithmetic then applies to the leads; and a wide span is not always wrong, since experienced people on familiar work need less coaching. What is always wrong is a wide span assumed to be coachable, because that is a commitment the arithmetic says cannot be met.

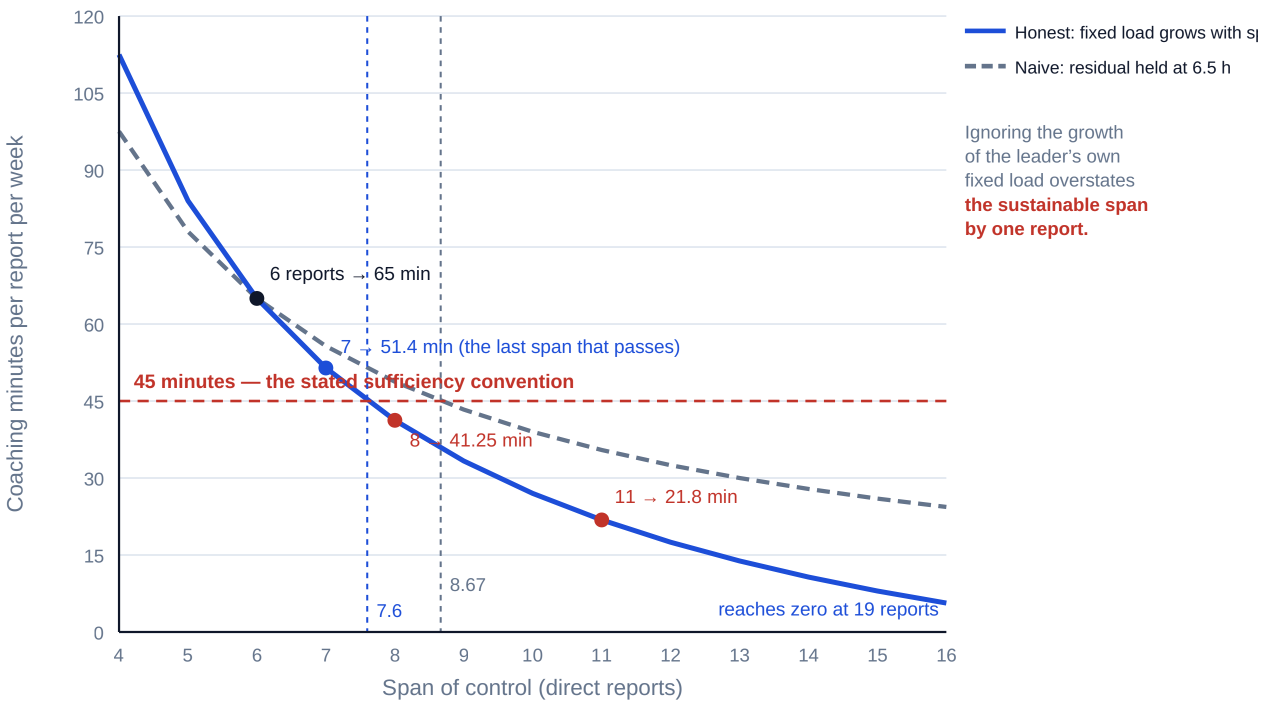


Fig 12.3.1 — Coaching minutes per report against span of control

12.3.3 The difficult conversation

Definition. A difficult conversation here is one in which the leader must tell someone something about their work, conduct or prospects that they will not want to hear, in a way that leaves the working relationship and the person's dignity intact. It is the highest value-per-hour act available to a delivery leader and the one most reliably postponed, and both facts have the same cause: the cost of holding it is immediate, personal and certain, while the cost of not holding it is deferred, diffuse and borne mostly by other people. Correcting that requires either changing the asymmetry — which is what pricing the deferral does (Case study B computes it) — or removing the structural excuse, which is what the span arithmetic does, since a leader with 21.8 minutes a report per week has a genuine reason the conversation never happens.

The structure that works, in five parts; the failure mode is losing control of the second minute. **The factual account, assembled beforehand** — specific, dated, verifiable observations, not characterisations: "three of the last five interface specifications were submitted after the agreed date, on the 4th, the 18th and the 2nd" is a fact, while "you have a problem with deadlines" is a judgement dressed as one, and the conversation will be spent litigating it. **The consequence, in the project's terms** — what it costs the work and the other people, using the numbers of Domain 4 and Domain 9 rather than the leader's displeasure. **The enquiry, genuinely open**, because a material proportion of under-performance has causes the leader does not know about — an unstated dependency, an absent skill nobody provided, a health or caring situation, a conflicting instruction — and a conversation with no room for this is a verdict, resented in proportion to how right it was. **The specific expectation and support**: what will be different, by when, measurably, and what the leader will supply. **The record and the follow-up date**, since Domain 3's decision-record discipline (KA 3.3.4) applies unchanged — an unrecorded conversation was, for every practical purpose afterwards, a chat.

The timing rule. Hold it at the **first clear instance**, not the third. The first is a short, low-stakes, factual conversation about one event; the third is about a pattern, which is about the person, which is the one that fails. That is a claim about what the available evidence lets you discuss, not about psychology. **What not to do**, each common and each with a consequence: **the sandwich** teaches that compliments precede bad news and devalues both; **feedback by proxy** tells the person the leader would not say it to them; **the deferred aggregate** — a year of observations saved for an annual review — denied them any chance to correct, so any consequence attached is unfair on its face; and **the written-only conversation** removes the enquiry entirely.

12.3.4 Performance management honestly

Formal performance-management systems vary enormously across organisations and jurisdictions, and their legal requirements are jurisdiction-specific: what constitutes fair process, adequate notice, reasonable adjustment or lawful termination differs by country and by employment relationship, and **nothing in this domain should be read as advice on any jurisdiction's employment law.** Where a performance issue may lead to a formal consequence, involve the organisation's human-resources and legal functions early, follow the applicable process precisely, and document contemporaneously.

Four things generalise, and they are the leader's obligations rather than the system's. **Contemporaneous, specific records** — the difference between a defensible process and an indefensible one is almost always whether the record was made at the time (Domain 3, KA 3.3.4). **The distinction between capability, conduct and circumstance**: a person who cannot do the work needs training or a different role, one who will not needs a consequence, one temporarily unable needs support — and treating any as another is both unjust and ineffective. **A stated standard the person knew about before they were measured against it**, the same fairness principle Domain 9 applies to acceptance criteria. And **proportionality**: the first response to an isolated lapse by a good performer is a conversation, not a process.

On rating and ranking systems this book takes no position, and notes two consequences. Ratings that determine reward draw effort toward the measured behaviours — Domain 9's measurement problem applied to people, and why a scheme ignoring collaboration produces less of it. And a distribution imposed across small teams produces outcomes the team can see are arbitrary, a psychological-safety cost paid for an administrative convenience. Two habits belong to the leader's own performance: ask for feedback in a form that makes it easy to give — a specific question about a specific recent decision — and **act visibly on one piece of it**, because the second time you ask depends on what happened the first time.

AI in this KA

Where it earns its place. Assembling from the project record the factual chronology a difficult conversation requires — dates, submissions, agreed commitments — the part leaders skimp and the part that decides whether the conversation holds. Rehearsal: generating likely responses and testing the leader's answers, on the situation rather than the person. Computing the delegation and span arithmetic across options. Checking written objectives for specificity, measurability and a date. Drafting the reallocation plan for the hours a delegation releases, so released time is committed rather than absorbed.

Where it must not go. No AI-generated assessment of a person's performance, potential or conduct in any record, and no AI recommendation on a promotion, reward or termination decision. The prohibition is not squeamishness: these decisions carry consequences that differ by jurisdiction, they require the accountable human's judgement on evidence largely not in the system, and an organisation that lets a model's output stand as the basis of one has created accountability without a holder — Domain 1's defect at the highest stakes. Nor should the words of a difficult conversation be model-authored: the person will be able to tell, and what they learn is that the leader would not compose it themselves.

Verification, concretely. Every fact in an AI-assembled chronology is checked against the source record line by line, because one wrong date hands the conversation to the other party. The delegation and span arithmetic is reproduced by hand. Where AI has helped draft objectives, the leader confirms each is something they would hold themselves to. And no personal data about a named individual is entered into any tool outside the organisation's stated and lawful boundary for it (Domain 3, KA 3.A.2).

KEY TERMS — KA 12.3

Term	Meaning
Delegation	Transfer of authority to decide and act, with the information and criteria to do so; accountability is retained.
Abdication	Transferred authority without the information or criteria to exercise it.
Released leader hours	Status-quo leader hours less retained oversight; the actual product a delegation buys.
Breakeven value of a released hour	Incremental direct cost ÷ released hours — USD 12.65 on Auriga, 6.836 % of the leader's rate.
Coaching time per report	`(R − e(n − b))/n`; 65.00 minutes at 6 reports on Auriga, 17.50 at 12.
Sufficiency convention `T`	The minimum weekly coaching time an organisation states as meaningful; 45 minutes on Auriga, a convention not a measurement.
Maximum sustainable span	`n ≤ (R + eb)/(T + e)`; 7.6 on Auriga, so 7 reports.
First-instance rule	Hold the conversation at the first clear instance, when it is about an event rather than a person.
Capability / conduct / circumstance	The three causes of under-performance, requiring training, consequence and support respectively.

SAMPLE MCQS — KA 12.3

MCQ 12.3-A `[12.3.2 · Application]` A leader has 6.5 hours a week of people-work capacity at six direct reports, and each additional report adds 0.5 hours to the leader's fixed load. Against a 45-minute weekly coaching convention, the maximum sustainable span is: - A. 6 reports - B. 7 reports - C. 8 reports - D. 19 reports

Rationale: `n ≤ (6.5 + 0.5 × 6)/(0.75 + 0.5) = 9.5/1.25 = 7.6`, so 7 (12.3.2), giving 51.43 minutes each; 8 gives 41.25 and fails. C is the answer from the naive curve `6.5/n`, which ignores the growth of the leader's fixed load and overstates the span by one; D is the span at which coaching capacity reaches zero.

MCQ 12.3-B `[12.3.1 · Analysis]` Delegating a task costs USD 430 more in direct labour over 12 weeks and releases 34.0 hours of the leader's time. The breakeven value of a released leader hour is: - A. USD 10.24 - B. USD 12.65 - C. USD 185.00 - D. USD 430.00

Rationale: `430.00/34.0 = 12.65` (12.3.1). A divides by the leader's original 42.0 hours rather than the 34.0 released; C is the leader's charge rate, which is the value the released hour may have, not the breakeven; D reads the incremental cost as an hourly figure.

MCQ 12.3-C `[12.3.1 · Evaluation]` The strongest argument for delegating in MCQ 12.3-B, given that a mishandled interface would cost USD 6,000 to re-verify and the released hours are worth USD 6,290 at the leader's rate, is that: - A. the senior engineer is capable - B. the leader is too busy - C. the delegation fails only if the probability of a 6,000-dollar incident exceeds 97.67 % - D. delegation develops people

Rationale: The decisive argument is the breakeven error probability, `(6,290 − 430)/6,000 = 97.67 %` (12.3.1). A and D are assertions, however true; B is a capacity claim, real but not the comparison.

MCQ 12.3-D `[12.3.3 · Analysis]` A leader raises a recurring lateness issue for the first time at its third occurrence. The principal cost of the delay is that: - A. the person will be more defensive - B. the conversation is now about a pattern, and therefore about the person, rather than about a single event - C. the leader has lost credibility with the team - D. the record is now incomplete

Rationale: The first-instance rule is about what the available evidence permits you to discuss: one event supports a short factual conversation, a pattern forces a conversation about the person, which is the one that fails (12.3.3). A is a likely symptom of B rather than the cause.

SELF-CHECK — KA 12.3

1. What does a delegation actually buy, and what is the breakeven? — Released leader hours; on Auriga 34.0 hours for USD 430, breaking even at USD 12.65 per released hour, 6.836 % of the leader's charge rate.
2. Why does omitting the leader's growing fixed load matter in span arithmetic? — It overstates the sustainable span by one report — 8 instead of 7 on Auriga's figures — and one report is the usual margin of error.
3. Why hold a difficult conversation at the first instance? — Because one event supports a factual conversation about the event; a pattern forces a conversation about the person.

KNOWLEDGE AREA 12.4

Remote and multicultural teams, ethical leadership

Topics: 12.4.1 distributed teams and the arithmetic of overlap · 12.4.2 culture without stereotype · 12.4.3 ethical leadership and the leader's own conduct · 12.4.4 leading people alongside AI.

12.4.1 Distributed teams and the arithmetic of overlap

The definition that matters. A distributed team is not one that is "remote"; it is one whose members' working hours do not fully coincide. Distance in space is mostly a solved problem. Distance in time is not, and it governs how long a decision takes, because a question asked outside the other party's working window waits. Of the two failure modes only one is discussed: the discussed one is communication quality, the undiscussed one is arithmetic. A pair of teams with **zero** overlapping working hours cannot complete an exchange in less than a working day, so a clarification requiring three exchanges takes three working days whatever anyone's intentions are. This is the same latency logic as Domain 3's `E[wait]` — a periodic opportunity to be heard, imposing a wait — and it should be computed at design time for the same reason.

WORKED EXAMPLE

Worked example 12.4.1 — the cost of Meridian's time-zone design, and the price of fixing it.

1. **Setup.** Meridian's programme runs across three groups, each working **09:00–17:00 local** — eight productive hours: the **core programme team** at **UTC+3**, the **development supplier** at **UTC+5:30**, and the **clinical-informatics advisers** at **UTC−5**. The first programme year logged **26** clarifications between the core team and the advisers that sat on the critical path of a clinic go-live and required **three** exchanges each. Meridian's cost of delay is **USD 14,280 per week**, or **USD 2,856** per working day over a five-day week.
2. **Formula.** Convert every window to a common reference (UTC), then overlap `= max(0, min(end) − max(start))`. For a zero-overlap pair, round-trip latency is one working day per exchange. Cost = exchanges × days × cost of delay per day.
3. **Substitution.** Core `09:00 − 3 =` **06:00–14:00 UTC**; supplier `09:00 − 5.5 =` **03:30–11:30 UTC**; advisers `09:00 + 5 =` **14:00–22:00 UTC**. Core ∩ supplier `min(14:00, 11:30) − max(06:00, 03:30)`. Core ∩ advisers `min(14:00, 22:00) − max(06:00, 14:00)`.
4. **Result.** Core and supplier overlap **5.5 hours** a day. Core and advisers overlap **0.0 hours**; supplier and advisers **0.0 hours**; all three together **0.0 hours**. Each core–adviser clarification therefore costs **3 working days — USD 8,568 —** and the 26 of them cost **USD 222,768** in the first year.
5. **Interpretation.** Two hundred and twenty-two thousand dollars was spent on a fact about the Earth's rotation that nobody had written down. **The fix and its exact cost are the valuable part.** Shifting the core team's day **two hours later** (11:00–19:00 local, 08:00–16:00 UTC) and the advisers' **one hour earlier** (08:00–16:00 local, 13:00–21:00 UTC) creates a **3.0-hour** daily overlap, inside which a three-exchange clarification completes in **one** working day. The 26 then cost **USD 74,256**, a saving of **USD 148,512** a year, or **USD 5,712** each. **But the redesign is not free, and the arithmetic shows where the cost lands.** Moving the core team two hours later reduces its overlap with the supplier from 5.5 hours to **3.5** — a consequence nobody would have noticed until it appeared as supplier friction, and the reason a distributed design must be computed as a whole rather than fixed one pair at a time. Here 3.5 hours still supports one substantial daily synchronisation, so the trade is worth making; the point is that it must be **a decision, priced, rather than a side effect**. **And there is a human cost that must be allocated explicitly, which is where this topic meets 12.4.3.** The 3.0 hours of overlap are bought with 3.0 hours a day of unsocial working. Split as designed — two hours on the core team, one on the advisers — the burden is **10 hours a week** for the core team and **5** for the advisers. Pushed entirely onto whichever group has least power, that group carries **15 hours a week** indefinitely, which is the default outcome when a supplier or an offshore team is the junior party. **Rotating the shift monthly equalises it at 7.5 hours a week each**, and the choice between those three allocations is an ethical decision made by a leader, not a scheduling detail. A leader who takes the 148,512 and does not decide who pays the 15 hours has not finished the calculation.

What else follows for practice. Design the team so that **the dependencies with the most exchanges sit inside a single time zone** — the same principle as Domain 4's interface minimisation and KA 12.2.2's pods: put the boundary where the traffic is thin. Convert exchanges into **asynchronous artefacts** wherever the exchange is informational rather than deliberative, since a complete written specification with the anticipated questions already answered removes exchanges rather than accelerating them, and removing one is worth a full working day at zero overlap. Make the **handover artefact** formal where work passes between zones — what was done, what is in flight, what is blocked, what the next party must decide. And publish the overlap map, because the commonest cause of a three-day answer is that the asker did not know the other party had already gone home.

12.4.2 Culture without stereotype

Cross-cultural difference in delivery teams is real, consequential and routinely handled badly in both directions — ignored, or reduced to a table of national generalisations applied to individuals. This book's position, stated as a position rather than a finding: **national-culture dimension frameworks are useful for anticipating the existence of differences and useless for predicting the behaviour of the person in front of you.** They earn their place as a checklist of what may vary; they fail when a leader uses a country average to predict an individual, because within-group variation in every dimension exceeds the between-group differences described.

Four differences matter most in project work. **How disagreement is expressed** — in some working cultures a direct "no, that will not work" is expected; in others the same message is carried by a qualified yes, a silence or a change of subject, and the leader who hears only the words hears agreement. The countermeasure is not a national rule but a **changed question**: replace "does everyone agree?" — which almost always returns agreement — with "what is the strongest argument against this?", asked of a named person, plus written confirmation so a misread silence surfaces within a day. **How hierarchy is treated** — whether a junior engineer will contradict a senior one varies enormously and interacts directly with psychological safety, so a leader who concludes there is no safety problem because nobody objected may be observing a hierarchy norm; the countermeasure is structural, asking for input in a form that does not require public contradiction and never punishing the first person who does. **How commitment is talked about** — whether "we will try to have it by Thursday" is a commitment or an aspiration is not universal, and many cross-boundary schedule failures are this ambiguity rather than any failure of intent; the countermeasure is Domain 4's discipline, commitments written, dated and confirmed by the committing party in their own words. **How much is left implicit** — some working cultures carry context implicitly and find exhaustive specification faintly insulting, others treat anything unwritten as unagreed; make the convention explicit for the project, and on a distributed programme make it written-and-explicit, because that is the one that survives being read at 03:00 by someone who cannot ask a follow-up until tomorrow.

Where the working language is a second language for part of the team: slow the pace and reduce idiom rather than raising the volume; put the substance in writing, so reading — easier than listening — carries the load; and never mistake fluency for competence in either direction, since the most competent engineer on a team is frequently not the most fluent speaker in the room. **Inclusion is a delivery variable as well as an ethical one:** a team member who cannot contribute — for reasons of language, time zone, hierarchy norm, disability, or simply never being asked — is capacity the project is paying for and not receiving. That is the honest business case, and it does not replace the ethical case, which stands on its own.

12.4.3 Ethical leadership and the leader's own conduct

Definition. Ethical leadership in delivery is the consistent subordination of the leader's own short-term interest — reputation, comfort, bonus, the appearance of control — to the obligations owed to the sponsor, the team, the users and the public. Domain 1 established those obligations (KA 1.2.2); this topic is about the leader's conduct when honouring them is costly.

Reporting what is true when it is unwelcome. The pressure is to report amber, absorb the variance personally and hope; the response is Domain 1's standard of care and Domain 3's record —

report the forecast you believe, with its basis, in writing, and keep the paper, because a leader who reports optimism they do not hold has transferred a risk to people who do not know they are carrying it. **Being asked to commit to a date you do not believe in** is the most common ethical situation in delivery and is rarely recognised as one; the response is neither refusal nor silent acceptance but stating the date you can commit to, the conditions under which the requested date becomes possible, and the risk of the requested date, in writing — then letting the accountable person decide with the information (Domain 3, KA 3.2.1). **Protecting the team from unreasonable demand**: sustained overtime is a loan against future capacity at a poor rate, and KA 12.2.4's 66,845 per departure is the argument. **Safety and harm** — where the work can hurt someone, the obligation to stop is absolute and personal and is not subject to the delegation schedule. **Fairness in the things people watch**: who gets the visible work, who is in the room, who is credited, who is believed when two accounts conflict — the decisions from which a team infers what the leader actually values, made quickly and observed closely. And **the small compromises** — supplier hospitality, a friend's firm on a shortlist, a favour that would help the schedule — for which the reliable test is whether you would be comfortable if the whole decision, including your reasoning, were in the decision record with your name on it.

Two closing observations. Ethical failure in delivery is rarely a single dramatic choice; it is an accumulation of small deferrals — an unreported slippage, an unraised concern, a number rounded favourably — each defensible alone and collectively a misrepresentation, and the countermeasure is the cumulative test Domain 3 applies to changes below a threshold. And the leader's conduct is the only reliable instrument of culture available: values statements are read, decisions are studied, and a leader who wants candour and punishes the first candid person has settled the matter whatever the poster says.

12.4.4 Leading people alongside AI

Four leadership problems are immediate. **Skill formation**: where a tool performs the tasks on which junior practitioners used to learn judgement, the pipeline of competence is affected; the honest position is that the effect is plausible and not yet measurable, so act conservatively — deliberately reserve some work for human development even where a tool would be faster, and make the review of AI output a taught skill rather than an assumed one, since reviewing a plausible wrong answer is harder than producing a rough right one. **Honest attribution**: a team member presenting model output as their own analysis has misled colleagues about the basis of a recommendation, which matters because the verification obligation attaches to the basis — state the norm before it is needed: **say what was AI-generated, say what you verified and how**. **Fear, and the conversation nobody is having**: where people believe a tool threatens their role they will not say so, and they will adopt it visibly while routing around it privately, which is the worst outcome because the organisation gets neither the productivity nor the truth; the response is a direct, early, honest conversation about what will and will not change, in which "I don't know yet" is acceptable and false reassurance is not. And **accountability, unchanged**: **AI proposes; the professional verifies, decides and remains accountable.** Applied here, a leader may not attribute a decision about a person to a system, may not accept a model's assessment of a person as evidence, and may not delegate to a tool a conversation that is theirs to hold.

AI in this KA

Where it earns its place. Time-zone and overlap computation across a large distributed programme — every pair, every proposed shift pattern, every consequence — the deterministic arithmetic of 12.4.1 at a scale nobody would attempt by hand, and where the second-order effect on the core–supplier overlap is easily missed. Producing the rotation roster that equalises unsocial hours, with the fairness constraint stated as an input. Translation and plain-language support for second-language colleagues, with the caution that a translated commitment must be confirmed by the committing party in their own words. Summarising a long asynchronous thread into decisions taken and questions still open.

Where it must not go. No cultural inference about individuals — a model asked "how should I communicate with a colleague from country X" returns a confident national generalisation, precisely the error 12.4.2 identifies, and receiving it from a system makes it feel like data. No monitoring of remote workers' activity as a proxy for productivity: the measure is poor, the jurisdictional position varies, and the effect on trust is severe and permanent. And no automated allocation of unsocial hours without a human deciding the fairness rule, because the fairness rule is the whole ethical content of the decision.

Verification, concretely. Recompute the overlaps that matter by hand, in UTC, and check each proposed shift against **every** pair rather than the one being fixed. Confirm any AI-summarised decision list against the thread with the people who took the decisions. And where a rotation roster is generated, verify that the unsocial-hours totals per group are equal to the stated tolerance — a one-line check, and the only part of the output that carries the ethics.

KEY TERMS — KA 12.4

Term	Meaning
Distributed team	A team whose members' working hours do not fully coincide; time distance, not spatial distance, is the governing variable.
Working-hour overlap	`max(0, min(end) − max(start))` with all windows converted to a common reference.
Zero-overlap pair	Two groups with no common working hours; each exchange costs at least one working day.
Overlap window	A deliberately created shared period, bought with unsocial hours that must be explicitly allocated.
Unsocial-hours burden	The hours per group per week spent outside normal local working time; equalised by rotation.
Asynchronous artefact	A written product complete enough to remove an exchange rather than accelerate it.
Handover artefact	What was done, what is in flight, what is blocked, what the next party must decide.
Within-group variation	The reason national-culture averages cannot predict an individual's behaviour.
Honest attribution (AI)	Stating what was AI-generated and what was verified, and how.

SAMPLE MCQS — KA 12.4

MCQ 12.4-A `[12.4.1 · Application]` Team A at UTC+3 and team B at UTC−5 both work 09:00–17:00 local. Their daily overlap is: - A. 0.0 hours - B. 2.0 hours - C. 3.0 hours - D. 8.0 hours

Rationale: In UTC, A works 06:00–14:00 and B works 14:00–22:00, so `min(14:00, 22:00) − max(06:00, 14:00) = 0` (12.4.1). B results from taking the offset as 6 hours rather than 8; C is the overlap after the shift redesign; D compares local clock times without converting to a common reference — the error the topic exists to prevent.

MCQ 12.4-B `[12.4.1 · Analysis]` Twenty-six critical-path clarifications a year each need three exchanges. At zero overlap each exchange costs one working day; the cost of delay is USD 2,856 per working day. Creating a three-hour overlap lets each clarification complete in one day. The annual saving is: - A. USD 5,712 - B. USD 74,256 - C. USD 148,512 - D. USD 222,768

Rationale: `26 × 3 × 2,856 = 222,768` before and `26 × 1 × 2,856 = 74,256` after, so the saving is **148,512** (12.4.1). A is the saving per clarification; B is the post-fix cost; D is the pre-fix cost.

MCQ 12.4-C `[12.4.1 · Evaluation]` Shifting the core team's day two hours later to create the overlap above reduces its overlap with a supplier from 5.5 hours to 3.5. The correct professional response is to: - A. abandon the redesign, since it damages an existing relationship - B. proceed, having computed the effect on every pair and judged 3.5 hours sufficient for one daily synchronisation - C. proceed without recomputing, since the adviser problem is larger - D. ask the supplier to shift instead, since they are the supplier

Rationale: A distributed design must be computed as a whole, not fixed one pair at a time, and the trade must be a priced decision rather than a side effect (12.4.1). D allocates the burden by power rather than by reason, the failure 12.4.1 and 12.4.3 both identify.

MCQ 12.4-D `[12.4.2 · Analysis]` A leader asks a distributed team "does everyone agree?" and receives agreement, then discovers a fortnight later that two members had serious objections. The best countermeasure is to: - A. learn the national communication norms of the members concerned - B. ask a named person for the strongest argument against the proposal, and confirm decisions in writing - C. require objections to be raised in the meeting - D. reduce the team's size

Rationale: Changing the question and confirming in writing works regardless of which norm is operating, whereas a national generalisation cannot predict an individual (12.4.2). C restates the requirement that has just failed.

SELF-CHECK — KA 12.4

1. How is working-hour overlap computed, and what is the trap? — Convert every window to a common reference, then `max(0, min(end) − max(start))`; the trap is comparing local clock times, which invents overlap that does not exist.
2. What does a three-hour overlap window actually cost, and who decides? — Three hours a day of unsocial working; the allocation between groups — 10/5, 15/0 or 7.5/7.5 rotated — is an ethical decision the leader must make explicitly.
3. What are national-culture frameworks good for, and not good for? — Anticipating that differences exist; not predicting an individual's behaviour, because within-group variation exceeds the between-group differences described.

Advanced topics — Domain 12

12.A.1 Leading a team through loss of confidence

Domain 3 designed recovery governance — raised authority, short cadence, narrowed membership, a single information source, a log kept in the room (KA 3.A.1). That structure does nothing about the condition that determines whether a recovery works: a team that has stopped believing the work can succeed.

The behavioural signature is specific and often mistaken for its opposite. Effort does not fall — it frequently rises and becomes defensive. People work longer on their own scope and stop attending to the whole. Estimates lengthen and stop being negotiable, because the estimate has become a shield rather than a forecast. Escalations become formal and documentary, a rational response where blame is expected, and Domain 3's escalation-lead-time indicator will be shortening at the same time. And the best people leave first: they have the most options and the least need to tolerate it. At KA 12.2.4's USD 66,845 per departure, a recovery that loses three people has spent **USD 200,535** before any recovery action, and has lost the three people whose judgement it most needed.

What works, as professional judgement rather than measured effect. **Re-establish a credible plan before asking for effort**, because the broken link is effort → performance and no exhortation repairs it — an honest re-baseline is a leadership act, not an admission. **Shorten the horizon** to something the team can succeed at within a fortnight, and let them succeed visibly. **Reduce the count of things in flight** rather than adding people, since KA 12.2.2's arithmetic says the second worsens coordination exactly when coordination is failing. **Be present and specific about what is not known**, since ambiguity filled by the team is filled pessimistically. **Protect the escalation route**, because a recovery in which bad news stops arriving cannot be steered. And **decide team changes early and transparently**, since a recovery in which people suspect unannounced changes spends the whole period on that question. Two cautions: raised authority must remain **bounded and time-limited** or the directive style it licenses becomes permanent, which is how an organisation loses the people it kept through the crisis; and replacing the project leader mid-recovery imposes a ramp-up cost on **every** member of the team simultaneously, not on one post — KA 12.2.4's ramp-up component multiplied by the team.

12.A.2 Team design as an architecture decision

The organisational and technical architectures are not independent. There is a long-standing observation in systems engineering that a delivered system tends to resemble the communication structure of the organisation that built it. This book treats that as an **observation worth designing against**, not a law, and notes that Domain 4's arithmetic gives it teeth: if the team structure and the interface structure disagree, one of them will be paid for. The design rule follows from KA 12.2.2 and KA 4.2.3 together: **put the team boundary where the interface traffic is thin.** A pod owning a cluster of clinics end to end has few external interfaces and needs few external relationships; a pod defined by technical speciality — all the testers, all the integrators — has an interface with every other pod on every deliverable, and will generate exactly the coordination load the pod structure was meant to remove.

The cost of double membership is where this fails in practice. Meridian's five pods of eight carry **150** paths, **4.6875 %** of capacity, **1.875 FTE**. Put **ten** of the forty people in a second pod as well, and each acquires the seven relationships of the additional pod: **70** extra paths, **220** in total, **110.0** hours a week, **6.875 %** of capacity, **2.75 FTE**. The extra **35.0** hours a week is **0.875 FTE** — nearly one full-time person consumed by a staffing convenience — and the resulting share is exactly that of a **12-person fully connected team**, which is a useful sentence in a resourcing discussion: a quarter of our people double-hatted means our forty-person programme coordinates like a twelve-person one, and pays for the privilege twice.

Three further consequences. **Long-lived teams are cheaper than re-formed ones**, because the relationships in `n(n − 1)/2` have a formation cost as well as a running cost, and a team reassembled each phase pays it repeatedly while passing through the conflict stage again. **Ownership must include the consequence**: a pod that delivers a clinic but does not own its defects will optimise for delivery, which is Domain 9's problem arriving through an organisational door. And **the representative in the coordination forum is a role, not a courtesy** — it carries both pods' context and is the single path the layered design depends on, so it needs allocated time and a named deputy, or the pods quietly revert to a mesh and the 7.875 FTE returns as unexplained slowness.

12.A.3 The reviewer's leadership eye

Invariants to test on any delivery organisation's people design; each cheap, each countable, each diagnostic.

The project leader's week has been **audited from observed data**, not intention, and people-work capacity is a stated number. Every team has a **computed coordination share** at a stated link cost, presented across at least three values of c. No team above roughly a dozen people works as a **single fully connected group** without a stated reason. Team boundaries sit where the **interface traffic is thin**, and the double-membership count is known and priced. Every leader's span is **within the sustainable span** implied by their own people-work capacity and the organisation's stated coaching convention — and the convention is written down. Coaching time per report is computed with the leader's **growing fixed load** included, not from a constant residual. Every recurring task consuming more than a stated share of a leader's discretionary capacity has a **delegation decision on record**, with the released hours **explicitly reallocated**. Turnover cost is computed on the three-component build-up with the **saved pay of a vacant post excluded**, and the resulting value at risk is what any retention proposal is measured against — with no claimed effect size attached to the proposal. Psychological safety is tracked through **countable signals** — issues found by assurance versus reported by the team, escalation lead time — not a survey score alone. Every distributed pair has a **computed overlap in a common time reference**, every proposed shift is checked against **every** pair, and the unsocial-hours burden is **allocated explicitly and rotated**. Difficult conversations are held at the **first instance** and recorded contemporaneously. And no AI system is inferring anything about a **named individual's** state, engagement, personality or likelihood of leaving — the single test in this domain that admits no proportionality argument.

Industry variations — Domain 12

- **Construction and infrastructure.** The leader's span includes people they do not employ, so coaching capacity applies mainly to a supervisory tier and the coordination arithmetic must count paths across contractual boundaries. Span is frequently fixed externally by safety supervision ratios — the one context where the number is not the leader's to design — and safety leadership is a personal, non-delegable obligation that overrides schedule pressure absolutely.
- **Healthcare.** Clinical professional autonomy means a project leader has authority over the project and none over clinical judgement: Domain 3's point about Meridian's clinical sign-off (KA 3.1.2) reappears as a leadership constraint — influence, not instruction. Multi-professional teams with parallel hierarchies put KA 12.4.2's hierarchy-norm problem inside one building.
- **Technology and product organisations.** Long-lived small teams with end-to-end ownership make 12.A.2's design rule native rather than a reform. Retention economics dominate because ramp-up is long and largely tacit: KA 12.2.4's eight-week curve is optimistic in a complex codebase, and the ramp-up component should be expected to exceed recruitment by a wide margin.
- **Public sector.** Staff rotation is often a feature of the career system rather than a failure, so the problem is **knowledge continuity** rather than retention — handover artefacts, written decision records and deliberate overlap between outgoing and incoming post-holders carry more value than any retention effort. Pay levers are usually unavailable, making KA 12.2.1's condition design the only approach on offer.

- **Energy and utilities.** Rotational and remote-site working — Auriga's commissioning crew is a mild version — makes shift handover a formal artefact with safety consequences and KA 12.4.1's unsocial-hours fairness question continuous rather than occasional. Turnover of specialist commissioning staff mid-campaign lands on zero-float activities, which is why KA 12.2.4's delay comparison matters more here than its labour figure.
- **Financial and professional services.** Utilisation targets compete directly with KA 12.1.4's people-work capacity, since coaching time is by construction non-chargeable: the span arithmetic collides with the business model and the collision must be made explicit rather than resolved by pretending. Matrix double-membership is endemic, which makes 12.A.2's 0.875 FTE the most useful number in this domain for these organisations.

Case study — Domain 12: the programme that tripled (health, Meridian)

Situation. Meridian Care Records moved from a 12-clinic pilot to the 40-clinic national rollout, and the programme team grew from **12** people to **40** over two quarters. No structural change accompanied the growth: the team continued as a single group with a weekly all-hands, a shared backlog and an expectation that anyone would coordinate with anyone. Nine months in, the rollout was running **six weeks** behind, individual productivity was reported as falling, and the quarterly report attributed the slippage to "clinic readiness and team ramp-up". Five people had left during the year. The steering committee, whose latency is 4.0 weeks (Domain 3, KA 3.2.3), was asked to approve eight additional staff.

What the arithmetic showed. The programme director computed three numbers before the paper went in. **Coordination.** At a link cost of 0.5 hours of team time per relationship per week on 40-hour weeks, the 12-person team had carried `12 × 11/2 =` **66** paths, **33.0** hours a week, **6.875 %** of capacity — **0.825** of a person. The 40-person team carried **780** paths, **390.0** hours, **24.375 %** — **9.75** full-time people, none visible in any cost line, all charged to the work packages people were coordinating about. **Structure.** Five pods of eight, each owning a cluster of eight clinics end to end with one representative in a weekly forum, would carry `5 × 28 + 10 =` **150** paths, **75.0** hours, **4.6875 %** — **1.875** people. The restructure would release **7.875 FTE**, more capacity than the eight staff being requested, at no additional cost. **Turnover.** On KA 12.2.4's three-component build-up with the same parameter set — a 6-week vacancy covered at a 50 % overtime premium, 2.80 person-weeks of ramp-up loss, 32 hours of mentoring, 0.8 person-weeks of leaver disengagement, 12 hours of handover — but Meridian's own figures, a USD 15,000 agency fee and a person-week of **USD 3,840** at USD 96 an hour, each departure had cost **USD 46,488**: recruitment 16,920, ramp-up 13,824, lost productivity 15,744. The five departures had therefore cost **USD 232,440**, of which **USD 147,840 — 63.6 %** — was ramp-up and lost productivity, recorded nowhere, and **USD 69,120** of it incurred after the last vacancy was filled and the recruitment problem considered closed.

What changed, and how it resolved. The recruitment request was withdrawn and replaced by a restructure into five pods, with the representative role given time, a named deputy and a place in its own pod's workload. The all-hands moved from weekly to fortnightly, the intervening slot given to pods — a small change that removed the main mechanism by which everyone stayed connected to everyone, which was precisely the point. And the five exit conversations were re-read against KA 12.2.1: three cited the same broken **effort → performance** link, a rollout plan nobody in the team believed, which the programme had been treating as a communication problem. The programme then delivered the remaining rollout **six weeks earlier** than the re-forecast made at restructure, worth `6 × 14,280 =` **USD 85,680** at Meridian's cost of delay. Read that alongside the theoretical figure: 7.875 FTE at a person-week of 3,840 is **USD 30,240** a week, or **USD 1,451,520** over a 48-week year — **16.94 times** what the programme actually banked. The gap is not an arithmetic error but the difference between **capacity released** and **value realised**: released capacity becomes value only where it is redeployed onto work that matters and where the rest of the system — clinic availability, training slots, the 4.0-week governance latency — does not immediately become the binding constraint instead. It did.

What the domain teaches here. The cost of an unstructured team is real, large and invisible, and it presents as declining individual productivity, the one explanation that guarantees the wrong response. **The instinctive remedy makes it worse**: eight more people in a fully connected 40-person team would have added `8 × 40 =` 320 hours of capacity and, from `h − c(n − 1)` summed across the additions, `0.5 × (40 + 41 + … + 47) =` **174.0** hours of new coordination — and the request was made in the same paper that reported falling productivity. And **a structural saving must be traced to a realised outcome or it is not a saving**: the honest claim was 85,680, not 1,451,520.

Case study B — Domain 12: the conversation that cost sixty-two times its price (rail systems design, transport)

Situation. A design consultancy was producing the systems-design package for a rail re-signalling scheme. The design manager had **11** direct reports. One of them, a lead engineer, had been producing **9** drawings a week against a plan of **12**, with **18 %** of drawings requiring rework against a **6 %** allowance. The manager was aware from week 2. The conversation happened in week **16**.

What the arithmetic showed — computed afterwards in a lessons review, which is when this arithmetic is usually done and far too late to be useful. Over the **14** weeks between the first clear instance and the conversation: an output shortfall of `3 × 14 =` **42** drawings, each taking **4.5** hours at a design rate of **USD 115** an hour — `42 × 4.5 × 115 =` **USD 21,735** of work displaced onto other people. Excess rework on the 126 drawings actually delivered: `(0.18 − 0.06) × 126 =` **15.12** drawings reworked at 2.0 hours each — `15.12 × 2 × 115 =` **USD 3,477.60**. And the package sat on the critical path of the client's design programme, so the 42-drawing shortfall converted at the planned rate of 12 a week into **3.5 weeks** of delay, exposed at **USD 28,000** a week under the consultancy's delay provisions — **USD 98,000**. Total consequence **USD 123,212.60**. The conversation itself, held in week 2, would have cost **3 hours** of the manager's time plus **1.5 hours a week for 6 weeks** of supported improvement — 12 hours at **USD 165** an hour, or **USD 1,980**. The ratio is **62.23 to 1**.

Why it did not happen, which is the instructive part. The manager was not avoidant by disposition; the structure had made the conversation unavailable. At 11 direct reports, with a people-work capacity of 6.5 hours a week measured at six reports and 0.5 hours of additional fixed load per report above that, coaching capacity was `6.5 − 0.5 × 5 =` **4.0** hours a week across 11 people — **21.82 minutes** each (KA 12.3.2). Twenty-two minutes a week is enough to ask how someone is and not enough to hold a documented, evidenced, five-part conversation about their performance. The manager had also never been given a sufficiency convention, a factual chronology, or any figure for what deferral cost.

How it resolved. Two structural changes and one conversation. The eleven reports were split into two teams of five under newly created team-lead roles, taking the design manager's span to **6** — **65.00** minutes a report a week — and giving the two leads spans of five. The consultancy adopted a written **45-minute** weekly coaching convention and required every span above the resulting sustainable maximum to be approved with a stated reason, which is the span arithmetic turned into a governance rule. And the conversation was held in week 16 on KA 12.3.3's five-part structure with a dated factual account. Its outcome was neither a dismissal nor a recovery to plan: the engineer's difficulty was substantially capability — a modelling technique the package needed and nobody had provided — and they moved to a role where their strengths applied, with the technique taught to two others. That is a common and under-described ending, and it was available in week 2 at a fraction of the price.

What the domain teaches here. First, the honest caveat, because a 62-to-1 ratio is the kind of number that gets misused: it attributes the **entire** consequence to the absent conversation, which overstates it. Halve the attribution and the deferral still cost **USD 61,606.30** against a conversation costing 1,980 — a ratio of **31.11 to 1** — and the conclusion is unchanged, which is what makes it worth quoting with the caveat. Second, **the asymmetry that causes deferral is neutralised by pricing it**: the manager was trading an hour of certain personal discomfort against a diffuse cost borne by other people, and only a number on the other side changes that trade. Third, **this was a design failure before it was a leadership failure**: an organisation that sets spans without the coaching arithmetic has decided that difficult conversations will be late — it has simply not written the decision down.

Executive perspective — Domain 12

What a programme director cannot delegate in this domain:

- **Your own attention budget, audited from observed data.** Recurring commitments, discretionary attention, people-work capacity and the interruption line — 82.2 %, 17.8 %, 6.5 hours and 9.0 hours on Auriga. A leadership commitment made without this number is a resource assumption nobody has checked (12.1.4).
- **The coordination cost of every team you own, at a stated link cost and as a range.** 24.375 % for 40 people in one group against 4.6875 % in five pods, a difference of 7.875 FTE. Adding people to an unstructured team is the one intervention the arithmetic says will not work (12.2.2, and D4 KA 4.2.3).
- **The span of every leader reporting to you, tested against a written coaching convention.** Not the chart's convention — the arithmetic: 7 reports at Auriga's capacity, not 8, and the way to widen it is to raise the leader's capacity, not divide a fixed budget more thinly (12.3.2).
- **The value at risk in turnover, computed honestly.** USD 66,845 per departure on Auriga's team of six and USD 46,488 on Meridian's, with the vacant post's saved pay excluded. Approve retention spending against that number, and refuse any proposal claiming an effect size it cannot evidence (12.2.4).
- **The allocation of the unsocial hours a distributed design requires.** Whether that is 10/5, 15/0 or 7.5/7.5 rotated is a decision you make, and if you do not make it, relative power will (12.4.1, 12.4.3).
- **The prohibition on AI inference about individuals.** No model output about a named person's state, engagement, personality or likely departure enters a decision, a record or a conversation — the one line in the domain with no proportionality argument, because the trust it protects cannot be re-baselined once spent.

Calculation exercises — Domain 12

Exercise 12.1 A 25-person team works as a single group. Each pairwise relationship consumes 0.5 hours of total team time per week; people are productive 40 hours a week. Compute the paths, the coordination hours and the share of capacity; then the same for five pods of five with one representative each, and the capacity released; then re-run both shares at a link cost of 0.8 hours. Solution. Mesh: paths `25 × 24/2 =` **300**; overhead `0.5 × 300 =` **150.0** hours; capacity `40 × 25 =` **1,000** hours; share **15.0 %** (equal to `(25 − 1)/160`). Pods: `5 × (5 × 4/2) + (5 × 4/2) = 50 + 10 =` **60** paths; overhead **30.0** hours; share **3.0 %**. Released **120.0** hours a week, **3.0 FTE**, a reduction of **12.0** percentage points. At `c` = 0.8: mesh **24.0 %**, pods **4.8 %** — the conclusion is unchanged across the range, which is the point of running it. Common error: using `n²` (625) instead of `n(n − 1)/2` (300), or omitting the division by two, both of which double the answer; and quoting a single link cost as though it were measured.

Exercise 12.2 A leader has 7.0 hours a week of people-work capacity measured at 5 direct reports, and each report above five adds 0.4 hours to their fixed load. The stated sufficiency convention is 40 minutes of weekly coaching per report. Compute coaching minutes per report at 5, 8 and 9 reports, the maximum sustainable span, and the span at which coaching capacity reaches zero. Solution. Per report `= (7.0 − 0.4(n − 5))/n = (9.0 − 0.4n)/n`. At 5: `7.0/5 = 1.4` h = **84.00** minutes. At 8: `5.8/8 = 0.725` h = **43.50** minutes. At 9: `5.4/9 = 0.6` h = **36.00** minutes. Threshold `n ≤ (7.0 + 0.4 × 5)/(0.6667 + 0.4) = 9.0/1.0667 =` **8.4375**, so the maximum sustainable span is **8**. Capacity reaches zero where `9.0 − 0.4n = 0`, at `n =` **22.5** — so from 23 reports the leader's fixed load exceeds their entire people-work capacity. Common error: dividing the whole 7.0-hour budget by the report count and ignoring the growth of fixed load, which gives `7.0/9 =` **46.67** minutes at nine reports and wrongly passes the 40-minute test.

Exercise 12.3 A five-person team loses one member. The person-week is 40 hours at USD 92 an hour. Recruitment: a USD 12,500 fee and 16 hours of selection time. The post is vacant 4 weeks, covered by overtime at a 40 % premium. The replacement ramps linearly from 40 % to 100 % effectiveness over 6 weeks with 3 hours a week of mentoring; the leaver's last 3 weeks run at 70 %; handover consumes 10 hours. Compute the total, its equivalent in person-weeks, and its share of the post's annual charge-out value over 46 weeks. Then compare the overtime decision with accepting the delay, given a cost of delay of USD 22,000 a week and a five-person zero-float activity. Solution. Person-week `40 × 92 =` **USD 3,680**. Recruitment `12,500 + 16 × 92 =` **USD 13,972**. Ramp effectiveness `0.40 + 0.52 + 0.64 + 0.76 + 0.88 + 1.00 =` **4.20**, so lost `6 − 4.20 =` **1.80** person-weeks; ramp-up `1.80 × 3,680 + 18 × 92 = 6,624 + 1,656 =` **USD 8,280**. Lost productivity: disengagement `0.9 × 3,680 = 3,312`, handover `10 × 92 = 920`, overtime premium `0.40 × 4 × 3,680 = 5,888` — **USD 10,120**. **Total USD 32,372**, or **8.80** person-weeks, or **19.12 %** of the post's annual charge-out value of `46 × 3,680 =` 169,280. Uncovered, 4 person-weeks missing from a five-person crew is `4/5 =` **0.8** weeks of delay costing **USD 17,600** — against a premium of 5,888, so covering is right by a wide margin, and the breakeven premium is `17,600/(4 × 3,680) =` **119.57 %**. Common error: counting the vacant post's four person-weeks of pay (14,720) as a cost. The pay is saved; the cost is the work not done and the 5,888 premium paid to get it done anyway. Including it inflates the total to 41,204.

Exercise 12.4 Three groups each work 09:00–17:00 local: P at UTC+2, Q at UTC+8, R at UTC−6. Compute all three pairwise overlaps. For the P–R pair, an average clarification needs 4 exchanges and 18 such clarifications a year sit on the critical path; the cost of delay is USD 22,000 a week over a five-day week. Compute the annual cost, then the cost and saving if P shifts 1.5 hours later

> and R 1.5 hours earlier. State the effect on the other pairs and the unsocial-hours burden. Solution. In UTC: P **07:00–15:00**, Q **01:00–09:00**, R **15:00–23:00**. Overlaps: P–Q **2.0** hours; P–R **0.0**; Q–R **0.0**. Cost of delay per working day `22,000/5 =` **USD 4,400**. As-is each P–R clarification costs `4 × 4,400 =` **USD 17,600**, and 18 of them **USD 316,800**. Shifted: P **08:30–16:30 UTC**, R **13:30–21:30 UTC**, overlap **3.0** hours, so a clarification completes in one day at **USD 4,400**; 18 cost **USD 79,200**, a saving of **USD 237,600**. The consequence elsewhere: P–Q falls from 2.0 hours to **0.5** — the shift very nearly destroys the only other overlap in the programme, which must be decided rather than discovered. Unsocial burden: 1.5 hours a day each, **7.5** hours a week per group, already equal without rotation. Common error: computing overlap from local clock times — both groups "work nine to five", so they must overlap — which invents eight hours of overlap that does not exist; and fixing the target pair without recomputing every other pair.

> **Exercise 12.5** A recurring task consumes 4.0 hours of a leader's week for the remaining 15 weeks. Delegated, the delegate needs 6.0 hours a week for 5 weeks and 4.0 thereafter, with leader oversight of 1.5 hours a week for 5 weeks and 0.5 thereafter. The leader is charged at USD 175 an hour, the delegate at USD 120. Compute both options, the released leader hours, the breakeven value of a released leader hour, and — valuing released hours at the leader's rate — the breakeven probability of a single USD 9,000 error incident. Solution. Status quo `4.0 × 15 = 60.0` leader hours `× 175 =` **USD 10,500.00**. Delegated: delegate `6.0 × 5 + 4.0 × 10 = 70.0` h `× 120 =` 8,400.00; oversight `1.5 × 5 + 0.5 × 10 = 12.5` h `× 175 =` 2,187.50; total **USD 10,587.50**. Incremental cost **USD 87.50**; released hours **47.5**. Breakeven value of a released leader hour `87.50/47.5 =` **USD 1.8421**, which is **1.053 %** of the leader's charge rate. Valuing the released hours at 175 gives `47.5 × 175 =` **USD 8,312.50**, a net **USD 8,225.00**, so the delegation fails only if the probability of a 9,000 incident exceeds `8,225.00/9,000 =` **91.39 %**. Common error: comparing only the two labour totals and concluding that delegation costs 87.50 more and is therefore worse — which ignores the 47.5 released hours that are the entire product being bought. The paired error is claiming the 8,312.50 without reallocating the released hours to anything, in which case none of the value exists.

Practitioner's toolkit — Domain 12

Adoption-ready artefacts; adapt headings to your organisation, then keep them stable.

Toolkit 12.T.1 — Leader's attention and span sheet

One page per leader, completed from **four weeks of observed data** and reviewed quarterly. Top block: the working-week assumption, then each recurring commitment with its hours — governance and reporting, external stakeholders, supplier and commercial, technical review, team ceremonies, observed interruption and escalation — the total, the discretionary residual, the personal reserve, and the resulting **people-work capacity** `R`. Second block: direct reports `n`; the stated **sufficiency convention** `T`; the per-report fixed load `e`; computed coaching time per report `(R − e(n − b))/n`; and the maximum sustainable span `(R + eb)/(T + e)`. Third block: the three largest recurring tasks consuming discretionary capacity, each with a delegation decision — delegate, retain with a reason, or eliminate — and, where delegated, the **released hours and their committed reallocation**. A span approved without this sheet is a commitment the arithmetic may not permit.

Toolkit 12.T.2 — Team structure and coordination sheet

One page per team or programme, completed at formation and whenever headcount changes by more than a stated threshold. Columns per candidate design: team size `n`; sub-team structure; required paths (computed, and separately the **needed** paths from the dependency map — the honest count is of required relationships, not the formula's maximum, exactly as Domain 4's interface register is built from need); link cost `c` with its basis; coordination hours; share of capacity; FTE consumed; and the same three columns at `c` × 0.5 and `c` × 1.6 as a stated sensitivity. Below: the double-membership count with its extra paths and FTE; each sub-team's external interface count against the dependency map (the test of whether the boundary is in the right place); and the named representative and deputy for each sub-team, with time allocated. A headcount increase proposed without this sheet should be refused.

Toolkit 12.T.3 — Difficult-conversation preparation and record

Two sides of one page, used for every performance, conduct or expectation conversation. **Before:** the factual account — dated, specific, verifiable observations only, each traceable to a source in the project record; the consequence in the project's terms with a figure where one exists; the three questions you will ask and genuinely want answered; the specific expectation, its measure and its date; the support you will provide; and the anticipated responses with your answers, rehearsed. **After, within 24 hours:** what was said, the other party's account, what was agreed, the review date, and their confirmation that the record is accurate. **Standing checks**, reported monthly: the interval between the first clear instance and the conversation, per case — the single metric that tells an organisation whether its leaders can afford to have these conversations; conversations held without a contemporaneous record; and expectations past their review date. Where a formal consequence is possible, the applicable process and the human-resources and legal functions are engaged before the conversation, and the jurisdiction's requirements govern.

Exam preparation — Domain 12

What is assessed. Leadership against management by what each supplies; the theory families with their honest uses and limits; emotional intelligence as four practised operations; **the attention audit and people-work capacity**; the motivation frameworks with their evidential status, and condition design over incentive design; team formation stages and the leader's task at each; **the arithmetic of team size — paths, coordination share, net capacity, marginal capacity and structured designs**; psychological safety, the safety–standards pairing and its countable signals; **turnover cost on the three-component build-up with the saved-pay exclusion**; **delegation economics and its breakevens**; **span of control and coaching capacity**; the structure and timing of a difficult conversation; performance management's generalisable obligations and jurisdictional limits; **working-hour overlap, round-trip latency and the fairness of an overlap window**; culture without stereotype; the ethical situations of delivery leadership; and the prohibitions on AI inference about individuals.

The calculations to be able to do under time pressure. `n(n − 1)/2` paths and the coordination share `c(n − 1)/(2h)`; net capacity `hn − c·n(n − 1)/2` and marginal capacity `h − c(n − 1)` with the size at which it reaches zero; the path count, share and released FTE for a pod design; the attention audit's totals and shares; coaching time per report `(R − e(n − b))/n` and the maximum sustainable span `(R + eb)/(T + e)`; the three-component turnover cost, in person-weeks and as a share of a post's annual value, plus the breakeven overtime premium against a delay; delegation cost comparison, released hours, the breakeven value of a released hour and the breakeven error probability; and working-hour overlaps in a common time reference with the latency, annual cost, saving and unsocial-hours allocation that follow.

The traps. Using `n²` or omitting the division by two in the path count, both of which double coordination (Exercise 12.1) · counting only one side of a communication link, which halves it (MCQ 12.2-A) · attributing all coordination hours to the leader (MCQ 12.2-A) · confusing the size at which marginal capacity reaches zero (81) with the size at which coordination consumes all capacity (161) (MCQ 12.2-B) · quoting a single link cost as measured rather than presenting a range (12.2.2) · counting a vacant post's saved pay as a turnover cost (MCQ 12.2-C, Exercise 12.3) · reporting a turnover total without the ramp-up and lost-productivity components, which are 60.2 % of it (12.2.4) · dividing a leader's whole discretionary budget by the report count and ignoring the growth of fixed load, which overstates the sustainable span by one (Exercise 12.2, MCQ 12.3-A) · confusing the zero-coaching span with the sustainable span (MCQ 12.3-A) · comparing only the labour totals in a delegation decision and missing the released hours (Exercise 12.5, MCQ 12.3-B) · claiming the value of released hours without reallocating them (12.3.1) · computing time-zone overlap from local clock times without converting to a common reference (MCQ 12.4-A, Exercise 12.4) · fixing one distributed pair without recomputing the others (MCQ 12.4-C, Exercise 12.4) · quoting an effect size for a leadership or culture intervention that the evidence does not support (12.2.1, 12.2.3, 12.2.4) · and using a national-culture average to predict an individual (MCQ 12.4-D).

How the domain connects. Domain 1 supplies the accountability principle, the benefits chain that makes the work's purpose truthful, and Meridian's cost of delay at which every latency here is priced. Domain 3 supplies the bounded envelope this domain's interruption arithmetic depends on, the `E[wait]` latency logic KA 12.4.1 reuses, the threshold economics KA 12.3.1 parallels rather than repeats, the decision-record discipline behind the conversation record, and the recovery structure 12.A.1 completes. Domain 4 supplies `n(n − 1)/2` and the layered-versus-mesh choice this domain applies to people, plus the interface re-verification cost that prices the delegation risk. Domain 6's resource levelling is the schedule form of the same capacity problem. Domain 7 supplies the blended rate and engineer-week every figure here is denominated in. Domain 9 supplies the countable psychological-safety signal — issues found versus reported — and the Auriga commissioning team this domain prices a resignation from. Domain 11 supplies the stakeholder and communication skills this domain assumes when the audience is outside the team. Domain 13's adaptive delivery depends on the safety and bounded discretion designed here. Domain 14 carries the systematic responsible-AI treatment whose people-specific prohibitions this domain states. And Domain 15 applies KA 12.2.2 and 12.A.2 at enterprise scale, where the coordination arithmetic is the dominant cost.

Domain 12 summary

Leadership in delivery supplies what a plan cannot: a truthful reason the work matters, interpretation under ambiguity, the conditions in which people tell the truth, and named bounded discretion. The theory families are lenses with honest uses and real limits — trait and personality approaches describe differences without predicting outcomes; situational diagnosis is the most useful discipline among them; emotional intelligence is four practised operations, not a score. **This domain refuses to quote effect sizes it cannot evidence, and finds instead the parts of the subject that are genuinely computable.**

The leader's attention is the binding constraint and is never baselined. Auriga's leader carries **37.0 hours** of recurring commitments in a 45.0-hour week — **82.2 %** — leaving **8.0 hours** of discretionary attention and **6.5 hours** of people-work capacity, **14.4 %** of the week, from which every coaching conversation, delegation and team-design decision is paid. The largest single line, **9.0 hours** of interruption, is **20.0 %** of the week and nobody decided it.

Team size is arithmetic. Communication paths grow as `n(n − 1)/2` (Domain 4, KA 4.2.3), and at a link cost of 0.5 hours of team time per week on 40-hour weeks the coordination share is the identity `(n − 1)/160`: **3.125 %** for Auriga's six, **6.875 %** for Meridian's original twelve, and **24.375 %** — **9.75 full-time people** — for the forty it became. Five pods of eight carry **150** paths instead of **780**, **4.6875 %** instead of 24.375 %, releasing **7.875 FTE** at no cost. Net capacity `hn − c·n(n − 1)/2` is **1,620 hours** at both 80 and 81 people, so the 81st member of a fully connected team adds exactly nothing — a model, not a law, whose conclusion (structure your teams) is robust across every plausible link cost while its specific ceiling is not. Meridian banked **USD 85,680** of realised acceleration against **USD 1,451,520** of theoretical released capacity, a ratio of **16.94**: the honest distance between capacity released and value realised.

Turnover is arithmetic. One resignation from Auriga's team of six costs **USD 66,845** — recruitment 26,612.50, ramp-up 18,810.00, lost productivity 21,422.50 — **12.79 engineer-weeks** and **26.65 %** of the post's annual charge-out value, of which recruitment is only **39.8 %** and the **28.1 %** that is ramp-up falls after the vacancy is filled and the problem thought closed. The vacant post's pay is **not** a cost: it is saved, and counting it inflates the figure to 82,520. The overtime cover that avoided a **1.0-week** delay worth **USD 45,000** cost **USD 15,675**, and would have been worth paying up to a **143.54 %** premium. That number sets the retention budget a leader may argue for; it does not establish that any intervention earns it.

Span of control is arithmetic. Coaching time per report is `(R − e(n − b))/n` — **65.00** minutes at six reports, **41.25** at eight, **21.82** at eleven — and against a stated 45-minute convention the maximum sustainable span is **7**, not the **8** the naive `6.5/n` curve gives, because the leader's own fixed load grows with the team; the way to widen a span is to raise `R`, and recovering 3.0 hours of interruption moves it from 7 to **10**. Delegation is arithmetic too: handing over Auriga's interface coordination costs **USD 430** more in labour and releases **34.0** leader hours, **5.23 weeks** of discretionary capacity, so it pays if a leader hour is worth more than **USD 12.65** — **6.836 %** of its own charge rate — and fails only if a 6,000-dollar error is more than **97.67 %** likely. Structurally Domain 3's threshold result, for the same reason.

Distance is arithmetic. Meridian's core team and clinical advisers, both working nine to five, overlap for **0.0 hours**, so each three-exchange clarification takes three working days and the 26 on the critical path cost **USD 222,768** a year; a **3.0-hour** window created by a two-hour and a one-hour shift saves **USD 148,512**, costs 2.0 hours of the supplier's overlap, and is bought with 3.0 hours a day of unsocial working whose allocation — 10/5, 15/0, or 7.5/7.5 rotated — is an ethical decision a leader must make explicitly. Culture varies in ways worth anticipating and not in ways that predict an individual, and the ethical obligations — reporting what is true, refusing a date you do not believe, protecting the team, stopping for safety, being fair in the decisions people watch — are the ones from which a team infers what a leader actually values.

The through-line: **the human capacity of a delivery organisation is a designed quantity with a computable cost.** Case study B is the domain in one sentence — a conversation worth USD 1,980 deferred fourteen weeks at a cost of USD 123,212.60, by a manager whose eleven reports left them 21.82 minutes each per week. Compute it, and what looked like a culture problem becomes a design decision — the only kind of problem an organisation reliably fixes.

13

DOMAIN 13

Agile, Adaptive and Hybrid Delivery

Group: Leading people and organisations (Domain 3 of 3 in Part Three — the part's closing domain). **Target:** ~70 pages. **Binds to:** the PCI Book Pattern Specification and the shared registries (`docs/books/registries/`). This domain closes Part Three by making adaptive delivery **arithmetic** rather than allegiance: it consumes Domain 3's governance latency formula, Domain 5's scope controls, Domain 6's flow-across-modes translation, Domain 10's risk-allocation frame and Domain 12's coordination overhead, and supplies the flow measures that Domains 14–16 report on. British English; USD (+SAR where useful, indicative `USD 1 ≈ SAR 3.75`).

Why this domain exists

Part Three has been about people: who is engaged (Domain 11), who leads and how a team is built (Domain 12). Both assumed a **way of working** — a cadence, a unit of delivery, a rhythm at which decisions arrive — and neither specified it. Part Two specified one: a network, a baseline, a variance. That specification is right for a great deal of project work and wrong for a great deal more, and the gap between the two is where this domain lives.

The gap is usually discussed as a preference, which is why the discussion goes nowhere. Practitioners declare for a method, defend it, and attribute failures to the other camp's contamination. The professional position is narrower and duller: **adaptive delivery is a set of mechanisms with measurable properties, and so is predictive delivery, and a leader's job is to know which properties each supplies and what each costs.** That is the domain's central claim. It has a consequence a candidate should expect to be tested on: almost every complaint made about agile delivery in large organisations — "the team keeps changing the date", "we can't get a commitment", "governance is constantly bypassed", "the contract doesn't fit" — is not a cultural complaint at all. Each is an arithmetic statement about throughput, work in progress, latency or commercial exposure, and each has a number attached that nobody has computed.

This domain computes them. KA 13.1 strips product ownership back to the decision right it is and prices the sequencing decision that prioritisation is — worth **USD 155,040** on Meridian's release, and routinely taken by whoever argues best. KA 13.2 builds the flow arithmetic and applies it to the commonest management instruction in delivery, start more work, which on Meridian buys nothing and costs **USD 180,379**; it then replaces the forecast date with a forecast range. KA 13.3 quantifies the mismatch Domain 3 named — a monthly committee governing a fortnightly cadence parks **20.0 %** of a team's work in progress and makes the affected items take **2.667 times** as long as the rest. KA 13.4 compares a capacity-based and a scope-based commercial model on the only test that discriminates between them — what a change costs the buyer — and closes on metrics that survive incentives and the anti-patterns that do not.

The through-line: **adaptive delivery is not a licence to stop measuring; it is a different set of things to measure, and every one of them is a number.**

Learning objectives. After this domain a candidate can: state the conditions under which adaptive delivery outperforms predictive delivery and the conditions under which it does not, in terms of feedback cost and requirement volatility rather than preference; specify product ownership as a decision right with a named holder, and apply the decidability test of Domain 3 to it; rank a backlog by **delay-cost density** and compute the delay cost of a chosen sequence, its saving against the intuitive ordering, and the breakeven estimate error that would change the ranking; state and apply **Little's Law** in both directions — deriving cycle time from work in progress and throughput, and deriving the throughput implied by a cycle-time change at fixed work in progress; compute what raising work in progress does to throughput, cycle time, flow efficiency and the delivery date, and price it; forecast from throughput history **as a range with a stated meaning**, adjusted for discovery arrivals, and test a committed date against the best rate the team has ever sustained; compute governance latency against iteration length using Domain 3's `E[wait] = M/2 + L`, express the mismatch as blocked work in progress and as a cycle-time decomposition, and price the remedy; design hybrid delivery with a named boundary, two honest control regimes and a translation rule for reporting; compare capacity-based and scope-based commercial models on buyer exposure under a change, including the breakeven number of changes; select flow metrics that resist gaming and explain why velocity is a planning input and never a performance measure; diagnose the standard anti-patterns with their arithmetic signature; and govern AI-assisted backlog work and forecasting without letting a model's output become a commitment.

The master programme. Meridian Care Records continues from Domains 1–4 and 11–12: the clinical-records rollout to **40 clinics**, approved cost **USD 2,400,000**, benefit **USD 685,440** a year at the realistic 70 % adoption (against **USD 979,200** at full potential), and the **cost of delay of USD 14,280 per week** derived in Domain 1 from 28 adopting clinics at 6 hours a week and USD 85 an hour. Meridian's shape is exactly the shape this domain is about: an **iteratively developed records application**, a **sequentially rolled-out** clinic programme with immovable estate and training dependencies, and a regulatory approval that is a genuine gate (Domain 3, KA 3.1.3). The application build carries a **240-item** release backlog delivered by a team whose measured throughput is **6 items a week**, governed by a steering committee whose expected decision latency Domain 3 computed at **4.0 weeks** from `M = 4`, `L = 2`. Those five numbers — 240, 6, 4.0, 14,280 and the 2-week iteration — generate almost every result in this domain. Project Auriga, the 25-week control-systems upgrade of Domains 6–8, reappears in the exercises where a physical commissioning queue makes the same flow arithmetic behave differently.

KNOWLEDGE AREA 13.1

Agile principles and product ownership

Topics: 13.1.1 what adaptive delivery actually is · 13.1.2 product ownership as a decision right · 13.1.3 prioritisation as a priced sequencing decision · 13.1.4 the increment and the definition of done.

13.1.1 What adaptive delivery actually is

Definition. Adaptive delivery is a delivery mode in which **capacity and cadence are fixed and scope is variable**, direction is corrected from working output at short intervals, and commitment to detail is deliberately deferred until the cost of deciding is lowest. Predictive delivery fixes scope and varies capacity and time to achieve it. Neither is a philosophy; each is a choice about **which variable absorbs uncertainty**.

That framing does the work that value statements cannot. The historical literature of agile methods — the collaborative statement of values published by a group of software practitioners in 2001, and the named frameworks that followed it (Scrum, Kanban, DSDM, and the various scaling frameworks) — is worth reading and is not reproduced here; what a leader needs is the decision rule underneath it. And the decision rule is economic. Adaptive delivery outperforms predictive delivery when three conditions hold together, and a leader should be able to test each:

1. **Requirements are genuinely volatile or genuinely unknown.** Not "poorly documented" — unknowable in advance, because they depend on what users do with something that does not exist yet. Where requirements are stable and knowable, deferring commitment buys nothing and costs coordination.
2. **Feedback is cheap and fast.** An increment can be put in front of someone who can judge it, within the iteration, at a cost small relative to the work. Where feedback requires a commissioned plant, a regulatory submission or a construction season, the iteration cannot close its loop and the method loses its engine (13.A.1).
3. **The cost of a wrong decision falls with delay, or the cost of reversal is low.** Deferring commitment is only valuable if information arrives; if the decision is irreversible and the information will not improve, deferring it is procrastination with a vocabulary.

What adaptive delivery does not do. It does not remove the baseline; it moves it onto capacity, cadence and the value envelope (Domain 5, KA 5.A.1). It does not remove governance; it changes the governance question from variance to continuation (Domain 3, KA 3.1.3). It does not remove the date; it changes what an honest date looks like, from a point to a range with a stated basis (13.2.4). And it does not make a project cheaper by itself: the case for it rests on **not building the wrong thing**, which is a value argument — so a programme that adopts adaptive delivery while retaining a fixed scope has kept the cost and discarded the benefit.

The honest sentence. Every method is a rule about which of scope, time and cost floats, and the useful professional habit is to say out loud which is floating on this work: "capacity and date fixed, scope floats" (adaptive); "scope and date fixed, cost floats" (a crash decision, Domain 6 KA 6.4.2); "scope and cost fixed, date floats" (most infrastructure). An organisation that cannot say the sentence has not chosen a method; it has chosen to be surprised.

13.1.2 Product ownership as a decision right

Definition. The product owner is the single individual accountable for the **order** of the backlog — for deciding what is built next, and therefore what is not built next — within a value envelope set by the sponsor. It is a decision right, not a facilitation role, and Domain 3's decidability test (KA 3.1.1) applies to it unchanged: exactly one accountable holder, with sufficient authority, or the decision is not made but taken by whoever is present.

The role fails in four recognisable ways, and each has a test a leader can run in an afternoon.

The proxy owner must consult a committee before ordering the backlog, so the effective authority and the latency are the committee's — the whole subject of KA 13.3.2. Test: count the ordering decisions made alone in the last quarter, and the elapsed time on the rest.

The unfunded owner holds authority over order but not over the value envelope, so they can sequence but not decline, and backlogs grow monotonically because refusal has no owner. Test: count items removed in the last quarter. A backlog from which nothing is ever removed is a wish list with a burndown chart.

The absent owner is available for ceremonies and unavailable between them, so the team decides by default and the ordering drifts to whatever is technically convenient. Test: the team's median wait for a product-owner decision — a number worth putting on the wall.

The specifier owner writes solutions rather than ordering outcomes, becoming a bottleneck on design and destroying the team's ability to propose. Test: the share of items stating a required outcome and an acceptance criterion rather than an implementation.

What the role must be able to do. Four capabilities, and the absence of any one is a design defect rather than a personal failing: state the value of an item in the terms of the benefits map (Domain 2), so that ordering is arguable on evidence; decline an item and record the refusal; accept or reject an increment against pre-agreed criteria (Domain 5, KA 5.4.2); and escalate a decision that exceeds their envelope, within a stated latency (Domain 3, KA 3.3.3). Meridian's product owner holds an envelope of **USD 25,000** per change — the threshold Domain 3's KA 3.2.3 arithmetic recommended — and the 15 % of items that exceed it are what KA 13.3.2 prices.

13.1.3 Prioritisation as a priced sequencing decision

The definition and the point. Prioritisation is not ranking by importance; it is choosing a **sequence**, and a sequence has a computable cost because value forgone accrues per unit of time. The instrument is **delay-cost density**: the cost of delay of an item divided by the effort required to deliver it.

```
delay-cost density = cost of delay per week ÷ effort in team-weeks
```

Sequencing in decreasing order of that ratio minimises the total cost of delay across a set of items delivered one at a time by a single team. This is not a framework opinion but a classical result in single-machine scheduling theory — the weighted-shortest-processing-time ordering, which minimises total weighted completion time — and ratio rules of this shape appear under various names in the scaling frameworks. What matters for a leader is that the rule is derivable, so its assumptions are visible: one delivery resource, items independent, each item releasable on completion, and effort and cost of delay both estimated. Every one of those assumptions is a place the rule can fail, and 13.1.3 closes by pricing the most important failure.

WORKED EXAMPLE

Worked example 13.1.3 — sequencing Meridian's four release epics.

1. **Setup.** Meridian's 240-item release comprises four epics. Domain 2's benefits map apportions all six hours per clinic per week of clinical time saved across the four capabilities, so the four epics between them carry the whole of the programme's **USD 14,280 per week** cost of delay (Domain 1). One team delivers them sequentially; total effort is **34 team-weeks**.

Epic	Cost of delay (USD/week)	Share of programme	Effort (team-weeks)	Density (USD/week per team-week)
E2 clinical notes	6,120	42.9 %	14	437.14
E1 appointment booking	4,080	28.6 %	6	680.00
E3 regulatory reporting	2,720	19.0 %	3	906.67
E4 patient portal	1,360	9.5 %	11	123.64

1. **Formula.** Density = cost of delay ÷ effort. Total delay cost of a sequence = `Σ (cost of delay of each epic × the week in which that epic completes)`.
2. **Substitution.** Density order is E3, E1, E2, E4, completing in weeks 3, 9, 23 and 34: `2,720 × 3 + 4,080 × 9 + 6,120 × 23 + 1,360 × 34`. The intuitive order — largest cost of delay first — is E2, E1, E3, E4, completing in weeks 14, 20, 23 and 34: `6,120 × 14 + 4,080 × 20 + 2,720 × 23 + 1,360 × 34`.
3. **Result.** Density order **USD 231,880**. Cost-of-delay-first order **USD 276,080**. The density order saves **USD 44,200**. The worst available order (E4, E2, E1, E3 — lowest density first) costs **USD 386,920**, so the spread between the best and worst sequences of the same scope, same effort and same team is **USD 155,040**.
4. **Interpretation.** The first thing to take from this is that sequencing is a **USD 155,040 decision**, and in most organisations it is taken in a room by whoever advocates most effectively, with no number in front of anyone. The second is that the intuitive rule — do the biggest thing first — is wrong by **USD 44,200**, and wrong for a reason worth internalising: E2 carries the largest benefit but also fourteen weeks of effort, during which the three smaller epics each accrue their own cost of delay. Doing the cheap, high-value work first is not opportunism; it is arithmetic. Third, the ranking is **robust to moderate estimation error**, which is what makes it usable with imperfect inputs: E2 would have to carry a cost of delay above `680 × 14 =` **USD 9,520** per week — a **55.6 %** increase on its estimate — before it belonged at the front, so a leader can defend the order without defending the third decimal place. Fourth, and most important, the entire **USD 155,040** is created by **releasability**. If nothing can go live until all four epics are complete, every epic completes at week 34 and the total delay cost is `14,280 × 34 =` **USD 485,520** regardless of order — so incremental release is worth `485,520 − 231,880 =` **USD 253,640** on this release, which is a larger number than the sequencing decision it enables and is usually treated as a technical detail. Two cautions. The rule assumes independence, and where E1 must precede E3 technically, the dependency binds and the sequence is constrained — price the constraint rather than ignoring the rule. And a cost of delay estimated from advocacy rather than from the benefits map will produce a confidently wrong order; the density is only as good as its numerator, which is why Domain 2's value attribution is a prerequisite and not a nicety.

13.1.4 The increment and the definition of done

Definition. An increment is a slice of the product that is **complete against a pre-agreed standard** — the definition of done — and therefore capable of being judged, released or rejected on its own. The definition of done is that standard, written once, applied unchanged to every item.

The increment is the unit that makes adaptive governance possible, because it is the only thing a governance body can inspect that is not a claim. A percentage complete is a claim; a working, tested, documented slice of service is evidence. This is why Domain 3's KA 3.1.3 puts the increment review at the centre of iterative governance and why the definition of done is a **governance artefact** rather than a team convention: it defines what the word "done" means in every report the programme issues, and a definition that is quietly relaxed under schedule pressure corrupts every subsequent number.

What belongs in it. The standard should be short, binary and testable: the acceptance criteria met (Domain 5, KA 5.4.2); the tests written and passing; the security and privacy checks completed, which for Meridian's clinical records is a regulatory obligation and not a preference; the documentation and operational handover material produced (Domain 16); no known defects above an agreed severity; and the item demonstrable in an environment representative of production. Each is either true or it is not.

The two failure modes. The **relaxed definition** — items counted as done with tests or documentation deferred — creates a debt that Domain 9's KA 9.4 quantifies as a rework share of capacity, degrading throughput at the multiplier `1/(1 − r)` established there. The **inflated definition** — a standard so demanding that no item ever meets it — produces a team with a full board and no throughput, which the flow measures of KA 13.2 detect immediately as a cycle time rising with no change in work in progress.

AI in this KA

Where it earns its place. Splitting a large item into candidate thin slices that each satisfy the definition of done, for the team to accept, amend or reject — a generative task with cheap verification, because a bad slice is obvious to the people who would build it. Drafting acceptance criteria from an outcome statement, then being checked against Domain 5's criteria tests. Computing delay-cost density and the total delay cost of alternative sequences across a large backlog, with the sensitivity analysis of 13.1.3 swept automatically — deterministic arithmetic over many combinations, which is exactly what a machine is for and no human has time to do. Detecting backlog items that duplicate one another or that carry no stated value, which is a text-comparison task humans perform badly at scale.

Where it must not go. It must not set the order. Ordering a backlog allocates the organisation's capacity between competing claims on it, and that is a decision right held by a named accountable person under 13.1.2; a model that produces an order has produced a recommendation, and a product owner who adopts it unexamined has vacated the role rather than automated it. It must not estimate a cost of delay from plausibility: asked for one, a model will produce a confident number with no provenance, and because the density ranking is proportional to that numerator the fabrication propagates straight into a sequencing decision worth six figures. And it must not write the definition of done, which encodes the organisation's quality and regulatory obligations.

Verification, concretely. Every AI-produced sequence is re-derived by hand for the top five items — five divisions — and the cost-of-delay inputs are traced to a benefits-map line with an owner, not to a model output. Every AI-drafted acceptance criterion is tested against a real example and a real counter-example before it is accepted. Where a model has split an item, the team confirms that each slice independently satisfies the definition of done, because a slice that cannot be released alone has re-created the big-bang release the arithmetic above shows to be worth **USD 253,640** to avoid.

KEY TERMS — KA 13.1

Term	Meaning
Adaptive delivery	A delivery mode fixing capacity and cadence and varying scope, correcting direction from working output at short intervals.
Hybrid delivery	A deliberate combination of predictive and adaptive methods under one governance frame, with a named boundary between them.
Product owner	The single individual accountable for the order of the backlog within a value envelope set by the sponsor.
Value envelope	The bounded authority within which a product owner may commit capacity without escalation.
Delay-cost density	Cost of delay per week ÷ effort in team-weeks; sequencing in decreasing order minimises total delay cost.
Increment	A slice of product complete against the definition of done and therefore judgeable, releasable or rejectable alone.
Definition of done	The pre-agreed, binary, testable standard that gives the word "done" one meaning across every report.
Releasability	The property that an increment can go live alone; it is what converts sequencing into value.

SAMPLE MCQS — KA 13.1

MCQ 13.1-A `[13.1.3 · Application]` Four epics carry costs of delay of 6,120, 4,080, 2,720 and 1,360 per week and efforts of 14, 6, 3 and 11 team-weeks respectively. The sequence that minimises total delay cost is: - A. 6,120 first, then 4,080, 2,720, 1,360 — largest cost of delay first - B. 2,720 first, then 4,080, 6,120, 1,360 — decreasing cost of delay per team-week - C. 2,720 first, then 4,080, 1,360, 6,120 — shortest effort first - D. 1,360 first, then 6,120, 4,080, 2,720 — lowest density first

Rationale: Densities are 906.67, 680.00, 437.14 and 123.64, so the order is 2,720 → 4,080 → 6,120 → 1,360, costing **231,880** (13.1.3). A is the intuitive largest-first error, costing **276,080** — 44,200 worse. C ranks on effort alone, which promotes the low-value 1,360 epic ahead of the 6,120 one and costs **280,160**. D reverses the density rule and is the worst available order at **386,920**.

MCQ 13.1-B `[13.1.3 · Evaluation]` For the same four epics, if no epic can be released until all four are complete at week 34, the total delay cost is: - A. USD 231,880, unchanged — sequencing still helps - B. USD 485,520, and sequencing is worth nothing - C. USD 155,040 - D. USD 253,640

Rationale: With a single release every epic completes at week 34, so the cost is `14,280 × 34 =` **485,520** whatever the order — the whole value of sequencing comes from releasability (13.1.3). C is the best-to-worst sequencing spread; D is the value of releasability itself, `485,520 − 231,880`.

MCQ 13.1-C `[13.1.2 · Analysis]` A named product owner must obtain a steering committee's agreement before changing the backlog order. The most accurate description is that: - A. the arrangement strengthens governance by adding scrutiny - B. the effective decision right, and therefore the latency, belongs to the committee - C. the product owner remains accountable and the arrangement is sound - D. the team should escalate less often

Rationale: Authority is where the binding agreement sits, so the product owner is a proxy and the committee's latency applies to every ordering decision (13.1.2, and Domain 3 KA 3.1.1's decidability test). A confuses scrutiny with authority; C mistakes the title for the right.

MCQ 13.1-D `[13.1.1 · Analysis]` A programme adopts adaptive delivery but keeps its scope fixed by contract. The predictable consequence is that: - A. delivery becomes faster because iterations are short - B. it retains the coordination cost of the method and forgoes its benefit, which comes from not building the wrong thing - C. quality improves because increments are tested - D. the cost of delay falls

Rationale: The economic case for adaptive delivery rests on varying scope in response to feedback; fixing scope removes the mechanism and leaves the overhead (13.1.1).

MCQ 13.1-E `[13.1.4 · Comprehension]` A team, under schedule pressure, begins counting items as done with tests deferred. The measure that detects this soonest is: - A. velocity, which will fall - B. the count of items completed per iteration, which will rise - C. the rework share of capacity in later iterations, which degrades throughput at `1/(1 − r)` - D. stakeholder satisfaction

Rationale: A relaxed definition of done initially raises reported completion and later shows as rework consuming capacity, at Domain 9 KA 9.2.3's multiplier (13.1.4). A and B move the wrong way; D is a lagging and non-specific signal.

SELF-CHECK — KA 13.1

1. State the three conditions under which adaptive delivery outperforms predictive delivery. — Requirements genuinely volatile or unknowable; feedback cheap and fast enough to close inside the iteration; and deferral genuinely valuable because information arrives or reversal is cheap.
2. Why does ranking by cost of delay alone give the wrong sequence? — Because a large, slow item blocks several smaller ones that are each accruing their own cost of delay; the correct ranking is cost of delay per unit of effort.
3. What single property creates the value of sequencing, and what happens without it? — Releasability. Without it every item completes at the end and the total delay cost is the same for every order — on Meridian, **USD 485,520** instead of **USD 231,880**.

KNOWLEDGE AREA 13.2

Backlogs, iteration planning, flow and Kanban

Topics: 13.2.1 the backlog as an instrument · 13.2.2 iteration planning and the commitment fallacy · 13.2.3 flow: throughput, work in progress and cycle time · 13.2.4 forecasting from throughput history.

13.2.1 The backlog as an instrument

Definition. A backlog is an **ordered** list of items, each **attributed** to a benefit, each **sized**, and each carrying an acceptance criterion. Remove any one of those four properties and it stops being an instrument: an unordered backlog is a wish list, an unattributed one is prioritised on advocacy (Domain 5, KA 5.3.1), an unsized one cannot be forecast, and one without criteria cannot be accepted.

Three disciplines keep it usable, and each is cheap.

Refinement is a capacity cost and should be budgeted as one. Items near the front must be understood well enough to start; items far back must not be, because elaborating them wastes effort on work that may never be built and creates a sunk-cost pull towards building it — Domain 6's rolling-wave principle, applied to a backlog: elaborate to an explicit horizon.

The value-envelope reconciliation is the creep control. A requirement count means nothing where the count is supposed to move, so the adaptive equivalent of Domain 5's creep test is the reconciliation Domain 5 KA 5.A.1 specifies — attributed benefit of delivered items against the business case's commitment, period by period. It detects the failure adaptive programmes actually suffer: delivering steadily against a backlog that has drifted from the benefits it was funded to produce. Case study B is that failure, costed.

Discovery arrivals must be counted. Items found necessary during build are the method working, not scope creep. The defect is failing to count them, because an uncounted arrival rate makes every forecast optimistic by exactly its magnitude. Meridian's measured rate is **1.5 items per week**, and KA 13.2.4 shows what ignoring it does to a date.

13.2.2 Iteration planning and the commitment fallacy

Iteration planning answers one question — what will the team take on in the next iteration — and it is routinely corrupted by being asked to answer a different one: what will the team promise. The difference matters because the two have different arithmetic.

Capacity is an observation, not an aspiration. Meridian's team completes **6 items a week**, so a 2-week iteration has a capacity of **12 items**. That is a measurement, and it is the only defensible input to a plan.

The commitment fallacy. Committing to the mean is committing to fail about half the time. Meridian's last twelve weeks, paired into six iterations, produced **12, 10, 13, 13, 13 and 11** items. The mean is exactly **12.00**, and **2 of the 6 iterations — 33.3 %** — delivered fewer than 12. A team held to the mean therefore misses its commitment in a third of iterations while performing exactly to its own average, and the organisational response is invariably to question the team rather than the arithmetic. Committing instead to the **lowest observed total, 10 items**, would have been met in every observed iteration, at the cost of forecasting less throughput than the team actually has. The professional resolution is to stop conflating the two numbers: **plan with the mean and commit with the low end**, and report both. A commitment is a statement about reliability; a forecast is a statement about expectation; and an organisation that has one number is using it for both purposes and being misled at least once.

What the iteration boundary is actually for. Three things, none of which is a status report: a **re-ordering opportunity**, the point at which new information legitimately changes the sequence — the mechanism that makes KA 13.4.1's commercial comparison come out as it does; an **acceptance point**, where the increment meets the definition of done or does not; and a **continuation decision** in Domain 3 KA 3.1.3's sense — is the next iteration's cost worth its expected value? A programme whose iteration boundaries produce reports rather than decisions has bought the cadence and left the value on the table.

13.2.3 Flow: throughput, work in progress and cycle time

This is the quantitative core of the domain, and it rests on one relationship.

Definitions. Throughput (`T`) is completed items per unit of time. **Work in progress** (`W`) is the number of items started and not finished. **Cycle time** (`C`) is the elapsed time from an item starting to its finishing. For a system at a steady state, **Little's Law** relates the three:

```
W = T × C        equivalently        C = W / T        and        T = W / C
```

Little's Law is a theorem, not a heuristic — it holds for any queueing system in steady state, independent of the arrival distribution, the service distribution or the queue discipline. Two consequences follow that make it the most useful single relationship in delivery flow. First, **cycle time is a consequence of a management decision, not an attribute of the work**: at a given throughput, the elapsed time an item takes is determined by how much work is in progress alongside it. Second, because the relationship can be read in either direction, a reduction in cycle time at unchanged work in progress is **arithmetically identical** to an increase in throughput — which is the identity KA 13.3.2 uses to price a governance change.

Flow management, and where Kanban fits. A flow-managed system — the approach the Kanban method names — does not commit to an iteration's contents at all. Work is made visible, work in progress is explicitly limited at each stage, new work is **pulled** when capacity frees rather than pushed on a cadence, and the system is managed by the three measures above rather than by iteration commitments. It is therefore the natural regime wherever arrivals are genuinely unpredictable — support, operations, regulatory response — and it can be combined with an iteration cadence, in which case the cadence serves review and release while the work-in-progress limit governs flow. The arithmetic below applies to both: Little's Law does not care whether the work was committed or pulled.

Meridian's team at its working limit. With `W = 18` items and `T = 6` items a week, `C = 18 / 6 =` **3.00 weeks**. If a typical item requires about **0.9 weeks** of active work, the **flow efficiency** — active time as a share of elapsed time — is `0.9 / 3.00 =` **30.0 %**, meaning seven-tenths of an item's life is spent waiting. That is not unusual and it is the number most worth knowing, because it says where the improvement is: not in working faster, but in waiting less.

WORKED EXAMPLE

Worked example 13.2.3 — the instruction to start more work.

1. **Setup.** Three clinics are waiting on the reporting module. Meridian's steering committee instructs the team to start it in parallel with existing work, raising work in progress from **18** to **30** items. Nothing else changes: same nine people, same cadence, same definition of done. The team's product owner objects and cannot price the objection. Assume the stated switching model, calibrated from the team's own records and to be recalibrated in any other setting: at or below the capacity point `W* = 18` the team is the constraint, so throughput is `6W/18` and cycle time sits on its floor; above it, **each additional concurrent item consumes 2 % of the team's productive capacity** in switching, handover and re-familiarisation, so `T(W) = 6 × (1 − 0.02 × (W − 18))`. Meridian's cost of delay is **USD 14,280 per week** (Domain 1) and the remaining release backlog is **240 items**.
2. **Formula.** `T(W)` as stated; `C = W / T` (Little's Law); time to clear the backlog = backlog ÷ throughput; cost = additional weeks × cost of delay.
3. **Substitution.** `T(30) = 6 × (1 − 0.02 × 12) = 6 × 0.76`. `C(30) = 30 / 4.56`. `240 / 4.56` against `240 / 6.00`.
4. **Result.** Throughput falls from **6.00** to **4.56** items a week (**−24.0 %**). Cycle time rises from **3.00** to **6.58 weeks** (**+119.3 %**, a multiple of **2.193**). Flow efficiency falls from 30.0 % to **13.7 %**. The 240-item backlog goes from **40.00 weeks** to **52.63 weeks**, and the extra **12.63 weeks** cost **USD 180,379**.
5. **Interpretation.** The headline is the one every delivery leader should be able to say in a sentence: **raising work in progress by 66.7 % bought 0 % more throughput and more than doubled the time any individual item takes.** Nothing arrived sooner. The reporting module the committee wanted accelerated arrives later than it would have if the team had finished its current work first, because it is now competing with that work for the same capacity — and so is everything else. The arithmetic also explains why the instruction is so persistently attractive: starting work is **visible and immediate**, while the cost lands weeks later, distributed across every item, and is attributed to the team's pace rather than to the decision. Three professional cautions, each of which a reviewer should press on. The 2 % switching coefficient is a **stated assumption calibrated locally**, not a law; the shape of the result — throughput flat then falling, cycle time rising more than proportionally — is robust to the coefficient, but the 180,379 is not, and a paper quoting it must quote the assumption beside it. The model is linear and therefore predicts zero throughput at `W = 68`, which is obviously false and is a useful reminder that the relationship is only calibrated near the observed range: **calibrate, do not extrapolate.** And Little's Law itself assumes a steady state, so it describes a team over several iterations and not a single week; a leader who applies it to one week's numbers will find it noisy and conclude wrongly that it does not work. The practical control that follows is a **work-in-progress limit** — an explicit number, visible, changed only deliberately — and the case for it is not tidiness. It is USD 180,379.

Fig 13.2.1 — Little's Law and the cost of starting more work

13.2.4 Forecasting from throughput history

The principle. A forecast produced from throughput history is a **range**, and the range means something precise that must be stated: it is the span of completion times that the team's **own recent behaviour** would have produced, on the assumptions that the future resembles the sampled past, that item sizes are drawn from the same distribution, that the team composition is unchanged and that the arrival of new work continues at the observed rate. It is **not** a confidence interval, it carries no probability, and presenting it as though it did is a misrepresentation that will be discovered.

Why a range and not a date. A single date drawn from an average is wrong roughly half the time by construction (13.2.2), and its wrongness is not symmetric in its consequences: the organisation plans irreversible things — training, clinic closures, communications — against it. A range with a stated basis lets those things be planned against the slow end and celebrated at the fast end, which is the only sequence that does not require an apology.

WORKED EXAMPLE

Worked example 13.2.4 — Meridian's release date, honestly.

1. **Setup.** Twelve weeks of completions: **5, 7, 4, 6, 8, 5, 7, 6, 5, 8, 6, 5** — total **72**, mean **6.00** items a week. Remaining release backlog **240 items**. Measured discovery arrivals **1.5 items a week**. The programme plan carries a committed date **34 weeks** away. Cost of delay **USD 14,280** per week.
2. **Formula.** Smooth the history into **four-week rolling totals** to remove single-week noise, and take the observed minimum and maximum of those as the sustained-rate bounds. Net drain = gross throughput − arrival rate. Forecast = remaining backlog ÷ net drain. Required rate = remaining backlog ÷ target weeks, plus the arrival rate.
3. **Substitution.** Rolling four-week totals: **22, 25, 23, 26, 26, 23, 26, 25, 24**, i.e. sustained rates of **5.50** to **6.50** items a week with a median of **6.25**. Net drains `6.50 − 1.5 = 5.00` and `5.50 − 1.5 = 4.00`. Forecasts `240 / 5.00` and `240 / 4.00`. Naive figure `240 / 6.00`. Required rate `240 / 34 + 1.5`.
4. **Result.** The naive forecast — mean rate, arrivals ignored — is **40.00 weeks**. At the mean rate net of arrivals it is **53.33 weeks**, so ignoring discovery understates the date by **13.33 weeks** and **USD 190,400**. The honest range is **48.00 to 60.00 weeks**. The plan's 34 weeks requires a net drain of **7.06** items a week and therefore a gross throughput of **8.56** items a week — **31.7 %** above the highest four-week rate the team has ever sustained and **42.6 %** above its twelve-week mean. Against the fast end of the range the plan is **14.00 weeks** and **USD 199,920** optimistic; against the slow end, **26.00 weeks** and **USD 371,280**.
5. **Interpretation.** The decisive sentence is not "the plan is optimistic" but **"the plan requires a throughput the team has never achieved, by 31.7 %, in its best four weeks on record"** — and that sentence changes a conversation, because it is not an opinion and it does not accuse anyone. Note the structure of the error, which is general: two independent optimisms compound. Using the mean rather than a range hides a **12-week** spread; ignoring arrivals adds **13.33 weeks** and **USD 190,400** on its own, and it is the more damaging of the two because it is invisible — the arrival rate appears in no burndown chart, and a team whose gross throughput is perfectly stable will still miss a date built on it. Three cautions belong in any paper carrying this range. The range is **empirical, not probabilistic**: it says what the last twelve weeks would have produced, and if the remaining items are systematically larger than the delivered ones the range is optimistic (KA 13.3.4 quantifies exactly that). Four-week smoothing is a **stated choice**; a two-week window widens the range and a six-week window narrows it, so the window must be declared and held constant rather than selected after seeing the answer. And a range is only honest if the organisation is allowed to receive one: where the culture demands a date, the range will be reduced to its midpoint by someone, and the professional protection is to publish the required rate — 8.56 items a week — alongside it, because that number cannot be averaged away.

What to do with the answer. The 34-week date is not recoverable by exhortation, and the arithmetic says so precisely: even the governance remedy of KA 13.3.2, which lifts gross throughput to **7.06** items a week, leaves **43.17 weeks** once arrivals are subtracted. Meeting 34 weeks would require arrivals of zero, which for a clinical records build is not a credible commitment. A **40-week** forecast, by contrast, is achievable and can be stated as a conditional undertaking: it requires the governance remedy and discovery arrivals held at or below **1.06 items a week**, a **29.4 %** reduction on the observed rate, with a named owner for each condition. That is what a defensible re-plan looks like — not a new date, but a date with its two conditions attached and both of them measurable weekly.

AI in this KA

Where it earns its place. Computing flow measures from a work-item history and flagging the anomalies a human eye misses — items with cycle times several times the median, items that have been started and abandoned, and the aged work in progress that no report shows. Sweeping the throughput history across window lengths to show how the range depends on the smoothing choice, which is the honest way to present 13.2.4's stated choice. Running the arrival-rate arithmetic across scenarios. Detecting a relaxed definition of done from the pattern of items reopened after acceptance — a correlation search over a large log, which is genuinely beyond manual capacity.

Where it must not go. It must not produce the forecast that goes to the board without the range's meaning attached, and this is the specific and serious hazard in this KA: a model asked "when will this finish?" will answer with a date, in confident prose, and that date will be quoted. The failure mode is not inaccuracy — the arithmetic may be perfect — it is the **silent loss of the basis**, so a range with four stated assumptions becomes a number with none. It must not be given a target date and asked to justify it, which is the same fabrication risk in a more flattering costume. And it must not size items, because sizing is the team's judgement about work it will do and a model's size estimate carries the authority of a measurement with none of the substance.

Verification, concretely. Every forecast that leaves the team carries its window length, its arrival rate, its assumption list and the required rate implied by any date it is compared against — and the arithmetic behind all four is reproducible on one page. Little's Law results are checked in both directions (`C = W/T` and `T = W/C`) as a consistency test, since the two must agree. And any model-supplied flow anomaly is confirmed against the underlying items before it is raised, because a false positive here accuses a person.

KEY TERMS — KA 13.2

Term	Meaning
Throughput (`T`)	Completed items per unit of time — a measurement, never an aspiration.
Work in progress (`W`)	Items started and not finished; a managed quantity with an explicit limit.
Cycle time (`C`)	Elapsed time from an item starting to finishing; a consequence of `W` at a given `T`.
Little's Law	`W = T × C` for a system in steady state, independent of arrival and service distributions.
Flow efficiency	Active work time as a share of cycle time; typically low, and where the improvement lies.
Work-in-progress limit	An explicit cap on `W`, changed only deliberately; the control the flow arithmetic justifies.
Discovery arrivals	New items found necessary during build; subtracted from throughput to give the net drain.
Net drain	Gross throughput minus arrival rate — the rate at which a backlog actually shrinks.
Range forecast	Backlog ÷ the observed high and low sustained net drains, with its assumptions stated.
Required rate	Backlog ÷ target weeks, plus arrivals — the throughput a committed date presupposes.

SAMPLE MCQS — KA 13.2

MCQ 13.2-A `[13.2.3 · Application]` A team completes 6 items a week with 18 items in progress. Its average cycle time is: - A. 0.33 weeks - B. 3.00 weeks - C. 6.00 weeks - D. 108 weeks

Rationale: `C = W/T = 18/6 = 3.00` weeks (13.2.3). A inverts the ratio; C reads the throughput as a duration; D multiplies instead of dividing.

MCQ 13.2-B `[13.2.3 · Analysis]` The same team is instructed to raise work in progress from 18 to 30 items. Each concurrent item above 18 costs 2 % of capacity. Throughput and cycle time become: - A. 6.00 items a week and 5.00 weeks - B. 4.56 items a week and 6.58 weeks - C. 4.56 items a week and 3.00 weeks - D. 7.20 items a week and 4.17 weeks

Rationale: `T = 6 × (1 − 0.24) = 4.56` and `C = 30/4.56 = 6.58` weeks (13.2.3). A is the common error of holding throughput constant, giving `30/6 = 5.00`; C forgets that cycle time depends on both; D assumes work in progress raises throughput.

MCQ 13.2-C `[13.2.4 · Application]` A 240-item backlog, sustained rates of 5.5 to 6.5 items a week, and measured discovery arrivals of 1.5 items a week. The honest forecast range is: - A. 36.9 to 43.6 weeks - B. 48.0 to 60.0 weeks - C. 40.0 weeks - D. 24.0 to 30.0 iterations, reported as weeks

Rationale: Net drains are `6.5 − 1.5 = 5.0` and `5.5 − 1.5 = 4.0`, giving `240/5.0 = 48.0` and `240/4.0 = 60.0` (13.2.4). A ignores arrivals — the single most common forecasting error here, and it understates the date by 11 weeks. C is the naive mean-rate point estimate. D is the correct range expressed in two-week iterations and mislabelled as weeks, which halves it.

MCQ 13.2-D `[13.2.4 · Evaluation]` For the same team, a plan commits to 34 weeks. The most useful single statement to put in front of the sponsor is that the plan requires: - A. more effort from the team - B. a gross throughput of 8.56 items a week, 31.7 % above the team's best sustained four-week rate - C. a throughput of 7.06 items a week - D. the backlog to be reduced

Rationale: `240/34 = 7.06` net, plus 1.5 arrivals gives **8.56** gross, which is 31.7 % above the 6.5 best observed rate (13.2.4). C omits arrivals and so understates the requirement; A and D are responses, not the diagnosis.

MCQ 13.2-E `[13.2.2 · Analysis]` Six iterations delivered 12, 10, 13, 13, 13 and 11 items, a mean of exactly 12. A team committed to 12 items an iteration will: - A. meet the commitment, since 12 is the mean - B. miss it in 33.3 % of iterations while performing exactly to its own average - C. miss it in 50 % of iterations - D. exceed it in every iteration

Rationale: Two of the six totals are below 12, so the commitment fails a third of the time at unchanged performance (13.2.2). C assumes a symmetric distribution the data does not have; the resolution is to plan with the mean and commit with the low end of 10.

MCQ 13.2-F `[13.2.3 · Comprehension]` A team's cycle time is 3.00 weeks and a typical item takes 0.9 weeks of active work. Flow efficiency is: - A. 30.0 % - B. 70.0 % - C. 3.33 % - D. 333 %

Rationale: `0.9/3.00 = 30.0 %` (13.2.3). B is the waiting share; C and D invert the ratio. The low figure is normal and locates the improvement in waiting, not in working faster.

SELF-CHECK — KA 13.2

1. State Little's Law in all three forms and say what each is used for. — `W = T × C`, `C = W/T`, `T = W/C`: the second derives the elapsed time a management decision about work in progress has caused; the third converts a cycle-time improvement into a throughput improvement at unchanged work in progress, which is how KA 13.3.2 prices governance.

2. Why is ignoring discovery arrivals worse than using a mean rather than a range? — Because it is invisible: a team with perfectly stable gross throughput still misses the date, and on Meridian the error is **13.33 weeks** and **USD 190,400** on its own.
3. What must accompany a range forecast for it to be honest? — The smoothing window, the arrival rate, the four assumptions (sampled past, item-size distribution, team composition, arrival continuity) and the statement that it is empirical and carries no probability.

KNOWLEDGE AREA 13.3

Scaling considerations and hybrid governance

Topics: 13.3.1 what scaling actually costs · 13.3.2 governance latency against iteration length · 13.3.3 designing hybrid delivery honestly · 13.3.4 reporting flow to a predictive governance body.

13.3.1 What scaling actually costs

The principle. Adaptive delivery scales badly by default and well by design, and the difference is entirely a matter of how many things must agree. Two costs rise as team count rises, and both have already been computed in this book, so they are cited rather than re-derived.

Coordination overhead within and between teams rises with the square of the number of people working as one group: Domain 12's KA 12.2.2 computes the fully connected case as `(n − 1)/160` of team capacity on its stated link-cost assumption, giving **6.875 %** at Meridian's original twelve people and **24.375 %** if forty were run as one team — against **4.6875 %** for the same forty in five structured groups of eight, a release of **7.875 full-time equivalents**. **Interface count between delivery streams** behaves the same way: Domain 4's KA 4.2.3 gives `n(n−1)/2` possible pairwise interfaces against `n` to an integration layer, so five streams carry **10** potential interfaces against **5**, and nine carry **36** against **9**.

What this implies for scaling design is short, because the arithmetic is decisive. Prefer **fewer, larger increments of independence** to more teams: a team that can release without agreement costs `n`, one that must agree with every other costs `n(n−1)/2`. Where teams must integrate, buy the **integration layer** — an architecture, an interface contract, a shared platform — rather than paying the mesh, and price it against the interface count it removes (Domain 4's method). And accept the honest consequence: **scaling frameworks are largely mechanisms for reducing the number of things that must agree**, and any scaling design should be assessed on exactly that question. A framework adopted without reducing agreement requirements has added ceremony to a mesh.

The dependency that matters most is on a shared scarce resource. Domain 6's KA 6.A.1 drum applies unchanged: where several streams need the same constrained capability — a security review, one test environment, one clinical safety officer — that capability sets programme throughput, and every stream's local flow measures look healthy while the programme's do not. The diagnostic is to compute cycle time for items **requiring** the shared resource separately, which is the same decomposition KA 13.3.2 performs for governance decisions.

13.3.2 Governance latency against iteration length

Domain 3 named this mismatch and left it as a design defect: a monthly steering committee governing two-week sprints is always behind, so the team either waits or proceeds and seeks retrospective approval (Domain 3, KA 3.1.3). This topic quantifies it, and the quantification is what converts the complaint into a proposal.

The two numbers, and their ratio. Domain 3's KA 3.2.3 gives the expected wait for a committee decision as `E[wait] = M/2 + L`: half the meeting interval plus the whole paper lead time. Meridian's steering committee has `M = 4` and `L = 2`, so `E[wait] =` **4.0 weeks**. The iteration is **2 weeks**. The ratio is therefore **2.00** — every escalated decision costs **two entire iterations of waiting**, which is the precise sense in which the governance body cannot keep up with the work it governs. Note that this is not a criticism of the committee's diligence: the number is a property of two administrative parameters and would be the same if every member were faultless.

WORKED EXAMPLE

Worked example 13.3.2 — what governance latency does to Meridian's flow.

1. **Setup.** Meridian's team runs at `T = 6` **items a week** with `W = 18` and therefore `C = 3.00` **weeks** (13.2.3). Reviewed history shows **15 %** of items require a decision above the product owner's **USD 25,000** envelope and therefore go to the steering committee at `E[wait] = 4.0` **weeks**. What does that cost in flow terms, and what would a one-week out-of-cycle written-resolution route (Domain 3, KA 3.3.3) be worth?
2. **Formula.** Blocked arrival rate = share × throughput. **Blocked work in progress = blocked arrival rate ×** `E[wait]` — Little's Law applied to the blocked sub-population. Decompose the average cycle time: `C = (1 − s) × Cu + s × (Cu + E[wait])`, so `Cu = C − s × E[wait]`. Then recompute `C` at the shorter wait and convert to throughput by `T = W / C`.
3. **Substitution.** Blocked arrivals `0.15 × 6 = 0.90` items a week; blocked work in progress `0.90 × 4.0`. Unblocked cycle time `3.00 − 0.15 × 4.0`. New average `0.85 × 2.40 + 0.15 × 3.40`. New throughput `18 / 2.55`.
4. **Result. 1.80** of every 12-item iteration requires a committee decision. **3.60 items** of work in progress are permanently parked waiting for one — **20.0 %** of the team's entire work in progress. Unblocked items take **2.40 weeks**; blocked items take **6.40 weeks**, a multiple of **2.667**. With a one-week out-of-cycle route, parked work in progress falls to **0.90 items** (**5.0 %**), average cycle time falls from 3.00 to **2.550 weeks** (**−15.0 %**), and at unchanged work in progress throughput rises from 6.00 to **7.0588 items a week** — **+17.6 %**.
5. **Interpretation.** Three results are worth stating separately because they persuade different audiences. To a delivery manager, **one item in five that the team is holding is not being worked on at all** — it is waiting for a room, and no report shows it, because a blocked item and a work-in-progress item look identical on a board. To a product owner, **the items that need a decision take 2.667 times as long as the items that do not**, which is why the roadmap's high-value items — the ones large enough to exceed the envelope — are systematically the late ones; the governance design is quietly selecting against value. And to a sponsor, the remedy is a **17.6 % throughput increase for the cost of an administrative procedure**, achieved without hiring anyone, working faster, or removing any scrutiny that anyone can name — the decision is still made by the same authority, merely in writing and within five working days. Note the identity that makes the last claim exact rather than rhetorical: at fixed work in progress, a cycle-time reduction is a throughput increase, because Little's Law can be read in either direction (13.2.3), and `18 / 2.55 = 7.0588` is the same operation as `240 / 34`. Two cautions. The 15 % share must come from a counted history, not an impression; the whole result scales linearly with it, so a paper quoting 17.6 % must show how the share was counted. And the remedy is only available where the decision class can legitimately be decided out of cycle: a clinical safety approval or a regulatory submission cannot be written-resolved, and Domain 3's KA 3.3.1 rule stands — keep the gates that carry real optionality, remove the ones that re-approve a decision already taken. The general form of the lesson: **governance latency is a flow parameter, and it belongs in the flow arithmetic rather than in the culture chapter.**

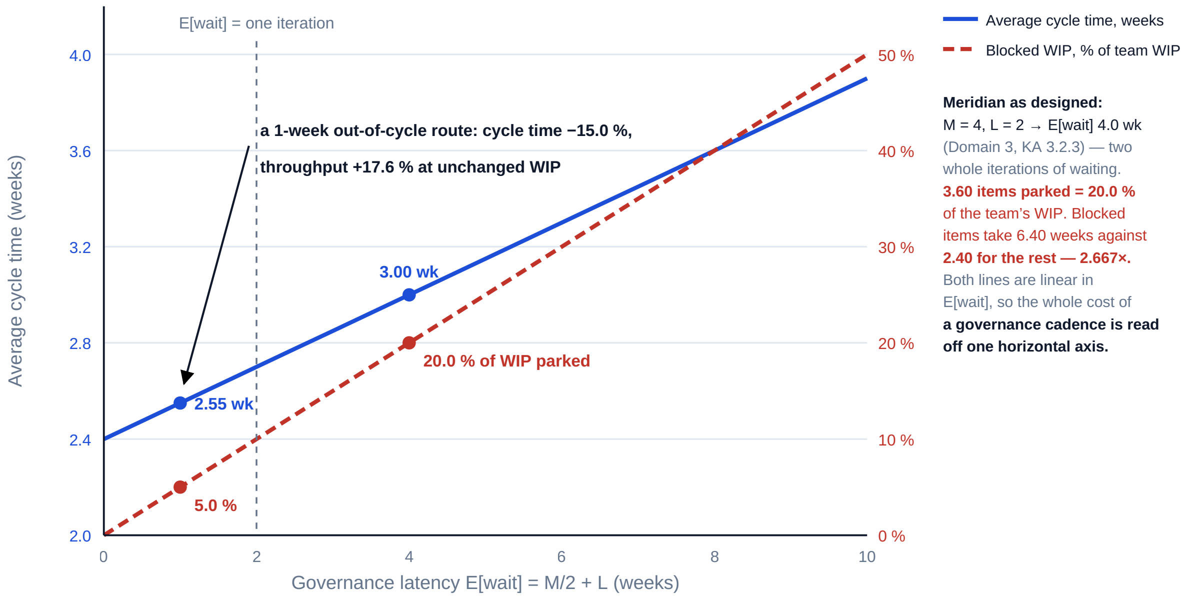


Fig 13.3.1 — Governance latency as a flow parameter

13.3.3 Designing hybrid delivery honestly

Definition. Hybrid delivery is a deliberate combination of predictive and adaptive methods under one governance frame, **with a named boundary**. The words that matter are "deliberate" and "named": most programmes are hybrid by circumstance and manage the whole of it with one control regime, which means they have silently chosen a method for work it does not suit.

The three design decisions. First, **where the boundary runs**. It runs where the three conditions of 13.1.1 change — where feedback stops being cheap and fast. Meridian's boundary is unusually clear: the records application is adaptive (feedback within an iteration from clinicians using a working increment), the clinic rollout is predictive (estate works, training cohorts, immovable dependencies, feedback measured in months), and the regulatory approval is a genuine gate. Second, **what crosses the boundary and in what form**. Domain 6's KA 6.4.1 gives the translation rule and it is not repeated here: a throughput-based forecast enters the network as a **ranged duration on an integration activity**, and the network returns the **latest acceptable delivery** read from the backward pass. Third, **which control regime governs which side**, stated explicitly, so that neither side is measured by the other's instruments — the adaptive side on flow and value delivered, the predictive side on variance against baseline, and the boundary itself on the integration milestone's date range.

The four hybrid failure modes, each with its test. **One regime for both** — a single percentage-complete report covering an adaptive stream, forcing the team to invent a denominator (13.3.4); test: ask what it is derived from. **A boundary nobody owns** — an integration milestone with no accountable holder on either side, so it slips without a decision, which is Domain 4's interface failure in a new costume; test: name the owner. **Cadence imposed across the boundary** — the predictive side's monthly reporting cycle becomes the adaptive side's decision cadence, re-creating 13.3.2's latency; test: compare `E[wait]` for each side against its own cycle. **Hybrid as a euphemism** — a predictive programme that holds stand-up meetings, carrying the cost of both regimes and the benefit of neither; test: ask which variable floats on the adaptive side.

13.3.4 Reporting flow to a predictive governance body

A governance body accustomed to variance against baseline will ask an adaptive stream for a percentage complete, and the team will supply one by dividing items delivered by items known. That number is almost always wrong in the same direction, and the error is computable.

WORKED EXAMPLE

Worked example 13.3.4 — the count view and the size view.

1. **Setup. 96** of Meridian's 240 release items are complete, delivered over **16 weeks**. On re-sizing, the delivered items average **1.2** size units and the remaining **144** average **2.0** — the later items being the integration-heavy and regulatory ones deliberately deferred.
2. **Formula.** Percentage complete by count = items done ÷ total items. By size = size delivered ÷ total size. Remaining duration on the count view = remaining items ÷ item throughput; on the size view = remaining size ÷ delivered size rate.
3. **Substitution.** `96/240` against `(96 × 1.2) / (96 × 1.2 + 144 × 2.0)`. Delivered size rate `115.2 / 16`. Remaining duration `144 / 6.00` against `288.0 / 7.20`.
4. **Result.** By count the release is **40.0 %** complete; by size, **28.6 %** — an overstatement of **11.4 percentage points**. The count view says **24.0 weeks** remain; the size view says **40.0 weeks**, an understatement of **16.0 weeks** and **USD 228,480**.
5. **Interpretation.** The count view is not a lie and the team is not manipulating anything; it is simply the wrong denominator, and it fails in a **predictable direction** because teams rationally sequence the tractable work first — which is precisely what 13.1.3's density rule tells them to do. So the two disciplines of this domain interact: **optimal sequencing guarantees that an item count will overstate progress.** A leader who understands that will report count and size together, or report size alone, and will never let an item count travel to a board unlabelled. The defensive habit that follows is to report **three numbers and no percentage**: items and size delivered against items and size remaining; the net drain and the range it implies (13.2.4); and the conditions under which the range holds. Two cautions. Re-sizing remaining work is legitimate and must be **dated and recorded**, because a re-size that quietly moves a denominator is indistinguishable from a re-baseline without governance (Domain 6, KA 6.A.2). And size units are a team-relative measure with no meaning across teams, so they must never be aggregated across streams into a programme total — which is the single most common misuse of adaptive metrics at scale, and the reason KA 13.4.2 insists on throughput and cycle time as the portfolio-level measures.

AI in this KA

Where it earns its place. Sweeping governance-design options — cadences, paper lead times, delegation envelopes — and reporting the resulting `E[wait]`, blocked work in progress and throughput for each, which is deterministic arithmetic over a large combination space. Classifying a decision history to establish the 15 % share honestly from records rather than impression. Detecting cross-stream dependencies from work-item text and commit metadata, producing a candidate interface list for human confirmation against Domain 4's interface register. Reconciling a count view against a size view automatically each period, so that 13.3.4's gap is a standing number rather than a discovery.

Where it must not go. It must not decide which decisions may be taken out of cycle: that is a risk-appetite and, for clinical or regulatory classes, a legal judgement, and it belongs to the accountable authority. It must not aggregate size units across teams, however plausible the resulting programme total looks — the operation is meaningless and a model will perform it without complaint. And it must not be the source of the hybrid boundary: where feedback is cheap and fast is a matter of fact about the work, established by people who know the work.

Verification, concretely. `E[wait]` is two operations and is reproduced by hand for every design that reaches a paper. The blocked share is traced to counted decisions with dates in and out. Any cross-stream dependency a model proposes is confirmed with both stream leads before it enters the interface register, because a false dependency creates real coordination cost — at Domain 12's rates.

KEY TERMS — KA 13.3

Term	Meaning
Blocked work in progress	Blocked arrival rate × `E[wait]` — the items a team holds but cannot work on, invisible on a board.
Cycle-time decomposition	Splitting average cycle time into its blocked and unblocked populations to locate the delay.
Out-of-cycle route	Domain 3's mechanism (KA 3.3.3) applied where governance latency exceeds the iteration length — the specific mismatch KA 13.3 quantifies.
Integration layer	An architecture or interface contract that replaces `n(n−1)/2` pairwise agreements with `n`.
Hybrid boundary	The named line where feedback stops being cheap and fast, and the control regime changes.
Integration milestone	The boundary event where an adaptive stream's ranged forecast enters a predictive network.
Count view / size view	Progress by item count against progress by size delivered; the first overstates when sequencing is optimal.

SAMPLE MCQS — KA 13.3

MCQ 13.3-A `[13.3.2 · Application]` A team runs at 6 items a week with 18 in progress; 15 % of items need a committee decision with `E[wait] = 4.0` weeks. The work in progress parked waiting for a decision is: - A. 1.80 items - B. 3.60 items, 20.0 % of the team's work in progress - C. 0.90 items - D. 2.70 items

Rationale: Blocked arrivals are `0.15 × 6 = 0.90` a week, and by Little's Law the blocked population is `0.90 × 4.0 = 3.60` items (13.3.2). A is the blocked items per iteration (`0.15 × 12`), not the parked population; C is the weekly arrival rate; D applies the 15 % share to the work in progress of 18 instead of to the arrival rate, which omits the wait entirely.

MCQ 13.3-B `[13.3.2 · Analysis]` For the same team, average cycle time is 3.00 weeks. The cycle times of the unblocked and blocked populations are: - A. 3.00 and 3.00 weeks - B. 2.40 and 6.40 weeks - C. 3.00 and 7.00 weeks - D. 2.55 and 6.55 weeks

Rationale: `Cu = 3.00 − 0.15 × 4.0 = 2.40`, and blocked items add the whole 4.0-week wait, giving 6.40 — a multiple of 2.667 (13.3.2). C adds the wait to the average rather than to the unblocked figure, double-counting the 0.60 weeks the blocked items already contribute; D uses the post-remedy average.

MCQ 13.3-C `[13.3.2 · Evaluation]` Replacing the 4.0-week committee wait with a 1.0-week written-resolution route changes cycle time to 2.550 weeks. At unchanged work in progress of 18, the throughput effect is: - A. none — throughput depends on capacity, not on cycle time - B. an increase from 6.00 to 7.06 items a week, +17.6 % - C. an increase of 15.0 %, matching the cycle-time reduction - D. an increase from 6.00 to 7.20 items a week

Rationale: `T = W/C = 18/2.55 = 7.0588`, an increase of 17.6 % (13.3.2). A forgets that Little's Law reads both ways; C confuses the proportional fall in `C` with the proportional rise in `T`, which are not equal because the relationship is reciprocal (`1/0.85 = 1.176`, not 1.15); D rounds the cycle time to 2.50 before dividing, and rounding before the division moves the answer by 0.14 items a week.

MCQ 13.3-D `[13.3.4 · Analysis]` 96 of 240 items are done, averaging 1.2 size units, and the 144 remaining average 2.0. Reporting 40.0 % complete: - A. is correct, since 96/240 = 40 % - B. overstates progress by 11.4 percentage points, because optimal sequencing delivers the tractable items first - C. understates progress - D. is correct provided the team's velocity is stable

Rationale: By size the release is `115.2/403.2 =` **28.6 %** complete (13.3.4). The direction of the error is systematic, not accidental: the density rule of 13.1.3 tells teams to do the cheap high-value work first, which guarantees that a count overstates.

MCQ 13.3-E `[13.3.1 · Application]` Nine delivery streams must integrate. The reduction in potential pairwise interfaces from adopting an integration layer is: - A. from 36 to 9 - B. from 81 to 9 - C. from 45 to 9 - D. from 36 to 18

Rationale: `n(n−1)/2 = 36` against `n = 9` (Domain 4, KA 4.2.3, cited at 13.3.1). B squares `n`; C uses `n(n+1)/2`.

SELF-CHECK — KA 13.3

1. How is blocked work in progress computed, and why does no board show it? — Blocked arrival rate × `E[wait]`. A board shows an item as in progress whether it is being worked on or waiting for a committee, so 20.0 % of Meridian's work in progress is invisible.
2. Why does an item count systematically overstate progress on an adaptive stream? — Because optimal sequencing delivers the cheap, high-value items first, leaving the larger ones — on Meridian, 40.0 % by count against 28.6 % by size.
3. What is the test for whether a programme is genuinely hybrid rather than nominally so? — Name the boundary, name the person accountable for the integration milestone, and state which variable floats on the adaptive side. If nothing floats, it is not adaptive.

KNOWLEDGE AREA 13.4

Contracting for adaptive delivery, metrics and anti-patterns

Topics: 13.4.1 capacity-based against scope-based commercial models · 13.4.2 what the buyer must control instead of price · 13.4.3 metrics that survive contact with incentives · 13.4.4 the anti-patterns and their arithmetic signatures.

13.4.1 Capacity-based against scope-based commercial models

The two models. A **scope-based** contract fixes a price for a defined body of work; the supplier bears the risk that the work takes more effort than expected. A **capacity-based** contract buys a funded, stable delivery capability for a period — a team, at a rate, for an agreed cadence — and the buyer bears the risk of how much gets done. Domain 10's KA 10.3.1 establishes the general frame: a contract is a risk allocation, and the only useful question about a mechanism is what it rewards at the margin. This topic applies that frame to the one test that actually discriminates between the two models in adaptive delivery, because on expected price at award they can be made identical: **what does a change cost the buyer?**

WORKED EXAMPLE

Worked example 13.4.1 — Meridian's build, priced two ways, then changed.

1. **Setup.** The 240-item release, two commercial offers designed to be equivalent at plan. Scope-based: **USD 1,200,000** firm for the 240 items — **USD 5,000** an item. Capacity-based: **USD 60,000** per 2-week iteration for the supplier's squad; the plan of 20 iterations (40 weeks) totals **USD 1,200,000**, which at the planned 12 items an iteration is also **USD 5,000** an item. At week 14 a regulatory reporting requirement emerges. The buyer wants **30 items added** and 30 lower-value items **dropped** — a net-zero change in item count. The scope contract's change schedule prices additions at **USD 7,500** an item (a 50 % premium on the bid rate, because change work is not competed) and credits omissions at **USD 3,000** an item (60 % of the bid rate, the supplier retaining margin and recovered overhead). A scope change must go to the change board at Domain 3's `E[wait] =` **4.0 weeks**; a capacity re-prioritisation happens at the next iteration boundary, an average wait of **1.0 week**. Cost of delay **USD 14,280** per week.
2. **Formula.** Scope exposure = (items added × change rate) − (items dropped × omission credit) + `E[wait]` × cost of delay. Capacity exposure = boundary wait × cost of delay. Then the buyer's quantity exposure under the capacity model = (worst-case iterations × iteration price) − the fixed price. Breakeven change count = quantity exposure ÷ the per-change difference.
3. **Substitution.** Additions `30 × 7,500 = 225,000`; credit `30 × 3,000 = 90,000`; latency `4.0 × 14,280 = 57,120`. Capacity `1.0 × 14,280`. Duration range from 13.2.4's honest forecast, 48 to 60 weeks, i.e. **24 to 30 iterations**: `24 × 60,000` to `30 × 60,000`.
4. **Result.** The net-zero change moves the scope-based price by **USD 135,000** and costs a further **USD 57,120** in decision latency: total buyer exposure **USD 192,120** for a change that adds no items at all. Under the capacity model the same change costs **USD 14,280** — the one week to the next boundary — a difference of **USD 177,840** per change. Against that, the capacity model exposes the buyer to duration: **USD 1,440,000 to USD 1,800,000** across the honest range, so worst-case quantity exposure above the fixed price is **USD 600,000**. The breakeven is `600,000 / 177,840 =` **3.37**, so from the **fourth** material re-prioritisation the capacity model is cheaper even at the worst end of the observed throughput range: 4 changes cost the scope buyer `1,200,000 + 4 × 192,120 =` **USD 1,968,480** against the capacity model's worst case of **USD 1,800,000**. Measured against the capacity model's expected cost — 53.33 weeks, 26.67 iterations, **USD 1,600,000** — the breakeven falls to **2.25** changes.
5. **Interpretation.** The result to carry away is the shape, not the numbers: **a scope-based contract converts every change into a priced commercial event with a governance latency attached, and a capacity-based contract converts every change into a free re-ordering at the next boundary — while transferring the quantity risk to the buyer.** Neither is better; they price different uncertainties, and the choice follows from which uncertainty the work actually has. The decisive diagnostic is therefore a count: how many material re-prioritisations does this work expect? Below about three, the fixed price is the cheaper instrument; above it, the capacity model is, and the gap widens by **USD 177,840** a change. Four cautions, and they matter more than the arithmetic. First, the **asymmetry in the change schedule** is where scope-based exposure really lives: additions at 7,500 against credits at 3,000 means a net-zero change moves the price by 135,000, and a buyer who negotiates the addition rate without negotiating the omission credit has secured half a protection. Second, the **fixed price presupposes specifiability**. Where the 240 items genuinely cannot be specified at bid — which is the premise of adaptive delivery — a competent supplier prices the ambiguity into the bid, so the premium is paid whether or not

a change occurs; the observed 1,200,000 is then not a bargain but a contingency the buyer cannot see or reclaim. Third, the capacity model's **600,000 of quantity exposure is real** and it is not managed by the contract; it is managed by the continuation decision of 13.4.2, and a buyer who adopts the model without that control has bought an open-ended commitment, which is Case study B. Fourth, none of this is a legal analysis: the enforceability of change schedules, omission credits and termination-for-convenience rights varies by jurisdiction and by contract form, and the arithmetic above informs a negotiation rather than settling one.

Test	Scope-based (firm price)	Capacity-based (funded team)
Expected price at award	USD 1,200,000	USD 1,200,000 at plan
Buyer exposure to a net-zero change	**USD 192,120** (135,000 price + 57,120 latency)	**USD 14,280** (one week to the boundary)
Buyer exposure to throughput risk	none — supplier bears it	**up to USD 600,000** across the 48–60 week range
What the buyer must control	change control and the change schedule's asymmetry	the continuation decision and the value envelope
Breaks even against the other at	fewer than ~3 material changes	4 or more material changes (2.25 against expected cost)

13.4.2 What the buyer must control instead of price

Under a capacity model the price is no longer the control, and a buyer who does not replace it has replaced a bounded commitment with an unbounded one. Three controls do the work, and all three already exist in this book.

The continuation decision at a stated cadence. Domain 3's KA 3.1.3 posture — is the next increment's cost worth its expected value — applied at a real interval with a real authority to stop. The mechanism is a **bounded funding envelope**: authority to spend a stated number of iterations, after which continuation requires an explicit decision. Meridian's would be four iterations — **USD 240,000** at 60,000 an iteration — which is small enough to be a genuine decision and large enough not to be a ceremony.

The value-envelope reconciliation. Domain 5's KA 5.A.1 test, run every funding cycle: attributed benefit of delivered items against the business case's commitment. This is the control whose absence Case study B costs at **USD 3,360,000**, and its price there is **USD 36,000**.

The exit provision that is actually usable. A termination or step-down right is only a control if exercising it leaves the buyer with something: the code, the data, the environments, the documentation and a supplier obligation to support transition, all of which belong in the definition of done (13.1.4) rather than in a schedule nobody reads. An exit right without a transferable asset is a theoretical right, and Domain 10's KA 10.3.3 supplier-governance treatment applies.

13.4.3 Metrics that survive contact with incentives

The principle. Goodhart's law — that a measure adopted as a target tends to cease being a good measure — is not a reason to avoid measurement; it is a reason to choose measures whose gaming is either impossible or visible. Four flow measures satisfy that test and one popular measure does not.

The four that hold. Throughput — completed items per period, where "completed" means the definition of done. **Cycle time**, ideally as a distribution rather than a mean, because the tail is where the failures are and the mean conceals it. **Work in progress**, including its aged tail, which is where blocked items hide (13.3.2). And **flow efficiency**, which is the only one of the four that points directly at the improvement. All four are counts of real events with timestamps, and all four are cross-team comparable, which is what makes them the right measures at portfolio level.

The one that does not: velocity as a performance measure. Velocity is throughput weighted by team-relative size units, and its weights are set by the team being measured. The arithmetic of gaming it is trivial: Meridian's team delivers 12 items an iteration at an average size of 1.2, so **14.4 size units**; inflate the sizing by 25 % and the same 12 items become **18.0 size units**, a reported 25 % improvement with no change whatever in delivery. The inflation is not usually dishonest — it is the predictable response to being measured on a number one controls — and it is undetectable from the metric itself. Velocity is therefore a legitimate and useful **planning input inside a team** and an illegitimate performance measure across teams, and the two uses must be separated explicitly rather than by convention. Size units must never be aggregated into a programme total (13.3.4).

What the governance body should actually receive, and it fits on one page: throughput and its trend; the cycle-time distribution with its tail; work in progress against its limit, and the blocked share; the net drain and the range forecast with its conditions (13.2.4); and the value-envelope reconciliation. Five numbers, none of them a percentage complete, all of them auditable.

13.4.4 The anti-patterns and their arithmetic signatures

Each of the recurring failures of adaptive delivery has a signature in the measures above, which is what makes them diagnosable rather than debatable.

Adaptive delivery with fixed scope. The overhead of the method without its benefit (13.1.1). Signature: a backlog from which nothing is ever removed; zero refusals by the product owner.

Starting more to go faster. Signature: work in progress rising, throughput flat or falling, cycle time rising more than proportionally — worth **USD 180,379** on Meridian (13.2.3). This is the most expensive and most common instruction in delivery.

The proxy product owner. Signature: a median product-owner decision wait of the same order as the committee's `E[wait]`, and blocked work in progress well above the 5.0 % a delegated envelope produces (13.1.2, 13.3.2).

Governing a fortnightly cadence monthly. Signature: **20.0 %** of work in progress parked; blocked items at **2.667×** the cycle time of the rest (13.3.2).

Reporting an item count as progress. Signature: a widening gap between the count view and the size view — **11.4 percentage points** and **16.0 weeks** on Meridian (13.3.4).

Velocity as a target. Signature: size units per iteration rising while throughput in items is flat (13.4.2).

The relaxed definition of done. Signature: completion rising, then reopened items rising, then throughput falling as rework consumes capacity at Domain 9's `1/(1 − r)` (13.1.4).

Hybrid as a euphemism. Signature: no named boundary, no floating variable, and an integration milestone with no accountable owner (13.3.3).

A capacity contract without a continuation decision. Signature: healthy velocity reports and no value-envelope reconciliation — Case study B, at **169.2 %** of planned cost for **41.0 %** of committed benefit.

AI in this KA

Where it earns its place. Modelling the commercial comparison of 13.4.1 across change counts, change rates, omission credits and throughput ranges, which is exactly the deterministic sweep a machine should perform and a human should not. Producing the cycle-time distribution and its tail from a work-item log, including the aged work in progress that no standard report surfaces. Reconciling the count and size views each period automatically. Screening a supplier's iteration reports for the signatures above — sizing inflation against flat item throughput, for example — and raising a candidate finding for a human to confirm.

Where it must not go. It must not select the commercial model, set a rate, or draft a change schedule: those are risk-appetite and legal judgements belonging to the accountable buyer, and Domain 10's KA 10.3 position stands. It must not evaluate individual or team performance from flow data. The measures of 13.4.2 are properties of a **system**, and applying them to people converts every one of them into a target and destroys it — the gaming arithmetic above is the demonstration, and a model that ranks teams on velocity has industrialised the error. And it must not be the source of the anti-pattern finding itself: a signature is evidence for a conversation, not a verdict.

Verification, concretely. Every figure in a commercial comparison is reproduced by hand at one change count, and the change schedule's rates are read from the contract rather than from a summary. Every flow anomaly is traced to the underlying items before it is reported. And any model-produced comparison states its assumed throughput range and its source, because the whole of 13.4.1's conclusion turns on the 48-to-60-week range and a fabricated range produces a confident and wrong contracting recommendation.

KEY TERMS — KA 13.4

Term	Meaning
Scope-based contract	A firm price for a defined body of work; the supplier bears effort risk, the buyer pays for change.
Capacity-based contract	A funded team at a rate for a cadence; the buyer bears quantity risk, change is free at the boundary.
Omission credit	The rate at which removed scope is credited back — typically below the bid rate, and the asymmetry that prices a net-zero change.
Bounded funding envelope	Authority to spend a stated number of iterations before continuation requires an explicit decision.
Continuation decision	The periodic judgement that the next increment's cost is worth its expected value; the capacity model's brake.
Velocity	Throughput weighted by team-relative size units — a planning input inside a team, never a performance measure.
Flow metric set	Throughput, cycle-time distribution, work in progress with its blocked and aged shares, flow efficiency.

SAMPLE MCQS — KA 13.4

MCQ 13.4-A `[13.4.1 · Application]` A firm-price contract of 1,200,000 for 240 items prices additions at 7,500 and credits omissions at 3,000. The buyer adds 30 items and drops 30. The price movement is: - A. zero — the change is net-zero in item count - B. an increase of USD 135,000 - C. an increase of USD 225,000 - D. a decrease of USD 90,000

Rationale: `30 × 7,500 − 30 × 3,000 = 135,000` (13.4.1). A is the error the item count invites; C counts only the additions; D only the credit. The asymmetry between the two rates is where the exposure lives.

MCQ 13.4-B `[13.4.1 · Evaluation]` The same change costs 57,120 in change-board latency, against 14,280 under a capacity model; the capacity model's worst-case quantity exposure is 600,000. The number of material changes at which the capacity model becomes cheaper is: - A. 2 - B. 3 - C. 4 - D. it never becomes cheaper

Rationale: The per-change difference is `192,120 − 14,280 = 177,840`, and `600,000/177,840 = 3.37`, so the fourth change tips it: `1,200,000 + 4 × 192,120 = 1,968,480` against 1,800,000 (13.4.1). A and B are below the breakeven; against the expected capacity cost of 1,600,000 the breakeven is 2.25, which is a different and clearly-labelled comparison.

MCQ 13.4-C `[13.4.2 · Analysis]` A buyer adopts a capacity-based contract and keeps its existing change-control process as the principal control. The defect is that: - A. change control is unnecessary in adaptive delivery - B. price is no longer the binding constraint, so the control must be the continuation decision and the value-envelope reconciliation - C. the iteration price should be renegotiated - D. the supplier will slow down

Rationale: A capacity contract has no per-change price to control, so retaining change control leaves the quantity exposure unmanaged (13.4.2). A overstates — the change record still matters (Domain 3, KA 3.3.4); the missing controls are the brake and the value test.

MCQ 13.4-D `[13.4.2 · Application]` A team delivers 12 items an iteration at an average size of 1.2 units. Sizing is inflated by 25 %. The reported velocity and the actual throughput become: - A. 18.0 units and 15 items - B. 18.0 units and 12 items — a 25 % reported improvement with no change in delivery - C. 14.4 units and 12 items - D. 14.4 units and 15 items

Rationale: `12 × 1.5 = 18.0` units against an unchanged 12 items (13.4.2). This is why velocity is a planning input inside a team and not a performance measure: the team being measured sets the weights, and the inflation is invisible in the metric itself.

MCQ 13.4-E `[13.4.4 · Analysis]` A supplier's reports show size units per iteration rising steadily while items completed per iteration are flat. The most likely diagnosis is: - A. the team is improving - B. sizing inflation under a velocity target - C. the definition of done has been relaxed - D. work in progress has been raised

Rationale: Rising units with flat item throughput is the specific signature of sizing inflation (13.4.2, 13.4.4). A relaxed definition of done would show completion rising then reopened items rising; raised work in progress would show cycle time rising.

SELF-CHECK — KA 13.4

1. What single question discriminates between a capacity-based and a scope-based model? — How many material re-prioritisations the work expects. Below about three the fixed price is cheaper; above it the capacity model is, by **USD 177,840** a change.
2. What replaces price as the buyer's control under a capacity contract? — The continuation decision at a bounded funding envelope, the value-envelope reconciliation, and an exit right that leaves a transferable asset.
3. Why must size units never be aggregated across teams? — They are team-relative weights set by the team being measured, so the sum has no meaning; throughput and cycle time are the portfolio-level measures.

Advanced topics — Domain 13

13.A.1 Adaptive delivery where the increment cannot be released

The honest limit of this domain's methods is reached when condition two of 13.1.1 fails: feedback is neither cheap nor fast, because the increment cannot be put in front of anyone. A partially commissioned substation, a half-migrated clinical dataset, a drug trial at interim, an aircraft modification awaiting airworthiness sign-off — in each case the loop the iteration exists to close cannot close inside it, and a team running two-week iterations against such work is performing the ceremony of adaptive delivery without its engine.

What survives, and it is a great deal, is the flow arithmetic. Little's Law does not require releasability; it requires a steady state. Project Auriga's commissioning queue is the illustration worked in Exercise 13.1: test packs flow through a single commissioning team, work in progress can be limited, cycle time can be measured, and raising work in progress damages throughput exactly as it does on a software team — Auriga's numbers give **4.94 weeks** of delay worth **USD 222,353**. So the practitioner's rule is to separate the two families of technique rather than adopting or rejecting them together: **flow control travels everywhere; iterative release travels only where feedback closes.**

What must be replaced is the feedback mechanism: a **representative environment** (a simulator, a pilot clinic, a digital twin — Domain 14, KA 14.2), a **partial-scope release to a limited population**, or an explicit acceptance that the work is predictive and should be controlled as such. The failure to avoid is the third option adopted implicitly — keeping the cadence and the vocabulary while quietly abandoning the feedback, which is 13.1.1's warning in its most expensive form.

13.A.2 AI-assisted adaptive delivery, and the forecast that becomes a commitment

Where AI is used inside the delivery loop — generating candidate backlog items, drafting acceptance criteria, producing code, forecasting throughput — three obligations arise that existing adaptive practice does not express, and they must be added deliberately (the same conclusion Domain 3's KA 3.A.2 reaches for governance).

The definition of done must extend to AI-produced work. An AI-generated item is not done because it compiles; it is done when it meets the same binary standard as anything else, including review by a person who can be answerable for it. The specific hazard is **volume**: a model can produce candidate items and code faster than a team can review them, which raises work in progress without raising capacity — and 13.2.3 has already computed what that does. AI-assisted generation without a matched increase in review capacity is a work-in-progress decision in disguise.

Throughput history must be segmented when the tooling changes. A team adopting a substantial new tool has changed its own service-time distribution, so the twelve-week history of 13.2.4 is no longer a sample of one system. State the change date, hold both histories, and widen the range until enough post-change observations exist — rather than claiming the improvement in advance, which is the commonest error and the one that produces the next missed date.

The forecast must not become the commitment. This is the most serious hazard in the domain and it is sociological rather than technical. A machine-produced forecast arrives with more apparent authority than a human one: it is precise, it is fast, and it does not hedge unless asked to. A range with four stated assumptions becomes a date; the date is quoted upward; and by the time it is challenged it has been repeated in three papers. The countermeasure is procedural and cheap: **no forecast leaves the team without its required rate attached** (13.2.4's 8.56 items a week), because a required rate cannot be rounded into a promise — it either exceeds the team's demonstrated capability or it does not, and the question is answerable by anyone.

13.A.3 The reviewer's flow eye

Invariants to test on any adaptive or hybrid delivery design, each cheap and each diagnostic.

Every backlog is **ordered, attributed, sized and criterion-bearing**, and items have been removed from it within the last quarter. Exactly **one** named product owner holds the ordering right, with a stated value envelope, and the median wait for their decisions is measured. Prioritisation is on **delay-cost density**, not on cost of delay or effort alone, and the cost-of-delay numerator traces to a benefits-map line with an owner. The **definition of done is written, binary and unchanged** under schedule pressure, and reopened-item counts are reported. A **work-in-progress limit exists as a number**, is visible, and every change to it is a recorded decision. **Throughput, cycle-time distribution, work in progress with its blocked and aged shares, and flow efficiency** are all measured; velocity, if used at all, is confined to planning inside one team and never aggregated. Every date is a **range with its window, its arrival rate and its assumptions stated**, and every committed date is accompanied by its **required rate**. The **net drain** is computed — arrivals are counted, not assumed away. `E[wait]` is computed for **each** decision body against **its own** cycle, the blocked share of work in progress is reported, and out-of-cycle routes exist for every class that may legitimately use one. Hybrid designs name the **boundary**, name the **integration milestone's owner**, and state which variable floats on each side. Progress reaches governance as **count and size**, never as an unlabelled percentage. And the commercial model is matched to the expected **change count**, with the capacity model's continuation decision and value-envelope reconciliation actually in place — because a capacity contract without a brake is the most expensive single arrangement in this domain.

Industry variations — Domain 13

- **Healthcare.** Clinical safety and information-governance approvals are decision classes that cannot be written-resolved, so 13.3.2's remedy applies to the commercial and functional classes only; Meridian's design must delegate those it may and place clinical sign-off with clinical authority (Domain 3's healthcare variation). The compensating move is a standing clinical reviewer inside the cadence rather than a committee outside it.
- **Financial services and other regulated releases.** Release is gated by change-management and audit requirements, so the increment can be complete far more often than it can be released; the honest design separates a fortnightly definition of done from a monthly or quarterly release train, and the flow measures must then track both cycle time to done and cycle time to live — the gap between them is the regulatory cost, and it should be a stated number rather than a grievance.
- **Construction and infrastructure.** Physical works are predictive and the design and digital workstreams around them are frequently adaptive, so these programmes are hybrid by nature and fail at 13.3.3's boundary rather than inside either regime. Flow control nonetheless travels: snagging, commissioning and inspection queues are pure Little's Law problems, and work-in-progress limits on them are among the cheapest interventions available (13.A.1).
- **Public sector.** Statutory approvals and annual appropriation cycles set `E[wait]` and the funding cadence from outside, so the leader's available levers are the paper lead time and the delegated envelope, exactly as Domain 3's public-sector variation concludes. The characteristic risk is the

capacity contract without a continuation decision, because appropriation encourages spend-to-budget — which is Case study B's mechanism.

- **Energy and utilities.** Control-systems and operational-technology work of Auriga's kind cannot release increments into a live network at will, so iterations produce tested, staged increments that are cut over in windows; the flow measure that matters is the cycle time from "ready to cut over" to "cut over", which is usually dominated by outage-window availability rather than by the team.
- **Software, digital and product organisations.** The natural home of the methods and the natural home of their anti-patterns: the characteristic failures are velocity as a performance measure across teams (13.4.2), scaling by adding teams rather than reducing the number of things that must agree (13.3.1), and light governance on genuinely irreversible architectural and data-model choices, which do not present as gates (Domain 3's technology variation).

Case study — Domain 13: the parallel start (health, Meridian)

Situation. At week 26 of Meridian's application build, three clinics were waiting on the regulatory reporting module and the programme was under visible pressure. The steering committee, meeting on its four-week cycle, instructed the team to start the reporting module in parallel with existing work, taking work in progress from **18** to **30** items. The team's product owner objected. Asked why, they said the team would slow down, which was heard as a preference. The instruction stood.

What the arithmetic showed. The delivery manager computed four numbers before the following governance review, all from records the programme already held. **The parallel start.** On the team's own switching coefficient of 2 % of capacity per extra concurrent item, throughput fell from **6.00** to **4.56** items a week and cycle time rose from **3.00** to **6.58 weeks**; the 240-item backlog moved from **40.00** to **52.63 weeks**, a cost of **USD 180,379** at the programme's 14,280 a week — and the reporting module the committee wanted accelerated arrived later than it would have if the team had finished its current work first. **Governance latency.** With `E[wait] = 4/2 + 2 =` **4.0 weeks** (Domain 3, KA 3.2.3) and 15 % of items exceeding the product owner's envelope, **3.60 items — 20.0 %** of the team's work in progress — were parked waiting for the committee at any moment, and those items took **6.40 weeks** against **2.40** for the rest. **The forecast.** The plan's 34-week date required a gross throughput of **8.56** items a week, **31.7 %** above the best four-week rate the team had ever sustained. **The combined position.** Left as instructed — work in progress at 30, arrivals at 1.5 a week, governance at four weeks — the net drain was **3.06** items a week and the release would take **78.43 weeks**. On the corrected design it would take **40.00**. The difference was **38.43 weeks** and **USD 548,800**.

How it resolved. Three changes, none of which reduced any scrutiny anyone could name. The work in progress limit was set at **18** and made visible, with any change to it recorded as a decision — which also settled the reporting module, now sequenced rather than parallelised. A **written-resolution route with a five-working-day turnaround** was adopted for commercial and functional decision classes, excluding clinical safety and information governance, taking blocked work in progress from 3.60 to **0.90 items** and average cycle time from 3.00 to **2.55 weeks** — a **17.6 %** throughput increase at unchanged staffing. And the 34-week date was replaced with a **40-week conditional forecast**: conditional on the governance route and on discovery arrivals being held at or below **1.06 items a week**, a **29.4 %** reduction, with a named owner for each condition and both reported weekly.

What the domain teaches here. The product owner's objection was correct and unpriced, and an unpriced objection loses to an urgent instruction every time. The same sentence in two forms:

"starting the reporting module in parallel will slow us down" is a preference, while "it costs USD 180,379, delivers the reporting module later than sequencing it would, and takes the release from 40 to 52.63 weeks" is a proposal — and the second was accepted at the first meeting at which it was made. Note also which of the four numbers moved the room: not the 180,379, which was contested, but the observation that **one item in five the team was holding was not being worked on at all**. Flow arithmetic persuades because it describes something the audience can verify by walking to the board.

Case study B — Domain 13: the capacity contract that lost its brake (public sector)

Situation. A public-service digital programme contracted two supplier squads on a capacity basis at **USD 60,000** per squad per two-week iteration — **USD 120,000** an iteration — with a planned twelve months of **26 iterations**, or **USD 3,120,000**. The commercial rationale was sound and matched 13.4.1's analysis: the requirement was genuinely unspecifiable at award and material re-prioritisation was expected. The governance body received an iteration report every cycle showing velocity, items completed and a burn-up chart. Velocity was stable and slightly improving throughout.

What had happened. The contract had been let for capacity and controlled as though it had been let for scope. There was no bounded funding envelope, so continuation was never a decision — each iteration followed the last because nothing stopped it. There was no value-envelope reconciliation, so nobody compared what had been delivered against what the business case had promised. And the reports the body received measured **activity that was genuinely healthy**: the squads were delivering, the definition of done was holding, and no anti-pattern of the delivery kind was present. By **iteration 44** the programme had spent **USD 5,280,000 — 169.2 %** of the planned cost, an overrun of **USD 2,160,000** — when an assurance review finally ran Domain 5's KA 5.A.1 reconciliation. Against a business-case benefit commitment of **USD 1,850,000** a year, the attributed benefit of everything delivered was **USD 758,500** a year: **41.0 %**. The programme had bought **169.2 %** of its cost for **41.0 %** of its benefit, and had done so while every delivery metric it monitored was green.

How it resolved. The reconciliation was made a standing control on a four-iteration cycle, alongside a **bounded funding envelope of four iterations — USD 480,000** — requiring an explicit continuation decision by a named authority at each boundary (13.4.2). The divergence between delivered value and committed benefit had first become computable at around iteration 12; had the two controls been in place from then, the natural stop point was **iteration 16** at **USD 1,920,000**, so the controls were worth **USD 3,360,000**. Their cost was three days of analyst time per cycle — about **USD 4,500**, or **USD 36,000** across the eight cycles — a protection ratio of **93.3 times**. The programme was re-scoped to the two capabilities carrying most of the attributed benefit and closed nine iterations later.

What the domain teaches here. A capacity-based contract is the right instrument for unspecifiable work and it has no brake of its own: the price does not stop rising when the value stops arriving, because price is a function of time and not of scope. Whatever else changes, **a capacity model must be paired with a continuation decision and a value test, or it is an open-ended commitment with a delivery report attached.** Note the specific trap, which is the reason this failure is so hard to see from inside: every metric the governance body received was accurate, favourable and about the wrong thing. Velocity, items completed and a burn-up chart measure whether a team is delivering; none of them can measure whether what is being delivered is worth buying, and no amount of improvement in the first answers the second. That is the whole case for the fifth number on 13.4.2's one-page report.

Executive perspective — Domain 13

What a programme director cannot delegate in this domain:

- **The work-in-progress limit, and the authority to instruct past it.** "Start it in parallel" is the most expensive sentence available to an executive in adaptive delivery — **USD 180,379** on Meridian, for nothing delivered sooner. If you may override the limit, you own the arithmetic (13.2.3).
- **The product owner's authority, in writing.** One named holder, a stated value envelope, and a measured decision wait. A product owner who must consult a committee is a proxy, and the committee's latency is now your delivery cycle time (13.1.2, 13.3.2).
- **Your own governance cadence, computed against the work's cadence.** `E[wait] = M/2 + L` against the iteration length. Four weeks against a fortnight parks **20.0 %** of a team's work in progress and makes the high-value items the late ones (13.3.2).
- **Whether a date is a forecast or a commitment, and which one you are quoting.** Ask for the range, the arrival rate and the **required rate**. A committed date requiring 31.7 % more throughput than the team's best ever four weeks is not ambitious; it is unevidenced (13.2.4).
- **The commercial model against the expected change count.** Below about three material changes take the fixed price; above it take capacity — and if you take capacity, install the continuation decision and the value-envelope reconciliation before the first iteration, not after the forty-fourth (13.4.1, 13.4.2, Case study B).
- **The refusal to measure people with flow data.** Throughput, cycle time and work in progress are properties of a system you designed. Used as individual or team targets they become useless within a quarter, and velocity fastest of all — a 25 % sizing inflation reports a 25 % improvement and delivers nothing (13.4.2).

Calculation exercises — Domain 13

Exercise 13.1 Project Auriga's commissioning team completes **5** test packs a week with **20** in progress. The site manager adds 10 packs, taking work in progress to 30; each pack in progress above 20 costs **1.5 %** of the team's capacity. The remaining queue is **140** packs and Auriga's schedule value is **USD 45,000** a week (Domain 6, KA 6.4.2). Compute cycle time before and after, and the cost of the decision. Solution. Before: `C = 20/5 =` **4.00 weeks**. After: `T = 5 × (1 − 0.015 × 10) = 5 × 0.85 =` **4.25 packs a week**, so `C = 30/4.25 =` **7.06 weeks**. Queue: `140/5 =` **28.00 weeks** against `140/4.25 =` **32.94 weeks** — a delay of **4.94 weeks** costing **USD 222,353**. Note that the physical setting changes nothing about the arithmetic: releasability is what does not travel to Auriga, not Little's Law. Common error: holding throughput constant and reporting cycle time as `30/5 =` **6.00 weeks** with no effect on the queue, which treats work in progress as free and understates both the cycle time and the whole of the cost.

Exercise 13.2 A team's last ten weeks of completions are **7, 5, 8, 6, 9, 6, 7, 5, 8, 9**. The remaining backlog is **180** items, discovery arrivals run at **1.25** items a week, and the cost of delay is **USD 9,500** a week. Using four-week rolling totals, compute the honest range, the mean-rate forecast, and the error made by ignoring arrivals. Solution. Total **70**, mean **7.00** items a week. Four-week rolling totals **26, 28, 29, 28, 27, 26, 29** — sustained rates **6.50** to **7.25**. Net drains

`7.25 − 1.25 = 6.00` and `6.50 − 1.25 = 5.25`, giving **30.00** to **34.29 weeks**. At the mean rate net of arrivals, `180/5.75 =` **31.30 weeks**. Ignoring arrivals gives `180/7.00 =` **25.71 weeks**, understating by **5.59 weeks** and **USD 53,106**. Common error: forecasting from the single best week — `180/9 =` **20.00 weeks** — which uses a rate the team has sustained for one week in ten and ignores arrivals as well, compounding two optimisms into a figure 11 weeks below the fast end of the honest range.

Exercise 13.3 A board meets every **6** weeks with a **2**-week paper lead time and governs a team running **3**-week iterations at **10** items a week with **40** items in progress; **12 %** of items require a board decision. Compute `E[wait]`, the blocked work in progress, the cycle-time decomposition, and the effect of a one-week written-resolution route. Solution. `E[wait] = 6/2 + 2 =` **5.0 weeks** — **1.67** iterations. `C = 40/10 =` **4.00 weeks**. Blocked arrivals `0.12 × 10 = 1.20` a week, so blocked work in progress `1.20 × 5.0 =` **6.00 items**, **15.0 %** of the total. Unblocked `4.00 − 0.12 × 5.0 =` **3.40 weeks**; blocked **8.40 weeks**, a multiple of **2.471**. With a one-week route, `C = 0.88 × 3.40 + 0.12 × 4.40 =` **3.52 weeks**, a **12.0 %** reduction, and at fixed work in progress `T = 40/3.52 =` **11.36 items a week**, up **13.6 %**. Common error: computing blocked work in progress from the iteration length rather than from `E[wait]` — `1.20 × 3 =` **3.60 items**, **9.0 %** — which understates the parked population by **40.0 %** and hides most of the problem.

Exercise 13.4 Three features carry costs of delay of **3,600**, **5,400** and **1,800** a week and efforts of **4**, **9** and **6** team-weeks. Compute the delay cost of the density order, of the cost-of-delay-first order and of the shortest-first order, and the estimate error that would change the ranking. Solution. Densities **900.00**, **600.00**, **300.00**, so the density order is F1, F2, F3, completing in weeks 4, 13 and 19: `3,600 × 4 + 5,400 × 13 + 1,800 × 19 =` **USD 118,800**. Cost-of-delay-first (F2, F1, F3), completing in weeks 9, 13, 19: **USD 129,600** — **USD 10,800** worse. Shortest-first (F1, F3, F2), completing in weeks 4, 10, 19: **USD 135,000** — **USD 16,200** worse. The worst order (F3, F2, F1) costs **USD 160,200**, a spread of **USD 41,400**. F2 would need a cost of delay above `900 × 9 =` **USD 8,100** a week — **50.0 %** above its estimate — to belong first, so the ranking is robust to moderate error. Common error: ranking on effort alone, which looks like a flow heuristic ("smallest first") and is **USD 16,200** worse here than ranking on the ratio; the ratio is the rule and either component alone is a special case of it.

Exercise 13.5 A firm price of **USD 900,000** covers **180** items; the alternative is capacity at **USD 45,000** per two-week iteration over a planned 20 iterations. The change schedule prices additions at **USD 7,500** and credits omissions at **USD 3,000**; the change board's `E[wait]` is **4.0** weeks against a **1.0**-week iteration boundary, and the cost of delay is **USD 9,500** a week. A net-zero change swaps **24** items. The honest duration range is **22 to 27 iterations**. Compute both buyer exposures and the breakeven change count. Solution. Both models price at **USD 900,000** (`900,000/180 =` **5,000** an item; `45,000 × 20 =` **900,000**). Scope-based change: `24 × 7,500 = 180,000` less `24 × 3,000 = 72,000`, a net **USD 108,000**, plus latency `4.0 × 9,500 =` **USD 38,000** — exposure **USD 146,000**. Capacity-based: `1.0 × 9,500 =` **USD 9,500**. Difference **USD 136,500** a change. Capacity cost range `22 × 45,000 =` **USD 990,000** to `27 × 45,000 =` **USD 1,215,000**, so worst-case quantity exposure is **USD 315,000** and the breakeven is `315,000/136,500 =` **2.31** — from the **third** change the capacity model is cheaper even at the worst end (`900,000 + 3 × 146,000 =` **USD 1,338,000** against 1,215,000). Common error: comparing the change order's **USD 108,000** against zero and concluding the capacity model is free. It is not free: it transfers up to **USD 315,000** of quantity risk to the buyer, and that risk is controlled by the continuation decision, not by the contract.

Practitioner's toolkit — Domain 13

Adoption-ready artefacts; adapt headings to your organisation, then keep them stable.

Toolkit 13.T.1 — Flow sheet (one page, one team, weekly)

Rows, all counts or timestamps and none of them a judgement: **throughput** this week and its four-week rolling total; **work in progress** against its stated limit, split into being worked, blocked (with what each is waiting for) and aged beyond twice the median cycle time; **cycle time** as a distribution — median, 85th percentile and longest — never as a mean alone; **flow efficiency** from a sampled touch-time estimate; **discovery arrivals** this week and their four-week rate; and the derived **net drain**. Below the rows, three computed lines: the **range forecast** from the highest and lowest observed four-week net drains, with the window length stated; the **required rate** implied by any date the organisation is currently quoting; and `E[wait]` for each decision body the team depends on, with the blocked share it produces. The sheet's purpose is that every one of these is a number the team can verify by walking to the board, and a delivery design that cannot fill it in is not being measured.

Toolkit 13.T.2 — Backlog ordering sheet with delay-cost density

Columns: item or epic · outcome statement · **benefits-map line and owner** (the numerator's provenance — without it the density is advocacy with arithmetic on top) · cost of delay per week · effort in team-weeks · **density** · dependency constraints · **releasable alone? (yes/no)** · acceptance criterion · product-owner decision (order / defer / decline) with date. Two standing checks: the **breakeven test** for the top three items — the cost-of-delay increase that would reorder them, which tells you whether the order is robust or fragile; and the **releasability count**, because items that cannot be released alone forfeit the sequencing value the sheet exists to capture (on Meridian, **USD 253,640**). One rule of use: the "decline" cell must be used. A sheet with no declined items records a backlog that only grows, which is 13.4.4's first anti-pattern.

Toolkit 13.T.3 — Adaptive commercial and continuation control sheet

Part A, **the model choice**: expected count of material re-prioritisations · addition rate and omission credit under a scope model, with the net movement on a representative net-zero change · `E[wait]` for the change board · the resulting per-change buyer exposure for each model · the honest duration range and the quantity exposure it implies under capacity · the **breakeven change count** · the recommendation and the one assumption it is most sensitive to. Part B, **the brake** (mandatory whenever the model is capacity): the **bounded funding envelope** in iterations and currency · the named authority for each continuation decision and the date of the next one · the **value-envelope reconciliation** — attributed benefit of delivered items against the business-case commitment, with its cycle and its owner · and the exit provision with the list of assets that must transfer for it to be exercisable. Part B is the artefact whose absence cost Case study B **USD 3,360,000** at a control cost of **USD 36,000**; it should be completed before the first iteration is funded, and reviewed at every envelope boundary.

Exam preparation — Domain 13

What is assessed. The conditions under which adaptive delivery outperforms predictive delivery, and which variable floats in each mode; product ownership as a decision right with its four failure modes; **delay-cost density and the cost of a sequence**, including the role of releasability; the backlog's four properties and the value-envelope reconciliation; capacity as an observation and the commitment fallacy; **Little's Law in all three forms**, work-in-progress effects on throughput, cycle time and flow efficiency, and the price of raising work in progress; **range forecasting from throughput history, net of discovery arrivals, and the required rate implied by a committed date**; the coordination and interface arithmetic of scaling (cited from Domains 12 and 4); **governance latency as a flow parameter — blocked work in progress, the cycle-time decomposition, and the throughput identity at fixed work in progress**; hybrid design with a named boundary and its four failure modes; the count view against the size view; **capacity-based against scope-based commercial models on buyer exposure under a change, and the breakeven change count**; what replaces price as the buyer's control; flow metrics that resist gaming and why velocity is not one; and the anti-patterns with their arithmetic signatures.

The calculations to be able to do under time pressure. `C = W/T`, `T = W/C` and `W = T × C`. Throughput under a stated switching coefficient, and the resulting cycle time, flow efficiency and backlog duration, priced at a cost of delay. Delay-cost density, the total delay cost of a sequence (`Σ` cost of delay × completion week), the saving against an alternative order, and the breakeven cost-of-delay change that reorders it. Four-week rolling throughput, the net drain, a range forecast, and the required gross rate implied by a target date. `E[wait] = M/2 + L`, blocked arrival rate, blocked work in progress, the cycle-time decomposition into blocked and unblocked populations, and the throughput uplift from a shorter wait at fixed work in progress. Percentage complete by count against by size, and the remaining duration each implies. Buyer exposure per change under both commercial models, the quantity exposure of the capacity model, and the breakeven change count.

The traps. Holding throughput constant when work in progress rises, so cycle time is computed as W/T_0 and the backlog looks unaffected (Exercise 13.1) · forecasting from a mean rather than a range, and from gross throughput rather than the net drain — the second is worth **13.33 weeks** and **USD 190,400** on Meridian on its own (13.2.4) · forecasting from the single best week (Exercise 13.2) · computing blocked work in progress from the iteration length instead of `E[wait]`, understating it by 40 % (Exercise 13.3) · adding `E[wait]` to the average cycle time rather than to the unblocked population, double-counting the blocked contribution (MCQ 13.3-B) · assuming a proportional cycle-time fall gives an equal proportional throughput rise, when the relationship is reciprocal (MCQ 13.3-C) · ranking a backlog on cost of delay alone or on effort alone rather than on the ratio (Exercise 13.4) · forgetting that sequencing value requires releasability, so a big-bang release costs **USD 485,520** whatever the order (MCQ 13.1-B) · reading a net-zero change as a zero-cost change, when the asymmetry between addition rate and omission credit moves the price by **USD 135,000** (MCQ 13.4-A) · treating a capacity contract as free of change cost while leaving its quantity exposure unmanaged (Exercise 13.5, Case study B) · reporting an item count as progress when optimal sequencing guarantees it overstates (13.3.4) · and aggregating team-relative size units into a programme total (13.4.2).

How the domain connects. Domain 1 supplies the cost of delay every result here is priced at and the accountability principle that keeps a forecast owned by a person; Domain 2 the benefits map that makes delay-cost density defensible and the kill criteria the continuation decision re-tests; Domain 3 `E[wait] = M/2 + L`, the bounded envelope and the out-of-cycle route, which this domain quantifies rather than re-derives; Domain 4 the interface arithmetic and the change-control machinery the scope-based model runs on; Domain 5 the acceptance criteria, value attribution and value-envelope reconciliation; Domain 6 the translation rule that lets a throughput forecast enter a CPM network, and the drum that explains why a shared scarce resource sets programme throughput; Domain 9 the rework multiplier that prices a relaxed definition of done; Domain 10 the risk-allocation frame; and Domain 12 the coordination arithmetic that makes scaling expensive. Forward: Domain 14 takes the flow measures into dashboards and the AI-use register, Domain 15 aggregates them at portfolio level

— throughput and cycle time, never size units — and Domain 16 receives the definition of done as the handover standard.

Domain 13 summary

Adaptive delivery fixes capacity and cadence and lets scope float; predictive delivery does the reverse; hybrid delivery does both, on named sides of a named boundary. None of that is a preference, and the whole of this domain's contribution is to show that the arguments practitioners have about it are arithmetic arguments in disguise.

Prioritisation is a sequencing decision with a price. Ranked by **delay-cost density** — cost of delay per team-week — Meridian's four release epics cost **USD 231,880** in delay; ranked by cost of delay alone, **USD 276,080**; in the worst available order, **USD 386,920**. The sequencing decision is therefore worth **USD 155,040**, and the ranking is robust to a **55.6 %** error in the largest epic's estimate. All of that value depends on **releasability**: a single release at week 34 costs **USD 485,520** whatever the order, so incremental release is worth **USD 253,640** on this release alone.

Flow rests on one theorem. **Little's Law** — `W = T × C` — says that cycle time is a consequence of a management decision about work in progress, and that at fixed work in progress a cycle-time reduction is a throughput increase. Meridian's team at `W = 18` and `T = 6` has a cycle time of **3.00 weeks** and a flow efficiency of **30.0 %**. Instructed to raise work in progress to 30, throughput falls to **4.56** (**−24.0 %**), cycle time rises to **6.58 weeks** (**+119.3 %**), flow efficiency falls to **13.7 %**, and the 240-item backlog moves from **40.00** to **52.63 weeks** at a cost of **USD 180,379** — a **66.7 %** increase in work in progress that bought **0 %** more throughput and delivered nothing sooner.

Forecasts are ranges with stated meanings. Meridian's twelve-week history of **72** completions smooths into four-week sustained rates of **5.50 to 6.50** items a week; net of **1.5** items a week of discovery arrivals the honest range is **48.00 to 60.00 weeks**. Ignoring arrivals alone understates the date by **13.33 weeks** and **USD 190,400**. The plan's committed 34 weeks requires a gross throughput of **8.56** items a week — **31.7 %** above the best four weeks the team has ever had — which is why the defensible answer was not a new date but a **40-week** forecast with two named, measured conditions.

Governance latency is a flow parameter. Domain 3's `E[wait] = M/2 + L` gives Meridian's committee **4.0 weeks**, which is **two entire iterations**; with 15 % of items exceeding the product owner's envelope, **3.60 items — 20.0 % of the team's work in progress — are parked at any moment**, and blocked items take **6.40 weeks** against **2.40** for the rest, a multiple of **2.667**. A one-week written-resolution route takes average cycle time to **2.55 weeks** and, at unchanged work in progress, throughput to **7.06** items a week: a **17.6 %** increase for the price of an administrative procedure. Reported to a predictive body, the same release is **40.0 %** complete by item count and **28.6 %** by size, an overstatement of **11.4 percentage points** and **16.0 weeks** — and the direction is systematic, because optimal sequencing delivers the tractable items first.

Commercially, the two models price different uncertainties. On Meridian's build both cost **USD 1,200,000** at plan; a net-zero change of 30 items exposes the scope-based buyer to **USD 192,120** (a **USD 135,000** price movement from the asymmetry between a 7,500 addition rate and a 3,000 omission credit, plus **USD 57,120** of decision latency) against **USD 14,280** under capacity — but capacity carries up to **USD 600,000** of quantity exposure across the honest duration range. The breakeven is **four** material changes against the worst case and **2.25** against the expected cost. And a capacity model has no brake of its own: Case study B's programme reached **169.2 %** of planned cost for **41.0 %** of committed benefit while every delivery metric it monitored was green, and the two controls that would have stopped it at **USD 1,920,000** cost **USD 36,000** against **USD 3,360,000** protected — a ratio of **93.3 times**.

The through-line: **adaptive delivery does not remove the numbers, it changes which numbers.** Throughput, work in progress, cycle time, net drain, blocked share, required rate, delay-cost density, buyer exposure per change — every complaint about agile delivery in a large organisation is one of these eight quantities in ordinary language, and a leader who can compute them stops having the argument and starts making the proposal.

04

PART FOUR

Enterprise delivery and the digital future

Domains 14–16 — digital delivery and responsible AI, programmes and portfolios, and transition and benefits realisation: delivery at enterprise scale.

14

DOMAIN 14

Digital Delivery, Data and Responsible AI

Group: Enterprise delivery and the digital future (Domain 14 of 3 in Part Four). **Target:** ~76 pages. **Binds to:** the PCI Book Pattern Specification and the shared registries (`docs/books/registries/`). This is the volume's **systematic** treatment of digital delivery, data and artificial intelligence. Domains 1 to 13 each carried an AI in this KA section covering the use and misuse of AI inside their own subject matter; this domain **consolidates** those positions into one governable system and supplies the arithmetic the earlier domains asserted but did not derive. It runs **Meridian Care Records** at programme scale and **Project Auriga** at single-project scale. British English; USD (+SAR where useful, indicative `USD 1 ≈ SAR 3.75`).

Registry note. This domain uses the registered **cost of delay** (PML-AI D1/D3) and the registered **governance latency** identity `E[wait] = M/2 + L` (PML-AI D3, KA 3.2.3) unchanged, applying the second to information rather than to decisions — the same two operations on a reporting period and a production lag, cited rather than re-derived. It cites Domain 9's sampling arithmetic (`p* = c/u` and the clean-sample bound, KA 9.3.2 and 9.4.4) and Domain 9's composite data fitness (KA 9.4.3) rather than restating them. It submits four candidate registry rows: **consequence-weighted defect exposure** $\Sigma\ n_i d_i u_i$ with the per-record exposure $d_i u_i$ that ranks remediation; **quality-adjusted automation breakeven volume** $n^* = F / [(m + e_m u) - (a + e_a u)]$; the **verification tier threshold** $u^*_k = \Delta v_k / (p \cdot \Delta q_k)$ — the derived form of the verification standard proportional to consequence that Domain 3, KA 3.A.2 asserted; and the **escape-cost asymmetry** `(1 − q_p)u_p ÷ (1 − q_o)u_o`. Each carries a verified golden example below. Every error rate, detection rate, defect rate and unit cost in them is a **locally calibrated planning or measured figure, not a constant** — the domain says so at each use, and says what must be measured to replace it.

Why this domain exists

Thirteen domains have now used AI in the way a professional actually uses it: as an assistant to a specific task, bounded by a specific prohibition. Domain 3 refused it a governance decision; Domain 4 refused it a baseline; Domain 5 refused it the authorship of a requirement; Domain 9 refused it a test result and an inspection signature; Domain 10 refused it a quality score and a liability cap; Domains 11 and 12 refused it inference about people. Each refusal was correct and each was local. What none of them could supply, standing alone, is the thing an organisation actually needs: **one system in which those refusals are consistent, resourced, auditable and priced.**

That is what is missing in practice, and its absence has a characteristic shape. Organisations adopt digital tools and AI assistance tool by tool, each justified on a productivity claim, each verified by whoever happens to be nervous, none priced. The result is an estate in which the verification effort is simultaneously excessive on low-consequence outputs and absent on high-consequence ones; in which a data defect rate is reported as a single percentage that no decision can use; in which an automation is celebrated for the labour it removed and never charged for the errors it lets through; and in which the most dangerous artefact in the whole system — a fluent, plausible, wrong number — passes every review that was designed to catch an obviously wrong one.

Hence the domain's central claim: **digital delivery, data quality and AI assistance are all economic quantities with computable breakevens, and the professional obligation is to compute them rather than to adopt or resist on principle.** A leader who can state the consequence at which verification becomes worth its cost, the volume at which an automation starts paying, the exposure that each class of data defect carries and the ratio by which a plausible error out-damages an obvious one has a governable digital estate. A leader who cannot will oscillate between two equally indefensible postures — blanket prohibition, which forfeits real value, and blanket adoption, which transfers an unmeasured cost to whoever inherits the output.

This is therefore the most rigorous domain in the volume rather than the softest, which is the opposite of how responsible-AI material is usually written. KA 14.1 builds the substrate — the digital environment, the governance of data, the common data environment, and the defect rate by data class that determines whether anything built on top of it is usable. KA 14.2 covers what organisations build on that substrate: dashboards, analytics, digital twins and automation, each tested against the decision it claims to serve and the volume at which it pays. KA 14.3 is the domain's core: where AI earns its place across the lifecycle, how a prompt becomes professional practice, and the derivation of the verification standard proportional to consequence — followed by the arithmetic of why the plausible wrong number is the one to fear. KA 14.4 closes with the four obligations that cannot be computed away: explainability that supports a decision, differential error, human accountability, and the security and privacy of a delivery estate that now holds everything.

Learning objectives. After this domain a candidate can: describe a digital delivery environment as a set of systems with a countable interface problem and apply Domain 4's interface arithmetic to it; specify data governance as ownership, definition, lineage and access rather than as a policy document; state what a common data environment must guarantee and test whether a claimed one does; **compute a defect rate by data class, convert it into consequence-weighted exposure, rank remediation by exposure per record, and demonstrate arithmetically that a uniform data-quality target both reduces defect counts and increases expected cost**; test a dashboard against the decision it claims to inform and **compute the expected age of a fact at the moment of decision using the registered `M/2 + L` identity, priced at the cost of delay**; place an analytics initiative on the descriptive-to-prescriptive ladder and state what each rung requires and returns; define what makes a digital twin a twin and **compute the fidelity at which it breaks even**; **compute an automation breakeven volume, adjust it for the differential error-escape rate, and identify the cases in which no volume justifies the automation**; state where AI earns its place at each stage of the lifecycle and cite the domain that governs each; write and govern a prompt as a professional artefact with grounding, provenance and a recorded version; **derive a tiered verification standard from the marginal cost of a review tier, its marginal detection rate and the consequence of an escaped error, and price the tiered standard against uniform alternatives**; **compute why a plausible wrong number costs an order of magnitude more than an obviously wrong one, and price the reperformance step that closes the gap**; specify explainability as a decision requirement rather than a model property; **compute differential error rates across groups, price the remediation, and state where expected-value reasoning stops applying**; maintain an AI use register that satisfies Domain 1's accountability test; and **compute the economics of security controls, distinguishing probability reduction from impact reduction and demonstrating why control benefits are sub-additive**.

The master threads. Meridian Care Records supplies the programme case: the clinical-records rollout to **40 clinics**, approved cost **USD 2,400,000**, full-potential benefit **USD 979,200** a year, realistic benefit **USD 685,440** a year at 70 % adoption, **cost of delay USD 14,280 per week** (Domain 1) — **USD 2,040 per day** — and a steering committee whose latency `E[wait] = 4/2 + 2` is **4.0 weeks** (Domain 3, KA 3.2.3). Internal effort is valued at the blended programme rate of **USD 110 per hour** established in Domain 11. This domain audits Meridian's common data environment across

14,100 records in six classes, derives its verification standard across **941** AI-assisted outputs a month, and prices its security controls. Project Auriga supplies the project case: the 25-week control-systems upgrade, `BAC` **USD 4,000,000**, cost of delay **USD 45,000 per week**, a blended engineering rate of **USD 130.625 per hour** (Domain 7, KA 7.4.1), and at week 13 `PV` **2,080,000** / `EV` **1,920,000** / `AC` **2,120,000**, giving `CPI` **0.91** and `SPI` **0.92**. This domain prices Auriga's data-check automation, its control system's digital twin, and the forecast error that its week-13 numbers make possible.

KNOWLEDGE AREA 14.1

Digital project environments, data governance and the common data environment

Topics: 14.1.1 the digital delivery environment · 14.1.2 data governance · 14.1.3 the common data environment · 14.1.4 defect rate by data class and the exposure it carries.

14.1.1 The digital delivery environment

Definition. A project's **digital delivery environment** is the set of systems that hold its authoritative information and mediate its work: the schedule tool, the cost ledger, the requirements and change registers, the document and model repository, the test and defect systems, the procurement and contract records, the risk register, and the reporting layer assembled over them.

The environment is not a technology question, and treating it as one is the first error. It is an **information architecture** question with a shape Domain 4 already priced. Each system holds facts that other systems need, so each pair of systems is a candidate interface, and the count of possible pairwise interfaces over `n` systems is the registered `n(n − 1)/2` (Domain 4, KA 4.2.3). Meridian's environment has eleven systems of record; the mesh count is `11 × 10 / 2 =` **55** possible interfaces against **11** connections to a single integration layer, and the practical consequence is the one Domain 4 states: the mesh is cheaper for the first two or three systems and catastrophically more expensive by the eleventh, because each interface is a thing to build, test, version, secure and re-verify after every change on either side.

Three properties determine whether the environment helps or hinders, and all three are testable before anything is procured. **Authority** — for each material fact, exactly one system is the source of record, and the others hold copies marked as copies. An environment in which two systems each believe they own the baseline dates has not integrated anything; it has arranged for a disagreement. **Lineage** — for any number on a report, the path from source record to displayed value is reconstructable. **Latency** — the age of the information at the point of use, which 14.2.1 computes.

What the leader owns here. Not the tool selection, which is usually an enterprise decision, but four things that are never in a procurement document: which system is authoritative for which fact; who may change each class of record and under what authority (which is Domain 3's delegation schedule expressed in system permissions); what the environment must produce for the governance body and by when; and what happens to the information at closeout, which Domain 16, KA 16.4 governs and which is decided far too late in almost every programme.

14.1.2 Data governance

Definition. Data governance is the allocation of ownership, definition, quality accountability and access rights over an organisation's data — the same four questions Domain 3 asked about decisions, asked about facts.

It is routinely written as a policy and therefore routinely ineffective. The workable form is a **data class schedule** with one row per class of data the project relies on and five columns that are each individually testable:

Column	The test it must pass
Owner	A named person, not a function, accountable for the class being fit for its stated use.
Definition	One written definition per field, with units and permitted values, agreed across every system that holds it.
Source of record	The one system whose value prevails; every other holding is a copy.
Quality standard	The conformance rate required, stated **per class and per use**, with the consequence that sets it (14.1.4).
Access and retention	Who may read, who may change, how long it is kept, and on what basis it is destroyed.

The **definition** column is the one that repays attention out of all proportion to its cost. A field whose definition differs between two systems produces a reconciliation problem that presents as a data quality problem and is actually a governance problem: nothing is wrong with either value. PFL-AI's treatment of `CFADS` makes the same point in a financial register — a defined term whose striking point changes every ratio built on it. In delivery the classic instances are "forecast completion date" (contractual? current best estimate? with or without float consumption?), "committed cost" (purchase order raised? goods received? invoiced?), and "percentage complete" (Domain 7's `PoC` conventions). Each has more than one defensible meaning and exactly one permissible meaning per organisation, and an organisation that has not chosen will reconcile the difference for the life of the programme.

International reference points may be named accurately: the ISO 8000 series on data quality and ISO/IEC 25012's data quality model give vocabulary and assessment structure for dimensions and characteristics; ISO 19650's information-management series is the widely used frame in construction and infrastructure; and ISO/IEC 38507 addresses the governance implications for an organisation of using AI. None of them supplies a target number. The target is a project decision, and 14.1.4 derives it.

14.1.3 The common data environment

Definition. A **common data environment (CDE)** is a single agreed information store, with defined states and transitions, through which project information is shared — such that every party draws on the same version of the same fact, and the state of each item (work in progress, shared, published, archived) is explicit.

The concept comes from information management in construction and infrastructure, where ISO 19650 frames it, and it generalises cleanly to any multi-party delivery. Its value is not storage; it is the elimination of a specific failure that costs real money and is otherwise almost undetectable: **two parties working correctly from different versions.** Domain 3's decision record made the same move for decisions with its versioned information reference; the CDE makes it for the information itself.

A claimed CDE is worth testing against five guarantees, and most fail at least one. **Single instance** — there is one copy, referenced, not distributed as attachments. **Explicit state** — an item's status is a property of the item, so nobody has to ask whether a drawing is issued. **Controlled transition** — moving an item between states is an authorised act with a record, which is what makes the environment auditable. **Access by role** — read and change rights follow the data class schedule of 14.1.2, not convenience. **Retained history** — superseded versions remain retrievable, because the question asked afterwards is always what was current on a given date.

Two cautions belong here because they are the ones that bite. A CDE is where a **project's** shared information lives, and it should not become a repository for information that has a stricter home: Meridian deliberately holds no patient clinical data in its delivery CDE, keeping the delivery estate and the clinical estate separate, which is both a privacy decision (14.4.4) and a simplification of the security problem. And a CDE **enforces nothing about quality** — it enforces consistency of version. A single authoritative copy of a defective record is still a defective record, propagated more efficiently, which is the subject of the rest of this KA.

14.1.4 Defect rate by data class and the exposure it carries

Domain 9, KA 9.4.3 established that a record satisfying six quality dimensions independently is less conforming than any dimension suggests, because the dimensions compound — Auriga's asset register scored **96.08 %** on the arithmetic mean of its dimensions and **78.43 %** on the product, so roughly **690** of **3,200** records were unfit for use. That result is about one class of record. The question this KA answers is the next one, and it is the question a leader actually faces: **across many classes of data, where should remediation effort go, and what quality target should each class carry?**

The wrong answer, and the one almost universally adopted, is a **single organisational target** — "98 % data quality" — applied uniformly. It is wrong for a reason that is arithmetic rather than philosophical, and the reason is worth stating precisely before the example demonstrates it: a uniform rate target treats a defect as a defect, while the organisation's actual loss is `defects × consequence`, and consequence varies across classes by two or three orders of magnitude.

The two quantities. For each data class i holding n_i records with observed defect rate d_i and a consequence per defect u_i (the internally assessed cost of one defect reaching use):

```
class defects       = nᵢ · dᵢ
class exposure      = nᵢ · dᵢ · uᵢ
total exposure      = Σ nᵢ · dᵢ · uᵢ
exposure per record = dᵢ · uᵢ          ← the quantity that ranks remediation
```

u_i is an internally assessed remediation-and-consequence cost, calibrated locally and stated as an assumption; it is not a market price and it is not comparable between organisations.

WORKED EXAMPLE

Worked example 14.1.4 — Meridian's common data environment, by class.

1. **Setup.** Meridian's delivery CDE holds **14,100** records in six classes. Sampling gives the defect rate for each, and the programme has assessed the cost of one defect of each class reaching use:

Data class	Records n_i	Defect rate d_i	Cost per defect u_i
Clinician accounts and role assignments	4,800	1.5 %	USD 250
Device and network asset records	1,600	4.0 %	USD 180
Interface field-mapping records	900	6.0 %	USD 1,200
Training and competency records	3,600	2.5 %	USD 90
Clinic readiness checklist records	1,200	3.0 %	USD 400
Migration reconciliation records	2,000	0.8 %	USD 3,500

1. **Formula.** Class defects $n_i d_i$; class exposure $n_i d_i u_i$; total exposure $\Sigma\ n_i d_i u_i$; weighted mean defect rate $\Sigma\ n_i d_i \div \Sigma\ n_i$; exposure per record $d_i u_i$. For the uniform-target comparison, exposure at a common rate $\bar{d}$ is $\bar{d} \cdot \Sigma\ n_i u_i$.
2. **Substitution.** Defects `72, 64, 54, 90, 36, 16`. Exposures `18,000 · 11,520 · 64,800 · 8,100 · 14,400 · 56,000`. Uniform 2 % exposure `0.02 × 10,372,000`.
3. **Result.** **332** defective records — a weighted mean defect rate of **2.3546 %** — carrying a total exposure of **USD 172,820** (indicatively SAR 648,075). Exposure per record, which is the ranking that matters: interface mappings **USD 72.00**, migration reconciliation **USD 28.00**, readiness checklists **USD 12.00**, device records **USD 7.20**, accounts **USD 3.75**, training records **USD 2.25**. A uniform **2 %** target across all six classes produces **282** defects — **15.06 % fewer** — and an exposure of **USD 207,440**, which is **USD 34,620** or **20.03 % worse**.
4. **Interpretation.** The last line of the result is the whole KA, and it should be read twice: the uniform target **reduces the defect count by 15 % and increases the expected cost by 20 %**. It does so by mechanism, not by accident. It loosens the two classes whose defects are expensive — migration reconciliation from 0.8 % to 2 %, mappings implicitly permitted at 2 % — and tightens the classes whose defects are cheap, spending effort on training records at USD 2.25 of exposure per record while permitting reconciliation defects at USD 3,500 each. Any target expressed as a rate, with no consequence attached, has this property.

The remediation ranking inverts the rate ranking, and that is the practical payoff. Ranked by defect rate the priority order is mappings (6.0 %), devices (4.0 %), readiness (3.0 %), training (2.5 %), accounts (1.5 %), reconciliation (0.8 %) — which puts the most expensive class last. Ranked by exposure per record it is mappings, reconciliation, readiness, devices, accounts, training. Test the two candidates that the two rankings disagree about most. Re-verifying all **900** interface mappings at **USD 22** per record costs **USD 19,800** and cuts the rate from 6.0 % to 1.0 %, removing 45 defects worth **USD 54,000** — a net **USD 34,200**, and it would remain worthwhile up to a remediation cost of **USD 60.00** per record. Re-reconciling the **2,000** migration records at **USD 12** each costs **USD 24,000** and cuts 0.8 % to 0.2 %, removing 12 defects worth **USD 42,000** — a net **USD 18,000**, breakeven at **USD 21.00** per record, from the class with the lowest defect rate in the estate. Now the class the rate ranking would have reached fourth: re-checking all **3,600** training records at **USD 8** each costs **USD 28,800** to remove 72 defects worth **USD 6,480** — a net **loss of USD 22,320**, and it only becomes worthwhile at a remediation cost below **USD 1.80** per record. Three interventions, two of them strongly positive and one strongly negative, and nothing in the defect rates alone distinguishes them.

Four professional cautions, each of which changes the answer if ignored. The consequence figures u_i are **assessed, not observed**, and the whole ranking is proportional to them — so they belong in the data class schedule with a named owner and a basis, and a class whose u_i nobody will own is a class whose quality target is arbitrary. This model treats a defect as **one discrete event with one cost**, which understates classes where a single defect propagates: a wrong interface mapping corrupts every record that passes through it, so its true u_i should be assessed per mapping, not per record that suffers, and Meridian's USD 1,200 is exactly such a propagated figure. Where a consequence is not cost-compensable — patient safety, a licence condition, a statutory notification — **expected value is the wrong test entirely**, the class carries an absolute standard, and the arithmetic here is used only to size the effort of meeting it, never to trade it away (the position Domain 9, KA 9.1 takes on safety-related requirements). And a defect rate is only meaningful against a **defined pass criterion** for the class, which is Domain 5, KA 5.4.2's testability requirement arriving as a prerequisite: rates measured against an undefined standard record the sampler's judgement.

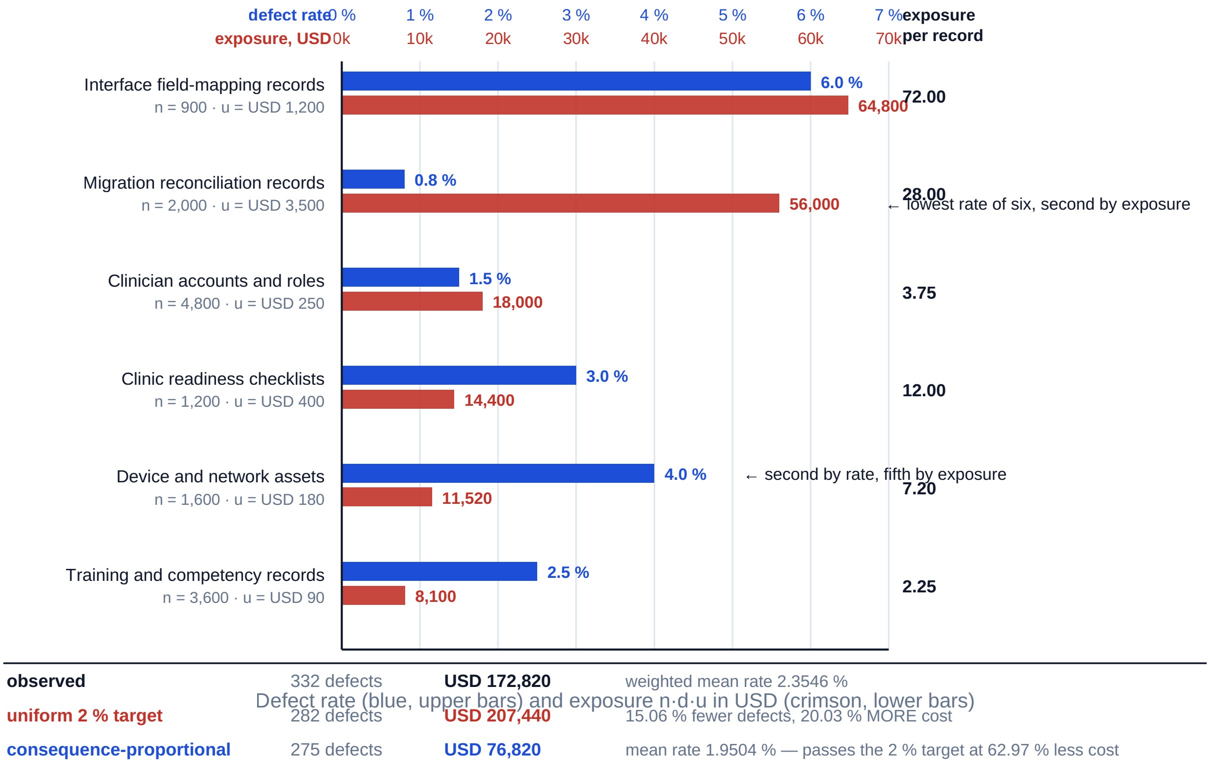


Fig 14.1.1 — Defect rate against exposure: why a uniform data-quality target costs more

AI in this KA

Where it earns its place. Profiling a data class against its defined pass criteria at full population scale, which is mechanical, high-volume and exactly checkable — Domain 9, KA 9.4's strongest AI application, and the input this KA's arithmetic consumes. Reconciling field definitions across systems and producing the list of fields whose definitions differ, which is a document-comparison task humans do badly because it is tedious and enormous. Proposing the remediation rule that would fix the largest number of records in a class, for human ruling. Detecting duplicate and near-duplicate records, where fuzzy matching genuinely outperforms exact rules. Assembling the lineage path for a reported number across systems, so that a challenged figure can be traced in minutes rather than days.

Where it must not go. It must not set a class's quality target or its consequence figure u_i — those are risk-appetite judgements owned by the data owner and the sponsor, and a model asked for them returns confident numbers with no provenance that then drive real remediation spending, the defect Domain 3, KA 3.2.3 and Domain 10, KA 10.1 both identify. It must not **remediate silently**: an inferred value that fills a completeness gap improves the metric and can degrade the register, which is the worst available combination, and Domain 9, KA 9.4's prohibition applies unchanged here. It must not be the source of record for anything — a generated value is a copy with no provenance. And it must not decide which system is authoritative for a fact, which is a governance allocation.

Verification, concretely. Any machine profiling result is confirmed on a human-checked sample with the sample size and its basis stated (Domain 9, KA 9.3.2). Where records have been machine-remediated, an untouched control sample is held and compared, because a plausible wrong fill is detectable only against ground truth — the general principle 14.3.4 prices. Definition discrepancies are ruled on by the named field owner, not accepted as found. And the exposure arithmetic is reproduced by hand: every figure in this KA is a multiplication of three numbers.

KEY TERMS — KA 14.1

Term	Meaning
Digital delivery environment	The set of systems holding a project's authoritative information and mediating its work.
Source of record	The one system whose value for a fact prevails; all other holdings are copies.
Lineage	The reconstructable path from source record to a displayed value.
Data class schedule	One row per data class: owner, definition, source of record, quality standard, access and retention.
Common data environment (CDE)	A single agreed information store with explicit item states and authorised, recorded transitions.
Defect rate by class (d_i)	The share of records in a class failing that class's defined pass criteria.
Consequence per defect (u_i)	The internally assessed cost of one defect of a class reaching use; assessed, owned and stated.
Consequence-weighted exposure	$\Sigma\ n_i d_i u_i$ — the expected cost of a data estate's defects.
Exposure per record	$d_i u_i$ — the quantity that ranks remediation effort across classes.
Uniform-target defect	Setting one rate target across classes of unequal consequence, which can cut defect counts while raising expected cost.

SAMPLE MCQS — KA 14.1

MCQ 14.1-A [14.1.4 · Application] A data class holds 900 records with a 6.0 % defect rate and an assessed consequence of USD 1,200 per defect. Its consequence-weighted exposure is: - A. USD 54.00 - B. USD 72.00 - C. USD 64,800 - D. USD 1,080,000

Rationale: $900 \times 0.06 \times 1{,}200 = 64{,}800$ (14.1.4). A is the defect count; B is the exposure per record ($d_i u_i$), which is the remediation ranking key, not the class exposure; D omits the defect rate and prices every record as defective.

MCQ 14.1-B [14.1.4 · Evaluation] Six data classes with a weighted mean defect rate of 2.3546 % carry a total exposure of USD 172,820. Imposing a uniform 2 % target on all six would produce 282 defects and an exposure of USD 207,440. The correct conclusion is that the uniform target: - A. improves quality and reduces cost, since 282 is fewer than 332 - B. reduces the defect count by 15.06 % and raises expected cost by 20.03 %, because it loosens the low-rate, high-consequence classes and tightens the cheap ones - C. is acceptable because it is simpler to administer - D. is equivalent to the observed position, since the mean rate is close to 2 %

Rationale: Cost is `defects × consequence`, and a rate target is blind to consequence (14.1.4). A counts defects and ignores their price; D confuses a mean rate with an expected cost.

MCQ 14.1-C `[14.1.4 · Analysis]` Across six classes, remediation effort should be ranked by: - A. defect rate d_i - B. record count n_i - C. exposure per record $d_i u_i$ - D. total class exposure $n_i d_i u_i$

Rationale: Remediation cost scales with records touched, so the return per record touched is $d_i u_i$ (14.1.4). D ranks by size of prize without regard to the effort of claiming it, and would direct effort to a large cheap class ahead of a small expensive one; A puts the lowest-rate, highest-consequence class last.

MCQ 14.1-D `[14.1.2 · Comprehension]` Two systems report different "committed cost" for the same purchase order, and both values are correct within their own system. The defect is: - A. a data accuracy failure - B. a definition failure — the field has more than one meaning across systems - C. an interface failure - D. a completeness failure

Rationale: Nothing is wrong with either value; the field has no single agreed definition (14.1.2), which is why the definition column of the data class schedule repays attention out of proportion to its cost. It will present as a reconciliation problem indefinitely until a meaning is chosen.

MCQ 14.1-E `[14.1.3 · Analysis]` A programme stores every document in one repository with full version history, but item status is recorded in a separate spreadsheet and changing status requires no authorisation. Against the five CDE guarantees this arrangement fails on: - A. single instance and retained history - B. explicit state and controlled transition - C. access by role only - D. nothing — it is a valid CDE

Rationale: Status is not a property of the item and transitions are unauthorised and unrecorded (14.1.3). Single instance and retained history are satisfied; the two failures are precisely the ones that make an environment unauditable.

SELF-CHECK — KA 14.1

1. Why can a uniform data-quality target reduce defect counts and increase expected cost? — Because loss is `defects × consequence` and a rate target is blind to consequence: it loosens low-rate, high-consequence classes and tightens cheap ones. Meridian's case: 15.06 % fewer defects, 20.03 % more cost.
2. What ranks remediation across classes, and why is it not the defect rate? — Exposure per record, $d_i u_i$, because remediation cost scales with records touched. Meridian's lowest-rate class (0.8 %) is its second-highest by exposure per record (USD 28.00).
3. What does a common data environment guarantee, and what does it not? — Consistency of version through explicit states and authorised, recorded transitions; it guarantees nothing about quality, so a defective record is simply propagated more efficiently.

KNOWLEDGE AREA 14.2

Dashboards, analytics, digital twins and automation

Topics: 14.2.1 dashboards and the age of a fact · 14.2.2 the analytics ladder · 14.2.3 digital twins and the fidelity that pays · 14.2.4 automation economics.

14.2.1 Dashboards and the age of a fact

Definition. A **dashboard** is a presentation of measures selected to support a defined set of decisions at a defined cadence. Every word in that constrains it, and the standard failure is to build a presentation of the measures that are available to an audience with no decision to make.

Domain 11, KA 11.2 governs what an executive report must contain and prohibits the smoothing of its warnings. This topic addresses the property Domain 11 did not price and that determines whether a dashboard can inform anything at all: **the age of the information at the moment of decision.**

The arithmetic is already in the registry. Domain 3, KA 3.2.3 established that a decision arising at a uniformly random point in a cycle of length `M`, with a paper lead time `L`, waits `E[wait] = M/2 + L`. A fact arising at a uniformly random point in a reporting cycle of period `R`, where production takes a lag `G` from data cut-off to publication, is by the identical argument expected to be `R/2 + G` old when it is published. The derivation is Domain 3's and is not repeated; only the interpretation changes — the quantity is now the staleness of information rather than the wait for a decision, and its two levers behave in the same asymmetric way.

WORKED EXAMPLE

Worked example 14.2.1 — how old is the fact on Meridian's report?

1. **Setup.** Meridian's rollout status is reported monthly: data cut-off at month end, then extraction, validation, review and commentary before publication, a production lag of **9 days**. The programme is considering weekly reporting with a compressed production lag of **4 days**. Cost of delay **USD 14,280 per week**, i.e. **USD 2,040 per day**.
2. **Formula.** Expected age of a fact at publication `= R/2 + G`, the registered `E[wait]` identity (Domain 3, KA 3.2.3) applied to information. Cost of acting that late on a detectable slip `= age × cost of delay per day`.
3. **Substitution.** Monthly `30/2 + 9`. Weekly `7/2 + 4`. Costs `24 × 2,040` and `7.5 × 2,040`.
4. **Result.** On the monthly cycle a fact is on average **24 days** old when the report is published — **3.4286 weeks** — worth **USD 48,960** at the programme's cost of delay. On the weekly cycle with a 4-day lag it is **7.5 days** old, worth **USD 15,300**. The change is worth **USD 33,660** per detectable slip.
5. **Interpretation.** Three things follow, and the third is the one that changes behaviour. First, the monthly report that everybody treats as current describes a world nearly three and a half weeks gone, which is why a slip is so often reported as new when the people doing the work have known about it for a month — the report is not concealing it, the cycle is. Second, the lever asymmetry carries over exactly from Domain 3: a one-day cut in the **production lag** `G` saves a full day of staleness, while a one-day cut in the **reporting period** `R` saves only half a day. Compressing the production process is twice as effective per day as reporting more often, and it is the cheaper of the two to change — the same result Domain 3 obtained for paper lead times, and for the same reason. Third, and decisively: **a dashboard whose information age exceeds the interval within which the decision must change cannot inform that decision, at any level of presentation quality.** Meridian's clinic readiness interventions must be triggered at least two weeks before a scheduled go-live to be effective; a 24-day-old fact arrives after the window has closed, so the monthly report is, for that decision, decoration. The professional response is not a better chart. It is either a faster cycle for that specific measure, or an exception trigger that fires on the source system rather than waiting for the report — and the honest test to apply to any dashboard is to name the decision, name the window, compute `R/2 + G`, and compare.

Two cautions. `R/2 + G` assumes facts arise uniformly through the cycle, which is a reasonable default and wrong where events cluster at period boundaries (invoicing, sprint ends, gate submissions) — for those, model the actual arrival pattern rather than the average. And a shorter cycle imposes real production cost: at the blended rate of USD 110 per hour, a report taking 14 person-hours to produce costs **USD 1,540** each time, so moving from monthly to weekly adds **3.2857** productions a month (`30/7 − 1`) and **USD 5,060** of monthly effort. That is worth paying against USD 33,660 of avoided lateness on a single slip, and it must still be paid rather than assumed away.

14.2.2 The analytics ladder

Analytics initiatives are usually described by their technology and should be described by the question they answer, because the four questions differ in what they require and what they are worth. **Descriptive** — what happened? Requires only clean data and correct aggregation, and is where the great majority of realised value in delivery analytics sits, unglamorously. **Diagnostic** — why did it happen? Requires the ability to decompose a measure into contributions, which in turn requires the data class schedule of 14.1.2 to be coherent. **Predictive** — what is likely to happen? Requires history that is genuinely comparable to the future being predicted, which is a strong assumption on projects and a fatal one on novel work. **Prescriptive** — what should be done? Requires a model of the levers and of their costs, and it is where the accountability question becomes acute, because a recommendation presented in a system's voice is very hard for a governance body to interrogate (Domain 3, KA 3.A.2).

The test each rung must pass, and the ordering rule. For each rung: is there a named decision that would change if this output changed? An analytics product with no such decision is a cost. And the ordering rule is unglamorous and reliable: **the rungs must be climbed in order, because each depends on the one below being right.** A predictive model built on a register with a 6 % mapping defect rate predicts the defects. The standard sequencing error in digital delivery, stated plainly, is to buy the analytics before fixing the data — which is why 14.1 precedes 14.2 in this domain and should precede it in a programme's plan.

14.2.3 Digital twins and the fidelity that pays

Definition. A **digital twin** is a model of a physical or process asset that is kept synchronised with the real asset through a data connection, such that questions can be asked of the model instead of the asset. The synchronisation is what distinguishes a twin from a design model or a simulation; a model that is not updated from the asset is a drawing, whatever it is called, and much of what is marketed as a twin is a drawing.

Twins earn their place in delivery for three uses: **rehearsal** (testing a sequence off-line before committing an irreversible operation), **handover** (a validated as-built information set the receiving organisation can operate from, which is Domain 16, KA 16.1's readiness problem), and **training** (operators competent before go-live rather than during it). Their value in all three depends on one property, and it is the property least often measured: **fidelity**, the share of questions on which the twin's answer matches the asset's behaviour within a stated tolerance.

Auriga's control-system twin, priced. The twin costs **USD 180,000** to build and **USD 3,500** a month to keep synchronised over the **9** months to handover — **USD 211,500** all in. Its two benefit streams, both assessed locally: avoiding **3** of the 8 live outage windows required for sequence testing at **USD 62,000** each, worth **USD 186,000**; and operator training plus the as-built information handover, assessed at **USD 120,000**. Gross benefit at perfect fidelity is therefore **USD 306,000**, and the **breakeven fidelity** is `211,500 / 306,000 =` **69.12 %**. Measured against the commissioned plant on **40** test cases, the twin agreed within tolerance on **34** — a measured fidelity of **85.00 %**, giving an expected net value of `0.85 × 306,000 − 211,500 =` **USD 48,600**.

That is a positive answer with a thin margin, and the thinness is the lesson. Measured fidelity sits only **15.88 percentage points** above breakeven, and it was measured on 40 cases. If the next 40 cases produce 10 failures, pooled fidelity is `64/80 =` **80.00 %** and expected net value falls to **USD 33,300**; a pooled fidelity of 69.12 % takes it to zero. Two structural cautions must accompany any such calculation. Scaling benefit linearly with fidelity assumes the twin's failures are **distributed across decisions in proportion to their value**, and they are not: a twin typically fails on the novel, coupled, poorly instrumented cases, which are exactly the cases worth rehearsing, so linear scaling is optimistic and the failures should be examined individually rather than counted. And a twin's fidelity **decays** — the asset is modified, the model is not, and the synchronisation that defines a twin is the first budget line cut after handover, which is why 14.A.2's revalidation cadence applies to twins as much as to models.

14.2.4 Automation economics

Definition. Automation here means replacing a repeated human task with a machine process — rule-based scripting, robotic process automation, or an AI-assisted step. The decision is an economic one with a well-known form and a routinely omitted term.

The familiar form is a breakeven volume. With a fixed build-and-maintain cost F over the horizon considered, a manual unit cost m and an automated unit cost a:

```
n* = F / (m − a)
```

The omitted term is quality. An automated step and a manual step do not have the same **error escape rate**, and errors that escape cost money. Writing e_m and e_a for the share of units on which each route lets a material error through, and u for the cost of one escaped error, the honest unit costs are $m + e_m u$ and $a + e_a u$, and the **quality-adjusted breakeven volume** is:

```
n* = F / [ (m + eₘu) − (a + eₐu) ]
```

Three consequences follow immediately and are worth stating before the example. If automation **raises** the escape rate, the breakeven volume rises and may exceed any volume the organisation will ever process. If automation **lowers** it — which deterministic rule-checking often does, since machines do not tire — the breakeven volume falls below the naive figure, and the naive calculation understates the case. And if $(a + e_a u) > (m + e_m u)$, **no volume justifies the automation**: the breakeven volume does not exist, and a business case expressed in visible unit costs will nonetheless show a healthy one, which is the arithmetic behind Case study B.

WORKED EXAMPLE

Worked example 14.2.4 — automating Auriga's handover data checks.

1. **Setup.** Auriga's **60** handover packages each require **12** data-consistency checks against the asset register — **720** checks on this project. A check takes an engineer **24 minutes** at the blended engineering rate of **USD 130.625** per hour. An automated checker costs **USD 46,000** to build and **USD 1,400** a month to maintain, and processes a check for **USD 4.25** of exception-handling time. Consider a **24-month** horizon. Measured on a held-back set, the manual route lets a material error through on **1.5 %** of checks and the automated route on **2.5 %**; an escaped error costs **USD 1,800** (a re-issued package, a repeated site visit, a corrected handover record). The utility runs **6** comparable projects a year.
2. **Formula.** Naive $n^* = F/(m - a)$; quality-adjusted $n^* = F/[(m + e_m u) - (a + e_a u)]$, with F = build + maintenance over the horizon.
3. **Substitution.** $m = 130.625 \times 24/60 = 52.25$; $a = 4.25$. $F = 46{,}000 + 1{,}400 \times 24 = 79{,}600$. Naive $79{,}600/48.00$. Adjusted $m + 0.015 \times 1{,}800 = 79.25$; $a + 0.025 \times 1{,}800 = 49.25$; $79{,}600/30.00$.
4. **Result.** Build cost alone breaks even at **958.33 → 959** checks. Including 24 months of maintenance the naive breakeven is **1,658.33 → 1,659** checks. The **quality-adjusted** breakeven is **2,653.33 → 2,654** checks — exactly **1.60 times** the naive figure, since the unit saving falls from USD 48.00 to USD 30.00. Auriga alone generates **720** checks: below every one of those thresholds. Across the portfolio the volume is **4,320** a year, **8,640** over 24 months, giving a quality-adjusted net of **USD 179,600** against a naive claim of **USD 335,120**.
5. **Interpretation.** The decision reverses depending on the boundary drawn, and both answers are correct within their boundary — which is why the boundary is the first thing to state. **For Auriga alone the automation is indefensible**: 720 checks against a build-only breakeven of 959, so the project would spend USD 46,000 to save USD 34,560 of visible effort even before quality is considered. **For the portfolio it is strongly positive**, and the case must therefore be made, funded and owned at portfolio level (Domain 15), with the tool treated as an asset with a maintainer rather than as a project deliverable. A project-funded automation with no portfolio owner is the standard way an organisation acquires an unmaintained tool that quietly stops being verified.

The quality term is the professional content of this example. A 1.0-percentage-point increase in the escape rate — from 1.5 % to 2.5 %, a difference no business case would notice — raises the breakeven volume by 60 % and removes **USD 155,520** from the 24-month net. Two design responses follow. If the automated route can be made better than the human one rather than worse, the arithmetic transforms: at an automated escape rate of **0.5 %** the adjusted unit cost is **USD 13.25**, the unit saving is USD 66.00, and the breakeven falls to **1,206.06 → 1,207** checks, below the naive figure. Machines are genuinely better than tired humans at exhaustive rule-checking, so this is often achievable and almost never measured. Alternatively, place a cheap containment layer after the automated step so that u itself falls — Domain 9, KA 9.4.4's result that the economics reward **building a containment chain around machine output** rather than reviewing harder. Three cautions: the escape rates must be **measured against a held-back set**, never asserted, and re-measured on any change of rule set, model or artefact type (14.A.2); maintenance is a real recurring cost that is systematically omitted, and omitting it here would have made the breakeven 959 instead of 1,659; and the automated route's exception-handling cost a must include the human time actually spent on exceptions, which in practice is where an automation's promised saving most often disappears.

AI in this KA

Where it earns its place. Generating the exception logic for a dashboard — the rules that decide what is worth surfacing — for human ruling, which is a genuine strength because the rules are testable against history. Producing the first draft of an aggregation or transformation, with the lineage stated, for review. Detecting anomalies in high-volume operational data where the anomaly class is defined and the model's output is a prompt to investigate rather than a finding. Reconciling a twin's predictions against measured behaviour across a large test set and clustering the disagreements by type, which is how a fidelity measure becomes diagnostic rather than a percentage. Modelling breakeven volumes across dozens of automation candidates, which is deterministic and verifiable.

Where it must not go. It must not select the measures on a governance dashboard: that is a statement about what the organisation will manage, and it belongs to the accountable body (Domain 11, KA 11.2). It must not supply the escape rates, defect rates, consequence figures or fidelity tolerances that these calculations consume — asked for any of them a model will return a confident, well-formatted number with no provenance, and it will then anchor a real capital decision. It must not write the commentary that accompanies a status measure, for the smoothing reason Domain 11 established: a model asked to make a report clearer reliably produces a calmer one, and the qualifier carrying the warning is the first casualty. And it must not be the **only** step between a generated output and an irreversible operation, which is what a twin-based rehearsal exists to prevent.

Verification, concretely. Every automation carries a **measured** escape rate against a held-back set, re-measured on change, and a named owner for the measurement. Every dashboard measure carries its $R/2 + G$ information age on the specification, so the age is a published property rather than a discovery. A twin carries its fidelity, the test-case count it was measured on, the date of measurement and the tolerance used. And the breakeven arithmetic in any automation business case is reproduced by hand — it is one division — with the maintenance term and the quality term both visible, since a case presenting neither has not been reviewed.

KEY TERMS — KA 14.2

Term	Meaning
Information age	$R/2 + G$ — the expected staleness of a fact at publication, for reporting period R and production lag G; the registered `E[wait]` identity applied to information.
Production lag (G)	Elapsed time from data cut-off to publication; the more powerful of the two staleness levers, day for day.
Analytics ladder	Descriptive → diagnostic → predictive → prescriptive; each rung depends on the one below being right.
Digital twin	A model kept synchronised with a real asset through a data connection; without synchronisation it is a drawing.
Twin fidelity	The share of test questions on which the twin's answer matches the asset within a stated tolerance.
Breakeven fidelity	Twin cost ÷ gross benefit at perfect fidelity — the fidelity below which the twin destroys value.
Error escape rate (e)	The share of units on which a route lets a material error through; differs between manual and automated routes and must be measured.
Quality-adjusted breakeven volume	$F / [(m + e_m u) - (a + e_a u)]$ — the volume at which an automation pays once escaped errors are priced.

SAMPLE MCQS — KA 14.2

MCQ 14.2-A `[14.2.1 · Application]` A programme reports monthly with a 9-day production lag from data cut-off to publication. The expected age of a fact at publication is: - A. 9 days - B. 15 days - C. 24 days - D. 39 days

Rationale: `R/2 + G = 30/2 + 9 = 24` days (14.2.1, applying Domain 3's `E[wait]` identity). A counts only the lag; B only half the period; D adds the whole period to the lag.

MCQ 14.2-B `[14.2.1 · Analysis]` For that report, which change reduces information age more, and by how much: cutting the production lag by 3 days, or cutting the reporting period by 3 days? - A. the reporting period, by 3.0 days - B. the production lag, by 3.0 days — twice the 1.5-day saving from the period - C. both equally, by 3.0 days - D. both equally, by 1.5 days

Rationale: Age is `R/2 + G`, so a cut of `x` in `G` saves `x` while a cut of `x` in `R` saves `x/2` — the same asymmetry Domain 3, KA 3.2.3 found for paper lead times, and the lag is usually the cheaper lever to move.

MCQ 14.2-C `[14.2.4 · Application]` An automation costs USD 79,600 over the horizon. Manual unit cost is USD 52.25 and automated USD 4.25; escape rates are 1.5 % manual and 2.5 % automated with an escaped error costing USD 1,800. The quality-adjusted breakeven volume is closest to: - A. 959 units - B. 1,659 units - C. 2,654 units - D. 1,524 units

Rationale: Adjusted unit costs are `52.25 + 27.00 = 79.25` and `4.25 + 45.00 = 49.25`, so `79,600/30.00 = 2,653.33 → 2,654` (14.2.4). A uses the build cost alone with unadjusted units; B is the naive figure including maintenance; D divides the fixed cost by the manual unit cost instead of by the saving.

MCQ 14.2-D `[14.2.4 · Evaluation]` An automation's quality-adjusted automated unit cost exceeds its quality-adjusted manual unit cost. The correct conclusion is that: - A. the breakeven volume is very large but attainable at portfolio scale - B. no volume justifies the automation; the breakeven volume does not exist - C. the automation is justified if the visible unit saving is positive - D. the escape rates should be excluded as they are estimates

Rationale: With a negative unit saving every additional unit adds loss, so there is no crossover (14.2.4) — the position Case study B describes. C is exactly the error that produces it; D discards the term that decides the answer.

MCQ 14.2-E `[14.2.3 · Application]` A digital twin costs USD 211,500 and would deliver USD 306,000 of benefit at perfect fidelity. Its breakeven fidelity is: - A. 58.82 % - B. 69.12 % - C. 85.00 % - D. 113.71 %

Rationale: `211,500/306,000 = 69.12 %` (14.2.3). A divides the build cost alone by the benefit, omitting nine months of synchronisation; C is the measured fidelity, not the breakeven; D divides by one benefit stream only.

SELF-CHECK — KA 14.2

1. How old is a fact on a monthly report with a 9-day production lag, and what is it worth at Meridian's cost of delay? — 24 days (`30/2 + 9`), worth USD 48,960 at USD 2,040 a day; a weekly cycle with a 4-day lag gives 7.5 days and USD 15,300.
2. What distinguishes a digital twin from a design model, and what determines its value? — Synchronisation with the real asset; and fidelity, measured against the asset on a stated number of test cases with a stated tolerance, against a computed breakeven — 69.12 % for Auriga's twin.
3. Which term is systematically omitted from automation business cases, and what does it do? — The differential error escape rate. One percentage point of extra escape raised Auriga's breakeven volume by 60 %, and a negative quality-adjusted saving removes the breakeven altogether.

KNOWLEDGE AREA 14.3

AI use across the lifecycle, prompting and verification

Topics: 14.3.1 where AI earns its place across the lifecycle · 14.3.2 prompting, grounding and provenance · 14.3.3 verification-effort economics and the standard it derives · 14.3.4 the plausible wrong number.

14.3.1 Where AI earns its place across the lifecycle

The preceding thirteen domains have each stated, in their own subject matter, what AI may and may not do. Restating those positions here would be a defect. What this topic contributes instead is the **pattern** they collectively describe, because once the pattern is visible it can be applied to a use case nobody has yet written a rule for — which is the situation a leader is actually in.

Across every domain in this volume, the permitted uses fall into five classes and the prohibited uses into four. **The five permitted classes:** extraction (pulling structure out of unstructured material — a decision register from minutes, Domain 3, KA 3.3; requirements candidates from interview notes, Domain 5, KA 5.1); comparison at scale (reading many documents against each other for inconsistency — decision rights across a contract and a terms of reference, Domain 3, KA 3.1; field definitions across systems, 14.1.2); enumeration (generating candidates a human will rule on — risks, test cases, failure modes, improvement actions, Domains 8 and 9); deterministic modelling (running arithmetic a human specified across many combinations — latency and threshold models in Domain 3, breakeven volumes in 14.2.4); and drafting for review (producing a first version of a document whose author remains the named human, Domains 4, 5, 10, 11). Each is characterised by a verifiable output and a human ruling.

The four prohibited classes, and this is the more useful list because it generalises: attributable acts — anything that confers authority or discharges an obligation: approving a change, accepting a deliverable, signing a test record, granting a concession, authoring a decision record (Domains 3, 4, 5, 9). Judgements of consequence and appetite — thresholds, liability caps, target costs, quality scores, risk appetites, kill criteria; these are risk-appetite decisions belonging to accountable people (Domains 2, 3, 9, 10). Unprovenanced parameters — supplying the probability, detection rate, critical-path share, escape rate, reservation value or benefit attribution that a calculation then consumes; the model will supply a confident number and it will enter a real decision as though it were data (Domains 3, 5, 8, 10, 11 and every worked example in this domain). Inference about people — emotional state, engagement, personality, likely departure, cultural disposition, performance or potential, whether from telemetry, text or images; Domain 12 states the prohibition in its strongest form in this volume, and this domain does not soften it.

The single test that generates all four prohibitions, and the one to apply to a novel case: **could a named human be held to answer for this output as their own judgement, on evidence they can produce?** If yes, AI assistance is a drafting aid and the human remains the author. If no — because the output is the exercise of an authority, or because it rests on a parameter nobody can source, or because the subject is a person's interior state — the use is not permitted at any level of verification, because no amount of checking creates an accountable author.

Across the lifecycle, the same pattern maps to stages. In definition and selection, extraction and enumeration are strong and benefit attribution is prohibited (Domain 2; Domain 5, KA 5.3). In planning, deterministic modelling and scenario enumeration are strong while estimates and durations are not (Domains 6 and 7). In execution, extraction, anomaly detection and drafting are strong while approvals, acceptances and classifications are prohibited (Domains 4, 5, 9, 10). In closeout and benefits, clustering and summarisation are strong while the benefit judgement and the archive decision remain human (Domain 16). The prohibitions do not weaken as a project matures; the volume of permitted use grows, which is why the verification standard of 14.3.3 has to be a standing instrument rather than a mobilisation exercise.

14.3.2 Prompting, grounding and provenance

A prompt is not a conversational convenience; where its output informs a professional judgement, it is a **method**, and it acquires the obligations of a method: it is written down, versioned, reviewed and reproducible. That single reframing removes most of the practical problems organisations report with AI assistance, and it is resisted mainly because prompts feel like typing.

The four properties of a professional prompt. Grounding — the material the output must be derived from is supplied and identified, and the instruction states that nothing outside it may be used. An ungrounded prompt asks for recall, and recall is where fabrication lives. Constraint — the output's form is specified (fields, units, allowed values, "state 'not stated' where the source is silent"), because an unconstrained output cannot be checked mechanically. Refusal permission — the instruction explicitly permits, and expects, an answer of "the source does not say", which is the single most valuable line in a professional prompt because its absence guarantees that gaps are filled with plausible content. Provenance — each material assertion in the output cites the supplied source location it came from, which is what converts verification from re-reading everything to checking citations.

What must be recorded. For any AI-assisted output that will be relied on: the prompt version, the model and version, the grounding material and its version, the date, the named human who reviewed it, and the verification tier applied (14.3.3). This is not bureaucracy; it is the minimum that lets the question asked afterwards be answered. Domain 3, KA 3.3.4's finding applies unchanged: the retrospective question is never "was the output right?" but "was the decision reasonable on what was known at the time?", and only a versioned record answers it.

Two failure modes specific to prompting, both common and both cheap to prevent. **Instruction drift**: a prompt refined over many sessions in a chat window is not the prompt anyone thinks it is, and its output is not reproducible; the fix is that the operative prompt lives in a file with a version, not in a history. **Context contamination**: material from an unrelated matter remains in a session and influences an output, which is both a correctness problem and, where the material is confidential or personal, a disclosure problem — the boundary Domain 3, KA 3.A.2 requires to be stated before use rather than after an incident.

14.3.3 Verification-effort economics and the standard it derives

Domain 3, KA 3.A.2 required a **verification standard proportional to consequence** and asserted it without deriving it. Domain 9, KA 9.4.4 supplied one axis of the derivation: given a population of machine-produced items, how large a sample must be checked, from the breakeven error fraction $p^* = c/u$ and the bound a clean sample supports. This topic supplies the other axis, and it is the one an organisation must answer first: **for a given class of output, how deep should the check be?**

The structure. Verification is available at increasing depths. Each depth k has a cost per item v_k and a **detection rate** q_k — the share of material errors present that it finds. For an output class whose items contain a material error with probability p, and where an escaped error costs u:

```
expected cost at tier k  =  vₖ + p (1 − qₖ) u
```

Moving from tier $k - 1$ to tier k is worth doing when the extra cost is less than the extra expected loss avoided, $\Delta v_k < p \cdot \Delta q_k \cdot u$, which rearranges into the quantity a standard is written from — **the consequence at which the next tier begins to pay**:

$$u^*_k = \Delta v_k / (p \cdot \Delta q_k)$$

Three properties of that expression should be read before any numbers. The thresholds depend on Δv and Δq, the **increments**, not on the totals — so the natural but wrong comparison of a tier's whole cost against the whole consequence over-verifies, because every tier looks cheap against a large consequence. The thresholds are **inversely proportional to** p, so a class's required depth falls as the producing process improves, which is where the productivity gain of AI assistance actually lives. And p and q are **measured quantities**: a standard built on asserted ones is a standard built on nothing, which is why 14.A.2 makes their measurement a standing obligation.

WORKED EXAMPLE

Worked example 14.3.3 — deriving Meridian's verification standard, and pricing it.

1. **Setup.** Meridian's AI-assisted outputs are reviewed at one of five depths, costed at the blended programme rate of **USD 110 per hour**, with detection rates measured against a held-back set of seeded errors: tier 0 accept unchecked (v = 0, q = 0); tier 1 author scan, 6 minutes (**USD 11.00**, q = **0.35**); tier 2 independent review, 24 minutes (**USD 44.00**, q = **0.70**); tier 3 independent review with source trace, 66 minutes (**USD 121.00**, q = **0.90**); tier 4 two independent reviewers with reperformance, 150 minutes (**USD 275.00**, q = **0.97**). The measured material-error rate of the current model-and-prompt configuration is p = **0.12**. Monthly output volumes and assessed consequences per escaped error: internal meeting summaries **620** at **USD 150**; clinic readiness assessments **180** at **USD 480**; configuration field-mapping proposals **95** at **USD 1,200**; migration reconciliation scripts **40** at **USD 3,500**; baseline change impact assessments **6** at **USD 28,560** (two weeks of programme delay at the cost of delay of USD 14,280 per week).
2. **Formula.** $u^*_k = \Delta v_k / (p \cdot \Delta q_k)$ for each tier step; then, for any assignment of tiers to classes, total cost $= \Sigma\, n_i v_{k(i)} + \Sigma\, n_i\, p\, (1 - q_{k(i)})\, u_i$.
3. **Substitution.** $11/(0.12 \times 0.35)$; $33/(0.12 \times 0.35)$; $77/(0.12 \times 0.20)$; $154/(0.12 \times 0.07)$.
4. **Result — the standard.**

Step	Δv	Δq	Consequence at which it pays, u*	Applies from
0 → 1 author scan	USD 11.00	0.35	**USD 261.90**	any output whose escaped error costs more than 261.90
1 → 2 independent review	USD 33.00	0.35	**USD 785.71**	" more than 785.71
2 → 3 review with source trace	USD 77.00	0.20	**USD 3,208.33**	" more than 3,208.33
3 → 4 two reviewers, reperformance	USD 154.00	0.07	**USD 18,333.33**	" more than 18,333.33

Applying it: summaries (150) → **tier 0**; readiness assessments (480) → **tier 1**; mapping proposals (1,200) → **tier 2**; reconciliation scripts (3,500) → **tier 3**; change impact assessments (28,560) → **tier**

4. Monthly verification cost **USD 12,650**; expected escaped loss **USD 24,300.10**; **total USD 36,950.10**.

The three alternatives, on the same 941 outputs. **No verification:** USD 0 + **USD 72,571.20** = **USD 72,571.20**. **Uniform tier 2 on everything** — the intuitively "consistent" policy: USD 41,404 + USD 21,771.36 = **USD 63,175.36**. **Uniform tier 4 on everything** — the intuitively "safe" policy: USD 258,775 + USD 2,177.14 = **USD 260,952.14**. The tiered standard is the cheapest of the four, beating uniform tier 2 by **USD 26,225.26 (41.51 %)** and uniform tier 4 by a factor of **7.06**. 5. **Interpretation.** Start with the comparison that changes minds. Uniform tier 2 spends **3.27 times** as much on verification as the tiered standard — USD 41,404 against USD 12,650 — and buys a reduction in escaped loss of **USD 2,528.74**, or **10.41 %**. That is the arithmetic signature of an undifferentiated review policy: most of the money goes to items whose errors are cheap, because that is where most of the items are. The tiered standard is not a relaxation of uniform tier 2; it is a **reallocation** — it removes review from 620 low-consequence items and adds review depth to 46 high-consequence ones, and it is better on both cost and loss because the consequences differ by a factor of **190.4** (28,560 against 150) while the review costs differ by a factor of **25.0** (275.00 against 11.00).

The uncomfortable feature, stated plainly, because a governance body will find it. Under the tiered standard, **45.93 %** of all expected escaped loss — USD 11,160 of USD 24,300.10 — sits in the one class deliberately left unverified. That looks like negligence and it is not: putting the 620 summaries on tier 1 costs **USD 6,820** and saves **USD 3,906** of escaped loss, a **net loss of USD 2,914**, which is exactly what a consequence of USD 150 sitting below the USD 261.90 threshold means. The correct response to a large aggregate of cheap errors is not to check them; it is to **reduce `p` or `u`** — a better prompt, a template, a structural constraint, a cheap containment layer downstream (Domain 9, KA 9.4.4). Checking is the most expensive of the available remedies and the only one organisations reliably reach for.

The sensitivity that matters most is `p`, and it points somewhere unexpected. If the configuration improves so that the measured error rate falls from 0.12 to **0.04**, every threshold triples — to **785.71 · 2,357.14 · 9,625.00 · 55,000.00** — and the assignment shifts down: summaries and readiness assessments to tier 0, mappings to tier 1, scripts to tier 2, impact assessments to tier 3. Total cost falls from USD 36,950.10 to **USD 16,036.44**, a **56.60 %** reduction, of which the **verification bill** falls by **72.09 %** (USD 12,650 → USD 3,531). This is the honest form of the AI productivity claim: **the return on a better model is mostly a reduction in checking, not a reduction in drafting time** — and it is realisable only by an organisation that measures `p`, because an unmeasured error rate leaves the verification standard frozen at whatever it was set to on the day someone was nervous.

Four cautions. Every threshold scales with `1/p`, so `p` must be **measured per output class and per configuration**, and a single organisational `p` is as wrong as a single organisational quality target (14.1.4). Detection rates `q` must be measured against **seeded or held-back errors**, and Domain 9, KA 9.2.2's independence rule binds: two passes by the same model are one pass, and a model reviewing its own output is not a tier. Where the consequence is not cost-compensable — a safety-related output, a regulatory submission, a clinical decision support artefact — the class carries an **absolute** standard and this arithmetic sizes the effort rather than choosing the depth. And the standard must be **published and owned**, because its value lies in being applied consistently by people who are not doing the calculation each time; an unpublished standard collapses into individual nervousness, which is the condition it exists to replace.

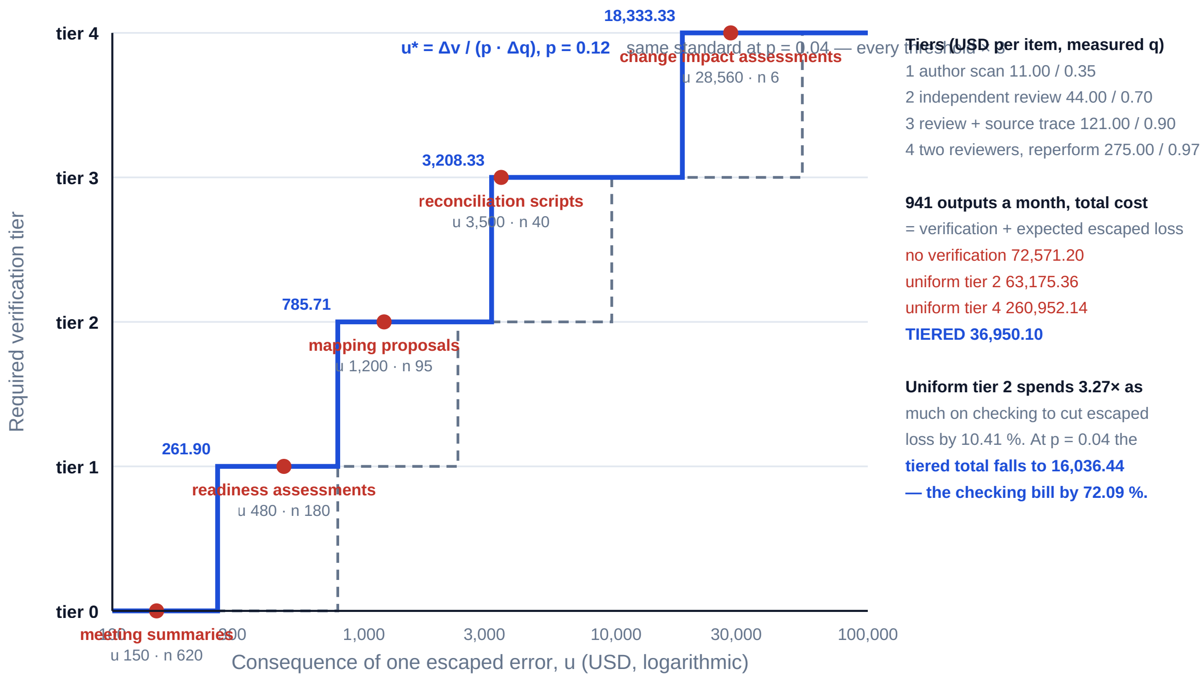


Fig 14.3.1 — The verification standard proportional to consequence

14.3.4 The plausible wrong number

Everything in 14.3.3 assumed a detection rate q. This topic is about what determines it, because the answer is counter-intuitive and it changes how review effort should be aimed: q **depends far more on the plausibility of the error than on its size.**

An obviously wrong number — a misplaced decimal, a units confusion, a total that fails to add — is caught by almost anyone who looks, because human review of quantitative material is largely a plausibility check. A plausible wrong number is caught by almost nobody, because there is nothing to notice. And machine-produced output is characteristically plausible: fluent regardless of correctness, internally consistent, correctly formatted, and derived by a process that produces well-shaped answers rather than hesitant ones. Domain 9, KA 9.4.4 made the point qualitatively — an unsure human writes hesitantly and a wrong model does not. Here it is priced.

WORKED EXAMPLE

Worked example 14.3.4 — two wrong forecasts on Auriga, one of them dangerous.

1. **Setup.** At week 13 Auriga reports `PV` **2,080,000**, `EV` **1,920,000**, `AC` **2,120,000** against a `BAC` of **4,000,000**, giving `CPI` **0.91** and `SPI` **0.92** (Domain 7). The CPI-based forecast is `EAC = BAC/CPI =` **USD 4,416,666.67**, a `VAC` of **(USD 416,666.67)**. Governance rule: a forecast overrun above **USD 350,000** obliges the sponsor to request supplementary funding. An AI-assisted forecasting step can fail in two ways. Obvious: a decimal slip reporting `EAC` **USD 43,200,000**, whose consequence if it escaped is assessed at **USD 900,000** of misdirected commitment. Plausible: a mis-weighted index producing `EAC` **USD 4,320,000** — an overrun of **USD 320,000**, below the threshold, so no funding request is made and the request is instead made **6 weeks** later at a cost of delay of **USD 45,000** per week, i.e. **USD 270,000**. Measured on a held-back set, each failure type occurs on **10 %** of forecast cycles; review detects the obvious error with probability **0.98** and the plausible one with probability **0.25**.
2. **Formula.** Expected escaped cost `= p (1 − q) u` per failure type. The asymmetry ratio is `[(1 − q_p) u_p] ÷ [(1 − q_o) u_o]`.
3. **Substitution.** Obvious `0.10 × 0.02 × 900,000`. Plausible `0.10 × 0.75 × 270,000`.
4. **Result.** The obvious error carries an expected escaped cost of **USD 1,800**. The plausible error carries **USD 20,250 — 11.25 times** as much, from a consequence **3.33 times smaller**. At equal consequence the ratio would be the ratio of escape probabilities, `0.75/0.02 =` **37.50**.
5. **Interpretation.** The generalisable statement is worth memorising in this form: **the expected damage of an error is its consequence multiplied by its probability of not being caught, and plausibility drives the second term across two orders of magnitude while magnitude drives only the first.** A 10× decimal slip is 3.33 times more consequential and 37.5 times more likely to be caught, so it is 11.25 times less dangerous. Everything a reviewer's instinct rewards — noticing the number that looks wrong — is aimed at the cheap failure mode.

Three practical consequences follow, and they are what this topic is for. **Review must be aimed at plausibility, not at magnitude.** A reasonableness check is a tier-1 control against the obvious class and close to worthless against the plausible one; the control that works is **reperformance** — recomputing the number from its inputs by an independent route. On Auriga, a reperformance step that raises `q_p` from **0.25** to **0.85** takes about **40 minutes** at the blended engineering rate, **USD 87.08** per forecast cycle, and cuts the expected escaped cost from USD 20,250 to **USD 4,050** — a saving of **USD 16,200**, a return of **186 times** its cost. Its breakeven is a detection improvement of **0.3225 percentage points**, so on this class of output reperformance pays if it works at all. **Presentation determines reperformability.** A forecast published as a single number cannot be reperformed; the same forecast published with `EV`, `AC`, `BAC` and the stated method can be reperformed in two minutes, so how a number is presented is a control, and Domain 11, KA 11.2's reporting standard should be read as carrying this requirement. **And the small, plausible error near a threshold is the most dangerous object in a reporting pack**, because its consequence is not proportional to its size: Auriga's plausible error is USD 96,666.67 of forecast — **2.42 %** of `BAC` — and it costs USD 270,000 because it lands on the wrong side of a governance trigger. Any measure with a threshold attached should be reperformed within a band around that threshold regardless of the tier its consequence would otherwise attract.

A related trap worth flagging, since it manufactures plausible errors from nothing: **rounding before dividing.** Computing `EAC` from the reported `CPI` of 0.91 rather than the underlying 0.905660 gives **USD 4,395,604.40** instead of USD 4,416,666.67, understating the overrun by **USD 21,062.27** —

5.05 % of the variance — from an operation that looks like tidiness. Full precision internally, rounding only at display, is not pedantry; it is the difference between a reperformable number and a plausible one.

AI in this KA

Where it earns its place. Building the tier table itself across many output classes, which is one division repeated and exactly checkable. Maintaining the measured error-rate and detection-rate tables from a seeded-error programme, and flagging classes whose measurements have gone stale (14.A.2). Producing a reperformance of a computed figure by an **independent route** — recomputing `EAC` from `EV`, `AC` and `BAC` where the original came from a different path — which is a legitimate and valuable second pair of eyes precisely because the arithmetic is deterministic. Checking that every material assertion in an AI-assisted document carries a provenance citation to the supplied grounding material, and listing those that do not, which is the mechanical part of verification and the part humans skip. Auditing prompt and model versions against the record.

Where it must not go. It must not set the verification tier for its own output — the decision whose independence the entire argument of 14.3.3 rests on, and Domain 9, KA 9.4's prohibition verbatim. It must not supply `p`, `q` or `u`; those are measurements and assessments with owners, and a model asked for them returns numbers that look exactly like data. It must not be the reviewer of record at any tier: a tier is a human act with a name attached, and a model-performed check is a tool used within a tier, whose contribution is included in the measured `q` of that tier and never credited separately. And a model's own expression of confidence must not be read as a detection rate or an error rate — it is neither, and treating it as either silently replaces a measurement with a sentiment.

Verification, concretely. Error rates and detection rates come from a **seeded-error programme** with a stated method, sample size and date, re-run on any change of model, prompt, grounding material or output class, and owned by a named person. Every tier-3 and tier-4 review records the reviewer, the date, the prompt and model versions, and what was reperformed. Reperformance is by an **independent route**, not a re-reading. And the tier table is recomputed by hand whenever `p` changes materially, since a standard nobody can reproduce will not survive its first challenge.

KEY TERMS — KA 14.3

Term	Meaning
Grounding	Supplying and identifying the material an output must be derived from, with nothing outside it permitted.
Refusal permission	An explicit instruction that "the source does not say" is an acceptable and expected answer.
Provenance citation	Each material assertion in an output referenced to the supplied source location it came from.
Detection rate (q)	The share of material errors present that a given review depth finds; a measured quantity, never asserted.
Material error rate (p)	The share of items of a class containing a material error before verification; measured per class and per configuration.
Verification tier	A defined review depth with a cost per item v and a measured detection rate q, performed by a named human.
Verification tier threshold	$u^*_k = \Delta v_k/(p \cdot \Delta q_k)$ — the escaped-error consequence at which the next tier begins to pay.
Reperformance	Recomputing a result from its inputs by an independent route; the only effective control against plausible error.
Escape-cost asymmetry	$(1 - q\_p)u\_p \div (1 - q\_o)u\_o$ — why a plausible wrong number out-damages an obviously wrong one.
Attributable act	An output that confers authority or discharges an obligation, and therefore cannot be produced by a tool at any verification depth.

SAMPLE MCQS — KA 14.3

MCQ 14.3-A `[14.3.3 · Application]` Tier 2 costs USD 44.00 per item with a detection rate of 0.70; tier 3 costs USD 121.00 with 0.90. The measured material-error rate is 0.12. The consequence at which tier 3 begins to pay is: - A. USD 712.96 - B. USD 1,120.37 - C. USD 3,208.33 - D. USD 5,041.67

Rationale: $u^* = \Delta v/(p \cdot \Delta q) = 77/(0.12 \times 0.20) = 3{,}208.33$ (14.3.3). A divides the increment by the total detection rate; B divides the total cost by the total detection rate; D divides the total cost by the increment. Only the increment-over-increment form answers "is the next tier worth it".

MCQ 14.3-B `[14.3.3 · Evaluation]` A tiered standard costs USD 12,650 in verification with USD 24,300 of expected escaped loss. Uniform tier 2 on the same outputs costs USD 41,404 with USD 21,771 of escaped loss. The best characterisation is that uniform tier 2: - A. is safer, since escaped loss is lower - B. spends 3.27 times as much on verification to reduce escaped loss by 10.41 %, and is USD 26,225 worse in total - C. is equivalent, since both are defensible policies - D. is cheaper, because one policy is simpler to administer

Rationale: Compare totals: 36,950 against 63,175 (14.3.3). A looks at one term of two; the extra review lands mostly on items whose errors are cheap, which is the signature of an undifferentiated policy.

MCQ 14.3-C `[14.3.3 · Analysis]` The measured material-error rate for a class falls from 0.12 to 0.04. Every verification threshold: - A. falls by a factor of three - B. rises by a factor of three - C. is unchanged, since the tiers' costs and detection rates are unchanged - D. rises by a factor of nine

Rationale: `u* = Δv/(p·Δq)` is inversely proportional to `p` (14.3.3), so thresholds triple and more classes fall into lighter tiers — which is where the productivity gain of a better configuration is actually realised. C confuses the tier definitions with the threshold.

MCQ 14.3-D `[14.3.4 · Analysis]` An obvious error occurs on 10 % of cycles, costs USD 900,000 if it escapes and is detected with probability 0.98. A plausible error occurs equally often, costs USD 270,000 and is detected with probability 0.25. Which is more dangerous, and by how much? - A. the obvious error, by 3.33 times, because its consequence is larger - B. the plausible error, by 11.25 times - C. the plausible error, by 37.50 times - D. they are equally dangerous once probability is taken into account

Rationale: Expected escaped costs are `0.10 × 0.02 × 900,000 = 1,800` and `0.10 × 0.75 × 270,000 = 20,250` (14.3.4). C is the escape-probability ratio alone, which would be the answer only at equal consequence; A ranks by consequence and ignores detectability.

MCQ 14.3-E `[14.3.4 · Application]` The control that most raises the detection rate for plausible numerical error in an AI-assisted forecast is: - A. a reasonableness check against expectation - B. reperformance from the inputs by an independent route - C. asking the model to check its own output - D. increasing the number of reviewers reading the same document

Rationale: A plausible error offers nothing to notice, so a plausibility check is near-worthless against it (14.3.4). C is not an independent layer (Domain 9, KA 9.2.2); D multiplies the same ineffective check.

SELF-CHECK — KA 14.3

1. State the verification tier threshold and what it is proportional to. — $u^*_k = \Delta v_k/(p \cdot \Delta q_k)$: the increment of cost over the increment of detection and the error rate. It is inversely proportional to `p`, so a better configuration raises every threshold.
2. Why is a plausible wrong number more dangerous than an obviously wrong one? — Expected damage is consequence × probability of not being caught, and plausibility moves the second term by two orders of magnitude. On Auriga the plausible error costs 11.25 times the obvious one despite a consequence 3.33 times smaller.
3. What single test generates every prohibition on AI use in this volume? — Whether a named human could be held to answer for the output as their own judgement on evidence they can produce. If not — an attributable act, an appetite judgement, an unprovenanced parameter, or inference about a person — no depth of verification makes it permissible.

KNOWLEDGE AREA 14.4

Explainability, bias, human accountability, cybersecurity and privacy

Topics: 14.4.1 explainability as a decision requirement · 14.4.2 bias and differential error · 14.4.3 human accountability and the AI use register · 14.4.4 cybersecurity and privacy in a digital delivery estate.

14.4.1 Explainability as a decision requirement

Definition. Explainability, for the purposes of delivery decisions, is the property that a named human can state why an output says what it says, in terms the affected party can engage with, sufficiently to defend the decision and to permit it to be contested.

The definition deliberately locates explainability in the **decision**, not in the model. Technical interpretability — which features drove an output, how sensitive it is to inputs — is useful and is not the requirement. The requirement is set by what the decision must survive: an appeal, a debrief, an audit, a regulator's question, a court. Domain 10, KA 10.2 already applies the strict form of this to tender evaluation, where an AI-generated quality score is prohibited precisely because nobody can explain it in the terms the criterion was written in. That is the general rule: **the explanation must be in the vocabulary of the obligation, not of the model.**

The three levels a leader should distinguish, because they have different costs and different triggers. No explanation required — the output informs an internal, reversible, low-consequence choice; most permitted AI use sits here, and demanding explainability of all of it is how organisations make responsible AI unaffordable. Explanation required — a person affected by the decision, or a body reviewing it, is entitled to know its basis: a supplier not shortlisted, a change not approved, a clinic prioritised for support ahead of another. Here the requirement is satisfied by the human's reasoning, evidenced, with the AI output as one input — which is the Domain 1 accountability position expressed as a documentation duty. Contestability required — the affected party must be able to challenge the decision and have the challenge examined by a human with authority to change it. This is the level at which model-derived scoring becomes untenable, because a challenge that cannot be answered in the criterion's own terms cannot be examined.

Explainability obligations in law and regulation are **jurisdiction-specific and moving**, and nothing in this book should be read as stating them. Regimes may be named accurately as reference points — the European Union's AI Act as a risk-tiered regulatory approach, data-protection regimes such as the General Data Protection Regulation as sources of rights concerning automated decisions, ISO/IEC 42001 as a management-system standard for AI, ISO/IEC 23894 as guidance on AI risk management, and the NIST AI Risk Management Framework as a voluntary framework — each applying on its own terms, in its own territory, to its own defined scope. The professional discipline is to establish the applicable obligations for the specific decision in the specific place with qualified advice, and to design to the **strictest** applicable level where a programme spans jurisdictions, because the alternative is designing twice.

14.4.2 Bias and differential error

Definition. In this context **bias** means a **systematic difference in error rates between identifiable groups**, such that the burden of the model's mistakes falls unevenly. It is a measurable property of a deployed system, and it is measurable only if someone measures error rates by group rather than in aggregate.

That last clause is the whole practical difficulty. An aggregate accuracy figure cannot reveal differential error, and it is the figure that is always reported.

WORKED EXAMPLE

Worked example 14.4.2 — Meridian's readiness prioritisation, by clinic type.

1. **Setup.** Meridian scores its **40** clinics for risk of failing go-live readiness and sends a support team to those flagged. Of **24 urban** clinics, **9** turned out to be genuinely at risk and the model flagged **8** of them. Of **16 rural** clinics, **11** were at risk and the model flagged **6**. Planned support costs **USD 1,900** per clinic; a clinic that is at risk and not flagged needs emergency remediation at **USD 6,400**.
2. **Formula.** Recall by group = flagged at-risk ÷ actual at-risk. Excess cost per missed clinic = emergency − planned. Group burden = missed × excess, and per clinic of the group.
3. **Substitution.** Urban `8/9`; rural `6/11`; overall `14/20`. Excess `6,400 − 1,900 = 4,500`. Missed: urban `1`, rural `5`.
4. **Result.** Overall recall **70.00 %**. Urban recall **88.89 %**; rural recall **54.55 %** — a gap of **34.34 percentage points**. Total excess cost **USD 27,000**, of which rural clinics bear **83.33 %** while being **40.00 %** of the estate. Per clinic of each group the excess is **USD 187.50** urban and **USD 1,406.25** rural — a ratio of **7.50**.
5. **Interpretation.** The aggregate figure — 70 % recall — is the only number that would normally have been reported, and it describes neither group. It is the arithmetic equivalent of Domain 9, KA 9.4.2's mean-of-yields problem: a summary statistic that no member of the population experiences. The measurement that matters is the **disaggregated** one, and the only way to obtain it is to define the groups in advance and measure by group as a standing practice, because the comparison cannot be reconstructed from an aggregate afterwards.

Correcting it is also, here, economically obvious, which is worth showing because the debate is usually conducted as though fairness and cost were opposed. Lowering the flagging threshold for rural clinics raises rural recall to **10 of 11 (90.91 %)** at the cost of two additional false positives — two clinics receiving planned support they did not need. Four fewer missed clinics avoids `4 × 4,500 =` **USD 18,000** of emergency remediation for **USD 3,800** of extra planned support: a net **USD 14,200** and a return of **4.74 times** the cost. The differential was not a price the programme was paying for efficiency; it was simply a defect nobody had measured.

Four cautions, and the first is the most important in the book. **These units are clinics.** Where the units are people, differential error rates raise legal and ethical questions that are not expected-value questions at all, the applicable obligations differ by jurisdiction and by the characteristic concerned, and Domain 12's prohibition on individual inference applies in full — nothing in this arithmetic licenses building such a model about people, and this volume does not endorse it. Second, the **base rates differ** between the groups here — 37.50 % of urban clinics were at risk against 68.75 % of rural ones — and where base rates differ it is mathematically impossible to equalise every error measure at once: equalising recall and equalising the share of flags that prove correct are different targets that cannot both be met, so the choice of which measure to equalise is a **value judgement that must be made explicitly by an accountable person** and recorded, not settled by whichever metric the tool happens to display. Third, the groups must be **defined in advance**: choosing the grouping after seeing the results permits any conclusion, and the honest practice is to name the groupings that matter — geography, size, deprivation, language, service type — before deployment. And fourth, **the cause is usually the data, not the model**: rural clinics were under-represented in the historical records the model learned from, which is a data-coverage defect (14.1.4) presenting as a model defect, and re-tuning a threshold treats the symptom while leaving the coverage gap to reappear in the next model built on the same register.

14.4.3 Human accountability and the AI use register

Domain 1 established the principle and Domain 3, KA 3.A.2 required four governance instruments. This topic makes the first of them operational, because the register is what turns a principle into something an auditor can test.

The AI use register, one row per use, with the columns that make it testable rather than descriptive:

Column	Why it is there
Use and output class	What the tool produces, specifically enough to attach a consequence to.
Decision informed	The named decision the output feeds. A use that informs no decision is a cost.
Accountable person	The named human answerable for the output — Domain 1's test, applied to tools.
Consequence per escaped error u	Assessed, owned; the input that sets the verification tier.
Measured p **and** q**, with date and method**	The measurements that make the tier defensible, and that go stale.
Verification tier applied	From 14.3.3's standard, with the reviewer recorded per item at tiers 3–4.
Model, prompt and grounding versions	Reproducibility, and the retrospective question of 14.3.2.
Data and confidentiality boundary	What may be supplied to the tool, stated before use, not after an incident.
Prohibited adjacent uses	The uses that must not creep in — the scope creep of AI adoption, and it is real.

Three accountability failures the register exists to prevent. The **unattributed output**: a document, register or analysis in circulation with no author who can answer for it, which is Domain 1's accountability-without-a-holder defect and which Domain 3 prohibits in anything relied upon. **Scope creep of a permitted use**: a tool approved to extract requirements candidates that begins drafting requirements, or one approved to summarise minutes that begins drafting the decision record — a change of class, not of degree, and the register's last column is the control. And **accountability laundering**, the most corrosive: presenting a model's output as a neutral input in order to avoid owning the judgement it embodies. Its diagnostic is simple and worth asking in a governance meeting: if this output were wrong, whose judgement was wrong? If no one can answer, the output is not governed, and Domain 3's duty on the receiving body applies — ask what the input was, who verified it, and what would change the conclusion.

Two obligations that sit above the register. Where a person is materially affected by a decision an AI output informed, the affected person's route of challenge must reach a human with authority to change it (14.4.1). And the **workforce obligation**: staff must be told what tools are in use on their work, must not be evaluated by inference from telemetry (Domain 12), and must be able to decline to sign for an output they have not been given the time or the information to verify — the last being the point at which a verification standard either is real or is a document, because a standard that cannot be met in the time allowed simply relocates accountability onto whoever signs.

14.4.4 Cybersecurity and privacy in a digital delivery estate

A programme's digital estate now holds its commercial position, its supplier terms, its vulnerabilities, its people's details and, increasingly, whatever has been supplied to third-party tools. Domain 8 treats risk generally and Domain 10 supplier risk; the specific obligations here are the ones a delivery leader owns rather than delegates to a security function.

Four that are non-delegable. Data minimisation by design — the estate holds what the project needs, not what happened to be available; Meridian's decision to keep patient clinical data out of the delivery CDE (14.1.3) is the model, and it reduces the security problem rather than defending it. Access on the data class schedule — 14.1.2's access column implemented as permissions, reviewed on movers and leavers, which is where real breaches begin. A stated boundary for third-party tools — what may be supplied to which tool, decided before use; the commonest breach in AI adoption is not an attack but a disclosure by a well-meaning person into a service whose terms nobody read. Continuity of the environment itself — the delivery estate is now a single point of failure for the delivery, and its loss is a delivery risk with a cost of delay attached, not merely an IT incident.

Reference points may be named: ISO/IEC 27001 for information security management, ISO/IEC 27701 for privacy information management, IEC 62443 for industrial automation and control system security (directly relevant to Auriga's operational technology), and data-protection regimes that apply on their own terms in their own territories. None of them decides the spend. The arithmetic below does part of it, and its limits matter as much as its results.

WORKED EXAMPLE

Worked example 14.4.4 — sizing two controls on Meridian's CDE.

1. **Setup.** Meridian assesses the annual probability of a credential-compromise incident affecting its CDE at **0.08**, with an assessed impact of **USD 480,000** (incident response, remediation, delivery disruption, notification effort). Two controls are proposed. **A — identity hardening** (single sign-on, multi-factor authentication, privileged access review): **USD 26,000** a year, assessed to reduce the probability to **0.02**. **B — segmentation and field-level minimisation**: **USD 14,000** a year, assessed to reduce the impact to **USD 260,000** with the probability unchanged.
2. **Formula.** Expected annual loss `EAL = P × impact`. Total position = control cost + residual `EAL`. For two controls together, the reductions apply to their respective terms and therefore **multiply** rather than add.
3. **Substitution.** None `0.08 × 480,000`. A `26,000 + 0.02 × 480,000`. B `14,000 + 0.08 × 260,000`. Both `40,000 + 0.02 × 260,000`.
4. **Result.**

Option	Control cost	Residual EAL	Total	Loss avoided
Do nothing	0	38,400	**38,400**	—
A identity hardening	26,000	9,600	**35,600**	28,800
B segmentation and minimisation	14,000	20,800	**34,800**	17,600
A and B together	40,000	5,200	**45,200**	33,200

B alone is the lowest total position, at USD 34,800. Buying both is **worse than buying either**, and worse than doing nothing. 5. **Interpretation.** Three results, each of which is routinely got wrong. First, **control benefits are sub-additive.** Summing the two avoided losses gives USD

46,400; the true combined avoided loss is **USD 33,200**, because the second control acts on a residual the first has already reduced. A business case that adds them overstates the combined benefit by USD 13,200 and, here, converts a value-destroying package into an apparently attractive one. Second, **the impact-reducing control is better value than the probability-reducing one** at these parameters — USD 1.26 of avoided loss per dollar spent against USD 1.11 — and this is the general tendency worth carrying: probability reduction is bought against an uncertain frequency, while impact reduction (segmentation, minimisation, tested recovery) works on every incident that does occur, including the ones the probability estimate did not anticipate. A programme that buys only probability reduction has bought the more speculative half.

Third, and this is the professional caution rather than the result: **the answer is fragile in exactly the place the arithmetic is weakest.** Setting the two totals equal gives a crossover at an incident probability of **8.57 %** — above it, identity hardening wins; below it, segmentation wins. The assessed probability is 8.00 %, which sits **0.57 percentage points** below a crossover that no organisation can estimate to that precision. The honest conclusion is therefore not "buy B"; it is that **the two controls are economically indistinguishable at any probability anyone can defend**, and the choice should be made on grounds the arithmetic cannot see: implementation risk, effect on the ability to detect an incident at all, obligations under applicable regimes, and the fact that minimisation reduces the scope of a future incident rather than its price. Doing nothing, by contrast, is beaten by segmentation above an incident probability of **6.36 %** and by identity hardening above **7.22 %**, so at the assessed 8.00 % both dominate it and inaction is not a serious option — which is the robust part of this analysis, and the part worth taking to the board. And the frame itself has a boundary: expected annual loss is a legitimate test where consequences are cost-compensable, and it is the **wrong test** where the consequence is a statutory penalty, a licence condition, a duty of confidence or harm to a person — there the control is a requirement whose cost is sized, not an investment whose return is computed, and presenting such a control as failing a cost test is a category error a leader should be able to name in the room.

AI in this KA

Where it earns its place. Measuring error rates by group across a large deployment, once the groups are defined by a human, and flagging where a differential exceeds a stated tolerance — mechanical, high-volume, exactly checkable, and the measurement nobody has time to do. Classifying data holdings against the data class schedule to find minimisation opportunities and misplaced records. Detecting anomalous access patterns as a **prompt to investigate**, never as a finding about a person. Drafting the AI use register's first pass from a tool inventory, for human completion of the accountability and consequence columns. Testing a set of decisions for whether the recorded reasoning would satisfy 14.4.1's explanation level.

Where it must not go. It must not generate the explanation for a decision after the fact — a model asked why a decision was made will produce a fluent, plausible rationalisation, and a retrospective rationalisation presented as the decision's basis is worse than no explanation because it is evidence of a reasoning process that did not occur. It must not choose which fairness measure to equalise, which is a value judgement with an accountable owner. It must not make an access, security or privacy decision, and it must not be given the data whose exposure is the risk being managed — the boundary must be stated before use. It must not perform inference about individuals in any of the forms Domain 12 prohibits, and monitoring staff activity as a productivity proxy remains prohibited here as there. And it must not be the route through which an affected person's challenge is answered: contestability requires a human with authority to change the decision.

Verification, concretely. Group definitions and the fairness measure chosen are recorded with the accountable person's name and date, before deployment. Differential error measurements state the group sizes, because a 16-clinic group supports a much weaker inference than a 24-clinic one and the gap must be read with that in mind. Explanation levels are assigned per decision class and tested by asking a reviewer to reconstruct the reasoning from the record alone. Security and privacy assessments are performed by qualified people against the applicable regime, with the arithmetic here used to size effort rather than to authorise an exception. And the register is audited on a stated cadence against the actual tool estate, since the failure it exists to catch — a permitted use quietly becoming a prohibited one — is invisible from inside the team using it.

KEY TERMS — KA 14.4

Term	Meaning
Explainability (decision sense)	The property that a named human can state why an output says what it says, in the vocabulary of the obligation.
Contestability	The affected party's ability to challenge a decision and have it examined by a human with authority to change it.
Differential error	A systematic difference in error rates between identifiable groups; measurable only by disaggregated measurement.
Recall (by group)	Flagged at-risk cases ÷ actual at-risk cases within a group.
Base-rate difference	Unequal prevalence between groups, which makes simultaneous equalisation of all error measures impossible.
AI use register	The row-per-use record of use, decision informed, accountable person, consequence, measurements, tier, versions and boundary.
Accountability laundering	Presenting a model's output as a neutral input to avoid owning the judgement it embodies.
Expected annual loss (`EAL`)	Incident probability × assessed impact; the residual after a control is the basis of comparison.
Sub-additivity of controls	Combined avoided loss is less than the sum of individual avoided losses, because reductions multiply.
Data minimisation	Holding only what the project needs, which reduces the security and privacy problem rather than defending it.

SAMPLE MCQS — KA 14.4

MCQ 14.4-A `[14.4.2 · Application]` Of 24 urban clinics, 9 were at risk and 8 were flagged; of 16 rural clinics, 11 were at risk and 6 were flagged. Rural recall is: - A. 37.50 % - B. 54.55 % - C. 68.75 % - D. 70.00 %

Rationale: `6/11 = 54.55 %` (14.4.2). A divides flagged by all rural clinics; C is the rural base rate `11/16`; D is the aggregate recall `14/20`, which describes neither group.

MCQ 14.4-B `[14.4.2 · Evaluation]` Lowering the rural flagging threshold would raise rural recall from 6 of 11 to 10 of 11, adding two false positives at USD 1,900 each and avoiding four emergency remediations whose excess cost is USD 4,500 each. The correct conclusion is: - A. it should not be done, because false positives rise - B. it is worth USD 14,200 net, a return of 4.74 times the extra cost - C. it is worth USD 18,000, the emergency cost avoided - D. it cannot be evaluated economically

Rationale: `4 × 4,500 − 2 × 1,900 = 14,200`, and `18,000/3,800 = 4.74` (14.4.2). C omits the extra support cost; the differential here was a defect, not a price paid for efficiency.

MCQ 14.4-C `[14.4.4 · Analysis]` Control A cuts incident probability from 0.08 to 0.02 at USD 26,000 a year; control B cuts impact from USD 480,000 to USD 260,000 at USD 14,000 a year. Which option gives the lowest total position? - A. both controls, total USD 45,200 - B. control A alone, total USD 35,600 - C. control B alone, total USD 34,800 - D. neither, total USD 38,400

Rationale: Totals are control cost plus residual `EAL` (14.4.4). Both controls cost more than they avoid together because their reductions multiply — the sub-additivity result — so A is the trap the example exists to expose.

MCQ 14.4-D `[14.4.4 · Evaluation]` For those two controls, the individual avoided losses are USD 28,800 and USD 17,600. The combined avoided loss is: - A. USD 46,400 - B. USD 33,200 - C. USD 28,800 - D. USD 38,400

Rationale: Combined residual `EAL` is `0.02 × 260,000 = 5,200`, so avoided is `38,400 − 5,200 = 33,200` (14.4.4). A adds the two individual figures and overstates by USD 13,200, which is the error that converts a value-destroying package into an attractive one.

MCQ 14.4-E `[14.4.1 · Analysis]` A supplier not shortlisted asks why. The applicable explainability requirement is best met by: - A. disclosing the model's feature importances - B. the evaluator's recorded reasoning against the published criteria, with any AI output identified as one input - C. a statement that the process was automated and consistent - D. re-running the model with the supplier present

Rationale: The explanation must be in the vocabulary of the obligation — the published criteria — and the accountable human's reasoning is what is owed (14.4.1, with Domain 10, KA 10.2). A explains a model, not a decision; C is the answer that converts a substantive objection into a legitimacy grievance.

SELF-CHECK — KA 14.4

1. Why can an aggregate accuracy figure not establish that a model is unbiased? — Because differential error is invisible in aggregate: Meridian's 70.00 % recall concealed 88.89 % urban and 54.55 % rural, a 34.34-point gap, with rural clinics bearing 83.33 % of the excess cost while being 40.00 % of the estate.
2. Why is buying two security controls sometimes worse than buying one? — Because their reductions multiply rather than add: Meridian's combined avoided loss is USD 33,200 against a naive sum of USD 46,400, and the two together total USD 45,200 against USD 34,800 for the cheaper control alone.
3. What does the AI use register have to contain to satisfy Domain 1's accountability test? — A named accountable human per use, the decision informed, the assessed consequence, the measured error and detection rates with their dates, the verification tier applied, the model, prompt and grounding versions, the data boundary, and the adjacent uses that are prohibited.

Advanced topics — Domain 14

14.A.1 The consolidated position across Domains 1 to 13

The volume's per-domain AI treatments are not thirteen policies; they are one policy applied thirteen times, and consolidating them makes both the consistency and the two genuine tensions visible. The table below is the map, by KA, and it is the artefact to put in front of a governance body that asks "what is our position on AI?"

Domain	Permitted class, and where	Prohibited, and why
1 The profession	— (establishes the principle)	Any delegation of the obligation to answer (KA 1.2, 1.4)
2 Strategy and selection	Enumeration of options; scoring arithmetic	Benefit attribution; kill criteria (KA 2.3, 2.4)
3 Governance	Comparison of authority documents; latency modelling (KA 3.1, 3.2)	Decisions; thresholds; authorship of the decision record (KA 3.1, 3.3)
4 Integration	Interface enumeration; register reconciliation (KA 4.2, 4.3)	Tailoring approval; baseline entry; schedule impact estimates (KA 4.1, 4.4)
5 Scope and requirements	Extraction of requirement candidates; traceability gap detection (KA 5.1, 5.3)	Authoring requirements; boundary decisions; acceptance (KA 5.2, 5.4)
6 Planning	Scenario enumeration; deterministic network arithmetic	Durations and logic without a planner's ruling
7 Cost and commercial	Blended-rate and forecast arithmetic	Estimates; contract terms
8 Risk	Risk candidate enumeration; sensitivity modelling	Probabilities and impacts from plausibility (KA 8.2)
9 Quality	Data profiling; test-case drafting; yield arithmetic (KA 9.4)	Test results; inspection records; concessions; root-cause conclusions; its own sample size (KA 9.2, 9.3, 9.4)
10 Procurement	Bid comparison mechanics; clause comparison (KA 10.1, 10.2)	Quality scores; liability caps; entitlement opinions; post-opening reweighting (KA 10.2, 10.3, 10.4)
11 Stakeholders	Register structuring; document comparison (KA 11.1)	Attitude and influence inference; smoothing a report; synthetic voice or likeness (KA 11.2, 11.4)
12 Leadership and teams	Aggregate structural analysis only (KA 12.2)	Any individual inference, assessment or people decision (KA 12.1–12.4)
13 Adaptive delivery	Backlog and flow analytics; metric computation (KA 13.2, 13.4)	Prioritisation as a decision; velocity as a performance judgement (KA 13.4)

Two tensions are real and should be named rather than smoothed. The first is between **the value of enumeration and the risk of anchoring**: a model's list of candidate risks, requirements or options genuinely improves coverage, and it also anchors the human list that follows, so the professional practice is to have the human enumerate first and the model second, which costs nothing and is almost never done in that order. The second is between **verification cost and the productivity claim**: 14.3.3's arithmetic shows the gain is real but lands mostly as reduced checking on low-consequence classes, so an organisation whose AI-assisted work is concentrated in high-consequence classes should expect a modest gain and should say so, rather than promising the gain observed in low-consequence work.

14.A.2 Drift, revalidation and the lifecycle of a model in a delivery organisation

Every quantity in this domain is measured, and measurements go stale. Three things drift, and each has a different trigger for re-measurement. **The configuration** drifts when the model, its version, the prompt, or the grounding material changes — an event, not a period, and any of those changes invalidates p and q for every class using it. **The population** drifts when the work changes: the same prompt applied to a new clinic type, a new contract form or a new plant area is a new output class, and its error profile is unknown until measured. **The environment** drifts when the downstream containment changes — remove a review step somewhere else and u rises, which moves every threshold in 14.3.3 without anything about the model having changed.

A workable revalidation régime is therefore **event-triggered with a periodic floor**: re-measure on any configuration change, on any new output class, and on any change to downstream containment; and in the absence of events, re-measure each class on a stated cadence with the cadence set by consequence — the tier-4 classes most often, the tier-0 classes least. The seeded-error method should be specified once and reused, because the comparability of measurements over time is what makes drift detectable at all; a re-measurement by a different method produces a different number and no information.

Two lifecycle obligations close the loop. **Retirement**: a use that is no longer needed is removed from the register and from the estate, because an unmaintained tool whose measurements have expired is worse than no tool — it produces output nobody is verifying to a standard nobody has revisited. And **retention**: the model version, prompt, grounding material and verification record for anything relied on must survive the project, which is Domain 16, KA 16.4's responsible archive and model/data retention obligation, and which must be designed at mobilisation because it cannot be reconstructed at closeout.

14.A.3 The reviewer's digital and AI eye

Invariants to test on any digital or AI-assisted delivery arrangement. Each is cheap, and each is diagnostic.

Every material fact has **exactly one source of record**, and every other holding is marked as a copy. Every data class has a **named owner, a written definition, and a quality standard expressed with its consequence** — never a single organisational rate target across classes of unequal consequence (14.1.4). Remediation is ranked by **exposure per record** $d_i u_i$, not by defect rate. Every dashboard measure carries its **information age** $R/2 + G$ on its specification, and the age is compared against the window of the decision it serves (14.2.1). Every claimed digital twin has a **measured fidelity**, a test-case count, a date and a tolerance, and a computed breakeven (14.2.3). Every automation business case shows **maintenance** and a **measured differential escape rate**, and its breakeven volume is compared against the volume at the stated organisational boundary (14.2.4). Every AI use appears in the **register** with a named accountable human, an assessed consequence and a verification tier (14.4.3). p and q are **measured, dated and owned**, with a re-measurement trigger (14.A.2). The verification standard is **published**, and the thresholds are reproducible by hand as $\Delta v/(p\cdot\Delta q)$ (14.3.3). Outputs whose consequences sit near a **governance threshold** are reperformed regardless of tier (14.3.4). Error rates for anything that allocates support, priority or resource are **measured by group**, with the groups defined in advance and the fairness measure chosen by a named person (14.4.2). Security control cases **do not add** avoided losses (14.4.4). And the question that subsumes several of the others: for any AI-informed output in circulation, if this were wrong, whose judgement was wrong? — if nobody can answer, the output is not governed.

Industry variations — Domain 14

- **Public sector and government.** Data residency, procurement rules for algorithmic systems and transparency expectations are set externally and vary sharply by jurisdiction; the leader's design freedom is mostly in where the human decision sits, not in whether it does. Publication duties can make the explanation level (14.4.1) higher than the internal consequence would justify, and the verification standard must be set to the disclosure obligation rather than to the arithmetic.

- **Regulated life sciences and medical devices.** Computerised systems supporting regulated activity carry validation expectations, and a change to a model or a prompt is a change to a validated system — so 14.A.2's event-triggered revalidation is not a good practice but a condition of use. The practical consequence is that configuration stability has a value here that it does not have elsewhere, and frequent model updates are a cost rather than a benefit.
- **Construction and infrastructure.** The common data environment is the mature instance of 14.1.3, framed by ISO 19650's information-management series and usually contractual; information obligations sit in the contract, so a data class schedule inconsistent with the contract loses to the contract (the general rule Domain 3 states for governance). Digital twins are furthest advanced here and fidelity decay after handover is the characteristic failure.
- **Energy, utilities and process industries.** Operational technology converges with information technology, and the security frame is different: IEC 62443's industrial control-system security applies alongside enterprise standards, availability and safety usually outrank confidentiality, and a control that reduces impact may be unavailable because the plant cannot be segmented as freely as an office network. Auriga's twin and its handover data are the domain's own example.
- **Financial services.** Model risk management is an established discipline with model inventories, independent validation and periodic review — much of 14.4.3's register already exists under another name, and the leader's task is to bring delivery-side AI use into it rather than to invent a parallel regime. Explainability obligations attached to decisions about customers are stricter than anything internal delivery requires.
- **Healthcare.** Clinical safety obligations run in parallel with project governance and are not delegable to a delivery body (the position Domain 3 takes for Meridian's clinical sign-off). Minimisation is the dominant design move — Meridian's exclusion of clinical data from the delivery CDE — and any output touching clinical decision support crosses from delivery governance into clinical governance, with a different accountable authority and a different standard.

Case study — Domain 14: the quality target that cost 34,620 (health, Meridian)

Situation. Eighteen weeks into the clinic rollout, Meridian's assurance function reported that the programme's data quality was "below standard at 97.6 % conformance against a 98 % target" and recommended a remediation programme across all six data classes of the delivery CDE. The programme board was minded to approve it. The estate held **14,100** records: **332** were defective, a weighted mean defect rate of **2.3546 %**.

What the arithmetic showed. The project leader computed the consequence-weighted exposure by class before the next board. Total exposure **USD 172,820**, and it was distributed nothing like the defect rates: interface field-mapping records, **900** records at a 6.0 % defect rate, carried **USD 64,800**; migration reconciliation records, at the lowest defect rate in the estate (**0.8 %**), carried **USD 56,000**. Together, two classes holding **20.57 %** of the records (2,900 of 14,100) carried **69.90 %** of the exposure. Meanwhile training and competency records, **3,600** of them at 2.5 %, carried **USD 8,100** — an exposure of **USD 2.25** per record.

Then the leader priced the recommendation itself. Bringing every class to the **2 %** target would produce **282** defects — **15.06 % fewer** than the observed 332 — at an exposure of **USD 207,440**, which is **USD 34,620** or **20.03 %** worse than the position it was correcting. The target achieved its own metric by loosening the two classes whose defects were expensive and tightening the four whose defects were cheap.

What changed. The board approved a **consequence-proportional schedule** instead. Interface mappings were re-verified at **USD 22** per record — **USD 19,800** — cutting the rate to 1.0 % and removing **USD 54,000** of exposure. Migration reconciliation records were re-reconciled at **USD 12** each — **USD 24,000** — cutting the rate to 0.2 % and removing **USD 42,000**. The other four classes were left where they were, and the proposed re-check of all 3,600 training records was declined on the arithmetic: it would have cost **USD 28,800** to remove **USD 6,480** of exposure, a net loss of **USD 22,320**, and would only have paid at a remediation cost below **USD 1.80** per record.

The outcome, and the part that mattered. Total remediation spend **USD 43,800** against **USD 96,000** of exposure removed — a net **USD 52,200** — leaving the estate at **275** defects and **USD 76,820** of exposure. The number the assurance function found most uncomfortable is the one worth dwelling on: the new weighted mean defect rate was **1.9504 %**, which passes the original 2 % target — the schedule reached a better rate and a **62.97 %** lower expected cost than the uniform target would have produced, for a fraction of the effort, by ignoring the target as a design instruction and using it only as an outcome check.

What the domain teaches here. A quality target expressed as a rate, with no consequence attached, is not a weak standard — it is a standard that can be met while making the organisation worse off, and Meridian's recommendation would have spent money to increase expected cost by USD 34,620. Compute exposure by class, rank by exposure per record, and be willing to leave a high-rate, low-consequence class alone. The uncomfortable corollary, which a leader must be prepared to defend, is that the resulting schedule deliberately tolerates the most defects where they are cheapest — and that defending it requires the arithmetic, because without it the position is indistinguishable from complacency.

Case study B — Domain 14: the automation that saved 79,200 a month and lost 86,688 (financial services)

Situation. A payments-modernisation programme automated the triage of reconciliation exceptions. The manual process took an analyst **5 minutes** per exception at a blended rate of **USD 112.80** an hour — **USD 9.40** each — across **9,600** exceptions a month. The automation cost **USD 128,000** to build and **USD 2,600** a month to maintain, and handled an exception for **USD 1.15** of residual human exception-handling time. On the visible unit costs the case was overwhelming: a saving of **USD 8.25** per exception, **USD 79,200** a month, a naive breakeven of **21,188** units reached in **2.21 months** against fixed costs of **USD 174,800** over an 18-month horizon. It was approved in one meeting and reported as a success for two quarters.

What had happened. A quarterly control review measured the two routes' **error escape rates** for the first time — against a held-back sample of exceptions with known correct dispositions. The manual route mis-triaged **0.8 %**; the automated route mis-triaged **3.5 %**. A mis-triaged exception cost an assessed **USD 640** in customer remediation and rework. Quality-adjusted, the manual unit cost was `9.40 + 0.008 × 640 =` **USD 14.52** and the automated unit cost was `1.15 + 0.035 × 640 =` **USD 23.55**. The automation was **USD 9.03 per exception more expensive than the process it replaced** — **USD 86,688** a month — so no volume justified it and none ever would: the quality-adjusted breakeven volume did not exist. Against a claimed saving of USD 79,200 a month, the true position was a loss of USD 86,688 a month, a swing of **USD 165,888** a month that had gone unmeasured for eighteen months and cost **USD 1,560,384** in escaped errors on top of the **USD 174,800** of build and maintenance.

How it resolved. The programme did not abandon the automation, which would have been the intuitive response and the wrong one. The escape analysis showed the automated errors were **systematic and concentrated**: they clustered on the **12 %** of exceptions above a value threshold, where the exception patterns were most varied — the characteristic shape of machine error that Domain 9, KA 9.4.4 describes. A rule-based containment check was placed on that 12 % only, at **USD 2.10** per checked item, i.e. **USD 0.25** per exception averaged across the population, and it cut the automated escape rate to **0.9 %**. The automated unit cost became `1.15 + 0.252 + 0.009 × 640 =` **USD 7.16**, against the manual **USD 14.52** — a genuine saving of **USD 7.36** per exception, **USD 70,636.80** a month, with a quality-adjusted breakeven of **23,756** units reached in **2.47 months** and an 18-month net of **USD 1,096,662.40**.

What the domain teaches here. Three things, in order of how often they are got wrong. An automation business case computed on **visible unit costs alone** is not a conservative estimate of a good decision; it is a different calculation that can have the opposite sign, and the missing term — the differential escape rate — is measurable in a day against a held-back sample and is almost never measured before approval. Where the quality-adjusted unit cost of the automated route exceeds the manual one, **no volume rescues it**, and scale makes it worse rather than better, which is the opposite of the intuition scale usually supports. And the remedy for machine error is rarely to review the output harder — it is to **place a cheap, targeted containment layer where the errors actually concentrate**, which here converted a value-destroying automation into a strongly positive one for USD 0.25 per exception, because machine errors are systematic and therefore locatable in a way human errors are not.

Executive perspective — Domain 14

What a programme director cannot delegate in this domain:

- **The consequence figures.** Every calculation here — remediation ranking, verification tiers, automation breakevens, control sizing — is proportional to an assessed cost per escaped error. Own the assessments and their owners, or accept that the whole standard is arbitrary (14.1.4, 14.3.3).
- **The verification standard, published and owned.** A tier table derived from measured `p` and `q`, applied consistently by people who are not recomputing it. Unpublished, it collapses into individual nervousness — which is expensive in both directions at once (14.3.3).
- **That `p` and `q` are measured, dated and re-triggered.** A seeded-error programme with a named owner. An unmeasured error rate freezes the standard and forfeits the entire productivity gain, which arrives as reduced checking rather than faster drafting (14.3.3, 14.A.2).
- **The AI use register, and the question it answers.** For every AI-informed output in circulation: if this were wrong, whose judgement was wrong? No answer means no governance, and it is Domain 1's accountability-without-a-holder defect in a new costume (14.4.3).
- **Disaggregated measurement wherever an output allocates support, priority or resource.** Groups defined in advance, error rates measured by group, the fairness measure chosen explicitly and recorded. Aggregate accuracy conceals exactly the failure that will be raised publicly (14.4.2).
- **The refusal to accept a productivity claim without its verification cost.** A saving on generation is a saving only if verification is cheaper than producing the output correctly first time; and an automation's business case that shows neither maintenance nor a measured escape rate has not been reviewed (14.2.4, Case study B).

Calculation exercises — Domain 14

Exercise 14.1 A common data environment holds four classes: **A** 2,500 records at a 5.0 % defect rate, consequence USD 260 per defect; **B** 900 records at 1.0 %, USD 4,800; **C** 1,800 records at 3.0 %, USD 150; **D** 600 records at 8.0 %, USD 90. Compute total defects, weighted mean defect rate and total exposure; rank the classes for remediation; and compute the exposure that a uniform 2 % target would produce. Solution. Defects $125 \cdot 9 \cdot 54 \cdot 48 =$ **236** of **5,800** records — a weighted mean rate of **4.0690 %**. Exposures $32{,}500 \cdot 43{,}200 \cdot 8{,}100 \cdot 4{,}320 =$ **USD 88,120**, of which class **B** alone is **49.02 %** despite having the lowest defect rate and the fewest records. Exposure per record — the remediation ranking — is **B USD 48.00 · A USD 13.00 · D USD 7.20 · C USD 4.50**. A uniform 2 % target gives $0.02 \times 5{,}800 =$ **116** defects, a 50.8 % reduction in count, at an exposure of $0.02 \times 5{,}294{,}000 =$ **USD 105,880** — **USD 17,760** or **20.15 % worse** than the observed position. Common error: ranking remediation by defect rate, which puts class D first — the class carrying **USD 4,320** of total exposure, less than a tenth of class B's — and leaves the 1.0 % class, which carries half the estate's exposure, until last.

Exercise 14.2 An automation costs USD 62,400 to build and USD 900 a month to maintain over a 24-month horizon. The manual unit cost is USD 31.20 and the automated unit cost USD 3.60. Measured escape rates are 1.0 % manual and 4.0 % automated; an escaped error costs USD 720. Volume is 500 units a month. Compute the naive and quality-adjusted breakeven volumes, decide, then re-decide with a containment layer costing USD 1.80 per unit that cuts the automated escape rate to 1.0 %. Solution. $F = 62{,}400 + 900 \times 24 =$ **USD 84,000**. Naive $84{,}000/(31.20 - 3.60) = 84{,}000/27.60 =$ **3,043.48 → 3,044** units. Quality-adjusted: manual $31.20 + 7.20 =$ **38.40**; automated $3.60 + 28.80 =$ **32.40**; $84{,}000/6.00 =$ **14,000** units — **4.60 times** the naive figure. Volume over the horizon is $500 \times 24 =$ **12,000**, which is below 14,000, so **as designed the automation should be rejected**, despite a naive breakeven it passes four times over. With containment: automated $3.60 + 1.80 + 7.20 =$ **USD 12.60**, unit saving **USD 25.80**, breakeven $84{,}000/25.80 =$ **3,255.81 → 3,256** units, and a 24-month net of $12{,}000 \times 25.80 - 84{,}000 =$ **USD 225,600**. Common error: computing the breakeven on visible unit costs only. Here it produces a breakeven the volume clears comfortably for a proposal that in fact destroys value, and it is the error Case study B cost a programme USD 1,560,384 to discover.

Exercise 14.3 Verification tiers for an output class cost USD 14.00 (detection 0.40), USD 42.00 (0.75) and USD 126.00 (0.92) per item, over tier 0 (accept, detection 0). The measured material-error rate is 0.08. Derive the consequence thresholds; assign tiers to outputs whose escaped errors cost USD 900, USD 4,000 and USD 20,000; and state what happens to the thresholds if the error rate doubles to 0.16. Solution. $u^*_k = \Delta v_k/(p \cdot \Delta q_k)$. Tier 0→1: $14/(0.08 \times 0.40) =$ **USD 437.50**. Tier 1→2: $28/(0.08 \times 0.35) =$ **USD 1,000.00**. Tier 2→3: $84/(0.08 \times 0.17) =$ **USD 6,176.47**. Assignments: USD 900 → **tier 1**; USD 4,000 → **tier 2**; USD 20,000 → **tier 3**. If p doubles to 0.16 every threshold halves — **218.75 · 500.00 · 3,088.24** — so the USD 900 output moves up to tier 2 and the USD 4,000 output to tier 3: a worse configuration requires deeper checking of the same outputs, which is the same result as 14.3.3's improvement case read backwards. Common error: comparing a tier's **total** cost against the **total** consequence rather than the increment of cost against the increment of detection. Doing so here would justify tier 3 for the USD 900 output (126 < 900), which over-verifies: the step from tier 2 to tier 3 buys 0.17 of detection for USD 84 and is worth only $0.08 \times 0.17 \times 900 =$ **USD 12.24**.

> **Exercise 14.4** Two error types occur on 15 % of cycles each. An obvious error costs USD 620,000 if it escapes and is detected with probability 0.99; a plausible error costs USD 180,000 and is detected with probability 0.30. Compute each expected escaped cost and the asymmetry ratio, then evaluate a reperformance step costing USD 145 per output that raises plausible-error detection to 0.88. Solution. Obvious `0.15 × 0.01 × 620,000 =` **USD 930**. Plausible `0.15 × 0.70 × 180,000 =` **USD 18,900** — a ratio of **20.32**, which decomposes as an escape-probability ratio of `0.70/0.01 =` **70.00** divided by a consequence ratio of `620,000/180,000 =` **3.4444**. Reperformance saves `0.15 × (0.88 − 0.30) × 180,000 =` **USD 15,660** for **USD 145** — a return of **108 times** — and its breakeven detection improvement is `145/(0.15 × 180,000) =` **0.5370 percentage points**. Common error: prioritising review by the size of the number. The obvious error is 3.44 times more consequential and 20.32 times less dangerous, so review effort aimed at magnitude is aimed at the cheap failure mode — and the effective control is reperformance, not scrutiny.

> **Exercise 14.5** An incident has an assessed annual probability of 0.12 and an impact of USD 750,000. Control **X** costs USD 34,000 a year and cuts the probability to 0.03; control **Y** costs USD 21,000 a year and cuts the impact to USD 400,000. Evaluate all four options, compute the combined avoided loss, and find the probability at which X and Y are equally attractive. Solution. Baseline `EAL = 0.12 × 750,000 =` **USD 90,000**. **Nothing** USD 90,000. **X** `34,000 + 0.03 × 750,000 = 34,000 + 22,500 =` **USD 56,500**. **Y** `21,000 + 0.12 × 400,000 = 21,000 + 48,000 =` **USD 69,000**. **X and Y** `55,000 + 0.03 × 400,000 = 55,000 + 12,000 =` **USD 67,000**. X alone is best; buying both is **USD 10,500 worse** than X alone. Combined avoided loss is `90,000 − 12,000 =` **USD 78,000**, not the naive `67,500 + 42,000 =` **USD 109,500** — an overstatement of USD 31,500. X and Y are equal at `(34,000 − 21,000)/(400,000 − 187,500) =` **6.1176 %**, so at the assessed 12 % X is robustly preferred, unlike the marginal case in 14.4.4. Common error: adding the two controls' avoided losses. Reductions compound rather than accumulate, and the naive sum here overstates the combined benefit by 40 % and makes a package that is worse than one control alone look like the best option available.

Practitioner's toolkit — Domain 14

Adoption-ready artefacts; adapt headings to your organisation, then keep them stable.

Toolkit 14.T.1 — AI use register with its verification standard

Two linked tables on two pages, owned by one named person and reviewed at every gate. **The register**, one row per use: use and output class · decision informed · accountable human · assessed consequence per escaped error `u` · measured `p` and `q` with method, sample size and date · verification tier applied · model, prompt and grounding versions · data and confidentiality boundary · prohibited adjacent uses · re-measurement trigger and next review date. **The standard**, one row per tier: tier · description · cost per item `v` · measured detection rate `q` · the derived threshold `u* = Δv/(p·Δq)` · who may perform it. Two integrity checks run monthly, each a count: uses whose measurements are past their re-measurement trigger, and outputs in circulation with no register row. The register's value is that it makes the question whose judgement was wrong? answerable in advance rather than after an incident.

Toolkit 14.T.2 — Data class schedule and quality exposure sheet

One row per data class: class · records n_i · owner (a named person) · written definition with units and permitted values · source of record · defined pass criteria · measured defect rate d_i with sample size and date · assessed consequence per defect u_i with its basis · **exposure** $n_i d_i u_i$ · **exposure per record** $d_i u_i$ · target rate for this class · remediation cost per record and its breakeven. Sort by exposure per record, not by defect rate. A footer carries the three totals a board needs: records, defects, total exposure — plus the exposure a uniform target would produce, which is the number that stops a uniform target being adopted by default. Where a class carries a non-compensable consequence, the target cell records an absolute standard and the arithmetic is used only to size the effort.

Toolkit 14.T.3 — Quality-adjusted digital business case sheet

One page per automation, analytics product, dashboard or twin, and it refuses to be completed dishonestly. Scope: the decision or task, and the **organisational boundary** at which volume is counted (project, portfolio, enterprise). Costs: build · maintenance per period × horizon · residual human handling per unit a. Manual comparison: unit time and rate m. Quality: measured escape rate manual e_m and automated e_a, with method, sample size and date · assessed cost per escaped error u. Results: naive breakeven $F/(m - a)$ · **quality-adjusted breakeven** $F/[(m + e_m u) - (a + e_a u)]$ · volume at the stated boundary · net over the horizon · and, for dashboards, the information age $R/2 + G$ against the decision window. Decision: recommendation, named owner of the tool as an asset, re-measurement trigger, and retirement condition. A sheet returning "no breakeven exists" is a valid and valuable output, and the sheet is designed so that this answer is reached before approval rather than after eighteen months.

Exam preparation — Domain 14

What is assessed. The digital delivery environment and its interface arithmetic; data governance as ownership, definition, source of record, quality standard and access; the common data environment and its five guarantees; **defect rate by class, consequence-weighted exposure, exposure per record and the uniform-target defect**; dashboards tested against a decision, with **information age** $R/2 + G$; the analytics ladder and its ordering rule; digital twins and **breakeven fidelity**; **automation breakeven volume, naive and quality-adjusted**; the five permitted and four prohibited classes of AI use and the single test that generates the prohibitions; prompting as a method with grounding, constraint, refusal permission and provenance; **the derivation of the verification standard** $u^*_k = \Delta v_k/(p \cdot \Delta q_k)$ and the pricing of tiered against uniform regimes; **the escape-cost asymmetry between plausible and obvious error** and reperformance as its control; explainability as a decision requirement at three levels; **differential error by group** and the base-rate constraint; the AI use register; and **security control economics with sub-additivity**.

The calculations to be able to do under time pressure. Class exposure $n_i d_i u_i$, total exposure, weighted mean defect rate, exposure per record, and the exposure a uniform rate target would produce. Remediation net value and its breakeven cost per record. Information age $R/2 + G$ priced at a cost of delay per day, and the two levers' asymmetry. Breakeven fidelity for a twin. Naive and quality-adjusted automation breakeven volumes, and the recognition that no breakeven exists. Verification thresholds $\Delta v/(p \cdot \Delta q)$, tier assignment from a consequence, total cost of a regime as verification plus expected escaped loss, and the effect on thresholds of a change in p. Expected escaped cost $p(1 - q)u$ and the asymmetry ratio. Recall by group and the economics of correcting a differential. `EAL`, residual `EAL` after a control, combined `EAL` for two controls, and a crossover probability.

The traps. Applying a uniform data-quality target across classes of unequal consequence, which can cut defect counts while raising expected cost (Exercise 14.1, Case study A) · ranking remediation by defect rate rather than by exposure per record (14.1.4) · computing an automation breakeven on visible unit costs, omitting maintenance and the differential escape rate (Exercise 14.2, Case study B) · assuming a breakeven volume exists when the quality-adjusted saving is negative (14.2.4) · taking information age as the production lag alone or as half the period alone (MCQ 14.2-A) · comparing a verification tier's total cost against the total consequence instead of increment against increment (Exercise 14.3) · forgetting that verification thresholds scale with `1/p`, so a better configuration raises them (MCQ 14.3-C) · prioritising review by the magnitude of a number rather than by its plausibility (Exercise 14.4) · treating a model's self-reported confidence as an error or detection rate (14.3.3) · reading an aggregate accuracy figure as evidence of no differential error (14.4.2) · adding two security controls' avoided losses (Exercise 14.5) · and applying expected-value reasoning to a consequence that is not cost-compensable (14.1.4, 14.3.3, 14.4.4).

How the domain connects. Domain 1 supplies the accountability principle every prohibition here rests on, and the cost of delay at which staleness, verification and automation are all priced. Domain 3 supplies the `E[wait]` identity this domain applies to information, the delegation logic that becomes system permissions, and the four AI governance instruments this domain makes operational. Domain 4 supplies the interface arithmetic for the systems estate. Domain 5 supplies the testability requirement that data pass criteria depend on. Domain 7 supplies the blended rates and the `CPI`-based forecast that 14.3.4 corrupts two ways. Domain 9 supplies the sampling arithmetic, the composite data fitness, the independence rule for layers and the containment-chain result this domain builds on rather than repeats. Domain 10 supplies the strictest explainability case in the volume. Domain 11 supplies the reporting standard the information-age calculation belongs to. Domain 12 supplies the prohibitions on inference about people, which this domain does not soften. Domain 13 supplies the delivery cadence that a dashboard's information age must keep up with. Domain 15 owns the portfolio boundary at which most automation cases actually pay. And Domain 16 owns the retention of models, prompts, data and verification records after the project that created them has gone.

Domain 14 summary

Thirteen domains each drew a local line around AI use. This domain makes those lines one system, and its instrument is arithmetic rather than principle — because a principle cannot tell you how deeply to check, and arithmetic can.

The substrate comes first. Meridian's common data environment holds **14,100** records in six classes with **332** defects — a weighted mean rate of **2.3546 %** — carrying **USD 172,820** of consequence-weighted exposure, distributed nothing like the defect rates: the lowest-rate class in the estate (0.8 %) is the second-highest by exposure per record at **USD 28.00**, behind interface mappings at **USD 72.00**. A uniform **2 %** target produces **282** defects, **15.06 % fewer**, at **USD 207,440 — 20.03 % worse** — which is the whole case against rate targets in one comparison. The consequence-proportional schedule that replaced it spent **USD 43,800** to remove **USD 96,000** of exposure and landed at a **1.9504 %** mean rate: it passed the original target while costing **62.97 %** less than the target's own remedy.

What is built on the substrate must be tested against a decision and a volume. A fact on a monthly report with a 9-day production lag is **24 days** old — `R/2 + G`, Domain 3's identity applied to information — worth **USD 48,960** at Meridian's cost of delay, against **7.5 days** and **USD 15,300** on a weekly cycle with a 4-day lag; and the production lag is twice the lever the reporting period is, day for day. Auriga's control-system twin costs **USD 211,500** against **USD 306,000** of benefit at perfect fidelity, so it needs **69.12 %** fidelity and has **85.00 %** — a positive **USD 48,600** on a margin of under sixteen points measured on forty cases. Auriga's data-check automation breaks even at **959** checks on build cost, **1,659** with maintenance and **2,654** once a one-point difference in escape rates is priced — **1.60 times** the naive figure — which rejects it for a 720-check project and approves it for an 8,640-check portfolio. Case study B's payments programme shows the other end: an automation reporting **USD 79,200** a month of saving and destroying **USD 86,688** a month, with no breakeven volume at any scale, rescued by a containment layer costing **USD 0.25** per exception.

The core is the verification standard Domain 3 asserted and this domain derives. $u^*_k = \Delta v_k/(p \cdot \Delta q_k)$ gives Meridian's thresholds as **USD 261.90 · 785.71 · 3,208.33 · 18,333.33**, assigning its five output classes to tiers 0 to 4 at a monthly total of **USD 36,950.10** — verification USD 12,650 plus expected escaped loss USD 24,300.10. Uniform tier 2, the intuitively consistent policy, spends **3.27 times** as much on verification to cut escaped loss by **10.41 %** and costs **USD 63,175.36**; uniform tier 4, the intuitively safe policy, costs **USD 260,952.14**, **7.06 times** the tiered standard; doing nothing costs **USD 72,571.20**. **45.93 %** of the tiered standard's escaped loss sits in the class deliberately left unverified, and checking that class would lose **USD 2,914** — the correct remedy for a large aggregate of cheap errors is to reduce `p` or `u`, not to check. And when the measured error rate falls from 0.12 to 0.04, thresholds triple and the total falls **56.60 %** while the verification bill falls **72.09 %**: **the return on a better configuration arrives as less checking**, which is the honest form of the productivity claim and is unavailable to anyone who does not measure `p`.

Then the asymmetry that determines where checking should point. On Auriga's week-13 numbers a plausible wrong forecast — **USD 4,320,000** against the correct **USD 4,416,666.67**, landing on the wrong side of a **USD 350,000** funding trigger — carries an expected escaped cost of **USD 20,250**, while a 10× decimal slip with **3.33 times** the consequence carries **USD 1,800**: the plausible error is **11.25 times** more dangerous, and would be **37.50 times** at equal consequence, because expected damage is consequence times the probability of not being caught. Reperformance from the inputs raises detection from 0.25 to 0.85 for **USD 87.08** and returns **186 times** its cost, with a breakeven of **0.3225** percentage points of detection. Rounding `CPI` to 0.91 before dividing manufactures a plausible error of **USD 21,062.27** — 5.05 % of the variance — from an operation that looks like tidiness.

The obligations that arithmetic cannot discharge close the domain. Explainability is a property of the decision, in the vocabulary of the obligation, at one of three levels. Differential error is invisible in aggregate: Meridian's **70.00 %** recall concealed **88.89 %** urban and **54.55 %** rural, a **34.34**-point gap, with rural clinics carrying **83.33 %** of the excess cost while being **40.00 %** of the estate — and correcting it returned **4.74 times** its cost, so the differential was a defect and not a price. Because the base rates differ (**37.50 %** against **68.75 %**), not every error measure can be equalised at once, and choosing which is a named person's judgement. Security controls are sub-additive: Meridian's two options total **USD 35,600** and **USD 34,800** alone and **USD 45,200** together, so buying both is worse than buying either, the naive sum of avoided losses overstates by **USD 13,200**, and the choice between them flips at an incident probability of **8.57 %** that nobody can estimate to that precision — which is the honest answer, not a preference. And where a consequence is not cost-compensable, every expected-value test in this domain stops applying and the control becomes a requirement to be sized rather than an investment to be justified.

The through-line: **AI proposes; the professional verifies, decides and remains accountable — and this domain is what makes the middle verb affordable.** Verification without arithmetic is either theatre or negligence, and usually both at once in the same organisation: excessive on the outputs that do not matter, absent on the ones that do. Compute the consequence, derive the tier, measure the rates, reperform anything near a threshold, and keep a register that can answer the only question that finally matters — if this were wrong, whose judgement was wrong?

15

DOMAIN 15

Programmes, Portfolios and Enterprise Delivery

Group: Enterprise delivery and the digital future (Domain 15 of 3 in Part Four). **Target:** ~72 pages. **Binds to:** the PCI Book Pattern Specification and the shared registries (`docs/books/registries/`). This domain lifts the **Meridian Care Records** programme of Domains 1 to 4 to the tier above it — the portfolio that funds it and the enterprise that governs it — and supplies the dependency-product, multi-period allocation, benefits-bridge and enterprise-latency arithmetic that Domain 16 consumes when it measures whether any of it landed. British English; USD (+SAR where useful, indicative `USD 1 ≈ SAR 3.75`).

Why this domain exists

Every domain before this one has, quite deliberately, held one thing constant: the boundary of the thing being led. Domain 4 integrated the parts of a project into a whole. Domain 6 scheduled that whole. Domain 8 put ranges round it. Domain 13 chose the delivery method for it. Domain 14 gave it a data and AI estate. In all of them the project had an outside, and the outside was assumed to cooperate.

It does not. The outside is where the other twelve initiatives live, where the same six integration engineers are already committed, where four business cases have each independently promised the same administrative post reduction, and where the decision the project needs will pass through tiers of governance that no one has ever added up. **The central claim of this domain is that a portfolio is not a bigger project, because the arithmetic that governs it is different in kind: at project scale quantities add, and at enterprise scale they multiply, compete and double-count.** A leader who carries project intuitions upward will be confidently wrong in four specific ways, each of which this domain computes. Milestone confidence multiplies down rather than averaging out. Capacity that is sufficient in aggregate is routinely infeasible period by period. Benefits summed across components overstate what the enterprise can bank. And governance latency accumulates tier by tier into a bill that appears in no budget line.

The domain proceeds outward. KA 15.1 builds programme architecture and shows that a programme milestone with several individually likely predecessors is not likely at all — the same multiplicative penalty Domain 8 found at a schedule merge point, now applied to whole components — and identifies **decoupling**, not better estimating, as the lever that pays. KA 15.2 handles portfolio balancing: the benefits register with its double counts eliminated, and allocation under a constraint that binds per period, which is where the ranking heuristic Domain 2 warned about fails expensively. KA 15.3 treats enterprise capacity as the real constraint it is, applies Domain 13's flow arithmetic to a portfolio's work in progress, prices protective capacity honestly (it does not always pay), and subjects the enterprise PMO to the same value test it applies to everyone else. KA 15.4 governs transformation: enterprise decision latency summed across tiers using Domain 3's formula, and strategic reporting — where the commonest portfolio report in circulation aggregates performance in a way that reverses its sign.

Learning objectives. After this domain a candidate can: distinguish a project, a programme and a portfolio by the decisions each exists to make rather than by size; design a programme architecture in components, tranches and a target operating state, and state what each tranche buys; **compute the probability that a programme milestone is met from the on-time probabilities of its independent predecessors, and show how quickly that collapses as predecessors accumulate**; derive the per-dependency reliability a stated milestone confidence demands, and use it to demonstrate when better estimating cannot possibly deliver the target; rank the three structural levers — reduce the count, decouple, buffer — and price each; build a portfolio benefits bridge that eliminates shared, same-pool, cascade and already-committed double counts, and show what elimination does to payback; **allocate a portfolio under a multi-period capacity constraint, demonstrate that aggregate feasibility does not imply period feasibility, and quantify what a ranking heuristic leaves on the table**; apply Domain 13's flow arithmetic to portfolio work in progress and price the benefit deferral that excess concurrency causes; compute the survival probability of a capacity plan and the breakeven failure cost at which protective capacity becomes worth its forgone value; state an enterprise PMO's recurring value test and distinguish it from one-off gains; **sum enterprise decision latency across programme and portfolio tiers, price it at a cost of delay, and rank the available redesigns**; aggregate portfolio performance correctly and detect the unweighted-average trap; and govern AI-assisted portfolio analysis without letting a model select a portfolio.

The master programme, one tier up. Meridian Care Records continues from Domains 1 to 4 — the clinical-records rollout to **40 clinics**, approved cost **USD 2,400,000**, full-potential benefit **USD 979,200** a year, realistic benefit **USD 685,440** a year at 70 % adoption, and the cost of delay of **USD 14,280 per week** derived in Domain 1. Domain 3 established Meridian's steering-body latency at `E[wait] = M/2 + L = 4/2 + 2 =` **4.0 weeks**. This domain places Meridian inside the health group's **five-component portfolio** (records, referrals interoperability, clinician scheduling, analytics platform, pharmacy stock automation) with a combined approved cost of **USD 4,880,000**, and inside a **five-tier enterprise decision architecture**. Every figure in the chapter descends from those.

KNOWLEDGE AREA 15.1

Programme architecture and dependency management

Topics: 15.1.1 what a programme is, and is not · 15.1.2 architecture — components, tranches and the target state · 15.1.3 dependency arithmetic at programme scale · 15.1.4 decoupling as the primary lever.

15.1.1 What a programme is, and is not

Definition. A programme is a temporary organisation created to deliver a **coherent change in outcomes** through a set of components — projects and enabling activities — that cannot deliver the outcome individually. Its unit of decision is the **component set**: what to start, what to stop, what to sequence, and what to fund next.

The definition that matters is comparative, and the comparison is by decision rather than by scale. A **project** decides how to produce a defined output within a mandate. A **programme** decides which components produce the outcome, and repeatedly re-decides it as evidence arrives. A **portfolio** decides which programmes and projects the organisation should be doing at all, given a finite capacity and a strategy — Domain 2's territory, revisited each cycle rather than once.

Three consequences follow, and each is a test a leader can apply on Monday morning.

A large project is not a programme. If the components cannot be usefully re-sequenced or stopped independently, the programme layer is adding governance without adding optionality, and the workable structure is one project with a good work breakdown structure (Domain 4, KA 4.2.1). The diagnostic question is direct: name a component this programme could stop next quarter without abandoning the outcome. If there is none, dissolve the layer.

A collection of projects is not a programme. If the components deliver independent outcomes, the programme layer is a reporting convenience, and the honest structure is a portfolio grouping. The cost of the pretence is real: components acquire a false interdependence, and each waits for programme-level decisions it did not need.

A programme's benefit is not the sum of its components' benefits. This is the single most consequential difference and KA 15.2 computes it. Component business cases are written to be approved, and they are written separately, so they claim overlapping benefits as though they were additive. At programme and portfolio level that claim has to be reconciled against reality, and the reconciliation reliably removes a double-digit percentage of the total.

15.1.2 Architecture — components, tranches and the target state

The target operating state. A programme needs a stated description of the organisation as it will be when the change has landed: the processes, roles, systems, data, locations and measures. Without it the programme has an output list and no test of coherence, and every component optimises locally. The description is a governance artefact, not a design document: its purpose is to make the question "does this component still contribute?" answerable at a gate.

Components. Each component carries a named owner, an output, a contribution to the target state, its dependencies in and out, and — this is the part usually missing — a statement of what happens to the outcome if the component is cancelled. A component whose cancellation has no describable consequence is not part of the programme.

Tranches. A tranche is a group of components delivered together to reach a **usable intermediate state** — one at which benefit begins and at which the programme could stop with something of value already in operation. Tranche design is therefore the programme's principal option-creating decision, and it is governed by one rule: **a tranche boundary belongs where the organisation can genuinely operate.** A boundary placed for reporting convenience produces the worst of both worlds, a gate with nothing to decide and a partial state nobody can use.

Meridian's tranches illustrate the rule. Tranche 1 delivers the records application and the national patient-index interface to **10** clinics in one region — a genuinely usable state, because those ten clinics can operate on the new record and the remaining thirty continue as they are. Tranche 2 adds the second and third regions and the migration of historical records. Tranche 3 completes the fourth region and retires the legacy system, which is the only irreversible step in the programme and therefore the only place a hard gate is warranted (Domain 3, KA 3.3.1).

Programme-level dependencies. Components depend on each other, and they also depend on things outside the programme: an estate refurbishment, a regulatory approval, a supplier's product release, another programme's platform. A programme dependency register records, per dependency, the giver, the receiver, what is needed, the date needed, the date promised, the owner on both sides, and the consequence of a breach. Domain 2's assumption and dependency management (KA 2.3.4) supplies the discipline at project level; the programme addition is that dependencies now cross accountability boundaries, so the register's most important column is the owner on the giving side, and a dependency with no owner there is not a dependency, it is a hope.

15.1.3 Dependency arithmetic at programme scale

Here the domain earns its quantitative keep, and the result is one every programme leader should be able to produce on a whiteboard.

The multiplication rule. A milestone that requires **all** of `k` independent predecessors to be met is met only if every one of them is met. Its probability is therefore the **product** of theirs, not their average and emphatically not their minimum:

```
P(milestone) = p1 × p2 × … × pk
```

Domain 8 established the schedule form of this at a merge point (8.A.2: two paths each 80 % likely give the merged event 64 %, which is why deterministic critical-path analysis is systematically optimistic at convergence). The programme form is the same theorem applied to whole components rather than to activity paths, and it is more dangerous, because at programme level the predecessors are owned by different people, each of whom reports honestly on their own item and none of whom is looking at the product.

WORKED EXAMPLE

Worked example 15.1.3 — Meridian's Region A go-live, and the four-region programme milestone.

1. **Setup.** Meridian's Region A go-live requires six independent predecessors, each assessed by its own owner at the following probability of being met on the committed date: records application release **0.90**; national patient-index interface **0.88**; data migration and reconciliation **0.85**; clinic estate and network works **0.92**; clinician training and competency **0.95**; information-governance approval **0.90**. Every one of them is "on track" by the programme's own reporting convention. The programme milestone committed externally is the **simultaneous** go-live of all four regions, each with its own instance of the same six-dependency structure — 24 dependencies in total. Independence is assumed and is discussed below.
2. **Formula.** `P(milestone) = Π pᵢ` over the predecessors. For the four-region milestone, $P = (\Pi\ p_i)^4$, the four regions being structurally identical and assumed independent.
3. **Substitution.** `0.90 × 0.88 × 0.85 × 0.92 × 0.95 × 0.90`; then that result raised to the fourth power.
4. **Result.** Region A: **0.52953912**, that is **52.95 %**. The four-region programme milestone: **7.86 %**. Expected number of regions on time: `4 × 0.52953912 =` **2.1182** — so **1.8818** regions are expected to be late.
5. **Interpretation.** Six components, none worse than 0.85, produce a coin flip; four such regions produce a milestone that is **92.1 % likely to be missed** while every single component is reported as on track. That gap between component reporting and milestone probability is the defining pathology of programme delivery, and it is arithmetic rather than culture. Three consequences a leader should draw. First, **the product is the number to report**: a programme status report that lists twenty-four green components and a committed date is internally inconsistent, and the fix is one multiplication. Second, **better estimating cannot rescue the design.** To reach an 80 % confidence on the four-region milestone every one of the 24 dependencies would have to be met with probability `0.80^(1/24) =` **99.07 %** — a standard no estate contractor, training programme or regulator has ever offered, and one the programme has no mechanism to enforce. Even across the six dependencies of a single region, 80 % confidence needs **96.35 %** each. Third, and following directly, **the lever is structural**: change k, not the p_i. To see why, note the sensitivity. Improving any single dependency by an absolute 0.03 moves the Region A probability by between **1.67** percentage points (training, 0.95 → 0.98) and **1.87** points (migration, 0.85 → 0.88) — the improvement is worth marginally most where the probability is lowest, but the whole range is under two points, and there is no combination of plausible estimating improvements that reaches 80 %. Removing one dependency from the milestone, by contrast, multiplies the probability by $1/p_i$, which for the 0.85 migration dependency is a gain of **9.34** points in one decision. **Two professional cautions.** The independence assumption is doing real work here and is usually optimistic: shared causes — one supplier, one scarce team, one regulator, one budget round — correlate the predecessors, and Domain 8's KA 8.A.1 treatment of common drivers explains why a register with many entries and few drivers is concentrated rather than diversified. Correlation makes the product less pessimistic than shown for the good outcome and makes clustered failure far more likely, so the honest statement is that **52.95 % is a planning figure derived under an explicit assumption, and the assumption belongs in the report next to the number.** And the p_i are owner assessments, not measurements; their calibration is a data-quality question

(Domain 9, and Domain 14's KA 14.1 on defect rates by class), which is why the register records who assessed each one and on what evidence.

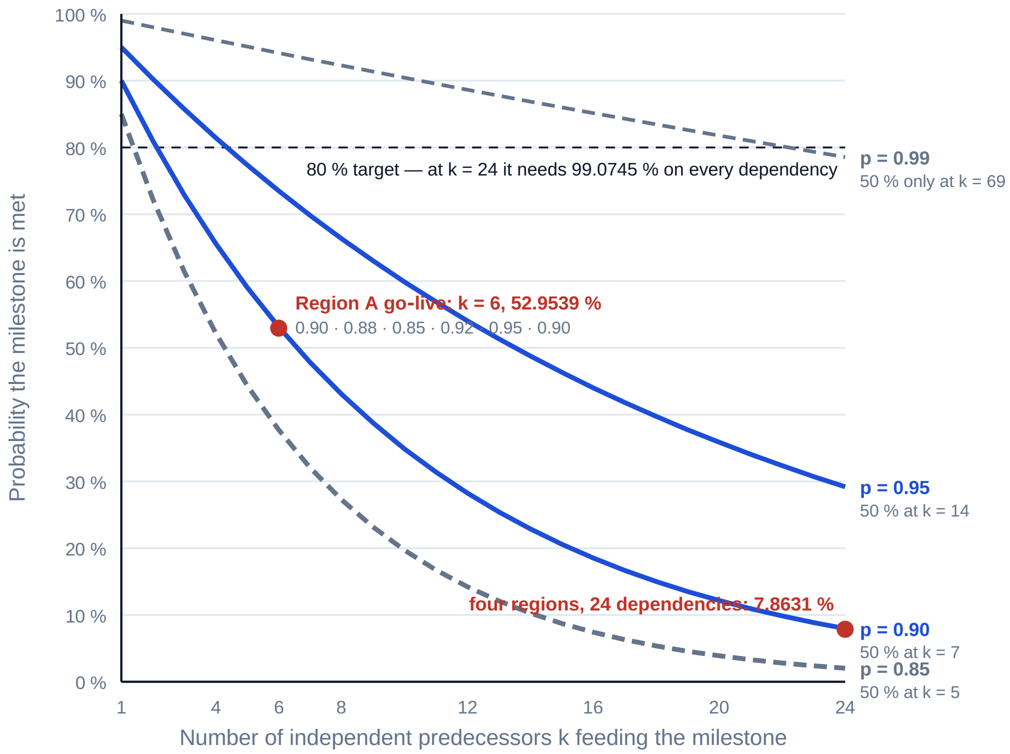


Fig 15.1.1 — Why a programme milestone becomes improbable

15.1.4 Decoupling as the primary lever

If the milestone probability is a product, the only interventions that materially change it are interventions on the structure of the product. Three exist, in descending order of value per unit of effort.

Reduce the count. Ask of every predecessor whether the milestone genuinely requires it. Many entries are there because someone drew an arrow, not because the outcome depends on the item. Each removal multiplies the probability by `1/pᵢ`, and the removals compound.

Decouple. Where a predecessor is genuinely needed but not needed by that date, move it off the milestone by designing an operable interim arrangement. Meridian did this twice. Historical **data migration and reconciliation** (0.85) was removed from go-live by operating a documented two-week paper-fallback for historical lookups, at a priced cost of **USD 18,000** in temporary clerical cover — taking Region A from 52.95 % to **62.30 %**. **Information-governance approval** (0.90) was obtained once at programme level for all four regions rather than four times regionally, removing it from each regional milestone and taking Region A to **69.22 %**. The four-region milestone rose from **7.86 %** to **22.96 %**, and the expected number of late regions fell from **1.8818** to **1.2312**.

Buffer. Where neither reduction nor decoupling is available, the honest response is to commit to a date that includes a buffer, sized by Domain 8's quantitative methods, and to hold internal dates that are tighter. What is not available is committing to the unbuffered date and calling the components green.

Why the cost comparison settles the argument. Programmes under milestone pressure reliably reach for money instead of structure, and the two can be priced against each other. Meridian's alternative proposal was **USD 240,000** of recovery spend — additional trainers and an accelerated estate package — assessed to raise training from 0.95 to 0.98 and estate works from 0.92 to 0.96. That raises Region A to **57.00 %** and the four-region milestone to **10.56 %**: a gain of **2.69** percentage points on the committed milestone for 240,000, or **USD 89,093** per percentage point. The structural changes gained **15.10** points for 18,000 — **USD 1,192** per point, or **1.34 %** of the cost of the money route, a ratio of **74.7 to one**. The general result is worth stating plainly because it is counter-intuitive to sponsors: **on a milestone whose probability is a product, structure is roughly two orders of magnitude cheaper than effort.** Money buys movement in one factor; decoupling removes a factor.

AI in this KA

Where it earns its place. Building and reconciling a programme dependency register from component plans, contracts and interface documents, and listing dependencies with no owner on the giving side, no date, or a promised date later than the date needed — a document-comparison task that is tedious, high-volume and has a checkable right answer. Computing the milestone product across hundreds of dependency configurations, and searching for the smallest set of removals that reaches a target confidence, which is a deterministic combinatorial problem and exactly the sort of thing to delegate. Detecting shared causes across component registers by clustering dependency descriptions, which surfaces the correlation that invalidates the independence assumption. Drafting the interim operating arrangement options for a decoupling decision, for human evaluation.

Where it must not go. Assessing the p_i. A model asked how likely an estate contractor is to meet a date will produce a confident number with no provenance, and that number will then be multiplied into a board commitment. Probabilities come from owners with evidence, or from calibrated history, and their source is recorded. Nor may a model decide a decoupling: an interim paper-fallback in a clinical setting is a patient-safety decision belonging to clinical authority, and the domain's arithmetic informs it without ever making it.

Verification, concretely. Reproduce the product by hand — six multiplications, and a leader quoting 52.95 % who cannot do so should not be quoting it. Check every AI-extracted dependency against its source document on a stated sample, and confirm each flagged defect with a human before reporting it as a finding, per Domain 14's verification tier for outputs of this consequence (KA 14.3.3). State the independence assumption every time the product is printed.

KEY TERMS — KA 15.1

Term	Meaning
Programme	A temporary organisation delivering a coherent outcome through components that cannot deliver it individually.
Portfolio	The set of programmes and projects an organisation chooses to run within a finite capacity and a strategy.
Target operating state	The stated description of the organisation once the change has landed; the test of component coherence.
Component	A project or enabling activity within a programme, with a named owner and a stated consequence of cancellation.
Tranche	A group of components delivered together to reach a usable intermediate state at which benefit begins.
Programme dependency register	The record of cross-boundary dependencies, whose critical column is the owner on the giving side.
Multiplication rule	The probability a milestone requiring all of `k` independent predecessors is met is the product of their probabilities.
Decoupling	Removing a predecessor from a milestone by designing an operable interim arrangement, multiplying the probability by `1/pᵢ`.
Required per-dependency reliability	The uniform probability each of `k` dependencies must carry to reach a target milestone confidence: the `k`-th root of the target.

SAMPLE MCQS — KA 15.1

MCQ 15.1-A `[15.1.3 · Application]` A programme milestone requires four independent predecessors, assessed at 0.95, 0.90, 0.90 and 0.85. The probability the milestone is met is closest to: - A. 0.90 - B. 0.85 - C. 0.65 - D. 0.60

Rationale: `0.95 × 0.90 × 0.90 × 0.85 = 0.654075`, so 0.65 (15.1.3). A is the arithmetic mean of the four, the commonest error; B is the minimum, the second commonest; D subtracts the summed shortfalls from one (`1 − 0.40`), which treats the shortfalls as additive.

MCQ 15.1-B `[15.1.3 · Analysis]` Meridian's Region A go-live has six independent predecessors giving a milestone probability of 52.95 %. Four structurally identical regions must go live simultaneously. The programme milestone probability is: - A. 52.95 %, because the regions are identical - B. 13.24 %, being 52.95 % divided by four - C. 7.86 % - D. 28.04 %

Rationale: `0.52953912⁴ = 7.86 %` (15.1.3). A ignores that all four must succeed; B divides instead of taking the fourth power; D squares instead of raising to the fourth power (`0.52953912² = 28.04 %`), the error of treating four regions as two.

MCQ 15.1-C `[15.1.3 · Evaluation]` A programme needs 80 % confidence in a milestone that depends on 24 independent predecessors. The required probability on each is closest to: - A. 80.0 % - B. 96.3 % - C. 99.1 % - D. 99.9 %

Rationale: `0.80^(1/24) = 99.0745 %` (15.1.3). A applies the target to each component; B is the answer for six dependencies, not 24; D over-corrects. The teaching point is that C is unattainable, which is why the lever is structural.

MCQ 15.1-D `[15.1.4 · Evaluation]` Removing the 0.85 data-migration dependency from Meridian's regional go-live milestone raises the regional probability from 52.95 % to 62.30 %. The mechanism is best described as: - A. improving the migration team's performance - B. multiplying the remaining product by `1/0.85` - C. adding 0.85 of buffer to the milestone - D. averaging the remaining five probabilities

Rationale: Decoupling divides the product by the removed factor, which is the same as multiplying by its reciprocal — `0.52953912/0.85 = 0.6229872` (15.1.4). A describes the money route, which bought 2.69 points on the programme milestone for 240,000; C and D misstate the arithmetic.

MCQ 15.1-E `[15.1.2 · Analysis]` The soundest test that a proposed tranche boundary is real is whether: - A. it falls at the end of a financial quarter - B. the organisation can genuinely operate in the intermediate state - C. a governance body is available to meet on that date - D. it divides the components into groups of similar size

Rationale: A tranche exists to reach a usable intermediate state at which benefit begins and the programme could stop with value in operation (15.1.2). A, C and D are reporting conveniences that produce gates with nothing to decide.

SELF-CHECK — KA 15.1

1. State the multiplication rule and the two wrong answers it displaces. — The probability a milestone requiring all of `k` independent predecessors is met is the product of their probabilities; the wrong answers are the average and the minimum.
2. Why can better estimating not deliver an 80 % four-region milestone at Meridian? — It would require 99.07 % on each of 24 dependencies, which no supplier, trainer or regulator offers and the programme cannot enforce; the lever is to reduce or decouple dependencies.
3. What distinguishes a programme from a large project? — A programme's components can be usefully re-sequenced or stopped independently; if none can, the programme layer adds governance without optionality.

KNOWLEDGE AREA 15.2

Benefits and portfolio balancing

Topics: 15.2.1 the portfolio benefits register and the four double counts · 15.2.2 balancing under a binding constraint · 15.2.3 the multi-period constraint and the aggregate-capacity illusion · 15.2.4 rebalancing, staging and stopping.

15.2.1 The portfolio benefits register and the four double counts

Definition. A portfolio benefits register is the single authoritative statement of what the portfolio will deliver, by benefit rather than by component, with each benefit carrying a measure, a baseline, a receiving-organisation owner and the components that contribute to it. It is organised by benefit precisely so that two components contributing to one benefit cannot each claim it.

Component business cases, written separately to be approved, overstate the portfolio's total for four distinct and separately detectable reasons. Naming them matters because each has a different correction.

The shared benefit. Two components claim the same saving because both genuinely contribute to it. Neither case is dishonest; the sum is wrong. Correction: count the benefit once, in the register, with both components recorded as contributors.

The same-pool over-claim. Components claim releases from a pool that cannot supply them all — most often headcount, but equally floor space, licences or clinical session time. Correction: cap the aggregate claim at the pool's size and allocate the cap.

The cascade, or enabler, double count. A platform or data component claims a share of the benefits that flow through it, which the consuming components have already claimed. Correction: an enabler's own benefit is only what does not pass through another component's claim; the rest is recorded as an enabling contribution with no independent value.

The already-committed benefit. A component claims against a baseline that already assumes a saving committed elsewhere — an operational efficiency target, a prior programme's residual, a contractual price reduction. Correction: reset the baseline to the committed position.

WORKED EXAMPLE

Worked example 15.2.1 — the Meridian portfolio benefits bridge.

1. **Setup.** Five components, each claiming annual benefit at its own **full potential**: Meridian records **979,200**; referrals interoperability **264,000**; clinician scheduling replacement **318,000**; analytics platform upgrade **180,000**; pharmacy stock automation **240,000**. Review establishes four adjustments. (i) **96,000** of clinician time released from chasing paper referrals is claimed in both records and referrals. (ii) Records claims **5.0** and scheduling **3.4** administrative posts from one pool of **6.0**, valued at **42,000** per post. (iii) **108,000** of the analytics platform's 180,000 is "improved decision-making" realised through records and scheduling, already counted there. (iv) **60,000** of the pharmacy claim rests on a baseline that already assumes savings committed in an operational programme outside the portfolio. Portfolio-wide adoption, per Domain 2's realistic assumption, is **70 %**. Approved portfolio cost is **USD 4,880,000** (records 2,400,000, referrals 640,000, scheduling 890,000, analytics 520,000, pharmacy 430,000). The investment committee applies a **four-year** simple payback rule.
2. **Formula.** Net benefit = Σ claimed at full potential − Σ eliminations, then × adoption. Elimination for the same-pool case = (Σ claims − pool cap) × unit value. Simple payback = portfolio cost ÷ net annual benefit.
3. **Substitution.** Gross `979,200 + 264,000 + 318,000 + 180,000 + 240,000`. Eliminations `96,000 + (8.4 − 6.0) × 42,000 + 108,000 + 60,000`. Then `× 0.70`, and `4,880,000 ÷` the result.
4. **Result.** Gross claimed at full potential **USD 1,981,200**. Eliminations **USD 364,800** — 96,000 shared, **100,800** same-pool (2.4 posts), 108,000 cascade, 60,000 already committed — an **elimination rate of 18.41 %**. Net full potential **USD 1,616,400**; at 70 % adoption, net realistic benefit **USD 1,131,480** a year. The unreconciled figure, `1,981,200 × 0.70 =` **USD 1,386,840**, overstates that by **USD 255,360**, which is **22.57 %** of the honest number. Simple payback moves from **3.5188 years** on the unreconciled figure to **4.3129 years** on the honest one — a difference of **0.7941 years**, or **9.53 months**.
5. **Interpretation.** The decision, not the number, is what moved: the unreconciled portfolio passes a four-year payback rule and the reconciled one fails it. That is the whole professional case for doing the bridge before approval rather than discovering it in Domain 16's benefits measurement, and it produces a threshold worth carrying: **the maximum elimination rate this portfolio can absorb and still clear four years is 12.03 %**, so an observed 18.41 % breaks the rule with room to spare, and a portfolio whose bridge has never been done should be assumed to be somewhere in that range. Three further points. First, an identity that makes the bridge auditable: because adoption scales the eliminations too, the **elimination rate is invariant to the adoption assumption** — 364,800/1,981,200 and 255,360/1,386,840 are both **18.41 %** — so the two adjustments can be argued independently and in either order, and a reviewer can check one without settling the other. Second, eliminating a double count **creates no value**; it prevents a wrong decision. A portfolio report that presents 364,800 of eliminations as a PMO saving is committing the same category error in the opposite direction, and KA 15.3 keeps it out of the PMO's value case for exactly that reason. Third, the same-pool correction is the one most often resisted, because both claimants are right about their own component; the register's answer is that the pool is an enterprise asset and its cap is an enterprise fact, which is why benefits are owned by the receiving organisation and not by the delivering component (Domain 2, KA 2.3.2; Domain 16 measures them).

15.2.2 Balancing under a binding constraint

The principle. A portfolio decision is an allocation under a constraint, and the first professional act is to **name the constraint honestly**. It is rarely money. It is usually a scarce capability — the integration engineers, the clinical safety reviewers, the one architect who understands the interface layer — and a portfolio balanced against a budget while a capability binds will be approved and will not be delivered.

Domain 2 established the base result at KA 2.2.3, and it is not re-derived here: **ranking candidates by value per unit of the binding constraint is a heuristic, not an optimum**, and with lumpy candidates the feasible sets must be enumerated — the Beta-plus-Gamma outcome that neither the raw value ranking nor the scoring model selected. Domain 15's contribution is the extension that makes this bite at enterprise scale: real constraints bind **per period**, not in aggregate.

Balance across dimensions, not only value. Before the arithmetic, a portfolio also has to be balanced against exposures that no single ratio expresses: risk (Domain 8), time to benefit, component novelty, dependence on one supplier or one team, and regulatory exposure. Domain 2 noted that a set of individually sound investments can be collectively unbalanced and that no scoring model detects it. The portfolio-level countermeasure is a small set of stated exposure limits — no more than a given share of the portfolio dependent on one platform, one supplier, or one scarce team — applied as feasibility constraints alongside capacity, so the enumeration returns only sets the organisation can survive.

15.2.3 The multi-period constraint and the aggregate-capacity illusion

The defect. Portfolio capacity is almost always assessed annually — "we have 24 engineer-quarters next year and the approved set needs 24, so it fits". That statement is not a feasibility test. Work arrives when the components need it, and a set that fits in aggregate can be grossly infeasible in a single period, which is the period in which the portfolio then fails.

WORKED EXAMPLE

Worked example 15.2.3 — allocating Meridian's portfolio against a quarterly constraint.

1. **Setup.** One scarce integration team supplies **6 units per quarter** for four quarters — **24 units** in aggregate. Six candidates compete, each with a quarterly demand profile and a net present value computed on PFL-AI Domain 4's methods:

Candidate	Q1	Q2	Q3	Q4	Units	NPV (USD)	NPV per unit
A records rollout continuation	2	3	3	2	10	1,700,000	170,000
B referrals interoperability	2	2	2	2	8	1,360,000	170,000
C pharmacy stock automation	5	1	0	0	6	1,140,000	**190,000**
D clinician scheduling replacement	0	1	2	3	6	1,020,000	170,000
E analytics platform upgrade	1	1	1	1	4	640,000	160,000
F estate network refresh	2	0	1	2	5	750,000	150,000

1. **Formula.** Three procedures compared. **Aggregate ranking:** take candidates in descending NPV per unit while total units remain. **Per-unit greedy with a period check:** take in the same order, skipping any candidate that breaches capacity in any quarter. **Enumeration:** evaluate all $2^6 - 1$

`= 63` non-empty subsets, retain those satisfying `Σ demand ≤ 6` in **every** quarter, and select the maximum NPV.

2. **Substitution and result.**
3. **Aggregate ranking** takes C (6 units), A (10) and B (8) — exactly **24 of 24 units**, NPV **USD 4,200,000**, and full utilisation. It is **infeasible**: Q1 demand is `5 + 2 + 2 =` **9 units against a capacity of 6**.
4. **Per-unit greedy** takes C first on its 190,000 ratio, leaving `(1, 5, 6, 6)`; A and B are then both blocked in Q1 and skipped; D and E fit; F is blocked. Result C + D + E, NPV **USD 2,800,000**, using **16 of 24** units.
5. **Enumeration** finds **26 feasible sets of 63** and selects **A + B + F**: quarterly demand `(6, 5, 6, 6)`, **23 of 24** units, NPV **USD 3,810,000**. The runner-up, A + B + E at **3,700,000** with demand `(5, 6, 6, 5)`, is the set to hold in reserve.
6. **Interpretation.** Two errors, two prices, and they point in opposite directions — which is what makes them hard to see together. The **aggregate illusion** overstates achievable value by `4,200,000 − 3,810,000 =` **USD 390,000**, or **10.24 %** of the true optimum, and it does so while reporting 100 % utilisation, which is why it survives review: a fully committed annual plan looks like good stewardship. The **greedy heuristic** is feasible but leaves **USD 1,010,000** on the table — **26.51 %** of the optimum, capturing only **73.49 %** — and it fails for a nameable reason: the highest-ratio candidate, C, consumes **five of the six Q1 units** and thereby excludes both of the two largest-value candidates. **A ratio cannot see when a demand profile is concentrated in the scarce period, because the ratio has already averaged the period away.** The professional conclusion is uncomfortable and correct: the portfolio-selection method most widely used in practice — rank by a value ratio, take until the budget is full — is wrong twice over, and the correction is not sophisticated. Enumerating 63 subsets is a spreadsheet exercise; at 25 candidates it is 2^{25} `=` **33,554,432** subsets, which is a fraction of a second of computation and is the point: **this is not a hard computation, it is an ungoverned one**, and KA 15.3 makes it the enterprise PMO's job. Two cautions. The optimum uses **23 of 24 units — 95.83 % planned utilisation with slack of** `(0, 1, 0, 0)` — and KA 15.3 shows what that does to the plan's survival probability, so 3,810,000 is the value of a plan, not a forecast. And the NPVs are themselves estimates; a difference of 110,000 between A + B + F and A + B + E is inside anyone's estimating error, so the honest output of an enumeration is the **feasible frontier** — the top few sets and what distinguishes them — rather than a single winner presented as a computation.

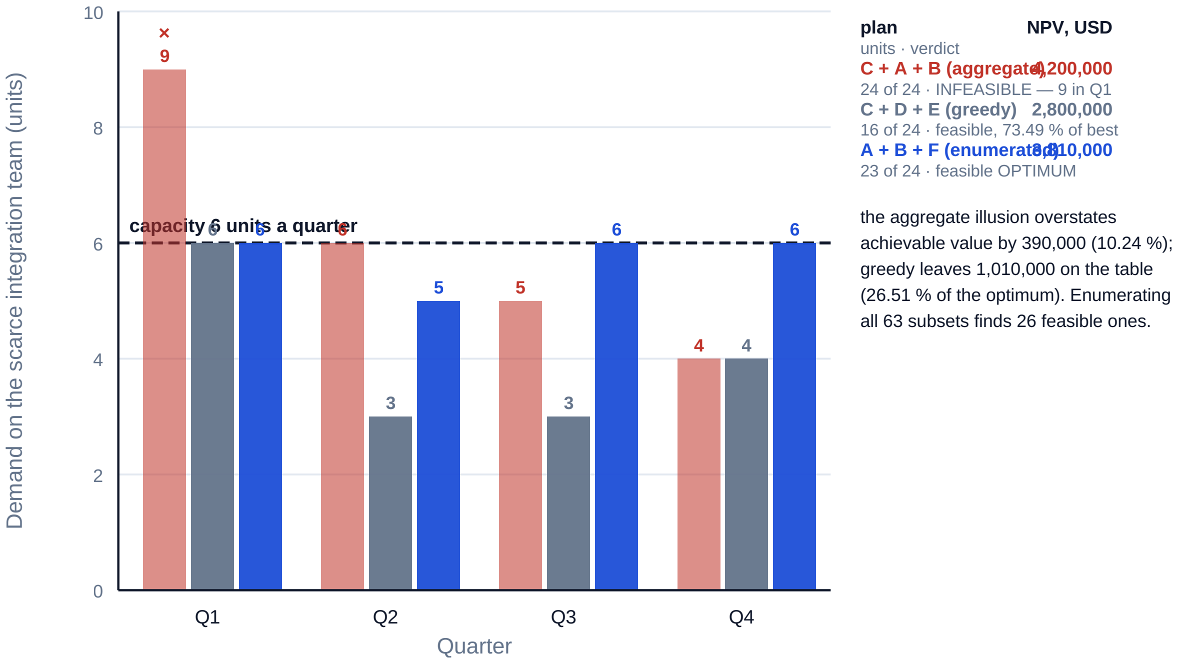


Fig 15.2.1 — Aggregate capacity feasible, period capacity not

15.2.4 Rebalancing, staging and stopping

Rebalancing is the normal state. A portfolio decided once is a budget, not a portfolio. The enumeration above is a snapshot; capacity, value estimates and strategy all move, so the allocation is re-run on a stated cycle with the components already in flight treated correctly — their remaining demand and remaining value, never their original figures, which is the portfolio form of Domain 2's sunk-cost warning (KA 2.4.2).

Staging creates option value. A component admitted as a small first tranche with a decision point attached consumes less of the scarce period and preserves the right to stop. Meridian's tranche design (15.1.2) is this device applied within a programme; at portfolio level the same move lets a constrained period carry more candidates at lower commitment.

Stopping is a portfolio capability, not an admission. The portfolio's hardest arithmetic is that stopping a component releases capacity in the scarce period, and that released capacity has a value equal to the best displaced candidate's contribution. Domain 2's kill criteria (KA 2.4.3) supply the test; the portfolio addition is that the case for stopping is strengthened, not weakened, by naming what the released capacity will do next — because a portfolio board that stops something without reallocating has taken a loss without buying anything.

AI in this KA

Where it earns its place. Running the enumeration, including the exposure-limit constraints, and returning the feasible frontier with the binding period identified for each rejected set — a deterministic search that scales far past human patience. Reconciling component business cases against a benefits register and proposing candidate double counts by matching benefit descriptions, measures and receiving owners across cases, which is the highest-yield document task in this domain and one nobody has time to do by hand. Testing whether a same-pool claim exceeds the pool by summing claims against establishment data. Re-running the allocation under alternative capacity, value and adoption assumptions to produce the sensitivity a board should be given.

Where it must not go. Selecting the portfolio. The enumeration returns sets; the choice among them involves strategic weight, risk appetite, political feasibility and obligations that no objective function contains, and it belongs to the accountable body. Nor may a model confirm a double count: it proposes candidates, and a human with access to the establishment data and the component owners decides which are real, because a wrongly eliminated benefit removes a component's case as surely as a wrongly counted one inflates it. And AI must not generate the value estimates it then optimises over — a closed loop of plausible numbers is the specific failure Domain 14 calls the plausible wrong number (KA 14.3.4).

Verification, concretely. Re-compute the chosen set's per-period demand by hand against capacity, which is a dozen additions and catches every infeasibility. Reconcile the benefits bridge with a signed statement from each receiving-organisation owner that they accept the capped figure — the elimination is only real when the person who must deliver it agrees. Publish the elimination rate and the breakeven elimination rate together, so the board sees whether its decision rule is robust or marginal.

KEY TERMS — KA 15.2

Term	Meaning
Portfolio benefits register	The authoritative statement of portfolio benefits, organised by benefit rather than by component.
Benefits bridge	The reconciliation from gross claimed benefit through eliminations and adoption to net realistic benefit.
Shared benefit	One benefit claimed by two components that both genuinely contribute to it; counted once.
Same-pool over-claim	Aggregate claims on a resource pool exceeding the pool; capped at the pool and allocated.
Cascade (enabler) double count	An enabling component claiming benefit already claimed by the components it enables.
Already-committed benefit	Benefit claimed against a baseline that already assumes a saving committed elsewhere.
Elimination rate	Eliminations ÷ gross claimed benefit; invariant to the adoption assumption.
Breakeven elimination rate	The elimination rate at which a portfolio just satisfies its investment decision rule.
Binding constraint	The resource that actually limits the portfolio — usually a scarce capability, rarely money.
Aggregate-capacity illusion	Judging feasibility on total capacity when the constraint binds period by period.
Feasible frontier	The top few feasible sets and what distinguishes them — the honest output of an enumeration.

SAMPLE MCQS — KA 15.2

MCQ 15.2-A `[15.2.3 · Analysis]` A portfolio's scarce team supplies 6 units a quarter for four quarters. Three candidates demanding 24 units in total are selected on the basis that total capacity is

24 units. The most accurate criticism is that: - A. the portfolio is under-committed - B. aggregate feasibility does not imply period feasibility; the set demands 9 units in Q1 - C. the team's capacity should have been expressed annually - D. net present value is the wrong selection measure

Rationale: Constraints bind per period; the set is aggregate-feasible and Q1-infeasible by 3 units (15.2.3). A inverts the problem; C names the cause of the error as its cure; D is a different debate.

MCQ 15.2-B `[15.2.3 · Evaluation]` In the Meridian portfolio allocation, ranking by NPV per unit yields 2,800,000 while enumeration yields 3,810,000. The reason the ranking fails is that: - A. the ratios were computed incorrectly - B. enumeration uses a different objective function - C. the highest-ratio candidate consumes five of the six units in the scarce quarter, excluding the two largest-value candidates - D. the candidates have equal ratios, so the ranking is arbitrary

Rationale: A ratio averages the period away and cannot see a demand profile concentrated in the binding period (15.2.3). Both methods maximise NPV, so B is wrong; three candidates do share a 170,000 ratio, but that is not what causes the 1,010,000 shortfall.

MCQ 15.2-C `[15.2.1 · Application]` Five components claim 1,981,200 of annual benefit at full potential. Eliminations total 364,800 and portfolio adoption is 70 %. The net realistic annual benefit is: - A. USD 1,616,400 - B. USD 1,386,840 - C. USD 1,131,480 - D. USD 792,036

Rationale: `(1,981,200 − 364,800) × 0.70 = 1,131,480` (15.2.1). A omits the adoption adjustment; B omits the eliminations; D applies adoption twice (`1,131,480 × 0.70`), the error of taking a component figure that is already adoption-adjusted and adjusting it again.

MCQ 15.2-D `[15.2.1 · Analysis]` Two components claim a combined 8.4 administrative posts from a pool of 6.0, valued at 42,000 each. The elimination is: - A. USD 352,800 - B. USD 252,000 - C. USD 100,800 - D. USD 42,000

Rationale: `(8.4 − 6.0) × 42,000 = 100,800` (15.2.1). A is the gross claim; B is the pool's total value, which is the retained benefit, not the elimination; D is one post.

MCQ 15.2-E `[15.2.1 · Evaluation]` A portfolio's benefits bridge eliminates 18.41 % of gross claimed benefit, and the breakeven elimination rate for its four-year payback rule is 12.03 %. The correct conclusion is that: - A. the portfolio clears the rule with a 6.4-point margin - B. the portfolio fails the rule, and would still fail it at any elimination rate above 12.03 % - C. the rule should be relaxed to accommodate the eliminations - D. the eliminations of 364,800 should be reported as a portfolio saving

Rationale: An observed rate above the breakeven rate means the rule is not met (15.2.1). A reads the comparison backwards; C is a decision for the investment committee, not a conclusion from the arithmetic; D commits the error the topic explicitly warns against — eliminating a double count prevents a wrong decision and creates no value.

SELF-CHECK — KA 15.2

1. Name the four portfolio double counts and one correction each. — Shared benefit (count once, record both contributors); same-pool over-claim (cap at the pool and allocate); cascade or enabler (an enabler's own benefit is only what does not pass through a consumer's claim); already committed (reset the baseline).
2. Why is the elimination rate invariant to the adoption assumption? — Adoption scales the eliminations and the gross claim identically, so their ratio is unchanged — which lets the two adjustments be argued and audited independently.

3. What is the honest output of a portfolio enumeration? — The feasible frontier: the top few sets, the binding period for each rejected one, and what distinguishes them — not a single winner, since the value estimates cannot support that precision.

KNOWLEDGE AREA 15.3

Capacity and enterprise PMOs

Topics: 15.3.1 enterprise capacity as the real constraint · 15.3.2 portfolio work in progress · 15.3.3 protective capacity and its price · 15.3.4 the enterprise PMO and its value test.

15.3.1 Enterprise capacity as the real constraint

Definition. Enterprise delivery capacity is the rate at which an organisation can complete change, measured in the unit of its binding capability, at a stated quality. It is a **measured rate**, not a budget and not an aspiration, and the measurement is historical: completions per period over a period long enough to include the organisation's normal disruptions.

Three properties are routinely misunderstood, and each one costs money.

Capacity is a rate, not a stock. An organisation with 40 people does not have "40 people of capacity"; it has a completion rate that its structure, dependencies and interruption load produce. Adding people changes the rate by less than proportionally, and by less again where coordination cost rises — Domain 12's coordination arithmetic (KA 12.2.2 and 12.A.2) and Domain 4's interface count (`n(n−1)/2` against `n` to an integration layer, KA 4.2.3) both bear on this and neither is re-derived here. The practical rule: **treat the completion rate as the primitive and the headcount as one of its inputs.**

Capacity is capability-specific. A portfolio does not consume generic effort. It consumes integration engineering, clinical safety review, data migration, change management and testing in component-specific mixes, and the constraint is whichever of those binds first. A capacity model with one number in it will approve a portfolio that is 60 % utilised on average and 190 % utilised on the thing that matters.

Capacity is consumed by the portfolio's own overheads. Governance attendance, reporting, onboarding, rework and the interruption load of work already in flight all draw on the same people. Capacity measured from completions already includes these; capacity estimated from headcount and availability does not, which is why the estimated figure is always higher and always wrong in the same direction.

15.3.2 Portfolio work in progress

The mechanism. Domain 13 established the flow arithmetic and it is applied, not re-derived, here: **Little's Law**, `W = T × C`, relates work in progress `W`, cycle time `T` and throughput `C` for any system in steady state, independent of arrival and service distributions (KA 13.2.3). Its portfolio reading is uncomfortable and exact: **starting more initiatives does not increase throughput, which capacity sets; it increases cycle time, which is when benefit arrives.**

WORKED EXAMPLE

Worked example 15.3.2 — the price of starting everything.

1. **Setup.** The health group's delivery organisation completed **15** portfolio initiatives in the last **three** years, a measured throughput of **5.0 a year**. The portfolio board has approved and started **12**. The portfolio's net realistic benefit run rate, from KA 15.2.1, is **USD 1,131,480** a year across five components — **USD 226,296** an initiative.
2. **Formula.** `T = W / C` (Little's Law, KA 13.2.3). Benefit forgone from excess concurrency = annual benefit run rate × excess cycle time, where excess cycle time = `(W_actual − W_limit) / C`.
3. **Substitution.** `T = 12/5` against `T = 5/5`; excess `(12 − 5)/5`; then `1,131,480 × 1.4`.
4. **Result.** Average cycle time **2.40 years** at 12 in flight, against **1.00 year** at 5 — an excess of **1.40 years**. The portfolio's entire benefit stream therefore arrives 1.40 years later than it need, worth **USD 1,584,072** — cross-checked per initiative as `226,296 × 1.4 × 5 =` the same figure.
5. **Interpretation.** The number is large, and its character matters more than its size: this is **forgone once and never recovered**. Throughput is unchanged — five initiatives finish per year under either policy — so no annual report will show a loss, no variance will be raised, and the only visible symptom is that everything takes two and a half years. The identity behind it is worth memorising because it prices any concurrency argument in one line: **cost of excess work in progress = annual benefit run rate × excess work in progress ÷ throughput.** Three professional qualifications. The 1,584,072 is undiscounted and is therefore a **floor**; the rigorous treatment discounts the deferred stream (PFL-AI Domain 3), which reduces the headline and does not change the sign. Little's Law requires a steady state, so a portfolio in the middle of a step change in capacity is outside its scope — the law is a theorem about a system, not a forecast for a transition. And a work-in-progress limit is not a decision to do less; it is a decision to **finish** in a sequence, which requires the portfolio board to choose an order and defend it, and that is the reason the limit is resisted: `W = 12` is what a board does when it will not choose. The board that says "all twelve are priorities" has, arithmetically, said that none of them is.

15.3.3 Protective capacity and its price

The problem. KA 15.2.3's optimum plans **23 of 24 units — 95.83 % utilisation — with quarterly slack of** `(0, 1, 0, 0)`. A plan with no slack in three of four quarters is a plan that survives only if nothing goes wrong in those quarters, and the same multiplicative penalty KA 15.1.3 found for milestones applies here to periods.

WORKED EXAMPLE

Worked example 15.3.3 — does protective capacity pay?

1. **Setup.** Measured over the last twelve quarters, the integration team loses units to unplanned support and absence with the following frequencies: **0 units 0.70**, **1 unit 0.22**, **2 units 0.06**, **3 or more 0.02**. The aggressive plan A + B + F demands `(6, 5, 6, 6)` against a capacity of 6, so slack is `(0, 1, 0, 0)`. The alternative is to **reserve one unit a quarter** — plan to 5 — and re-run the enumeration at that capacity, which selects **A + E + F** with demand `(5, 4, 5, 5)`, slack against the true 6 of `(1, 2, 1, 1)`, and NPV **USD 3,090,000**. A quarter's capacity breach slips the affected work one quarter: **13 weeks at the programme cost of delay of 14,280**, or **USD 185,640**.
2. **Formula.** A plan survives a quarter if the loss is no greater than that quarter's slack; the plan survives the year with probability $\prod P(\text{loss} \le \text{slack}_t)$ over the four quarters — the multiplication rule of KA 15.1.3 applied to periods. Expected slippage cost = `(1 − P(survive)) ×` breach cost. Breakeven breach cost = NPV forgone ÷ the improvement in failure probability.
3. **Substitution.** Aggressive `0.70 × 0.92 × 0.70 × 0.70`; protective `0.92 × 0.98 × 0.92 × 0.92`. Then `(1 − P) × 185,640` for each, and `(3,810,000 − 3,090,000) ÷ (0.76311424 − 0.31556)`.
4. **Result.** The aggressive plan survives all four quarters with probability **31.56 %**; the protective plan, **76.31 %**. Protective capacity costs **USD 720,000** of NPV — **18.90 %** of the optimum. Expected slippage cost falls from **USD 127,059** to **USD 43,975**, a saving of **USD 83,084**. The breakeven breach cost is **USD 1,608,744**, equivalent to **112.66 weeks** of programme delay, or **8.67 quarters**.
5. **Interpretation.** On these numbers **protective capacity does not pay**, and a leader should be able to say so, because the received advice — always leave slack — is not a theorem. Spending 720,000 of value to avoid an expected 83,084 of slippage is a poor trade by a factor of nearly nine, and the discipline the arithmetic imposes is to state the condition under which the conclusion reverses: **a breach must cost more than about 1.61 million for the reservation to be worth it.** That condition is not exotic. A breach that lands on a regulatory submission date, a clinical safety approval, a contractual liquidated-damages milestone or a go-live already announced to patients is easily worth more than eight quarters of ordinary delay, and in that case the answer flips decisively. So the professional output is not "reserve capacity" or "do not"; it is: **price the breach on the specific work in the zero-slack periods, then compare.** Two cautions. The 31.56 % figure is itself the product of four period probabilities, so it inherits KA 15.1.3's independence assumption — correlated losses (one absence wave, one support incident spanning two quarters) make clustered breaches likelier than the product suggests, and where the loss driver is shared the calculation should be run on the pessimistic assumption as well. And a 31.56 % survival probability is not a 68 % chance of disaster: it is a high chance of some re-planning, which is normal, and the reason to compute it is to know in advance which quarters will need it, not to be alarmed.

15.3.4 The enterprise PMO and its value test

Definition. An enterprise portfolio management office is the function that maintains the portfolio's authoritative data, runs the allocation and prioritisation process, measures capacity and delivery performance, and provides second-line assurance (Domain 3, KA 3.3.2). It has no delivery accountability and no decision rights; it exists to make the accountable bodies' decisions better.

The role has three honest variants and one dishonest one. A **reporting** office collects and publishes; a **standards** office defines and assures method; a **delivery** office also supplies delivery capability. The dishonest variant is the office that reports without improving any decision, which is Domain 3's governance-artefact failure at enterprise scale (KA 3.1.1) — the organisation has the portal, the template and the monthly pack, and no allocation decision has ever changed as a result.

The value test, and why it is usually failed on the second reading. An enterprise PMO should be subject to the arithmetic it applies to everyone else, and the test that matters distinguishes **recurring** from **one-off** value.

WORKED EXAMPLE

Worked example 15.3.4 — does the enterprise PMO pay?

1. **Setup.** The health group's enterprise PMO costs **8** staff at a blended **132,000** plus **124,000** of tooling and other costs. Its claimed year-one contributions: **half** of KA 15.3.2's work-in-progress deferral recovered by imposing a limit; the **517,650** of annual latency saving computed in KA 15.4.1; **186,000** a year of duplicate tooling eliminated across three components; and the **364,800** of benefits eliminations from KA 15.2.1.
2. **Formula.** Cost = staff × rate + other. Separate the claimed value into recurring and one-off, and test the recurring value alone against the annual cost.
3. **Substitution.** `8 × 132,000 + 124,000`. Recurring `517,650 + 186,000`. One-off `1,131,480 × 0.5`. The eliminations are excluded — see below.
4. **Result.** Cost **USD 1,180,000** a year, which is **24.18 %** of the portfolio's 4,880,000 approved cost. Recurring value **USD 703,650**. One-off value **USD 565,740**. Year one totals **USD 1,269,390** — a surplus of **USD 89,390**. Year two, with the one-off gone, is **703,650** against 1,180,000 — a **deficit of USD 476,350**.
5. **Interpretation.** The office pays in year one and fails in year two, and it will be defunded in year three by a finance director who has noticed. That pattern is the single commonest fate of enterprise PMOs and it is a **reporting** failure before it is a value failure: the year-one case was built on a one-off, and one-offs do not repeat. Note also what has been kept out. The **364,800** of benefits eliminations is excluded because eliminating a double count creates no value (KA 15.2.1) — it prevented a portfolio from being approved on a false payback, which is decision quality of the highest order and is not a cash benefit, and an office that books it as one has falsified its own case. The constructive output is a target rather than a verdict: the office needs **USD 476,350** of additional recurring value, and KA 15.2.3 identified exactly where it is available — the **1,010,000** allocation gap between the ranking heuristic and the enumerated optimum, which recurs at every allocation cycle. The office therefore breaks even recurringly if it captures **47.16 %** of that gap. That is a testable, monitorable commitment of the right shape: it names the mechanism, it is measurable at the next cycle by comparing the selected set against the enumerated frontier, and it fails visibly. **An enterprise PMO whose value case cannot be written in that form should be resized to the reporting it actually does.**

AI in this KA

Where it earns its place. Measuring capacity honestly from completion history rather than from availability assumptions, including the interruption load that headcount models omit. Maintaining a capability-specific capacity model and flagging the constraint that binds first, per period, across a whole portfolio — high-volume bookkeeping with a right answer. Computing plan survival probabilities across candidate plans and loss distributions, and returning the breakeven breach cost, which is the number a board needs. Detecting duplicate tooling and overlapping capability across component architectures. Producing the enumeration and the feasible frontier each cycle, which is precisely the recurring work the PMO's value case above depends on capturing.

Where it must not go. Setting the work-in-progress limit or choosing the sequence, which is a prioritisation decision belonging to the portfolio board — and the decision the board is avoiding when it starts twelve initiatives. Estimating the loss distribution from plausibility rather than from twelve quarters of records. Assessing an enterprise PMO's own value, which is a self-interested judgement no function should automate. And no AI-produced capacity number should reach a board without its measurement window and its exclusions stated, because a capacity figure without those is not a measurement.

Verification, concretely. Recompute throughput from the raw completion list, not from a dashboard, and state the window — five a year from fifteen in three years is one division and it is the foundation of every figure in this KA. Reproduce the survival product by hand for the chosen plan; it is four multiplications. Require the PMO's value case to separate recurring from one-off in the statement itself, and to name the mechanism and the measurement for each recurring item, since that separation is the check that catches the year-two failure a year early.

KEY TERMS — KA 15.3

Term	Meaning
Enterprise delivery capacity	The measured rate at which an organisation completes change, in the unit of its binding capability.
Capability-specific capacity	Capacity modelled per scarce capability, since the portfolio consumes them in component-specific mixes.
Portfolio work in progress	The number of initiatives in flight; by Little's Law it sets cycle time, not throughput.
Excess work in progress cost	Annual benefit run rate × excess work in progress ÷ throughput — forgone once and never recovered.
Protective capacity	Capacity deliberately withheld from allocation so that a period's plan can absorb variance.
Plan survival probability	The product over periods of the probability that each period's loss does not exceed its slack.
Breakeven breach cost	Value forgone by reserving capacity ÷ the improvement in failure probability it buys.
Enterprise PMO	The function that maintains portfolio data, runs allocation, measures capacity and gives second-line assurance; no decision rights.
Recurring versus one-off value	The distinction that decides whether a support function survives its second year.

SAMPLE MCQS — KA 15.3

MCQ 15.3-A `[15.3.2 · Application]` A delivery organisation completes 5 initiatives a year and has 12 in flight. Average cycle time is: - A. 0.42 years - B. 1.00 year - C. 2.40 years - D. 5.00 years

Rationale: `T = W/C = 12/5 = 2.40` years (Little's Law, cited from KA 13.2.3). A inverts the ratio; B is the cycle time at a work-in-progress limit of 5; D is the throughput read as a duration.

MCQ 15.3-B `[15.3.2 · Evaluation]` Reducing portfolio work in progress from 12 to 5 at an unchanged throughput of 5 a year, with a net benefit run rate of 1,131,480, is best described as: - A.

increasing throughput by 140 % - B. bringing the whole benefit stream forward by 1.40 years, worth 1,584,072 once - C. saving 1,584,072 every year thereafter - D. reducing the portfolio's benefit, since fewer initiatives are in flight

Rationale: Throughput is capacity-set and unchanged; cycle time falls by `(12−5)/5 = 1.40` years and the whole stream shifts forward once — `1,131,480 × 1.4 = 1,584,072` (15.3.2). A misreads Little's Law; C converts a one-off shift into an annuity; D confuses starting with finishing.

MCQ 15.3-C `[15.3.3 · Analysis]` A four-quarter plan has slack of 0, 1, 0 and 0 units. Losses of 0, 1 and 2 or more units occur with probabilities 0.70, 0.22 and 0.08. The probability the plan survives all four quarters is closest to: - A. 70.0 % - B. 41.5 % - C. 31.6 % - D. 24.0 %

Rationale: `0.70 × 0.92 × 0.70 × 0.70 = 31.56 %` (15.3.3). A applies the single-quarter figure to the year; B uses 0.92 in two quarters rather than one; D applies 0.70 to all four quarters and ignores the slack quarter's tolerance.

MCQ 15.3-D `[15.3.3 · Evaluation]` Reserving capacity costs 720,000 of NPV and lifts plan survival from 31.56 % to 76.31 %. A quarter's breach costs 185,640. The correct conclusion is: - A. reserve, because the survival probability more than doubles - B. do not reserve unless a breach would cost more than about 1.61 million - C. reserve, because the expected saving of 83,084 is positive - D. the comparison cannot be made without a discount rate

Rationale: `720,000 / (0.76311424 − 0.31556) = 1,608,744` is the breach cost at which the trade breaks even (15.3.3). A argues from a ratio without a price; C is true and irrelevant, since 83,084 is far below 720,000; D is a refinement, not an obstacle.

MCQ 15.3-E `[15.3.4 · Evaluation]` An enterprise PMO costing 1,180,000 a year claims 703,650 of recurring value and 565,740 of one-off value. The soundest assessment is that it: - A. pays, with a year-one surplus of 89,390 - B. pays, because total claimed value exceeds cost - C. fails from year two by 476,350 and needs a named recurring mechanism - D. should book the 364,800 of benefits eliminations to close the gap

Rationale: One-offs do not repeat, so the recurring test is the one that matters (15.3.4). A and B are true of year one and irrelevant thereafter; D books a prevented wrong decision as a cash benefit, which the domain rejects at KA 15.2.1.

SELF-CHECK — KA 15.3

1. Why is capacity a rate rather than a stock? — Completion rate is produced by structure, dependencies and interruption load; headcount is one input to it, and adding people changes the rate by less than proportionally.
2. State the cost of excess portfolio work in progress in one line. — Annual benefit run rate × excess work in progress ÷ throughput; at Meridian, `1,131,480 × 7/5 =` 1,584,072, forgone once.
3. What makes an enterprise PMO's value case credible? — Recurring value separated from one-off, each recurring item with a named mechanism and a measurement — for example capturing 47.16 % of the 1,010,000 allocation gap each cycle.

KNOWLEDGE AREA 15.4

Transformation governance and strategic reporting

Topics: 15.4.1 enterprise decision latency · 15.4.2 strategic reporting and the aggregation trap · 15.4.3 the portfolio view and what it must not hide.

15.4.1 Enterprise decision latency

The extension. Domain 3 established that a decision body's expected latency is `E[wait] = M/2 + L` — half the meeting interval plus the whole paper lead time — that latency sums across escalation tiers, and that a one-week cut in the paper lead time saves a full week while a one-week cut in the meeting interval saves only half of one (KA 3.2.3, KA 3.3.3). None of that is re-derived. The enterprise extension is that a transformation portfolio does not send **one** decision up a path; it sends a **distribution** of decisions up different paths, and the bill is the weighted sum.

WORKED EXAMPLE

Worked example 15.4.1 — the annual cost of an enterprise decision architecture.

1. **Setup.** The health group's five tiers, each with its meeting interval `M` and paper lead time `L` in weeks: **component board** `M` = 2, `L` = 1; **programme board** `M` = 4, `L` = 2; **portfolio board** `M` = 6, `L` = 2; **investment committee** `M` = 12, `L` = 3; **executive board** `M` = 13, `L` = 4. Decisions are routed by value class: class 1 (≤ 25,000) **48** a year through 1 tier; class 2 (25,001–150,000) **22** through 2; class 3 (150,001–500,000) **9** through 3; class 4 (500,001–2,000,000) **4** through 4; class 5 (> 2,000,000, reserved matters) **2** through all 5. Reviewed history, per Domain 3's convention, shows **25 %** of escalated decisions sit on the critical path and convert their wait into delay. Cost of delay **USD 14,280** a week.
2. **Formula.** Tier latency `M/2 + L`; cumulative latency for a `t`-tier path is the sum over the first `t` tiers (Domain 3, KA 3.3.3). Gross latency-weeks = Σ over classes of count × cumulative latency. Delaying weeks = gross × critical-path share. Cost = delaying weeks × cost of delay.
3. **Substitution.** Tier latencies `2/2+1 = 2.0`, `4/2+2 = 4.0`, `6/2+2 = 5.0`, `12/2+3 = 9.0`, `13/2+4 = 10.5`; cumulative `2.0, 6.0, 11.0, 20.0, 30.5`. Then `48(2.0) + 22(6.0) + 9(11.0) + 4(20.0) + 2(30.5)`, `× 0.25`, `× 14,280`.
4. **Result.** A decision travelling all five tiers waits **30.5 weeks** and costs **USD 435,540**. Across the portfolio's **85** decisions: **468 gross latency-weeks**, an average of **5.51 weeks** a decision; **117.0** delaying weeks at the 25 % share; a bill of **USD 1,670,760 a year**.
5. **Interpretation.** One million six hundred and seventy thousand dollars a year is the running cost of a committee timetable, and it appears in no budget, is attributed to no decision, and is described in year-end reviews as the portfolio having been slow. Three results follow, ranked by value per unit of organisational pain, and the ranking is the useful part. **First, the paper deadline.** One week off every tier's paper lead time saves one week on every tier traversal, and the portfolio makes **145** traversals a year: `145 × 0.25 × 14,280 =` **USD 517,650**, or **30.98 %** of the entire bill, from an administrative change that costs nothing and removes no scrutiny anyone can name. This is Domain 3's lever result at enterprise scale, and the scale is what makes it a board-level item rather than a housekeeping one. **Second, the out-of-cycle route.** A written-resolution procedure completing classes 4 and 5 in 1.5 weeks saves **USD 471,240**, or **28.21 %** — nearly as much as the first lever, from a mechanism affecting only **6** of the 85 decisions. That is because those 6 decisions — **7.06 %** of the volume — carry **141 of the 468 gross weeks**, or **30.13 %** of the latency. **Seven per cent of the decisions carry thirty per cent of the delay**, which is why "add the executive board to the approval path" is such an expensive sentence and why the out-of-cycle mechanism belongs at the top of the architecture, not at the bottom. **Third, tier removal**, which is the intervention organisations reach for first and which is worth least here: removing the component board from classes 3 to 5 (raised directly, component board informed) saves 30 gross weeks — **USD 107,100**, or **6.41 %** — because the tier removed is the fast one. **The general rule: remove latency where the latency is, which is the slow tier and the paper deadline, not the tier that is easiest to remove.** Two cautions. The 25 % critical-path share is an assumption from history and the whole bill is proportional to it, so it must be stated as an assumption and re-estimated; and latency is not the only property of a governance tier — a reserved matter exists because someone must be answerable for it, and the arithmetic prices the tier without ever deciding whether it is legitimate. What the arithmetic does is force the trade to be explicit, which is a conversation governance reviews almost never have because until the latency is added up there is nothing to weigh the tier against.

15.4.2 Strategic reporting and the aggregation trap

The principle. A portfolio report exists to support three decisions and no others: **continue, change or stop** each component; **reallocate** capacity between them; and **escalate** what the portfolio cannot resolve. Every element that serves none of the three is decoration, and Domain 11's executive-communication discipline (KA 11.2) applies unchanged — the portfolio addition is that aggregation itself can invert a conclusion.

The trap. Performance indices are **ratios**, and ratios do not average. Averaging component indices without weighting them by the quantity in their denominator gives an answer that is not merely imprecise but frequently of the opposite sign — and this is the most common defect in circulating portfolio reports, because the unweighted average is what a spreadsheet produces by default.

WORKED EXAMPLE

Worked example 15.4.2 — the portfolio index that reversed its sign.

1. **Setup.** At the end of Q2 the five components report, in USD:

Component	PV	EV	AC	BAC	CPI	Share of portfolio AC
Meridian records	1,300,000	1,196,000	1,352,000	2,400,000	0.8846	**54.67 %**
Referrals interoperability	380,000	361,000	372,000	640,000	0.9704	15.04 %
Clinician scheduling	250,000	262,500	245,000	890,000	1.0714	9.91 %
Analytics platform	300,000	306,000	291,000	520,000	1.0515	11.77 %
Pharmacy automation	220,000	224,400	213,000	430,000	1.0535	8.61 %
Portfolio	**2,450,000**	**2,349,900**	**2,473,000**	**4,880,000**		

1. **Formula.** Portfolio `CPI = ΣEV / ΣAC` and `SPI = ΣEV / ΣPV` (the registered EVM definitions, Domain 7 KA 7.3). `EAC = BAC / CPI` by the index method. The defective alternative is the unweighted mean of the component `CPI`s.
2. **Substitution.** `2,349,900/2,473,000` and `2,349,900/2,450,000`; the mean of 0.8846, 0.9704, 1.0714, 1.0515 and 1.0535; then `4,880,000` divided by each index.
3. **Result.** Portfolio `CPI` **0.95** (0.950222) and `SPI` **0.96** (0.959143). The unweighted mean of the component indices is **1.01** (1.006308). `EAC` at the portfolio `CPI` is **USD 5,135,639.81**, giving `VAC` of **(USD 255,639.81)**; at the unweighted mean it is **USD 4,849,408.40**, giving `VAC` of **USD 30,591.60**. The reported variance at completion swings by **USD 286,231** and changes sign. (Cents are carried here only so the swing reconciles exactly; the reporting convention remains whole units.)
4. **Interpretation.** One arithmetic choice moves the portfolio from 30,592 favourable to 255,640 adverse, and the direction of the error is predictable: **the unweighted mean flatters whenever the large components are the troubled ones**, which is the normal case, because scale and difficulty correlate. Here one component holds **54.67 %** of portfolio cost incurred and is the only one below 0.95; four small components each slightly ahead outvote it four to one in an unweighted average and hold 45.33 % of the money. The corrections are two, and both belong in the standard. **Aggregate by summing the numerators and denominators, never by averaging the ratios** — a rule that generalises to every ratio a portfolio reports, including adoption, defect and utilisation rates. And **publish the concentration alongside the**

aggregate: the share of portfolio cost sitting in components below the alert threshold, here **54.67 %** below `CPI` 0.95, which is the figure that tells a board whether one component is carrying the portfolio's problem or the problem is general. Two cautions. A single portfolio `CPI` is an aggregate and therefore always masks distribution — it is a headline for the concentration line beneath it, not a substitute for the component table. And `EAC = BAC/CPI` assumes performance to date continues, which is one of several defensible forecasts; Domain 7 (KA 7.2) governs the selection, and a portfolio report that presents one `EAC` without naming its method has reported an opinion as a measurement.

15.4.3 The portfolio view and what it must not hide

What the one-page view carries. Per component: the continue/change/stop recommendation with its owner; benefit at stake and benefit realised to date (Domain 16 owns the measurement); performance aggregated correctly with the concentration line; the binding-capability demand in the next two periods; the top dependency with an owner on the giving side; and the decisions required with their dates. At portfolio level: net benefit against the bridge, capacity utilisation by capability and period, work in progress against its limit, and the open decisions with their expected latency.

Four things it must not hide, each of which this domain has priced. The **milestone product**, rather than a list of green components against a committed date (15.1.3). The **binding period**, rather than annual capacity (15.2.3). The **elimination rate** on claimed benefits, rather than the gross claim (15.2.1). And the **latency of the open decisions**, rather than a list of items awaiting approval (15.4.1). A portfolio report that omits all four is not a weak report; it is a report that cannot support any of the three decisions it exists for.

The information-age discipline. Domain 14 established that a fact's age governs the decisions it can support (KA 14.2.1). At portfolio level the constraint is sharper than at project level, because a portfolio report is assembled from component reports that were themselves assembled earlier: a monthly portfolio pack routinely presents facts six to eight weeks old as current. The remedy is to **date every figure at its own source**, not at the pack, and to refuse to make a reallocation decision on a capacity figure older than the period it constrains.

AI in this KA

Where it earns its place. Computing the enterprise latency bill across every decision class and candidate architecture, and ranking the redesigns by saving per unit of change — deterministic and tedious in exactly the right proportions. Assembling the portfolio report from component data with each figure carrying its own source date, and flagging figures older than a stated tolerance. Recomputing aggregate indices correctly from the underlying `EV`, `PV` and `AC` sums and detecting where a submitted report has averaged ratios — a mechanical check that catches the KA 15.4.2 defect before a board sees it. Drafting the narrative around a computed exception set, for review.

Where it must not go. Writing the continue/change/stop recommendation. That is a judgement with an accountable owner, and a portfolio board receiving a machine-written recommendation is being invited to ratify rather than to decide — Domain 3's decidability failure (KA 3.1.1) reproduced at enterprise scale. Nor may a model set a tolerance, a threshold or a decision class, all of which are risk-appetite decisions. And no AI-assembled portfolio pack goes to a board without the register of AI uses Domain 14 requires (KA 14.4.3), naming for each output what the tool did, whose decision it informs and who verified it.

Verification, concretely. Recompute the aggregate indices from the sums by hand — two divisions — and compare against whatever the submitted pack contains; a discrepancy is the averaged-ratio defect and it is present more often than not. Reproduce the latency arithmetic for the two largest decision classes. Confirm every figure's source date against its source system on a stated sample. And require that each recommendation in the pack carries a named human owner, because a recommendation without one is not a recommendation, it is a suggestion nobody has made.

KEY TERMS — KA 15.4

Term	Meaning
Enterprise decision architecture	The set of governance tiers, the decision classes routed through them, and the resulting latency distribution.
Gross latency-weeks	Σ over decision classes of count × cumulative tier latency; the portfolio's total waiting.
Tier traversal	One decision passing one tier; the unit on which a paper-lead-time saving accrues.
Out-of-cycle route	Domain 3's mechanism (KA 3.3.3) applied at enterprise scale, where it is worth most: the saving is largest at the slowest tier, because `E[wait] = M/2 + L` grows with `M`.
Aggregation trap	Averaging component ratios instead of summing their numerators and denominators; frequently reverses the sign.
Concentration line	The share of portfolio cost sitting in components below the alert threshold; the companion to any aggregate index.
Portfolio one-page view	The report supporting exactly three decisions: continue/change/stop, reallocate, escalate.
Source-dated figure	A reported value carrying the date of its own source rather than the date of the pack.

SAMPLE MCQS — KA 15.4

MCQ 15.4-A `[15.4.1 · Application]` Five tiers have `M` and `L` of (2, 1), (4, 2), (6, 2), (12, 3) and (13, 4) weeks. The expected latency of a decision that must pass all five is: - A. 18.5 weeks - B. 30.5 weeks - C. 37.0 weeks - D. 20.0 weeks

Rationale: `(2/2+1) + (4/2+2) + (6/2+2) + (12/2+3) + (13/2+4) = 2.0+4.0+5.0+9.0+10.5 = 30.5` (15.4.1, formula from KA 3.3.3). A halves the intervals and omits the paper lead times entirely (1.0+2.0+3.0+6.0+6.5); C sums the meeting intervals alone without halving them or adding the lead times (2+4+6+12+13); D omits the executive tier.

MCQ 15.4-B `[15.4.1 · Evaluation]` A portfolio's 85 decisions generate 468 gross latency-weeks and 145 tier traversals; 25 % sit on the critical path and delay costs 14,280 a week. Cutting one week from every tier's paper lead time saves: - A. USD 517,650 - B. USD 1,670,760 - C. USD 303,450 - D. USD 2,070,600

Rationale: A one-week cut saves one week per traversal: `145 × 0.25 × 14,280 = 517,650`, which is 30.98 % of the 1,670,760 total (15.4.1). B is the whole bill; C applies the saving to the 85 decisions rather than the 145 traversals; D omits the critical-path share.

MCQ 15.4-C `[15.4.1 · Analysis]` In that architecture, 6 of the 85 decisions carry 141 of the 468 gross latency-weeks. The strongest implication is that: - A. those 6 decisions should be delegated downward - B. an out-of-cycle route at the slow tiers is worth far more per decision affected than removing a fast tier - C. the executive board should meet more often - D. the 6 decisions are the most important and their latency is justified

Rationale: 7.06 % of decisions carry 30.13 % of latency, so a written-resolution route for them saves 471,240 against 107,100 from removing the fast component-board tier (15.4.1). A may breach the reserved-matters rationale; C is the weaker lever, since a week off `M` saves half a week; D is a legitimate position but not an implication of the arithmetic.

MCQ 15.4-D `[15.4.2 · Analysis]` A portfolio has `ΣEV` 2,349,900 and `ΣAC` 2,473,000. The unweighted mean of its five component `CPI`s is 1.006. The portfolio `CPI` is: - A. 1.01, the mean of the components - B. 0.95 - C. 0.96 - D. 1.05

Rationale: `CPI = ΣEV/ΣAC = 0.950222` (15.4.2). A averages ratios, the defect the topic exists to prevent; C is the portfolio `SPI` (`ΣEV/ΣPV`); D inverts the ratio.

MCQ 15.4-E `[15.4.2 · Evaluation]` The unweighted-average `CPI` flatters this portfolio because: - A. there are five components and five is too few to average - B. the component with 54.67 % of cost incurred is the only one below 0.95, and four small ahead-of- budget components outvote it - C. the components have different budgets at completion - D. `EAC = BAC/CPI` is the wrong forecasting method

Rationale: An unweighted mean gives each component equal voice regardless of the money behind it, so it flatters whenever the large component is the troubled one (15.4.2). A is not the mechanism; C is true but not the cause, since the weighting that matters is cost incurred; D is a separate question governed by Domain 7.

MCQ 15.4-F `[15.4.3 · Comprehension]` A portfolio report should support exactly three decisions. They are: - A. plan, monitor and control - B. continue/change/stop, reallocate capacity, escalate - C. approve, reject and defer - D. report, review and assure

Rationale: Those three are what a portfolio body can actually do; anything serving none of them is decoration (15.4.3).

SELF-CHECK — KA 15.4

1. Which enterprise latency lever pays most, and why? — One week off every paper lead time: it saves a full week on each of 145 traversals, 517,650 or 30.98 % of the bill, at no cost and with no scrutiny removed.
2. How is a portfolio performance index aggregated correctly? — By summing numerators and denominators (`ΣEV/ΣAC`), never by averaging component ratios — which here reverses the sign of the reported variance at completion.
3. Name the four things a portfolio view must not hide. — The milestone product, the binding period, the elimination rate on claimed benefits, and the latency of the open decisions.

Advanced topics — Domain 15

15.A.1 Where the boundary belongs, and what it costs to get wrong

The project/programme/portfolio boundary is treated as a naming question and is in fact a cost question, because each layer added imposes three recurring charges that can be estimated before the layer exists. It adds **coordination cost** — Domain 4's interface arithmetic and Domain 12's coordination results apply, and neither is re-derived. It adds **latency**, at exactly KA 15.4.1's rates: a tier inserted into a path used by 22 decisions a year at 4.0 weeks of expected wait, with a 25 % critical-path share, costs `22 × 4.0 × 0.25 × 14,280 =` **USD 314,160** a year before it has improved a single decision. And it adds **reporting load**, which consumes the delivery capacity KA 15.3 has just shown to be the binding constraint.

Against those charges a layer must supply something nameable: optionality (components that can be stopped or re-sequenced), a benefit that no component owns alone, or an authority that no component holds. The test is therefore symmetrical and it can be applied to an existing structure as easily as a proposed one: **name what this layer decides that no other layer can, and compare it with the layer's latency bill.** Layers that fail this test are common, and they are usually created by an organisational event rather than a delivery need — a reorganisation, a merger, an audit finding — which is why nobody remembers what they were for.

The inverse error deserves equal attention. Running a genuine portfolio problem as a programme produces the failure Case study B records: components acquire false interdependence, the allocation is never re-run, and capacity is committed once and never rebalanced. The diagnostic is whether the components' benefits are separable. If they are, it is a portfolio, and it needs the arithmetic of KA 15.2, not the coherence machinery of KA 15.1.

15.A.2 Rolling horizons and the option value of deferral

KA 15.2.3's enumeration is a single-shot allocation over four quarters, and real portfolios allocate on a **rolling horizon**: re-optimise each period over the remaining horizon, with committed work treated as fixed and everything else free. Three consequences matter and none of them requires new mathematics.

Re-optimisation dominates a good one-shot plan, because information arrives. A plan optimal on January's estimates is not optimal on April's, and the value of re-running the enumeration is the difference — which is precisely the allocation gap KA 15.3.4 asks the enterprise PMO to capture, now recurring quarterly rather than annually.

Deferral has option value that the enumeration does not price. A candidate deferred one period is not a candidate lost; it is a candidate whose value estimate will be better next period, and whose demand profile may fit a period that is not yet committed. Staging a candidate into a small first tranche (15.2.4) buys the same option more cheaply. The honest position is that this domain's enumeration maximises value under stated estimates and does **not** value the option; a portfolio board should be told that explicitly, because the un-priced option is the strongest available argument for admitting a candidate at low commitment rather than rejecting it.

Commitment discipline is what makes a rolling horizon work. A portfolio that re-optimises freely each period will churn: components stopped and restarted, teams re-formed, and the coordination cost of Domain 12 paid repeatedly. The countermeasure is a stated commitment horizon — typically one to two periods in which the allocation is fixed absent a material event — beyond which everything is provisional and is described as provisional in every report.

15.A.3 The reviewer's portfolio eye

Invariants to test on any programme or portfolio, each cheap and each diagnostic.

Every committed milestone has a stated **product** of its predecessors' probabilities, with the independence assumption disclosed, rather than a list of green components. Every dependency has an owner **on the giving side**, a date needed and a date promised. Every tranche boundary corresponds to a state the organisation can operate in. The benefits register is organised **by benefit**, not by component, and carries a bridge showing the elimination rate and the breakeven elimination rate for the applicable investment rule. Every same-pool claim reconciles to establishment data with the receiving owner's signature. The binding constraint is named as a **capability**, measured as a **rate** from completions, and modelled **per period** — and the selected portfolio's feasibility is demonstrated in the tightest period, not in aggregate. The selected set is compared against an enumerated frontier, and the gap is reported. Work in progress has a stated limit with the excess priced. Every zero-slack period has its breach cost estimated and compared against the value of reserving capacity. Every escalation path has its latency computed as `Σ (M/2 + L)` and priced, and the paper lead times are the first thing examined. Aggregate ratios are computed from summed numerators and denominators and are published with a concentration line. Every reported figure carries its own source date. Every AI-produced portfolio output appears in the AI use register with its verifier named. And the one test that subsumes several others: **the portfolio's approved benefit total reconciles, line by line, to the receiving organisations' own budgets** — because a benefit no budget holder has accepted is a benefit nobody will deliver.

Industry variations — Domain 15

- **Public sector and government.** Annual appropriation cycles impose a hard multi-period constraint that cannot be smoothed by borrowing forward, so KA 15.2.3's period feasibility is binding in law rather than in practice; benefits are frequently non-cash and non-fungible, so the bridge eliminates against policy outcomes rather than budgets, and the same-pool correction is politically the hardest because establishment reductions are announced separately from the programmes that enable them.
- **Regulated industries (pharmaceutical, nuclear, aviation, financial services).** Some milestone predecessors are external approvals whose probabilities the portfolio cannot influence at any price, which makes decoupling the only structural lever available and makes KA 15.1.4's cost comparison decisive; reserved-matters classes are larger, so the latency concentration of KA 15.4.1 is more extreme and the out-of-cycle route correspondingly more valuable.
- **Construction and infrastructure.** The binding capability is often a physical resource — a tunnelling machine, a crane, a rail possession — whose period constraint is absolute and whose reallocation cost is large, so protective capacity usually does pay on KA 15.3.3's test because a breach cascades into a possession window that cannot be re-booked within the year.
- **Healthcare.** Clinical session time is the binding capability and it is not fungible with money; clinical authority is not delegable to a portfolio body, so Meridian's decoupling decisions run through clinical governance in parallel, and the same-pool over-claim is endemic because clinician time released is the benefit almost every digital component claims.
- **Energy and utilities.** Regulatory price-control periods create hard multi-period envelopes and a benefits register dominated by avoided costs, which double-count with unusual ease — Case study B is this pattern; commissioning crews are the classic capability constraint and their peak-month demand is where portfolios fail.
- **Technology and product organisations.** Capacity is measured in team-quarters and the portfolio is genuinely re-optimisable, so the rolling-horizon discipline of 15.A.2 is the dominant practice; the characteristic failure is unlimited work in progress, because starting is cheap and visible while finishing is neither — which makes KA 15.3.2's arithmetic the single highest-value conversation available.

Case study — Domain 15: the milestone that was never going to happen (health, Meridian)

Situation. Meridian's programme board had committed, to its funder and to a patient-facing communications campaign, that all four regions would go live on the same date. Eleven weeks out, the programme's status report showed 24 component dependencies, every one rated green or amber-improving, and a green overall milestone. The programme director asked a question nobody had asked: what is the probability of the committed date?

What the arithmetic showed. The six dependencies of a single region, at the owners' own assessments — 0.90, 0.88, 0.85, 0.92, 0.95 and 0.90 — multiply to **52.95 %**. Four structurally identical regions on one date give **7.86 %**: the committed milestone was **92.1 % likely to be missed**, and the expected number of on-time regions was **2.1182** of four. Nothing in the component reporting was wrong. Every owner had assessed their own item honestly, and no one had multiplied.

The two proposals, priced against each other. The delivery directorate proposed **USD 240,000** of recovery spend — additional trainers and an accelerated estate package — assessed to lift training from 0.95 to 0.98 and estate works from 0.92 to 0.96. That takes a region to **57.00 %** and the programme milestone to **10.56 %**: **2.69** percentage points for 240,000, or **USD 89,093 per point**. The programme office proposed two structural changes instead. Historical data migration (0.85) came off the go-live milestone, operated for two weeks on a documented paper fallback for historical lookups at a priced **USD 18,000** of temporary clerical cover — taking a region to **62.30 %**. Information-governance approval (0.90) was obtained **once at programme level** for all four regions rather than four times regionally, taking a region to **69.22 %** and the four-region milestone to **22.96 %**: **15.10** points for 18,000, or **USD 1,192 per point — 1.34 %** of the cost per point of the money route, a ratio of **74.7 to one**.

How it resolved. The board took both structural changes and declined the recovery spend. It then did the thing the arithmetic made unavoidable: at 22.96 % the simultaneous commitment was still not defensible, so the four regional dates became **internal** dates and the external commitment moved to a single programme milestone set after the fourth region with a buffer sized by Domain 8's methods. The paper fallback went through clinical governance before it went into the plan. Expected late regions fell from **1.8818** to **1.2312**, and the programme reported a milestone probability, an independence assumption and a buffer, in place of twenty-four green squares.

What the domain teaches here. A programme status report that lists green components against a committed date is internally inconsistent, and the inconsistency is one multiplication away from being visible. Once it is visible, the cheap lever is structural: **on a milestone whose probability is a product, removing a factor is worth roughly seventy times what buying improvement in a factor is worth.** And note what the board could not do — it could not estimate its way to 80 %, because that would have required 99.07 % on each of 24 dependencies, a standard nobody in the supply chain had offered or could.

Case study B — Domain 15: the portfolio that was fully funded and could not deliver (energy and utilities)

Situation. A regional transmission utility, anonymised, approved **11** grid-connection projects for a three-year regulatory period. The scarce capability was high-voltage commissioning crews, of which the utility had **9 crew-weeks a month — 324** crew-weeks over 36 months. The approved portfolio demanded **311** crew-weeks, **95.99 %** of aggregate capacity, and the investment committee approved it on precisely that basis: the work fitted. Claimed benefits were **USD 24,600,000** a year.

What had happened. Two defects, both detectable before approval. **The constraint was never tested by period.** Demand peaked at **17 crew-weeks a month for six consecutive months** in months 14 to 19 — **188.89 %** of capacity — against which the annual view was silent. Unmet demand in that window was `(17 − 9) × 6 =` **48 crew-weeks**, and clearing 48 crew-weeks at the observed throughput of 9 a month takes **5.3333 months** of pure catch-up. **The portfolio was therefore a**

minimum of five and a third months late in its approved form, before any project encountered a single problem. And **the benefits were never bridged.** Three projects each claimed the same **3,100,000** of avoided constraint payments and two each claimed the same **1,850,000** of deferred reinforcement: eliminations of **USD 8,050,000** against a claimed 24,600,000, an elimination rate of **32.72 %** and a net portfolio benefit of **USD 16,550,000**.

How it resolved. Six projects completed; five slipped past the regulatory window and carried **USD 7,400,000** a year of the net benefit. Benefit forgone over the 5.3333-month structural catch-up was `7,400,000 × 5.3333/12 =` **USD 3,288,889**, and the portfolio delivered **USD 9,150,000** of its 16,550,000 on time — **55.29 %**. A re-sequencing exercise then did what the approval process had not: it enumerated feasible sets against the **monthly** constraint and deferred two projects to the following period. Nine projects, carrying **USD 15,000,000**, landed inside the window — **90.63 %** — with the deferred two carrying 1,550,000. **Re-sequencing alone moved USD 5,850,000 of annual benefit from late to on time, and it changed no project's scope, budget or team.**

What the domain teaches here. Aggregate capacity feasibility is not feasibility; a portfolio at 95.99 % of annual capacity was at 188.89 % of monthly capacity for half a year, and one afternoon of period-by-period arithmetic before approval would have found it. A benefits register organised by component rather than by benefit will let the same avoided cost be claimed three times, and a 32.72 % elimination rate is not an outlier. And the largest single improvement available to this portfolio was not more crews, more money or better project management: it was **sequence**, which is free.

Executive perspective — Domain 15

What a portfolio director cannot delegate in this domain:

- **The product behind every committed milestone.** Insist on $\Pi\ p_i$ with its independence assumption stated, not a count of green components. A programme that cannot produce it has not assessed its own commitment (15.1.3).
- **The named binding constraint, measured as a rate and modelled by period.** Not a budget, not an annual total, not a headcount. Feasibility is demonstrated in the tightest period or it is not demonstrated (15.2.3, 15.3.1).
- **The benefits bridge before approval.** The elimination rate, the breakeven elimination rate for your investment rule, and a receiving owner's signature against every capped figure. Meridian's 18.41 % elimination rate broke a four-year payback rule whose breakeven was 12.03 % (15.2.1).
- **The work-in-progress decision, which is a prioritisation decision you are avoiding.** Twelve initiatives in flight at a throughput of five costs 1.40 years of the entire benefit stream — 1,584,072, forgone once. Saying that all twelve are priorities is arithmetically saying none is (15.3.2).
- **Your own decision architecture's annual bill.** 468 gross latency-weeks, 1,670,760 a year, and a free 517,650 available from one week off every paper deadline. Examine the paper deadlines and the slow tiers, not the tier that is easiest to remove (15.4.1).
- **How your portfolio's performance is aggregated.** Sums of numerators and denominators, with a concentration line. An averaged index moved this portfolio from 30,592 favourable to 255,640 adverse, and the flattering direction is the normal one (15.4.2).

Calculation exercises — Domain 15

Exercise 15.1 A programme milestone requires five independent predecessors, assessed at 0.96, 0.93, 0.90, 0.88 and 0.80. (a) Compute the milestone probability. (b) Compute it again with the 0.80 dependency decoupled. (c) If instead every dependency were assessed at 0.93, how many could the milestone carry and remain at or above 0.50? (d) What uniform per-dependency probability would deliver 0.85 across five dependencies? Solution. (a) `0.96 × 0.93 × 0.90 × 0.88 × 0.80 =` **56.5678 %**. (b) Divide by 0.80: **70.7098 %** — a gain of **14.1420** percentage points from one structural change. (c) 0.93^9 = **52.0411 %** and 0.93^{10} = **48.3982 %**, so at most **nine**. (d) `0.85^(1/5) =` **96.8019 %**. Common error: answering (a) with the minimum, **0.80**, or with the mean of the five, **89.40 %** — both of which are above the true figure and both of which are what a component-by-component status report implicitly asserts.

Exercise 15.2 A portfolio's scarce capability supplies **5 units a period** for three periods (15 units in aggregate). Four candidates: P demands (2, 2, 2) for an NPV of 1,020,000; Q demands (2, 2, 1) for 850,000; R demands (4, 1, 0) for 950,000; S demands (1, 1, 2) for 640,000. Compute the selection under (a) ranking by NPV per unit with a period feasibility check, (b) full enumeration, and (c) ranking by NPV per unit against aggregate capacity only. State the cost of each error. Solution. Ratios: R **190,000**, P **170,000**, Q **170,000**, S **160,000**. (a) Take R, leaving (1, 4, 5); P and Q are both blocked in period 1 and skipped; S fits. **R + S = USD 1,590,000**, using 9 of 15 units. (b) Enumerating all $2^4 - 1 = 15$ subsets, the maximum feasible set is **P + Q + S = USD 2,510,000** with demand exactly (5, 5, 5). The heuristic leaves **920,000**, **36.65 %** of the optimum. (c) Aggregate-only ranking takes R, P and S for 15 of 15 units and **2,610,000** — apparently the best answer and fully utilised — but period 1 demands `4 + 2 + 1 =` **7 units against 5**, so it is infeasible and overstates achievable value by 100,000. Common error: reporting (c) as the answer because it uses 100 % of capacity; full utilisation of an annual total is evidence of nothing.

Exercise 15.3 Four components claim **1,240,000** of annual benefit at full potential. Review finds: 85,000 of benefit claimed twice; claims of 4.8 posts against a pool of 3.5 valued at 38,000 each; 64,000 of enabler benefit already counted downstream; and 45,000 already committed outside the portfolio. Adoption is 65 %; portfolio cost is 3,150,000; the investment rule is a four-year simple payback. Compute the elimination rate, the net realistic benefit, the payback on both the reconciled and unreconciled figures, and the breakeven elimination rate. Solution. Eliminations `85,000 + (4.8 − 3.5) × 38,000 + 64,000 + 45,000 = 85,000 + 49,400 + 64,000 + 45,000 =` **USD 243,400**, an elimination rate of **19.6290 %**. Net full potential **996,600**; at 65 % adoption, **USD 647,790**. The unreconciled figure is `1,240,000 × 0.65 =` **806,000**, overstating by **158,210** — **24.42 %** of the honest number. Payback: honest **4.8627 years**; unreconciled **3.9082 years**, which clears the rule. Breakeven: the rule needs `3,150,000/4 =` 787,500, so `787,500/0.65 =` 1,211,538.46 of net full potential and at most **28,461.54** of eliminations — a breakeven elimination rate of **2.2953 %**. Common error: computing the elimination on the same-pool claim as the pool's value (3.5 × 38,000 = 133,000) rather than as the excess over it; the excess is 1.3 posts.

Exercise 15.4 Four governance tiers have `M` and `L` in weeks of (2, 1), (4, 1), (8, 2) and (12, 4). Decision volumes: **40** through one tier, **18** through two, **7** through three, **3** through four. 30 % of decisions sit on the critical path; cost of delay is **11,500** a week. Compute the gross latency-weeks, the annual delay bill, and the saving from cutting one week from every paper lead time. Then state what proportion of the latency the top class carries. Solution. Tier latencies `2/2+1 = 2.0`, `4/2+1 = 3.0`, `8/2+2 = 6.0`, `12/2+4 = 10.0`; cumulative **2.0, 5.0, 11.0, 21.0**. Gross `40(2.0) + 18(5.0) + 7(11.0) + 3(21.0) = 80 + 90 + 77 + 63 =` **310 latency-weeks**. Delaying weeks `310 × 0.30 =` **93.0**;

bill `93 × 11,500 =` **USD 1,069,500**. Tier traversals `40 + 36 + 21 + 12 =` **109**, so a one-week cut in every paper lead time saves `109 × 0.30 × 11,500 =` **USD 376,050 — 35.1613 %** of the bill. The top class is **3 of 68** decisions (**4.4118 %**) carrying **63 of 310** weeks (**20.3226 %**). Common error: adding meeting intervals and omitting paper lead times, which gives cumulative 1.0, 3.0, 7.0 and 13.0 and a total of **182** weeks — an understatement of **41.2903 %**.

Exercise 15.5 A delivery organisation completes **8** initiatives a year at steady state and has **20** in flight. Its portfolio's net benefit run rate is **2,240,000** a year. Compute the average cycle time, the cycle time under a work-in-progress limit of 8, the benefit consequence of the excess, and the limit that would deliver a 1.5-year cycle time. Solution. `T = W/C = 20/8 =` **2.50 years**; at `W` = 8, `T =` **1.00 year**; excess **1.50 years**. The whole benefit stream therefore arrives 1.50 years later than it need, worth `2,240,000 × 1.5 =` **USD 3,360,000**, forgone once and never recovered. For `T` = 1.5, `W = C × T = 8 × 1.5 =` **12**. Common error: assuming that cutting work in progress from 20 to 8 cuts throughput in proportion — Little's Law says throughput is set by capacity, not by how much work is started (KA 13.2.3), so the same eight initiatives finish each year under either policy and the only thing that changes is when.

Practitioner's toolkit — Domain 15

Adoption-ready artefacts; adapt headings to your organisation, then keep them stable.

Toolkit 15.T.1 — Programme dependency register with a computed milestone probability

One row per dependency: reference · milestone it feeds · what is needed · **owner on the giving side** (name, not function) · owner on the receiving side · date needed · date promised · assessed probability of being met on the date needed · **evidence for that assessment** · shared cause or driver · consequence of breach · decoupling option and its priced cost. A footer per milestone computes $\Pi\ p_i$ across its predecessors, states the independence assumption, lists the shared drivers that violate it, and shows the probability after each candidate decoupling with its cost — so the board sees the product, the assumption and the priced structural options on one page. The register's purpose is to make the difference between component status and milestone probability **impossible to overlook**; a programme whose report shows green components and no product has not filled it in.

Toolkit 15.T.2 — Multi-period portfolio allocation sheet

A grid: candidates down the side; the binding capability's demand **per period** across the top, plus total units, value, value per unit, and the exposure tags (supplier, platform, team, regulatory) used as feasibility limits. Below it: capacity per period as a measured rate with its measurement window; per-period demand and slack for the selected set; and the **feasible frontier** — the top three to five feasible sets, their values, and the binding period for each rejected higher-value set. Three mandatory lines: the value of the selected set, the value of the enumerated optimum, and the gap between them; the aggregate-only answer and its infeasible period, recorded so the illusion is visible rather than merely avoided; and the plan's survival probability from the period slacks with the breach cost estimated. Re-run every cycle with committed work fixed and remaining demand and remaining value used for work in flight.

Toolkit 15.T.3 — Portfolio benefits bridge

A single reconciliation, top to bottom, published with every investment paper: gross claimed benefit at full potential by component; then a line per elimination, each tagged **shared / same-pool / cascade / already-committed**, with the components affected, the basis, and the receiving owner who has accepted it; then net full-potential benefit and the **elimination rate**; then the adoption adjustment with its source; then net realistic benefit. Beneath it: the investment rule being applied, the result, and the **breakeven elimination rate** at which the rule just holds. Two integrity checks, both counts: same-pool claims reconciled against establishment data, and benefits accepted in writing by the receiving budget holder. A bridge whose eliminations no receiving owner has signed is a calculation, not a reconciliation.

Exam preparation — Domain 15

What is assessed. The project/programme/portfolio distinction by decision rather than scale; programme architecture in components, tranches and a target operating state; the dependency multiplication rule and the structural levers; the four portfolio double counts and the benefits bridge; portfolio balancing under a constraint that binds per period; enterprise capacity as a measured rate; portfolio work in progress and its cost; protective capacity and its breakeven; the enterprise PMO's recurring value test; enterprise decision latency and its redesigns; and correct aggregation of portfolio performance.

The calculations to be able to do under time pressure. $\Pi\ p_i$ for a milestone, and the k-th root of a target confidence to get the required per-dependency probability. The effect of decoupling, as a multiplication by $1/p_i$. Per-period feasibility of a candidate set, and enumeration over a small candidate list. A benefits bridge with all four elimination types, the elimination rate, and simple payback before and after. $T = W/C$ and the excess-work-in-progress cost as benefit run rate × excess ÷ throughput. Plan survival as a product over periods, and the breakeven breach cost. Gross latency-weeks from decision classes and cumulative tier latency $\Sigma\ (M/2 + L)$, priced at a cost of delay, and the saving from a one-week cut per traversal. Portfolio `CPI` and `SPI` from summed `EV`, `AC` and `PV`, and $EAC = BAC/CPI$.

The traps. Taking a milestone probability as the average or the minimum of its predecessors' rather than their product (Exercise 15.1) · treating aggregate capacity feasibility as feasibility (Exercise 15.2c, Case study B) · ranking by value per unit of constraint and assuming the result is optimal when demand is concentrated in the binding period (15.2.3, Exercise 15.2) · computing a same-pool elimination as the pool's value rather than the excess over it (Exercise 15.3) · omitting the adoption adjustment or applying it twice (MCQ 15.2-C) · booking eliminated double counts as a saving (15.2.1, 15.3.4) · assuming a work-in-progress cut reduces throughput (Exercise 15.5) · converting a one-off benefit shift into an annuity (MCQ 15.3-B) · justifying a support function on one-off value (15.3.4) · adding meeting intervals without paper lead times in a multi-tier path (Exercise 15.4) · applying a per-traversal saving to the decision count instead of the traversal count (MCQ 15.4-B) · averaging component ratios instead of summing numerators and denominators (15.4.2) · reporting a single aggregate index with no concentration line · omitting the independence assumption when a product is printed.

How the domain connects. Domain 1 supplies the accountability principle and the cost of delay every figure here is priced at. Domain 2 supplies the business case, the benefits map, the kill criteria and the greedy-versus-enumeration result this domain extends to multiple periods. Domain 3 supplies $E[wait] = M/2 + L$, the paper-deadline lever and the tier-summation this domain scales to a whole decision architecture. Domain 4 supplies the interface arithmetic and the integrated baseline a programme's components must reconcile to. Domain 6 consumes programme dependency dates as schedule inputs. Domain 7 supplies the EVM definitions aggregated in KA 15.4.2. Domain 8 supplies merge bias — the schedule form of this domain's multiplication rule — the common-driver treatment that qualifies independence, and the methods that size a programme buffer. Domain 12 supplies the coordination cost of every layer added. Domain 13 supplies Little's Law, applied here to portfolio work in progress. Domain 14 supplies the AI use register, the verification tier and the information-age discipline. Domain 16 measures whether the benefits this domain reconciled were realised. PFL-AI Domain 4 supplies the appraisal methods behind the candidate values.

Domain 15 summary

A portfolio is not a bigger project, because at enterprise scale quantities stop adding. They multiply, they compete for the same period, and they double-count — and each of those has an arithmetic this domain computes.

They multiply. A milestone requiring all of k independent predecessors is met with the **product** of their probabilities. Meridian's Region A go-live, with six predecessors none worse than 0.85, stands at **52.95 %**; four such regions on one date stand at **7.86 %** — a commitment **92.1 %** likely to be missed while all twenty-four components report green. Better estimating cannot fix it: 80 % across 24 dependencies demands **99.07 %** on each. Structure can: decoupling migration and centralising the information-governance approval took a region to **69.22 %** and the programme milestone to **22.96 %** for **USD 18,000**, against **USD 240,000** of recovery spend that bought 2.69 points — **USD 1,192** a point against **USD 89,093**, a ratio of **74.7 to one**.

They compete for the same period. Meridian's portfolio has 24 units of scarce integration capacity a year, and the annual view approves a set worth **4,200,000** that demands **9 units in a 6-unit quarter**. Ranking by value per unit is feasible but takes **2,800,000**, because the highest-ratio candidate consumes five of six units in the binding quarter and excludes the two largest; enumerating 63 subsets finds **26 feasible** and an optimum of **3,810,000**. The aggregate illusion overstates by **390,000 (10.24 %)**; the heuristic leaves **1,010,000 (26.51 %)** on the table. That optimum runs at **95.83 %** planned utilisation and survives all four quarters with probability **31.56 %**; reserving a unit a quarter lifts survival to **76.31 %** at a cost of **720,000** of value, which does **not** pay unless a breach costs more than **USD 1,608,744** — a condition a regulatory or go-live date meets easily and an ordinary quarter does not. And starting twelve initiatives against a measured throughput of five gives a **2.40-year** cycle time instead of **1.00**, deferring the portfolio's whole **1,131,480** benefit stream by **1.40 years — USD 1,584,072**, forgone once, invisible in every annual report.

They double-count. Five components claiming **1,981,200** at full potential contain **364,800** of shared, same-pool, cascade and already-committed benefit — an **18.41 %** elimination rate, invariant to the adoption assumption — leaving **1,131,480** a year net at 70 % adoption. That moves simple payback from **3.5188** to **4.3129 years** and fails a four-year rule whose breakeven elimination rate was **12.03 %**. Eliminating a double count creates no value; it prevents a wrong decision, which is why it cannot appear in the enterprise PMO's value case — a case that here shows a **89,390** year-one surplus on a one-off and a **476,350** year-two deficit, closable only by capturing **47.16 %** of the recurring allocation gap.

And the decisions themselves have a price. Five tiers, 85 decisions and **468 gross latency-weeks** cost **USD 1,670,760** a year at Meridian's cost of delay. One week off every paper deadline saves **517,650 (30.98 %)** and costs nothing; a written-resolution route for the **6** decisions that carry **30.13 %** of the latency saves **471,240**; removing the fast component-board tier saves **107,100**. Remove latency where the latency is. Then report it correctly: the same portfolio reads **CPI 0.95** when aggregated as `ΣEV/ΣAC` and **1.01** as an unweighted mean of component indices, swinging variance at completion by **USD 286,231** and reversing its sign, because the component holding **54.67 %** of the cost incurred is the only one in trouble.

The through-line: **project intuitions fail upward in four specific, computable ways — probabilities multiply, constraints bind per period, benefits overlap, and governance tiers accumulate — and a leader who computes all four can defend a portfolio to a board, while one who computes none of them will be asked to commit to a milestone that is 7.86 % likely and to call it green.**

16

DOMAIN 16

Transition, Closeout and Benefits Realization

Group: Enterprise delivery and the digital future (Domain 3 of 3 in Part Four). **Target:** ~74 pages. **Binds to:** the PCI Book Pattern Specification and the shared registries (`docs/books/registries/`). This domain closes the book and closes the **Meridian Care Records** account opened in Domain 1 and appraised in Domain 2, and takes **Project Auriga** through commissioning and final account. It consumes the cost of delay from Domain 1, the benefits map and baseline discipline from Domain 2, the decision record from Domain 3, the change and configuration baseline from Domain 4, the acceptance machinery from Domain 9, the contract mechanisms from Domain 10 and the AI-use register from Domain 14. British English; USD (+SAR where useful, indicative `USD 1 ≈ SAR 3.75`). Percentages are given to two decimals where a threshold turns on them; money to whole units unless cents are material.

Why this domain exists

Fifteen domains have described how to choose work, structure it, plan it, price it, staff it, lead it and control it. Every one of them assumed something that only this domain can supply: **that the project ends, that something else takes it over, and that somebody eventually finds out whether it was worth doing.** Until that happens, a project's entire claim to value rests on an estimate, and the organisation's memory of it rests on whoever is still in post.

Closeout is, on this book's argument, the least respected phase in the delivery lifecycle and the one in which value is most easily lost beyond recovery. The failure modes below are the ones a reviewer meets repeatedly, and they are not primarily about paperwork. Handover happens on a date rather than on a condition, so a receiving organisation that is 90 per cent ready absorbs a system it cannot run. Contracts are left open because closing them requires a conversation nobody wants, and the carrying cost of that avoidance is never added up. Lessons are captured diligently and never retrieved, so the same defect is bought again at full price on the next programme. And benefits — the entire justification for the expenditure — are measured, if at all, against a baseline that was never taken, by an owner who was never appointed, over a period after everyone who could explain the variance has moved on.

That is the domain's central claim: **the closing account is the only honest one, and it is settled after the project ends, not at it.** A leader who cannot state what was realised, against what was promised, decomposed into the reasons for the difference, has not completed the work — they have merely stopped doing it. This domain makes that account computable. KA 16.1 replaces the handover date with a handover condition and shows arithmetically why partial readiness is not proportional readiness. KA 16.2 takes the receiving organisation's run cost seriously and prices the final account, the retention mechanism and the specific cost of leaving a claim open. KA 16.3 treats knowledge as an asset with a retrieval rate, and shows that the value of a post-project review is destroyed at the retrieval step, not the capture step. KA 16.4 settles Meridian's account, decomposes the variance, and sets the retention and deletion rules for the evidence, data and models the programme leaves behind.

Learning objectives. After this domain a candidate can: specify handover as a set of conditions rather than a date, and distinguish what handover transfers from what it cannot; design commissioning and service-acceptance tests that test the service rather than the works; **compute the probability of a clean go-live as the conjunction of its readiness conditions, demonstrate why an averaged readiness dashboard systematically overstates it, and rank remediation by proportional rather than absolute shortfall**; price a hold-or-go decision against remediation cost under both independent and correlated failure assumptions; structure hypercare with exit criteria and a reversion plan; transfer a service to a run organisation with its total cost of ownership stated; operate the retention mechanism and strike a final account including variations, claim settlement and defect recovery; **compute the monthly carrying cost of an open claim and the breakeven settlement premium**; run a post-project review whose lessons are retrievable, and compute the breakeven reuse rate that makes it worth holding; build a benefits measurement plan with a named owner, defined measures, a pre-change baseline and a stated duration; **decompose realised benefit against the business case into adoption, benefit-per-unit and timing components, state the interaction term and the convention that assigns it**; test attribution against a comparison cohort established at baseline; set a class-by-class retention and deletion schedule for evidence, personal data and the models used in delivery decisions; and govern AI use in measurement, review and archive without letting a tool author the conclusion it is measuring.

The master threads, and the figures this domain settles. Meridian Care Records returns for the last time: the clinical-records rollout to **40 clinics**, an approved cost of **USD 2,400,000**, a business case claiming **USD 979,200** a year against Domain 1's honest **USD 685,440** at 70 per cent adoption, a cost of delay of **USD 14,280** a week, and Domain 2's appraisal over eight years at 7 per cent giving a present value of **USD 3,732,898** on the honest ramped profile and an NPV of **+USD 1,332,898**. This domain measures what actually happened: **27 of 40 clinics** in daily use at **5.4 clinician-hours** released a week, worth **USD 594,864** a year — and decomposes the **USD 90,576** shortfall against the honest case, prices the operating cost the case omitted, and computes the realised NPV of **(USD 304,827)** and the **USD 180,000** of post-project work that turns it positive. Project Auriga returns from Domains 6 to 8 — the 25-week control-systems upgrade for a regional utility, **BAC USD 4,000,000**, `CPI` **0.91** and `SPI` **0.92** at week 13 — to have its commissioning test and final account settled in Case study B. The benefits measure used throughout is the benefits register's benefit measure (`EVA(benefit)` in the shared symbol table, written in words here to avoid the earned-value clash).

KNOWLEDGE AREA 16.1

Handover, commissioning and readiness

Topics: 16.1.1 what handover transfers · 16.1.2 commissioning and service acceptance · 16.1.3 readiness as a conjunction of conditions · 16.1.4 the go-live decision, hypercare and reversion.

16.1.1 What handover transfers

Definition. Handover is the transfer of a deliverable, and of the accountability for operating it, from the delivering organisation to a named receiving organisation, on stated conditions, with the supporting obligations of both parties recorded.

Three things transfer, and they are habitually confused. **Custody** of the asset or service — who holds it, runs it, pays for it. **Accountability for its operation** — who answers when it fails at 02:00. And **the residual obligations of the deliverer** — defects liability, warranty, documentation, training, transition support. Each has a different date, and a handover that names one date for all three has hidden two of them.

Two things do **not** transfer, and pretending otherwise is the commonest and most expensive handover error. **Accountability for the benefit does not transfer at handover, because it was never the project's to begin with** — Domain 2 placed it with a benefits owner in the receiving organisation from the outset, and handover is where that appointment is tested rather than where it is made. And **the deliverer's accountability for what it built does not transfer either**: a defect present at handover is a defect the deliverer owns, and the retention and defects-liability mechanisms of 16.2.2 exist precisely because signature is not absolution.

Handover on a condition, not a date. The single structural choice that determines whether a transition works is whether the handover trigger is a calendar date or a satisfied condition set. A date creates an incentive to declare readiness; a condition set creates an incentive to achieve it. The condition set must be agreed early enough that it can still influence the work — which in practice means at the same gate that approves the delivery plan (Domain 3, KA 3.3.1) — and it must be written so that each condition is objectively assessable by someone who is not the person delivering it.

16.1.2 Commissioning and service acceptance

Definition. Commissioning is the progressive proving of a deliverable in its operating environment, from component test through integrated test to performance demonstration under representative load; service acceptance is the receiving organisation's formal agreement that the service, not merely the works, meets the agreed specification.

The distinction between **works tests** and **service tests** is where most acceptance regimes fail. A works test asks whether the thing was built as specified. A service test asks whether the organisation can run it: whether the runbook is correct, the alerts reach a person, the second-line team can restore from backup within the stated window, and the fallback works when the primary does not. A programme that has passed every works test and no service test has proved that it built the right object and knows nothing about whether it delivered a service. Domain 9's acceptance and nonconformance machinery supplies the discipline for both; the addition here is that the **service test must be witnessed and run by the receiving organisation**, because a service test performed by the project team tests the project team.

Three commissioning disciplines carry disproportionate weight. **Test on production configuration**, since a test in a staging environment that differs from production tests the staging environment — the defect most often found on the first live day is a configuration difference nobody listed. **Test the failure paths, not only the success paths**: reversion, degraded mode, manual fallback, and the restoration of service after an outage, because those are the paths that will be used under stress by people who have never rehearsed them. And **record the evidence in a form that survives the team** — witnessed test records, versioned configuration baselines (Domain 4, KA 4.3.2), and the as-built documentation that 16.3.1 shows costs seven times as much to reconstruct later.

16.1.3 Readiness as a conjunction of conditions

Definition. Readiness is the joint satisfaction of every condition necessary for a clean transition. It is therefore a **conjunction**, and its probability is the product of the conditions' probabilities — not their average, not their weighted average, and not the percentage of conditions that are green.

This is the most consequential piece of arithmetic in the domain, because the artefact that organisations actually use — a readiness dashboard showing a weighted percentage — is not a measurement of readiness at all. The product rule itself is Domain 15's (KA 15.1.3, where a six-predecessor programme milestone with no predecessor assessed worse than 0.85 comes to 52.95 per cent); what this KA adds is the contrast with the averaged dashboard, the remediation ranking that follows from it, and the two correlation bounds the go-live decision must be tested against. With k conditions each holding with probability p_i, and treating their failures as independent,

$$P(\text{clean transition}) = \prod p_i \quad \text{for } i = 1 \ldots k$$

whereas a dashboard reports $(\Sigma\ p_i)/k$ or some weighted variant of it. Those two numbers diverge quickly and in one direction: the product is always at or below the average, and the gap widens with both the number of conditions and their dispersion. A programme with many conditions, most of them nearly complete, will show a comfortable average and carry a poor probability of a clean transition — which is the precise mechanism by which "we were 93 per cent ready" and "the go-live failed" are both true statements about the same event.

WORKED EXAMPLE

Worked example 16.1.3 — the readiness dashboard that said 90.71 per cent.

1. **Setup.** Meridian's clinic go-live gate carries **seven** conditions. The transition manager's assessment of the probability that each is fully met at the planned go-live date, for a representative clinic, is: clinician training completed to the roster threshold **0.96**; data migration reconciled and signed off **0.90**; interfaces verified in production **0.94**; network and devices installed and load-tested **0.98**; workflow redesign signed off by the clinical lead **0.85**; clinical champion in post and trained **0.80**; fallback and business-continuity procedure rehearsed **0.92**. These are locally calibrated planning figures from the four pilot clinics, not constants. The programme dashboard reports readiness as the equal-weight average.
2. **Formula.** Dashboard reading $= (\Sigma\ p_i)/k$. Probability of a clean go-live $= \prod\ p_i$.
3. **Substitution.** Average $= 6.35/7$. Product $= 0.96 \times 0.90 \times 0.94 \times 0.98 \times 0.85 \times 0.80 \times 0.92$.
4. **Result.** Dashboard reading **90.71 %**. Probability of a clean go-live **49.79 %**. The gap is **40.92 percentage points**.
5. **Interpretation.** The dashboard is not optimistic; it is measuring a different quantity, and the quantity it measures has no operational meaning — no one is interested in the average completeness of the conditions, only in whether all of them hold. Four consequences follow, and all four are actionable. **First, the running product is the useful display.** Reading the conditions in sequence, the conjunction falls 96.00 → 86.40 → 81.22 → 79.59 → 67.65 → 54.12 → **49.79 %**: the two conditions that are also the two least likely to be reported as blockers — workflow sign-off and the clinical champion, both owned outside the project, both part of Domain 2's enabling change — together remove **25.47 percentage points**. **Second, remediation must be ranked by proportional shortfall, not absolute gap**, because lifting a condition from p to p' multiplies the conjunction by p'/p. Lifting the champion condition from 0.80 to 0.98 multiplies by 1.2250 and gains **11.20 points**; lifting training from 0.96 to 0.98 multiplies by 1.0208 and gains **1.04**. The weakest link is worth almost eleven times the strongest, and a remediation plan that works through the conditions in dashboard order will spend its budget on the wrong ones. **Third, certainty cannot be bought.** With all seven conditions at 0.98 the conjunction is only **86.81 %** — so **13.19 %** of go-lives still fail, which is the arithmetic case for the reversion plan of 16.1.4 rather than for more assurance. A **95 %** chance of a clean go-live would require **99.27 %** on every one of the seven, and a programme that claims that has almost certainly not assessed its conditions honestly. **Fourth, and the professional caution: independence is an assumption, and it must be stated rather than assumed.** If the failures share a root cause — an under-resourced change team that starves training, workflow and champion recruitment alike — then the conditions fail together and the conjunction rises towards $\min(p_i) =$ **80.00 %**. That is the optimistic bound, not the expected case, and 16.1.4 tests the go-live decision against both bounds rather than choosing between them.

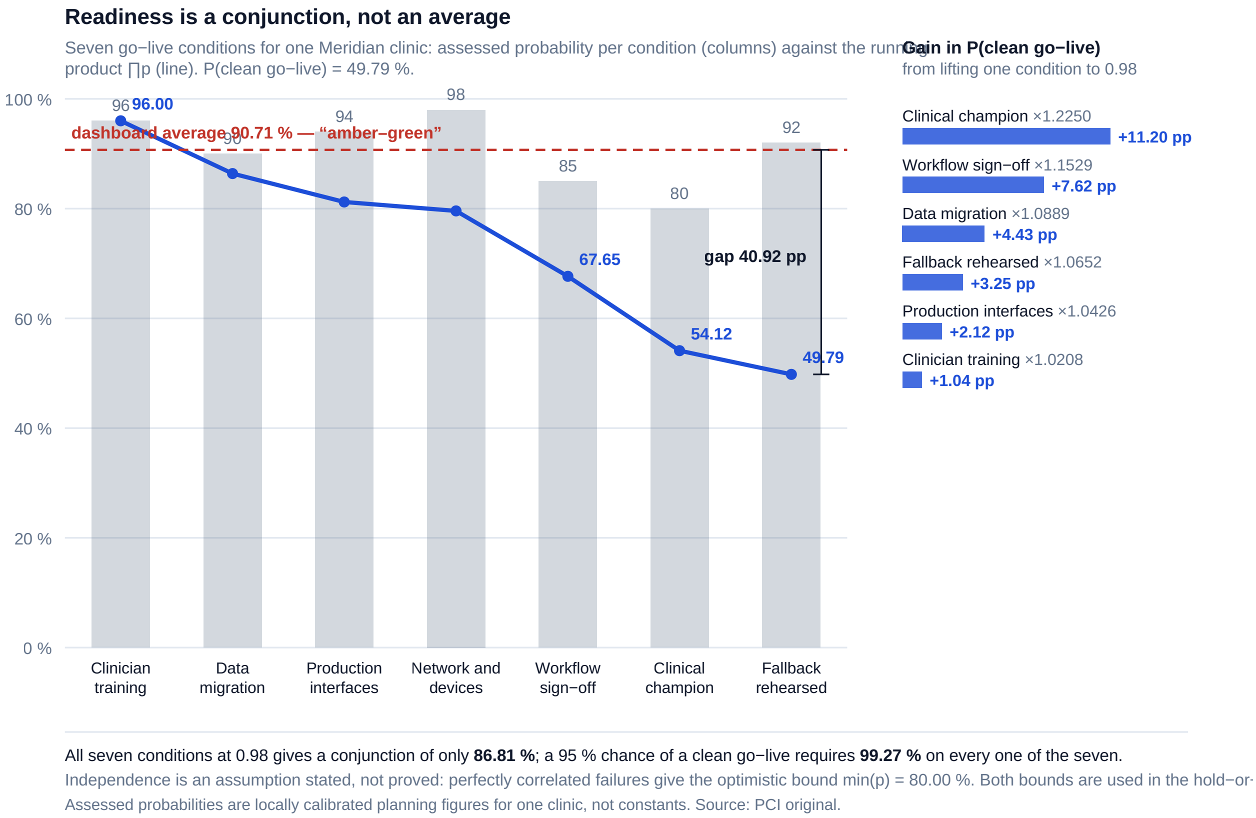


Fig 16.1.1 — Readiness is a conjunction, not an average

16.1.4 The go-live decision, hypercare and reversion

Definition. The go-live decision is a gate in Domain 3's sense — a named authority, pre-set criteria, and the power to hold as well as to proceed — whose distinguishing feature is that it is substantially irreversible for the users, even where it is technically reversible for the system.

The decision has three real options, not two: proceed, hold and remediate, or proceed with a reduced scope of users. Pricing them requires the conjunction of 16.1.3 and a cost per failed transition, and it is worth doing because the intuitive answer is usually wrong in the expensive direction.

WORKED EXAMPLE

Worked example 16.1.4 — hold or go, tested against both correlation bounds.

1. **Setup.** All **40** Meridian clinics face the readiness profile of 16.1.3. A clinic that goes live unready reverts to paper: remediation costs **USD 24,000** in re-training, re-migration and support, and the clinic loses **6 weeks** of benefit at $6 \times 85 =$ **USD 510** a week, so a failed go-live costs **USD 27,060**. Three options: **(1)** go now; **(2)** a targeted two-week uplift costing **USD 38,000** that lifts the two weakest conditions (champion and workflow) to 0.98; **(3)** a blanket three-week uplift costing **USD 96,000** that lifts all conditions to 0.98. Deferred benefit is priced at Domain 1's cost of delay, **USD 14,280** a week. Evaluate each option under both the independent conjunction and the perfectly correlated bound $\min(p_i)$.
2. **Formula.** Total cost $= (\text{weeks deferred} \times \text{cost of delay}) + \text{uplift spend} + 40 \times (1 - P(\text{clean})) \times 27{,}060$, with $P(\text{clean}) = \prod p_i$ in the independent case and $\min(p_i)$ in the correlated case.
3. **Substitution.** Option 1 independent: $40 \times (1 - 0.4979) \times 27{,}060$. Option 3 independent: $3 \times 14{,}280 + 96{,}000 + 40 \times (1 - 0.98^7) \times 27{,}060$. Correlated cases substitute $\min(p_i) =$ 0.80, 0.90 and 0.98 respectively.
4. **Result.**

Option	Independent ($\prod p_i$)	Correlated bound ($\min p_i$)
1 — go now	**USD 543,445** (20.08 failed clinics)	**USD 216,480** (8.00 failed)
2 — targeted uplift, 2 weeks, 38,000	**USD 387,766** (11.87 failed)	**USD 174,800** (4.00 failed)
3 — blanket uplift, 3 weeks, 96,000	**USD 281,581** (5.27 failed)	**USD 160,488** (0.80 failed)

1. **Interpretation.** The blanket uplift is the **dominant** option — best under both bounds — which is the result a leader wants, because the correlation structure is exactly what nobody knows at the time of the decision. It saves **USD 261,864** against going now if the failures are independent and **USD 55,992** if they share a single root cause, for a total commitment of **USD 138,840** (three weeks of deferred benefit at 42,840 plus the 96,000 spend). Three points generalise beyond Meridian. **The delay cost is small relative to the failure cost, and it is the only one that is visible.** Forty-two thousand dollars of deferred benefit appears in a report as a slipped milestone; half a million dollars of remediation appears as twenty separate operational incidents attributed to change resistance. That asymmetry is a large part of why go-live decisions tend to be taken too early. **The breakeven uplift spend is enormous**: holding remains worthwhile up to **USD 357,864** of uplift cost under the independent case, which is 3.7 times what the blanket remediation actually costs — so the decision is not close, and a leader arguing it should say so rather than presenting it as a fine balance. **And the residual matters more than the improvement.** Even after the blanket uplift, **5.27** clinics are expected to fail, because 0.98 across seven conditions is 86.81 per cent and no amount of readiness work makes a conjunction certain. The professional response is not further assurance but a rehearsed reversion plan and a wave structure that limits the exposure of any single go-live — which is what the remaining cost buys.

Hypercare is the bounded period of elevated support immediately after go-live, and it needs three things a project rarely gives it: a **stated duration**, **exit criteria expressed as a measurement rather than a date**, and a **named owner in the receiving organisation from day one** rather than at the end. Meridian's design used an incident rate per clinic per week with a threshold of **0.8** sustained for two consecutive weeks; the observed series over weeks 4 to 7 ran 1.4, 0.9, 0.6, 0.5, so the two-consecutive-week condition was first satisfied at **week 7** and hypercare exited then rather than at the planned week 6 — which is the point of a measured exit criterion. Hypercare that ends on a date ends by transferring an unstable service.

The reversion plan is the option the go-live arithmetic proves is needed. It must state the decision-maker, the trigger, the maximum time to revert, what happens to data created since go-live, and — the element most often missing — the **latest point at which reversion is still possible**, because after that point the plan is a comfort document rather than a control. A reversion plan that has never been rehearsed is an assumption, and 16.1.2's failure-path testing is where it stops being one.

AI in this KA

Where it earns its place. Assembling the readiness picture, which is a data-collection problem across many sources — training records, migration reconciliation reports, device inventories, test results — and reconciling them into one condition-by-condition status with the source of each assertion named. Computing the conjunction and the proportional-shortfall ranking across dozens of sites, which is deterministic and tedious. Reading test evidence packs against the agreed acceptance criteria and listing criteria for which no evidence exists — a genuinely valuable gap-finding task, and one humans do badly at scale. Drafting runbooks and reversion procedures from configuration baselines, for verification by the people who will use them.

Where it must not go. It must not set the readiness probabilities. Those are judgements about specific organisations and specific people, and a model asked for them will supply confident, provenance-free numbers that then drive a real go-live decision — precisely the failure Domain 14 KA 14.3 names. It must not issue a service acceptance, which is an accountable act by the receiving organisation. And it must not author a test result: a generated evidence record cannot evidence that a test was witnessed, and an acceptance regime resting on unwitnessable evidence is worse than none, because it looks complete.

Verification, concretely. Every probability carries a named human assessor and a stated basis (pilot data, comparable site, judgement). Every AI-assembled status line carries a link to its source record and is sampled by the transition manager at a stated rate before the gate. The conjunction is reproduced by hand for the go-live paper — it is six multiplications — and the paper states both correlation bounds rather than one, so the decision body can see the robustness rather than a point estimate.

KEY TERMS — KA 16.1

Term	Meaning
Handover	Transfer of custody and operational accountability for a deliverable to a named receiving organisation, on stated conditions.
Condition-based handover	Handover triggered by a satisfied condition set rather than a calendar date.
Commissioning	Progressive proving of a deliverable in its operating environment, from component test to performance demonstration.
Works test / service test	Whether the thing was built as specified / whether the organisation can actually run it.
Service acceptance	The receiving organisation's formal agreement that the service meets specification.
Readiness conjunction	$P(\text{clean transition}) = \prod p_i$ over the k necessary conditions; always at or below the averaged dashboard figure.
Proportional shortfall	The ratio p'/p by which lifting a condition multiplies the conjunction — the correct remediation ranking.
Hypercare	A bounded period of elevated support after go-live, with measured exit criteria and a receiving-organisation owner.
Reversion plan	The rehearsed route back, with a trigger, a decision-maker, a time limit and a latest-possible-reversion point.

SAMPLE MCQS — KA 16.1

MCQ 16.1-A [16.1.3 · Application] Seven readiness conditions, whose failures are assessed as independent, hold with probabilities 0.96, 0.90, 0.94, 0.98, 0.85, 0.80 and 0.92. All seven are necessary for a clean go-live. The probability of a clean go-live is closest to: - A. 90.71 % - B. 80.00 % - C. 49.79 % - D. 47.83 %

Rationale: The conjunction is the product, $0.96 \times 0.90 \times 0.94 \times 0.98 \times 0.85 \times 0.80 \times 0.92 = 49.79\ \%$ (16.1.3). A is the equal-weight average — the dashboard figure, and a different quantity. B is $\min(p_i)$, the perfectly correlated bound, which the stem's independence assumption excludes and which is in any case the optimistic case. D rounds the dashboard reading to 0.90 and raises that to the seventh power, $0.90^7 = 47.83\ \%$ — an answer that both rounds and double-counts the averaging.

MCQ 16.1-B [16.1.3 · Analysis] Seven independent readiness conditions, holding at 0.96, 0.90, 0.94, 0.98, 0.85, 0.80 and 0.92, give a conjunction of 49.79 %. The gain in the probability of a clean go-live from lifting one condition is governed by: - A. the absolute gap to the target, with an equal gain per percentage point of gap closed - B. the ratio p'/p, so the same gap closed at a lower p yields a larger gain - C. the condition's weight on the readiness dashboard - D. $1/k$, equally across all seven conditions

Rationale: Lifting a condition from p to p' multiplies the conjunction by p'/p, so the gain is $\prod p_i \times (p'/p - 1)$ (16.1.3). Closing two points of gap at 0.80 is worth **1.24 pp** while closing the same two points at 0.96 is worth **1.04 pp**, which is exactly what A denies — a common target ranks the conditions in the same order either way, but the magnitudes are set by the ratio. B is therefore the statement that generalises. C substitutes an artefact for the arithmetic. D confuses "a product" with "symmetric in its factors".

MCQ 16.1-C `[16.1.3 · Evaluation]` A programme wants a 95 % probability of a clean go-live across seven independent readiness conditions. The minimum probability required on each condition is closest to: - A. 95.00 % - B. 99.27 % - C. 98.00 % - D. 99.90 %

Rationale: `p = 0.95^(1/7) = 99.27 %` (16.1.3). A applies the target to each condition rather than to their conjunction. C gives only $0.98^7 = 86.81$ %. D is stricter than required and would be rejected as unachievable, losing the argument the arithmetic wins.

MCQ 16.1-D `[16.1.4 · Application]` Forty sites each face seven independent readiness conditions giving a 49.79 % probability of a clean go-live; a failed go-live costs 27,060. A three-week hold costs 42,840 in deferred benefit plus 96,000 of remediation and lifts all seven conditions to 0.98. The expected saving from holding is closest to: - A. USD 138,840 - B. USD 261,864 - C. USD 404,605 - D. USD 543,445

Rationale: Going now costs `40 × (1 − 0.4979) × 27,060 = 543,445`; holding costs `42,840 + 96,000 + 40 × (1 − 0.8681) × 27,060 = 281,581`; the saving is **261,864** (16.1.4). A is the cost of holding, not the saving. C omits the residual remediation after the uplift and so credits the hold with the whole 543,445, giving `543,445 − 138,840 = 404,605` — the error that makes readiness work look better than it is. D is the cost of going now with no credit for the hold at all.

MCQ 16.1-E `[16.1.1 · Comprehension]` Which of the following does **not** transfer at handover? - A. custody of the asset - B. accountability for operating the service - C. accountability for realising the benefit - D. responsibility for paying the running costs

Rationale: The benefits owner sits in the receiving organisation from the outset (Domain 2, KA 2.3.1), so handover tests that appointment rather than creating it; a benefit "handed over" at closeout has no owner and no baseline (16.1.1).

SELF-CHECK — KA 16.1

1. Why does an averaged readiness dashboard overstate readiness, and in which direction is the error always signed? — Readiness is a conjunction, so its probability is the product of the conditions' probabilities; a product is always at or below the average, so the dashboard error is always optimistic, and it grows with the number of conditions and their dispersion.
2. A works test has passed and a service test has not been run. What has been proved? — That the thing was built as specified, and nothing whatever about whether the organisation can run it.
3. What must a hypercare period state that a date cannot? — A measured exit criterion, so that the period ends when the service is stable rather than when the calendar says so.

KNOWLEDGE AREA 16.2

Operational transition and contract closeout

Topics: 16.2.1 the run organisation and total cost of ownership · 16.2.2 defects liability, warranty and retention · 16.2.3 the final account · 16.2.4 the cost of leaving a claim open.

16.2.1 The run organisation and total cost of ownership

Definition. Operational transition is the establishment of a permanent capability to run, support, fund and improve what the project delivered, with the running cost recognised in the receiving organisation's budget before the project closes.

The last clause is where business cases go to die. A project's cost is a capital number, visible, approved and controlled; the service's running cost is an operating number that appears in someone else's budget, in a later year, and is therefore very frequently **not in the business case at all**. Domain 2's whole-life requirement (KA 2.3.3) exists to prevent this, and Meridian's approved case failed it: it carried no operating-cost line whatever. The measured run cost of the records service is **USD 108,000** a year — hosting **42,000**, licences **39,000**, second-line support **27,000** — and over the case's own eight-year appraisal at 7 per cent that is a present value of **USD 644,900**, which is 48.4 per cent of the honest case's entire NPV. A programme that hands over a service without a funded run line has not transferred it; it has abandoned it in a place where someone will find it.

What the run organisation needs at transition, stated as a checklist because its items are countable and its gaps are cheap to find: a named service owner and a funded support model with stated hours and response targets; runbooks and as-built documentation at the configuration version actually deployed; a licence and contract register with renewal dates and the notice periods that govern them; an access and privilege model with the project team's own access **removed** on a stated date; a capacity and obsolescence view stating when the platform will next need investment; and the benefits measurement duties of 16.4.1, which are operating duties, not project ones.

16.2.2 Defects liability, warranty and retention

Definition. Retention is a proportion of the contract sum withheld from payment as security for the contractor's completion and defect-rectification obligations, released in stages against defined events. A **defects-liability period** is the interval after acceptance during which the contractor must rectify defects that appear; a **warranty** is a broader promise about the deliverable's condition or performance, typically longer and narrower in scope.

The mechanism is simple and it is misused in two opposite directions. Held too long or released too loosely, retention becomes either a commercial grievance that poisons the closing relationship or a security that was never there when it was needed. The disciplines are: **release against events, not dates** (takeover, end of defects liability, completion of outstanding documentation); **state what the retention secures**, since a retention held against defects cannot be applied to a performance shortfall unless the contract says so; and **track outstanding items as a priced list from the day of takeover**, because a retention release argued from an unpriced snagging list is argued from nothing. Domain 10's contract mechanisms supply the drafting; the closeout discipline is that the retention release is a decision with a record (Domain 3, KA 3.3.4), not an accounts-payable event.

The jurisdictional caution is unavoidable and it is real: retention practice, its permissibility, any requirement to hold it in trust, and the statutory limitation periods on latent defects vary by jurisdiction and by contract family. Nothing in this domain states a legal position; it states a management discipline that must be operated within whatever legal frame applies.

16.2.3 The final account

Definition. The final account is the single agreed statement of everything owed under a contract at its conclusion: the original sum, all approved variations, the settlement of any claims, all deductions and recoveries, and the resulting final payment — signed by both parties as full and final.

WORKED EXAMPLE

Worked example 16.2.3 — Meridian's final account and retention release.

1. **Setup.** The systems integrator's contract sum was **USD 1,680,000**. Approved variations total **USD 96,000**. Retention was **5 %** of the contract sum, half released at takeover of the last clinic. A prolongation claim was settled at **USD 74,000** (16.2.4). Three clinics' interface defects were rectified by another supplier at a cost of **USD 18,600**, recoverable from the integrator. Strike the final account, the final retention release and the final payment.
2. **Formula.** Final account = contract sum + variations + claim settlement − recoveries. Final payment = final account − amounts already paid.
3. **Substitution.** Retention `1,680,000 × 0.05 = 84,000`; certified works `1,680,000 + 96,000 = 1,776,000`; paid on certification `1,776,000 − 84,000 = 1,692,000`; plus the takeover release of `84,000/2 = 42,000`, so **1,734,000** paid to date. Final account `1,680,000 + 96,000 + 74,000 − 18,600`.
4. **Result.** Final account **USD 1,831,400 — 9.01 %** above the original sum. Final retention release `42,000 − 18,600 =` **USD 23,400**. Final payment `1,831,400 − 1,734,000 =` **USD 97,400**, which reconciles as `42,000 + 74,000 − 18,600 = 97,400`.
5. **Interpretation.** The reconciliation in the last line is the whole point of the example, and it is the check that a final account is actually closed rather than merely totalled: **the final payment must be explainable as the sum of the specific movements since the last certificate**, and a final payment that cannot be so explained conceals either an unrecorded variation or a double-counted recovery. Three professional observations. The **recovery is only collectable because it was made against the retention that was still held** — had the full 84,000 been released at takeover, the 18,600 would have become a debt to be pursued from a contractor with no remaining incentive, which is the practical argument for staged release and the reason "release it, they have been good to work with" is a decision and not a courtesy. The **9.01 % growth over the original sum is the number the organisation should carry into its next estimate**, not the change-control record of variations alone: variations were **5.71 %** of the original sum, the claim settlement added **4.40 %** and the defect recovery returned **1.11 %**, netting to 9.01 %, so an organisation that benchmarks only its approved variations will understate contract growth by roughly the net settlement rate every time. And the account must be closed with a **written full and final statement**, because the alternative — a balance agreed in correspondence and never formalised — is the state in which 16.2.4's carrying cost accrues.

16.2.4 The cost of leaving a claim open

Definition. The carrying cost of an open claim is the recurring cost, per period, of it remaining unresolved: external advisory fees, internal management attention, and the opportunity cost of the cash locked in retention and provision — none of which appears as a claim cost in any ledger.

Open items are left open because closing them requires accepting a number, and accepting a number feels like a loss while carrying an item feels like prudence. The arithmetic reverses that intuition reliably, and it is worth being able to do in a meeting.

WORKED EXAMPLE

Worked example 16.2.4 — what Meridian's open claim cost per month, and the premium worth paying.

1. **Setup.** The integrator claimed **USD 148,000** for prolongation. Meridian's quantity assessment put the defensible value at **USD 62,000**, and a provision of the full claimed amount was held. Carrying the claim costs: external commercial advice **USD 3,250** a month; internal management **1.5 days** a month at **USD 700** a day; and the opportunity cost of the **USD 190,000** locked up (42,000 of retention plus the 148,000 provision) at **6 %** a year. Left to formal determination, the claim would close at **month 14**, with outcomes assessed at **USD 40,000** (probability 0.25), **USD 62,000** (0.50) and **USD 110,000** (0.25). The alternative is to settle at **month 2**. What settlement price is worth paying?
2. **Formula.** Carrying cost `c = advisory + internal + locked cash × rate/12`. Expected cost of determination `= EMV + c × 14`. Breakeven settlement price at month 2 `= EMV + c × 14 − c × 2`. Breakeven premium over the assessed value `= c × Δmonths + (EMV − assessed)`.
3. **Substitution.** `c = 3,250 + 1.5 × 700 + 190,000 × 0.06/12 = 3,250 + 1,050 + 950`. `EMV = 0.25 × 40,000 + 0.50 × 62,000 + 0.25 × 110,000`.
4. **Result.** Carrying cost **USD 5,250 a month — USD 73,500** over fourteen months, more than the claim's own assessed value. `EMV` **USD 68,500**. Expected cost of fighting to determination **USD 142,000**. Breakeven settlement price at month 2 **USD 131,500**, a breakeven **premium** over the assessed 62,000 of **USD 69,500** — which the identity `12 × 5,250 + (68,500 − 62,000) = 63,000 + 6,500` reproduces exactly. Meridian settled at **USD 74,000**, a premium of **12,000**, or **17.27 %** of the premium that would still have been worth paying, saving **USD 57,500** against the expected cost of determination.
5. **Interpretation.** The identity is the transferable result: **the breakeven early-settlement premium is the carrying cost times the months saved, plus the difference between the expected determination and your own assessment.** It says that time, not merit, dominates small and medium claims — twelve months of carrying cost at 5,250 is 63,000, almost exactly the assessed value of the claim itself, so an organisation that fights a 62,000 claim for a year has spent the claim to win the claim. It also identifies which lever to pull first: of the 5,250, **USD 4,300** is advisory and management attention that stops the moment the item closes, and USD 950 is opportunity cost that stops when the cash releases. Four cautions keep this honest. The arithmetic is **not an argument for settling everything** — where a claim raises a point of principle that will recur across a supplier's other contracts, the precedent value can exceed the carrying cost, and that must be quantified rather than asserted. The **internal management figure is real cost and is almost never counted**, which is exactly why the carrying cost is invisible; 1.5 days a month of a senior commercial manager is a modest estimate and doubling it doubles the case for settling. The **assessment must precede the negotiation**, because a settlement premium can only be judged against a defensible assessed value, and an organisation that settles without one is not paying a premium, it is guessing. And a settlement must be **documented as full and final for defined matters** — drafting that belongs with qualified legal advice, since what a release covers and what it cannot reach is a legal question and not a management one — or the carrying cost simply resumes under a different heading.

AI in this KA

Where it earns its place. Reconciling a final account against the change register, the payment certificates and the variation approvals, and listing every item that appears in one and not the others

— a document-reconciliation task at which it is fast, thorough and genuinely better than a tired human at month-end. Extracting a licence and contract register with renewal and notice dates from a contract set, which is the single most useful artefact the run organisation receives and the one most often missing. Building the carrying-cost model above across a portfolio of open items and ranking them by monthly cost, which turns "we have eleven open claims" into a prioritised list. Summarising a claim's correspondence history for a settlement paper, with every assertion referenced.

Where it must not go. It must not value a claim. A claim valuation is an expert commercial and technical judgement with contractual and evidential foundations, it will be tested by people who are paid to test it, and a model's plausible number carries no provenance and no accountable author. It must not agree a final account or authorise a retention release — both are accountable decisions with a record. And its summary of a correspondence trail must never be relied on in a dispute without verification against the source documents: the summary is a convenience, the versioned original is the evidence (Domain 3, KA 3.3.4).

Verification, concretely. Every reconciliation exception is confirmed against the source certificate before it is reported. The carrying-cost model's components are each traceable to an invoice, a timesheet or a stated cost-of-capital assumption, and the assumption is named in the paper. The valuation on which any settlement rests is signed by the accountable commercial lead, not by the analysis that supported it. And the final account carries a human signature on both sides — the one part of closeout where the signature is the deliverable.

KEY TERMS — KA 16.2

Term	Meaning
Operational transition	Establishment of a funded, staffed, permanent capability to run and improve the delivered service.
Total cost of ownership	Whole-life cost of the service, including the operating cost that must appear in the receiving organisation's budget before closeout.
Retention	A proportion of the contract sum withheld as security, released against defined events.
Defects-liability period	The interval after acceptance in which the contractor must rectify defects that appear.
Final account	The single agreed statement of the original sum, variations, settlements, deductions and final payment, signed as full and final.
Contract growth	Final account against original sum — variations and settlements, not variations alone.
Carrying cost of an open item	Advisory + internal management + opportunity cost of locked cash, per period; invisible in every ledger.
Breakeven settlement premium	`carrying cost × months saved + (EMV of determination − own assessed value).`

SAMPLE MCQS — KA 16.2

MCQ 16.2-A `[16.2.3 · Application]` A contract sum of 1,680,000 carries 96,000 of approved variations, a claim settled at 74,000 and an 18,600 recovery for defects rectified by others. The final account is: - A. USD 1,776,000 - B. USD 1,831,400 - C. USD 1,850,000 - D. USD 1,757,400

Rationale: `1,680,000 + 96,000 + 74,000 − 18,600 = 1,831,400` (16.2.3). A is the certified works value, omitting both the settlement and the recovery. C omits the recovery. D omits the settlement.

MCQ 16.2-B `[16.2.4 · Application]` Carrying an open claim costs 5,250 a month. Determination at month 14 has an expected value of 68,500; the alternative is settlement at month 2. The breakeven settlement price is: - A. USD 68,500 - B. USD 131,500 - C. USD 142,000 - D. USD 73,500

Rationale: `68,500 + 14 × 5,250 − 2 × 5,250 = 131,500` (16.2.4). A ignores carrying cost entirely. C is the expected cost of fighting, without crediting the two months still carried under the settlement option. D is fourteen months of carrying cost alone.

MCQ 16.2-C `[16.2.4 · Analysis]` A claim costing 5,250 a month to carry, with an expected determination of 68,500 at month 14 and settlement available at month 2, is assessed by the client at 62,000 against the contractor's claimed 148,000. The breakeven **premium** over the assessed value is 69,500. The most defensible reading is that: - A. the claim should be conceded in full at 148,000 - B. up to 69,500 above the assessed value can be paid to close twelve months earlier, because carrying cost times months saved plus the gap between the expected determination and the assessment is 63,000 + 6,500 - C. the claim should be fought, since 62,000 is the defensible figure - D. the premium equals the provision held

Rationale: The identity `c × Δm + (EMV − assessed)` gives 69,500 (16.2.4). A ignores the assessment altogether. C is the intuition the arithmetic overturns — winning at 62,000 after fourteen months costs 135,500. D confuses a provision with a price.

MCQ 16.2-D `[16.2.1 · Evaluation]` A business case with no operating-cost line hands over a service whose measured run cost is 108,000 a year. Over an eight-year appraisal at 7 % (`AF = 5.971299`), the omission is worth: - A. USD 864,000 - B. USD 644,900 - C. USD 108,000 - D. USD 540,000

Rationale: `108,000 × 5.971299 = 644,900` (16.2.1). A is the undiscounted eight-year total, the commonest error and an overstatement of 34.0 %. C is a single year. D applies a five-year horizon undiscounted.

MCQ 16.2-E `[16.2.2 · Analysis]` A client holds 84,000 of retention, releases half of it at takeover, and later recovers 18,600 for defects that another supplier had to rectify. Releasing the whole retention at takeover rather than in stages would most directly have: - A. improved the contractor's cash position at no cost to the client - B. converted an 18,600 recovery into a debt to be pursued from a contractor with no remaining incentive - C. breached the defects-liability period - D. reduced the final account by 18,600

Rationale: Retention is security, and its value is precisely that it is still held when a recovery arises (16.2.2). C confuses a payment mechanism with a liability period; D reverses the direction of the recovery.

SELF-CHECK — KA 16.2

1. Why is the operating cost the business-case line most often missing, and what does that cost Meridian? — It falls in another budget in a later year, so nobody in the approval chain owns it; at 108,000 a year over eight years at 7 % it is a present value of 644,900, or 48.4 % of the honest case's whole NPV.
2. What are the three components of the carrying cost of an open claim, and which is largest? — External advisory, internal management attention and the opportunity cost of locked cash; the first two together are 4,300 of Meridian's 5,250 a month, and both stop the day the item closes.
3. What check proves a final account is closed rather than merely totalled? — The final payment must reconcile to the specific movements since the last certificate — here 42,000 of retention plus 74,000 of settlement less 18,600 of recovery equals the 97,400 due.

KNOWLEDGE AREA 16.3

Knowledge transfer and post-project review

Topics: 16.3.1 transferring knowledge to the run organisation · 16.3.2 the post-project review · 16.3.3 the economics of a lesson · 16.3.4 closing the project organisation.

16.3.1 Transferring knowledge to the run organisation

Definition. Knowledge transfer is the deliberate conversion of what the project team knows into forms the receiving organisation can use without them: documentation at the deployed configuration version, trained people, and rehearsed procedures.

The failure is rarely a refusal to document; it is that documentation is treated as a deliverable rather than as a **test**. A runbook is adequate when someone who was not on the project can execute it under supervision — and nothing short of that observation proves it. Two measurements make the gap visible cheaply. **Key-person concentration**: of Meridian's 34 operational procedures, **11** were executable by exactly one named individual at handover, a **32.35 %** single-point-of-knowledge rate, and each of those 11 is an availability risk that no risk register had recorded because it was not a project risk. And **documentation currency against the as-built baseline** (Domain 4, KA 4.3.2), since documentation written against the design and never updated against the deployment describes a system that does not exist.

The economics are one-sided and worth stating because they are always resisted at the moment they matter. Three of Meridian's clinics had interface configurations that were changed during commissioning and never recorded. Reconstructing each afterwards — reading the live configuration, re-testing, re-documenting — cost **USD 8,400**, a total of **USD 25,200**, against **USD 1,200** each to document at the time, or **USD 3,600**: a ratio of **7.00**. The multiple is Meridian's own and is not a constant, but its direction follows from a mechanism that does not change with the context — at handover the knowledge is in someone's head and the cost is a note; afterwards it must be re-derived from the artefact by someone who never had it. An organisation that wants the multiple for its own estimating has to measure it, and the measurement is cheap because both costs are already invoiced.

16.3.2 The post-project review

Definition. A post-project review is a structured examination, after delivery, of what happened and why, conducted to improve the organisation's future decisions rather than to assess its people.

The design faults are predictable and each has a countermeasure. **Held too late**, when the team has dispersed and only the outcome is remembered, not the reasoning; the countermeasure is to hold it within a stated window of handover and to run interim reviews at gates rather than one review at the end. **Held as an attribution exercise**, which guarantees defensive testimony; the countermeasure is an explicit and enforced separation from performance assessment, and the psychological-safety conditions Domain 12 sets out. **Focused on events rather than on decisions**, so it records that the schedule slipped rather than which decision, taken on which information, made the slip likely; the countermeasure is to start from the decision record (Domain 3, KA 3.3.4) and ask of each material decision whether it was reasonable on what was known — a question that produces transferable findings where "what went wrong" produces narrative. And **producing recommendations with no owner**, which is how a review generates activity and no change; every finding needs a named owner, a date and a place in a standing process that will be looked at again.

The most valuable single addition to a review agenda is the **estimate-versus-actual comparison on the organisation's own parameters** — the contract growth of 16.2.3, the adoption ramp of 16.4.2, the readiness probabilities of 16.1.3 against what actually happened. Those numbers are what calibrate the next business case, and they are the only output of a review that improves an estimate rather than a behaviour.

16.3.3 The economics of a lesson

Definition. A lesson's value is the cost it avoids when it is retrieved and applied, weighted by the probability that it is retrieved and applied at all. Capture without retrieval has a value of zero, and this is not a rhetorical point — it is where almost all of the value is lost.

WORKED EXAMPLE

Worked example 16.3.3 — what Meridian's post-project review was worth.

1. **Setup.** The review cost **USD 46,000** in workshop time, analysis and write-up, and produced **34** lessons. From the organisation's own history, a lesson that is retrieved before a comparable decision avoids an average of **USD 26,000** of rework or delay. In the current, unindexed repository, **12 %** of captured lessons are retrieved before a comparable decision. A proposal would spend a further **USD 18,000** indexing the lessons against the specific decision points at which they apply, raising retrieval to an estimated **35 %**.
2. **Formula.** Expected recovery = lessons × retrieval rate × avoided cost per applied lesson. Breakeven retrieval rate = review cost ÷ (lessons × avoided cost).
3. **Substitution.** At 12 %: `34 × 0.12 × 26,000`. At 35 %: `34 × 0.35 × 26,000`. Breakeven: `46,000/(34 × 26,000) = 46,000/884,000`.
4. **Result.** Expected recovery **USD 106,080** at the current retrieval rate and **USD 309,400** with indexing — an increment of **USD 203,320** for **USD 18,000**, a ratio of **11.30**. The review breaks even at a retrieval rate of **5.20 %**; with the index it must reach **7.24 %**. Net value of review plus index at 35 % retrieval: **USD 245,400**.
5. **Interpretation.** Three results, in ascending order of importance. The review is worth holding — at 2.31 times its cost even with a retrieval rate most organisations would be embarrassed to report. **The breakeven retrieval rate of 5.20 % is the number that should end the recurring argument about whether reviews are worth the time**: they are, comfortably, provided anyone ever reads them, and the honest reading of the widespread belief that they are not is that in many organisations the retrieval rate really is near zero, at which point the belief is correct and the remedy is not to stop reviewing. **And the highest-return intervention available is not better capture but better retrieval** — an 11.30-to-one return on indexing, against which any proposal to improve the workshop is not competitive. This inverts where effort normally goes: organisations invest in the review event, which is visible and social, and not in the retrieval mechanism, which is unglamorous infrastructure. The professional cautions are that the 26,000 avoided cost and both retrieval rates are **organisational parameters that must be measured, not assumed** — the honest way to establish the retrieval rate is to instrument the repository and count, and an organisation that cannot state its retrieval rate is not in a position to argue about the value of its reviews; and that a lesson only avoids cost if it is **specific enough to act on**, since "communicate better" has an avoided cost of zero at any retrieval rate, which is why lesson quality and lesson count are different measurements.

16.3.4 Closing the project organisation

Definition. Closing the project organisation is the deliberate dissolution of the temporary structure: its authority, its accounts, its access rights, its records and its people.

Four items are routinely left undone, and each has a specific consequence. **Authority is not withdrawn**, so a change board with no project continues to approve changes to a live service that now belongs to someone else — the governance defect of Domain 3 KA 3.1.1, arriving after the project has ended. **Cost accounts are left open**, so late charges land in a closed period, the outturn moves after it was reported, and the estimate-versus-actual comparison of 16.3.2 is calibrated on a wrong number. **Access rights persist**, which is a security exposure with a named cause and a trivial fix (Domain 14, KA 14.4). And **people are released without a record of what they did**, which is both an individual injustice and the reason the organisation cannot find, two years later, the person who knows why the interface was configured that way.

The leadership content of this topic is not administrative. A team that has spent two years on a programme experiences its dissolution as an ending, and the manner of it materially affects whether those people will volunteer for the next difficult programme. Recognition that is specific and attributable, honest conversations about what each person did well and what they should develop (Domain 12, KA 12.3), and a visible route to their next role are not courtesies — they are the mechanism by which delivery capability survives the project that consumed it.

AI in this KA

Where it earns its place. Indexing lessons against the decision points at which they apply, which is exactly the intervention 16.3.3 prices at an 11.30-to-one return, and is a classification and retrieval problem of the kind the technology is genuinely good at. Clustering findings across many reviews to expose the recurring cause that no single review can see — the organisational analogue of Domain 3's re-decision count. Drafting as-built documentation from configuration baselines and change records for human verification. Generating candidate review questions from the decision record so the review starts from decisions rather than from recollection. Producing the first draft of a transition-out handbook from existing artefacts.

Where it must not go. It must not conduct the review. A post-project review is partly a conversation about judgement under uncertainty, and its value comes from people saying things they would not write down; a transcript-and-summarise exercise gets the record and loses the content. It must not attribute cause, because causal attribution in a delivery failure is contested, consequential for named individuals, and not a text-processing task. And it must not be the author of a lesson that is then relied on: an unattributable lesson cannot be interrogated, and a repository of them decays into plausible advice.

Verification, concretely. Every indexed lesson is checked against its source finding on a sampled basis, and the retrieval rate is instrumented and reported rather than estimated. Every AI-drafted document is executed by someone who was not on the project before it is accepted as adequate — the same test as any other runbook (16.3.1). And every finding that names a cause is confirmed by the accountable human reviewer before it leaves the room.

KEY TERMS — KA 16.3

Term	Meaning
Knowledge transfer	Conversion of project knowledge into documentation, trained people and rehearsed procedures usable without the project team.
Runbook adequacy test	Whether someone who was not on the project can execute it under supervision — the only proof documentation works.
Key-person concentration	The share of procedures executable by exactly one named individual; an availability risk no project register records.
Post-project review	Structured examination of what happened and why, to improve future decisions, separated from performance assessment.
Retrieval rate	The proportion of captured lessons retrieved before a comparable decision; the term in which almost all lesson value is lost.
Breakeven retrieval rate	review cost ÷ (lessons × avoided cost per applied lesson) — the threshold above which reviewing pays.
Transition-out	Dissolution of the project organisation: authority withdrawn, accounts closed, access revoked, records placed, people released well.

SAMPLE MCQS — KA 16.3

MCQ 16.3-A `[16.3.3 · Application]` A review costing 46,000 produces 34 lessons; an applied lesson avoids 26,000 on average. The breakeven retrieval rate is closest to: - A. 5.20 % - B. 12.00 % - C. 43.36 % - D. 52.04 %

Rationale: `46,000/(34 × 26,000) = 46,000/884,000 = 5.20 %` (16.3.3). B is the organisation's current retrieval rate, not the breakeven. C is `46,000/106,080` — the share of the expected recovery consumed by the review, which is a cost-recovery ratio and not a rate. D misplaces the decimal, a common slip on a small ratio.

MCQ 16.3-B `[16.3.3 · Evaluation]` A review costing 46,000 produces 34 lessons, an applied lesson avoids 26,000 on average, and 12 % of captured lessons are currently retrieved before a comparable decision. The highest-return improvement is to: - A. run a longer, better-facilitated review workshop - B. spend 18,000 indexing the lessons against the decision points where they apply, raising retrieval from 12 % to 35 % - C. capture more lessons per project - D. shorten the review to reduce its 46,000 cost

Rationale: Indexing yields `34 × 0.23 × 26,000 = 203,320` for 18,000 — a ratio of 11.30 (16.3.3). A and C invest in capture, where the value is not being lost. D reduces a cost that is already recovered 2.31 times over.

MCQ 16.3-C `[16.3.1 · Application]` Three undocumented interface configurations cost 8,400 each to reconstruct after handover, against 1,200 each to document at the time. The ratio of avoidable to incurred cost is: - A. 7.00 - B. 3.00 - C. 21,600 to 1 - D. 2.33

Rationale: `25,200/3,600 = 7.00`, which is also the per-item ratio `8,400/1,200` (16.3.1). B counts the three items rather than the cost ratio. C is the absolute saving, `25,200 − 3,600 = 21,600`, mislabelled as a ratio. D divides the per-item reconstruction cost by the total documentation cost, `8,400/3,600 = 2.33`, mixing a unit figure with a total.

MCQ 16.3-D `[16.3.2 · Analysis]` The review practice most likely to produce transferable findings rather than narrative is to: - A. begin from the schedule variance and work backwards - B. begin from the decision record and ask of each material decision whether it was reasonable on what was known at the time - C. invite every stakeholder to describe their experience - D. rank the events of the project by impact

Rationale: Decisions are the unit at which an organisation can improve; events are consequences (16.3.2, and Domain 3 KA 3.3.4 on the retrospective question). A and D produce narrative; C produces testimony without a comparison.

SELF-CHECK — KA 16.3

1. What is the only proof that a runbook is adequate? — That someone who was not on the project can execute it under supervision.
2. Where is the value of a post-project review actually lost, and what does that imply about where to invest? — At retrieval, not at capture; so indexing lessons against the decisions they bear on returns far more than improving the workshop — 11.30 to one on Meridian's figures.
3. Name two consequences of failing to close the project organisation. — A change board with no project continues to authorise changes to someone else's live service, and open cost accounts let late charges move an already-reported outturn, corrupting the estimating calibration.

KNOWLEDGE AREA 16.4

Benefits measurement, responsible archive and model/data retention

Topics: 16.4.1 the benefits measurement plan · 16.4.2 the closing account · 16.4.3 attribution and the comparison cohort · 16.4.4 responsible archive and retention · 16.4.5 model and data retention for AI-assisted decisions.

16.4.1 The benefits measurement plan

Definition. A benefits measurement plan states, for each benefit in the register, the measure and its exact definition, the baseline value and the date it was taken, the target and its profile over time, the frequency and method of measurement, the named owner accountable for realisation, the reporting route, and the date the measurement obligation ends.

Every element earns its place by a failure it prevents, and three deserve emphasis because they are the ones that make a measured number arguable.

The measure's exact definition, including its denominator. Meridian registered "clinics in daily use" as its adoption measure, with "daily use" defined as at least one clinical episode recorded in the system on at least four days of a working week, averaged over a month. At the twelve-month benefits review the figure was **27 of 40 clinics, 67.50 %**. An informal figure of **68 %** had circulated — the number Domain 1's case study records — computed on a clinician count rather than a clinic count; the two measures differ because the non-adopting clinics are smaller than average. Neither number is wrong; only one is the registered measure, and the discipline that matters is that the definition was fixed before measurement and the review reports the registered measure with the variant identified. A programme that discovers at measurement time that its measure has two readings will end up reporting whichever reading is more comfortable.

The baseline, taken before the change, at the date recorded. Domain 2 KA 2.3.2 required this and Meridian complied: clinician time on records tasks was sampled in all 40 clinics in the quarter before the first go-live. Without that, every number in 16.4.2 would be a reconstruction, and a reconstruction is an estimate shaped by the result it is used to justify.

The end date of the measurement obligation. Benefits measurement that runs indefinitely is not measured at all; the plan should state the period — Meridian's is the eight years of the appraisal, with formal reviews at 12, 24 and 48 months — and what happens at the end, which is normally that the measure either becomes a permanent operational indicator or is retired with a reason.

16.4.2 The closing account

Definition. The closing account is the reconciliation of realised value against the value the business case promised, decomposed into the components that explain the difference, and reported to the body that approved the case.

This is the account the whole book has been keeping. Meridian's business case claimed **USD 979,200** a year on an output basis; Domain 1 corrected that to **USD 685,440** at 70 per cent adoption, and Domain 2 appraised the honest ramped profile at a present value of **USD 3,732,898** and an NPV of **+USD 1,332,898**. The measurement is now in.

WORKED EXAMPLE

Worked example 16.4.2 — the closing account of Meridian Care Records.

1. **Setup.** At the twelve-month benefits review, with all 40 clinics live: **27 clinics** meet the registered daily-use definition (**67.50 %** adoption), and time sampling in the adopting clinics shows **5.4 clinician-hours** released a week rather than the **6.0** the case assumed. The valuation rate is unchanged at **USD 85** an hour over **48** operating weeks, and the comparison cohort of 16.4.3 shows no material change from other causes. Measured adoption by year, against the case's 40 % / 60 % / 70 % ramp, was **15 %**, **40 %** and **60 %**, reaching the measured steady state of 67.50 % only in year 4 — the ramp arrived a full year later than the case assumed. The run cost is **USD 108,000** a year (16.2.1) against no operating-cost line in the case, and the capital outturn was **USD 2,514,000** against **USD 2,400,000** approved.
2. **Formula.** Annual benefit `= U × a × h`, where `U` `=` clinics × rate × weeks `= 40 × 85 × 48 =` **163,200** is the benefit per unit of adoption-hours, `a` is adoption and `h` is hours released per adopting clinic per week. Two-factor variance decomposition: adoption effect `= U × h_plan × (a_actual − a_plan)`; hours effect `= U × a_actual × (h_actual − h_plan)`.
3. **Substitution.** Case `163,200 × 0.70 × 6.0`; measured `163,200 × 0.675 × 5.4`. Adoption effect `163,200 × 6.0 × (0.675 − 0.700)`; hours effect `163,200 × 0.675 × (5.4 − 6.0)`.
4. **Result.** Measured steady-state benefit **USD 594,864** a year (≈ SAR 2.23 million indicatively) against the honest case's **USD 685,440** — a shortfall of **USD 90,576**, or **13.21 %**. The realised figure is **86.79 %** of the honest case and exactly **60.75 %** of the flat claim the case actually made. Decomposed, taking adoption first: adoption **(USD 24,480)**, hours per adopting clinic **(USD 66,096)**, total **(USD 90,576)**.

The full present-value bridge, eight years at 7 % (`AF = 5.971299`):

Bridge line	NPV (USD)
Flat claim, as approved	**+3,447,096**
Adoption term corrected (Domain 1's term, valued in Domain 2)	(2,114,198)
Honest ramped case (Domain 2)	**+1,332,898**
Level variance — measured hours 5.4 and steady adoption 67.50 %	(465,015)
Timing variance — the ramp arrived one year late	(413,809)
Operating cost omitted from the case (108,000 a year)	(644,900)
Cost outturn variance (2,514,000 against 2,400,000)	(114,000)
Realised NPV	**(304,827)**
Post-project improvement plan	+340,156
Post-plan NPV	**+35,329**

Rows are rounded to the nearest dollar; subtotals are struck at full precision, so a row-by-row addition may differ by USD 1. The two variance lines are counterfactuals and the convention is the same one used on the annual decomposition — first factor at plan, second at actual. The level line values the measured 5.4 hours and 67.50 % steady adoption on the case's own adoption timing (40 % in year 1, 60 % in year 2, steady from year 3), and the timing line is the further loss from the measured ramp of 15 % / 40 % / 60 % reaching steady state only in year 4. Reversing the order would report timing at (479,771) and level at (399,053) — the same total, with USD 65,962 of level-and-timing interaction moved between the lines.

1. **Interpretation.** Six readings, and they are the closing lessons of the book. **First, the largest single line is not a delivery failure at all.** The **2,114,198** correction is the adoption term the business case omitted — money that was never available to be realised, and which Domain 1's arithmetic removed before a single clinic was installed. An organisation that measures realisation against the approved claim will record a **USD 384,336** annual shortfall and conclude the programme failed; measured against the honest case the shortfall is **USD 90,576**, and the difference between those two numbers is a defect in the case, not in the delivery. **Second, the decomposition's order matters and the convention must be stated.** Taking adoption first gives (24,480) adoption and (66,096) hours; taking hours first gives (68,544) hours and (22,032) adoption. Both sum to (90,576); the **USD 2,448** that moves between the lines is the interaction term, `163,200 × (−0.025) × (−0.6)`, and it exists because both factors moved. The convention used here — first factor at plan, second at actual — assigns it to the second line, and a report that does not say which convention it used has published two of its three numbers ambiguously. **Third, the hours error is nearly three times the adoption error**, which is the opposite of what the programme's own narrative said: everyone discussed adoption, because adoption is visible and Domain 1's case study had made it the headline. The benefit per adopting unit was never re-measured until this review, and it was wrong by 10 per cent — a reminder that a benefits case has two quantitative assumptions and organisations habitually monitor one. **Fourth, timing cost almost as much as level**: (413,809) against (465,015), from nothing more than the ramp arriving a year late. A benefit deferred is a benefit partly destroyed, and this is the same arithmetic as the cost of delay from Domain 1 seen from the other end. **Fifth, the realised NPV is negative — and it is negative because of the line the project never controlled.** Strip the operating cost out and the realised position is **+USD 340,073**; the run cost the case never carried is what turns the account. That is the single most transferable finding in this domain: **the omission of an operating cost line is not a presentational defect, it is frequently the**

whole margin. Sixth, the account is not closed by the project. The improvement plan — **USD 180,000** at the end of year 3 to complete the workflow redesign in the thirteen non-adopting clinics and re-run the champion programme, lifting adoption to **80 %** and restoring the full **6.0** hours from year 5 — is worth `188,496 × 2.584087 =` **USD 487,090** in present value against a present cost of **USD 146,934**: an NPV of **+USD 340,156** and a benefit-cost ratio of **3.32**. It converts the programme's realised NPV from **(304,827)** to **+35,329**. The whole value of a USD 2.4 million programme came to rest on USD 180,000 of post-project adoption work that appeared in no project plan, and the professional caution attached is exact: that plan needs an owner, a budget line and a decision, and at the moment the project closes there is no project to provide any of the three. The breakeven is worth stating for the board — with the plan's USD 180,000 spent, the account needs an effective realisation level of **78.60 %** of full potential in years 5 to 8 to reach zero, against **60.75 %** on the current trajectory and the **80 %** the plan is designed to deliver.

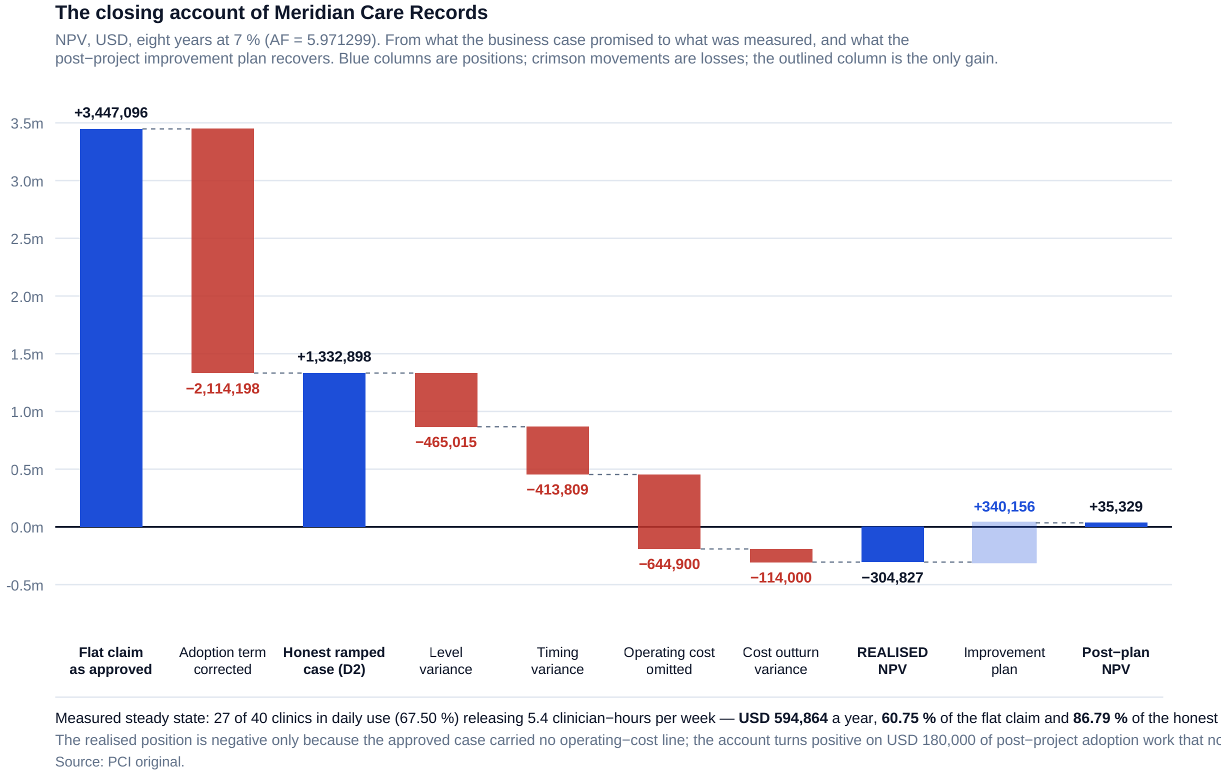


Fig 16.4.1 — The closing account of Meridian Care Records

16.4.3 Attribution and the comparison cohort

Definition. Attribution is the determination of how much of an observed change was caused by the project rather than by anything else happening at the same time. A **comparison cohort** is a set of comparable units, measured at baseline and at review but not receiving the change (or receiving it later), against which the observed change is netted.

Attribution is the argument that destroys benefits claims, and it is unwinnable after the fact. Meridian handled it correctly and cheaply: because the rollout ran in waves, the **8 clinics** in the final wave were measured at the same baseline and again at the twelve-month review while still on the old process. Their records-task time changed by **0.0 hours a week**, within the sampling tolerance of **±0.3 hours**, so no attribution adjustment was made and the full 5.4 hours is claimed as attributable.

The value of that cohort is best seen by supposing it absent. A reviewer could then assert that a concurrent national workflow initiative accounted for, say, **0.6** of the 5.4 hours. That assertion is unfalsifiable without a comparison, and it is expensive: at **4.8** attributable hours the steady-state benefit falls to **USD 528,768** a year, the realised present value of benefits falls to **USD 2,536,954**, and the realised NPV falls from **(304,827)** to **(621,946)** — the claim more than doubles the measured loss. A comparison cohort that costs a few days of sampling therefore protects a figure of that order, and the professional rule follows directly: **the attribution question must be settled by design at baseline, not by argument at review**, because at review the only available instruments are assertion and seniority.

Two honest limitations. A wave-based comparison cohort is not a randomised comparison, and the later waves may differ systematically from the earlier ones — Meridian's did, being smaller — so the cohort bounds the attribution question rather than closing it, and the review should say so. And where no comparison is possible at all, the correct treatment is to report the benefit in its physical unit with the attribution uncertainty stated, not to convert an uncertain quantity into a confident monetary claim (Domain 2, KA 2.3.2 on cash-releasing and non-cash-releasing benefits).

A correction the closing account forces on the rest of the book. Every delay calculation in Domains 3, 4 and 6 was priced at Domain 1's cost of delay of **USD 14,280** a week, derived from 28 adopting clinics releasing 6 hours at 85. The realised rate is `27 × 5.4 × 85 =` **USD 12,393** a week — **86.79 %** of it, meaning the figure used throughout was **15.23 %** higher than realisation justifies. The professional obligation is to re-test the conclusions, not to restate them, and the result is reassuring: Domain 3's delegation-threshold analysis saved **USD 171,360** a year at 14,280 and saves **USD 148,716** at 12,393, moving the breakeven value-destruction rate on the delegated band from 84 % to **72.90 %** — still far beyond any plausible delegate. **A conclusion that survives a 15 per cent error in its principal input is a robust conclusion, and one that does not survive should never have been presented as a point estimate.** That test is the most useful thing a closing account gives back to the organisation's future decisions.

16.4.4 Responsible archive and retention

Definition. The archive is the retained record of the project: its decisions, contracts, evidence, configurations and data, held for a stated period against stated purposes, with a defined disposal or de-identification action at the end. Retention is a **schedule by class**, never a single rule.

Two forces pull in opposite directions and both are legitimate. Evidence must be kept, because the questions that arrive later — a dispute, an audit, a latent defect, an inquiry — can only be answered from records that exist, at the versions relied on. And personal data must **not** be kept, because holding it creates obligations and exposure that grow with time and volume while its usefulness falls. A single retention rule cannot serve both, which is why the schedule is class-by-class. Meridian's 6.4 TB archive resolves into: contract and commercial evidence **1.9 TB**; the decision and governance record **0.4 TB**; design, configuration and as-built **1.6 TB**; test and commissioning evidence **1.4 TB**; and personal and patient-linked migration logs **1.1 TB**.

WORKED EXAMPLE

Worked example 16.4.4 — retention as insurance, and retention as liability.

1. **Setup.** Storing the archive costs **USD 21** per terabyte per month. The programme's assessed probability that a dispute, audit or latent-defect question will require the contractual and technical evidence within seven years is **0.18**, and the assessed cost of being unable to answer it — an adverse determination or an unrecoverable rectification — is **USD 310,000**. Separately, the **1.1 TB** of patient-linked migration logs can be de-identified for a one-off **USD 34,000**, removing an assessed breach exposure of probability **0.04** and consequence **USD 1,250,000**.
2. **Formula.** Cost of retention = volume × unit rate × months. Value of retention = `P(need) × cost of not having it`. Value of de-identification = exposure removed + storage avoided − de-identification cost. Breakeven need probability = retention cost ÷ consequence.
3. **Substitution.** `6.4 × 21 = 134.40` a month, `× 12 × 7`. `0.18 × 310,000`. De-identification: `0.04 × 1,250,000 + 1.1 × 21 × 12 × 7 − 34,000`.
4. **Result.** Seven years of retention costs **USD 11,290**; the expected value of holding the evidence is **USD 55,800** — a ratio of **4.94**, with a breakeven need probability of **3.64 %**. De-identifying the patient-linked logs removes **USD 50,000** of expected exposure and **USD 1,940** of storage for **USD 34,000**, a net gain of **USD 17,940**.
5. **Interpretation.** The two halves of the result point in opposite directions and that is the whole lesson. **Retaining evidence is extraordinarily cheap insurance** — it pays at a need probability above 3.64 % — roughly one chance in twenty-seven — and no serious programme can claim its evidence is less likely than that to be wanted — so the reflex to purge storage for cost reasons is almost always wrong, and the correct argument for deleting is never cost. **Retaining personal data is the reverse trade**: the same volume that costs almost nothing to store carries an exposure that dwarfs its storage cost by a factor of 26 to one, and de-identification pays for itself several times over even before any regulatory dimension is considered. Three cautions. The breakeven is so low that the **decision is really about the consequence estimate**, and a programme that cannot articulate what it would lose by being unable to answer has not thought about the risk. Retention periods for personal and health data are **set by law and by contract in each jurisdiction and by data class**, they differ substantially, and nothing here states a legal minimum or maximum — the schedule must be agreed with whoever holds the data-protection accountability in the organisation, and the figures above are an illustration of the economics, not of the law. And **de-identification must be tested, not asserted**: a log that can be re-identified by joining it to another retained dataset has not been de-identified, and the test belongs in the schedule.

16.4.5 Model and data retention for AI-assisted decisions

Definition. Model and data retention is the retained record of what an AI system was asked, what it returned, which model version returned it, who verified the output and what decision relied on it — the minimum that permits an AI-informed decision to be explained after the fact.

Domain 14 established the AI-use register and the verification standard proportional to consequence; this topic is the retention consequence, and it is the one closeout usually discovers too late. Of the **214** AI-assisted outputs that Meridian's programme relied on in recorded decisions, **61** cannot be reproduced because the model identifier and version were not captured at the time — **28.50 %** of the AI-informed decision base is therefore unexplainable, and re-deriving those outputs by hand at **USD 1,850** each would cost **USD 112,850**, which nobody will spend. The fix costs nothing and is three fields on a record that was being written anyway: **model identifier, model version, date**, alongside the prompt or input, the output, and the named human verifier.

Retention here has its own class distinctions. Prompts and outputs relied on in a decision belong with the decision record and inherit its retention period (Domain 3, KA 3.3.4). Training or fine-tuning data, where the organisation created any, carries the personal-data considerations of 16.4.4 and usually the strictest treatment in the schedule. Model artefacts themselves may be unretainable — a third-party service's version may simply cease to exist — which is precisely why the record of what it produced and who verified it is the durable evidence, and why a retention plan that assumes the model will still be available is not a plan. The closing obligation is a short one, and it is assessable: **for every AI-informed decision in the record, a reader two years later can identify the model, the input, the output, the verifier and the decision-maker.** Anything less means the organisation cannot explain its own decisions, which is the accountability defect Domain 1 named and Domain 14 quantified, arriving at the archive.

AI in this KA

Where it earns its place. Assembling the measurement data — extracting usage counts against the registered definition, reconciling them to the source systems, and flagging units whose classification changed between periods, which is where measured adoption series usually break. Computing the variance decomposition across many benefits and both decomposition orders, and reporting the interaction term explicitly. Classifying an archive by data class against a retention schedule and listing records whose class cannot be determined — the highest-value output, because unclassified records are the ones that get kept by default. Detecting records missing the model-identifier fields of 16.4.5 across a large decision log.

Where it must not go. It must not decide what the measured benefit was. The measure's definition, the treatment of edge cases and the attribution judgement are accountable determinations that will be read as the organisation's own statement about its performance. It must not generate the explanation of a variance: a model asked why realisation fell short will produce a fluent, plausible causal account that is indistinguishable in tone from an evidenced one, and a benefits report containing one has substituted narrative for measurement. And it must not authorise a deletion — disposal under a retention schedule is an accountable, usually irreversible act, and it must carry a human authorisation and a record of what was destroyed and on what authority.

Verification, concretely. Every measured figure traces to a source-system query that a second person can re-run, and the query is retained with the figure. The decomposition is reproduced by hand for the report — it is four multiplications — and both orders are computed so the interaction term is visible rather than absorbed. Attribution rests on the comparison cohort's measured data, not on an argument. And every disposal action generates a record identifying the class, the volume, the authority and the date, which is itself retained under the longest applicable period.

KEY TERMS — KA 16.4

Term	Meaning
Benefits measurement plan	Measure definition, baseline and date, target profile, method, frequency, owner, reporting route and end date, per benefit.
Registered measure	The single definition of a benefit measure fixed before measurement; variants are identified, not substituted.
Closing account	Reconciliation of realised value against the promise, decomposed into its causes and reported to the approving body.
Benefit per unit of adoption-hours (`U`)	clinics × valuation rate × operating weeks — the constant in `benefit = U × a × h`.
Level variance	The value effect of a wrong steady-state adoption or benefit-per-unit assumption.
Timing variance	The value effect of the benefit profile arriving later than the case assumed.
Interaction term	The product of both factor movements; the amount that shifts between decomposition lines when the order changes.
Comparison cohort	Comparable units measured at baseline and review without receiving the change, used to net out other causes.
Retention schedule	Class-by-class statement of what is kept, for how long, on what authority, and what disposal or de-identification follows.
Model retention record	Model identifier, version and date alongside input, output, verifier and decision — the minimum that makes an AI-informed decision explainable.

SAMPLE MCQS — KA 16.4

MCQ 16.4-A `[16.4.2 · Application]` Twenty-seven of 40 clinics are in daily use, each releasing 5.4 hours a week at USD 85 over 48 weeks. The measured annual benefit is: - A. USD 685,440 - B. USD 594,864 - C. USD 881,280 - D. USD 660,960

Rationale: `27 × 5.4 × 85 × 48 = 594,864` (16.4.2). A is the honest business-case figure at 70 % adoption and 6.0 hours. C omits the adoption term, counting all 40 clinics at the measured 5.4 hours. D keeps the correct 27 clinics but the case's 6.0 hours, omitting the benefit-per-unit correction.

MCQ 16.4-B `[16.4.2 · Analysis]` With `U = 163,200`, planned adoption 0.700 and 6.0 hours, and measured adoption 0.675 and 5.4 hours, the decomposition taking **adoption first** gives: - A. adoption (24,480); hours (66,096) - B. adoption (22,032); hours (68,544) - C. adoption (24,480); hours (68,544) - D. adoption (66,096); hours (24,480)

Rationale: Adoption at planned hours is `163,200 × 6.0 × (−0.025) = (24,480)`; hours at measured adoption is `163,200 × 0.675 × (−0.6) = (66,096)`; both sum to (90,576) (16.4.2). B is the reverse order — correct arithmetic, wrong to the question asked. C mixes the two orders and over-states the total by the 2,448 interaction term. D transposes the labels.

MCQ 16.4-C `[16.4.2 · Evaluation]` Meridian's realised NPV is (304,827). Which single line is most responsible, and what does that imply? - A. the adoption shortfall; the rollout under-delivered - B. the operating cost omitted from the case, at (644,900); without it the realised position is +340,073, so the whole margin turned on a line the project never controlled - C. the cost outturn variance of (114,000); the programme overspent - D. the timing variance; benefits arrived late

Rationale: The operating-cost line is the largest realised-period movement and exceeds the whole negative position (16.4.2, 16.2.1). A and D are real but smaller — (465,015) and (413,809). C is the smallest line at 4.75 % of approved cost.

MCQ 16.4-D `[16.4.3 · Analysis]` A reviewer claims a concurrent national initiative accounts for 0.6 of the 5.4 measured hours. The measurement design element that answers the claim is: - A. a larger time-sampling exercise in the adopting clinics - B. the comparison cohort of clinics measured at baseline and review without the change - C. a sensitivity analysis on the valuation rate - D. the benefits owner's professional judgement

Rationale: Only a comparison can net out other causes, and it must exist from baseline (16.4.3); without it the assertion is unfalsifiable and, at 4.8 attributable hours, would move the realised NPV from (304,827) to (621,946). A measures the same population more precisely and answers a different question.

MCQ 16.4-E `[16.4.4 · Application]` A 6.4 TB archive costs USD 21 per TB per month. If the probability that the evidence is needed within seven years is 0.18 and the cost of being unable to answer is 310,000, the breakeven need probability is closest to: - A. 18.00 % - B. 3.64 % - C. 0.52 % - D. 36.42 %

Rationale: `6.4 × 21 × 12 × 7 = 11,290`; `11,290/310,000 = 3.64 %` (16.4.4). A is the assessed probability, not the breakeven. C uses one year of storage. D misplaces the decimal.

MCQ 16.4-F `[16.4.5 · Comprehension]` The minimum retention that makes an AI-informed decision explainable two years later is: - A. the model artefact itself, retained locally - B. the prompt and the output - C. model identifier and version, date, input, output, named verifier and decision-maker - D. a summary of the analysis in the decision record

Rationale: A third-party model version may cease to exist, so the durable evidence is the record of what it produced and who verified it (16.4.5). B omits the version and the verifier, which is exactly the omission that made 28.50 % of Meridian's AI-informed decision base unexplainable.

SELF-CHECK — KA 16.4

1. Why must the measure's denominator be fixed before measurement? — Because a measure with two readings will be reported at whichever reading is more comfortable; Meridian's 67.50 % of clinics and the informally circulated 68 % of clinicians are both defensible arithmetic and only one is the registered measure.
2. Which decomposition line moves when the order is reversed, and by how much? — The interaction term, `163,200 × (−0.025) × (−0.6) =` 2,448, which is why the convention must be stated.
3. State the opposite economics of retaining evidence and retaining personal data. — Evidence pays as insurance above a 3.64 % need probability, so cost is never the argument for deleting it; personal data carries exposure that dwarfs its storage cost, so de-identification pays for itself — 17,940 net on Meridian's 1.1 TB.

Advanced topics — Domain 16

16.A.1 Benefits decay and sustainment

Benefits are not a stock, they are a flow that depends on a behaviour continuing, and behaviours revert. Where the enabling change is not maintained — champions move on, new staff are trained on the old workflow because that is what the induction pack says, a system update degrades a step nobody owns — realisation decays. Meridian's post-review assessment was a **4 % relative decline a year** in the effective realisation level from year 4 in the absence of active sustainment: 0.6075 of full potential falling to 0.5832, 0.5599, 0.5375, 0.5160 and 0.4953 across years 4 to 8. The present value of that decay is **USD 216,824**, against a sustainment programme — refresher training, champion succession, a quarterly workflow review — costing **USD 22,000** a year for those five years, a present value of **USD 73,634**. Net value **USD 143,190**, a ratio of **2.94**, and the breakeven sustainment spend is **USD 64,782** a year, almost three times what the programme proposed.

Two structural points follow. Sustainment is an **operating** activity with a **capital-programme** justification, which is exactly the kind of expenditure that organisations fail to fund because the benefit accrues to a case that is already closed; the closing account of 16.4.2 is the instrument that makes the argument, and it must be handed to the benefits owner with the number attached. And decay is measurable long before it is visible in the money — the leading indicator is the registered adoption measure, monitored monthly against its own trend rather than against its target, because a measure at 95 per cent of target and falling is a different situation from one at 95 per cent and steady, and a report that shows only the variance against target cannot distinguish them.

16.A.2 Closing a project that failed, and closing one that was cancelled

Both are legitimate closures and both are done badly for the same reason: the organisation wants the episode over, and speed is mistaken for decisiveness.

A cancelled project — Domain 2 KA 2.4 made the case for stopping — must still be closed properly, and the specific obligations are: secure and value whatever is salvageable, including work in progress that another project can use, licences that can be redeployed and, most often overlooked, **requirements and design work that retains value independently of the delivery**; settle contracts on a termination basis rather than abandoning them, since an unterminated contract accrues the carrying cost of 16.2.4 indefinitely — termination rights, their notice mechanics and the compensation they trigger are contractual and jurisdictional, so take qualified legal advice before serving anything; account for the sunk cost honestly and separately from the decision to stop, because conflating them is how the next cancellation gets delayed; and hold the review, which is the highest-value review the organisation will ever run and the one it is least likely to hold, because the findings are uncomfortable and the participants have dispersed.

A project that delivered but failed to realise is the harder case, because there is no moment at which anyone is obliged to say so. Meridian is close to this shape: outputs delivered, adoption short, operating cost unbudgeted, realised NPV negative until a post-project intervention nobody had planned. The closing discipline is to **report the closing account to the body that approved the case**, with the variance decomposed and the improvement plan priced, rather than to close the project and report delivery. That is an uncomfortable paper to write and it is the one that makes the next business case honest — which is the whole return on writing it.

16.A.3 The reviewer's closeout eye

Invariants to test on any transition and closeout, each cheap and each diagnostic.

The handover trigger is a **condition set**, not a date, and each condition is assessable by someone who is not delivering it. The readiness figure presented to the go-live decision is the **conjunction** $\prod p_i$, not an average, and the paper shows both the independent and the $\min(p_i)$ bounds. A **service test** exists and was run by the receiving organisation. Hypercare has a **measured** exit criterion. A **rehearsed reversion plan** exists with a latest-possible-reversion point. The receiving organisation has a **funded run line** and the whole-life cost appears in the case. Retention releases against **events**, and outstanding items are a priced list from the day of takeover. Every open claim has a **stated monthly carrying cost** and a settlement authority. The final payment **reconciles** to the movements since the last certificate, and the account is signed as full and final. Documentation has passed the **execution test** by someone who was not on the project, and the key-person concentration is counted. The lesson **retrieval rate is instrumented**, not assumed. Every benefit has a **registered measure with a stated denominator**, a **pre-change baseline with a date**, a named owner outside the project, and an end date for the obligation. The closing account is **decomposed** into level, timing, per-unit and cost components with the **interaction term and its convention stated**. A **comparison cohort** was established at baseline. The retention schedule is **class-by-class** with a disposal action and a recorded authority. And for every AI-informed decision, the **model identifier, version, verifier and decision-maker** are all in the record — the test that 28.50 % of Meridian's decision base failed.

Industry variations — Domain 16

- **Healthcare.** Clinical safety governs the go-live decision and is not delegable to a project authority: a clinical safety case and its hazard log must be closed by clinical governance before transition, and a reversion plan must be viable at any point because the fallback is patient care, not a batch job. Benefits are overwhelmingly capacity rather than cash (Domain 2, KA 2.3.2), which makes the measure definition of 16.4.1 the whole argument.
- **Public sector and government.** Closure reporting is frequently statutory and published, benefits are audited by an external body against the approved case rather than against the honest one, and the archive's retention period is set by public-records legislation rather than by the economics of 16.4.4. The practical consequence is that the adoption-term correction of 16.4.2 must be made and approved before delivery, because after it the published claim is the benchmark.
- **Construction and infrastructure.** Commissioning, takeover, defects liability and the final account are contractually formal, with certificates, dated periods and a defined dispute route; the closeout risk is concentrated in latent-defect exposure that outlives everyone involved, so the as-built documentation and test evidence of 16.1.2 are the retained assets that matter most.
- **Energy and utilities.** Performance demonstration under representative load is the acceptance event, and an availability or efficiency shortfall accepted at commissioning becomes a permanent operating annuity — the arithmetic of Case study B. Regulatory witnessing constrains when a retest can occur, so the retest window is a schedule constraint, not a contingency.
- **Technology and product organisations.** There is often no closeout at all, because the product continues: the risk inverts from a botched transition to an **absent** one, with the delivery team becoming a permanent unfunded support function and the run cost of 16.2.1 never surfacing as a decision. The discipline that substitutes for closeout is a funded service-ownership transfer with a stated date and an operating budget.
- **Financial services and regulated industries.** Model inventory, change evidence and decision auditability drive the archive: the retention schedule of 16.4.4 and the model records of 16.4.5

are supervisory expectations rather than good practice, and an unexplainable AI-informed decision is a finding rather than an inconvenience.

Case study — Domain 16: the account nobody asked for (health, Meridian)

Situation. Meridian Care Records closed twenty-six months after mobilisation with all 40 clinics live, the systems integrator's final account agreed at **USD 1,831,400**, and a capital outturn of **USD 2,514,000** against **USD 2,400,000** approved — a 4.75 % overspend that the programme board accepted without discussion. The closure report ran to nine pages, recorded delivery of every output, and was noted. No one asked what had been realised, and the governance body that had approved the business case had, by then, been reconstituted twice.

What the project leader did anyway. Twelve months after the last go-live, and with no obligation to do so, the outgoing project leader and the clinical directorate's benefits owner produced a closing account against the registered measures. Adoption: **27 of 40** clinics meeting the daily-use definition, **67.50 %**. Time released, by sampling in the adopting clinics: **5.4** hours a week against the case's 6.0. Comparison cohort — the final wave of 8 clinics, measured at the same baseline and still on the old process: **no material change**, so no attribution adjustment. Measured annual benefit **USD 594,864**: **86.79 %** of Domain 1's honest figure and **60.75 %** of the claim the approved case had actually made.

The decomposition, and what it changed. The **USD 90,576** shortfall against the honest case split into **(24,480)** of adoption and **(66,096)** of hours per adopting clinic, with the **2,448** interaction term assigned to the second line under the stated convention. That split reversed the programme's own narrative: eighteen months of governance attention had gone to adoption, because adoption was visible and had already been the subject of a public failure, while the benefit per adopting clinic — the other half of the case's arithmetic — had never been re-measured and was wrong by 10 per cent. Extending the account to present value produced the harder number: level variance **(465,015)**, timing variance **(413,809)** from a ramp that arrived a year late, the omitted operating cost of 108,000 a year worth **(644,900)**, the cost outturn **(114,000)**, and a realised NPV of **(USD 304,827)** against the **+1,332,898** the honest case had promised.

How it resolved. The account was sent, unrequested, to the board that had approved the case, with one proposal attached: **USD 180,000** to complete the workflow redesign in the thirteen non-adopting clinics and re-run the champion programme, lifting adoption to 80 % and restoring the full 6.0 hours from year 5. Its present value was **USD 487,090** against a present cost of **USD 146,934** — an NPV of **+340,156**, a benefit-cost ratio of **3.32**, and enough to move the programme's realised NPV to **+USD 35,329**. It was approved in one meeting, funded from operating budget, and given to the benefits owner with a monthly adoption report against the **78.60 %** effective level the account identified as the programme's breakeven. Two further changes followed, both about the next programme rather than this one: every business case in the portfolio acquired a mandatory operating-cost line, and the standard benefits register template acquired a **benefit-per-unit** measure alongside the adoption measure.

What the domain teaches here. The account nobody asks for is the only one that changes anything. Meridian's largest single bridge line — **2,114,198** — was the adoption term the case omitted, money that never existed, and an organisation that had measured realisation against the approved claim would have recorded a 384,336 annual failure and learned nothing about its own case-writing. The second largest was an operating cost the project could not have controlled and should have insisted on. And the entire value of a 2.4 million dollar programme finally rested on 180,000 dollars of adoption work that appeared in no project plan, had no owner at the moment the project closed, and would not have been funded had nobody computed what it was worth.

Case study B — Domain 16: the 0.8 percentage point that was worth a million (energy, Project Auriga)

Situation. Project Auriga, the 25-week control-systems upgrade for a regional utility, reached its witnessed performance demonstration two weeks late. Domain 7's week-13 position had given `CPI` **0.9057** and an estimate at completion of `BAC/CPI =` **USD 4,416,667**; the outturn was **USD 4,412,000**, **0.11 %** below that forecast and **10.30 %** above the **USD 4,000,000** budget — a useful reminder that a CPI-based forecast made twelve weeks from the end was accurate to within five thousand dollars while the optimistic recovery narrative running alongside it was not.

What happened at commissioning. The contract required demonstrated system availability of **98.0 %** over the witnessed test period. The system achieved **97.2 %**. The contractor offered the contractual reduced-performance deduction of **USD 240,000** in lieu of remediation, which was within its liability cap and was recommended for acceptance by the project team on the grounds that it closed the contract, released the **USD 160,000** of retention held on the 3,200,000 control-systems subcontract, and avoided further delay.

What the operations director asked for instead. One number: what does 0.8 of a percentage point of availability cost to operate. The utility's own switching-restriction model priced restricted operation at **USD 1,900** an hour; 0.8 % of 8,760 hours is **70.08** hours a year, or **USD 133,152** annually, and over the asset's 12-year life at 7 % (`AF = 7.942686`) a present value of **USD 1,057,585**. The remediation-and-retest alternative cost **USD 74,000** plus three weeks of deferred cutover at the utility's calibrated **USD 18,000** a week — **USD 128,000** in total. Accepting the deduction was therefore worth `240,000 − 1,057,585 =` **(USD 817,585)**; retesting was worth **USD 689,585** more than accepting. The breakeven deduction — the payment at which acceptance would have been the better trade — was **USD 1,185,585**, nearly five times what was offered; equivalently, acceptance would only have been right if the restriction cost were below **USD 661** an hour rather than 1,900.

How it resolved. The contractor remediated two controller configurations and one network path and retested, achieving **98.3 %**; the retention was released against the passed test rather than against a deduction; the cutover ran three weeks later than the failed test date. The readiness conjunction for the cutover itself, computed on the five conditions the utility's own gate required (0.97, 0.93, 0.99, 0.88, 0.96), stood at **75.45 %** against an averaged dashboard reading of **94.60 %** — and the remaining exposure was carried by a rehearsed reversion to manual switching rather than by further assurance.

What the domain teaches here. A performance shortfall accepted at commissioning is not a one-off concession, it is a permanent annuity of operating cost paid by an organisation that was not in the room. The contractual deduction prices the **contractor's** risk, capped by the contract; the shortfall prices the **operator's** loss, uncapped and lasting as long as the asset — and the two are not comparable quantities, which is why the acceptance decision belongs to the receiving organisation and not to the project that is trying to finish. The transferable discipline is to convert every proposed acceptance concession into an annual operating cost and then into a present value over the asset life before answering, and to state the breakeven — here USD 661 an hour — so that the decision can be argued on its actual sensitivity rather than on the attractiveness of closing the contract.

Executive perspective — Domain 16

What a programme director cannot delegate in this domain:

- **The handover condition set, and the arithmetic of readiness.** Insist on the conjunction $\prod p_i$ with both correlation bounds, never an averaged dashboard. Meridian's dashboard read 90.71 % where the probability of a clean go-live was 49.79 %, and no other single number in closeout is so routinely misreported (16.1.3).
- **The acceptance of any performance concession.** Convert it to an annual operating cost and a present value over the asset life, and state the breakeven, before agreeing. A deduction prices the supplier's capped risk, not the operator's uncapped loss (Case study B).
- **The funded run line before the project closes.** No transition without a named service owner and an operating budget. Meridian's omitted 108,000 a year was worth 644,900 in present value and turned the whole account negative (16.2.1, 16.4.2).
- **The carrying cost of every open item.** A monthly number per open claim, and a settlement authority. At 5,250 a month, twelve months of avoidance cost Meridian more than the claim's assessed value (16.2.4).
- **The closing account, reported to the body that approved the case.** Decomposed into level, timing, per-unit and cost, with the interaction term and its convention stated, and the improvement plan priced. Nobody will ask for it, and it is the only document that makes the next business case honest (16.4.2, 16.A.2).
- **The retention schedule and the model record.** Class-by-class retention with disposal authority, and the model identifier, version and verifier on every AI-informed decision. Twenty-eight and a half per cent of Meridian's AI-informed decision base is unexplainable for want of three fields (16.4.4, 16.4.5).

Calculation exercises — Domain 16

Exercise 16.1 A transition gate carries six conditions, all necessary and assessed as independent, at 0.95, 0.88, 0.92, 0.99, 0.86 and 0.90. Compute the averaged dashboard reading and the probability of a clean transition; identify which single condition, lifted to 0.99, gains most and by how much; and state the per-condition probability needed for a 90 % chance of a clean transition. Solution. Average $= (0.95+0.88+0.92+0.99+0.86+0.90)/6 = 5.50/6 =$ **91.67 %**. Conjunction $= 0.95 \times 0.88 \times 0.92 \times 0.99 \times 0.86 \times 0.90 =$ **58.93 %** — a gap of **32.73 percentage points**. Lifting the 0.86 condition to 0.99 multiplies the conjunction by $0.99/0.86 = 1.1512$, giving **67.84 %**, a gain of **8.91 points** — against **2.48** for the 0.95 condition, **7.37** for 0.88, **4.48** for 0.92 and **5.89** for 0.90. For 90 % clean, $p = 0.90^{(1/6)} =$ **98.26 %** on every condition. Common error: treating the gain as proportional to the gap closed — an equal return per percentage point — which understates the value of fixing the weak conditions, since the conjunction is multiplied by p'/p, so the gain per point closed is proportional to $1/p$ and a point closed at 0.86 is worth $0.99/0.86 =$ **1.15** times a point closed at 0.99. Reading the dashboard average of 91.67 % as a probability loses the ranking entirely.

Exercise 16.2 A contract sum of USD 2,450,000 carries approved variations of USD 138,000. Retention was 5 % of the contract sum, half released at takeover. A claim settled at USD 55,000 and USD 26,400 of defects were rectified by others and are recoverable. Compute the retention, the amount paid to date, the final account, the final retention release and the final payment.

Solution. Retention `2,450,000 × 0.05 =` **USD 122,500**. Certified works `2,450,000 + 138,000 =` **2,588,000**; paid on certification `2,588,000 − 122,500 =` 2,465,500; plus the takeover release of `122,500/2 =` **61,250**, so **USD 2,526,750** paid to date. Final account `2,450,000 + 138,000 + 55,000 − 26,400 =` **USD 2,616,600**. Final retention release `61,250 − 26,400 =` **USD 34,850**. Final payment `2,616,600 − 2,526,750 =` **USD 89,850**, which reconciles as `61,250 + 55,000 − 26,400`. Common error: stopping at the certified works value of 2,588,000 and calling it the final account, which omits both the settlement and the recovery and understates contract growth — here by 28,600, or 1.17 % of the original sum.

Exercise 16.3 An open claim costs USD 2,900 a month in external advice, 1.2 days a month of internal management at USD 750 a day, and the opportunity cost of USD 240,000 of locked cash at 6 % a year. Determination at month 11 has assessed outcomes of 30,000 (probability 0.30), 55,000 (0.45) and 96,000 (0.25); the client's own assessment is 55,000. Settlement is available at month 1. Compute the monthly carrying cost, the expected cost of determination, the breakeven settlement price and the breakeven premium over the client's assessment. Solution. Carrying cost `= 2,900 + 1.2 × 750 + 240,000 × 0.06/12 = 2,900 + 900 + 1,200 =` **USD 5,000 a month**. `EMV = 0.30 × 30,000 + 0.45 × 55,000 + 0.25 × 96,000 = 9,000 + 24,750 + 24,000 =` **USD 57,750**. Expected cost of determination `57,750 + 11 × 5,000 =` **USD 112,750**. Breakeven settlement price at month 1 `112,750 − 5,000 =` **USD 107,750**. Breakeven premium `107,750 − 55,000 =` **USD 52,750**, which the identity `10 × 5,000 + (57,750 − 55,000)` reproduces. Common error: omitting the internal management time — 900 of the 5,000, or 18 % — on the grounds that the staff are paid anyway, which is the reasoning that makes the carrying cost invisible and, at ten months, understates the breakeven premium by 9,000.

Exercise 16.4 A programme's case assumed 60 sites at 75 % adoption releasing 4.0 hours a week at USD 92 an hour over 46 weeks. Measured: 66 % adoption and 3.6 hours. Compute the planned and measured annual benefits and the total variance, then decompose it both ways and state the interaction term. Solution. `U = 60 × 92 × 46 =` **USD 253,920** per unit of adoption-hours. Planned `253,920 × 0.75 × 4.0 =` **USD 761,760**; measured `253,920 × 0.66 × 3.6 =` **USD 603,313.92** — **79.20 %** of plan, a variance of **(USD 158,446.08)**. Adoption first: adoption `253,920 × 4.0 × (−0.09) =` **(91,411.20)**; hours at measured adoption `253,920 × 0.66 × (−0.4) =` **(67,034.88)**. Hours first: hours `253,920 × 0.75 × (−0.4) =` **(76,176.00)**; adoption at measured hours `253,920 × 3.6 × (−0.09) =` **(82,270.08)**. Both sum to (158,446.08); the interaction term `253,920 × (−0.09) × (−0.4) =` **USD 9,141.12** is what moves between the lines. Common error: computing both variances at plan — `(91,411.20)` and `(76,176.00)` — which sums to (167,587.20) and over-states the shortfall by exactly the interaction term, because the same movement has been counted twice.

Exercise 16.5 A post-project review costs USD 62,000 and produces 41 lessons; an applied lesson avoids USD 19,500 on average; the current retrieval rate is 9 %. Compute the expected recovery and the breakeven retrieval rate, then evaluate spending a further USD 24,000 to raise retrieval to 30 %. Solution. Addressable value `41 × 19,500 =` **USD 799,500**. At 9 %: `799,500 × 0.09 =` **USD 71,955** — the review already returns 1.16 times its cost. Breakeven retrieval `62,000/799,500 =` **7.75 %**. At 30 %: **USD 239,850**, an increment of `799,500 × 0.21 =` **USD 167,895** for 24,000, a ratio of **7.00**. Combined breakeven `86,000/799,500 =` **10.76 %**, and the net value at 30 % retrieval is **USD 153,850**. Common error: comparing the review's cost with the addressable value of 799,500 rather than with the value weighted by retrieval, which makes every review look overwhelmingly worthwhile and removes the reason to fix retrieval — the one intervention that actually pays here at seven to one.

Practitioner's toolkit — Domain 16

Adoption-ready artefacts; adapt headings to your organisation, then keep them stable.

Toolkit 16.T.1 — Transition readiness certificate

One page per receiving unit, signed by the receiving organisation. Rows: each readiness condition, its owner (who must not be the person delivering it), its objective assessment method, its assessed probability p_i with the basis of the assessment named, and its evidence reference. Footer, computed and printed rather than described: the **conjunction** $\prod p_i$, the averaged figure alongside it so the gap is visible, the $\min(p_i)$ correlated bound, the expected number of failed transitions across the remaining units at the stated cost per failure, and the residual after any proposed uplift. Below that: the hypercare duration with its **measured** exit criterion, the reversion trigger, the reversion decision-maker, the time to revert and the latest point at which reversion remains possible. A certificate that cannot be completed is itself the finding.

Toolkit 16.T.2 — Closeout and final account pack

Four sections, each a table. **Commercial:** per contract — original sum, approved variations, claims open with their **monthly carrying cost** and settlement authority, claims settled, recoveries, retention held and released with the event that released it, final account, amounts paid, final payment with its reconciliation to the movements since the last certificate, and the date the full and final statement was signed. **Operational:** service owner, funded run line with its annual amount, support model, licence and contract register with renewal and notice dates, access revocation date for the project team, and the obsolescence horizon. **Knowledge:** documentation items with the date each passed the execution test and by whom, key-person concentration count, and the outstanding documentation list with owners and dates. **Organisational:** authority withdrawn on, cost accounts closed on, records placed under the retention schedule, and people released with their contribution recorded.

Toolkit 16.T.3 — Benefits closing account and retention schedule

Two linked artefacts, owned by the benefits owner and reported to the body that approved the case. The **closing account**: per benefit — registered measure with its exact definition and denominator, baseline value and date, target profile, measured value and date, and the variance decomposed into adoption (or volume), benefit per unit, and timing, with the interaction term shown and the decomposition convention stated in a footnote. Then the present-value bridge from the approved case to the realised position, with an explicit line for any operating cost the case omitted, and the priced improvement plan with its owner and funding source. The **retention schedule**: per data class — volume, retention period, legal or contractual basis, disposal or de-identification action, the authority who may execute it, and the test that de-identification actually worked; plus, for every AI-informed decision, the required fields of 16.4.5 — model identifier, version, date, input, output, named verifier, decision-maker — with a monthly count of records missing any of them.

Exam preparation — Domain 16

What is assessed. What handover transfers and what it cannot; condition-based handover; the distinction between works tests and service tests; **readiness as a conjunction and its arithmetic**; proportional-shortfall ranking of remediation; the hold-or-go decision under both correlation assumptions; hypercare exit criteria and the reversion plan; operational transition with a funded run line and whole-life cost; retention, defects liability and warranty; **the final account and its reconciliation**; **the carrying cost of an open item and the breakeven settlement premium**; knowledge transfer and the runbook execution test; the post-project review and **the economics of retrieval**; the benefits measurement plan and the registered measure; **the closing account and its decomposition**, including the interaction term and the convention that assigns it; attribution and the comparison cohort; class-by-class retention economics; and the model-retention record.

The calculations to be able to do under time pressure. $\prod p_i$ against $(\Sigma p_i)/k$, the gain p'/p from lifting one condition, and `p = target^(1/k)`. Expected failed transitions and the hold-or-go comparison. Final account, retention release and final payment with its reconciliation. Monthly carrying cost, `EMV` of determination, breakeven settlement price and the identity `c × Δm + (EMV − assessed)`. Annual benefit `U × a × h`, the two-factor decomposition in both orders, and the interaction term. Present value of an operating cost over an appraisal horizon. Breakeven retrieval rate. Retention cost against `P(need) × consequence`, and the breakeven need probability.

The traps. Reporting an averaged readiness figure as a probability (16.1.3, Exercise 16.1) · ranking readiness remediation by absolute gap rather than by the ratio p'/p (Exercise 16.1) · omitting the residual failure rate after an uplift, which makes readiness work look better than it is (MCQ 16.1-D) · treating $\min(p_i)$ as the expected case rather than the correlated bound (16.1.3) · calling the certified works value the final account (Exercise 16.2) · ignoring internal management time in a carrying cost (Exercise 16.3) · comparing an early-settlement premium with the claim's assessed value rather than with the expected cost of determination (16.2.4) · computing both benefit variances at plan and double-counting the interaction term (Exercise 16.4) · using the undiscounted total for an operating-cost omission (MCQ 16.2-D) · comparing a review's cost with the addressable rather than the retrieval-weighted value of its lessons (Exercise 16.5) · measuring realisation against a claim the case should never have made rather than against the honest figure (16.4.2) · and arguing attribution at review instead of designing a comparison cohort at baseline (16.4.3).

How the domain connects. Domain 1 supplies the accountability principle, the outputs-to-benefits chain and the cost of delay that this domain re-tests at the realised rate. Domain 2 supplies the benefits map, the baseline discipline, the appraisal this account is measured against and the kill criteria that 16.A.2 executes. Domain 3 supplies the gate at which the go-live decision sits and the decision record the post-project review reads. Domain 4 supplies the configuration baseline that as-built documentation must match. Domain 9 supplies the acceptance and nonconformance machinery commissioning uses. Domain 10 supplies the contract mechanisms of retention, defects liability and claim settlement. Domain 12 supplies the conditions under which a review produces candour and a team dissolves well. Domain 14 supplies the AI-use register and verification standard that 16.4.5 retains. Domain 15 supplies the portfolio benefits register that must not double-count what this domain measures. And PFL-AI Domain 4 supplies the appraisal machinery — equivalent annual value and unequal lives — behind the whole-life comparison of 16.2.1 and Case study B.

Domain 16 summary

The project ends; the account does not. This domain replaces the handover date with a handover condition, and shows that readiness is a **conjunction**: Meridian's seven go-live conditions, averaging **90.71 %** on the programme dashboard, gave a probability of a clean go-live of **49.79 %** — a gap of nearly forty-one percentage points between the number that was reported and the number that mattered. The remediation ranking follows the ratio p'/p, so the clinical champion at 0.80 was worth **11.20 points** against **1.04** for training at 0.96; even all seven conditions at 0.98 leaves **13.19 %** of go-lives failing, which is why the answer to the residual is a rehearsed reversion plan and not more assurance. Holding three weeks and spending **USD 96,000** dominated going now under both the independent conjunction and the perfectly correlated bound, saving **USD 261,864** and **USD 55,992** respectively against a commitment of **USD 138,840**.

Closeout is commercial as well as operational. Meridian's final account came to **USD 1,831,400 — 9.01 %** above the original sum, being 5.71 % of variations plus 4.40 % of claim settlement less 1.11 % of defect recovery — and the final payment of **USD 97,400** reconciled exactly to the movements since the last certificate, which is the check that an account is closed rather than totalled. The open claim cost **USD 5,250 a month** to carry, so fourteen months of it would have cost **USD 73,500**, more than the claim's own assessed value of 62,000; the breakeven early-settlement premium was **USD 69,500**, given by the transferable identity **carrying cost × months saved + (expected determination − own assessment)**, and settling at 74,000 saved **USD 57,500**. Knowledge behaves the same way: three undocumented configurations cost **7.00** times as much to reconstruct as to record, and a post-project review that breaks even at a **5.20 %** retrieval rate returns **11.30 to one** on the indexing that raises retrieval — which is where the value is lost, and where nobody spends.

The closing account settles what Domains 1 and 2 opened. Measured: **27 of 40** clinics in daily use and **5.4** clinician-hours released, worth **USD 594,864** a year — **86.79 %** of Domain 1's honest **685,440** and exactly **60.75 %** of the **979,200** the approved case claimed. The **USD 90,576** shortfall against the honest case decomposes into **(24,480)** of adoption and **(66,096)** of benefit per adopting clinic, with **USD 2,448** of interaction assigned by a stated convention — and the larger of the two was the assumption nobody was monitoring. In present value the bridge runs **+3,447,096** as claimed, **(2,114,198)** for the adoption term the case omitted, **(465,015)** of level variance, **(413,809)** of timing, **(644,900)** for an operating cost of 108,000 a year that the case never carried, and **(114,000)** of cost outturn, to a realised NPV of **(USD 304,827)**. Strip out the run cost and the realised position is **+340,073**: the omission of an operating-cost line was the entire margin. And **USD 180,000** of post-project adoption work, worth **+USD 340,156** at a benefit-cost ratio of **3.32**, moved the account to **+USD 35,329** — the whole value of a 2.4 million dollar programme, resting on a plan no project document contained. Auriga's closeout makes the same point in a different currency: 0.8 of a percentage point of availability, offered for a **240,000** deduction, was worth **USD 1,057,585** over the asset life, and the retest that cost **128,000** was worth **689,585** more than accepting the money.

What survives the project is what was written down. Retaining the evidence pays as insurance above a **3.64 %** need probability; retaining personal data is the reverse trade, with de-identification worth **USD 17,940** net on 1.1 TB; and **28.50 %** of Meridian's AI-informed decision base is permanently unexplainable because three fields — model identifier, version, date — were not recorded on records that were being written anyway. Benefits decay at **4 %** a year without sustainment, worth **USD 216,824** in present value against a **USD 73,634** sustainment programme, so the account has to be handed on rather than closed.

The through-line, and the last of the book: **outputs are delivered, benefits are realised, and the two are separated by an organisation, a period of time and a measurement nobody is obliged to take.** Take it anyway. Decompose it honestly, state the convention, price the recovery, and send it to the people who approved the promise — because the closing account is the only document in the delivery lifecycle that improves the next decision rather than defending the last one.

Appendix A — Formula and symbol sheet

23 symbols, of which **22 carry a verified worked example** in this volume or its companion. A symbol marked pending is registered and used but does not yet have a golden-answer check behind a printed result — treat its appearances with the same care you would any unchecked number. Units are stated because a unit error is the commonest arithmetic failure in practice and the hardest to see in a finished sentence.

SYMBOL	MEANING	UNIT	FIRST HOME	VERIFIED
PV (BCWS)	Planned Value (EVM context)	currency	inherited (PCP-AI master table)	yes
EV (BCWP)	Earned Value	currency	inherited (PCP-AI master table)	yes
AC (ACWP)	Actual Cost	currency	inherited (PCP-AI master table)	yes
BAC / EAC / ETC / VAC	Budget / Estimate at Completion, Estimate to Complete, Variance at Completion	currency	inherited (PCP-AI master table)	yes
CV = EV − AC · SV = EV − PV	Cost / Schedule Variance	currency	inherited (PCP-AI master table)	yes
CPI = EV/AC · SPI = EV/PV	Performance indices	ratio	inherited (PCP-AI master table)	yes
TCPI	To-Complete Performance Index	ratio	inherited (PCP-AI master table)	yes
PoC	Percentage of completion	%	inherited (PCP-AI master table)	yes
r, n	Discount rate per period; number of periods	ratio; count	inherited (PCP-AI master table)	yes
PV(x)	Present value of amount x	currency	inherited (PCP-AI master table)	yes
ES	Earned Schedule (time-based schedule measure)	time	PML-AI D6	yes
SPI(t)	Schedule Performance Index (time) = ES/AT	ratio	PML-AI D6	yes
TF, FF	Total float, free float	time	PML-AI D6	yes
EMV	Expected monetary value = Σ(probability × impact)	currency	PML-AI D8	yes
E[wait]	Governance latency = M/2 + L (meeting interval M, paper lead time L) — the expected wait for a committee decision; sums across escalation tiers	time	PML-AI D3	yes
Cost of delay	Value forgone per unit of elapsed time; the price at which governance latency, gate duration and escalation paths are all evaluated	currency/ period	PML-AI D1/D3	yes
Gate net value	P(defect) × build-fix cost − [review + elapsed × cost of delay + P(defect) × (P(detect) × design-fix + P(miss) × build-fix)]	currency	PML-AI D3	yes
Committee capacity	meetings per year × substantive items per meeting; utilisation = demand ÷ capacity	count	PML-AI D3	yes
Mesh / layered interfaces	n(n−1)/2 possible pairwise interfaces against n to an integration layer — the architecture choice, priced	count	PML-AI D4	yes

SYMBOL	MEANING	UNIT	FIRST HOME	VERIFIED
Hundred-per-cent rule	Σ children − parent = 0 at every WBS level; a non-zero result is an omission or a duplication	currency	PML-AI D4	yes
Assessed total impact	direct + (schedule weeks × cost of delay) + rework + interface re-verification + regression + documentation (+ risk, + benefit) — the basis a delegation threshold must read on	currency	PML-AI D4	yes
Baseline drift	(change count × average direct cost) + (affected count × average weeks × cost of delay); cumulative-test threshold derived from the observed change rate	currency	PML-AI D4	yes
`EVA(benefit)`	Benefit measure in benefits register (named in words to avoid EV clash)	currency	PML-AI D2/ D16	pending

Conventions that apply to every formula above. Financial arithmetic is decimal, never binary floating point. Intermediate values carry full precision; rounding happens only at display, which is why a printed figure may not reproduce if you round as you go. Rates and periods must agree — an annual rate against monthly periods is the error the golden suite catches most often. Datetimes are `YYYY-MM-DD HH:MM:SS` in UTC and are compared as instants, never lexically.

Appendix B — Notation, terminology and conventions

Terms fixed at programme level, which the chapters use without redefining:

TERM	AS USED IN BOTH VOLUMES
PML-AI	PCI Project Management Leader – AI: the certification; its BoK is this programme's Book One
PFL-AI	PCI Project Finance Leader – AI: the certification; its BoK is Book Two
Responsible AI principle	"AI proposes; the professional verifies, decides and remains accountable" — the suite-wide restatement of PCP-AI's "AI proposes, the professional disposes"
Domain / Knowledge Area / Topic	The three-level content hierarchy `D.K.T`, identical to PCP-AI
Sponsor	The accountable executive owner of the business case (PML-AI); in PFL-AI project-finance contexts, an equity investor promoting the project — the books flag the context at each use
Special-purpose vehicle (SPV)	The ring-fenced legal entity created to own, finance and operate a project
Bankability	The degree to which a project's contracts, risks and cash flows support limited-recourse financing on acceptable terms
CFADS	Cash flow available for debt service, as defined in the formula registry
Cash waterfall	The contractually agreed priority order in which project cash is applied each period
Benefits realization	The identification, planning, measurement and sustainment of the outcomes a programme exists to deliver
Decision rights	The explicit allocation of which role may take which decision at which threshold
Psychological safety	A shared belief that the team is safe for interpersonal risk-taking — candour without penalty
Hybrid delivery	A deliberate combination of predictive and adaptive methods under one governance frame
Model risk	The risk of loss from decisions based on flawed, misused or misunderstood models (financial or AI)

Conventions

The hierarchy. Content is addressed as `D.K.T` — Domain, Knowledge Area, Topic. A cross-reference of the form KA 7.3 or 10.2.2 is precise and is the authority; a page number is not used, because the corpus is regenerated.

Context flags. Where one word carries two genuinely different concepts, the later use states the collision explicitly rather than relying on the reader to infer it — tailoring of method against tailoring of a message, `PV` as present value against `PV` as planned value, `A` as annuity payment against `A` as assets. A flag is a signal that both meanings are live in the corpus, not a warning that one is wrong.

Citation over restatement. A result established in one domain is cited from the others rather than re-derived. If two domains appear to derive the same thing, that is a defect and worth reporting.

Currency and units. USD throughout, with SAR shown where it aids a Gulf reader at an indicative `USD 1 ≈ SAR 3.75`. That rate is illustrative and is not a forecast or a quotation. Every worked example states its units, its currency, its rounding and its date basis; where one is missing, treat it as a drafting defect.

Fictitious entities. Every organisation, project, person and case in this volume is invented. Any resemblance to a real entity is coincidental, and no real project's data is used.

The responsible-AI principle, which governs every AI in this KA section: AI proposes; the professional verifies, decides and remains accountable.

Appendix C — Verification record

7,351 golden-answer checks stand behind this volume. Every number printed as a result — in worked examples, in-text calculations, multiple-choice options (all of them, not only the correct one), exercise solutions, case studies and figure specifications — is recomputed independently with decimal arithmetic at 28-digit precision and compared to the printed value. Each domain's module also pins the invariants the domain claims: an identity, a breakeven, a bound, an inequality, so that a claim cannot silently stop being true.

DOMAIN		CHECKS
1	The Project Leadership Profession	405
2	Strategy, Selection and Business Alignment	450
3	Governance, Organization and Decision Rights	515
4	Integration and Delivery Architecture	1,947
5	Scope, Requirements and Value Definition	326
6	Planning, Scheduling and Delivery Flow	545
7	Cost, Resources and Commercial Awareness	383
8	Risk, Uncertainty and Resilience	534
9	Quality, Assurance and Continuous Improvement	302
10	Procurement, Contracts and Supply Networks	299
11	Stakeholders, Communication and Influence	236
12	Leadership, Teams and Organizational Behaviour	263
13	Agile, Adaptive and Hybrid Delivery	236
14	Digital Delivery, Data and Responsible AI	295
15	Programmes, Portfolios and Enterprise Delivery	274
16	Transition, Closeout and Benefits Realization	341
	Total	**7,351**

Reproduce it with `python3 _build/verify_formulas.py`, or check one domain with `python3 _build/run_checks.py checks/<module>.py`. Both loaders are fail-closed: a check module that raises is a failure, not a skip.

What this record does and does not establish. It establishes that the arithmetic is right — that each number is the one its stated method produces from its stated inputs. It does **not** establish that the method chosen was the right method for the situation, that the professional judgements are ones an experienced practitioner would endorse, that the emphasis across topics is well calibrated, or that nothing important is missing. Those are questions for editorial and technical review, and a reader should not read a large number in this appendix as an answer to them.

Glossary

Consolidated from every Knowledge Area's key-terms table. The Knowledge Area reference after each entry is the authority — the gloss points to the treatment and does not replace it.

A

Abdication — Transferred authority without the information or criteria to exercise it. (KA 12.3)

Abnormally low tender — A bid materially below its benchmark; the benchmark must be stated, since the bid spread and the buyer's estimate can disagree. (KA 10.2)

AC — Actual Cost of the work performed, including period accruals. (KA 7.2)

Acceptance — A formal decision by a named authority that a deliverable meets its criteria and may pass on. (KA 9.3)

Acceptance by silence — Deemed acceptance through the expiry of a review period; not a decision. (KA 9.3)

Acceptance sampling — Accepting or rejecting a population on the evidence of a sample. (KA 9.3)

Acceptance-criteria expansion — Creep entering through a growing criterion while the requirement text is unchanged. (KA 5.4)

Accountability — The obligation to answer; single-holder, non-delegable. (KA 1.2)

Accountability laundering — Presenting a model's output as a neutral input to avoid owning the judgement it embodies. (KA 14.4)

Accrual — Cost of work received but not yet invoiced, recognised in the period. (KA 7.2)

Accuracy class — The definition-linked range accompanying an estimate; never optional. (KA 7.1)

Adaptive delivery — A delivery mode fixing capacity and cadence and varying scope, correcting direction from working output at short intervals. (KA 13.1)

Adequacy ratio — Remaining reserve ÷ the confidence-level requirement of the register still open; below 1.00 is a governance trigger. (KA 8.3)

Adoption — The outcome measure linking an output to any benefit at all. (KA 1.3)

Aggregate-capacity illusion — Judging feasibility on total capacity when the constraint binds period by period. (KA 15.2)

Aggregation illusion — Treating a framework call-off as competitively priced because the framework was competed; the rate may have drifted from the market. (KA 10.2)

Aggregation trap — Averaging component ratios instead of summing their numerators and denominators; frequently reverses the sign. (KA 15.4)

AI use register — The row-per-use record of use, decision informed, accountable person, consequence, measurements, tier, versions and boundary. (KA 14.4)

Alignment decay — The erosion of strategic fit after approval, as strategy and understanding move. (KA 2.1)

Alignment half-life — `ln(0.5) ÷ ln(1 − h)` — the delivery duration beyond which alignment is more likely lost than kept. (KA 2.1)

Alignment index — `Σ min(declared weight, funded share)` — the overlap between stated priorities and funded spend. (KA 2.1)

Already-committed benefit — Benefit claimed against a baseline that already assumes a saving committed elsewhere. (KA 15.2)

Amber compression — The collapse of traffic-light status into a single uninformative colour, caused by author discretion over undefined thresholds. (KA 11.2)

Analogous / parametric / bottom-up — Scale an analogue · rate × parameter · sum the packages. (KA 7.1)

Analytics ladder — Descriptive → diagnostic → predictive → prescriptive; each rung depends on the one below being right. (KA 14.2)

Approval percentile — The position of the approved number inside its own declared range; Auriga's baseline is a P33. (KA 7.1)

Assessed total impact — Direct + schedule + rework + interface re-verification + regression + documentation + risk + benefit. (KA 4.4)

Assumption — Something taken as true without being controlled, carrying an owner and a resolution date. (KA 5.1)

Assumption exposure ratio — `Σ EMV ÷ NPV`; as it approaches one, the case is a bet rather than a value proposition. (KA 2.3)

Assumption register — Statement, basis, impact-if-false, owner, test date, falsifying trigger. (KA 2.3)

Assurance capture — An assurance function assuring work it helped produce, and therefore unable to challenge it. (KA 3.3)

Assurance map — Risks and controls against assurance lines, marking coverage, gaps and duplication. (KA 3.3)

Asynchronous artefact — A written product complete enough to remove an exchange rather than accelerate it. (KA 12.4)

Attitude — Supportive, neutral, sceptical or opposed, assessed on evidence; predicts what an engagement contact will achieve. (KA 11.1)

Attributable act — An output that confers authority or discharges an obligation, and therefore cannot be produced by a tool at any verification depth. (KA 14.3)

Attributable improvement — Raw improvement less the counterfactual improvement; the difference of the two differences. (KA 2.3)

Attribution — The portion of a measured improvement plausibly caused by this project. (KA 2.3)

Authority bounds — The spend, scope, resource and commitment limits within which the leader decides alone. (KA 4.1)

Avoid / reduce / transfer / accept — The threat-response families; transfer is a price, not a disappearance. (KA 8.3)

B

BAC — Budget at Completion — the time-phased cost baseline's total. (KA 7.1)

Back-translation — Independent re-translation to the source language as a check on anything carrying a commitment, instruction or safety implication. (KA 11.4)

Backward traceability — Design or test → approved requirement; detects unrequested scope, and costs one query. (KA 5.2)

Base rate — The prevalence of the condition a monitor looks for; with the saving-to-investigation ratio it decides how a monitor should be tuned. (KA 8.4)

Base-rate difference — Unequal prevalence between groups, which makes simultaneous equalisation of all error measures impossible. (KA 14.4)

Baseline — The measured pre-change position; must be measured before, not reconstructed after. (KA 2.3)

Baseline drift — Cumulative movement through individually authorised small changes, with no decision on the total. (KA 4.3)

Baseline maintenance — Applying changes by accumulation, with traceability to authority, preserving the original. (KA 4.3)

Benefit per unit of adoption-hours (U) — clinics × valuation rate × operating weeks — the constant in `benefit = U × a × h`. (KA 16.4)

Benefits bridge — The reconciliation from gross claimed benefit through eliminations and adoption to net realistic benefit. (KA 15.2)

Benefits map — Output → enabling change → outcome → benefit → objective, with owners and measures. (KA 2.3)

Benefits measurement plan — Measure definition, baseline and date, target profile, method, frequency, owner, reporting route and end date, per benefit. (KA 16.4)

Benefits owner — The accountable person outside the project who will realise the benefit. (KA 2.1)

Benefits profile — The time-phased benefit stream; ramps in reality, flat in most business cases. (KA 2.2)

Best alternative — What happens absent agreement, priced fully — invoice cost plus delay at the cost of delay plus internal effort plus risk as an `EMV`. (KA 11.3)

Binding constraint — The resource that actually limits the portfolio — usually a scarce capability, rarely money. (KA 15.2)

Binding step — The step whose yield improvement raises `RTY` most; identifiable from the yields alone. (KA 9.4)

Blended rate — Weighted average cost per hour of a mixed resource pool. (KA 7.4)

Blocked work in progress — Blocked arrival rate × `E[wait]` — the items a team holds but cannot work on, invisible on a board. (KA 13.3)

Boundary exposure — The priced consequence of an unstated extent: additional units × unit cost. (KA 5.1)

Bounded envelope — Stated limits within which a delivery team decides freely, reviewed rather than approved decision by decision. (KA 3.1)

Bounded funding envelope — Authority to spend a stated number of iterations before continuation requires an explicit decision. (KA 13.4)

Breakeven adoption — The adoption level at which NPV is zero — a more useful board sentence than NPV. (KA 2.2)

Breakeven breach cost — Value forgone by reserving capacity ÷ the improvement in failure probability it buys. (KA 15.3)

Breakeven defective fraction (p) — `c/u` — verification cost ÷ escaped-defect cost; above it, full verification pays. Independent of `n`. (KA 9.3)

Breakeven dispute probability — The dispute rate at which writing testable criteria exactly pays for itself; **2.67 %** at Meridian against an observed 25 %. (KA 5.4)

Breakeven elicitation spend — The investment at which the saving from earlier detection is exactly recovered; expressed either as a sum or as a tolerable delay. (KA 5.2)

Breakeven elimination rate — The elimination rate at which a portfolio just satisfies its investment decision rule. (KA 15.2)

Breakeven error probability (P) — `C_v ÷ [(1 − q) × L]` — the error rate above which a verification pays for itself. (KA 1.4)

Breakeven fidelity — Twin cost ÷ gross benefit at perfect fidelity — the fidelity below which the twin destroys value. (KA 14.2)

Breakeven retrieval rate — review cost ÷ (lessons × avoided cost per applied lesson) — the threshold above which reviewing pays. (KA 16.3)

Breakeven rework probability — `p* = good outcome ÷ (good − bad)`; the probability at which an overlap stops paying. (KA 6.4)

Breakeven settlement premium — `carrying cost × months saved + (EMV of determination − own assessed value)`. (KA 16.2)

Breakeven value of a released hour — Incremental direct cost ÷ released hours — USD 12.65 on Auriga, 6.836 % of the leader's rate. (KA 12.3)

Breakeven volume (Q) — The volume at which two options cost the same: `(F_make − F_buy)/(v_buy − v_make)`. (KA 10.1)

C

Capability / conduct / circumstance — The three causes of under-performance, requiring training, consequence and support respectively. (KA 12.3)

Capability-specific capacity — Capacity modelled per scarce capability, since the portfolio consumes them in component-specific mixes. (KA 15.3)

Capacity-based contract — A funded team at a rate for a cadence; the buyer bears quantity risk, change is free at the boundary. (KA 13.4)

Capture-recapture estimate — Register completeness inferred from the overlap of two independent identification methods: $\hat{N} = n_1 n_2 / m$. (KA 8.1)

Carrying cost of an open item — Advisory + internal management + opportunity cost of locked cash, per period; invisible in every ledger. (KA 16.2)

Cascade (enabler) double count — An enabling component claiming benefit already claimed by the components it enables. (KA 15.2)

Cash-releasing / non-cash-releasing — Benefits that reduce spend versus those that release capacity. (KA 2.3)

Causation — The demonstrated link from the contractual basis to the effect claimed; where most claims are decided. (KA 10.4)

Cause concentration — The share of nonconformances attributable to the single largest root cause. (KA 9.3)

Cause → event → consequence — The three-part risk statement; each part maps to a response type. (KA 8.1)

Ceiling price — The cap on the buyer's total exposure under an incentive contract. (KA 7.4)

Change — An assessed, authorised, recorded movement of the scope baseline (Domain 4, KA 4.4). (KA 5.4)

Change board — The body authorised to approve baseline changes within limits, including a technical authority. (KA 4.4)

Change vs clarification vs defect — Baseline moves · baseline already covered it · work fails to meet the baseline. (KA 4.4)

Channel load — Channels × hours per channel per period; compared with engagement capacity to give the sustainable party count. (KA 11.2)

Charter — The document authorising the project and conferring bounded authority on its leader. (KA 4.1)

Closing account — Reconciliation of realised value against the promise, decomposed into its causes and reported to the approving body. (KA 16.4)

Coaching time per report — `(R − e(n − b))/n`; 65.00 minutes at 6 reports on Auriga, 17.50 at 12. (KA 12.3)

Commensurability — Whether two figures may be compared directly; a capital variance and an annual benefit may not, without a horizon and a rate. (KA 1.1)

Commissioning — Progressive proving of a deliverable in its operating environment, from component test to performance demonstration. (KA 16.1)

Commitment — A contractual obligation (e.g. a PO); a funding fact, not yet a cost. (KA 7.2)

Commitment coverage — (`AC` + open commitments) ÷ `EAC` — the share of the forecast whose price is already fixed; 74.72 % on Auriga. (KA 7.2)

Commitment exposure — Range width × cost of delay — what a date quoted from a ranged package risks. (KA 6.3)

Commitment lead-time rule — The rolling-wave horizon must be at least as long as the longest commitment lead time inside it. (KA 4.1)

Committee capacity — Meetings per year × substantive items per meeting; compared against decision demand — average, peak and off-peak. (KA 3.2)

Common data environment (CDE) — A single agreed information store with explicit item states and authorised, recorded transitions. (KA 14.1)

Communication architecture — The designed set of channels between parties — mesh, routed or hybrid — with its load computed against capacity. (KA 11.2)

Communication path — One pairwise relationship in a team; counted by `n(n − 1)/2` (Domain 4, KA 4.2.3). (KA 12.2)

Community stakeholder — An unbounded, heterogeneous group with a legitimate interest in a project's effects, often with statutory engagement rights that vary by jurisdiction. (KA 11.4)

Comparison cohort — Comparable units measured at baseline and review without receiving the change, used to net out other causes. (KA 16.4)
· Also read at KA 2.3: A group that did not receive the change, used to measure the counterfactual improvement.

Component — A project or enabling activity within a programme, with a named owner and a stated consequence of cancellation. (KA 15.1)

Composite fitness — The product of dimension conformance rates; the share of records fit for use. (KA 9.4)

Concentration line — The share of portfolio cost sitting in components below the alert threshold; the companion to any aggregate index. (KA 15.4)

Concession (use as is) — Acceptance of a nonconformance as it stands, authorised by the party bearing the consequence. (KA 9.3)

Condition-based handover — Handover triggered by a satisfied condition set rather than a calendar date. (KA 16.1)

Condition-closure rate (r) — The share of a gate's conditions verified closed by the following gate. (KA 3.3)

Conditional acceptance — Acceptance with open items; priced by moving each item up the correction ladder **by severity class**. (KA 5.4)

Conditional pass — Proceeding subject to conditions — a real instrument only if the conditions have owners, dates and consequences. (KA 3.3)

Confidence bound (p_upper) — `1 − α^(1/n)` — the largest defective fraction consistent with a clean sample of `n`. (KA 9.3)

Configuration audit — Verification that the actual state matches the recorded state. (KA 4.3)

Configuration management — Knowing which version of every controlled item is approved, and how it came to be. (KA 4.3)

Confirmation rule — The provision that stops a tolerance trigger firing on one period's measurement noise. (KA 3.3)

Conflict source — Structural, resource, information, interest or interpersonal — each with a different remedy. (KA 11.3)

Consent risk — Whether the party holds a veto, approval, licence, signature or practical ability to withhold cooperation; a property of position, not disposition. (KA 11.1)

Consent risk pricing — EMV of a consent failure, presented with its breakeven cost and required probability reduction, never as a point estimate alone. (KA 11.4)

Consent-risk floor — A minimum engagement allocation to a party that can stop the work, justified by breakeven probability rather than by benefit rate. (KA 11.1)

Consequence per defect (u_i) — The internally assessed cost of one defect of a class reaching use; assessed, owned and stated. (KA 14.1)

Consequence threshold — A point where the consequence function steps rather than scales; where resilience is bought. (KA 8.4)

Consequence-weighted exposure — $\Sigma\, n_i d_i u_i$ — the expected cost of a data estate's defects. (KA 14.1)

Consistency check — A specific question testing whether two subsidiary plans agree. (KA 4.1)

Consolidation time (C) — Elapsed time from data cut-off to pack issue. (KA 11.2)

Constraint — An externally imposed date; modelled as logic, not as a pin. (KA 6.3)

Containment layer — A review, test or inspection with a stated coverage, detection rate and unit correction cost. (KA 9.2)

Contestability — The affected party's ability to challenge a decision and have it examined by a human with authority to change it. (KA 14.4)

Contingency draw index — (Contingency drawn ÷ contingency) ÷ percent complete; above 1.00 the reserve is depleting faster than the work is being done. (KA 7.1)

Contingency reserve — Inside the baseline; funds identified risks; PM-controlled. (KA 7.1)

Continuation decision — The periodic judgement that the next increment's cost is worth its expected value; the capacity model's brake. (KA 13.4)

Contract growth — Final account against original sum — variations and settlements, not variations alone. (KA 16.2)

Control account — The level at which performance is measured and reported (Domain 7's earned value attaches here). (KA 4.2)
· Also read at KA 7.1: The WBS × organisation point where scope, budget and actuals integrate.

Control schedule — The L3 CPM network under change control; home of float and progress. (KA 6.1)

Convergence point — A node where paths merge; slippage compounds there. (KA 6.2)

Coordination share — $c(n - 1)/(2h)$ — coordination overhead as a fraction of team capacity; $(n - 1)/160$ at $c = 0.5$, $h = 40$. (KA 12.2)

Correction asymmetry — The observed tendency of corrections to propagate less well than the original claim; real, and not reduced here to a multiplier. (KA 11.4)

Correction ladder — The cost of correcting one requirement defect by the stage found; Meridian's observed ladder rises ×4 per stage. (KA 5.2)

Correction-cost ladder — The rising cost of correcting a defect by the layer at which it is found. (KA 9.2)

Correlation — Shared drivers that raise aggregate variance; independence is usually optimistic. (KA 8.2)

Cost of conformance — Prevention + appraisal — money spent so defects do not occur or do not escape. (KA 9.1)

Cost of delay — The benefit forgone per period of lateness. (KA 1.3)

Cost of nonconformance — Internal + external failure — money lost because they did. (KA 9.1)

Cost of quality (CoQ) — Prevention + appraisal + internal failure + external failure. (KA 9.1)

Cost per crew-week of capacity — The priced rungs of Domain 6's capacity ladder, from 15,200 at straight time to 45,000 for accepting the delay (7.4.1c). (KA 7.4)

Cost per productive hour — Burdened pay ÷ (1 − non-productive share) × (1 + overhead); USD 100.79 against a USD 95.00 charge-out rate. (KA 7.4)

Cost slope — An activity's crash cost per week, with a technical limit on weeks available. (KA 6.4)

Count view / size view — Progress by item count against progress by size delivered; the first overstates when sequencing is optimal. (KA 13.3)

Counterfactual cost — Cost incurred only in the do-nothing case, and therefore value created by avoiding it. (KA 2.2)

Counterfactual test — What happens to the benefit if this bundle is not delivered? A zero answer means zero attributed value. (KA 5.3)

Counterparty risk (priced) — The secondary EMV of a transfer: $p \times P(\text{cannot pay}) \times \text{amount recovered}$, usually correlated with the event transferred. (KA 8.3)

Coverage — Distinct risks identified as a share of the estimated population; an optimistic figure, since shared blind spots deflate the population and so inflate it. (KA 8.1)

CPI SPI — `EV/AC` · `EV/PV`, as ratios. (KA 7.3)

Crashing / fast-tracking — Buying duration with money / with overlap risk. (KA 6.4)

Criterion influence — `(score range) × weight` — the most a criterion can move a total, and therefore what the model cannot decide. (KA 2.2)

Critical chain — The binding sequence once logic and capacity are honoured; need not be the critical path. (KA 6.3)

Critical path — The zero-float longest path; the project's shortest possible duration. (KA 6.2)

Crossover horizon — `R × (1 + d/g)` — the remaining work below which reinforcement finishes later, not earlier. (KA 1.3)

Crossover weighting (w) — The price weight at which two bidders' weighted scores are equal. (KA 10.2)

Cumulative test — The delegation-schedule provision that related decisions aggregating above `X` within period `P` require the authority appropriate to the aggregate. Setting `X` and `P` is arithmetic, not judgement, and is derived in Domain 4, KA 4.3.3 — a round number chosen without reference to the observed decision rate provides the appearance of a control and none of the function. (KA 3.3)
· Also read at KA 4.3: A threshold on aggregated related changes over a stated period, set from the observed change rate.

Cumulative-effect test — The rule that related concessions aggregating above a threshold require the authority appropriate to the aggregate. (KA 9.3)

CV SV — `EV − AC` · `EV − PV`, in currency. (KA 7.3)

Cycle time (C) — Elapsed time from an item starting to finishing; a consequence of `W` at a given `T`. (KA 13.2)

Cycle-time decomposition — Splitting average cycle time into its blocked and unblocked populations to locate the delay. (KA 13.3)

D

Dangle — An activity missing a predecessor or successor; a network defect. (KA 6.1)

Data class schedule — One row per data class: owner, definition, source of record, quality standard, access and retention. (KA 14.1)

Data minimisation — Holding only what the project needs, which reduces the security and privacy problem rather than defending it. (KA 14.4)

Data quality dimensions — Completeness, accuracy, validity, consistency, timeliness, uniqueness. (KA 9.4)

Date constraint — A pinned date overriding logic; each one weakens the model. (KA 6.1)

Daylight test — Would every party, seeing the full picture, still regard this as impartial? (KA 1.4)

Decidability — The property that every decision has exactly one accountable decision-maker with sufficient authority. (KA 3.1)

Decidable scope — Enumerated deliverables, quantified extent, named interfaces, explicit exclusions. (KA 5.1)

Decision action window — `(remaining duration − escalation latency) ÷ remaining duration`; negative means the escalation is ceremonial. (KA 3.3)

Decision breakeven adoption — The adoption at which one priced decision stops paying — 36.76 % for Meridian's acceleration. Distinct from Domain 2's **breakeven adoption**, which is the level at which the whole investment's NPV is zero (41.05 %). (KA 1.3)

Decision convention — Whether a meeting decides or ratifies, and whether authority is individual or collective. (KA 11.4)

Decision record — The versioned, attributable log that converts a decision from a memory into an institutional fact. (KA 3.3)

Decision right — The authority to make a specified class of decision, bounded by value, scope and reversibility. (KA 3.1)

Decision tree — Explicit rollback of decisions and chance events to a present choice. (KA 8.2)

Decoupling — Removing a predecessor from a milestone by designing an operable interim arrangement, multiplying the probability by $1/p_i$. (KA 15.1)

Defect rate by class (d_i) — The share of records in a class failing that class's defined pass criteria. (KA 14.1)

Defects-liability period — The interval after acceptance in which the contractor must rectify defects that appear. (KA 16.2)

Definition of done — The pre-agreed, binary, testable standard that gives the word "done" one meaning across every report. (KA 13.1)

Delay-cost density — Cost of delay per week ÷ effort in team-weeks; sequencing in decreasing order minimises total delay cost. (KA 13.1)

Delegation — Transfer of authority to decide and act, with the information and criteria to do so; accountability is retained. (KA 12.3)

Delegation schedule — The statement of authority by decision class, on value, reversibility and externality. (KA 3.2)

Delivery verdict — The project-level judgment: was the output delivered to time, cost and quality? A one-off variance. (KA 1.1)

Depth and kind — Two separate verification choices: how hard to look, and what sort of looking finds this class of error. (KA 1.4)

Derived audit tolerance — The item-level tolerance obtained by inverting `p^k` from an interface-level target: `p = q^(1/k)`. (KA 4.3)

Descope exchange rate — `1/CPI` — the forecast cash saved per unit of budgeted scope removed; USD 1.10 per dollar on Auriga. (KA 7.3)

Detection rate (d_i) — The share of defects reaching a layer that the layer finds. (KA 9.2)

Detection rate (q) — The share of material errors present that a given review depth finds; a measured quantity, never asserted. (KA 14.3)

Deterministic bias — $t_e - m = (o - 2m + p)/6$ — one sixth of the estimate's skew. (KA 6.4)

Deterrence cap vs compensation cap — The service-credit cap needed to exceed the supplier's cost of compliance, against the cap needed to cover the buyer's loss; they differ. (KA 10.3)

Differential error — A systematic difference in error rates between identifiable groups; measurable only by disaggregated measurement. (KA 14.4)

Digital delivery environment — The set of systems holding a project's authoritative information and mediating its work. (KA 14.1)

Digital twin — A model kept synchronised with a real asset through a data connection; without synchronisation it is a drawing. (KA 14.2)

Discovery arrivals — New items found necessary during build; subtracted from throughput to give the net drain. (KA 13.2)

Discretionary attention — The leader's working week less recurring commitments; the resource all leadership acts are paid from. (KA 12.1)

Displaced judgment — Ceasing to think because a tool is watching. (KA 8.4)

Disruption / measured mile — Lost productivity on unchanged work, established by comparing an unimpacted period's productivity with the impacted period's. (KA 10.4)

Distributed team — A team whose members' working hours do not fully coincide; time distance, not spatial distance, is the governing variable. (KA 12.4)

Do-minimum option — The cheapest compliant alternative where inaction is not permissible; the honest comparator, and an option in its own right. (KA 2.2)

Do-nothing baseline — The comparison against which all benefits are measured; without it value is unmeasurable. (KA 2.2)

Double-counting — The same benefit claimed by more than one case; one claimant per benefit. (KA 2.3)

Draw protocol — The published rules for releasing contingency. (KA 8.3)

Driver — The reason the work exists; determines what success means and how it is measured. (KA 2.1)

Duration multiplier — `1/(1 − r)` — the hyperbolic effect of rework on elapsed time. (KA 9.2)

E

EAC ETC — Estimate at / to complete; `EAC = AC + ETC`. (KA 7.3)

Early-warning indicator — A leading measure tied to a risk, firing before the event. (KA 8.3)

Earning rule — The convention by which physical progress becomes `EV`; fixed before the period it measures. On Auriga's ten-package group the choice moved `EV` by 15.00 % of budget. (KA 7.3)

Effective detection (q_eff) — A gate's detection rate multiplied by its condition-closure rate: `q_eff = q × r`. (KA 3.3)

Elicitation — Deliberate discovery of need, as distinct from collection of requests. (KA 5.2)

Elimination rate — Eliminations ÷ gross claimed benefit; invariant to the adoption assumption. (KA 15.2)

Emergency change route — A named authority, a short deadline and mandatory retrospective ratification. (KA 4.4)

Emergency-route share — Emergency changes ÷ total requests — a latency diagnostic, so set a review trigger on it rather than a target. (KA 4.4)

Emotional intelligence — Four practised operations — self-awareness, self-regulation, social awareness, relationship management — skills, not a trait score. (KA 12.1)

EMV — probability × impact; the average of outcomes that will not individually occur. (KA 8.2)

Enabling change — The non-project work (training, workflow, adoption support) without which no benefit occurs. (KA 2.3)

Engagement objective — A stated future state of a relationship with a date and a test, never an activity count. (KA 11.1)

Enterprise decision architecture — The set of governance tiers, the decision classes routed through them, and the resulting latency distribution. (KA 15.4)

Enterprise delivery capacity — The measured rate at which an organisation completes change, in the unit of its binding capability. (KA 15.3)

Enterprise PMO — The function that maintains portfolio data, runs allocation, measures capacity and gives second-line assurance; no decision rights. (KA 15.3)

Entitlement — The contractual basis for a claim; without it, merits are irrelevant. (KA 10.4)

Enumeration — Evaluating every feasible subset under the constraint; the optimum where the candidate set is small enough. (KA 5.3)

Envelope coverage — The share of a stream's above-authority decisions the delegated envelope contains. (KA 3.1)

Error escape rate (e) — The share of units on which a route lets a material error through; differs between manual and automated routes and must be measured. (KA 14.2)

ES EF LS LF — Earliest/latest start and finish per activity. (KA 6.2)

Escalation class — A defined trigger, destination, decision, latency and out-of-cycle route. (KA 3.3)

Escalation duty — The obligation to raise, with options, a conflict beyond one's authority. (KA 1.2)

Escalation lead time — How far before impact an escalation arrives; a shortening trend signals a decaying mechanism. (KA 3.3)

Escalation of commitment — Continuing because of sunk cost; countered by forward-looking framing. (KA 8.4)
· Also read at KA 2.4: Continuing because of what is invested; countered structurally.

Escalation-grade message — Exception, cause, impact priced, options, recommendation by name, decision required by whom and by when, and the date the leader knew. (KA 11.2)

Escape — A defect reaching the next stage, or the customer, undetected. (KA 9.1)

Escape fraction — $\prod(1 - d_i)$ across the chain — the share of introduced defects reaching the customer. (KA 9.2)

Escape-cost asymmetry — $(1 - q_p)u_p \div (1 - q_o)u_o$ — why a plausible wrong number out-damages an obviously wrong one. (KA 14.3)

Evaluation model — Weights, normalisation, scoring scale and moderation rules; a piece of arithmetic that materially selects the winner. (KA 10.2)

Excess work in progress cost — Annual benefit run rate × excess work in progress ÷ throughput — forgone once and never recovered. (KA 15.3)

Exclusion — A written statement of what is outside the boundary; the highest-return line in the document. (KA 5.1)

Expected annual loss (EAL) — Incident probability × assessed impact; the residual after a control is the basis of comparison. (KA 14.4)

Expected containment cost per introduced defect — (Internal + external failure) ÷ defects introduced; the breakeven price of prevention. (KA 9.2)

Expected cost of disruption — Σ P(state) × consequence(state); compared against the certain cost of the structure that reduces it. (KA 10.4)

Explainability (decision sense) — The property that a named human can state why an output says what it says, in the vocabulary of the obligation. (KA 14.4)

Exposed loss — $(1 - q) \times L$ — the loss a verification is actually protecting against, after any downstream control. (KA 1.4)

Exposure per record — $d_i u_i$ — the quantity that ranks remediation effort across classes. (KA 14.1)

External-failure unit cost — The average cost of one escaped defect in the operating context; set by context, not the project. (KA 9.1)

F

Factual-claims register — Each claim in an external communication with its source document, retained with the issued item. (KA 11.4)

False stop / false continue — Stopping a project that would have created value; continuing one that will not. The two errors a criterion trades. (KA 2.4)

Feasible frontier — The top few feasible sets and what distinguishes them — the honest output of an enumeration. (KA 15.2)

Feedback and delay — Systems properties that make interventions look wrong before they look right. (KA 1.3)

FFP / FPIF / CPFF / CPIF / T&M — Contract models ordered by who carries cost risk. (KA 7.4)

Final account — The single agreed statement of the original sum, variations, settlements, deductions and final payment, signed as full and final. (KA 16.2)

First-instance rule — Hold the conversation at the first clear instance, when it is about an event rather than a person. (KA 12.3)

First-time-right yield (y_i) — The share of items passing a step without rework. (KA 9.4)

Fitness for purpose — Whether conforming, correctly graded output serves the use it was built for. (KA 9.1)

Flat-equivalent basis — Breakeven stated as the level a profile flat from year one would need (Meridian 41.05 %). (KA 2.2)

Flip point — The smallest weight shift that reverses a ranking; report it with the ranking or the ranking is unqualified. (KA 2.2)

Flow efficiency — Active work time as a share of cycle time; typically low, and where the improvement lies. (KA 13.2)

Flow metric set — Throughput, cycle-time distribution, work in progress with its blocked and aged shares, flow efficiency. (KA 13.4)

Forward / backward pass — Left-to-right ES/EF computation; right-to-left LS/LF computation. (KA 6.2)

Forward breakeven — The adoption or benefit level at which the remaining cost is just repaid; the only defensible level for a gate criterion. (KA 2.4)

Forward exit cost — The cost of stopping — termination, demobilisation, disposal; a forward cost, never sunk. (KA 2.4)

Forward traceability — Requirement → design → test → accepted deliverable; detects unmet scope. (KA 5.2)

Forward-looking NPV — Remaining benefit PV less remaining cost; the only basis for continuation. (KA 2.4)

Free float FF — Slip available without delaying any successor; `FF ≤ TF`. (KA 6.2)

FS / SS / FF / SF — The four dependency types. (KA 6.1)

Funding absorbed — Cash paid out − cash received; equals client exposure − payables + loss to date. Best stated in days of spend (44.80 on Auriga). (KA 7.4)

G

Governance — The decision rights, accountabilities and information flows through which an organisation directs and controls a project. (KA 3.1)

Governance artefact — A report, portal or template — evidence of governance, never a substitute for decision rights. (KA 3.1)

Governance calendar — The meeting and paper-deadline dates an approval event must be scheduled against, rather than approximated by a lag. (KA 6.1)

Governance latency — The expected wait for a committee decision: `E[wait] = M/2 + L`. (KA 3.2)

Grade — The level of capability or specification the requirements ask for; independent of quality. (KA 9.1)

Greedy ranking — Taking candidates in ratio order while they fit; a heuristic that strands capacity when candidates are lumpy. (KA 5.3)

Gross latency-weeks — Σ over decision classes of count × cumulative tier latency; the portfolio's total waiting. (KA 15.4)

Gross residual — The sum of the reconciliation's error classes before netting — the figure that carries the finding. (KA 4.3)

Grounding — Supplying and identifying the material an output must be derived from, with nothing outside it permitted. (KA 14.3)

H

Hallucination — Fluent, confidently-stated content that is false. (KA 1.4)

Handover — Transfer of custody and operational accountability for a deliverable to a named receiving organisation, on stated conditions. (KA 16.1)

Handover artefact — What was done, what is in flight, what is blocked, what the next party must decide. (KA 12.4)

Hard / soft constraint — Physics, law, contract vs preference and convention; only soft ones are tradeable. (KA 2.1)

Honest attribution (AI) — Stating what was AI-generated and what was verified, and how. (KA 12.4)

Honest gate — Real stop option, pre-set evidence standard, independent deciders, recorded reasons. (KA 2.4)

Honesty asymmetry — Bad news costs the messenger and helps the project; leaders must make it safe. (KA 1.2)

Hundred-per-cent rule — The children of any element sum to exactly that element — an arithmetic invariant, and checkable. (KA 4.2)

Hybrid boundary — The named line where feedback stops being cheap and fast, and the control regime changes. (KA 13.3)

Hybrid delivery — A deliberate combination of predictive and adaptive methods under one governance frame, with a named boundary between them. (KA 13.1)

Hypercare — A bounded period of elevated support after go-live, with measured exit criteria and a receiving-organisation owner. (KA 16.1)

I

Implied ETC efficiency — The efficiency an `EAC` assumes for the remaining work; equal to the `TCPI` computed to that `EAC`. (KA 7.3)

Increment — A slice of product complete against the definition of done and therefore judgeable, releasable or rejectable alone. (KA 13.1)

Influence — Capacity to affect the project's decisions, resources or permissions; predicts the damage a disagreement does. (KA 11.1)

Influence test — A risk should be allocated to the party that can influence it; a risk allocated elsewhere is priced, not managed. (KA 10.3)

Information age — `R/2 + G` — the expected staleness of a fact at publication, for reporting period `R` and production lag `G`; the registered `E[wait]` identity applied to information. (KA 14.2)

Integrated change control — Assessing and authorising a change across scope, schedule and cost together. (KA 4.4)

Integrating point (hub) — The function through which routed communication passes; carries its own overhead and its own distortion risk. (KA 11.2)

Integration layer — An architecture or interface contract that replaces `n(n−1)/2` pairwise agreements with `n`. (KA 13.3)

Integration milestone — The boundary event where an adaptive stream's ranged forecast enters a predictive network. (KA 13.3)
· Also read at KA 6.4: The CPM node where an agile stream's delivery enters the network.

Interaction term — The product of both factor movements; the amount that shifts between decomposition lines when the order changes. (KA 16.4)

Interest — Attention paid, not stake held; predicts how quickly a party notices. (KA 11.1)

Interface — A defined relationship across a boundary at which something must be agreed and verified. (KA 4.2)

Interface agreement — The parties, content, format, direction, timing, exceptions, verifier and version of one interface. (KA 4.2)

Interruption load — Observed unplanned escalation and interruption time; the largest and most designable line in the attention audit. (KA 12.1)

Irrecoverable cost — The share of a formal dispute's cost that cannot be recovered whatever the outcome; the figure that usually decides whether fighting can pay. (KA 10.4)

Irreversibility — The rising cost of correction as commitments harden — projects' defining failure mode. (KA 1.1)

Issue — A risk that has occurred; managed, not analysed. (KA 8.1)

J

Joint gain — Σ(value to one party) − Σ(cost to the other) over the issues traded; negative for issues that destroy value. (KA 11.3)

K

Key-person concentration — The share of procedures executable by exactly one named individual; an availability risk no project register records. (KA 16.3)

Kill criteria — Pre-agreed stop conditions; powerful only because agreed in advance, and only useful if calibrated. (KA 2.4)

Knowledge transfer — Conversion of project knowledge into documentation, trained people and rehearsed procedures usable without the project team. (KA 16.3)

L

Lag / lead — Imposed delay / overlap on a dependency. (KA 6.1)

Last responsible moment — The latest point a decision can be taken without foreclosing a needed option. (KA 1.3)

Latency-to-cycle ratio — Governance latency ÷ the delivery cadence it governs; above 1.0 the answer arrives after the increment that asked. (KA 3.1)

Leadership (delivery context) — The production of willing, aligned and informed effort under conditions the plan did not anticipate. (KA 12.1)

Least-cost duration — The duration maximising net saving; found where the marginal week's cost first exceeds the value of a week. (KA 6.4)

Legal liability — Who must compensate whom, set by contract and jurisdiction; distinct from professional accountability and capable of sitting elsewhere. (KA 1.2)

Legitimately product-free work — Integration, testing, migration, training delivery, commissioning: real work with no product, and the class a product-derived WBS omits by construction. (KA 4.2)

Level of effort — Earning by calendar (`EV ≡ PV`); can never show schedule variance. (KA 7.3)

Level variance — The value effect of a wrong steady-state adoption or benefit-per-unit assumption. (KA 16.4)

Level-of-effort distortion — `s × (1 − SPI_d)` — the amount a level-of-effort share `s` adds to a reported `SPI`; 0.28 at a 70 % share with discrete `SPI` 0.60. (KA 7.3)

Lineage — The reconstructable path from source record to a displayed value. (KA 14.1)

Link cost c — Hours of total team time consumed per active pairwise relationship per week; a locally calibrated planning figure. (KA 12.2)

Little's Law — `W = T × C` for a system in steady state, independent of arrival and service distributions. (KA 13.2)

Local optimisation — Function-level efficiency that degrades whole-system performance. (KA 1.3)

Logic density — Links ÷ activities; the surplus over `n − 1` counts the network's convergences. (KA 6.1)

Logrolling (trading whole issues) — Conceding entirely on issues valued asymmetrically, rather than partially on all of them. (KA 11.3)

Long-lead item — A component whose procurement chain exceeds the time available after baseline approval, and must therefore be committed earlier. (KA 10.1)

M

Management (delivery context) — The allocation and control of resources against a plan. (KA 12.1)

Management reserve — Outside the baseline; unknown-unknowns; sponsor-controlled. (KA 7.1)

Marginal capacity of the nth member — `h − c(n − 1)`; zero at `n` = 81 under the same parameters. (KA 12.2)

Marginal value of the constraint — `V(n) − V(n−1)` from enumeration; lumpy, non-monotone, and a property of the candidate set. (KA 2.2)

Marginal-dollar allocation identity — For any payment mechanism, buyer share + supplier share of the next dollar of cost = 1; a contract allocates cost, it does not remove it. (KA 10.3)

Material error rate (p) — The share of items of a class containing a material error before verification; measured per class and per configuration. (KA 14.3)

Maximum blindness — `P + C + L` — how long a problem arising just after a cut-off stays invisible to the decision body. (KA 11.2)

Maximum sustainable span — `n ≤ (R + eb)/(T + e)`; 7.6 on Auriga, so 7 reports. (KA 12.3)

Meeting interval (M) — The period between meetings of a decision body. (KA 3.2)

Member vs attendee — Members decide; attendees inform. Conflating them enlarges the body and reduces its authority. (KA 3.2)

Mesh vs layered — `n(n−1)/2` possible interfaces against `n` — the architecture choice priced in WE 4.2.3. (KA 4.2)

Milestone — Zero-duration event with an owner and a done-definition; its date is computed. (KA 6.3)

Model retention record — Model identifier, version and date alongside input, output, verifier and decision — the minimum that makes an AI-informed decision explainable. (KA 16.4)

Monte Carlo simulation — Repeated sampling to produce an outcome distribution rather than a point. (KA 8.2)

Multiplication rule — The probability a milestone requiring all of `k` independent predecessors is met is the product of their probabilities. (KA 15.1)

Must-have inflation — Originator-assigned categories collapsing into near-universal "must have", destroying the scheme's information content. (KA 5.3)

N

Named owner — The human who verified an AI-assisted artefact and answers for it. (KA 1.4)

Near-critical band — The float threshold below which a path joins the monitored set; a decision, not a convention. (KA 6.2)

Need-by time — The point beyond which a decision no longer changes what the team does; the test of whether a body can serve it. (KA 3.1)

Net drain — Gross throughput minus arrival rate — the rate at which a backlog actually shrinks. (KA 13.2)

Net team capacity — `hn − c·n(n − 1)/2`; maximised at 80–81 people under the illustrative parameters. (KA 12.2)

Nonconformance — Output that does not meet its specified requirement. (KA 9.3)

Notice provision — The required form and period for notifying a claim; in some contracts and jurisdictions a condition precedent. (KA 10.4)

Notice-bar exposure — The value of the heads of claim that depend on valid notice, grossed up for overhead and profit. (KA 10.4)

O

Omission credit — The rate at which removed scope is credited back — typically below the bid rate, and the asymmetry that prices a net-zero change. (KA 13.4)

Operational transition — Establishment of a funded, staffed, permanent capability to run and improve the delivered service. (KA 16.2)

Original-to-current reconciliation — Original baseline + Σ approved changes = current baseline, exactly, every reporting period. (KA 4.3)

Orphan — A design element or test case tracing to no approved requirement. (KA 5.2)

Orphan work package — Budgeted work producing no product anyone asked for — waste that still reports 100 % complete. (KA 4.2)

Out-of-cycle mechanism — A written-resolution or delegated-authority route used when the ordinary cadence is too slow. (KA 3.3)

Out-of-cycle route — Domain 3's mechanism (KA 3.3.3) applied at enterprise scale, where it is worth most: the saving is largest at the slowest tier, because `E[wait] = M/2 + L` grows with `M`. (KA 15.4)
· Also read at KA 10.3: Domain 3's mechanism (KA 3.3.3) applied across a contract boundary: named authorities on **both** sides, since a route that exists for only one party is not a route.
· Also read at KA 13.3: Domain 3's mechanism (KA 3.3.3) applied where governance latency exceeds the iteration length — the specific mismatch KA 13.3 quantifies.

Outcome / deliverable / activity registers — The three levels routinely conflated; scope is the deliverable register, bounded by outcomes and decomposed into activities. (KA 5.1)

Output / outcome / benefit / value — Delivered thing · change in how things work · measurable improvement · benefit against cost and risk. (KA 1.3)

Output vs input specification — Specifying what must be achieved against how it must be built; determines who is answerable for fitness for purpose. (KA 10.1)

Outturn ratio — Final cost ÷ approved cost for a comparable completed project; the unit of a reference class. (KA 8.4)

Over-claim share — `counterfactual ÷ raw` — invariant to the valuation rate, the operating year and the number of adopters. (KA 2.3)

Over-trust through fluency — Polished output suppressing the scrutiny rough output would attract. (KA 1.4)

Overlap window — A deliberately created shared period, bought with unsocial hours that must be explicitly allocated. (KA 12.4)

P

P50 / P80 — Confidence levels — amounts covering the aggregate outcome with stated probability. (KA 8.2)

Paid-to-productive multiplier — `1/(1 − n)` — 1.17647 at a 15 % non-productive share; the commonest silent under-funding in a labour estimate. (KA 7.4)

Pairwise configuration integrity — `p^k` — the conformance of a relationship spanning `k` controlled items, always worse than the item rate `p`. (KA 4.3)

Paper lead time (L) — How far before a meeting submissions close. (KA 3.2)

Partial horizon — One element planned to work-package level ahead of the wave because its lead time demands it, labelled as such. (KA 4.1)

Party readiness (p) — The probability that a party arrives at a cycle able to approve; countable as the share of items approved at first presentation. (KA 3.1)

Path float — `PD` − path length; the slip a whole chain can absorb, shared by every activity on it. (KA 6.2)

Path migration — The critical path moving as durations change; why passes are re-run. (KA 6.4)

People-work capacity — Discretionary attention less the personal planning reserve: 6.5 h/week on Auriga. (KA 12.1)

Performance measurement baseline — The authorised scope, arranged in time, with cost attached — one statement in three documents. (KA 4.3)

Plan of plans — The integrated set of subsidiary plans and the baselines they produce. (KA 4.1)

Plan survival probability — The product over periods of the probability that each period's loss does not exceed its slack. (KA 15.3)

Plan-set consistency audit — Reading the subsidiary plans against one another and pricing the findings, before baseline approval (WE 4.1.2). (KA 4.1)

Planning package — A far-horizon element planned above work-package level, with its uncertainty stated. (KA 4.1)
· Also read at KA 6.3: A far-wave summary activity with ranged duration awaiting elaboration.

Pod (structured team) — A sub-team with end-to-end ownership and one named path outward — the human form of Domain 4's layered architecture. (KA 12.2)

Point of total assumption (PTA) — The cost above which the supplier bears 100 % of further overrun (Domain 7, KA 7.4.3); read here as the point at which behaviour turns from efficiency to entitlement. (KA 10.3)
· Also read at KA 7.4: The cost above which the seller bears 100 % of further overrun.

Portfolio — The set of programmes and projects an organisation chooses to run within a finite capacity and a strategy. (KA 15.1)
· Also read at KA 1.1: The funded, balanced set of projects and programmes.

Portfolio benefits register — The authoritative statement of portfolio benefits, organised by benefit rather than by component. (KA 15.2)

Portfolio one-page view — The report supporting exactly three decisions: continue/change/stop, reallocate, escalate. (KA 15.4)

Portfolio process — Comparable description, explicit criteria, funding to capacity, periodic re-decision. (KA 2.1)

Portfolio work in progress — The number of initiatives in flight; by Little's Law it sets cycle time, not throughput. (KA 15.3)

Post-project review — Structured examination of what happened and why, to improve future decisions, separated from performance assessment. (KA 16.3)

Pre-decision consultation — Consultation capable of changing the design — engagement; as distinct from post-decision notification. (KA 11.4)

Pre-mortem — Imagining failure before committing, to license dissent and surface missed risk. (KA 8.4)

Precedence rule — The pre-agreed rule for which authority prevails in a matrix conflict. (KA 3.1)

Precision — True alerts ÷ all alerts; worth investigating while precision exceeds investigation cost ÷ saving. (KA 8.4)

Price normalisation — The formula converting prices into scores; the ratio and linear conventions give different scores and different crossovers. (KA 10.2)

Price of confidence — The cost per percentage point of moving a reserve between confidence levels. (KA 8.2)

Priority rule — The heuristic (commonly minimum total float first) used to break resource conflicts; beatable, and known to be. (KA 6.3)

Procurement — Converting a need the project cannot meet internally into an enforceable obligation held by another organisation, and managing it to closure. (KA 10.1)

Procurement lead time — Process legs + governance latency + physical legs; a schedule predecessor, not an administrative overhead. (KA 10.1)

Product breakdown structure — Decomposition of the thing delivered; drives configuration management and acceptance. (KA 4.2)

Product owner — The single individual accountable for the order of the backlog within a value envelope set by the sponsor. (KA 13.1)

Product-to-work cross-match — The traceability test that every product has work and every work package a product — a candidate list, not a finding. (KA 4.2)

Production lag (G) — Elapsed time from data cut-off to publication; the more powerful of the two staleness levers, day for day. (KA 14.2)

Programme — A temporary organisation delivering a coherent outcome through components that cannot deliver it individually. (KA 15.1)
· Also read at KA 1.1: Related projects managed together for outcomes no one project delivers.

Programme dependency register — The record of cross-boundary dependencies, whose critical column is the owner on the giving side. (KA 15.1)

Programme verdict — The programme-level judgment: did the outcome and its benefit materialise? An annual flow. (KA 1.1)

Progressive elaboration — Adding detail as it becomes knowable, at a stated horizon and cadence. (KA 4.1)

Project — Temporary endeavour producing a defined result. (KA 1.1)

Prolongation — Time-related costs of extended duration; requires a demonstrated critical-path effect. (KA 10.4)

Proportional shortfall — The ratio p'/p by which lifting a condition multiplies the conjunction — the correct remediation ranking. (KA 16.1)

Protective capacity — Capacity deliberately withheld from allocation so that a period's plan can absorb variance. (KA 15.3)

Provenance citation — Each material assertion in an output referenced to the supplied source location it came from. (KA 14.3)

Psychological safety — A shared belief that the team is safe for interpersonal risk-taking — candour without penalty. (KA 12.2)

PTA headroom — The distance from the seller's current cost forecast to the `PTA`; 5.59 % on Auriga, and the figure to trend monthly. (KA 7.4)

PV EV AC — Budgeted cost of work scheduled · performed · the cost actually incurred. (KA 7.3)

Q

Qualified alternate — A second supplier approved and kept exercisable, receiving no volume; the structure that actually buys resilience. (KA 10.4)

Quality — The degree to which delivered work conforms to its stated requirements. (KA 9.1)

Quality assurance — Process-directed, preventive activity: is the way we work capable of producing conforming output? (KA 9.2)

Quality control — Product-directed, detective activity: does this output conform? (KA 9.2)

Quality premium test — Comparing a higher-quality bid's price premium with the expected cost it demonstrably avoids. (KA 10.2)

Quality-adjusted breakeven volume — $F / [(m + e_m u) - (a + e_a u)]$ — the volume at which an automation pays once escaped errors are priced. (KA 14.2)

Quantification threshold — The impact above which a screened risk is quantified individually; a screen without one has replaced quantification. (KA 8.2)

Quantum — The claim's money, built up by head of cost and evidenced contemporaneously. (KA 10.4)

Quoted direct cost — The figure on the form — typically a minority of the true cost, and the wrong basis for a threshold. (KA 4.4)

R

Ramp-up loss — Weeks of a replacement's tenure minus the sum of their weekly effectiveness, valued at a person-week. (KA 12.2)

Ramped basis — Breakeven stated as the steady state a proportionally ramped profile must reach (Meridian 45.01 %). (KA 2.2)

Range forecast — Backlog ÷ the observed high and low sustained net drains, with its assumptions stated. (KA 13.2)

Range of optimality — The band of an uncertain input over which the recommended option stays recommended. (KA 9.1)

Re-baselining — Domain 4's instrument (KA 4.3.3) seen from earned value: it resets the `PV` curve every index is measured against, which is exactly why it cannot be used to retire an adverse variance. (KA 7.2) · Also read at KA 4.3: Replacing a baseline that no longer conveys information — legitimate, and the most abused instrument in project control.

Re-decision tax — The committee slots and delay cost consumed by questions that had already been decided. (KA 3.3)

Readiness conjunction — `P(clean transition) = ∏ pᵢ` over the `k` necessary conditions; always at or below the averaged dashboard figure. (KA 16.1)

Reallocation distance — The money that must move to close the gaps: `(1 − index) × denominator`; deficits equal surpluses. (KA 2.1)

Recall (by group) — Flagged at-risk cases ÷ actual at-risk cases within a group. (KA 14.4)

Recovery plan — Target date, ranked moves, costs, risks, decision authority. (KA 6.4)

Recurrence rate — Repeat nonconformances as a share of all; the test of whether causes are being removed. (KA 9.3) · Also read at KA 9.4: Repeat nonconformances as a share of all; the honest measure of an improvement loop.

Recurring versus one-off value — The distinction that decides whether a support function survives its second year. (KA 15.3)

Reference-class forecasting — Estimating from comparable completed projects to counter optimism bias. (KA 8.4)

Refusal permission — An explicit instruction that "the source does not say" is an acceptable and expected answer. (KA 14.3)

Registered measure — The single definition of a benefit measure fixed before measurement; variants are identified, not substituted. (KA 16.4)

Regrade — Use of nonconforming output for a lesser purpose for which it conforms, with traceability. (KA 9.3)

Reinforcement trough — The transient period after adding people in which progress is worse than it would have been; exists when supervision load per newcomer exceeds their ramp productivity. (KA 1.3)

Rejection entry — The log record of a change not approved — the entry most often missing and the reason requests recur. (KA 4.4)

Relatedness class — The set over which a cumulative test sums; widening it multiplies false trips and re-centralises the delegation. (KA 3.3)

Releasability — The property that an increment can go live alone; it is what converts sequencing into value. (KA 13.1)

Released hours — Capacity freed, not cash; a benefit only once redeployed to a stated use. (KA 1.3)

Released leader hours — Status-quo leader hours less retained oversight; the actual product a delegation buys. (KA 12.3)

Reperformance — Recomputing a result from its inputs by an independent route; the only effective control against plausible error. (KA 14.3)

Reporting period (P) — The interval of activity a report covers. (KA 11.2)

Required draw efficiency — Remaining contingency ÷ (demonstrated draw rate × remaining progress) — the `TCPI` of the reserve. (KA 7.1)

Required per-dependency reliability — The uniform probability each of `k` dependencies must carry to reach a target milestone confidence: the `k`-th root of the target. (KA 15.1)

Required rate — Backlog ÷ target weeks, plus arrivals — the throughput a committed date presupposes. (KA 13.2)

Required throughput — Backlog ÷ (successor's `LS` − stream start): what the network asks a cadence team to sustain. (KA 6.4)

Requirement bundle — The smallest set of requirements that together delivers a measurable benefit — the unit of value attribution. (KA 5.3)

Requirement classes — Functional · non-functional · constraint · data · transition · regulatory — reported separately because they fail differently. (KA 5.2)

Requirement inflation — Elaborating need into a count that impresses and disperses meaning, defeating traceability and prioritisation. (KA 5.2)

Requirement-count reconciliation — Baseline count + approved additions = traced count; the only cheap control that detects creep. (KA 5.4)

Reservation value — The worst terms a party would rationally accept, derived from its priced alternative; a maximum for a buyer, a minimum for a seller. (KA 11.3)

Reserved matter — A decision class that requires the highest (often unanimous) authority; kept deliberately short. (KA 3.1)

Residual exposure — $\Sigma\, p_i\, \prod(1 - q_{ij})\, u_i$ across an assurance map; the only quantity that ranks its rows. (KA 3.3)

Resilience — Capability to absorb unidentified risk; buffers, optionality, modularity, redundancy, fast detection. (KA 8.4)

Resolution route — The chosen mode, its cost, and the escalation path with its latency if the mode fails. (KA 11.3)

Resource histogram — Demand per period from the loaded schedule. (KA 6.3)

Resource-feasible schedule — A schedule in which no period's demand exceeds any resource cap. (KA 6.3)

Responsibility — The obligation to do; shareable and delegable. (KA 1.2)

Responsible-AI principle — AI proposes; the professional verifies, decides and remains accountable. (KA 1.1)

Restated criterion — A criterion that repeats the requirement instead of stating a measurable condition — worse than an absent one, and commoner. (KA 5.4)

Retention — A withheld percentage of payment, released on completion criteria; Auriga's 5 % is 55.00 % of the bid margin at completion. (KA 7.4) · Also read at KA 16.2: A proportion of the contract sum withheld as security, released against defined events.

Retention schedule — Class-by-class statement of what is kept, for how long, on what authority, and what disposal or de-identification follows. (KA 16.4)

Retrieval rate — The proportion of captured lessons retrieved before a comparable decision; the term in which almost all lesson value is lost. (KA 16.3)

Reversal-cost ratio (ρ) — Cost to undo a decision ÷ its value; authority reads on `max(value, cost to undo)`. (KA 3.2)

Reversion plan — The rehearsed route back, with a trigger, a decision-maker, a time limit and a latest-possible-reversion point. (KA 16.1)

Rework allowance — The capacity a plan reserves for rework; a forecast, and a decision. (KA 9.2)

Rework share of capacity (r) — The fraction of a team's available effort consumed correcting output. (KA 9.2)

Risk / opportunity — Uncertain event affecting an objective, adversely / favourably. (KA 8.1)

Risk appetite / threshold — Accepted exposure, expressed operably enough to apply consistently. (KA 8.1)

Risk loading — The premium a risk-averse supplier adds above the risk-neutral price; the arguable, visible price of certainty. (KA 10.3)

Risk retirement — Closing a risk whose window has passed; frees contingency for return — often far more than the risk's `EMV`. (KA 8.3)

Risk-neutral fixed price — Expected cost plus the required margin; at this price, transferring risk costs the buyer nothing in expectation and buys variance reduction only. (KA 10.3)

Risk-tiered diligence — Concentrating supply-chain assurance effort where breach probability is highest, rather than screening uniformly. (KA 10.4)

Rolled throughput yield (RTY) — $\prod\, y_i$ — the share of items passing every sequential step first time. (KA 9.4)

Rolling wave — Near-term detail, far-term packages, elaborated under change control. (KA 6.3)

Route to market — The competitive structure chosen for a procurement; each route buys a different amount of price discovery at a different cost in time. (KA 10.2)

Runbook adequacy test — Whether someone who was not on the project can execute it under supervision — the only proof documentation works. (KA 16.3)

S

Safety–standards pairing — Safety governs whether people can speak; standards govern what they are held to. Both are required. (KA 12.2)

Salience — Influence × interest — a useful first cut and an inadequate allocation rule. (KA 11.1)

Same-pool over-claim — Aggregate claims on a resource pool exceeding the pool; capped at the pool and allocated. (KA 15.2)

Scenario analysis — Complete network re-runs under coherent stated assumptions. (KA 6.4)

Schedule hierarchy (L1–L4) — Nested plans for different audiences, derived from one logic. (KA 6.1)

Scope baseline — The approved scope statement + WBS + WBS dictionary — one of Domain 4's three baseline dimensions. (KA 5.1)

Scope creep — Movement of delivered content with no baseline movement and no record; invisible to every change-log instrument. (KA 5.4)

Scope statement — The authorised description of what will and will not be delivered, specific enough to be tested against. (KA 5.1)

Scope-based contract — A firm price for a defined body of work; the supplier bears effort risk, the buyer pays for change. (KA 13.4)

Screening ruling — The recorded classification: the baseline clause relied on, the outcome, and a named technical authority — the only artefact that makes a classification contestable later. (KA 4.4)

Secondary risk — Risk created by a response; a register without them is incomplete. (KA 8.3)

Self-consistent sampling plan — One whose confidence bound lies below its own breakeven defective fraction. (KA 9.3)

Service acceptance — The receiving organisation's formal agreement that the service meets specification. (KA 16.1)

Share ratio — The agreed buyer/seller split of over- and under-run. (KA 7.4)

Shared benefit — One benefit claimed by two components that both genuinely contribute to it; counted once. (KA 15.2)

Silent measure — An outcome measure reported by two parties, owned by neither, and never accompanied by a proposed action. (KA 1.2)

Simple payback — Cost ÷ annual benefit, in periods; scales as the reciprocal of adoption. (KA 1.3)

Single-A test — The check that each decision class has exactly one Accountable role. (KA 3.3)

Situational diagnosis — Deciding what direction and support this person needs on this task before choosing a behaviour. (KA 12.1)

Smoothing / levelling — Flattening demand within float / accepting a later date under a hard cap. (KA 6.3)

Solution capture — Recording a stakeholder's proposed solution as the requirement, locking the design before the problem is understood. (KA 5.2)

Source of record — The one system whose value for a fact prevails; all other holdings are copies. (KA 14.1)

Source-dated figure — A reported value carrying the date of its own source rather than the date of the pack. (KA 15.4)

Sponsor — The individual accountable for the project's business outcome and mandate. (KA 3.2)

Sponsor load — The annual diary hours the sponsor obligations imply; a portfolio allocation constraint, not a personal one. (KA 3.2)

Sponsor turnaround — The measured interval from a decision reaching the sponsor to its return; read as a distribution, not a mean. (KA 3.2)

Stage gate — A continuation decision by a named authority against pre-set criteria, with power to stop, hold or redirect. (KA 3.3)

Stakeholder — A party whose decisions, consent, resources, behaviour or objection can materially affect the project's outputs, outcomes or legitimacy, or who is materially affected by them. (KA 11.1)

Stakeholder system — The stakeholder set together with the relationships between its members — who influences whom, and which pairs align under pressure. (KA 11.1)

Standard of care — The care a competent practitioner would exercise in the circumstances. (KA 1.2)

Status accounting — The record of the current version of every controlled item. (KA 4.3)

Steering committee — The body that makes decisions exceeding the project leader's authority, in time for them to matter. (KA 3.2)

Stranded capacity — Unused constrained resource that no remaining candidate is small enough to use — the signature of greedy failure. (KA 5.3)

Straw options — An options set built so the preferred option cannot lose. (KA 2.2)

Sub-additivity of controls — Combined avoided loss is less than the sum of individual avoided losses, because reductions multiply. (KA 14.4)

Sub-tier concentration — A shared dependency below the first tier that correlates suppliers' failures and destroys a dual-sourcing case. (KA 10.4)

Sufficiency convention T — The minimum weekly coaching time an organisation states as meaningful; 45 minutes on Auriga, a convention not a measurement. (KA 12.3)

Sum test — Attributed benefit across bundles must not exceed the business case, and any shortfall is explained, not distributed. (KA 5.3)

Sunk cost — Unrecoverable spend carrying no information beyond the forecast; irrelevant to a continuation decision. (KA 2.4)

Supersession hazard — The annual probability that a stated strategic priority is replaced, merged or de-prioritised. (KA 2.1)

Surplus — Own reservation value less the price agreed (buyer), or price less own reservation value (seller). (KA 11.3)

Synthetic content risk — Convincing reproduction of a programme's identity, spokespeople, voice or documents at negligible cost. (KA 11.4)

Systematic error — An error repeating consistently across similar items — the characteristic shape of AI error, requiring stratified sampling. (KA 9.4)

T

T&M/fixed-price breakeven — `fixed price ÷ hourly rate`; the premium as a percentage of the estimate equals the hours overrun at which the two options cross. (KA 7.4)

Tailoring — A recorded, owned decision to adapt method — never a silent omission. (KA 4.1)

Tailoring (of a message) — Varying depth and vocabulary by audience while holding the facts identical. **Context flag:** distinct from tailoring of method (Domain 4, KA 4.1.3), which is a recorded decision to adapt process; the shared word carries two different concepts and neither reading substitutes for the other. (KA 11.2)

Target fee vs expected fee — The fee at the target cost, against the probability-weighted fee actually expected; any real overrun risk makes the second lower than the first. (KA 10.3)

Target operating state — The stated description of the organisation once the change has landed; the test of component coherence. (KA 15.1)

TCPI — Efficiency the remaining work must achieve to hit a stated target. (KA 7.3)

Temporary organisation — A team with no accumulated trust or shared norms; both must be designed. (KA 1.1)

Test-order rule — Rank assumptions by `EMV ÷ cost to test`, so the free tests are cleared before the paper is written. (KA 2.3)

Three lines of assurance — Management controls; independent oversight inside management; independent audit outside it. (KA 3.3)

Three-point / PERT estimate — $t_e = (o+4m+p)/6$, $\sigma = (p-o)/6$. (KA 6.4)

Threshold mismatch — Client and supplier change-approval thresholds set independently, so changes between them arrive committed on one side and unapproved on the other. (KA 10.1)

Throughput (T) — Completed items per unit of time — a measurement, never an aspiration. (KA 13.2)

Tier traversal — One decision passing one tier; the unit on which a paper-lead-time saving accrues. (KA 15.4)

Time-phased cost baseline — The `PV` curve; it must move whenever the schedule does. (KA 4.3)

Timing value of an escalation — Weeks of earliness × the run rate the remedy recovers — what the timing alone is worth, separate from the remedy. (KA 1.2)

Timing variance — The value effect of the benefit profile arriving later than the case assumed. (KA 16.4)

Tolerance — The amount by which an objective may be exceeded before appetite is breached; the room inside which contingency sits. (KA 8.3)
· Also read at KA 9.1: The range within which output is conforming; a specification without one is untestable.

Total cost of ownership — Whole-life cost of the service, including the operating cost that must appear in the receiving organisation's budget before closeout. (KA 16.2)

Total cost of ownership (TCO) — Transition-in + operating cost over the period of need + transition-out; the only valid basis for make-or-buy. (KA 10.1)

Total float TF — `LS − ES`; slip available without delaying the project. (KA 6.2)

Total funding requirement — `BAC` + management reserve — the cash the sponsor must have available; Auriga's is USD 4,240,000. (KA 7.1)

Tranche — A group of components delivered together to reach a usable intermediate state at which benefit begins. (KA 15.1)

Transition-in / transition-out — The cost of entering and of leaving an arrangement; the exit column is the one most often scored at zero. (KA 10.1)

Transition-out — Dissolution of the project organisation: authority withdrawn, accounts closed, access revoked, records placed, people released well. (KA 16.3)

Turnover cost — Recruitment + ramp-up + lost productivity, excluding the saved pay of a vacant post. (KA 12.2)

Twin fidelity — The share of test questions on which the twin's answer matches the asset within a stated tolerance. (KA 14.2)

Two-envelope evaluation — Scoring quality before price is revealed, to remove price anchoring from quality judgement. (KA 10.2)

Two-person rule — A named drafter and a named verifier for every external communication. (KA 11.4)

U

Uncertainty — Conditions that cannot be meaningfully enumerated with probabilities. (KA 8.1)

Uncommitted forecast balance — `EAC` − `AC` − open commitments; its excess over the baseline's uncommitted balance equals \ (KA 7.2)

Uniform yield requirement — `RTY^(1/k)` — the per-step yield an end-to-end target implies over `k` steps. (KA 9.4)

Uniform-target defect — Setting one rate target across classes of unequal consequence, which can cut defect counts while raising expected cost. (KA 14.1)

Unsocial-hours burden — The hours per group per week spent outside normal local working time; equalised by rotation. (KA 12.4)

V

VAC — `BAC - EAC` — forecast variance at completion. (KA 7.3)

Validation — Does what we built produce the outcome we needed? A project can pass the first completely and fail the second completely. (KA 5.4)

Value envelope — The bounded authority within which a product owner may commit capacity without escalation. (KA 13.1)

Value of information — The reduction in expected cost that knowing enables; zero if it changes no action. (KA 8.2)

Value of perfect information — The most any signal could be worth: expected cost without information less expected cost knowing the truth free. (KA 8.2)

Value per unit of constrained effort — Attributed benefit ÷ effort in the scarce resource — the ranking that engages the constraint. (KA 5.3)

Value per unit of constraint — Ranking basis when a scarce resource binds; a heuristic, not an optimum. (KA 2.2)

Value-led core — Engagement effort sized from the benefit arithmetic rather than from the salience score. (KA 11.1)

Variance contribution — $p(1-p)I^2$ for one risk; quadratic in impact, so it ranks a register differently from `EMV`. (KA 8.2)

Velocity — Throughput weighted by team-relative size units — a planning input inside a team, never a performance measure. (KA 13.4)

Verification — Did we build what we specified? (KA 5.4)

Verification proportionality — Depth of checking matched to the stakes of reliance. (KA 1.4)

Verification standard proportional to consequence — A rule setting review depth from the cost of an escaped error, not from convenience. (KA 9.4)

Verification tier — A defined review depth with a cost per item `v` and a measured detection rate `q`, performed by a named human. (KA 14.3)

Verification tier threshold — $u^*_k = \Delta v_k/(p \cdot \Delta q_k)$ — the escaped-error consequence at which the next tier begins to pay. (KA 14.3)

W

WBS — Deliverable-oriented hierarchical decomposition of the total scope of work. (KA 4.2)

WBS dictionary — Per work package: work, deliverable, acceptance basis, owner, estimate, dependencies — where scope actually lives. (KA 5.1)

Weighted scoring — Ranking by Σ(weight × ordinal score); forces criteria into the open, steerable by whoever sets weights. (KA 2.2)

Within-cell span — The `EMV` range a single matrix cell contains: the product of its two band ratios. (KA 8.2)

Within-group variation — The reason national-culture averages cannot predict an individual's behaviour. (KA 12.4)

Work in progress (W) — Items started and not finished; a managed quantity with an explicit limit. (KA 13.2)

Work package — The lowest WBS level: one owner, a verifiable outcome, an endorsed estimate, an assessable duration. (KA 4.2)

Work-in-progress limit — An explicit cap on `W`, changed only deliberately; the control the flow arithmetic justifies. (KA 13.2)

Working-hour overlap — `max(0, min(end) - max(start))` with all windows converted to a common reference. (KA 12.4)

Works test / service test — Whether the thing was built as specified / whether the organisation can actually run it. (KA 16.1)

Z

Zero-fee cost — `target cost + target fee ÷ seller share` — where the incentive fee is exhausted; on Auriga USD 2,500,000, above the `PTA`, which is why the ceiling binds first. (KA 7.4)

Zero-overlap pair — Two groups with no common working hours; each exchange costs at least one working day. (KA 12.4)

Zone of possible agreement (ZOPA) — The interval between the two reservation values; empty means no deal at any level of skill. (KA 11.3)

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